# 8 Introduction to the Synthesis of Peptidomimetics

C. Toniolo and M. Goodman

The discovery of a multitude of naturally occurring, bioactive peptides has generated a rich source of pharmacophores from which medicinal chemists are developing new useful therapeutic drugs. After binding to an enzyme, or a membrane receptor, peptide-based inhibitors, neurotransmitters, immunomodulators, and hormones influence cell-to-cell communications and control a variety of vital functions such as metabolism, immune defense, digestion, respiration, sensitivity to pain, reproduction, and behavior.

As a result of intense research, a wealth of protection strategies and coupling methods is available for synthesis in solution and on solid supports. Peptides of up to almost fifty amino acid residues can now be prepared in sufficient quantities for pharmacological and clinical assays, and for clinical drugs and diagnostics. Thus, in the last few years new peptide-based treatments for a series of diseases have been established.[1,2]

Though naturally occurring peptides based on coded amino acids have been widely used as drugs, there are problems with the use of peptides as therapeutic agents. The problems mainly arise from their rapid metabolism by proteolysis, and their interactions at multiple receptors. As a result, peptide researchers have sought modifications of peptide structures to create more stable and bioavailable molecules.[3–27]

Given the extensive biology associated with peptides, it is not surprising that considerable effort has been devoted to the synthesis of *peptidomimetics* to overcome the unfavorable properties and therapeutic deficiencies of peptides. In particular, since the advent of solid-phase synthesis and, more recently, combinatorial chemistry, interest in peptidomimetic design and preparation has exploded. In theory, the peptide chemist is only limited by their imagination in the novel modifications that can be tailored for synthetic peptide analogues.

Peptidomimetics have found wide application as biostable, bioavailable, and often potent surrogates of naturally occurring peptides. They form the basis of important families of enzyme inhibitors and they act as receptor agonists and antagonists. Peptide chemists are also gaining a deeper understanding of the structural features of this class of compounds that influence their ability to permeate biological membranes and their pharmacokinetic properties.

Nanomolar affinity for a protein receptor can be achieved by a peptide having a few, appropriately oriented side chains. This implies that the peptide backbone is not an absolute requirement for this high affinity, thus justifying the search for peptidomimetics. These compounds imitate the topology of a peptide, but may partly, or completely, lack its amide backbone characteristics. It is the rationale for the use of peptidomimetics that the backbone scaffold places the amino acid side chains in a particular 3D-position to allow contact with the enzyme or receptor protein. The 3D-relationship of the side-chain groups in the peptide (topology) determines its binding and biological messages.

However, it was immediately recognized by peptide chemists that, even in the cases where a direct (backbone)peptide$\cdots$protein(backbone) interaction is not operative, the backbone conformation may dramatically influence the biological response. It is evident that the introduction of new, promising peptidomimetics is based primarily on the combined knowledge of the complementary conformational, topochemical, and electronic properties of the native peptide and of its address (in other words, of the receptor or the active site of the enzyme with which it interacts). Then, the design of peptidomimetics as potential bioactive compounds must take into particular account two structural factors: (i) a favorable fit (tertiary structure) with respect to the corresponding complementary spatial situation at the active site; (ii) the placement of structural elements (e.g., functional groups, polar and

*for references see p 2*

hydrophobic regions) in well-defined positions so that the required interactions (e.g., hydrogen bonds, electrostatic or hydrophobic interactions) can occur.

A major problem in this area of research is represented by the conformational flexibility of most peptides, the high dependence of their conformation on environment, and the relationship between the conformation in solution and the receptor-bound conformation. To overcome these difficulties, a successful approach involves the preparation of conformationally constrained analogues. Synthetic, bioactive peptidomimetics specifically designed for this approach are characterized either by cyclization (main-chain-to-main-chain; side-chain-to-main-chain; or, side-chain-to-side-chain) or by the incorporation of amino acid or dipeptide building blocks that can explore only a very limited portion of conformational space. Corollary, useful properties of these systems often include prolonged biological activity as a result of an increased stability to proteolytic enzymes. By combining computer-assisted molecular modeling, modern spectroscopic and NMR techniques, and X-ray diffraction analysis it is now possible to design peptidomimetic drugs, the structural spectrum of which lies between that of slightly modified peptides and that of pure nonpeptide molecules. The preparation of such compounds requires the utilization of the complete arsenal of modern, synthetic organic chemistry in addition to the methodologies developed by peptide chemists.

This volume brings together most of these critical issues by highlighting recent advances in a number of core areas of peptidomimetic synthesis. Section 9 focuses on side-chain-modified peptides, Section 10 describes the preparation and use of a variety of peptide main-chain modifications. Combined side-chain- and main-chain-modified peptides are covered in Section 11. Finally, Section 12 provides chemistry leading to peptides incorporating secondary structure (β- and γ-turns, helices, β-sheets) mimetics and inducers.


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# 9 Side-Chain-Modified Peptides

A. M. Felix

The biological function of peptides and proteins depends on their native conformation. The side-chain functionalities of the α-amino acids that comprise peptides and proteins have profound effects on their properties. These functionalities reside in the 20 naturally occurring α-amino acids, which have different propensities for formation of the three major secondary structural conformations.[1] In addition to these naturally occurring α-amino acids whose primary structure enables the polypeptide to fold into a predictable secondary and tertiary structure, the incorporation of unnatural amino acids has opened important areas of research.

The large number of theoretically possible conformational states for a linear peptide is significantly reduced when the side chain of the peptide is modified with conformationally restricted α-amino acids. Conformationally restricted analogues of biologically active peptides may be more resistant to enzymatic degradation, or may be locked into a biologically active preferred conformation. Conformationally restricted peptides may result in other potential advantages including enhanced receptor selectivity, prolonged receptor binding, or increased hydrophobicity that may improve absorption and serve as important models for peptide design.[2]

The introduction of β-substituted analogues of natural amino acids into biologically important peptides enables their use as templates for peptidomimetics. Procedures for the synthesis of β-substituted analogues have been devised for several naturally occurring amino acids including β-substituted phenylalanines, 2′,6′-dimethyltyrosines, naphthylalanines, and cysteines, 1,2,3,4-tetrahydroisoquinoline-3-carboxylic acids, and substituted phenylglycines, and glutamic acids (Section 9.1).

The pyrrolidine ring structure of proline is unique and places it in a special category among the natural α-amino acids since it leads to conformational restriction and a high probability of a *cis*-peptide bond preceding the proline residue in a polypeptide.[3] Proline and analogues of proline can play important roles in the de novo design of biologically important peptides and proteins. Procedures have been reported for the synthesis of nitrogen-containing three-membered rings (aziridine-2-carboxylic acids), four-membered rings (azetidine-2-carboxylic acids), five-membered rings (prolines), and six-membered rings [pipecolic acids (piperidine-2-carboxylic acids)], as well as those containing other heteroatoms (Section 9.2).

The incorporation of photoreactive amino acids into peptides (either during stepwise synthesis of the peptide or postsynthetically) enables the generation of a photoaffinity reagent that can bind to a receptor through carbene or nitrene intermediates. These peptides also contain tags (fluorophores, radionuclei, or biotin functions) which enable the identification of ligand-contact sites.[4] The synthesis of a variety of side-chain-modified phenylalanine- and benzophenone-containing photophores, and their use in the synthesis of the corresponding peptides (substance P, angiotensins, vasopressins, enkephalins), are reported (Section 9.3). The preparation of benzophenone-containing peptides using solid-phase methods either by direct insertion during synthesis or post-insertion after assemblage is described (Section 9.3.5.1).

Synthetic peptides containing side-chain modification have also been used as molecular scaffolds for the preparation of multiple receptors and molecular devices.[5] These include the use of crown ethers, cyclodextrins, porphyrins, and peptides with metal-binding sites (including ferrocenyl and EDTA side chains) (Section 9.4). Cyclization procedures have been developed to prepare biologically active cycloisodityrosine peptides which contain 14- or 17-membered rings (Section 9.5). The use of tryptathionine, a cross-linking dipeptide consisting of side-chain-to-side-chain linked L-Trp-L-Cys that is present in phallotoxins,[6] a family of cyclic heptapeptides, is also described (Section 9.6).

*for references see p 4*

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# 9.1 Synthesis of Side-Chain Conformationally Restricted α-Amino Acids

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The side-chain groups of the 20 naturally occurring α-amino acids, which are found in peptides and proteins that are the products of genes, have a wide range of chemical functionalities which make them readily adaptable as acids, bases, nucleophiles, electrophiles, chelators, charged groups, hydrophobes, hydrophiles, proton acceptors, proton donators, etc. These also tend to be quite conformationally flexible with energy barriers to rotation about their torsional angles $\chi^1$ (α–β bond), $\chi^2$ (β–γ bond), etc. generally less than 8 kcal·mol$^{-1}$ and, therefore, they readily interconvert at physiological temperatures. The exception is proline which has a five-membered ring which restricts the conformation; even so, proline readily interconverts between envelope-like conformations rapidly on the NMR timescale. Why nature has chosen 19 amino acid side-chain groups (Gly has no side-chain group) that are so flexible is not known, but it has great significance.

In order to understand peptide–protein, peptide–nucleic acid, peptide–peptide, peptide–lipid, and peptide–sugar interactions, binding, and information transduction in biological systems, side-chain conformational restriction can provide a very powerful tool for design.[1–3] Of course, of the 20 naturally occurring amino acids found in proteins, three have β-disubstitutions; namely, valine which has β-dimethyl substitution and hence is prochiral at the β-carbon atom, threonine which can be viewed as β-methylserine and hence is chiral at the β-carbon atom, and isoleucine which has β-methyl and β-ethyl substitution, and hence has a β-chiral center, and can be viewed as a β-methyl derivative of α-aminopentanoic acid. The presence of a β-substitution for these amino acids does lead to additional torsional constraint about the $\chi^1$ ($C_\alpha$–$C_\beta$) torsional angle. Undoubtedly this has been exploited by nature in some way, but surprisingly little has been undertaken by peptide and protein chemists to examine this question, and to systematically evaluate the effect of β-substitution on protein folding or molecular recognition, until quite recently.

It has been known for many years, however, that the β-branched amino acids, especially valine and isoleucine, cause problems in synthesis,[4,5] and special care and additional reaction time are required when β-substituted amino acids are added to a growing peptide chain in synthesis. For example, in the synthesis of [2,4-diisoleucine]oxytocin efforts to couple the isoleucine to isoleucine by the azide method failed and only the rearranged product was obtained.[6] Also, it is much more difficult to hydrolyze peptide bonds formed between two or more contiguous β-substituted amino acids using standard 6 M HCl conditions. For example, in the hydrolysis of [2,4-diisoleucine]oxytocin (3 isoleucine residues adjacent to each other) complete hydrolysis takes 60 hours.

## Synthetic Side-Chain Conformationally Constrained α-Amino Acids

Because β-substituted analogues of the naturally occurring amino acids and their analogues can lead to novel amino acids with conformationally restricted side-chain groups, they provide unique opportunities to explore (a) the topographical requirements [g(−), *trans*, or g(+)] for molecular recognition with a particular conformational scaffold (α-helix, β-turn, etc.), (b) the stereostructural requirements for information transduction for specific pharmacophore elements, (c) the dynamic requirements for binding and induction of conformational changes accompanying binding, (d) the effects of topographical constraint on peptide stability against proteolysis on bioavailability, and on peptide–membrane permeability, and (e) their use as templates for the preparation of peptidomimetics. Thus, their design, synthesis, and incorporation into bioactive peptides and/or peptides designed to

*for references see p 49*

explore topographical properties is an increasingly important area of research. In this section we examine a few of the various modes of side-chain topographical constraints which have been examined to date, with specific applications to the design of biologically active peptides.

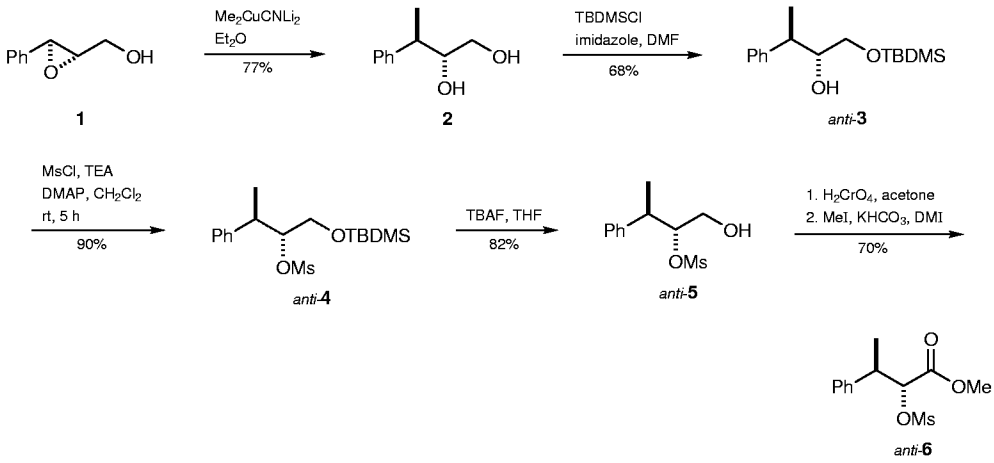
## 9.1.1 β-Methylphenylalanine

There is increasing evidence[3,7–19] demonstrating the important impact on bioactivity that β-methylphenylalanines can have when incorporated into bioactive peptides. Asymmetric synthesis of these compounds with well-established chemistry and economic starting materials has become well developed. Among the reported syntheses, the chiral auxiliary assisted asymmetric pathway[20–22] remains the most well-developed approach, while other pathways have only been briefly explored, because either too many steps were involved[23] or poor stereoselectivity was obtained.[24]

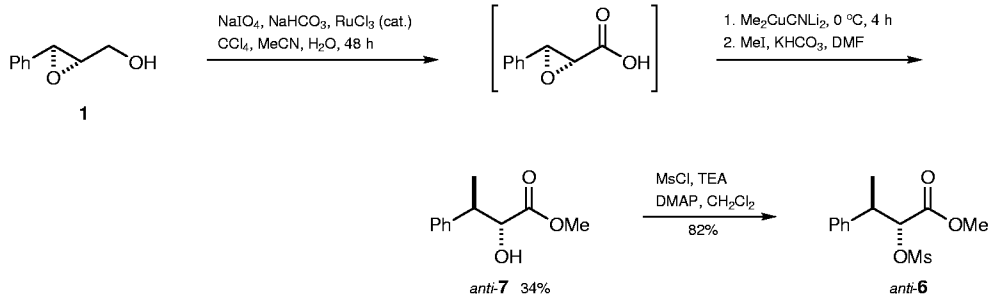
### 9.1.1.1 Asymmetric Epoxide Ring Opening

$N^{\alpha}$-Boc-protected β-methylphenylalanines can be prepared from diastereoselective ring opening of epoxide derivatives, followed by selective protection, oxidation, and functional group substitutions (Schemes 1–5). Low yields in some steps limit the applications of this method.[23]

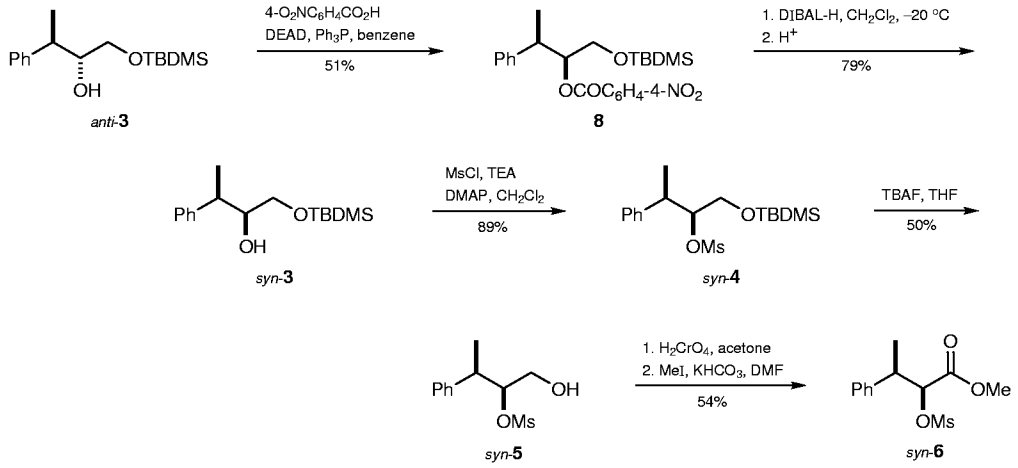
**Scheme 1** Synthesis of the *anti*-Intermediate[23]



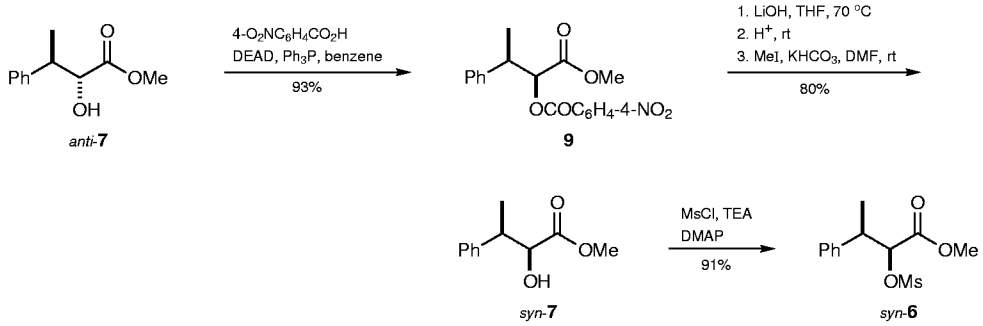

**Scheme 2** Improved Synthesis of the *anti*-Intermediate[23]



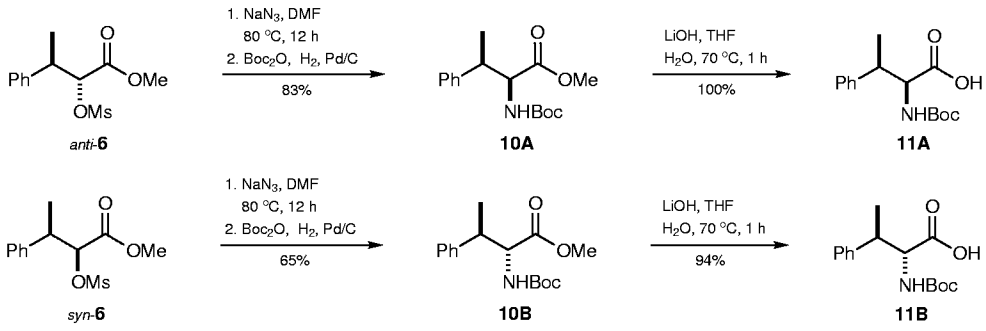
**Scheme 3** Synthesis of the *syn*-Intermediate[23]




**Scheme 4** Improved Synthesis of the *syn*-Intermediate[23]




**Scheme 5** Synthesis of $N^{\alpha}$-Boc-β-methylphenylalanines[23]




**(2*R*,3*R*)-3-Phenylbutane-1,2-diol (2):**[23]

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

To a stirred slurry of CuCN (2.69 g, 0.03 mol) in anhyd $Et_2O$ at −68 °C was added 1.6 M MeLi in $Et_2O$ (37.5 mL, 0.06 mol), and the mixture was stirred at this temperature for 1 h. A soln of (2*S*,3*R*)-2,3-epoxy-3-phenylpropan-1-ol (**1**)[25,26] (≥99% ee; 1.5 g, 10 mmol) in $Et_2O$ (36 mL) was added via cannula at −68 °C, and the mixture was allowed to warm to 0 °C over 3 h under $N_2$. When no starting material could be detected by TLC, sat. aq $NH_4Cl$ (80 mL) and $NH_4OH$ (10 mL) were added to the mixture, and the

*for references see p 49*

organic layer was extracted with $Et_2O$. The combined organic phases were dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by flash chromatography to give **2** as an oil; yield: 1.28 g (77%); $[\alpha]_D$ +16.3 (*c* 0.82, $CHCl_3$).

**(2*R*,3*R*)-1-(*tert*-Butyldimethylsiloxy)-3-phenylbutan-2-ol (*anti*-3):**[23]
To a soln of **2** (1.26 g, 7.6 mmol) in DMF (37 mL) was added TBDMSCl (1.72 g, 11.4 mmol) and imidazole (1.14 g, 16.7 mmol), and the reaction progress was monitored by TLC. After 24 h, the mixture was diluted with $Et_2O$ (40 mL) and washed with sat. aq $NH_4Cl$. The organic layer was separated, dried ($MgSO_4$), and concentrated under reduced pressure. The residue (2.5 g) was purified by column chromatography; yield: 1.45 g (68%); $[\alpha]_D$ +7.25 (*c* 1.05, $CHCl_3$).

**(2*R*,3*R*)-1-(*tert*-Butyldimethylsiloxy)-2-[(methylsulfonyl)oxy]-3-phenylbutane (*anti*-4):**[23]
To a soln of *anti*-**3** (1.45 g, 5.2 mmol) in $CH_2Cl_2$ (8 mL) at 0 °C were added TEA (1.8 mL, 12.8 mmol), DMAP (30 mg), and MsCl (0.9 mL, 11.4 mmol). The mixture was stirred for 5 h at rt, the reaction progress being monitored by TLC. When no starting material could be detected, $H_2O$ (10 mL) was added, the aqueous phase was extracted with $CH_2Cl_2$ (3 × 15 mL), and the combined organic phases were dried ($MgSO_4$), concentrated under reduced pressure, and purified by column chromatography to afford *anti*-**4** as an oil; yield: 1.7 g (90%); $[\alpha]_D$ +19.3 (*c* 0.83, $CHCl_3$).

**(2*R*,3*R*)-2-[(Methylsulfonyl)oxy]-3-phenylbutan-1-ol (*anti*-5):**[23]
To a soln of *anti*-**4** (1.67 g, 4.66 mmol) in THF (47 mL) at 0 °C was added 1.1 M TBAF in THF (8.5 mL, 9.35 mmol), and the reaction progress was monitored by TLC. After stirring for 20 min, the starting material could no longer be detected. The mixture was then concentrated under reduced pressure, diluted with $CH_2Cl_2$, and washed with sat. aq $NH_4Cl$. The organic phase was dried ($MgSO_4$) and concentrated under reduced pressure, and the residue was purified by column chromatography to give *anti*-**5** as a white solid; yield: 0.93 g (82%); mp 74–75 °C; $[\alpha]_D$ +21.9 (*c* 1.43, $CHCl_3$).

**Methyl (2*R*,3*R*)-2-[(Methylsulfonyl)oxy]-3-phenylbutanoate (*anti*-6):**[23]
*Method A (From* anti-**5**, *see Scheme 1):* Freshly prepared Jones reagent (7.35 mL) was added dropwise to a soln of *anti*-**5** (0.87 g, 3.56 mmol) in acetone (75 mL). The mixture was stirred at rt until no starting material could be detected by TLC (ca. 4 h). iPrOH (5 mL) was added, and acetone was removed under reduced pressure. The residue was partitioned between $H_2O$ and $CH_2Cl_2$; the aqueous phase was extracted with $CH_2Cl_2$. The combined organic phases were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure to afford (2*R*,3*R*)-2-[(methylsulfonyl)oxy]-3-phenylbutanoic acid as an oil. To a soln of the crude acid in DMF (6.2 mL) were added $KHCO_3$ (0.77 g, 7.7 mmol) and MeI (0.4 mL, 6.42 mmol), and the mixture was stirred at rt for 5 h. $H_2O$ was then added, and the organic layer was extracted with EtOAc. The combined organic phases were successively washed with sat. aq $Na_2SO_3$ and brine, and dried ($MgSO_4$). The solvents were removed under reduced pressure, and the residue was purified by column chromatography to give *anti*-**6** as a colorless oil; yield: 0.68 g (70%); $[\alpha]_D$ +23.5 (*c* 1.24, $CHCl_3$).

*Method B (From* anti-**7**, *see Scheme 2):* The procedure described above for the preparation of *anti*-**4** from *anti*-**3** was followed with hydroxy ester *anti*-**7** (8.0 mg, 0.04 mmol) as starting material. After purification by column chromatography, *anti*-**6** was obtained as a colorless oil; yield: 9.0 mg (82%).

**Methyl (2*R*,3*R*)-2-Hydroxy-3-phenylbutanoate (*anti*-7):**[23]

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

To a vigorously stirred mixture of the epoxy alcohol **1** (0.41 g, 2.73 mmol) and $CCl_4$/MeCN/$H_2O$ (1:1:1.5; 20 mL) were added $NaHCO_3$ (1.16 g, 13.8 mmol), $NaIO_4$ (1.77 g, 8.27 mmol), and $RuCl_3 \cdot 3H_2O$ (21 mg, 0.08 mmol). The mixture was stirred for 48 h, the acidic material was carefully extracted into $Et_2O$ at 0 °C, and the ethereal soln was briefly dried ($Na_2SO_4$). The solvent was concentrated under reduced pressure, and the resulting residue was dissolved in $Et_2O$ (10 mL) and added via cannula to a stirred suspension of $Me_2CuCNLi_2$ [prepared in situ from CuCN (0.74 g, 8.26 mmol) in $Et_2O$ (80 mL) and 1.6 M MeLi in hexanes (10 mL, 16 mmol) at 0 °C]. After 4 h at 0 °C, the mixture was filtered through a pad of Celite, and the filtrate was extracted with $Et_2O$. The organic layer was dried ($Na_2SO_4$) and the solvent was removed under reduced pressure. The oily residue was dissolved in DMF (4.5 mL), and $KHCO_3$ (0.55 g, 5.54 mmol) and MeI (0.27 mL, 4.33 mmol) were added. This mixture was stirred at rt for 12 h, $H_2O$ was added, and the organic layer was extracted with EtOAc (3 × 10 mL). The combined organic phases were successively washed with sat. aq $Na_2SO_3$ and brine, then dried ($MgSO_4$). Solvents were removed and the resulting oily residue was purified by column chromatography to give *anti*-**7** as an oil; yield: 0.18 g (34%); $[\alpha]_D$ +8.4 (*c* 2.21, $CHCl_3$).

**(2*S*,3*R*)-1-(*tert*-Butyldimethylsiloxy)-2-[(4-nitrobenzoyl)oxy]-3-phenylbutane (8):**[23]

| **CAUTION:** *Benzene is carcinogenic!* |
|---|

To a soln of *anti*-**3** (0.17 g, 0.61 mmol) in benzene (12 mL) were added $Ph_3P$ (0.77 g, 2.93 mmol), 4-nitrobenzoic acid (0.44 g, 2.63 mmol), and DEAD (0.46 mL, 2.92 mmol). The mixture was stirred at rt for 12 h and then concentrated under reduced pressure. The residue was purified by column chromatography to afford **8** as an oil; yield: 0.135 g (51%); $[\alpha]_D$ −27.5 (*c* 1.25, $CHCl_3$).

**(2*S*,3*R*)-1-(*tert*-Butyldimethylsiloxy)-3-phenylbutan-2-ol (*syn*-3):**[23]
To a soln of **8** (0.06 g, 0.14 mmol) in $CH_2Cl_2$ (1 mL) at −20 °C was added 20% DIBAL-H in hexanes (0.4 mL, 0.4 mmol). The mixture was stirred at −20 °C until no starting material was observed by TLC (ca. 1 h). $CH_2Cl_2$ (0.5 mL) was then added, and the resulting soln was treated with ca. 4 M HCl (0.8 mL). The aqueous phase was extracted with $CH_2Cl_2$, and the combined organic extracts were dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by column chromatography to afford *syn*-**3**; yield: 0.03 g (79%); $[\alpha]_D$ +11.8 (*c* 1.25, $CHCl_3$).

**(2*S*,3*R*)-1-(*tert*-Butyldimethylsiloxy)-2-[(methylsulfonyl)oxy]-3-phenylbutane (*syn*-4):**[23]
In the same manner used to prepare *anti*-**4**, the reaction of alcohol *syn*-**3** (0.05 g, 0.18 mmol) afforded *syn*-**4** as a colorless oil; yield: 0.06 g (89%); $[\alpha]_D$ −5.9 (*c* 2.2, $CHCl_3$).

**(2*S*,3*R*)-2-[(Methylsulfonyl)oxy]-3-phenylbutan-1-ol (*syn*-5):**[23]
In the same manner used to prepare *anti*-**5**, the reaction of *syn*-**4** (0.05 g, 0.14 mmol) gave *syn*-**5** as a colorless oil; yield: 18 mg (50%); $[\alpha]_D$ −30.4 (*c* 0.89, $CHCl_3$).

**Methyl (2*S*,3*R*)-2-[(Methylsulfonyl)oxy]-3-phenylbutanoate (*syn*-6):**[23]
*Method A (From* syn-**5**, *see Scheme 3):* In the same manner used to prepare *anti*-**6**, the reaction of *syn*-**5** (18 mg, 0.07 mmol) afforded *syn*-**6** as a colorless oil; yield: 11 mg (54%); $[\alpha]_D$ −11.5 (*c* 1.93, $CHCl_3$).

*Method B (From* syn-**7**, *see Scheme 4):* In the same manner used to prepare *syn*-**4**, *syn*-**6** was obtained from the reaction of *syn*-**7** (0.12 g, 0.61 mmol), as a colorless oil; yield: 0.15 g (91%).

**Methyl (2*S*,3*R*)-2-[(4-Nitrobenzoyl)oxy]-3-phenylbutanoate (9):**[23]

| **CAUTION:** *Benzene is carcinogenic!* |
|---|

To a soln of *anti*-**7** (0.2 g, 1.03 mmol) in benzene (20 mL) were added $Ph_3P$ (1.29 g, 4.9 mmol), 4-nitrobenzoic acid (0.73 g, 4.37 mmol), and DEAD (0.8 mL, 5 mmol). The mixture was stirred at rt for 12 h, concentrated under reduced pressure, and the residue was purified by column chromatography to give **9** as an oil; yield: 0.33 g (93%); $[\alpha]_D$ −72.9 (*c* 1.46, $CHCl_3$).

**Methyl (2*S*,3*R*)-2-Hydroxy-3-phenylbutanoate (*syn*-7):**[23]
To a soln of **9** (0.33 g, 0.96 mmol) in THF (10 mL) was added 2 M aq LiOH (2 mL, 4 mmol). The mixture was warmed to 70 °C and stirred at that temperature for 35 min, then cooled to rt and acidified with 10% HCl. The phases were separated, the aqueous phase was extracted with $CH_2Cl_2$, and the combined organic phases were dried ($Na_2SO_4$) and concentrated under reduced pressure. The crude hydroxy acid (0.29 g) was dissolved in DMF (3 mL) and treated with $KHCO_3$ (0.39 g, 3.9 mmol) and MeI (0.19 mL, 3.05 mmol) at rt for 5 h. $H_2O$ (5 mL) was then added, and the aqueous layer was extracted with EtOAc. The combined organic phases were successively washed with sat. aq $Na_2SO_3$ and brine, and dried ($MgSO_4$). Solvent was removed under reduced pressure and the resulting residue was purified by column chromatography to give *syn*-**7** as a white solid; yield: 0.15 g (80%); $[\alpha]_D$ +34.7 (*c* 1.09, $CHCl_3$).

**Methyl (2*S*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-phenylbutanoate (10A):**[23]

| **CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.* |
|---|

To a soln of *anti*-**6** (0.36 g, 1.3 mmol) in DMF (2 mL) was added $NaN_3$ (0.17 g, 2.6 mmol), and the mixture was heated to 80 °C while stirring. After 12 h the mixture was cooled to rt, $H_2O$ (5 mL) was added, and the mixture was extracted with EtOAc (3 × 10 mL). The combined organic phases were dried ($MgSO_4$) and concentrated under reduced pressure to give an oil. The oil was dissolved in EtOAc (4.4 mL). To this soln was added $Boc_2O$ (0.38 g, 1.74 mmol) and 10% Pd/C (35 mg), and the mixture was hydrogenated at

*for references see p 49*

atmospheric pressure. After 12 h, the mixture was filtered through a pad of Celite, concentrated, and submitted to column chromatography to give **10A** as a white solid; yield: 0.32 g (83%); mp 58–60 °C; $[\alpha]_D$ +33.0 (*c* 1.03, $CHCl_3$).

**(2*S*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-phenylbutanoic Acid ($N^{\alpha}$-Boc-β-methylphenylalanine, 11A):**[23]

To a soln of **10A** (0.04 g, 0.14 mmol) in THF (1.5 mL) was added 3.5 M aq LiOH (0.3 mL, 1.05 mmol). The mixture was heated to 70 °C and stirred for 60 min. The mixture was then cooled to rt and acidified with 5% HCl. The phases were separated, and the aqueous phase was extracted with $CH_2Cl_2$ (3 × 10 mL). The combined organic phases were dried ($Na_2SO_4$) and concentrated to give **11A** as a white solid; yield: 0.04 g (100%); mp 117 °C; $[\alpha]_D$ +17.0 (*c* 1.7, $CHCl_3$); $^1H$ NMR (200 MHz, $CDCl_3$/TMS, δ): 1.2 (br s, 3H), 1.4 (s, 9H), 3.3 (m, 1H), 4.4 (m, 1H), 4.6 (m, 1H, rotamer), 5.1 (m, 1H, NH), 6.5 (m, 1H, NH, rotamer), 7.1–7.3 (m, 5H), 9.2–9.4 (br m, 1H); $^{13}C$ NMR (50 MHz, $CDCl_3$, δ): Rotamer 1: 16.1 ($CH_3$), 28.2 ($CH_3$), 42.1 (CH), 58.6 (CH), 80.1 (C), 127.0 (CH), 127.7 (CH), 128.4 (CH), 141.3 (C), 155.0 (C), 176.2 (C); Rotamer 2: 14.5 ($CH_3$), 27.8 ($CH_3$), 41.8 (CH), 60.1 (CH), 81.0 (C), 142.0 (C), 156.0 (C), 176.0 (C).

**Methyl (2*R*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-phenylbutanoate (10B):**[23]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

To a soln of *syn*-**6** (0.10 g, 0.37 mmol) in DMF (0.6 mL) was added $NaN_3$ (0.05 g, 0.77 mmol), and the mixture was heated to 80 °C while stirring. After 12 h the mixture was cooled to rt, $H_2O$ (0.5 mL) was added, and the mixture was extracted with EtOAc (3 × 5 mL). The combined organic phases were dried ($MgSO_4$) and concentrated under reduced pressure to yield an oil that was dissolved in EtOAc (1.6 mL). To this soln was added $Boc_2O$ (0.10 g, 0.46 mmol) and 10% Pd/C (10 mg), and the mixture was hydrogenated at atmospheric pressure. After 12 h, the mixture was filtered through a pad of Celite, the volatiles were removed, and the residue was subjected to column chromatography to give **10B** as an oil; yield: 0.07 g (65%); $[\alpha]_D$ −32.0 (*c* 1.72, $CHCl_3$).

**(2*R*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-phenylbutanoic Acid ($N^{\alpha}$-Boc-β-methylphenylalanine, 11B):**[23]

In the same manner used to prepare **11A**, the reaction of **10B** (36 mg, 0.12 mmol) gave **11B** as a colorless oil; yield: 32 mg (94%); $[\alpha]_D$ −9.5 (*c* 1.58, $CHCl_3$); $^1H$ NMR (200 MHz, $CDCl_3$/TMS, δ): 1.2 (br s, 3H), 1.4 (s, 9H), 3.3–3.5 (m, 1H), 4.4–4.6 (m, 1H), 4.8 (m, 1H), 7.2–7.4 (m, 5H), 9.0–9.4 (br m, 1H); $^{13}C$ NMR (50 MHz, $CDCl_3$/TMS, δ): 17.8 ($CH_3$), 28.2 ($CH_3$), 41.6 (CH), 58.8 (CH), 80.0 (C), 127.3 (CH), 127.7 (CH), 128.6 (CH), 141.0 (C), 156.0 (C), 176.6 (C).

### 9.1.1.2 Asymmetric Aziridine Ring Opening

The experimental procedures used for the preparation of β-methylphenylalanine via this synthetic approach are the same as those described in Section 9.1.3.1.[27]

### 9.1.1.3 Through Chiral Auxiliary Oxazolidinone

For the synthesis of β-methylphenylalanine, the same procedures can be used as those described in Section 9.1.2.1. This synthetic approach has proven to be very reliable and can be readily scaled up.[21,28]

## 9.1.2 β,2,6-Trimethyltyrosine

As a topographically constrained analogue of tyrosine, β,2,6-trimethyltyrosine (TMT) offers many advantages for topographical design of peptide and peptidomimetic analogues due to its very well-defined conformational and dynamic properties. Examination of the $\chi^1/\chi^2$ en-

ergy map[3,29] for all four isomers [(α*S*,β*S*), (α*S*,β*R*), (α*R*,β*R*), and (α*R*,β*S*)] by computational chemistry demonstrated that there are significant energy barriers for rotation about both the $\chi^1$ and $\chi^2$ torsional angles. Dynamic NMR spectroscopic measurement[30] experimentally determined that the torsional energy barrier about $\chi^1$ is between 15 and 20 kcal·mol$^{-1}$ for the compounds and various derivatives consistent with the 17–18 kcal·mol$^{-1}$ determined by calculations using a modified force field.[3,29] Furthermore, it was shown that for each isomer [(α*S*,β*S*), (α*S*,β*R*), (α*R*,β*R*), and (α*R*,β*S*)] there is a distinctly preferred gauche conformation [g(+), *trans*, or g(−)] with an energy difference for the different conformers of 3–7 kcal·mol$^{-1}$. Thus these compounds can be used to explore, with some confidence, the preferred bioactive topography when tyrosine (or phenylalanine) side-chain moieties are pharmacophore elements in a bioactive peptide or protein, and we have used them very successfully for this purpose (*vide infra*).[31] In view of their importance, details of the asymmetric synthesis of all four isomers of β,2,6-trimethyltyrosine[32] are described. A similar asymmetric synthesis of all four isomers of β,2,6-trimethylphenylalanine has been reported,[33] and will not be discussed here. The tyrosine synthesis has been chosen because it is more difficult as a result of the presence of a 4-hydroxy group on the aromatic ring.

## Incorporation of β,2′,6′-Trimethyltyrosine into Bioactive Peptides Used in De Novo Design of Peptidomimetics and Nonpeptide Agonist Peptide Mimics

The four isomers of TMT and its derivatives have been used to explore a number of structure–activity relationships in peptide hormones and neurotransmitters where tyrosine is an important pharmacophore element. For example, exploration of the topographical requirements of the Tyr$^1$ residue in cyclic enkephalins such as *c*[-D-Pen$^2$,D-Pen$^5$-]enkephalin (DPDPE)[29,34] and the linear deltorphin[29] has been undertaken. It was determined that only the (α*S*,β*R*)-compound gives highly potent and selective analogues.[29,34] This finding was used in conjunction with extensive computational studies[35] to design a nonpeptide peptide mimetic δ-opioid receptor agonist ligand.[36] In addition, these compounds were shown to have very interesting in vivo analgesic activities.[37] As an added bonus it was found that the (α*S*,β*R*)-analogue is an antagonist at the μ-opioid receptor,[29,37] thus demonstrating unequivocally the different topographical requirements the μ- and δ-opioid receptors have for agonist and antagonist opioid activity. In addition, the four isomers of TMT have been incorporated into the highly topographically constrained dipeptide TMT-Tic[31] (a cyclic phenylalanine analogue, *vide infra*), and these analogues were shown to have unique biological activity profiles at δ- and μ-opioid receptors. One of the analogues, (α*S*,β*R*)-TMT$^1$-Tic, is a potent and highly selective inverse agonist at the δ-opioid receptor. This highly unique biological activity is undoubtedly the result of its unique topographical properties. The 4-methoxy derivatives of TMT have been incorporated into oxytocin antagonists and the unique topographical requirements for oxytocin antagonists have been determined.[38]

## Other Related Amino Acids

The similarly constrained four isomers of β,2′,6′-trimethylphenylalanine[33] have been prepared by asymmetric synthesis and incorporated into bioactive peptides.[39,40]

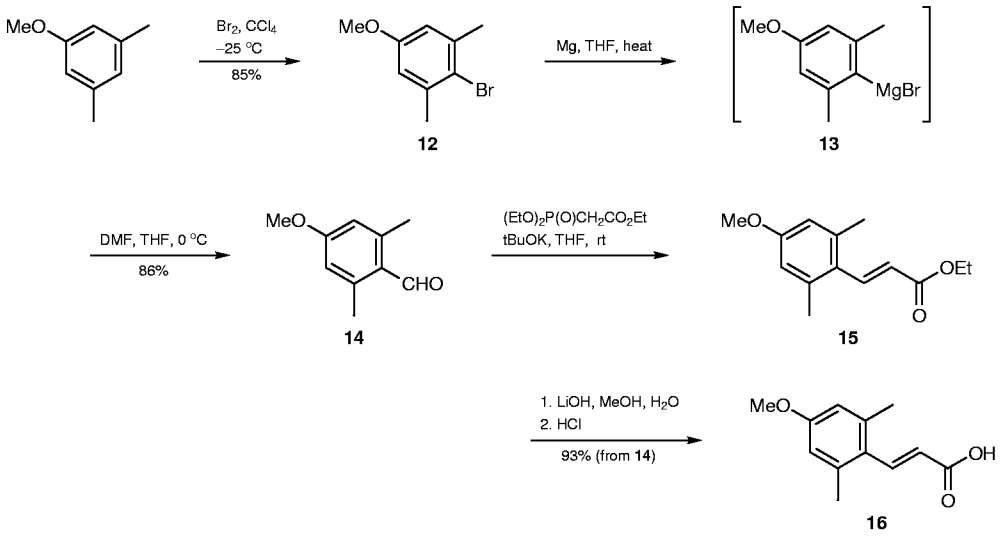
Closely related in topographical structure, but with different and quite unique torsional properties, are the β-isopropyl-substituted analogues of tyrosine[41,42] and phenylalanine.[43,44] All four isomers of each amino acid analogue have been prepared by asymmetric synthesis. In addition, the highly topographically constrained amino acid β-isopropyl-2′,6′-dimethyltyrosine has been prepared by asymmetric synthesis.[45]

*for references see p 49*

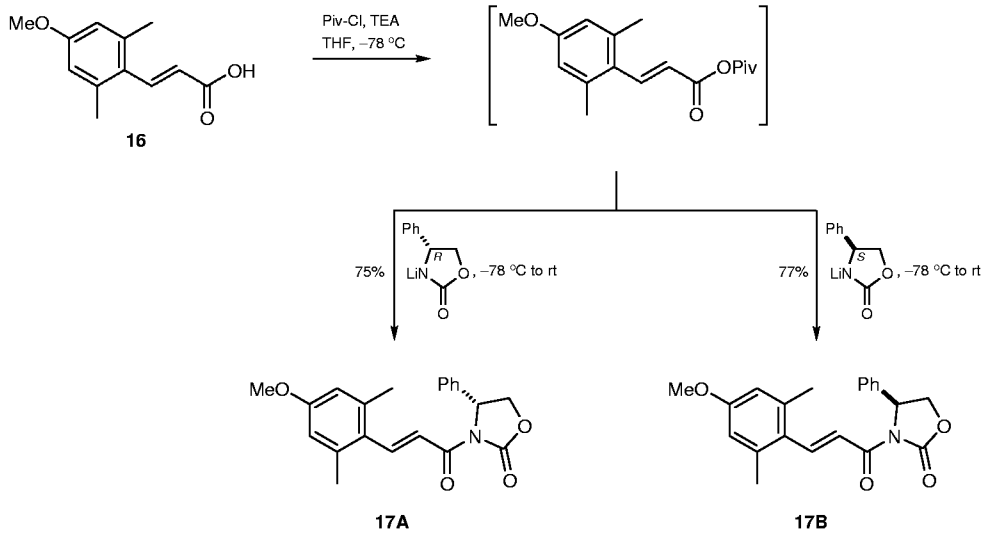
### 9.1.2.1 Through Chiral Auxiliary Oxazolidinone

The synthesis of β,2′,6′-trimethyltyrosine from chiral auxiliary oxazolidinones is a very effective procedure for preparing all four isomers (Schemes 6–12).[32] The two chiral auxiliaries, (4*S*)- and (4*R*)-4-phenyloxazolidin-2-one, are prepared according to modified literature procedures.[46–48] Compounds **17A** and **17B** are prepared by a procedure for coupling of the precursor acids **16** to the chiral auxiliaries using mixed anhydrides (Scheme 7). Compounds **19A** and **19B** are synthesized using a procedure for coupling of (*E*)-crotonyl chloride (**18**) to the optically pure lithiated auxiliaries (Scheme 8). Conjugate additions (Scheme 9), conversion into the azides **22** (Scheme 10), hydrolysis and hydrogenation to the amines **24** (Scheme 11), followed by further hydrolysis provides the individual isomers **25A–D** of β,2,6-trimethyltyrosine (Scheme 12).

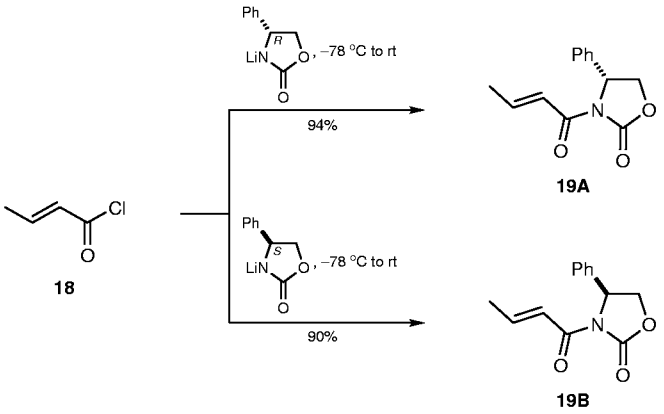
**Scheme 6** Synthesis of the Acrylic Acid Intermediate[32]



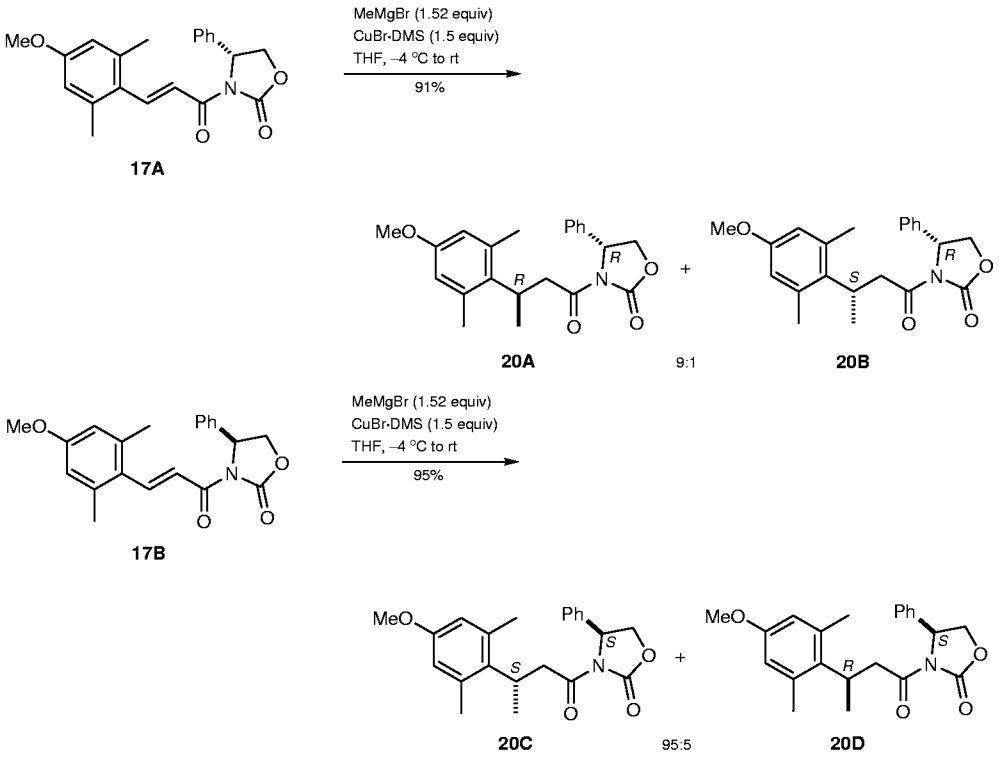

**Scheme 7** Synthesis of the *N*-Acryloyloxazolidin-2-one Intermediates[32]



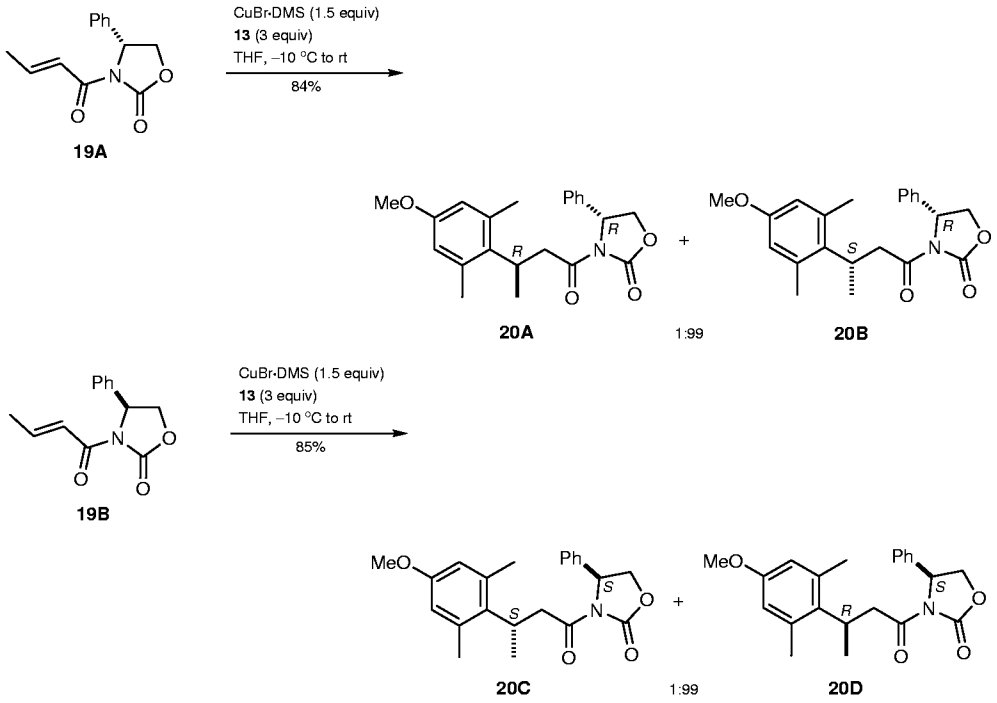
**Scheme 8** Synthesis of the *N*-Butenoyloxazolidin-2-one Intermediates[32]



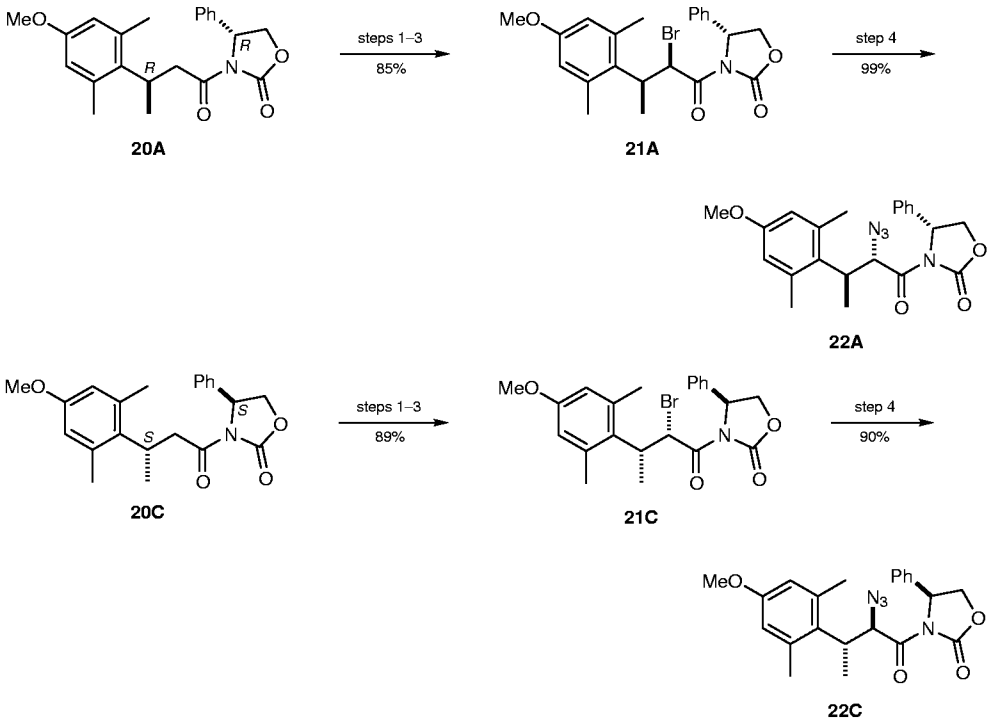

**Scheme 9** Michael Additions[32]

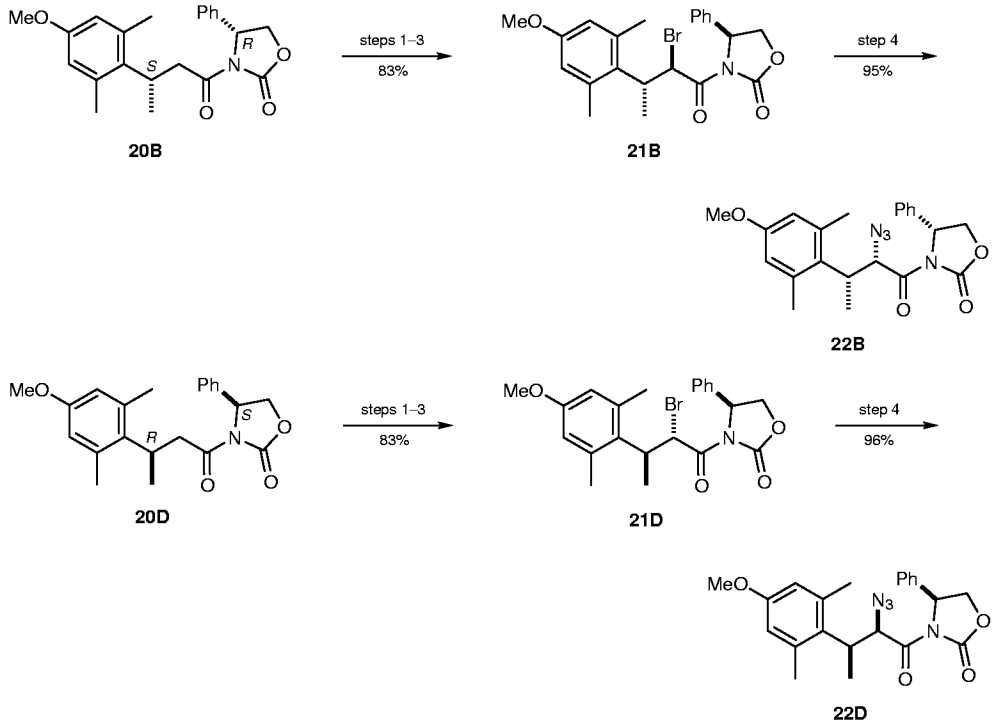



*for references see p 49*



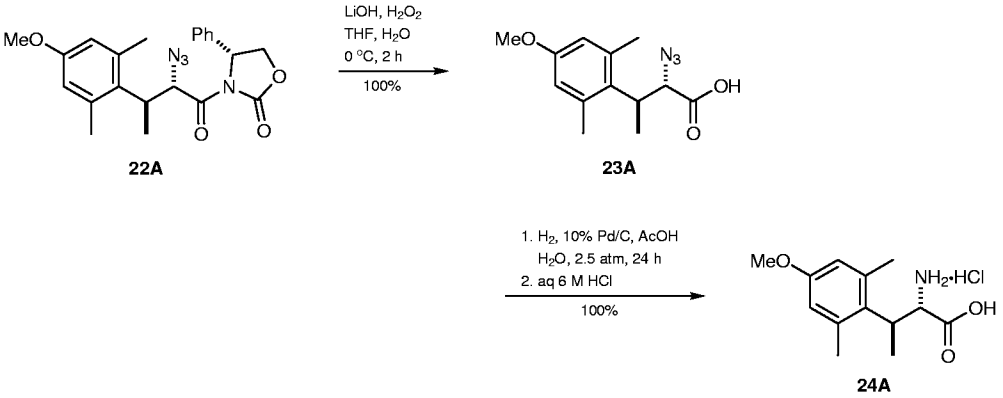

**Scheme 10** Synthesis of the α-Azide Derivatives[32]



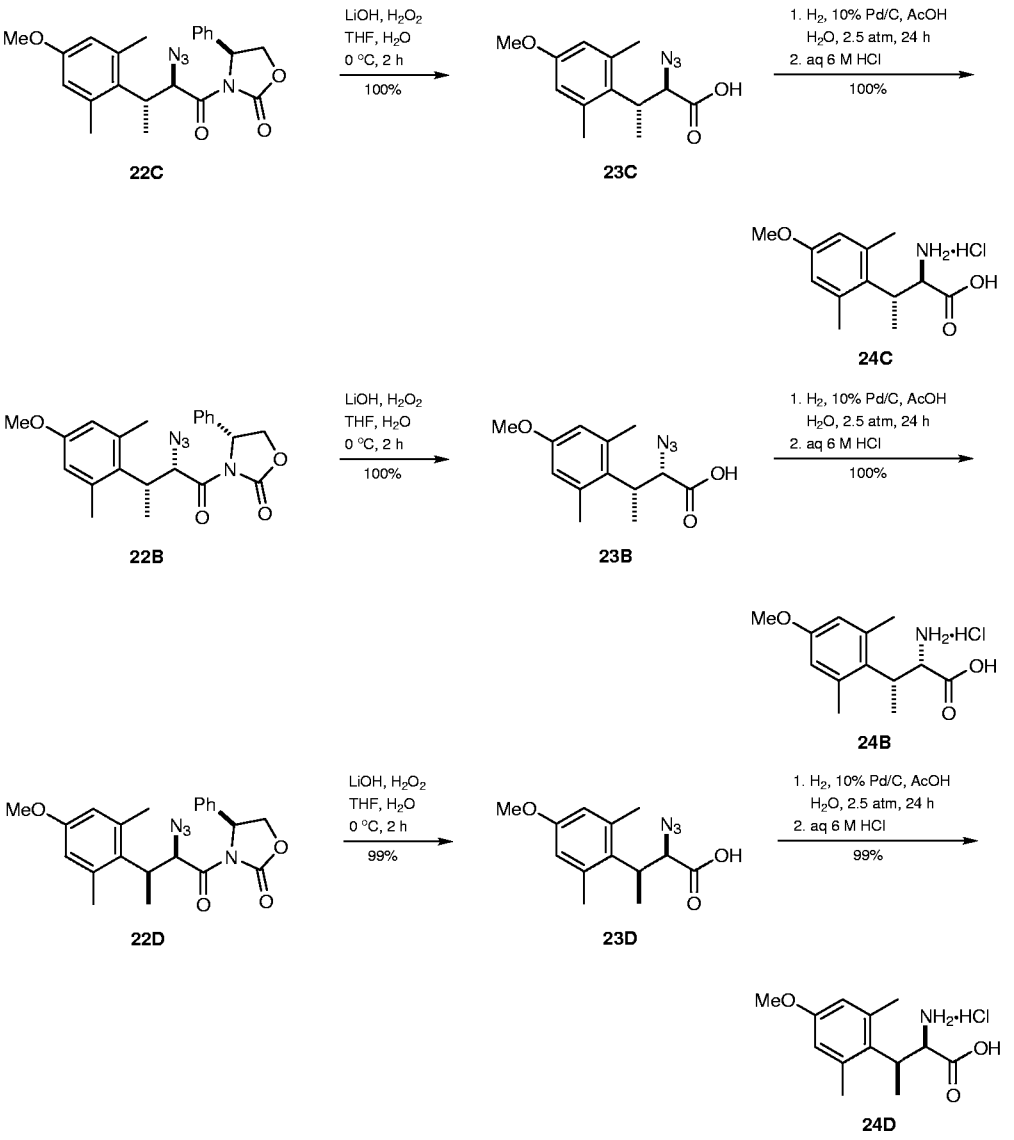



steps 1–3: 1. 1 M $Bu_2BOTf$ (1.05 equiv), DIPEA, $CH_2Cl_2$, –78 °C; 2. NBS (1.1 equiv), $CH_2Cl_2$, –78 °C; 3. column chromatography
step 4: Amberlite IR-400 azide-exchange resin, MeCN, rt, 9 d

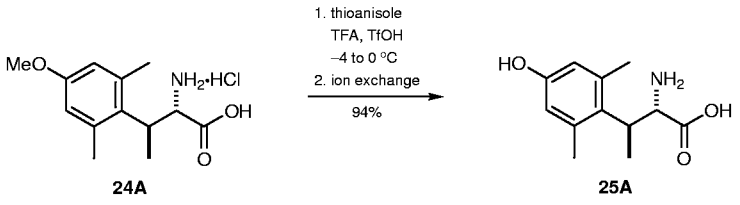
**Scheme 11** Hydrolysis and Reduction To Give the β,2′,6′-Trimethyltyrosine Derivatives[32]

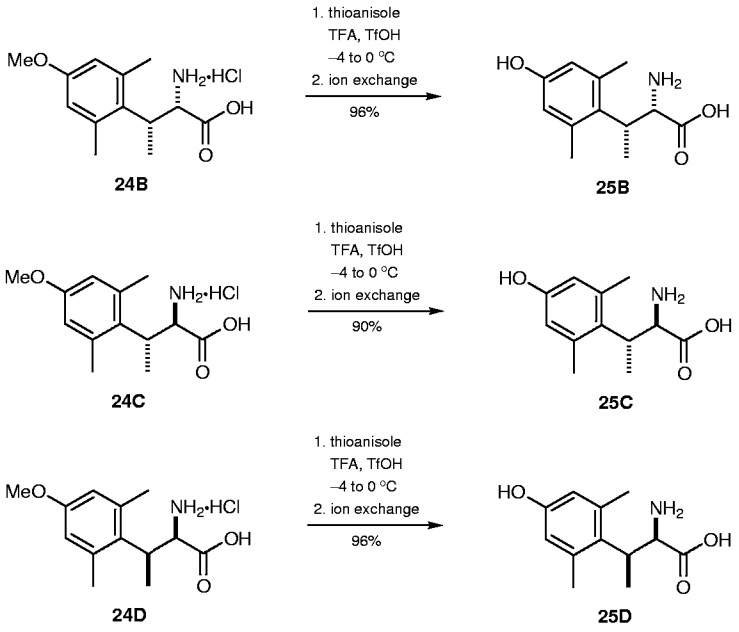



*for references see p 49*




**Scheme 12** Hydrolysis To Give the Four Isomers of β,2′,6′-Trimethyltyrosine[32]






**4-Bromo-3,5-dimethylanisole** (12):[32]

**CAUTION:** *Carbon tetrachloride is toxic by inhalation and skin contact, and is a possible human carcinogen.*

3,5-Dimethylanisole (87.5 mL, 619 mmol) was dissolved in $CCl_4$ (1.0 L). The mixture was cooled to −25 °C, and a soln of $Br_2$ (35.0 mL, 679 mmol) in $CCl_4$ (200 mL) was added over 8 h until the bromine color persisted. The soln was allowed to warm to rt, poured into $H_2O$ (1.0 L) and stirred for 1.5 h. The aqueous and organic phases were separated, and the aqueous phase was extracted with $Et_2O$ (3 × 200 mL). The combined organic phases were dried ($MgSO_4$) and filtered, the solvents were removed by rotary evaporation, and the residual oil was vacuum distilled to give a clear colorless liquid; yield: 113.5 g (85%); bp 64 °C/2.1 Torr.

**4-Methoxy-2,6-dimethylbenzaldehyde** (14):[32]

A mixture of 4-bromo-3,5-dimethylanisole (**12**; 68.0 g, 316 mmol), polished Mg strips (9.0 g, 375 mmol), and anhyd THF (1.0 L) was gently heated. The heating source was removed until the reaction ceased to reflux. The black colloidal suspension was reheated to reflux for 4 h and then recooled to 0 °C. A soln of anhyd DMF (28.7 mL, 371 mmol) in anhyd THF (30 mL) was added to the suspension dropwise over 20 min. The gray suspension was allowed to warm to rt and stirred for 1 h. The reaction was quenched by decanting the soln into aq $NH_4Cl$ (1.0 L). The reaction flask was washed with additional THF (2 × 30 mL) which was decanted into the above $NH_4Cl$ soln. The aqueous and organic phases were separated, and the aqueous phase was extracted with $Et_2O$ (3 × 200 mL). The combined organic phases were dried ($MgSO_4$) and filtered, and the solvents were removed by rotary evaporation. The residue, a yellow oil, was further dried under reduced pressure to give a yellow solid (yield: 96%). The crude product was recrystallized (hexanes at −20 °C) to give **14**; yield: 44.9 g (86%); bp 104 °C/5 Torr; 2,4-Dnp derivative: mp 44.0–45.5 °C. The product was of satisfactory purity (pure by $^1H$ NMR spectroscopy) for use in the next reaction.

**Ethyl (2*E*)-3-(4-Methoxy-2,6-dimethylphenyl)acrylate** (15):[32]

To a soln of tBuOK (53.1 g, 450 mmol) and triethyl phosphonoacetate (93 mL, 450 mmol) in anhyd THF (1.2 L), was added aldehyde **14** (41.4 g, 252 mmol) in one portion at rt. The soln thickened considerably after a few min and the reaction was stirred at rt for 2 h. A small amount of this material, **15**, was removed from the mixture and purified by column chromatography for characterization; mp 68.0–69.0 °C; TLC: $R_f$ 0.27 (EtOAc/hexane 1:9).

*for references see p 49*

**(2*E*)-3-(4-Methoxy-2,6-dimethylphenyl)acrylic Acid (16):**[32,49]

To a slurry containing crude ester **15** was added MeOH (400 mL) and $H_2O$ (400 mL). $LiOH \cdot H_2O$ (75 g, 1.8 mol, 4 equiv) was added and the reaction was stirred overnight. Volatiles were removed by rotary evaporation and the aqueous layer, now containing a white precipitate, was diluted to 1.5 L with $H_2O$ and gently heated to dissolve solids. The warm soln was washed with $CHCl_3$ (3 × 200 mL) and acidified to pH 1 with 6 M HCl. The white solid was filtered, washed with $H_2O$ (1.0 L), and dried under reduced pressure in the presence of $P_2O_5$ to give an off-white solid; yield: 48.6 g (93% from **14**); mp 174.5–175.5 °C. The product was used for the next reaction without further purification.

**(4*R*)- and (4*S*)-3-[(2*E*)-3-(4-Methoxy-2,6-dimethylphenyl)acryloyl]-4-phenyloxazolidin-2-one (17A and 17B):**[32]

To a stirred, −78 °C precooled soln of acid **16** (6.50 g, 31.5 mmol) in anhyd THF (650 mL) was added TEA (5.3 mL, 37.8 mmol) via syringe, followed by Piv-Cl (4.3 mL, 34.9 mmol). The suspension was stirred for 15 min at −78 °C and 45 min at 0 °C, and then recooled to −78 °C. The suspension was then transferred via cannula to a stirred slurry of lithiated (4*S*)-4-phenyloxazolidin-2-one at −78 °C [prepared 20 min in advance at −78 °C by the addition of 1.6 M BuLi in hexanes (18.0 mL, 28.8 mmol) to a soln of (4*S*)-4-phenyloxazolidin-2-one (4.70 g, 28.8 mmol) in anhyd THF (120 mL)]. The resulting suspension was stirred at −78 °C for 20 min and at rt for 2 h. The reaction was quenched by the addition of sat. aq $NH_4Cl$ (70 mL). Volatiles were removed by rotary evaporation and the residual aqueous slurry was extracted with $CHCl_3$ (3 × 110 mL). The combined extracts were washed with dil aq $NaHCO_3$ (3 × 70 mL), dried ($MgSO_4$), filtered, rotary evaporated to dryness, and stored under reduced pressure. The crude product was chromatographed (EtOAc/hexane 2:8) to give (4*S*)-3-[(2*E*)-3-(4-methoxy-2,6-dimethylphenyl)acryloyl]-4-phenyloxazolidin-2-one (**17B**); yield: 77%; mp 157.5–158.0 °C (EtOAc/hexane); $[\alpha]_D^{25}$ +32.0 (*c* 0.29, $CHCl_3$).

Similarly prepared by addition of lithiated (4*R*)-4-phenyloxazolidin-2-one was (4*R*)-3-[(2*E*)-3-(4-methoxy-2,6-dimethylphenyl)acryloyl]-4-phenyloxazolidin-2-one (**17A**); yield: 75%; mp 158.5–160.0 °C (EtOAc/hexane); $[\alpha]_D^{22}$ −26.1 (*c* 1.05, $CHCl_3$).

**(4*R*)- and (4*S*)-3-[(2*E*)-But-2-enoyl]-4-phenyloxazolidin-2-one (19A and 19B):**[32]

Freshly distilled (*E*)-crotonyl chloride (**18**; 21.08 mL, 220 mmol) was dissolved in anhyd THF (80 mL), and the soln was cooled to −78 °C. In a separate 2-L round-bottomed flask with a magnetic stirrer bar, (4*R*)-4-phenyloxazolidin-2-one (32.64 g, 200 mmol) was dissolved in anhyd THF (500 mL), then cooled to −78 °C, and 1.6 M BuLi in hexanes (150 mL, 240 mmol) was added via syringe. The mixture was stirred for 20 min and the crotonyl chloride soln was transferred to the lithiated oxazolidinone via cannula. The resulting soln was stirred for 30 min at −78 °C and 1.5 h at rt before quenching the reaction by the addition of aq $NH_4Cl$ (150 mL). Volatiles were removed by rotary evaporation. The slurry was diluted with $H_2O$ (200 mL) and extracted with EtOAc (3 × 150 mL). The combined organic layers were washed with aq $NaHCO_3$ (2 × 150 mL) and brine (150 mL), dried ($MgSO_4$), filtered, and concentrated to dryness by rotary evaporation. The product, (4*R*)-3-[(2*E*)-but-2-enoyl]-4-phenyloxazolidin-2-one (**19A**), was purified by column chromatography (EtOAc/hexanes 3:7); yield: 94%; mp 77–79 °C (hexanes/EtOAc); $[\alpha]_D^{22}$ +111.8 (*c* 1.08, $CHCl_3$); ratio of *E*/*Z*-isomers 96:4 (by $^1H$ NMR spectroscopy).

Similarly prepared by addition of lithiated (4*S*)-4-phenyloxazolidin-2-one was (4*S*)-3-[(2*E*)-but-2-enoyl]-4-phenyloxazolidin-2-one (**19B**); yield: 90%; mp 74.5–75.5 °C (hexanes/EtOAc); $[\alpha]_D^{22}$ +113.4 (*c* 1.085, $CHCl_3$); no *Z*-isomer observed (by $^1H$ NMR spectroscopy).

**Preparation of the Organocopper Reagents:**[32]

The organocopper reagent was first prepared as follows: To a suspension of Mg turnings (2.91 g, 120 mmol) in anhyd THF (50 mL) was added **12** (23.5 g, 109 mmol) and EtBr (0.80 mL, 11 mmol). The mixture was heated with stirring to reflux for 4 h, cooled to −4 °C, and then transferred via cannula to a −40 °C stirring slurry of CuBr·DMS complex (11.23 g, 53.8 mmol) in anhyd THF (130 mL) and anhyd DMS (65 mL). The gray mixture (mixture A) was warmed to −10 °C ready for the conjugate additions. In the case of MeMgBr, a commercial soln (3.0 M in $Et_2O$; 7.5 mL, 22.5 mmol) was added by syringe to a soln of CuBr·DMS complex (4.58 g, 22.0 mmol) in anhyd THF (50 mL) and anhyd DMS (20 mL) at −4 °C. The resulting yellow-greenish mixture (mixture B) was stirred for 10 min at −4 °C and used for the conjugate addition at the given temperature.

**Conjugate Additions (Scheme 9):**[32]

The procedure for conjugate additions was slightly different depending on the substrate used. For **20A** and **20C**, a soln of the α,β-unsaturated acyloxazolidinone **17A** or **17B** (5.20 g, 14.8 mmol) in anhyd THF (25 mL) was added dropwise to mixture B over 60 min at 0 °C, and the resulting mixture was stirred for an additional 90 min at −4 °C. Then, the mixture was warmed to rt and stirred overnight. The reaction was quenched by slow addition of sat. $NH_4Cl$ (70 mL) and stirred for 30 min. The phases were separated, and the aqueous phase was extracted with $Et_2O$ (3 × 50 mL). The combined organic phases were washed

with sat. aq $NH_4Cl$ (3 × 30 mL), $H_2O$ (3 × 30 mL), and brine (30 mL), and dried (anhyd $MgSO_4$). The drying agent was filtered off, the solvent was removed, and the diastereomeric mixture was purified by column chromatography (EtOAc/hexanes 3:7) to give **20A** or **20C**.

(4*R*)-3-[(3*R*)-3-(4-Methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**20A**); yield: 91%; 9:1 mixture of diastereomers (by $^1H$ NMR spectroscopy). The diastereomeric product was recrystallized (EtOAc and hexanes) to obtain **20A** as a single isomer; yield: 75%; mp 99.5–101.0 °C; $[\alpha]_D^{22}$ −88.5 (*c* 1.07, $CHCl_3$).

(4*S*)-3-[(3*S*)-3-(4-Methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**20C**); yield: 95%; 95:5 mixture of diastereomers (by $^1H$ NMR spectroscopy). The material was recrystallized (EtOAc and hexanes) to obtain **20C** as a single isomer; yield: 85%; mp 104.0–104.5 °C; $[\alpha]_D^{25}$ +95.6 (*c* 0.39, $CHCl_3$).

For the preparation of **20B** and **20D**, a soln of α,β-unsaturated acyloxazolidinone **19A** or **19B** (8.5 g, 36.6 mmol) in anhyd THF (65 mL) was added dropwise to the organocopper soln mixture A at −10 °C, within a period of 1 h. The suspension was slowly warmed to rt over 1.5 h and stirred at rt overnight. The quenching, workup, and purification procedures were similar to those for the preparation of **20A** and **20C**.

(4*R*)-3-[(3*S*)-3-(4-Methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**20B**); yield: 84%; single diastereomer (by $^1H$ NMR spectroscopy). The product was recrystallized (EtOAc and hexanes) to obtain **20B** as a single isomer; mp 99.5–101.0 °C; $[\alpha]_D^{22}$ −25.6 (*c* 1.20, $CHCl_3$).

(4*S*)-3-[(3*R*)-3-(4-Methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**20D**); yield: 85%; single diastereomer (by $^1H$ NMR spectroscopy). The product was recrystallized (EtOAc and hexanes) to obtain **20D** as a single isomer; mp 135.0–136.0 °C; $[\alpha]_D^{22}$ +16.8 (*c* 1.038, $CHCl_3$).

**(4*R*)-3-[(2*R*,3*S*)-2-Bromo-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (21A); Typical Procedure for Asymmetric Bromination of *N*-Acyloxazolidinones 20 (Scheme 10):**[32]

A soln of **20A** (3.87 g, 10.53 mmol) in dry $CH_2Cl_2$ (50 mL) was cooled to −78 °C, then dry DIPEA (2.20 mL, 12.64 mmol) and 1 M $Bu_2BOTf$ in $CH_2Cl_2$ (11.06 mL, 11.05 mmol) was added via syringe. Meanwhile, in another dry 3-neck round-bottomed flask (100 mL), a suspension of recrystallized NBS (2.06 g, 11.58 mmol) in dry $CH_2Cl_2$ (35 mL) was cooled to −78 °C. The boron enolate soln was transferred to the NBS suspension at −78 °C via cannula. The resulting mixture was stirred at −78 °C for 2 h and warmed to −4 °C. Then, it was quenched with 0.5 M $NaHSO_4$ (20 mL). The mixture was warmed to rt and stirred for 30 min. The phases were separated, and the aqueous layer was extracted with $CH_2Cl_2$ (2 × 25 mL). The combined organic phases were washed with 0.5 M $NaHSO_4$ (50 mL), 1 M $Na_2S_2O_3$ (3 × 50 mL), $H_2O$ (50 mL), and brine (50 mL). The organic layer was then dried ($MgSO_4$), filtered, and concentrated to give the crude bromide as a yellow solid which was further purified by column chromatography (EtOAc/hexanes 2:8) to give **21A**; yield: 85%; mp 172.0–173.5 °C; $[\alpha]_D^{22}$ −94.0 (*c* 1.01, $CHCl_3$); TLC: $R_f$ 0.50 (EtOAc/hexanes 3:7).

Similarly prepared were:

(4*S*)-3-[(2*S*,3*R*)-2-Bromo-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**21C**) from **20C**: yield: 89%; mp 150.0–151.0 °C; $[\alpha]_D^{25}$ +87.9 (*c* 1.16, $CHCl_3$); TLC: $R_f$ 0.24 (EtOAc/hexanes 1:4).

(4*R*)-3-[(2*R*,3*R*)-2-Bromo-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**21B**) from **20B**; yield: 83%; mp 186.0–187.5 °C; $[\alpha]_D^{22}$ −1.5 (*c* 1.03, $CHCl_3$); TLC: $R_f$ 0.48 (EtOAc/hexanes 3:7).

(4*S*)-3-[(2*S*,3*S*)-2-Bromo-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**21D**) from **20D**; yield: 83%; mp 192.0–193.0 °C; $[\alpha]_D^{25}$ +1.75 (*c* 1.14, $CHCl_3$); TLC: $R_f$ 0.17 (EtOAc/hexanes 1:4).

**(4*R*)-3-[(2*S*,3*S*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (22A); Typical Procedure for Azide Formation (Scheme 10):**[32]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

Bromide **21A** (2.20 g, 4.93 mmol) was dissolved in MeCN (dried over $MgSO_4$, 50 mL). Amberlite IR-400 azide-exchange resin (11.03 g, ca. 41.90 mmol $N_3$) was added and the mixture was stirred gently for 9 d. The resin was filtered and washed with MeCN (4 × 25 mL). The filtrate was concentrated to provide an off-white solid which was then chromatographed (EtOAc/hexanes 3:7) to give **22A**; yield: 99%. A small sample was recrystallized (hexanes/EtOAc) for characterization; mp 121.0–121.5 °C; $[\alpha]_D^{22}$ −48.7 (*c* 1.24, $CHCl_3$).

Similarly prepared were:

(4*S*)-3-[(2*R*,3*R*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**22C**) from **21C**; yield: 90%; mp 111.5–112.0 °C; $[\alpha]_D^{22}$ +49.1 (*c* 1.24, $CHCl_3$).

*for references see p 49*

(4*R*)-3-[(2*S*,3*R*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**22B**) from **21B**; yield: 95%; mp 102.5–103.5 °C; $[\alpha]_D^{22}$ −41.9 (*c* 1.01, $CHCl_3$).

(4*S*)-3-[(2*R*,3*S*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoyl]-4-phenyloxazolidin-2-one (**22D**) from **21D**; yield: 96%; mp 105.5–106.0 °C; $[\alpha]_D^{23}$ +40.6 (*c* 1.01, $CHCl_3$).

**(2*S*,3*S*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoic Acid** (23A)**; Typical Hydrolysis Procedure (Scheme 11):**[32]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

The 3-(α-azidoacyl)oxazolidinone **22A** (2.0 g, 4.9 mmol) was dissolved in THF (60 mL) and $H_2O$ (20 mL). The mixture was cooled to −4 °C and 30% $H_2O_2$ (3.4 mL, 30 mmol) was added via syringe over 5 min. A soln of LiOH [LiOH·$H_2O$ (0.44 g, 10.5 mmol) in $H_2O$ (3 mL)] was then added dropwise over 10 min. The reaction was stirred at 0 °C for 2 h, then quenched with 1.3 M $Na_2SO_3$ (30 mL). The mixture was warmed to rt and stirred for 30 min. Volatiles were removed by rotary evaporation and the aqueous phase was extracted with $CH_2Cl_2$ (3 × 40 mL) to remove the chiral auxiliary. The remaining aqueous phase was cooled to 0 °C, acidified to ca. pH 1.5 with 6 M HCl, and extracted with $CH_2Cl_2$ (3 × 40 mL). The colorless extracts were dried ($Na_2SO_4$), filtered, and concentrated to give **23A** as an off-white solid; yield: 100%; mp 55.0–57.0 °C; $[\alpha]_D^{22}$ −17.0 (*c* 0.82, $CHCl_3$).

Similarly prepared were:

(2*R*,3*R*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoic acid (**23C**) from **22C**; yield: 100%; mp 77.0–77.5 °C; $[\alpha]_D^{25}$ +16.6 (*c* 1.10, $CHCl_3$).

(2*S*,3*R*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoic acid (**23B**) from **22B**; yield: 100%; oil; $[\alpha]_D^{22}$ +84.8 (*c* 1.66, $CHCl_3$).

(2*R*,3*S*)-2-Azido-3-(4-methoxy-2,6-dimethylphenyl)butanoic acid (**23D**) from **22D**; yield: 99%; oil; $[\alpha]_D^{23}$ −84.3 (*c* 1.065, $CHCl_3$).

**(2*S*,3*S*)-2-Amino-3-(4-methoxy-2,6-dimethylphenyl)butanoic Acid Hydrochloride** (24A)**; Typical Procedure for the Reduction of Azido Acids (Scheme 11):**[32]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

Azide **23A** (1.00 g, 3.8 mmol) was dissolved in glacial AcOH (48 mL) and $H_2O$ (24 mL). The soln was bubbled with argon for 45 min, and then 10% Pd/C (0.1 g) was added, and the mixture was bubbled with argon for another 20 min. The reaction flask was flushed with $H_2$ (3 ×) and was charged with $H_2$ (2.5 atm) and shaken for 24 h. The catalyst was filtered off, and the volatiles were removed by rotary evaporation. To the remaining aqueous phase was added 6 M HCl (6 mL). It was then concentrated, frozen, and lyophilized to give **24A** as an off-white solid; yield: 1.04 g (100%); mp 109.5–111.5 °C; $[\alpha]_D^{22}$ −31.9 (*c* 0.63, MeOH).

Similarly prepared were:

(2*R*,3*R*)-2-Amino-3-(4-methoxy-2,6-dimethylphenyl)butanoic acid hydrochloride (**24C**) from **23C**; yield: 100%; oil; $[\alpha]_D^{25}$ +32.1 (*c* 0.95, MeOH).

(2*S*,3*R*)-2-Amino-3-(4-methoxy-2,6-dimethylphenyl)butanoic acid hydrochloride (**24B**) from **23B**; yield: 100%; mp 110.0–113.0 °C; $[\alpha]_D^{22}$ +62.5 (*c* 0.84, MeOH).

(2*R*,3*S*)-2-Amino-3-(4-methoxy-2,6-dimethylphenyl)butanoic acid hydrochloride (**24D**) from **23D**; yield: 99%; mp 118.5–120.0 °C; $[\alpha]_D^{23}$ −63.7 (*c* 1.02, MeOH).

**(2*S*,3*S*)-2-Amino-3-(4-hydroxy-2,6-dimethylphenyl)butanoic Acid [β,2′,6′-Trimethyltyrosine (TMT),** 25A**]; Typical Procedure for the Hydrolysis of Methyl Ethers and Ion-Exchange Purification of the Final Amino Acids (Scheme 12):**[32]

**CAUTION:** *Trifluoroacetic acid is corrosive and severely irritating to skin, eyes, and mucous membranes. Inhalation can destroy tissue and be fatal. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

Methyl ether **24A** (1.15 g, 4.85 mmol) was dissolved in TFA (50 mL). The soln was cooled to −4 °C, thioanisole (3.98 mL, 33.92 mmol) was added, and the mixture was stirred for 10 min. Finally, TfOH (6.43 mL, 72.69 mmol) was added via syringe and the light yellow, cloudy soln was stirred at 0 to −4 °C

for 30 min. Volatiles were removed by rotary evaporation. The residue, a dark red-brownish tar, was dissolved in $H_2O$ (100 mL). The soln was loaded on an ion-exchange column (2.5 cm × 46 cm) with Amberlite IR-120 ($H^+$) resin (150 g). The column was washed with $H_2O$ until the eluant was neutral. The amino acid was washed out with 5% $NH_4OH$ soln; this process was monitored by TLC. Fractions containing the product were combined, concentrated to remove excess $NH_4OH$, frozen, and lyophilized to give an off-white solid **25A**; yield: 1.02 g (94%); mp 153.0–156.0 °C (dec); $[\alpha]_D^{22}$ −35.36 (*c* 0.51, MeOH); $^1H$ NMR ($D_2O$, δ): 1.14 (d, *J* = 7.4 Hz, 3H), 2.08 (s, 3H), 2.16 (s, 3H), 3.36 (m, 1H), 4.17 (d, *J* = 11.0 Hz, 1H), 6.41 (s, 2H); $^{13}C$ NMR ($D_2O$/1,4-dioxane, δ): 14.6, 18.4, 34.4, 57.4, 113.2, 117.5, 127.9, 131.3, 138.4, 143.9, 154.5, 171.4.

Similarly prepared were:

(2*R*,3*R*)-2-Amino-3-(4-hydroxy-2,6-dimethylphenyl)butanoic acid (**25C**) from **24C**; yield: 90%; mp 168.0–170.0 °C (dec); $[\alpha]_D^{22}$ +35.5 (*c* 0.71, MeOH); $^1H$ NMR ($D_2O$, δ): 1.12 (d, *J* = 7.3 Hz, 3H), 2.06 (s, 3H), 2.14 (s, 3H), 3.23 (m, 1H), 3.86 (d, *J* = 11.6 Hz, 1H), 6.32–6.39 (br s, 2H); $^{13}C$ NMR ($D_2O$/1,4-dioxane, δ): 17.4, 18.3, 21.6, 21.8, 38.1, 59.9, 116.9, 118.7, 128.8, 139.0, 140.6, 157.0, 174.4.

(2*S*,3*R*)-2-Amino-3-(4-hydroxy-2,6-dimethylphenyl)butanoic acid (**25B**) from **24B**; yield: 96%; hygroscopic solid; $[\alpha]_D^{22}$ +46.6 (*c* 0.79, MeOH); $^1H$ NMR ($D_2O$, δ): 1.23 (d, *J* = 7.2 Hz, 3H), 2.02 (s, 3H), 2.15 (s, 3H), 3.50 (m, 1H), 4.11 (d, *J* = 11.3 Hz, 1H), 6.36 (s, 2H); $^{13}C$ NMR ($D_2O$/1,4-dioxane, δ): 15.2, 20.6, 20.8, 35.8, 56.1, 115.1, 116.7, 128.7, 138.9, 153.9, 171.4.

(2*R*,3*S*)-2-Amino-3-(4-hydroxy-2,6-dimethylphenyl)butanoic acid (**25D**) from **24D**; yield: 96%; mp 175.0–178.0 °C (dec); $[\alpha]_D^{22}$ −42.8 (*c* 0.50, MeOH); $^1H$ NMR ($D_2O$, δ): 1.20 (d, *J* = 7.2 Hz, 3H), 2.11 (br s, 6H), 3.44 (m, 1H), 3.85 (d, *J* = 11.0 Hz, 1H), 6.36 (s, 2H); $^{13}C$ NMR ($D_2O$/1,4-dioxane, δ): 15.1, 20.6, 20.8, 35.8, 58.4, 115.6, 116.7, 130.2, 138.9, 139.0, 153.5, 174.0.

## 9.1.3 β-Methylnaphthylalanine

β-Methylnaphthylalanines are interesting topographically constrained amino acids that may be viewed as phenylalanine analogues but with a bulkier and more hydrophobic aromatic moiety. As such, their energy barrier to rotation about $\chi^2$ would be expected to be more hindered and provide insight into the stereostructure required for the hydrophobic binding site. Since 2′-naphthylalanine and 1′-naphthylalanine have been used extensively in peptide design,[50,51] and often the 2′- and 1′-derivatives give quite different bioactivities,[52] the more constrained β-methyl analogues should be very interesting in such studies. The asymmetric synthesis of β-methylnaphthylalanines has been accomplished by two different methods: an asymmetric aziridine ring opening[27] and a chiral auxiliary approach.[53]

### Other Related Amino Acids

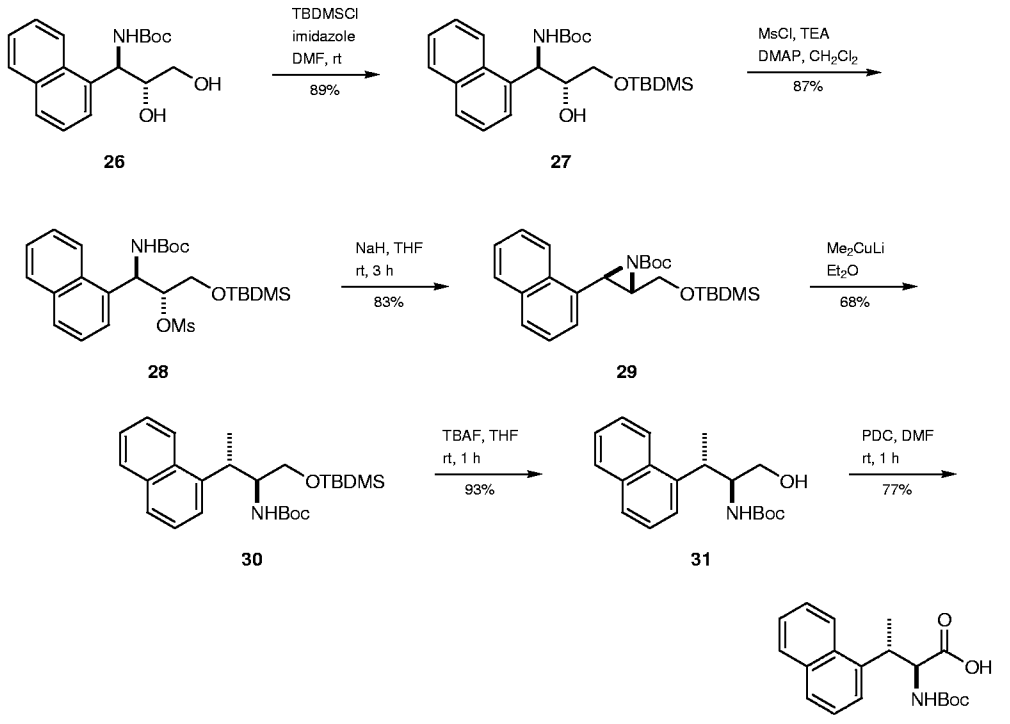
A closely related amino acid system, but of a naturally occurring amino acid, are the four isomers of β-methyltryptophan. An asymmetric synthesis of all four isomers has been developed.[54] Incorporation into a truncated cyclic α-MSH analogue, Ac-Nle$^4$-*c*[-Asp$^5$,D-Phe$^7$,Lys$^{10}$-]α-MSH(4–10)-$NH_2$ (MT-II) gave a series of four diastereomeric β-MeTrp$^9$-containing derivatives of MT-II[55,56] which had very unique biological activity profiles, especially in prolonged bioactivities, and provided the first evidence that prolonged biological activity at a seven transmembrane G-protein-coupled receptor was a function not of molecular recognition per se, but rather of the topographical relationships of key pharmacophore groups.

### 9.1.3.1 Asymmetric Aziridine Ring Opening

$N^{\alpha}$-Boc-β-methylnaphthylalanine (**32**) has been synthesized by stereo- and regioselective ring opening of aziridine **29** (Scheme 13).[27] This method cannot provide all four isomers of β-methylnaphthylalanine.

*for references see p 49*

**Scheme 13** Synthesis of $N^{\alpha}$-Boc-β-methylnaphthylalanine via Aziridine Ring Opening[27]




**(1*R*,2*R*)-1-(*tert*-Butoxycarbonylamino)-3-(*tert*-butyldimethylsiloxy)-1-(1-naphthyl)propan-2-ol (27):**[27]
To a soln of (2*R*,3*R*)-3-(*tert*-butoxycarbonylamino)-3-(1-naphthyl)propane-1,2-diol (**26**; 930 mg, 2.93 mmol) in DMF (13 mL) were added TBDMSCl (487 mg, 3.21 mmol) and imidazole (440 mg, 6.46 mmol). The reaction was monitored by TLC. After 24 h, $H_2O$ (20 mL) was added, and the aqueous phase was extracted with $Et_2O$ (3 × 15 mL). The combined organic phases were washed with sat. aq $NH_4Cl$ and dried ($MgSO_4$). The solvents were removed under reduced pressure. The crude product was purified by chromatography (EtOAc/hexane 0–5%) affording **27** as a white solid; yield: 1.13 mg (89%); mp 71–72 °C; $[\alpha]_D$ +15 (*c* 1.0, $CHCl_3$).

**(1*R*,2*R*)-1-(*tert*-Butoxycarbonylamino)-3-(*tert*-butyldimethylsiloxy)-2-[(methylsulfonyl)oxy]-1-(1-naphthyl)propane (28):**[27]
To a soln of alcohol **27** (1.0 g, 2.31 mmol) in $CH_2Cl_2$ (5 mL) at −15 °C were added TEA (0.36 mL, 2.55 mmol), DMAP (14.2 mg, 0.11 mmol), and MsCl (0.19 mL, 2.50 mmol). The mixture was allowed to warm to rt with stirring, and the reaction progress was monitored by TLC. When no starting material could be detected (ca. 5 h), $H_2O$ (5 mL) was added and the aqueous phase was extracted with $CH_2Cl_2$ (3 × 15 mL). The combined organic phases were washed with cold 10% HCl, sat. aq $NaHCO_3$, and $H_2O$, dried ($MgSO_4$), concentrated under reduced pressure, and purified by column chromatography (EtOAc/hexanes 10–15%) to afford **28** as an oil; yield: 1.02 g (87%); $[\alpha]_D$ +38.3 (*c* 1.0, $CHCl_3$).

**(2*R*,3*R*)-1-(*tert*-Butoxycarbonyl)-2-[(*tert*-butyldimethylsiloxy)methyl]-3-(1-naphthyl)aziridine (29):**[27]
To a suspension of NaH [3.61 mmol; from an 80% mixture with paraffin (108.5 mg) washed with anhyd hexane under $N_2$] in THF (2 mL) was added a soln of **28** (460 mg, 0.9 mmol) in THF (3 mL). After 3 h of stirring at rt, EtOAc (5 mL) and some drops of MeOH were added to the mixture to remove any trace of unreacted hydride. The solvent was removed under reduced pressure, and the crude mixture was treated with $H_2O$/EtOAc (1:1). The aqueous phase was extracted with EtOAc (3 × 7 mL), and the combined organic phases were dried ($MgSO_4$) and concentrated. Purification of the crude mixture by column chromatography (EtOAc/hexanes 97:3) gave aziridine **29** as a white solid; yield: 0.309 g (83%); mp 63–65 °C; $[\alpha]_D$ −67.2 (*c* 0.97, $CHCl_3$).

**(2*S*,3*S*)-2-(*tert*-Butoxycarbonylamino)-1-(*tert*-butyldimethylsiloxy)-3-(1-naphthyl)butane** (30):[27]
To a stirred slurry of CuI (114 mg, 0.60 mmol) in anhyd $Et_2O$ (2 mL) at 0 °C was added 1.6 M MeLi in $Et_2O$ (0.75 mL, 1.2 mmol), and the mixture was stirred for several min. A soln of aziridine **29** (100 mg, 0.24 mmol) in anhyd $Et_2O$ (2.4 mL) was added via cannula and the reaction was stirred under $N_2$ and monitored by TLC. When no starting material could be detected (ca. 2 h), a mixture of sat. aq $NH_4Cl$ and $NH_4OH$ (8:1, 4 mL) was added to the reaction, and the organic layer was extracted with $Et_2O$ (3 × 4 mL). The combined organic phases were dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by flash chromatography (EtOAc/hexanes 99.5:0.5) to give **30**; yield: 70 mg (68%); $[\alpha]_D$ +12.4 (*c* 1.1, $CHCl_3$).

**(2*S*,3*S*)-2-(*tert*-Butoxycarbonylamino)-3-(1-naphthyl)butan-1-ol** (31):[27]
To a soln of **30** (115 mg, 0.27 mmol) in THF (1.4 mL) at 0 °C was added a soln of TBAF·$H_2O$ (0.14 g, 0.53 mmol) in THF (0.82 mL). The reaction was stirred at rt and monitored by TLC. After 60 min, the soln was washed with $H_2O$ and the aqueous layer was extracted with $Et_2O$. The combined organic phases were dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by chromatography (EtOAc/hexane) to afford **31** as a colorless oil; yield: 79 mg (93%); $[\alpha]_D$ +29.1 (*c* 1.1, $CHCl_3$).

**(2*S*,3*S*)-2-(*tert*-Butoxycarbonylamino)-3-(1-naphthyl)butanoic Acid** (32):[27]
To a soln of alcohol **31** (53 mg, 0.17 mmol) was added a soln of PDC (318 mg, 0.85 mmol) in DMF (0.64 mL) under $N_2$ at rt. After stirring for 24 h, $H_2O$ (8.5 mL) and $Et_2O$ (8.5 mL) were added. The aqueous phase was extracted with $Et_2O$ (3 × 6 mL), and the combined organic phases were dried ($MgSO_4$) and concentrated. The crude product was dissolved in EtOAc and extracted with sat. aq $NaHCO_3$. The aqueous layer was washed with EtOAc, acidified with 2 M HCl, and extracted with EtOAc. The organic phases were then dried ($MgSO_4$) and concentrated to give **32**; yield: 43 mg (77%); $[\alpha]_D$ +66.7 (*c* 0.9, MeOH); methyl ester: >97% ee [determined by chiral HPLC analysis; CHIRALCEL OD (25 cm) column, 30 °C, hexane/iPrOH 99:1, 0.45 mL·$min^{-1}$, λ 254 nm; major peak at 31.7 min].

### 9.1.3.2 Through Chiral Auxiliary Oxazolidinone

The methods used are similar to those described in Section 9.1.2.1. See ref[53] for details.

## 9.1.4 1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid

Another way to topographically constrain aromatic amino acids, such as phenylalanine, tyrosine, tryptophan, and histidine, in the χ space[3] is through cyclization of their α-amino groups with the aromatic ring. A typical example is 1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid (Tic) in which a methylene group serves to link the α-amino group to the 2′-carbon of the aromatic ring. Thus, these analogues can also be viewed as *N*-methyl- or 2′-methyl-substituted aromatic amino acids depending upon where the disconnection is made. The topographical consequence is that only a gauche(−) and gauche(+) conformation is possible at $\chi^1$, which usually can be determined by simple 2-D NMR methods. This amino acid has proven a very useful one in structure–activity relationships of enzymes, hormones, neurotransmitters, cytokins, and other bioactive peptides. For example, in the case of peptide hormones and neurotransmitters, it has been incorporated into opioids,[31,56–62] μ-opioid receptor antagonists based on a modified somatostatin-based template,[63–69] deltorphins,[58,70] oxytocins,[71] vasopressins,[72] and melanotropins,[73] often with dramatic effects on biological activity including selectivity, potency, and efficacy.

Synthesis of Tic analogues has utilized cyclization with formaldehyde (Section 9.1.4.1) since it involves only one step; however, as discussed below, other methods can also be used.

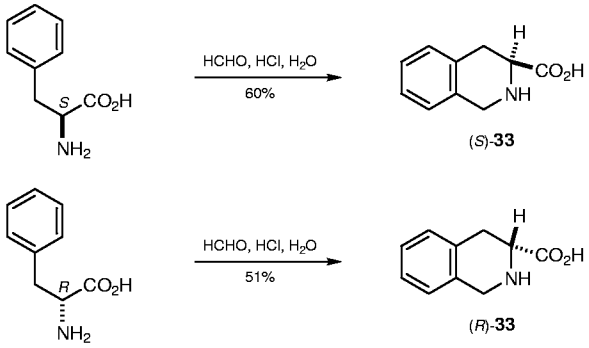
Similar cyclic analogues of substituted phenylalanines[74,75] and histidine[28] have been prepared and incorporated into biologically active peptides.

*for references see p 49*

### 9.1.4.1 Cyclization of Phenylalanine with Formaldehyde

Synthesis of Tic (**33**) from phenylalanine and formaldehyde is a very effective route (Scheme 14), which has been used for large-scale production.[72,76–78]

**Scheme 14** Synthesis of Tic through Cyclization of Phenylalanine and Formaldehyde[72,76–78]



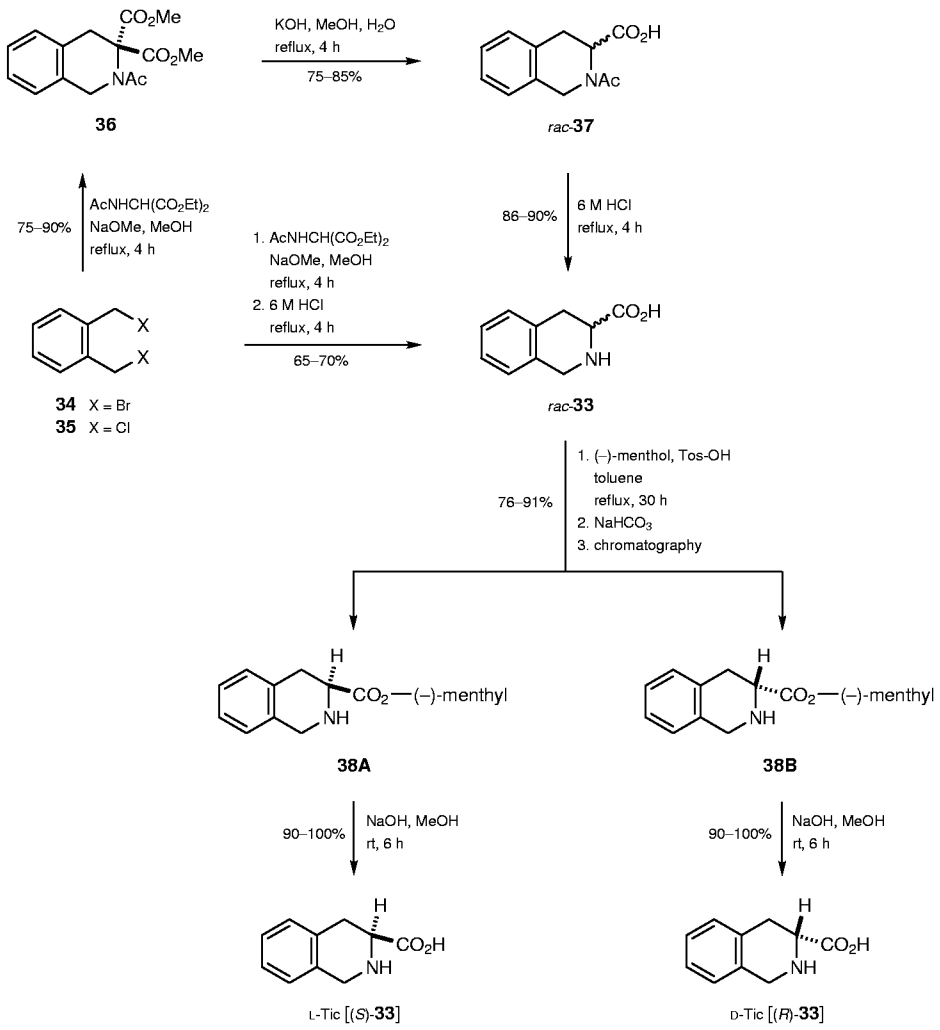

**(*S*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid [L-Tic, (*S*)-33]:**[72]
A mixture of (*S*)-phenylalanine (82.5 g, 0.5 mol), 35–40% aq HCHO (200 mL), and concd aq HCl (600 mL) was intensively stirred at 60 °C for 1 h. After addition of a further portion of HCHO (80 mL) and concd aq HCl (160 mL), stirring was continued for an additional 3 h at 60 °C. The mixture was then left to stand overnight in a refrigerator at 5 °C; precipitated L-Tic hydrochloride was filtered off, washed with a small amount of ice-cold $H_2O$, and dried under reduced pressure over solid KOH. The crude product was dissolved in refluxing 60% aq EtOH (3 L). The hot soln was then neutralized with aq $NH_3$ to pH 6–7 and left to stand for 5 h in a refrigerator at 5 °C. Precipitated L-Tic [(*S*)-**33**] was filtered off, washed with aq EtOH, and dried in a desiccator; yield: 53.0 g (60%); mp >280 °C (dec). After recrystallization (aq EtOH); overall yield: 51%; $[\alpha]_D^{25}$ −172 (*c* 1.0, 1 M NaOH); TLC: $R_f$ 0.55 (Chiralplate, MeCN/MeOH/$H_2O$ 4:1:1); *R*-isomer <2%.

**(*R*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid [D-Tic, (*R*)-33]:**[72]
The procedure described above was followed, with (*R*)- rather than (*S*)-phenylalanine; yield: 51%; $[\alpha]_D^{25}$ +172 (*c* 1.0, 1 M NaOH); TLC: $R_f$ 0.51 (Chiralplate, MeCN/MeOH/$H_2O$ 4:1:1); *S*-isomer <2%.

### 9.1.4.2 Alkylation Followed by Resolution Using Menthol

The synthesis of racemic Tic (*rac*-**33**) can be accomplished by alkylation of acetamidomalonates in a reasonable yield (Scheme 15). Racemic Tic can then be subjected to resolution using menthol.[79] This route is a good alternative for synthesizing both enantiomers of Tic.

**Scheme 15** Synthesis of Tic via Resolution Using (−)-Menthol[79]




**Dimethyl 2-Acetyl-1,2,3,4-tetrahydroisoquinoline-3,3-dicarboxylate (36):**[79]
To a well-stirred suspension of 1,2-bis(bromomethyl)benzene (**34**; 26.4 g, 0.1 mol) and diethyl 2-(acetylamino)malonate (21.7 g, 0.1 mol) in MeOH (175 mL) was added 30% NaOMe in MeOH (18 mL, 0.1 mol) dropwise over 10 min at rt. The mixture was heated to reflux, treated with additional 30% NaOMe in MeOH (18 mL, 0.1 mol) over 2 h, and stirred for a further 2 h at reflux temperature. The soln was cooled to rt, concentrated, and the solid residue was distributed between $H_2O$ (100 mL) and EtOAc (150 mL). The aqueous phase was extracted with EtOAc (2 × 75 mL), and the combined organic phases were washed with brine (100 mL), dried ($Na_2SO_4$), and concentrated. The crude residue was recrystallized (iPrOH/*tert*-butyl methyl ether 1:1) to give **36**; yield: 21.5 g (75%); mp 141–143 °C.

**(±)-2-Acetyl-1,2,3,4-tetrahydroisoquinoline-3-carboxylic Acid (*rac*-37):**[79]
KOH (4.55 g, 81 mmol) was added portionwise to a well-stirred suspension of **36** (10.65 g, 37 mmol) in a mixture of MeOH (100 mL) and $H_2O$ (20 mL). The mixture was heated to reflux over 4 h, allowed to cool, and concentrated. The residue was treated with EtOAc and 2 M HCl to pH 1 (**CAUTION:** *decarboxylation*). After separation, the aqueous phase was extracted with EtOAc (3 × 50 mL), and the combined organic phases were dried ($Na_2SO_4$) and concentrated. The residue was stirred with *tert*-butyl methyl ether (10 mL) over 10 min and *rac*-**37** was collected by suction; yield: 6.84 g (85%); mp 171–173 °C.

*for references see p 49*

**(±)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid (*rac*-33):**[79]
*rac*-**37** (7.6 g, 35 mmol) was heated to reflux in 6 M aq HCl (50 mL) over 4 h and then cooled to 0 °C. The mixture was treated with 25% aq $NH_4OH$ to neutral pH and stirred for an additional 1 h at 0 °C to complete the precipitation. The residue was filtered, washed with $H_2O$ (2 × 10 mL), and dried at 60 °C under reduced pressure; yield: 5.3 g (86%); mp 321–325 °C.

**(±)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid (*rac*-33) from 1,2-Bis(chloromethyl)benzene (35):**[79]
To a mixture of 1,2-bis(chloromethyl)benzene (**35**; 5.00 kg, 28.6 mol) and diethyl 2-(acetylamino)-malonate (6.2 kg, 28.6 mol) in MeOH (49 L), a soln of 30% NaOMe in MeOH (5.14 kg, 28.6 mol) was added dropwise over 30 min with simultaneous heating to reflux. The mixture was boiled for 10 min before additional 30% NaOMe soln (5.14 kg, 28.6 mol) was added dropwise to the heated mixture over 2 h, and refluxed altogether 4 h. The suspension was concentrated to dryness and the remaining residue was treated with 6 M aq HCl (43 L) and heated to reflux for 4 h (**CAUTION:** *decarboxylation*). The mixture was cooled to 0 °C, buffered to pH 6.5 with 25% $NH_4OH$ (6 L), and stirred for a further 30 min. The precipitate *rac*-**33** was filtered, washed twice with small portions of $H_2O$ and toluene, and dried at 60 °C under reduced pressure; yield: 3.3 kg (65%); mp 320–325 °C.

**(−)-Menthyl (*R*)- and (*S*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylate (**38B **and** 38A**):**[79]
A soln of TosOH · $H_2O$ (10 g, 53 mmol) in toluene (150 mL) was heated to reflux for 1 h in a Dean–Stark trap to separate toluene (50 mL) and $H_2O$ (1 mL). The mixture was cooled to 80–90 °C, (±)-1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid (*rac*-**33**; 6.3 g, 36 mmol) and (−)-menthol (8.3 g, 53 mmol) were added portionwise, and the mixture was heated to reflux for an additional 30 h to separate the theoretical amount of $H_2O$. The mixture was then cooled to rt and washed with 2 M aq $NaHCO_3$ (3 × 75 mL). The combined organic phases were washed with brine (50 mL), dried ($Na_2SO_4$), and concentrated to dryness; yield: 10.3 g (91%). The residual crude product was separated into its diastereomers **38B** and **38A** by chromatography (silica gel, 300 g, 30–70 μm, cyclohexane/EtOAc 7:3); yield of **38B**: 4.2 g (76%); $[\alpha]_D^{20}$ −10 (*c* 0.5, MeOH); yield of **38A**: 4.6 g (83%); $[\alpha]_D^{20}$ −117 (*c* 1, MeOH).

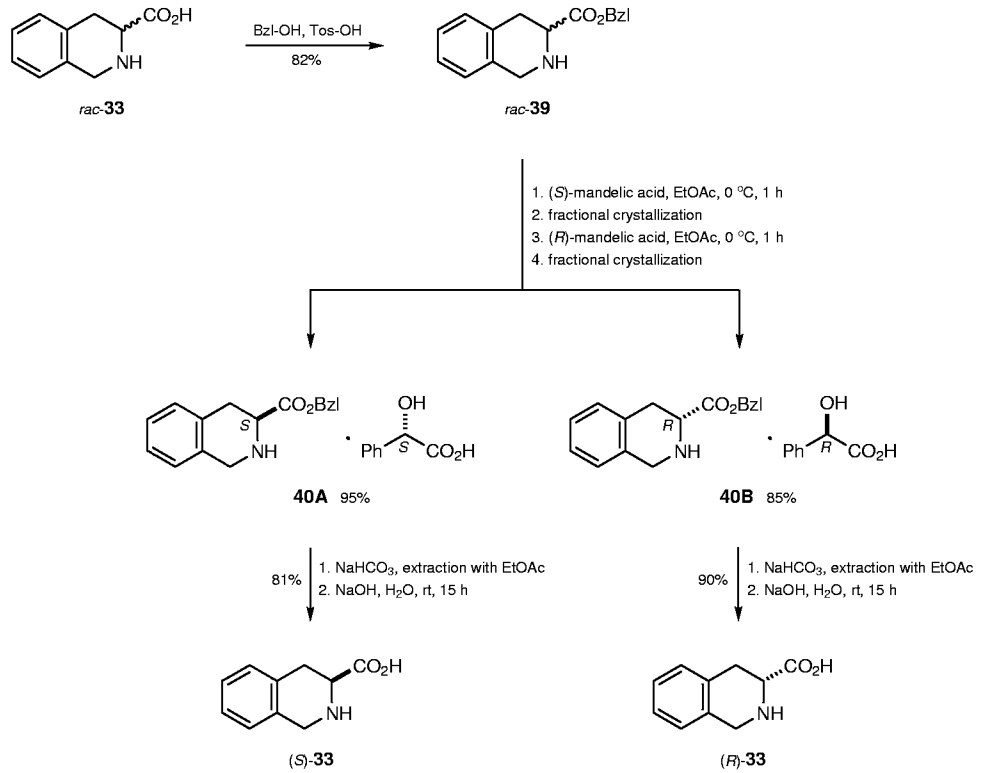
**(*R*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid [D-Tic, (*R*)-33]:**[79]
A mixture of (−)-menthyl (*R*)-1,2,3,4-tetrahydroisoquinoline-3-carboxylate (**38B**; 1.1 g, 3.5 mmol) and NaOH pellets (0.21 g, 5.3 mmol) in MeOH (15 mL) was stirred for 6 h at rt and then buffered to neutral pH with 2 M HCl. To complete the precipitation of the title compound the mixture was allowed to stand overnight at rt and was then filtered, washed with small portions of $H_2O$, and dried under reduced pressure; yield: 0.564 g (90%); mp 321–323 °C; $[\alpha]_D^{20}$ +138 (*c* 1, 0.1 M HCl); >99% ee (by GC).

**(*S*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid [L-Tic, (*S*)-33]:**[79]
This compound was prepared analogously to (*R*)-Tic, using (−)-menthyl (*S*)-1,2,3,4-tetrahydroisoquinoline-3-carboxylate (**38A**) as the starting material; yield: 0.590 g (94%); mp 320–323 °C; $[\alpha]_D^{20}$ −139 (*c* 1, 0.1 M HCl); >99% ee (by GC).

## 9.1.4.3 Resolution Using Mandelic Acid

(*R*)- and (*S*)-Mandelic acid can also be used to resolve the benzyl esters **39**[77,80] of racemic Tic (Scheme 16).[79] Both isomers of Tic are released by hydrolysis of their respective benzyl esters. This also constitutes a valuable synthetic route.

**Scheme 16** Synthesis of Tic by Resolution Using Mandelic Acid[79]




**Benzyl (*S*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylate (*S*)-Mandelic Acid Salt** (40A) **and Benzyl (*R*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylate (*R*)-Mandelic Acid Salt** (40B)**:**[79]
Oily, racemic benzyl 1,2,3,4-tetrahydroisoquinoline-3-carboxylate[77,80] (*rac*-**39**; 67.0 g, 0.25 mol) was dissolved in EtOAc (500 mL). (*S*)-Mandelic acid (38.0 g, 0.245 mol) was added in portions and the mixture was stirred for 1 h at 0 °C. The precipitate was collected by filtration, washed with small portions of EtOAc and iPrOH, and dried under reduced pressure. The mother liquor, containing enriched (*R*)-**39**, was liberated from residual (*S*)-mandelic acid by treatment with sat. aq $NaHCO_3$ and extraction into EtOAc to give a yellowish oil, which was subsequently treated with (*R*)-mandelic acid (18.0 g, 0.116 mol), and stirred for an additional 1 h at 0 °C. The colorless precipitate was filtered, washed with EtOAc and iPrOH, and dried under reduced pressure. Yield of (*S*)-Tic·(*S*)-mandelic acid (**40A**): 50.0 g (95%); mp 98–100 °C; $[\alpha]_D^{20}$ +13.3 (*c* 1, MeOH); >99% de. Yield of (*R*)-Tic·(*R*)-mandelic acid (**40B**): 44.6 g (85%); mp 99–101 °C; $[\alpha]_D^{20}$ +13.3 (*c* 1, MeOH); >99% de.

**(*R*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid** [D-Tic, (*R*)-33]**:**[79]
Benzyl (*R*)-1,2,3,4-tetrahydroquinoline-3-carboxylic acid, released quantitatively from its (*R*)-mandelic acid salt (**40B**; 2.7 g, 6.4 mmol) by treatment with sat. aq $NaHCO_3$ (100 mL) and extraction with EtOAc (60 mL), was stirred over 15 h at rt in a soln of NaOH pellets (310 mg, 7.8 mmol) in $H_2O$ (25 mL) and then buffered to pH 4.5 with 2 M HCl. Stirring was continued for 1 h, and the precipitate was collected by filtration then dried under reduced pressure; yield of (*R*)-**33**: 1.02 g (90%); mp 321–325 °C; $[\alpha]_D^{20}$ +140 (*c* 1, 0.1 M HCl); >99% ee (by GC).

**(*S*)-1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid** [L-Tic, (*S*)-33]**:**[79]
This compound was prepared in the same manner as (*R*)-Tic, using benzyl (*S*)-1,2,3,4-tetrahydroisoquinoline-3-carboxylate (*S*)-mandelic acid salt (**40A**) as starting material; yield of (*S*)-**33**: 920 mg (81%); mp 322–327 °C.

*for references see p 49*

## 9.1.5 Substituted Phenylglycine

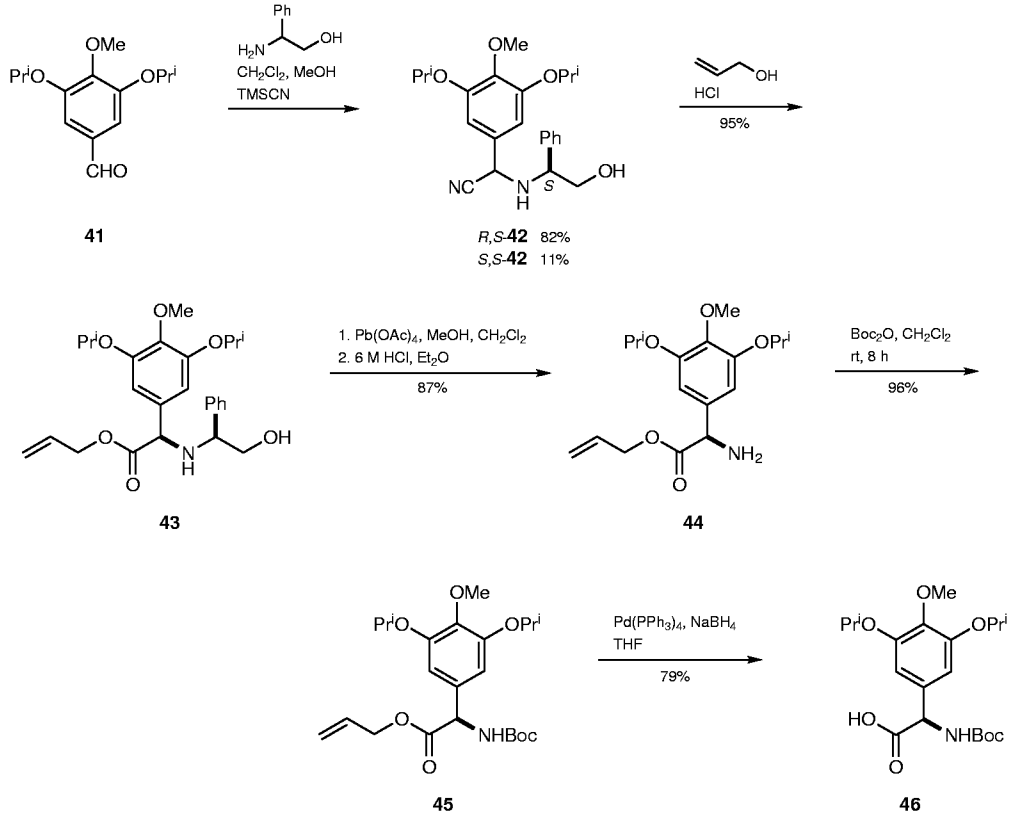
Phenylglycine derivatives are key intermediates in the synthesis of vancomycin.[81–85] This amino acid is also very useful for incorporation into oxazolidinones as a chiral auxiliary[45,47,48,53,86,87] and for modulating bioactivities of peptides and peptidomimetics.[85,88]

There are several possible approaches to the synthesis of phenylglycines, including the Strecker reaction and the Evans electrophilic azidation reaction.[81]

### 9.1.5.1 The Chiral Strecker Reaction

The chiral Strecker reaction has been used in an effective synthesis of the substituted phenylglycine **46** (Scheme 17).[81] α-Phenylglycinol was used as the chiral amine supply.

**Scheme 17** Synthesis of a Substituted Phenylglycine via the Chiral Strecker Reaction[81]




**(*R*)-(3,5-Diisopropoxy-4-methoxyphenyl)[(*S*)-2-hydroxy-1-phenylethylamino]acetonitrile (*R,S*-42):**[81]

**CAUTION:** *Hydrogen cyanide (HCN) is produced as a toxic byproduct in the quenching of this reaction. Appropriate safety precautions and procedures should be employed.*

To a soln of aldehyde **41**[89] (1.00 g, 3.97 mmol) in $CH_2Cl_2$ (27 mL) was added (*S*)-2-phenylglycinol (0.65 g, 4.76 mmol) at rt. After being stirred at rt for 2 h, the mixture was cooled to 0 °C, and MeOH (3.3 mL) and TMSCN (0.79 mL, 5.96 mmol) were then introduced successively. After being stirred at 0 °C for 1 h and at rt for 12 h, the mixture was carefully quenched by the addition of aq citric acid soln (**CAUTION:** *HCN; perform in an efficient fume hood*) and stirring was continued for another 30 min. The aqueous phase was then neutralized with pH 7 phosphate buffer and extracted with $CH_2Cl_2$. The combined

organic extracts were washed with brine, dried, and concentrated to dryness. Purification by flash chromatography ($Et_2O$/heptane 5:6 and 5:4) afforded compound **42A** {yield: 1.30 g (82%); $[\alpha]_D$ +49 (*c* 0.7, $CHCl_3$)} and its (*S,S*)-diastereomer **42B** {yield: 170 mg (11%); $[\alpha]_D$ +32 (*c* 0.6, $CHCl_3$)}.

**Allyl (*R*)-(3,5-Diisopropoxy-4-methoxyphenyl)[(*S*)-2-hydroxy-1-phenylethylamino]acetate (43):**[81]
To a soln of amino nitrile *R,S*-**42** (1.01 g, 2.54 mmol) in a minimum amount of allyl alcohol was added allyl alcohol saturated with gaseous HCl at 0 °C. After being stirred at 0 °C for 2 h and at rt for 8 h, the volatiles were removed, and the residue was redissolved in $H_2O$, neutralized first with aq $NaHCO_3$ (to pH ca. 5–6) then with phosphate buffer (pH 7), and extracted with $CH_2Cl_2$. The combined organic extracts were washed with brine, dried, and concentrated to dryness. Purification by flash chromatography (silica gel, EtOAc/heptane from 1:5 to 1:2) afforded compound **43**; yield: 1.10 g (95%); $[\alpha]_D$ −6.0 (*c* 0.3, $CHCl_3$).

**Allyl (*R*)-(3,5-Diisopropoxy-4-methoxyphenyl)glycinate (44):**[81]

**CAUTION:** *Lead and inorganic lead compounds are classified as possible human carcinogens. Appropriate safety precautions and procedures should be taken when handling and disposing of lead compounds.*

To a soln of $Pb(OAc)_4$ (1.01 g, 2.33 mmol) in MeOH (20 mL) cooled at 0 °C was added a soln of ester **43** (866.7 mg, 1.90 mmol) in $CH_2Cl_2$ precooled at 0 °C. After being stirred at 0 °C for 30 min, volatiles were removed under reduced pressure. The residue was redissolved in $H_2O$, neutralized with phosphate buffer (pH 7), and extracted with $CH_2Cl_2$. The combined organic extracts were washed with brine, dried, and concentrated to give the crude imine which was used directly for the next step. The crude imine, dissolved in $Et_2O$ (4 mL), was treated with 6 M HCl for 90 min at rt. The volatiles were removed and the residue was redissolved in $H_2O$ and extracted with $Et_2O$/heptane to remove the neutral compounds. The aqueous phase was then neutralized with phosphate buffer and extracted with $CH_2Cl_2$. The combined organic extracts were washed with brine, dried, and concentrated to give **44**; yield: 556.0 mg (87%); $[\alpha]_D$ −48 (*c* 0.2, $CHCl_3$). Compound **44** is not very stable and is better protected directly as a carbamate.

**Allyl (*R*)-*N*-(*tert*-Butoxycarbonyl)(3,5-diisopropoxy-4-methoxyphenyl)glycinate (45):**[81]
To a soln of amino ester **44** (330 mg, 0.98 mmol) in $CH_2Cl_2$ was added $Boc_2O$ (321.0 mg, 1.47 mmol). After being stirred at rt for 8 h, the volatiles were removed under reduced pressure and the crude product was purified by flash chromatography (EtOAc/heptane 1:5) to afford **45**; yield: 411.0 mg (96%); $[\alpha]_D$ −63 (*c* 0.2, $CHCl_3$).

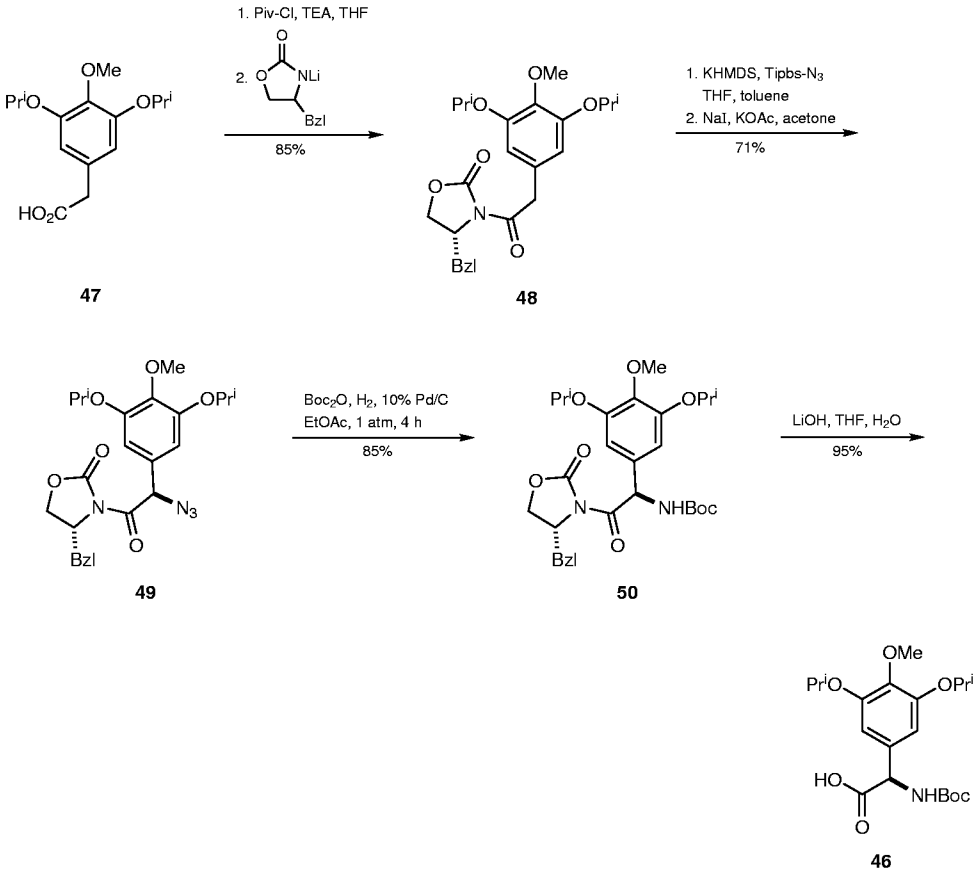
**(*R*)-*N*-(*tert*-Butoxycarbonyl)(3,5-diisopropoxy-4-methoxyphenyl)glycine (46):**[81]
A soln of compound **45** (172 mg, 0.39 mmol) and $Pd(PPh_3)_4$ (9.1 mg, 7.87 μmol) in THF (5 mL) was stirred at rt for 10 min. The mixture was cooled to 0 °C and $NaBH_4$ (30 mg, 0.78 mmol) was introduced. After being stirred at 0 °C for 1 h, the mixture was diluted with aq $NH_4Cl$, acidified with 1 M HCl and extracted with EtOAc. The combined organic extracts were washed with brine, dried ($Na_2SO_4$), and concentrated. Purification by flash chromatography (EtOAc/heptane 1:3, then $CH_2Cl_2$/MeOH 20:1) afforded **46**; yield: 124.0 mg (79%); $[\alpha]_D$ −92 (*c* 1.5, $CHCl_3$).

## 9.1.5.2 Evans Electrophilic Azidation

Diastereoselective α-azidation via the chiral oxazolidinone auxiliary provides a valuable means of synthesizing highly substituted phenylglycine **46** (Scheme 18).[81]

*for references see p 49*

**Scheme 18** Synthesis of a Substituted Phenylglycine via Chiral Auxiliary Assisted Azidation[81]




Tipbs-$N_3$ = triisopropylbenzenesulfonyl azide

**(4*R*)-4-Benzyl-3-[2-(3,5-diisopropoxy-4-methoxyphenyl)acetyl]oxazolidin-2-one (48):**[81]

A soln of **47** (5.35 g, 18.98 mmol) in anhyd THF was cooled to −78 °C, whereupon TEA (3.5 mL, 24.90 mmol) and redistilled Piv-Cl (2.6 mL, 21.11 mmol) were added successively via a syringe with stirring. The resulting slurry was stirred at −78 °C for 15 min and at 0 °C for 45 min, and then cooled again to −78 °C. In a separate flask, (4*R*)-4-benzyloxazolidin-2-one (4.10 g, 23.16 mmol) was dissolved in anhyd THF (40 mL) and cooled to −78 °C, a soln of 1.6 M BuLi in hexane (16.6 mL, 26.56 mmol) was added slowly, and the mixture was stirred at −78 °C for another 5 min after completion of the addition. The metalated oxazolidinone (yellow colored) was transferred via a cannula to the flask containing the mixed anhydride. The resulting slurry was stirred for 15 min at −78 °C and then warmed to rt over 90 min. The reaction was quenched by addition of aq $NH_4Cl$, the volatiles were removed under reduced pressure, and the aqueous soln was extracted with EtOAc. The combined organic phases were washed with brine, dried ($Na_2SO_4$), and concentrated under reduced pressure. The crude product was purified by flash chromatography (EtOAc/heptane 1:2) to afford **48** as a yellow oil; yield: 7.12 g (85%); $[\alpha]_D$ −40 (*c* 1.0, $CHCl_3$).

**(4*R*)-3-[(2*R*)-2-Azido-2-(3,5-diisopropoxy-4-methoxyphenyl)acetyl]-4-benzyloxazolidin-2-one (49):**[81]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

To a soln of **48** (6.50 g, 14.70 mmol) in THF (30 mL), stirred at −78 °C under argon, was added 0.5 M KHMDS in toluene (40 mL). The soln was stirred for 30 min at −78 °C, then a precooled (−78 °C) soln of triisopropylbenzenesulfonyl azide (Tipbs-$N_3$; 5.50 g, 17.80 mmol) in THF (30 mL) was added via a cannula. After being stirred for 5 min, the reaction was quenched with AcOH, warmed to 0 °C, and

diluted with sat. aq $NH_4Cl$. The volatiles were removed under reduced pressure and the aqueous phase was extracted with EtOAc. The combined organic layers were washed with brine, dried ($Na_2SO_4$), and concentrated. The resulting residue was dissolved in acetone (30 mL), then NaI (11.00 g, 73.39 mmol) and KOAc (4.50 g, 45.91 mmol) were added. The mixture was stirred overnight at rt, then filtered through a short pad of Celite. The volatiles in the filtrate were removed, and the residue was dissolved in aq $Na_2S_2O_3$ and extracted with EtOAc. The combined organic extracts were washed with brine, dried ($Na_2SO_4$), and concentrated. Purification by flash chromatography (silica gel, EtOAc/heptane 1:5 then 1:4) afforded the azide **49**; yield: 5.04 g (71%); $[\alpha]_D$ −162 (*c* 1.5, $CHCl_3$).

***tert*-Butyl (1*R*)-2-[(4*R*)-4-Benzyl-2-oxooxazolidin-3-yl]-1-(3,5-diisopropoxy-4-methoxyphenyl)-2-oxoethylcarbamate (50):**[81]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

A suspension of azide **49** (3.70 g, 7.66 mmol), $Boc_2O$ (2.50 g, 11.45 mmol), and a catalytic amount of 10% Pd/C in EtOAc was hydrogenated at 1 atm for 4 h. The mixture was filtered through a short pad of Celite, the filtrate was concentrated under reduced pressure, and the residue was purified by flash chromatography (EtOAc/heptane 1:4) to afford **50** as white crystals; yield: 3.62 g (85%); mp 50–52 °C; $[\alpha]_D$ −129 (*c* 0.5, $CHCl_3$).

**(*R*)-*N*-(*tert*-Butoxycarbonyl)(3,5-diisopropoxy-4-methoxyphenyl)glycine (46):**[81]
To a soln of **50** (3.02 g, 5.43 mmol) in THF (60 mL) and $H_2O$ (20 mL) was added $LiOH \cdot H_2O$ (0.45 g, 10.71 mmol) at 0 °C. After being stirred at 0 °C for 3 h, the solvent was removed and the aqueous phase was extracted with $Et_2O$. The combined organic layers were washed with brine, dried, and concentrated under reduced pressure to recover the oxazolidinone. The aqueous phase was then acidified with citric acid and extracted with EtOAc. The combined organic extracts were washed with brine, dried, and concentrated under reduced pressure to afford compound **46** as a white film; yield: 2.05 g (95%); $[\alpha]_D$ −92 (*c* 1.5, $CHCl_3$).

## 9.1.6 Substituted Glutamic Acid

Substituted glutamic acids have tremendous potential as derivatives in their own right for incorporation into biologically active peptides to provide topographically constrained glutamic acid and glutamine-containing analogues with constraints at $\chi^1$, $\chi^2$, and $\chi^3$ (α,β; β,γ; γ,δ) side-chain torsional angles. This, of course, requires well-chosen substitutions at the $C^\beta$ and $C^\gamma$ (and perhaps $C^\alpha$ as well) side-chain carbons. With such substitutions there is the possibility of 16 different chiral isomers and each of these isomers has 27 ($3^3$) different topographical arrangements, providing a rich and largely unexplored topographical space.

In addition, glutamic acid is readily converted into pyroglutamic acid, which in turn can be converted into proline analogues. Hence, topographically constrained glutamic acids provide access to topographically constrained glutamine and proline amino acids as well. The development of efficient syntheses of glutamic acids would therefore be an important contribution to providing derivatives that can be used for a better understanding of peptide conformation–biological activity relationships, as well as to produce important templates for peptidomimetic design. Furthermore, glutamic acid plays a significant role in the aging process and in a number of other biological activities including interactions with the NMDA receptor.[20,90–93] Thus, developing efficient asymmetric syntheses of glutamic acid derivatives opens the way to a new chemistry in peptide and peptidomimetic design.

The synthesis of substituted glutamic acid analogues has been pursued by many routes[94–106] using asymmetric alkylation, Simmons–Smith reactions, Diels–Alder reactions, and nickel complexes of glycine or alanine equivalents.

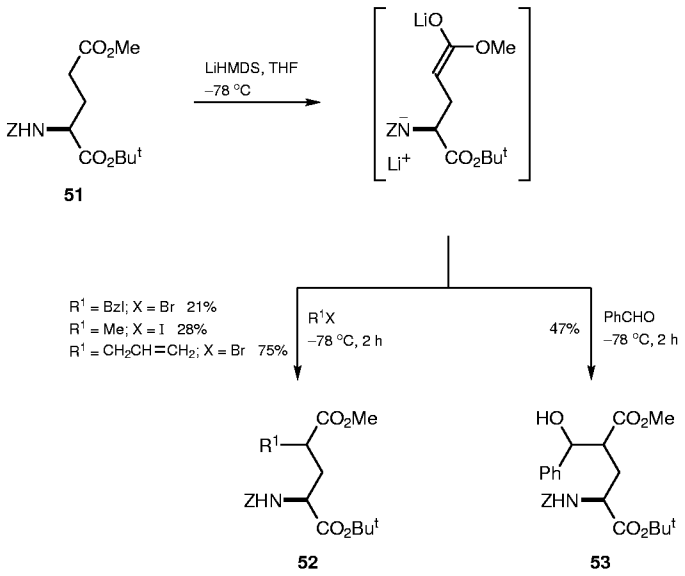
*for references see p 49*

## 9.1.6.1 γ-Monosubstituted Glutamic Acid

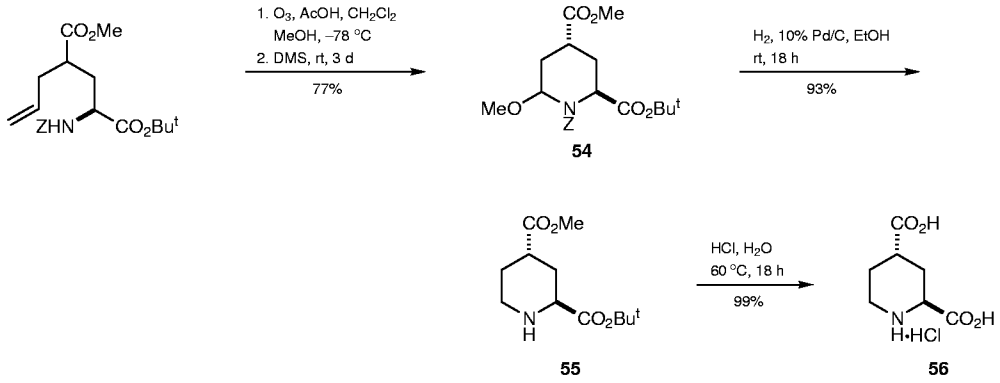
### 9.1.6.1.1 Alkylation of Glutamic Acid

Dianions of $N^{\alpha}$-monoprotected glutamic acid esters are very useful intermediates for γ-monoalkylation leading to γ-monosubstituted glutamic acids (such as **52**, **53**; Scheme 19).[94] Among them, γ-allyl-substituted glutamic acid **52** ($R^1 = CH_2CH{=}CH_2$) can be derivatized into γ-substituted homoproline **56** (Scheme 20).

**Scheme 19** Asymmetric Synthesis of γ-Substituted Glutamic Acid Derivatives via a Glutamic Acid γ-Enolate Synthon[94]




**Scheme 20** Synthesis of a γ-Carboxy-Substituted Homoproline Derivative[94]




**1-*tert*-Butyl 5-Methyl (2*S*)-4-Benzyl-*N*-(benzyloxycarbonyl)glutamate (52, $R^1$ = Bzl):**[94]
1-*tert*-Butyl 5-methyl (*S*)-*N*-(benzyloxycarbonyl)glutamate[100] (**51**; 0.5 g, 1.4 mmol) was dissolved in THF (4 mL), cooled to −78 °C, and 1.0 M LiHMDS in THF (3.6 mL, 3.6 mmol) was added. The resulting soln was stirred for 1 h at −40 °C, and recooled to −78 °C. BzlBr (0.33 mL, 2.8 mmol) was added, and the soln was stirred for 2 h at −78 °C. 1 M HCl (20 mL) was added, and the soln was then extracted with $Et_2O$ (3 × 20 mL). The combined organic phases were dried ($MgSO_4$), and the solvent was removed under reduced pressure. The product was purified by flash chromatography (EtOAc/petroleum ether 1:4) to give **52** ($R^1$ = Bzl) as a cloudy white oil; yield: 0.13 g (21%); $[\alpha]_D^{21}$ −3.0 (*c* 1.0, $CHCl_3$).

**1-*tert*-Butyl 5-Methyl (2*S*)-*N*-(Benzyloxycarbonyl)-4-methylglutamate (52, $R^1$ = Me):**[94]
Compound **51** (0.5 g, 1.4 mmol) was dissolved in THF (4 mL), cooled to −78 °C, and 1.0 M LiHMDS in THF (3.6 mL, 3.6 mmol) was added. The resulting soln was stirred for 1 h at −40 °C, and recooled to −78 °C. MeI (0.35 mL, 5.64 mmol) was added, and the soln was stirred for 2 h at −78 °C. 1 M HCl (20 mL) was added, and the soln was extracted with $Et_2O$ (3 × 20 mL). The combined organic phases were dried ($MgSO_4$) and the solvent was removed under reduced pressure to give a reddish oil. The oil was subjected to flash chromatography ($Et_2O$/petroleum ether 3:7) to give **52** ($R^1$ = Me) as an oil; yield: 150 mg (28%).

**1-*tert*-Butyl 5-Methyl (2*S*)-4-Allyl-*N*-(benzyloxycarbonyl)glutamate (52, $R^1 = CH_2CH{=}CH_2$):**[94]
Compound **51** (0.5 g, 1.4 mmol) was dissolved in THF (4 mL), cooled to −78 °C, and 1.0 M LiHMDS in THF (3.6 mL, 3.6 mmol) was added. The resulting soln was stirred for 1 h at −40 °C, and recooled to −78 °C. Allyl bromide (0.5 mL, 5.64 mmol) was added, and the soln was stirred for 2 h at −78 °C. 1 M HCl (20 mL) was added and the soln was extracted with $Et_2O$ (3 × 20 mL). The combined organic phases were dried ($MgSO_4$) and the solvent was removed under reduced pressure to give a reddish oil. The oil was subjected to flash chromatography (EtOAc/petroleum ether 1:4) to give **52** ($R^1 = CH_2CH{=}CH_2$) as an oil; yield: 0.42 g (75%).

**1-*tert*-Butyl 5-Methyl (2*S*)-*N*-(Benzyloxycarbonyl)-4-[hydroxy(phenyl)methyl]glutamate (53):**[94]
Compound **51** (0.5 g, 1.4 mmol) was dissolved in anhyd THF (4 mL), cooled to −78 °C, and 1.0 M LiHMDS in THF (3.6 mL) was added. The resulting soln was stirred for 1 h at −40 °C, then recooled to −78 °C. Benzaldehyde (0.5 mL, 4.9 mmol) was added, and the soln was stirred for 2 h at −78 °C. 1.0 M HCl (20 mL) was added and the soln was extracted with $Et_2O$ (3 × 20 mL). The combined organic phases were dried ($MgSO_4$), and the solvent was removed under reduced pressure. Purification by flash chromatography (EtOAc/petroleum ether 1:4) gave **53** as an oil; yield: 0.30 g (47%).

**1-Benzyl 2-*tert*-Butyl 4-Methyl (2*S*,4*R*)-6-Methoxypiperidine-1,2,4-tricarboxylate (54):**[94]
1-*tert*-Butyl 5-methyl (2*S*)-4-allyl-*N*-(benzyloxycarbonyl)glutamate (**52**, $R^1 = CH_2CH{=}CH_2$; 1.2 g, 3.1 mmol) was dissolved in MeOH/$CH_2Cl_2$ (10:1, 11 mL), and AcOH (0.2 mL, 3.1 mmol) was added. The resulting soln was cooled to −78 °C, and $O_3$ was bubbled through the soln until a persistent blue color was observed. The soln was allowed to warm to rt. DMS (0.7 mL, 9.3 mmol) was added, and the soln was stirred at rt for 3 d. $CH_2Cl_2$ (30 mL) was then added, and the resulting soln was washed with 10% $Na_2CO_3$ (2 × 20 mL) and then with $H_2O$ (2 × 20 mL). The organic phase was dried ($MgSO_4$) and concentrated under reduced pressure to give **54** as a yellow oil; yield: 0.96 g (77%); $[\alpha]_D^{21}$ −21.1 (*c* 1.0, $CHCl_3$).

**2-*tert*-Butyl 4-Methyl (2*S*,4*S*)-Piperidine-2,4-dicarboxylate (55):**[94]
Compound **54** (0.7 g, 1.7 mmol) was dissolved in EtOH (20 mL), and added to a suspension of 10% Pd/C (0.1 g) in EtOH (5 mL). The soln was then stirred under $H_2$ atmosphere at rt for 18 h. The mixture was filtered through Celite, and the filtrate was concentrated under reduced pressure to give **55** as an oil; yield: 0.38 g (93%); $[\alpha]_D^{21}$ +7.4 (*c* 1.0, MeOH).

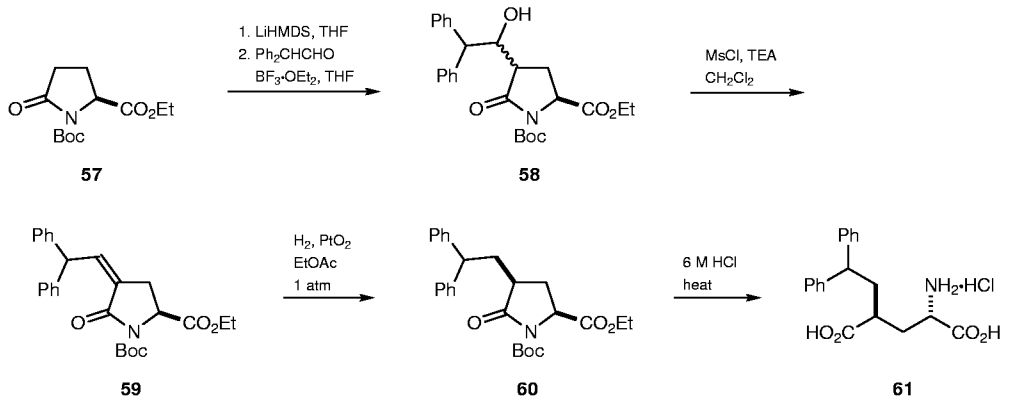
**(2*S*,4*S*)-Piperidine-2,4-dicarboxylic Acid Hydrochloride (56):**[94]
Compound **55** (97 mg, 0.4 mmol) was dissolved in 6 M HCl (4 mL), and stirred at 60 °C for 18 h. The solvent was removed under reduced pressure, leaving **56** as a white solid; yield: 83 mg (99%); $[\alpha]_D^{20}$ −5 (*c* 0.1, $H_2O$/DMSO; note: solvent ambiguous, no further details given); $^1H$ NMR ($D_2O$, δ): 1.7–2.0 (m, 3H), 2.11 (dt, *J* = 15.0, 4.9 Hz, 1H), 2.69 (pentet, *J* = 5.0 Hz, 1H), 2.9–3.0 (m, 1H), 3.1–3.2 (m, 1H), 4.00 (dd, *J* = 9.7, 4.1 Hz, 1H); $^{13}C$ NMR ($D_2O$, δ): 25.81, 29.02, 38.10, 43.55, 56.41, 173.29, 179.36.

### 9.1.6.1.2 Alkylation of Pyroglutamic Acid

Pyroglutamic acid can be used as an alternative for glutamic acid during the synthesis of γ-substituted glutamic acids.[95] Selective γ-alkylation of protected pyroglutamic acid **57** has been utilized for the synthesis of (2*S*,4*S*)-2-amino-4-(2,2-diphenylethyl)pentanedioic acid hydrochloride (**61**, Scheme 21).

*for references see p 49*

**Scheme 21** Asymmetric Synthesis of a γ-Substituted Glutamic Acid from Pyroglutamic Acid[95]




**(2*S*,4*S*)-4-(2,2-Diphenylethyl)glutamic Acid Hydrochloride [(2*S*,4*S*)-2-Amino-4-(2,2-diphenylethyl)-pentanedioic Acid Hydrochloride, 61]:**[95]

To a soln of ethyl $N^{\alpha}$-(*tert*-butoxycarbonyl)pyroglutamate (**57**; 15.6 mmol) in THF (50 mL) stirred at −78 °C was added 1 M LiHMDS in THF (18.7 mL, 18.7 mmol). The mixture was stirred for 1 h at −78 °C prior to the addition of a soln of diphenylacetaldehyde (17.2 mmol) and $BF_3 \cdot OEt_2$ (17.2 mmol) in THF (50 mL). The mixture was stirred for 1 h at −78 °C, quenched with sat. $NH_4Cl$ (100 mL), and extracted with $Et_2O$ (3 × 50 mL). The combined organic phases were dried ($Na_2SO_4$), filtered, and concentrated to dryness. Purification of the crude material by flash chromatography (hexane/EtOAc 2:1) gave a mixture of aldol products **58**, which were dissolved in $CH_2Cl_2$ (30 mL) and treated with MsCl (1.34 mL, 17.2 mmol) and TEA (18 mL, 172 mmol). After stirring this soln for 2 d at rt, it was quenched with $H_2O$ and extracted with $CH_2Cl_2$ (3 × 50 mL). The organic layer was dried ($Na_2SO_4$) and concentrated under reduced pressure affording **59**.

$PtO_2$ (0.65 mmol) was added to **59** (6.46 mmol) in EtOAc (50 mL). $H_2$ (1 atm) was introduced and the heterogeneous mixture was stirred at rt for 4 h. The mixture was filtered through a pad of Celite and concentrated to give **60** exclusively.

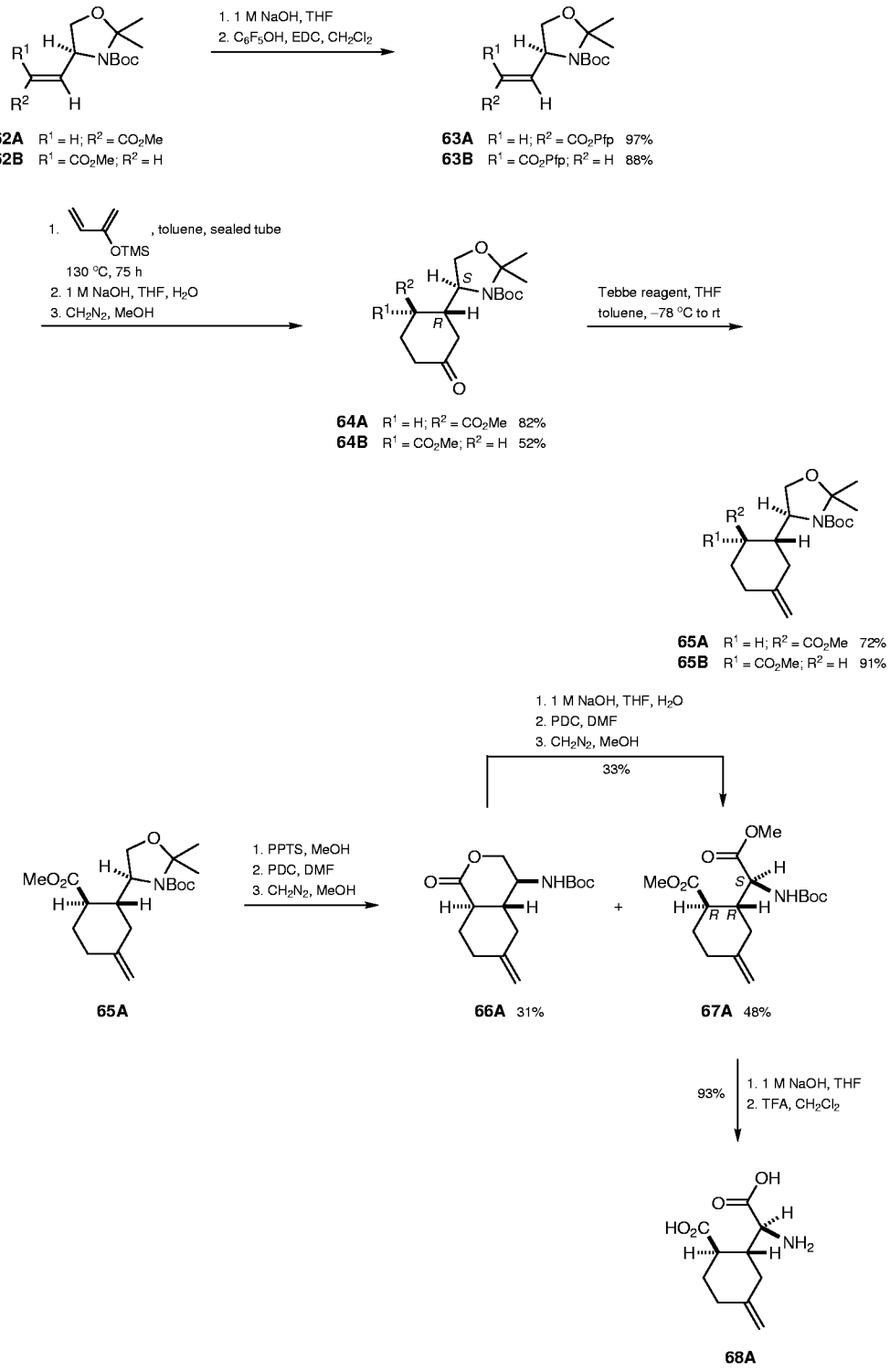
Finally, a soln of **60** (2 mmol) and 6 M HCl (25 mL) was refluxed overnight. The resulting soln was concentrated to dryness. The residue was triturated with $Et_2O$ (3 × 20 mL), giving **61** as a white solid; yield: not reported; mp 160–162 °C; $[\alpha]_D$ +7.4 (*c* 1.6, MeOH).

## 9.1.6.2 β,γ-Disubstituted Glutamic Acid

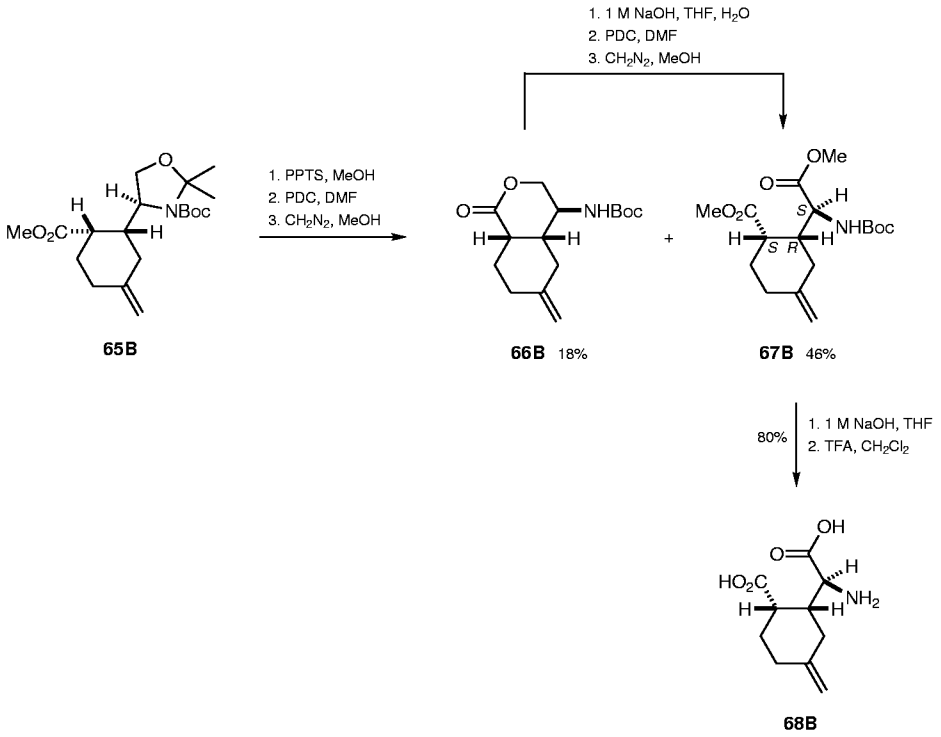
### 9.1.6.2.1 Asymmetric Diels–Alder Reaction

The Diels–Alder reaction can be used for asymmetric synthesis of β,γ-disubstituted glutamic acids.[98] Fully protected dehydroglutaminol **63** serves as a dienophile and, after a series of deprotection and oxidation of the Diels–Alder reaction product, disubstituted glutamic acids **68A** and **68B** (Scheme 22) can be obtained.

**Scheme 22** Asymmetric Synthesis of β,γ-Disubstituted Glutamic Acids[98]




*for references see p 49*




**Pentafluorophenyl (2*E*)-3-[(4*S*)-3-(*tert*-Butoxycarbonyl)-2,2-dimethyloxazolidin-4-yl]acrylate** (63A)**:**[98] A soln of **62A**[107] (6.00 g, 21.0 mmol) in THF (100 mL) and 1 M NaOH (25 mL) was stirred at 0 °C for 3 h. After additional stirring at rt for 15 h, THF was removed under reduced pressure. The aqueous soln was acidified with 1 M HCl to pH 2, saturated with NaCl, and extracted several times with $Et_2O$. The combined extracts were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure to give an oily residue. The residue was used for the subsequent esterification without chromatographic purification. To a soln of the crude residue in $CH_2Cl_2$ (100 mL) was added pentafluorophenol (4.50 g, 24.2 mmol) and EDC (5.00 g, 26.4 mmol). After stirring at rt for 3 h, the mixture was poured into $H_2O$ and extracted with $Et_2O$. The combined extracts were washed with sat. aq $NaHCO_3$ and brine, dried ($MgSO_4$), and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, $Et_2O$/hexane 1:6) to give crude crystals. Recrystallization (pentane) gave **63A** as colorless crystals; yield: 8.90 g (97%); mp 58 °C; $[\alpha]_D^{28}$ +63.9 (*c* 1.15, $CHCl_3$).

***tert*-Butyl (4*S*)-4-[(1*R*,2*R*)-2-(Methoxycarbonyl)-5-oxocyclohexyl]-2,2-dimethyloxazolidine-3-carboxylate** (64A)**:**[98]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate safety precautions and procedures should be used when handling this reagent (e.g., special glassware, blast shield, well-ventilated fume hood, etc.).*

A soln of **63A** (1.50 g, 3.43 mmol) and 2-(trimethylsiloxy)buta-1,3-diene (3.0 mL) in toluene (3.0 mL) was heated at 130 °C in a sealed tube for 75 h. The mixture was filtered through a short pass column of silica gel ($Et_2O$) and the resulting soln was concentrated under reduced pressure. The residue was dissolved in THF (50 mL) and $H_2O$ (10 mL). To this soln was added 1 M NaOH (10 mL), and the soln was stirred at 0 °C for 1 h and at rt for 15 h. After removal of THF under reduced pressure, the aqueous soln was acidified with 1 M HCl to pH 2, saturated with NaCl, and extracted with $Et_2O$. The combined extracts were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure. The residue was dissolved in MeOH (50 mL) and treated with $CH_2N_2$ in $Et_2O$. The resulting soln was concentrated under reduced pressure to give a crude residue which, upon purification by column chromatography (silica gel, $Et_2O$/hexanes 1:4), gave **64A** as a colorless oil; yield: 1.00 g (82%); $[\alpha]_D^{23}$ +16.6 (*c* 3.54, $CHCl_3$).

***tert*-Butyl (4*S*)-4-[(1*R*,2*R*)-2-(Methoxycarbonyl)-5-methylenecyclohexyl]-2,2-dimethyloxazolidine-3-carboxylate** (65A):[98]

To a soln of **64A** (2.50 g, 7.04 mmol) in THF (100 mL) at −78 °C was added 0.5 M μ-chloro-μ-methylene[bis(cyclopentadienyl)titanium]dimethylaluminum (Tebbe reagent) in toluene (15 mL, 7.5 mmol). The reaction was stirred at −78 °C for an additional 1 h, and then gradually warmed to 0 °C over 1 h. The mixture was poured into sat. aq $NaHCO_3$, filtered through a Celite pad, and the filtrate was extracted several times with $Et_2O$. The combined extracts were washed with $H_2O$ and brine, dried ($MgSO_4$), and concentrated under reduced pressure. The crude product was purified by column chromatography (silica gel) to give **65A** as a colorless oil; yield: 1.80 g (72%); $[\alpha]_D^{23}$ +15.4 (*c* 0.42, $CHCl_3$).

**Methyl (2*S*)-*N*-(*tert*-Butoxycarbonyl)-2-[(1*R*,2*R*)-2-(methoxycarbonyl)-5-methylenecyclohexyl]glycinate** (67A):[98]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate safety precautions and procedures should be used when handling this reagent (e.g., special glassware, blast shield, well-ventilated fume hood, etc.).*

To a soln of **65A** (960 mg, 2.72 mmol) in MeOH (300 mL) was added PPTS (400 mg) at rt. After stirring at 60 °C for 30 min, the solvent was removed under reduced pressure and the residue was filtered through a short-pass column of silica gel ($Et_2O$) to give crude residue. To a soln of the residue in DMF (40 mL) was added 4-Å molecular sieves (10 g) and PDC (5.10 g, 14.0 mmol). The reaction was stirred at rt for 15 h, poured into $Et_2O$ (200 mL), and filtered through a Celite pad. The filtrate was concentrated under reduced pressure to give a crude residue, which was dissolved in MeOH (50 mL). The soln was treated with $CH_2N_2$ in $Et_2O$. After removal of the solvent under reduced pressure, the residue was dissolved in $Et_2O$ (300 mL), washed with $H_2O$ and brine, dried ($MgSO_4$), and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, $Et_2O$/hexanes 1:4 then 1:1) to give **67A** [yield: 450 mg (48%)] and the δ-lactone **66A** [yield: 245 mg (31%); mp 154.5–155.0 °C ($Et_2O$); $[\alpha]_D^{23}$ −24.0 (*c* 1.16, $CHCl_3$); FTIR (film) $\tilde{\nu}_{max}$: 3351, 2978, 2942, 1742, 1753 $cm^{-1}$], respectively, as colorless crystals.

To a soln of lactone **66A** (500 mg, 1.80 mmol) in THF (20 mL) and $H_2O$ (10 mL) was added 1 M NaOH (10 mL). The mixture was stirred at 0 °C for 3 h and at rt for 15 h. After removal of THF under reduced pressure, the mixture was washed with $Et_2O$, and the aqueous layer was acidified with 1 M HCl to pH 2, saturated with NaCl, and extracted several times with $Et_2O$. The combined extracts were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure to give crude glycinol. This was subjected to the same sequence of reactions as described above (PDC oxidation, followed by esterification with $CH_2N_2$) to give the recovered lactone **66A** (100 mg, 16%) and the desired product **67A**; yield: 200 mg (33%); mp 89.0–89.5 °C (hexane); $[\alpha]_D^{23}$ +59.9 (*c* 1.16, $CHCl_3$); FTIR (film) $\tilde{\nu}_{max}$: 3366, 2947, 2880, 1736, 1715 $cm^{-1}$.

**(2*S*)-2-[(1*R*,2*R*)-2-Carboxy-5-methylenecyclohexyl]glycine** (68A):[98]

**CAUTION:** *Trifluoroacetic acid is corrosive and severely irritating to skin, eyes, and mucous membranes. Inhalation can destroy tissue and be fatal. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

To a soln of **67A** (100 mg, 0.29 mmol) in THF (2.8 mL) was added 1 M NaOH (0.7 mL, 0.7 mmol). The soln was stirred at rt for 48 h and concentrated under reduced pressure. The residue was dissolved in $H_2O$ and washed with $Et_2O$. The aqueous layer was acidified with 1 M HCl to pH 2, saturated with NaCl, and extracted with EtOAc. The extracts were dried ($MgSO_4$) and concentrated under reduced pressure to give an oily residue. A soln of the residue in $CH_2Cl_2$ (2 mL) and TFA (2 mL) was stirred at 0 °C for 30 min and at rt for 30 min. The mixture was concentrated under reduced pressure to give a crude crystalline solid. The crude crystals were dissolved in $H_2O$ (5 mL) and the solution was concentrated under reduced pressure. This process was repeated several times to remove trace amounts of TFA. A soln of the crude product in $H_2O$ was subjected to a column of ion-exchange resin [Dowex 50WX4, $H_2O$ then 1% $NH_4OH$] to give **68A** as colorless crystals; yield: 58 mg (93%); mp 109–110 °C; $[\alpha]_D^{23}$ +35.3 (*c* 1.05, $H_2O$/TFA 1:1).

**Pentafluorophenyl (2*Z*)-3-[(4*S*)-3-(*tert*-Butoxycarbonyl)-2,2-dimethyloxazolidin-4-yl]acrylate** (63B):[98]

In a manner similar to that used to prepare **63A**, the reaction of **62B** (26.0 mmol) gave **63B** as colorless crystals; yield: 10.3 g (88%); mp 89.3 °C; $[\alpha]_D^{23}$ −52.1 (*c* 1.17, $CHCl_3$).

*for references see p 49*

***tert*-Butyl (4*S*)-4-[(1*R*,2*S*)-2-(Methoxycarbonyl)-5-oxocyclohexyl]-2,2-dimethyloxazolidine-3-carboxylate** (64B):[98]
In a manner similar to that used to prepare **64A**, the reaction of **63B** (2.00 g, 4.58 mmol) with 2-(trimethylsiloxy)buta-1,3-diene (5.0 mL) in toluene (5.0 mL) gave the mixture of the cycloproduct **64B** and recovered **62B**. The mixture was subjected to column chromatography (silica gel, $Et_2O$/hexane 4:1 then 1:1) to give **62B** [yield: 250 mg (22%)] and **64B** as an oil; yield: 840 mg (52%); $[\alpha]_D^{23}$ +2.28 (*c* 1.84, $CHCl_3$).

***tert*-Butyl (4*S*)-4-[(1*R*,2*S*)-2-(Methoxycarbonyl)-5-methylenecyclohexyl]-2,2-dimethyloxazolidine-3-carboxylate** (65B):[98]
In a manner similar to that used to prepare **65A**, the reaction of **64B** (2.00 g, 5.60 mmol) gave **65B** as a colorless oil; yield: 1.80 g (91%); $[\alpha]_D^{23}$ +8.3 (*c* 0.31, $CHCl_3$).

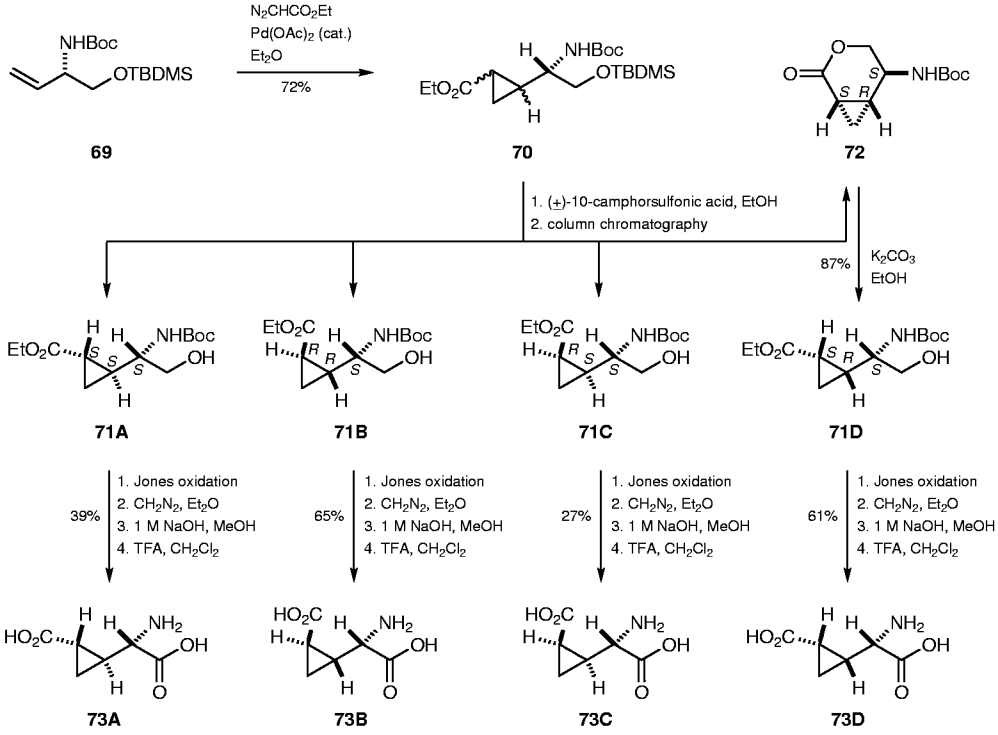
**Methyl (2*S*)-*N*-(*tert*-Butoxycarbonyl)-2-[(1*R*,2*S*)-2-(methoxycarbonyl)-5-methylenecyclohexyl]glycinate** (67B):[98]
In a manner similar to that used to prepare **67A**, the reaction of **65B** (2.1 g, 5.95 mmol) gave **67B** [yield: 1.0 g (46%); mp 104.5–105.0 °C (hexane); $[\alpha]_D^{23}$ −16.8 (*c* 1.16, $CHCl_3$); FTIR (film) $\tilde{\nu}_{max}$: 3364, 2984, 2955, 1745, 1703 $cm^{-1}$] and the δ-lactone **66B** [yield: 300 mg (calcd: 18%, reported: 29%); mp 154.5–155.0 °C ($Et_2O$); $[\alpha]_D^{23}$ +44.5 (*c* 0.41, $CHCl_3$); FTIR (film) $\tilde{\nu}_{max}$: 3340, 2978, 2939, 1748, 1715 $cm^{-1}$] as colorless crystals.

**(2*S*)-2-[(1*R*,2*S*)-2-Carboxy-5-methylenecyclohexyl]glycine** (68B):[98]
In a manner similar to that used to prepare **68A**, the reaction of **67B** (100 mg, 0.29 mmol) gave **68B** as colorless crystals; yield: 50 mg (calcd: 80%, reported: 90%); mp 167 °C; $[\alpha]_D^{23}$ −30.7 (*c* 0.55, $H_2O$/TFA 1:1).

### 9.1.6.2.2 Simmons–Smith Cyclization

The acid group in the side chain of substituted glutamic acids can be introduced by ethyl diazoacetate via the Simmons–Smith reaction (Scheme 23).[99] This reaction is not stereospecific and four isomers are obtained, which can be separated and further converted into the derivatives **73A–D**.

**Scheme 23** Synthesis of (Carboxycyclopropyl)glycine Derivatives via the Simmons–Smith Reaction[99]



**(2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)-2-[2-(ethoxycarbonyl)cyclopropyl]glycinol** (71A–D) **and (1*S*,5*S*,6*R*)-5-(*tert*-Butoxycarbonylamino)-3-oxabicyclo[4.1.0]heptan-2-one** (72):[99]

**CAUTION:** *Many organic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

To a soln of $Pd(OAc)_2$ (168 mg, 0.75 mmol) and alkene **69** (4.34 g, 14.4 mmol) in $Et_2O$ (150 mL) at rt was simultaneously added, drop by drop, a soln of ethyl diazoacetate (17.1 g, 150 mmol) in $Et_2O$ (300 mL) and a soln of $Pd(OAc)_2$ (168 mg, 0.75 mmol) in $Et_2O$ (200 mL) over 3 h. The mixture was filtered. The filtrate was concentrated under reduced pressure to give a crude oil which, upon column chromatography (silica gel, $Et_2O$/hexane 1:4), gave a mixture of the four diastereomers **70** as a colorless oil; yield: 4.00 g (calcd: 72%, reported: 88%). The mixture was dissolved in EtOH (100 mL) and treated with (*RS*)-10-camphorsulfonic acid (5 mg) at rt for 16 h. Then the solvent was removed under reduced pressure. The residue was purified by medium-pressure column chromatography (silica gel, $Et_2O$/hexane 3:1) to give **71B** [932 mg; mp 88.0–89.0 °C; $[\alpha]_D^{23}$ −47.2 (*c* 0.55, $CHCl_3$); IR (neat) $\tilde{\nu}_{max}$: 3460, 3028, 1712 $cm^{-1}$], **71C** [175 mg; mp 94.0–95.0 °C; $[\alpha]_D^{23}$ −56.0 (*c* 0.48, $CHCl_3$); IR (neat) $\tilde{\nu}_{max}$: 3392, 2984, 1716 $cm^{-1}$], a mixture of **71A** and **71D** (363 mg), and a mixture of **71A–71D** (197 mg).

The mixture of **71A** and **71D** was treated with (*RS*)-10-camphorsulfonic acid (5 mg) in $CH_2Cl_2$ (10 mL) at rt for 18 h. The mixture was then washed with aq $NaHCO_3$. The organic phase was dried and the solvent was removed under reduced pressure to give an oily residue. This was purified by medium-pressure column chromatography (silica gel, $Et_2O$/hexane 3:1) to give unchanged **71A** as a colorless oil [190 mg; $[\alpha]_D^{25}$ +72.90 (*c* 0.50, $CHCl_3$); IR (neat) $\tilde{\nu}_{max}$: 3372, 2984, 1712 $cm^{-1}$] and (1*S*,5*S*,6*R*)-δ-lactone **72** as colorless crystals [80 mg; mp 152.0–154.0 °C; $[\alpha]_D^{23}$ −59.4 (*c* 0.46, $CHCl_3$); IR (neat) $\tilde{\nu}_{max}$: 3328, 2984, 1734, 1710 $cm^{-1}$].

**(2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)-2-[(1*R*,2*S*)-2-(ethoxycarbonyl)cyclopropyl]glycinol** (71D):[99]
A mixture of **72** (514 mg, 2.26 mmol), $K_2CO_3$ (20 mg), and EtOH (30 mL) was stirred at rt for 24 h. The mixture was then extracted several times with EtOAc. The combined extracts were washed with $H_2O$, dried ($MgSO_4$), and concentrated under reduced pressure. The oily residue was purified by column chromatography (silica gel, $Et_2O$/hexane 3:1) to give **71D** as colorless crystals; yield: 540 mg (87%); mp 109.0–110.5 °C; $[\alpha]_D^{23}$ −49.3 (*c* 1.04, $CHCl_3$); IR (neat) $\tilde{\nu}_{max}$: 3380, 2984, 2940, 1726 $cm^{-1}$.

**(2*S*)-2-[(1*S*,2*S*)-2-Carboxycyclopropyl]glycine** (73A):[99]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate safety precautions and procedures should be used when handling this reagent (e.g., special glassware, blast shield, well-ventilated fume hood, etc.).*

**CAUTION:** *Trifluoroacetic acid is corrosive and severely irritating to skin, eyes, and mucous membranes. Inhalation can destroy tissue and be fatal. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

To a soln of **71A** (1.00 g, 3.66 mmol) in acetone (10 mL) at 0 °C was added a 1.5-fold excess of Jones reagent drop by drop over 5 min. The mixture was stirred at 0 °C for 3 h and then at rt for 1.5 h. The excess Jones reagent was destroyed with iPrOH. The mixture was then extracted several times with EtOAc. The combined extracts were washed with $H_2O$, dried ($MgSO_4$), and concentrated under reduced pressure. The residual oil was dissolved in $Et_2O$ and was treated with an $Et_2O$ soln of $CH_2N_2$. The solvent was removed under reduced pressure. The residue was purified by column chromatography (silica gel, $Et_2O$/hexane 3:1). A soln of the crude product and 1 M aq NaOH (9 mL, 9 mmol) in MeOH (5 mL) was stirred at 0 °C for 19 h. The mixture was then acidified to pH 2 with 1 M HCl, saturated with solid NaCl, and extracted several times with EtOAc. The combined extracts were dried ($MgSO_4$) and concentrated under reduced pressure to give an oily residue. To a soln of the residue (crude $N^{\alpha}$-Boc-**73A**) in $CH_2Cl_2$ (2 mL) at 0 °C was added TFA (2 mL). The mixture was stirred for 30 min at rt and then was concentrated under reduced pressure. The residue was passed through a column of Dowex 50WX4 (100–200 mesh) ion-exchange resin ($H_2O$, then 1 M $NH_4OH$) to give a soln of the ammonium salt of **73A**. The eluate was concentrated under reduced pressure and then was dissolved in $H_2O$ (2 mL). The pH of the soln was adjusted to 3 with 1 M aq HCl. The crystals that precipitated from soln were collected by filtration. They were recrystallized ($H_2O$) to give **73A** as colorless crystals; yield: 228 mg (39% from **71A**); mp 243–247 °C (dec); $[\alpha]_D^{21}$ +102.00 (*c* 0.50, $H_2O$); $^1H$ NMR (360 MHz, $D_2O$, δ): 1.23 (ddd, *J* = 5.1, 6.2, 8.7 Hz, 1H), 1.32 (ddd, *J* = 5.0, 5.1, 9.1 Hz, 1H), 1.68 (dddd, *J* = 4.1, 6.2, 9.1, 10.2 Hz, 1H), 1.76 (ddd, *J* = 4.1, 5.0, 8.7 Hz, 1H), 3.23 (d, *J* = 10.2 Hz, 1H).

*for references see p 49*

**(2*S*)-2-[(1*R*,2*R*)-2-Carboxycyclopropyl]glycine** (73B):[99]
In a manner similar to that used to prepare **73A**, **73B** was prepared from **71B** (200 mg, 0.73 mmol) as colorless crystals; yield: 75.2 mg (65%); mp 255–258 °C (dec); $[\alpha]_D^{25}$ −20.20 (*c* 0.51, $H_2O$); $^1H$ NMR (360 MHz, $D_2O$, δ): 1.10 (ddd, *J* = 5.0, 6.5, 9.0 Hz, 1H), 1.25 (ddd, *J* = 5.0, 5.0, 9.0 Hz, 1H), 1.72 (dddd, *J* = 4.0, 6.5, 9.0, 9.0 Hz, 1H), 1.84 (ddd, *J* = 4.0, 5.0, 9.0 Hz, 1H), 3.39 (d, *J* = 9.0 Hz, 1H).

**(2*S*)-2-[(1*S*,2*R*)-2-Carboxycyclopropyl]glycine** (73C):[99]
In a manner similar to that used to prepare **73A**, **73C** was prepared from **71C** (100 mg, 0.37 mmol) as colorless crystals; yield: 16.1 mg (27%); mp 192–197 °C (dec); $[\alpha]_D^{22}$ +20.80 (*c* 0.52, $H_2O$); $^1H$ NMR (360 MHz, $D_2O$, δ): 1.26 (ddd, *J* = 5.0, 6.0, 7.0 Hz, 1H), 1.39 (ddd, *J* = 5.0, 9.0, 9.0 Hz, 1H), 1.61 (dddd, *J* = 7.0, 8.0, 9.0, 11.0 Hz, 1H), 1.86 (ddd, *J* = 6.0, 8.0, 9.0 Hz, 1H), 3.89 (d, *J* = 11.0 Hz, 1H).

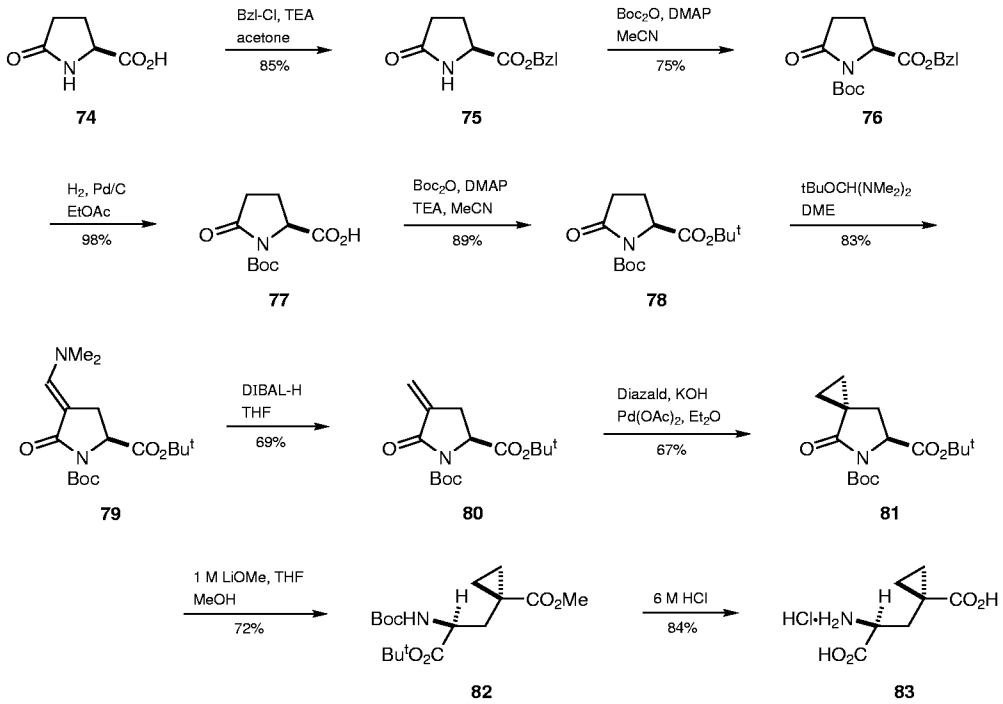
**(2*S*)-2-[(1*R*,2*S*)-2-Carboxycyclopropyl]glycine** (73D):[99]
In a manner similar to that used to prepare **73A**, **73D** was prepared from **71D** (450 mg, 1.65 mmol) as colorless crystals; yield: 159 mg (61%); mp 178–180 °C; $[\alpha]_D^{26}$ +103.4 (*c* 0.50, $H_2O$); $^1H$ NMR (360 MHz, $D_2O$, δ): 1.05 (ddd, *J* = 5.0, 6.0, 7.0 Hz, 1H), 1.20 (ddd, *J* = 5.0, 8.5, 8.5 Hz, 1H), 1.60 (dddd, *J* = 7.0, 8.5, 9.0, 9.0 Hz, 1H), 1.93 (ddd, *J* = 6.0, 8.5, 9.0 Hz, 1H), 3.89 (d, *J* = 9.0 Hz, 1H).

## 9.1.6.3 γ,γ-Disubstituted Glutamic Acid

### 9.1.6.3.1 Asymmetric Simmons–Smith Cyclization

By applying Bredereck's reagent [*tert*-butoxybis(dimethylamino)methane], a regioselective exocyclic γ-enamine **79** is obtained when fully protected pyroglutamic acid **78** is used as the starting material. Reduction of enamine **79** leads to **80**, with an exocyclic double bond, which undergoes stereoselective Simmons–Smith reaction with Diazald (*N*-methyl-*N*-nitroso-4-toluenesulfonamide). Subsequent ring opening of the pyroglutamic acid derivative **81** and deprotection affords γ,γ-disubstituted glutamic acid **83** in good yield (Scheme 24).[97]

**Scheme 24** Synthesis of a γ,γ-Disubstituted Glutamic Acid via an Asymmetric Simmons–Smith Reaction[97,108]



**Benzyl (2*S*)-Pyroglutamate (75):**[108]
This was prepared by the following variation of a reported method.[109] (2*S*)-Pyroglutamic acid (**74**; 100 g, 0.775 mol) was dried by heating at 55 °C and 0.1 Torr for 1 h and then added to acetone (1 L). TEA (86.99 g, 0.861 mol) was added to the stirred slurry under $N_2$, and the mixture was stirred at rt until a colorless soln was obtained. Bzl-Cl (107.84 g, 0.852 mol) was added, and the mixture was heated at reflux for 7 d. A yellow soln, containing a white crystalline deposit, was obtained on cooling. The mixture was filtered and the solvents were removed under reduced pressure to give a brown oil, which was dissolved in EtOAc (300 mL). The organic layer was washed with sat. aq $NaHCO_3$ (3 × 100 mL). The resultant aqueous layer was extracted with EtOAc (3 × 100 mL). The organic layers were combined, and washed successively with 1 M aq HCl (3 × 100 mL) and 10% NaCl (200 mL), and then dried ($Na_2SO_4$) and filtered. The solvents were removed under reduced pressure to afford **75** as a pale yellow oil; yield: 144.75 g (85%); IR (film) $\tilde{\nu}_{max}$: 3224 (br, NH), 1744 (ester), 1703 $cm^{-1}$ (lactam).

**Benzyl (2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)pyroglutamate (76):**[108]
Benzyl (2*S*)-pyroglutamate (**75**; 144.75 g, 0.661 mol) was dissolved in stirred MeCN (900 mL) at 0 °C under $N_2$ and DMAP (8.07 g, 0.066 mol) was added, followed by a soln of $Boc_2O$ (187.43 g, 0.859 mol) in MeCN (100 mL). The soln effervesced and was stirred at 0 °C for 2 h and at rt overnight. A dark orange soln was obtained. The solvent was removed under reduced pressure, and the brown crude solid was purified by column chromatography (silica gel, $Et_2O$/light petroleum ether). The resultant pale yellow solid was recrystallized [EtOAc/petroleum ether (60–80 °C)] to afford **76** as a crystalline solid; yield: 158.5 g (75%); mp 72–74 °C; $[\alpha]_D^{25}$ −35.0 (*c* 0.98, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 1785, 1741, 1703 $cm^{-1}$.

**(2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)pyroglutamic Acid (77):**[108]
Ester **76** (20 g, 0.063 mol) was dissolved in EtOAc (150 mL) and 10% Pd/C (2 g) was added. The mixture was stirred for 5 d at rt under $H_2$, then filtered. The solvent was removed under reduced pressure to afford a pale yellow oil, which was crystallized [EtOAc/petroleum ether (60–80 °C)] to give **77** as a crystalline solid; yield: 14.13 g (98%); mp 118–119 °C (115–116 °C[110]); $[\alpha]_D^{29}$ −36.9 (*c* 1.11, AcOH) {$[\alpha]_D^{25}$ −35.3 (*c* 1.0, AcOH)[110]}; IR (KBr) $\tilde{\nu}_{max}$: 1790, 1741, 1690 $cm^{-1}$.

***tert*-Butyl (2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)pyroglutamate (78):**[108]
*Method A:* (2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)pyroglutamic acid (**77**; 40 g, 0.175 mol) was dissolved in MeCN (500 mL), and DMAP (2.14 g, 0.018 mol) and TEA (26.51 g, 0.262 mol) were added at 0 °C to the stirred soln under $N_2$. A soln of $Boc_2O$ (57.18 g, 0.262 mol) in MeCN (100 mL) was added dropwise at 0 °C over a period of 45 min. Effervescence occurred, and the resultant light brown soln was stirred at rt overnight. The solvent was removed under reduced pressure to afford a red-brown oil, which was purified by column chromatography (silica gel, $Et_2O$). The resultant pale yellow oil was crystallized ($Et_2O$/light petroleum ether) to give **78** as a solid; yield: 44.30 g (89%); mp 54–56 °C (oil[111]); $[\alpha]_D^{21}$ −36.6 (*c* 0.93, $CHCl_3$) {$[\alpha]_D^{22}$ −35.1 (*c* 0.9, $CHCl_3$)[111]}; IR (KBr) $\tilde{\nu}_{max}$: 1775 (imide), 1719 $cm^{-1}$ (ester).

*Method B:* *tert*-Butyl (2*S*)-pyroglutamate (5.0 g, 0.027 mol) was dissolved in MeCN (100 mL) and the mixture was cooled to 0 °C. DMAP (0.33 g, 2.70 mmol) was added to the stirred mixture, followed by addition of $Boc_2O$ (8.85 g, 0.041 mol) in MeCN (30 mL). The soln effervesced and was stirred at 0 °C for 2 h and for 12 h at rt. A dark orange soln was obtained. The solvent was removed under reduced pressure to afford a brown solid, which was purified by column chromatography (silica gel, $Et_2O$/light petroleum ether) to afford a pale yellow solid, which was recrystallized ($Et_2O$/light petroleum ether) to give **78** as a solid; yield: 5.50 g (72%); mp 54–56 °C; $[\alpha]_D^{25}$ −36.6 (*c* 0.93, $CHCl_3$) {$[\alpha]_D^{22}$ −35.1 (*c* 0.9, $CHCl_3$)[111]}.

***tert*-Butyl (2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)-4-[(*E*)-(dimethylamino)methylene]pyroglutamate (79):**[108]
Compound **78** (81.94 g, 0.288 mol) was dissolved in DME (700 mL) and *tert*-butoxybis(dimethylamino)methane (Bredereck's reagent; 75.08 g, 0.431 mol) was added. The mixture was heated under $N_2$ at 75 °C for 16 h. The solvent was removed under reduced pressure to afford a red-brown oil, which was crystallized ($Et_2O$/light petroleum ether) to give **79** as a pale yellow solid; yield: 80.94 g (83%); mp 126–127 °C; $[\alpha]_D^{26}$ −47.5 (*c* 1.31, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 1759, 1738, 1687, 1631 $cm^{-1}$; UV (MeOH) $\lambda_{max}$ (ε): 313 nm (31 700).

***tert*-Butyl (2*S*)-$N^{\alpha}$-(*tert*-Butoxycarbonyl)-4-methylenepyroglutamate (80):**[108]
Enamine **79** (10.0 g, 29.4 mmol) was dissolved in THF (300 mL) and the soln was cooled to −78 °C. 1 M DIBAL-H in THF (44 mL, 44.0 mmol) was added dropwise to the stirred mixture during 10 min, and the resulting mixture was stirred at −78 °C for 1 h. The mixture was allowed to warm to rt and was stirred for a further 2 h. It was quenched with sat. aq $NH_4Cl$ (ca. 50 mL) at rt and stirred overnight at rt to afford a pale yellow soln containing a white slurry. The mixture was decanted, and the slurry was washed with EtOAc (5 × 40 mL). The organic layer was washed in turn with 10% aq citric acid (40 mL), aq NaCl (40 mL), sat. aq $NaHCO_3$ (40 mL), and 10% NaCl (40 mL), and dried ($Na_2SO_4$). The solvent was removed under reduced pressure to afford a pale yellow oil consisting of crude alkene **80** [yield: 6.45 g

*for references see p 49*

(74%)], which was repeatedly crystallized ($Et_2O$/light petroleum ether) to give the major product, **80**, as a crystalline solid; yield: 6.01 g (69%); mp 108–110 °C; $[\alpha]_D^{28}$ −3.5 (*c* 1.37, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 1777, 1736, 1698, 1662 $cm^{-1}$; UV (MeOH) $\lambda_{max}$ (ε): 228 nm (8900).

**Di-*tert*-butyl (6*S*)-4-Oxo-5-azaspiro[2.4]heptane-5,6-dicarboxylate (81):**[97]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate safety precautions and procedures should be used when handling this reagent (e.g., special glassware, blast shield, well-ventilated fume hood, etc.).*

Diazald (20 g, 93 mmol) in $Et_2O$ (250 mL) was added dropwise to KOH (10 g, 178 mmol) in 2-ethoxyethanol (20 mL), $H_2O$ (15 mL), and $Et_2O$ (10 mL) at 60 °C. The yellow ethereal soln of $CH_2N_2$ generated by distillation was added dropwise to **80** (4.55 g, 15 mmol) and $Pd(OAc)_2$ (172 mg, 0.766 mmol) in $Et_2O$ (100 mL) at 0 °C. A black precipitate was observed. The mixture was allowed to warm to rt and stirred overnight at rt. The black precipitate was removed by filtration and the solvent was removed under reduced pressure to afford **81** as a white crystalline solid; yield: 3.186 g (67%); mp 89–94 °C; $[\alpha]_D^{19}$ −6.2 (*c* 1, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 1782, 1741 (ester), 1701 $cm^{-1}$ (imide).

**Methyl 1-[(2*S*)-3-*tert*-Butoxy-2-(*tert*-butoxycarbonylamino)-3-oxopropyl]cyclopropanecarboxylate (82):**[97]

Compound **81** (1.00 g, 3.22 mmol) was dissolved in THF (16 mL) and cooled to −12 °C under argon. 1 M LiOMe in MeOH (3.86 mL, 3.86 mmol) was added dropwise with stirring, which was continued at −12 °C for 30 min. The mixture was quenched with sat. aq NaCl and $Et_2O$ (20 mL) and allowed to warm to rt. The aqueous layer was extracted with $Et_2O$ (20 mL). The combined organic layers were dried ($Na_2SO_4$) and the solvent was removed under reduced pressure to afford a pale yellow oil, which was purified by column chromatography [silica gel, petroleum ether (bp 40–60 °C)/$Et_2O$ 2:1] to afford **82** as a colorless oil; yield: 796 mg (72%); $[\alpha]_D^{18}$ +4.25 (*c* 1, $CHCl_3$); IR (film) $\tilde{\nu}_{max}$: 3377, 1719 $cm^{-1}$.

**1-[(2*S*)-2-Amino-2-carboxyethyl]cyclopropanecarboxylic Acid Hydrochloride (83):**[97]

Ester **82** (661 mg, 1.93 mmol) was heated to reflux with 6 M HCl (10 mL) for 3 h. The solvent was removed under reduced pressure to afford an off-white solid. Traces of residual HCl were removed by azeotropic distillation with $Et_2O$ and the residue was recrystallized (EtOH/$Et_2O$) to give **83** as a white crystalline solid; yield: 340 mg (84%); mp 57–61 °C; $[\alpha]_D^{22}$ +14.8 (*c* 1, 1 M HCl); IR (film) $\tilde{\nu}_{max}$: 2900–3600 (br, NH and OH), 1733 $cm^{-1}$ (acid).

## 9.1.7 Substituted Cysteine

Cysteine is a key amino acid in many proteins, serving as an important catalytic moiety in e.g. cysteine proteases, but most importantly serving as a stabilizer of secondary structures by formation of disulfide bonds. Cysteine side-chain-to-side-chain cyclization is also widely used to stabilize secondary structures important for the bioactivity of peptides.[112]

Substituted cysteine derivatives have very important applications for both conformational constraints via cyclization and topographical constraint via $\chi^1$ and $\chi^2$ constraints. In addition, the disulfide bond has inherent chirality (right handed or left handed) providing yet another level of topographical constraint. The synthesis of substituted cysteine derivatives has turned out to be a very serious challenge due to the fact that alkylsulfanyl groups resulting from protection of the thiol group in β-substituted cysteine precursors are very good leaving groups.[113] Thus, other strategies need to be adopted for successful preparation of substituted cysteines. Nevertheless, pursuit of the challenge is very worthwhile due to the many potential applications already demonstrated for substituted cysteines.[114–121]

Fortunately, nature has provided a very useful cysteine derivative, D-penicillamine (β,β-dimethylcysteine), which is readily available and inexpensive (the L-isomer is also available, but is much more expensive). The usefulness of this amino acid in biologically active peptide analogues was shown by Schulz and du Vigneaud[114] who incorporated it into oxytocin and

obtained an antagonist. Subsequently, Hruby and co-workers[115] demonstrated that this change in bioactivity was a result of a conformational and dynamic change in the structure of oxytocin, a proposal which accounted for the antagonist activity. Incorporation of L-penicillamine into a cyclic somatostatin analogue was of key importance in converting a somatostatin analogue selective for somatostatin receptors into one highly potent and selective for μ-opioid receptors;[112,122,123] again, conformational effects were shown to be of critical significance.[67,124]

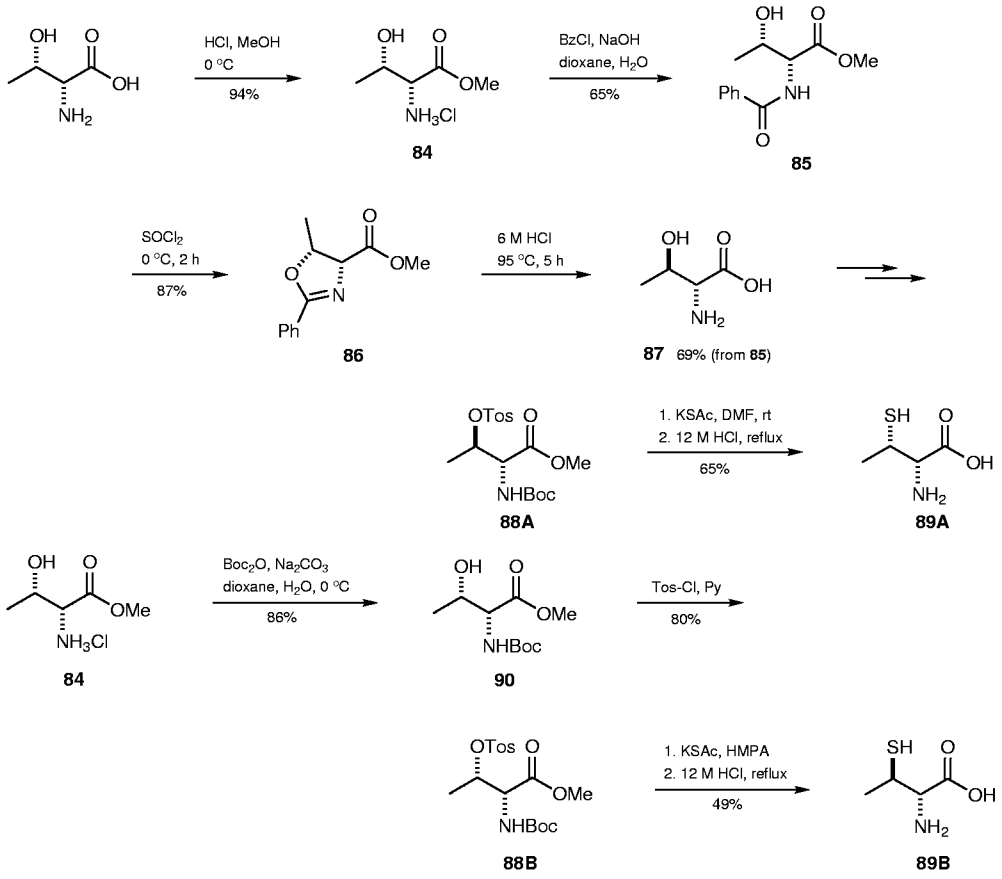
The synthesis of substituted cysteines can be accomplished via Michael addition reactions,[67,124–126] by nucleophilic displacement,[127] from racemic thiazolines,[128] via aziridine ring opening,[129] and by asymmetric synthesis using a chiral auxiliary.[130] The details for some of these methods are described.

## 9.1.7.1 β-Methylcysteine

### 9.1.7.1.1 Nucleophilic Substitution

Nucleophilic substitution of the *O*-tosylthreonine derivatives **88** by the thioacetate anion provides a very convenient pathway for synthesizing β-methylcysteine (**89**, Scheme 25).[127]

**Scheme 25** Synthesis of β-Methylcysteine via Nucleophilic Substitution[127]




*for references see p 49*

**Methyl *N*-Benzoyl-D-threoninate** (85):[127]

| **CAUTION:** *Benzene is carcinogenic!* |
|---|

The methyl ester **84**[131] was benzoylated according to the procedure of Elliot[132] by adding BzCl dropwise to ester **84** dissolved in dioxane/$H_2O$ while maintaining the pH at 8.8–9.0 with 5 M NaOH. After removal of the dioxane, the aqueous phase was extracted with EtOAc. After drying ($MgSO_4$), the EtOAc was removed to afford **85**, which was recrystallized (benzene) to give pure material; yield: 65%; mp 93–94 °C.

**Methyl (4*R*,5*R*)-5-Methyl-2-phenyl-4,5-dihydrooxazole-4-carboxylate** (86):[127]
Treatment of compound **85** with $SOCl_2$ at 0 °C for 2 h gave, after removal of excess $SOCl_2$ under reduced pressure, an oily residue. This residue was dissolved in $CHCl_3$ and poured slowly with stirring into 10% $Na_2CO_3$; care was taken to maintain a basic pH of the soln. The $CHCl_3$ layer was separated, dried, and the solvent was removed under reduced pressure to give **86** which solidified on standing. Crystallization of a portion (petroleum ether) gave a pure sample; yield: 87%; mp 75–76 °C (74–75 °C[132]).

**D-Allothreonine** (87):[127]
Crude **86** was hydrolyzed by heating in 6 M HCl at 95 °C for 5 h followed by cooling to rt, filtering off the benzoic acid precipitate, and concentrating the aqueous filtrate. The resulting syrup was purified by cation-exchange chromatography to give **87**; yield: 69% (based on **85**); mp 274–276 °C; $[\alpha]_D^{25}$ +32.0 (*c* 8, 1 M HCl) {mp 276–280 °C; $[\alpha]_D^{27}$ +32.5 (*c* 8.2, 1 M HCl)[133]}.

**Methyl *N*-(*tert*-Butoxycarbonyl)-*O*-tosyl-D-allothreoninate** (88A):[127]
Compound **88A**, an oil (yield not reported), was prepared from **87** by the methods described for the preparation of the diastereomer, **88B**.

**(3*S*)-3-Methyl-D-cysteine** (89A):[127]
Treatment of compound **88A** with KSAc in DMF overnight at rt, followed by hydrolysis of the crude *S*-acetyl amino acid product in refluxing 12 M HCl for 5 h, gave the crude amino acid. The compound was further purified via cation-exchange chromatography to give pure **89A**; yield: 65%; mp 204–206 °C (206–208 °C[134]); $[\alpha]_D^{25}$ −45 (*c* 1, 1 M HCl) {$[\alpha]_D^{19}$ −41.4 (*c* 1, 1 M HCl)[133]}.

**Methyl *N*-(*tert*-Butoxycarbonyl)-D-threoninate** (90):[127]
The methyl ester **84** was dissolved in a $Na_2CO_3$ soln (dioxane/$H_2O$ 2:1) at 0 °C and treated with $Boc_2O$ according to standard literature protocol[135] to give the threonine derivative **90** as an oil (yield: 86%), which was used without further purification in the next step.

**Methyl *N*-(*tert*-Butoxycarbonyl)-*O*-tosyl-D-threoninate** (88B):[127]
To compound **90** dissolved in pyridine and cooled to 0 °C was added solid Tos-Cl in portions. The reaction was allowed to stand overnight and then poured into ice $H_2O$. The oil that formed was extracted with $Et_2O$. The combined $Et_2O$ fractions were washed with 0.5 M HCl, $H_2O$, and sat. $Na_2CO_3$. Removal of the $Et_2O$, after drying ($MgSO_4$), afforded crude **88B**; yield: 80%. The crude tosylate was purified (silica gel, hexane/EtOAc 9:1 and 7:3); concentration of the fractions containing the tosylate gave **88B** as a clear, viscous oil; yield: 70%.

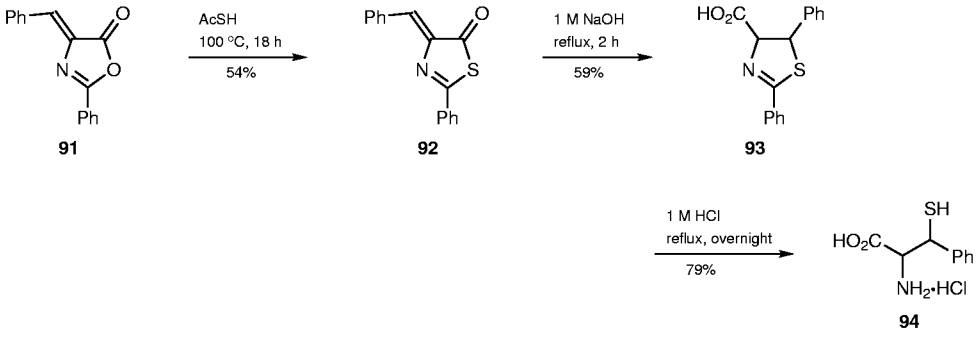
**(3*R*)-3-Methyl-D-cysteine** (89B):[127]
Compound **89B** was prepared from **88B** in a manner similar to that for **89A**, except HMPA was used as solvent. The methyl (*Z*)-2-(*tert*-butoxycarbonylamino)crotonate impurity (ca. 40% of a 90% total yield) was removed from the *S*-acetyl derivative by chromatography (silica gel) to give **89B** after hydrolysis; yield: 49% (based on **88B**); mp 146–149 °C (mp 147–153 °C[134]); $[\alpha]_D^{25}$ +13.0 (*c* 1, 1 M HCl) {$[\alpha]_D^{25}$ +13.5 (*c* 1, 1 M HCl)[134]}.

## 9.1.7.2 β-Phenylcysteine

### 9.1.7.2.1 Oxazoline Ring Opening

(±)-β-Phenylcysteine hydrochloride (**94**) can be prepared by ring opening of oxazoline **91** in a very short and unique synthetic pathway (Scheme 26).[128]

**Scheme 26** Synthesis of β-Phenylcysteine by Oxazoline Ring Opening[128]




**91** **92** **93** **94**

**4-Benzylidene-2-phenylthiazol-5(4*H*)-one (92):**[136]

**CAUTION:** *Benzene is carcinogenic!*

4-Benzylidene-2-phenyloxazol-5(4*H*)-one (**91**; 5 g, 20 mmol) was heated with AcSH (4 g, 52.5 mmol) at 100 °C for 18 h, followed by cooling and addition of EtOH (100 mL) to give **92**; yield: 54%; mp 132–133 °C (EtOH/benzene).

**2,5-Diphenyl-4,5-dihydrothiazole-4-carboxylic Acid (93):**[128]

**CAUTION:** *Benzene is carcinogenic!*

A suspension of 4-benzylidene-2-phenylthiazol-5(4*H*)-one (**92**; 2.66 g, 10.0 mmol) and 1 M NaOH (30 mL) was refluxed for 2 h. The bright yellow soln was allowed to cool to rt, extracted with $Et_2O$ (2 × 15 mL), acidified to pH 1 (pH paper) with concd HCl, and extracted with $Et_2O$ (3 × 25 mL). The combined organic extracts were dried ($MgSO_4$) and concentrated under reduced pressure to give a light yellow solid. Two recrystallizations (benzene) provided **93**; yield: 1.6 g (59%); mp 138–140 °C (136–137 °C[136]); IR (KBr) $\tilde{\nu}_{max}$: 2100–3400 ($CO_2$), 1740 (acid carbonyl), 1590 $cm^{-1}$ (C=N).

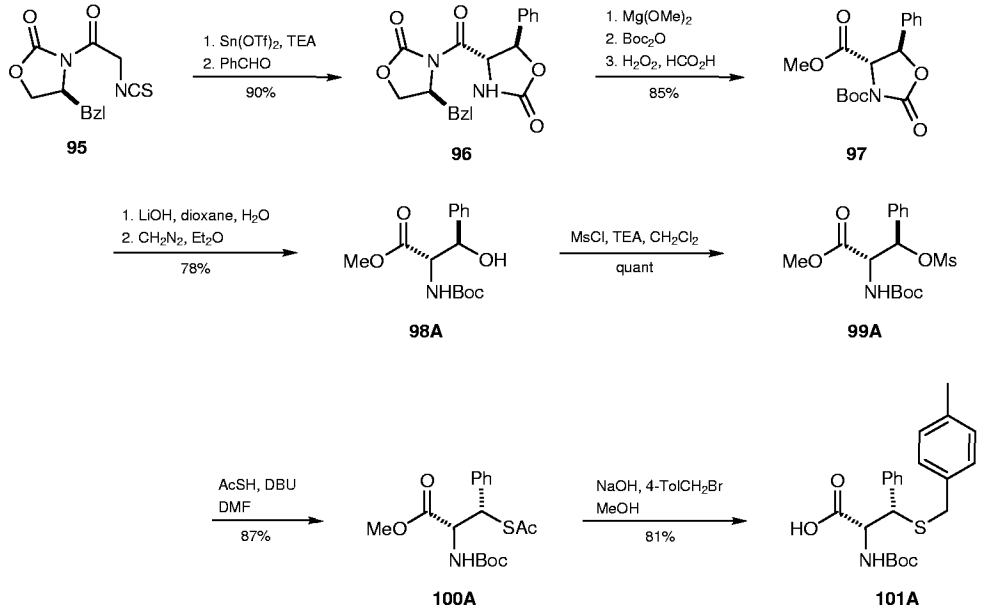
**β-Phenylcysteine Hydrochloride (94):**[128]
A suspension of **93** (100 mg, 0.35 mmol) and 1 M HCl (3 mL) was refluxed overnight. After cooling to rt, the aqueous soln was diluted with $H_2O$ (ca. 7 mL), extracted with $Et_2O$ (3 × 5 mL), concentrated, and dried under reduced pressure to provide crude **94**; yield: 65 mg (79%).

### 9.1.7.2.2 Through Auxiliary Oxazolidinone

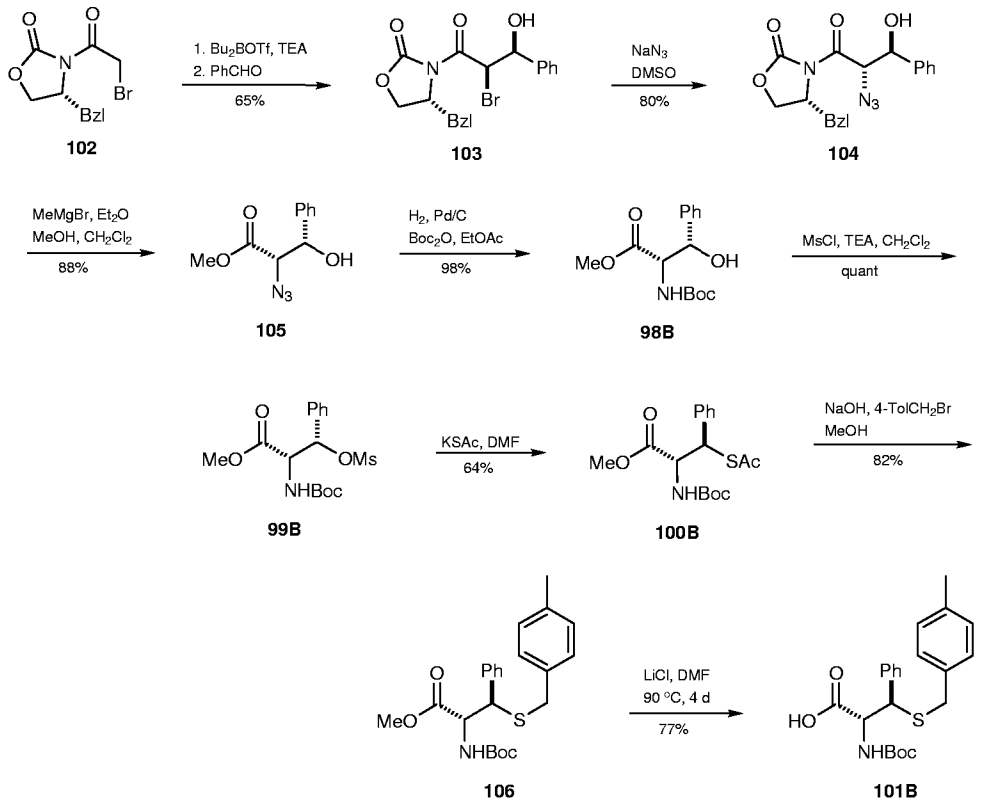
Mesylates of β-hydroxyphenylalanine derivatives, **99A** and **99B**, can be converted into cysteine derivatives **100A** and **100B** by nucleophilic substitution using the thioacetate anion. Selective hydrolysis of protecting groups leads to $N^{\alpha}$-Boc orthogonally protected diastereomers of β-phenylcysteine, **101A** and **101B** (Schemes 27 and 28).[130]

*for references see p 49*

**Scheme 27** Synthesis of a Derivative of (2*R*,3*S*)-3-Phenylcysteine[130]




**Scheme 28** Synthesis of a Derivative of (2*R*,3*R*)-3-Phenylcysteine[130]



**Methyl (2*S*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-hydroxy-3-phenylpropanoate (98A):**[130]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate safety precautions and procedures should be used when handling this reagent (e.g., special glassware, blast shield, well-ventilated fume hood, etc.).*

To a soln of methyl ester **97** (Note: no synthetic procedure or characterization was given for compounds **95–97**; see the related ref[137]) (5.6 g, 17.4 mmol) in dioxane (300 mL) at rt was added freshly prepared 2 M aq LiOH soln (44 mL, 88 mmol). The resultant suspension was stirred at rt overnight. Volatiles were removed under reduced pressure. The residue was then redissolved in 1 M aq $NaHSO_4$ (300 mL) and extracted with $CH_2Cl_2$ (3 ×). The combined organic phases were dried ($Na_2SO_4$) and concentrated under reduced pressure; the residue was then redissolved in $Et_2O$ (300 mL) and, after cooling to 0 °C, a soln of ethereal $CH_2N_2$ was added until a pale yellow color persisted. The soln was stirred at 0 °C for 15 min and at rt for 36 min. The solvent was removed and the residual foamy oil was purified by flash column chromatography (EtOAc/hexanes 3:7) to give the Boc-protected amino alcohol **98A**; yield: 4.01 g (78%). The amino alcohol **98A** was obtained as a white solid which was recrystallized; mp 101–102 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ −15.10 (*c* 0.93, $CHCl_3$).

**Methyl (2*S*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-[(methylsulfonyl)oxy]-3-phenylpropanoate (99A):**[130]
To a soln of alcohol **98A** (0.94 g, 3.18 mmol) and TEA (0.73 g, 7.18 mmol) at 0 °C in $CH_2Cl_2$ was added MsCl (418 mg, 3.65 mmol). The mixture was stirred at 0 °C for 45 min, and then quenched with aq dil HCl and extracted with $CH_2Cl_2$. The organic layer was washed with aq $NaHCO_3$, followed by brine, and dried ($Na_2SO_4$). The solvent was removed under reduced pressure to give the crude mesylate **99A** as a foamy white solid that was used in the next step without further purification.

**Methyl (2*R*,3*S*)-3-(Acetylsulfanyl)-2-(*tert*-butoxycarbonylamino)-3-phenylpropanoate (100A):**[130]
To a soln of mesylate **99A** (1.18 g, 3.16 mmol) in DMF (10 mL) was added a preformed soln of the DBU salt of AcSH [formed by the addition of AcSH (1.2 g, 15.8 mmol) to DBU (1.68 g, 11.07 mmol) in DMF (5 mL)]. The mixture was stirred at rt for 30 h and was then quenched with $H_2O$, diluted with $CH_2Cl_2$, and washed with $H_2O$ (4 ×). The organic layer was washed with brine and dried ($Na_2SO_4$). The solvent was removed under reduced pressure and the crude oil was purified by flash column chromatography (EtOAc/toluene 8:92) to give acetylated thiol **100A** as a colorless liquid, followed by a small amount of elimination compound [yield: 50 mg (6%)]; yield of **100A**: 0.97 g (87%); $[\alpha]_D^{20}$ +143.60 (*c* 2.50, $CHCl_3$); IR (neat) $\tilde{\nu}_{max}$: 3380, 1750, 1690 $cm^{-1}$.

**(2*R*,3*S*)-2-(*tert*-Butoxycarbonylamino)-3-[(4-methylbenzyl)sulfanyl]-3-phenylpropanoic Acid (101A):**[130]
To acetylated thiol **100A** (520 mg, 1.47 mmol) dissolved in MeOH (8 mL) was added 1 M aq NaOH (1.62 mL, 1.62 mmol) followed by 4-methylbenzyl bromide (300 mg, 1.62 mmol). After 20 min, additional base (2.5 equiv) was added. The reaction was stirred for 3 h at rt (monitored by TLC). The soln was then carefully neutralized with dil HCl. The volatiles were removed under reduced pressure and the residue was extracted with $CHCl_3$ (3 ×). The organic layer was washed with brine, dried ($Na_2SO_4$), and concentrated under reduced pressure. The crude product was purified by flash column chromatography (AcOH/EtOAc/hexane 2:20:78) to give acid **101A** as a white solid which was recrystallized; yield: 480 mg (81%); mp 131–132 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ +211.70 (*c* 0.99, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 3380, 1700 $cm^{-1}$.

**(4*R*)-3-[(2*S*,3*S*)-2-Azido-3-hydroxy-3-phenylpropanoyl]-4-benzyloxazolidin-2-one (104):**[130]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

A soln of bromide **103** (Note: no synthetic procedure or characterization was given for compounds **102** and **103**; see the related ref[137]) (3.1 g, 7.67 mmol) and $NaN_3$ (0.99 g, 15.3 mmol) in DMSO (26 mL) was stirred at rt for 5 h. The resultant dark soln was diluted with hexane/$CHCl_3$ (1:1), then washed with $H_2O$ (4 ×), dried ($Na_2SO_4$), and concentrated under reduced pressure to give a dark oil which crystallized. Purification by recrystallization ($Et_2O$/hexane) gave **104** as a white solid; yield: 2.2 g (80%); mp 115–116 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ −6.60 (*c* 2.2, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 3420, 2100, 1760, 1700 $cm^{-1}$.

*for references see p 49*

**Methyl (2*S*,3*S*)-2-Azido-3-hydroxy-3-phenylpropanoate (105):**[130]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

To a soln of the azide **104** (1.52 g, 4.15 mmol) in anhyd MeOH (8 mL) and $CH_2Cl_2$ (8 mL) at 0 °C was added, via cannula, a suspension formed by the addition of 0.88 M MeMgBr in $Et_2O$ (5.2 mL, 4.5 mmol) to anhyd MeOH (5 mL). After stirring for 3 min, the reaction was quenched by the addition of 1 M aq $NaHSO_4$ (20 mL) and extracted with $CH_2Cl_2$ (3 ×). The combined organic phases were dried ($Na_2SO_4$) and concentrated under reduced pressure. The residue was purified by flash column chromatography ($CH_2Cl_2$) to afford **105** as a white solid which was recrystallized; yield: 807 mg (88%); mp 40–41 °C ($Et_2O$/hexane); IR (KBr) $\tilde{\nu}_{max}$: 3500, 2120, 1740 $cm^{-1}$.

**Methyl (2*S*,3*S*)-2-(*tert*-Butoxycarbonylamino)-3-hydroxy-3-phenylpropanoate (98B):**[130]

**CAUTION:** *Many organic and inorganic azides are unstable or explosive under appropriate conditions of initiation. Appropriate safety precautions and procedures should be adopted when handling these substances.*

10% Pd/C (80 mg) in EtOAc (4 mL) was vigorously stirred under $H_2$ for 15 min. To this suspension was added a mixture of the azido alcohol **105** (800 mg, 3.6 mmol) and $Boc_2O$ (943 mg, 4.32 mmol) in EtOAc (4 mL). The resulting mixture was stirred under $H_2$ at rt for 2 h (monitored by TLC) and then filtered through a plug of Celite. The filtrate was concentrated under reduced pressure and the resulting white solid was purified by short column flash chromatography (EtOAc/hexane 3:7) to give Boc-protected amino alcohol **98B** as a white solid; yield: 1.04 g (98%); mp 101–102 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ +83.30 (*c* 1.06, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 3450, 3390, 1760, 1710 $cm^{-1}$.

**Methyl (2*S*,3*S*)-2-(*tert*-Butoxycarbonylamino)-3-[(methylsulfonyl)oxy]-3-phenylpropanoate (99B):**[130]
To a soln of alcohol **98B** (1.94 g, 6.58 mmol) and TEA (1.25 g, 12.4 mmol) at 0 °C in $CH_2Cl_2$ (15 mL) was added MsCl (0.96 g, 8.4 mmol). The mixture was stirred at 0 °C for 20 min. It was then quenched with cold dil HCl and extracted into $CH_2Cl_2$. The organic layer was washed with aq $NaHCO_3$, followed by brine, and dried ($Na_2SO_4$). The solvent was removed under reduced pressure, and the crude mesylate **99B** was obtained as a foamy white solid that was used in the next step without further purification.

**Methyl (2*R*,3*R*)-3-(Acetylsulfanyl)-2-(*tert*-butoxycarbonylamino)-3-phenylpropanoate (100B):**[130]
To a soln of mesylate **99B** (2.72 g, 7.29 mmol) in DMF (7 mL) was added KSAc (3 g, 26.3 mmol) followed by additional DMF (5 mL). The mixture was stirred at rt for 20 h. The product was partitioned between $H_2O$ and $CH_2Cl_2$. The organic layer was washed with $H_2O$ (4 ×) and dried ($Na_2SO_4$). The solvent was removed under reduced pressure and the crude product was purified by flash column chromatography (EtOAc/hexane 5:95 to 1:1) to give acetylated thiol **100B** as a tan solid. The acetylated thiol was recrystallized to produce a white solid; yield: 1.65 g (64%); mp 87–88 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ −97.20 (*c* 1.0, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 3380, 1745, 1690 $cm^{-1}$.

**Methyl (2*R*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-[(4-methylbenzyl)sulfanyl]-3-phenylpropanoate (106):**[130]
The acetylated thiol **100B** (500 mg, 1.4 mmol) was dissolved in MeOH (2 mL), and 1 M aq NaOH (1.6 mL, 1.6 mmol) was added, followed by 4-methylbenzyl bromide (300 mg, 1.62 mmol). After 20 min, most of the starting material was consumed (monitored by TLC). The soln was then carefully neutralized with dil HCl. The volatiles were removed under reduced pressure and the residue was extracted with $CH_2Cl_2$ (3 ×). The organic layer was washed with brine, dried ($Na_2SO_4$), and concentrated under reduced pressure. The crude product was purified by flash column chromatography (EtOAc/hexane 5:95 to 2:8) to give ester **106** as a white solid; yield: 480 mg (82%); mp 97–98 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ −171.80 (*c* 1.05, $CHCl_3$).

**(2*R*,3*R*)-2-(*tert*-Butoxycarbonylamino)-3-[(4-methylbenzyl)sulfanyl]-3-phenylpropanoic Acid (101B):**[130]
The methyl ester **106** (380 mg, 0.91 mmol) and anhyd LiCl (376 mg, 8.89 mmol) were dissolved in dry DMF (10 mL). The mixture was heated at 90 °C for 4 d, cooled to rt, then quenched with dil HCl and extracted several times with EtOAc. The organic layer was washed with brine, dried ($Na_2SO_4$), and concentrated under reduced pressure. The crude product was purified by flash column chromatography (EtOAc/hexane 2:8) to recover starting material **106** (75 mg, 20%); elution with EtOAc/AcOH/hexane (25:5:70) gave carboxylic acid **101B** as a white solid; yield: 77%; mp 107–108 °C ($Et_2O$/hexane); $[\alpha]_D^{20}$ −143.70 (*c* 1.66, $CHCl_3$).

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# 9.2 Synthesis of Peptides Containing Proline Analogues

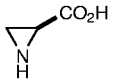
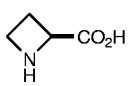
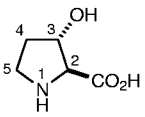
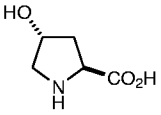
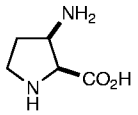
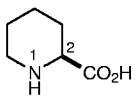
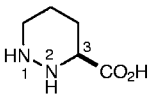
P. THAMM, H.-J. MUSIOL, and L. MORODER

Proline occupies a special place among the proteinogenic amino acids. On one hand, proline residues restrict the conformational space of the peptide chain, but on the other hand, both *cis* and *trans* Xaa-Pro bond isomers occur in solution since the free enthalpy difference between the two isomeric states is small (2.0 vs 10.0 $kJ \cdot mol^{-1}$ for Xaa-nonPro bonds).[1] Nevertheless, only 5.2% of Xaa-Pro bonds in all known X-ray structures of proteins are in *cis* configuration versus the 30% expected from the thermodynamic equilibrium at 293 K.[2] Since the rotational barrier of the *cis/trans* isomerization is high ($85 \pm 10$ $kJ \cdot mol^{-1}$), this interconversion is slow and thus it represents one of the rate limiting steps in protein folding.[3] Nature uses peptidyl-proline *cis/trans* isomerases to accelerate this folding step.[4,5] Comparative studies performed with the proline analogues thiazolidine-4-carboxylic acid (**11**) and oxazolidine-4-carboxylic acid (**9**),[6] (4*R*)-hydroxyproline (**4**) and the related (4*R*)-fluoro derivative[7–9] clearly reveal that the presence of electron-withdrawing substituents on/in the five-membered ring affects the kinetics of the Xaa-Pro bond isomerization by lowering the free energy of activation via pyramidalization of the prolyl nitrogen. A similar shift of hybridization from $sp^2$ toward $sp^3$ is responsible for the lower rotational barrier of the four-membered ring in azetidine-2-carboxylic acid (**2**),[6,10] whereas for the six-membered analogue piperidine-2-carboxylic acid (**6**) increased steric hindrance might be responsible for a destabilization of the planar peptide amide bond.[6,11]

## Natural Proline Analogues

Besides proline itself, *trans*-3- and *trans*-4-hydroxy-L-proline (**3** and **4**, respectively) are found in proteins as a result of posttranslational hydroxylation, particularly in collagens.[12] Whilst various alkylated derivatives of proline and hydroxyproline, as well as 3-aminoproline, have been identified as constituents of natural products (Table 1),[13,14] natural occurrence of halogenated proline derivatives, as well as of the thiazolidine- and oxazolidine-4-carboxylic acids (**11** and **9**, respectively), has not been reported. Some of these proline analogues are, however, incorporated into natural products when administered under particular conditions to *S. antibioticus*.[13] Even bioexpression of proteins containing thiazolidine-4-carboxylic acid (**11**) has successfully been performed, whilst incorporation of the related oxa- and selena-analogues **9** and **13** into proteins failed.[15] The proline analogues with ring restriction, i.e. azetidine-2-carboxylic acid (**2**), and ring expansion, i.e. piperidine-2-carboxylic acid (**6**) and related dehydro-, mono-, and bis-hydroxylated or oxo derivatives, as well as piperazic acid (**7**) and the related hydroxy and chloro derivatives are found in natural products.[13]

**Table 1** Natural Proline Analogues and Homologues

| Structure | Nomenclature | Symbols, Abbreviations[a] |
|---|---|---|
|  1 | azyline<br>(2*S*)-aziridine-2-carboxylic acid | Azy |
|  2 | (2*S*)-azetidine-2-carboxylic acid | Aze |
|  3 | *trans*-3-hydroxy-L-proline<br>(2*S*,3*R*)-3-hydroxypyrrolidine-2-carboxylic acid | (2*S*,3*R*)-3-Hyp<br>(2*S*,3*R*)-Pro(3-OH) |
|  4 | *trans*-4-hydroxy-L-proline<br>(2*S*,4*R*)-4-hydroxypyrrolidine-2-carboxylic acid | (2*S*,4*R*)-Hyp<br>(2*S*,4*R*)-Pro(4-OH) |
|  5 | *cis*-3-amino-L-proline<br>(2*S*,3*S*)-3-aminopyrrolidine-2-carboxylic acid | (2*S*,3*R*)-Pro(2-$NH_2$) |
|  6 | L-pipecolic acid<br>(2*S*)-piperidine-2-carboxylic acid | Pip |
|  7 | L-piperazic acid<br>(3*S*)-piperidazine-3-carboxylic acid<br>(3*S*)-[1,2]diazinane-3-carboxylic acid | Piz |

[a]Ring substituents in parentheses.

## Synthetic Proline Analogues

Conformational studies of the Xaa-Pro imide bond geometries of bioactive peptides have been extensively undertaken, but it is difficult to define whether the *cis* or *trans* conformations are associated with particular bioactive structures.[16,17] Therefore, restriction of the Xaa-Pro imide conformations would provide one promising means for studying relationships between imide bond geometry and bioactivity of peptides and for tuning the biological, pharmaceutical, or physicochemical properties of naturally occurring, as well as of de novo designed, linear and cyclic peptides. For this purpose, various 3-, 4-, and 5-alkylproline derivatives have been synthesized to induce preference for *trans*- or *cis*-isomer populations, respectively.[18–28] Related synthetic procedures have been comprehensively reviewed.[29] Among the large variety of proline mimetics based on ring substitutions with alkyl and aromatic groups, various derivatives represent side-chain conformationally restricted α-amino acids and, thus, the reader is directed to Section 9.1 for selected examples.

*for references see p 81*

Alternatively, the use of dipeptide isosters mimicking the *cis*- or *trans*-imide geometry[30] or replacement of the imide bond with *Z*- or *E*-alkene units[31–36] (see also Section 10.5) have been proposed.

Regarding halogenated proline derivatives, NMR spectroscopic analysis of model compounds has clearly revealed a high preference of (4*R*)- and (4*S*)-fluoroproline for the *trans* and *cis* conformation of the related imide bonds as well as for the *exo*- and *endo*-pucker, respectively.[7–9] This property was then efficiently exploited to stabilize the *trans* conformation in collagen peptides by replacing hydroxyproline with (4*R*)-fluoroproline,[37–40] while bioincorporation of (4*S*)-fluoroproline into barstar leads to a significant stabilization of the protein by favoring the native *cis*-imide geometry at Pro[48].[9]

Following the lesson from nature, great attention has also been paid to the reduction or expansion of the proline ring size and to the incorporation of heteroatoms into the ring to further tune the conformational properties of proline peptides. The following section is mainly focused on proline analogues with restricted or expanded ring-size and those containing additional heteroatoms in the five- and six-membered ring (Table 2).

**Table 2** Nonnatural Proline Analogues and Homologues

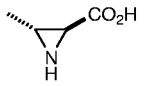
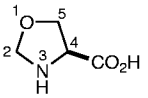
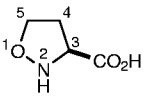
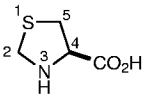
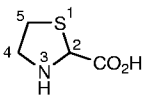
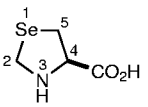
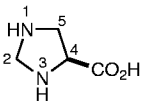
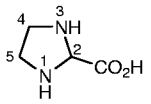
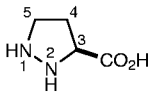
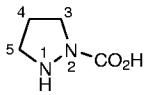
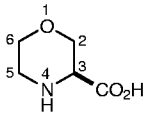
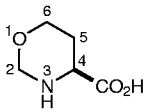
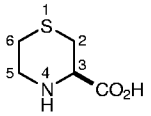
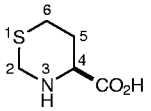
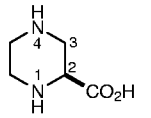
| Structure | Nomenclature | Symbols, Abbreviations |
|---|---|---|
|  **8** | (2*S*,3*R*)-3-methylaziridine-2-carboxylic acid | (2*S*,3*R*)-Azy(3-Me) |
|  **9** | 4-oxa-L-proline[a]<br>(4*S*)-oxazolidine-4-carboxylic acid[b] | Oxa[b] |
|  **10** | 5-oxa-L-proline[a]<br>(3*S*)-isoxazolidine-3-carboxylic acid[b] | 3-Oxa[b] |
|  **11** | 4-thia-L-proline[a]<br>(4*R*)-thiazolidine-4-carboxylic acid[b] | Thz[b] |
|  **12** | 3-thia-L-proline[a]<br>thiazolidine-2-carboxylic acid[b] | 2-Thz[b] |
|  **13** | 4-selena-L-proline[a]<br>(4*R*)-1,3-selenazolidine-4-carboxylic acid[b] | |
|  **14** | 4-aza-L-proline[a]<br>(4*S*)-imidazolidine-4-carboxylic acid[b] | |

**Table 2** (cont.)

| Structure | Nomenclature | Symbols, Abbreviations |
|---|---|---|
|  **15** | 3-aza-L-proline[a]<br>imidazolidine-2-carboxylic acid[b] | |
|  **16** | 5-aza-L-proline[a]<br>(3*S*)-pyrazolidine-3-carboxylic acid[b] | |
|  **17** | 2-azaproline[a]<br>pyrazolidine-2-carboxylic acid[b] | |
|  **18** | 4-oxa-L-pipecolic acid[a]<br>(3*S*)-morpholine-3-carboxylic acid[b] | |
|  **19** | 5-oxa-L-pipecolic acid[a]<br>(4*S*)-tetrahydro-2*H*-1,3-oxazine-4-carboxylic acid[b] | |
|  **20** | 4-thia-L-pipecolic acid[a]<br>(3*R*)-1,4-thiazinane-3-carboxylic acid[b] | |
|  **21** | 5-thia-L-pipecolic acid[a]<br>(4*S*)-1,3-thiazinane-4-carboxylic acid[b] | |
|  **22** | 4-aza-L-pipecolic acid[a]<br>(2*S*)-piperazine-2-carboxylic acid[b] | |

[a] Trivial names, numbering of substituents in and on the rings are based on azyline, proline, and pipecolic acid.
[b] Names and numbering according to the IUPAC rules.

## 9.2.1 Aziridine-2-carboxylic Acid

(2*S*)-Aziridine-2-carboxylic acid (**1**), also named azyline (Azy),[41,42] exhibits an electron-withdrawing or quasi-aromatic effect on the nitrogen and is therefore susceptible to an attack by nucleophiles.[41,43] Unprotected aziridine-2-carboxylic acid (**1**) is very labile, but the

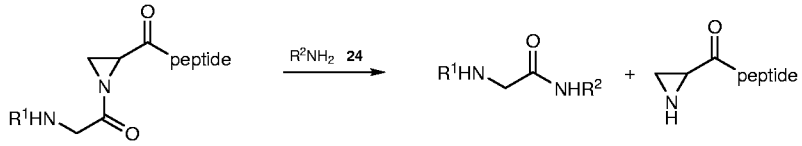
*for references see p 81*

alkaline metal salts are stable.[41,44] At low temperatures (0–10 °C) aziridine-2-carboxylic acid derivatives with a free N-terminus are generally inert toward reagents such as acetic acid, trifluoroacetic acid and formic acid. However, N-acylated derivatives are very susceptible to these acids, since the carboxylate anion attacks as nucleophile the β-carbon of the aziridine ring to produce the corresponding *O*-acyl-β-hydroxy amino acid derivatives.[41] Similarly, *N*-tosyl- and *N*-(2-nitrophenylsulfonyl)aziridine-2-carboxylic acid esters are ring-opened with primary alkyl amines to $N^{\beta}$-alkylated 2,3-diaminopropanoic acid[45] and with amino acid derivatives to produce useful intermediates for β-lactam synthesis.[45,46]

In contrast, N-alkylated derivatives are significantly more stable.[47] Regarding reaction with hydrogen halides, in the case of small substituents at the nitrogen or at C3 the attack by the halide anion takes place predominantely at C2 with formation of α-halo-β-amino acids. However, with larger substituents the nucleophilic attack occurs at C3 with production of β-halo-α-amino acids. Additionally, ring opening is directed by the polarity of the solvent, e.g. water promotes the attack at C3.[48] With bases such as triethylamine or 1,4-diazabicyclo[2.2.2]octane, ring opening to dehydroalanine may occur.[41,49,50] Isomerization to dehydroalanines can also be effected on treatment with sodium iodide in acetone.[51] The stability of unprotected 2-carbonylaziridine-substituted peptides is, in most cases, sufficient for peptide synthesis and purification procedures, but not for storage over longer periods as lyophilized materials.[47] Whilst with primary amines **24** aminoacyl(glycyl)-2-carbonylaziridine-substituted peptides **23** are cleaved to produce aminoacylglycine amides **25** and 2-carbonylaziridine-substituted peptides **26** (Scheme 1),[52,53] such transacylation was found to proceed very slowly with amino acid esters.[49] Except for (glycyl)-2-carbonylaziridine-substituted peptides, the aziridine moiety is stable under the conditions of catalytic hydrogenation.[54]

Aziridine-2-carboxylic acid derivatives have gained considerable interest since they can be considered as a versatile synthetic equivalent of the "β-alanyl cation" synthon suitable for transformation into β-substituted *S*-amino acids.[55–57,262]

**Scheme 1** Transacylation of Aminoacyl(glycyl)-2-carbonylaziridine Peptides with Primary Amines[52,53]




**23** **25** **26**

### Analytical Characterization

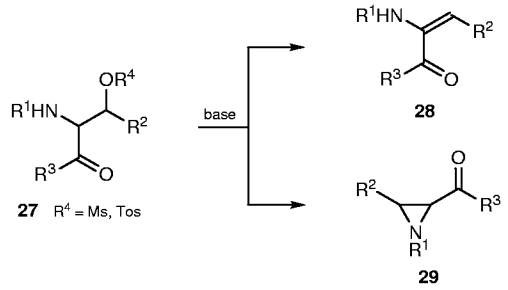
Aziridines can be detected by a color reaction with 4-(4′-nitrobenzyl)pyridine that was originally developed for the detection and/or determination of a wide range of alkylating substances.[58] It can be adapted to TLC.[59,60] Upon acid hydrolysis of aziridine-2-carboxylic acid containing peptides, as required for amino acid analysis, this amino acid is recovered as serine in varying yields.[47] Configurational characterization of aziridine-2-carboxylates is performed by NMR spectroscopy[47,51,61–63] and X-ray analysis.[61] In addition, optical rotation of the related benzyl *N*-tritylaziridine-2-carboxylate is also used[64] with $[\alpha]_D$ −95.5 (*c* 1, THF) for the 2*S*-enantiomer[44] as reference.

## 9.2.1.1 Synthesis of Aziridine-2-carboxylic Acid

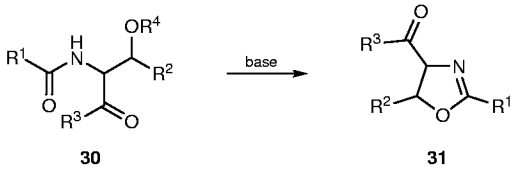
For the enantioselective syntheses of stereochemically pure (2*S*)-aziridine-2-carboxylic acid and (2*S*,3*S*)-3-methylaziridine-2-carboxylic acid, so far the precursors of choice are L-serine or L-threonine. Ring closure of these β-hydroxy-α-amino acids via β-elimination is per-

formed by treatment of the respective *O*-tosyl[44,65,66] or *O*-mesyl derivatives **27**[41,47,67] with tertiary bases (Scheme 2). The course of the reaction is governed by the substituents at the amino and carboxy groups and by the type of β-hydroxy-α-amino acid. Aziridine-2-carboxylic acid (**1**) is produced from the serine precursor and its 3-methyl derivative **8** from the threonine precursor, and dehydroalanine and α-amino crotonic acid **28**, respectively, in varying amounts. Cyclization of threonine derivatives is favored relative to serine derivatives. Moreover, $N^{\alpha}$-alkylation of the precursors favors cyclization, whilst their $N^{\alpha}$-acylation to **30** normally leads to oxazolines **31** (Scheme 3) instead of alkenes **28** or aziridines **29**.[63] Esters of the precursors facilitate alkene formation, whereas their amides direct cyclization. Based on these results, the $N^{\alpha}$-trityl-β-hydroxy-α-amino acid benzyl esters represent the best compromise between aziridine ring formation and easy deprotection of the resulting aziridine-2-carboxylic acid derivative **29**.

**Scheme 2** Synthesis of Aziridine-2-carboxylic Acids via the Serine and Threonine *O*-Sulfonates[41,44,47,65–67]




**Scheme 3** Formation of Oxazolines[63]




Ring closure of β-hydroxy-α-amino acids with sulfuryl chloride/triethylamine[68] is accompanied by formation of β-chloroalanine,[61] whereas cyclization of urethane-protected serine and threonine by the Mitsunobu reaction[54,69,70] leads to oxazoline and dehydroalanine formation as side products.[47,71] Formation of dehydroalanine can be prevented by bulky carboxy protecting groups such as *tert*-butyl esters.[69]

Attempts to cyclize *N*-trityl-β-chloroalanine esters in the presence of strong bases (such as NaH, DBU, tBuOK, or tBuOLi) lead to dehydroalanine; however with weak bases (such as $KHSO_3$, $K_2SO_3$, $K_2HPO_4$, $K_2S_2O_5$) or weak organic amines (such as TEA or DIPEA) upon refluxing in alkyl nitriles (e.g., MeCN, EtCN, etc.) for 24 hours almost quantitative conversion into the related *N*-tritylaziridine-2-carboxylic acid esters takes place.[72] Using other solvents, such as THF, DMSO, DMF, $CH_2Cl_2$, or $Et_2O$, the starting material is recovered. Under optimal conditions methyl (2*S*,3*S*)-*N*-trityl-3-chloro-2-aminobutyrate, prepared from L-Thr, is cyclized quantitatively to the related (2*S*,3*R*)-3-methylaziridine-2-carboxylate.[72]

An alternative route starting from serine[73] or threonine[68,74] makes use of diethoxytriphenylphosphorane. Attempts to asymmetrically synthesize (2*S*)-aziridine-2-carboxylic acid (**1**) by treating α,β-dibromopropanoates with chiral amines[75] or by the Staudinger reaction from oxirane-2-carboxylic acid ester[70,76] leads to optically impure products, whereas 3-alkyl derivatives of *tert*-butyl aziridine-2-carboxylates can be prepared with high *cis*-selectivity from α-halo ester enolates and *N*-trimethylsilyl imines.[77] Moreover, when optically

*for references see p 81*

active oxirane-2-carboxylic esters, prepared from allylic alcohols employing the Sharpless epoxidation, are treated with sodium azide the resulting azido alcohols are subsequently converted into aziridine-2-carboxylic esters by reaction with triphenylphosphine in good yields and with high optical purity; however, the stereochemical configuration at C2 and C3 is inverted.[78] This method allows for the preparation of 3-substituted aziridine-2-carboxylic acids.

The Gabriel–Cromwell reaction of amines with chiral α,β-unsaturated α-bromo carbonyl compounds was exploited for the synthesis of aziridine-2-carboxylic acid derivatives.[79] This procedure was optimized for a solid support synthesis in which the peptide resin was acylated with 2,3-dibromopropanoic acid active ester in the presence of 3 equivalents of NMM to produce directly on resin the α-bromoacrylamide for the addition of amines to produce the aziridine ring.[80]

Conversion of *N*-tritylaziridine-2-carboxylic acid benzyl esters into the free amino acid is somewhat cumbersome due to the formation of byproducts.[47,51] In fact, exposure to trifluoroacetic acid[41,44,81] for removal of the trityl group leads to impure benzyl ester and treatment with hydrogen chloride in methanol to chloroalanine benzyl ester.[47] Similarly, cleavage with formic acid[68] or 4-toluenesulfonic acid[47] under most of the investigated conditions resulted in impure products. Among the various salts used for the isolation of the N-unprotected benzyl ester the dibenzosulfimide salt proves to be the most suited, as it is obtained in a pure form without additional purification steps.[47] Cleaner N-detritylation products are expected from treatment with TFA in the presence of reducing agents capable of trapping the trityl cations, such as triethylsilane or trimethylamine/$BH_3$; these conditions are successfully used in the case of more sensitive *N*-tritylaziridines.[82]

Finally, saponification of the benzyl ester leads to the aziridine-2-carboxylic acid lithium salt as a stable compound.[44]

**Cyclization of β-Hydroxy-α-amino Acid *O*-Sulfonates: Trt-Azy-OBzl; Typical Procedure:**[44,65]
To a soln of Trt-Ser-OBzl (65.5 g, 0.15 mol) in dry pyridine (200 mL) was added a soln of Tos-Cl (85.8 g, 0.45 mol) (or the corresponding amount of Ms-Cl[65]) in dry pyridine (50 mL) at −10 °C over a period of 1 h. After stirring for 24 h at 0 °C, the solvent was removed and the residue partitioned between EtOAc and $H_2O$. The organic layer was washed with 10% citric acid and $H_2O$, and dried ($Na_2SO_4$). The solvent was removed to give the *O*-tosylate as an oily product; yield: 84.3 g (95%).

The *O*-tosylate was dissolved in dry THF (100 mL), and TEA (39.2 mL, 0.28 mol) was added. After refluxing the soln at 75 °C for 24 h, the mixture was concentrated and the residue dissolved in EtOAc. The organic layer was washed with 10% citric acid, 1 M $NaHCO_3$, and $H_2O$, and dried ($Na_2SO_4$). The solvent was removed and the residue recrystallized (EtOAc/$Et_2O$/hexane); yield: 31.4 g (57.8%); mp 114–116 °C; $[\alpha]_D^{20}$ −95.5 (*c* 1, THF).[44] Purification was by chromatography (silica gel, petroleum ether/EtOAc 10:1); yield: 57%; mp 117–118 °C; $[\alpha]_D^{20}$ −98.3 (*c* 1, THF).[65]

**Cyclization of β-Hydroxy-α-amino Acids with Sulfuryl Chloride: Trt-Azy-OBzl; Typical Procedure:**[61]
To a soln of Trt-Ser-OBzl (142.5 g, 326 mmol) and TEA (135.8 mL, 977 mmol) in dry toluene (1.1 L), $SO_2Cl_2$ (31.69 mL, 391 mmol) dissolved in dry toluene was added dropwise within 20 min under stirring at −50 °C. After additional stirring for 1 h at −50 °C and 2 h at rt, the mixture was concentrated and the residue partitioned between EtOAc and sat. NaCl (× 4); the combined organic layers were washed with $H_2O$ (× 3) and dried ($Na_2SO_4$). The solvent was removed and the resulting oily crude product was chromatographed in aliquots of 4.5 g (silica gel, $CH_2Cl_2$) and recrystallized (hexane); yield: 79.8 g (56%); mp 117 °C; $[\alpha]_D^{20}$ −98.3 (*c* 1, THF). As side product, Trt-Ala(Cl)-OBzl was isolated; yield: 38.0 g (27%).

**Cyclization of β-Chloro-α-amino Acids: Trt-Azy-OBzl; Typical Procedure:**[72]
To a stirred suspension of $KHSO_3$ (2.5 mmol) in anhyd MeCN (15 mL) was added *N*-Trt-β-chloroalanine benzyl ester (0.5 mmol) and the mixture was refluxed for 48 h. After cooling to rt, 5% $NaHCO_3$ (5 mL) was added and the product was extracted with $CH_2Cl_2$ (2 × 15 mL). The combined organic layers were dried ($MgSO_4$) and concentrated. The residue was purified by preparative TLC (silica gel); yield: 98%.

**H-Azy-OBzl·Dibenzosulfimide Salt; Typical Procedure:**[47]
To a stirred soln of Trt-Azy-OBzl (2.98 g, 7.1 mmol) in $CHCl_3$ (40 mL) a soln of Tos-OH·$H_2O$ (13.5 g, 71 mmol) in MeOH (30 mL) was added. After 2 h, the solvent was removed and the residue partitioned

between EtOAc and $H_2O$. The aqueous phases were combined and neutralized with $NaHCO_3$. The product was re-extracted with EtOAc, the organic phase dried ($Na_2SO_4$), and concentrated. The residue was dissolved in MeOH, and after addition of dibenzosulfimide (DBSI; 1 equiv) the product was precipitated with $Et_2O$; yield: 1.66 g (72%); mp 115–116 °C; $[\alpha]_D^{23}$ −10.8 (*c* 1, MeOH).

**H-Azy-OH·Lithium Salt:**[44]

To a soln of H-Azy-OBzl (540 mg, 3.05 mmol) in MeOH (3 mL) was added a soln of LiOH (115.2 mg, 2.74 mmol) in $H_2O$ (1 mL) with stirring at −10 °C. After keeping the mixture at −5 °C for 6 h, it was taken to dryness and the residue was crystallized (EtOH); yield: 100%; mp 260 °C; $[\alpha]_D^{23}$ −39.5 (*c* 1, $H_2O$).

## 9.2.1.2 Protected Aziridine-2-carboxylic Acid Derivatives

The key intermediate in the preparation of derivatives suitable for peptide synthesis (Table 3) is the benzyl (2*S*)-*N*-tritylaziridine-2-carboxylate, since it is smoothly converted into the N-protected imino acid by catalytic hydrogenolysis.[47] Similarly, the related methyl ester is saponified by lithium hydroxide to produce *N*-tritylaziridine-2-carboxylate.[83,84] Detritylation to benzyl (2*S*)-aziridine-2-carboxylate is more difficult, but the dibenzosulfimide salt is found to be perfectly stable on storage as a solid for longer periods of time (see Section 9.2.1.1).[47] In solution, upon addition of bases the benzyl ester is sufficiently stable to allow for peptide syntheses.[47]

**Table 3** Derivatives of (2*S*)-Aziridine-2-carboxylic Acid and (2*S*,3*S*)-3-Methylaziridine-2-carboxylic Acid[41,44,47,68,69,81,83,84,256,257]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| H-Azy-OLi | 260 (dec) | $[\alpha]_D^{23}$ −39.5 (*c* 1, $H_2O$) | [44] |
| H-Azy-OBzl | [b] | $[\alpha]_D^{23}$ −56.8 (*c* 1, THF) | [44] |
| H-Azy-OBzl·DBSI[a] | 115–116[a] | $[\alpha]_D^{23}$ −10.8 (*c* 1, MeOH)[a] | [47] |
| H-Azy-$NH_2$ | 141–142 | $[\alpha]_D^{23}$ −44.8 (*c* 1.07, DMF) | [44] |
| Trt-Azy-OH | > 175(dec) | $[\alpha]_D^{23}$ −59.6 (*c* 1, DMF) | [47] |
| Ac-Azy-OBzl | [b] | $[\alpha]_D^{23}$ −39.6 (*c* 1, MeOH) | [44] |
| Boc-Azy-OMe | | $[\alpha]_D^{23}$ −68.8 (*c* 0.6, MeOH) | [81] |
| Tos-Azy-OtBu | 97–99 | $[\alpha]_D^{25}$ −52.3 (*c* 1, $CH_2Cl_2$) | [69] |
| Trt-Azy-OBzl | 117–118 | $[\alpha]_D^{20}$ −98.3 (*c* 1, THF) | [44,47,68] |
| Z-Azy-OMe | [b] | $[\alpha]_D^{23}$ −95.4 (*c* 1.1, MeOH) | [81,83] |
| Z-Azy-OBzl | [b] | $[\alpha]_D^{23}$ −19.0 (*c* 1, MeOH) | [81,256,257] |
| Trt-Azy-OPfp | 79–80 | $[\alpha]_D^{23}$ −127.8 (*c* 1, MeOH) | [47] |
| H-Azy(3-Me)-OMe | [b] | | [41] |
| Trt-Azy(3-Me)-OH | 173–175 | $[\alpha]_D^{23}$ −77.3 (*c* 1, MeOH) | [84] |
| Trt-Azy(3-Me)-OMe | 103–107 | $[\alpha]_D^{23}$ −97.7 (*c* 1.1, THF) | [41] |
| Trt-Azy(3-Me)-OMe | 109–110 | $[\alpha]_D^{23}$ −102.8 (*c* 1.1, $CHCl_3$) | [84] |
| Trt-Azy(3-Me)-OBzl | 99–101 | $[\alpha]_D^{23}$ −108.9 (*c* 1.54, $CHCl_3$) | [68] |
| Z-Azy(3-Me)-OBzl | [b] | $[\alpha]_D^{23}$ −66.2 (*c* 1, MeOH) | [256,257] |

[a] As dibenzosulfimide (DBSI) salt.
[b] Product isolated as an oil.

**Trt-Azy-OH:**[47]

The derivative Trt-Azy-OBzl (5.8 g, 13.8 mmol) was hydrogenated in dioxane (100 mL) over Pd/C. After 2 h, the catalyst was filtered off and the solvent removed. The residue was chromatographed (silica gel, EtOAc); yield: 3.6 g (79%); mp >175 °C (dec); $[\alpha]_D^{23}$ −59.6 (*c* 1, DMF).

*for references see p 81*

## 9.2.1.3 Synthesis of Aziridine-2-carboxylic Acid Peptides

### 9.2.1.3.1 Acylation with N-Protected Aziridine-2-carboxylic Acid

Unfortunately, *N*-(9-fluorenylmethoxycarbonyl)aziridine-2-carboxylic acid cannot be used in peptide synthesis, since N-deprotection of the respective peptides with secondary amines leads to oxazoline or dehydroamino acid side products. Similarly, *N*-(*tert*-butoxycarbonyl)aziridine-2-carboxylic acid is inappropriate due to the instability of the aziridine moiety to TFA treatment. Attempts to convert *N*-tritylaziridine-2-carboxylic acid into homogenous and stable active esters as useful intermediates in peptide synthesis leads to positive results only in the case of the pentafluorophenyl ester.[47] Consequently, this active ester seems to be the method of choice for acylating peptides. The related *N*-hydroxysuccinimide and *N*-3-hydroxy-4-oxo-3,4-dihydro-1,2,3-benzotriazine ester could not be isolated in pure form and have therefore been used as crude products.[47] Access to 2-carbonylaziridine peptides is also possible by carbodiimide-mediated coupling. Additionally, alkylamides of *N*-tritylaziridine-2-carboxylic acid are prepared by the azide method,[85] yet this method fails in peptide coupling steps.[85]

**Trt-Azy-OPfp:**[47]
To an ice-cold soln of Trt-Azy-OH (1.0 g, 3.04 mmol) and Pfp-OH (0.62 g, 3.34 mmol) in DMF (5 mL), DCC (0.69 g, 3.34 mmol) was added. The mixture was then diluted with EtOAc (15 mL) and stirred for 1 h at 0 °C and for further 48 h at rt. DCU was filtered off, the solvent removed and the residue purified [dry silica gel (120 °C, 24 h), petroleum ether/EtOAc 11:1]. The solvent was removed and the solid lyophilized from dioxane; yield: 770 mg (51%); mp 79–80 °C; $[\alpha]_D^{23}$ −127.8 (*c* 1, MeOH).

**Trt-Azy-Leu-Arg-OH (Active Ester Method):**[47]
H-Leu-Arg-OH (260 mg, 0.89 mmol) was treated with Trt-Azy-OPfp (0.4 g, 0.81 mmol) in DMF (10 mL) for 24 h. The soln was concentrated and the residue partitioned between EtOAc and $H_2O$. The organic layer was dried ($Na_2SO_4$) and concentrated. The residue was precipitated with petroleum ether; yield: 401 mg (83%); mp >185 °C (dec).

**Trt-Azy-D-Val-Pro-OtBu (Carbodiimide Method):**[83]
To a soln of Trt-Azy-OH (3.17 g, 10 mmol) and H-D-Val-Pro-OtBu (100 mmol) in $CH_2Cl_2$ (60 mL) was added DCC (2.06 g, 10 mmol) in $CH_2Cl_2$ (30 mL) at −10 °C. The mixture was stirred for 3 h at −10 °C, and then kept overnight in a refrigerator. DCU was filtered off, and the filtrate was concentrated, diluted with EtOAc, and washed with 10% citric acid, 1 M $NaHCO_3$, and $H_2O$. The soln was dried ($Na_2SO_4$) and concentrated; yield: 5.34 g (93.8%); $[\alpha]_D^{23}$ −90.8 (*c* 1, MeOH).

### 9.2.1.3.2 2-Carbonylaziridine Peptides from N-Terminal Serine/Threonine Peptides

The cyclization of a peptide with an N-terminal β-hydroxy-α-amino acid is generally performed via the *O*-mesyl or *O*-tosyl derivatives. Again the course of the cyclization reaction is governed by the N-terminal protecting group and best results are obtained with *N*-trityl protected seryl/threonyl peptides.

**Trt-Azy(3-Me)-Gly-OMe:**[63]
To a soln of Trt-Thr-Gly-OBzl (16.5 g, 32.4 mmol) in anhyd pyridine (50 mL) was added dropwise a soln of Tos-Cl (18.5 g, 97.2 mmol) in pyridine at −10 °C with stirring. After 3 d at −10 °C, the soln was concentrated and the residue dissolved in EtOAc. The organic layer was washed with $H_2O$, dried ($Na_2SO_4$), and the solvent removed. The oily residue was crystallized (EtOAc/hexane); yield: 13.6 g (88%); mp 175.0–176.5 °C; $[\alpha]_D^{23}$ −70.4 (*c* 1.3, DMF).

### 9.2.1.3.3 Acylation of 2-Carbonylaziridine Peptides

Access to peptides with internally positioned aziridine-2-carboxylic acid is only possible by coupling N-protected amino acids to 2-carbonylaziridine amino acids or peptides,[63] since cyclization of $N^{\alpha}$-aminoacylated β-hydroxy α-aminoacyl peptides leads to oxazolines, as

shown for Z-Gly-Thr(Tos)-Gly-OH.[63] Attempts to acylate aziridine-2-carboxylic acid as the lithium salt with N-protected amino acid active esters failed, as did the use of the in situ prepared trimethylsilyl ester. Even when the active esters are replaced with the much more reactive NCAs of Fmoc-protected amino acids no dipeptides are produced.[47] Better results are obtained with mixed anhydrides[41] or DCC without[44,63,73] or with *N*-hydroxysuccinimide as additive.[47] Acylation with N-protected amino acid chlorides works as well, but the reaction has to be performed in the cold and with only 0.9 equivalents of acylating agent.[54]

Employing the Mitsunobu reaction with diisopropyl azodicarboxylate and triphenylphosphine, aminoacylserine (in contrast to aminoacylthreonine) is converted into (aminoacyl)oxazolidine-2-carboxylic acid.[54] By the same procedure *allo*-threonine peptides are converted into the oxazoline derivatives, whilst the threonine peptides are apparently converted into the aziridines in good yields.[86]

**Z-Ala-Azy-OBzl:**[73]
To a soln of H-Azy-OBzl (0.69 g, 3.88 mmol) and Z-Ala-OH (0.89 g, 4.0 mmol) in $CH_2Cl_2$ (40 mL) was added DCC (0.87 g, 4.2 mmol) at −20 °C. After 7 h at −20 °C, the DCU was removed by filtration. $CH_2Cl_2$ (60 mL) was added and the mixture was washed with 1 M $NaHCO_3$ and with $H_2O$. The organic layer was dried ($MgSO_4$), concentrated, and the crude product was purified by chromatography (silica gel, hexane then EtOAc gradient 20–100%); yield: 1.36 g (92%).

**Z-Gly-Azy(3-Me)-Gly-OBzl:**[63]
A soln of DCC (227 mg, 1.1 mmol) in $CHCl_3$ was added dropwise to a soln of H-Azy(3-Me)-Gly-OBzl (248 mg, 1.0 mmol) and Z-Gly-OH (209 mg, 1.0 mmol) in $CHCl_3$ at 0 °C with stirring. After 12 h in the cold, the DCU was filtered off and the filtrate was washed with 5% tartaric acid, 1 M $NaHCO_3$ and $H_2O$, and dried ($Na_2SO_4$). The solvent was removed and the residue crystallized (EtOAc/$Et_2O$/hexane); yield: 381 mg (89%); mp 123.5–124.5 °C; $[\alpha]_D^{23}$ −58.2 (*c* 1, EtOAc).

### 9.2.1.3.4 Deprotection of Aziridine-2-carboxylic Acid Peptides

The hydrogenolytic cleavage of benzyl-based protecting groups from aminoacyl-substituted aziridine-2-carboxylic acid peptides is problematic. In fact, attempts to deprotect Z-Phe-Azy-Bzl and similar peptides by hydrogenolysis over Pd/C in several solvents, with and without concomitant titration at pH 6.5 with perchloric acid, led to complex mixtures from which the desired peptides could not be isolated.[47] Similarly, saponification of the C-terminal benzyl ester group of N-protected peptidyl-aziridine-2-carboxylic acid with sodium or lithium hydroxide is unsuccessful.[47] According to the known sensitivity of aziridines to strong acids, acidolytic cleavage of the *N*-trityl group usually generates more or less impure products. Appropriate procedures are reported for TFA,[47] 4-toluenesulfonic acid,[47] or formic acid.[63] Again addition of trityl cation scavengers such as triethylsilane could improve the acidolytic deprotection step.

**H-Azy-Leu-OH, Deprotection with TFA:**[47]
To a stirred soln of Trt-Azy-Leu-OtBu (0.48 g, 0.96 mmol) in $CH_2Cl_2$ (3.6 mL), TFA/$H_2O$ (9:1, 2.5 mL) was added at −10 °C. After 6 h stirring at −10 °C, the mixture was concentrated and then partitioned between $Et_2O$ and $H_2O$. The aqueous layer was neutralized with Dowex 44 (OH form), the resin filtered off and the filtrate lyophilized. The crude product was purified (silica gel, $CHCl_3$/MeOH 12:1). After two further purification steps an oil was obtained; yield: 26 mg (14%).

**H-Azy(3-Me)-Gly-OBzl, Deprotection with Formic Acid:**[63]
To a soln of Trt-Azy(3-Me)-Gly-OBzl (4.76 g, 10 mmol) in $CHCl_3$ (15 mL) and MeOH (20 mL) at 0 °C was added dropwise formic acid (20 mL) with stirring. After 3 d at 0 °C, the mixture was concentrated at 0 °C, the precipitated Trt-OMe (mp 82–84 °C) was filtered off and washed with MeOH, the soln concentrated and the oily residue dissolved in EtOAc. This soln was washed with 1 M $NaHCO_3$ and $H_2O$, dried ($Na_2SO_4$), and concentrated. The residue was recrystallized (EtOAc/$Et_2O$/hexane); yield: 1.85 g (74%); mp 54–55 °C; $[\alpha]_D^{23}$ −44.0 (*c* 1, MeOH).

*for references see p 81*

## 9.2.2 Azetidine-2-carboxylic Acid

L-Azetidine-2-carboxylic acid (L-Aze) (**2**) was first isolated from *Convallaria majalis* (lily-of-the-valley).[87] It is reported to inhibit the proliferation of *E. coli*[88] and is believed to be an antagonist of proline.[89] The contraction of the pyrrolidine ring of proline to the four-membered azetidine ring of azetidine-2-carboxylic acid (**2**) has effects on the bond angles and internal torsion angle of angiotensin II analogues according to calculations[90] and NMR conformational analysis.[91] The conformational effects of this imino acid residue on model peptides such as poly(azetidine-2-carboxylic acid) homopolymers, poly(Gly-Pro-Aze), and dipeptides have been extensively reviewed.[90,92,93]

### Stability and Analytical Characterization

The four-membered azetidine ring remains unaffected by sodium in liquid ammonia,[94] methanolic ammonia,[95] 5 M barium hydroxide at 100 °C for 24 hours,[87] sodium hydroxide under ester saponification conditions,[87] HCl at room temperature,[96,97] catalytic hydrogenation under normal conditions,[87,95] and ozonolysis.[87] Correspondingly, as observed for proline, the azetidine-2-carboxylic acid (**2**) raises no particular difficulties in the synthesis of related peptides.

Azetidine-2-carboxylic acid (**2**) like proline gives an intense blue color with sodium nitroprusside in 10% acetaldehyde solution in the presence of sodium carbonate.[98,99] Upon usual acid hydrolysis (6 M HCl, 110 °C, 24 h or more) as required for amino acid analysis, azetidine-2-carboxylic acid is completely decomposed, yielding mainly homoserine lactone, as well as other ninhydrin-positive compounds.[87,89,99] To enable an accurate quantification of this imino acid, azetidine-2-carboxylic acid peptides should be hydrolyzed by alkali (5 M barium hydroxide, at 100 °C for 18 h[89] or 2 M sodium hydroxide at 110 °C for 22 h[100]). There are extensive $^1$H NMR spectroscopic data available[100–104] and the absolute configurations of *N*-acetyl-L-azetidine-2-carboxylic acid[105] and *N*-*tert*-butoxycarbonyl-L-azetidine-2-carboxylic acid[106] have been determined by X-ray analysis.

### 9.2.2.1 Synthesis of Azetidine-2-carboxylic Acid

Azetidine-2-carboxylic acid (**2**) is commercially available. It is readily prepared as the racemate by refluxing 2,4-dibromobutyric acid ester with benzhydrylamine in acetonitrile. If benzyl 2,4-dibromobutyrate is treated with benzhydrylamine, the resulting benzyl *N*-benzhydryl-D,L-azetidine-2-carboxylate is hydrogenolytically processed to D,L-azetidine-2-carboxylic acid in a one-step reaction.[101,107] Resolution of the racemate can be performed by the method of Vogler[108] via fractional crystallization of the Z-D,L-Aze-OH · H-Tyr-$N_2H_3$ salt; thereby the salt of the D-imino acid precipitates first from methanol.[96] A stereoselective synthesis of *N*-tosyl-L-azetidine-2-carboxylic acid can be achieved by a two-step reaction from *N*-tosyl-L-homoserine lactone.[94]

L-Aze with functionalized side chains have been synthesized as rigidified congeners of glutamic acid[109] and a stereoselective synthesis of related 3-alkyl derivatives has been reported.[110]

### 9.2.2.2 Synthesis of Azetidine-2-carboxylic Acid Peptides

*N*-(*tert*-Butoxycarbonyl)azetidine-2-carboxylic acid is obtained by reaction of azetidine-2-carboxylic acid with Boc-$N_3$,[111,112] $Boc_2O$,[113,114] or 2-(*tert*-butoxycarbonyloxyimino)-2-phenylacetonitrile.[115] In a similar manner, the Fmoc derivative is prepared by the standard procedure using Fmoc-OSu as acylating agent.[116] For derivatives of azetidine-2-carboxylic acid, see Table 4.

**Table 4** Azetidine-2-carboxylic Acid Derivatives[94,96,99,101,111,113,114,116,117,121,263]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| H-L-Aze-OH | 210–211 | $[\alpha]_D^{20}$ −120.0 (*c* 0.5, $H_2O$) | [94,99,101,111,121] |
| Boc-L-Aze-OH | 104–106 | $[\alpha]_D^{20}$ −118.5 (*c* 1, MeOH) | [111,113,121] |
| Boc-L-Aze-OH·DCHA | 156–157 | $[\alpha]_D^{24}$ −60.6 (*c* 1.2, MeOH) | [114] |
| Fmoc-L-Aze-OH | 139 | $[\alpha]_D^{20}$ −73.6 (*c* 1, DMF) | [116] |
| Z-L-Aze-OH | [a] | $[\alpha]_D^{25}$ −99.0 (*c* 3.9, $CHCl_3$) | [96,263] |
| Z-L-Aze-OH·DCHA | 162–174 | $[\alpha]_D^{20}$ −71.0 (*c* 1, $CHCl_3$) | [121] |
| Boc-L-Aze-OSu | 115–116 | $[\alpha]_{578}^{25}$ −117.2 (*c* 1, EtOAc) | [111] |
| Boc-L-Aze-OPcp | 98–99 | $[\alpha]_{578}^{25}$ −72 (*c* 1, EtOAc) | [111] |
| Z-L-Aze-OPcp | 107–108 | $[\alpha]_D^{20}$ −72.6 (*c* 1, $CHCl_3$) | [117] |

[a] Product isolated as an oil.

In solution-phase peptide synthesis, acylation of amino acids or peptides with N-protected azetidine-2-carboxylic acid is performed via the active esters, e.g. *N*-hydroxysuccinimide[100,111,112] or pentachlorophenyl ester,[111,117] as well as by the mixed anhydride[101,114] or carbodiimide[118] methods. An attempt to prepare the *N*-carbonic acid anhydride by cyclization of *N*-(chloroformyl)azetidine-2-carboxylic acid with silver oxide in acetone or by addition of triethylamine in situ failed, presumably due to steric hindrance.[111] In SPPS, activation of the Fmoc-protected imino acid by HBTU[119,120] is reported. In solution-phase peptide synthesis, coupling of N-protected amino acids or peptides to C-protected azetidine-2-carboxylic acid or related peptides may be performed by active esters,[100,118,121] mixed anhydrides,[95] or similar methods. It may be worth mentioning that the probability of piperazine-2,5-dione formation from azetidine-2-carboxylic acid dipeptides is significantly reduced compared to proline dipeptides.[111]

Removal of acid-labile protecting groups with TFA[117] or HCl[100,114] is performed by standard protocols. Similarly, Fmoc cleavage by piperidine[120] and hydrogenolysis of the Z-protected derivative[117,118] are carried out under standard conditions. The benzyl ester group of benzyl *N*-benzhydrylazetidine-2-carboxylate can be selectively hydrogenated over 1% Pd/C catalyst to yield *N*-benzhydrylazetidine-2-carboxylic acid.[101] Deprotection of benzyl *N*-benzylazetidine-2-carboxylate requires aggravated hydrogenolysis conditions and results in substantial losses.[107] Hydrogenolysis of methyl *N*-benzhydrylazetidine-2-carboxylate hydrochloride yields 90% of diphenylmethane together with a complex mixture of products, most likely resulting from ring cleavage concomitant with the intended hydrogenolysis reaction.[107] An attempt to saponify the ester group of methyl *N*-benzhydrylazetidine-2-carboxylate failed.[107]

*for references see p 81*

**Boc-Aze-OH:**[113]
To a soln of H-Aze-OH (1.01 g, 10 mmol) in $H_2O$ (100 mL) at pH 11.0 BuOH (40 mL) and $Boc_2O$ (2.40 g, 11 mmol) were added. The mixture was stirred for 90 min and diluted with $H_2O$. The excess $Boc_2O$ was extracted with pentane (3 × 15 mL). The aq soln was cooled to 0 °C, acidified to pH 2 with 6 M HCl and extracted with EtOAc (4 × 20 mL). The organic layers were pooled, washed to neutrality, dried ($Na_2SO_4$), and concentrated. The residue was recrystallized (petroleum ether); yield: 1.9 g (93%); mp 85 °C; $[\alpha]_D^{20}$ −118.5 (*c* 1, MeOH).

**Fmoc-Aze-OH:**[116]
A soln of Fmoc-OSu (5.810 g, 17.2 mmol) in warm dioxane (15 mL) was added dropwise to a soln of H-L-Aze-OH (1.740 g, 17.2 mmol) and $Na_2CO_3$ (3.646 g, 34.4 mmol) in $H_2O$ (30 mL). After precipitation $H_2O$ was added. The mixture was extracted twice with $Et_2O$, acidified with 6 M HCl and extracted with EtOAc. The organic phase was dried ($Na_2SO_4$) and diluted with $iPr_2O$. The precipitate was collected and dried; yield: 4.69 g (85%); mp 139 °C; $[\alpha]_D^{20}$ −73.6 (*c* 1, DMF).

## 9.2.3 Five-Membered Proline Analogues

As mentioned in the introduction to Section 9.2, the most commonly used five-membered proline analogues are listed in Tables 1 and 2. In addition to the *trans*-4- and *trans*-3-hydroxy-L-proline (**3** and **4**, respectively) as naturally occurring constituents of collagens, various 3-, 4-, and 5-alkylproline derivatives have been synthesized and related synthetic procedures have been comprehensively reviewed.[29] Moreover, halogenated proline derivatives, particularly the fluorinated compounds, have found increasing application in collagen model peptides,[38–40,122,123] and 4-aminoproline has been used as a functionalized proline derivative for synthesis of peptidomimetics and related scaffolds as well as branched peptides.[124–127] Similarly, the five-membered proline analogue thiazolidine-4-carboxylic acid (thiaproline, **11**) has often been incorporated into collagen model compounds[128,129] or bioactive molecules such as thrombin inhibitors,[130] somatostatin,[131,132] dipeptidyl peptidase IV substrates,[133] angiotensin II,[134] HIV inhibitors,[135] ACE inhibitors,[136] and oxytocin.[137] Conversely, the related selena-proline analogue **13**[138] has so far not been incorporated into synthetic peptides despite its potential application as a heavy atom analogue for X-ray crystallography.[15] The oxazolidine-4-carboxylic acid (oxaproline, **9**) is rarely used[133,136] because of the acid lability of the oxazolidine ring and related synthetic difficulties.[139] More attention has been paid to the pyrazolidine-2-carboxylic acid (azaproline, **17**).

### 9.2.3.1 Hydroxyproline

The name hydroxyproline (Hyp) is used to refer to the most common naturally occurring form, i.e. (2*S*,4*R*)-4-hydroxyproline (**4**). Because of the two chiral C-atoms, this amino acid exists in four epimeric forms. The 2*S*,4*R*- and 2*R*,4*S* forms are referred to as *trans*-hydroxyprolines and the 2*S*,4*S*- and 2*R*,4*R* forms as *cis*-hydroxyprolines or *allo*-hydroxyprolines. As expected from their configuration, the *cis*-4-hydroxyprolines can form lactones.

*cis*-4-Hydroxyprolines can be obtained by several methods: by inversion at the C4 of the respective, more readily available, *trans*-compound, e.g. via intramolecular Mitsunobu reactions;[26,140,141] via the synthesis of *cis*-4-hydroxyproline lactones by an internal displacement reaction of N,C-protected *O*-tosyl-*trans*-4-hydroxyprolines with potassium carbonate;[143] and via a highly efficient transesterification with MeOH in the presence of sodium azide.[144,145] *cis*-4-Hydroxyproline lactones are also produced by reacting N-protected, O,C-unprotected *cis*-4-hydroxyprolines with 4-toluenesulfonyl chloride in pyridine (presumably by a mixed anhydride intermediate) or with *N*,*N*′-dicyclohexylcarbodiimide.[143] Peptides containing C-terminal O-unprotected *trans*-4-hydroxyproline are converted into the corresponding pep-

tidyl-*cis*-4-hydroxyproline lactones by similar procedures.[141] The lactones are stable toward anhydrous acids, whereas in aqueous acids a slow ring opening to the hydroxy compounds takes place.[143] An alternative route for the production of *cis*-4-hydroxyprolines is based on oxidation of *trans*-4-hydroxyprolines to the 4-oxo derivatives followed by stereoselective reduction.[143] Diastereomeric mixtures of *N*-(alkyloxycarbonyl)-4-hydroxyproline alkyl esters can be resolved using the lipase fraction B of *Candida antarctica* to produce *cis*-4-hydroxy-D-proline or *trans*-4-hydroxy-L-proline.[146]

*trans*-3-Hydroxyproline (**3**) is commercially available. Conversion of N-protected *trans*-3-hydroxyproline ester into the related *cis*-3-hydroxyproline derivative is carried out under Mitsunobu conditions with formation of the *cis*-3-hydroxyproline 4-nitrobenzoate, which is selectively hydrolyzed with sodium azide in MeOH.[145] Additionally, stereoselective syntheses have been reported for the naturally occurring 2,3-*trans*-3,4-*cis*-L-dihydroxyproline[147–149] and *trans*-3-hydroxy-3-methylproline.[150]

Peptide syntheses with *trans*- or *cis*-4-hydroxyprolines are usually performed by standard procedures as applied for ω-hydroxy-α-amino acids. However, protection of the hydroxy group in many cases is not required due to the low reactivity of the secondary hydroxy group of hydroxyprolines. Correspondingly, O-unprotected hydroxyprolines have been widely used in peptide synthesis both as carboxy or amino components.[151] In case *O*-acylation has occurred, this reaction can readily be reversed by base treatment, e.g. with piperidine under the conditions of Fmoc cleavage. However, $N^{\alpha}$-deprotection of *O*-acyl hydroxyproline derivatives with bases bears the risk of O→N migration of the acyl groups, and, although the extent of O-aminoacylation under usual coupling conditions may be rather low, it does lead to contamination. Therefore, protection of the hydroxy group is recommended for N-terminal hydroxyprolines in the chain elongation steps, especially when applying the Fmoc/tBu strategy, whilst an O-protection of internally positioned hydroxyproline residues is not required. For useful derivatives of hydroxyproline in peptide synthesis see Table 5. The synthesis of O-protected hydroxyproline derivatives is described in Vol. E 22a, Section 2.6.4.

**Table 5** Hydroxyproline Derivatives[141,143,152,258–261]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| Z-(2*S*,3*R*)-3-Hyp-OH | 123–124 | $[\alpha]_D$ −15 (MeOH) | [258] |
| H-(2*S*,3*R*)-3-Hyp-OMe·HCl | 161–164 | $[\alpha]_D$ −3 ($H_2O$) | [258] |
| Fmoc-(2*S*,4*R*)-Hyp-OH | 187–189 | $[\alpha]_D^{28}$ −60.1 (*c* 1, MeOH) | [141,261] |
| Fmoc-(2*S*,4*R*)-Hyp(tBu)-OH | 84–87 | | [152] |
| Boc-(2*S*,4*R*)-Hyp(Bzl)-OH | 67–69 | $[\alpha]_D^{20}$ −40.5 (*c* 2.2, EtOH) | [259] |
| Z-(2*S*,4*R*)-Hyp-OH | 106–107 | $[\alpha]_D^{20}$ −72 (*c* 1, $CHCl_3$) | [143] |
| Z-(2*R*,4*R*)-Hyp-OH | 110.5–111.5 | $[\alpha]_D^{20}$ +26.3 (*c* 1, $CHCl_3$) | [143] |
| Z-(2*S*,4*S*)-Hyp-OH | 110–111 | $[\alpha]_D^{20}$ −23.7 (*c* 1, $CHCl_3$) | [143] |
| H-(2*S*,4*R*)-Hyp-OMe·HCl | 171–172 | | [260] |
| H-(2*S*,4*S*)-Hyp-lactone·HBr | 193–194 (dec) | | [143] |
| H-(2*R*,4*S*)-Hyp-lactone·HBr | 188 (dec) | | [143] |
| Z-(2*S*,4*S*)-Hyp-lactone | 102–103 | $[\alpha]_D$ +33.6 (*c* 1, $CHCl_3$) | [143] |
| Z-(2*R*,4*R*)-Hyp-lactone | 102–103 | $[\alpha]_D$ −30.9 (*c* 1, $CHCl_3$) | [143] |

N-Protected *cis*-4-hydroxyproline lactones react with nucleophiles to generate the corresponding *cis*-4-hydroxyproline derivatives, e.g. the methyl ester with methanol,[141] the amide with ammonia,[141] or peptides with sterically unhindered amino acid esters.[143]

**Fmoc-(2*S*,4*R*)-Hyp-OH:**[141]
Powdered crystalline (2*S*,4*R*)-4-hydroxyproline (**4**; 2.23 g, 17 mmol) was suspended in dry $CH_2Cl_2$ (35 mL) and treated with TMSCl (7.6 mL, 60 mmol) and DIPEA (10.2 mL, 60 mmol). The resulting

*for references see p 81*

mixture was refluxed, with exclusion of moisture, for 30 min to ensure dissolution. Fmoc-Cl (4.66 g, 18 mmol) was then added in a single portion at rt and the resulting mixture was stirred for 1.5 h prior to treatment with MeOH (3 mL) for 15 min. The solvents were removed and the oily residue was taken up in 2.5% aq $NaHCO_3$ (50 mL) and washed twice with $Et_2O$. The aqueous phase was cooled to 0 °C and acidified with 1 M HCl to pH 2. The mixture was extracted twice with EtOAc, the combined organic layers were washed with sat. NaCl, dried ($Na_2SO_4$), and concentrated to dryness to give a white powder; yield: 5.5 g (92%); mp 187–189 °C; $[\alpha]_D^{28}$ −60.1 (*c* 1, MeOH).

**Trt-(2*S*,4*S*)-Hyp-Lactone from Trt-(2*S*,4*R*)-Hyp-OH:**[140]

The derivative Trt-(2*S*,4*R*)-Hyp-OH·DEA (11.2 g, 25 mmol) was partitioned between precooled (0 °C) 5% aq citric acid (200 mL) and EtOAc (100 mL), and the aqueous layer was extracted with EtOAc (100 mL). The combined organic layers were washed with brine, dried ($MgSO_4$), and concentrated at ca. 4 °C to yield Trt-(2*S*,4*R*)-Hyp-OH. To a soln of this solid in dry THF (70 mL) was added $Ph_3P$ (7.9 g, 30 mmol) and the resulting mixture was treated dropwise with DEAD (4.7 mL, 30 mmol) at 0 °C. After 5 min, the solvent was removed and the residual oil was flash chromatographed (silica gel, benzene). The fractions containing the lactone were pooled and taken to dryness; yield: 5.3 g (60%); the product can be recrystallized from acetone.

An alternative workup procedure takes advantage of the solubility of the lactone in MeOH. Upon completion of the reaction, the residual oil was dissolved in EtOAc and washed with 5% aq citric acid (50 mL), $H_2O$ (2 × 50 mL), 5% $NaHCO_3$ (50 mL), and $H_2O$ (2 × 50 mL). The organic phase was dried and concentrated, and the oil triturated with MeOH to afford the crystalline product; yield: 4.9 g (55%). The latter workup procedure, while very simple and economic, cannot be applied to other N-protected lactones, e.g. to the *N*-Boc-4-hydroxyproline lactone, where careful flash chromatography was required for separation of other polar byproducts including $Ph_3PO$.

**Z-(2*S*,4*S*)-Hyp-OH from Z-(2*S*,4*R*)-Hyp-OH via 4-Oxo-(2*S*)-proline:**[143]

To Z-(2*S*,4*R*)-Hyp-OH (7.58 g, 28.6 mmol) in acetone (420 mL) 8 M $CrO_3$ in $H_2SO_4$ (29 mL) was added under stirring over a period of 5 min. Stirring was continued for an additional 30 min, and the excess of the oxidant was then destroyed by adding MeOH (5 mL). Most of the chromium salts were removed by filtration with the aid of HyFlo Supercel. The filtrate was diluted with $CHCl_3$ and extracted several times with brine, dried ($MgSO_4$), and concentrated. The *N*-Z-4-oxo-(2*S*)-proline was crystallized from $Et_2O$/petroleum ether (bp 65 °C) as colorless crystals; yield: 5.08 g (68%); mp 99–101 °C (analytical sample mp 101–102 °C); $[\alpha]_D$ +18.5 (*c* 1, $CHCl_3$).

A soln of $NaBH_4$ (0.575 g) in $H_2O$ (2 mL) was added to *N*-Z-4-oxo-(2*S*)-proline (1.00 g, 3.8 mmol) in MeOH (30 mL) at 0 °C. The mixture was left overnight at 5 °C and the MeOH was then removed. The residue was treated with 3 M NaOH (15 mL) at rt for 30 min. After cooling to 0 °C, the soln was acidified with concd HCl and extracted with EtOAc. The organic layer was washed twice with brine and dried ($MgSO_4$). The soln was decolored with Darco and concentrated. Upon addition of petroleum ether (bp 65 °C) a precipitate was collected which was recrystallized (EtOAc/petroleum ether); yield: 0.76 g (75%); mp 110–111 °C; $[\alpha]_D^{20}$ −23.7 (*c* 1, $CHCl_3$).

**Z-Pro-(2*S*,4*R*)-Hyp-OH:**[152]

A soln of H-(2*S*,4*R*)-Hyp-OH (11.5 g, 87.6 mmol) in 2 M benzyltrimethylammonium hydroxide in MeOH (43 mL) was taken to dryness, the residue redissolved in DMF and treated for 3 h with Z-Pro-OSu (28.8 g, 83.4 mmol). The bulk of the solvent was removed and the residue partitioned between 5% $NaHCO_3$ and $Et_2O$. The aqueous phase was acidified with 2 M HCl (pH 2.5) and the resulting precipitate was collected by filtration and recrystallized (MTBE); yield: 27.7 g (92%); mp 216–217 °C.

**Z-Pro-(2*S*,4*R*)-Hyp-Gly-OBzl:**[152]

To an ice-cold soln of Z-Pro-(2*S*,4*R*)-Hyp-OH (8.14 g, 50.08 mmol), H-Gly-OBzl·HCl (11.62 g, 57.2 mmol), and HOBt (6.8 g, 50.1 mmol) in DMF, DIPEA (20 mL, 55 mmol) was added, followed by EDCI (10.5 g, 55 mmol). After 3 h stirring at rt, the solvent was removed and the residue washed in EtOAc with 5% $NaHCO_3$, 5% $KHSO_4$, and $H_2O$. The organic layer was dried ($Na_2SO_4$) and concentrated to a small volume. Upon addition of $Et_2O$, the solid was collected by filtration; yield: 25 g (96%); mp 55–59 °C.

### 9.2.3.2 Proline Analogues Derived from Hydroxyproline

The commercially available (2*S*,4*R*)- and (2*S*,4*S*)-4-hydroxyprolines are ideal intermediates for the stereoselective synthesis of 4-azido derivatives with inversion of the chirality.[264] This is achieved by the classical displacement of the 4-tosyl[125] or 4-mesyl[153] of N,C-protected hydroxyproline with sodium azide, or in one pot using N,C-protected 3- or 4-hydroxyproline derivatives under Mitsunobu conditions with diphenylphosphoryl azide (DPPA).[145] Interconversion of the protecting groups is carried out before reduction of the azides to aminoprolines with $Ph_3P$[125,153] in order to allow for orthogonal protection of the $N^{\alpha}$- and $N^{\omega}$-amino groups. The aminoprolines may be used for further derivatizations[124,126,127] or can be transformed into the 4-guanidinoprolines as arginine mimetics.[125]

Boc- or Fmoc-protected *trans*-4- and *cis*-4-hydroxyproline esters are readily converted with DAST or morpholinosulfur trifluoride in dry $CH_2Cl_2$ into the corresponding *cis*-4- and *trans*-4-fluoroproline derivatives, respectively.[26,35,123,142] In an identical manner, the *cis*-3- and *trans*-3-fluoroprolines are prepared from the suitably protected hydroxyproline precursors.[154] Similarly, peptidyl-4-hydroxyproline is converted into the fluorinated derivative with concomitant activation of the carboxy group as fluoride for subsequent peptide couplings.[155] Conversely, 4,4′-difluoroproline is obtained by oxidation of N-protected 4-hydroxyproline ester to the 4-oxo compound followed by reaction with DAST or morpholinosulfur trifluoride.[26,156] Among the various methods reported for the oxidation of 4-hydroxyproline derivatives to the corresponding 4-oxo derivative, the use of tetrapropylammonium perruthenate[157] appears to be superior to others such as chromic acid in sulfuric acid [see Section 9.2.3.1, preparation of Fmoc-(2*S*,4*R*)-Hyp-OH].[143] This 4-oxo proline derivative serves as a useful intermediate for ring alkylations with Grignard reagents.[157]

**(2*S*,4*R*)-Z-Hyp(Tos)-OBzl:**[125]
To a soln of Z-Hyp-OBzl (16 g, 45 mmol) in pyridine (70 mL) at 0 °C was added, portionwise, Tos-Cl (9.45 g, 50 mmol). After 12 h at rt, the solvent was removed and the residue treated with ice (50 g) and extracted with EtOAc (3 × 100 mL). The extracts were washed with 0.5 M HCl (0.5 mL), $H_2O$, 10% $Na_2CO_3$, $H_2O$ and dried ($Na_2SO_4$). The solvent was removed and the residue chromatographed (silica gel, hexane/EtOAc 7:3) to give an oil; yield: 19.4 g (86%); $[\alpha]_D^{30}$ −35.3 (*c* 1.18, $CHCl_3$).

**(2*S*,4*S*)-*N*-Z-4-Azidoproline Benzyl Ester:**[125]
To a soln of (2*S*,4*R*)-Z-Hyp(Tos)-OBzl (6.46 g, 12.7 mmol) in DMF (30 mL) was added sodium azide (3.30 g, 51 mmol). The mixture was heated at 65–70 °C for 8 h, diluted with $H_2O$ and extracted with EtOAc. The extract was washed with brine, dried ($Na_2SO_4$), and concentrated to an oil which was purified on silica gel (hexane/EtOAc, 8:2). The product was isolated as an oil which solidified from $Et_2O$/hexane; yield: 4.42 g (87%); mp 66–68 °C; $[\alpha]_D^{25}$ −47.3 (*c* 0.96, $CHCl_3$).

**(2*S*,4*S*)-*N*-Z-4-Bromoproline Benzyl Ester:**[125]
DEAD (10.3 g, 59 mmol) was added dropwise with stirring to a soln of $Ph_3P$ (15.7 g, 60 mmol) in anhyd THF (100 mL) under argon. After 20 min, LiBr (10.4 g, 120 mmol) was added followed by Z-Hyp-OBzl (4.27 g, 12 mmol) in anhyd THF (50 mL). The mixture was stirred at rt for 20 h, concentrated, poured into $H_2O$, and extracted with $Et_2O$ (× 2). The extracts were washed with brine, dried ($Na_2SO_4$), and concentrated, and the residue was purified by flash chromatography (silica gel, hexane/EtOAc 8:2) to give a white solid; yield: 4.52 g (90%); mp 82–83 °C; $[\alpha]_D^{25}$ −42.5 (*c* 1.12, $CHCl_3$).

*for references see p 81*

**(2*S*,4*R*)-*N*-Z-4-Azidoproline Benzyl Ester:**[125]
Displacement of the bromo derivative with sodium azide was performed at rt for 14 h as described for the synthesis of (2*S*,4*S*)-*N*-Z-4-azidoproline benzyl ester; yield: 94%; $[\alpha]_D^{25}$ −51.7 (*c* 1.63, $CHCl_3$).

***tert*-Butyl $N^{\alpha}$-*tert*-Butoxycarbonyl-4-oxoprolinate:**[157]
Tetrapropylammonium perruthenate (63 mg) was added in one portion to a stirred mixture of Boc-Hyp-OtBu (1 g, 3.47 mmol), NMO (0.62 g, 15.6 mmol), and powdered 4-Å molecular sieves (1.78 g) in $CH_2Cl_2$ (7 mL) at rt under argon. The mixture was stirred for 3 h, filtered, and concentrated. The product was purified by flash chromatography ($CH_2Cl_2$/EtOAc, 1:1) and recrystallized ($Et_2O$/hexane); yield: 0.9 g (90%); mp 66-67 °C; $[\alpha]^{25}_D$ +11.2 (*c* 1.2, $CHCl_3$).

**$N^{\alpha}$-9-Fluorenylmethoxycarbonyl-(2*S*,4*R*)-4-fluoroproline:**[123]
To an ice-cold suspension of Fmoc-Hyp-OH (1 g, 2.83 mmol) in $CH_2Cl_2$ (50 mL) was added DAST (1.48 mL, 11.32 mmol). Upon dissolution, the soln was stirred for additional 5 h at rt, concentrated, and partitioned between EtOAc and 5% $NaHCO_3$. The aqueous layer was extracted with EtOAc, acidified with 5% $NaHSO_4$, and extracted with EtOAc. The extracts were washed with brine, dried ($Na_2SO_4$), and taken to dryness; yield: 1 g (quant).

**Z-Pro-Pro(4*S*-F)-Gly-OBzl:**[155]
To an ice-cold suspension of Z-Pro-Hyp-OH (5 g, 12.6 mmol) in $CH_2Cl_2$ (30 mL) DAST (7.23 mL, 50.4 mmol) was added and the mixture was stirred for 4 h at rt. Then H-Gly-OBzl·Tos-OH (5.26 g, 15.1 mmol) was added, followed by DIPEA (5.2 mL, 30.2 mmol) and the soln was stirred for a further 6 h at rt. The soln was poured onto ice and the organic layer was extracted with 5% $NaHCO_3$, 5% $KHSO_4$, and $H_2O$. After drying ($MgSO_4$) the soln was concentrated and the crude product was chromatographed (silica gel, $CHCl_3$ then 0.2% AcOH in 5–15% MeOH in $CHCl_3$); yield: 4 g (62%).

## 9.2.3.3 Azaprolines

Replacement of carbon atoms in the cyclic aliphatic side chain of proline with nitrogen leads to the azaprolines where incorporation of an additional imino group may affect molecular recognition processes via electrostatic interactions. While the proline analogues imidazolidine-2-carboxylic acid (3-azaproline, **15**) and pyrazolidine-3-carboxylic acid (5-azaproline, **16**) allow for additional side-chain modifications in peptidomimetic structures, with pyrazolidine-2-carboxylic acid (2-azaproline, **17**) (Table 2), where the α-CH is replaced by nitrogen, the peptide backbone is modified with consequently strong conformational effects.

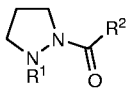
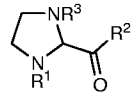
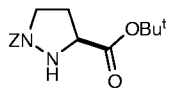
### 9.2.3.3.1 Pyrazolidine-2-carboxylic Acid

The exchange of α-CH for nitrogen in proline is more than a simple isoelectronic substitution as it induces significant structural perturbations in the proline peptide backbone. Therefore, pyrazolidine-2-carboxylic acid (2-azaproline, **17**) inserts are of great interest in the design of bioactive peptides. The most striking feature is the nonplanar, prochiral character of both nitrogens in the 2-azaproline residue.[158,159] Both nitrogens are accessible to proton donors. The aminoacyl-2-azaproline bond assumes a rigid *cis* conformation whilst the corresponding Xaa-Pro bond prefers the *trans* conformation.[159] Moreover, the 2-azaproline residue prevents a β-turn formation centered on the 2-azaproline-Xaa bond, in contrast to the corresponding Xaa-Pro bond that favors it.[158] $^1H$ and $^{13}C$ NMR spectroscopic data[160] and X-ray structures of 2-azaproline-containing peptides[159,160] have been reported.

### 9.2.3.3.1.1 Synthesis of Pyrazolidine-2-carboxylic Acid Derivatives

Carboxy-unprotected 2-azaproline or its $N^{\beta}$-substituted derivatives cannot be isolated because of the immediate decarboxylation that takes place to form the corresponding pyrazolidine derivatives. Therefore, in peptide synthesis this building block must be protected at its carboxy group, e.g. as an ester. Reaction of 1,3-dibromopropane, 1-benzyloxycarbonyl-2-(*tert*-butoxycarbonyl)hydrazine and sodium hydride in dimethylformamide leads to *N*-(benzyloxycarbonyl)pyrazolidine-2-carboxylic acid *tert*-butyl ester in high yield.[161] By treatment with HCl pyrazolidine-2-carboxylic acid benzyl ester is obtained as the hydrochloride and by hydrogenolysis as the Boc-protected pyrazolidine-2-carboxylic acid (Table 6). 2-Azaproline derivatives may also be prepared by ring closure of *N*-(but-3-enyl)glycines through radical intermediates.[161]

**Table 6** Derivatives of Azaprolines[160–162,177]

| Compounds | | mp (°C) | Ref | Compounds | | | mp (°C) | Ref |
|---|---|---|---|---|---|---|---|---|
|  | | | |  | | | | |
| $R^1$ | $R^2$ | | | $R^1$ | $R^2$ | $R^3$ | | |
| Boc | OBzl | a | [161] | Boc | H | Z | a | [162] |
| Z | OtBu | a | [161] | Fmoc | H | Boc | 168–169 | [162] |
| Boc | ONp | a | [160] | Z | H | Boc | a | [162] |
| H | OBzl·HCl | 162–164 | [161] | | | | 55–57 | [162] |
| H | OtBu | a | [177] |  | | | | |

[a] Product isolated as an oil.

***tert*-Butyl $N^1$-(Benzyloxycarbonyl)pyrazolidine-2-carboxylate:**[161]
To a stirred suspension of NaH (4.8 g, 200 mmol) in DMF (100 mL) under $N_2$, a soln of 1-(benzyloxycarbonyl)-2-(*tert*-butoxycarbonyl)hydrazine[161] (26.6 g, 100 mmol) in DMF (250 mL) was added slowly. The mixture was further stirred under $N_2$ for 30 min. Then a soln of 1,3-dibromopropane (20.2 g, 100 mmol) was added and the reaction was allowed to proceed overnight. The solvent was removed, the residue dissolved in EtOAc, and washed with $H_2O$, aq citric acid, sat. aq $NaHCO_3$, and $H_2O$. After removal of the solvent, Z-[2N]Pro-OtBu (formally identical to Boc-[2N]Pro-OBzl) was obtained as an oil; yield: 25.5 g (83%).

**Benzyl Pyrazolidine-2-carboxylate·HCl:**[161]
Benzyl $N^1$-(*tert*-butoxycarbonyl)pyrazolidine-2-carboxylate (45.9 g, 150 mmol) was treated with 2 M HCl in EtOAc (200 mL, 400 mmol) for 45 min at rt. The solvent was removed, the residue triturated with $Et_2O$ and recrystallized ($MeOH/Et_2O$); yield: 16 g (50%); mp 162–164 °C.

***tert*-Butyl Pyrazolidine-2-carboxylate:**[161]
*tert*-Butyl $N^1$-(benzyloxycarbonyl)pyrazolidine-2-carboxylate (30.6 g, 100 mmol) was hydrogenated in $MeOH/H_2O$ (200 mL., 4:1) over 5% Pd/C (4 g) for 5 h. The filtrate from the catalyst was concentrated to give an oil; yield: 10.2 g (59%).

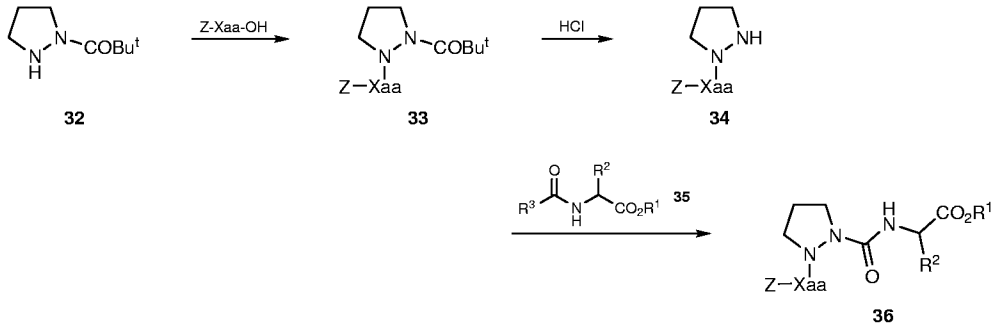
### 9.2.3.3.1.2 Synthesis of Pyrazolidine-2-carboxylic Acid Peptides

The synthesis of pyrazolidine-2-carboxylic acid containing peptides is carried out following the standard procedures established for azaamino acid peptides as described in Section 10.4. Accordingly, 1-(*tert*-butoxycarbonyl)pyrazolidine (which is the equivalent of 2-azaproline *tert*-butyl ester) is treated with 4-nitrophenyl chloroformate to generate (*tert*-butoxycarbonyl)pyrazolidine-2-carboxylic acid 4-nitrophenyl ester.[160] This active ester is then used

*for references see p 81*

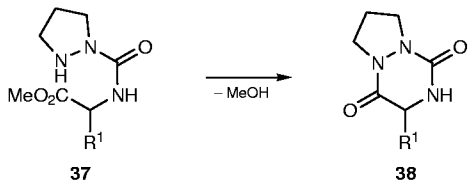
to acylate $N^{\alpha}$-unprotected amino acids and peptides to produce the related Boc-protected 2-azaprolyl amino acid or peptide derivatives.[160] Conversely, aminoacylation of 2-azaproline esters can be carried out by standard methods, such as active esters[161] or DCC,[160] or by reaction of 1-urethane-protected pyrazolidine with *N*-carbonyl amino acid esters.[161] A typical procedure for the synthesis of *endo*-2-azaproline peptides is outlined in Scheme 4. *tert*-Butyl 2-azaprolinate (**32**) is acylated with an N-protected amino acid. The *tert*-butyl ester of the resulting dipeptide **33** is cleaved by acidolysis to produce the (aminoacyl)pyrazolidide **34** which in turn is reacted with a *N*-carbonyl amino acid ester **35** to generate the tripeptide **36**.

**Scheme 4** Synthesis of Pyrazolidine-2-carboxylic Acid Peptides[160,165]

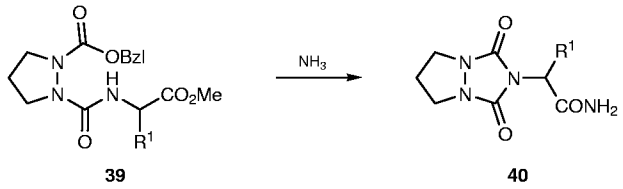



In analogy to proline dipeptides, $N^1$-unprotected 2-carbonylpyrazolidine amino acid dipeptide esters **37** on storage at ambient temperature are prone to conversion into the related triazines **38** that correspond to the aza analogues of piperazine-2,5-diones (Scheme 5).[160,161] An additional side reaction is reported for the 2-azaproline peptides[161] which leads to formation of the cyclic imide **40** from methyl $N^1$-(benzyloxycarbonyl)-2-carbonylpyrazolidine glycinate (**39**) upon treatment with methanolic ammonia (Scheme 6).

**Scheme 5** Side Reactions Reported for Pyrazolidine-2-carboxylic Acid Dipeptides[160,161]




**Scheme 6** Cyclic Imide Formation with Ammonia[161]




### 9.2.3.3.2 Imidazolidine-2-carboxylic Acid

Imidazolidine-2-carboxylic acid (**15**) is also referred to as 3-azaproline. Procedures for the synthesis of protected derivatives have been reported: racemic 1-(benzyloxycarbonyl)-3-(*tert*-butoxycarbonyl)imidazoline-2-carboxylic acid [formally identical to 1-(*tert*-butoxycarbonyl)-3-(benzyloxycarbonyl)imidazoline-2-carboxylic acid], a useful intermediate for

the solution synthesis of 3-azaproline peptides; racemic 1-(9-fluorenylmethoxycarbonyl)-3-(*tert*-butoxycarbonyl)imidazoline-2-carboxylic acid, an intermediate for synthesis on solid supports via the related pentafluorophenyl (no experimental details) (Table 6).[162]

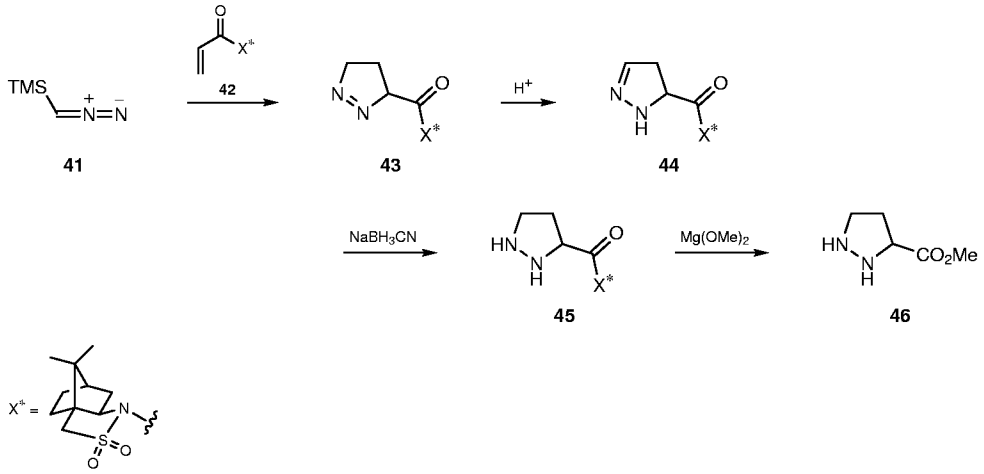
### 9.2.3.3.3 Pyrazolidine-3-carboxylic Acid

Esters of racemic pyrazolidine-3-carboxylic acid (5-azaproline, **16**) are obtained by a two-step reaction via addition of diazomethane to acrylic acid esters to form the corresponding dihydropyrazoline derivatives, which are converted into the desired pyrazolidine-3-carboxylic acid derivative by hydrogenation over palladium on charcoal,[163] or by reduction with sodium cyanoborohydride,[164] or with zinc in acetic acid. Details are given in Table 6.[165]

#### 9.2.3.3.3.1 Synthesis of Pyrazolidine-3-carboxylic Acid and Related Derivatives

A practicable strategy to provide access to *chiral* pyrazolidine-3-carboxylic acid (**16**) makes use of asymmetric dipolar cycloaddition of diazoalkanes to α,β-unsaturated carboxylic acid derivatives. For this purpose a chiral auxiliary of the alkene component is used, e.g. Oppolzer's[166,167] (1*R*)-2,10-camphorsultam.[164] As shown in Scheme 7, by reaction of (trimethylsilyl)diazomethane (**41**) with *N*-(acryloyl)camphorsultam (**42**), the *N*-(4,5-dihydropyrazoline-5-carbonyl)camphorsultam (**43**) is obtained. Reduction of **44** with sodium cyanoborohydride leads to *N*-(pyrazolidine-3-carbonyl)camphorsultam (**45**) as the 3*S*-diastereoisomer (ee 9:1) in 65 to 80% yields.[164] The camphorsultam **45** is then converted into the methyl ester **46** by reaction with magnesium methylate without racemization.[164]

**Scheme 7** Synthesis of Methyl (3*S*)-Pyrazolidine-3-carboxylate[164]




***rac*-1-Benzyl 3-*tert*-Butyl Pyrazolidine-1,3-dicarboxylate:**[163]

**CAUTION:** *Diazomethane is a detonator and is highly toxic. Appropriate safety precautions should be taken when using this reagent.*

A soln of $CH_2N_2$ in $Et_2O$ was added to *tert*-butyl acrylate (17.2 g, 134 mmol) until the yellow color persisted for at least 2 min. The solvent was removed and the residue dissolved in EtOH (300 mL) and hydrogenated over 10% Pd/C. The filtrate from the catalyst was concentrated and the residue dissolved in EtOAc (260 mL). After addition of $NaHCO_3$ (24 g, 286 mmol) in $H_2O$ (260 mL) and cooling to 0 °C, a soln of Z-Cl (18.76 g, 110 mmol) in EtOAc (50 mL) was added dropwise with stirring. After 1 h, the organic layer was separated, washed with 2 M HCl and brine, dried ($MgSO_4$), and concentrated. The

*for references see p 81*

resulting oil was chromatographed (silica gel, EtOAc/hexane 1:1) and the residue from the homogeneous fractions was recrystallized ($Et_2O$/hexane); yield: 10.8 g (25%); mp 55–57 °C.

#### 9.2.3.3.3.2 Synthesis of Pyrazolidine-3-carboxylic Acid Peptides

Esters of pyrazolidine-3-carboxylic acid (**16**) are selectively $N^1$-protected by simple reaction with suitable acylating agents, e.g. benzyl chloroformate[163] or $Boc_2O$,[165] due to the peculiar property of α-hydrazino fatty acids of being more readily acylated at $N^\gamma$ than at $N^\beta$ (Table 6). This difference in reactivity may be enhanced by steric effects. Since the reactivity of the $N^1$ moiety toward acylating agents exceeds by far that of the $N^2$ moiety, it has to be protected for selective acylation of 5-azaproline esters. For coupling reactions both DCC[164,165] and DCC/HOBt[168] are used. Analogously, for C-terminal chain elongation of suitably protected $N^1$-(aminoacyl)pyrazolidine-3-carboxylic acid standard coupling reagents are used, such as DCC.[165] The alkyl esters of pyrazolidine-3-carboxylic acid and related peptides are saponified with 4 M NaOH in a dioxane/water mixture[165,168] and *tert*-butyl ester and *tert*-butoxycarbonyl groups are cleaved by treatment with HCl in an organic solvent[165] or with trifluoroacetic acid.[168]

### 9.2.3.4 Oxaprolines

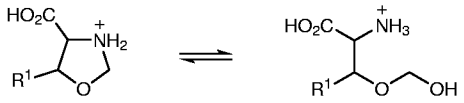
#### 9.2.3.4.1 Oxazolidine-4-carboxylic Acid

Oxazolidinecarboxylic acids and, in particular, oxazolidine-4-carboxylic acid (4-oxaproline, Oxa, **9**) are used for synthetic and conformational purposes. In fact, conversion of serine and threonine into 4-oxaproline derivatives serves for reversible protection of their hydroxy groups (see Vol. E 22a, Section 2.6.4.1.4) and of the backbone amide as well (see Vol. E 22a, Section 2.3.2.4) with significant effects on the solubility of sparingly soluble peptides. This is triggered by the conformational effects of the oxaproline residues due to their high preference for the *cis* configuration of the Xaa-Oxa bond.[169–172] Moreover, oxazolidine-4-carboxylic acid containing peptides exhibit enhanced polar character compared to the related serine or threonine peptides generally protected at the side chain as *tert*-butyl or benzyl ethers.[139] Acidolysis of oxazolidine-4-carboxylic acids, particularly when disubstituted at C2 of the proline ring, allows for facile regeneration of the serine and threonine residues as described in Vol. E 22a, Sections 2.3.2.4 and 2.6.4.1.4.

[Note: The oxazolidine-4-carboxylic acids as precursors of serine and threonine residues are also named pseudoprolines, i.e. Ser(ΨPro) and Thr(ΨPro),[139,169] whereby oxazolidine-4-carboxylic acids with substituents at C2 are designed as Ser($\Psi^{R',R''}$Pro) and Thr($\Psi^{R',R''}$Pro), respectively.]

As shown in Scheme 8, in acidic aqueous solution oxazolidine-4-carboxylic acid (**47**, $R^1 = H$) exists in equilibrium with the protonated open-ring form **48**.[173,174]

**Scheme 8** Equilibrium of Cyclic and Open Forms of Oxaprolines[173,174]




**47** **48**

The acid lability of oxazolidine-4-carboxylic acids strongly depends on the C2 substituents. Whilst dilute trifluoroacetic acid is sufficient to open the ring of 2,2-dimethyloxazolidine-4-carboxylic acids, strong acids such as trifluoromethanesulfonic acid require longer periods of time to open the unsubstituted ring.[139] Since in neutral or alkaline solutions oxazolidine-4-

carboxylic acid is stable, alkaline saponification, e.g. with 1 M NaOH in dioxane/water,[170] as well as Fmoc-removal steps in peptide synthesis on solid support by piperidine can be carried out.[139] After N-acylation or N-aminoacylation, the acid stability of oxazolidine-4-carboxylic acid derivatives and peptides is significantly enhanced.[173] Moreover, oxazolidine-4-carboxylic acids are stable toward catalytic hydrogenation, thus allowing their use even in synthesis in solution using Z protection.[171]

#### 9.2.3.4.1.1 Synthesis of Oxazolidine-4-carboxylic Acid Peptides

Oxazolidine-4-carboxylic acids are synthesized by reacting β-hydroxy-α-amino acids with aldehydes or by acid-catalyzed (e.g., 4-toluenesulfonic acid[175] or pyridinium 4-toluenesulfonate[139,176]) reaction with ketals with full retention of chirality at the α-carbon. Due to the instability of oxazolidine-4-carboxylic acids, their direct aminoacylation to the N-protected dipeptides is the preferred synthetic route.[169] Thereby, acid fluorides or *N*-carboxyanhydrides are used in the acylation step because of the low nucleophilicity of the imino group that results from the electron-withdrawing effect of the oxygen, as well as from steric effects in the case of oxaprolines that are mono- or disubstituted at C2.[139,177] Efficient aminoacylation has also been obtained by the mixed anhydride method.[170] For incorporation of N-protected aminoacyl-4-oxaproline dipeptides into peptides a free choice among the coupling methods is allowed due to the absence of racemization via the oxazol-5(4*H*)-one mechanism. Activation as acid fluoride with cyanuric fluoride, as active esters, e.g. *N*-hydroxysuccinimide or 1-hydroxybenzotriazole esters, or by the mixed anhydride method[169,171] has been reported as well as the use of coupling reagents such as PyBOP.[169]

**Fmoc-Ala-Oxa-OH:**[169]
Aq HCHO (37%; 15.4 mL, 206 mmol) was added to H-Ser-OH (10.8 g, 103 mmol) in aq 2.5 M $Na_2CO_3$ (32 mL) at rt. After overnight stirring at 4 °C, aminoacylation was performed by dropwise addition of 1 M Fmoc-Ala-F (4.0 g, 12.8 mmol) in acetone over a period of 70 min while adjusting the pH between 7.8 and 8.2 by addition of $Na_2CO_3$ (approx. 1 g). The mixture was stirred for another 30 min at rt, and then cooled to 0 °C. The soln was acidified with concd HCl to pH 3–4 and extracted with EtOAc (4 × 50 mL). The organic phases were combined, washed with brine (3 × 80 mL), and dried ($MgSO_4$). The solvent was removed and the crude product was flash chromatographed (silica gel, $CHCl_3$/MeOH/AcOH 100:3:3). The product was isolated by lyophilization from MeCN/$H_2O$; yield: 3.94 g (75%); mp 115–120 °C.

#### 9.2.3.4.1.2 Synthesis of Side-Chain-Substituted Oxazolidine-4-carboxylic Acid Peptides

Access to the imino group of oxazolidine-4-carboxylic acid is even more sterically hindered when substitutions are performed at the C2 position. For such cases, the typical procedure is based on the conversion of the desired serine or threonine dipeptides with unprotected hydroxy groups into the oxazolidine rings by reaction with aldehydes or ketones,[139,169] as described in Vol. E 22a, Section 2.3.2.4.

### 9.2.3.4.2 Isoxazolidine-3-carboxylic Acid

Chiral isoxazolidine-3-carboxylic acid (5-oxaproline, **10**) is synthesized by 1,3-dipolar cycloaddition of *N*-glycosylnitrones to ethylene.[178,179] The resulting (3*S*)- or (3*R*)-2-(2,3:5,6-di-*O*-isopropylidene-α-D-mannofuranosyl)isoxazolidine-3-carboxylic *tert*-butyl esters[179] were treated with HCl to obtain the N-unprotected compounds.[179] The *tert*-butyl ester can be N-protected by reaction with benzyl chloroformate, and upon acidolytic cleavage of the *tert*-butyl ester group with trifluoroacetic acid the related $N^{\alpha}$-Z-derivative is obtained.[179] This can be treated with diazomethane to produce the methyl ester.[179] The *N*-Z-protecting group is cleaved by HBr in AcOH, whilst hydrogenation leads to the N-unprotected form, which is

*for references see p 81*

then cleaved at the N–O bond to form the related homoserine derivative.[179] Thus isoxazolidine-3-carboxylic acid is stable toward acid treatments and to hydrogenation only in the N-acylated form. Peptides containing this amino acid are inactivating agents of prolyl-4-hydroxylase[180] and have been used to identify the mechanism of hydroxylation of this enzyme.[181]

#### 9.2.3.4.2.1 Synthesis of Isoxazolidine-3-carboxylic Acid Peptides

Couplings with N-protected isoxazolidine-3-carboxylic acid can be performed without particular restrictions of the method, and the use of a mixed anhydride for acylation of amino acids or amide formation has been reported.[179] A similarly free choice of coupling methods is available for the acylation of isoxazolidine-3-carboxylic acid derivatives; so far the use of carbodiimide and mixed anhydrides have been reported.[168,179] Isoxazolidine-3-carboxylic acid derivatives are listed in Table 7.

**Table 7** Derivatives of Isoxazolidine-3-carboxylic Acid (5-Oxaproline)[178,179]

| Compound | mp (°C) | Specific Rotation | Ref. |
|---|---|---|---|
| H-L-3-Oxa-OtBu | 52–53 | $[\alpha]_D^{25}$ –25.5 (*c* 1, $CHCl_3$) | [178,179] |
| Z-L-3-Oxa-OH | | $[\alpha]_D^{25}$ –9.3 (*c* 1, aq 95% AcOH) | [178,179] |
| Z-L-3-Oxa-OtBu | 48.5–49 | $[\alpha]_D^{25}$ –85.3 (*c* 1.1, $CHCl_3$) | [178,179] |
| Z-L-3-Oxa-OMe | [a] | $[\alpha]_D^{25}$ –97.8 (*c* 1, $CHCl_3$) | [179] |
| Z-L-3-Oxa-$NH_2$ | 232–234 | $[\alpha]_D^{25}$ –126.5 (*c* 1, $CHCl_3$) | [179] |

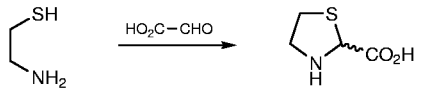
[a] Product isolated as an oil.

### 9.2.3.5 Thiaprolines

#### 9.2.3.5.1 Thiazolidine-2-carboxylic Acid

Access to racemic thiazolidine-2-carboxylic acid (3-thiaproline, **12**) is obtained by reacting cysteamine (**49**) with glyoxylic acid ester (Scheme 9),[165,182,183] whilst the reaction of (*R*)-cysteine with glyoxylic acid[184,185] leads to (2*R*/*S*,5*R*)-thiazolidine-2-carboxylic acid.[185] The diastereomers of thiazolidine-2-carboxylic acid (**12**) are rapidly interconverting and therefore cannot be separated.[185] In the presence of (2*R*,3*R*)- and (2*S*,3*S*)-tartaric acid, reaction of cysteamine with glyoxylic acid leads to the enantiomerically pure (2*R*)- and (2*S*)-thiazolidine-2-carboxylic acid salts.[186] The acids undergo fast racemization in acetic acid.[186]

**Scheme 9** Synthesis of Thiazolidine-2-carboxylic Acid




**49** **12**

#### 9.2.3.5.1.1 Synthesis of Thiazolidine-2-carboxylic Acid Peptides

So far, only a limited use of thiazolidine-2-carboxylic acid (**12**) in peptide synthesis has been reported. This amino acid has been acylated as the alkyl ester with suitably N-protected amino acids via the HOBt/DCC method.[165] Subsequent saponification of peptidyl-thiazolidine-2-carboxylic acid ethyl ester proceeds smoothly with 1 M NaOH in dioxane/water mixtures.[165]

**Ethyl (2*R*,2*S*)-Thiazolidine-2-carboxylate:**[165]
2-Sulfanylethylamine hydrochloride (22.6 g, 190 mmol) and ethyl glyoxylate (20.4 g, 261 mmol) were heated on a water bath for 45 min in the presence of 3-Å molecular sieves (100 g). The filtrate was concentrated and the residue was flash chromatographed (silica gel). Fractions were pooled, concentrated, and the residue crystallized from $CH_2Cl_2$/*tert*-butyl methyl ether; yield: 27 g (88%).

### 9.2.3.5.2 Thiazolidine-4-carboxylic Acid

The aims of inserting a thiazolidine-4-carboxylic acid (4-thiaproline, Thz, **11**) residue into peptide chains are similar to those discussed for the 4-oxaproline analogue in Section 9.2.3.4.1. Thiazolidine-4-carboxylic acid (**11**), particularly when disubstituted at the C5 position of the proline ring, can act as a cysteine precursor and concomitantly as an amide backbone protecting group (see Section 2.3.2.4). Its solubilizing properties derive from its preference for the *cis* configuration of the Xaa-Thz bond.[169–172]

[Note: 4-Thiaproline is also named as pseudoproline, Cys(ΨPro),[169] whereby the related disubstituted derivatives at the C5 position of the proline ring are defined as Cys($\Psi^{R',R''}$Pro).]

Thiazolidine-4-carboxylic acid containing peptides exhibit an enhanced polar character when compared with most of the side-chain protected cysteine peptides.[139] The anomalously low nucleophilicity of its imino group ($pK_a = 6.24$) in comparison to that of proline ($pK_a = 10.60$) is due to the resonance stabilization of its unprotonated form.[187,188] This supposed resonance stabilization is supported by the X-ray crystal structure of 4-thiaproline.[188]

The somewhat unusual electronic structure of thiazolidine-4-carboxylic acid (**11**) is also revealed by unusual positions of the corresponding NMR signals.[189] These facts may explain the difficult N-acylation of 4-thiaproline in peptide synthesis.[189]

The relative stability of the thiazolidine ring toward acids is about tenfold higher than that of the corresponding oxazolidine ring.[139] Conditions and reagents that favor ring opening have been compiled.[139,190] The acid lability of thiazolidine-4-carboxylic acid (**11**) strongly depends upon the degree of substitution at the C5 position and the nature of the substituents.[139,191] Trifluoroacetic acid is sufficient to open the ring of 5-(2,4-dimethoxyphenyl)thiazolidine-4-carboxylic acid residues within minutes, whilst the related 5,5-dimethyl derivative is resistant to trifluoroacetic acid for hours. Conversely, unsubstituted thiazolidine-4-carboxylic acid is fully stable to strong acids such as trifluoromethanesulfonic acid[139] or HBr in acetic acid.[121] Ring opening and hydrolysis can easily be explained by a ring–chain tautomerism in the 1,3-thiazolidine system.[192] The extent of cysteine liberation may be determined by color reactions, e.g. with ferric chloride or sodium nitritoprussiate.[191]

Thiazolidine-4-carboxylic acid is not stable in the presence of reducing agents, e.g. sodium in liquid ammonia[193] and, in contrast to N-protected Thz, the N-unprotected forms are unstable under oxidizing conditions, e.g. slow ring opening occurs in air at pH 10 in the presence of traces of ferric chloride, or, more rapidly, with hydrogen peroxide, bromine, or iodine in aqueous solution.[190]

#### 9.2.3.5.2.1 Synthesis of Thiazolidine-4-carboxylic Acid and Related Analogues

The ring closure of L-cysteine by reaction with aldehydes or ketals[190,194] involving Schiff's base intermediates[195] leads to (4*R*)-thiazolidine-4-carboxylic acids. Except for formaldehyde and symmetrical ketones, the condensation of L-cysteine with aldehydes or unsymmetrical ketones generates a mixture of epimers, i.e. (2*S*,4*R*)- and (2*R*,4*R*)-thiazolidine-4-carboxylic acid derivatives that can be separated by RP-HPLC or fractional crystallization.[139]

The reaction of cysteine with aliphatic aldehydes,[196] e.g. acetaldehyde, or with aromatic aldehydes,[196,197] e.g. benzaldehyde, leads to the corresponding 5-monosubstituted thiazolidine-4-carboxylic acid derivatives, whereas condensation of cysteine with ketones, such as

*for references see p 81*

acetone,[196] or with ketals, e.g. 2,2-dimethoxypropane,[198] yields 5,5-disubstituted thiazolidine-4-carboxylic acids. In an analogous manner selenazolidine-4-carboxylic acids are obtained from selenocysteine.[199]

**Thiazolidine-4-carboxylic Acid, H-Thz-OH:**[185]
To a stirred soln of L-cysteine hydrochloride hydrate (15.8 g, 0.1 mol) and AcOK (9.8 g, 0.1 mol) in a mixture of $H_2O$ and EtOH (150 mL, 2:1) a soln of 37% aq HCHO (17 mL, 0.1 mol) was added. The mixture was stirred for 1 h at rt and for 1 h at 90 °C. After cooling, the precipitate obtained was collected by filtration. The crude product was recrystallized (hot $H_2O$); yield: 91%; mp 203 °C; $[\alpha]_D^{20}$ −141 (*c* 1.3, $H_2O$).

### 9.2.3.5.2.2 Synthesis of Thiazolidine-4-carboxylic Acid Peptides

Due to the extremely low nucleophilicity of the imino group,[187,188] acylation of thiazolidine-4-carboxylic acid (**11**) is not a trivial procedure. In fact, N-protected Thz derivatives can be prepared by standard procedures, but strong acylating conditions are required. The related *N*-Boc derivative is obtained only by prolonged reaction times with Boc-$N_3$[112,189] or with $Boc_2O$.[113] For preparation of the *N*-Z derivative, silylation of Thz with, for example, chlorotrimethylsilane is recommended prior to the reaction benzyl chloroformate.[200] Due to the stability of thiazolidine-4-carboxylic acid to acids its methyl ester is obtained by HCl catalyzed reaction with methanol,[190] whereas the amide is formed by reacting Thz *N*-carboxyanhydride with ammonia.[189] Derivatives of thiazolidine-4-carboxylic acid are listed in Table 8.

**Table 8** Derivatives of Thiazolidine-4-carboxylic Acid (4-Thiaproline)[112,113,185,189,190,198,200,201]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| H-L-Thz-OH | 203 (dec) | $[\alpha]_D^{20}$ −141 (*c* 1.3, $H_2O$)[a] | [185,190,198,201] |
| Boc-L-Thz-OH | 127–129 | $[\alpha]_D^{25}$ −118.7 (*c* 0.9, MeOH) | [112,113,189] |
| Z-L-Thz-OH | [b] | | [200] |
| H-L-Thz-OMe | [b] | $[\alpha]_D^{21}$ −94.7 (MeOH) | [190] |
| H-L-Thz-OEt | [b] | $[\alpha]_D^{20}$ −161 (*c* 8, $CHCl_3$) | [185] |
| H-L-Thz-$NH_2$ | 97–99 | $[\alpha]_D^{27}$ −112.6 (*c* 1.0, $H_2O$) | [189] |
| H-[L-Thz-NCA] | 46 | | [201] |

[a] The optical rotation is strongly dependent on pH.[190]
[b] Product isolated as an oil.

As mentioned above, thiazolidine-4-carboxylic acid is characterized by an anomalously low basicity and thus difficult acylation in peptide synthesis.[189] Therefore, the incorporation of this amino acid residue into a growing peptide chain is preferentially preformed via dipeptide derivatives.[139] Suitably N-protected amino acids are coupled directly to the thiazolidine-4-carboxylic acid by the acid fluoride[139] or *N*-carboxyanhydride[139,201] methods. The resulting dipeptides are used as building blocks without risk of racemization[139] and standard coupling procedures are applied as pentachlorophenyl esters prepared by the mixed anhydride procedure[121] or PyBOP.[171]

**Fmoc-Xaa-Thz-OH:**[139]
H-Thz-OH (1.33 g, 10 mmol) was dissolved in dry DMF (100 mL) in the presence of DIPEA (3.23 mL, 19 mmol). A soln of Fmoc-protected amino acid fluoride (9 mmol) in dry DMF (100 mL) was added dropwise, and the mixture stirred for 2 h at rt. The soln was then concentrated, the residue taken up in EtOAc, washed with aq 5% citric acid and brine, and dried ($Na_2SO_4$). The solvent was removed and the residue purified by flash chromatography (silica gel, EtOAc/AcOH/petroleum ether); yields: 60–95%.

## 9.2.4 Six-Membered Proline Analogues

As in the case of the five-membered proline analogues, besides pipecolic acid (**6**) as the six-membered homologue of proline, related aza-, oxa-, and thia-analogues are used to mimic the conformationally restricting properties of proline in the peptidomimetic design of bioactive compounds.[202]

### 9.2.4.1 Pipecolic Acid

The chemistry of pipecolic acid (piperidine-2-carboxylic acid, homoproline, **6**) and thus the protecting groups and coupling methods closely resemble those of proline, although steric hindrance due to the larger ring size makes its use in standard peptide chemistry more difficult.

#### 9.2.4.1.1 Synthesis of Pipecolic Acid

Racemic pipecolic acid (**6**) is obtained by ring closure of *N*-alkylglycines by ionic[203] or radical[204] mechanisms. It also may be obtained by conversion of suitable substituents at the C2 of piperidine into the 2-carboxy group, e.g. hydrolysis of a nitrile group[205] or oxidation of a 1,2-dihydroxyethyl group.[206] Resolution of the racemic mixture can be carried out by fractional crystallization.[207–209] Enzymatic resolution of racemic pipecolic acid[210–213] or of synthetic intermediates[214] has been reported.

Enantiomerically pure pipecolic acid (**6**) is accessible essentially by two well-established synthetic routes: (i) cyclization of L- or D-lysine by reaction with disodium nitrosylpentacyanoferrate(II) with preservation of configuration at C2;[215,216] (ii) ring closure of $N^{\alpha}$,$N^{\varepsilon}$-bis(*N*-nitroso-*N*-tosyl) derivatives of L- or D-lysine, again with retention of chirality at C2.[217] Stereoselective synthesis of pipecolic acid derivatives, substituted in position 4, is achieved using the aza-Diels–Alder reaction of imines with dienes[218–220] or via an ene–iminium cyclization.[221,222]

In addition, stereoselective syntheses of pipecolic acid derivatives, such as the stereoisomers of 4-hydroxypipecolic acid,[223–225] 3-hydroxypipecolic acid,[226] and variously ring-substituted analogues,[227–234] have been reported.

#### 9.2.4.1.2 Synthesis of Pipecolic Acid Peptides

$N^{\alpha}$-Protection of pipecolic acid (**6**) as its Boc derivative is carried out by standard procedures with Boc-$N_3$,[177,208] $Boc_2O$,[113,114] or with 2-(*tert*-butoxycarbonyloxyimino)-2-phenylacetonitrile.[235] Similarly, the Fmoc derivative is prepared in a standard manner (Table 9).[200,207]

*for references see p 81*

**Table 9** Pipecolic Acid Derivatives[113,114,121,177,203,206–208,214,215]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| H-L-Pip-OH | 263–264 | $[\alpha]_D^{25}$ −26 (*c* 5, $H_2O$) | [207,208,215] |
| H-L-Pip-OH·HCl | | $[\alpha]_D^{25}$ −20.4 (*c* 2, $H_2O$) | [177] |
| H-D-Pip-OH | 263 | $[\alpha]_D^{25}$ +26 (*c* 5, $H_2O$) | [208] |
| Boc-L-Pip-OH | 124 | $[\alpha]_D^{25}$ −56 (*c* 1, AcOH) | [113,114,177,208,214,215] |
| Boc-D-Pip-OH | 123 | $[\alpha]_D^{25}$ +56 (*c* 1, AcOH) | [208] |
| Boc-L-Pip-OPcp | 136–138 | | [215] |
| Fmoc-L-Pip-OH | | $[\alpha]_D$ −21.7 (*c* 1, MeOH) | [207] |
| Fmoc-D-Pip-OH | | $[\alpha]_D$ +21.4 (*c* 1, MeOH) | [203] |
| Z- L-Pip-OH | 112–113 | $[\alpha]_D^{25}$ −57 (*c* 5.3, AcOH) | [121,208] |
| Z- D-Pip-OH | 112–113 | $[\alpha]_D^{25}$ +57.2 (*c* 5.3, AcOH) | [121,206,208] |
| Z- L-Pip-ONp | [a] | | [208] |
| Z- D-Pip-ONp | [a] | | [208] |
| Z- L-Pip-OPcp | 110–111 | $[\alpha]_D^{25}$ −38.5 (*c* 1, DMF) | [121,208] |
| Z- D-Pip-OPcp | 110–111 | $[\alpha]_D^{25}$ +38.5 (*c* 1, DMF) | [208] |

[a] Product isolated as an oil.

The Z-protected derivative, again prepared by standard methods using benzyl chloroformate,[208] may serve in the case of racemic pipecolic acid for resolution into the pure enantiomers by fractional crystallization with L-tyrosine hydrazide.[208] Acylation with N-protected pipecolic acid or of pipecolyl peptides is performed by standard procedures via the active ester methods, e.g. *N*-hydroxysuccinimide ester,[121] by the mixed anhydride method, e.g. with isobutyl chloroformate[95,114] or pivalic acid chloride,[121] as well as by DCC/HOBt.[118] In the synthesis on solid support, longer coupling times are required when compared to N-protected proline.[235]

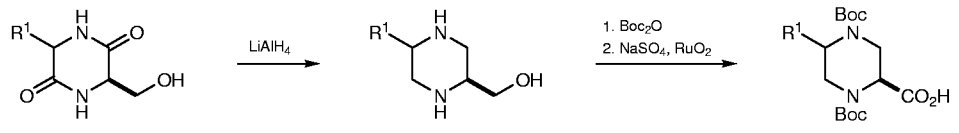
## 9.2.4.2 Azapipecolic Acids

### 9.2.4.2.1 Piperazine-2-carboxylic Acid

Racemic piperazine-2-carboxylic acid (4-azapipecolic acid, **22**) is obtained by hydrogenation of pyrazine carboxylic acid salts.[236] For its enantiomeric resolution several methods have been described, e.g. fractional crystallization of its *S*-camphor-10-sulfonic acid salt,[236,237] fractional crystallization of 1,4-dibenzylpiperazine-2-carboxylic acid menthyl ester,[238] or selective digestion of racemic piperazine-2-carboxylic acid amide by leucine aminopeptidase.[239]

Optically pure piperazine-2-carboxylic acid (**22**) and related derivatives, substituted at position 5, are synthesized from piperazine-2,5-diones **50** derived from Xaa-Ser dipeptides as educts, where the substitution $R^1$ at position 5 depends on the side group of the aminoacyl moiety (Scheme 10). After reduction of the piperazine-2,5-diones **50** to 5-alkyl-2-(hydroxymethyl)piperazines **51** and urethane-type protection of both the imino groups, the desired *N,N'*-bis-protected piperazine-2-carboxylic acid or related 5-alkyl derivatives **52** are obtained by selective oxidation of the hydroxymethyl group.[240]

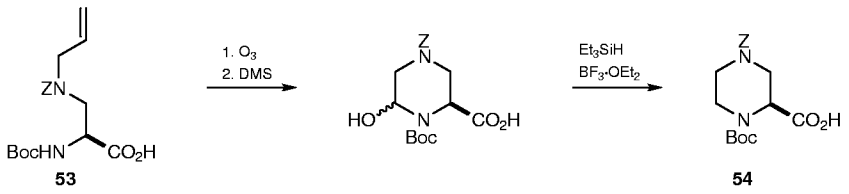
**Scheme 10** Synthesis of Enantiomerically Pure Piperazine-2-carboxylic Acid[240]




**50** **51** **52**

The enantiomerically pure (*S*)-4-(benzyloxycarbonyl)-1-(*tert*-butoxycarbonyl)piperazine-2-carboxylic acid derivative **54** is also prepared by a multistep cyclization of chiral (*S*)-3-[(benzyloxycarbonyl)(prop-2-enyl)amino]-2-[(*tert*-butoxycarbonyl)amino]propanoic acid (**53**) as shown in Scheme 11.[241]

**Scheme 11** Synthesis of (*S*)-4-(Benzyloxycarbonyl)-1-(*tert*-butoxycarbonyl)piperazine-2-carboxylic Acid[241]




### 9.2.4.2.1.1 Synthesis of Piperazine-2-carboxylic Acid Peptides

Reaction of piperazine-2-carboxylic acid (**22**) with Z-OSu at pH 11 leads to the $N^4$-(benzyloxycarbonyl) monoprotected derivative,[242] which is then converted with 2-(*tert*-butoxycarbonyloxyimino)-2-phenylacetonitrile at pH 9.5 into the $N^1$-*tert*-butoxycarbonyl-$N^4$-benzyloxycarbonyl bis-protected derivative.[242] For further steps in peptide synthesis standard protocols are applied.

### 9.2.4.2.2 Piperidazine-3-carboxylic Acid (Piperazic Acid)

The chemistry of piperidazine-3-carboxylic acid (piperazic acid, Piz, **7**) is characterized by the divergent nucleophilicity of both nitrogens with NH in position 1 being more readily acylated than that in position 2. This difference in the nucleophilicity may be further enhanced by the steric effects of ring substituents. Another remarkable aspect that has to be kept in mind is the reducing effect of the hydrazino group. Moreover, piperazic acid is susceptible to cleavage by hydrogenolysis, leading to ornithine, but only under drastic hydrogenolysis conditions.[243]

#### Analytical Characterization

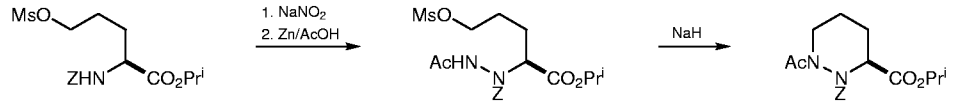
Piperazic acid (**7**) develops a yellow color with 4-(dimethylamino)benzaldehyde, while with ninhydrin a red color is formed.[243] Under standard conditions of acid hydrolysis piperazic acid is largely decomposed to ornithine and ammonia, but under optimized conditions (6 M HCl, 3 h, 95 °C in sealed tubes) it can be recovered.[243] Piperazic acid can also be identified by IR or NMR spectroscopy.[243,244]

### 9.2.4.2.2.1 Synthesis of Piperidazine-3-carboxylic Acid (Piperazic Acid) Peptides

A number of methods for the synthesis of piperazic acid (**7**) and related derivatives are currently available as a result of growing interest in natural product chemistry and in their potential in medicinal chemistry. Their chemistry and conformational properties have been comprehensively reviewed.[245] Racemic piperazic acid is obtained by condensation of penta-2,4-dienoic acid with phthalazinedione and subsequent reductive deprotection of the resulting *N*,*N*′-bis(phthaloyl)-1,2,3,6-tetrahydropyridazine-3-carboxylic acid.[243] Resolution of racemic piperazic acid is achieved by fractional crystallization of the ephedrine salt of $N^1$-(benzyloxycarbonyl)piperazic acid from ethyl acetate.[246,247] A typical route to enantiomerically pure (3*S*)-piperazic acid **56** starts from chiral 2-amino-5-hydroxyvaleric acid **55** as shown in Scheme 12.[248] Convenient stereoselective syntheses have been reported for 5-hydroxy- and 5-chloropiperazic acids as important constituents of natural cyclic peptides and depsipeptides.[249,252]

*for references see p 81*

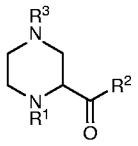
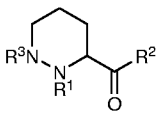
**Scheme 12** Synthesis of Optically Pure Piperazic Acid[248]




55 56

Although piperazic acid, so far, has found only limited use in peptide synthesis, the different reactivity of NH-1 and NH-2 can be exploited for the selective protection of the NH-1, which then allows for a subsequent selective protection of NH-2 according to the overall protection scheme (Table 10). Exploiting these characteristic properties $N^1$-(benzyloxycarbonyl)-$N^2$-(trifluoroacetyl)piperazic acid[246] and $N^1$-(benzyloxycarbonyl)-$N^2$-(9-fluorenylmethoxycarbonyl)piperazic acid[265] were prepared and used for peptide synthesis.[250,251,266] To overcome the reduced reactivity of both the amino and carboxy group of piperazic acid the acid chloride method has been used for acylation of this amino acid, as well as for coupling it to amino components.[246] The acid fluoride method is expected to have similar efficiency.

**Table 10** Azapipecolic Acid Derivatives[236,238,241,246]

| Compound | | | D/L | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|---|---|---|
| $R^1$ | $R^2$ | $R^3$ | | | | |
|  | | | | | | |
| H | OH | H | L | | $[\alpha]_D^{20}$ −3.9 (*c* 6, $H_2O$) | [236] |
| H | OH | H | D | | $[\alpha]_D^{20}$ +3.9 (*c* 6, $H_2O$) | [236] |
| H | OH·2HCl | H | D | 238–243 (dec) | $[\alpha]_D^{20}$ +5.9 (*c* 1.25, $H_2O$) | [238] |
| Boc | OH | Z | L | 50–53 | $[\alpha]_D^{20}$ −17.5 (*c* 1, $CHCl_3$) | [241] |
|  | | | | | | |
| Z | OH | H | L | 168–171 | $[\alpha]_D^{20}$ −35.6 (*c* 0.5, MeOH) | [246] |
| Z | OH | H | D | 168–172 | $[\alpha]_D^{20}$ +35.0 (*c* 0.5, MeOH) | [246] |
| Z | OtBu | H | L | [a] | | [246] |

[a] Product isolated as an oil.

### 9.2.4.3 Oxapipecolic Acids

Racemic morpholine-3-carboxylic acid (4-oxapipecolic acid, **18**) is prepared by a multistep synthesis, starting from 3-methoxymorpholine.[205] It can, however, be obtained as optically pure (3*S*)-morpholine-3-carboxylic acid (L-4-oxapipecolic acid) by base-mediated cyclization of *O*-(2-chloroethyl)-L-serine.[253] Similarly, (4*S*)-tetrahydro-2*H*-1,3-oxazine-4-carboxylic acid (L-5-oxapipecolic acid, **19**) is obtained from L-homoserine by reaction with aqueous formaldehyde under alkaline conditions.[173]

#### 9.2.4.3.1 Synthesis of Oxapipecolic Acid Peptides

Both 4- and 5-oxapipecolic acids (**18** and **19**, respectively) are protected by standard procedures and no particular precautions are required in peptide synthesis (Table 11).

**Table 11** Morpholine-3-carboxylic Acid[253]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| L-morpholine-3-carboxylic acid | 218 (dec) | $[\alpha]_D^{25}$ −14.4 (*c* 2, 1 M HCl) | [253] |
| D-morpholine-3-carboxylic acid | 216–218 (dec) | $[\alpha]_D^{25}$ +13.8 (*c* 1, 1 M HCl) | [253] |

### 9.2.4.4 Thiapipecolic Acids

(3*R*)-1,4-Thiazinane-3-carboxylic acid (4-L-thiapipecolic acid, **20**) is obtained by base-assisted cyclization of (*S*)-(2-chloroethyl)-L-cysteine.[198,253,254] (4*S*)-Tetrahydro-2*H*-1,3-thiazine-4-carboxylic acid (L-5-thiapipecolic acid) is prepared from L-homocysteine and aqueous formaldehyde,[255] or from L-homocysteine thiolactone hydrochloride with aqueous formaldehyde under acid catalysis with HCl.[198] For protection and peptide synthesis standard procedures are applied. Data for 4-thiapipecolic acid are listed in Table 12.

**Table 12** 4-Thiapipecolic Acid[198,253]

| Compound | mp (°C) | Specific Rotation | Ref |
|---|---|---|---|
| L-4-thiapipecolic acid | 250–251 (dec) | $[\alpha]_D^{25}$ −51.7 (*c* 1, $H_2O$) | [198,253] |
| D-4-thiapipecolic acid | 248–249 (dec) | $[\alpha]_D^{25}$ +50.6 (*c* 1, $H_2O$) | [253] |


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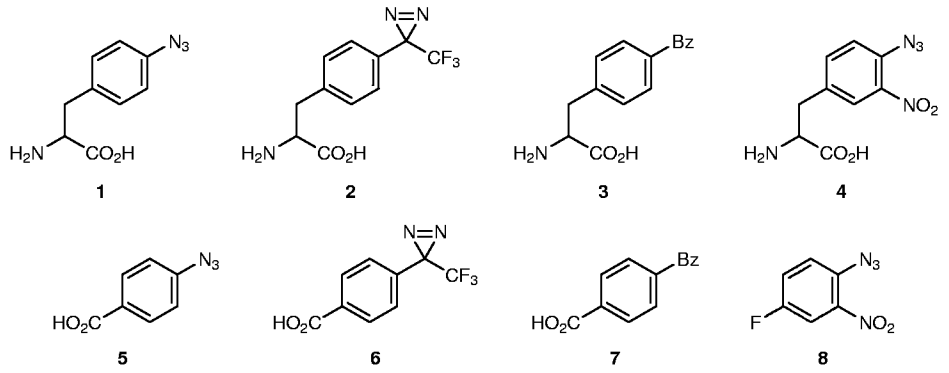
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# 9.3 Synthesis of Photoreactive Peptides

A. Bisello and M. Chorev

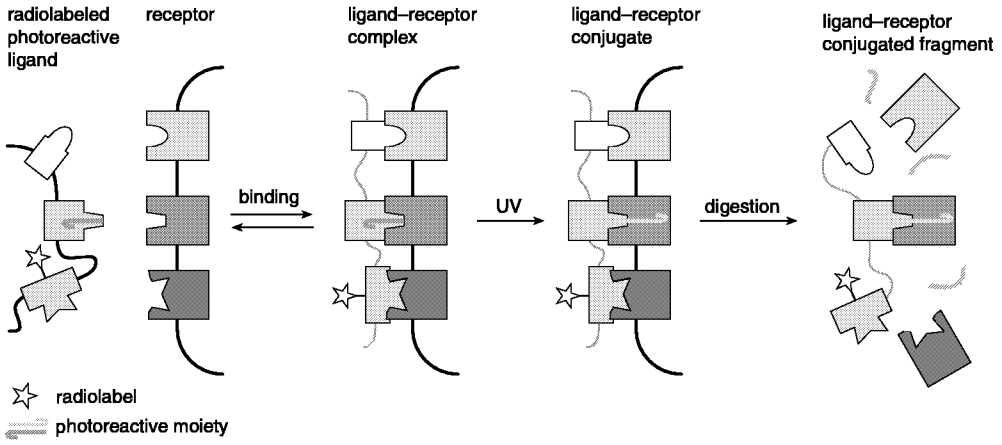
Photoreactive peptides are peptides modified with a photophore, a photolabile group. Irradiation of these peptides generates a highly reactive intermediate, usually a carbene or nitrene, which will react with an acceptor molecule to form an intermolecular covalent bond. In order for a photoreactive ligand to become a photoaffinity reagent, which specifically labels the acceptor molecule, it must display high affinity toward the molecular target. In general, a photoaffinity reagent will carry a tag such as a radionuclei, a fluorophore, or a biotin moiety. Scheme 1 shows typical amino acids (compounds **1–4**) and congeners (compounds **5–9**) that have been developed as photophores for incorporation into photoaffinity peptides.

**Scheme 1** Examples of Amino Acids and Congeners Developed as Photophores




Photoreactive peptides are used in photoaffinity labeling to study bimolecular interactions. In addition to identifying the molecules that are in contact, one can probe the bimolecular interface by carrying a detailed photoaffinity scanning study (Scheme 2). Photoaffinity cross-linking generates a photoconjugate in which the bimolecular contact between the two molecular entities is "frozen", through a photo-insertion reaction, in the form of a covalent bond. Analysis of the photoconjugate employing chemical and enzymatic degradations, and physicochemical methods, may identify a bimolecular "contact site".

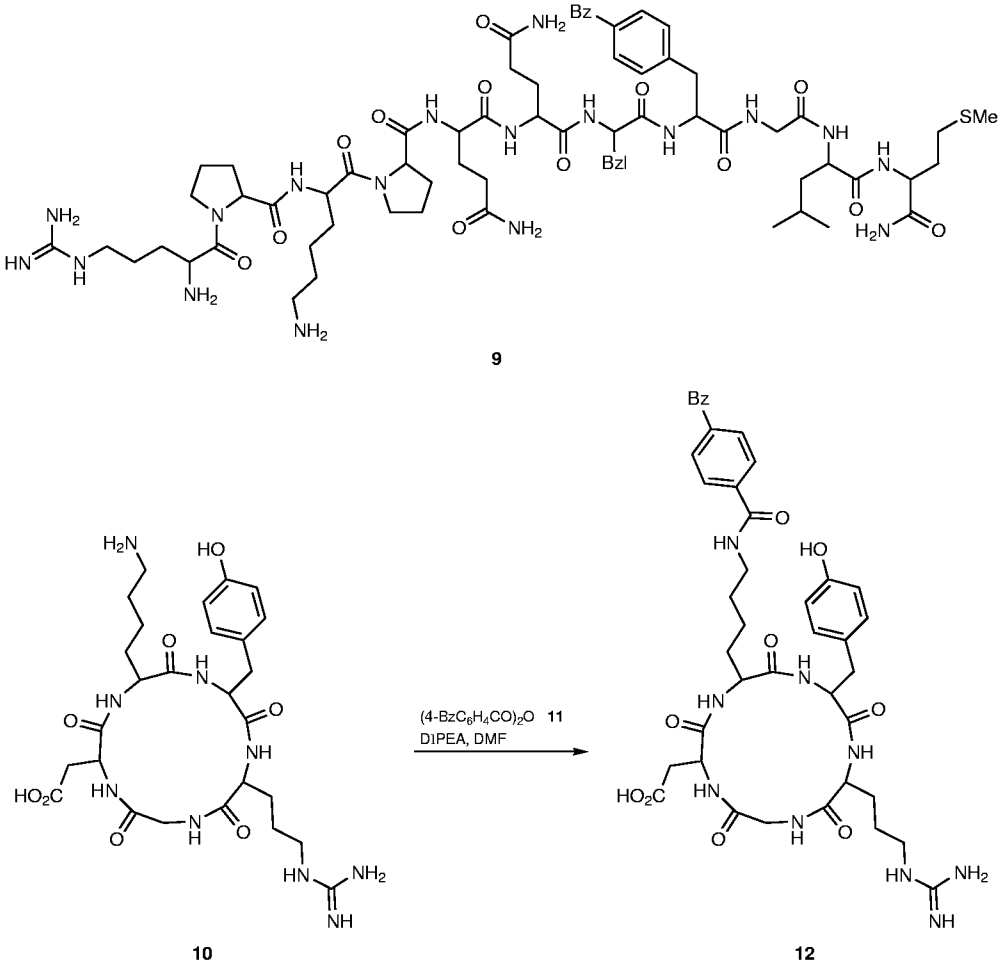
**Scheme 2** Photoaffinity Labeling and Identification of a Ligand-Contact Site




*for references see p 147*

There are two principal modes for introducing the photophore into a peptide: (1) Coupling a photoreactive N-protected amino acid during the stepwise assembly of the peptide; e.g. in Scheme 3, the amino acid 4-benzoylphenylalanine (**3**) is incorporated into position 8 of substance P (**9**). (2) Post-synthetic modification of a fully assembled peptide, at either a free N-terminal α-amino group or a side-chain functionality, by the photophore in a site-specific or nonspecific manner; e.g. in Scheme 3, treatment of cyclic R-G-D-containing peptide **10** with 4-benzoylbenzoic anhydride (**11**) gives the modified peptide **12**.

**Scheme 3** Two General Modes for Synthesis of Photoreactive Peptides[202,128]

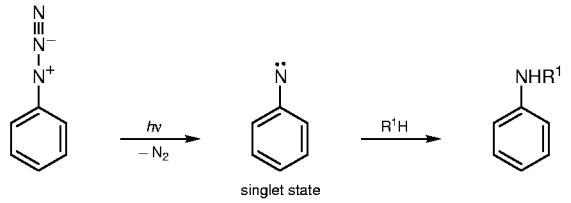



This section summarizes the synthetic methodologies related to the preparation of photophores that are amino acids or congeners, their incorporation into photoreactive peptides, and their use in the site-directed photoaffinty labeling of macromolecular targets. This section does not cover the extensive and very important work related to the chemistry and application of bifunctional cross-linking reagents. Some excellent general reviews have been written on photoaffinity labeling and are recommended for further reading.[1–7]

## 9.3.1 Peptides Containing Phenyl Azides (Azidoarenes)

Phenyl azides (azidoarenes), introduced by Knowles and co-workers,[8,9] are the most abundantly used class of photophores. Examples include 4-azidophenylalanine (**1**) and 4-azido-3-nitrophenylalanine (**4**) (Scheme 1). Irradiation (≤300 nm) of phenyl azide (**13**) generates nitrene **14**, electrophilic in nature, which prefers insertion into O–H and N–H bonds over C–H bonds. Nitrenes are considerably less reactive and, therefore, more selective than carbenes. Nevertheless, due to their short life span (0.1–1 ms) they react indiscriminately with virtually any amino acid residue in the target protein.[10] Intramolecular rearrangements do not compete effectively with intermolecular proton abstraction and insertion reactions (Scheme 4).

**Scheme 4** Photoactivation of Phenyl Azide[8,9]




**13** **14**

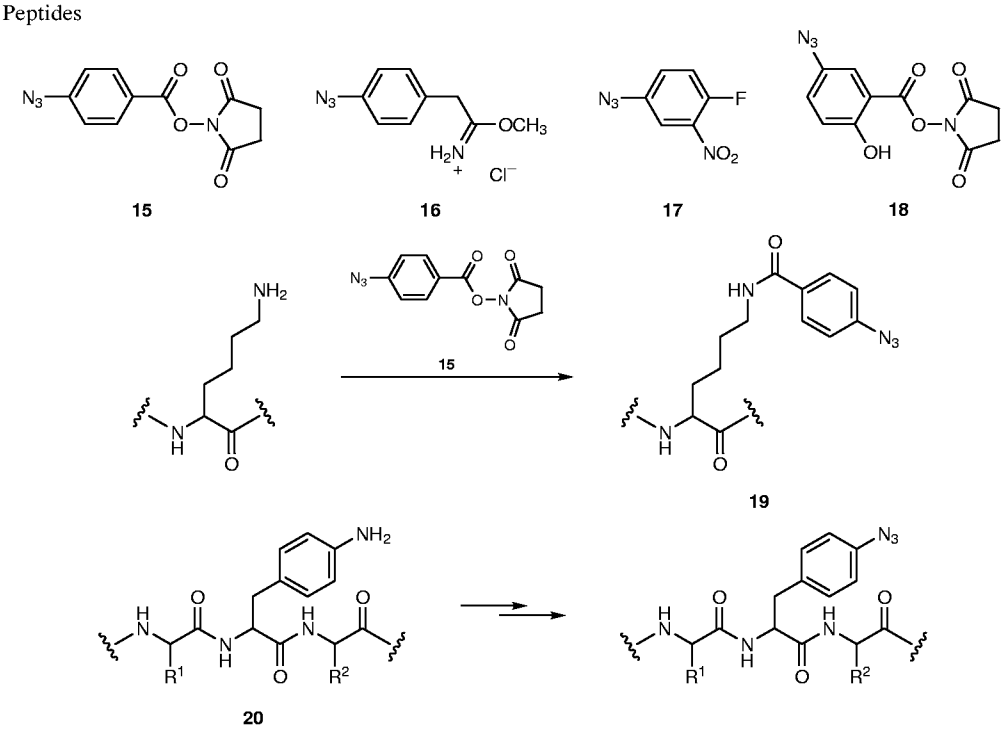
The photochemistry of aryl azides is quite complex, suggesting that the nitrene **14** may not be the only reactive intermediate and that insertion reactions may not be the only route to form photoconjugates.[9,11,12] Although aryl nitrenes are much less susceptible to rearrangements than acyl nitrenes, they may still occur and lead to the formation of reactive intermediates such as azepines, which may go on to react with nucleophiles.[9,11–14] Addition of nitrenes to double bonds will generate azirines, while dimerization will produce azobenzenes.[11] Aryl azides are stable to most of the procedures used in the course of peptide synthesis except for reduction reactions. Non-photochemical reduction of aryl azides to the primary amines by thiols has been reported by Staros et al.[15]

The introduction of electron-withdrawing groups such as a nitro group, e.g. as in 4-azido-3-nitrophenylalanine (**4**), shifts the photolysis to longer wavelengths (>350 nm) that are less harmful to proteins.[8] Consequently, the generated nitrene is more reactive than the one generated from phenyl azide, leading also to higher rates of reaction with water.

The first attempts to form photo-cross-linking ligand–acceptor complexes used heterobifunctional linkers containing such aryl azide photophores as compounds **15–18** (Scheme 5). This approach leads to a high degree of nonspecificity, affecting multiple membrane proteins and cross-linking the ligand–acceptor complex at sites unrelated to the functional bimolecular interface. Alternatively, the same hetero-bifunctional linkers can be used to modify a peptide site, specifically generating a monofunctional photoreactive reagent, e.g. **19** (Scheme 5). De novo synthesis of peptides containing either an aryl azide moiety or an aryl amine as a precursor (e.g., peptide **20**, containing a 4-aminophenylalanine residue) offers an attractive mode to position a photophore at or in close proximity to the binding domain (Scheme 5).

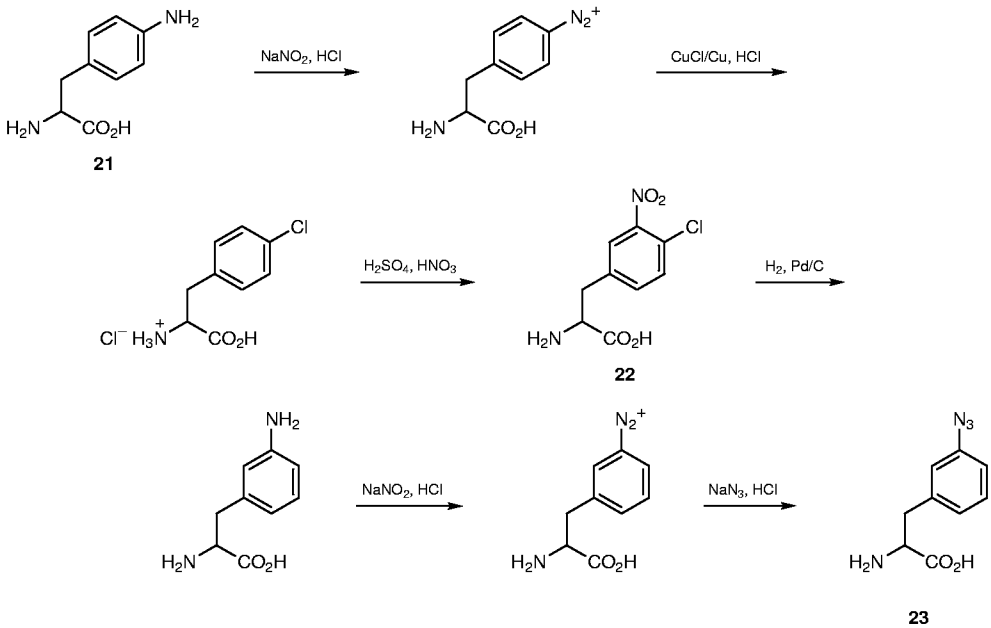
*for references see p 147*

**Scheme 5** Structures of Some Hetero-bifunctional Photo-cross-linking Reagents and Their Incorporation into Peptides




In an effort to better characterize the topochemical relationship between the "hydrophobic pocket" in the chymotrypsin binding site and the aromatic side chain in the substrate, 3-azidophenylalanine has been prepared.[16] Conversion of 4-aminophenylalanine (**21**) into 3-azidophenylalanine (**23**) was carried out via 4-chloro-3-nitrophenyl intermediate **22** as outlined in Scheme 6.

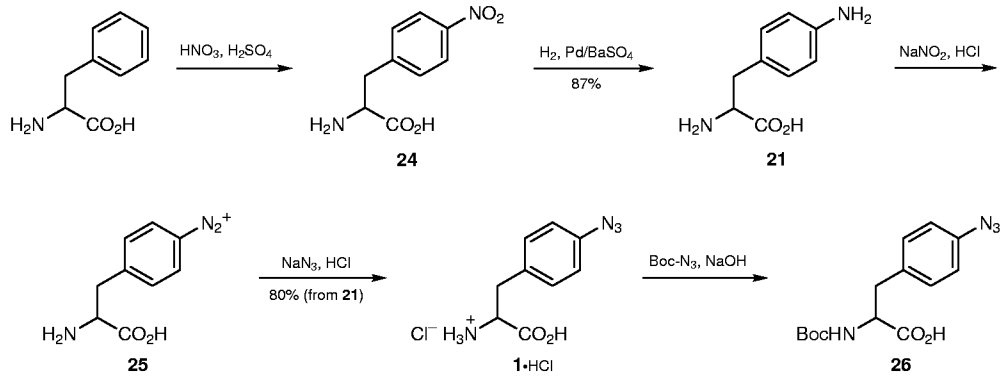
**Scheme 6** Synthesis of 3-Azidophenylalanine from 4-Aminophenylalanine[16]



### 9.3.1.1 Synthesis of 4-Azidophenylalanine and Its Protected Derivatives

Preformed N-protected 4-azido-L-phenylalanine was originally used as a building block in the synthesis of peptides. Wieland and Schwyzer and their co-workers report the introduction of 4-azido-L-phenylalanine into peptide chemistry in the preparation of [Phe$^3$(4-$N_3$)]-antamanide,[17] and substrates and inhibitors of chymotrypsin.[18] Nitration of L-Phe generates 4-nitro-L-phenylalanine (**24**), the precursor of 4-azido-L-phenylalanine (**1**), in high yield.[19] The nitro compound is catalytically hydrogenated in the presence of Pd/$BaSO_4$ to generate 4-amino-L-phenylalanine (**21**).[17,20] The diazotization of the latter generates the diazonium salt **25** in situ that, upon reaction with sodium azide, yields the anticipated 4-azido-L-phenylalanine hydrochoride salt (**1**·HCL).[17] Preparation of Boc-L-Phe(4-$N_3$)-OH (**26**) is carried out by the standard procedure employing Boc-$N_3$ (Scheme 7).[17] Alternatively, Boc-L-Phe(4-$NO_2$)-OH and Phth-L-Phe(4-$NO_2$)-OH can be generated by the standard methods employing Boc-$N_3$ and *N*-(ethoxycarbonyl)phthalimide, respectively.[21] These N-protected 4-nitrophenylalanine derivatives are transformed in the same manner described above to the corresponding N-protected 4-aminophenylalanine derivatives, and subsequently to the succinimidyl and 4-aminophenylalanine active esters.[21]

**Scheme 7** Synthesis of Boc-Protected 4-Azidophenylalanine from Phenylalanine[17,19,20]




**H-L-Phe(4-$NH_2$)-OH** (21):[20]

L-H-Phe(4-$NO_2$)-OH (**24**; 8 g, 44 mmol) in $H_2O$ (150 mL) was hydrogenated at rt and atmospheric pressure in the presence of Pd/$BaSO_4$ and the theoretical amount of $H_2$ was absorbed rapidly. The filtered soln was concentrated at 50 °C in vacuo until crystallization occurred. Upon cooling, the amino acid was obtained in the form of fine needles; yield: 5.9 g (87%).

**H-L-Phe(4-Cl,3-$NO_2$)-OH** (22):[16]

A stirred soln of H-L-Phe(4-$NH_2$)-OH (**21**; 54 g, 300 mmol) in 4 M HCl (600 mL) was cooled in a MeOH/ice bath and treated slowly with aq $NaNO_2$ (20.7 g, 300 mmol). The mixture was kept for 1 h under vacuo (until negative to $HNO_2$ in the starch–iodine paper test). The soln of the diazonium salt was mixed slowly into 20% HCl (400 mL) containing CuCl (10 g) and Cu powder (3 g). After the cessation of gas release, the mixture was left overnight at 4 °C, the precipitate was filtered off (53 g of hydrochloride) and the filtrate was concentrated under vacuo and filtered again (9.2 g); combined yield: 61.2 g (86%).

A stirred cold (MeOH/ice bath) soln of free 4-chlorophenylalanine (4.5 g, 22.5 mmol) (obtained by desalting) in concd $H_2SO_4$ (15 mL) was treated with 100% $HNO_3$ (1 mL, 25.5 mmol). The mixture was stirred for 2 h and then added to 5 volumes of ice. The resultant precipitate was left overnight, filtered, and the nonstoichiometic hydrogen sulfate salt was used directly in the following step. The product obtained in a separate preparation was characterized.

**H-L-Phe(3-$NH_2$)-OH:**[16]

The wet hydrosulfate obtained in the previous step was dissolved in 50% AcOH (250 mL), treated with Pd/C (100 mg), and hydrogenated overnight at 22 °C under $H_2$ at atmospheric pressure. The catalyst was filtered off and the fitrate was concentrated in vacuo. The residue obtained was dissolved in $H_2O$ and desalted with Dowex 50. Addition of iPrOH to the aqueous soln formed a precipitate, which was dried; yield: 2.25 g (56%).

*for references see p 147*

**H-L-Phe(3-$N_3$)-OH (23):**[16]
A soln of H-L-Phe(3-$NH_2$)-OH (3.0 g, 16.7 mmol) in 6 M HCl (200 mL) was treated with $NaNO_2$ (1.27 g, 18.5 mmol) as described above for H-L-Phe(4-$NH_2$)-OH and with aq $NaN_3$ (1.24 g, 1.95 mmol). The mixture was concentrated in vacuo. The residue was dissolved in $H_2O$ and desalted. The azide was crystallized ($H_2O$ following the addition of iPrOH) and dried; yield: 1.34 g (39%).

**H-L-Phe(4-$N_3$)-OH·HCl (1·HCl):**[17]
To a soln of H-L-Phe(4-$NH_2$)-OH (**21**; 36 g) in 6 M HCl (100 mL) at <5 °C (ice bath) was added dropwise, over a period of 20 min, aq $NaNO_2$ (14 g in 10 mL $H_2O$). A second portion of 6 M HCl (166 mL) was then added and the mixture cooled for 10 min followed by dropwise addition aq $NaN_3$ (12.28 g in 10 mL $H_2O$) over a period of 20 min. After the addition, the mixture was stirred at 5 °C for 15 min and then at 20 °C for 1 h. The precipitated H-L-Phe(4-$N_3$)-OH·HCl (**1**·HCl) was filtered off and washed with a small amount of cold $H_2O$ followed by $Et_2O$. The solid material was dissolved in EtOH and precipitated by the addition of $Et_2O$ (3 volumes). The aqueous filtrate was concentrated in vacuo at 40 °C. The residue obtained was dissolved in EtOH and filtered. The clear filtrate was treated with $Et_2O$ to form a precipitate (ca. 9 g) that was filtered off, dissolved in MeOH (50 mL), and the soln stirred with charcoal (20 g) for 30 min. Treatment of the filtrate with $Et_2O$ yielded a precipitate of the title product; yield: 35.5 g (80%); mp 195 °C (dec); IR (KBr) $\tilde{\nu}_{max}$: 2140 $cm^{-1}$.

**Boc-Phe(4-$NO_2$)-OH:**[21]
H-Phe(4-$NO_2$)-OH (**24**; 4.2 g, 20 mmol)[19] and distilled Boc-$N_3$ (3.16 g, 22 mmol) were dissolved in a mixture of dioxane (15 mL) and $H_2O$ (15 mL) and kept at pH 10.1 using a pH-stat. Workup was carried out in the standard manner; yield: 4.3 g (69%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1339 (s), 1485 (m), 1598 (m), 1695 (s) $cm^{-1}$.

**Boc-Phe(4-$N_3$)-OH·DCHA and the Free Acid 26:**[21]
Boc-Phe(4-$NO_2$)-OH (6.2 g, 20 mmol) was dissolved in a mixture of MeOH (150 mL) and glacial AcOH (1.5 mL) and hydrogenated at 0–5 °C using as the catalyst 10% Pd/C (340 mg). After 3 h, the hydrogenation was complete, as indicated by TLC (silica gel, $CHCl_3$/MeOH/AcOH 95:5:3). The product, Boc-Phe(4-$NH_2$)-OH (approx. 20 mmol), after filtration from the catalyst and evaporation of the solvent, was dissolved in a mixture of 2 M HCl (10 mL) and $H_2O$ (2 mL) at 0–2 °C. The diazonium salt was formed by treating the mixture with an ice-cold soln of $NaNO_2$ (1.45 g, 21 mmol) in $H_2O$ (5 mL) for 30 min. Filtration removed the solid byproducts. To remove the excess of $HNO_2$, the filtrate was degassed in vacuo while maintaining the soln temperature at 0–2 °C throughout. A soln of $NaN_3$ (1.30 g, 20 mmol) in $H_2O$ (5 mL) was then added slowly with stirring and the mixture was kept at approximately 0 °C for 15 min. The product was extracted with EtOAc and isolated by the standard procedure. A soln of the azide in EtOAc/petroleum ether (1:1) was treated with DCHA (3.6 g, 20 mmol) to convert it into the salt, which crystallized at 0 °C. The solid was recrystallized ($Et_2O$/hexanes); yield: 8.6 g (89%); mp 151.5 °C; $[\alpha]_D^{25}$ +40.8 (*c* 1.00, MeOH); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1575 (m), 1620 (s), 2100 $cm^{-1}$ (s).

The free acid Boc-Phe(4-$N_3$)-OH (**26**) was obtained practically quantitatively using 0.05 M $H_2SO_4$ and EtOAc as a solvent. The solid was recrystallized (cyclohexane); mp 85.4 °C; $[\alpha]_D^{25}$ +21.5 (*c* 2.75, MeOH); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1575 (m), 1620 (s), 2100 $cm^{-1}$ (s).

**Boc-Phe(4-$N_3$)-$OC_6H_4$-4-$NO_2$:**[21]
A soln of Boc-Phe(4-$N_3$)-OH [obtained by the standard procedure from Boc-Phe(4-$N_3$)-OH·DCHA; 2.44 g, 5 mmol] and 4-nitrophenol (765 mg, 5.5 mmol) in dry EtOAc (15 mL) in an ice bath was treated with an ice-cold soln of DCC (1.13 g, 5.5 mmol) in EtOAc (5 mL). The mixture was kept at 0 °C for 2 h and then at rt for ca. 15 h. The filtrate obtained after removing the DCU was treated in the standard manner, phenol was removed by washing with a diluted (1:1) ice-cold soln of sat. $NaHCO_3$. The residue obtained after removal of the organic solvent in vacuo was crystallized (EtOAc/iPrOH); yield: 1.7 g (80%); mp 141.4 °C; $[\alpha]_D^{25}$ +0.923 (*c* 1.17, MeOH).

**Boc-Phe(4-$N_3$)-ONSu:**[21]
The procedure was the same as that used for the prepapration of the corresponding 4-nitrophenyl ester. A soln of Boc-Phe(4-$N_3$)-OH [obtained by the standard procedure from Boc-Phe(4-$N_3$)-OH·DCHA; 2.44 g, 5 mmol] and HOSu (547 mg, 5 mmol) in dry EtOAc (15 mL) in an ice bath was treated with an ice-cold soln of DCC (1.13 g, 5.5 mmol) in EtOAc (20 mL). The mixture was kept at 0 °C for 2 h and then at rt for ca. 15 h. DCU was removed by filtration and the residue obtained after removal of the solvent in vacuo was crystallized (benzene/$iPr_2O$/hexanes); yield: 1.65 g (82%). The TLC (alumina, $CHCl_3$/acetone 7:3) indicated pure product, whereas chromatography on silica gel with the same solvent mixture or with $CHCl_3$/MeOH/AcOH (95:5:3) always resulted in gradual hydrolysis of the active ester. The residue obtained after removal of the organic solvent in vacuo was crystallized (EtOAc/iPrOH); mp 140.1 °C; $[\alpha]_D^{25}$ −26.6 (*c* 1.05, MeOH).

**Phth-Phe(4-$NO_2$)-OH:**[21]
To a stirred suspension of H-Phe(4-$NO_2$)-OH (**24**; 2.0 g, 10 mmol) in DMF (30 mL) under $N_2$, TEA (1.2 g, 12 mmol) was added slowly, thus creating a thick suspension of the salt. Stirring was continued for another 0.5 h. After cooling the mixture to 0–5 °C, it was treated with *N*-(ethoxycarbonyl)phthalimide (2.6 g, 12 mmol). The mixture was stirred for 12 h at rt. The solvent was evaporated at 0.01 Torr and the oily residue was treated with warm 0.1 M HCl. Crystallization occurred upon cooling. The crude solid was recrystallized (MeOH/$H_2O$); yield: 2.8 g (82%); mp 204.7 °C; $[\alpha]_D^{25}$ −232.5 (*c* 1.55, MeOH).

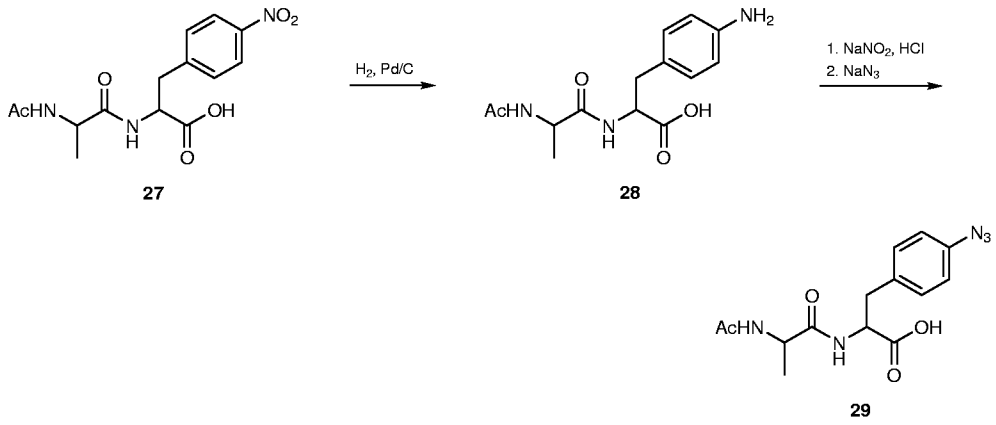
**Phth-Phe(4-$N_3$)-OH:**[21]
Preparation from Phth-Phe(4-$NO_2$)-OH (3.1 g, 9.9 mmol) follows the same procedure as described for the preparation of Boc-Phe(4-$N_3$)-OH except that the dried concentrated EtOAc phase, obtained in the standard manner, was treated at this stage with DCHA to form a precipitate. The salt was crystallized (EtOAc/$Et_2O$) to give the product; yield: 3.7 g (76%); mp 153.8 °C; $[\alpha]_D^{25}$ −150.5 (*c* 1.03, MeOH); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1622 (s), 1697 (s), 2100 $cm^{-1}$ (s).

## 9.3.1.2 Strategies for the Synthesis of Peptides Containing 4-Azidophenylalanine

Several approaches have been developed to prepare 4-azidophenylalanine containing peptides. The direct approach uses standard solution-phase methodology to incorporate either the $N^{\alpha}$-protected Phe(4-$N_3$)-OH to the free amino terminus[17,22] or the free acid to carboxy-activated peptides (e.g., succinimidyl ester).[16] This early incorporation of the light-susceptible azidoarene moiety into a peptide complicates handling and restricts certain reactions used routinely in peptide synthesis. Nevertheless, following the above strategy, peptide derivatives of antamanide have been prepared.[17]

The azido function is unstable during cleavage of side-chain protecting groups by hydrogenation and addition of Na/liq $NH_3$, and detachment from the solid support by liquid HF.[23] Therefore, incorporation of $N^{\alpha}$-protected Phe(4-$NO_2$)-OH into the peptide chain is carried out in a standard manner. The 4-nitro group in the free Phe(4-$NO_2$)-containing peptide **27** is reduced to the 4-amino analogue **28**, which is then converted into the 4-azido analogue **29** (Scheme 8).[24] Many 4-azidophenylalanine-containing peptides are prepared by this synthetic pathway. These include substance P,[25] LHRH,[26] and BK.[27]

**Scheme 8** Conversion of a 4-Nitrophenylalanine-Containing Dipeptide into a 4-Azidophenylalanine-Containing Dipeptide[24]




**27** **28** **29**

Catalytic hydrogenation of peptides containing a Phe(4-$NO_2$) residue, to be subsequently converted into Phe(4-$N_3$), that also contain cystine and/or methionine residues, requires extended reaction times. This may lead to desulfuration. Therefore, a Phe(4-$NH_2$) residue is used as a precursor for Phe(4-$N_3$) in the synthesis of AVP to give Phe(4-$N_3$) residues at either position 2 or 3.[28] In this solution-phase synthesis, the 4-amino group in the 4-amino-

*for references see p 147*

phenylalanine moiety is protected by either benzyloxycarbonyl or *tert*-butoxycarbonyl groups.[29] In this example, during the transformation of Phe(4-$NH_2$) into Phe(4-$N_3$) there are no side reactions and the preformed disulfide bridge is stable.[29]

Interestingly, 4-aminophenylalanine allows the easy introduction of iodine into the aromatic ring.[27,30–32] Electrophilic iodination of H-Phe(4-$NH_2$)-OH with iodine monochloride generates the free amino acid H-Phe(3,5-$I_2$,4-$NH_2$)-OH (**30**), or the corresponding residue **33** when carried on a 4-aminophenylalanine-containing peptide. Dehalogenation of **30** and **33** has become a routine procedure for radio-tagging by replacing the two iodine atoms with tritium (see Section 9.3.1.4, Scheme 9).[27,30,31] These tritiated peptides serve as radioligands for radioreceptor assays and to tag the ligand–acceptor photoconjugate. Alternatively, diazotization of the 4-amino group generates the diazonium salt, which can be transformed via a Sandmeyer-type reaction into the corresponding radiolabeled derivatives **32** and **35**. This synthetic route allowed Escher to convert [Sar$^1$,Phe$^8$(4-$NH_2$)]-AT II into a potential imaging agent [Sar$^1$,Phe$^8$(4-$^{131}$I)]AT II.[32]

Table 1 compiles studies in which 4-azidoarene photophores are incorporated into peptides either as congeners or in the form of the 4-azidophenylalanine residue.

**Table 1** Photoreactive Peptides Modified with 4-Azido-Based Photophores[16–18,22,24–28,31,33–91]

| Peptide | Comments on the Synthesis | Ref |
|---|---|---|
| *angiotensin II* | | |
| $N^{\alpha}$-[Bz(2-$NO_2$,5-$N_3$)]-[Asn$^1$,Val$^5$]-AT II | post-SPPS: with Bz(2-$NO_2$,5-$N_3$)-OSu | [33] |
| [Sar$^1$,Phe$^4$(4-$N_3$)]-AT II | post-SPPS: from the corresponding Phe(4-$NO_2$)-containing peptides by reduction and azidation | [34] |
| [Sar$^1$,Phe$^8$(4-$N_3$)]-AT II | | [34] |
| [Sar$^1$,Phe$^4$(4-$N_3$),Phe$^8$(4-$N_3$)]-AT II | | [34] |
| [Sar$^1$,D-Phe$^8$(4-$N_3$)]-AT II | | [35] |
| [Sar$^1$,Val$^5$,D-Phe$^8$(4-$N_3$)]-AT II | | [36] |
| [Sar$^1$,Phe$^4$(3,5-$^3H_2$,4-$N_3$)]-AT II | post-SPPS: with Boc-Phe(4-$NH_2$,3,5-$I_2$)-OH | [34] |
| [Sar$^1$,Phe$^8$(3,5-$^3H_2$,4-$N_3$)]-AT II | | [34] |
| [Phe$^6$(4-$N_3$)]-AT IV | SPPS | [37] |
| *antamanide* | | |
| [Phe$^6$(4-$N_3$)]-antamanide | SPPS: with Boc-Phe(4-$N_3$)-OH | [17] |
| *atrial natriuretic factor (ANF)* | | |
| $N^{\alpha}$-[Bz(4-$N_3$)]-ANF | post-SPPS: with Bz(4-$N_3$)-OSu | [38] |
| *antigenic peptides* | | |
| *immunogenic hen egg white lysozyme (HEL)* | | |
| $N^{\alpha}$-[Bz(2-OH,3-I,4-$N_3$)]-HEL(52–61) | post-SPPS: with Bz(2-OH,3-I,4-$N_3$)-OSu | [39] |
| $N^{\alpha}$-Ac-[Cys$^{62}${*S*-Bz(2-OH,3-I,4-$N_3$)}]-HEL(49–62) | | [39] |
| $N^{\alpha}$-[Bz(2-OH,3-I,4-$N_3$)]-HEL(46–61) | | [39] |
| *K$^d$-restricted plasmodium berghei circumsporozoite peptide (253–260)* | | |
| $N^{\alpha}$-[Bz(2-OH,3-$^{125}$I,4-$N_3$)]-Tyr-Ile-Pro-Ser-Ala-Glu-Lys($N^{\varepsilon}$-biotinyl)-Ile-OH | post-SPPS: with Bz(2-OH,3-I,4-$N_3$)-OSu | [40] |
| $N^{\alpha}$-[Bz(2-OH,3-I,4-$N_3$)]-Tyr-Ile-Pro-Ser-Ala-Glu-Lys-Ile-OH | | [40] |
| $N^{\alpha}$-[Bz(2-OH,3-$^{125}$I,4-$N_3$)]-Tyr-Ile-Pro-Ser-Ala-Glu-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-Ile-OH | post-SPPS: on the $N^{\alpha}$-Fmoc-protected peptides, followed by Bz(4-$N_3$)-OSu, deprotection, and either by Bz(2-$NO_2$,5-$N_3$)-OSu, Bz(2-OH,4-$N_3$)-OSu or Bz(2-OH,3-I,4-$N_3$)-OSu | [41,42] |
| $N^{\alpha}$-[Bz(2-OH,3-$^{125}$I,4-$N_3$)]-Tyr-Ile-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-Ser-Ala-Glu-Lys-Ile-OH | | [41] |
| Dap{$N^{\beta}$-[Bz(2-OH,3-$^{125}$I,4-$N_3$)]}-Tyr-Ile-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-Ser-Ala-Glu-Lys-Ile-OH | | [41] |
| *D$^b$-restricted adenovirus 5 early region 1A peptide (234–243)* | | |
| $N^{\alpha}$-[Bz(2-OH,3-$^{125}$I,4-$N_3$)]-Ser-Gly-Pro-Ser-Asn-Thr-Pro-Pro-Glu-Ile-OH | | [40] |

**Table 1** (cont.)

| Peptide | Comments on the Synthesis | Ref |
|---|---|---|
| *herpes simplex virus type 1 glycoprotein D peptide (292–301) [HSV1-GPD(292–301)]* | | |
| $N^{\alpha}$-[Bz(4-$N_3$)]-Glu-Leu-Ala-Pro-Glu-Asp-Pro-Glu-Asp-Ser-OH | post-SPPS: with Bz(4-$N_3$)-OSu | [43] |
| *human leukocyte antigen-Ai-restricted MAGE-1 peptide (161–169)* | | |
| H-Glu-Ala-Asp-Pro-Thr-Gly-Dap{$N^{\beta}$-[Bz(2-OH,3-$^{125}$I,4-$N_3$)]}-Ser-Tyr-OH | post-SPPS: with Bz(2-OH,4-$N_3$)-OSu on the Tyr-phosphorylated peptide followed by iodination and enzymatic dephosphorylation | [44] |
| *N-terminal dodecapeptide from EDP208 pili* | | |
| $N^{\alpha}$-Ac-Thr-Asp-Leu-Leu-Ala-Gly-Gly-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-Asp-Val-Asp-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-$NH_2$ | post-SPPS: with Bz(4-$N_3$)-OSu | [45] |
| *bradykinin* | | |
| desArg$^9$[Phe$^5$(4-$N_3$)]-BK | post-SPPS: 1. $H_2$, Pd/C; 2. HCl, $NaNO_2$ | [27] |
| desArg$^9$[Phe$^8$(4-$N_3$)]-BK | | [27] |
| [Phe$^5$(4-$N_3$)]-BK | | [27] |
| [Phe$^8$(4-$N_3$)]-BK | | [27] |
| desArg$^9$[Tyr$^5${3-(4-$N_3C_6H_5$)$N_3$}]-BK | post-SPPS: 4-$N_3C_6H_4NH_2$, HCl, $NaNO_2$ | [46] |
| *cholecystokinin* | | |
| $N^{\alpha}$-[Bz(2-OH,4-$N_3$)-NH$(CH_2)_2S_2(CH_2)_2$CO]-[Thr$^{28}$,Nle$^{31}$]-CCK(25–33) | post-SPPS: Bz(4-$N_3$,2-OH)-NH$(CH_2)_2S_2(CH_2)_2CO_2$Su | [47] |
| $N^{\alpha}$-[Bz(2-$NO_2$,5-$N_3$)]-Gly-Trp-Met-Asp-Phe-$NH_2$ | post-SPPS: Bz(2-$NO_2$,5-$N_3$)-OSu | [33] |
| $N^{\alpha}$-[4-$N_3C_6H_4CH_2$CONH$(CH_2CH_2)_n$CO]-D-Tyr-Gly-[(Nle)$_2$]$^{[28,31]}$-CCK(26–33) | post-SPPS: 4-$N_3C_6H_4CH_2$COCl | [48] |
| *corticotropin* | | |
| [Trp$^9$]{2,4-$(O_2N)_2$,5-$N_3C_6H_2S$}]-ACTH (DNAPS-ACTH) | post-SPPS: A: 2,4-$(O_2N)_2$,5-$N_3C_6H_2$SCl or B: 1. 2,4-$(O_2N)_2C_6H_4$SCl, 2. β-ME, 3. $NaN_3$ | [49] |
| [Trp$^9$(2-$O_2$N,4-$N_3C_6H_3S$)]-ACTH (2,4-NAPS-ACTH) | post-SPPS: with 2-$O_2$N,4-$N_3C_6H_3$SCl | [49] |
| [Trp$^9$(2-$O_2$N,5-$N_3C_6H_3S$)]-ACTH (2,5-NAPS-ACTH) | post-SPPS: with 2-$O_2$N,5-$N_3C_6H_3$SCl | [50] |
| H-Gly-Phe(2-$NO_2$,4-$N_3$)-OH | solution: H-Phe(2-$NO_2$,4-$N_3$)-OH | [51] |
| *enkephalin (EK)* | | |
| [D-Ala$^2$,Phe$^4$(4-$N_3$),Met$^5$]-EK | solution: Boc-Phe(4-$N_3$)-OH | [52,53] |
| Tyr-D-Ala-Gly-Phe-Leu-Lys[$N^{\varepsilon}$-(2-$O_2$N,4-$N_3C_6H_3$)]-OH | solution: $N^{\alpha}$-Boc-peptide + 4-F,3-$O_2NC_6H_3N_3$ | [54] |
| Tyr-D-Ala-Gly-Phe-Leu-NH$(CH_2)_2$NH(2-$O_2$N,4-$N_3C_6H_3$) ([D-Ala$^2$,Leu$^5$]-EK deriv.) | | [55] |
| Tyr-D-Ala-Gly-Phe-Leu-NH$(CH_2)_2$NHCO$(CH_2)_2$NH(2-$O_2$N,4-$N_3C_6H_3$) ([D-Ala$^2$,Leu$^5$]-EK deriv.) | | [55] |
| Tyr-D-Ala-Gly-Phe-Leu-NH$(CH_2)_2$NHCOCH$_2$[$^3$H-CH$_2$]NH(2-$O_2$N,4-$N_3C_6H_3$) [D-Ala$^2$,Leu$^5$]-EK | | [55] |
| Tyr-D-Ala-Gly-Phe-Met-Tyr-NH$(CH_2)_2$NH(2-$O_2$N,4-$N_3C_6H_3$) | solution: H-Tyr-NH$(CH_2)_2$NH(2-$O_2$N,4-$N_3C_6H_3$) | [56] |
| Tyr-D-Ala-Gly-NH$(CH_2)_3$(4-$N_3C_6H_4$) | solution: from the corresponding nitro precursors by reduction and azidation | [57] |
| Tyr-D-Ala-Gly-Phe-NH$(CH_2)_3$(4-$N_3C_6H_4$) | | [57] |
| Tyr-D-Ala-Gly-D,L-Phe(3-$N_3$)-Leu-$NH_2$ | | [57] |
| *enzyme substrates and inhibitors* | | |
| Z-Ala-Ala-Phe(3-$N_3$)-OH chymotrypsin inhibitor | solution: with the corresponding N-protected Phe($N_3$)-OH | [16,18] |
| Z-Ala-Ala-Phe(3- or 4-$N_3$)-OH chymotrypsin inhibitor | | [16] |
| Z-Ala-Ala-Phe(2-$NO_2$,3-$N_3$)-OH chymotrypsin inhibitor | | [16] |
| Ac-Phe(4-$N_3$)-OEt chymotrypsin substrate | | [22] |
| Phe(4-$N_3$)-Phe-OH leucine aminopeptidase substrate | | [22] |
| Phe(4-$N_3$)-Phe-Pro-OH leucine aminopeptidase substrate | | [22] |
| Thr(O-tBu)-Phe(4-$N_3$)-Pro-OH leucine aminopeptidase inhibitor | | [22] |
| *FMRFamide receptor ligand* | | |
| $N^{\alpha}$-[Bz(4-$N_3$)]-Phe-Nle-Arg-Phe-$NH_2$ | post-SPPS: with Bz(4-$N_3$)-OSu | [58] |
| *gastrin* | | |
| $N^{\alpha}$-[Bz(4-$N_3$)]-Gly-Trp-Met-Asp-Phe-$NH_2$ | solution: Bz(4-$N_3$)-Gly-OSu | [59] |
| $N^{\alpha}$-[Bz(2-$NO_2$,5-$N_3$)]-Gly-Trp-Met-Asp-Phe-$NH_2$ | solution: Bz(2-$NO_2$,5-$N_3$)-Gly-OSu | [59] |
| $N^{\alpha}$-[Bz(2-$NO_2$,5-$N_3$)]-Gly-Gly-Met-Asp-Phe-$NH_2$ | post-SPPS: Bz(2-$NO_2$,5-$N_3$)-OSu | [33] |

*for references see p 147*

**Table 1** (cont.)

| Peptide | Comments on the Synthesis | Ref |
|---|---|---|
| *glucagon* | | |
| [$Trp^{25}$(2-$O_2N$,4-$N_3C_6H_3S$)]-glucagon | post-SPPS: 2-$O_2N$,4-$N_3C_6H_3SCl$ | [60] |
| [Lys{$N^{\varepsilon}$-(2-$O_2N$,4-$N_3C_6H_3$)}]-glucagon | post-SPPS: 3-$O_2N$,4-$FC_6H_3N_3$ | [61] |
| *growth hormone releasing hormone* | | |
| [$Lys^{12\ and/or\ 21}${$N^{\varepsilon}$-[Bz(2-$NO_2$,5-$N_3$)]}]-GHRH | post-SPPS: Bz[2,5-$(NO_2)_2$]-OSu | [62] |
| *insulin* | | |
| [Bz(4-$N_3$)]$^{125}$I-insulin | post-SPPS: Bz(4-$N_3$)-OSu | [63] |
| [2-$O_2N$,4-$N_3C_6H_3$]$^{125}$I-Insulin | post-SPPS: 3-$O_2N$,4-$FC_6H_3N_3$ | [64,65] |
| B1-[Bz(2-OH,4-$N_3$)]-[B1-biocytin,B2-Lys]-insulin | post-semisynthesis: Bz(2-OH,4-$N_3$)-OSu | [66] |
| Bz(2-OH,4-$N_3$)-NH$(CH_2)_2S_2(CH_2)_2$CO-insulin | post-SPPS: Denny–Jaffe reagent | [67,68] |
| *integrin ($\alpha_4\beta_1$) inhibitor* | | |
| $N^{\alpha}$-[4-(2-$MeC_6H_4$NHCONH)$C_6H_4CH_2$CO]-Lys[$N^{\varepsilon}$-{2-$O_2N$,5-$N_3$-$C_6H_3$CONH$(CH_2)_5$CO}]-Asp-Val-Pro-OH | post-SPPS: Bz(2-$NO_2$,5-$N_3$)-OSu | [69] |
| *luteinizing hormone releasing hormone* | | |
| [D-$Ala^6$,$Phe^7$(4-$N_3$)]-LHRH | post-SPPS: modification of the Phe(4-$NO_2$) residue | [26] |
| [(D-Phe)$_3^{2,3,6}$,$Phe^7$(4-$N_3$)]-LHRH | | [26] |
| [D-$Phe^6$(4-$N_3$)]-LHRH | | [26] |
| [(D-Phe)$_2^{2,6}$,$Phe^7$(4-$N_3$)]-LHRH | | [26] |
| [(Ac)$Pro^1$,D-$Phe^2$(4-$N_3$),(D-2-Nal)$_2^{3,6}$]-LHRH | | [26] |
| [D-$Lys^5$($N^{\varepsilon}$-biotinyl),$Phe^7$(4-$N_3$)]-LHRH | | [26] |
| [D-$Lys^5$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-LHRH | post-SPPS: Bz(4-$N_3$)-OSu | [26] |
| *melanocyte stimulating hormone (MSH)* | | |
| [$Nle^4$,D-$Phe^7$,$Trp^9$(2-$O_2N$,4-$N_3C_6H_3S$)]-α-MSH | post-SPPS: with 2-$O_2N$,4-$N_3C_6H_3SCl$ | [70] |
| $N^{\alpha}$-Ac-[D-$Ala^1$,$Phe^2$(3,5-$^3H_2$,4-$N_3$),$Nva^4$]-α-MSH | solution: Ac-D-Ala-Phe(3,5-$^3H_2$,4-$N_3$)-OH | [24] |
| [$Phe^{13}$(4-$N_3$)]-α-MSH | solution: fragment condensation | [71] |
| $^{125}$I-Bz(4-$N_3$)-β-MSH | succinimidyl 4-azidobenzimidate | [72] |
| *parathyroid hormone (PTH)* | | |
| [(Nle)$_2^{8,18}$,$Lys^{13}${$N^{\varepsilon}$-(2-$O_2N$,4-$N_3$-$C_6H_3$)},$Tyr^{34}$]-bPTH(1–34)-$NH_2$ | post-SPPS: $C_6H_3N_3$ | [73] |
| [(Nle)$_2^{8,18}$,$Lys^{13}${$N^{\varepsilon}$-(2-$O_2N$,3,5-$^3H_2$,4-$N_3C_6H$)},$Tyr^{34}$]-bPTH(1–34)-$NH_2$[a] | post-SPPS: 2,6-$^3H_2$,3-$O_2N$,4-$FC_6HN_3$ | [74] |
| *protein kinase inhibitor (cAMP-dependent, PKI)* | | |
| [$Phe^{10}$(4-$N_3$)]-PKI(6–22) | post-SPPS: from Phe(4-$NO_2$)-containing peptides | [75] |
| [$Phe^{10}$(3,5-$^3H_2$,4-$N_3$)]-PKI(6–22) | post-SPPS: from Phe(4-$NH_2$,3,5-$I_2$)-containing peptides | [75] |
| *somatostatin* | | |
| *c*[-Phe(4-$NH_2$)-Thr-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-Trp-Phe-D-Pro-] | post-SPPS: Bz(4-$N_3$)-OSu | [76] |
| *c*[-Phe[4-NH(1-$^{14}$C)Ac]-Thr-Lys{$N^{\varepsilon}$-[Bz(4-$N_3$)]}-Trp-Phe-D-Pro-] | | [76] |
| *substance P* | | |
| D-Tyr-[$Phe^8$(4-$N_3$)]$Nle^{11}$]-substance P | post-SPPS: from the Phe(4-$NO_2$)-containing peptide | [25] |
| [$Phe^7$(4-$N_3$),$Nle^{11}$]-substance P | | [25] |
| [$Phe^8$(4-$N_3$),$Nle^{11}$]-substance P | | [25] |
| *vasoactive intestinal peptide (VIP)* | | |
| [$Arg^{14}${$N^G$-(4-$N_3$-phenylglyoxal)}]-VIP | post-synthesis: with 4-$N_3C_6H_5COCHO$ | [77] |
| [$Arg^{12}$-{$N^G$-(4-$N_3$-phenylglyoxal)},$Arg^{14}$-{$N^G$-(4-$N_3$-phenylglyoxal)}]-VIP | | [77] |
| $N^{\alpha}$-[Bz(4-$N_3$)]-porcine VIP | post-SPPS: Bz(4-$N_3$)-OSu | [78] |
| *vasopressin* | | |
| [Mpa$^1$,$Pro^4$,$Lys^8$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VP[a] | post-SPPS: succinimidyl 4-azidobenzoate | [79] |
| [Mpa$^1$,$Hyp^4$,$Lys^8$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VP[a] | | [79] |
| [Mpa$^1$,$Hyp^4$,$Lys^8$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VP[a] | | [79] |
| [Mpa$^1$,$Lys^4$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VT[a] | | [79] |
| [Mpa$^1$,$Lys^4$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]},$Hyp^7$]-VT[a] | | [79] |
| [Mpa$^1$,$Dab^4$-{$N^{\gamma}$-[Bz(4-$N_3$)]}]-AVP[a] | | [80] |
| [Mpa$^1$,$Lys^8$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VP[a] | | [80] |
| [Mcha$^1$,D-$Tyr^2$,(Me)$Ala^7$,$Arg^8$,$Lys^9$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VP[a] | | [81] |
| [Mpa$^1$,$Lys^8$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VT[a] | | [82] |
| [Mcha$^1$,D-$Tyr^2$,(Me)$Ala^7$,$Arg^8$,D-$Lys^9$-{$N^{\varepsilon}$-[Bz(4-$N_3$)]}]-VP[a] | | [81] |

**Table 1** (cont.)

| Peptide | Comments on the Synthesis | Ref |
|---|---|---|
| [Mpa$^1$,Lys$^8${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | post-SPPS: methyl 4-azidobenzimidate | [80] |
| [Mpa$^1$,Tyr$^2$(3,5-$^3H_2$),Lys$^8${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | | [83] |
| [Mcha$^1$,D-Tyr$^2$,(Me)Ala$^7$,Lys$^8${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | | [81] |
| [Mcha$^1$,D-Tyr$^2$(3-I),(Me)Ala$^7$,Lys$^8${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | | [81] |
| [Mcha$^1$,D-Tyr$^2$,(Me)Ala$^7$,Arg$^8$,Lys$^9${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | | [81] |
| [Mcha$^1$,D-Tyr$^2$(3-I),(Me)Ala$^7$,Arg$^8$,Lys$^9${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | | [81] |
| [Mcha$^1$,D-Tyr$^2$,(Me)Ala$^7$,Arg$^8$,D-Lys$^9${$N^\varepsilon$-[4-$N_3C_6H_4C$(NH)]}]-VP[a] | | [81] |
| [Mpa$^1$,Dab$^4${$N^\gamma$-[Bz(2-$NO_2$,4-$N_3$)-βAla]}]-AVP[a] | post-SPPS: Bz(2-$NO_2$,4-$N_3$)-βAla-OH, DCC/HOBt, NMM | [80] |
| [Mpa$^1$,Lys$^8${$N^\varepsilon$-[Bz(2-$NO_2$,4-$N_3$)-βAla]}]-VP[a] | | [80] |
| [Mpa$^1$,Phe$^2$(4-$N_3$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | post-SPPS: 1. $H_2$, Pd/C; 2. HCl, $NaNO_2$ | [84] |
| [Depa$^1$,Phe$^2$(4-$N_3$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | | [84] |
| [Ppa$^1$,Phe$^2$(4-$N_3$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | | [84] |
| [Dmpa$^1$,Phe$^2$(4-$N_3$),Val$^4$,Lys$^8$,D-Tyr$^9$(3-$^{125}$I)]-VP[a] | | [84] |
| [Phe$^3$(4-$N_3$)]-VP | | [84] |
| [D-Tyr$^2$,Phe$^3$(4-$N_3$)]-VP | | [84] |
| [Phe$^3$(4-$N_3$)]-AVP | SPPS: from the Phe(4-NHZ)[b]- or Phe(4-$NO_2$)-containing peptides[c] | [84,85] |
| [Mpa$^1$,Phe$^3$(4-$N_3$),Lys$^8$($N^\varepsilon$-Rhod)]-VP[a] | | [86] |
| [Phe$^2$(4-$N_3$)]-AVP | solution: with Boc-Phe(4-NHZ)-OH | [28] |
| $N^\alpha$-Dns-[Phe$^2$(4-$N_3$)]-AVP | | [28] |
| [Phe$^2$,Phe$^3$(4-$N_3$)]-AVP | | [28] |
| [(Asu)$_2^{1,6}$,Phe$^3$(4-$N_3$)]-AVP | | [87] |
| [(Asu)$_2^{1,6}$,Phe$^3$(3,5-$^3H_2$,4-$N_3$)]-AVP | | [88] |
| desamino[(Asu)$_2^{1,6}$,Phe$^2$(4-$N_3$)]-AVP | solution: with Nps-Phe(4-NHBoc)-OSu | [31] |
| desamino[(Asu)$_2^{1,6}$,Phe$^2$(3,5-$^3H_2$,4-$N_3$)]-AVP | | [31] |
| $N^\alpha$-[4-$HOC_6H_4(CH_2)_2CO$]-[Phe$^2$(4-$N_3$)]-AVP | solution: Boc-Cys(Acm)-Phe(4-NHZ)-OH | [89] |
| $N^\alpha$-[4-$HOC_6H_4(CH_2)_2CO$]-[Phe$^2$,Phe$^3$(4-$N_3$)]-AVP | | [89] |
| $N^\alpha$-[4-$N_3C_6H_4(CH_2)_nCO$]-D-Tyr(OMe)-Phe-Gln-Asn-Arg-Pro-Arg-Tyr(3-I)-$NH_2$, n = 0–3 | post-SPPS: 3- or 4-$N_3C_6H_4(CH_2)_nCO_2H$, PyBOP | [90] |
| $N^\alpha$-[3-$N_3C_6H_4(CH_2)_nCO$]-D-Tyr(OMe)-Phe-Gln-Asn-Arg-Pro-Arg-Tyr(3-I)-$NH_2$, n = 0–2 | | [90,91] |
| $N^\alpha$-[3-I,4-$HOC_6H_3(CH_2)_2CO$]-D-Tyr(OMe)-Phe-Gln-Asn-Arg-Pro-Lys[$N^\varepsilon$-{3-$N_3C_6H_4(CH_2)_2CO$}]-$NH_2$ | | [91] |

[a] Mcha = (1-sulfanylcyclohexyl)acetic acid; Mpa = 2-sulfanylpropanoic acid; Dmpa = 3-methyl-3-sulfanylbutanoic acid; Depa = 3-ethyl-3-sulfanylpentanoic acid; Ppa = (1-sulfanylcyclopentyl)acetic acid; Rhod = rhodamine; VT = vasotocin; bPTH = bovine parathyroid hormone; Dab = 2,4-diaminobutyric acid.

[b] See Refs[85,86].

[c] See Ref[84].

**[Phe$^7$(4-$NH_2$)]-Substance P:**[25]

[Phe$^7$(4-$NO_2$)]-substance P (30 mg, 22 μmol) in a microvial was dissolved in 50% AcOH (0.5 mL) and treated with 10% Pd/C (ca. 2 mg). Hydrogenation was carried out at rt at 8 atm for 90 min.[27] The catalyst was filtered off, washed with 10% β-ME (2 mL), and lyophilized twice to give pure [Phe$^7$(4-$NH_2$)]-substance P; yield: 28.4 mg (90%).

**[Phe$^7$(4-$N_3$)]-Substance P:**[25]

[Phe$^7$(4-$NH_2$)]-substance P (12 mg, 8.8 μmol) was dissolved in 100 mM HCl (1.3 mL) at 0 °C. 1 M $NaNO_2$ (25 μL) was added with stirring for 10 min. After a positive iodine–starch test, 1 M soln of sulfamic acid (30 μL) was added in order to destroy any excess of nitrite. After 10 min and a negative iodine–starch test, 1 M $NaN_3$ (20 μL) was added, and 5 min later the pH was increased to 7 by adding solid $NH_4OAc$. The product was lyophilized and subjected to a partition chromatography on Sephadex G-25 in a Teflon tube (6 mm × 50 cm), eluted with the upper phase of the solvent system BuOH/AcOH/$H_2O$ (4:1:5). The product was collected and lyophilized, and [Phe$^7$(4-$N_3$)]-substance P was obtained; yield: 8.0 mg (66%); IR (KBr pellet) $\tilde{\nu}_{max}$: 2110 ($N_3$) cm$^{-1}$.

*for references see p 147*

**H-Cys(*S*-Acm)-Phe(4-$NH_2$)-Phe-Gln-Asn-Cys(*S*-Acm)-Pro-Arg-Gly-$NH_2 \cdot 3CF_3SO_3H$:**[28]
The protected nonapeptide Boc-Cys(*S*-Acm)-Phe(4-NHZ)-Phe-Gln-Asn-Cys(*S*-Acm)-Pro-Arg($N^G$-Tos)-Gly-$NH_2$ (500 mg, 370 μmol) was dissolved in anhyd TFA (10 mL) containing DMS (310 μL, 5 mmol) and anisole (540 μL, 5 mmol). The soln was kept for 10 min at 20 °C and then TfOH (1–1.5 mL) was added dropwise until the soln had a violet color. After 4 h at 20 °C, the soln was poured into anhyd $Et_2O$ (100 mL) and the precipitate was either collected by filtration or by decantation, washed with $Et_2O$, and dried in vacuo over $P_2O_5$. Paper electrophoresis at pH 1.6 showed one product with electrophoretic mobility of 0.64 (ratio of the distance from the origin for the sample and the reference arginine at the same pH).

**[$Phe^2$(4-$NH_2$)]-AVP:**[28]
The partially deblocked nonapeptide H-Cys(*S*-Acm)-Phe(4-$NH_2$)-Phe-Gln-Asn-Cys(*S*-Acm)-Pro-Arg-Gly-$NH_2 \cdot 3CF_3SO_3H$ (370 μmol) was dissolved in AcOH (165 mL) and $H_2O$ (45 mL) and added dropwise over a period of 10 min to a well-stirred soln of $I_2$ (5.4 g, 21 mmol) in AcOH (900 mL) and $H_2O$ (500 mL) containing 1 M HCl (2.25 mL). After stirring for another 10 min, the soln was concentrated by evaporation at 30 °C to a volume of 150 mL, which removed most of the $I_2$. The remaining $I_2$ was reduced by addition of 1 N ascorbic acid soln (5 mL). The soln was desalted on a Sephadex G-10 column (6 × 2.5 cm, elution with 50% AcOH) and fractions containing the peptide were collected and lyophilized. The crude peptide was purified by partition chromatography on a Sephadex G-25 column (fine) (80 × 3 cm) (BuOH/EtOH/pyridine/0.1 M AcOH 4:1:1:7). Aliquots (0.1 mL) were used for Folin–Lowry determination. The fractions eluting with $R_f$ 0.17 containing most of the peptide were pooled, diluted with $H_2O$, and repeatedly lyophilized. The material was further purified by ion-exchange chromatography on a CM-Sephadex C-25 column (30 × 0.8 cm) (linear gradient from 0.05–0.5 M of $NH_4OAc$, pH 6.25). The main product eluted at a salt concentration of 0.13 M. The corresponding fractions were pooled, lyophilized, and chromatographed on a Sephadex G-25 column (260 × 1.8 cm) with 20% AcOH as eluant giving only one peak ($V_e$ = 490 mL). Pure peptide [$Phe^2$(4-$NH_2$)]-AVP was obtained by lyophilization from $H_2O$; yield: 135 mg (30%); $[\alpha]_D^{20}$ −48.6 (*c* 0.3, $H_2O$).

**[$Phe^2$(4-$N_3$)]-AVP:**[28]
A soln of [$Phe^2$(4-$NH_2$)]-AVP (25 mg, 20 μmol) in 0.1 M HCl (1 mL) was treated with 1 M $NaNO_2$ (25 μL) for 15 min at 0 °C (pH was below 2). After degassing the soln in vacuo, 1 M $NaN_3$ soln (25 μL) was added and the mixture stirred for 30 min at 0 °C. The mixture was then chromatographed on a Sephadex G-25 column (260 × 1.8 cm) with 20% AcOH as eluant. The material eluting with $V_e$ = 480 mL was homogeneous on TLC and electrophoresis at pH 1.6; yield: 19 mg (80%); $[\alpha]_D^{20}$ −25.2 (*c* 1, $H_2O$); IR (KBr) $\tilde{\nu}_{max}$: 2120 $cm^{-1}$ ($N_3$).

**[$Phe^2$,$Phe^3$(4-$NH_2$)]-AVP:**[28]
A soln of the protected nonapeptide Boc-Cys(*S*-Acm)-Phe-Phe(4-NHBoc)-Gln-Asn-Cys(*S*-Acm)-Pro-Arg(Tos)-Gly-$NH_2$ (330 mg, 0.21 mmol) was dissolved in anhyd TFA (10 mL) containing DMS (310 μL, 5 mmol) and anisole (540 μL, 5 mmol); the soln was kept for 10 min at 20 °C, and then TFMSA (1–1.5 mL) was added dropwise until the soln had a violet color. After 4 h at 20 °C, AcOH (70 mL) and $H_2O$ (40 mL) were added and this mixture was added dropwise within 8 min into a well-stirred soln of $I_2$ (2.82 g, 11.1 mmol) in AcOH (420 mL) and $H_2O$ (140 mL). After 10 min, the excess of $I_2$ was reduced by addition of 1 M soln of $H_2NNH_2 \cdot xH_2O$ in $H_2O$ (10 mL). The soln was desalted and the mixture was fractionated by partition chromatography on Sephadex G-25 (fine) column (80 × 3 cm) (BuOH/EtOH/pyridine/0.1 M AcOH 4:1:1:7). Aliquots (0.1 mL) were used for Folin–Lowry determination. The fractions eluting with $R_f$ 0.13–0.16 containing most of the peptide were pooled, diluted with $H_2O$ and repeatedly lyophilized. The material was further purified by ion-exchange chromatography on a CM-Sephadex C-25 column (30 × 0.8 cm) (linear gradient from 0.05–0.5 M of $NH_4OAc$, pH 6.25). The product eluting at a salt concentration of 0.24 M was identified as the linear Cys(*S*-Acm)-protected nonapeptide, the material eluting with 0.29 M $NH_4OAc$ was the anticipated [$Phe^2$,$Phe^3$(4-$NH_2$)]-AVP. The corresponding fractions were pooled, lyophilized, and chromatographed on a Sephadex G-25 column (260 × 1.8 cm) (20% AcOH). Pure peptide [$Phe^2$,$Phe^3$(4-$NH_2$)]-AVP was obtained by lyophilization from $H_2O$; yield: 50 mg (20%); $[\alpha]_D^{21.5}$ −30 (*c* 0.23, $H_2O$).

**[$Phe^2$,$Phe^3$(4-$N_3$)]-AVP:**[28]
[$Phe^2$,$Phe^3$(4-$N_3$)]-AVP was prepared from [$Phe^2$,$Phe^3$(4-$NH_2$)]-AVP (12 mg, 10 μmol) by the same procedure described for [$Phe^2$(4-$N_3$)]-AVP; yield: 9.9 mg (83%); $[\alpha]_D^{21.5}$ −34.8 (*c* 0.1, $H_2O$); IR (KBr) $\tilde{\nu}_{max}$: 2120 $cm^{-1}$ ($N_3$).

### 9.3.1.3 Synthesis of Protected 4-Aminophenylalanine Derivatives

Sensitivity to light presents a problem when the 4-azidophenylalanine moiety is introduced in early stages of the synthesis and is then carried through further synthetic procedures. Ideally, the azido function is introduced as late as possible into the synthesis. $N^4$-Protected 4-aminophenylalanine can be introduced as a non-photoreactive precursor for the 4-azido form. The different $pK_a$ values of the α-amino and 4-amino groups (9.1 and 4.25, respectively) allow selective acylation of the 4-amino group at pH 4.0.[29] The following procedures describe the synthesis of $N^4$-, $N^\alpha$- and C-protected derivatives of 4-aminophenylalanine, as well as its activated esters.

**H-Phe(4-NHZ)-OH:**[29]
H-Phe(4-$NH_2$)-OH (18 g, 100 mmol) was dissolved in 1 M sodium citrate buffer (500 mL) which was adjusted to pH 4.6 with 1 M HCl. Benzyl chloroformate (15.74 mL, 110 mmol) was added and the soln was stirred at 20 °C for 2 d. The precipitate was collected by filtration and recrystallized (AcOH/$H_2O$); yield: 18.5 g (59%); mp 250–253 °C; $[\alpha]_D^{20}$ −3 (*c* 1, AcOH).

**H-L-Phe(4-NHZ)-OMe·HCl:**[29]
Thionyl chloride (4 mL, 55 mmol) was added dropwise to MeOH (250 mL) at −15 °C. After 10 min, H-Phe(4-$NH_2$)-OH (15.7 g, 87 mmol) was added, keeping the temperature at −5 °C. After 16 h the solvent was removed. If the reaction was incomplete, the procedure was repeated by the addition of thionyl chloride (1 mL, 14 mmol) in MeOH. The residue was dissolved in a small amount of MeOH, 5% $KHCO_3$ (150 mL) was added, and the free ester was extracted into EtOAc (3 × 200 mL). The organic layers were washed with $H_2O$, dried in vacuo, and concentrated (to about 150 mL). After addition of 2 M HCl in EtOAc until pH 3–4 (spotting on wet indicator paper) was reached and cooling the soln, the precipitated ester hydrochloride was collected by filtration, washed with $Et_2O$, and dried in vacuo; yield: 15.4 g (85%); mp 186 °C; $[\alpha]_D^{20}$ +8 (*c* 1, MeOH).

**H-Phe(4-NHBoc)-OH:**[29]
To a soln of H-L-Phe(4-$NH_2$)-OH (18 g, 100 mmol) in 1 M sodium citrate buffer (250 mL), pH 4.6, was added di-*tert*-butyl dicarbonate (24 g, 110 mmol) dissolved in dioxane (250 mL) (adjusted to pH 4.6 with 1 M HCl) over a period of 30 min. The soln was stirred at 20 °C for 24 h and the resulting precipitate collected by filtration and washed with 10% AcOH (400 mL), to remove any starting material, and with EtOAc (400 mL), to remove some of the bis-Boc derivative of Phe(4-$NH_2$); yield: 28 g (68%); mp 225–226 °C; $[\alpha]_D^{20}$ −2.5 (*c* 1, AcOH).

**H-Phe(4-NHNps)-OH:**[29]
The pH of a stirred suspension of H-L-Phe(4-$NH_2$)-OH (2.25 g, 12 mmol) in $H_2O$ (100 mL) and dioxane (20 mL) was adjusted to 9.6 by the addition of 1 M NaOH. Nps-Cl (2.6 g, 14 mmol) in dioxane (20 mL) was added slowly, keeping the pH constant with 1 M NaOH. After 1.5 h, the soln was concentrated in vacuo to remove the dioxane and then acidified to pH 4 by the addition of 5% $KHSO_4$/5% $K_2SO_4$ (1:2). The resulting precipitate was filtered off, washed with $H_2O$, and dissolved in hot MeOH. Addition of EtOAc followed by concentration in vacuo to remove MeOH resulted in the precipitation of H-Phe(4-NHNps)-OH; yield: 2.2 g (56%); mp 196 °C; $[\alpha]_D^{25}$ −11 (*c* 1, 0.5 M NaOH).

**Boc-Phe(4-NHZ)-OH·DCHA from H-Phe(4-NHZ)-OH:**[29]
A suspension of H-Phe(4-NHZ)-OH (1.2 g, 0.38 mmol) in DMF (20 mL) was treated at 40 °C with di-*tert*-butyl dicarbonate (0.83 g, 0.42 mmol, and after 20 h with an additional 0.16 g, 0.08 mmol). After 24 h, no starting material could be detected by TLC (BuOH/AcOH/$H_2O$ 4:1:1). The soln was diluted with $H_2O$ (20 mL) and was extracted twice with petroleum ether (bp 60–80 °C). Acidification of the aqueous phase with 5% $KHSO_4$/5% $K_2SO_4$ (1:2) to pH 3 gave a precipitate. This was extracted into EtOAc, and the organic layer was dried ($Na_2SO_4$) and concentrated by removing most of the solvent. Addition of DCHA (0.76 mL, 0.42 mmol) generated a precipitate; yield: 1.6 g (93%).

**Boc-Phe(4-NHZ)-OH·DCHA from Boc-Phe(4-$NH_2$)-OH:**[29]
To a stirred soln of Boc-Phe(4-$NH_2$)-OH (5.74 g, 20 mmol) in dioxane (100 mL) and 1 M $NaHCO_3$ (40 mL, 40 mmol) was added benzyl chloroformate (4 mL, 28 mmol) dropwise at 0 °C (maintaining pH 7 by addition of 1 M $NaHCO_3$ soln). After stirring at 0 °C for 3 h and at 20 °C for 20 h and addition of $H_2O$ (100 mL), the dioxane was removed and the suspension acidified with 5% $KHSO_4$ to pH 3. The resulting precipitate was collected by filtration, washed with $H_2O$, dried, and crystallized (EtOAc/hexanes). This was converted into its DCHA salt as described above; yield: 4.9 g (41%); mp 155–156 °C; $[\alpha]_D^{20}$ +22 (*c* 1, MeOH).

*for references see p 147*

**Ddz-Phe(4-NHZ)-OH·DCHA:**[29]
To a suspension of H-Phe(4-NHZ)-OH (9.6 g, 30.5 mmol) in DMF (100 mL) were added TEA (19 mL, 67 mmol) and Ddz-$N_3$ (12.6 g, 47.5 mmol). After stirring at 40 °C for 20 h the turbidity was removed by filtration. The filtrate was diluted with $H_2O$ (100 mL) and extracted twice with $Et_2O$. The aqueous phase was acidified at 0 °C to pH 4 with 5% $KHSO_4$/5% $K_2SO_4$ (1:2) and the resulting oil extracted into $CH_2Cl_2$. The organic phase was washed with $H_2O$, dried ($MgSO_4$), and the solvent was removed in vacuo. The residue was converted into its DCHA salt by dissolving it in $CHCl_3$ and adding DCHA until pH 7–8 (wet indicator paper). The salt (which precipitated as oil) was purified by column chromatography (silica gel, $CHCl_3$/MeOH 117:3); yield: 14.8 g (68%); mp 137–138 °C; $[\alpha]_D^{20}$ +55.4 (*c* 1, MeOH).

**Nps-Phe(4-NHBoc)-OH·DCHA:**[29]
A suspension of H-L-Phe(4-NHBoc)-OH (7 g, 25 mmol) in dioxane (300 mL) and $H_2O$ (100 mL) was treated with 1 M NaOH to adjust and maintain pH at 8–8.5 during the reaction (using a pH-stat). To the stirred mixture, Nps-Cl (5.2 g, 27.5 mmol) was added in portions over a period of 15 min. Additional stirring for 15 min gave a clear soln which was diluted with ice water (1:1) and acidified in the cold with 0.5 M $H_2SO_4$ to pH 4. The resulting oil was separated from the aqueous phase and dissolved in EtOAc. The soln was washed with $H_2O$, dried ($Na_2SO_4$), concentrated, and diluted with $Et_2O$. Addition of DCHA (4.9 mL, 25 mmol) resulted in the precipitation of salt. After 1 h at 4 °C, the solid was collected by filtration and washed with $Et_2O$; yield: 13.95 g (91%); mp 177–178 °C; $[\alpha]_D^{20}$ +30.7 (*c* 2, THF).

**Nps-Phe(4-NHBoc)-ONSu:**[29]
A soln of Nps-Phe(4-NHBoc)-OH·DCHA (6.15 g, 10 mmol) in MeOH (100 mL) and EtOAc (300 mL) was acidified at 0 °C with 5% $KHSO_4$/5% $K_2SO_4$ (1:2) to pH 4. After separating the organic phase it was washed with $H_2O$, dried ($Na_2SO_4$), and the solvent removed in vacuo. The residue and HOSu (1.18 g, 10 mmol) were dissolved in THF (20 mL), treated at 0 °C with DCC (2.06 g, 10 mmol), and stirred at 0 °C for 1 h and at 20 °C for 16 h. The mixture was filtered, the filtrate was evaporated to dryness, and the residue crystallized (iPrOH); yield: 5 g (95%); mp 175–176 °C; $[\alpha]_D^{20}$ +12 (*c* 2, THF).

**Z-Phe(4-NHBoc)-OH·DCHA:**[29]
To a well-stirred suspension of H-Phe(4-NHBoc)-OH (7 g, 25 mmol) and $KHCO_3$ (5.5 g, 55 mmol) in dioxane (200 mL) and $H_2O$ (100 mL) was added benzyl chloroformate (3.55 mL, 27.5 mmol) dropwise at 0 °C. After stirring 3 h at 0 °C a clear soln was obtained, which was then stirred for 48 h at 20 °C. After addition of $H_2O$ (200 mL), the dioxane was evaporated and the suspension acidified with 5% $KHSO_4$/5% $K_2SO_4$ (1:2) to pH 3. The resulting precipitate was extracted into EtOAc, and the organic layer was washed with $H_2O$ and dried ($Na_2SO_4$). Addition of DCHA (4.98 mL, 25 mmol) to the concentrated soln gave an oil, which was dissolved in $CHCl_3$ and purified by column chromatography (silica gel, $CHCl_3$/MeOH 100:3). The material obtained from the column was triturated with $Et_2O$ and crystallized [MeOH/EtOAc/petroleum ether (bp 60–80 °C)]; yield: 6.7 g (45%); mp 161.8 °C; $[\alpha]_D^{20}$ +24.6 (*c* 1, MeOH).

**Ddz-Phe(4-NHBoc)-OH·DCHA:**[29]
To a stirred suspension of H-Phe(4-NHBoc)-OH (2.8 g, 10 mmol) in DMF (30 mL), TEA (2.8 mL, 22 mmol) and Ddz-$N_3$ (4 g, 15 mmol) were added. After 16 h at 40 °C, the soln was kept 1 h at 4 °C, and then filtered and diluted with $H_2O$ (30 mL). The mixture was extracted with $Et_2O$, acidified at 0 °C with 5% $KHSO_4$/5% $K_2SO_4$ (1:2) to pH 4, and extracted with EtOAc. The organic layer was washed to neutrality with ice water, dried ($MgSO_4$), and the solvent removed in vacuo. The remaining oil was dissolved in $Et_2O$, DCHA was added until pH 7–8 was reached (wet indicator paper), and the soln was poured with vigorous stirring into petroleum ether (bp 60–80 °C). The precipitate was collected by filtration and washed with petroleum ether; yield: 4.9 g (71%); mp 81–83 °C; $[\alpha]_D^{20}$ +56 (*c* 1, EtOAc).

### 9.3.1.4 Synthesis of Tritiated 4-Azidophenylalanine and Its Protected Derivatives

Radiolabeled photoaffinity ligands, in which the radio-tag is an integral part of the photophore, are used to simplify the analysis of the ligand–acceptor conjugate.[92] Identification of the photo-insertion site requires that both the photophore and the radio-tag will remain bound to the fragmented acceptor. To this end, introduction of Phe(3,5-$^3H_2$,4-$NH_2$) offers an efficient and supposedly facile solution to this problem.

Although preparation of H-L-Phe(3,5-$^3H_2$,4-$NH_2$)-OH (**31**) and its incorporation into Z-Ala-Ala-Phe(3,5-$^3H_2$,4-$NH_2$)-OH has been reported, its general applicability is very limited.[30] Carrying out on a radioactive precursor the extensive protection schemes necessary to minimize side reactions, and the multiple elongation steps required in a standard peptide

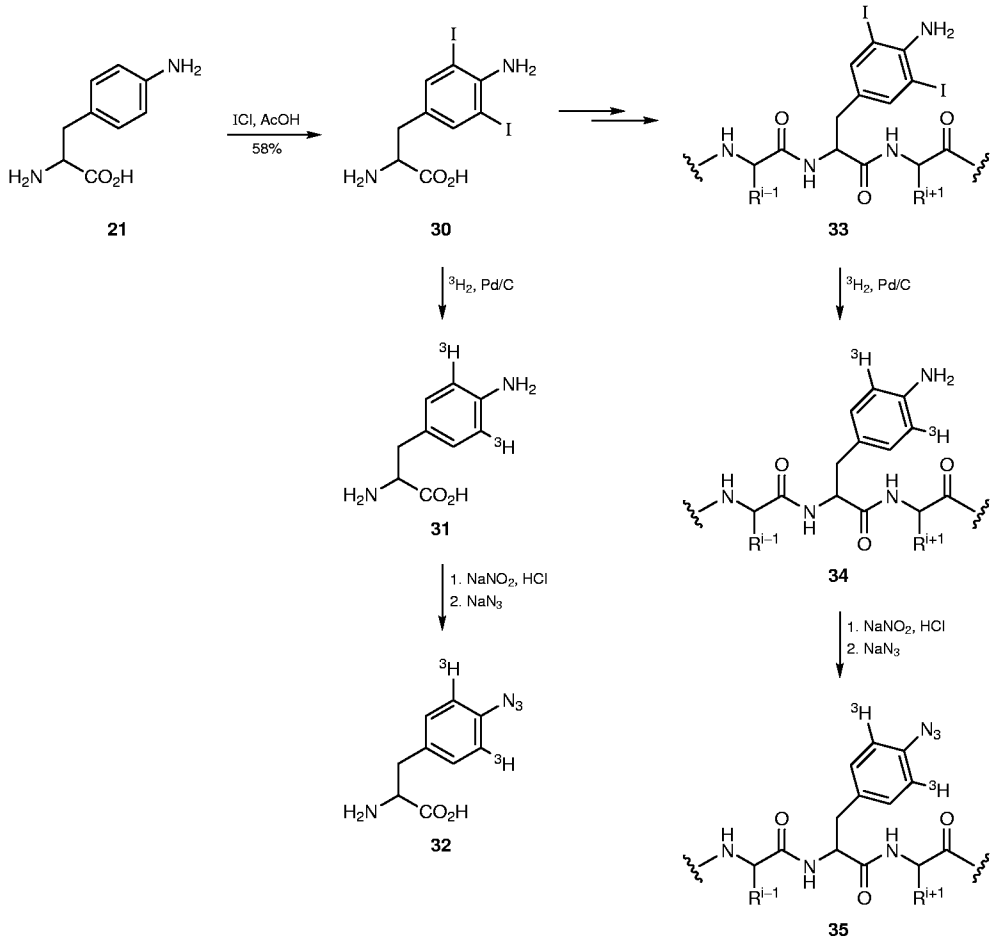
synthesis, presents serious issues. Therefore, the preferred strategy is one in which the designated photoreactive amino acid residue is incorporated in the form of a precursor and transformed into the radioactive photophore only after the completion of the peptide assembly (Scheme 9).

Alternatively, incorporation of Phe(4-NHBoc) into a peptide is followed by deprotection to regenerate the 4-amino group. Iodination generates the Phe(3,5-$I_2$,4-$NH_2$)-containing peptide, which is then converted into the tritium-labeled Phe(3,5-$^3H_2$,4-$NH_2$) for the formation of the [$(Asu)_2^{1,6}$,$Phe^2$(3,5-$^3H_2$,4-$NH_2$),$Arg^8$]-VP.[31]

The steric hindrance on the aromatic ring provides ample protection of the 4-amino group in H-Phe(3,5-$I_2$,4-$NH_2$)-OH (**30**) against acylation and allows it to be used without side-chain protection.[27] The chemical stability of **30** toward a broad spectrum of reagents employed in peptide synthesis (including liq HF) and its facile incorporation into peptides are also attractive features. Last, but not least, the conversion of the diiodo peptide **33** into the ditritiated compound **34** is accomplished by a straightforward, high yielding heterogenous catalytic tritiation.

Disulfide-containing peptides still present a difficulty, in that there is the potential for the catalytic desulfuration of the disulfide bridge during the dehalogenation of the Phe(3,5-$I_2$,4-$NH_2$) residue.[93] Replacement of the disulfide bridge by a hydrocarbon bridge, when tolerated, represents a plausible solution.[31]

**Scheme 9** Approaches for Radio-tagging 4-Azidophenylalanine-Containing Peptides[30,31,34]




*for references see p 147*

**H-Phe(3,5-$I_2$,4-$NH_2$)-OH (30):**[34]
4-Aminophenylalanine (**21**; 5 g, 27.77 mmol) was dissolved in glacial AcOH (100 mL), ICl (17.99 g, 111.0 mmol) was added, and the reaction was stirred at rt for 2 h. The solvent and excess of ICl were removed and the residue was extracted several times with MeOH until a white powder was obtained. Recrystallization (neutralized 2 M $NH_3$) gave **30**; yield: 7.03 g (58%); mp 185 °C (dec).

**Boc-Phe(3,5-$I_2$,4-$NH_2$)-OH:**[34]
H-Phe(3,5-$I_2$,4-$NH_2$)-OH (**30**; 5 g, 11.6 mmol) was treated with $Boc_2O$ (5.1 g, 23.2 mmol). The reaction and purification were carried out according to the standard procedure; yield: 5.74 g (93%); mp 168–171 °C. $[\alpha]_D^{25}$ +60.03 (*c* 1, EtOH).

**[Sar$^1$,Val$^5$,Phe$^8$(3,5-$^3H_2$,4-$NH_2$)]-Angiotensin II:**[94]
A soln of [Sar$^1$,Val$^5$,Phe$^8$(3,5-$I_2$,4-$NH_2$)]-angiotensin II (5 mg, 4 μmol) in DMA (0.5 mL) was treated with 10% Pd/C (2 mg) and tritiated for 30 min at 600 Torr with $^3H_2$. After removal of volatile $^3H_2$ and catalyst, the peptide was resuspended in $H_2O$ and evaporated several times to remove any exchangeable radioactivity. The crude product was filtered through a column of Sephadex G-15 with 0.1 M formic acid; yield: 1.2 mg (30%); $1.93 \times 10^{11}$ dpm or 87.6 mCi. On TLC (with autoradiography), the tritiated peptide was homogeneous and identical with [Sar$^1$,Val$^5$,Phe$^8$(4-$NH_2$)]-angiotensin II; the product was dissolved in $H_2O$ (15 mL), divided into 30 equal aliquots, and stored in liq $N_2$. After 2 years of storage, only minor impurities were detectable.

**[Sar$^1$,Val$^5$,Phe$^8$(3,5-$^3H_2$,4-$N_3$)]-Angiotensin II:**[94]
An aqueous soln of [Sar$^1$,Val$^5$,Phe$^8$(3,5-$^3H_2$,4-$NH_2$)]-angiotensin II (40 μg, 40 nmol, 2.92 mCi or $6.5 \times 10^9$ dpm) was filtered over Sephadex G-15 and the central peak fractions were combined and lyophilized. The resulting pure [Sar$^1$,Val$^5$,Phe$^8$(3,5-$^3H_2$,4-$NH_2$)]-angiotensin II (21 nmol) was dissolved in $H_2O$ (150 μL) at 0 °C, and 1 M HCl (150 μL) was added, followed by 0.1 M $NaNO_2$ (50 μL). After stirring for 10 min, the iodine–starch test was positive and 0.1 M sulfamic acid (300 μL) was added in order to destroy the excess of nitrite. 0.1 M $NaN_3$ (50 μL) was added and the mixture was brought to pH 3.5 with 1 M NaOAc. The product was lyophilized, immediately dissolved in 0.2 M AcOH (100 μL), and filtered over Sephadex G-15. The radioactive peptide fractions were pooled, partially lyophilized, and subjected to partition chromatography on a Sephadex G-25 (fine) column (0.2 × 50 cm, Teflon tube) equilibrated with the lower phase of BuOH/AcOH/$H_2O$ (4:1:5) and eluted by gravity with the upper phase. The peptide fractions were again pooled and their purity assessed by TLC and autoradiography which was identical with [Sar$^1$,Val$^5$,Phe$^8$(4-$N_3$)]-angiotensin II; $R_f$ 0.43 (BuOH/AcOH/$H_2O$ 10:2:3), 0.51 (BuOH/AcOH/$H_2O$/pyridine 15:3:12:10), 0.19 (BuOH/iPrOH/$H_2O$/$ClCH_2CO_2H$ 65:15:20:3). The soln was partially evaporated in vacuo and diluted with 1 mM H-Phe(4-$NH_2$)-OH; yield: 9 μg (43% of the prepurified [Sar$^1$,Val$^5$,Phe$^8$(3,5-$^3H_2$,4-$NH_2$)]-angiotensin II; $1.6 \times 10^9$ dpm).

**[(Asu)$_2$$^{1,6}$,Phe$^2$(3,5-$I_2$,4-$NH_2$),Arg$^9$]-Vasopressin:**[31]
To a soln of [(Asu)$_2$$^{1,6}$,Phe$^2$(4-$NH_2$),Arg$^9$]-vasopressin (14.5 g, 12.5 μmol) in 0.1 M $NH_4OAc$ buffer (6 mL), pH 5.7, at 0 °C was added dropwise under vigorous stirring a soln of 1.35 M ICl in dioxane (25 μL, 35 μmol). After 2 min, the reaction was quenched with aq 0.1 M $H_2NNH_2 \cdot xH_2O$ (0.2 mL) and the mixture loaded on a Sephadex G-25 column (2 × 200 cm). The peptide was eluted with 20% AcOH (430 mL). Detection was accomplished by a positive color reaction with $FeCl_3/K_3Fe(CN)_4/NaAsO_2$ and by the bathochromic shift of the UV [$\lambda_{max}$ (nm) 280 (starting material), 294 (product)]; yield: 11 mg (65%); UV ($M^{-1} \cdot cm^{-1}$ in 20% AcOH): $\varepsilon_{244}$ 5100, $\varepsilon_{294}$ 1850.

## 9.3.1.5 Biochemical, Site-Specific Incorporation of $N^{\varepsilon}$-(5-Azido-2-nitrobenzoyl)lysine into Proteins

Chemical side-chain modification of lysyl-tRNA$^{Lys}$ by a bromoacetyl moiety allows the incorporation of bromoacetyl-lysyl into proteins and the subsequent photo-cross-linking of ribosomes,[95] elongation factor Tu,[96] or the corresponding eukaryotic elongation factors.[97] The incorporation of 5-azido-2-nitrobenzoyl, a potent photophore, onto the ε-amino of [$^3$H]lysyl-tRNA$^{Lys}$ allows the incorporation of $N^{\varepsilon}$-[Bz(2-$NO_2$,5-$N_3$)]-Lys into the nascent chain of preprolactin and its photo-cross-linking to the signal recognition particle.[98] Apparently, the photophore-based cross-linking is more specific than the earlier chemical cross-linking with $N^{\varepsilon}$-bromoacetyl-[$^3$H]lysyl-tRNA$^{Lys}$ and, therefore, allows the mapping of the translation through the ribosome and translocation through the endoplasmic reticulum.[99]

**$N^{\varepsilon}$-[Bz(2-$NO_2$,5-$N_3$)]-Lys-$tRNA^{Lys}$:**[98]
Modification of [$^3$H]lysyl-$tRNA^{Lys}$ was carried out in a dark room illuminated with a Na vapor lamp equipped with UV cutoff filters.[96,97] Bz(2-$NO_2$,5-$N_3$)-OSu (8.8 mg) dissolved in DMSO (1.75 mL) was added with stirring at 0 °C to a soln of [$^3$H]lysyl-$tRNA^{Lys}$ (0.75 mL, 55–140 $A_{260}$ units) in 50 mM potassium phosphate (pH 7.0); immediately thereafter 4 M KOH (15 µL) was added to the reaction vessel. After 20 s, 4 M AcOH (15 µL) was added, followed by 2.0 M KOAc (2.5 mL, pH 5.0) and cold EtOH (15 mL). The modified [$^3$H]lysyl-$tRNA^{Lys}$ was purified on a benzoylated DEAE–cellulose (Boeringer Mannheim) column; the material that was eluted in buffer A (1.0 M NaCl, 0.01 M NaOAc, pH 4.5/0.01 M $MgCl_2$) was discarded. The $N^{\varepsilon}$-[Bz(2-$NO_2$,5-$N_3$)]-Lys-tRNA was then eluted in buffer A containing 25% EtOH. After concentration and dialysis into 1 mM KOAc (pH 5.0), the $N^{\varepsilon}$-[Bz(2-$NO_2$,5-$N_3$)]-Lys-tRNA was stored at −80 °C in the dark. The extent of modification of [$^3$H]lysyl-$tRNA^{Lys}$ was determined by paper electrophoresis at pH 3.6 for 150 min at 28 $V \cdot cm^{-1}$.

## 9.3.2 Nitrophenyl-Containing Peptides

4-Nitrophenylalanine was originally used as a non-photolabile precursor for the photoreactive 4-azidophenylalanine,[21] to be converted into the light-labile compound at the completion of peptide assembly. In 1975, Escher and Schwyzer reported that it also acts as a photophore.[16] Its advantages as a photophore are due to its stability under ambient light conditions and in the presence of liquid HF, which are conditions used in many solid-phase peptide syntheses.

Photodecomposition of a 4-nitrophenyl moiety is a multiphoton process: absorption of one photon leads to the singlet, which relaxes immediately into a long-lived triplet state. Capturing another photon transforms the singlet through a dissociative decay process into reactive intermediates, probably radicals, which react with nucleophilic sites. In the absence of a reactive species, these reactive intermediates will relax into the initial ground state.[23]

The low level of photo-incorporation into BSA of peptides containing Phe(4-$NO_2$) at 350 nm suggests that this method may not be of general applicability.[43]

Table 2 compiles studies in which 4-nitroaryl-containing photophores are incorporated into peptides either as congeners or in the form of the amino acid residue, 4-nitrophenylalanine.

*for references see p 147*

**Table 2** Photoreactive Peptides Modified with 4-Nitroaryl-Based Photophores[16,25,27,34,43,48,84,100–108]

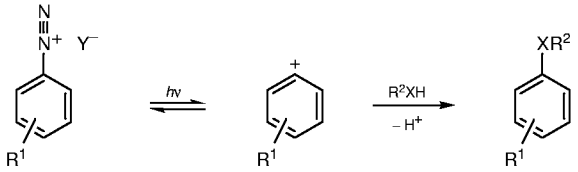
| Peptide | Comments on the Synthesis | Ref |
|---|---|---|
| *angiotensin II* | | |
| [Sar$^1$,Phe$^4$(4-$NO_2$)]-AT II | SPPS: Boc chemistry | [34] |
| [Sar$^1$,Phe$^8$(4-$NO_2$)]-AT II | | [34] |
| [Sar$^1$, Phe$^4$(4-$NO_2$),Phe$^8$(4-$NO_2$)]-AT II | | [34] |
| [Asn$^1$,Phe$^4$(4-$NO_2$)]-AT II | | [100] |
| *bradykinin* | | |
| desArg$^9$-[Phe$^5$(4-$NO_2$)]-BK | SPPS: Boc chemistry | [27] |
| desArg$^9$-[Phe$^8$(4-$NO_2$)]-BK | | [27] |
| [Phe$^5$(4-$NO_2$)]-BK | | [27] |
| [Phe$^8$(4-$NO_2$)]-BK | | [27] |
| *cholecystokinin* | | |
| D-Tyr-Gly-[$(Nle)_2^{28,31}$,Phe$^{33}$(4-$NO_2$)]-CCK(26–33) | solid phase: | [101] |
| D-Tyr-[$^{125}$I]-[$(Nle)_2^{28,31}$,Phe$^{33}$(4-$NO_2$)]-CCK(26–33) | | [102] |
| desamino-Tyr-Gly[$(Nle)_2^{28,31}$,Phe$^{33}$(4-$NO_2$)]-CCK(26–33) | | [103] |
| D-Tyr-Gly[$(Nle)_2^{28,31}$,Trp$^{30}$(6-$NO_2$)]-CCK(26–33) | solid phase: Fmoc/DIC–HOBt | [104] |
| 4-$O_2NC_6H_4CO$-NH$(CH_2CH_2O)_n$CO-D-Tyr(3-$^{125}$I)-Gly-[$(Nle)_2^{28,31}$]-CCK(26–33) n = 2, 4, 7, 10 | post-synthesis: with the corresponding acid chloride | [48] |
| *chymotrypsin inhibitors* | | |
| Z-Ala-Ala(β-$^3$H)-Phe(4-$NO_2$)-OH | solution chemistry | [16] |
| *enkephalin (EK)* | | |
| [Phe$^4$(4-$NO_2$),Leu$^5$]-EK | solution: Boc chemistry | [105] |
| *herpes simplex virus type 1 glycoprotein D peptide (292–301) [(HSV1-GPD(292–301)]* | | |
| Ac-Phe(4-$NO_2$)-Glu-Leu-Ala-Pro-Glu-Asp-Pro-Glu-Asp-Ser-OH | solid phase: Boc/DCC/HF | [43] |
| *secretin* | | |
| [Tyr$^{10}$(3-$^{125}$I),Phe$^{22}$(4-$NO_2$)]-rSecretin | solid phase: Boc chemistry | [106] |
| [Phe$^6$(4-$NO_2$),Tyr$^{10}$(3-$^{125}$I)]-rSecretin | | [106] |
| *substance P* | | |
| [Phe$^7$(4-$NO_2$),Nle$^{11}$]-substance P | solid phase: Boc/Fmoc/DCC, Boc/DCC | [25] |
| [Phe$^8$(4-$NO_2$),Nle$^{11}$]-substance P | | [25] |
| D-Tyr-[Phe$^8$(4-$NO_2$),Nle$^{11}$]-substance P | | [107] |
| *vasopressin* | | |
| [Phe$^3$(4-$NO_2$)]-VP | SPPS: Boc chemistry | [84] |
| [D-Tyr$^2$,Phe$^3$(4-$NO_2$)]-VP | | [84] |
| [Mpa$^1$,Phe$^2$(4-$NO_2$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | | [84] |
| [Dmpa$^1$,Phe$^2$(4-$NO_2$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | | [84] |
| [Depa$^1$,Phe$^2$(4-$NO_2$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | | [84] |
| [Ppa$^1$,Phe$^2$(4-$NO_2$),Val$^4$,Lys$^8$,D-Tyr$^9$]-VP[a] | | [84] |
| [Depa$^1$,Phe$^2$(4-$NO_2$),Val$^4$,Lys$^8$,D-Tyr$^9$(3-$^{125}$I)]-VP[a] | | [108] |

[a] Mpa = 2-sulfanylpropanoic acid; Dmpa = 3-methyl-3-sulfanylbutanoic acid; Depa = 3-ethyl-3-sulfanylpentanoic acid; Ppa = (1-sulfanylcyclopentyl)acetic acid.

### 9.3.3 Peptides Containing Arenediazoniums

Arenediazonium salts substituted with electron-donating groups are good candidates for photoaffinity probes.[109] Photoactivation of these salts leads to the corresponding aryl cations (Scheme 10), which can be stabilized by electron-donating groups at the 2- and 4-positions.[109]

**Scheme 10** Photoactivation Mechanism of Arenediazonium Salts[109]




Arenediazonium-containing reagents are used to photolabel acceptor sites, which bind cationic groups such as acetylcholine and nicotine in acetylcholine sterase and the nicotinic receptor, respectively.[110–112]

Several angiotensin II (AT II) analogues substituted with Phe(4-$N_2^+$) (Dip) in positions 4 and/or 8 have been prepared from their Phe(4-$NO_2$)-containing precursors and have been biologically characterized.[34] The highly positively charged diazonium compound is able under physiological pH to undergo azo coupling to Tyr, His, and other nucleophilic residues in proteins; thus, cross-linking with the acceptor macromolecule.[113] Although all Dip-containing AT II analogues behave like reversible agonists, and therefore it is suggested that they do not act as affinity labels, the very low bioactivity of [$Dip^4$]- and [$(Dip)_2^{4,8}$]-AT II analogues is attributed to the spontaneous decay of the Dip to the corresponding Tyr under physiological pH.[114] Nevertheless, this does not explain the much higher potency of [$Dip^8$]-AT II.[114] Both spontaneous decay and nonspecific photo-independent cross-linking might result in the lack of photo-inactivation of bradykinin receptors by Dip-containing analogues.[115]

In summary, the inherent chemical susceptibility of the arenediazonium-based photoaffinity ligands does not make them a popular tool for routine photo-cross-linking studies. Examples of arenediazonium-based photophores that have been incoporated into peptides are given in Table 3.

**Table 3** Photoreactive Peptides Modified with Benzenediazonium-Based Photophores[25,27,34,115]

| Peptide | Comments on the Synthesis | Ref |
|---|---|---|
| *angiotensin II* | | |
| [$Sar^1$,$Phe^4$(4-$N_2^+$)]-AT II | post-SPPS: Boc chemistry, from the corresponding Phe(4-$NO_2$)-containing peptides | [34] |
| [$Sar^1$,$Phe^8$(4-$N_2^+$)]-AT II | | [34] |
| [$Sar^1$,$Phe^4$(4-$N_2^+$),$Phe^8$(4-$N_2^+$)]-AT II | | [34] |
| *bradykinin* | | |
| desArg$^9$[$Phe^5$(4-$N_2^+$)]-BK | post-SPPS: Boc chemistry, from the corresponding Phe(4-$NO_2$)-containing peptides | [27,115] |
| desArg$^9$[$Phe^8$(4-$N_2^+$)]-BK | | [27,115] |
| [$Phe^5$(4-$N_2^+$)]-BK | | [27,115] |
| [$Phe^8$(4-$N_2^+$)]-BK | | [27,115] |
| *substance P* | | |
| [$Phe^7$(4-$N_2^+$),$Nle^{11}$]-substance P | post-SPPS: Boc chemistry, from the corresponding Phe(4-$NO_2$)-containing peptides | [25] |
| [$Phe^8$(4-$N_2^+$),$Nle^{11}$]-substance P | | [25] |

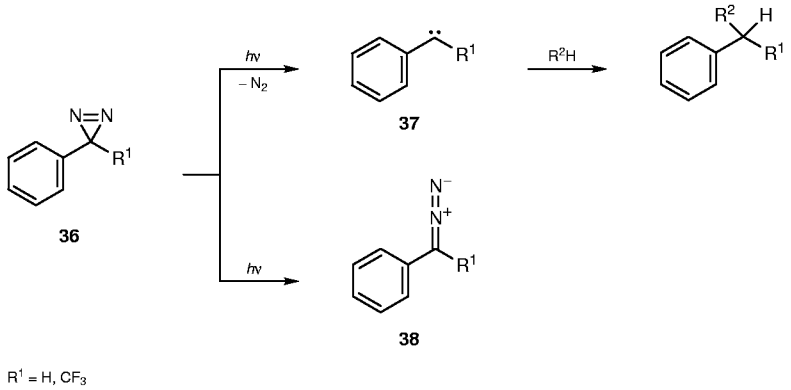
## 9.3.4 [3-(Trifluoromethyl)-3*H*-diazirin-3-yl]phenyl-Containing Peptides

[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]arenes, e.g. **36** ($R^1 = CF_3$), were introduced as photophores by Brunner et al.[116] to provide a more favorable carbene than the one produced from diazoesters[117] or from aryldiazirines.[118] Both 3*H*-diazirin-3-ylarenes[118–120] and the [(3-trifluoromethyl)-3*H*-diazirin-3-yl]arenes (e.g., **36**, $R^1$ = H, $CF_3$, respectively)[116] are more stable in acid than most diazoesters, and photolyze at around 360 nm to generate highly reactive carbenes such as **37**. However, these diazirines can photo-isomerize to some extent to the

*for references see p 147*

corresponding (diazomethyl)arenes, e.g. **38**.[12] Nevertheless, in acid solution (1-diazo-2,2,2-trifluoroethyl)benzene (**38**, $R^1 = CF_3$) is more stable than (diazomethyl)benzene (**38**, $R^1 = H$). Protonation of the latter generates undesired side products obtained through nucleophilic-type reactions, thus leading to photochemically irrelevant labeling.[116] Therefore, (1-diazo-2,2,2-trifluoroethyl)benzene (**38**, $R^1 = CF_3$), generated photochemically from **36** ($R^1 = CF_3$), may be considered inert under the photolabeling conditions (Scheme 11).[7]

**Scheme 11** Photolysis Reactions of [3-(Trifluoromethyl)-3*H*-diazirin-3-yl]arenes and 3*H*-Diazirin-3-ylarenes[7]

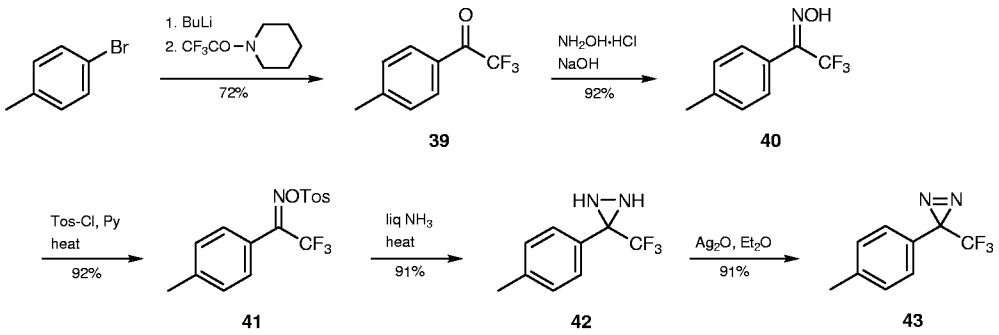


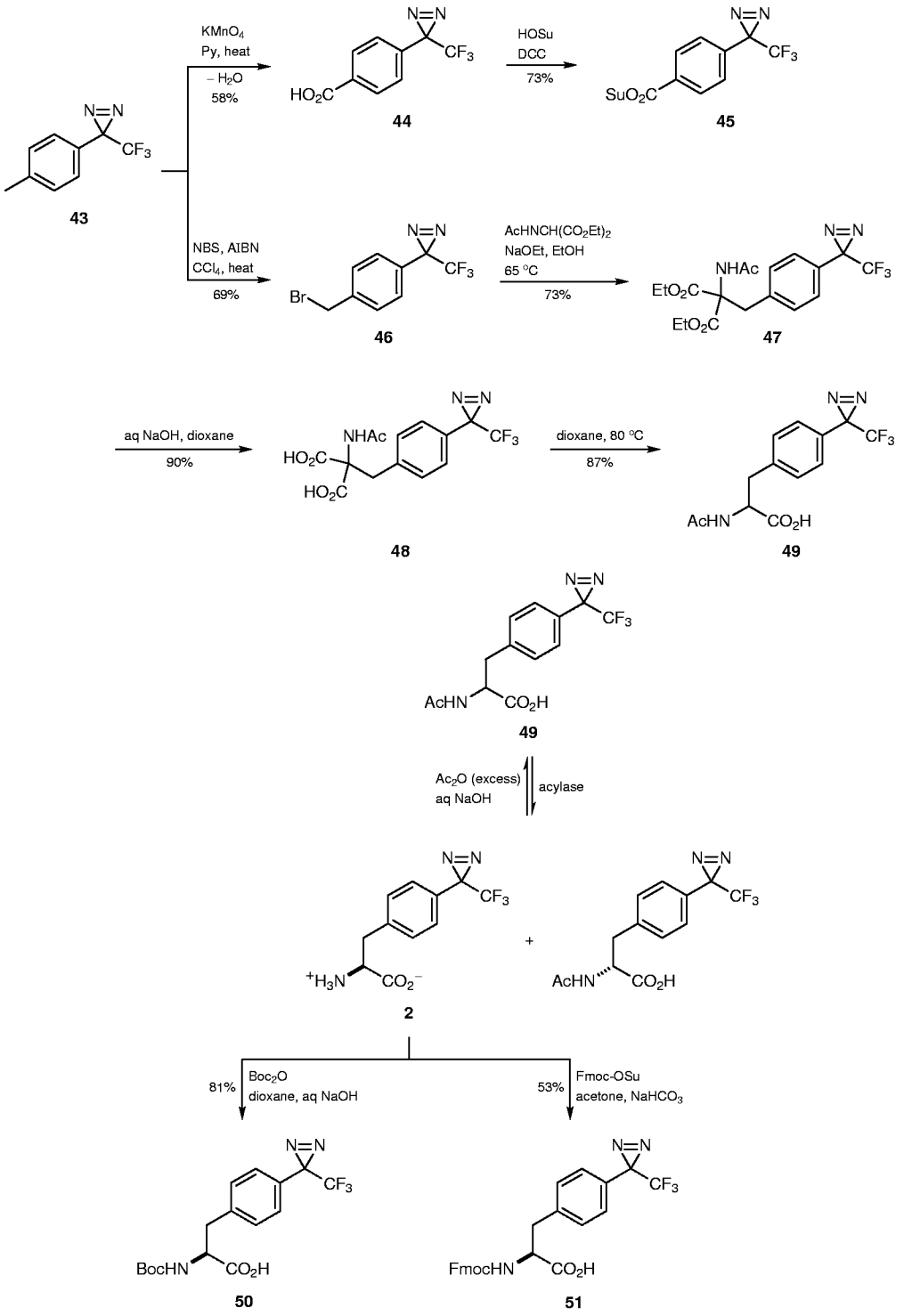

The chemical stability of the [3-(trifluoromethyl)-3*H*-diazirin-3-yl]arenes is remarkable. This type of photophore is stable to strong bases (such as KOH), oxidizing reagents (such as NBS and $KMnO_4$), heat (refluxing dioxane, 80 °C), strong acids (such as 50% TFA), acid anhydrides (such as acetic anhydride and di-*tert*-butyl dicarbonate), and mild reducing agents (such as $NaBH_4$ and thiols).[121,122] Nevertheless, the electrophilic nature of the corresponding highly reactive carbene results in high reactivity toward water. Therefore, [3-(trifluoromethyl)-3*H*-diazirin-3-yl]arenes could be the photophores of choice when accessibility to the binding pocket is not perturbed by water molecules.[123]

### 9.3.4.1 Synthesis of N-Protected 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]-L-phenylalanine

The synthesis reported by Nassal is based on the $C^{\alpha}$-alkylation of diethyl 2-(acetylamino)malonate by 3-[4-(bromomethyl)phenyl]-3-(trifluoromethyl)-3*H*-diazirine (**46**) (Scheme 12).[122] The precursor for the diazirine-containing alkylating agent is 2,2,2-trifluoro-1-(4-tolyl)ethanone (**39**).[124]

**Scheme 12** Synthesis and Optical Resolution of $N^{\alpha}$-Boc- and $N^{\alpha}$-Fmoc-Protected 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]-L-phenylalanine According to Nassal[122,124,129]

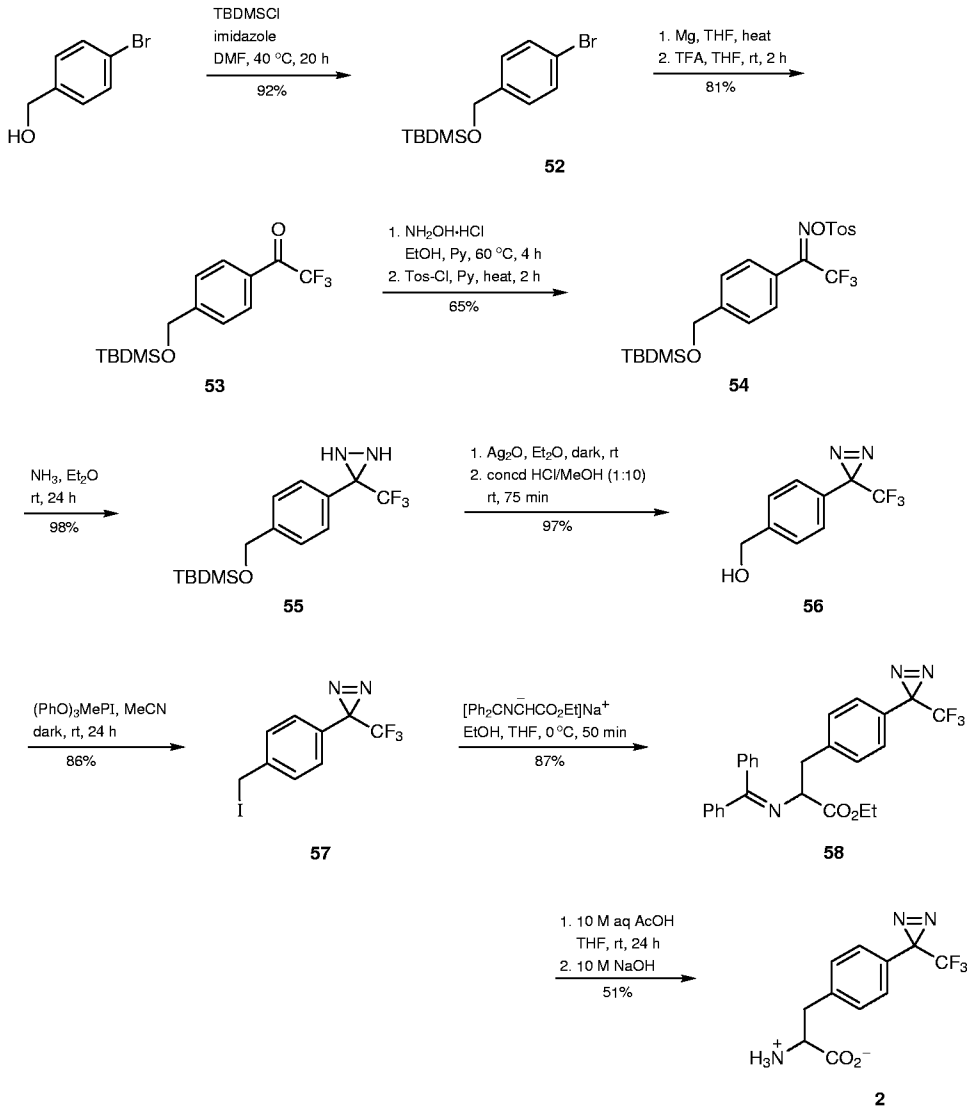





On the other hand, the synthesis of Shih and Bayley is based on the $C^{\alpha}$-alkylation of ethyl *N*-(diphenylmethylene)glycinate by 3-[4-(iodomethyl)phenyl]-3-(trifluoromethyl)-3*H*-diazirine (**57**) (Scheme 13). The precursor for the diazirine-containing alkylating reagent is the O-protected benzyl alcohol **53**.[125]

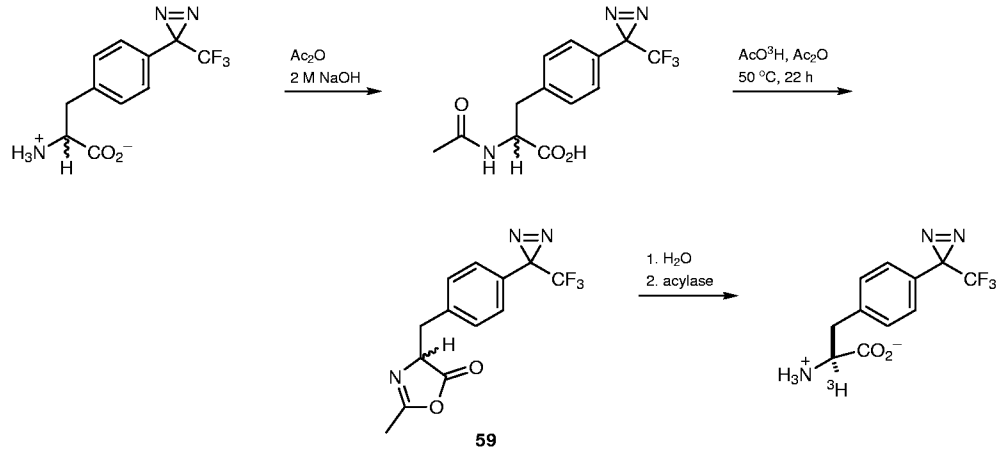
*for references see p 147*

**Scheme 13** Synthesis of 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]-D,L-phenylalanine According to Shih and Bayley[125]




The incorporation of tritium into the photophore H-Phe(4-Tmd)-OH [Tmd = 3-(trifluoromethyl)-3*H*-diazirin-3-yl] is of great importance since it introduces a radioactive tag. This will allow the monitoring of the separation of the photoconjugate and its photoconjugated fragments without the risk of dissociating the tag from the cross-linking site.[125] This tritiation is accomplished by the in situ formation of oxazolone **59** from Ac-Phe(4-Tmd)-OH in the presence of $AcO^3H/Ac_2O$ and is accompanied by tritium exchange with the hydrogen in the ring (Scheme 14).

**Scheme 14** Oxazolone-Mediated Tritiation of 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]phenylalanine[125]




59

**2,2,2-Trifluoro-1-(4-tolyl)ethanone (39):**[124]
To a stirred soln of 4-bromotoluene (3.42 g, 20 mmol) in dry $Et_2O$ (100 mL) was added dropwise under argon 15% BuLi in hexanes (1.1 equiv) at −30 °C. The mixture was allowed to warm to 0 °C over a period of 2 h and was then cooled down again to −50 °C. *N*-(Trifluoroacetyl)piperidine[126,127] (3.62 g, 20 mmol) dissolved in dry $Et_2O$ (20 mL) was added. After 3 h at −50 °C, the cooling bath was removed and the mixture was hydrolyzed with sat. aq $NH_4Cl$. The solvent was removed in vacuo. The crude oily residue was purified by column chromatography (silica gel, petroleum ether/$CH_2Cl_2$ 2:1). The ketone **39** was obtained as a colorless oil; yield: 2.7 g (72%).

**2,2,2-Trifluoro-1-(4-tolyl)ethanone Oxime (40):**[124]
To a stirred mixture of $NH_2O \cdot HCl$ (2.77 g, 39.9 mmol) and NaOH (1.60 g, 39.9 mmol) in boiling abs EtOH was added a soln of ketone **39** (2.5 g, 13.29 mmol) in EtOH (20 mL). After refluxing for 16 h, the solvent was removed in vacuo, and the residue was partitioned between $Et_2O$ and $H_2O$. The organic layer was washed with 0.01 M aq HCl (× 3) followed by $H_2O$ (× 3), dried ($Na_2SO_4$), and the solvent was removed under reduced pressure. The residue was crystallized (hexanes, −20 °C); yield: 92%; mp 63–86 °C.

**2,2,2-Trifluoro-1-(4-tolyl)ethanone *O*-Tosyloxime (41):**[124]
Oxime **40** (1.8 g, 8.86 mmol), dissolved in dry pyridine (30 mL), was refluxed with Tos-Cl (2.53 g, 13.29 mmol). After 3 h, the solvent was removed in vacuo, and the residue was subjected to column chromatography (silica gel, $CH_2Cl_2$/petroleum ether 3:1). Crystallization ($Et_2O$/hexanes) afforded colorless needles; yield: 2.90 g (92%); mp 104–105 °C.

**3-(4-Tolyl)-3-(trifluoromethyl)diaziridine (42):**[124]
*O*-Tosyloxime **41** (1.7 g, 4.76 mmol) in precooled $Et_2O$ (10 mL) was added to liq $NH_3$ (2 mL) in a thick-walled screw glass tube. After stirring the mixture at rt for ca. 12 h in the closed tube, the soln was cooled down to −40 °C, poured into a beaker in order to evaporate the ammonia, and subsequently concentrated under reduced pressure. The residue was partitioned between $Et_2O$ and $H_2O$; the organic layer was dried ($Na_2SO_4$), the solvent removed in vacuo, and the residue purified by column chromatography (silica gel, $CHCl_3$/EtOAc 19:1). Crystallization (hexanes, −20 °C) afforded the diaziridine **42** as colorless needles; yield: 0.88 g (91%); mp 45–46 °C.

**3-(4-Tolyl)-3-(trifluoromethyl)-3*H*-diazirine (43):**[124]
Diaziridine **42** (600 mg, 2.97 mmol), dissolved in dry $Et_2O$ (30 mL), was stirred with freshly prepared $Ag_2O$ (1.4 g). The progress of the slightly exothermic reaction was followed by TLC ($CHCl_3$). After 1 h, Ag and $Ag_2O$ were filtered off, washed with $Et_2O$, and the combined filtrates concentrated in vacuo. The residue crystallized upon storage at −20 °C, but melted while warming up to rt; yield: 540 mg (91%).

**4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]benzoic Acid (44):**[124]
Diazirine **43** (1.0 g, 4.99 mmol) was dissolved in pyridine (20 mL) and $H_2O$ (maximum amount that allows soln to stay homogenous). Powdered $KMnO_4$ (4 equiv) was added and the soln stirred at 50 °C. After 20 h, the mixture was diluted with $H_2O$ (150 mL), acidified with 0.5 M $H_2SO_4$ to pH 2–3, and aq

*for references see p 147*

$NaHSO_4$ was added until the precipitated $MnO_2$ had dissolved. The colorless soln was extracted with $Et_2O$ (3 × 100 mL) and washed with $H_2O$. The product was extracted into the aqueous layer with 0.1 M KOH. The aqueous phase was washed with $Et_2O$, acidified, and again extracted with $Et_2O$. The crude acid **44** was purified on a Sephadex LH-20 column with MeOH. The product was crystallized ($Et_2O$/hexanes or MeOH/$H_2O$); yield: 660 mg (58%); mp 123–125 °C (dec).

**Succinimidyl 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]benzoate (45):**[124]
Acid **44** (230 mg, 1 mmol) was mixed with HOSu (115 mg, 1 mmol) in dioxane/EtOAc (1:1, 20 mL), cooled to −10 °C, and a soln of DCC (206 mg, 1 mmol) in dioxane (10 mL) was added. After stirring at rt for 6 h, the precipitated DCU was removed by filtration and the filtrate concentrated in vacuo. The residue was crystallized (iPrOH); yield: 240 mg (73%); mp 106 °C.

**3-[4-(Bromomethyl)phenyl]-3-(trifluoromethyl)-3*H*-diazirine (46):**[122]
To a stirred soln of diazirine **43** (5.0 g, 24.98 mmol) in $CCl_4$ (100 mL) was added powdered NBS (4.44 g, 24.98 mmol) and the mixture warmed to 70 °C. AIBN (100 mg) was added and the temperature was gradually raised to reflux. After 20 min, the mixture was cooled, the succinimide separated by filtration, and the filtrate concentrated in vacuo. Purification of the residue on a column (silica gel, petroleum ether/$CH_2Cl_2$ 20:1) afforded the product as a colorless oil, which crystallized at −30 °C, but melted on warming to rt; yield: 4.8 g (69%).

**Diethyl 2-(Acetylamino)-2-{4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]benzyl}malonate (47):**[122]

**CAUTION:** *Sodium reacts violently or explosively with a wide range of substances! Appropriate safety precautions and procedures should be adopted when handling and disposing of sodium metal.*

Diethyl 2-(acetylamino)malonate (3.85 g, 17.74 mmol) was added portionwise to Na (408 mg, 17.74 mmol) in absolute EtOH (30 mL). A soln of diazirine **46** (4.5 g, 16.1 mmol) in EtOH (10 mL) was added and the mixture was heated to 65 °C for 2 h and then concentrated in vacuo. The residue was partially purified on a column (silica gel, $CHCl_3$/EtOH 20:1). The fractions containing the product were concentrated in vacuo and the residue was crystallized (EtOAc/$Et_2O$/petroleum ether); yield: 3.33 g. Rechromatography of the mother liquor followed by crystallization gave further product (1.53 g; total yield: 73%; mp 110–111 °C.

**2-(Acetylamino)-2-{4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]benzyl}malonic Acid (48):**[122]
Diester **47** (2.6 g, 6.26 mmol) was dissolved in dioxane (20 mL) and stirred with 2 M aq NaOH (12.5 mL) for 16 h. Most of the dioxane was removed in vacuo, the residual alkaline aqueous phase was extracted with EtOAc/$Et_2O$ (1:1, × 3), acidified with 1 M HCl to pH 2–3, and extracted with EtOAc. The organic layer was washed with brine, dried ($Na_2SO_4$), and concentrated in vacuo. The residue was crystallized (EtOAc/petroleum ether); yield: 2.03 g (90%); mp 113–114 °C.

***N*-Acetyl-4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]-D,L-phenylalanine [Ac-D,L-Phe(4-Tmd)-OH] (49):**[122]
A stirred soln of acid **48** (3.0 g, 8.35 mmol) in dioxane (30 mL) was warmed for 1 h in a preheated oil bath to 80 °C. The residue obtained after solvent removal in vacuo was partitioned between 5% aq $NaHCO_3$ and EtOAc/$Et_2O$ (1:1). The aqueous phase was acidified with 1 M HCl to pH 2–3 and extracted with EtOAc (× 3). The combined organic extracts were washed with brine, dried ($Na_2SO_4$), and concentrated in vacuo, and the residue crystallized (EtOAc/petroleum ether); yield: 2.3 g (87%); mp 131–132 °C.

**4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]-L-phenylalanine [H-L-Phe(4-Tmd)-OH] (2):**[122]
Ac-D,L-Phe(4-Tmd)-OH (**49**; 1.0 g, 3.17 mmol), neutralized with 1 M NaOH, was dissolved in 20 mM phosphate buffer (80 mL, pH 6.6, $5 \times 10^{-4}$ M $CaCl_2$) and incubated at 37 °C with acylase from *Aspergillus* (80 mg, specific activity 66 $U \cdot mg^{-1}$). The mixture, acidified after 8 h, was extracted with EtOAc and the aqueous phase was concentrated in vacuo. The residue was dissolved in a mixture of MeOH/aq TFA and subjected to a Lobar RP-8 column (size B; Merck, Darmstadt). Elution with 0.1% aq TFA/MeOH (9:1) removed a brown impurity derived from the enzyme preparation. Subsequently, the free amino acid was eluted using a linear gradient by changing the above solvent mixture from a 9:1 to a 1:9 ratio. The fractions containing the product were pooled and concentrated in vacuo, and the residue was crystallized ($H_2O$/MeOH); yield: 325 mg (75%); $[\alpha]_D$ −15.8 (*c* 0.176, $H_2O$).

***N*-*tert*-Butoxycarbonyl-4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]-L-phenylalanine [Boc-L-Phe(4-Tmd)-OH)] (50):**[122]
Introduction of the Boc group on H-L-Phe(4-Tmd)-OH (**2**) was performed with $Boc_2O$ in dioxane/aq NaOH. *N*-Boc-L-Phe(4-Tmd)-OH (**50**) was crystallized ($Et_2O$/petroleum ether) at −30 °C; yield: 81%; mp 101–102 °C (dec with foaming); $[\alpha]_D$ +18.6 (*c* 3.55, $CHCl_3$).

***N*-(9-Fluorenylmethoxycarbonyl)-4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]-L-phenylalanine [Fmoc-L-Phe(4-Tmd)-OH] (51):**[129]
H-L-Phe(4-Tmd)-OH (**2**; 140 mg, 0.51 mmol) was dissolved in acetone (5 mL) and 10% $NaHCO_3$ (3 mL). Fmoc-OSu (220 mg, 0.65 mmol) was added and the mixture stirred for 2 h. The mixture was acidified with 0.01 M HCl, pH 2, and the precipitate was collected and washed (0.01 M HCl) and dried. The product was dissolved in EtOAc, dried ($Na_2SO_4$), and crystallized after addition of heptane to turbidity; yield: 53%; mp 121–122 °C.

**4-Bromobenzyl *tert*-Butyl(dimethyl)silyl Ether (52):**[125]
A mixture of 4-bromobenzyl alcohol (21.90 g, 117 mmol), imidazole (19.5 g, 287 mmol), and TBDMSCl (18.76 g, 135 mmol) in DMF (46 mL) was stirred for 20 h at 40 °C. The mixture was diluted with pentane (150 mL) and washed with $H_2O$ (3 × 100 mL). The organic layer was dried ($MgSO_4$) and the solvent removed in vacuo. Vacuum distillation of the residual light yellowish liquid at 84.5–85 °C gave the silyl ether **52**; yield: 32.29 g (92%).

**1-{4-[(*tert*-Butyldimethylsiloxy)methyl]phenyl}-2,2,2-trifluoroethanone (53):**[125]
A soln of silyl ether **52** (26.19 g, 87 mmol) in anhyd THF (17.5 mL) was added dropwise (ca. 2 mL) to a mixture of Mg turnings (2.19 g, 90 mmol) and anhyd THF (27 mL). The reaction was initiated by the addition of a small amount of $I_2$. The remainder of the halide soln was added at a rate at which the mixture boiled gently. At the end of addition, the mixture was refluxed for 4 h. The cold (0 °C) mixture was then treated with a soln of TFA (2.92 g, 26 mmol) in THF (8 mL), and stirred at rt for 2 h. After dilution of the mixture with $Et_2O$ (100 mL), the organic phase was washed successively with sat. $NH_4Cl$ (2 × 50 mL) and $H_2O$ (2 × 50 mL). The organic layer was dried ($MgSO_4$) and evaporated in vacuo. Vacuum distillation of the yellowish liquid (92–94 °C/0.6 Torr) afforded the ketone **53**; yield: 6.59 g (81%); IR (KBr) $\tilde{\nu}_{max}$: 1720 $cm^{-1}$.

**1-{4-[(*tert*-Butyldimethylsiloxy)methyl]phenyl}-2,2,2-trifluoroethanone *O*-Tosyloxime (54):**[125]
A soln of the ketone **53** (11.90 g, 37 mmol) and $NH_2OH \cdot HCl$ (2.65 g, 37 mmol) in abs EtOH (50 mL) and dry pyridine (115 mL) was heated at 60 °C for 4 h. The dark residue obtained after removal of pyridine in vacuo was dissolved in $Et_2O$ (300 mL) and the soln was washed with $H_2O$. The organic layer was dried ($MgSO_4$) and the solvent removed under vacuum to yield the crude oxime; yield: 12.5 g (ca. 100%); $R_f$ 0.26 ($CH_2Cl_2$). The crude material was treated with Tos-Cl (9.70 g, 51 mmol) in dry pyridine (75 mL) at 120 °C for 2 h. The residue obtained after removal of pyridine under reduced pressure was dissolved in $Et_2O$ (300 mL). The soln was washed with $H_2O$ and the organic layer dried ($MgSO_4$) and evaporated to yield crude product **54**, which was purified by flash column chromatography ($CH_2Cl_2$/hexanes 3:2); yield: 11.74 g (65%).

***tert*-Butyl(dimethyl)silyl 4-[3-(Trifluoromethyl)diaziridin-3-yl]benzyl Ether (55):**[125]
The tosyloxime **54** (4.0 g, 8.2 mmol) was dissolved in anhyd $Et_2O$ (16 mL) and cooled to −78 °C in a sealed tube. Liq $NH_3$ (Na dried, 0.9 mL) was added and the sealed mixture was stirred at rt for 24 h. The precipitated 4-toluenesulfonamide, obtained after evaporation, was filtered and washed with $Et_2O$. The combined $Et_2O$ solns were dried ($MgSO_4$) and evaporated; yield: 2.66 g (98%).

**{4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]phenyl}methanol (56):**[125]
The diaziridine **55** (2.66 g, 8 mmol) was dissolved in dry $Et_2O$ (60 mL) and stirred with freshly prepared $Ag_2O$ (7.5 g, 32 mmol) in the dark. After the reaction was complete, as judged by TLC, the solids were filtered off and the filtrate evaporated to give the silylated alcohol [$\lambda_{max}$ (hexanes) 353 nm]. The silyl-protecting group was removed in concd HCl/MeOH (1:10, 15.4 mL). After stirring at rt for 75 min, the soln was diluted with $Et_2O$ (100 mL) and washed, first with 5% $NaHCO_3$ and then with $H_2O$. The organic layer was dried and evaporated, yielding a pale yellow liquid. Purification by flash column chromatography ($CH_2Cl_2$/$Et_2O$ 9.5:0.5) gave the pure alcohol; yield: 1.68 g (97%).

*for references see p 147*

**3-[4-(Iodomethyl)phenyl]-3-(trifluoromethyl)-3*H*-diazirine (57):**[125]
Methyltriphenoxyphosphonium iodide (3.0 g, 6.6 mmol) was added to the alcohol **56** (0.70 g, 3.2 mmol) in freshly distilled and dry MeCN (6 mL). The soln was kept in the dark at rt for 24 h before dilution with $Et_2O$ and extraction with 1 M NaOH. The organic layer was washed with $H_2O$ and dried ($MgSO_4$). The crude yellowish liquid obtained after solvent evaporation, which solidified after refrigeration, was purified by flash column chromatography ($CH_2Cl_2/Et_2O$ 9.5:0.5); yield: 0.91 g (86%).

**Ethyl 2-[(Diphenylmethylene)amino]-3-{4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]phenyl}propanoate (58):**[125]
Ethyl *N*-(diphenylmethylene)glycinate[1] (880 mg, 3.3 mmol) was dissolved in a mixture of abs EtOH (7.8 mL) and dry THF (3.3 mL). The soln was transferred into an argon-flushed, flame-dried flask containing Na (78.2 mg, 3.4 mmol). After stirring for 40 min at 0 °C, a soln of the diazirine **57** (931 mg, 2.9 mmol) in dry THF (5.5 mL) was added quickly. The soln was stirred for a further 50 min at 0 °C and then diluted with $Et_2O$ (100 mL). The mixture was washed consecutively with sat. $NH_4Cl$ (2 × 50 mL) and $H_2O$ (50 mL), dried ($MgSO_4$), and evaporated to give a yellowish residue, which was purified by flash column chromatography; yield: 1.15 g (87%).

**4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]phenylalanine (2):**[125]
A soln of the protected amino acid **58** (1.45 g, 3.12 mmol) in THF (83 mL) and 10 M aq AcOH (21 mL) was stirred for 24 h at rt. The solvents were removed, and $H_2O$ (15 mL) was added to the residue, which was then extracted with $Et_2O$ to remove benzophenone. With ice-bath cooling, the aqueous phase was adjusted to ca. pH 12 with 10 M NaOH and stirred at rt for 18 h to hydrolyze the ester group. Na ions were removed by adjusting the pH of the soln to ca. 3 with $HCO_2OH$, applying it (in six batches) to a CM-Sephadex ($NH_4$) column (1.5 × 25 cm) at 28 mL · $h^{-1}$, and eluting the amino acid with $H_2O$. After lyophilization, the white residue contained ammonium formate, which was removed by repeatedly extracting the solid with small volumes of $H_2O$; yield: 430 mg (51%); >95% purity.

## 9.3.4.2 Methods of Incorporating 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]arenes into Peptides

The syntheses of 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]arene-containing peptides are carried out in a straightforward manner. In solid-phase syntheses, H-Phe(4-Tmd)-OH [Tmd = 3-(trifluoromethyl)-3*H*-diazirin-3-yl] is incorporated as the Fmoc-Phe(4-Tmd)-OH (Table 4).[123,130–133] Solution-phase syntheses of [$Phe^4$(4-Tmd)]-Leu-enkephalin and [$Phe^2$(4-Tmd)]-aspartame employ Boc-Phe(4-Tmd)-OH and are carried out by the mixed anhydride method.[122] Apparently Phe(4-Tmd) is stable in TFA/$CH_2Cl_2$ (1:1) for 30 min at rt[122] or even in neat TFA for 15 min at 0 °C.[134] A post-synthetic modification of a free peptide has been carried out by the mixed anhydride mediated coupling of 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]benzoic acid [Bz(4-Tmd)-OH, **6**] to the free $N^{\varepsilon}$-amino group in [$Lys^6$]-antamanide.[135] Another post-synthetic modification involves the use of Bz(4-Tmd)-OSu to generate [Bz(4-Tmd)]-β-Ala-OH, which is then coupled via a mixed anhydride to the side chain in δ-aminophalloin.[136] We could not find out whether the 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]phenyl moiety would be stable in the liquid HF used for the final cleavage and deprotection in Boc-based solid-phase peptide synthesis.

Table 4 compiles studies in which 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]arenes are incorporated into peptides, either as congeners, or in the form of the amino acid residue, 4-[-3-(trifluoromethyl)-3*H*-diazirin-3-yl]-L-phenylalanine [H-L-Phe(4-Tmd)-OH].

**Table 4** Photoreactive Peptides Modified with 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]aryl-Based Photophores[31,39,122,123,129–137]

| Peptide[a] | Comments on the Synthesis | Ref |
|---|---|---|
| 2′(3′)-*O*-phenylalanyl(4-Tmd)-5′-*O*-phosphorylcytidylyl(3′-5′)adenosine | solution: Boc, CDI | [134,137] |
| *δ-aminophalloin* | | |
| $N^{\alpha}$-Bz(4-Tmd)-βAla-NH-$CH_2$-CH(OH)Me-$CH_2$ <br> c-[-Ala-D-Thr-Ala-Hyp-Ala-Trp-NH-CH-CO-] | solution: Bz(4-Tmd)-OSu, TEA, EtOH | [136] |
| *antamanide* | | |
| c[-Val-Pro-Pro-Ala-Phe-Lys{$N^{\varepsilon}$-[Bz(4-Tmd)]}-Pro-Pro-Phe-Phe-] | solution: Bz(4-Tmd)-OH, NMM, iBuOCOCl, $CH_2Cl_2$ | [135] |
| *aspartame* | | |
| H-Asp-Phe(4-Tmd)-OMe | solution: Boc, iBuOCOCl | [39] |
| *ovine corticotropin releasing factor (oCRF)* | | |
| $N^{\alpha}$-[Bz(4-Tmd)]-YSQEPPISLDLTFHLLREVLEMT-KADQLAQQAHSNRKLLDIA-$NH_2$ | solid phase | [132] |
| *enkephalin* | | |
| H-Tyr-Gly-Gly-Phe(4-Tmd)-Leu-OH | solution: Boc, iBuOCOCl | [122] |
| *neuropeptide Y* | | |
| [$N^{\alpha}$-biotinyl-Ahx$^2$,[$^3$H]propionyl-Lys$^4$,Ahx$^{5-24}$,Phe$^{27}$(4-Tmd)]NPY | solid phase: Fmoc/tBu, | [133] |
| [$N^{\alpha}$-biotinyl-Ahx$^2$,[$^3$H]propionyl-Lys$^4$,Ahx$^{5-24}$,Phe$^{36}$(4-Tmd)]NPY | | [133] |
| *thymopentin-[Phe$^{36}$(4-Tmd)]thymopoietin(32–36)* | | |
| H-Arg-Lys-Asp-Val-Phe(4-Tmd)-OH | solid phase: Fmoc/tBu, DIC/HOBt | [123] |
| *plasma membrane $Ca^{2+}$ pump-derived sequence, calmodulin-binding* | | |
| [$^3$H]Ac-LRRGQILW-Phe(4-Tmd)-RGLNRIQT-QIKVVVNAFSSS-OH (C28W) | solid phase: Fmoc/tBu, DCC/HOBt, NMP | [129] |
| *antagonists of urokinase-type plasminogen activator* | | |
| S-Phe(4-Tmd)-NFSQYLWS | solid phase: Fmoc/tBu, DIC/HOAt, DMF | [130,131] |
| SLN-Phe(4-Tmd)-SQYLWS | | [130,131] |
| SLNFSQ-Phe(4-Tmd)-LWS | | [130,131] |
| SLNFSQY-Phe(4-Tmd)-WS | | [130,131] |
| *vasopressin* | | |
| [(Asu)$_2^{1,6}$,Phe(4-Mda)($CH_2$)$_2$CONH)]-AVP | post-SPPS: with 3-methyl-3-diazirin-3-ylpropanoyl chloride[b] | [31] |

[a] Tmd = 3-(trifluoromethyl)-3*H*-diazirin-3-yl; Mda = 3-methyl-3*H*-diazirin-3-yl.
[b] See Ref[138].

**[Lys$^6$-[$N^{\varepsilon}$-{Bz(4-Tmd)}]]-Antamanide:**[135]
A soln of 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]benzoic acid (**6**; 20.4 mg, 88.6 mmol) and NMM (10 μL) in $CH_2Cl_2$ (10 mL) was cooled to −15 °C and treated with iBuOCOCl (9.3 μL, 70.9 mmol). After 8 min, a soln of [Lys$^6$]-antamanide (20 mg, 17.7 mmol) was added to the mixture and the cold bath removed. After 3 h at rt, the solvent was removed in vacuo and the residue was purified on a Sephadex LH-20 column and eluted with MeOH; yield: 19.8 mg (84%).

**SLN-Phe(4-Tmd)-SQYLWS:**[130]
Solid-phase synthesis was carried out manually using a polyethylene syringe as the reaction vessel. KA-resin (Novasyn KA) preloaded with Fmoc-Ser on an acid-labile linker. The Fmoc amino acid derivatives (5.0 equiv) activated by HOAt and DIC were used in the coupling steps and were allowed to react for >2 h. Fmoc-Phe(4-Tmd)-OH was coupled overnight. The Fmoc protecting group was removed with 20% piperidine in DMF for 15–20 min. For cleavage, the resin-bound peptide was treated for 1.5 h with 85% TFA containing 5% each of phenol, β-ME, and thioanisole. The filtrate was concentrated by $N_2$ flashing, the peptide was subsequently precipitated and washed with $Et_2O$ (× 4). Crude peptide was dissolved in 10% AcOH and lyophilized again. Analytical HPLC was performed on C-18 with a 25-min linear gradient of 0–70% B in A (A: 0.1% TFA in $H_2O$; B: 0.1% TFA, 9.9% $H_2O$, 90% MeCN). When necessary, purification was carried out on a preparative scale. ES-MS: 1352.5 Da (calcd 1351.58 Da).

*for references see p 147*

**[Phe$^4$(4-Tmd)]-Leu-Enkephalin:**[122]

*Ddz-Tyr(tBu)-Gly-Gly-OH:* To a soln of Ddz-Tyr(tBu)-OH (1.22 g, 2.65 mmol) activated by the mixed anhydride method (iBuOCOCl in $CH_2Cl_2$ at −15 °C for 8 min in the presence of NMM) was added H-Gly-Gly-OEt trifluoroacetate salt (1.0 g, 3.66 mmol); yield: 1.56 g (99%, crude product). The tripeptide ethyl ester was saponified with aq NaOH in dioxane (1.1 equiv); yield: 1.22 g (82%).

*Boc-Phe(4-Tmd)-Leu-OMe:* Boc-Phe(4-Tmd)-OH (234 mg, 0.627 mmol) was coupled to Leu-OMe·HCl (181 mg, 1.0 mmol) by the mixed anhydride method. The product was then crystallized (EtOAc/petroleum ether); yield: 283 mg (90%); mp 125–126 °C.

*Ddz-Tyr(tBu)-Gly-Gly-Phe(4-Tmd)-Leu-OMe:* Boc-Phe(4-Tmd)-Leu-OMe (250 mg, 0.5 mmol) was treated with 50% TFA in $CH_2Cl_2$ (1 mL). The product was coupled to Ddz-Tyr(tBu)-Gly-Gly-OH (423 mg, 0.75 mmol) by DCC/HOBt. After 16 h at rt, the mixture was concentrated in vacuo, taken up in a small amount of MeOH, and the precipitated DCU was removed by filtration. Purification of the pentapeptide was achieved by gel permeation chromatography on a Sephadex LH-20 column with MeOH as eluent. The product was crystallized (MeOH); yield: 355 mg (75%); mp 187–189 °C (dec); UV $\lambda_{max}$ 359 nm.

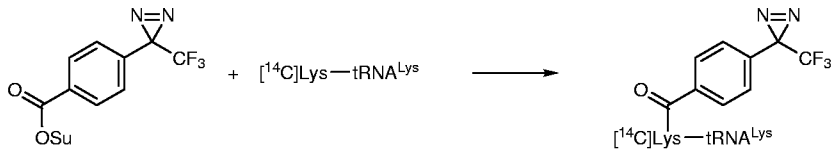
*H-Tyr-Gly-Gly-Phe(4-Tmd)-Leu-OH:* The protected pentapeptide was saponified with 2 M aq NaOH (200 µL) in MeOH (1 mL). After 1 h, the soln was neutralized with AcOH and subjected to column chromatography (silica gel, $CHCl_3$/MeOH/$H_2O$ 65:25:4); yield: 280 mg (81%). The pentapeptide was further deprotected by treatment with 5% TFA in $CH_2Cl_2$ for 10 min (removal of the Ddz group), followed by 50% TFA in $CH_2Cl_2$ (removal of the tBu). The residue, obtained after removing the solvent in vacuo, was purified by column chromatography (silica gel, $CHCl_3$/MeOH/$H_2O$ 65:25:4), followed by gel permeation chromatography (Sephadex LH-20, MeOH); yield: 140 mg (83%).

### 9.3.4.3 Biochemical, Site-Specific Incorporation of 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]phenylalanine into Proteins by Phe(4-Tmd)-tRNA$^{Phe}$

Structure–function studies of proteins by site-directed mutagenesis are complemented by the site-specific incorporation of noncoded amino acids into proteins carried out in a cell-free translation system. Photo-cross-linking of peptides or proteins containing site-specifically incorporated photoreactive amino acids is a better way of studying topological relationships than using cross-linking bifunctional reagents[139,140] and gives an insight into the intermolecular interactions. Moreover, biochemical, site-specific incorporation of a noncoded photoreactive amino acid into an acceptor macromolecule complements and synergizes the chemical, site-specific incorporation of photophores into peptide ligands.

The first approach to the introduction of a site-specific photoreactive group into a nascent preprolactin in a cell-free wheat germ translation system is shown in Scheme 15.[141,142] The tRNA$^{Lys}$ on the ε-amino of the Lys residue is loaded with succinimidyl 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]benzoate. Translation, followed by irradiation, generates photoconjugates, which allow identification of the signal sequence receptor, an integral protein of the rough endoplasmic reticulum membrane, as associated with the nascent preprolactin during the in vitro translocation.[143]

**Scheme 15** Loading the ε-Amine in [$^{14}$C]Lysyl-tRNA$^{Lys}$ with Succinimidyl 4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]benzoate[141,142]

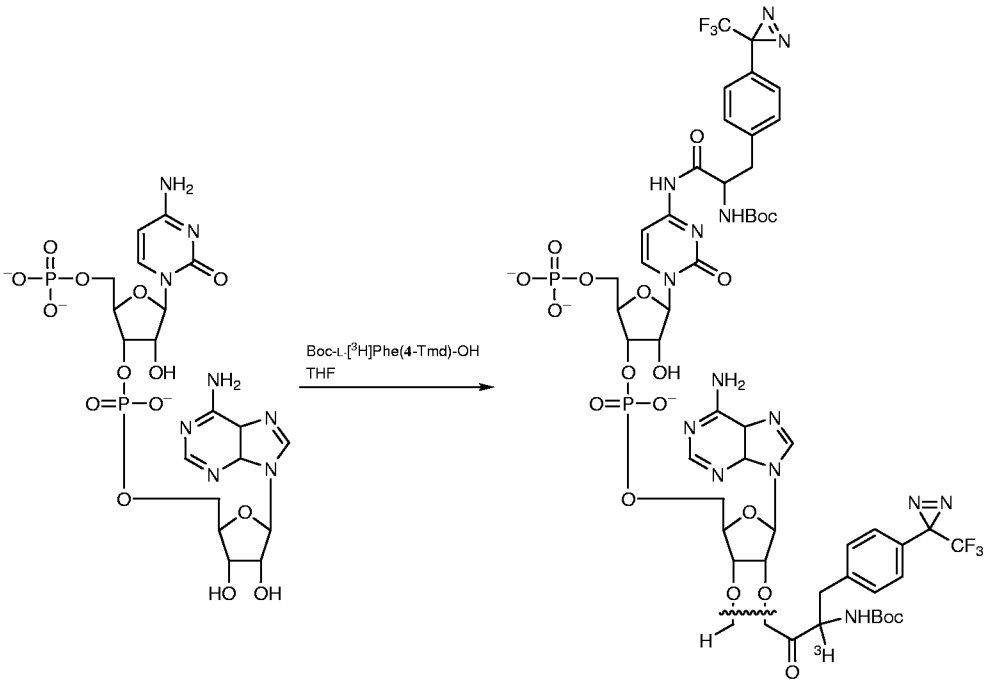



Chemical misacylation of *E. coli* with Phe(4-Tmd) is reported as the first step toward the biosynthesis of polypeptides containing nonnatural amino acids.[137] T4 RNA ligase mediated coupling of the synthetic 2′(3′)-*O*-[(4-Tmd)phenylalanyl]-5′-*O*-[phosphorylcytidylyl]phosphoryladenosine **62** [Phe(4-Tmd)-pCpA] to tRNA$^{Phe}$ missing the 3′-terminal cytidine and phosphoryladenosine moieties generates the misaminoacylated Phe(4-Tmd)-tRNA$^{Phe}$ (Scheme 16). Indirect evidence suggests that the Phe(4-Tmd)-tRNA$^{Phe}$ is active in an in vitro

protein biosynthesizing system. However, the amino acid is inserted into multiple sites, since both the chemically acylated and the wild-type tRNA compete for the same Phe codon.

The amber suppressor tRNA ($tRNA^{Sup}$) is introduced to overcome the limitation associated with uniform incorporation of the noncoded amino acid into the target protein. The amino acid-loaded $tRNA^{Sup}$ incorporates the nonnatural amino acid at a specific site in the protein, corresponding to the nonsense suppression site in the gene.[144] Phe(4-Tmd)-$tRNA^{Sup}$ is used to study the interactions of translocating preprolactin with lipids and proteins at the endoplasmic reticulum membrane.[134,145]

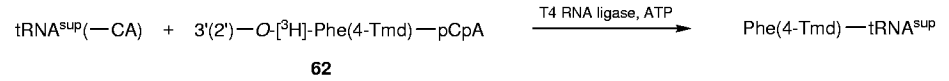
**Scheme 16** Synthesis of Phe(4-Tmd)-pCpA and Its Ligation to the Truncated (-CA) $tRNA^{Sup}$[137]




**60** **61**

**62** 3'(2')-*O*-[$^3$H]Phe(4-Tmd)-pCpA

*for references see p 147*



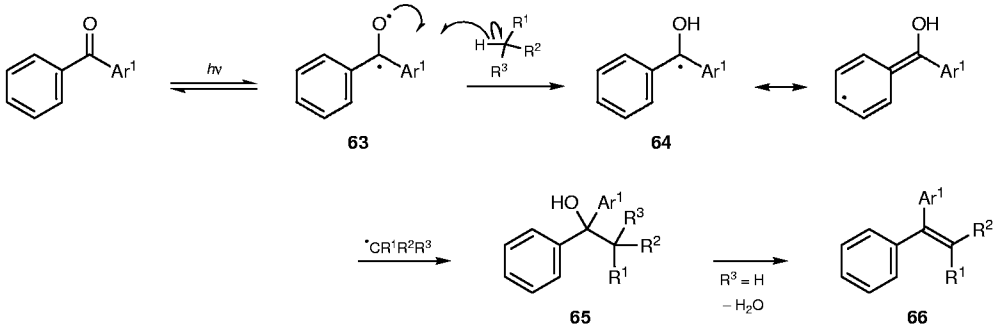

**2′(3′)-*O*-{2-[$^3$H]-4-[3-(Trifluoromethyl)-3*H*-diazirin-3-yl]phenylalanyl}-5′-*O*-[phosphorylcytidylyl]phosphoryladenosine {[$^3$H]Phe(4-Tmd)-pCpA} (62):**[137]

To a soln of Boc-L-[$^3$H]Phe(4-Tmd)-OH (35 mg, 94 μmol) in dry THF (160 μL) was added CDI (16 mg, 94 μmol). After activation of the acid for 15 min at rt, 5′-*O*-[phosphorylcytidylyl]phosphoryladenosine **60** (pCpA; 160 $A_{260nm}$ units) dissolved in a mixture of $H_2O$ (175 μL) and MeCN (190 μL) were added, and the mixture was allowed to react for 4 h at rt. The soln was extracted with EtOAc, and the aqueous phase, containing unreacted pCpA and acyl-pCpAs, was subjected to RP-HPLC using an Aquapore RP300 column (10 μm, 4 × 250 mm). Elution of the products was affected using a gradient of 0–100% MeCN in $NH_4OAc$ (50 mM), pH 4.5, over 50 min at 1 mL·min$^{-1}$. The fractions containing the acyl-pCpAs were pooled and lyophilized to give a mixture of 3′- and 2′-*O*-{Boc-[$^3$H]Phe(4-Tmd)}-pCpA **61**; combined yield: 19 $A_{260nm}$ units. To remove the Boc-protecting group, the lyophilized material was dissolved in dry TFA (10 μL per $A_{260nm}$ units), and the soln was kept on ice for 15 min. TFA was removed by a stream of $N_2$ and the deprotected **62** was precipitated by the addition of a small amount of $Et_2O$ and collected by centrifugation. The resulting pellet was dried in vacuo, dissolved in 50 mM $NH_4OAc$, pH 4.5, and lyophilized. This material was used for ligation without further purification.

## 9.3.5 Peptides Containing Aryl Ketones

Due to their photochemical properties, aryl ketones in general, and benzophenones in particular, are widely used in modern photo-cross-linking studies.[146] The first application of photoactivated benzophenones in biological systems was reported by Galardy et al. in the early 1970s.[59,147] The development of the photoreactive amino acid 4-benzoylphenylalanine (**3**) in 1986 by Kauer et al. gave new impetus to the use of benzophenone-containing peptides to study peptide–protein interaction by photoaffinity cross-linking methodologies.[148] The photochemistry of aryl ketones is well understood and schematically represented in Scheme 17.[146]

**Scheme 17** Mechanism of the Photoactivation of Benzophenone Derivatives[146]




For benzophenone, absorption of a photon at ca. 350 nm results in a n–π* transition of a nonbonding electron of the carbonyl oxygen to give **63** ($Ar^1$ = Ph). The electron-deficient oxygen, generated in the triplet state, interacts with weak C–H σ-bonds resulting in hydrogen abstraction to give **64**, which is followed by the formation of a new C–C bond in **65**. Importantly, the diradical triplet state reacts only with topologically accessible C–H bonds, thus providing considerable specificity for the hydrogen abstraction. On the basis of both experimental data and modeling, the reactive volume of benzophenone has been estimated to be a sphere of radius 3.1 Å centered on the carbonyl oxygen. The reactivity order for C–H

bonds is -$NCH_x$ > -$SCH_x$ > -$C{=}CCH_2$- > -$CH_2$- > -Me, although the actual photo-insertion site is highly dependent on the topology of the C–H–BP (BP = benzophenone) interaction. According to O'Neil and DeGrado,[149] particularly reactive sites include electron-rich tertiary centers such as $C^{\gamma}$–H of leucine, $C^{\beta}$–H of valine and those $CH_2$ groups adjacent to heteroatoms in lysine, arginine, and methionine. In addition, depending on the C–H bond involved in the photoreaction, several rearrangements of the conjugate **65** may occur. For example, following photo-insertion on a $C^{\alpha}$–H bond of Gly, conjugate **65** ($R^1$ = NH; $R^2$ = CO; $R^3$ = H) can eliminate water to give **66**.[3] In the absence of an accessible hydrogen donor, the triplet state readily relaxes to the ground state, so the same molecule can undergo several excitation–relaxation cycles prior to the occurrence of a productive photo-insertion event.

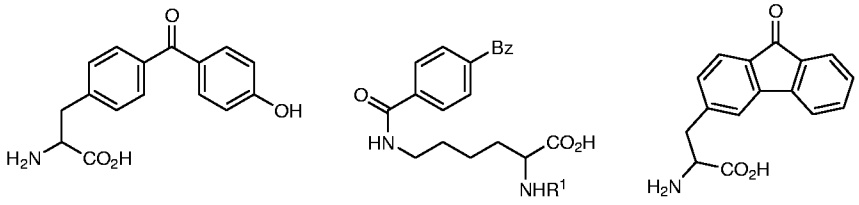
The widespread use of benzophenone derivatives in photoaffinity cross-linking can be attributed to several advantageous properties:[3] First, benzophenones are chemically more stable than diazo esters, aryl azides, and diazirine. 4-Benzoylphenylalanine is fully compatible with SPPS chemistry using the standard protocols of either Boc or Fmoc strategies (see, however, Section 9.3.5.2) and can be manipulated in ambient light. Second, benzophenones are activated at ca. 360 nm, thus avoiding protein-damaging wavelengths. Third, photoactivated benzophenones are poorly reactive toward water and nucleophiles, which allows their use in biologically compatible media. Fourth, the photochemical properties of benzophenones generally provide high efficiency and specificity of cross-linking. It should be noted, however, that benzophenones also introduce additional hydrophobicity and bulkiness. This might result in reduced photo-cross-linking in cases when the photo-insertion site is located in hydrophylic or sterically hindered regions of the acceptor molecule.

A couple of extensive reviews on the chemistry of benzophenone photoprobes and their applications in biochemistry have been published over the last decade.[3,6]

### 9.3.5.1 Strategies for the Synthesis of Peptides Containing Benzophenone Derivatives

Two basic strategies for introducing benzophenone photophores into peptides have been used. The first strategy is based on the introduction of benzophenone-containing amino acids during the stepwise assembly of the peptide. Several amino acids containing benzophenone derivatives have been described and used in solid-phase peptide synthesis: 4-benzoylphenylalanine (**3**),[148] *N*-methyl-4-benzoylphenylalanine,[150] 4-(4-hydroxybenzoyl)phenylalanine (**67**),[151] $N^{\varepsilon}$-(4-benzoylbenzoyl)lysine (**68**, $R^1$ = H),[152] and 3-(9-oxo-9*H*-fluoren-3-yl)-L-alanine (**69**) (Scheme 18).[153] Due to the chemical stability of benzophenones, these amino acids can be introduced into peptides following either Boc or Fmoc methodologies and standard SPPS protocols.

**Scheme 18** Structures of Amino Acids Containing Benzophenone Derivatives[150-153]

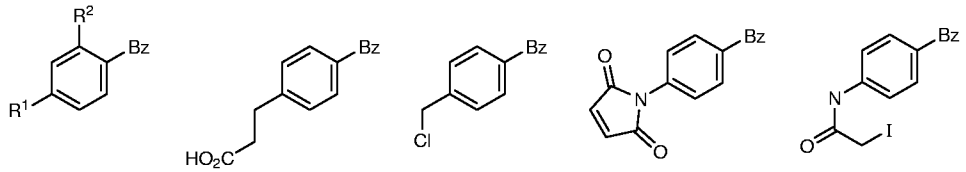



**67** **68** **69**

*for references see p 147*

The second strategy is based on the post-synthetic modification of functional groups in peptides with benzophenone-based photophores. Obviously, the more informative studies are those in which the introduction of the photophore is site-specific. Several benzophenone-containing reagents with different chemically reactive functionalities have been described (Scheme 19).[3,154] 4-Benzoylbenzoic acid (**7**),[155–157] 4-benzoylphenyl thiocyanate (**70**, $R^1$ = NCS, $R^2$ = H),[158] 4-benzoyl-3-iodobenzoic acid (**70**, $R^1$ = $CO_2H$, $R^2$ = I),[159] and 3-(4-benzoylphenyl)propanoic acid (**71**)[154] have been used to modify α- and ε-amino groups. 4-(Chloromethyl)benzophenone (**72**), 4-maleimidobenzophenone (**73**),[160–162] and *N*-(4-benzoylphenyl)-2-iodoacetamide (**74**)[163] have been used to introduce benzophenone in peptides by modification of cysteine thiols.

**Scheme 19** Structures of Benzophenone-Containing Reagents Used to Modify Amino Acids[154–163]




**70** **71** **72** **73** **74**

A third modality is based on the use of orthogonal protection during SPPS synthesis to achieve selective removal of a specific side-chain protecting group, followed by selective modification with an activated benzophenone moiety. At the conclusion of the side-chain modification, which is carried out on the resin-bound peptide, stepwise elongation of the peptide chain is resumed.[157] Table 5 provides a current summary of reported biologically active peptides substituted with a benzophenone moiety and includes some synthetic details about their preparation.

A methodology to introduce unnatural amino acids into proteins has been developed (Section 9.3.4.3).[141,142] This method is based on the use of amber suppressor tRNAs chemically modified with unnatural amino acids. Addition of this suppressor tRNA and a gene containing an amber nonsense codon, at the site of interest, to an in vitro transcription/translation system will generate the encoded protein containing the unnatural amino acid at a specific site. This methodology has been used to introduce 4-benzoylphenylalanine (**3**) into T4 lysozyme,[164] and 4-benzoylphenylalanine (**3**) and 2-amino-5-(4-benzoylphenyl)pentanoic acid at various positions throughout the cytochrome $b_2$-dihydrofolate reductase fusion protein.[165] In vitro suppression of the amber codon competes with termination of protein synthesis, since only two out of the three termination codons remain available. The use of minor codon AGG extends the possibility of introducing photoreactive nonnatural amino acids into proteins through *E. coli* in vitro protein biosynthesis.[166] Similarly, incorporation of 4-benzoylphenylalanine (**3**) into streptavidin has been achieved by translation of the four-base codon CGGG by chemically aminoacylated frame-shift suppressor tRNA containing the complementary four-base anticodon CCCG.[167] By using two pairs of four-base codons, CCCG and AGGU, translated into amino acids by chemically aminoacylated frame-shift suppressors tRNA containing complementary four-based anticodons, CGGG and ACCU, respectively, Hohsaka et al. have been able to incorporate two different nonnatural amino acids into two different sites of a single protein,[168] thus expanding remarkably the scope of nonnatural amino acid mutagenesis.

**Table 5** Photoreactive Peptides Modified with 4-Benzoylbenzoyl-Based Photophores[30,35,37,43,123,130,148–152,154–156,159–161,169–212]

| Photophore | Peptide | Comments on the Synthesis | Ref |
|---|---|---|---|
| Phe(4-Bz) | atrial natriuretic peptide | SPPS | [169–171] |
| | angiotensin II | SPPS: Boc | [35,172] |
| | angiotensin IV | SPPS | [37,173] |
| | α-factor | SPPS | [174] |
| | β-casomorphin | SPPS | [175] |
| | bombolitin | SPPS | [176] |
| | bradykinin | SPPS | [177,178] |
| | cAMP-dependent protein kinase substrate | SPPS: Boc | [179] |
| | cholecystokinin | SPPS: Fmoc | [180] |
| | calmodulin, binding peptides | SPPS: Boc | [148,149] |
| | c-Src-derived peptide | SPPS: Fmoc | [181] |
| | epidermal growth factor fragment | SPPS: Fmoc | [182,183] |
| | farnesyl-protein transferase (inhibitor of) | Solution: Boc | [184] |
| | formyl peptide | SPPS: Fmoc | [185] |
| | head-activator | SPPS | [186] |
| | hexarelin | SPPS: Fmoc | [187] |
| | insulin | Semisynthesis | [188] |
| | insulin receptor substrate-1 (IRS-1) | SPPS: Fmoc | [189] |
| | morphiceptin | SPPS | [190] |
| | melanin-concentrating hormone | SPPS: Fmoc | [191] |
| | myosin light-chain kinase (peptide substrate for) | SPPS: Boc | [192] |
| | pituitary adenylate cyclase activating polypeptide | SPPS | [193] |
| | parathyroid hormone related protein | SPPS: Fmoc | [194,195] |
| | parathyroid hormone | SPPS: Boc | [152,196] |
| | pro-factor IX | SPPS | [197] |
| | *c*RGD-containing peptides | SPPS and solution | [198] |
| | secretin | SPPS: Boc | [199,200] |
| | substance P | SPPS | [201–205] |
| | thymopentin | SPPS: Fmoc | [123] |
| | urokinase-type plasminogen activator (antagonist of ) | SPPS: Fmoc | [130] |
| $N^{\alpha}$-Me-Phe(4-Bz) | substance P antagonist | SPPS: Boc | [150] |
| [$^3$H]Phe(4-Bz) | luteinizing hormone subunit | SPPS: Boc | [206] |
| Phe(3-OH,4-Bz) | substance P | SPPS: Fmoc | [151] |
| $N^{\alpha}$-Bz(4-Bz) | calcitonin | SPPS: | [155] |
| | farnesyl-protein transferase (substrates of) | HOBt/DCC | [207] |
| | fertilinβ-derived peptide | post-SPPS: modification HBTU/HOBt | [208] |
| | glycoprotein D of herpes simplex virus type 1 (peptide fragment of) | SPPS | [43] |
| | RGD-containing peptides | SPPS: with Bz(4-Bz)-Cl | [209] |
| | substance P | SPPS | [210] |
| | *E. coli* heat-stable enterotoxin | post-SPPS: modification, Bz(4-Bz)-OSu | [156] |
| $^3$H-BP[a] | cholecystokinin | post-SPPS: modification, HBTU/HOBt | [211] |
| | farnesyl-protein transferase (substrates of) | | [207] |
| 4-Bz-$C_6H_4(CH_2)_2CO$ | H-Gly-Gly-Arg-OH | post-SPPS: modification on $N^{\alpha}$ or on Lys(ε-$NH_2$) with 4-Bz$C_6H_4(CH_2)_2CO_2$Su | [154] |
| | pheromone biosynthesis activating neuropeptide | | [154] |
| [*N*-(4-Bz$C_6H_4$)]-3-Suc | laminin fragment | post-SPPS: modification, on Cys with 4-maleimidobenzophenone | [160] |
| | glutathione | SPPS: Boc | [212] |
| | phosphodiesterase inhibitory γ-subunit | | [161] |

*for references see p 147*

**Table 5** (cont.)

| Photophore | Peptide | Comments on the Synthesis | Ref |
|---|---|---|---|
| Boc-Lys[$N^{\varepsilon}$-Bz(4-Bz)] | calcitonin | SPPS: Boc | [155] |
| | parathyroid hormone | SPPS: Boc | [152] |
| | substance P | post-SPPS: modification, Bz(4-Bz)-OSu | [210] |
| | glycoprotein D of herpes simplex virus type 1 (peptide fragment of) | SPPS: Boc | [43] |
| $N^{\alpha}$-Bz(4-Bz)-Gly-OH | glycoprotein D of herpes simplex virus type 1 (peptide fragment of) | SPPS: Boc | [43] |
| Lys[$N^{\varepsilon}$-Bz(3-I,4-Bz)] | parathyroid hormone | post-SPPS: modification, Bz(3-I,4-Bz)-OSu | [159] |
| $N^{\alpha}$-Bz(4-Bz)-L-biocytin | pepstatin | post-SPPS: modification | [30] |

[a] BP = benzophenone.

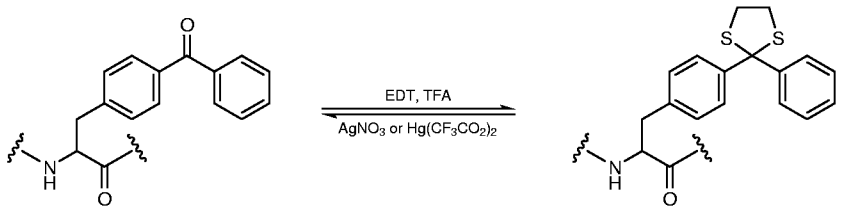
**Introduction of Benzophenone into Peptides on Solid Support:**[157]
Boc-Lys($N^{\varepsilon}$-Fmoc)-OH was loaded onto hydroxymethyl resin (0.5 g, 0.5 mmol) using the DMAP (0.006 equiv) catalyzed DCC condensation of the amino acid (4 equiv.). This procedure was repeated two additional times to assure complete reaction. After removal of the ε-Fmoc group by two consecutive treatments (3 and 20 min) with 20% piperidine in NMP, the resin was thoroughly washed with NMP. 4-Benzoylbenzoic acid (**7**) was then double-coupled (as the HOBt ester in NMP) to the lysine ε-amino group. Chain elongation was continued by the standard method employing Boc/Bzl protection strategy. The peptide was cleaved with HF/anisole (10:1) for 50 min at 0 °C. The cleaved peptide was washed with $Et_2O$ and extracted from the resin with 30% AcOH. The AcOH was removed by rotary evaporation, and the remaining aqueous peptide soln was diluted fourfold with $H_2O$, shell-frozen, and lyophilized.

**Post-Solid-Phase Modification of a Peptide by a Benzophenone-Containing Moiety:**[159]
The synthesis of $N^{\alpha}$-Fmoc-[$(Nle)_2^{8,18}$,L-$Nal^{23}$,$(Arg)_2^{26,27}$,$Tyr^{34}$]-bPTH(1–34)-$NH_2$ was carried out on pMeBHA·HCl resin (0.5 mmol, 0.57 mmol·$g^{-1}$) using $N^{\alpha}$-Boc-protected amino acid derivatives activated by HOBt/DCC before coupling. $Ala^1$ was coupled as the Fmoc derivative. The Fmoc-peptide-resin was cleaved with HF/anisole (10:1) at 5 °C for 75 min. The crude Fmoc-peptide was purified on RP-HPLC and lyophilized. $N^{\alpha}$-Fmoc-[$(Nle)_2^{8,18}$,L-$Nal^{23}$,$(Arg)_2^{26,27}$,$Tyr^{34}$]-bPTH(1–34)-$NH_2$ [30 mg ($3.6 \times 10^{-6}$ mol, peptide content 60%); bPTH = bovine parathyroid hormone] was dissolved in DMF (1 mL) in the presence of Bz(3-I,4-Bz)-OSu (3.68 mg, $8.5 \times 10^{-6}$ mol, 2.4 equiv) and DIPEA (10 μL). Upon completion of reaction (1 h), the mixture was treated for 30 min with piperidine (300 μL) to remove the $N^{\alpha}$-Fmoc protecting group. The mixture then was diluted with $H_2O$ (50 mL), and the soln lyophilized.

### 9.3.5.2 Side Reactions of the Benzophenone Moiety during Peptide Synthesis

The major side reaction involving benzophenone moieties in peptide synthesis is dithioketal formation in the presence of some dithiols in acidic media (Scheme 20). This side reaction is particularly important during standard resin cleavage procedures following Fmoc-based peptide assembly using the conventional TFA/$H_2O$/EDT mixture.[213] Formation of the dithioketal proceeds faster with EDT than with either butane-1,3-dithiol or butane-2,3-dithiol. In contrast, no side reaction is observed when dithiothreitol or 1,4-dithio-L-threitol are used as scavengers.[213] Regeneration of Phe(4-Bz) from the dithioketal derivative has been reported.[214] Treatment of Phe(4-Bz)(dithioketal)-containing peptides with DMSO at 140–160 °C is not recommended.[215] Under these conditions, significant decomposition of Phe(4-Bz)-containing α-factor analogues is observed.[214] Alternatively, treatment of Phe(4-Bz)(dithioketal) derivatives with mercury(II) trifluoroacetate (1.5 molar excess over the peptide dissolved in 50% $H_2O$/MeCN at a concentration of 0.5–2.0 mg·$mL^{-1}$) rapidly regenerates Phe(4-Bz). In some cases, however, degradation of α-factor analogues is observed. In contrast, treatment with large molar excess of $AgNO_3$ under the same conditions results in a slower, but highly specific, conversion of the dithioketal into Phe(4-Bz).[214]

**Scheme 20** Side Reaction of 4-Benzoylphenylalanine with Dithiols[213]

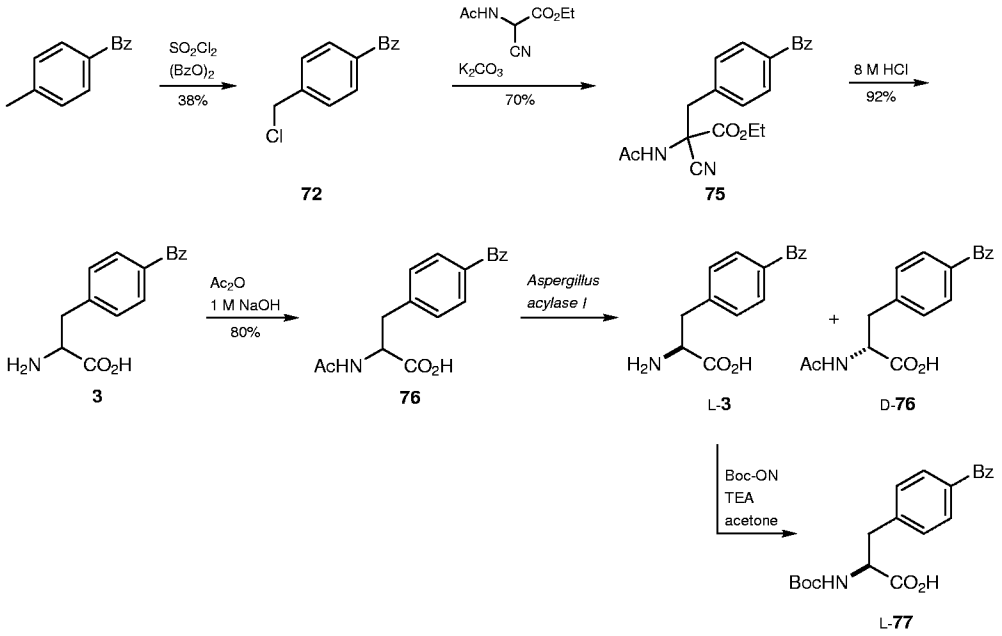



### 9.3.5.3 Synthesis of Protected Amino Acids Containing Benzophenone Derivatives

Several $N^{\alpha}$-protected amino acids containing benzophenone moieties have been described. Among them, the most widely used are protected derivatives of 4-benzoylphenylalanine (**3**), which was first synthesized as the racemic Boc derivative by DeGrado and co-workers in 1986 (Scheme 21).[148] In the following years, *N*-Fmoc-4-benzoylphenylalanine,[203] *N*-Boc-3-(9-oxo-9*H*-fluoren-3-yl)-L-alanine (**83**) (Scheme 22)[153], $N^{\alpha}$-Boc-$N^{\varepsilon}$-(4-benzoylbenzoyl)lysine (**68**, $R^1$ = Boc) (Scheme 23),[152] and *N*-Boc-*N*-methyl-4-benzoylphenylalanine[150] have been prepared and used in solid-phase peptide synthesis. The incorporation of these amino acids into peptides follows standard SPPS methodologies, although care should be taken in the deprotection/cleavage step (see Section 9.3.5.2). Additionally, the synthesis of *N*-Aloc-2-amino-5-(4-benzoylphenyl)-D,L-pentanoic acid [*N*-Aloc-5-(4-benzoylphenyl)norvaline] and *N*-Aloc-4-benzoyl-D,L-phenylalanine have been described.[165]

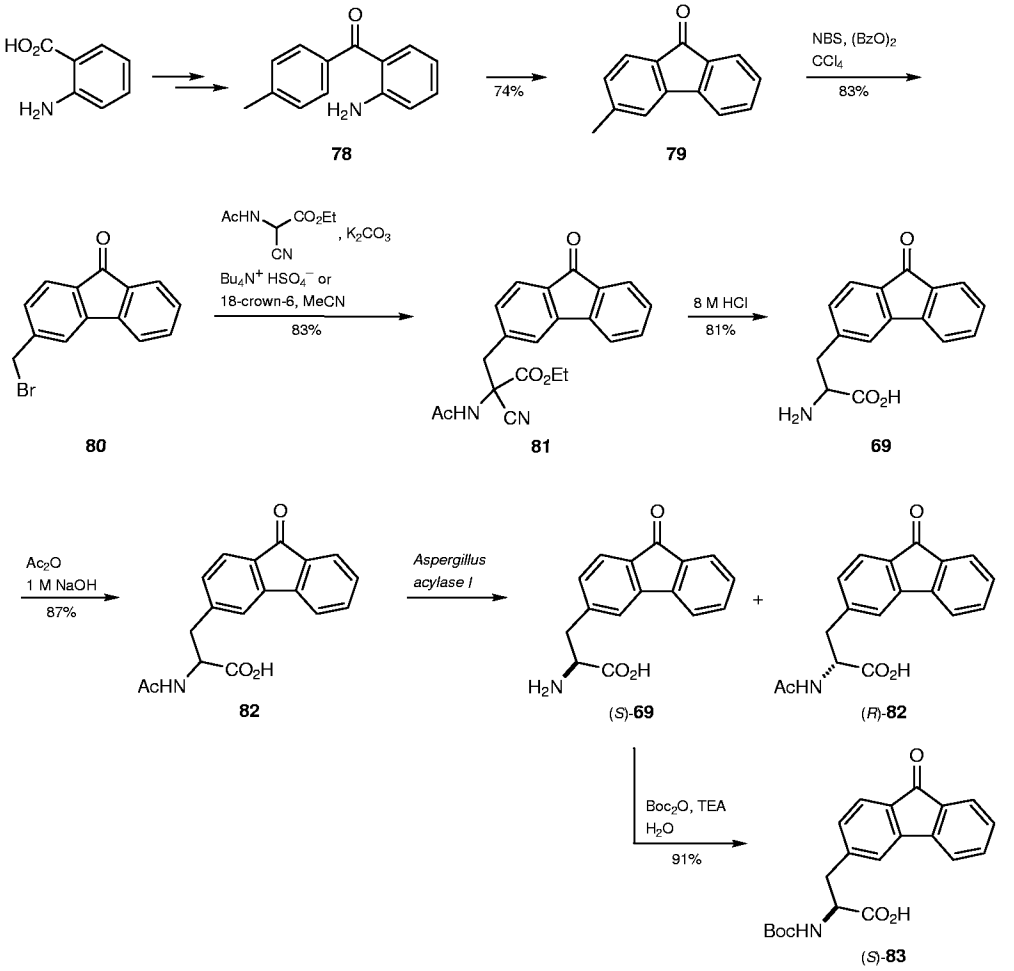
Some of the benzophenone-containing amino acids, e.g. protected (4-hydroxybenzoyl)-phenylalanine[151] and 4-(2,5-dibromobenzoyl)phenylalanine, [216] were specifically developed in order to allow the incorporation of radioactive tags on the benzophenone moiety (see Section 9.3.5.4). Presenting the radioactive tags on the photophore has the advantage of maintaining the linkage between the two throughout the digestions employed in the analysis of the photoconjugate. It also saves major effort spent in the development of a cleavage-resistant photoreactive analogues suitable for photoaffinity scanning analysis.

**Scheme 21** Synthesis of *N*-(*tert*-Butoxycarbonyl)-4-benzoylphenylalanine[148]

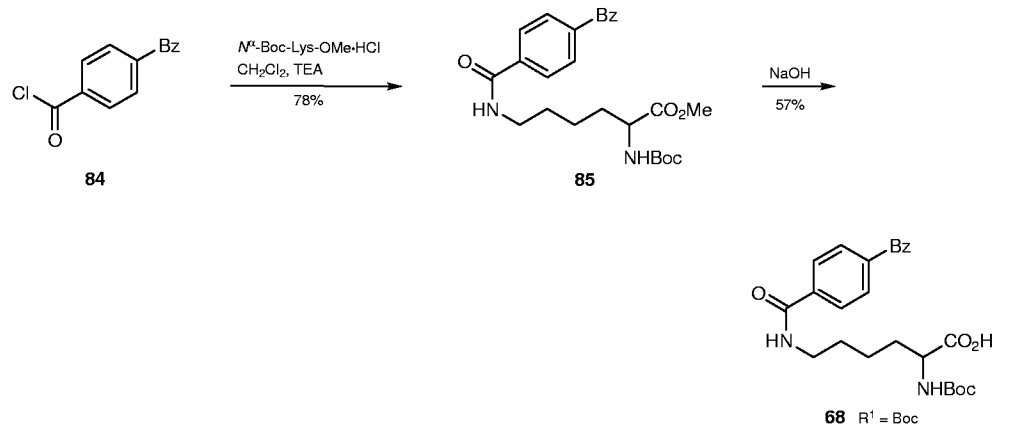



*for references see p 147*

**Scheme 22** Synthesis of *N*-(*tert*-Butoxycarbonyl)-3-(9-oxo-9*H*-fluoren-3-yl)-L-alanine[153]




**Scheme 23** Synthesis of $N^{\alpha}$-(*tert*-Butoxycarbonyl)-$N^{\varepsilon}$-(4-benzoylbenzoyl)lysine[152]




**4-(Chloromethyl)benzophenone (72):**[148]

A soln of 4-methylbenzophenone (100 g, 0.51 mol) in $CCl_4$ (130 mL) was treated with $SO_2Cl_2$ (98 g) and $(BzO)_2$ (55 mg), and refluxed under $N_2$. At 2-h intervals, three additional portions of $(BzO)_2$ (50 to

70 mg) were added. After 18 h a final portion of $(BzO)_2$ (50 mg) was added. After 1 h, unreacted $SO_2Cl_2$ was removed by codistillation with $CCl_4$ (bp 69 °C). The residue, containing $CCl_4$, was treated with $K_2CO_3$ (5 g) and Woelm basic alumina (5 g). The mixture was refluxed for 15 min with stirring, and then filtered while hot, giving crystals on cooling. Recrystallization (EtOH) gave colorless crystals; yield: 44.7 g (38%); mp 98 °C. The product contained about 2% of 4-methylbenzophenone and 0.7% of 4-(dichloromethyl)benzophenone.

**Ethyl 2-(Acetylamino)-3-(4-benzoylphenyl)-2-cyanopropanoate (75):**[148]

A stirred mixture of **72** (41.9 g, 181 mmol), ethyl (acetylamino)(cyano)acetate (30.9 g, 182 mmol), acetone (300 mL), anhyd $K_2CO_3$ (17.3 g), and KI (1.6 g) was refluxed overnight, cooled, and filtered. The solid was washed with acetone, and the combined filtrates were evaporated under reduced pressure. The residue was crystallized (EtOH, 70 mL) by cooling to 5 °C. The resulting product (59 g) was dissolved in hot EtOH (900 mL), treated with activated charcoal, filtered, and diluted with hexanes (900 mL) to give colorless crystals; yield: 45.6 g (70%); mp 151–152 °C.

**4-Benzoylphenylalanine (3):**[148]

A suspension of **75** (45.4 g) in 8 M HCl and (188 mL) was heated under $N_2$ at 100 °C for 20 h. The mixture was cooled, the solid was collected and washed with 8 M HCl, and then with EtOH, and dried to give **3**·HCl; yield: 36.1 g (95%); mp 202–204 °C. The crude hydrochloride was dissolved in boiling $H_2O$ (480 mL) and filtered while hot; the filtrate was diluted with hot $H_2O$ (480 mL) and immediately neutralized (to pH 7) with 1 M NaOH (120 mL). The mixture was cooled on ice and the resulting fine solid **3** was collected by filtration, washed with $H_2O$, and vacuum-dried; yield: 31.0 g (92%); mp 217 °C (dec).

***N*-Acetyl-4-benzoylphenylalanine (76):**[148]

To a stirred soln of **3** (41.2 g, 153 mmol) in 1 M NaOH (600 mL) was added ice (600 g) and $Ac_2O$ (52 mL). The mixture was stirred for 5 min and was then acidified to pH 3 by the slow addition of 1 M HCl (500 mL). The fine crystalline product was filtered, washed with $H_2O$ and dried. Recrystallization (EtOH/hexanes) gave **76**; yield: 38.0 g (80%); mp 174–176.5 °C.

**Enzymatic Resolution of *N*-Acetyl-4-benzoylphenylalanine (76) To Give 4-Benzoyl-L-phenyllanine (L-3) and *N*-Acetyl-4-benzoyl-D-phenylalanine (D-76):**[148]

A suspension of **76** (6.24 g, 20 mmol) in $H_2O$ (1 L) was stirred and the pH adjusted to 7.5 with 4.7 M $NH_4OH$ (3.5 mL). The soln was filtered and treated with *Aspergillus acylase I* (Sigma; 200 mg), and 5 drops of toluene. The soln was stirred at 37 °C for 18 h, then cooled to 25 °C and filtered. The damp solid product was dissolved in 0.5 M HCl (50 mL) at 70 °C, Celite was added, the suspension was filtered, and the cake was washed with hot $H_2O$ (50 mL). The clear filtrate was brought to pH 7 by the addition of 1 M NaOH (7.3 mL). The solid was isolated by filtration (see below for filtrate processing), washed with $H_2O$ and a small amount of EtOH, and dried to give 4-benzoyl-L-phenylalanine (L-**3**) as the sesquihydrate; yield: 2.36 g (80%); mp 178–179 °C; $[\alpha]_D^{25}$ 3.0 ± 0.8 (*c* 1.01, 1 M HCl).

The aqueous filtrate from the acylase hydrolysis described above was acidified to pH 2.4 with 1 M HCl. The crystals were isolated (2.91 g) and recrystallized from boiling EtOAc (500 mL) to give *N*-acetyl-4-benzoyl-D-phenylalanine (D-**76**); yield: 2.72 g (87%); mp 186.5–187 °C; $[\alpha]_D^{25}$ −49.0 ± 0.8 (*c* 1.05, EtOH). Reaction with $Ac_2O$ converted a small aliquot of 4-benzoyl-L-phenylalanine (L-**3**) into *N*-acetyl-4-benzoyl-L-phenylalanine (L-**76**); mp 187–188 °C; $[\alpha]_D^{25}$ +50.9 ±0.8 (*c* 0.98, EtOH).

**4-Benzoyl-*N*-(*tert*-butoxycarbonyl)-L-phenylalanine (L-77):**[148]

To a stirred soln of L-**3** sesquihydrate (11.95 g, 40.2 mmol) and TEA (8.7 mL, 62 mmol) in $H_2O$ (125 mL) was added a soln of 99% 2-[(*tert*-butoxycarbonyl)oxyimino]-2-phenylacetonitrile (Boc-ON) (11.3 g, 45.8 mmol) in acetone (210 mL). After 3 h, acetone was removed under reduced pressure and the aqueous soln was extracted with $Et_2O$ (× 3). The aqueous layer was separated and acidified with 1 M HCl (53 mL) to pH 2. The soln was extracted with EtOAc and the organic layer was dried ($MgSO_4$) and evaporated to give a yellow oil; yield: 14.76 g. This was triturated for 5 days with petroleum ether. The resulting amorphous solid (13.4 g) crystallized very slowly (EtOAc, 60 mL) by the addition of hexanes (160 mL) to give white crystalline L-**77**; yield: 7.15 g (48%); mp 91–92 °C; $[\alpha]_D^{25}$ +18.5 ±0.8 (*c* 1.03, EtOH; UV (EtOH) $\lambda_{max}$ 260, 331 nm ($\varepsilon_{260}$ 18,000, $\varepsilon_{332}$ 180 $M^{-1} \cdot cm^{-1}$).

**3-(Bromomethyl)-9*H*-fluoren-9-one (80):**[153]

A mixture of 3-methyl-9*H*-fluoren-9-one (**79**) [prepared from 2-aminophenyl-4-tolyl ketone (**78**);[217–220] 5.82 g, 30 mmol)], NBS (5.34 g, 30 mmol), and $(BzO)_2$ (0.5 mg, 2.1 mmol) in $CCl_4$ (50 mL) was stirred at rt for 2 h, then refluxed for 2 h. The hot soln was filtered and washed with hot $CCl_4$ until the solid became almost colorless. The yellow crystals (6.71 g) obtained from the concentrated filtrate (80 mL) were filtered and washed with cold $CCl_4$ (5 mL). This crude product **80** contained **79** (ca. 5%) and the dibromo

*for references see p 147*

derivative (ca. 2%) as impurities. Recrystallization ($CCl_4$) gave **80** as yellow crystals, which was used in the following synthesis without further purification; yield 5.57 g (83% recovery); mp 148–149 °C.

**Ethyl 2-(Acetylamino)-2-cyano-3-(9-oxo-9*H*-fluoren-3-yl)propanoate (81):**[153]
A mixture of ethyl (acetylamino)(cyano)acetate (5.10 g, 30 mmol), **80** (5.46 g, 20 mmol), 18-crown-6 (528 mg, 2 mmol), and finely powdered $K_2CO_3$ (8.28 g, 60 mmol) in MeCN (60 mL) was stirred overnight at rt. The mixture was filtered through a Celite bed and the cake was washed with $CHCl_3$. The residue obtained after rotary evaporation was dissolved in hot EtOH (100 mL), treated with activated charcoal, filtered, and diluted with hexanes (100 mL) affording **81** as bright yellow crystals; yield: 6.01 g (83%); mp 162–163 °C.

**3-(9-Oxo-9*H*-fluoren-3-yl)alanine (69):**[153]
A mixture of **81** (5.79 g, 16 mmol) and 8 M HCl (40 mL) was heated at 100 °C for 20 h, then cooled to rt and the precipitate was filtered, washed with 8 M HCl, and dried affording a yellow amorphous powder. This crude hydrochloride was dissolved in boiling $H_2O$ (60 mL), filtered, and immediately neutralized to pH 7 with 1 M NaOH. The fine precipitate of **69**, formed after cooling the filtrate in an ice–water bath, was filtered, washed with cold $H_2O$, and dried to give a yellow amorphous powder; yield: 3.47 g (81%); mp 195 °C (dec).

***N*-Acetyl-3-(9-oxo-9*H*-fluoren-3-yl)alanine (82):**[153]
To a soln of **69** (3.47 g, 13 mmol) in 1 M NaOH (60 mL) were added ice (60 g) and $Ac_2O$ (5 mL). The mixture was stirred for 30 min, then acidified to pH 3 by the slow addition of 1 M HCl. The fine crystalline product was filtered, washed with $H_2O$, and dried affording **82** as a yellow powder; yield: 3.50 g (87%); mp 211–212 °C.

**Enzymatic Resolution of *N*-Acetyl-3-(9-oxo-9*H*-fluoren-3-yl)alanine (82) To Give (2*S*)-3-(9-Oxo-9*H*-fluoren-3-yl)alanine [(*S*)-69] and (2*R*)-*N*-Acetyl-3-(9-oxo-9*H*-fluoren-3-yl)alanine [(*R*)-82]:**[153]
A suspension of **82** (3.09 g, 10 mmol) in $H_2O$ (500 mL) was stirred and neutralized with 3 M $NH_4OH$ to pH 7.5. The soln was filtered, and *Aspergillus acylase I* (Sigma; 100 mg) and 3 drops of toluene were added to the filtrate. The soln was incubated at 37 °C for 18 h, then cooled to 25 °C and filtered (see below for filtrate processing). The damp solid product was dissolved in 0.5 M HCl (25 mL) at 70 °C, Celite was added and the mixture was filtered and the cake was washed with hot $H_2O$ (30 mL). The clear soln was neutralized to pH 7 with 1 M NaOH. The solid was collected by filtration, washed with $H_2O$, and dried affording (*S*)-**69** as a yellow powder; yield: 1.12 g (42%): mp 195 °C (dec); $[\alpha]_D^{25}$ +3.4 (*c* 0.63, 0.1 M $NH_4OH$).

The aqueous filtrate obtained from the acylase-mediated hydrolysis above was acidified with 1 M HCl to pH 2.4. A solid formed, which was filtered and washed with $H_2O$ and dried in vacuo affording (*R*)-**82** as yellow crystals; yield: 1.38 g (46%); mp 203–204 °C; $[\alpha]_D^{25}$ −53.2 (*c* 0.74, AcOH).

**(2*S*)-*N*-(*tert*-Butoxycarbonyl)-3-(9-oxo-9*H*-fluoren-3-yl)alanine [(*S*)-83]:**[153]
To a suspension of (*S*)-**69** (267 mg, 1 mmol) and TEA (101 mg, 1 mmol) in $H_2O$ (20 mL) was added $Boc_2O$ (240 mg, 1.1 mmol). After being stirred at rt for 3 h, the soln was extracted with $Et_2O$ (5 × 20 mL). The combined extracts were washed with $H_2O$ (2 × 10 mL), dried, and concentrated in vacuo. Purification by flash column chromatography (EtOAc/hexanes/AcOH 20:79:1) afforded (*S*)-**83** as a yellow powder; yield: 334 mg (91%); mp 168 °C (dec); $[\alpha]_D^{25}$ +19.7 (*c* 0.94, MeOH).

**4-Benzoylbenzoyl Chloride (84):**[152]
A soln of 4-benzoylbenzoic acid (**7**; 4.37 g, 19.3 mmol) in $CH_2Cl_2$ (120 mL) was treated with $(COCl)_2$ (3.37 mL, 38.6 mmol) under $N_2$ at rt for 2 h. The solvent was removed under vacuum. The residue was redissolved in $CH_2Cl_2$ (50 mL) and the soln was evaporated to dryness, yielding the crude acid chloride **84**, which was used in the next step without further purification.

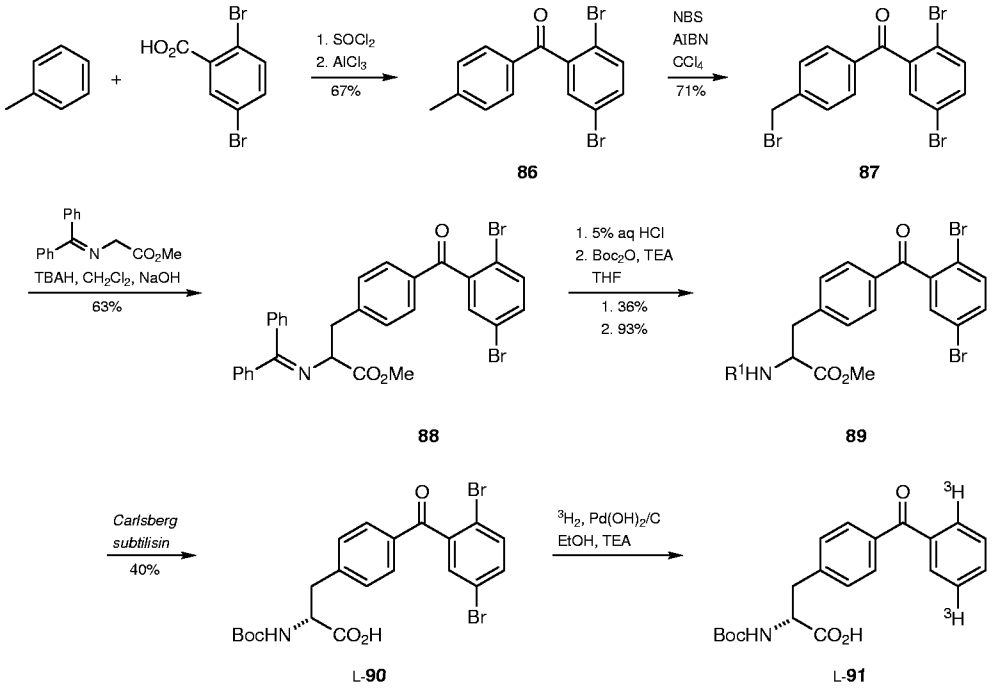
***N*$^{\varepsilon}$-(4-Benzoylbenzoyl)-*N*$^{\alpha}$-(*tert*-butoxycarbonyl)lysine Methyl Ester (85):**[152]
To a soln of *N*$^{\alpha}$-Boc-Lys-OMe · HCl (5.0 g, 16.8 mmol) in $CH_2Cl_2$ (80 mL) and TEA (4.7 mL, 33.6 mmol) was added a soln of the crude acid chloride in $CH_2Cl_2$ (30 mL) dropwise at 0 °C under $N_2$. The mixture was allowed to reach rt and left for 10 h, during which time the pH was maintained at 7–8 by dropwise addition of TEA. The mixture was washed with $H_2O$ (30 mL). The residue obtained after removal of $CH_2Cl_2$ under vacuum was dissolved in EtOAc and the organic soln was washed consecutively with sat. solns of $KHSO_4$ (3 × 30 mL), $NaHCO_3$ (3 × 30 mL), and brine (2 × 30 mL). The organic phase was dried ($MgSO_4$), filtered, and evaporated to dryness under vacuum. The solid residue obtained was recrystallized ($Et_2O$) to afford a white powder; yield: 6.1 g (78%).

**$N^{\varepsilon}$-(4-Benzoylbenzoyl)-$N^{\alpha}$-(*tert*-butoxycarbonyl)lysine** (68, **$R^1$ = Boc**):[152]
1 M NaOH (26 mL) was added dropwise to a stirred soln of **85** (6.1 g, 13.08 mmol) in MeOH/$H_2O$ (80 mL, 8:1) over a period of 1 h. After 1 h stirring at rt, the MeOH was removed under vacuum and the residual basic aqueous soln was further diluted with $H_2O$ (20 mL) and washed with EtOAc. The ice-cold aqueous phase was acidified to pH 4 with cold sat. $KHSO_4$ and extracted with EtOAc (3 × 30 mL). The combined organic phase was washed consecutively with $H_2O$ (3 × 30 mL) and brine (30 mL), dried ($MgSO_4$), and filtered, and the solvent was removed under vacuum. The pure product was obtained as a white powder following recrystallization ($Et_2O$/petroleum ether); yield: 3.4 g (57%); mp 67 °C.

## 9.3.5.4 Radiolabeled Benzophenone-Containing Photophores

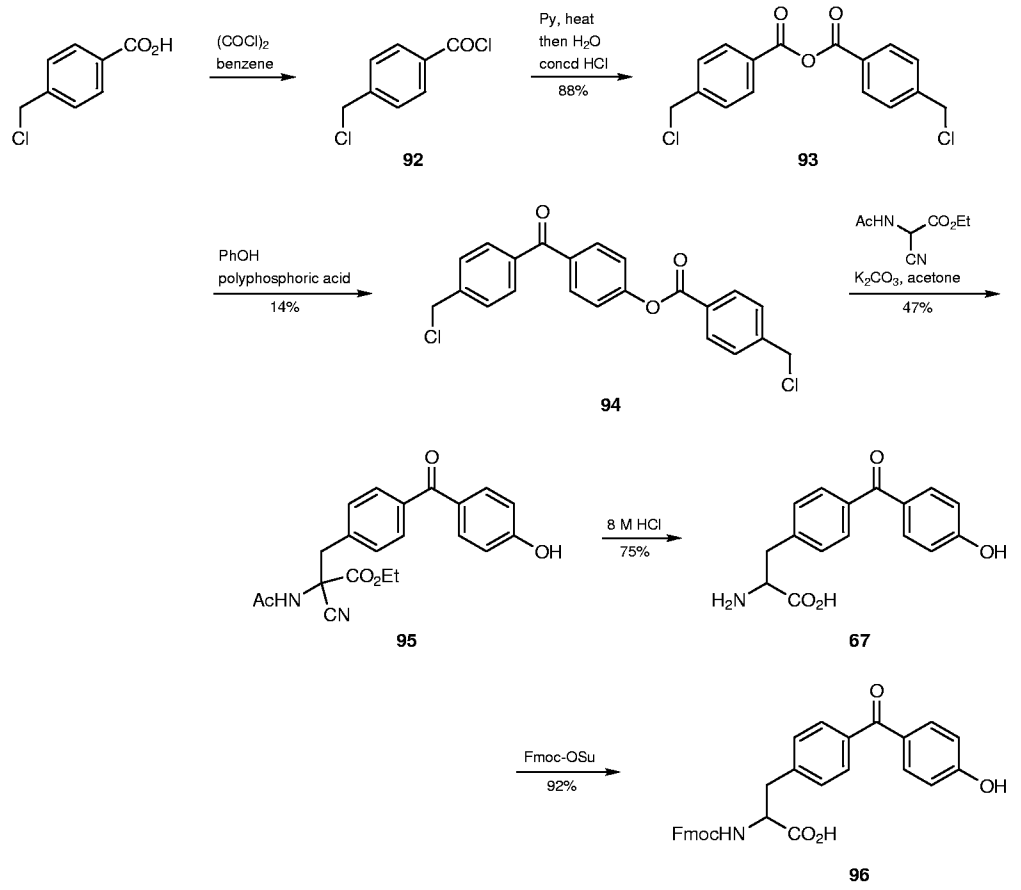
Several applications of photoreactive peptides require the presence of a radionuclide to allow specific and sensitive detection of the photo-cross-linked conjugates. In several cases, radioiodination of tyrosyl moieties and radiolabeled Bolton–Hunter reagents have been used. However, the presence of a radiolabel within the benzophenone photophore is desirable, particularly when the objective is to identify the site of photo-insertion of benzophenone. To this end some radiolabeled, benzophenone-based compounds have been developed and used in peptide synthesis, in particular tritiated Phe(4-Bz) (Scheme 24),[216] [1-$^{14}$C-carboxy]-4-benzoylbenzoic acid,[221] and 4-benzoyl-(2,3-$^3H_2$)-dihydrocinnamic acid.[154] In addition, 4-(4-hydroxybenzoyl)phenylalanine (Scheme 25) has been directly radioiodinated with $Na^{125}I$ and Chloramine-T.[151]

**Scheme 24** Synthesis of Tritiated *N*-(*tert*-Butoxycarbonyl)-4-benzoylphenylalanine[216]




*for references see p 147*

**Scheme 25** Synthesis of *N*-Fmoc-4-(4-Hydroxybenzoyl)phenylalanine[151]




**2,5-Dibromophenyl 4-Tolyl Ketone** (86):[216]
Refluxing a soln of 2,5-dibromobenzoic acid (950 mg, 3.35 mmol) in $SOCl_2$ (13 mL) generated 2,5-dibromobenzoyl chloride. The excess of the reagent was evaporated in vacuo, and the residue was dissolved in freshly distilled toluene (4.5 mL) at 45 °C. The soln was cooled to rt and anhyd $AlCl_3$ (1.09 g) was added in five portions. The mixture became dark green and the temperature was raised to 60 °C. After 2 h, the mixture was poured onto a mixture of crushed ice (14 g) and concd HCl (5 mL) and stirred for 30 min. The aqueous suspension was extracted with $Et_2O$ (2 × 10 mL). The ethereal extracts were washed with sat. $NaHCO_3$, $H_2O$, and brine, dried ($MgSO_4$), and concentrated. The isomers were separated by silica gel chromatography (25% $CH_2Cl_2$ in hexanes) to give pure **86** as a reddish white powder; yield: 770 mg (67%). Also isolated were some mixed fractions (50 mg) and pure 2-methyl isomer (65 mg).

**4-(2,5-Dibromobenzoyl)-*N*-(diphenylmethylene)phenylalanine Methyl Ester** (88):[216]
The benzylic bromination of **86** was carried out according to Itoh[222] in 71% yield. The crude material was partially purified by flash chromatography (5% EtOAc in hexanes), and the resulting 4-(bromomethyl)phenyl 2,5-dibromophenyl ketone (**87**) was used to alkylate the glycine methyl ester Schiff's base. The phase transfer reaction was carried out in a $CH_2Cl_2$/10% aq NaOH (5 mL of each) two-phase system containing glycine methyl ester Schiff's base (0.975 g, 3.85 mmol), **87** (2.0 g, 4.24 mmol, ca. 90%), and tetrabutylammonium hydrogen sulfate (TBAH) (1.30 g, 3.85 mmol). The more polar alkylated product was separated by column chromatography (silica gel, 75% $CH_2Cl_2$ in hexanes, then $CH_2Cl_2$) resulting in **88** as an oil; yield: 1.94 g (63%).

**4-(2,5-Dibromobenzoyl)phenylalanine Methyl Ester** (89, $R^1 = H$):[216]
Compound **88** was stirred in a mixture of THF (2 mL) and 5% aq HCl (6 mL) for 2 h. Sat. $NaHCO_3$ was added and the basic mixture was extracted with EtOAc (3 × 5 mL). The combined organic extracts were washed with $H_2O$ and brine (3 mL each), dried, and evaporated. The crude product was chromatographed (silica gel, EtOAc/hexanes 3:1) to give **89** ($R^1 = H$) as a white, oily powder; yield: 270 mg (36%).

***N*-(*tert*-Butoxycarbonyl)-4-(2,5-dibromobenzoyl)phenylalanine Methyl Ester (89, $R^1$ = Boc):**[216]
Compound **89** ($R^1$ = H; 315 mg, 0.72 mmol) was dissolved in dry THF (1.5 mL) and TEA (0.11 mL) was added at 0 °C. The mixture was treated with $Boc_2O$ (0.18 mL, 0.77 mmol). After being stirred for 2 h, the mixture was concentrated in vacuo and the residue taken up in EtOAc (10 mL). The organic phase was washed with $H_2O$ (3 mL) and brine (3 mL), dried ($MgSO_4$), and evaporated. The product was purified by chromatography (silica gel, EtOAc/hexanes 1:2) giving **89** ($R^1$ = Boc); yield: 315 mg (93%); mp 111–112 °C.

**Enzymatic Resolution of *N*-(*tert*-Butoxycarbonyl)-4-(2,5-dibromobenzoyl)phenylalanine Methyl Ester (89, $R^1$ = Boc) To Give *N*-(*tert*-Butoxycarbonyl)-4-(2,5-dibromobenzoyl)-L-phenylalanine (L-90) and *N*-(*tert*-Butoxycarbonyl)-4-(2,5-dibromobenzoyl)-D-phenylalanine Methyl Ester (D-89, $R^1$ = Boc):**[216]
The purified methyl ester **89** (100 mg, 0.185 mmol) was dissolved in acetone (0.3 mL) and *Carlsberg subtilisin* (5 mg) in 0.2 M $NaHCO_3$ buffer (1 mL, pH 8.16) was added. The clear soln was stirred vigorously at 37 °C, and soon the hydrolyzed product precipitated. The progress of the hydrolysis was monitored by TLC (10% MeOH in $CHCl_3$) until 50% of the substrate was hydrolyzed (ca. 8 h). The reaction was quenched by addition of 1 M HCl (1 mL), and the mixture was extracted with EtOAc (3 × 1.5 mL). The organic extracts were evaporated, and the products were separated by column chromatography (silica gel, 8% MeOH in $CHCl_3$) to give L-**90**; yield: 39 mg (40%); and unreacted D-**89** (R′ = Boc; 45 mg).

***N*-(*tert*-Butoxycarbonyl)-4-(2,5-ditritiobenzoyl)-L-phenylalanine (L-91):**[216]
A mixture of L-**90** (10 mg, 0.019 mmol) and 5% $Pd(OH)_2/C$ (4 mg) suspended in EtOH (2 mL) containing TEA (0.2 mL) was stirred under an atmosphere of carrier-free tritium gas for 90 min. Labile tritium was removed and the mixture filtered through a bed of Celite (2.5 × 0.6 cm), washing with MeOH (20 mL). The products were purified (>97%) by HPLC [Zorbax C8 column, MeCN/0.1% TFA in $H_2O$ (2:3)]. The specific activity of L-**91** was 36.6 Ci · $mmol^{-1}$ as determined by UV spectrometry using $\varepsilon_{265}$ 26568 $cm^{-1} \cdot M^{-1}$.

**4-(Chloromethyl)benzoyl Chloride (92):**[151]
A mixture of 4-(chloromethyl)benzoic acid (50 g, 0.29 mol) and $(COCl)_2$ (100 g, 0.79 mol) in anhyd benzene (100 mL) (**CAUTION:** *carcinogen!*) was stirred at 25 °C and then refluxed for 2 h. Unreacted $(COCl)_2$ and benzene were removed by distillation. Traces of the remaining reagents were removed under reduced pressure to give **92** as a colorless liquid. The acid chloride was used in the next step without further purification.

**4-(Chloromethyl)benzoic Anhydride (93):**[151]
A soln of **92** (14.2 g, 0.075 mol) in anhyd pyridine (36 mL, 0.45 mol) was warmed gently over a steam bath for 10 min. The mixture was poured over a slurry of ice (75 g) and concd HCl (37.5 mL). The solid residue was filtered under suction on a Buchner funnel and washed successively with MeOH (3 × 6 mL) followed by benzene (3 × 6 mL) (**CAUTION:** *carcinogen!*). The residue was dried under high vacuum to give **93**; yield: 10.6 g (88%).

**[4-(Chloromethyl)benzoyl]phenyl 4-(Chloromethyl)benzoate (94):**[151]
A mixture of **93** (32.3 g, 0.1 mol) and PhOH (3.76 g, 0.04 mol) in polyphosphoric acid (100 g) was stirred at 150 °C for 2 h. The mixture was cooled to 80 °C and poured over stirred crushed ice (500 g). The resulting precipitate was filtered on a Buchner funnel and thoroughly washed with $H_2O$. The solid was suspended in $H_2O$ (400 mL) and the pH adjusted to 7.0 under vigorous stirring with 10 M NaOH. After stirring for 4 h, the suspension was extracted with $CHCl_3$ (2 × 300 mL) (NaCl was added to separate the phases). The combined $CHCl_3$ extracts were taken to dryness on a rotary evaporator. The residue was stirred in benzene (50 mL) (**CAUTION:** *carcinogen!*) and filtered. The soln was evaporated on a rotary evaporator and the residue dried under high vacuum to give a solid; yield: 10.4 g. TLC analysis indicated the presence of two products, $R_f$ 0.14 and 0.41 (benzene), which were identified by NMR spectroscopy as **94** and phenyl 4-(chloromethyl)benzoate, respectively.

The crude solid was suspended in benzene/hexanes (1:1, 100 mL), heated, and decanted after cooling. The supernatant was applied to a column of silica gel G (70 g, benzene/hexanes 1:1) to separate the benzophenone derivative from the phenyl benzoate contaminant. The column was eluted with the same solvent (250 mL), followed by benzene/hexanes (3:1, 350 mL), and finally with benzene. Fractions (50 mL) were collected. Fractions containing the component with $R_f$ 0.14 (silica gel, benzene) were pooled and concentrated on a rotary evaporator. Recrystallization (abs EtOH) of the residue [2.4 g (15% based on starting phenol)] gave colorless needles of **94**; yield: 2.2 g (14%); mp 137–138 °C.

*for references see p 147*

**N-Acetyl-2-cyano-4-(4-hydroxybenzoyl)phenylalanine Ethyl Ester (95):**[151]
A stirred mixture of **94** (798 mg, 2 mmol), ethyl (acetylamino)(cyano)acetate (701 mg, 4.12 mmol), $K_2CO_3$ (382 mg, 2.76 mmol), and KI (8.3 mg, 0.05 mmol) in acetone was refluxed in the dark under argon for 20 h. After cooling, the solid precipitate was filtered and washed with acetone. The crude product was purified by preparative TLC ($R_f$ 0.24, $CHCl_3$/EtOH 94:6) to give **95** as a glassy solid; yield: 360 mg (47%). [TLC purification resulted in loss of material, apparently because the crude material also contained the phenolic ester of *N*-acetyl-2-cyano-4-(4-hydroxybenzoyl)phenylalanine.]

**4-(4-Hydroxybenzoyl)phenylalanine (67):**[151]
The crude reaction product **95** was suspended in 8 M HCl (5 mL) and refluxed under argon in the dark for 20 h. After concentration on a rotary evaporator, the residue was suspended in $H_2O$ (10 mL) and the pH adjusted to 7.0 with 1 M NaOH. After vigorous stirring for 1 h, the precipitate was filtered and washed with $H_2O$. The dried precipitate was resuspended in EtOH (15 mL) and refluxed. After cooling, the precipitate was filtered, washed with EtOH, and dried under high vacuum to give **67**; yield: 427 mg (75% yield based on **95**).

***N*-[(9*H*-Fluoren-9-ylmethoxy)carbonyl]-4-(4-hydroxybenzoyl)phenylalanine (96):**[151]
Compound **67** (1.4 mmol) was dissolved in 10% $NaHCO_3$ (5 mL) and treated with Fmoc-OSu (1.4 mmol) in acetone (5 mL) for 18 h. TLC ($R_f$ 0.3, $CH_2Cl_2$/MeOH/AcOH 93:5:2) showed the reaction to be ca. 95% complete. Acetone was removed by evaporation. The clear yellow aqueous soln was further diluted with $H_2O$ to a final volume of 25 mL and extracted with $Et_2O$ (2 × 20 mL). EtOAc was added, and the mixture was acidified to pH ca. 2. The organic phase was evaporated, and the residue was coevaporated with $CH_2Cl_2$ (2 × 50 mL). The solid obtained was crystallized ($CH_2Cl_2$, 30 mL) to give **96**; yield: 92%.

**[L-Phe$^8${4-Bz(4-OH)}]-Substance P:**[151]
The synthesis of substance P (SP) analogue H-Arg-Pro-Lys-Pro-Gln-Gln-Phe-L-Phe[4-Bz(4-OH)]-Gly-Leu-Met-$NH_2$ was performed using a standard Fmoc solid-phase synthetic strategy. The crude synthetic peptide was purified by HPLC using a Synchrom C18 column (250×4.6 mm). The crude racemic peptide ([D,L-Phe$^8${4-Bz(4-OH)}]-SP) had two major UV-absorbing (215 nm) peaks of approximately equal intensity eluting at 51.0 and 54.4% in MeCN. The [L-Phe$^8${4-Bz(4-OH)}]-SP elutes earlier than [D-Phe$^8${4-Bz(4-OH)}]-SP on RP-HPLC. The first peak was more active than the second in both a radioimmunoassay using SP specific antibody and and in radioligand binding to SP receptor, as expected for the L-diastereomer. The isolated [L-Phe$^8${4-Bz(4-OH)}]-SP was analyzed for purity and correct structure by amino acid analysis, mass spectroscopy, and sequencing. The peptide sequence was confirmed by Edman sequencing and mass spectrometry (MALDI-MS) gave *m/z* 1467 ($[M+1]^+$) as expected. The phenylthiohydantoin derivative of 4-(4-hydroxybenzyl)phenylalanine eluted from the sequencer column under standard conditions at 15.3 min. The extinction coefficient for the peptide ([L-Phe$^8${4-Bz(4-OH)}]-SP) was determined to be 16600 $cm^{-1} \cdot M^{-1}$ at 292 nm in MeCN/DMSO (6:1) by absorbance spectroscopy, using amino acid analysis for quantitation of the peptide. Like Phe(4-Bz), Phe[4-Bz(4-OH)] was not detected by conventional amino acid analysis (ionexchange using postcolumn ninhydrin detection), but the remaining amino acids gave the expected molar ratios for SP with Phe[4-Bz(4-OH)] substituted for Phe$^8$.

**Radioiodination of [Phe$^8${4-Bz(4-OH)}]-Substance P:**[151]
The dry peptide (13.5 μg, 10 nmol) was dissolved in 0.5 M borate buffer (50 μL), pH 8.0, and vortex-mixed with 1 mCi of Na$^{125}$I (Amersham IMS-30). Chloramine-T (10 μg, 44 nmol) was added, and the mixture was vortex-mixed for 90 s before the reaction was quenched with $Na_2S_2O_5$ (100 μg, 530 nmol). To separate the peptide from free $^{125}$I, the mixture was diluted and acidified with 60 mM TFA (0.525 mL), with the addition of 2% BSA (25 μL) soln to limit nonspecific adsorption. The mixture was then applied to an activated C18 Sep-Pak cartridge (Waters), and the iodide and peptide were eluted from the cartridge with a series of 0.5-mL portions of 10 mM TFA in increasing alcohol (EtOH/MeOH 1:1) content. The iodide eluted immediately, whereas the peptide was retained until the alcohol content reached about 60%. The peptide fractions (containing both oxidized and reduced methionine) eluting with 60–90% alcohol were pooled and reduced in volume under a $N_2$ stream. After the addition of 20% β-ME, the sample was heated at 90 °C for 2 h to reduce methionine sulfoxide to native methionine. Further purification was achieved by RP-HPLC on a Vydac C18 column. The eluate was collected in fractions during shallow MeCN gradient elution, and the fractions were counted for radioactivity. The reduced (methionine) monoiodinated Phe$^8$[4-Bz(4-OH)] peptide eluted in a well-resolved peak (at 32% MeCN), predictably later than the original unlabeled peptide (at 30% MeCN) or the oxidized products, but prior to the reduced diiodinated peptide. The reduced monoiodinated [$^{125}$I][Phe$^8${4-Bz(4-OH)}]-SP (specific activity ca. 2000 Ci·$mmol^{-1}$; 1 Ci = 37 GBq) was protected from oxidation with 0.5% β-ME added immediately after purification and stored at −20 °C until use. The amount of free $^{125}$I in the final tracer was determined by quantitative separation of free $^{125}$I and the $^{125}$I-peptide using an activated C18 Sep-Pak cartridge (Waters) as described above.

## 9.3.6 Photocaged Peptides

"Caged" compounds are synthetic molecules whose biological activity is controlled by light.[223] Caged ATP was the first example of this type of "photoprotected" substrate. A photolytic reaction, which cleaves a single bond, converts the inactive form of the molecule into an active one or vice versa. Light-activated caged peptides are potentially powerful tools for controlling the release of peptides into cells in order to overcome diffusional delays, circumvent microinjection artifacts, and target selected proteins in a time domain which is inaccessible by antisense or gene knockout techniques.[224] In addition, caged peptides can provide a means to spatially map out selected protein activities within living cells.

For this discussion, bioactive peptides will be defined as peptides which interact specifically with a target macromolecular acceptor or are derived from domains involved in a critical protein–protein interaction and, therefore, can compete effectively to mimic or disrupt this bimolecular interaction. Once the structure–activity relationship of a bioactive peptide is revealed, one can identify the termini and/or positions in which introduction of a caging group will be disruptive for target recognition. Alternatively, caging the peptide in an inactive conformation can be accomplished by end-to-end or end-to-side-chain cyclization.

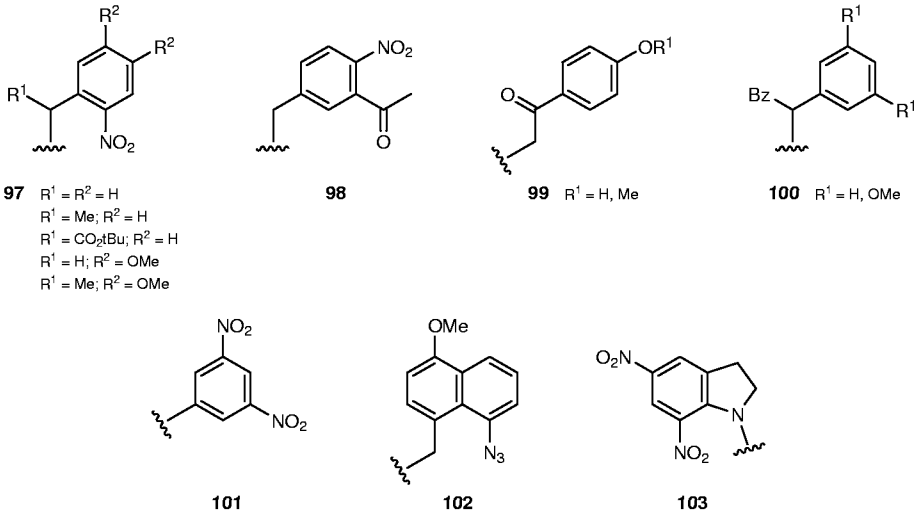
For biological applications, the optimal caging or phototrigger group must fulfill the following criteria: (1) The photoprotected compound should be photolabile at wavelengths that are nondetrimental to the biological system ($\lambda_{max}$ >300 nm), and photolized effectively (preferably with quantum efficiency >0.1) and rapidly to render the "naked" compound in the original active form. (2) The photoprotected compound must be soluble in aqueous buffered media. (3) The caging or phototrigger group must be stable to hydrolysis, especially at high ionic strength solutions. (4) The caged compound should be biologically benign, while the active form should display good selectivity and high affinity for its target. (5) The photoproducts should not absorb at the photolyzing wavelength to avoid a light-filtering effect by the product at high conversion. (6) The photolyzing reaction should be primarily photochemical, i.e. cleaving directly the bond between the caging moiety and the nascent bioactive compound from the reactive excited state, thus avoiding any long-lived intermediates prior to the release of the active compound.[225]

Caged peptides are synthesized by incorporation of an appropriately protected, caged amino acid. Selection of the caged amino acid will dictate the chemistry suitable for the introduction of the caging group and therefore the selection of the caging moiety. Scheme 26 summarizes most of the frequently used photocaging groups, **97–103**.

This section offers examples of syntheses of three major photocaged amino acid families: 2-nitrobenzyl **97** (Section 9.3.6.1), desyl **100** (Section 9.3.6.2), and 4-hydroxyphenacyl **99** (Section 9.3.6.3)-protected amino acids. Many of these photocaged amino acids are derivatives of important neuroexcitatory amino acids and have been developed as tools for studying mechanistic aspects of neurotransmission. This section does not cover the various photoreactive protecting groups developed for orthogonal protection schemes used in peptide synthesis. Only those protecting groups that are also used as photocaging moieties are included in this section. In addition, examples of chemical or biochemical syntheses of photocaged peptides and proteins are included (see Section 9.3.6.4).

*for references see p 147*

**Scheme 26** Structures of Photocaging Moieties




**97** $R^1 = R^2 = H$; $R^1 = Me$; $R^2 = H$; $R^1 = CO_2tBu$; $R^2 = H$; $R^1 = H$; $R^2 = OMe$; $R^1 = Me$; $R^2 = OMe$

**98**

**99** $R^1 = H, Me$

**100** $R^1 = H, OMe$

**101**

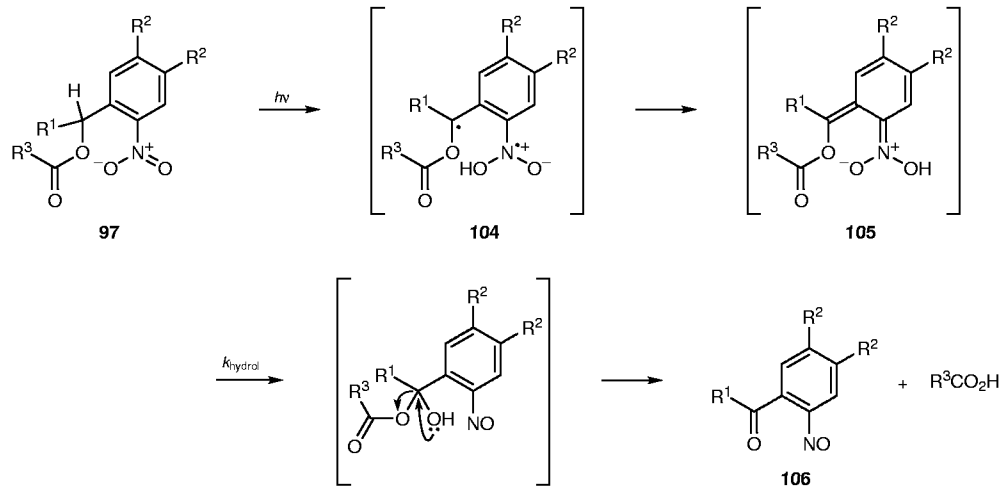
**102**

**103**

## 9.3.6.1 Synthesis of 2-Nitrobenzyl-Protected Amino Acids

The use of 2-nitrobenzyl esters **97** as photosensitive protecting groups for carboxylic acids in organic synthesis was first reported by Barltrop et al.[226] They suggest that a photoinduced intramolecular redox mechanism is responsible for the photolysis and concomitant release of 2-nitrosobenzaldehyde (**106**) and the corresponding carboxylic acid. Upon near-UV irradiation it is generally accepted that the primary photochemical event on the 2-nitrobenzyl group is an n→π* transition of the nitro group to give intermediate **104**, followed by intramolecular hydrogen transfer to form the *aci*-nitro intermediate **105**, which occurs within picoseconds (Scheme 27).[227]

**Scheme 27** Mechanism of the Photolysis of 2-Nitrobenzyl-Containing Photocaging Moieties[227]




**97** **104** **105**

**106**

$R^1 = H, Me$; $R^2 = H, OMe$

Patchornik et al. have developed the 2-nitrobenzyl-based amino protecting groups 4,5-dimethoxy-2-nitrobenzyloxycarbonyl (Nvoc) and 2-nitrobenzyloxycarbonyl [Z(2-$NO_2$)], which

can be removed by irradiation at $\lambda_{max}$ >320 nm.[228] However, reaction between the released aldehyde and the free α-amino group is blocked in the presence of aldehyde scavengers such as hydrazine, hydroxylamine hydrochloride, or semicarbazide hydrochloride. To avoid the formation of the aldehyde upon photocleavage of Nvoc and Z(2-$NO_2$) Patchornik et al. introduced bis(2-nitrophenyl)methoxycarbonyl, derived from an α-substituted benzyl alcohol, as an α-amino protecting group. Upon photolysis this yields the corresponding ketone, which is less reactive towards amino groups than the aldehyde moiety obtained from Nvoc and Z(2-$NO_2$).[228]

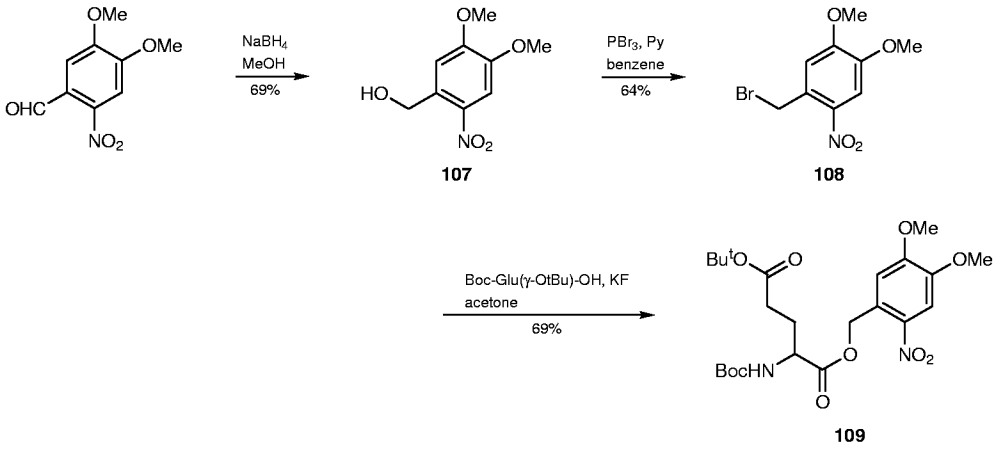
Kaplan et al. introduced the 1-(2-nitrophenyl)ethyl moiety as a more effective caging group[223] following Patchornik's suggestion that photolysis of α-substituted 2-nitrobenzyl alcohols will result in better yields. Indeed, the photogenerated acetophenone is less reactive and less toxic than 2-nitrosobenzaldehyde, particularly in the presence of scavengers such as thiols and semicarbazides.[223] However, both the 2-nitrobenzyl (**97**, $R^1 = R^2 = H$) and the 1-(2-nitrophenyl)ethyl (**97**, $R^1 = Me$; $R^2 = H$) groups absorb weakly at wavelengths >340 nm. The 4,5-dimethoxy-2-nitrobenzyl (**97**, $R^1 = H$; $R^2 = OMe$) has much higher absorbance in the convenient region 350–400 nm and faster release kinetics than the 1-(2-nitrophenyl)ethyl group but it also has lower quantum yields.[229] Combination of the 1-(2-nitrophenyl)ethyl and the 4,5-dimethoxy-2-nitrobenzyl groups resulted in the 1-(4,5-dimethoxy-2-nitrophenyl)ethyl (**97**, $R^1 = Me$; $R^2 = OMe$) group, which showed both improved quantum yield and absorption at $\lambda_{max}$ >340 nm, but failed to show fast release kinetics.[229,230] Introduction of a negatively-charged carboxylate group as in 1-carboxy-2-nitrobenzyl **97** ($R^1 = CO_2tBu$; $R^2 = H$) improves the release kinetics.[231,232] Both phenylethyl-based caging groups, the 1-(2-nitrophenyl)ethyl and 1-(4,5-dimethoxy-2-nitrophenyl)ethyl, are chiral, and therefore when coupled to chiral biomolecules they generate diastereomers. This presents a drawback, since the two resultant diastereomers may have different biological and photochemical properties.

The development of amino-photocaging groups by converting the carboxylate and phosphate photocaging groups into the corresponding oxycarbonyl-containing groups, as has been done in the Nvoc and Z(2-$NO_2$) protecting groups, is problematic.[228,233] The photolysis is followed by a reaction in the dark, in which carbon dioxide is released from the carbamate. This reaction could conceivably become the rate-determining step, which would compromise the sought burst kinetics required in many photocaging applications.

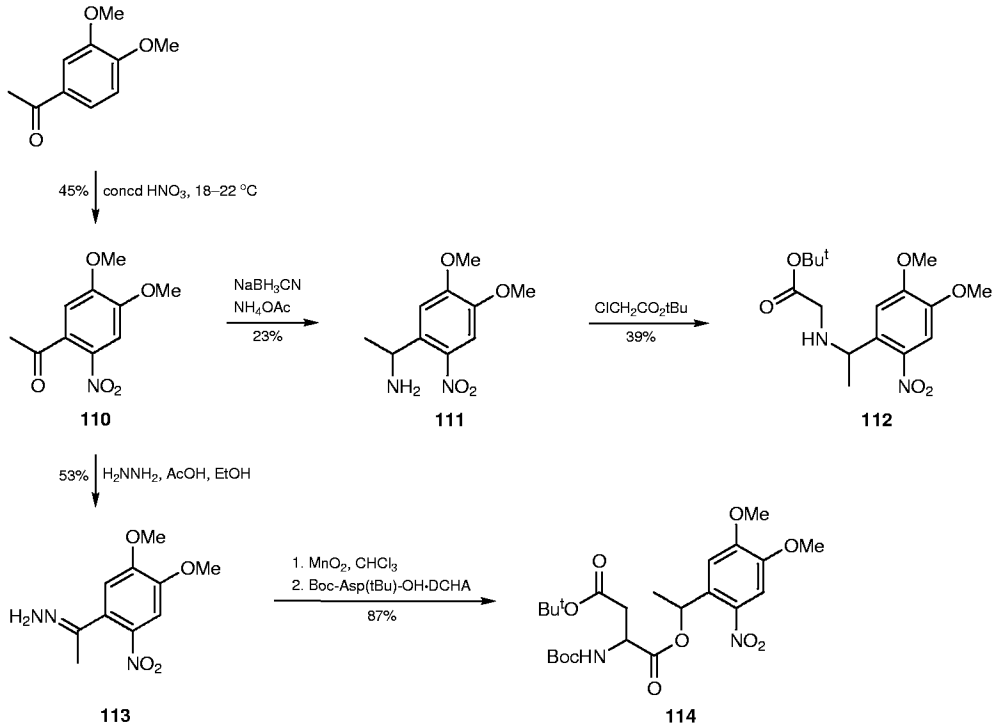
In this section we provide experimental procedures for the synthesis of various 2-nitrophenyl-containing amino acid derivatives; glutamic acid derivative **109**[230] (Scheme 28), glycine derivative **112**[230] and aspartic acid derivative **114**[230] (Scheme 29), Fmoc-protected tyrosine derivative **119**[234] (Scheme 30), and glycine derivatives **120**[235] and **122**[232] (Scheme 31).

*for references see p 147*

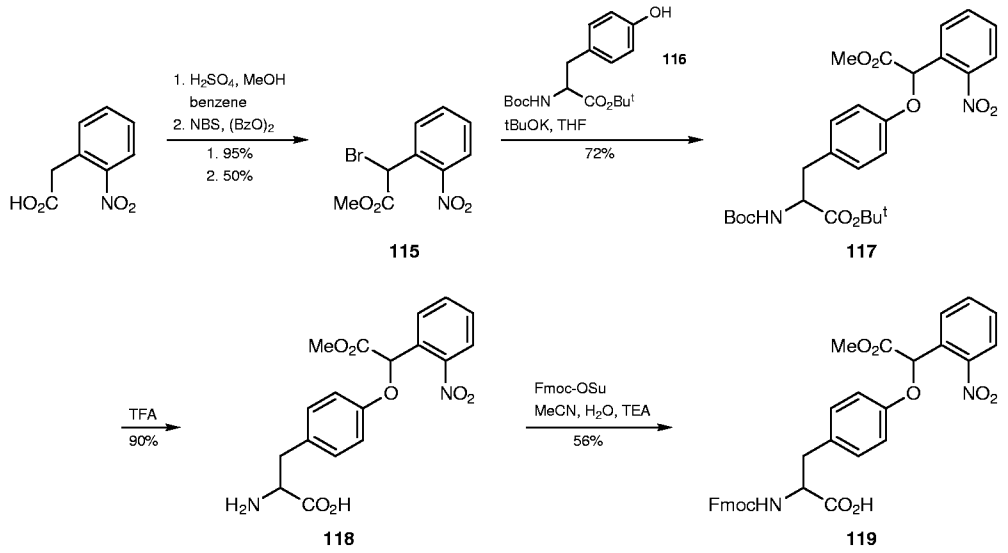
**Scheme 28** Preparation of 5-*tert*-Butyl 1-(4,5-Dimethoxy-2-nitrobenzyl) *N*-(*tert*-Butoxycarbonyl)-L-glutamate Diester[230]



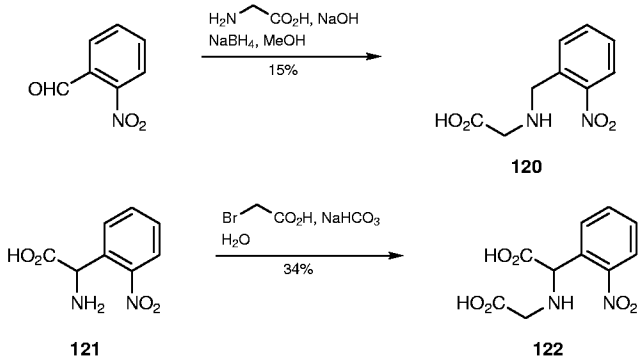

**Scheme 29** Preparation of *tert*-Butyl {[1-(4,5-Dimethoxy-2-nitrophenyl)ethyl]amino}acetate and 4-*tert*-Butyl 1-(4,5-Dimethoxy-2-nitrophenyl)ethyl *N*-(*tert*-Butoxycarbonyl)aspartate Diester[230]



**Scheme 30** *N*-[(9*H*-Fluoren-9-ylmethoxy)carbonyl]-4-[2-methoxy-1-(2-nitrophenyl)-2-oxoethoxy]phenylalanine[234]




**Scheme 31** Preparation of *N*-(2-Nitrobenzyl)glycine and *N*-[(2-Nitrophenyl)carboxymethyl]glycine[232,235]




**(4,5-Dimethoxy-2-nitrophenyl)methanol (107):**[230]

To a stirred mixture of 80% 4,5-dimethoxy-2-nitrobenzaldehyde (2.0 g, 9.5 mmol) and MeOH (40 mL) was added $NaBH_4$ (0.18 g, 4.8 mmol). After 1 h, $H_2O$ (10 mL) was added, and the mixture was extracted with $CHCl_3$ (3 ×). The combined $CHCl_3$ layers were dried and concentrated to an oil, which was purified by chromatography (silica gel, $CH_2Cl_2$); yield 1.4 g (69%); mp 145–147 °C.

**1-(Bromomethyl)-4,5-dimethoxy-2-nitrobenzene (108):**[230]

To a chilled soln of **107** (0.372 g, 1.74 mmol) in benzene (50 mL) (**CAUTION**: *carcinogen!*) and 3 drops of dry pyridine, was added, dropwise, $PBr_3$ (0.27 g, 0.093 mL, 1.0 mmol) in benzene (2 mL). After 24 h, $H_2O$ (5 mL) was added, and the soln extracted with $Et_2O$. The solvent was removed, and the crude product was recrystallized (toluene: pentane) to give **108**; yield: 0.313 g (64%); mp 131–132 °C.

**5-*tert*-Butyl 1-(4,5-Dimethoxy-2-nitrobenzyl) *N*-(*tert*-Butoxycarbonyl)-L-glutamate Diester (109):**[230]

**CAUTION:** *Fluorides are eye and respiratory tract irritants and chronic exposure may result in skeletal fluorosis. Appropriate safety precautions and procedures should be adopted when handling this substance.*

To **108** (355 mg, 1.27 mmol), 5-*tert*-butyl *N*-(*tert*-butoxycarbonyl)-L-glutamic acid (385 mg, 1.27 mmol), and anhyd KF (149 mg, 2.6 mmol) was added acetone (5 mL). The soln was heated to 50 °C for 28 h, diluted with $H_2O$ (60 mL), and then extracted with EtOAc (4 ×). The combined EtOAc layers were dried, and the solvent was removed to give a yellow oil. Purification by chromatography (silica gel, $CH_2Cl_2/Et_2O$ 1:1) gave **109** as a yellow oil; yield: 429 mg (69%).

*for references see p 147*

**4,5-Dimethoxy-2-nitroacetophenone (110):**[230]
4,5-Dimethoxyacetophenone (15 g, 83 mmol) was added over 1 h to concd $HNO_3$ (90 mL) with stirring, keeping the internal reaction temperature in the range 18–22 °C. After the addition was complete, the soln was stirred for an additional hour and then poured into $H_2O$ (1200 mL). After chilling, the product was collected and recrystallized from EtOH; yield: 8.4 g (45%); mp 130–132 °C.

**1-(4,5-Dimethoxy-2-nitrophenyl)ethylamine (111):**[230]
To **110** (8.25 g, 367 mmol) was added anhyd MeOH (100 mL), $NH_4OAc$ (38.5 g, 500 mmol), and $NaBH_3CN$ (2.2 g, 35 mmol). A drying tube was attached and the mixture was stirred at rt. After 4 days, MeOH (200 mL) was added, and the mixture heated to 60 °C for 24 h. The suspension was filtered, and the filtrate concentrated to a yellow oil. The oil was dissolved in 1 M HCl (200 mL) and then extracted with $Et_2O$ (4 × 50 mL). The $Et_2O$ layer was discarded and the pH of the aqueous phase was increased to 11 by addition of KOH. A second $Et_2O$ extraction (4 × 50 mL) afforded the amine product. The $Et_2O$ soln was dried ($MgSO_4$) and concentrated to yield the title product as a yellow oil; yield: 8.96 g (23%).

***tert*-Butyl {[1-(4,5-Dimethoxy-2-nitrophenyl)ethyl]amino}acetate (112):**[230]
Amine **111** (1.17 g, 5.18 mmol) was mixed with MeCN (5 mL) and added to a soln of *tert*-butyl chloroacetate (0.780 g, 5.18 mmol) and Proton-Sponge (1.11 g, 0.00518 mol) in MeCN (30 mL). The mixture was stirred and heated at 60 °C for 15 h. The dark soln was cooled and concentrated to remove MeCN. The residue was triturated with $Et_2O$ (3 ×), and the combined $Et_2O$ layers were concentrated to an oil. Purification of the oil by chromatography (silica gel, $Et_2O$/pentane 3:1) gave **112** as a yellow oil; yield: 0.67 g (39%).

**{[1-(4,5-Dimethoxy-2-nitrophenyl)ethyl]amino}acetic Acid Hydrochloride Salt:**[230]
To **112** (0.67 g, 2 mmol) was added anhyd 4 M HCl (15 mL) in EtOAc. The soln was stirred at 40 °C for 45 min to give, after recrystallization (EtOAc:MeOH:$Et_2O$), the title product; yield: 400 mg (63%).

**1-(4,5-Dimethoxy-2-nitrophenyl)ethanone Hydrazone (113):**[230]
To **110** (8.38 g, 37.2 mmol) in 95% EtOH (144 mL) was added 85% hydrazine hydrate (5.71 g, 97 mmol) and glacial AcOH (7.2 mL). The mixture was heated under reflux for 3 h and then poured into $H_2O$ (1 L) and chilled for 24 h. The yellow precipitate was collected and recrystallized (toluene) to give **113**; yield: 4.7 g (53%); mp 112–114 °C.

**4-*tert*-Butyl 1-(4,5-Dimethoxy-2-nitrophenyl)ethyl *N*-(*tert*-Butoxycarbonyl)aspartate Diester (114):**[230]
To a soln of **113** (280 mg, 1.3 mmol) in $CHCl_3$ (10 mL) was added $MnO_2$ (2.0 g). The mixture was stirred in the dark for 7 min and then filtered by gravity into a soln of the DCHA salt of *tert*-butyl-*N*-(*tert*-butoxycarbonyl)-L-aspartic acid (356 mg, 0.76 mmol) in $CHCl_3$ (10 mL). The soln was stirred in the dark for 2 d, diluted with $CH_2Cl_2$ (20 mL), and passed through a column of silica gel (3 g). The yellow eluent was concentrated to an oil and chromatographed [silica gel (30 g), $Et_2O$/pentane 2:1]. Fractions collected containing a component with $R_f$ 0.25 were pooled to give a yellow oil; yield: 0.405 g (87%).

**Methyl Bromo(2-nitrophenyl)acetate (115):**[234]

**CAUTION:** *Carbon tetrachloride is toxic by inhalation and skin contact and is a possible human carcinogen, and operations should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

To a stirred soln of (2-nitrophenyl)acetic acid (1 g, 5.5 mmol) in benzene (10 mL) (**CAUTION**: *carcinogen!*) was added anhyd MeOH (3.5 mL) followed by three drops of concd $H_2SO_4$. The mixture was refluxed for 8 h with an azeotropic distillation device. TLC ($CH_2Cl_2$) showed quantitative conversion into the corresponding methyl ester ($R_f$ 0.28). Purification by chromatography (silica gel, hexanes/EtOAc) gave the methyl ester as a white solid; yield: 0.95 g (95%).

Methyl (2-nitrophenyl)acetate (6.2 g, 31.8 mmol) was dissolved in $CCl_4$ (100 mL) in a 500-mL round-bottomed flask. While stirring magnetically, $(BzO)_2$ (0.31 g, 1.28 mmol) was added, followed by NBS (6.22 g, 34.9 mmol). The flask was fitted with a reflux condenser and the mixture was refluxed for 24 h. TLC ($CH_2Cl_2$) showed a new spot for **115** ($R_f$ 0.43) and some starting material ($R_f$ 0.28). Extending the reaction time did not increase the yield of **115**, but modest improvements were afforded by another 24 h of reflux, after filtering the reaction through Whatman #1 paper and recharging with fresh $(BzO)_2$ and NBS. In preparation for purification, the mixture was filtered through Whatman #1 paper and concentrated to near dryness. Aliquots (ca. 3-mL) of the crude mixture were chromatographed on a flash column [5-cm i.d., silica gel (400 g), hexanes then EtOAc/hexanes 1:18] to give **115**; 3.08 g (50%).

***tert*-Butyl *N*-(*tert*-Butoxycarbonyl)-4-[2-Methoxy-1-(2-nitrophenyl)-2-oxoethoxy]phenylalanine Ester (117):**[234]

To a 100-mL round-bottomed flask under $N_2$ was added *tert*-butyl *N*-(*tert*-butoxycarbonyl)tyrosine ester, (**116**; 2.5 g, 7.4 mmol) in THF (10 mL). Then 1 M tBuOK (7.6 mL; to enhance the reactivity of the phenolic group) in THF was added dropwise with stirring. The mixture immediately turned dark purple, then brown. Finally, **115** (2.48 g, 9 mmol) was added and left to stir overnight at rt. The yellow-brown suspension was centrifuged; TLC of the supernatant soln (petroleum ether/$Et_2O$ 2:1) showed **117** ($R_f$ 12) and traces of **116** ($R_f$ 0.01) and **115** ($R_f$ 0.31). The soln was concentrated and purified by column chromatography [5-cm i.d. column, silica gel (400 g), $CH_2Cl_2$/hexanes 2:1], to give **117**; yield: 1.8 g (72%).

**4-[2-Methoxy-1-(2-nitrophenyl)-2-oxoethoxy]phenylalanine (118):**[234]

Removal of α-amino and α-carboxy protecting groups from **117** was accomplished by treatment in neat TFA at rt. Lower levels of TFA do not remove the *tert*-butyl ester. The methyl ester on the caging moiety is stable in the presence of neat TFA, and for most applications this methyl ester is maintained until the final step of caged peptide preparation. The progress of deprotection of α-amines and α-carboxyls is readily monitored by analytical RP-HPLC using a Whatman ODS-3 column (1 × 25 cm, MeCN/$H_2O$/TFA 40:60:0.1). Using a flow rate of 1.5 mL·min$^{-1}$, **117** eluted at 7 min and **118** eluted at 4 min. Deprotection in neat TFA at rt was complete in 4 h. Caged tyrosine methyl ester **118** was purified by semipreparative RP-HPLC using a 2.5 × 25-cm ODS-3 column and the same solvent system at 3 mL·min$^{-1}$; yield: ca. 90%

***N*-[(9*H*-Fluoren-9-ylmethoxy)carbonyl]-4-[2-methoxy-1-(2-nitrophenyl)-2-oxoethoxy]phenylalanine (119):**[234]

Compound **118** (1.04 g, 1.95 mmol) was suspended in MeCN (7 mL), $H_2O$ (4 mL), and TEA (0.7 mL); Fmoc-OSu (0.65 g, 1.93 mmol) was added portionwise with stirring. The pH of the homogeneous mixture was initially ca. 8.5 but decreased as the reaction proceeded. Small aliquots of TEA were added to maintain the pH ca. 8 (using standard pH paper). Progress of the reaction was monitored by analytical RP-HPLC (MeCN/$H_2O$/TFA 60:40:0.1); and the reaction was complete within ca. 30 min. The mixture was washed with 0.1 M HCl (3 × 25 mL), the organic phase was separated and concentrated. Compound **119** was purified by semipreparative RP-HPLC (MeCN/$H_2O$/TFA 45:55:0.1, 3 mL·min$^{-1}$); yield: 0.58 g (56%). The product is stable for at least 6 months when stored as a solid at −20 °C.

***N*-(2-Nitrobenzyl)glycine (120):**[235]

A mixture of 2-nitrobenzaldehyde (0.15 g, 1 mmol), glycine (0.113 g, 1.507 mmol), and 2 M NaOH (0.75 mL) in MeOH (1.5 mL) was treated with $NaBH_4$ (0.03 g, 0.79 mmol) and purified by vacuum liquid chromatography [silica gel (10–40 μm, type H), 0.1% AcOH in EtOH, with increasing amounts of $H_2O$]; yield: 15%; mp 189–191 °C.

***N*-[(2-Nitrophenyl)carboxymethyl]glycine (122):**[232]

To a soln of **121**·HCl (500 mg, 2.15 mmol) in $H_2O$ (50 mL), $NaHCO_3$ (456 mg, 4.30 mmol) and bromoacetic acid (299 mg, 2.15 mmol) were added. The mixture was protected from light and was stirred at rt for 11 d. The pH was adjusted to 2.0 with 1 M HCl and the soln was washed with $CHCl_3$ (3 ×). The pH of the aqueous portion was adjusted to 7.0 with 1 M NaOH and then lyophilized. The residue was subjected to Sephadex LH-20 chromatography (4 × 50 cm) with $H_2O$ elution to yield three bands. The bands, in order of elution, consisted of an *N*,*N*-dialkylated side product, compound **122**, and unreacted **121**. Lyophilization of the product band gave **122** as a light yellow powder; yield: 220 mg (34%); mp 123–125 °C (dec).

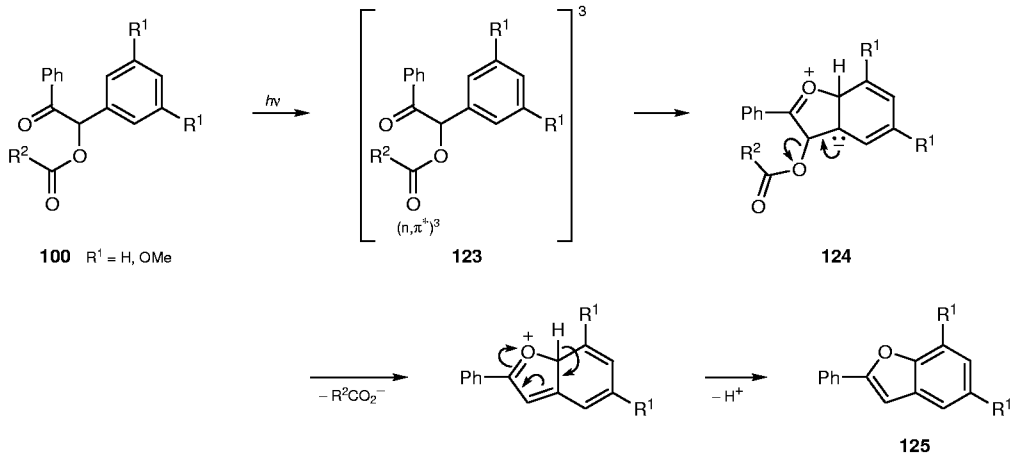
## 9.3.6.2 Synthesis of Desyl-Protected Amino Acids

Sheehan et al. report that benzoinyl (dansyl), and especially 3,5-dimethoxybenzoinyl acetates **100** ($R^1$ = OMe) undergo efficient photolysis using near-UV light.[236] The remarkably high efficiency and rate-constant for the photolytic reaction of the desyl derivatives compared to the nitrobenzyl analogues establishes them as photocaging moieties of major importance. Upon irradiation benzoinyl esters **100** are reported to yield, in addition to the solvent-dependent α-homolytic cleavage products, the intramolecular rearrangement product, 2-phenylbenzofuran (**125**).[237] Substitution with *meta*-methoxy results in photocyclization as the predominant photocleavage pathway.[236] Photolysis of 1-(3,5-dimethoxyphenyl)-2-oxo-2-phenylethyl acetate (**100**, $R^1$ = OMe; $R^2$ = Me) affords acetic acid and the corresponding furan **125** as the major products. This photolysis occurs via n→π* excitation of the

*for references see p 147*

carbonyl to the electrophilic triplet state **123**, attacking the benzyl group to generate the cyclic intermediate **124**, which, upon the sequential or simultaneous departure of the leaving group (caged moiety) and the proton forms the benzofuran **125** (Scheme 32).[225] Such an electrophilic attack is especially favorable on a 3,5-dimethoxybenzyl moiety due to the electron-donating contributions of the *meta*-directing methoxy groups. For the corresponding phosphate ester this photoreaction occurs with a high quantum yield (0.78) and fast first order release kinetics ($>10^8$ $s^{-1}$).[238–240] Both the quantum yield and product yields are essentially solvent independent.[225]

**Scheme 32** Mechanism of Cleavage of Desyl-Protected Photocaged Compounds[225]



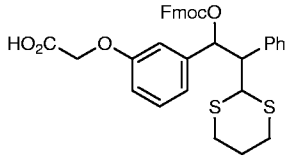

**100** $R^1$ = H, OMe **123** **124** **125**

The limitations of the desyl phototrigger include: (1) introduction of a new chiral center, which leads to the generation of diasteromeric mixtures whenever the caged moiety is chiral; (2) generation of highly absorbing, fluorescent benzofurans; (3) poor aqueous solubility of the reagent used to generate the photocaged compound and the resultant benzofurans.[241] The 3,5-dimethoxybenzoinyl (desyl) group has been used as a photolabile protecting group for carboxyl,[236,237,241] phosphate derivatives,[238,240,242] nucleotides,[240] carbamate esters,[243–246] and carbonates.[247]

Importantly, *N*-carbamoyl derivatives of primary amines obtained from photolabile benzoins exist in varying proportions as cyclic hydroxyoxazolidinone tautomers. Therefore, the preparation of *N*-carbamoyl derivatives of benzoins is applicable only for secondary amines and *N*-alkylated amino acids.[244] At basic pH unsymmetrical benzoins, such as 3,5-dimethoxybenzoin, their mixed carbonate and carbamate derivatives, tend to equilibrate to the isomeric forms. Nevertheless, in TFA and aqueous solutions at pH 8 the structural integrity is fully maintained.[244] Preparation of 3,5-dimethoxybenzoin-derived carbamates of secondary amines and amino acids can be mediated by either CDI/methyl triflate in nitromethane[246] or 4-nitrophenyl chloroformate/DMAP in dry THF.[244]

An interesting application of the 3-methoxybenzoin-based photolabile linker has been described in which the dithiane, (±)-(3-{2-(1,3-dithian-2-yl)-1-[(9-fluorenylmethoxycarbonyl)oxy]-2-phenylethyl}benzyl)oxy] acetic acid (**126**, Scheme 33), a transiently masked form of the caging moiety, is incorporated into a peptide chain generated by solid-phase synthesis.[248]

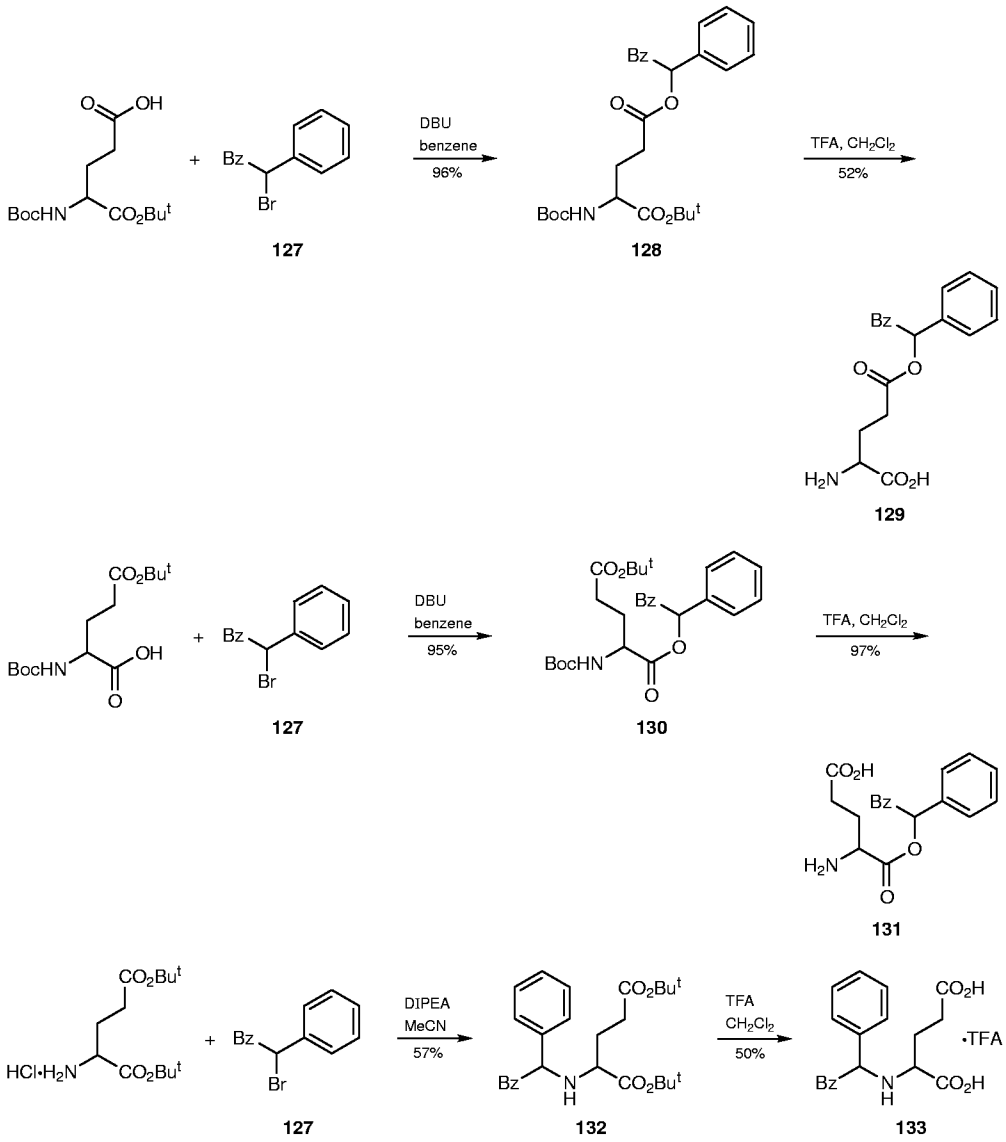
**Scheme 33** Structure of (±)-(3-{2-(1,3-Dithian-2-yl)-1-[(fluorenylmethoxycarbonyl)oxy]-2-phenylethyl}benzyl)oxy]acetic Acid[248]




**126**

The syntheses of γ-, α-, and *N*-benzoinyl (γ-, α-, and *N*-desyl) protected glutamic acid (**129**, **131**, and **133**, respectively, Scheme 34) and *tert*-butyl {[1-(3,5-dimethoxyphenyl)-2-phenylethoxy]carbonyloxy} protected sarcosine ester (**135**, Scheme 35) are described below.

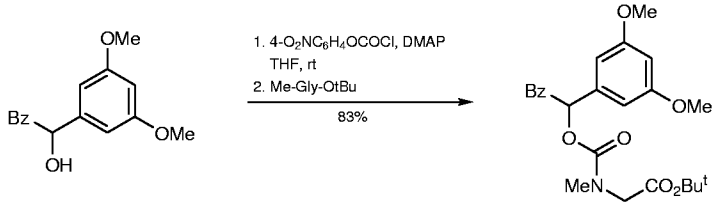
**Scheme 34** Synthesis of γ-, α-, and *N*-Benzoinyl Protected Glutamic Acid[241]




**127** **128** **129** **130** **131** **132** **133**

*for references see p 147*

**Scheme 35** Synthesis of *tert*-Butyl [{[1-(3,5-Dimethoxyphenyl)-2-oxo-2-phenylethoxy]carbonyloxy}(methyl)-amino]acetate[244]




134 135

**α-*tert*-Butyl γ-(2-Oxo-1,2-diphenylethyl) *N*-(*tert*-Butoxycarbonyl)-L-glutamate Diester (128):**[241]
A soln of 2-bromo-1,2-diphenylethanone (**127**; 605 mg, 2.2 mmol), α-*tert*-butyl *N*-(*tert*-butoxycarbonyl)-L-glutamate ester (672 mg, 2.2 mmol), and DBU (0.35 mL, 2.3 mmol) was heated at reflux in benzene (25 mL) (**CAUTION:** *carcinogen!)* and stirred overnight to give **128** as a colorless oil; yield: 1.06 g (96%).

**γ-(2-Oxo-1,2-diphenylethyl) L-Glutamate (129):**[241]
A soln of **128** (1.06 g, 2.13 mmol) and TFA (2.0 mL) was stirred at rt in $CH_2Cl_2$ (5 mL) and gave, after Sephadex LH-20 purification, **129** as a hygroscopic colorless powder. TFA was removed from the product salt during lyophilization; yield: 0.51 g (52%); mp 118–120 °C (dec).

**γ-*tert*-Butyl α-(2-Oxo-1,2-diphenylethyl) *N*-(*tert*-Butoxycarbonyl)-L-glutamate Diester (130):**[241]
A soln of γ-*tert*-butyl *N*-(*tert*-butoxycarbonyl)-L-glutamate (239 mg, 0.788 mmol), 2-bromo-1,2-diphenylethanone (**127**; 204 mg, 0.741 mmol), and DBU (0.12 mL, 0.80 mmol) in benzene (17 mL) (**CAUTION:** *carcinogen!*) was heated at reflux for 1 h. The soln was washed with $H_2O$ (25 mL) and brine (25 mL), dried ($Na_2SO_4$), and concentrated to a pale yellow oil. This oil was purified by flash chromatography (EtOAc/$CHCl_3$ 0–10%) to give **130** as a colorless solid, and as a mixture of diastereomers; yield: 0.37 g (95%); mp 87–90 °C.

**α-(2-Oxo-1,2-diphenylethyl) L-Glutamate (131):**[241]
A soln of **130** (0.17 g, 0.34 mmol) and TFA (2.0 mL, 26 mmol) in $CH_2Cl_2$ (4 mL) was stirred at rt for 3 h. The volatiles were removed in vacuo, and the residue was dissolved in $H_2O$ (15 mL). The resulting soln was frozen and lyophilized to give **131** as a mixture of diastereomers, and as a cream colored hygroscopic powder; yield: 151 mg (97%); mp 89–92 °C. Attempted further purification via RP-chromatography on Sephadex LH-20, with $H_2O$ as the mobile phase, resulted in extensive hydrolysis to free glutamic acid and benzoin, as determined by isolation and coelution with authentic samples on TLC.

**Di-*tert*-butyl *N*-(2-oxo-1,2-diphenylethyl)-L-glutamate Diester (132):**[241]
A soln of di-*tert*-butyl L-glutamate diester·HCl (2.03 g, 6.86 mmol), 2-bromo-1,2-diphenylethanone (**127**; 1.89 g, 6.86 mmol), and DIPEA (1.81 g, 14.0 mmol) in MeCN (30 mL) was stirred overnight at rt. The pale brown solid obtained after concentration in vacuo was purified by flash chromatography (EtOAc/$CHCl_3$ 0–5%) to give a mixture of diastereomers of **132** as a clear pale brown oil; yield: 1.77 g (57%).

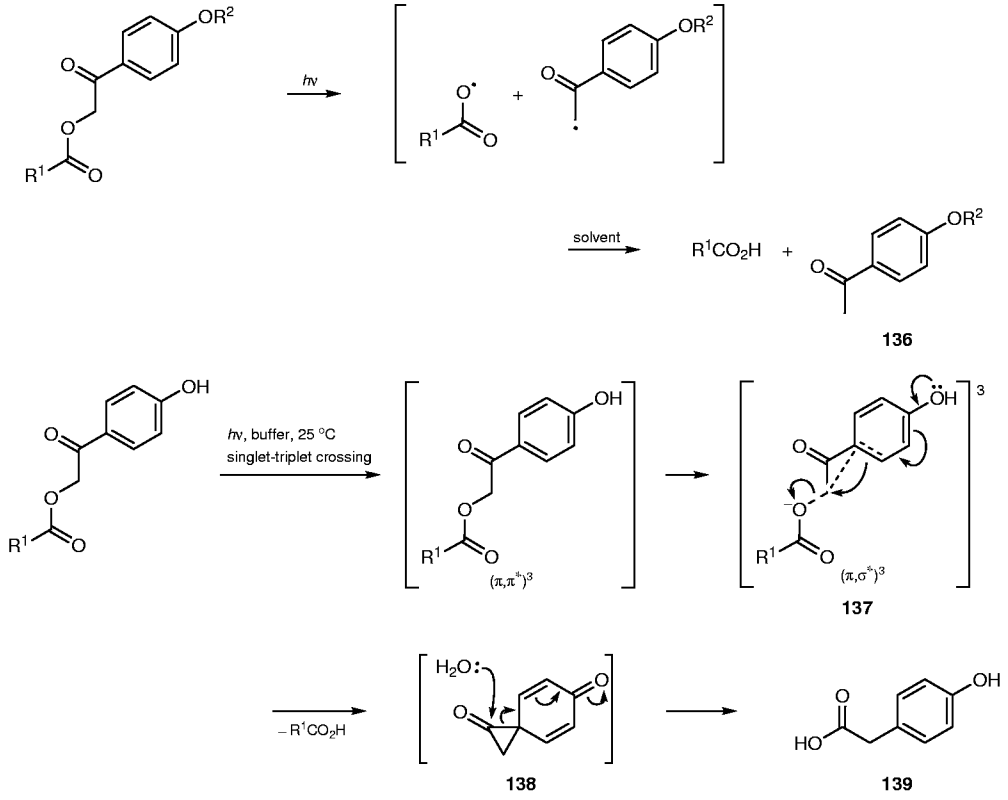
***N*-(2-Oxo-1,2-diphenylethyl)-L-glutamic Acid Trifluoroacetate Salt (133):**[241]
A soln of **132** (1.75 g, 3.86 mmol) and TFA (20 mL) in $CH_2Cl_2$ (80 mL) was stirred at rt overnight. The volatiles were removed in vacuo, and the residue was extracted with benzene (2 × 20 mL) (**CAUTION:** *carcinogen!*), and then evaporated to yield a residue of pale brown solid foam. Half of this crude material was purified on Sephadex LH-20 ($H_2O$), to give a mixture of diastereomers of **133** as a hygroscopic colorless powder after lyophilization; yield: 0.44 g (50%); mp 110–114 °C (dec).

***tert*-Butyl [{[1-(3,5-Dimethoxyphenyl)-2-oxo-2-phenylethoxy]carbonyl}(methyl)amino]acetate (135):**[244]
A soln of 3,5-dimethoxybenzoin (**134**; 1.5 mmol) in dry THF (40 mL) was treated with DMAP (244 mg, 2 mmol) and 4-nitrophenyl chloroformate (333 mg, 1.65 mmol). The soln was stirred at rt under $N_2$ for 3 h and treated with *tert*-butyl sarcosinate (2 mmol). The mixture was stirred at rt for 20 h, diluted with $Et_2O$, and washed with sat. aq. $K_2CO_3$ until the aqueous washes become colorless. The organic layer was washed with 1 M HCl and $H_2O$, dried and concentrated. The crude product was flash chromatographed (EtOAc/petroleum ether 3:7) and recrystallized from petroleum ether to give **135**; yield: 83%; mp 99–100 °C.

### 9.3.6.3 4-Hydroxyphenacyl-Protected Amino Acids

The phenacyl group has been proposed by Sheehan et al. as a carboxyl-photocaging group.[249] This phototrigger overcomes several of the limitations of the desyl group. It has improved aqueous solubility and lacks a chiral center. In the photocleavage reaction of a phenacyl ester, carried out at $\lambda_{max}$ >330 nm, the products obtained are the carboxylic acid and an acetophenone. The homolysis of the C–O bond is followed by hydrogen transfer from the solvent to the carboxyl and phenacyl radicals. 4-Hydroxy- and 4-methoxy-substituted phenacyl protecting groups **99** display improved stability towards acids, and are therefore frequently employed in synthetic procedures using orthogonal protection schemes. However, they also promote a competing photorearrangement process, to give (4-hydroxy or 4-methoxyphenyl)acetic acid **139** (Scheme 36).[250] Photolysis in polar solvents, such as alcohols, generates predominantly the rearranged product **139**,[242] and in water or aqueous buffers the phototrigger undergoes almost exclusively the rearrangement reaction at the expense of all other possible photoproducts.[251,252] The putative photorearrangement process involves a direct displacement of the leaving group from the electron-rich, $\pi^*\rightarrow\sigma^*$ triplet state **137** generating the dienedione intermediate **138**, which is quenched by nucleophilic ring opening of the spirocyclopropanone ring by water to give **139**.[225]

**Scheme 36** Mechanisms of Photolysis of 4-Hydroxyphenacyl-Protected Acids[225]

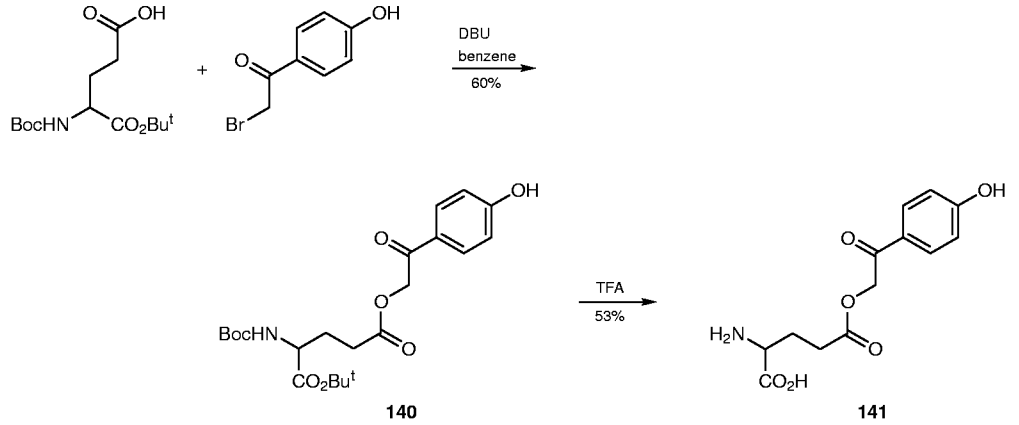



All of the α-amino-substituted carboxylic acid 4-hydroxyphenacyl phototriggers have proved to be unstable in aqueous environments, hydrolyzing to the free amino acid and the α-hydroxy phototrigger. Nevertheless, the photocaged dipeptide, 4-hydroxyphenacyl alanyl-alanylate is stable to aqueous and buffered media.[253]

*for references see p 147*

The synthesis of γ-*O*-(4-hydroxyphenacyl)glutamic acid is described below as an example of this class of photocaged amino acid derivatives (Scheme 37).

**Scheme 37** Synthesis of γ-*O*-(4-Hydroxyphenacyl)glutamic Acid[253]



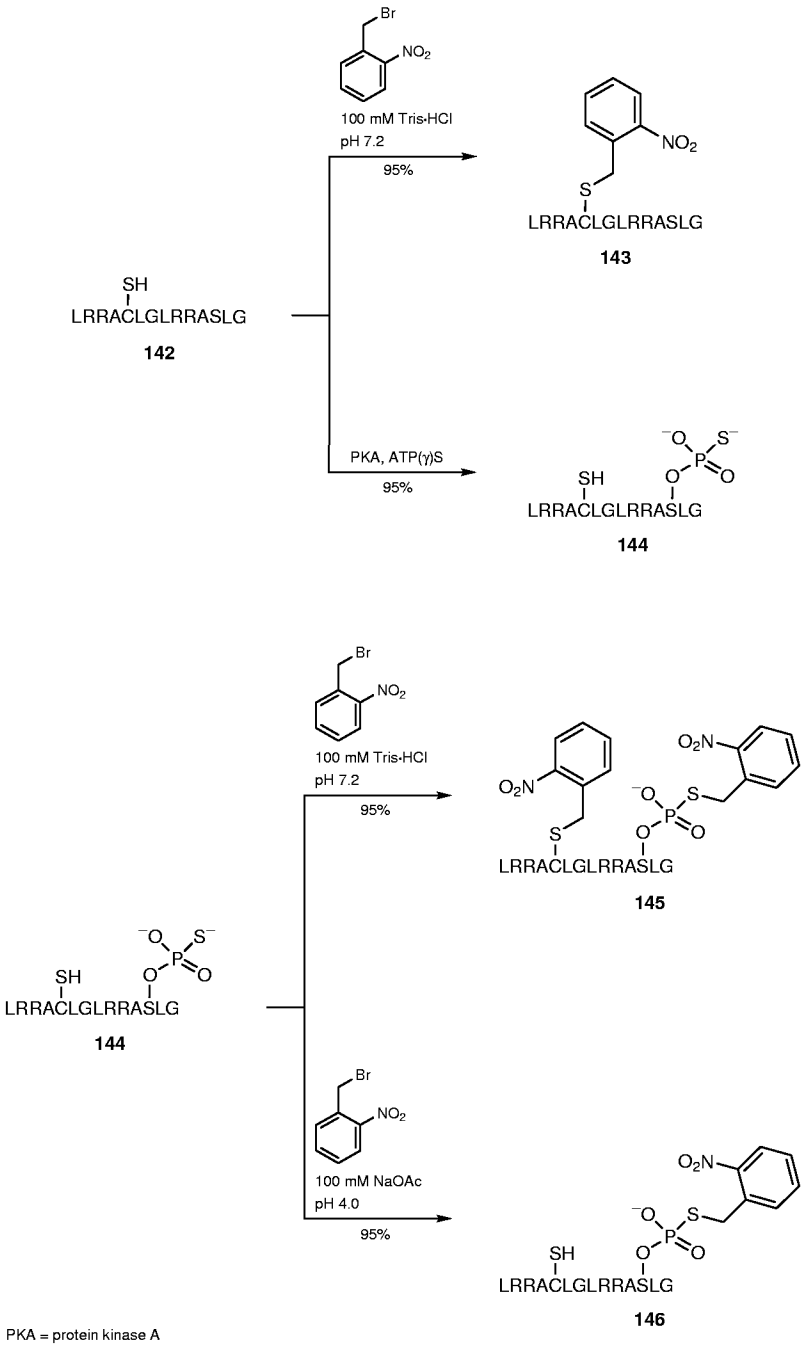

**α-*tert*-Butyl γ-*O*-(4-Hydroxyphenacyl) *N*-(*tert*-Butylcarboxy)-L-glutamate Diester** (140):[253]
A suspension of 4-hydroxyphenacyl bromide (1.24 g, 5.77 mmol) was stirred under reflux in benzene (20 mL) (**CAUTION:** *carcinogen!*) until dissolution. The soln was cooled to 6 °C and a soln of α-*tert*-butyl *N*-*tert*-butoxycarbonyl-L-glutamate (1.70 g, 5.60 mmol) in benzene (15 mL) was added. A cold soln of DBU (0.90 mL, 6.02 mmol) in benzene (10 mL) was added dropwise over a period of 15 min. During the addition of the base the mixture turned light yellow. The mixture was allowed to warm to rt and stirred for 17 h, then the solvent was removed in vacuo. Purification by column chromatography (silica gel, $CH_2Cl_2$/EtOAc 9:1) gave **140** as a yellow solid; yield: 1.48 g (60%).

**α-*O*-(4-Hydroxyphenacyl)glutamic Acid** (141):[253]
A soln of **140** (1.23 g, 3.02 mmol) was treated with TFA (10 mL) at 4 °C and stirred for 3 h at this temperature. Removal of the TFA in vacuo was followed by the addition of EtOAc (30 mL) and $H_2O$ (30 mL). The layers were separated and the aqueous layer was concentrated in vacuo. Treating the oily residue with MeCN-$H_2O$ (1:1, 25 mL) immediately caused a precipitation of colorless crystals of **141**; yield: 447 mg (53%); mp 187–188 °C (dec).

## 9.3.6.4 Synthesis of Photocaged Peptides

Two basic strategies for the site-selective introduction of photocage moieties into peptides have been used. The first strategy is based on the introduction of caged amino acids during the stepwise assembly of the peptide. For example, Fmoc-protected caged tyrosine [Tyr{*O*-2-(2-nitrophenylacetic acid)} and Tyr{*O*-Bzl(2-OH)}] and lysine [Lys{$N^{\varepsilon}$-Z(2-$NO_2$)}] derivatives have been used in standard solid phase peptide synthesis to generate caged analogues of neuropeptide Y,[254] angiotensin II,[254] myosin light chain autoinhibitory peptide,[234] calmodulin inhibitory peptide,[255] and autocamtide-2 related inhibitory peptide[256] (see Table 6). The second strategy is based on the post-synthetic modification of functional groups in peptides with photocaging reagents. For example, 2-bromo-4-hydroxyacetophenone has been used to modify the α-amino group of partially protected bradykinin.[257] Photocleavable affinity tags are useful for the affinity-based isolation and detection, and the subsequent phototriggered release of the target biomolecule.[258] The approaches described are illustrated in Schemes 38 and 39.

**Scheme 38** Synthesis of Cysteine and Serine 2-Nitrobenzyl-Protected Kemptide Dimer[259]




PKA = protein kinase A

*for references see p 147*

**Scheme 39** Synthesis of Photocleavable Biotinyl (PCB)-[Leu]-Enkephalin[260]

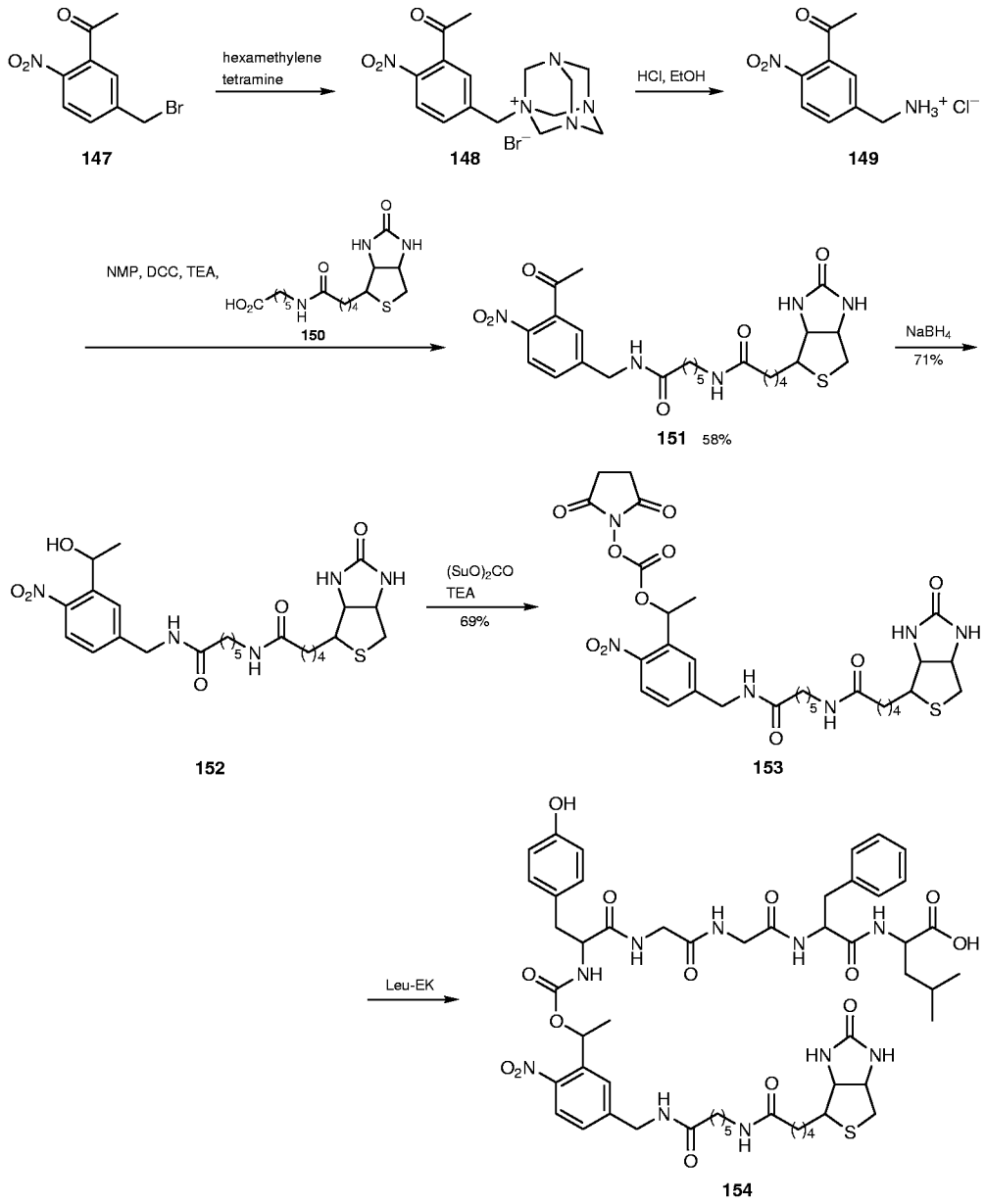


**Table 6** Selected Photocaged Peptides[234,254–257,259,260]

| Peptide | Photocage | Comments on the Synthesis | Ref |
|---|---|---|---|
| Bradykinin | 4-Hydroxyphenacyl on C-terminus | Fmoc SPPS: on chlorotrityl resin → HPLC →protected BK<br>solution: 2-bromo-4-hydroxacetophenone, DBU in DMF, TFA deprotection and HPLC purification | [257] |
| calmodulin inhibitory peptide RS-20 and variants | 2-nitrophenylacetyl on Tyr | Fmoc SPPS: Tyr(2-nitrophenylacetyl) | [255] |
| myosin light chain autoinhibitory peptide | 2-nitrophenylacetyl on Tyr | Fmoc SPPS: Tyr(2-nitrophenylacetyl) | [234] |
| kemptide | 2-nitrobenzyl<br>2-nitrophenylacetyl<br>4,5-dimethoxy-2-nitrobenzyl on Cys and thiophosphoryl Ser | solution: modifications of Cys and Ser(thiophosphoryl) with bromides of caging groups | [259] |
| NPY | 2-nitrobenzyl on Tyr | Fmoc SPPS: Tyr(2-nitrobenzyl) | [254] |
| angiotensin II | 2-nitrobenzyl on Tyr | Fmoc SPPS: Tyr(2-nitrobenzyl) | [254] |
| autocamtide-2 related inhibitory peptide | 2-nitrobenzyloxycarbonyl on ε-amino of Lys | Fmoc SPPS: Lys[$N^{\varepsilon}$-(Z(2-$NO_2$)) | [256] |
| Leu-enkephalin | 5-(6-biotinamidocaproamidomethyl)-1-(2-nitrophenyl)ethyl carbonate on α-$NH_2$ | solution: modification of α-$NH_2$ with hydroxysuccinimidyl ester | [260] |

Peach et al. developed a general, racemization free, and high yield procedure for the synthesis of *O*-desyl peptide esters from the poorly nucleophilic benzoin.[261] Employing the cesium salt of the model dipeptide Z-Gly-Phe-OH in acetone the intramolecular hydantoin formation obtained in polar aprotic solvents, such as DMF and DMSO, can be suppressed and a high yield of Z-Gly-Phe-O-desyl with ee >90% can be generated.[261]

Caged proteins have also been generated. In most cases, however, nonspecific modification of either lysyl or cysteinyl residues is employed. Reversible inactivation of G-actin is achieved with NVOC-Cl.[262,263] Various nitrophenyl-based photoreactive heterobifunctional reagents are prepared and used in the photocaging of G-actin and BSA.[264] Combination of an amino reactive NVOC-based reagent, that will generate a photolabile carbamate, with a thiol reactive oxirane group, substituting the benzylic carbon of the NVOC moiety, forms the foundation for this new class of heterobifunctional, photocleavable crosslinking reagents.[265] α-Thrombin and Factor Xa were acylated on a Ser residue at the active site with photocleavable biotin-substituted 2-hydroxycinnamamic acid[266] and 2-hydroxy-1-methylcinnamic acid[267] both as 4-amidinophenyl esters. Effective caging of heavy meromyosin on $Cys^{707}$ was achieved with the thiol reactive reagent 4,5-dimethoxy-2-nitrobenzyl bromide, generating a tool to study muscle contraction in vivo.[268] Photocleavable actin dimer was generated by cross-linking F-actin with *N*-succinimidyl 4-bromomethyl-2-nitrobenzoate through the ε-amino and thiol function on Lys and Cys residues, respectively.[269] Photocleavage of the caged dimer at the sulfide bond modulates the nucleating activity of dimer generating a polymerizable G-actin monomer.

Site-directed mutagenesis is employed to introduce a single Cys residue or eliminate an extra one, thus targeting site specific introduction of a caging moiety. This method was employed to generate α-hemolysin[R104C] and α-carbon subunit of mouse PKA[C343S], which were modified respectively on $Cys^{104}$ or $Cys^{199}$ by 2-bromo-2-(2-nitrophenyl)acetic acid.[270]

Biosynthetic site-specific incorporation of photocaging amino acids into proteins by amber suppressor tRNA, chemically acylated by the photocaging amino acid, was used to

*for references see p 147*

generate photocaged proteins (for details see also Section 9.3.4.3).[271] Employing this method Asp[β-Bzl(2-$NO_2$)] replaced $Asp^{20}$, at the active site of phage T4 lysozyme (pT4L), rendering an inactive enzyme.[272] Irradiation of the crude in vitro reaction supernatant at λ >315 nm restored lytic activity of the pT4L[$D^{20}${β-Bzl(2-$NO_2$)}]. A similar modification of $Asp^{38}$ in $p21^{ras}$ yielded a photocaged protein that did not interact with its downstream effector p120-GTPase activating protein (GAP).[273] Irradiation of the photocaged $p21^{ras}$[$D^{38}${β-Bzl(2-$NO_2$)}] restored roughly 50% of the p120-GAP-dependent GTPase activity relative to the wild type ras.

Okada et al. suggested the (pyren-1-ylmethoxy)carbonyl (Pmoc) as a fluorescent "caged" group for compounds containing an amino group.[274] Acylation of amino acids on their α-amino group is achieved by 4-nitrophenyl pyren-1-yl carbonate in moderate to good yiles. Fas and efficient photolysis is achieved due to the large absorption coefficient of Pmoc amino acids at around 340 nm. In addition to the free amino acids or peptides the other photolytic products include: pyren-1-ylmethanol (~70–95%), 1-methylpyrene (5–10%), and trace amounts of 1,2-di(pyren-1-yl)ethane (<5%). Nevertheless, the low solubility of Pmoc amino acid and peptides in phosphate buffer saline solution containing 2% DMSO is a major limitation in applying these Pmoc-containing compound to biological systems.

**Synthesis of Photocaged Carboxypeptidase A Substrate, *N*$^{\alpha}$-Z-Gly-Phe-*O*-Desyl:**[261]
2-Bromo-1,2-diphenylethanone (385 mg, 1.4 mmol) and $CsCO_3$ (228 mg, 1.4 mmol) were added to a stirred soln of *N*$^{\alpha}$-Z-Gly-Phe-OH (500 mg, 1.4 mmol) in acetone (20 mL). The flask was stoppered and the soln stirred overnight. After filtering, the solvent was removed to afford a 1:1 diastereomeric mixture of *N*$^{\alpha}$-Z-Gly-Phe-*O*-desyl as an oil; yield: 770 mg (100%); $[\alpha]_D^{20}$ −3.53 (*c* 1.5, acetone).

**Bradykinin 4′-Hydroxyphenacyl Ester:**[257]

***N*-Boc-Arg(*N*$^{G}$-Pbf)-Pro-Pro-Gly-Phe-Ser(tBu)-Pro-Phe-Arg(*N*$^{G}$-Pbf)-OH:**[257]
The partially protected bradykinin was synthesized by a solid-phase Fmoc strategy. The side-chain functional groups on bradykinin were introduced as the protected amino acids, i.e. Arg was protected as its Pbf derivative and Ser as its tBu derivative. The coupling of the protected amino acids was accomplished using standard Fmoc protocol on 2-chlorotrityl resin (nominal capacity: 1.2 mmol·g$^{-1}$). The loading of the resin was 0.6 mmol·g$^{-1}$ initial capacity as determined by the Fmoc assay. The couplings were carried out by using the protected amino acid (2 equiv), and HBTU (1.9 equiv) in NMP or DMF in the presence of DIPEA. When the Kaiser test was positive (usually after 30 min), the resin was washed and coupled again until no free amino groups were detected. The Fmoc group was then cleaved with 30% piperidine in DMF. This process was repeated for each amino acid. The N-terminal Arg was incorporated as the Fmoc-protected derivative, and the Fmoc group was cleaved as described above. The N-terminal amino group was acylated with $Boc_2O$ in the presence of DIPEA. Finally, the fully protected oligopeptide was cleaved from the resin by treatment with $CH_2Cl_2$/MeOH/AcOH (8:1:1) which gave complete conversion to *N*-Boc-Arg(*N*$^{G}$-Pbf)-Pro-Pro-Gly-Phe-Ser(tBu)-Pro-Phe-Arg(*N*$^{G}$-Pbf)-OH within 2 h.
The protected bradykinin carboxylic acid was purified by precipitation from $CHCl_3$ with MeOH, and the purity was established by RP-HPLC and TLC: $R_f$ 0.6 ($CHCl_3$/MeOH/AcOH 8:1:1); $R_f$ 0.6 (EtOAc/pyridine/AcOH/$H_2O$ 60:20:6:11) on air-dried Merck Kieselgel 60 F254 glass plates. The purity of the protected bradykinin carboxylic acid was >90% based on RP-HPLC analysis. The amino acid composition was confirmed by a FAB-MS spectrum.

***N*-Boc-Arg(*N*$^{G}$-Pbf)-Pro-Pro-Gly-Phe-Ser(tBu)-Pro-Phe-Arg(*N*$^{G}$-Pbf)-o[Pac(4-OH)]:**[257]
To a soln of DBU (23.1 mg, 0.15 mmol) dissolved in dry DMF (2 mL) and cooled to 0 °C with ice was added of *N*-Boc-Arg(*N*$^{G}$-Pbf)-Pro-Pro-Gly-Phe-Ser(tBu)-Pro-Phe-Arg(*N*$^{G}$-Pbf) (100 mg, 0.058 mmol) under argon. 2-Bromo-4-hydroxyacetophenone (37.5 mg, 0.17 mmol) was then added, and the mixture was cooled to 0 °C. The mixture was allowed to reach rt and was stirred overnight. TLC indicated complete reaction. The solvent was removed in vacuo, and the crude product was first purified by column chromatography (silica gel, $CHCl_3$/MeOH/AcOH 90:8:2) to give a light oil. This was further purified by HPLC on a 2.2 cm Vydac C-18 peptide/protein column [gradient: 0.1% TFA in MeCN (50–100% over 90 min); flow rate: 8 mL·min$^{-1}$ ($t_R$ 35.6 min)]. The solvent was concentrated to give the title ester as a white crystalline solid; yield: 78 mg (72%).

**Bradykinin 4-Hydroxyphenacyl Ester:**[257]
To a 25-mL conical glass flask was added the dry, protected bradykinin 4-hydroxyphenylacetic ester (54 mg, 0.0291 mmol), followed by the deprotection cocktail of 88% TFA, 7% thioanisole, 5% $H_2O$ (2 mL). The soln was stirred with a glass rod every 15 min for 2.5 h to effect the deprotection. The resulting mixture was concentrated at 30 °C on a rotary evaporator to yield a light yellow syrup. This was dissolved in TFA (400 mL) and the resulting soln was taken up into a Pasteur pipet and then squirted forcibly into *tert*-butyl methyl ether (12 mL) at −75 °C in a 12 mL centrifuge tube. A fine white precipitate formed immediately. After 5 min of centrifugation, the *tert*-butyl methyl ether was decanted, and the crude product was dispersed thoroughly with a glass rod in fresh *tert*-butyl methyl ether (10 mL) and centrifuged once again. This process was repeated six times. The crude product was dried overnight to give a white pellet (35.5 mg). The product was further purified by HPLC on a 2.2 cm Vydac C-18 peptide/protein column [gradient: 0.1% TFA in MeCN (5–50% over 45 min); flow rate: of 8 mL·min$^{-1}$ ($t_R$ 33.6 min)]. The solvent was concentrated on a rotary evaporator to remove the MeCN and the remaining material was lyophilized to give white crystals of bradykinin 4-hydroxyphenacyl ester; yield: 29.1 mg (84%).

**Calmodulin Inhibitory Peptide:**[255]

**Synthesis of ARRKY[Pac(2-$NO_2$]QKTGHAVRAIGRLSS-OH (5cgY-RS-20):**[255]
L-Tyr, protected on its amino and carboxyl groups, was coupled to methyl 2-bromo-2-(nitrophenyl)acetate through its phenolic group. Methylcarboxyl caged Tyr (MW = 374.1, *m/z* 375.1) was derivatized with an Fmoc moiety and incorporated into peptides during automated synthesis. Cleavage from the solid support and deprotection of side chains (90% TFA) was followed by HPLC purification of the peptide. This was followed by hydrolytic removal of the methyl ester from the cage moiety using 10% aqueous $K_2CO_3$. l-Carboxyl-2-(nitrobenzyl)tyrosine containing peptides were repurified by HPLC and lyophilized. Peptide composition was confirmed by amino acid analysis and MS: Cald: 2434.2. Found: 2434.5.

**Myosin Light Chain Kinase Autoinhibitory Peptide (LSM-1):**[234]

**Synthesis of LSKDRMKKY[*O*-{CH($CO_2$H)[Ph(2-$NO_2$)]}]MARR-OH (9cgY-LSM-1):**[234]
Peptide synthesis was carried out by the solid-phase methodology using Fmoc chemistry and Wang resin. Activation of α-carboxyls for peptide-bond formation was accomplished with a five-fold excess of HBTU, and selective removal of the Fmoc group was carried out in 10% piperidine in DMF. Side-chain protecting groups were *tert*-butyl for S; Boc for K; and Pmc for R. Double coupling was carried out for *N*-Fmoc-Tyr[*O*-{CH($CO_2$Me)[Ph(2-$NO_2$)]}]. Cleavage was performed with TFA/thioanisole/EDT (92.5:5:2.5, 1 mL) for 2–4 h. The crude peptide was precipitated from this mixture by filtration to remove the resin followed by dropwise addition of filtrate to ice-cold *tert*-butyl methyl ether (10 volumes). The precipitate was then washed with the same solvent (3 × 25 mL), dried in a vacuum desiccator for 15 min, suspended in $H_2O$ (2 mL), frozen, and lyophilized to a white powder.
The peptide was purified by analytical RP-HPLC in MeCN/$H_2O$/TFA. To generate the free α-carboxyl on the cage moiety, the purified peptide was demethylated by treatment with 10% $K_2CO_3$, neutralized with 1 M HCl, and repurified by RP-HPLC. The demethylated caged peptide, 9cgY-LSM-1, resolved into two HPLC peaks, which appeared to be stereoisomers. The peptide was double lyophilized to remove traces of TFA and characterized by amino acid analysis and mass spectrometry.

**[Cys$^5${*S*-Bzl(2-$NO_2$)}]Kemptide Dimer (143):**[259]
A soln of [Cys$^5$]kemptide dimer (**142**; 1 mg, 0.5 mM) in 100 mM Tris·HCl, pH 7.2, was treated in the dark at 25 °C for 1 h with 50 mM 1-(bromomethyl)-2-nitrobenzene (2 mmol) in 95% EtOH. The [Cys$^5${*S*-Bzl(2-$NO_2$)}]kemptide dimer **143** was purified by HPLC using a Vydac C-18 column (buffer A: 0.1% TFA in water; buffer B: 0.8% TFA in MeCN/$H_2O$ 7:3); yield: ca. 95%; MALDI-MS *m/z*: 1677.

**(Cys$^5${*S*-Bzl(2-$NO_2$)},Ser$^{12}$[*O*-{*S*-Bzl(2-$NO_2$)}thiophosphoryl])Kemptide Dimer (145):**[259]
[Cys$^5$]kemptide dimer (**142**; 1 mg) was dissolved in 100 mM 4-morpholinepropanesulfonic acid sodium salt, pH 7.0, containing 100 mM KCl and 10 mM $MgCl_2$ (300 µL). ATP(γ)S in $H_2O$ (300 µL) was added in 5-fold molar excess over peptide. Thiophosphorylation was initiated by addition of protein kinase A catalytic unit (1 µL, final concentration of 1.0 µg·mL$^{-1}$). Thiophosphorylated peptide **144** was purified by HPLC as described above; yield: ca. 95%; MALDI-MS *m/z*: 1638. Purified thiophosphorylated [Cys$^5$]kemptide dimer **144** was lyophilized and stored at −20 °C.
Exhaustive photocaging of thiophosphorylated [Cys$^5$]keptide dimer with 1-(bromomethyl)-2-nitrobenzene was carried out as described above gave the anticipated dimer (**145**); yield: ca. 95%; (MALDI-MS, *m/z*: 1909).

*for references see p 147*

**(Cys$^5$,Ser$^{12}$[*O*-{*S*-Bzl(2-$NO_2$)}thiophosphoryl])Kemptide** (146):[259]
Selective caging of the thiophosphoryl moiety in [Cys$^5$,Ser$^{12}$(*O*-thiophosphoryl)]kemptide dimer **144** (0.5 mM) was achieved in 100 mM NaOAc buffer, pH 4.0, at 25 °C in the dark for 100 min. The product **146** was purified by HPLC as described above; yield: 95%; (MALDI-MS, *m/z*: 1774).

**Neuropeptide Y and Angiotensin II; Synthesis of Peptides:**[254]
Caged peptides (NPY and angiotensin II) containing Tyr[Bzl(2-$NO_2$) were synthesized on an automated solid-phase peptide synthesizer using Fmoc chemistry and cleaved from the resin by treatment with TFA. Crude peptides were purified by semipreparative HPLC with a gradient phase of 0.1% TFA and MeCN. For fractions containing peptides, the eluant was not passed through the UV monitor to avoid potential photolysis. The collected peptide was examined by analytical HPLC and MS.

***N*$^\alpha$-(1-Pyrenylmethoxy)carbonyl Dipeptides (*N*$^\alpha$-Pmoc-Leu-Leu and *N*$^\alpha$-Pmoc-Val-Ala):**[274]
**4-Nitrophenyl 1-Pyrenylmethyl Carbonate:**[274]
To a THF (15 mL) soln of (1-pyrenyl)methanol (3.07 g, 13.2 mmol) and 4-nitrophenyl chloroformate (3.21 g, 16.0 mmol), pyridine (1.3 mL) was added dropwise at 0 °C. The mixture was stirred at 0 °C for 1 h and then at rt for 2 h. The solvent was distilled off and the residue dissolved in $CH_2Cl_2$, washed successively with 1 M HCl, 4% $NaHCO_3$, $H_2O$, and sat. NaCl, then dried ($Na_2SO_4$). After removal of the solvent, recrystallization of the crude product from $CHCl_3$/hexanes gave 4-nitrophenyl 1-pyrenylmethyl carbonate; yield: 4.79 g (91%); mp 130 °C.

***N*$^\alpha$-Pmoc-Amino Acids; General Procedure:**[274]
To a 10% $Na_2CO_3$ soln (50–70 mL) of an amino acid (Gly, Ala, Leu, Phe, Val, Trp, Met, and γ-aminobutyric acid) or dipeptide (Leu-Leu and Val-Ala) (1.0 mmol) 4-nitrophenyl 1-pyrenylmethyl carbonate (1 mmol) in dioxane (100–150 mL) was added dropwise and stirred for 5 h at rt. After removal of the solvent, the mixture was dissolved in $H_2O$ and adjusted to pH 2.0 with 1 M HCl and then extracted with EtOAc (3 ×). The organic layer was washed with $H_2O$ and sat. NaCl, then dried ($Na_2SO_4$). After removal of the solvent, the resulting solid was washed thoroughly with $Et_2O$ in order to remove by-products. Filtration gave almost pure *N*$^\alpha$-Pmoc-amino acid, which was recrystallized (EtOAc/hexanes).

***N*$^\alpha$-Pmoc-Amino Acid Esters: General Procedure:**[274]
To a benzene (200 mL) (**CAUTION:** *carcinogen!*) suspension of *N*$^\alpha$-Pmoc-amino acid (1 mmol), DBU (1 mmol), and haloalkane, such as BzlBr, (1 mmol) were added successively and the mixture was refluxed for 2 h. $Et_2O$ (200 mL) was added to the cooled soln at rt and the resulting precipitate was filtered off. The filtrate was washed with $H_2O$, 1 M HCl, 1 M $NaHCO_3$, then dried ($Na_2SO_4$). The crude ester was purified by chromatography (silica gel, EtOAc/hexanes) followed by recrystallization (EtOAc/hexanes).

***N*$^\alpha$-Pmoc-Dipeptide Esters; General Procedure:**[274]
To a soln of an amino acid ethyl (or methyl) ester·HCl (1.2 mmol) in DMF (30 mL), NMP (1.2 mmol), *N*$^\alpha$-Pmoc-amino acid (1.2 mmol) in DMF (50 mL), HOBt (1.8 mmol) in DMF (30 mL), and EDC (1.8 mmol) in DMF (50 mL) were added successively at −5 °C. The mixture was stirred at −15 °C for 3 h and at rt for 24 h, then EtOAc (200 mL) was added. The soln was washed successively with $H_2O$, 1 N HCl, 3 N $NaHCO_3$, and $H_2O$, then dried ($Na_2SO_4$) and concentrated. Chromatography (silica gel, EtOAc/hexanes 1:1) gave almost pure *N*$^\alpha$-Pmoc-dipeptide ester.

**Photocleavable Biotinyl (PCB)-[Leu]EK:**[260]

**5-(6-Biotinamidocaproamidomethyl)-2-nitroacetophenone** (151):[260]
5-Bromomethyl-2-nitroacetophenone (**147**; 2.0 g, 7.75 mmol) was added to a soln of hexamethylenetetramine (1.14 g, 8.13 mmol) in chlorobenzene (15 mL). The mixture was stirred overnight and the resulting precipitate was filtered off and washed with chlorobenzene (10 mL) and $Et_2O$ (20 mL). The intermediate **148** (2.93 g, 7.36 mmol) was suspended in 95% EtOH (35 mL) and then concd HCl (3.12 mL, 5 equiv) was added. The mixture was stirred overnight and then concentrated to dryness. DMF (10 mL) was added to the residue, containing 5-aminomethyl-2-nitroacetophenone·HCl (**149**) and $NH_4Cl$ (which remains as a precipitate), followed by 6-biotinamidocaproic acid (**150**; 3.29 g, 1.25 equiv) in NMP (35 mL), DCC (2.28 g, 1.5 equiv), and TEA (1.28 mL, 1.25 equiv). The soln was stirred overnight at rt, the precipitate was filtered off, and the filtrate was added to $Et_2O$ (700 mL). The resulting precipitate was dried and purified [silica gel, step gradient 5–20% MeOH in $CHCl_3$] to give **151**; yield: 2.27 g (58%).

**1-[5-(6-Biotinamidocaproamidomethyl)-2-nitrophenyl]ethanol** (152):[260]
5-(6-Biotinamidocaproamidomethyl)-2-nitroacetophenone (**151**; 1 g, 1.87 mmol) was dissolved in 70% EtOH (15 mL), and after the soln had been cooled to 0 °C, $NaBH_4$ (141 mg, 4 equiv) was added. The soln

was stirred to 0 °C for 30 min and at rt for a further 2 h. The reaction was quenched by addition of acetone (1 mL), and the mixture was neutralized with 0.1 M HCl and concentrated to about 5 mL. The aqueous layer was decanted and the residue was washed with $H_2O$ (3 × 5 mL) and dried to give **152**; yield: 0.71 g (71%).

**1-[5-(6-Biotinamidocaproamidomethyl)-2-nitrophenyl]ethyl *N*-Succinimidyl carbonate** (153):[260]
Alcohol **152** (1.07 g, 2 mmol) was dissolved in DMF (10 mL). To this soln, *N,N'*-disuccinimidyl carbonate (1 g, 1.5 equiv) was added, followed by TEA (0.81 mL, 3 equiv). After stirring at rt for 5 h, the solvents were removed. The residue was washed consecutively with 0.1 M $NaHCO_3$, $H_2O$, dioxane, and $Et_2O$, and then dried to give **153**; yield: 1.04 g (69%); mp 113–114 °C.

**$N^{\alpha}$-PCB-[Leu$^5$]-Enkephalin** (154):[260]
[Leu]-EK (3.1 μmol) in 0.1 M $NaHCO_3$, pH 8.0 (200 μL), and PCB-NHS ester (**153**; 3.4 μmol) in DMF (200 μL) were mixed and stirred overnight at rt. The formation of $N^{\alpha}$-PCB-[Leu]-enkephalin was analyzed by HPLC [mobile phase: A, 0.1% TFA in $H_2O$; B, 0.1% TFA in 3:1 MeCN/$H_2O$; linear gradient 30–45% B over 10 min followed by 45% B isocratic for 10 min].


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# 9.4 Synthesis of Peptides as Receptors and Devices

N. VOYER

The ability of polypeptides to fold into predictable, regular secondary, and tertiary structures in solution makes them attractive as molecular scaffolds for the development of functional molecular devices and receptors.[1,2] In addition, other features of peptide synthesis bring more incentives to their use. Indeed, the possibility of rapidly synthesizing large polypeptides by solid-phase strategies allows the preparation of several analogues and even libraries of peptidic molecular devices in a short period of time.[3] Coupled with powerful chromatographic purification methods and spectroscopic identification techniques (LC-MS, NMR, IR, circular dichroism, fluorescence, etc.), peptide synthesis is ideally suited for the preparation of multiple receptors and devices for their uses in supramolecular chemistry, molecular recognition, and molecular sensors. All these attractive features have stimulated research efforts in the past ten years to develop peptide-based molecular receptors and devices. This subject has been reviewed recently.[1,2] There are two construction strategies. One is to introduce into the peptide framework at specific positions amino acids bearing unnatural side chains that can serve as binding sites or effectors. The second route is to modify C- or N-terminal positions to incorporate molecular groups to be used as elements that precisely control the supramolecular aggregation of multiple peptidic structures. The synthesis of *de novo* designed proteins onto covalent templates and other strategies to induce stable peptidic secondary and tertiary structures are covered in Sections 12 and 13. This part of Section 9 discusses the two strategies described.

## 9.4.1 Crown Ether Peptides

Crown ethers are useful neutral macrocyclic ligands that bind cations, more specifically alkali metal ions and primary ammoniums, in a predictable manner.[4] One interesting feature of these ligands is the possibility of engineering their ion binding ability by selecting the "size" of the macrocycle. Numerous crown ether based receptors and devices were developed in the early days of host–guest chemistry,[5] but it was only in the late 1980s that crown ether peptides were described. Since then, several crown ether peptidic receptors and devices have been reported. Peptides bearing crown ether side chains have been used for the development of receptors for $Cs^+$,[6] diammoniums,[7] aromatic ammoniums,[8] and chiral amino acid esters,[9,10] of artificial ion channels,[11–13] and to control backbone conformations.[14,15]

Usually, the crown ether moiety is introduced into the peptide sequence by using a side-chain-modified amino acid. Either a nonchiral crown ether is attached to the side chain of lysine[15] or glutamic acid[10] or the crown is built on the catechol side chain of dihydroxy-L-Phe (L-DOPA).[10,14]

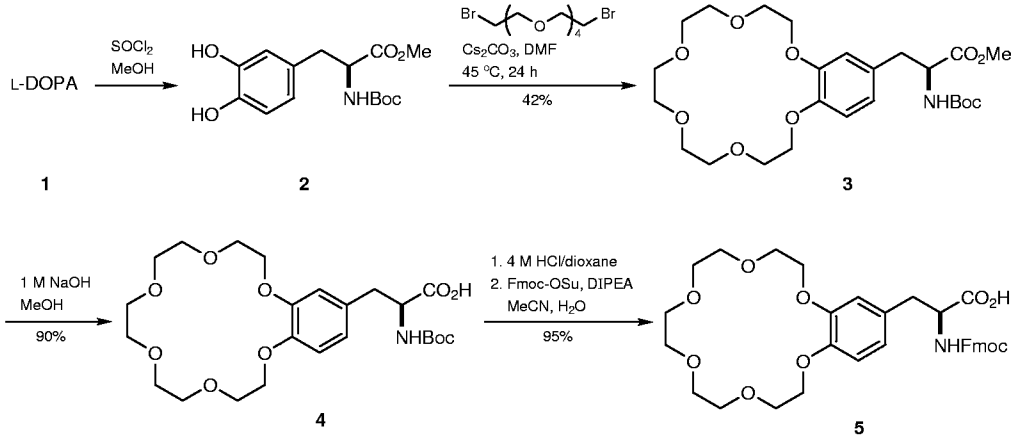
### 9.4.1.1 Synthesis of 3-(3,4-Dihydroxyphenyl)alanine Crown Ethers

L-DOPA (**1**) is a useful substrate for the synthesis of different crown ether modified amino acids. Both enantiomers are commercially available and the built-in catechol function can be used as the bis-nucleophile in macrocyclic ring-closure reactions. Therefore, different sized crown amino acids can readily be prepared by this strategy by using appropriate oligoethylene glycol bis-electrophiles. The synthesis was shown to be almost racemization free as long as the cyclization reaction and its workup are performed at a temperature lower than

*for references see p 191*

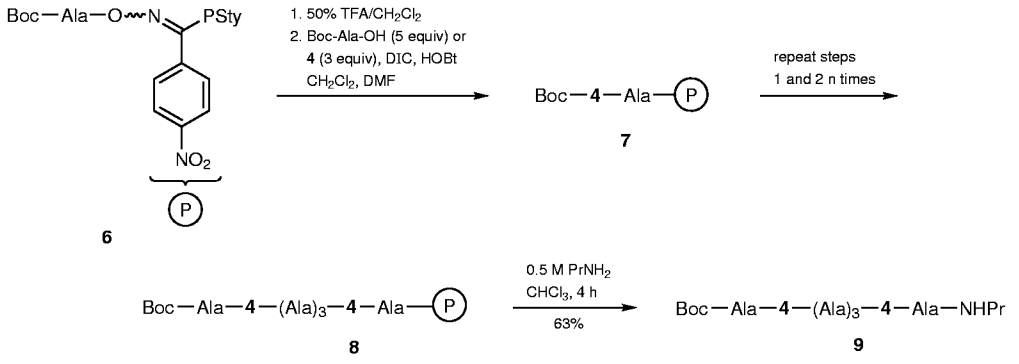
45 °C.[10,14] The 12-crown-4, 13-crown-4, 14-crown-4, 15-crown-5, 18-crown-6, and the 21-crown-7 derivatives were prepared using $Cs_2CO_3$ as the base and the yield of macrocyclization is higher with the largest ring, since $Cs^+$ serves as an excellent template for the 21-crown-7 macrocycle. The synthetic strategy is illustrated by synthesis of the 18-crown-6 Fmoc derivative **5** (Scheme 1).[14] Note that Z or Boc can be used as the protecting group in the synthesis. However, the Fmoc protecting group in **5** is not compatible with the basic conditions of the macrocyclization reaction and this derivative needs to be prepared, after ester hydrolysis of **3**, by a deprotection/reprotection step from the Boc ester derivative **4** (Scheme 1).

**Scheme 1** Synthesis of DOPA Crown Ether[14,17]




Boc-protected crown ether DOPAs have been used in the synthesis of crown peptide receptors[14] and artificial ion channels[11] through solid-phase peptide synthesis using the 4-nitrophenyl oxime resin **6**, e.g. Scheme 2, which shows the initial coupling of **4** to the resin **6** giving **7**, followed by coupling with either Boc-Ala-OH or **4** to give **8** and cleavage from the resin giving **9**. The Boc derivatives were coupled to the resin using the DIC/HOBt method and racemization with crown derivatives was low and comparable with that of Phe. Although the coupling of the first crown DOPA derivative was successfully achieved with three equivalents, the second crown ether amino acid was always more difficult to introduce and double couplings were performed as necessary.

**Scheme 2** Solid-Phase Synthesis of Crown Ether Modified Peptides[14]



**Boc-18-crown-6-L-DOPA (4):**[14]
Under argon, Boc-L-DOPA methyl ester[16] (**2**; 44 g, 142 mmol) and $Cs_2CO_3$ (46.23 g, 142 mmol) were dissolved in dry degassed MeOH (600 mL). The mixture was gently warmed on a water bath to 45 °C until complete dissolution. MeOH was then removed under reduced pressure at 45 °C. DMF (50 mL) was added and the solvent removed under high vacuum at the same temperature to remove the residual MeOH. The green product obtained was dissolved in DMF (300 mL) and transferred into a 3-L three-necked flask containing DMF (1 L) under argon at 60 °C to generate a deep green soln. 1,14-Dibromo-3,6,9,12-tetraoxatetradecane (51.7 g, 142 mmol) in DMF (300 mL) was then added dropwise over a 1 h period. The resulting soln was stirred at 60 °C overnight. The brown suspension was concentrated under high vacuum at 45 °C. The residue was taken up in $CH_2Cl_2$, washed with 5% $NaHCO_3$ (× 3) and $H_2O$ (× 3), and then dried ($MgSO_4$). After filtration and removal of the solvent, the crude brown oil was purified by flash chromatography (silica gel, $CH_2Cl_2$ then $CH_2Cl_2$ with 2% and 4% MeOH). Trituration with petroleum ether/$Et_2O$ (10:1) gave the ester **3** as a white powder; yield: 42% (isolated); mp 91–93 °C; $[\alpha]_D^{23}$ +5.5 (*c* 1, MeOH).

The above ester **3** (16.36 g, 32 mmol) was suspended in 1 M NaOH (50 mL) at 0 °C and stirred at rt for 1.5 h. The clear alkaline soln was washed with $Et_2O$, then acidified with ice cold 1 M HCl to pH 4 producing a milky soln. The aqueous layer was extracted with $CH_2Cl_2$ (3 × 300 mL). The organic phase was washed with $H_2O$, dried ($MgSO_4$), filtered, and the solvent removed to give a yellowish oil that was triturated three times with $Et_2O$ to give the desired amino acid **4** as a white fluffy solid; yield: 90%; mp 96–98 °C; $[\alpha]_D^{23}$ +15.5 (*c* 1, MeOH); FAB-MS *m/z*: 522 [M + $Na^+$], 499 [$M^+$].

**Fmoc-18-crown-6-L-DOPA (5):**[17]
Boc-18-crown-6-L-DOPA (**4**; 66.4 g, 0.133 mol) was treated for 2 h with 4 M HCl in dioxane. After concentration to dryness under vacuo, the amino acid hydrochloride obtained (57.96 g, 0.133 mol) was dissolved in a $H_2O$/MeCN (1:9) mixture. DIPEA (69 mL) and Fmoc-OSu (49.80 g, 0.148 mol) were then added portionwise to the soln which was stirred mechanically for 3 h. The solvents were removed under reduced pressure and the residual oil was dissolved in $CH_2Cl_2$. The organic phase was washed with 1 M HCl then water. After drying ($MgSO_4$), the solvent was removed and the residue was triturated with $Et_2O$ to give **5** as a white solid; yield: 95%; mp 110–114 °C; TLC $R_f$ 0.30 (silica gel, $CHCl_3$/MeOH/AcOH 70:15:15); $[\alpha]_D^{25}$ −14.8 (*c* 1, DMF).

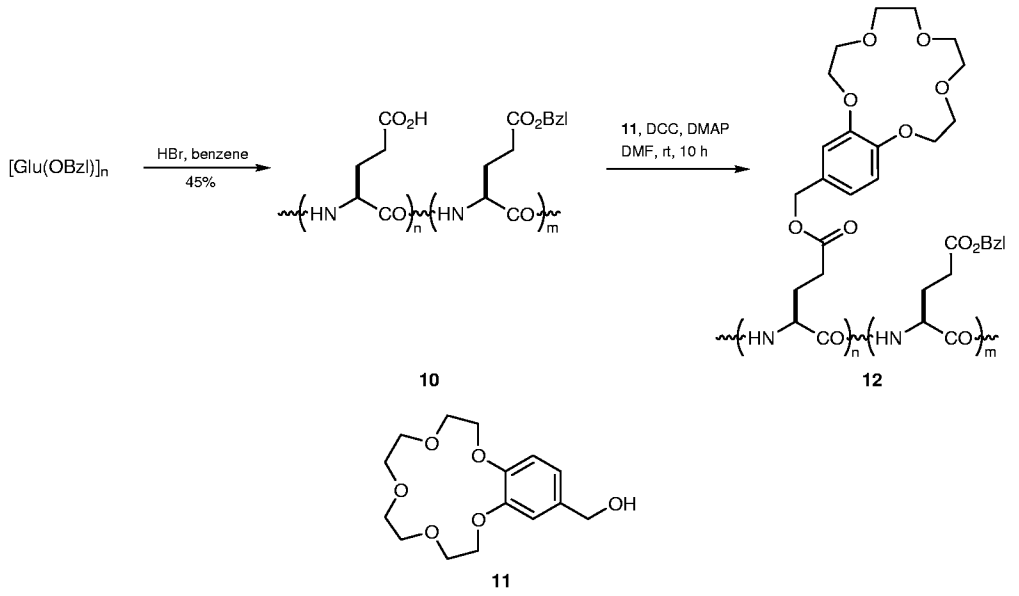
**Synthesis of Bis-Crown Peptide 9:**[14]
Peptide **9** was prepared using the Boc-Ala-oxime-resin **6** (5 g).[14] Deprotection of the Boc group was performed by treating the resin with 50% TFA in $CH_2Cl_2$ for 30 min. The washing steps in all cases were as follows: DMF (3 × 100 mL), MeOH (3 × 100 mL), DMF (3 × 100 mL), MeOH (3 × 100 mL). Boc-protected amino acids (5 equiv for Ala, 3 equiv for crown-DOPA) were activated at 0 °C for 30 min in $CH_2Cl_2$ using DCC and HOBt and coupled in DMF for 1 h in the presence of DIPEA (1 equiv). Completion of the coupling reactions was monitored by the ninhydrin test. The peptide was cleaved from the resin by a 4 h treatment using freshly prepared 0.5 M $PrNH_2$ in $CHCl_3$. The suspension was filtered and the filtrate was concentrated to give the crude peptide **9** that was crystallized ($CHCl_3$/MeOH) then purified by RP-HPLC; yield: 63%; FAB-MS ($C_{61}H_{96}O_{21}N_8$) *m/z*: 1278 [M + $H]^+$, 1300 [M + $Na]^+$.

## 9.4.1.2 Synthesis of Crown Ether Lysine and Glutamic Acid Derivatives

The crown ether moiety can be incorporated into peptidic structures during their synthesis by using side-chain-modified amino acids. Most often, the crown ether is attached as an amide linkage with the Lys ε-amino group[15] or as an ester with the γ-carboxy group of Glu.[10] In contrast to the crown DOPA derivatives, these compounds have more flexible spacers between the crown unit and the peptide backbone. Conventional peptide coupling methods were used to attach the crown unit to the side chains. For example, poly(L-glutamate) modified with a benzo-15-crown-5 derivative has been reported (Scheme 3).[10] The crown moiety **11** was linked as an ester on the Glu side chain of a partially deprotected poly[Glu(OBzl)] **10** using DCC and DMAP in DMF for 10 hours. The resulting crown peptide **12** was studied for its ion binding ability and conformational behavior in the presence of alkali metal cations. Even though **12** was shown to form 2:1 complexes, no conformational changes were observed upon addition of guests, even with ions forming sandwich 2:1 complexes.[10]

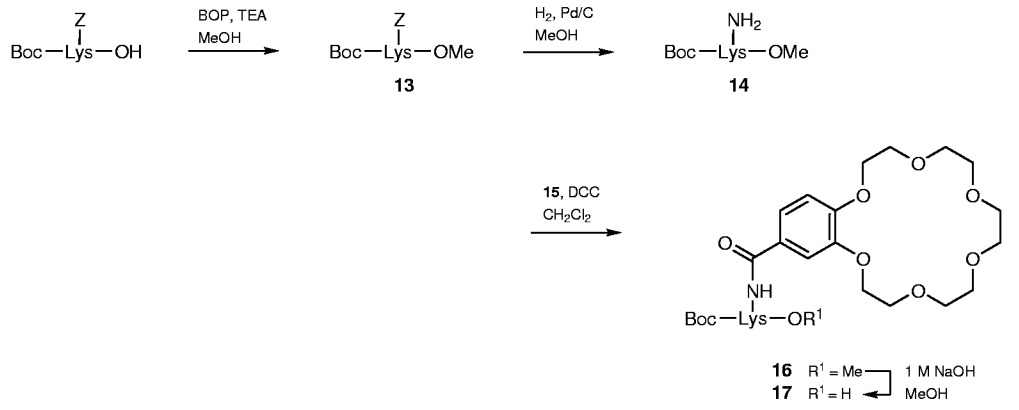
*for references see p 191*

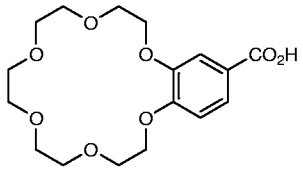
**Scheme 3** Crown Ether Modification of Poly(benzyl glutamate)[10]




On the other hand, the synthesis of a crown lysine derivative was described.[15] The macrocyclic ligand was attached to the ε-$NH_2$ group starting from Boc-Lys(Z)-OH by first esterification to give the methyl ester **13**, deprotection of the ε-$NH_2$ group giving **14** and then coupling using DCC in $CH_2Cl_2$ and a 4-carboxy derivative of benzo-18-crown-6 **15** to give **16** (Scheme 4). After deprotection of the methyl ester by saponification, the Boc-ε-(benzo-18-crown-6)-L-Lys-OH (**17**, $R^1 = H$) was used in conventional solution-phase peptide synthesis using DCC/HOBt as coupling reagent. Likewise, the crown ether **15** was also attached to the Lys side chain of a tripeptide under the same conditions, e.g. starting from **18**, deprotection of the ε-$NH_2$ group of Lys to give **19** and coupling of **19** with **15** to give **20** (Scheme 5). This strategy is quite useful since it allows the preparation of several analogues of a peptide by reaction with different crown ethers. Three positional isomeric peptides incorporating this crown ether amino acid as host side chain and Lys as guest side chain were prepared this way and tested for their ability to adopt specific conformations driven by intramolecular host–guest side-chain interactions.[15]

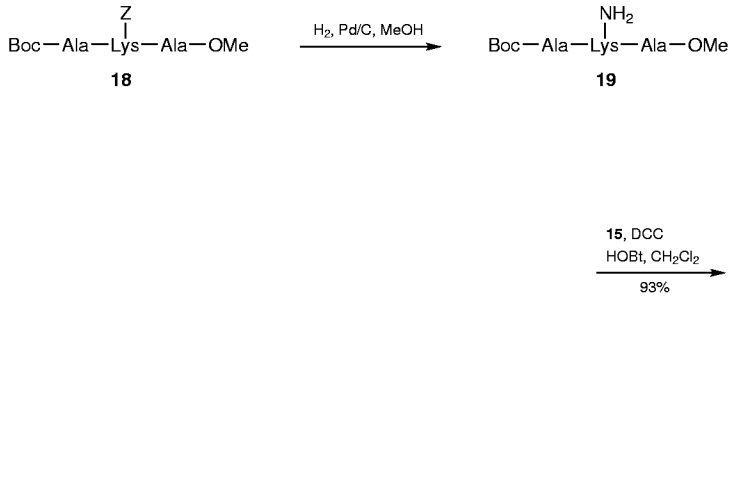
**Scheme 4** Synthesis of Crown Ether Lysine[15]






15

**Scheme 5** Attachment of Crown Ethers on a Lysine Tripeptide[15]




**Synthesis of Boc-Ala-Lys[$N^6$-(benzo-18-crown-6)-4-carbonyl]-Ala-OMe (20):**[15]
To a stirred soln of (benzo-18-crown-6)-4-carboxylic acid **15**[18] (0.25 g, 0.70 mmol) in the minimum amount of $CH_2Cl_2$ at 0 °C were added HOBt (0.10 g, 0.77 mmol) and DCC (0.16 g, 0.77 mmol). After 20 min, the DCU was filtered off. A soln of Boc-Ala-Lys-Ala-OMe (**19**; 0.032 g, 0.70 mmol) in $CH_2Cl_2$ was then added to the filtrate. After stirring 15 h at rt, the reaction was stopped and the crude mixture was washed with 0.5 M HCl, 5% $NaHCO_3$, and $H_2O$, then dried ($MgSO_4$), filtered, and concentrated. The product was precipitated from $CH_2Cl_2$ with $Et_2O$ to give a white hygroscopic solid; yield: 0.44 g (93%); TLC $R_f$ 0.18 ($CHCl_3$/MeOH 1:1); $[\alpha]_D^{20}$ −23 (*c* 1, AcOH).

## 9.4.1.3 Crown Ether at the C- or N-Terminal Position

The introduction at the C- or N-terminal position of a crown ether unit has been used as a strategy to control the aggregation of poly(benzyl glutamate) derivatives.[19] The incorporation of the crown unit at the C-terminal position is performed using (benzo-15-crown-5)-4-amine as initiator of the polymerization of L-Glu(OBzl)-NCA. Physical properties of such crown derivatives can be modulated by the formation of sandwich 2:1 complexes driven by the addition of specific alkali metal ions. In the reported case, the formation of $K^+$ sandwich complex between two C-terminal benzo-15-crown-5 modified helical polypeptides induced aggregation. In a similar approach,[20] addition of $Cs^+$ to 18-crown-6 terminated helical peptides results in the formation of supramolecular assemblies having membrane ion conductivity activities.

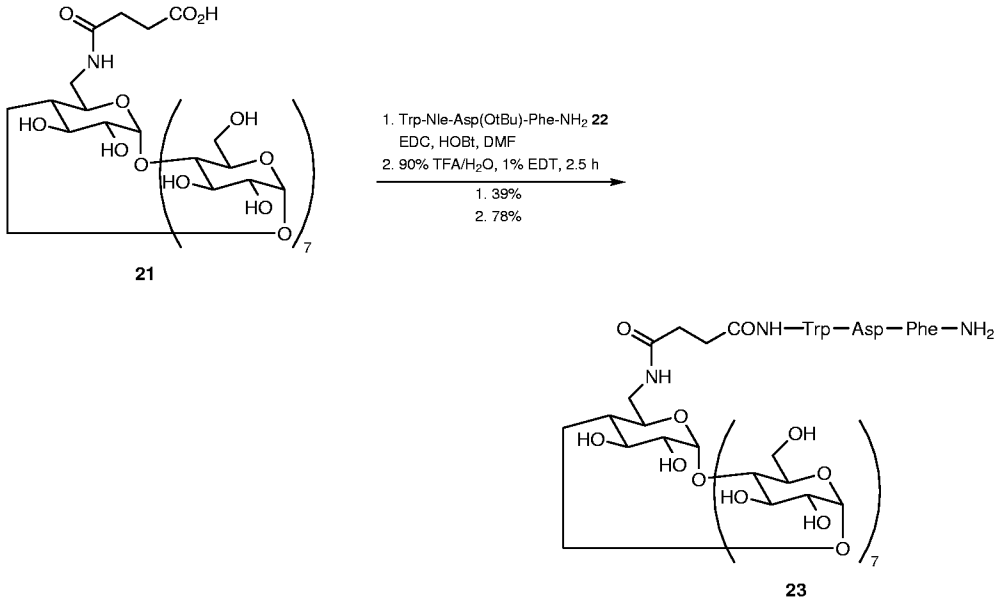
**Synthesis of C-Terminal Modified Benzo-15-crown-5 Poly[L-Glu(OBzl)]:**[19]
Benzyl L-glutamate-NCA was polymerized in dioxane at rt using (benzo-15-crown-5)-4-amine as initiator. The ratio of the NCA to the initiator was 50. The polymer with C-terminal crown ether was characterized by FTIR.

*for references see p 191*

## 9.4.2 Cyclodextrin Peptides

Cyclodextrins are cyclic oligosaccharides that have the ability to form inclusion complexes in water with a variety of hydrophobic guest molecules. They have been used extensively for several years as molecular receptors,[21] enzyme models,[21] and in the formulation of pharmaceutical ingredients, notably peptides to enhance their pharmacological profile.[22] Although specific side-chain modification of an amino acid with a cyclodextrin *per se* has not been reported so far, the covalent attachment of one or several peptides to the rim of a cyclodextrin has been described. The conjugation of bioactive peptides to cyclodextrin leads to potentially useful prodrugs with better pharmacokinetic properties than the usual inclusion complexes.[23] The peptide is usually linked to the 5′-hydroxymethyl rim of the cyclodextrin and the use of a spacer is required to retain full bioactivity. Recently, mono[6-deoxy-6-(3-carboxypropanoylamino)]-β-cyclodextrin (**21**)[23] was coupled to the C-terminal peptide from gastrin hormone **22** (Scheme 6).[24] The side-chain protecting groups were removed using 90% aqueous TFA and ethane-1,2-dithiol as scavenger. The conjugates with the succinyl spacer, e.g. **23**, were shown to be as active as the peptides alone.

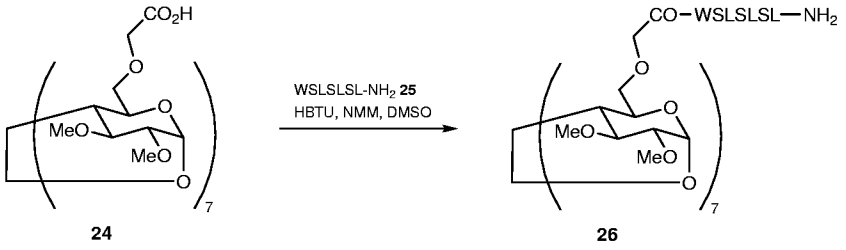
**Scheme 6** Synthesis of Peptide–Cyclodextrin Conjugates[24]




**21** **23**

On the other hand, the advantage of using a cyclodextrin as template for the construction of *de novo* artificial proteins is the inherent incorporation of a binding site, the cyclodextrin cavity, difficult to introduce otherwise. Seven units of a prototypical heptapeptide **25** designed to adopt a β-sheet structure were coupled to a β-cyclodextrin derivative **24** using HBTU as coupling reagent (Scheme 7).[25] Although the structure of **26** was characterized by MS, no further details were reported on the complexation ability of the hexapeptide–cyclodextrin hybrid compound.

**Scheme 7** Coupling of Peptide Segments on a Cyclodextrin Derivative[25]




**Synthesis of β-Cyclodextrin-Gastrin Peptide Conjugate** 23:[24]
Mono[6-deoxy-6-(3-carboxypropanoylamino)]-β-cyclodextrin[23] (**21**; 0.2 g, 0.16 mmol) and Trp-Nle-Asp(OtBu)-Phe-$NH_2$ (**22**; 0.1 g, 0.16 mmol) in DMF (2.5 mL) were reacted overnight at 25 °C with EDC/HOBt (1 equiv). The solvent was removed and the crude mixture was purified by RP-HPLC (C18 column, linear gradient MeCN/$H_2O$, 20–80% MeCN in 3 h) to afford 39% of the protected conjugate. The latter was then deprotected by dissolving it in a ice-cold 90% aq TFA soln containing 1% EDT. After stirring for 2 h at rt, the solvent was removed and the residue purified by RP-HPLC (C18, gradient of MeCN and $H_2O$, 5 min isocratic at 5% MeCN, then 5–18% in 5 min, and 18–60% in 80 min) to give compound **23**; yield: 78%; FAB-MS *m/z*: 1794.7 [M + $H^+$].

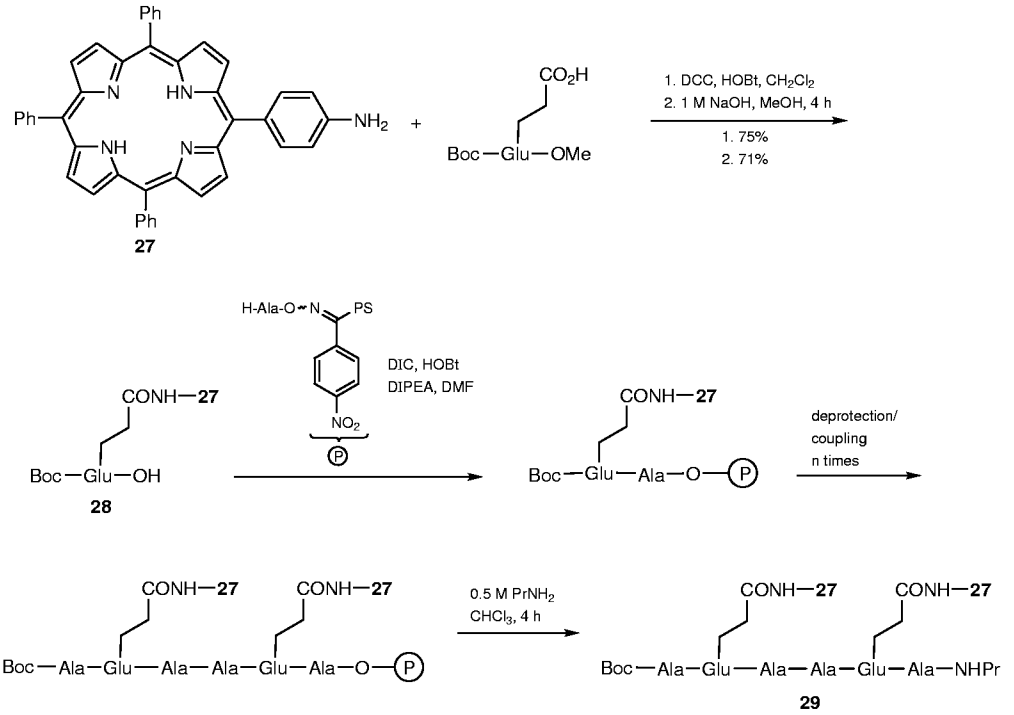
**Synthesis of Heptakis(WSLSLSL)-β-Cyclodextrin** (26):[25]
In an Eppendorf tube, NMM (7.17 mmol), HBTU (4.78 mmol), and WSLSLSL-$NH_2$ (**25**; 4.78 mmol) were added in sequence to the heptacarboxylic acid derivative **24** of β-cyclodextrin (0.593 mg, 0.341 mmol) in DMF (200 μL) and the resulting mixture was stirred for 19 h at rt. After concentration in vacuo, the crude product was purified by RP-HPLC (C18 column, gradient of MeCN and $H_2O$, 50–100% MeCN); FAB-MS *m/z*: 7365 ± 63 [M + $K^+$].

## 9.4.3 Porphyrin Peptides

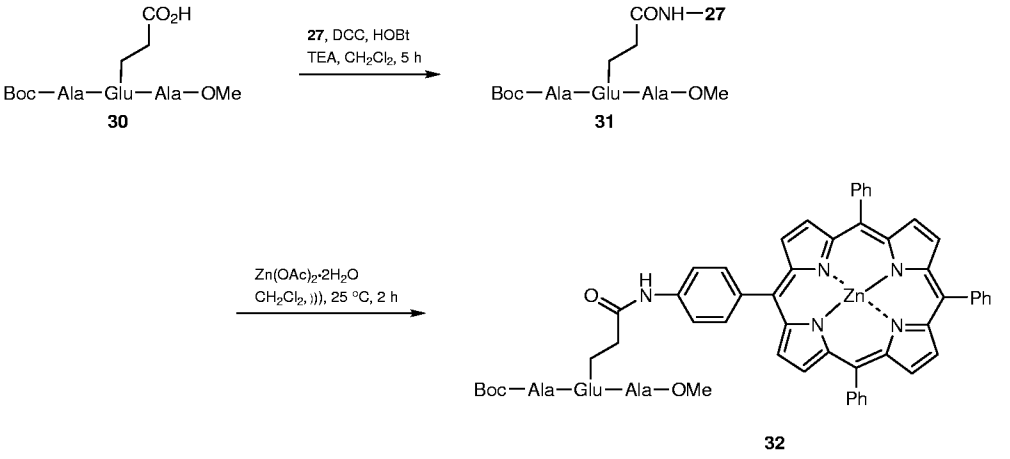
Metallo-heme proteins are ubiquitous in nature. Metalloporphyrins possess unique spectroscopic properties and the ability to form well-defined and predictable complexes with electron-donating axial ligands. Different complexes can be prepared having structures and labilities that vary depending on the metal used. These properties coupled with the tunable electrochemical behaviors of metalloporphyrins make them useful molecular tools for the construction of optical and recognition devices. Several porphyrin-based molecular receptors and devices have been reported.[26,27] However, few examples of peptide–porphyrin conjugates are found in the literature, despite the fact that these conjugates are reminiscent of heme proteins. A bis-porphyrin peptidic receptor **29** was shown to complex selectively to α,ω-diaminoalkanes.[28] The cooperative complexation of these guests in between the two porphyrin side chains induces a specific shift in the Soret absorption band. Therefore such porphyrin peptidic receptors could be used in optical devices. Porphyrin **27** was linked to a Glu side chain by an amide bond to give a suitably protected amino acid **28** that can be used in solid-phase peptide synthesis (Scheme 8). Likewise, porphyrin **27** can be added to γ-deprotected Glu-containing peptide **30** using DCC and HOBt as the coupling reagent (Scheme 9). The porphyrin-modified peptides can be easily metalated with $Zn^{2+}$. Indeed, complete metalation occurs, e.g. with peptide **31** using $Zn(OAc)_2$ and ultrasound in $CH_2Cl_2$ to give **32**.[28,29] The peptide backbone is stable under these mild conditions. Several groups have used metalloporphyrins to induce peptide aggregation or folding.[30,31]

*for references see p 191*

**Scheme 8** Solid-Phase Synthesis of a Porphyrin-Modified Peptide[28,36]

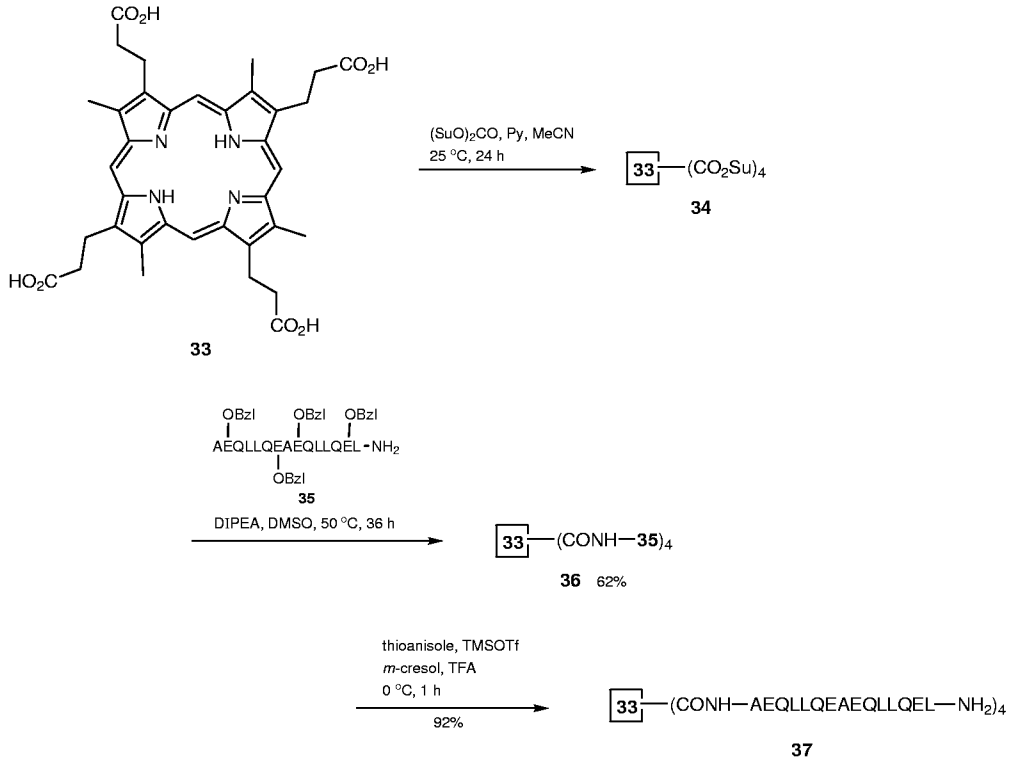



**Scheme 9** Coupling of a Porphyrin Derivative to a Glutamic Acid Tripeptide[36]




More relevant for this section is the use of porphyrins as template for the construction of *de novo* metalloproteins. Indeed, the attachment of helical peptide units to these templates creates four-helix bundle structures that have been used as an artificial ion channel[32] or a hydroxylase enzyme.[33,34] In these cases, the peptide units were coupled to the template by using the HOSu or the TBTU methods. As illustrated in Scheme 10 starting from **33**, formation of the tetrasuccimidyl ester **34** and attachment of the protected peptide unit **35** gives **36** and this is followed by deprotection to **37**.

**Scheme 10** Preparation of an Artificial Hydroxylase by Coupling of Peptide Segments on a Porphyrin Template[33,34]




**Synthesis of Boc-Glu(4-amidotetraphenylporphyrin) (28):**[28,36]
Boc-Glu-OMe (250 mg, 0.95 mmol) was dissolved in a minimum of $CH_2Cl_2$ at 0 °C and activated with HOBt (129 mg, 0.95 mmol) and DCC (197 mg, 0.95 mmol). After 30 min, the white DCU was filtered off and 4-aminotetraphenylporphyrin[35] (**27**; 500 mg, 0.79 mmol) in $CH_2Cl_2$ was added. The mixture was stirred for 24 h then washed with 0.05 M HCl, 5% $NaHCO_3$, and $H_2O$. The solvent was removed and the crude product purified by flash chromatography (5% MeOH/$CH_2Cl_2$) to give the methyl ester of **28** as a purple solid; yield: 518 mg (75%); mp >300 °C; UV–vis $\lambda_{max}$, nm (ε): 418 (630), 516 (12.2), 550 (6.8), 592 (5.3), 648 (5.1).

The above ester (1.72 g, 1.96 mmol) was saponified in MeOH (3 mL) with 1 M NaOH (4 mL). After 12 h of stirring $H_2O$ (25 mL) was added and MeOH was removed under reduced pressure. The resulting aqueous phase was extracted with $CH_2Cl_2$, acidified to pH 3 with 0.1 M HCl, then reextracted with $CH_2Cl_2$. The organic phase was separated, dried ($MgSO_4$), and concentrated to give **28** as a purple solid; yield: 1.20 g (71%); mp >300 °C.

**Metalation of the Porphyrin Ring in Peptides Containing Porphyrin 27:**[28,36]
Quantitative metalation of the porphyrin ring with $Zn^{2+}$ can be achieved by dissolving peptides incorporating the above amino acid **28** in a minimum of $CH_2Cl_2$ and sonicating them for 2 h with hydrated $Zn(OAc)_2$ (2 equiv). The metalation reaction is followed by TLC or UV–vis spectroscopy with two characteristic bands around 420 nm (589) and 548 nm (24.2) for the metalated species. The compounds are obtained after aqueous washings of the organic layer, drying ($MgSO_4$), and concentration.

**Preparation of Artificial Metalloprotein 37:**[33,34]
The free N-terminal protected peptide **35** ($4.52 \times 10^{-5}$ mol) was dissolved in hot DMSO (5 mL) with brief sonication. The apparent pH (wet pH paper) of the mixture was adjusted to 8.5 with DIPEA and the activated OSu ester porphyrin **34** (4.2 mg, $4.2 \times 10^{-6}$ mol) (prepared from the tetracarboxylic acid **33** and disuccinimido carbonate) was added. The resulting mixture was stirred at rt for 36 h and then applied to a Sephadex LH-60 column (2.5 cm i.d. × 87 cm) equilibrated with DMF. Elution with DMF gave the protected compound **36** after concentration; yield: 23.4 mg (62% based on the porphyrin); UV–vis (DMF) $\lambda_{max}$, nm: 621, 567, 530, 497, 398. The desired artificial protein **37** was obtained in a 92%

*for references see p 191*

yield by treating **36** with TFA, *m*-cresol, thioanisole, and TMSOTf for 1 h at 0 °C. The crude product was purified by RP-HPLC (C4 column) to give **37**; UV–vis (20 mM phosphate, 0.16 M KCl, pH 7.5) $\lambda_{max}$, nm: 616, 564, 534, 499, 397; MS ($^{252}$Cf ionization) *m/z*: 7598.9 $[M + H]^+$

## 9.4.4 Peptides with Metal Binding Sites

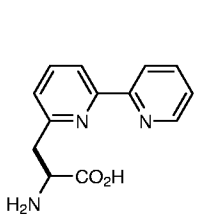
The incorporation of metal ligands onto peptidic structures is an active area of research. These compounds can be used as molecular devices to control the formation of supramolecular peptidic architectures, to induce and stabilize secondary structures, and as sensing devices for the detection of certain metals. The metal ligand group can be linked to the side chain of an amino acid and incorporated into a peptide chain, or it can be incorporated as a C- or N-terminal group. The strategy used depends on the application targeted.

### 9.4.4.1 Pyridyl, Bipyridyl, and Related Metal Ligand Peptides

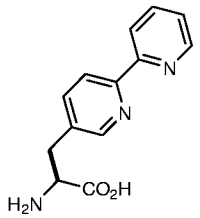
Pioneering work in this area was aimed at using specific metal ligand interactions to induce and stabilize secondary structures. This has been achieved by Ghadiri et al. for α-helical structures through the formation of transition metal and Ru(II) inert complexes with two imidazoles of His or one thiol of Cys and one imidazole of His in *i*, *i*+3 or *i*, *i*+4 relationships.[37,38] In almost all cases the helix content and stability increased upon metal complexation, especially with *i*, *i*+4 peptides. This work resembles the stabilization of helical structures using metal complexation by EDTA-like side chains discussed in Section 9.4.6.

The use of metal ligand modified amino acids allows more flexibility to the design principles. Several such amino acids have been prepared (**38–41**, Scheme 11). For example, Imperiali and Fisher[39–41] used an approach that involves the utilization of a β-turn structure of type II (**42**) to orient two bipyridyl side chains in a perfect manner to form stable chiral complexes with certain transition metals. The selective and intramolecular binding of $Cu^{2+}$ was demonstrated and the complexation induced a strong CD signal for the bipyridyl moieties showing that the metal is bound in a chiral environment.[39] The required Fmoc-protected bipyridyl-Ala derivative **45** was prepared according to Scheme 12.[41] In addition, ligand-modified amino acids **38–41** were prepared first as racemates by a modified version of the Sorenson method, then resolved by an enzymatic hydrolysis. This is illustrated in Scheme 13 by the reaction of diethyl acetamidomalonate with 2-(chloromethyl)-1,10-phenanthroline to give **46**, deprotection to give **47**, and resolution with alkaline protease and reprotection to give **48**.[42,43]

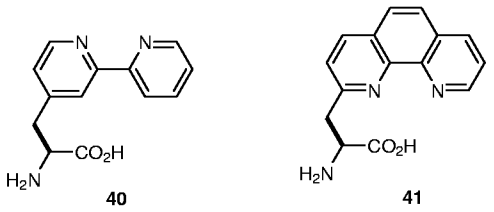
**Scheme 11** Metal Ligand Modified Amino Acids




**38**

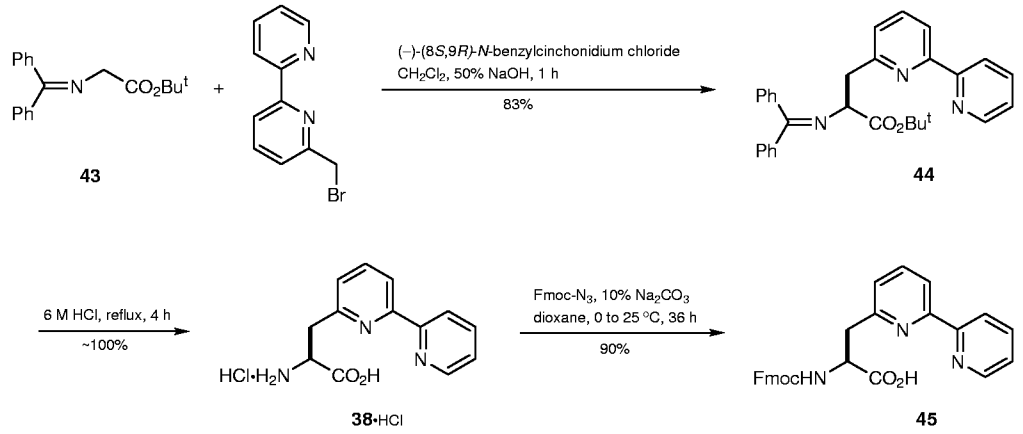



**39**

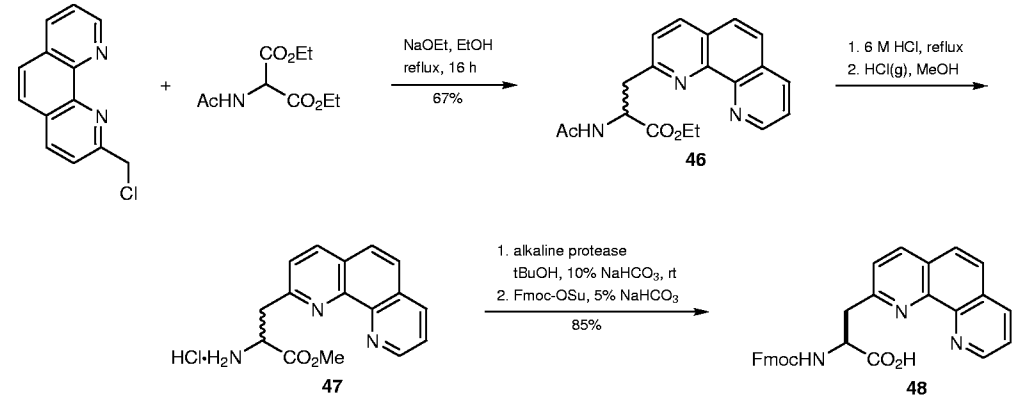



The same group extended their work recently to the synthesis of a peptide-based device detecting $Zn^{2+}$ in aqueous solutions.[44] The principle is to add a dansyl fluorophore on a Lys side chain of a synthetic Zn finger peptide. Upon complexation of $Zn^{2+}$, an important conformational change occurs that brings the fluorophore into a different environment and leads to a change in the fluorescence properties.

**Scheme 12** Synthesis of Bipyridylalanine[39–41]




**Scheme 13** Synthesis of Phenanthroline Alanine by Enzymatic Resolution[42,43]




***tert*-Butyl (*S*)-2-(Diphenylmethyleneamino)-3-(2,2-bipyridyl-6-yl)propanoate (44):**[41]
Commercially available *N*-diphenylmethylene-Gly-OtBu (**43**; 1.3 g, 4.4 mmol), *N*-benzylcinchonidinium chloride (0.31 g, 0.74 mmol), and 6-(bromomethyl)-2,2′-bipyridyl (0.95 g, 3.5 mmol) were added to a mixture of $CH_2Cl_2$ (20 mL) and 50% aq NaOH (7.04 mL) stirred vigorously for 1 h at rt. The organic

*for references see p 191*

phase was separated and the aqueous phase extracted with $CH_2Cl_2$. The combined organic phases were concentrated in vacuo and the residue taken up with $CH_2Cl_2$ (40 mL) and $H_2O$ (20 mL). The organic layer was separated, washed with $H_2O$ (× 2), dried ($Na_2SO_4$), and concentrated to afford the crude compound **44**. The latter was purified by flash chromatography (silica gel, hexane/EtOAc mixtures containing 0.5% TEA) to give chemically pure compound **44**; yield: 83%; 53% ee of the *S*-enantiomer. Optically pure material was obtained after crystallization from hexane at 4 °C of enantiomerically enriched **44**; mp (DL) 90–91 °C; $[\alpha]_D^{25}$ −327 (*c* 1, EtOAc).

**(−)-(*S*)-2-Amino-3-(2,2′-bipyridyl-6-yl)propanoic Acid (38):**[41]

A suspension of optically pure **44** (0.5 g) was refluxed in 6 M HCl (10 mL) for 4 h. The mixture was cooled, extracted with $Et_2O$, and concentrated to dryness. The residue obtained was lyophilized several times from $H_2O$ to afford a quantitative yield (0.3 g) of the amino acid **38** as its hydrochloride salt; mp 220 °C; $[\alpha]_D^{25}$ −18.6 (*c* 1, $H_2O$, pH 7), −12.9 (*c* 1, 5 M HCl); UV ($H_2O$) $\lambda_{max}$, nm (ε): 238 ($8.95 \times 10^3$), 284 ($13.42 \times 10^3$).

**(−)-3-(2,2′-Bipyridyl-6-yl)-2-(Fmoc-amino)propanoic Acid (45):**[41]

A soln of Fmoc-$N_3$ (0.83 g, 3.1 mmol) in dioxane (10 mL) was added dropwise over a 2 h period to a stirred soln of the free amino acid **38** in 10% aq $Na_2CO_3$ (10 mL) at 0 °C. The mixture was allowed to warm to rt and stirred 36 h. After dilution to 100 mL with $H_2O$, the crude mixture was extracted with $Et_2O$ (3 × 50 mL). The aqueous layer was cooled in an ice bath and brought to pH 2 with concd HCl. The suspension was centrifuged at 5000 rpm for 10 min, the aqueous phase was decanted, and the solid washed with $H_2O$ (× 2) the same way. The solid was taken up in MeOH and concentrated to give white powder; yield: 1.36 g (90%); $[\alpha]_D^{25}$ −61.3 (*c* 1, MeOH); HRMS *m/z*: 466.1781.

**Synthesis of Peptide 42 Incorporating 38:**[39,40]

Peptide **42** was synthesized using solid-phase Fmoc protection and BOP/HOBt activated ester chemistry on PAL resin. For commercially available amino acids, 4 equiv were used per coupling whereas, for amino acid **45**, only 2.5 equiv were used. Activated esters were formed in situ in DMF using BOP, HOBt, and 0.451 M NMM. Coupling times were 1 or 2 h depending on the amino acid. Fmoc group deprotection was performed by a 1 h treatment with a 20% piperidine/DMF soln. Peptide was acetylated using $Ac_2O$ (21 equiv) and TEA (5 equiv) in DMF (3 mL) for 2 h. The resin was washed with $CH_2Cl_2$, briefly air-dried, and cleaved using Reagent R (TFA/thioanisole/EDT/anisole 90:5:3:2). The crude peptide was obtained by lyophilization from $H_2O$ and purified using P2 gel filtration chromatography (1 M AcOH).

**Synthesis of Chiral Phenanthroline-Modified Ala 48:**[42,43]

Diethyl acetamidomalonate (5.68 g, 26.1 mmol) was added to a soln of Na (0.61 g, 26.4 mmol) in abs EtOH (60 mL) and stirred for 30 min. A soln of 2-(chloromethyl)-1,10-phenanthroline (3.03 g, 13.2 mmol) in abs EtOH (15 mL) was added dropwise to the soln. After 16 h of reflux, the mixture was cooled, concentrated, suspended in $H_2O$, and extracted with $CHCl_3$ (3×). The organic fractions were combined and concentrated to a red oil, which was suspended in 0.05 M HCl and washed with EtOAc. The aqueous layer was neutralized at 0 °C with sat. $Na_2CO_3$ and extracted with $CHCl_3$. The organic fractions were combined, dried ($Na_2SO_4$), and then concentrated to crude **46** obtained as a red oil (2.96 g, 67%). Purification on neutral alumina ($CH_2Cl_2$/EtOAc, 2:1) yielded pure **46** as a white solid; mp 182.5–184.3 °C (dec); HRMS *m/z*: 338.1504 $[M]^+$.

A soln of compound **46** (10.21 g, 30 mmol) in 6 M HCl (300 mL) was heated to 90 °C with stirring for 10 h. The reaction was cooled and concentrated in vacuo to a thick red oil. The oil was then resuspended in $H_2O$ and concentrated twice to remove excess HCl, then dried ($P_2O_5$). The solid mixture was then suspended in MeOH (250 mL) and anhyd HCl was bubbled through for 10 min. After stirring overnight, the suspension was concentrated, resuspended in MeOH, and the solvent removed twice to remove excess HCl. The solid residue was neutralized with sat. $Na_2CO_3$ and extracted with $CHCl_3$. The organic phase was dried ($Na_2SO_4$), and concentrated to give a quantitative yield of racemic **47** as a yellow oil used directly in the next step; HRMS *m/z*: 268.1101.

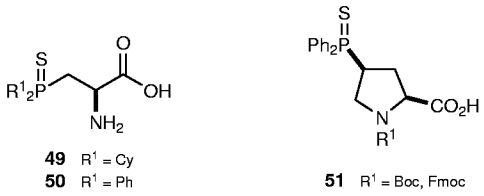
Racemic **47** (3.66 g, 16 mmol) was dissolved in tBuOH (45 mL) and added to 10% $NaHCO_3$ (445 mL). A sample of alkaline protease Type VIII (Subtilisin Carlsberg, CAS Registry #9014-01-1, 11.6 units·$mg^{-1}$ from Sigma Chemical Co.; 125 mg, 800–2000 units) in 0.2 M $NaHCO_3$ (120 mL) was added to the methyl ester soln in a 1-L Erlenmeyer flask. The mixture was rotated at 120–200 rpm and the reaction monitored by chiral HPLC at 20 min intervals. When the ratio of the D- to L-amino acid ester was >50:1, the mixture was extracted with $CHCl_3$ (6 × 150 mL). The aqueous phase was reduced in volume to remove any $CHCl_3$ then lyophilized to give the crude L-amino acid mixed with salts. This crude material was dissolved in 10% $Na_2CO_3$ (25 mL), then Fmoc-OSu (1.2 equiv) dissolved in dioxane (15 mL) was added dropwise to the amino acid soln. The resulting mixture was shaken periodically for 1.5 h, then diluted with $H_2O$ (100 mL), washed several times with $Et_2O$, and acidified to pH <2 at 0 °C with concd HCl. The precipitate was recovered by centrifugation and transferred into a round-bottomed

flask as a MeOH soln. The MeOH was removed and replaced by toluene (80 mL). The toluene mixture was concentrated and the residue triturated with $Et_2O$ to afford the desired compound **48** as white solid; yield: 85%; mp 187 °C (dec); TLC $R_f$ 0.55 (silica gel, $CHCl_3$/MeOH 4:1); $[\alpha]_D^{25}$ −67.9 (*c* 0.35, DMSO); UV (MeOH) $\lambda_{max}$, nm: 299, 268, 224; HRMS (*m/z*): 490.1749 $[M+H]^+$.

### 9.4.4.2 Peptides with Phosphine Ligands

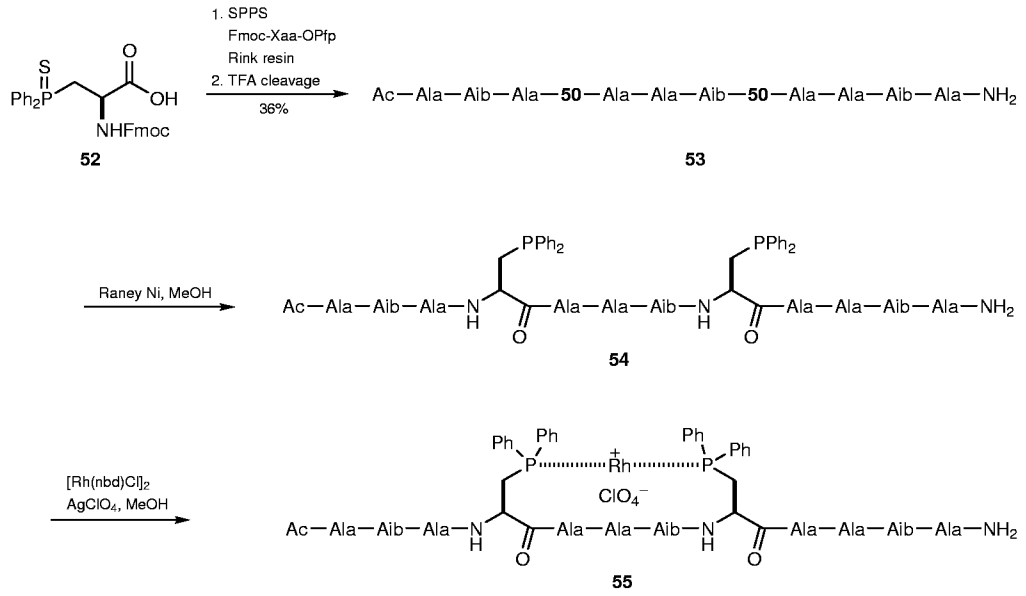
The development of chiral peptide-based metal catalysts has also been studied. The group of Gilbertson has synthesized several phosphine-modified amino acids and incorporated two of them into short peptide sequences.[45–47] They demonstrated the formation of several metal complexes, in particular Rh complexes, and reported their structure as well as their ability to catalyze enantioselectively certain hydrogenation reactions.[48] While the enantioselectivities observed are modest so far, optimization through combinatorial synthesis will probably lead to useful catalysts. The synthesis of the sulfide protected form of both Fmoc- and Boc-dicyclohexylphosphinoserine **49** and -diphenylphosphinoserine **50** has been reported, in addition to diphenylphosphino-L-proline **51** (Scheme 14).[49] To show their compatibility with solid-phase peptide synthesis, they were incorporated into hydrophobic peptides, such as dodecapeptide **53**, using the standard Fmoc protocol (Scheme 15).[45] For better results, the phosphine-modified amino acid **50** was coupled as a Fmoc-protected dipeptide **56**, rather than the usual Fmoc derivative **52**.[47] As an illustrative example, the synthesis of diphenylphosphinoserine **52** is depicted in Scheme 16.[45]

**Scheme 14** Sulfide Protected Phosphinoserine and Phosphino-L-proline[49]




**49** $R^1$ = Cy
**50** $R^1$ = Ph
**51** $R^1$ = Boc, Fmoc

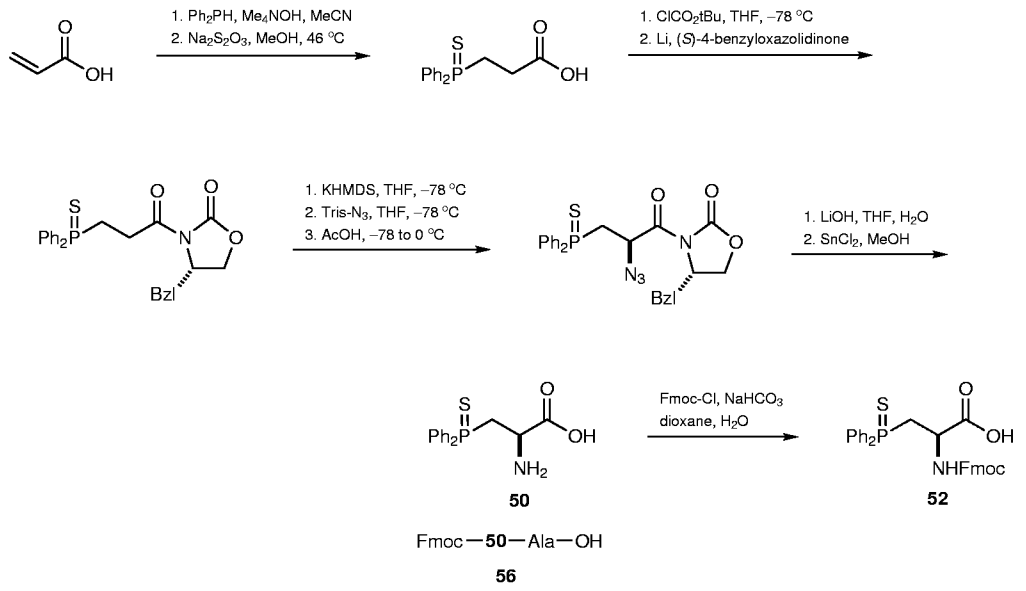
**Scheme 15** Solid-Phase Synthesis of a Bis-phosphine Modified Peptide[45,47]




**52** **53** **54** **55**

*for references see p 191*

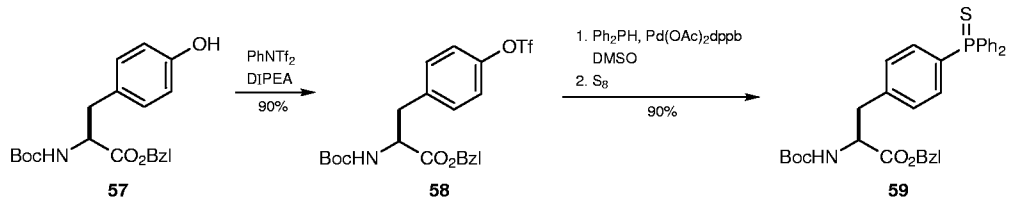
**Scheme 16** Synthesis of Diphenylphosphine Modified Alanine[45,47]




The synthesis uses Evans chemistry to introduce enantioselectively an azide group, which is then reduced to the α-amino group (Scheme 16). Note that in all cases, the phosphinyl group is protected as its sulfide during the synthesis to avoid complication (oxidation) observed with the phosphine. The phosphine sulfide is compatible with Fmoc- and Boc-peptide synthesis conditions. The free phosphine can later be regenerated by treating the sulfide with Raney nickel in MeOH (as shown in Scheme 15) or by methylation followed by treatment with HMPT.[50] This reduction was even applied successfully to sulfide-protected phosphine peptides still attached to the solid support.[51]

Amino acids having a phosphine ligand modified side chain can also be prepared conveniently by a catalytic conversion of phenols into arylphosphines, using a Stille-type reaction.[52] Hence, suitably protected Tyr and hydroxyphenylglycines can be converted readily into phosphine-modified amino acids, which are protected and stored as their sulfide analogues; for example, in Scheme 17 compound **59** is synthesized by conversion of the phenolic hydroxy group in **57** into its triflate **58** and reaction with diphenylphosphine followed by oxidation of the phosphinyl group with $S_8$. Several analogues such as **59** have been prepared by this strategy and extended to Tyr within peptides. The desired phosphines can also be regenerated by reduction with Raney nickel. The process however is sensitive to racemization. The degree of epimerization depends on the nature of the N-protecting group and is minimal with Boc groups and more important when an acetyl group is used.

**Scheme 17** Conversion of Phenolic Side Chains into Phosphine Ligands[52]



**Synthesis of Rh·Diphosphine Modified Peptide Complex 55:**[47]
Peptide **55** was synthesized using a segmental approach where the phosphine sulfide amino acid **50** was coupled as a dipeptide with Ala **56**, and the Aib as either a dipeptide Fmoc-Ala-Aib or a tripeptide Ac-Ala-Aib-Ala. Wang, RapidAmide, or Rink resins can be used depending on the C-terminal group desired.

**Fmoc-50-Ala-OH (56):**[47]
To Ala-ONbz·TosOH (1.80 g, 4.93 mmol) in $CH_2Cl_2$ (150 mL), was added TEA (3.2 mL, 24.6 mmol). The clear soln was washed with $H_2O$ and brine, dried ($MgSO_4$), then concentrated under reduced pressure. $CH_2Cl_2$ (100 mL) was added, the soln cooled to 0 °C, then **52** (3.79 mmol) in $CH_2Cl_2$ (100 mL) was introduced followed by EDC (872 mg, 4.55 mmol) and HOBt (614 mg, 4.55 mmol). The mixture was stirred 4 h at 0 °C and 20 h at rt before being washed with 2 M HCl, 1 M $NaHCO_3$, $H_2O$, and brine. The organic phase was dried ($MgSO_4$), filtered, and the solvent removed to afford the crude product which was crystallized (EtOAc/hexane). The Nbz ester of dipeptide **56** thus obtained (2.33 g, 84%) was used directly in the next step. HRFAB-MS *m/z*: 734.20900 $[M + H]^+$.

Zn powder (8.93 g, 137 mmol) was added to a soln of the above Nbz-protected dipeptide (2.0 g, 2.73 mmol) in AcOH/$H_2O$ (9:1, 150 mL). After stirring at rt for 1.5 h, the mixture was filtered through Celite. Most of the AcOH and $H_2O$ were removed under vacuum at 45 °C. The resulting yellowish slurry was triturated with $CHCl_3$ (500 mL). The $CHCl_3$ phase was filtered and washed with 1 M HCl, brine, and dried ($MgSO_4$). After filtration, the solvent was concentrated to give crude dipeptide **56**, which was purified by chromatography (silica gel, EtOAc/hexane/AcOH 50:49:1); yield: 1.48 g (91%); HRFAB-MS *m/z*: 599.17680 $[M + H]^+$.

**Ac-Ala-Aib-Ala-50-Ala-Ala-Aib-50-Ala-Ala-Aib-Ala-$NH_2$ (53):**[47]
The peptide was synthesized on Rink resin (0.2 mmol) using Pfp activated ester of the Fmoc-protected amino acids. All couplings were complete after 2 h. After acetylation, the peptide was cleaved by a 2 min treatment with a 50% TFA soln in $CH_2Cl_2$. After filtration, the TFA soln was concentrated by a stream of $N_2$ to 1–2 mL and the crude peptide was precipitated by adding $Et_2O$ (25 mL) and cooling to −78 °C. The $Et_2O$ layer was removed and the peptide dried under vacuum. After purification by RP-HPLC pure peptide **53** was obtained; yield: 101 mg (36%); HRFAB-MS *m/z*: 1408.55180 $[M + H]^+$.

**Desulfurization of 53:**[47]
Raney nickel (300 mg) was washed with degassed MeOH (3 × 2 mL) and then transferred, using MeOH (3 mL), into a Schlenk tube containing peptide **53** (21 mg, 15 μmol). The mixture was pump–thaw degassed (three cycles) and was stirred for 12 h at 48 °C. After cooling to rt, the mixture was filtered under $N_2$ into a two-necked, round-bottomed flask and the solvent concentrated. $^{31}P$ NMR was used to monitor the course of desulfurization and the reduced peptide **54** was used immediately in the next step. $^{31}P$ NMR (methanol-$d_4$, δ): −20.85, −23.79.

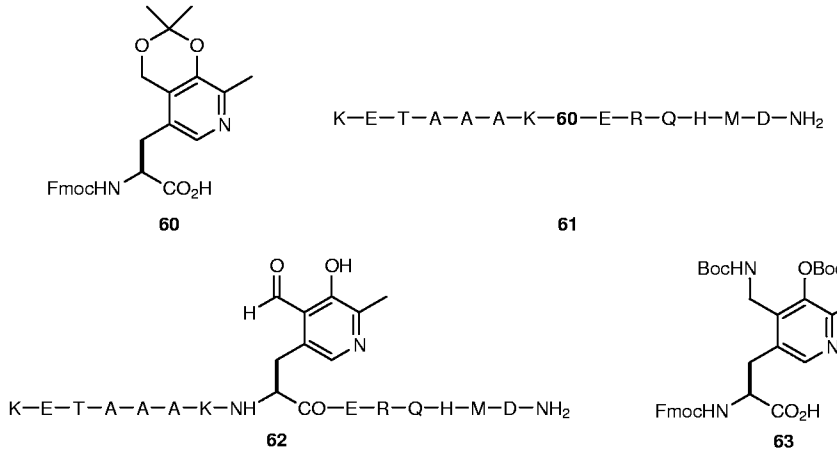
**Rh Peptide Complex 55:**[47]
$[Rh(nbd)Cl]_2$ (3.46 mg, 7.5 μmol) and $AgClO_4$ (3.11 mg, 15.0 μmol) were weighed into a vial. Degassed MeOH (0.5 mL) was added and the mixture was stirred for 15 min under $N_2$. The precipitated AgCl was filtered off and the filtrate was added to the above round-bottomed flask containing diphosphine peptide **54**. The soln was stirred under $N_2$ until the color changed from yellow to orange, then the solvent was removed. The complex was purified by flash chromatography (silica gel). After eluting with degassed $CH_2Cl_2$, the desired complex **55** was eluted with MeOH and contained in the yellow fraction; FAB-MS *m/z*: 1516.9 $[M + H]^+$; $^{31}P$ NMR (methanol-$d_4$, δ): 27.24, 23.85.

## 9.4.4.3 Peptides with Coenzyme Chimeras

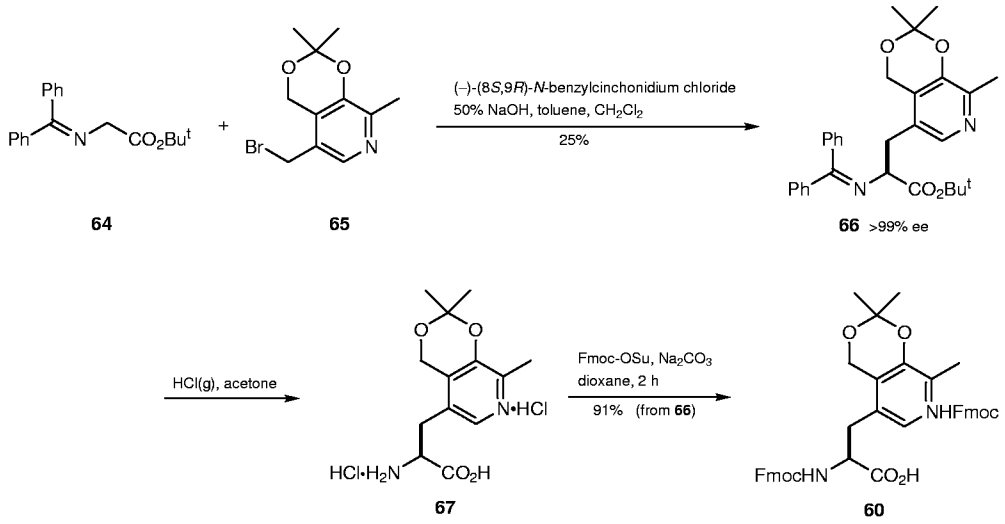
Toward the development of *de novo* designed functional proteins, Imperiali and co-workers prepared an unnatural amino acid **60** bearing a pyridoxal moiety on the side-chain (see Schemes 18 and 19).[53,54] When this coenzyme–amino acid chimera is introduced into peptidic structures, it gives them some transaminase activity. In a first semi-synthetic approach, pyridoxal-amino acid conjugate **60** was introduced at position 8 of analogues of the S-peptide of RNase A (e.g., **61**). The selective oxidation of **60** (within **61**) to the active form **62** was accomplished by a treatment with γ-$MnO_2$ in DMSO.[53] A mixture of reduced and oxidized peptides was obtained, but used directly to study their transaminase activity when complexed with the S-protein. One of the *de novo* enzymes showed notable transaminase activity. Importantly, the activity decreased in the presence of $Cu^{2+}$ suggesting the involvement of the unnatural side chain in the reaction mechanism.[54]

*for references see p 191*

**Scheme 18** Synthetic Approach to a Peptide Containing a Pyridoxal Moiety on a Side Chain[53,54] and a Synthetic Co-factor Amino Acid **63**[55]



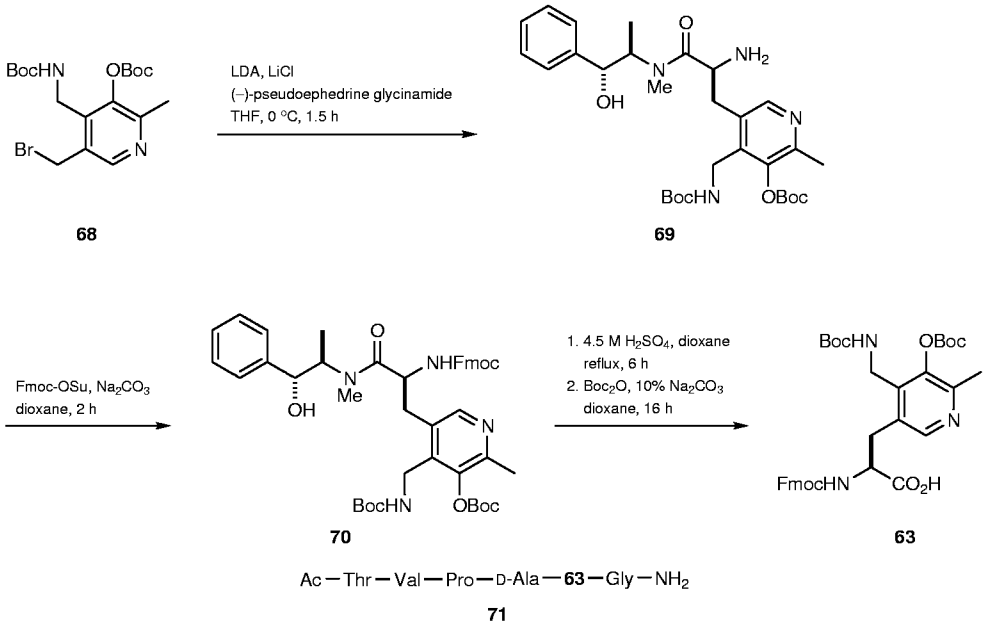

**Scheme 19** Synthesis of a Pyridoxal-Modified Alanine by Asymmetric Phase-Transfer Catalysis[53,54]




In an effort to prepare totally synthetic systems and to avoid post-synthesis transformation of the coenzyme–amino acid chimera, the same group reported the synthesis and the incorporation in peptide structures of a chimera **63** having the pyridoxamine oxidation state.[55] The latter concept involves the synthesis of Fmoc/Boc-protected amino acid **63** according to Scheme 20, from commercially available pyridoxamine.[55] The key step is the alkylation of (−)-pseudoephedrine glycinamide enolate with the bis-Boc-protected pyridoxamine **68** giving **69**. Diastereoselectivities larger than 10:1 were obtained. After protection with Fmoc-OSu by the usual procedure giving **70**, the chiral auxiliary was removed by refluxing for 6.5 hours with a 1:1 solution of 4.5 M $H_2SO_4$ and dioxane. The hydroxy and amino groups on the side chain were subsequently reprotected as Boc groups to afford a fully protected derivative **63**. To demonstrate the compatibility of **63** with solid-phase synthesis, hexapeptide **71** was prepared using Fmoc-PAL-PEG resin and DIC/HOBt as coupling reagent. An acid treatment released the desired peptide amides. Unexpectedly, the phenolic group of the side chain gets acetylated during the synthesis and a 12-hour treatment of the resin with saturated

$K_2CO_3/H_2O$/MeOH (1:4:5) prior to the final cleavage step was necessary to remove the undesired *O*-acetyl group.

**Scheme 20** Synthesis of a Pyridoxal-Modified Alanine by Stereoselective Alkylation and Structure of Hexapeptide **71**[55]




**Synthesis of Pyridoxal-amino Acid** 60:[54]
*tert*-Butyl *N*-(diphenylmethylene)glycinate (**64**; 1.66 g, 5.61 mmol) and (–)-(8*S*,9*R*)-*N*-benzylcinchonidinium (472 mg, 1.12 mmol) were dissolved in a mixture of toluene/$CH_2Cl_2$ (7:3, 20 mL) and 50% NaOH (8.2 mL) at 5 °C. The bromide **65** (2.29 g, 8.41 mmol) was added dropwise and the resulting mixture was stirred for 2 h. The organic layer was separated, dried ($Na_2SO_4$), and concentrated in vacuo. The residue was purified by flash chromatography ($Et_2O$/hexane 2:1). Compound **66** (1.86 g, 68%) was obtained as an oil which was crystallized from hexane. The supernatant was decanted and concentrated to give the (*S*)-enantiomer; yield: 687 mg (25%); >99% ee; $[\alpha]_D^{25}$ –143.9 (*c* 0.5, $CHCl_3$); HRFAB-MS *m/z*: 487.2599 $[M + H]^+$.

The above compound **66** (1.22 g, 2.51 mmol) was dissolved in anhyd acetone (20 mL) and cooled to 0 °C. The soln was saturated with anhyd HCl and stirred at rt for 30 min. After the solvent was removed, the dihydrochloride **67** was neutralized with 10% $Na_2CO_3$ and washed with hexane. A soln of Fmoc-OSu (931 mg, 2.51 mmol) in dioxane (26 mL) was added dropwise to the amino acid soln at 5 °C and the mixture was stirred at rt for 12 h. The mixture was diluted with $H_2O$ and washed with $Et_2O$. The aqueous phase was acidified to pH 3 with concd HCl and the suspension formed extracted with EtOAc. The organic phase was dried ($Na_2SO_4$) and concentrated to yield the desired Fmoc-protected amino acid **60**; yield: 1.11 g (91% yield for the two steps); TLC $R_f$ 0.14 (silica gel, $CHCl_3$/MeOH 5:1); $[\alpha]_D^{25}$ +18.8 (*c* 0.5, $CHCl_3$); HRFAB-MS *m/z*: 489.2014 $[M + H]^+$.

**Synthesis of Pyridoxal-Modified Peptide** 62:[53,54]
Peptide **61** incorporating pyridoxal-modified amino acid **60** was synthesized on a 0.124–0.146 mmol scale using Fmoc amino protection strategy on PAL-PEG-polystyrene resin. Amino acids were coupled using DIC/HOBt as reagent using commercially available Fmoc-protected amino acids (3 equiv) and **60** (2 equiv). Standard procedures were used for coupling and deprotection steps. The peptide was cleaved using reagent R (TFA/thioanisole/EDT/anisole 90:5:3:2, 5 mL). The crude peptide was purified by RP-HPLC (MeCN/$H_2O$ gradient); UV (0.1 M NaOH) $\lambda_{max}$, nm: 308; HRFAB-MS *m/z*: 693.3604 $[M + H]^+$.

The peptide was oxidized to the aldehyde active form **62** using freshly prepared γ-$MnO_2$ (50 mg) added to a soln of the above peptide (2 mg) in DMSO (120 μL). The mixture was stirred in the dark for 12 h, filtered through Celite, and the column washed with $H_2O$ for maximum recovery. The filtrate was filtered on a 1.2-μm filter and lyophilized to give the pyridoxal peptide **62** as a yellow oil (15–40%),

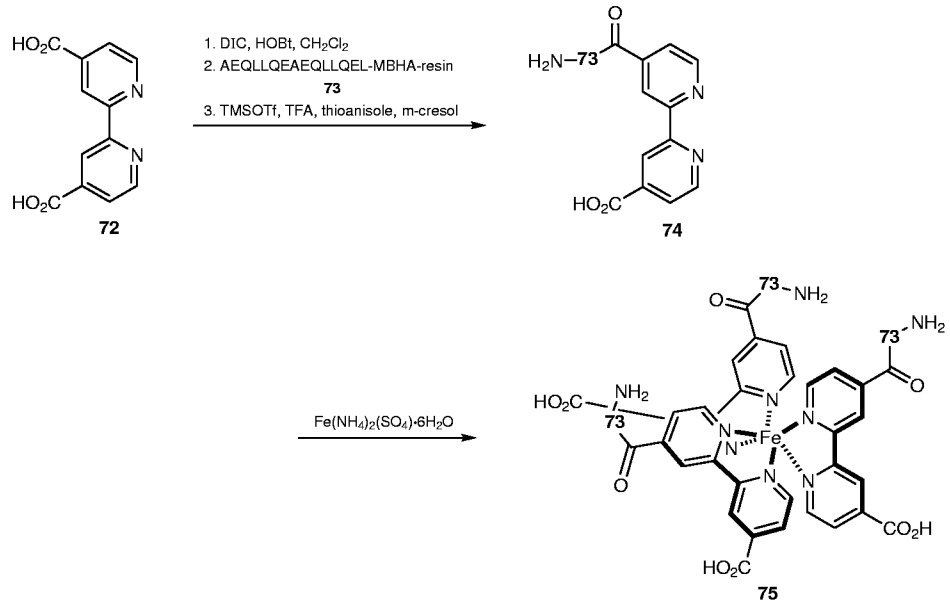
*for references see p 191*

which was used directly in transaminase reactions. The degree of oxidation was determined by UV–vis spectroscopy ($\varepsilon_{390\ nm}$ 6400 $cm^{-1}$ in 0.1 M NaOH); UV (0.1 M NaOH) $\lambda_{max}$, nm; 392; HRFAB-MS *m/z*: 691.3427 $[M + H]^+$.

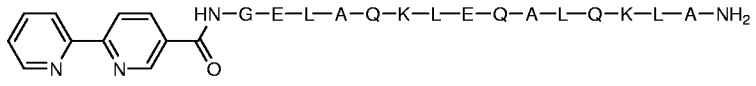
### 9.4.4.4 Peptides with a Ligand at Terminal Positions

The attachment of metal ligands at the C- or N-terminal position has been utilized to induce and control the formation of well-defined supramolecular peptidic structures. Sasaki and Lieberman attached a bipyridyl group **72** at the N-terminal of an amphiphilic peptide unit **73** attached to MBHA resin (Scheme 21).[56,57] After cleavage and purification giving **74**, addition of $Fe^{2+}$ resulted in the formation of four isomeric, three-helix-bundle superstructures such as **75**. Similarly, Ghadiri et al. used a 5-carboxy-2,2′-bipyridyl group at the N-terminal of peptide structure **76** to eliminate the possible stereoisomers (Scheme 22).[58] The bipyridyl unit was also coupled to the peptide still linked to Merrifield resin using the conventional DIC/HOBt method. They demonstrated that only one major diastereomer of a three-helix structure was formed upon addition of metal. As an extension to their work, the group of Ghadiri prepared a stable four-helix-bundle superstructure by self assembling around one Ru(II) four units of peptide **77** modified at the N-terminal position with a pyridine group.[59] The pyridine moiety was introduced by coupling nicotinic acid through a spacer at the N-terminal position of a 15-residue peptide.

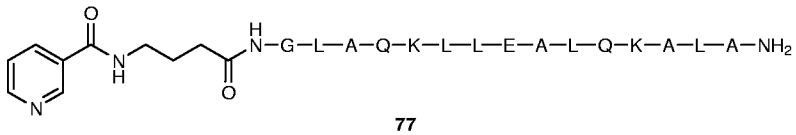
**Scheme 21** Coupling of Peptide Segments to a Bipyridyl Ligand and $Fe^{2+}$ Induced Self-Assembly of a Three-Helix Supramolecular Structure[56,57]



**Scheme 22** Metal Ligand N-Terminal Modified Peptides[58,59]




**76**




**77**

**Coupling of Bipyridylcarboxylic Acid (72):**[56,57]
Boc-protected peptide **73** (0.1 mmol) attached to MBHA resin was deprotected with TFA and washed with $CH_2Cl_2$. The 2,2′-bipyridyl-4,4′-dicarboxylic acid (**72**, 0.2 mmol) was dissolved in 5% DIPEA in $CH_2Cl_2$ (3 mL) and HOBt (0.5 mmol) and DIC (0.5 mmol) were added to form the HOBt diester. After 45 min of stirring, the diester soln was incorporated into the peptide suspension and the mixture was shaken for 8 h. The resin was washed with $CH_2Cl_2$ and 30% EtOH/$CH_2Cl_2$ then dried. Cleavage of the peptide from the resin was performed using TMSOTf in TFA in the presence of thioanisole and *m*-cresol at 0 °C for 2 h. The crude peptide was purified first by gel-permeation chromatography on a Sephadex G-15 column eluting with 2% $NH_4HCO_3$, followed by preparative RP-HPLC on a C4 column; UV–vis. ($H_2O$) $\lambda_{max}$, nm (ε, $M^{-1}$ $cm^{-1}$): 289 (11,170); FAB-MS *m/z*: 2003 $[M + Na]^+$.

## 9.4.5 Ferrocenyl Peptides

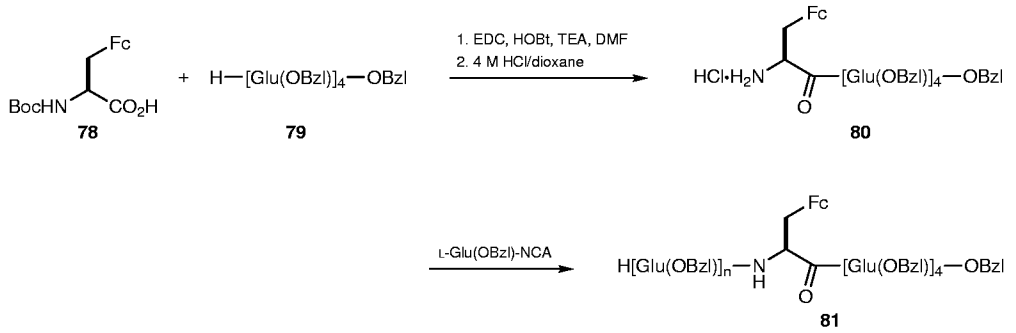
The ubiquitous electrochemical behavior of ferrocene and its relative chemical stability have made this organometallic complex a useful group for the preparation of redox-active devices. The incorporation of ferrocene-modified amino acids into larger polypeptide structures can therefore lead to electrochemically active *de novo* designed proteins. In addition, the attachment of ferrocene derivatives to peptides make them electroactive and eligible to electrochemical detection. Hence, it is not surprising that the first synthesis of a ferrocene-modified amino acid dates back to the 1950s.

### 9.4.5.1 Preparation of Ferrocenylalanine via Resolution

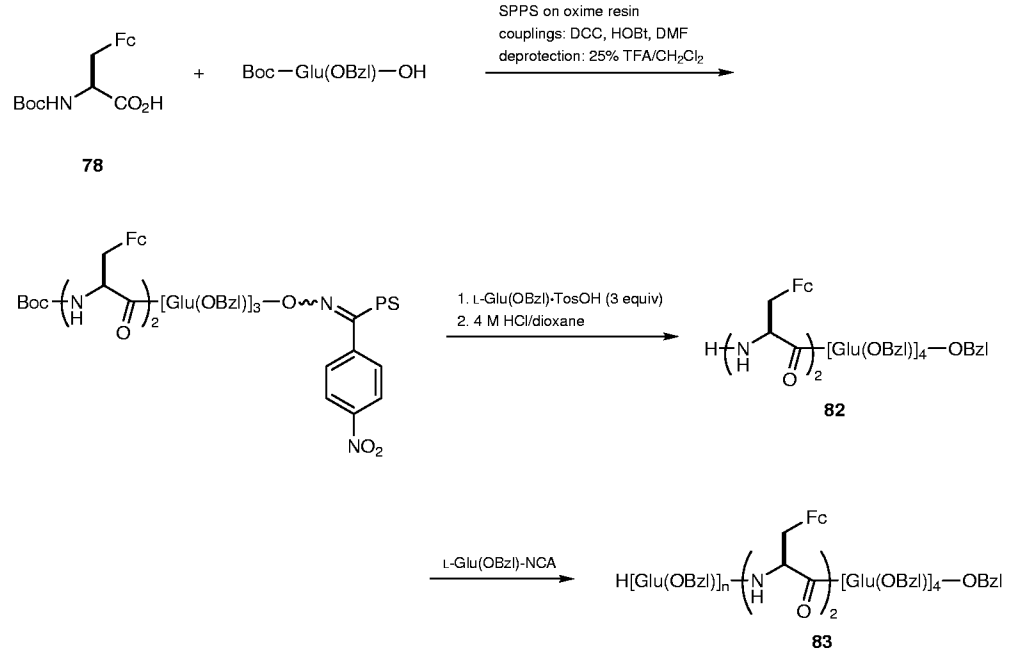
Toward the preparation of electroactive biomaterials, helical poly(benzyl-L-glutamate) incorporating one or two ferrocenyl-L-alanine at the C-terminal position, **81** and **83**, were prepared.[60] The synthesis was done by the classic polymerization of benzyl-L-glutamic acid-NCA using the ferrocenyl-modified peptides **80** or **82** (Schemes 23 and 24). In the preparation of **81**, the initiator pentapeptide **80** was synthesized in solution and the Boc-ferrocenyl-L-alanine **78** was coupled to a protected tetraGlu derivative **79** using EDC and HOBt in DMF. Interestingly, the other initiator hexapepeptide **82** bearing two ferrocenyl-L-alanines was prepared by solid-phase synthesis using the oxime resin[61] as depicted in Scheme 24. The necessary Boc-protected ferrocenyl-L-alanine **78** was synthesized by adapting the original procedure of Osgerby and Pauson[62] (Scheme 25) and resolving the racemic acetamido derivative **87** enzymatically.[60]

*for references see p 191*

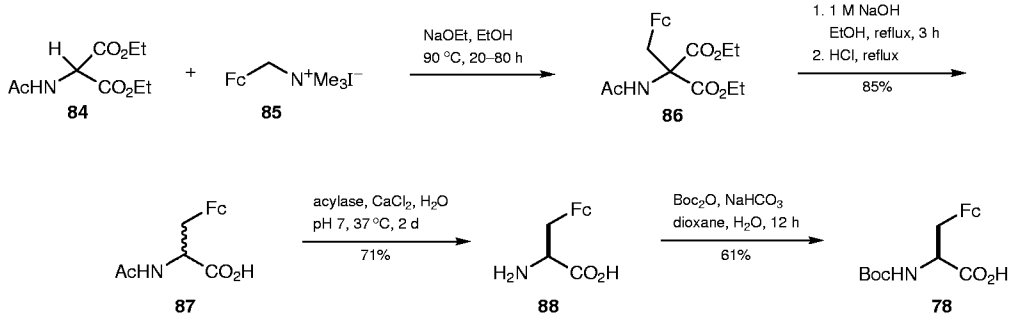
**Scheme 23** Coupling of Ferrocenylalanine and Its Use as NCA Polymerization Initiator[60]




**Scheme 24** Solid-Phase Synthesis of a Ferrocenyl-Modified Peptide and Its Use as NCA Polymerization Initiator[60]




**Scheme 25** Synthesis of Ferrocenylalanine by Enzymatic Resolution[60]




**Synthesis of Boc-ferrocenyl-L-alanine (78):**[60]

Na (0.6 g, 0.026 mmol) was dissolved in dry EtOH, then diethyl acetamidomalonate (**84**; 5.64 g, 0.026 mmol) and subsequently (ferrocenylmethyl)trimethylammonium iodide (**85**; 5.0 g, 0.013 mmol)

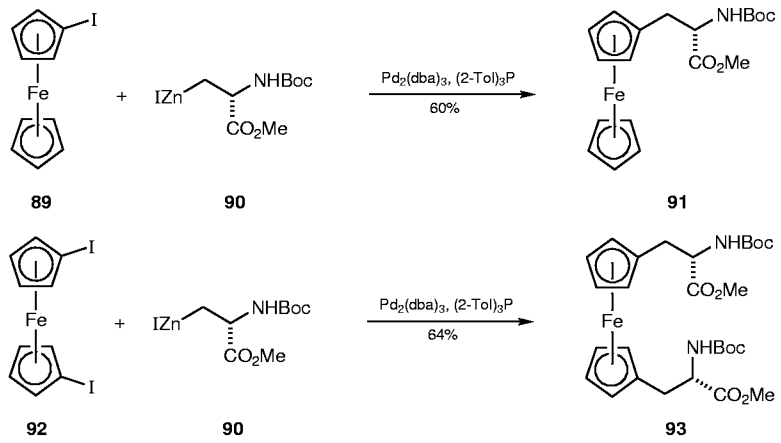
were added to the soln. The mixture was refluxed for 20–80 h, then EtOH was removed under reduced pressure, $H_2O$ was added, and extracted with $Et_2O$. After drying ($Na_2SO_4$), removal of the solvent gave the intermediate **86** as an oil. The latter (3.0 g, 7.2 mmol) was refluxed with an excess of 1 M NaOH for 3 h and then cooled to rt. HCl was added to adjust pH to 2 and the resulting mixture was again refluxed for 2 h. EtOH was removed and the desired *N*-acetyl-DL-ferrocenylalanine **87** was precipitated with $H_2O$; yield: 85%; mp 198 °C.

The racemic acetylamino acid **87** was dissolved in 1 M NaOH and the pH adjusted to 7 with HCl. Then an acylase soln [4 mL, from *Aspergillus* genus, 260 mg in $H_2O$ (10 mL) with $CoCl_2 \cdot 6H_2O$ (4 mg)] was added and the resulting mixture was stirred gently at 37 °C for 2 days. Optically pure L-ferrocenylalanine **88** was precipitated as a yellow powder and recrystallized (MeOH); yield: 0.55 g (71%). The solid (0.30 g, 1.1 mmol) was suspended in dioxane/$H_2O$ (2:1, 3 mL) and $NaHCO_3$ (0.23 g) in $H_2O$ (1.5 mL) was added. $Boc_2O$ (0.29 g, 1.32 mmol) in dioxane (1 mL) was incorporated and the mixture stirred for 12 h. Dioxane was concentrated and the pH of the aqueous phase adjusted to 2 with 5% $KHSO_4$. Extraction with EtOAc, drying ($Na_2SO_4$), and removal of the solvent gave Boc-L-ferrocenylalanine **78**; yield: 0.25 g (61%); UV $\lambda_{max}$, nm (ε): 203.5 ($5.7 \times 10^4$), 435.5 (103).

### 9.4.5.2 Enantioselective Synthesis of Ferrocenylalanine

An enantioselective synthesis of ferrocenyl-modified alanines **91** and **93** was also reported.[63] In this case, an organozinc derivative of serine **90** was coupled to iodoferrocene **89** or 1,1′-diiodoferrocene **92** using a palladium catalyst [$Pd_2(dba)_3$] to give the Boc-ferrocenyl-L-alanine methyl esters **91** and **93** in 60 and 64% yield, respectively (Scheme 26).

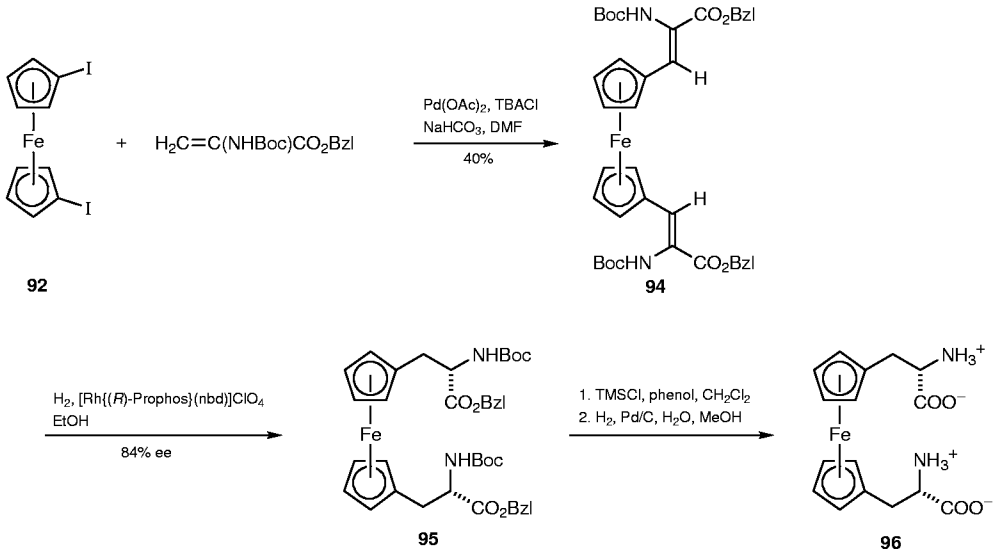
**Scheme 26** Synthesis of Optically Pure Ferrocenylalanines **91** and **93** via a Palladium Coupling[63]



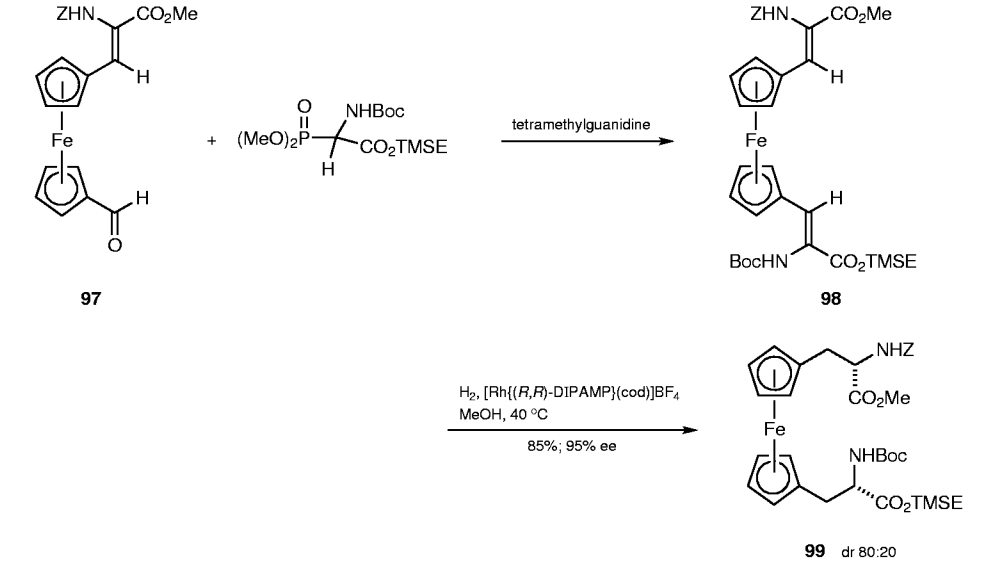

Another synthesis of the corresponding benzyl ester **95** taking advantage of well-known catalytic asymmetric hydrogenation processes has been described (Scheme 27).[64] The hydrogenation of **94** using [Rh{(*R*)-Prophos}(nbd)]$ClO_4$ gave **95** as a mixture of the (−)-isomer, the *meso*, and the (+)-isomer in a 5:38:57 ratio corresponding to 84% enantiomeric excess and a diastereomeric excess of only 23.5%. Fortunately, the same authors recently improved their synthetic strategy and prepared **95** in 90:10 diastereomeric ratio and >95% enantiomeric excess using [Rh{(*R*,*R*)-DIPAMP}(cod)]$BF_4$ as catalyst;[65] **95** could be deprotected to the corresponding free amino acid **96** (Scheme 27). They extended their work to the synthesis of a fully protected 1,1′-ferrocenylbis(L-alanine) **99** having a >95% enantiomeric excess and 80:20 diastereomeric ratio; starting from the ferrocenyl aldehyde **97**, Horner–Emmons reaction gives the diene **98** and asymmetric hydrogenation gives **99** (Scheme 28).[65] The orthogonal protection scheme in **99** allows its possible selective deprotection and incorporation into peptidic devices.

*for references see p 191*

**Scheme 27** Synthesis of Ferrocenylbis(alanine) by Asymmetric Hydrogenation[64]

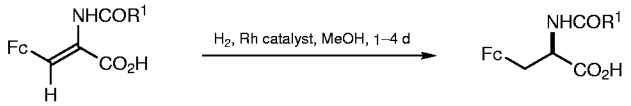



**Scheme 28** Synthesis of an Orthogonally Protected Ferrocenylbis(alanine)[65]




On the other hand, the preparation of *N*-acetyl- and *N*-benzoylferrocenylalanines **102** and **103** by enantioselective catalytic hydrogenation of the *Z*-alkenes **100** and **101** has been reported (Scheme 29).[66] In this case, a comparative study showed that among different chiral ligands for Rh, (+)- or (−)-Norphos were the ones giving the highest enantioselectivities (up to 94% ee) in the hydrogenation of **100**.

**Scheme 29** Preparation of *N*-Acylferrocenylalanines by Asymmetric Hydrogenation[66]




**100** $R^1$ = Ph
**101** $R^1$ = Me
**102** $R^1$ = Ph
**103** $R^1$ = Me

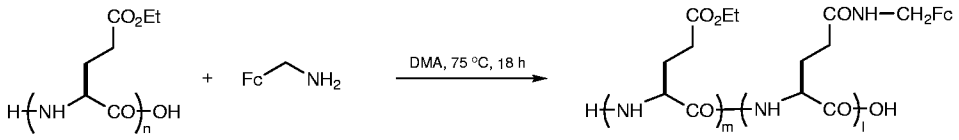
**Enantioselective Synthesis of (+)-(*S*,*S*)-1,1′-Bis[2-(benzyloxycarbonyl)-2-(Boc-amino)ethyl]ferrocene (95):**[65]

Compound **94** (0.85 mmol) and [Rh{(*R*,*R*)-DIPAMP}(cod)]$BF_4$ (0.01 mmol) were combined in degassed MeOH (5 mL) and hydrogenated at 4 atm of $H_2$ at 45 °C for 60 h. The completion of the reaction was monitored by TLC. The solvent was removed under reduced pressure and the residue chromatographed (silica gel, heptane/EtOAc 3:1); $[\alpha]_D^{22}$ +25 (*c* 0.75, $CHCl_3$). The diastereomeric (90:10 dr) and enantiomeric purity (>95% ee) were determined by HPLC using a Merck (*R*,*R*)-Whelk-01 column.

## 9.4.5.3 Ferrocenyl Modification of Peptide Side Chains

In the search for novel electroactive materials, ferrocenyl moieties have been incorporated into polypeptides by attachment to the side-chain functionality of lysine and glutamic acid.[67,68] For example, (aminomethyl)ferrocene **104** was linked to poly(glutamate) through an amide bond by reacting the amine with poly(ethyl glutamate) in DMA giving **105** (Scheme 30).[68] Applying this procedure, it was found that 21% of the side chains contained a ferrocenyl group. When coated as a film on a platinum electrode, the polymer thus formed displayed the typical redox behavior of ferrocene.

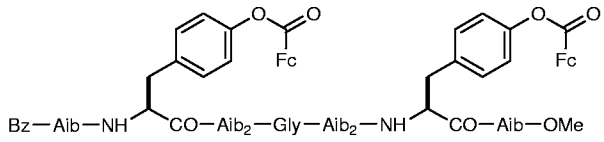
**Scheme 30** Ferrocenyl Modification of Poly(Glu) Side Chains[68]




**104** **105**

The preparation of a bis-*O*-ferrocenoyl-L-tyrosine peptide **106** has been reported (Scheme 31).[69] This compound was rationally designed to be used as a model peptide to study the binding ability of a fullerene-modified silica gel. The attachment of the ferrocenyl units was performed presumably in solution using the free hydroxy tyrosine peptide and ferrocenoyl chloride.

**Scheme 31** Bis-*O*-ferrocenoyl-L-tyrosine Peptide **106**[69]




**106**

**Synthesis of Ferrocene-Modified Poly(ethyl glutamate) 105:**[68]

Poly(γ-ethyl L-glutamate) (0.10 g), prepared by the polymerization of the corresponding NCA, was allowed to swell in DMA (0.8 mL) and ferrocenylmethylamine **104** (0.2 g) was added. The viscous orange soln was heated to 75 °C for 18 h, then cooled, and diluted with EtOH (5 mL) and $Et_2O$ (20 mL). The ferrocene-modified polypeptide **105** (0.10 g) formed a dark orange solid characterized by elemental analysis, which gave 6.0% of iron; IR (KBr) $\tilde{\nu}_{CO}$: 1725, 1640, 1530 $cm^{-1}$.

*for references see p 191*

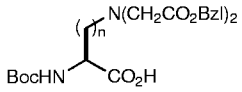
## 9.4.6 Peptides with Aminodiacetic Acid and Ethylenediaminetetraacetic Acid Side Chains

As discussed in Section 9.4.4, the incorporation of a metal ligand binding group in the side chain of peptidic structures can be used to induce and stabilize secondary structures and in the development metallopeptide molecular devices. Along these lines and based on the unique complexing ability toward metal cations of EDTA, several amino acids containing aminodiacetic acids have been reported.

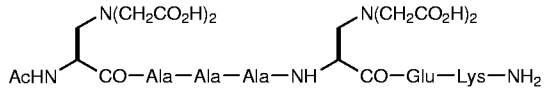
### 9.4.6.1 Peptides with Aminodiacetic Side Chains

Hopkins et al. described the synthesis of a series of amino acids **107–110** bearing one half of EDTA, an aminodiacetic acid group, on the side chain (Scheme 32).[70,71] Two of these amino acids were incorporated at different positions in several peptide structures and their conformational behavior was studied in the presence of metal ions. In most cases, the side-chain complexation of $Cd^{2+}$ or $Cu^{2+}$ leads to an increase of helicity demonstrating the involvement of the two distant ligand side chains. In the case of peptide **111**, addition of $Cd^{2+}$ changed the helical content from approximately 0 to 82%.

**Scheme 32** Aminodiacetic Acid Modified Boc-Protected Amino Acids and Peptides[70,71]



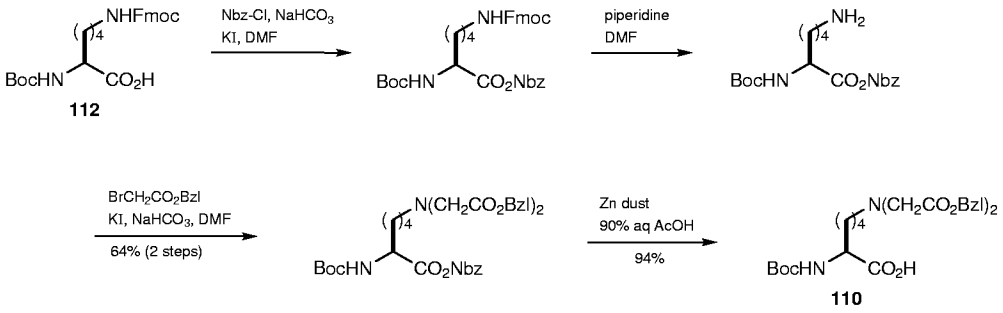

**107–110** n = 1–4



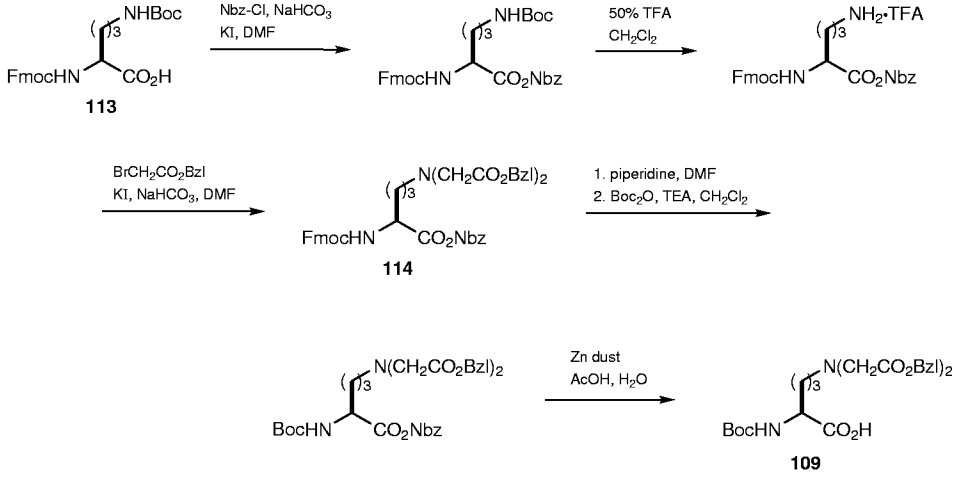

**111**

The modified amino acids containing the diaminoacetic acid moiety separated by one to four methylene units (**107–110**) were prepared by different synthetic routes from commercially available amino acid derivatives; **110** was prepared from Boc-Lys(Fmoc)-OH **112** (Scheme 33), **109** was prepared from Fmoc-Orn(Boc)-OH **113** via **114** (Scheme 34), **108** was prepared from **115** via Boc-$A_2$bu-$CH_2OH$ **116** (Scheme 35), and finally **121** (the alanine dipeptide form of **107**) was formed from Boc-Asn-OH **118** via Boc-$A_2$pr(Fmoc)-OH **119** and **120** (Scheme 36).[71] Although the use of **109** and **110** in the solid-phase synthesis of desired peptides using Merrifield resin gave the expected compounds, appreciable amounts of trifluoroacetyl-capped sequences were obtained during their couplings. Therefore, to avoid such side reactions with **107** and **108**, they were incorporated through their alanine dipeptide forms **117** and **121**. Note that the synthesis of amino acid **107** having only one methylene spacer shown in Scheme 36 is the best synthetic strategy since it avoids the undesirable formation of an oxopiperazine side product when the α-amino group is deprotected.[71] For all these derivatives, the stereochemical integrity was preserved and verified by NMR after the formation of diastereomeric mixtures with L- and DL-Ala. Finally, the preparation of the Fmoc analogues of **107–110**, **122–125** was also reported.[72]

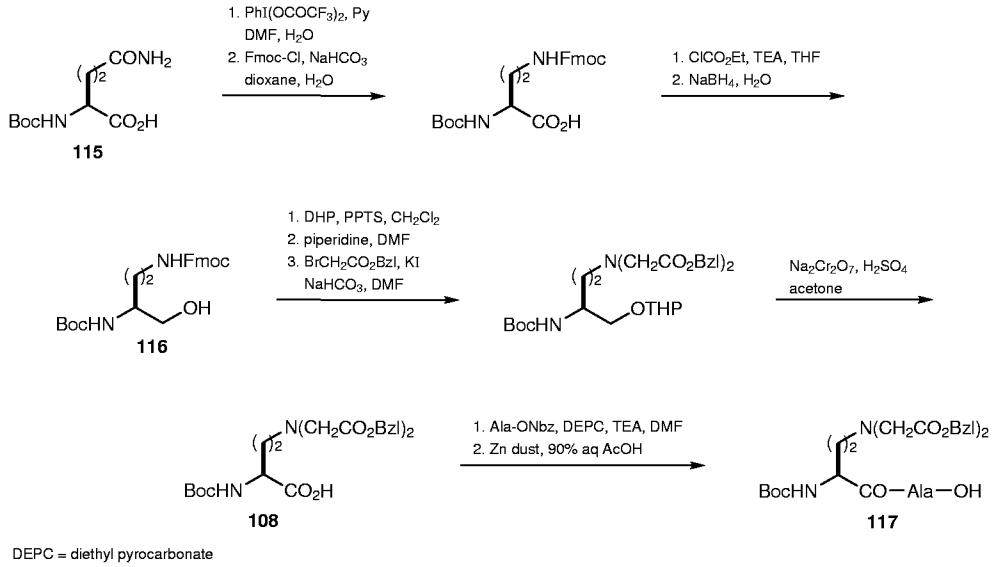
**Scheme 33** Synthesis of Aminodiacetic Acid Modified Lysine[71]




**Scheme 34** Synthesis of Aminodiacetic Acid Modified Ornithine[71]



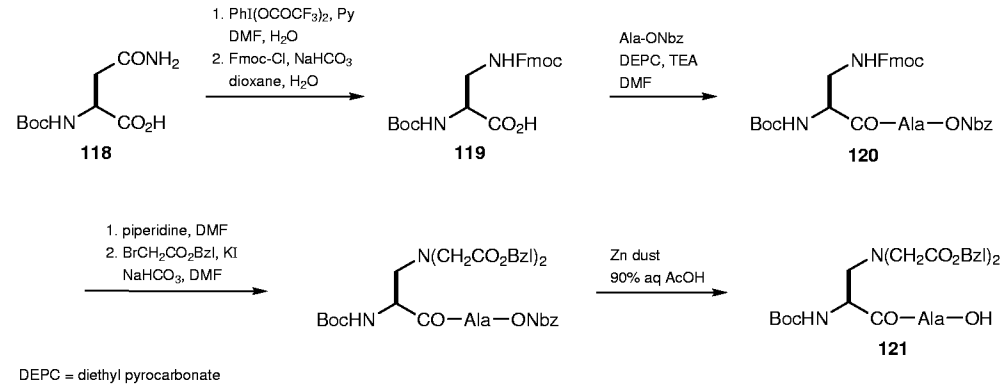

**Scheme 35** Synthesis of Aminodiacetic Acid Modified α-Aminobutryic Acid[71]




DEPC = diethyl pyrocarbonate

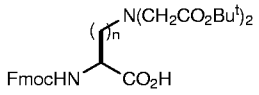
*for references see p 191*

**Scheme 36** Synthesis of Aminodiacetic Acid Modified α-Aminopropanoic Acid[71]




DEPC = diethyl pyrocarbonate

**Scheme 37** Aminodiacetic Acid Modified Fmoc-Protected Amino Acids[72]




**122–125** n = 1–4

**(*S*)-6-[Bis(benzyloxycarbonylmethyl)amino]-2-(Boc-amino)hexanoic Acid (110):**[71]
Piperidine (0.11 mL, 1.1 mmol) was added to a DMF soln of Boc-Lys(Fmoc)-ONbz (0.22 g, 0.37 mmol). The mixture was stirred at rt for 45 min, then concentrated under reduced pressure. The resulting mixture was dissolved in DMF (10 mL) and treated with KI (0.067 g, 0.40 mmol) and $NaHCO_3$ (0.46 g, 5.5 mmol). To this heterogeneous mixture was added dropwise benzyl bromoacetate (0.46 g, 2.9 mmol) and the reaction was stirred for 16 h at 25 °C. After being poured into $H_2O$ (30 mL), the mixture was extracted with $CH_2Cl_2$. The organic layer was worked up as usual and concentration gave the crude Nbz ester of **110**, which was purified by chromatography (silica gel, hexane/EtOAc mixtures) to afford a pale yellow oil; yield: 0.16 g (64%); FAB-MS *m/z*: 678 $[M + H^+]$.

A soln of the above ester (0.15 g, 0.22 mmol) in 90% aq AcOH (19 mL) was treated with Zn dust (0.73 g, 11 mmol). The mixture was stirred at rt for 2 h, then filtered through Celite, and dried in a vacuum dessicator over NaOH. The residue was taken up in EtOAc (30 mL) and washed sequentially with 10% aq $H_3PO_4$ and $H_2O$. The organic layer was dried ($MgSO_4$), filtered, and concentrated. The remaining traces of AcOH were eliminated by co-evaporation with $CHCl_3$ several times. The desired compound **110** was obtained as a yellow oil; yield: 0.11 g (94%); $[\alpha]_D^{25}$ +4.5 (*c* 4.5, $CHCl_3$); FAB-MS *m/z*: 543 $[M + H]^+$.

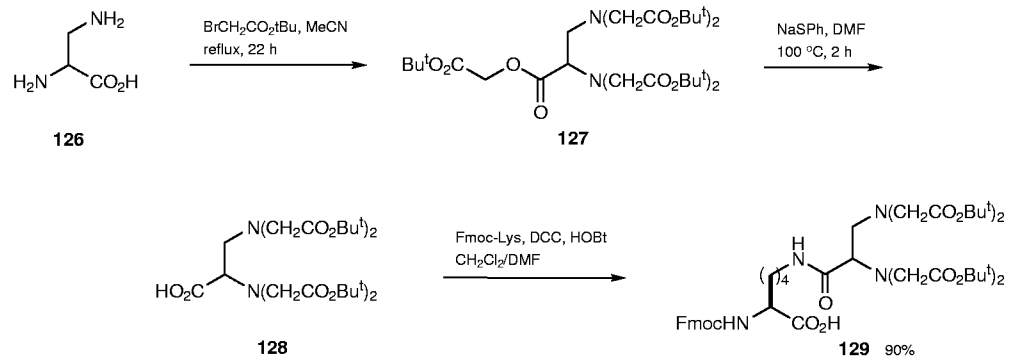
**Synthesis of Peptide 111:**[71]
Peptide **111** and analogues incorporating aminodiacetic acid modified amino acids **107–110** were prepared by SPPS using Boc-protected amino acids and MBHA resin. Deprotection steps were performed by a 30-min treatment with 50% TFA/$CH_2Cl_2$ and couplings were realized using the DIC or DCC/HOBt method. For commercially available acids 3–6 equiv were used and for the unnatural amino acids, 3 equiv were used. At the end, peptides were cleaved using HF in the presence of anisole. Peptides were recovered by the usual manner and purified by RP-HPLC using MeCN/$H_2O$ gradients. The structure of peptide **111** was confirmed by FAB-MS and amino acid analysis; FAB-MS *m/z*: 1218 $[M + H]^+$.

## 9.4.6.2 Peptides with Ethylenediaminetetraacetic Acid Side Chains

The incorporation of an EDTA group into peptidic structures is of interest for the development of protein and nucleic acid binding molecules capable of affinity cleavage, as well as radiopharmaceuticals. The incorporation of the chelating EDTA group can be done by linking a derivative at the N- or C-terminal position of peptides.[73] However, the incorporation in solid-phase synthesis of an EDTA-modified amino acid allows more flexibility in the design since the metal chelating group can be introduced at any position of a peptidic device. The selective synthesis of such suitably protected EDTA-modified amino acids has

been described.[74,75] Compound **129** was designed to be compatible with Fmoc solid-phase synthesis protocols and prepared from an EDTA derivative **127** according to Scheme 38.[74]

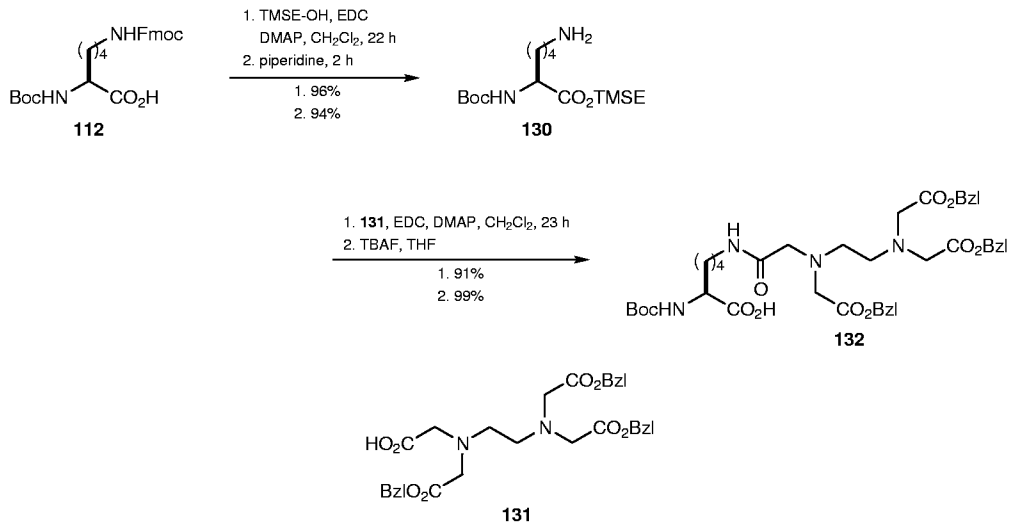
**Scheme 38** Synthesis of Fmoc-Protected EDTA-Modified Lysine[74]




On the other hand, compound **132** was developed to be used in SPPS using Boc strategies (Scheme 39).[75] In its synthesis, one carboxylic acid of EDTA is utilized as anchor to the lysine ε-$NH_2$. Both synthesis used Lys as the "carrier" amino acid.

In an earlier report, the EDTA intermediate **128** was coupled during solid-phase synthesis to the N-terminal position of bioactive peptides using DCC/HOBt.[76]

**Scheme 39** Synthesis of Boc-Protected EDTA-Modified Lysine[75]




**(2*S*)-6-{2,3-Bis[bis(*tert*-butoxycarbonylmethyl)amino]propanoylamino}-2-(Fmoc-amino)hexanoic Acid (129):**[74]

2,3-Diaminopropanoic acid (**126**; 5.0 g, 48 mmol) was dissolved in MeCN (100 mL) and mixed with *tert*-butyl bromoacetate and DIPEA (5 equiv). The mixture was refluxed for 22 h, cooled, and the solvent removed under reduced pressure. The residue was triturated with $Et_2O$ (350 mL) with stirring overnight and filtered. The filtrate was dried, dissolved in toluene (200 mL), and washed with 0.1 M phosphate buffer, pH 2. The organic phase was dried (molecular sieves) and the solvent removed under reduced pressure to give the crude intermediate **127** (30.8 g). The latter was dissolved in DMF (200 mL) and NaSPh (4.65 g, 35 mmol) was added. The mixture was heated at 100 °C for 2 h, then cooled and diluted with toluene (300 mL). After washing with the phosphate buffer, the solvent was concentrated to yield **128** (28.18 g) as a yellow oil, which could be purified by chromatography (silica gel, increasing amounts

*for references see p 191*

of EtOAc in hexane). The desired compound **128** was finally eluted with 1% AcOH in EtOAc; TLC $R_f$ 0.52 (silica gel, $CHCl_3$/MeOH/AcOH 9:1:0.004); FAB-MS *m/z*: 561 $[M + H]^+$.

The acid **128** (0.5 mmol) was dissolved in $CH_2Cl_2$ (3 mL) and HOBt (0.5 mmol dissolved in 1.5 mL of DMF) was added. The resulting soln was mixed with DCC (0.5 mmol) dissolved in $CH_2Cl_2$. The mixture was stirred at 0 °C for 1 h and at rt for an extra 2 h, then filtered. The filtrate was concentrated to dryness and the activated ester thus formed dissolved in DMF (5 mL) and $CH_2Cl_2$ (2 mL). Fmoc-L-Lys (0.6 mmol) partially dissolved in ethyleneglycol monoethyl ether (50 mL), was added to the activated ester, followed by the addition of DIPEA (100 μL) to bring the apparent pH to 8. The mixture was stirred overnight then the solvent was concentrated. The residue was taken up in EtOAc and extracted with 10% citric acid. The organic phase was dried (molecular sieves), filtered, and concentrated to provide **129** as a dark oil; yield: 90%; TLC $R_f$ 0.77 (silica gel, acetone/$H_2O$ 7:3); FAB-MS *m/z*: 911 $[M + H]^+$.

**(2*S*)-6-[2-([Benzyloxycarbonylmethyl]{2-[bis(benzyloxycarbonylmethyl)amino]ethyl}amino)acetyl]-2-(Boc-amino)hexanoic Acid** (132):[75]

To a soln of Boc-Lys(Fmoc) (**112**; 920 mg, 1.96 mmol), EDC (753 mg, 3.92 mmol), and a catalytic amount of DMAP in $CH_2Cl_2$ (10 mL) was added TMSE-OH (347 mg, 2.94 mmol). After 12 h of stirring at rt, the mixture was diluted with $CH_2Cl_2$ (40 mL) and washed with sat. $NaHCO_3$, brine, and $H_2O$. The organic phase was dried and concentrated. The residue was chromatographed (silica gel, 30% EtOAc/hexane) to afford Boc-L-Lys(Fmoc)-OTMSE as a colorless oil; yield: 1.07 g (96%); $[\alpha]_D^{25}$ −12 (*c* 1, MeOH). The Fmoc group was then removed by dissolving the oil (558 mg, 0.98 mmol) in piperidine (5 mL) and stirring for 2 h at rt. Piperidine was removed under vacuum and the crude product obtained was purified by chromatography (silica gel, 10% $NH_3$-saturated MeOH in $CH_2Cl_2$) to give **130** as a colorless oil; yield: 320 mg (94%); $[\alpha]_D^{25}$ −10 (*c* 1, MeOH).

A soln of **130** (39 mg, 113 μmol), the acid **131** (96 mg, 170 μmol), EDC (43.4 mg, 226 μmol), and a catalytic amount of DMAP in $CH_2Cl_2$ (1 mL) was stirred at rt for 23 h. The soln was diluted with $CH_2Cl_2$ (5 mL) and washed with 10% citric acid, sat. $NaHCO_3$, brine, and $H_2O$. The organic layer was dried and concentrated to yield a crude residue, which was chromatographed (silica gel, 8% MeOH/$CH_2Cl_2$) to afford the pure TMSE ester of **132** as a pale yellow oil; yield: 91.3 mg (91%); $[\alpha]_D^{25}$ −8 (*c* 1, MeOH). The latter (29.4 mg, 34 μmol) in THF (500 μL) was then treated with 1 M TBAF (200 μL) in THF. After 3 min, the mixture was cooled to 4 °C, diluted with $H_2O$ (500 μL), and the solvent removed. The residue was dissolved in EtOAc (5 mL) and washed as above. Removal of the solvent yielded the desired EDTA modified Lys **132** as a colorless oil; yield: 26.6 mg (99%); TLC $R_f$ 0.52 (silica gel, 10%MeOH/$CH_2Cl_2$); $[\alpha]_D^{25}$ −4 (*c* 1, MeOH).

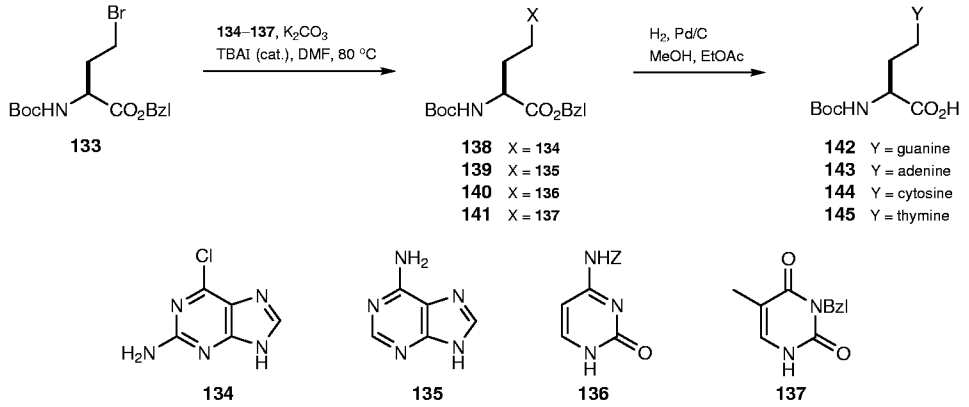
## 9.4.7 Peptides with Nucleobase Side Chains

The sequence-specific binding of oligonucleotides to RNA or DNA provides a potentially powerful way to control gene expression. In this regard, the preparation of several backbone-modified analogues of nucleic acids more resistant to nucleases and more membrane permeable has been reported.[77] Along these lines, the development of molecular devices without a sugar and/or phosphate backbone that could bind selectively specific DNA or RNA sequences has attracted considerable attention recently.[78] Pioneering work in this area was done by the group of Nielsen who developed the so-called peptide nucleic acids (PNAs) where the backbone nitrogens of an oligo (2-aminoethyl)glycine have been modified by nucleobases.[79] This work has been the subject of recent reviews [78,79] and will not be covered in this section, which will focus on the synthesis of "chiral", "real" peptides having side chains modified by nucleobases (chiral PNAs) inspired by the basic principles of PNAs. The premise underlying these chiral compounds is that the peptidic framework conformation will influence and possibly modulate the DNA/RNA binding (hybridization) properties.

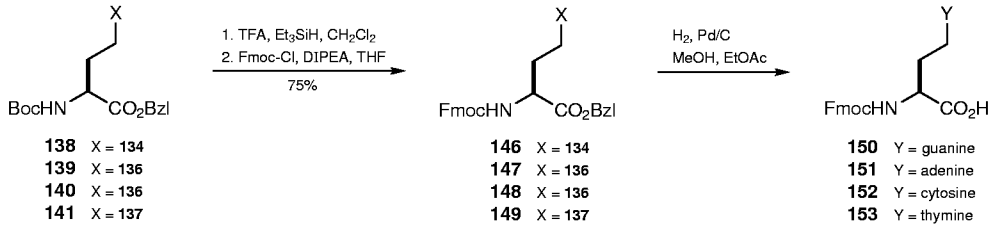
### 9.4.7.1 Nucleo Ethylglycines

Toward the preparation of chiral PNAs, the synthesis of Boc-protected L-ethylglycines with all four nucleobases (adenine, thymine, cytosine, guanine) at the γ-position was performed as illustrated in Scheme 40.[80,81] The strategy uses Boc-L-glutamic acid benzyl ester as the starting material and takes advantage of Barton's chemistry to prepare a key bromide intermediate **133**. The bases were added by a classic nucleophilic substitution reaction using $K_2CO_3$ as base in DMF. The overall yields of **138–141** ranged from 55–65% and it was shown that no racemization occurred during the coupling of the bases to **133**. Although the unprotected bases could be used directly in the synthesis, better yields and regioselectivity were obtained using the protected 3-benzylthymine **137**, $N^4$-Z-cytosine **136**, and 2-amino-6-chloropurine **134** as nucleophiles; benzyl deprotection gave the acids **142–145**. Unprotected adenine **135** was utilized successfully. The transformation of the Boc-protected amino acids **138–141** to the Fmoc analogues **150–153** via **146–149** was accomplished by the procedure shown in Scheme 41, but low to moderate yields were obtained (10–65%). To avoid side reactions, triethylsilane was used as *tert*-butyl cation scavenger in the Boc-deprotection steps.

**Scheme 40** Synthesis of Boc-Protected Nucleobase-Modified Ethylglycines[80]




**Scheme 41** Synthesis of Fmoc-Protected Nucleobase-Modified Ethylglycines[82]




The nucleobase amino acids **143** and **145** were used in the synthesis of a representative example of a chiral PNA **154**, designed to be complementary to the oligonucleotide sequence ATATATAT. The solid-phase synthesis was achieved using PAM resin with a Gly linker as illustrated in Scheme 42. The coupling steps were done using twofold excess of the Boc-protected amino acids and BOP/DIPEA as activating reagents. Usually, 30-minute couplings were performed. The Boc-thymine derivative **145** was coupled twice in a DMF/$CH_2Cl_2$ mixture and the adenine analogue **143** in a DMF/DMSO mixture due to its low solubility. The peptide was cleaved using standard HF conditions and purified by RP-HPLC. Chiral PNA **154** was shown to hybridize with the complementary DNA sequence, but less strongly than the corresponding achiral, more flexible PNA.[81]

*for references see p 191*

**Scheme 42** Solid-Phase Synthesis of Peptides Incorporating Nucleo Ethylglycines[80,82]



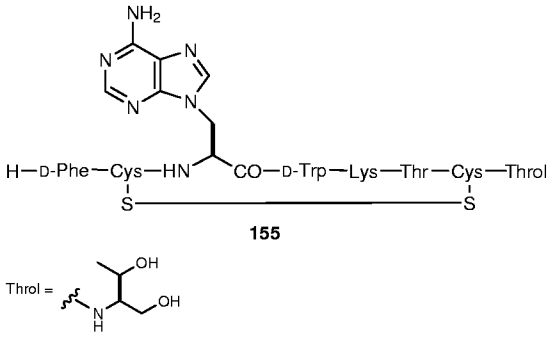

**Boc-γ-Adenyl-L-ethylglycine** (143):[82]
Bromide **133**[80] (5.2 g, 13.9 mmol), adenine (5.07 g, 37.5 mmol), $K_2CO_3$ (5.18 g, 7.5 mmol), and TBAI (0.5 g) were stirred in dry DMF (150 mL) for 24 h at rt. Water was added and the soln extracted several times with EtOAc. The organic layers were combined, dried ($Na_2SO_4$), filtered, and concentrated to give crude **139**, which was purified by flash chromatography (silica gel, EtOAc/EtOH 10:1); yield: 3.84 g (65%). The latter was hydrogenated with 1 atm of $H_2$ and Pd/C in EtOAc for 8 h. The mixture was filtered on Celite, which was Soxhlet extracted with MeOH. The organic phases were combined and concentrated to afford **143**; yield: 45–65%.

**Fmoc-γ-Adenyl-L-ethylglycine** (151):[82]
The Boc-protected amino acid **139** (2.6 g, 6.1 mmol) was dissolved in dry $CH_2Cl_2$ (13 mL) containing $Et_3SiH$ (2.4 mL). Then TFA (6.1 mL) was added and the resulting mixture was stirred at rt for 2 h. The solvents were concentrated and the oily residue was triturated with $Et_2O$ until a solid was obtained. The latter was immediately dissolved in dry THF and DIPEA (4.3 mL, 24.4 mmol) was added followed by FmocCl (1.58 g, 6.1 mmol) in THF (10 mL). After 12 h, sat. $NH_4Cl$ was added, followed by EtOAc. The organic layer was washed with brine, dried ($Na_2SO_4$), and concentrated. Purification of the crude residue by chromatography (silica gel, EtOAc/MeOH 10:1) gave pure benzyl ester **147** (2.0 g, 68%). Benzyl ester **147** (2–5 g) was dissolved in MeOH (10 mL) and added to a suspension of 10% Pd/C (1–2 g) in MeOH (40 mL). After three cycles of vacuum/$H_2$ activation, the mixture was hydrogenated for 2 h. The mixture was filtered on Celite and the solid cake reextracted with hot MeOH. The soln and the extracts were combined and concentrated to give the desired product **151** as a solid; yield: 65%; mp 184–186 °C (dec); $[\alpha]_D^{25}$ −17.5 (*c* 1, DMF).

## 9.4.7.2 Nucleo Alanines

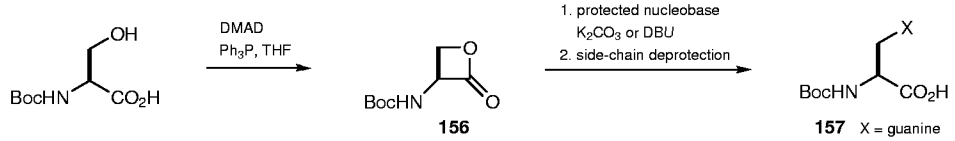
Toward the preparation of potential therapeutic agents such as the somatostatin analogue **155**, the synthesis of nucleobase L-alanine derivatives **157–160** and their incorporation into specific chiral PNAs were described.[83,86] The synthesis of the Boc-protected **157–160** was accomplished by the method developed by Vederas et al., which uses Boc-serine β-lactone **156** as key intermediate (Scheme 44).[84,85] It was shown that the lactone can be opened selectively at the β-carbon by a nucleophilic attack to introduce miscellaneous functionalities. By this synthetic strategy, both L- and D-isomers are therefore readily available. For the synthesis of **157–160**, the exocyclic amine of adenine and cytosine was protected by a benzyloxycarbonyl (Z) group, and the $O^6$ of guanine was protected as the *O*$^6$-(2-methoxyethoxy)methyl ether (Mem), although this proved to be deleterious for the $N^9$ regioselectivity. After deprotection of the nucleobase protecting groups, the Boc-protected nucleobase alanine **157–160** were used in solution and solid-phase peptide synthesis.

**Scheme 43** Somatostatin Analogue **155**[83]




**155**

**Scheme 44** Synthesis of Nucleobase-Modified Alanines[84–86]



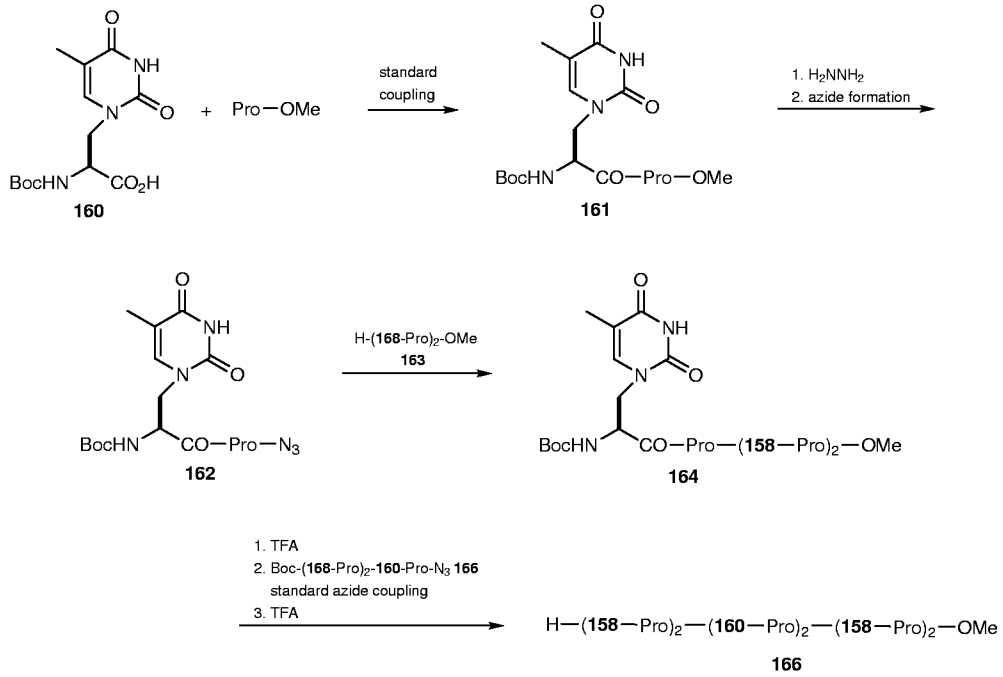

**156**

**157** X = guanine
**158** X = adenine
**159** X = cytosine
**160** X = thymine

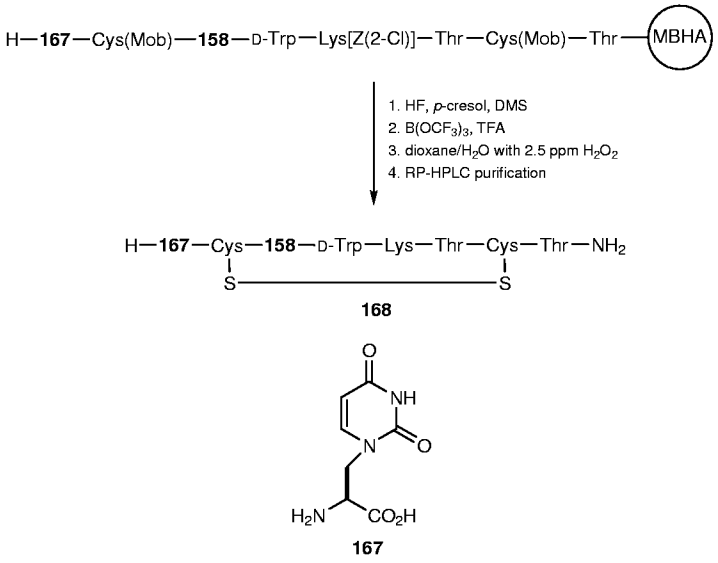
For example, in the solution-phase synthesis of **166**, a segment condensation approach was adopted and dipeptide fragments with proline were used (Scheme 45);[86] **160** and **158** were first coupled by standard procedures with Pro-OMe to give the corresponding dipeptide fragments (such as **161**). These and other segments were coupled together by the azide method, i.e. the azide **162** was coupled with **163** to give **164** which was again coupled by azide coupling with **165**, in order to give the therapeutically interesting chiral PNA **166**. No traces of epimerization were observed in the products of the segment couplings. In addition, the same authors reported the use of amino acids **157–160** in the SPPS of somatostatin analogues typified by **168** (Scheme 46).[83] MBHA resin was used and the Boc-protected amino acids were coupled by the DIC/HOBt method. Boc groups were removed with a 95:5 TFA/$H_2O$ solution. The final cleavage steps were realized with HF/*p*-cresol/DMS under standard conditions.

*for references see p 191*

**Scheme 45** Solution-Phase Synthesis of Peptides Incorporating Nucleo Alanines[83,86]




**Scheme 46** Cleavage of Peptides Containing Nucleo Alanines Prepared on MBHA Resin[83]




## 9.4.8 Peptides with Miscellaneous Side Chains

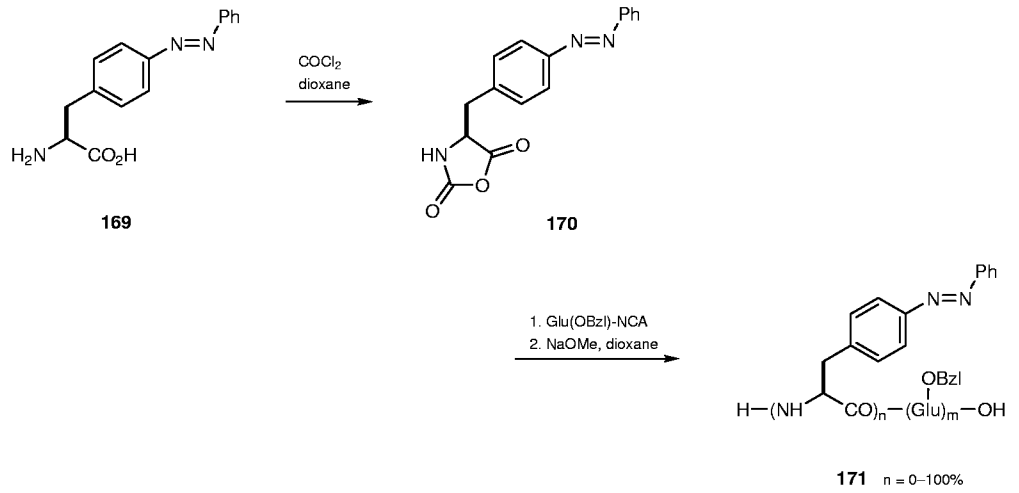
Several amino acids bearing other effectors or ligands than those described *vide supra* have been prepared and incorporated into peptidic structures with the objective of developing functional molecular devices. The targeted applications of such devices vary widely from

optical devices to DNA binding compounds. This section covers the preparation of unnatural amino acids bearing representative effector groups on the side chain and presents their potential uses.

### 9.4.8.1 Peptides with Diazobenzene Alanine

The uses of polypeptide frameworks in the development of chiroptical devices is an area that has been investigated using photochromic side chains.[2] The basic idea is to take advantage of the unique conformational properties of polypeptides to prepare polymeric films that could be light-engraved for applications, in particular, in data storage. Noteworthy of mention here is the seminal contribution of Goodman et al.[87,88] who synthesized and studied in the 1960s the conformational behavior of polypeptides **171** composed of diazobenzene-L-alanine [4-(phenylazo)-L-phenylalanine] **169** and γ-benzyl-L-glutamate (Scheme 47), probably before the term "optical data storage" was even coined. Polypeptides **171** with various proportions of diazo amino acid **169** were prepared by the NCA method from **170**. It was shown that the pure *trans*-poly **169** in TFA had a positive Cotton effect at 425 nm in ORD, whereas after irradiation the *cis*-isomer had a negative Cotton effect at the same wavelength.

**Scheme 47** Synthesis of Polypeptides Incorporating Diazobenzene Alanine by the NCA Method[87]



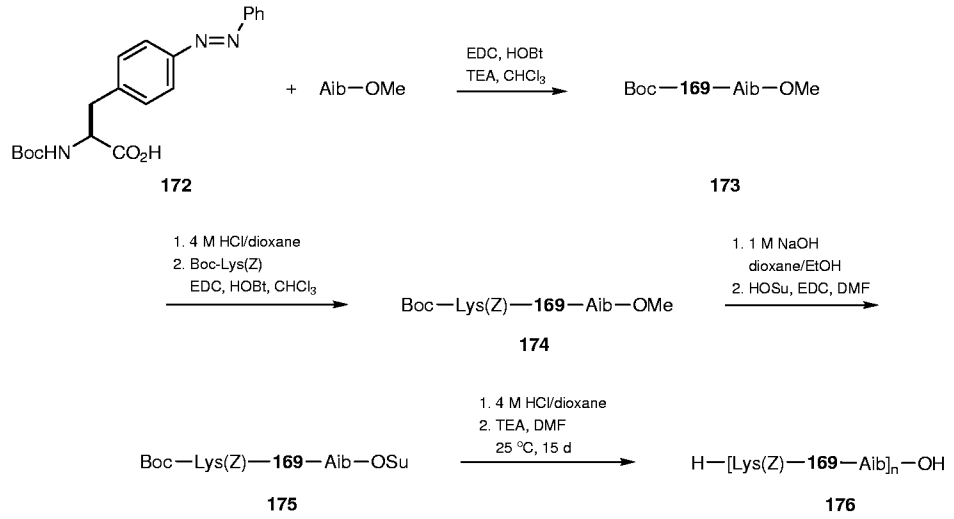

**169** **170**

**171** n = 0–100%

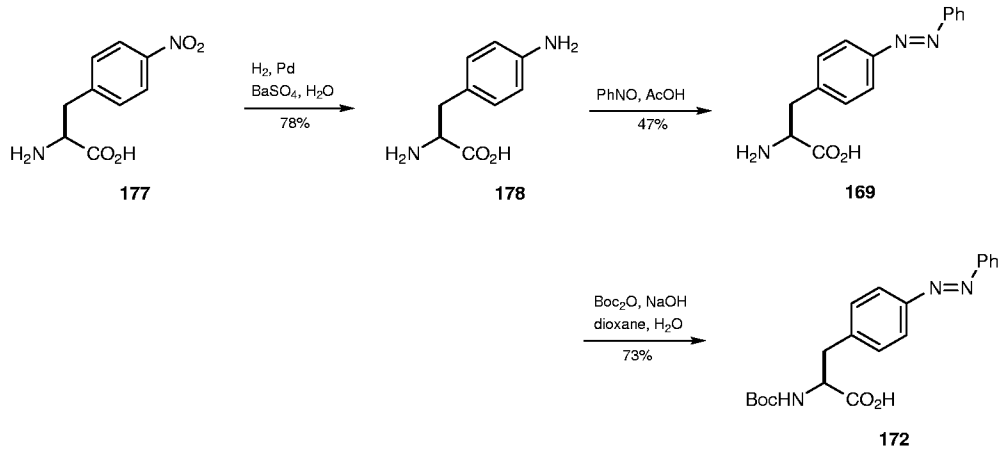
Along the same lines, the synthesis of a family of polypeptides incorporating diazobenzene-L-alanine, typified by **176**, was reported.[89,90] These peptidic devices were designed in a way that the diazobenzene chromophores are helically arranged along the polypeptide chain. As expected, irradiation of these photochromic peptides leads to important conformational changes and, in the case of **176**, in a change of helical sense. Noteworthy of mention is the fact that the backbone conformation of **176** could be modulated reversibly in trimethyl phosphate during several cycles by irradiating the solution at 340 nm (*trans* → *cis*), then at 425 nm (*cis* → *trans*) without loss of efficiency. The polypeptides were synthesized by the polymerization of the respective peptidic segments (**173** → **174** and deprotection/activation to **175**) by the HOSu method (Scheme 48).[89] The necessary Boc-protected diazobenzene-L-alanine **172** was prepared according to Scheme 49.[87,89] The starting L-Phe(4-$NO_2$)-OH (**177**) was first reduced to the amine, and coupled with nitrosobenzene. The diazobenzene alanine analogue **169** thus obtained was shown to be almost racemization free (>98% ee) by NMR using a chiral shift reagent. **172** was coupled in solution under classical conditions using EDC/HOBt in $CHCl_3$ and the diazo moiety is stable to the Boc deprotection conditions.

*for references see p 191*

**Scheme 48** Synthesis of Polypeptides Incorporating Diazobenzene Alanine by the *N*-Hydroxysuccinimide Method[89]




**Scheme 49** Synthesis of Diazobenzene Alanine[87,89]




**Synthesis of *N*-Boc-4-phenylazo-L-phenylalanine (172):**[87,89]

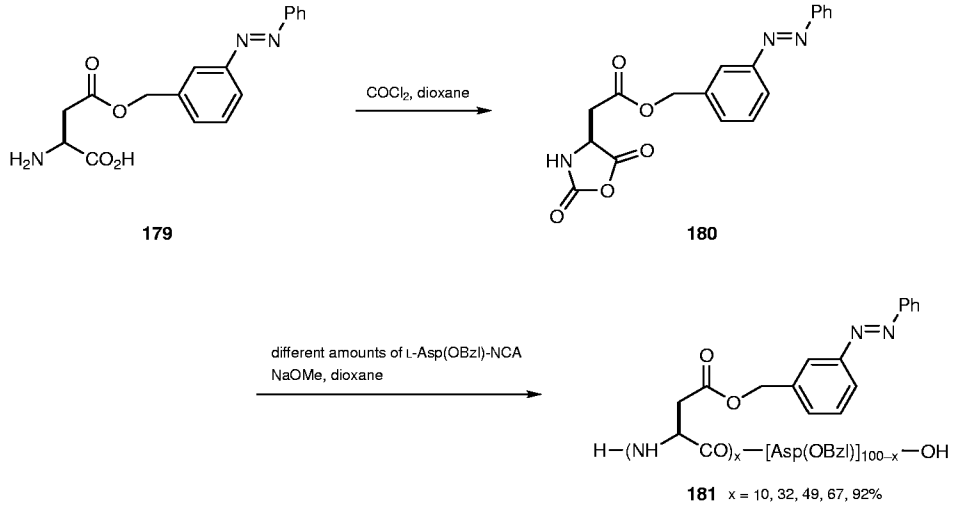
H-L-Phe(4-$NO_2$)-OH (**177**; 10 g, 0.048 mol) was reduced by catalytic hydrogenation in $H_2O$ following the procedure of Bergman.[91] After crystallization (acetone), a 78% yield of **178** was obtained [mp 235–237 °C; $[\alpha]_D^{25}$ −42.4 (*c* 0.45, $H_2O$)]. The amine **178** (4.4 g, 0.024 mol) was dissolved in glacial AcOH (50 mL) at 16 °C and a freshly prepared soln of nitrosobenzene (2.6 g, 0.024 mol) was added over a 40 min period. After precipitation, the mixture was diluted with $H_2O$ (250 mL) and left standing at −5 °C overnight. The yellow precipitate was filtered and dried under reduced pressure over $CaCl_2$ to give 4-phenylazo-L-phenylalanine **169**; yield: 3.1 g (47%); mp 219.7–220.5 °C; UV–vis (3 M HCl) $\lambda_{max}$, nm: 229, 324, 418.

The free amino acid **169** (1.97 g, 7.32 mmol) was dissolved in dioxane/$H_2O$ (2:1, 20 mL) and cooled to 0 °C. 1 M NaOH (7.43 mL) and $Boc_2O$ (1.78 g, 8.16 mmol) were then added simultaneously and the resulting mixture was stirred 1 h at 0 °C and overnight at rt. The mixture was concentrated, then EtOAc and 5% $KHSO_4$ were added. The organic layer was washed with brine and dried ($MgSO_4$). The solvent was concentrated to give the desired Boc-protected amino acid **172** as an oil; yield: 2.01 g (73%).

### 9.4.8.2 Peptides with Diazobenzene Aspartic Acid

Following up the previous work, the poly[Asp(OBzl)] **181** modified with different proportions of a diazobenzene moiety linked via an ester group was prepared by the co-polymerization of the respective NCAs of **179** (**180**) and L-Asp(OBzl) (Scheme 50).[92–94] Again, the idea was to use the well-known photoinduced *cis–trans* isomerization of the diazobenzene group to control the helical sense of the polypeptide framework. Impressive results obtained by CD showed that indeed the sense of the helix could be modulated by irradiation of the diazo group at 320–390 nm.[93] However, the changes observed are sensitive to the nature of the solvent. The photomodulation of the conformation was also observed with the *para* analogue of **181**.

**Scheme 50** Synthesis of Polypeptides Incorporating Diazobenzene-Modified Aspartic Acid[92,94]




### 9.4.8.3 Peptides with Pyrenylalanine

With the aim of preparing photoinduced electron transfer devices or simply to be used as fluorescent reporter groups, the synthesis and utility of pyrene-L-alanine have been reported on several occasions. Regarding the development of charge-transfer devices, it was shown that intramolecular electron transfer occurred in $3_{10}$- and α-helical peptidic systems, such as **182** and **185**, between a pyrene and a dimethylaminophenyl chromophore (Scheme 51).[95–101] The rates depend on the relative orientation of the chromophoric side chains and on the framework conformation. Therefore, the preparation of "tailor-made" charge-transfer devices with "tunable" properties can be envisaged.

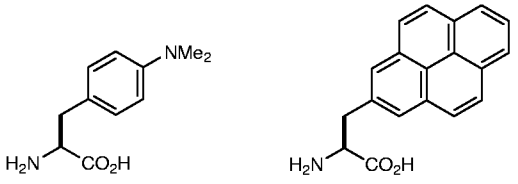
On the other hand, pyrenyl-L-alanine **184** has also been used as a conformational probe in the characterization of an artificial 4-α-helix bundle protein.[102] The 53-residue peptide **186** incorporating one residue of **184** in each of two different helical segments was synthesized by solid-phase synthesis using a segment condensation strategy and the oxime resin. Boc-pyrenyl-L-alanine **191** was coupled just like any other amino acid by the BOP/HOBt method in DMF. CD and fluorescence studies demonstrated that the two pyrene groups were in close proximity forming an excimer complex, which is possible only when the polypeptide chain folds into a 4-α-helix bundle structure.

*for references see p 191*

**Scheme 51** Amino Acids and Peptides Containing Chromophoric Side Chains[95–102]

Boc—Aib—**183**—Ala—Aib—Ala—Aib—Ala—**184**—Aib—OMe

**182**



**183** **184**

H—[Glu(OBzl)]$_n$—**183**—(Ala)$_n$—**184**—[Glu(OBzl)]$_4$—OBzl

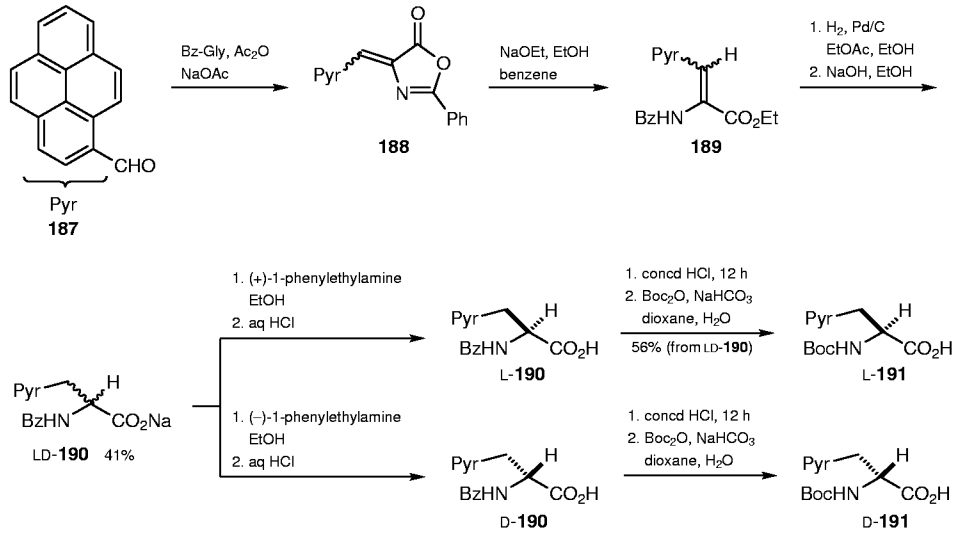
**185**

D-A—P—E—L—L—K—A—**184**—A—E—L—L—K—D-A—P—E—L—L—K—A—F—A—E—L—L—K—D-A—P—E
|
G—K—L—L—E—A—F—A—K—L—L—E—P—D-A—K—L—L—E—A—**184**—A—K—L—L

**186**

The synthesis of Boc-pyrenyl-L-alanine **191** was accomplished either by a classic strategy starting from pyrene-1-carbaldehyde (**187**) illustrated in Scheme 52,[98,100] or by asymmetric hydrogenation of the corresponding Boc dehydroamino acid.[97] In the former case, enantiomerically pure L- and D-**184** can be obtained on a small scale by an enzymatic resolution, or on a larger scale by selective crystallization of the diastereomeric salts formed with (*R*)- or (*S*)-1-phenylethylamine. The optical purity of **184** was found to be >97%, as determined by $^1$H NMR with the methyl ester using a Eu chiral shift reagent.[100] Boc-pyrenyl-L-alanine **191** was coupled to other amino acids and peptides using standard coupling reagents such as EDC/HOBt.[98]

**Scheme 52** Synthesis and Resolution of Pyrenylalanine[98,100]




**Synthesis of *N*-Boc-β-(pyren-1-yl)-L-alanine** 191:[98,100]
A mixture of pyrene-1-carbaldehyde (**187**; 5 g, 22 mmol), *N*-benzoylglycine (4.4 g, 24.6 mmol), and NaOAc (2.8 g, 34 mmol) was dissolved in $Ac_2O$ (100 mL) and heated at 120 °C for 3 h. The mixture was cooled, concentrated, and the residue poured into $H_2O$. The red precipitate **188** (9 g) was collected, dried, and added to a soln of Na (2 g) in EtOH (300 mL) and benzene (100 mL). After 1 h, AcOH was

added and the yellow precipitate **189** was collected. More material can be obtained by removing the solvent and by taking up the residue with benzene. A total of 6 g of **189** was afforded (mp 185–187 °C). The latter was hydrogenated in EtOH/EtOAc (1:1) with Pd/C for 2 days. After the usual workup, 5 g of hydrogenated **189** were obtained as a yellow solid; mp 168–170 °C.

To resolve pyren-1-yl-L-alanine, the above ester was first saponified with 2 M NaOH in EtOH to give, after neutralization, racemic acid **190** (3.6 g, mp 212–213 °C). The latter was suspended in EtOH (300 mL) and (+)-1-phenylethylamine (1.1 g) was added and dissolved with heating. Upon cooling, a precipitate was collected and recrystallized (EtOH) to afford the L-**190** diastereomeric complex (1.7 g). The filtrate was neutralized with HCl to recover D-rich **190** (2 g), which was resuspended in EtOH (200 mL) and treated with (−)-1-phenylethylamine (0.7 g). Upon cooling, the D-**190** diastereomeric complex (1.5 g) was collected and recrystallized (EtOH). The two complexes were dissolved separately in EtOH and 1 M HCl was added to precipitate the desired enantiomerically pure L- and D-*N*-benzoylamino acid **190** [mp 230–234 °C; for L-**190** $[\alpha]_D$ −232.5 (*c* 0.5, DMF); for D-**190** $[\alpha]_D$ +233 (*c* 0.5, DMF)]. Refluxing L-**190** with concd HCl in AcOH gave free L-**184**; mp 240–250 °C (dec).

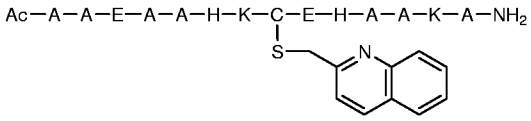
The acid **184** (0.34 g, 1.2 mmol) and $NaHCO_3$ (0.21 g, 2.49 mmol) were suspended in a dioxane/$H_2O$ mixture (3:1; 20 mL) and cooled to 0 °C. $Boc_2O$ (0.28 g, 1.3 mmol) was added dropwise and the resulting mixture was stirred 1 h at 0 °C and 12 h at rt. Dioxane was concentrated, 5% $KHSO_4$ was added and the aqueous layer was extracted with EtOAc. After drying ($MgSO_4$), the solvent was concentrated and the crude solid was obtained by trituration with hexane and recrystallized ($CHCl_3$/hexane) to give pure L-**191**; yield: 0.253 g (56%); mp 178–179 °C.

## 9.4.8.4 Peptides with Side-Chain-Modified Cysteine

Several interesting functional peptidic devices were constructed by attaching different effectors on the lateral thiol group of cysteine. One advantage of this strategy is the possibility of incorporating the effector moiety at the end of the synthesis. The disadvantage, however, is that unreacted materials with roughly the same properties as the desired compounds have to be removed by chromatography. Nevertheless, some interesting systems have been prepared in this way.

For example, a DNA binding device **192** was constructed by linking a quinoline intercalator to a peptidic helical framework via a cysteine side chain (Scheme 53).[103] Also, toward sequence selective DNA cleavage reagents, a peptide from the transcriptional activator protein GCN4 **193** was coupled to a terpyridine metal ligand by a disulfide bridge to yield the hybrid devices **195**.[104] The coupling was performed by reacting the free cysteine peptide with the activated mixed disulfide **194** (Scheme 54).

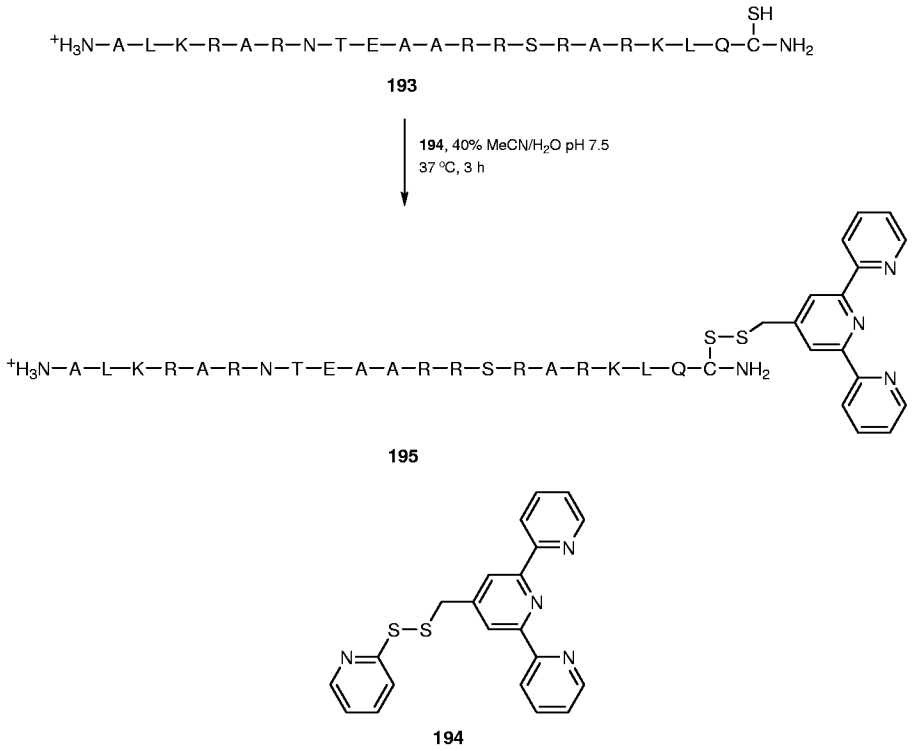
**Scheme 53** DNA Binding Device **192**[103]




**192**

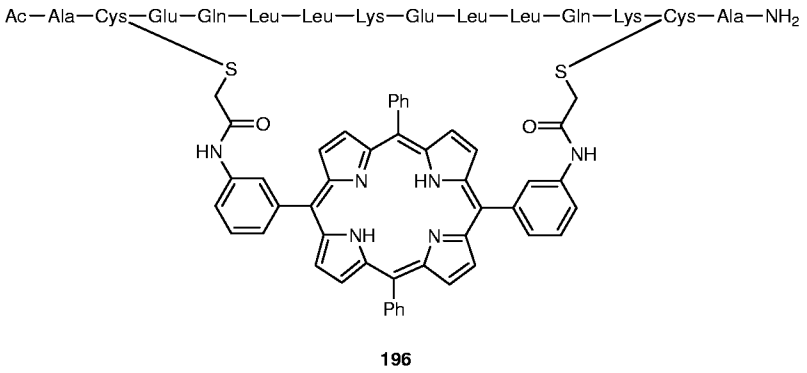
*for references see p 191*

**Scheme 54** Coupling of a Terpyridine Ligand to a Cysteine Side Chain[104]




In the development of *de novo* metalloproteins, helical peptidic structures were attached to tetraphenylporphyrins by the thiol group of cysteine residues to create structurally defined molecular systems, such as **196** (Scheme 55).[105,106] Potential applications of these devices include their use as biomimetic oxidation catalysts. The coupling of the effectors is achieved by classic nucleophilic substitution reactions using a base in a polar solvent and in the absence of oxygen. In the case of **196**, $Na_2CO_3$ was used as a base in DMF.[105]

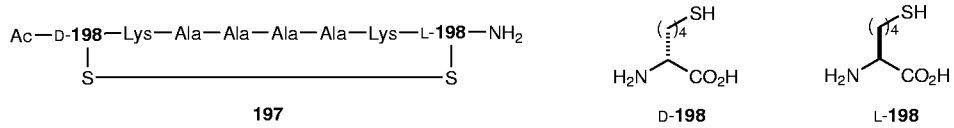
**Scheme 55** Intrahelical Disulfide Porphyrin Bridged Peptide **196**[105]




The use of an interhelical disulfide bridge to stabilize *de novo* designed helical proteins has also been reported.[107] In addition, the formation of intrahelical disulfide bridges to stabilize the helical conformation of a series of peptides, typified by **197**, was described (Scheme 56).[108] For that purpose, the unnatural thiol amino acid **198** was synthesized. The best result

was obtained with peptide **197** with a residue of **198** at each of the N- and C-terminal positions, seven residues apart. Indeed, the formation of the disulfide bridge enhanced the helical content from 16 to approximately 100% at 0 °C; at 60 °C, **197** retains 59% of its helical content. Noteworthy of mention is the fact that the D-isomer of **198** at the C-terminal position is necessary for the induction of helicity in **197**.

**Scheme 56** Intrahelical Disulfide-Bridged Peptide **197** and Thiol Amino Acids **198**[108]

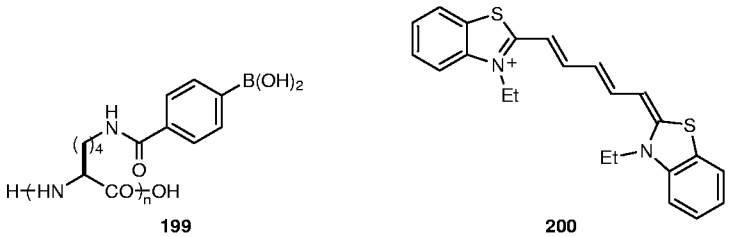



197 D-198 L-198

## 9.4.8.5 Peptides with Boronic Acid Side Chains

A phenylboronic acid modified poly-L-lysine **199** was developed for the selective detection of disaccharides.[109,110] The binding of disaccharides to adjacent boronic acid side chains creates two anionic $sp^3$-hybridized boron atoms which can serve as binding sites to a cationic dye **200** and alter its UV–visible absorption properties. The molecular recognition phenomenon results in a change of color, so the presence of disaccharides can be detected visually. The overall binding process induces important conformational changes in the helical poly-lysine framework that can be followed by CD in the infrared region (450–800 nm). The conformational and the color changes are specific to disaccharides.

**Scheme 57** Phenylboronic Acid Modified Poly-L-lysine **199** and Cationic Dye **200**[109,110]




199 200


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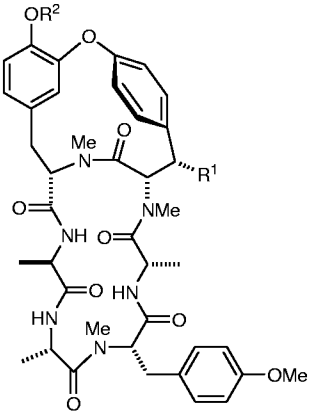
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# 9.5 Synthesis of Cycloisodityrosine Peptides

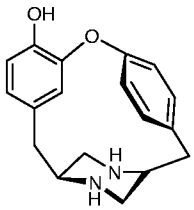
D. L. Boger and S. L. Castle

Cycloisodityrosines are cyclic peptides that contain the isodityrosine residue in a ring. The cycloisodityrosine subunit is present in several natural products, including bouvardin (**1**), deoxybouvardin (**2**), RA-VII (**3**), piperazinomycin (**4**), OF4949-I–OF4949-IV (**5–8**), and K-13 (**9**) (Scheme 1).[1] To date, all cycloisodityrosine-containing rings found in natural products are *either* 14- or 17-membered rings. Many of these naturally occurring cycloisodityrosine compounds possess potent biological activity; accordingly, they have been the targets of considerable synthetic effort.[1] This section summarizes and evaluates the methods used to form cycloisodityrosines, with an emphasis on ring-forming reactions.

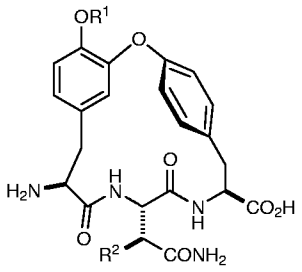
**Scheme 1** Cycloisodityrosine Natural Products[1]



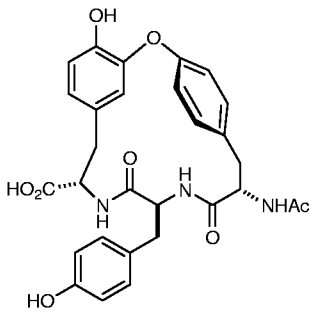
**1** bouvardin $R^1 = OH$; $R^2 = H$
**2** deoxybouvardin $R^1 = R^2 = H$
**3** RA-VII $R^1 = H$; $R^2 = Me$



**4** piperazinomycin



**5** OF4949-I $R^1 = Me$; $R^2 = OH$
**6** OF4949-II $R^1 = H$; $R^2 = OH$
**7** OF4949-III $R^1 = Me$; $R^2 = H$
**8** OF4949-IV $R^1 = H$; $R^2 = H$

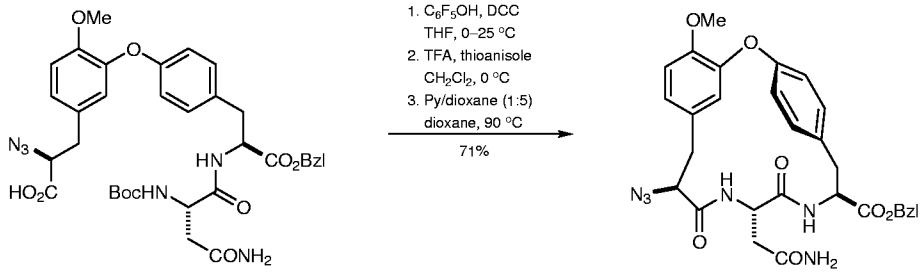


**9** K-13

## 9.5.1 Cycloamidation

Cycloamidation has been used extensively to prepare 17-membered cycloisodityrosines. The acyclic biaryl ether precursors were prepared by methods including the Ullmann reaction[2–5] and nucleophilic aromatic substitution ($S_NAr$).[6,7] Since these methods have all been used intramolecularly in cyclization reactions, they will be discussed in Sections 9.5.3 and 9.5.4. Evans and co-workers employed the pentafluorophenyl ester method of macrolactamization[8] to prepare **11**, an intermediate in their total synthesis of OF4949-III (**7**) (Scheme 2).[3] In this case, the acidic removal of a Boc group was employed to release the cyclization substrate, although hydrogenolysis of a Z group is also effective.[3]

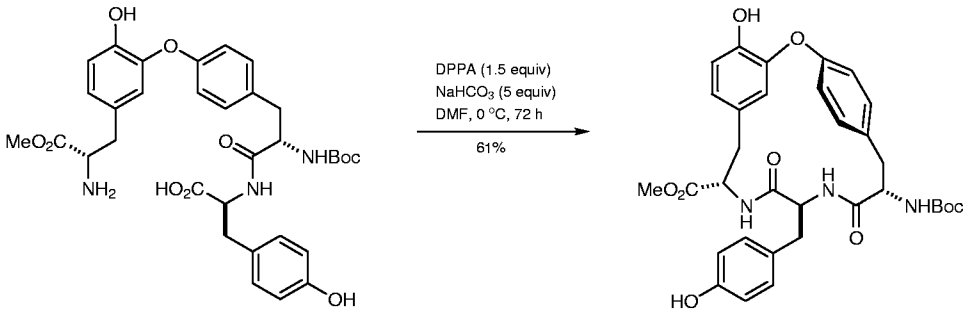
**Scheme 2** Evans Cycloamidation Reaction[3]




**10** **11**

Another cycloamidation reagent which has found wide use is diphenyl phosphorazidate (DPPA).[9] In their synthesis of K-13 (**9**), Boger and co-workers used DPPA to form the 17-membered cyclic peptide **13** (Scheme 3).[4,5]

**Scheme 3** Boger Cycloamidation[4,5]



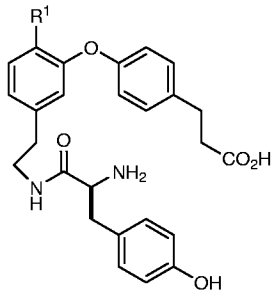
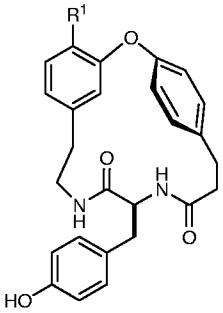
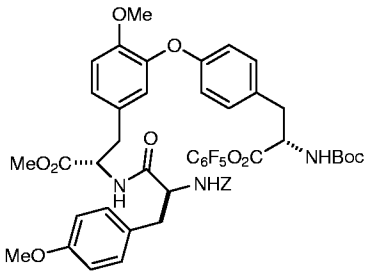
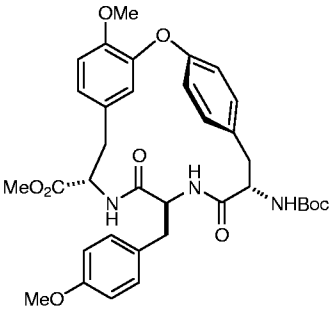
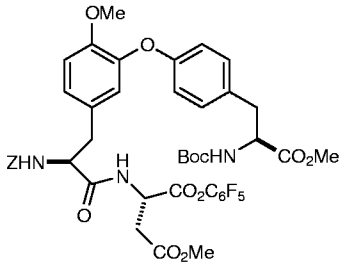
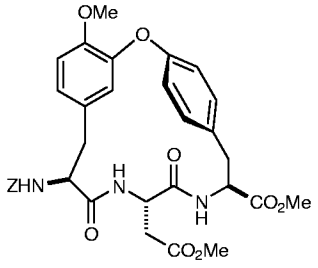
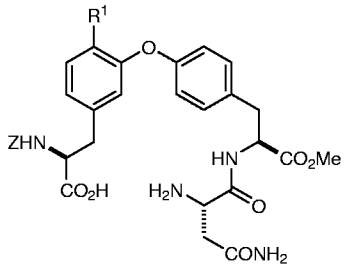
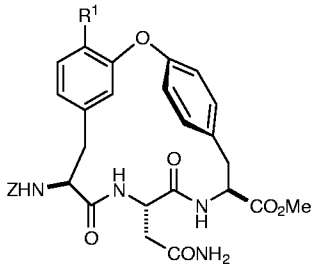

**12** **13**

Boger and co-workers performed a detailed study on the effect of substrate structural features on the 17-membered ring macrolactamization. It was found that the substituent at the remote C4 aryl position had a profound effect on the rate and yield of the cyclization. Substrates bearing either a free phenol or no substituent were found to cyclize more efficiently, whereas those bearing methyl or benzyl ethers cyclized considerably less efficiently (Table 1).[5]

Macrolactamization has limitations. All macrolactamization methods studied to date are ineffective at forming 14-membered cycloisodityrosines of the type found in deoxybouvardin (**2**) and RA-VII (**3**).[10] In addition, macrolactamization was not successful at preparing 16-membered rings that are structurally similar to cycloisodityrosines, such as the type found in the vancomycin family of glycopeptide antibiotics.[11,12]

*for references see p 205*

**Table 1** Cycloamidation[2–5]

| Starting Material | Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | DPPA |  | | |
| | | $R^1$ = H | 68 | [4,5] |
| | | $R^1$ = OH | 66 | [4,5] |
| | | $R^1$ = OMe | 12–25 | [4,5] |
|  | $H_2$, Pd(0), NMM, EtOH/ dioxane (2:98) |  | 67 | [3] |
|  | 1. TMSOTf 2. sat. aq $NaHCO_3$/ $CHCl_3$ (1:1) |  | 40 | [2] |
|  | DPPA |  | | |
| | | $R^1$ = OH | 65 | [5] |
| | | $R^1$ = OBzl | 36 | [5] |

**Benzyl (9*S*,12*S*,15*S*)-12-(2-Amino-2-oxoethyl)-9-azido-4-methoxy-10,13-dioxo-2-oxa-11,14-diazatricyclo[15.2.2.1$^{3,7}$]docosa-1(19),3(22),4,6,17,20-hexaene-15-carboxylate (11):**[3]
A soln of **10** (82 mg, 0.117 mmol) in freshly distilled THF (5 mL) at 0 °C was treated with pentafluorophenol (64.6 mg, 0.351 mmol), followed by DCC (38 mg, 0.176 mmol). The soln was stirred at 0 °C for 2 h and then at rt for 12 h. The resulting slurry was filtered through glass wool and concentrated in vacuo to give a pale yellow oil containing the activated ester. The oil was diluted with $CH_2Cl_2$ (8 mL), and after cooling the soln to 0 °C, thioanisole (274 μL, 2.34 mmol) and then TFA (6 mL) were added. The soln was stirred at 0 °C for 1 h and concentrated in vacuo to give the amine salt as a pale yellow oil. The amine salt was diluted with dioxane (10 mL) and added by syringe pump over 3.5 h to dioxane/pyridine (5:1) at 90 °C under $N_2$. After the addition was complete, the soln was stirred at 90 °C for 1 h, cooled to rt, and concentrated in vacuo. Chromatography (silica gel, $CH_2Cl_2$/MeOH 96:4) followed by recrystallization (MeCN/$H_2O$ 4:1) afforded **11**; yield: 50 mg (71%); mp 246–247 °C; $R_f$ 0.19 ($CH_2Cl_2$/MeOH 95:5).

**Methyl (9*S*,12*S*,15*S*)-15-(*tert*-Butoxycarbonylamino)-4-hydroxy-12-(4-hydroxybenzyl)-11,14-dioxo-2-oxa-10,13-diazatricyclo[15.2.2.1$^{3,7}$]docosa-1(19),3(22),4,6,17,20-hexaene-9-carboxylate (13):**[5]
A soln of **12** (61 mg, 0.096 mmol) in DMF (12 mL) at 0 °C was treated with $NaHCO_3$ (40 mg, 0.48 mmol) and DPPA (32 μL, 0.144 mmol). The mixture was stirred at 0 °C for 72 h, poured into $H_2O$ (20 mL), and extracted with EtOAc (6 × 5 mL). The combined extracts were washed with $H_2O$ (2 × 10 mL) and brine, dried ($Na_2SO_4$), and concentrated in vacuo. Chromatography (silica gel, EtOAc/hexane 3:2–4:1) afforded **13** as a tan, glassy solid; yield: 36 mg (61%); mp 214–216 °C; $[\alpha]_D^{22}$ +187 (*c* 1.2, MeOH).

## 9.5.2 Intramolecular Thallium Trinitrate Oxidative Coupling

The intramolecular thallium(III) nitrate (thallium trinitrate, TTN) oxidative coupling reaction[13,14] is employed to prepare both 14- and 17-membered cycloisodityrosines. In order to undergo the reaction, the two tyrosine residues in the acyclic peptide substrates must bear *o*-dihalo functionality. Conversion of the cyclized products into cycloisodityrosines requires reduction with Zn/AcOH followed by hydrogenolysis of the three remaining aryl halide moieties. The nature of the halide substituents can be varied in order to control the regioselectivity of the cyclization. The susceptibility of halides to displacement via TTN oxidative coupling decreases in the order I>Br>Cl, so a more reactive halide substituent is often placed at the desired site of displacement. Inoue and co-workers used Yamamura's TTN oxidative coupling to prepare **15**, an intermediate in total syntheses of RA-VII (**3**) and deoxybouvardin (**2**) (Scheme 4).[15,16] In addition to **15**, methoxy compound **16** is also formed in the reaction as an undesired, separable byproduct. Compound **15** was reduced, methylated, and subjected to hydrogenolysis to afford cycloisodityrosine **17**.

*for references see p 205*

**Scheme 4** Inoue Intramolecular Oxidative Coupling[15,16]

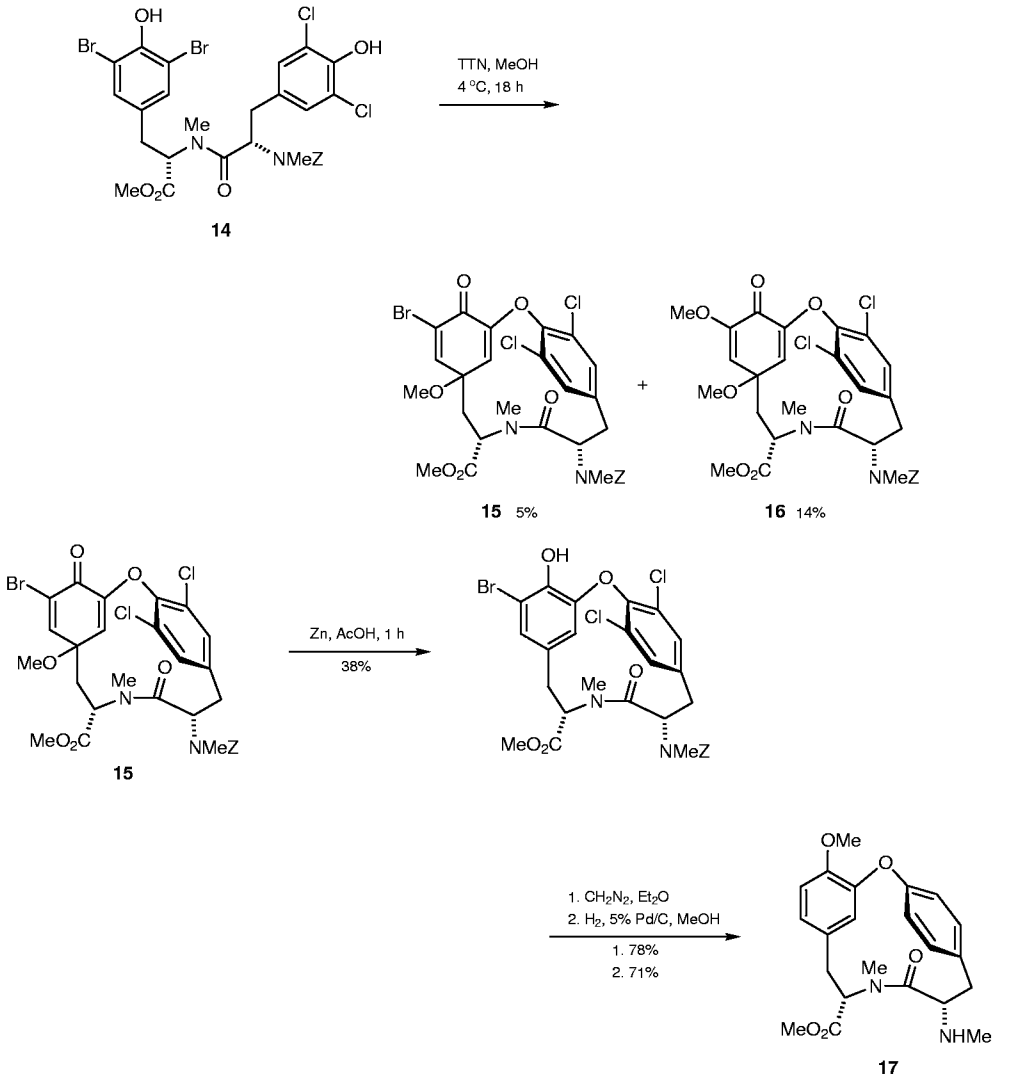
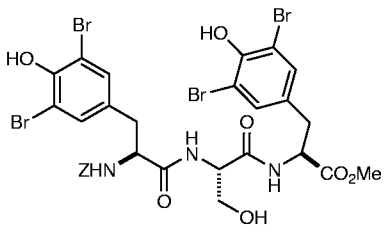
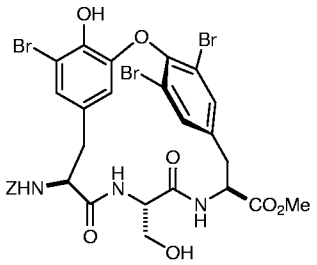
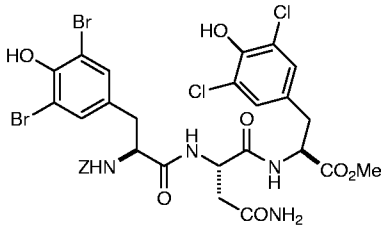
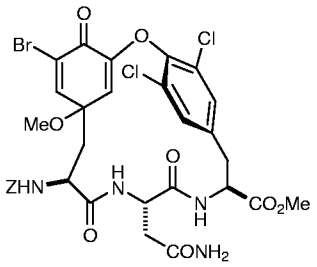
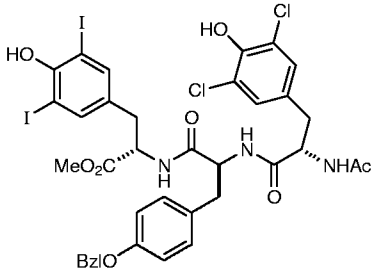
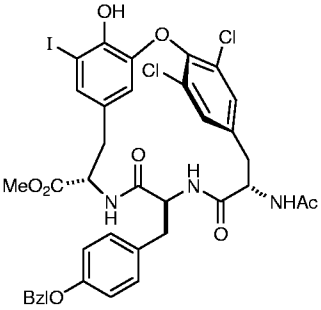



Table 2 contains some examples of TTN oxidative coupling reactions. These examples show that the reaction can often tolerate unprotected amino acid side chains, and that both 14- and 17-membered cycloisodityrosines can be obtained. In addition, the TTN oxidative coupling has been employed in the formation of 16-membered cyclic biaryl ethers of the type found in vancomycin.[20,21] The drawbacks of TTN oxidative coupling are the low yields, the formation of byproducts, and the need for additional transformations to arrive at the cycloisodityrosine target.

**Table 2** Intramolecular Thallium Trinitrate Oxidative Coupling[17–19]

| Starting Material | Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | 1. TTN, MeOH, 4 °C<br>2. Zn, AcOH, rt |  | 14 | [17] |
|  | TTN, MeOH, 0 °C to rt |  | 25 | [18] |
|  | 1. TTN, MeOH/THF (1:2.5), −22 °C to rt<br>2. Zn, AcOH, rt |  | 15 | [19] |

**Methyl (9*S*,12*S*)-12-[(Benzyloxycarbonyl)(methyl)amino]-5-bromo-16,17-dichloro-7-methoxy-10-methyl-4,11-dioxo-2-oxa-10-azatricyclo[12.2.2.1$^{3,7}$]nonadeca-1(16),3(19),5,14,17-pentaene-9-carboxylate (15):**[16] A soln of **14** (8.0 g, 10.7 mmol) in MeOH (5.5 L) at 4 °C was treated with TTN (14.3 g, 32.2 mmol). The mixture was stirred at 4 °C for 18 h. Pyridine (15 mL) was added, and the mixture was concentrated in vacuo. The residue was diluted with brine and extracted with EtOAc. The combined organic layers were washed with sat. aq $Na_2S_2O_3$ and brine, dried ($MgSO_4$), and concentrated in vacuo. Chromatography (silica gel, hexane/EtOAc gradient 3:1–1:1) afforded **15** as a colorless, amorphous mass; yield: 380 mg (5%); $[\alpha]_D^{29}$ −120.6 (*c* 0.45, $CHCl_3$); and **16** as a colorless, amorphous mass; yield: 1.0 g (14%); $[\alpha]_D^{29}$ −170.8 (*c* 0.69, $CHCl_3$).

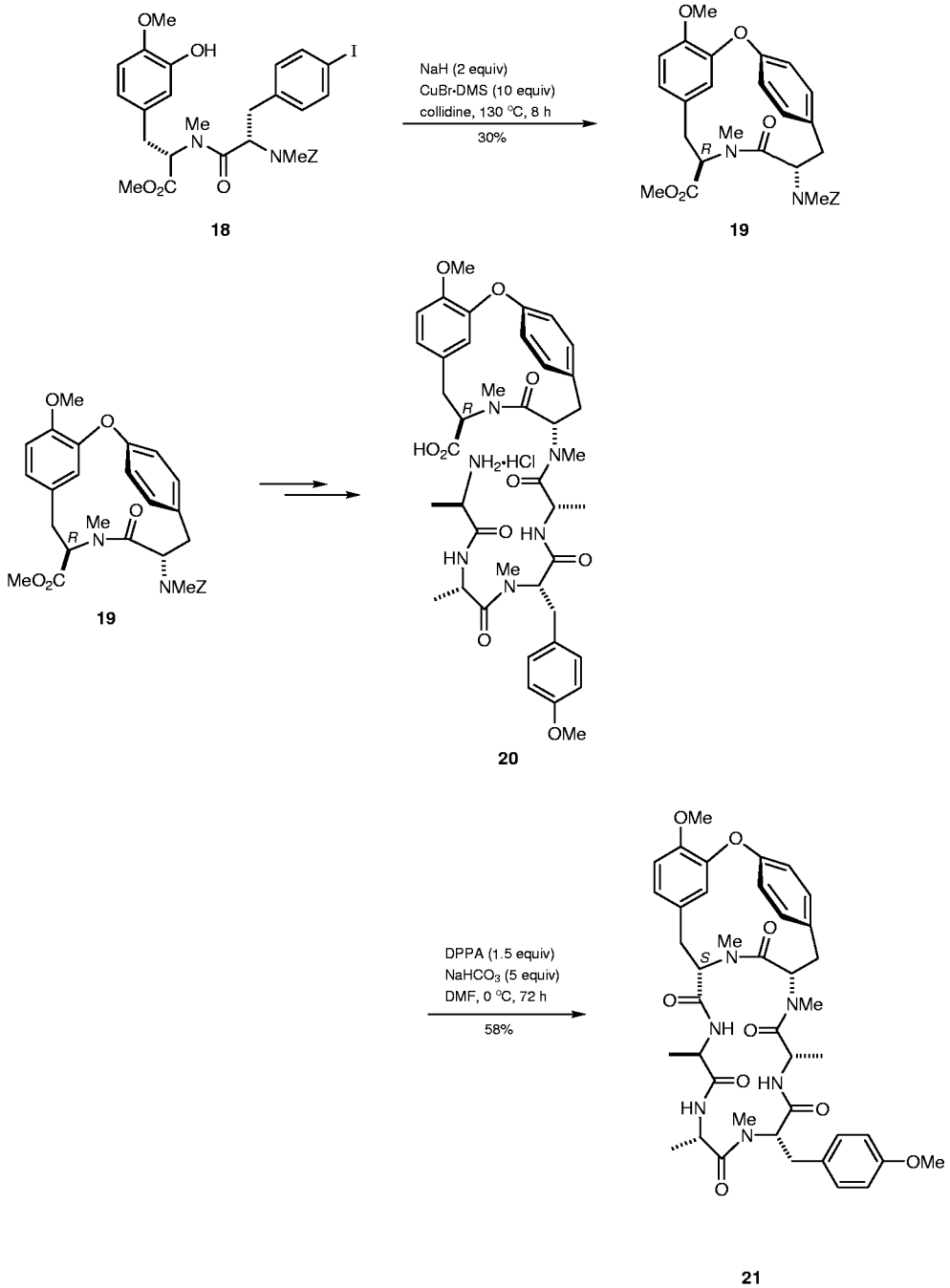
## 9.5.3 Intramolecular Ullmann Reaction

The intramolecular Ullmann reaction has been used by Boger and co-workers to prepare 14-membered cycloisodityrosines.[22] This reaction involves heating a dilute solution of the acyclic dipeptide substrate in pyridine, dioxane, or collidine at reflux in the presence of NaH and CuBr · DMS. One of the tyrosine residues must possess an iodide moiety at the desired site of displacement. Due to the harsh conditions that are necessary for cyclization, epimerization at the C9 center from the natural *S* to the unnatural *R* configuration occurs.[23,24] Boger and co-workers have studied this epimerization and have shown that it originates from destabilizing transannular interactions and a gauche/eclipsed relationship that the C9 substituent must endure when it is in the *S* configuration.[24,25] However, this epimerization can be advantageous. Boger and co-workers were able to employ 14-membered cycloisodityr-

*for references see p 205*

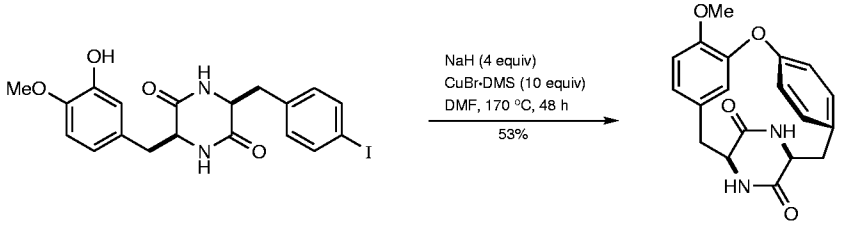
osines prepared by the intramolecular Ullmann reaction in the total syntheses of the bicyclic hexapeptides RA-VII (**3**), deoxybouvardin (**2**), and bouvardin (**1**).[10,26,27] When the cyclization of hexapeptide **20** was performed under conventional macrolactamization conditions (DPPA, $NaHCO_3$, DMF), the C9 center epimerized back to the natural *S* configuration before ring closure, thereby affording compound **21**, or RA-VII (**3**) (Scheme 5). Accordingly, the intramolecular Ullmann reaction provided an effective method of preparing bicyclic hexapeptide natural products.

**Scheme 5** Boger Intramolecular Ullmann Cyclization[10,26]



Boger and co-workers successfully used the intramolecular Ullmann reaction to prepare **23**, an intermediate in their total synthesis of piperazinomycin (**4**) (Scheme 6). In this case, the C9 center of the piperazine–2,5-dione substrate did not epimerize, and the cyclization proceeded smoothly and in high yield.[28]

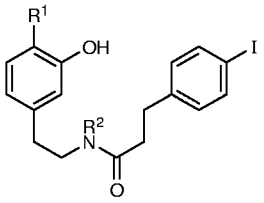
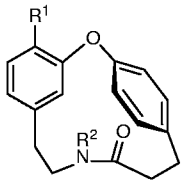
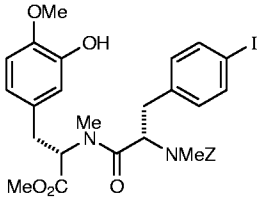
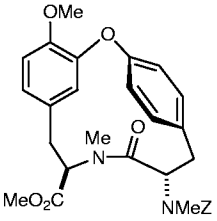
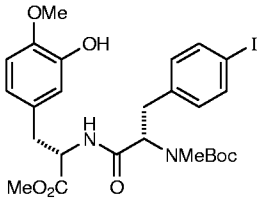
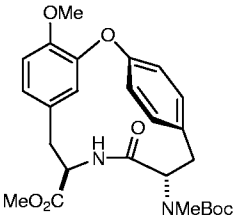
**Scheme 6** Boger Intramolecular Ullmann Reaction with Piperazine-2,5-dione[28]




**22** **23**

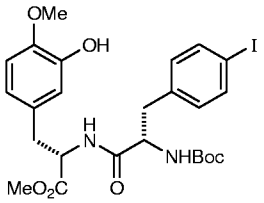
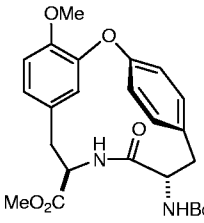
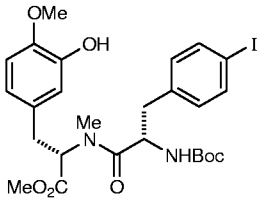
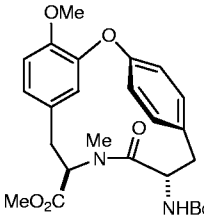
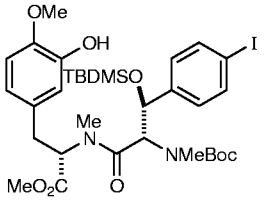
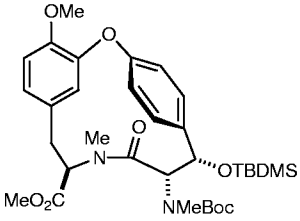
Boger and co-workers have studied the intramolecular Ullmann reaction extensively. Some representative examples are shown in Table 3. For substrates in which the $N^{10}$ site is not methylated, the reaction is complicated by intramolecular N-acylation of the $N^{10}$–$C^{11}$ amide by the $C^{12}$ carbamate. In some cases, the generation of this byproduct could be suppressed by using freshly dried and degassed solvent with purified CuBr·DMS complex.[10] However, in other cases, the byproduct was still produced and the yield of the desired product suffered accordingly.[27]

**Table 3** Intramolecular Ullmann Reaction[10,22,26–29]

| Starting Material | Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | NaH, CuBr·DMS, pyridine, 130 °C, 18 h |  | | |
| | | $R^1 = R^2 = H$ | 58 | [22] |
| | | $R^1 = H$; $R^2 = Me$ | 49 | [22] |
| | | $R^1 = OMe$; $R^2 = H$ | 46 | [22] |
| | | $R^1 = OMe$; $R^2 = Me$ | 45 | [22] |
|  | NaH, CuBr·DMS, collidine, 130 °C, 8 h |  | 24–30 | [10,26] |
|  | NaH, CuBr·DMS, collidine, 130 °C, 9 h |  | 30–39 | [10] |

*for references see p 205*

**Table 3** (cont.)

| Starting Material | Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | NaH, CuBr · DMS, pyridine, 130 °C, 9 h |  | 5–10 | [28] |
|  | NaH, CuBr · DMS, collidine, 130 °C, 10 h |  | 34 | [29] |
|  | NaH, CuBr · DMS, 2,6-lutidine, 130 °C, 9 h |  | 37 | [27] |

**Methyl (9*R*,12*S*)-12-[(Benzyloxycarbonyl)(methyl)amino]-4-methoxy-10-methyl-11-oxo-2-oxa-10-aza-tricyclo[12.2.2.1$^{3,7}$]nonadeca-1(16),3(19),4,6,14,17-hexaene-9-carboxylate (19):**[10]

A soln of **18** (0.178 g, 0.268 mmol) in anhyd collidine (1 mL) was added to a 0 °C suspension of NaH (60% dispersion in mineral oil, 22 mg, 0.54 mmol) in collidine (1 mL). This was followed by the addition of CuBr · DMS (0.522 g, 2.68 mmol). After stirring for 0.5 h at 25 °C, the mixture was diluted with collidine (70 mL) and heated at 130 °C for 9 h. The cooled mixture was poured over 10% aq HCl (30 mL) and extracted with EtOAc (30 mL). The organic layer was washed with 10% aq HCl (3 × 50 mL), sat. aq $NH_4Cl$ (2 × 30 mL), and brine (30 mL), dried ($MgSO_4$), and concentrated in vacuo. Chromatography (silica gel, EtOAc/hexane gradient 1:2–3:2) afforded **19** as a clear yellow oil; yield: 0.04 g (30%); $[\alpha]_D^{22}$ −22 (*c* 0.12, $CHCl_3$).

**(3*S*,6*S*)-11-Methoxy-5,19-dioxo-13-oxa-4,20-diazatetracyclo[12.2.2.2$^{3,6}$.1$^{8,12}$]hemicosa-1(16),8(21),9,11,14,17-hexaene (23):**[28]

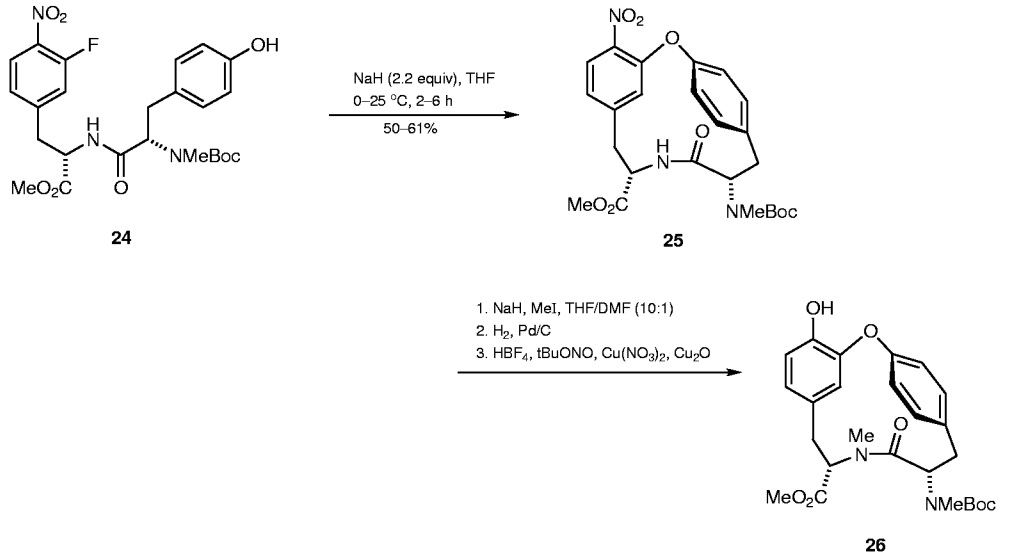
A soln of **22** (466 mg, 1.0 mmol) in anhyd DMF (5 mL) was added dropwise to a suspension of NaH (60% dispersion in mineral oil, 160 mg, 4.0 mmol) in anhyd DMF (2 mL) at 0 °C under argon. The mixture was stirred for 20 min, treated with CuBr · DMS (2.06 g, 10.0 mmol), and stirred at 25 °C for 1 h. The mixture was diluted with anhyd degassed DMF to 0.004 M (243 mL) and stirred at 170 °C for 48 h. The mixture was cooled to 25 °C, concentrated in vacuo, and the residue was treated with sat. aq $NH_4Cl$/concd $NH_4OH$ (9:1, 40 mL) and $CHCl_3$/MeOH (9:1, 40 mL). The mixture was stirred for 30 min before the layers were separated and the aqueous layer was extracted with $CHCl_3$/MeOH (9:1, 5 × 40 mL). The combined organic extracts were washed with sat. aq $NH_4Cl$ (2 × 20 mL), $H_2O$ (20 mL), and brine (2 × 30 mL), dried ($MgSO_4$), and concentrated in vacuo. Chromatography (silica gel, $CHCl_3$/MeOH gradient 49:1–9:1) afforded **23** as a white solid; yield: 0.180 mg (53%); mp 284–286 °C (dec); $[\alpha]_D^{25}$ +182 (*c* 0.05, MeOH).

## 9.5.4 Intramolecular Nucleophilic Aromatic Substitution

Intramolecular nucleophilic aromatic substitution ($S_NAr$) is a method for cycloisodityrosine formation that has recently gained popularity. This reaction employs intramolecular attack of a phenoxide on an electron-deficient aromatic ring for cyclization. To date, both *o*-fluoronitro systems[30] and arene–ruthenium π-complexes[31,32] have been used as the electron-deficient moiety. In the case of the fluoronitro substrates, the cyclic products can be converted into cycloisodityrosines by either transforming the nitro group into a phenol by means of arenediazonium chemistry,[25] or by reductively removing it,[33] depending on its position in the product. The ruthenium π-complexes can be demetalated by photolysis in acetonitrile to give cycloisodityrosines.[34] Typical bases that are used to promote cyclization include sodium hydride, potassium carbonate, and cesium fluoride for the fluoronitro substrates, and sodium 2,6-di-*tert*-butylphenoxide and potassium *tert*-butoxide for the arene–ruthenium complexes. The intramolecular $S_NAr$ reaction is quite versatile, as it allows for preparation of both 14- and 17-membered cycloisodityrosines.

Boger and co-workers used the intramolecular $S_NAr$ reaction on *o*-fluoronitro substrate **24** with NaH to prepare **25**. Compound **25** was then converted in three steps into phenol **26**, the cycloisodityrosine subunit of deoxybouvardin (**2**) and RA-VII (**3**) (Scheme 7).[24,25] Small amounts of the C9 epimer were also detected in the cyclization reaction, and when $K_2CO_3$ was employed as the base, this epimer was formed nearly exclusively.[25]

**Scheme 7** Boger Intramolecular $S_NAr$ Reaction[24,25]




Rich and co-workers employed the intramolecular $S_NAr$ reaction of arene–ruthenium π-complexes in their total syntheses of K-13 (**9**) and OF4949-III (**7**). Treatment of tripeptide complex **27** with potassium *tert*-butoxide gave 17-membered macrocycle **28**, which was demetalated in acetonitrile to give cycloisodityrosine **29**, a precursor to OF4949-III (**7**) (Scheme 8).[34]

*for references see p 205*

**Scheme 8** Rich Intramolecular $S_NAr$ Reaction[34]

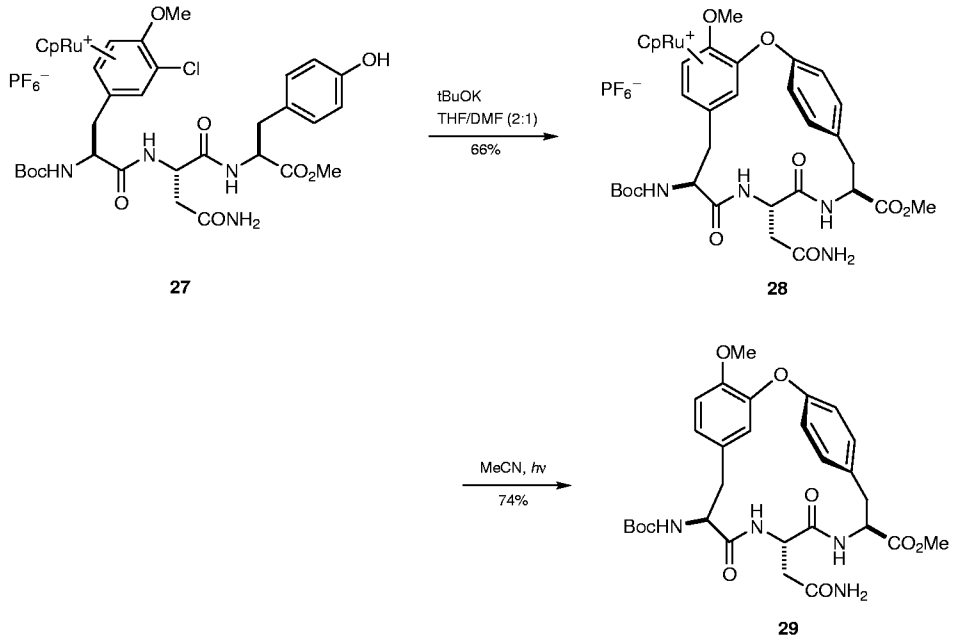



Table 4 lists some representative examples of macrocyclization via the intramolecular $S_NAr$ reaction. In addition to the 14- and 17-membered cycloisodityrosines shown in the table, a variety of mono-, bi-, and tricyclic systems from the vancomycin family of natural products including the orienticin C,[40] vancomycin,[44–47] and teicoplanin[48] aglycons have been prepared by this method.[38–48] This versatility, coupled with the high yields obtained, makes the intramolecular $S_NAr$ reaction currently the most widely applicable method for cycloisodityrosine formation.

**Table 4** Intramolecular $S_NAr$ Reaction[25,33,35–37]

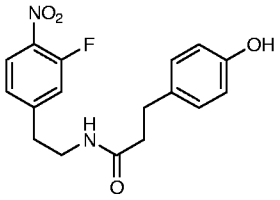
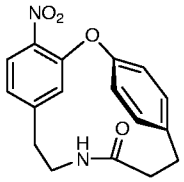
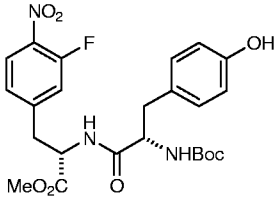
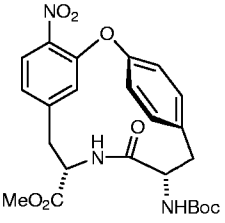
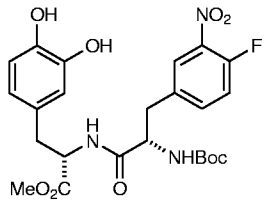
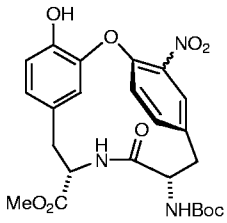
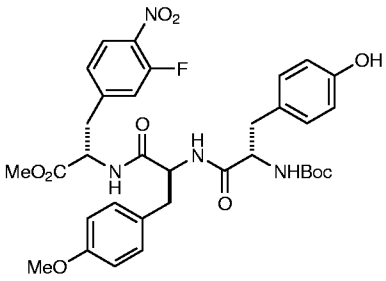
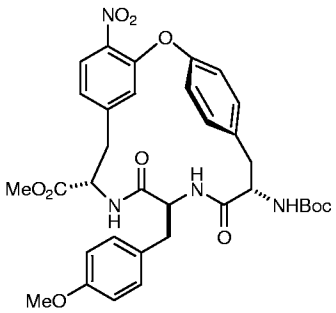
| Starting Material | Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | $K_2CO_3$, DMF, 90 °C, 2 h |  | 78 | [37] |
|  | NaH, THF, 0–25 °C, 6–12 h |  | 50–55 | [25,35] |

**Table 4** (cont.)

| Starting Material | Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | $K_2CO_3$, DMSO, rt |  | 55–65[a] | [33] |
|  | $K_2CO_3$, DMF, rt |  | 87 | [36] |

[a] Product was obtained as a mixture of atropisomers.

**Methyl (9*S*,12*S*)-12-[(*tert*-Butoxycarbonyl)(methyl)amino]-4-nitro-11-oxo-2-oxa-10-azatricyclo[12.2.2.1$^{3,7}$]-nonadeca-1(16),3(19),4,6,14,17-hexaene-9-carboxylate (25):**[25]

A soln of **24** (156 mg, 0.3 mmol) in anhyd THF (2 mL) was added dropwise to a suspension of NaH (60% dispersion in mineral oil, 26.4 mg, 0.66 mmol) in anhyd THF (73 mL) at 0 °C under argon. The mixture was stirred at 0 °C for 30 min and at 25 °C for 6 h. The reaction was quenched by the addition of $H_2O$ (5 mL) and EtOAc (50 mL). The aqueous soln was extracted with EtOAc (3 × 10 mL), and the combined organic layers were washed with $H_2O$ (15 mL) and brine (10 mL), dried ($MgSO_4$), and concentrated in vacuo. Chromatography (silica gel, hexane/EtOAc gradient 4:1–3:2) afforded **25** as a pale yellow oil which solidified upon standing; yield: 81.3 mg (54%); $[\alpha]_D^{25}$ −91 (*c* 0.5, $CHCl_3$). In addition, the C9 epimer was also isolated as a white solid; yield: 21.4 mg (14%); mp 228–230 °C; $[\alpha]_D^{25}$ −83 (*c* 0.36, $CHCl_3$).

**[*N*-(*tert*-Butoxycarbonyl)-3-tyrosyl(4-OMe)-asparagyl-tyrosine methyl ester](cyclopentadienyl)ruthenium Hexafluorophosphate Biphenyl Ether (28):**[34]

A soln of tBuOK (2.4 mg, 0.024 mmol) in THF (1 mL) at −78 °C was treated with a soln of **27** (20.0 mg, 0.021 mmol) in THF/DMF (2:1, 3 mL). The mixture was warmed to rt, stirred for 2 h, filtered, concentrated to 2 mL, and added to $Et_2O$ (50 mL). The precipitate was filtered, redissolved in $CH_2Cl_2$/MeOH, concentrated in vacuo, and dried to give **28** as a brown solid; yield: 12.4 mg (66%).


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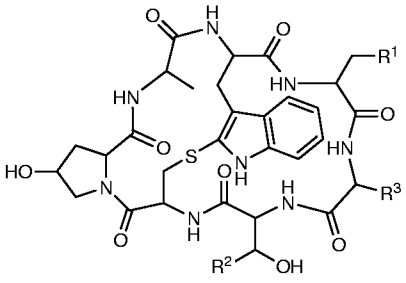
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# 9.6 Tryptathionine Peptides

A. Fontana

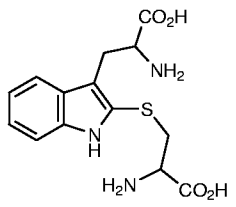
The poisonous components of the most deadly mushroom *Amanita phalloides* (the Death Cap) are bicyclic heptapeptides which have an additional covalent bond that connects the β-sulfur atom of an L-cysteine residue with the carbon atom in position 2 of the indole ring of the L-tryptophan. Phalloidin (or phalloidine) is the most abundant member of a whole family of related cyclic heptapeptides called phallotoxins (for a review, see Wieland[1]). These poisonous peptides, therefore, contain a cross-linking moiety consisting of L-tryptophan coupled to L-cysteine, designated tryptathionine (**1**), alternatively called *S*-(tryptophan-2-yl)cysteine or 2-(L-3-alanylsulfenyl)-L-tryptophan (Scheme 1).

**Scheme 1** Chemical Structures of Phallotoxins and Tryptathionine



phalloidin: $R^1$ = $CMe(OH)CH_2OH$; $R^2$ = $R^3$ = Me
phalloin: $R^1$ = $C(OH)Me_2$; $R^2$ = $R^3$ = Me

phallotoxins



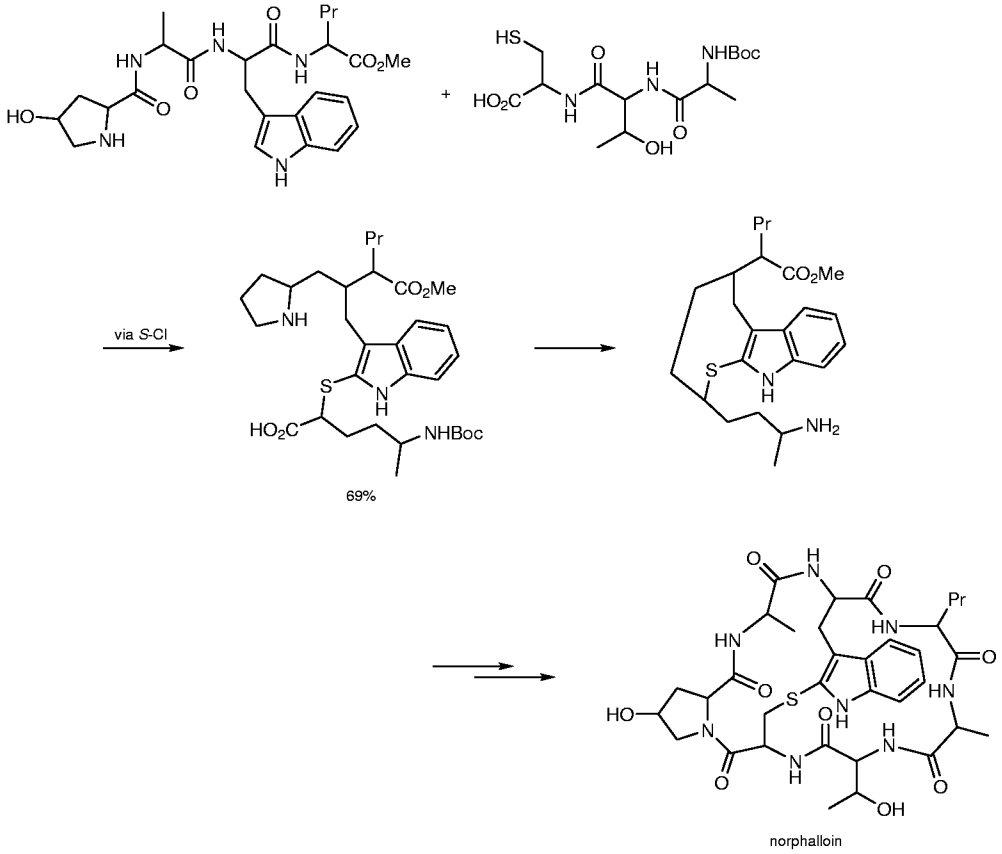
**1** tryptathionine

The chemical synthesis of the *Amanita* toxins has presented several problems, in particular those related to the formation of the sulfur bridge. The latter has been explored with model compounds.[2,3] It has been found that the synthesis of the (sulfanyl)indole moiety can be achieved by reacting an indole compound with an alkanesulfenyl chloride. A model tryptathionine compound has been prepared by reacting *N*-acyl-L-cysteine and *N*-acyl-L-tryptophan in the presence of *N*-chlorosuccinimide in glacial acetic acid at room temperature.[4] The sulfanylation reaction has been subsequently exploited for the selective chemical modification of tryptophan residues in proteins using 2-nitrophenylsulfenyl chloride.[5]

The first total synthesis of a phallotoxin was that of norphalloin, the norvaline analogue of phalloidin.[6] This synthesis, outlined in Scheme 2, initially involves the synthesis of two building blocks followed by their covalent binding via a tryptathionine moiety. Subsequent deprotection and coupling steps using classical methods of peptide synthesis results in the formation of norphalloin, which has proved to be even more toxic than the natural toxins. Since the synthesis of norphalloin, a number of synthetic analogues have been synthesized in order to obtain information on structure–activity relationships.[7,8]

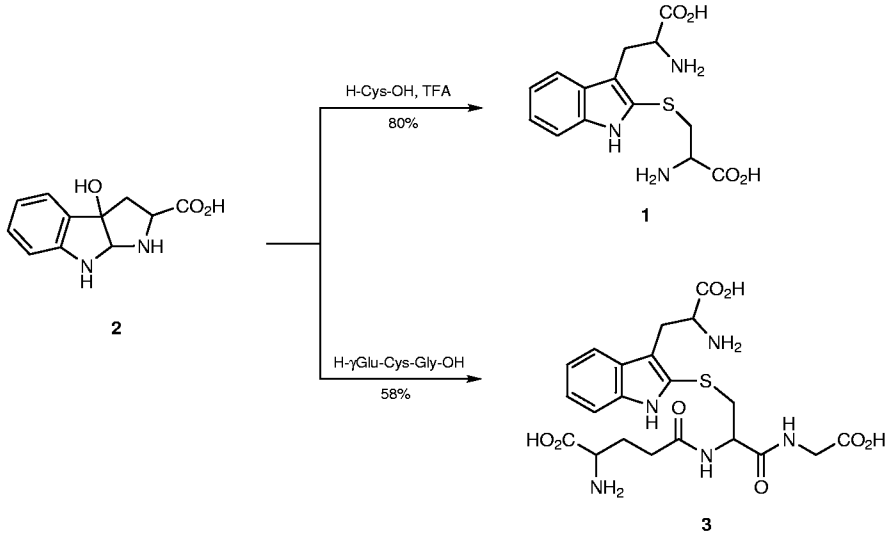
*for references see p 212*

**Scheme 2** Synthesis of Norphalloin[6]



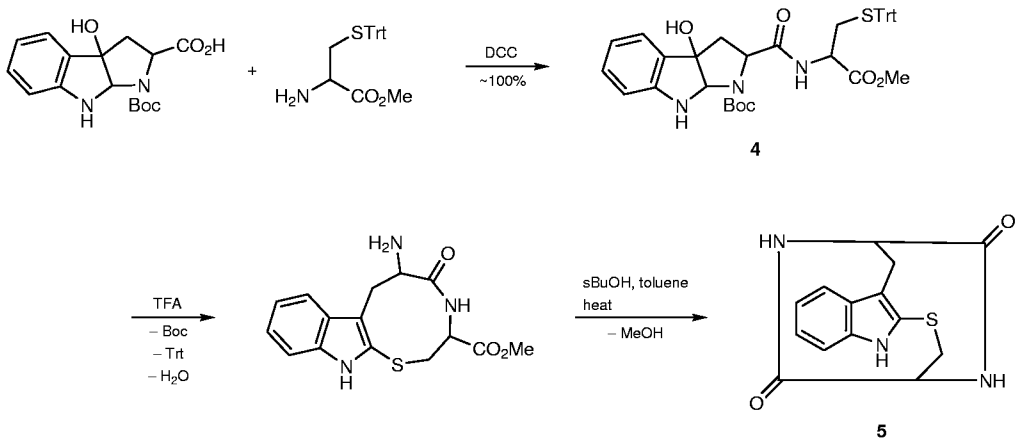

A second successful synthetic approach to tryptathionine (**1**) and its derivatives has been made possible by the observation that the oxidation product of L-tryptophan, L-3a-hydroxy-1,2,3,3a,8,8a-hexahydropyrrolo[2,3-*b*]indole-2-carboxylic acid (**2**),[9] smoothly reacts with thiols yielding 2-thioethers of tryptophan (Scheme 3). The reaction of **2** (1.2 equiv) with cysteine in 25% aqueous TFA produces almost quantitatively tryptathionine (**1**).[10] Similarly, the reaction of a moderate excess of **2** with the thiol group of reduced glutathione (H-γGlu-Cys-Gly-OH) gives the 2-(sulfanyl)indole derivative **3**.[10] It has been found that **2** reacts selectively with the thiol group of cysteine in aqueous TFA and that no other amino acid is modified by the reagent. This allows the selective and quantitative covalent binding of tryptophan to the eight thiol groups of reduced ribonuclease (*S*-tryptophanylation reaction).[10]

**Scheme 3** Reaction of L-3a-Hydroxy-1,2,3,3a,8,8a-hexahydropyrrolo[2,3-*b*]indole-2-carboxylic Acid with Thiol Compounds in Acid[10]




Compound **2** can be prepared in good yields from L-tryptophan by oxidation with peracetic acid[9] and thus constitutes an easily preparable intermediate of potential value in the synthesis of biologically important tryptophan derivatives, including the *Amanita* peptides. In fact, Wieland et al.[11] have taken advantage of the reaction of **2** with thiols[10,12] for synthesizing the smallest bicyclic thioether peptide **5**, called "mini-phallotoxin" (Scheme 4). In order to use compound **2** as the acyl component in peptide synthesis, it is protected at the pyrrolidine nitrogen with the Boc group and then coupled with H-Cys(Trt)-OMe with the aid of DCC to give **4**. The S- and N-protecting groups can then be removed from the dipeptide ester **4** by using 50% aqueous TFA, which leads to the simultaneous formation of the cyclic thioether; which on heating is converted into the "mini-phallotoxin" **5** (Scheme 4).

**Scheme 4** Synthesis of the Smallest Bicyclic Thioether Peptide[11]

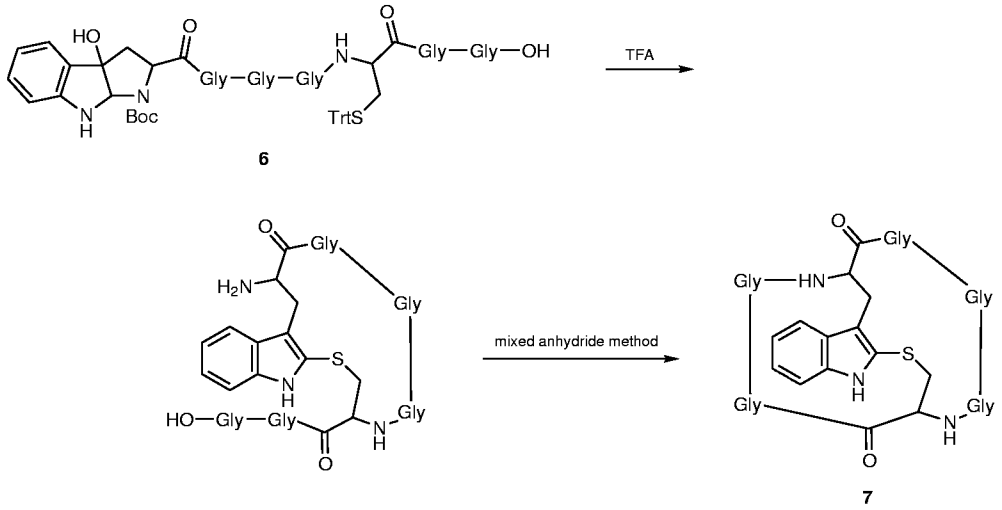



Another model phallotoxin **7** can be prepared from heptapeptide **6**. Dissolving heptapepide **6** in TFA leads to the removal of both *N*-Boc and *S*-Trt protecting groups and concomitant cyclization to form the tryptathionine moiety.[13] A second cyclization of the monocyclic

*for references see p 212*

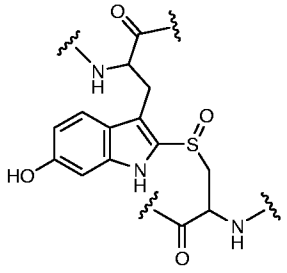
heptapeptide is achieved by formation of the peptide bond between the amino- and carboxy-terminal groups by using the mixed anhydride method to form phallotoxin **7** (Scheme 5).

**Scheme 5** Synthesis of the Simplest All-Gly Phallotoxin[13]

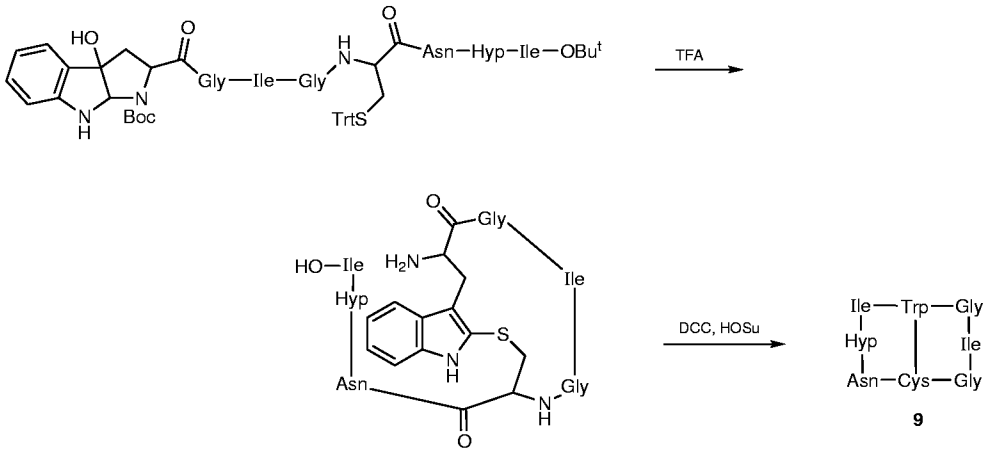



Amatoxins are toxic components of *Amanita* species and, at variance from phallotoxins, these are bicyclic octapeptides with a chain crosslink given by the 6′-hydroxytryptathionine (*R*)-sulfoxide group **8** (see Scheme 6).[1] However, among the amatoxins, amanin and amaninamide lack the 6′-hydroxy group. Moreover, since the sulfoxide oxygen may be absent without diminishing the toxicity, a variety of amaninamide analogues were synthesized by using the novel indolyl-thioether synthesis by means of compound **2**.[14,15] An outline of the synthesis of the $Ile^3$ amaninamide **9** is shown in Scheme 7.

**Scheme 6** Structure of the 6′-Hydroxytryptathionine (*R*)-Sulfoxide Moiety of Amatoxins



**Scheme 7** Synthesis of Ile$^3$ Amaninamide[13,14]




**L-3a-Hydroxy-1,2,3,3a,8,8a-hexahydropyrrolo[2,3-*b*]indole-2-carboxylic Acid (2):**[9]
A soln of H-L-Trp-OH (10.2 g, 50 mmol) in $H_2O$ (1 L) was formed by warming and was cooled to 0 °C. The soln was then treated with 2.75 M peracetic acid in AcOH (18.5 mL, 51 mmol) and kept at 0–5 °C for 18 h. TLC of the pale yellow soln indicated good conversion into **2** (mixed diastereomers), together with small amounts of Trp and formylkynurenine. Excess oxidant was destroyed by adding EDT (0.2 mL), so that after standing for 5 min the soln gave a negative starch/iodide reaction. The pH was adjusted to 3.0 with pyridine and then lyophilized. The residue was triturated with $H_2O$ (50 mL) and washed with ice water (3 × 10 mL) to give a crystalline product; which gave a single ninhydrin spot on TLC. Recrystallization ($H_2O$) gave the product as needles; yield: 2.5 g (23%); mp 237–238 °C (dec) after vacuum drying at 120 °C for 4 h; $[\alpha]_D^{20}$ +63 (*c* 0.38, $H_2O$); UV ($H_2O$) $\lambda_{max}$ (ε): 235 (6320), 292 nm (1960); IR $\tilde{\nu}_{max}$: 3400, 3270, 3130, 1610, 1471, 1380, 1318 $cm^{-1}$; $^1H$ NMR (δ): 7.52–6.75 (m, 4H, Ph), 5.47 (s, 1H, C(8a)H), 4.73 (t, 1H, C(2)H), 2.96 (d, $J$ = 7.0 Hz, 2H, C(3)$H_2$); MS *m/z*: 220 ($M^+$), 177, 175, 158, 148, 147, 146, 133, 132, 130. The solubility of the product in $H_2O$ is about the same as L-Trp, while it is also appreciably soluble in MeOH or aq EtOH, but insoluble in acetone. This preparation of **2** leads to the β-diastereomer, which on paper electrophoresis at pH 2.5 is faster moving[9] and bears the hydroxy group *trans* to the carboxy group.[16]

**Tryptathionine (1):**[10]
A soln of H-L-Cys-OH · HCl · $H_2O$ (17.6 mg, 0.1 mmol) and compound **2** (28.5 mg, 0.13 mmol) in 25% aq TFA (1 mL) was kept at rt for 2 d. Concentration of the mixture in vacuo at 37 °C gave a residue which was redissolved in $H_2O$ (1 mL), applied to a Sephadex LH-20 column (2.2 × 70 cm), and eluted with $H_2O$. Tryptathionine was located in the effluent by spectrophotometry and the fractions of the corresponding peak were combined and lyophilized; yield: 27 mg (80%).

***S*-Tryptophan Glutathione (3):**[10]
A soln of reduced glutathione (H-γGlu-Cys-Gly-OH; 5.2 mg, 0.017 mmol) and **2** (6.9 mg, 0.029 mmol) in 50% aq TFA (1 mL) was kept at rt for 24 h. The mixture was concentrated and the residue dissolved in 10% aq AcOH (1 mL). The soln was then applied to a Sephadex G-25 SF column (3 × 54 cm), equilibrated, and eluted with 10% aq AcOH. The product was located in the effluent by spectrophotometry, the UV absorption being similar to that of tryptathionine. The relevant fractions were combined and lyophilized. The product was further purified by a second gel filtration; yield: 5 mg (58%). Amino acid analysis of an acid hydrolyzate with TosOH[10] gave: Glu 1.00, Gly 0.98, Cys 0.70, oxindolylalanine 1.08, and traces of cystine.

*for references see p 212*

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# 10 Main-Chain-Modified Peptides

C. Toniolo

Peptidomimetics are peptide molecules characterized by modifications to the side chain (Section 9), to the α-carbon (Sections 10.3 and 10.4), or, to the components of a peptide bond itself either singularly (e.g., Sections 10.1 and 10.2), or, in combination. Peptidomimetics containing a modification to the nature of the peptide bond are often referred to as pseudopeptides or peptide bond surrogates.[1] These compounds are sometimes termed as peptide bond isosteres, although the isosteric nature of the replacement is often not obvious or may even be dubious.

The psi-bracket (ψ[ ]) symbology,[2,3] in which the structural and stereochemical integrities of the adjacent pair of α-carbons is assumed to remain intact, helps reveal the breadth of possible peptide bond replacements. The symbology specifies only the changes which occur between the two α-carbons. In this volume this symbology was used in particular through Sections 10.5–10.9.

The -NH- group of the secondary amide bond of peptides can be replaced by the -NR- (Section 10.1) or the -O- group (Section 10.2). With N-alkylation, a stereochemical problem generated by the onset of the *cis/trans* equilibrium about the tertiary amide group, must be taken into account in designing peptidomimetic analogues and in the interpretation of the conformational and biological data.[4–7] The increasingly popular class of peptoids (N-alkylated glycines) is also discussed (Section 10.1.2).[8–10] Depsipeptides, including important families of naturally occurring compounds, are treated in Section 10.2.[6,11]

$C^{\alpha}$-Tetrasubstitution (Section 10.3) is recommended when the stabilization of specific backbone conformations (β-turns, $\alpha/3_{10}$-helices, or the fully extended structure) is the target of modifications.[6,12] The central $C^{\alpha}$-trisubstituted atom of an α-amino acid may also be replaced by an -NH- or an -NR- group, thereby producing α-azapeptides (Section 10.4).[7,13]

Sections 10.5 through 10.8 envisage single or, more commonly, multiple replacements of the peptide bond functionality. In particular, substitutions characterized by single, double, or triple C–C bonds are covered in Section 10.5, while substitutions containing C/O, C/N, and C/N/O amide bond replacements are described in Sections 10.6, 10.7, and 10.8, respectively. Sections 10.9 and 10.10 focus on the known types of sulfur- or phosphorus-based peptide bond replacements,[1,2,6,7] including different oxidation states of the heteroatoms.

The topology of the peptide bond may be modified by synthesizing *retro-* or *retro-inverso* peptides (Section 10.11).[6,7,14,15] Currently, interest in amino acids in which the amino and carboxy functionalities are separated by more than one carbon atom is increasingly expanding. β-Amino acids and γ-amino acids are discussed in Sections 10.12 and 10.13, respectively.[16–19] Based on rigid members of these two classes of compounds, new types of helical structures and promising scaffolds for biochemical applications have been proposed. γ-Amino acids include statine analogues which are versatile building blocks for the construction of potent enzyme inhibitors. Expansion of the -CONH- function has been achieved by adding an extra amino group or an oxygen atom, thereby generating an ureine or an urethane (carbamate) linkage, respectively (Section 10.14), or by inserting an aromatic moiety (Section 10.15).

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# 10.1 Synthesis of N-Alkylated Peptides

C. Gilon, M. A. Dechantsreiter, F. Burkhart, A. Friedler, and H. Kessler

N-Alkylation is an important modification of the peptide bond. In particular, N-methylation is widespread and commonly occurs in natural peptides from plants, marine sources, and various microorganisms. Several of these compounds have important biological activity, such as antibiotic (e.g., enniatins,[1] monamycins,[2] echinomycin[3]), antitumor (e.g., bouvardin,[4] dolastatin[5,6]) and immunosuppressor activity (e.g., cyclosporin[7]). For reviews see refs[8–10].

The incorporation of *N*-methyl amino acids into biologically active peptides, replacing the naturally occurring amino acids, has resulted in analogues with improved pharmacological properties, such as metabolic stability, receptor selectivity, enhanced potency, and enhanced bioavailability. Therefore, structural modification of the peptide backbone by N-alkylation is a powerful approach to modify profiles of biological activity. Also, in addition to alanine-scan [Ala-scan of a given peptide consists of two operations: (1) synthesis of a series of peptides in which each amino acid residue is replaced by Ala, (2) biological screening of each peptide in the series; Ala-scan furnishes valuable information on the role of each side chain in the activity of the peptide] or D-amino acid scan (D-Xaa-scan) the systematic N-methylation of single amide bonds in a bioactive peptide (*N*-alkyl-scan) can be used as a routine tool to probe structure–activity relationships.[11]

The structural effects of N-alkylation have been intensively examined in model peptides[12–16] and on N-methylated analogues of biologically relevant peptides.[17,18] Data available for amides and peptides have outlined the following structural perturbations induced by N-methylation:

(1) The occurrence of the *cis*-isomer in a tertiary amide bond.[19–21] N-Alkylated amides normally exhibit a mixture of *cis*/*trans*-isomers that equilibrate in the order of seconds ($\Delta G^{\ddagger}$ ~18–20 kcal·mol$^{-1}$). The situation is shifted from a pure *trans* configuration in a secondary (–NHCO–) peptide bond to an equilibrium of both isomers of similar energy upon N-alkylation; the percentage of *cis*-isomer depends upon both the solvent and the sequence. The *cis*-peptide bond[22–27] is observed between the N-alkylated and the preceding residues in the chain. It is evident that Xaa-Pro peptides will behave similarly.

(2) Steric constraints introduced by the *N*-alkyl group.[12,17] The steric hindrance of the *N*-alkyl group is experienced both by the peptide backbone and the side chains of neighboring amino acids.[18]

(3) Decrease in the number of inter- and intramolecular hydrogen bonds due to the removal of the proton-donating NH group.

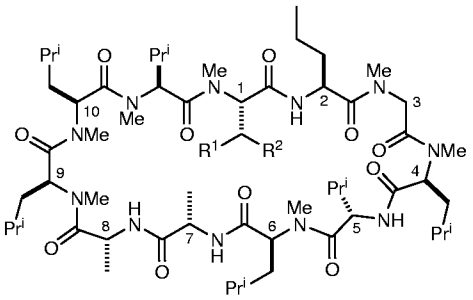
(4) Increased basicity and decreased polarity of the attached carbonyl group.

The only observed N-alkylations of the peptide bond in natural compounds are N-methylation and N-benzylation. The most prominent example of a N-methylated peptide is cyclosporin A (Scheme 1), a member of the group of cyclic peptide immunosuppressants, which was first isolated from *Trichoderma polysporum*.[7] It is a lipophilic cyclic undecapeptide with seven sterically hindered *N*-methyl amino acids including (Me)Leu, (Me)Val, Sar [(Me)Gly], and the unique amino acid, (4*R*)-4-[(*E*)-but-2-enyl]-*N*,4-dimethyl-L-threonine [(Me)Bmt]. The chemical synthesis was first performed by Wenger.[28] The solid-phase synthesis of cyclosporin O (Scheme 1) has been reported[29] using the highly efficient activating reagent bis(trichloromethyl) carbonate.[30] Structural studies in solution,[31,32] in the crystalline state,[31] and bound to its receptor cyclophilin[33,34] have been reported.

Cyclosporin A has various biological activities, but its immunosuppressive effects, combined with relatively low toxicity, and improved pharmacological properties has made it one of the most successful drugs (Sandimmun) to be used after organ transplantations. The biosynthesis of cyclosporin A has been investigated by Kobel et al.[35,36] Cyclosporin A and its homologues are synthesized by a single multifunctional enzyme from their precursor amino acids where *S*-adenosyl-L-methionine serves as the methyl group donor.[37]

*for references see p 265*

**Scheme 1** The Structures of Cyclosporin A and O

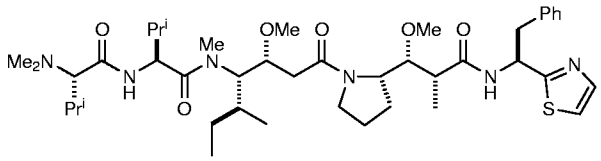


cyclosporin A (CsA) $R^1$ = OH; $R^2$ = $CH(Me)CH_2CH{=}CHMe$
cyclosporin O (CsO) $R^1$ = $R^2$ = Me

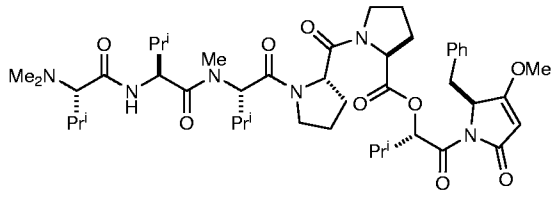
A highly N-methylated cyclic dodecapeptide, omphalotin, has been isolated from the basidiomycete *Omphalotus olearius*.[38,39] This peptide has selective activity against phytopathogenic nematodes such as *Meloidogyne incognita*. Jung and co-workers have synthesized omphalotin A by SPPS in 31% yield using bis(trichloromethyl) carbonate.[40]

Another group of N-alkylated peptides that has received great attention are the dolastatins. They were isolated from the sea hare *Dolabella auricularia* and the majority of them have been identified by Pettit et al.[41] Among them, dolastatins 10 and 15 (Scheme 2) show promising antitumoral activity. They are both highly potent inhibitors of the proliferation of murine and human leukemia cell lines. Dolastatin 10 and a derivative of the slightly less active dolastatin 15 are currently in clinical development as chemotherapeutic agents for cancer therapy.[42]

**Scheme 2** The Structures of Dolastatin 10 and 15



dolastatin 10



dolastatin 15

Biologically active linear peptides, due to their excessive flexibility and their susceptibility to enzymatic degradation, frequently have undesirable pharmacological properties such as metabolic instability, lack of receptor selectivity, and poor bioavailability.[43,44] These properties hampered their *in vivo* studies and clinical application. Due to the profound structural effects mentioned above, N-alkylation of peptide bonds is a powerful tool for structure–activity relationship studies that may lead to analogues with significantly improved pharmacological properties. Indeed, N-alkylated, and in particular N-methylated, analogues of many biologically active peptides have been prepared and the effect of N-alkylation on their

bioactivity, metabolic stability, selectivity, and bioavailability evaluated. It is not within the scope of this section to describe all these analogues, rather, the systematic method for the evaluation of the effect of N-methylation on the activity of a given peptide will be described and a few illustrative examples will be given.

The commercial availability of protected *N*-methyl amino acids [(Me)Xaa] of many proteinogenic amino acids (as well as other *N*-alkyl amino acids), the availability of procedures for the synthesis of protected *N*-alkyl analogues of all the protein amino acids, and the availability of synthetic procedures for site-selective alkylation during SPPS (see Section 10.1.2.1) allows the alkylation of nearly all peptide bonds in a given parent peptide. The synthesis of a series of *N*-alkyl peptide analogues based on the sequence of a given bioactive peptide (linear or cyclic) in which each peptide bond is successively alkylated and evaluation of the biological activity of this series will be called herein: *N*-alkyl-scan (for example Me-scan, Et-scan, etc.)

Applying *N*-alkyl-scan to a given peptide sequence may lead to the following results: (1) The alkylation of some peptide bonds will have no effect on its activity. If the alkylated peptide bond or its neighboring peptide bonds are also the site of proteolytic cleavage, this will result in metabolically stable analogues. If there is more than one peptide bond that is not affected by alkylation, preparation of a di- or even trialkylated analogue will improve metabolic stability and will enhance intestinal bioavailability (*vide infra*). (2) N-Alkylation results in an inactive peptide. These amide bonds are essential for biological activity and may not be alkylated. (3) N-Alkylation imposes selectivity on a nonselective parent peptide or augments the selectivity of a less selective analogue. (4) N-Alkylation converts an agonist into an antagonist.

The concept of *N*-alkyl-scan was first introduced by Sugano et al.[45] who synthesized a series of N-methylated analogues of the eledoisin related peptide H-Lys$^6$-Phe-Ile-Gly-Leu-Met$^{11}$-$NH_2$ in which each peptide bond was successively N-methylated, resulting in a series of five mono-N-methylated analogues. Screening this series on the depressor activity in rabbit blood revealed that the replacement of Ile$^8$ and Met$^{11}$ by their N-methylated analogues removed the depressor activity whereas the replacement of Phe$^7$ and Leu$^{10}$ did not affect the activity. Based on these results, the peptide H-Lys-(Me)Phe-Ile-Gly-(Me)Leu-Met-$NH_2$ was prepared. This peptide retained full depressor activity and displayed protracted resistance to chymotrypsin digestion (full stability after 30 min, as compared to a $t_{1/2}$ of 5 min for the unmethylated parent compound).[46]

The same approach was applied to the C-terminal hexapeptide analogue of substance P, pGlu$^6$-Phe-Phe-Gly-Leu-Met$^{11}$-$NH_2$.[47,48] Me-scan of this peptide[49] demonstrated that replacement of Gly$^9$ by Sar imposes selectivity to the NK-1 receptor which resulted in the discovery of the NK-1 selective analogues septide (pGlu-Phe-Phe-Pro-Leu-Met-$NH_2$)[50,51] and Ac-Arg-septide (Ac-Arg-Phe-Phe-Pro-Leu-Met-$NH_2$).[52] However, replacement of Phe$^8$ by (Me)Phe$^8$ imposed selectivity to the NK-3 receptor. This finding resulted in the discovery of senktide [Suc-Asp-Phe-(Me)Phe-Gly-Leu-Met-$NH_2$],[50,51,53] which is the most active and selective NK-3 agonist found so far ($EC_{50}$ GPI, nM: NK-1 = 35 000, NK-2 = >200 000, NK-3 = 0.5). These selective analogues also showed persistent action when incubated with rat parotid and hypothalamic slice systems.[54,55] The super-selective analogue senktide and its radiolabeled analogues [$^{125}$I-BH]senktide[53] and [$^3$H]senktide[56] were instrumental in the demonstration, characterization, and localization of the third neuronal neurokinin receptor NK-3.[49,53,57] Moreover, because of their high potency, metabolic stability, and high selectivity, Ac-Arg-septide and senktide were used to decipher the bizarre combination of behavioral symptoms produced by the intrathecal injection of the parent peptide substance P and its unmethylated analogues. Using the selective N-methylated analogues, it was shown that substance P elicits different and sometimes conflicting behavioral responses by acting simultaneously on different functional systems via different receptors. This effect arises from the lack of receptor selectivity of substance P (it activates the three neurokinin receptors

*for references see p 265*

NK-1, NK-2, and NK-3). Moreover, being metabolically unstable, substance P is proteolyzed rapidly upon *in vivo* administration, yielding various fragments that are responsible for the bizarre symptoms. The receptor-selective analogues produce dramatically different, and clear cut, behavioral effects when injected intrathecally into rats. Ac-Arg-septide caused profound pain symptoms, whereas senktide caused profound analgesia.[52,58]

Augmentation of the biological activity, as well as the selectivity of a cyclic peptide, is demonstrated by the RGD analogue *c*[-Arg-Gly-Asp-D-Phe-(Me)Val-], discovered by Me-scan of the $\alpha_V\beta_3$ antagonist *c*[-Arg-Gly-Asp-D-Phe-Val-]. The N-methylated analogue is one of the most active (0.5 nM) and selective compounds in inhibiting vitronectin binding to the $\alpha_V\beta_3$ integrin.[59,60]

Me-scan was applied to three agonists of the luteinizing hormone releasing hormone (LHRH, GnRH), pGlu$^1$-His-Trp-Ser-Tyr$^5$-Gly-Leu-Arg-Pro-Gly$^{10}$-$NH_2$: leuprolide (pGlu$^1$-His-Trp-Ser-Tyr$^5$-D-Leu-Leu-Arg-Pro$^9$-NHEt), deslorelin (pGlu$^1$-His-Trp-Ser-Tyr$^5$-D-Trp-Leu-Arg-Pro$^9$-NHEt), and nafarelin [pGlu$^1$-His-Trp-Ser-Tyr$^5$-D-Nal-Leu-Arg-Pro-Gly$^{10}$-$NH_2$]. The (Me)Ser, (Me)Leu, and Sar analogues of nafarelin had 2-, 20-, and 9-fold higher affinity to the LHRH receptor than the parent compound nafarelin. Moreover, for the first time, conversion of LHRH agonists into antagonists was observed as a result of N-methylation. For example, the peptide pGlu$^1$-(Me)His-Trp-Ser-Tyr$^5$-D-Trp-Leu-Arg-Pro$^9$-NHEt {[(Me)His$^2$]deslorelin} was an effective antagonist ($pA_2$ 8.26) of the LHRH receptor as opposed to its parent peptide deslorelin. Me-scan led also to the discovery of metabolically stable analogues either to chymotryptic digest and/or to intestinal degradation. Thus, N-methylation of Tyr$^5$ in all three parent agonists led to protracted intestinal activity.[61,62]

Together with the increase in metabolic stability, N-alkylation also improves the bioavailability of peptides. Cyclosporin is an orally bioavailable drug.[63] Being both N-methylated and cyclic, its oral bioavailability could not be attributed to either of these structural elements or to the enhanced metabolic stability. In order to determine the influence of N-alkylation on the intestinal permeability of linear peptides, detailed studies were performed on an all-D model peptide series (to eliminate the effect of metabolic stability on the rate of permeation). Thus, mono-, di-, and tri-N-methylations of the peptide bonds in the tripeptide Ac-D-Phe-D-Phe-D-Phe-$NH_2$ gradually increased the intestinal permeability.[64–66] This effect was mainly attributed to the reduction in the hydrogen bonding ability rather than to the increase in lipophilicity by N-methylation. An example is the application of Me-scan to the endothelin-A/endothelin-B receptor antagonist Ac-D-Bhg$^{16}$-Leu-Asp-Ile-Ile-Trp$^{21}$ (Bhg = dibenzoberanylglycine), which is an analogue of the C-terminal hexapeptide sequence of endothelin (-His$^{16}$-Leu-Asp-Ile-Ile-Trp$^{21}$-). The Me-scan led to the discovery of Ac-D-Bhg$^{16}$-Leu-Asp-Ile-(Me)Ile-Trp$^{21}$-OH, which retained full receptor affinity to both endothelin receptor subtypes along with enhanced proteolytic stability and cellular permeability.[67] N-Methylation also improves the oral bioavailability of a series of growth hormone stimulating peptides.[68]

## 10.1.1 Synthesis of *N*-Alkyl Amino Acids as Building Blocks for N-Alkylated Peptides

Various N-alkylated derivatives of amino acids are natural products [e.g., H-D-(Me)Tyr-OH (D-surinamine) and H-(Me)Trp-OH (abrin) were found in cabbage tree bark[69]] and many of them are used as enzyme inhibitors, receptor agonists and antagonists, building blocks for heterocyclic scaffolds in combinatorial chemistry, etc. In this section the preparation of *N*-alkyl amino acids in solution for their use in peptide synthesis is described. This implies that the synthetic procedures described in this section will ultimately result in *N*-alkyl amino acids appropriately protected for peptide synthesis.

Since *N*-methyl amino acids are the most abundant *N*-alkyl amino acid in naturally occurring N-alkylated peptides it is apparent that much attention has been devoted to the synthesis of these amino acids.

Innumerable papers have appeared dealing with the synthesis of *N*-alkyl amino acids, in particular protected derivatives for the synthesis of N-alkylated peptides. The development of synthetic methodologies for the preparation of *N*-alkyl amino acids correlates with the development of new methodologies for peptide synthesis. Thus, early methods were limited to amino acids without functionalized side chains and to simple alkyl moieties, predominately methyl. The introduction of SPPS, and particularly Fmoc chemistry, prompted the development of synthetic methods for the preparation of optically pure, orthogonally protected *N*-alkyl amino acids derived from all amino acids. The development of technologies such as backbone cyclization (Vol. E 22b, Sections 6.8.3.2.4 and 6.8.4.2.2),[70] PNA (see Section 12.6),[71] and peptoids[72] (*vide infra*) that depend on protected *N*-alkyl amino acids building blocks, led to fine tuning and optimization of the experimental procedures. As demonstrated in this section, methods are now available for convenient synthesis of almost any orthogonally protected *N*-alkyl amino acid. These methods usually utilize commercially available starting materials, the reactions are performed under mild conditions that do not interfere with sensitive protecting groups or cause racemization, and the products are easily purified in high yields.

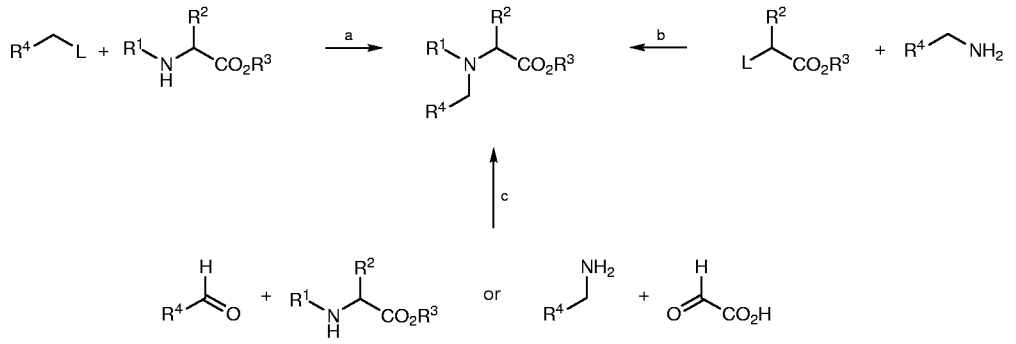
The three general routes for the preparation of *N*-alkyl α-amino acid derivatives are shown in Scheme 3. These are:

(a) Direct alkylation of N-protected amino acids or amino acid esters.

(b) Nucleophilic substitution of carboxylic acids with a leaving group in the α-position with amines.

(c) Reductive alkylation of either amino acids and amino acid esters with aldehydes or amines with glyoxylic acid.

**Scheme 3** General Routes for the Preparation of *N*-Alkyl Amino Acids




$R^1$ = H, $N^{\alpha}$ protecting group (e.g., Z, Boc, Fmoc, Tos, Bzl)
$R^2$ = amino acid side chain
$R^3$ = H, carboxy protecting group
$R^4$ = H, alkyl, functionalized alkyl
L = leaving group

Early attempts showed that during the synthesis of *N*-alkyl amino acid derivatives by routes a or b in homogenous solution the following problems must be solved:

(1) Functional groups on the $R^2$ side chain must be orthogonally protected by suitable protecting groups that are stable toward the alkylation conditions.

(2) The basic reaction conditions can lead to racemization.

(3) The isolation of the products from the reaction mixture is difficult, and frequently it is impossible to remove the nonalkylated compound from the desired compound.

The major side reaction of route c is overalkylation.

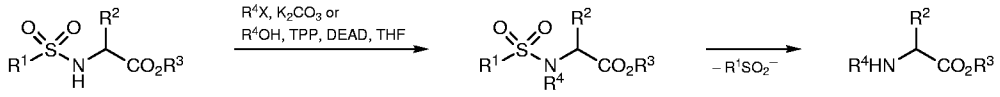
*for references see p 265*

## 10.1.1.1 N-Alkylation of Protected Amino Acids

### 10.1.1.1.1 N-Alkylation of $N^{\alpha}$-Arylsulfonyl Amino Acids

The first synthesis of optically pure N-methylated derivatives of Ala, Leu, Phe, and Tyr was published by Fischer and Lipschitz in 1915[73] using the sulfonamide method. Two main developments have ensured that this method remains useful for the preparation of *N*-alkyl amino acids both in solution and solid phase: (1) the introduction of the Mitsunobu reaction for the alkylation step; and (2) the introduction of replacements for Tos (such as the Fukuyama Nbs) that allow easy removal of the sulfonamide protecting group after the alkylation step. Sulfonamide-protected amino acid derivatives can be alkylated in two different ways. Because of the acidity of the sulfonamide hydrogen it is possible to introduce the N-substituent either by direct alkylation (e.g., alkyl halides) or by the Mitsunobu reaction[74] (Scheme 4).

**Scheme 4** Synthesis of *N*-Alkyl Amino Acids via Alkylation of Sulfonamide-Protected Amino Acid Derivatives




$R^1$ = Ph, Tol, 4-$O_2NC_6H_4$
$R^2$ = amino acid side chain
$R^3$ = H, carboxy protecting group
$R^4$ = alkyl, functionalized alkyl

Fischer and Bergmann[75] adapted the method of Hinsberg[76] for the N-methylation of benzenesulfonamides in the synthesis of Sar, 5-aminopentanoic acid, and racemic Orn. Two years later Fischer and Lipschitz[73] succeeded in the synthesis of optically active, N-methylated derivatives of Ala, Leu, Phe, and Tyr. *N*-Tosyl amino acids were N-methylated with two equivalents of iodomethane at 65–68 °C in a sealed pressure tube. The tosyl group was then deprotected either with sodium in liquid ammonia or with hot hydrochloric acid, but partially racemic products were obtained that were purified by crystallization with optically active bases.[77] Cheung and Benoiton did not observe racemization during deprotection with sodium in liquid ammonia.[78]

Weinreb and co-workers[79] were the first to use the Mitsunobu reaction, which offered an efficient route for the synthesis of *N*-alkyl amino acids. One of the great advantages of the Mitsunobu method is that it uses mild conditions thus allowing its application to SPPS. $N^{\alpha}$-Boc- and $N^{\alpha}$-Z-protected amino acids are not $N^{\alpha}$-alkylated under these mild reaction conditions.

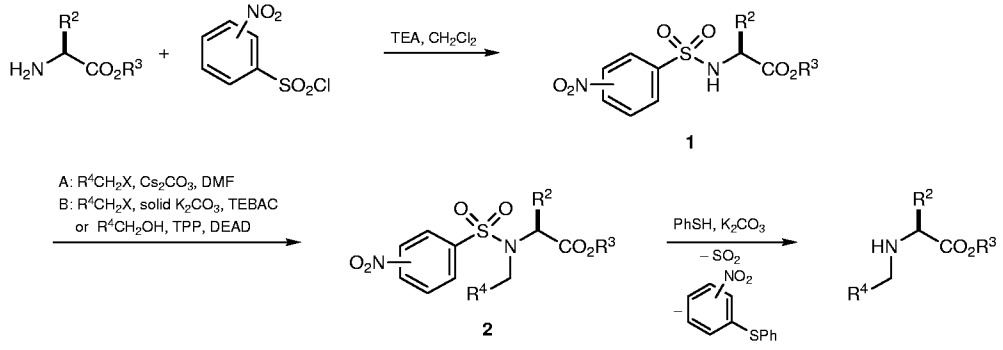
The $pK_a$ values of *N*-methyltrifluoromethanesulfonamide (TfNHMe = 7.5) and *N*-methyltoluenesulfonamide (TosNHMe = 11.7) have been examined and it was found that these sulfonamides are applicable to the Mitsunobu reaction conditions.[80] A modified Mitsunobu reaction has been used for the synthesis of $N^{\alpha}$-alkyl amino acid esters;[81,82] this method is only applicable to amino acid esters and not to the free acids. Thus, *N*-Tos amino acid esters are condensed with MeOH, EtOH, or iPrOH in the presence of TPP and DEAD. The Tos group is deprotected by sodium in liquid ammonia or with sodium amalgam.[83] The deprotection of the Tos group has also been achieved electrochemically under mild conditions and in good yields.[84]

To circumvent the problems of racemization during deprotection of the Tos group the acid labile Pmc group has been used,[85] the Pmc group is commonly used for the protection of the guanidine function in Arg derivatives.[86] The Pmc group is a compromise between the contradictory needs for sulfonamide proton acidity adequate for the Mitsunobu reaction and acid lability sufficient for deprotection. The electron-donating substituents, which are an essential prerequisite for the enhanced acid lability of the Pmc group, cause a reduction of the acidity of the Pmc-based sulfonamide proton. Indeed, the $pK_a$ value of Pmc-substituted

methylamide (12.05)[85] is only slightly above that obtained for Tos-substituted methylamide (11.69)[80,87] and therefore still adequate for the Mitsunobu reaction, where the $pK_a$ value is usually assumed to be <15.[88]

To avoid the relatively harsh deprotection conditions of the Tos group, a novel method for the synthesis of a wide range of secondary amines from primary amines using 2-nitrophenylsulfonyl (oNbs) and 4-nitrophenylsulfonyl (pNbs) amides has been reported.[89] The removal of the Nbs groups is simple via $S_NAr$ substitution using thiols such as benzenethiolate or 2-sulfanylethanol. The method allows synthesis without contamination with primary or tertiary amines. This method was utilized for the synthesis of *N*-alkyl amino acid esters.[90] Reaction of oNbs or pNbs chlorides (Scheme 5) with amino acid esters gives the corresponding sulfonamides **1** in high yields. There are two possibilities for N-alkylation: either direct alkylation with alkyl or aryl halides under mild basic conditions such as $K_2CO_3$ or $Cs_2CO_3$ in dry DMF, or alkylation under neutral Mitsunobu conditions with alcohols. Limitations of the former method appear when less electrophilic alkyl bromides are used. This is due to decreasing nucleophilicity of the anion caused by the extra electron-withdrawing effect of the α-ester. Elevated temperatures (~60 °C), longer reaction times (6 h) and exactly one equivalent of $Cs_2CO_3$ led to the optimized yield of 87% for the N-alkylation of pNbs-Val-OMe with 5-bromopent-1-ene. Removal of the Nbs groups from the intermediate N-alkylated sulfonamides **2** using benzenethiol/potassium carbonate proceeds without difficulty via a Meisenheimer analogue complex. Small differences only between the efficiency of the oNbs and pNbs were observed in either the alkylation or the removal steps. Notable is the fact that all three synthetic steps namely: protection, N-alkylation, and deprotection, occur almost without racemization. This method[90] has been improved and extended.[91] Instead of using $Cs_2CO_3$ to ionize the sulfonamides, solid–liquid phase-transfer catalysis with $K_2CO_3$ as the solid and TEBAC as the catalyst is used.[91] As can be seen from Table 1, this shortens the reaction time, gives pure products, and improves the yields. The phase-transfer catalysis method has also been applied to amino acids with functional side chains (e.g., Ser, Thr, Met) and to a variety of alkyl groups (e.g., Me, Al, Bu, $(CH_2)_4Me$, Bzl, $CH_2C{\equiv}CH$).[91] The phase-transfer catalysis method also found no difference between the oNbs and pNbs groups, and the three-step procedure did not lead to racemization [or epimerization in the case of Thr or D-allothreonine (aThr)]. A solid-phase method for the preparation of *N*-Fmoc-*N*-methyl amino acids for Fmoc SPPS using the Fukuyama/Mitsunobu methods has been developed [Scheme 5, $R^3$ = Trt(2-Cl)-PSty resin and Table 2].[92] Thus, preloaded (amino acid)-Trt(2-Cl)-PSty resins were reacted with oNbs-Cl and DIPEA in THF (or THF/$CH_2Cl_2$) to give the oNbs-Xaa-Trt(2-Cl)-PSty resin. These were N-methylated by the Mitsunobu reaction using MeOH in the presence of TPP. The oNbs group was removed and Fmoc group was introduced. Finally the Fmoc-(Me)Xaa-OH were removed from the polymeric support by 0.5% TFA in $CH_2Cl_2$ (5 min).

**Scheme 5** N-Alkylation of *N*-Nbs Amino Acid Esters[90,91]




*for references see p 265*

**Table 1** N-Alkylation of *N*-Nbs Amino Acid Esters[90,91]

| Amino Acid[a] | $R^2$ | $R^3$ | $R^4$ | Method | Temp, Time | Yield[b] (%) | Ref |
|---|---|---|---|---|---|---|---|
| Val | iPr | Me | $CH{=}CH_2$ | A | rt, 1.5 h | 80 | [90] |
| D-Val | iPr | Me | $CH{=}CH_2$ | A[c] | rt, 1.5 h | 63 | [90] |
| Val | iPr | Me | Ph | A[c] | rt, 1.5 h | 76 | [90] |
| Val | iPr | Me | Pr | A | 60 °C, 6 h | 77 | [90] |
| Val | iPr | Me | Bu | A | 60 °C, 6 h | 72 | [90] |
| Val | iPr | Me | Bu | A | 60 °C, 6 h | 87 | [90] |
| Val | iPr | Me | $CH{=}CH_2$ | B | rt, 2 h | 91 | [91] |
| Val | iPr | Me | Pr | B | rt, 12 h | 89 | [91] |
| Val | iPr | Me | Ph | B | rt, 1 h | 96 | [91] |
| Val | iPr | Me | Bu | B | rt, 8 h | 96 | [91] |
| Val | iPr | Me | Me | B | rt, 1.5 h | 87 | [91] |
| Ser | $CH_2OH$ | Me | $CH{=}CH_2$ | B[d] | rt, 7 h | 76 | [91] |
| Met | $(CH_2)_2SMe$ | Me | $CH{=}CH_2$ | B | rt, 1 h | 94 | [91] |
| Met | $(CH_2)_2SMe$ | Me | Ph | B | rt, 0.25 h | 91 | [91] |
| Thr | CH(Me)OH | Me | $CH{=}CH_2$ | B[d] | rt, 6 h | 84 | [91] |
| D-aThr | CH(Me)OH | Me | $CH{=}CH_2$ | B[d] | rt, 2 h | 84 | [91] |
| Gly | iPr | Me | Pr | B | rt, 1.5 h | 80 | [91] |
| Phe | Bzl | Me | $CH{=}CH_2$ | B | rt, 0.5 h | 88 | [91] |
| Phe | Bzl | Me | Me | B | rt, 0.15 h | 86 | [91] |
| Phg | Bzl | Me | Me | B | rt, 15 h | 86 | [91] |
| Val | iPr | Me | $CH{=}CH_2$ | B | rt, 2 h | 91 | [91] |
| Val | iPr | Me | iPr | B | rt, 6 h | 91 | [91] |

[a] The oNbs derivative was used in each case.
[b] Yields refer to the alkylation step from compounds **1** to **2** in Scheme 5.
[c] The alkylation reaction was performed from the oNbs amino acid ester without its isolation.
[d] Using anhyd DMF as solvent; purified by flash chromatography.

**Table 2** *N*-Fmoc-*N*-methyl Amino Acids Prepared by the Solid-Phase Fukayama/Mitsunobu Reaction[92]

| Amino Acid[a,b] | $t_R$ (min)[c] | Purity[d] | Yield (%)[d] | Ref |
|---|---|---|---|---|
| Ala | 6.65 | 98 | 86 | [92] |
| Asp(tBu) | 9.38 | 90 | 96 | [92] |
| Lys(Boc) | 9.45 | 90 | 100 | [92] |
| Phe | 9.34 | 90 | 80 | [92] |
| Ser(tBu) | 9.39 | 98 | 100 | [92] |
| Trp | 8.60 | 88 | 100 | [92] |
| Trp(Boc) | 13.11 | 96 | 98 | [92] |
| Val | 8.58 | 96 | 95 | [92] |

[a] See experimental procedures for details of conditions.
[b] Scheme 5.
[c] Partasil ODS-3 RAC II column (100 × 4.6 mm, 5 µm). Linear gradient of 40–80% MeCN in $H_2O$/0.1% TFA over 15 min at 1 mL·$min^{-1}$, 278 nm
[d] Purity and crude yield after cleavage.
[e] ESI in 82% MeCN/18% aq 1.3 mM TFA/1.3 mM ammonium formate.

**Nbs-Xaa$^1$-OMe 1 ($R^3$ = Me); General Procedure:**[91]

In a well-dried, round-bottomed flask HCl·H-Xaa$^1$-OMe (10 mmol) was dissolved in anhyd $CH_2Cl_2$ (80 mL). This soln was cooled to 0 °C, anhyd TEA (2.3 g, 22.7 mmol) was added under stirring in 5 min. Nbs-Cl (10 mmol) was added in portions over 10 min and the stirring was continued at 25 °C until no starting material was detectable (TLC). The mixture was washed with $H_2O$ (3 × 20 mL), and the organic phase dried ($Na_2SO_4$) and taken to dryness in vacuo. Purification was by flash chromatography (silica gel, 230–400 mesh, EtOAc/petroleum ether 1:3); yield: 70–90% (depends on HCl·H-Xaa$^1$-OMe).

**oNbs-(Al)-Val-OMe by Direct Alkylation (Method A); Typical Procedure:**[90]
$Cs_2CO_3$ (2.0 g, 6.1 mmol) was added to a soln of oNbs-Val-OMe (1.3 g, 4.1 mmol) in DMF (40 mL). The mixture was stirred at rt for 30 min before allyl bromide (0.7 mL, 8.1 mmol) was added. The mixture was stirred for an additional 90 min (in cases where $R^4$ = Bu the mixture was heated at 60 °C for 6 h) and then taken to dryness in vacuo. Purification by chromatography (silica gel, EtOAc/petroleum ether 2:3) gave oNbs-(Al)-Val-OMe as a pale yellow liquid; yield: 1.2 g (80%); $[\alpha]_D^{29}$ 62.4 (*c* 3.6, $CHCl_3$).

**Alkylation of Sulfonamides under Phase-Transfer Conditions (Method B); General Procedure:**[91]
In a dried flask connected to a $CaCl_2$ drying tube, anhyd $K_2CO_3$ (2.07 g, 15 mmol) was added to a MeCN (or DMF) soln (50 mL) of sulfonamide (10 mmol), alkyl halide (10.5–20 mmol), and TEBAC (0.23 g, 1 mmol). The heterogeneous mixture was stirred at 20–80 °C until no starting material was detectable (TLC analysis, EtOAc/petroleum ether 1:1) and then cooled to rt, diluted with $H_2O$ (50 mL), and extracted with $Et_2O$ or $CH_2Cl_2$ (3 × 20 mL). The organic layer was washed with $H_2O$ (2 × 20 mL), dried ($Na_2SO_4$), and taken to dryness in vacuo, to give pure *N*-alkylsulfonamides.

**H-(Al)Val-OMe; Typical Procedure for the Cleavage of the Nbs Protecting Group:**[90]
$K_2CO_3$ (0.36 g, 2.6 mmol) was added to a soln of oNbs-(Al)Val-OMe (0.30 g, 0.8 mmol) and PhSH (0.1 mL, 0.97 mmol) in MeCN (40 mL). The mixture was stirred at rt overnight. The resulting soln was taken to dryness in vacuo and the residue taken up in $Et_2O$. Then, 1 M HCl (20 mL) was added and the mixture stirred for 10 min. The organic layer was removed and washed with $H_2O$, the combined aqueous layers were washed with $Et_2O$ followed by solid $K_2CO_3$. The product was extracted from the basic layer with $Et_2O$. The $Et_2O$ layer was washed with brine, dried, and then taken to dryness in vacuo to give the crude product which was distilled in vacuo (15 Torr) using a Kugelrohr oven (150 °C) over glass wool to give H-(Al)Val-OMe as a colorless liquid; yield: 90 mg (65%); $[\alpha]_D^{23}$ −21.8 (*c* 1.6, $CHCl_3$).

**Synthesis of Fmoc-(Me)Xaa$^1$-OH by On-Resin Methylation Using the Fukuyama/Mitsunobu Reactions; General Procedure:**[92]
*Sulfonation:* H-Trp-Trt(2-Cl)-resin was swollen in THF (2 mL/100 mg resin) and DIPEA (6 equiv). 1 M oNbs-Cl in $CH_2Cl_2$ (4 equiv) was introduced slowly with agitation. The mixture was agitated at rt for 3 h until the Kaiser ninhydrin test was negative. Double coupling was carried out if necessary. With other amino acids, DMAP (1 equiv) in THF/$CH_2Cl_2$, (2:1) was most effective. The resin was drained and washed [THF (2 ×), MeOH (2 ×), $CH_2Cl_2$ (2 ×), THF (2 ×)].

*N-Methylation:* The sulfonated-Xaa$^1$-resin was mixed with 2 M TPP in THF (5 equiv) and MeOH (10 equiv) in THF (1.5 mL/100 mg resin). 1 M DEAD in THF (5 equiv) was introduced slowly and the resulting mixture agitated for 1 h. The resin was washed [THF (4 ×), $CH_2Cl_2$ (3 ×)].

*Removal of oNbs:* The resin-bound methyl sulfonamide was swollen in DMF (1 mL · 100 mg$^{-1}$ resin) and treated with 1 M PhSH in DMF (1 mL/100 mg resin) for 1 h and this procedure was repeated once more. The resin was drained and washed [DMF (4 ×), MeOH (4 ×), $CH_2Cl_2$ (4 ×)].

*Introduction of Fmoc:* The resin was swollen in $CH_2Cl_2$, and DIPEA (6 equiv) and Fmoc-Cl (4 equiv) were added. The mixture was agitated at rt for 4 h and drained and washed [$CH_2Cl_2$ (2 ×), MeOH (2 ×), $CH_2Cl_2$ (2 ×), $Et_2O$ (2 ×)] and dried under $N_2$.

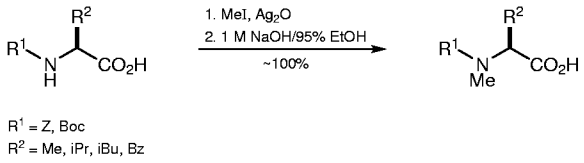
*Cleavage from the resin:* The resin was treated with 0.5% TFA in $CH_2Cl_2$, for 5 min. This was repeated twice. The resin was removed by filtration and washed with 0.5% TFA in $CH_2Cl_2$,. The combined filtrates and washings were taken to dryness to yield the product.

### 10.1.1.1.2 N-Alkylation of Benzyloxycarbonyl- and *tert*-Butoxycarbonyl-Protected Amino Acids

N-Methylation of Z- and Boc-protected amino acids was first described by Olsen[94] but was limited to hydrophobic aliphatic and aromatic amino acids. Thus, readily available Z and Boc amino acids are treated with a four- to eightfold molar excess of freshly distilled iodomethane and a three- to fourfold molar excess of silver oxide in DMF (Scheme 6). This reaction effects methylation of the amide nitrogen and affords the corresponding *N*-methyl amino acid derivatives in one step and in nearly quantitative yields. Unprotected carboxy groups also undergo O-methylation to give the corresponding methyl ester. The free carboxylic function is obtained after mild saponification of the methyl ester group with 1 M NaOH in 95% ethanol.

*for references see p 265*

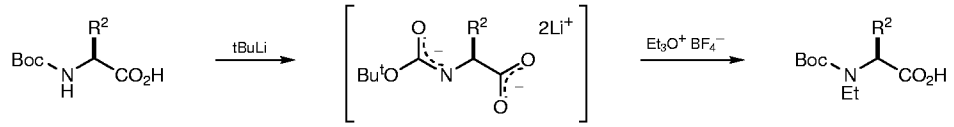
**Scheme 6** N-Methylation of Z- and Boc-Protected Amino Acids[94]




The disadvantage of this method is that saponification of the resulting ester group can result in racemization. A method for the synthesis of *N*-methyl amino acids without esterification has been presented by Benoiton and co-workers,[78,95] which uses sodium hydride as the base and iodomethane in THF at room temperature. This method is also applicable to the synthesis of side-chain-functionalized amino acid derivatives when the side chain is protected by a *tert*-butyl-type group.

For the synthesis of *N*-alkyl amino acid derivatives with longer alkyl groups, this method is not efficient because the highly basic reaction conditions can result in β-elimination of the alkylating agent. A method has been described for the ethylation of several Boc-protected amino acids.[96] The dianion **3** (Scheme 7) was initially generated using *tert*-butyllithium as a strong nonnucleophilic base. Treatment of **3** with the alkylating agent triethyloxonium tetrafluoroborate gives the desired ethylated product.

**Scheme 7** N-Ethylation of *N*-Boc Amino Acid Dianions[96]

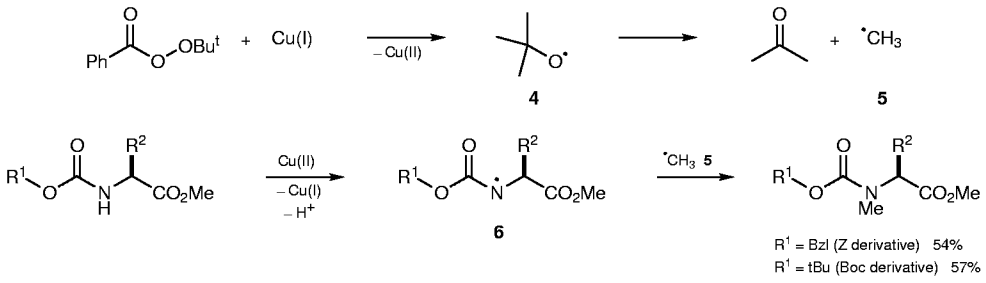



**3**

An efficient route for the synthesis of a variety of *N*-alkyl amino acids is provided by a modified version of the Benoiton method, which was used by for the synthesis of *N*-alkylglycines.[97] This method includes the use of 0.2 equivalents of TBAI as catalyst to avoid the β-elimination reaction of the alkylating agent.

An alternative route for the synthesis of *N*-methyl amino acids without racemization is shown in Scheme 8.[98] This method includes the use of TBPB in the presence of copper(I) octanoate. The proposed mechanism of this free radical reaction is given in Scheme 8. Electron transfer from copper(I) to TBPB affords the copper(II), benzoate, and tBuO radical **4**, which undergoes β-scission to acetone and methyl radical **5**. In turn, electron transfer from the urethane to the copper(II) ion, followed by proton transfer, affords the corresponding urethane radical **6**, which reacts with the methyl radical **5** to give the desired product in overall yields of 54% (Z derivative) or 57% (Boc derivative), respectively.

**Scheme 8** Free-Radical N-Methylation of Urethane-Protected Amino Acids[98]

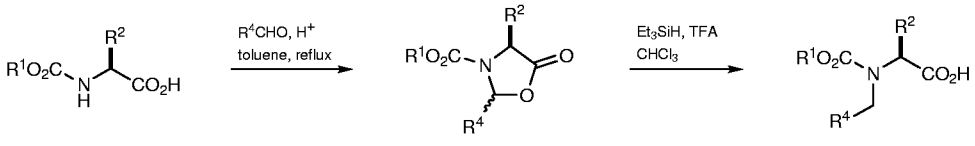


**Synthesis of *N*-Alkyl-*N*-Z- or *N*-Alkyl-*N*-Boc Amino Acids; General Procedure:**[97]
The protected amino acid ester (10 mmol), TBAI (2 mmol), and the alkylating agent (80 mmol) were dissolved in dry distilled THF (30 mL) and NaH dispersion (30 mmol) was added cautiously with gentle stirring to the soln at 0 °C. The suspension was stirred at rt for 3–4 d. The reaction was quenched with aq AcOH and taken to dryness in vacuo. The residue was diluted in EtOAc (50 mL), and the organic layer was washed twice with 5% $Na_2S_2O_3$, 5% $NaHCO_3$, aq HCl (to pH 3), and brine. The organic phase was dried ($Na_2SO_4$), taken to dryness in vacuo and the residue was purified by chromatography (silica gel, hexane/EtOAc) to give a colorless oil; yield: 80–90%.

### 10.1.1.2 N-Alkylation of Urethane-Protected Amino Acids with Aldehydes

Ben-Ishai[99] was the first to react Z-protected amino acids (Scheme 9, $R^1$ = Bzl; $R^2$ = H, Me, Bzl, iPr, iBu) with paraformaldehyde in refluxing benzene under acid catalysis (TosOH) and obtained their corresponding optically active oxazolidin-5-ones **7** ($R^4$ = H). It was also shown that catalytic hydrogenation of Z($CH_2OH$)N-Gly-NHBzl obtained by the reaction of 3-Z-oxazolidin-5-one **7** ($R^1$ = Bzl, $R^2$ = H, $R^4$ = H ) with benzylamine, yielded the *N*-methyl analogue of Gly, namely Sar-NHBzl. The oxazolidinones of Z-Asp-OH and Z-Glu-OH [Scheme 9, $R^1$ = Bzl; $R^2$ = $CH_2CO_2H$, $(CH_2)_2CO_2H$; $R^4$ = H] have also been prepared.[100] However, attempts to obtain H-(Me)Asp-OH and H-(Me)Glu-OH from their corresponding oxazolidinones by catalytic hydrogenation caused complete excision of the formyl methylene. An important development in attempts to obtain protected *N*-methyl amino acids via the oxazolidin-5-one method was the application of the $Et_3SiH$/TFA method[101] (originally introduced by Weinreb and co-workers[102]) to reduce 3-Fmoc-oxazolidin-5-ones of various amino acids to their corresponding *N*-alkyl-*N*-Fmoc derivatives in high yield.

**Scheme 9** Synthesis of Urethane-Protected *N*-Alkyl Amino Acids via Oxazolidin-5-ones[99–101]




**7**

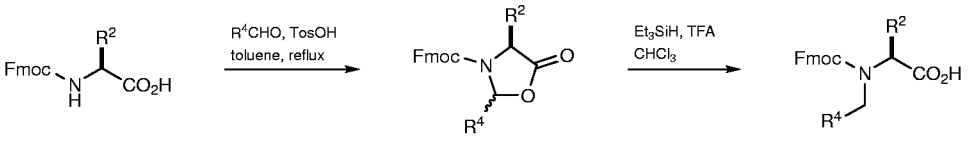
The acid-catalyzed preparation of oxazolidin-5-ones from protected amino acids and paraformaldehyde or trioxane[103] is operationally simple, the reaction conditions are mild and the oxazolidin-5-ones are obtained in high yield and high optical purity. The reductive cleavage of the oxazolidin-5-one by $Et_3SiH$/TFA also proceeds in high to excellent yield and high optical purity and without side reactions. Obviously, this general method is applicable only to amino acids that contain protecting groups stable to TFA (e.g., Z, and other Bzl type, Fmoc, Phth) since the reductive ring opening of the oxazolidin-5-one takes place under acidolytic conditions that removes acid sensitive side-chain protection (e.g., Boc, tBu, Trt). These must then be reintroduced in order to use the fully protected *N*-methyl amino acids in peptide synthesis. Attempts to use unprotected functionalized side chains led to side reactions that lowered the yields.[104] Attempts to cleave 3-Boc-oxazolidin-5-ones by non-acidolytic catalytic hydrogenation[105] led to excision of the formyl methylene[100] or yielded mixtures of the Boc amino acids and their corresponding *N*-methyl analogues.[104] Procedures have been described for the preparation of the *N*-Z-*N*-methyl derivatives of all the protein amino acids and some *N*-alkyl-*N*-Fmoc derivatives via the oxazolidin-5-one method.

*for references see p 265*

### 10.1.1.2.1 N-Alkylation of Fmoc-Protected α-Amino Acids

The $Et_3SiH$/TFA method introduced by Freidinger et al.[101] offers a useful route for the N-alkylation of Fmoc amino acids that do not have acid labile protecting groups on their functionalized side chains. This method consists of the two-step, acid-catalyzed reductive amination procedure described in Section 10.1.1.2. Thus, commercially available Fmoc amino acids **8** (Table 3, entries 1–8) are condensed with an aldehyde in the presence of a catalytic amount of TosOH in refluxing toluene to afford the corresponding oxazolidin-5-ones **9**. These oxazolidin-5-ones are stable compounds and generally are available in good to excellent yields. In the next step the oxazolidin-5-ones **9** are treated with a mixture of $Et_3SiH$/TFA and $CHCl_3$, which results in reductive ring opening to provide the *N*-alkyl-*N*-Fmoc amino acids **10** in good overall yields. Although this method is applicable to a variety of aldehydes, best results of oxazolidin-5-one formation are observed with paraformaldehyde (Table 3, entries 1, 5–7).

**Table 3** Preparation of *N*-Alkyl-*N*-Fmoc Amino Acids by the Oxazolidinone Method[101]



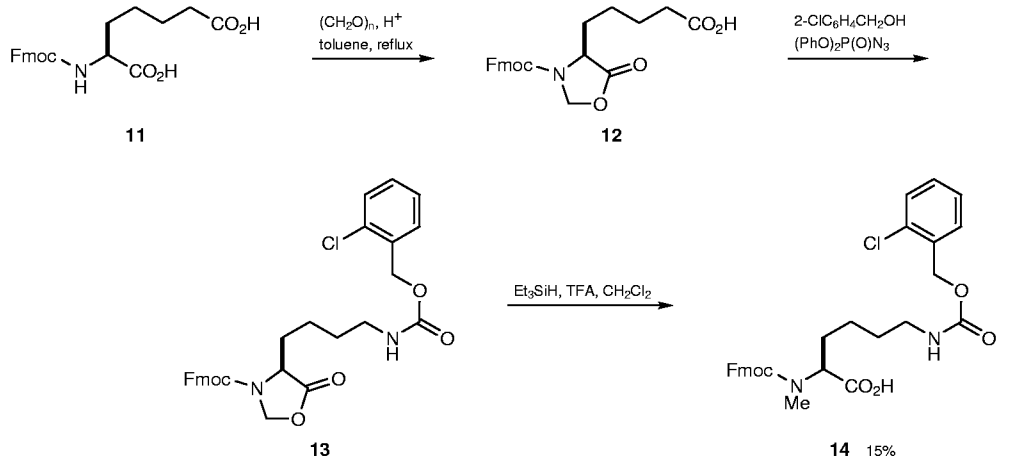

**8** **9** **10**

| Entry | $R^2$ | $R^4$ | Yield (%) | | Ref |
|---|---|---|---|---|---|
| | | | **9** | **10** | |
| 1 | Me | H | 96 | 98 | [101] |
| 2 | Me | Cy | 77 | 91 | [101] |
| 3 | Me | Me | 79 | 74 | [101] |
| 4 | Me | Bzl or Ph | 30 | 95 | [101] |
| 5 | iPr | H | 96 | 100 | [101] |
| 6 | $CH_2OBzl$ | H | 98 | 96 | [101] |
| 7 | $(CH_2)_4NPhth$ | H | 73 | 70 | [101] |
| 8 | $(CH_2)_2SMe$ | H | 88 | 22 | [101] |

This method[101] includes the isolation of the oxazolidinones **9** before reductive ring opening. A one-pot method without isolation of the oxazolidin-5-ones has also been developed[106] and mainly used to prepare a variety of *N*-Fmoc-*N*-methyl ω-carboxylic acids.

Side-chain protected Fmoc-(Me)Lys-OH can be prepared in good yield via the $N^{\alpha}$-Fmoc-$N^{\varepsilon}$-phthaloyl derivative[101] (Table 3, entry 7). This compound is adaptable to peptide synthesis since Fmoc can be removed selectively with $iPr_2NH$/DMF (1:1). In contrast, piperidine, more commonly used for removal of Fmoc, also attacks phthalimide. The method of Freidinger et al.[101] cannot be adapted to the $N^{\alpha}$-Fmoc-$N^{\varepsilon}$-Z derivative; condensation with formaldehyde gives only a mixture of products. Therefore, masking of both hydrogens on the $N^{\varepsilon}$-amino group is required. A suitably protected Fmoc-(Me)Lys-OH was synthesized by the route outlined in Scheme 10. Fmoc-Apm-OH (**11**) is converted into oxazolidin-5-one **12**, thus simultaneously $N^{\alpha}$-alkylation and protecting of the α-carboxy group occurs. The side-chain carboxy group is then subjected to the modified Curtius rearrangement using phosphorazidic acid diphenyl ester in the presence of 2-chlorobenzyl alcohol to provide the fully protected oxazolidin-5-one **13**. Finally, the oxazolidin-5-one is cleaved under the usual conditions to give Fmoc-(Me)Lys[Z(2-Cl)]-OH (**14**) in 15% overall yield from **11**. This route is applicable to the preparation of a variety of $N^{\alpha}$-alkyl α,ω-diamino acid derivatives.

**Scheme 10** Synthesis of Fmoc-(Me)Lys[Z(2-Cl)]-OH by the Oxazolidine-5-one Method[101]




**(4*S*)-3-(9-Fluorenylmethoxycarbonyl)-4-methyloxazolidin-5-one (9, $R^2$ = Me; $R^4$ = H); Typical Procedure:**[101]

Fmoc-Ala-OH (**8**, $R^2$ = Me; $R^4$ = H; 1.56 g, 5.0 mmol) was suspended in toluene (100 mL), and paraformaldehyde (1 g) and TosOH (100 mg) were added. The mixture was refluxed for 30 min with azeotropic $H_2O$ removal. The soln was cooled, washed with aq 1 M $NaHCO_3$ (2 × 25 mL), dried ($Na_2SO_4$), and taken to dryness in vacuo; yield: 1.56 g (96%); mp 142–144 °C; $[\alpha]_D^{25}$ +66.7 (*c* 1.0, $CH_2Cl_2$); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1790, 1705 $cm^{-1}$.

**Fmoc-(Me)Ala-OH (10, $R^2$ = Me; $R^4$ = H); Typical Procedure:**[101]

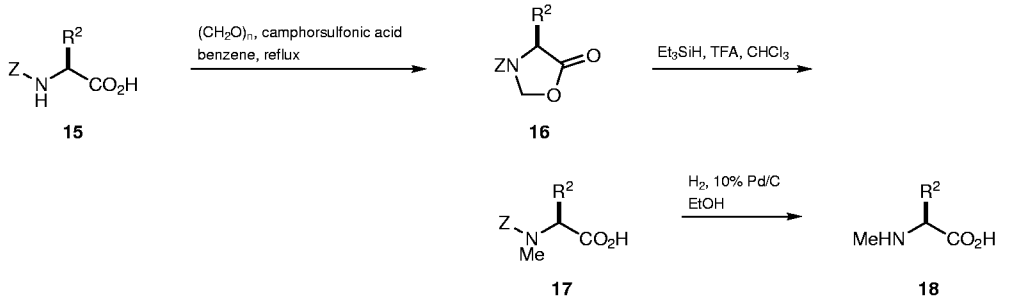
The oxazolidin-5-one from Fmoc-L-Ala-OH (**9**, $R^2$ = Me; $R^4$ = H; 0.97 g, 3.0 mmol) was dissolved in $CHCl_3$ (15 mL) and TFA (15 mL) and $Et_3SiH$ (1.43 mL, 1.04 g, 9.0 mmol) were added. The soln was stirred at rt for 22 h and taken to dryness in vacuo to give an oil. The oil was dissolved in $CH_2Cl_2$ and taken to dryness three times to remove residual $Et_3SiH$. The resultant oil crystallized on standing. The crystals were dissolved in $Et_2O$ and taken to dryness to a crystalline solid which was washed with 5% $Et_2O$/hexane and dried; yield: 0.96 g (98%); mp 154.5–155.5 °C; $[\alpha]_D^{25}$ −21.4 (*c* 1.0, $CH_2Cl_2$); $^1H$ NMR ($CDCl_3$) δ: 1.43 (d, 3H, *J* = 7.5 Hz), 2.92 (s, 3H), 4.2–5.0 (m, 4H), 7.2–7.8 (m, 8H), 9.78 (br, s); MS (70 eV) *m/z*: 325 [$M^+$].

### 10.1.1.2.2 N-Alkylation of Benzyloxycarbonyl-Protected α-Amino Acids

The general method for the preparation of *N*-methyl amino acids **18** by the oxazolidin-5-one method (Scheme 9, $R^4$ = H), which gives high yields of the corresponding *N*-Z-*N*-methyl amino acids **17** with nonfunctionalized side chains has been applied systematically[104,107] to Z amino acids having protected and unprotected functionalized side chains (Scheme 9, $R^1$ = Bzl; $R^2$ = protein amino acid side chains; $R^4$ = H). As can be seen from Table 4 the yields of the oxazolidin-5-ones **16** ranged from poor (Table 4, entry 10) to very good (e.g., entries 5, 11, 14). Unprotected side chains give a variety of side products shown in Scheme 11. Therefore, to obtain high yield, functionalized side chains must be protected and the protecting groups that come off in the acidolytic ring opening must be reintroduced.

*for references see p 265*

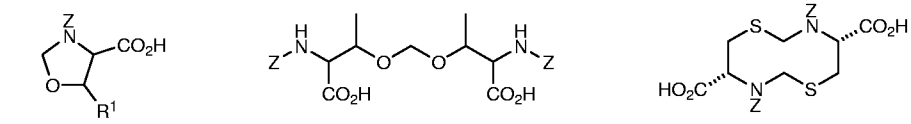
**Table 4** *N*-Z-*N*-Methyl Amino Acids and *N*-Methyl Amino Acids Prepared by the Oxazolidin-5-one Method[103,104,107]




| Entry | Z Amino Acid | $R^2$ | Config | Yield (%) | | | Ref |
|---|---|---|---|---|---|---|---|
| | | | | **16** | **17** | **18** | |
| 1 | Ser | $CH_2OH$ | L | 15 | 79 | 64 | [104] |
| 2 | Thr | CH(OH)Me | L | 21 | 35 | 62 | [104] |
| 3 | Tyr | 4-$HOC_6H_4CH_2$ | L | 37 | 60 | 47 | [104] |
| 4 | Gln(Z) | $(CH_2)_2CONHZ$ | L | 68 | 62[a] | 56 | [104] |
| 5 | Glu | $(CH_2)_2CO_2H$ | DL | 86 | 63 | 63 | [104] |
| 6 | Glu(OBzl) | $(CH_2)_2CO_2Bzl$ | L | 60 | 65 | 63 | [104] |
| 7 | Glu(OtBu) | $(CH_2)_2CO_2tBu$ | L | 82 | | | [103] |
| 8 | Asp(Bzl) | $CH_2CONHBzl$ | D | 58 | 65[b] | 65 | [107] |
| 9 | Phg | Ph | L | 70 | 64 | 72 | [107] |
| 10 | Cys | $CH_2SH$ | DL | 3 | | | [107] |
| 11 | Met | $(CH_2)_2SMe$ | L | 91 | | | [107] |
| 12 | Lys(Phth) | $(CH_2)_4NPhth$ | L | 81 | 79 | 67 | [107] |
| 13 | Lys(Z) | $(CH_2)_4NHZ$ | L | 60 | | | [107] |
| 14 | Trp(For) | 1-formyl-1*H*-indol-3-ylmethyl | L | 86 | 22 | | [107] |
| 15 | His(Dnp) | 1-(dinitrophenyl)imidazol-5-ylmethyl | L | 66 | 81 | | [107] |
| 16 | Asn(Trt) | $CH_2CONHTrt$ | L | 83 | 75[c] | | [107] |

[a] Product **17** is Z-(Me)Gln-OH and **18** is H-(Me)Gln-OH.
[b] Product **17** is Z-(Me)Asp-OH and **18** is H-(Me)Asp-OH.
[c] Product is Z-(Me)Asn-OH.

**Scheme 11** Side Products Obtained by the Application of the Oxazolidin-5-one Method to Z-Ser-OH, Z-Thr-OH, and Z-Cys-OH[104]




from Z-Ser-OH $R^1$ = H 55%
from Z-Thr-OH $R^1$ = Me 50%

from Z-Thr-OH

from Z-Cys-OH 65%

**3-(Benzyloxycarbonyl)oxazolidin-5-ones 16; General Procedure:**[104]

A mixture of the Z amino acid **15** (7.5 mmol), camphorsulfonic acid (88 mg, 0.38 mmol), and benzene (60 mL), was placed in a 250-mL round-bottomed flask and the mixture brought to reflux. To the refluxing soln was added paraformaldehyde (1.13 g, 38 mmol) in five equal portions, allowing for the soln to become clear before each subsequent addition. After the last addition the mixture was heated at reflux until the soln was clear. The benzene soln was taken to dryness and the residue was dissolved in EtOAc (100 mL). The organic layer was washed with 5% aq $NaHCO_3$, dried ($Mg_2SO_4$), filtered, and taken to dryness in vacuo. Residual solids were recrystallized ($CH_2Cl_2$/hexane) and oils were purified by chromatography (silica gel, 30–50% EtOAc/hexane).

***N*-Benzyloxycarbonyl-*N*-methyl Amino Acids 17; General Procedure:**[104]

The oxazolidin-5-one **16** (1 mmol) was dissolved in $CHCl_3$ (5 mL) in a round-bottomed flask followed by addition of TFA (5 mL) and $Et_3SiH$ (0.48 mL, 3.0 mmol) and the mixture was left to stir. Oxazolidinones

without functionalized side chains were stirred for 24 h, those with carboxylic acids in their side chains for 5 d, and those with alcohols in their side chains for 3 d. After stirring the solns were taken to dryness to give residues, which were dissolved in EtOH and taken to dryness. This was repeated three times. The residue was taken up in a mixture of $Et_2O$ (15 mL) and 5% $NaHCO_3$ (30 mL). The ethereal soln was extracted with 5% $NaHCO_3$ (2 × 10 mL). The combined aqueous layers were washed with $Et_2O$ (2 × 10 mL) and the aqueous layer was then acidified with 5 M HCl to pH 2 and extracted with EtOAc (3 × 15 mL). The combined EtOAc layers were dried ($MgSO_4$), filtered, and taken to dryness to give an oil. The oil was further purified by chromatography (silica gel, solvent systems ranged from $CHCl_3$/MeOH/AcOH 90:9.9:0.1 to $CHCl_3$/MeOH 95:5).

***N*-Methyl Amino Acids 18; General Procedure:**[104]
To a mixture of the *N*-benzyloxycarbonyl-*N*-methyl amino acid **17** (1 mmol) and EtOH (30 mL) in a 100-mL round-bottomed flask was added 10% Pd/C catalyst (20 mg). The mixture was stirred at rt in an atmosphere of $H_2$ until the required amount of $H_2$ had been absorbed. The mixture was then taken to dryness. The residue was taken up in hot distilled $H_2O$ (15 mL) and filtered through Celite. The filter cake was washed with hot $H_2O$ (2 × 10 ml) and the combined filtrates were taken to dryness in vacuo to afford a white solid. The solid was then dissolved in the minimum of hot distilled water and crystallized by addition of EtOH to afford the pure *N*-methyl amino acid.

### 10.1.1.3 Reductive Alkylation of Amino Acids and Their Derivatives

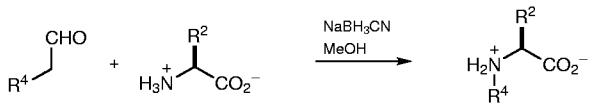
Reductive alkylation (also termed reductive amination) is one of the most useful and widely used reactions in organic chemistry for the construction of C–N bonds. For a general review on reductive amination see ref[108] and references therein. The very mild conditions of reductive alkylation (which do not employ acidic or basic media), its ease of operation (especially the use of sodium cyanoborohydride[109] and sodium triacetoxyborohydride[93]) and its versatility seem ideal for the synthesis of *N*-alkyl amino acids, particularly those with protected functionalized side chains. Historically, this method was neglected for many years. One reason for this neglect is side reactions such as over alkylation (see ref[108] and references therein). Another major reason is that while *N*-methyl amino acids are the most widely used of the *N*-alkyl amino acids, formaldehyde reacts with amino acids in the presence of borohydrides to give the *N*,*N*-dimethyl amino acid derivatives that could not be used in peptide synthesis; they could be used only as the N-terminal amino acid. New developments in the methodology of reductive alkylation have prompted its application to the synthesis of *N*-alkyl amino acids, both in solution and in SPPS, and it has thus become the most widely used method, in particular for the preparation of N-alkyl amino acids other than methyl and side-chain protected *N*-alkyl amino acids.

#### 10.1.1.3.1 Direct N-Monoalkylation of Zwitterionic Amino Acids

On reaction of concentrated mixtures of aldehydes and chiral zwitterionic amino acids in methanol with $NaBH_3CN$, the resulting fully unprotected mono-*N*-methyl amino acids **19** precipitate from the reaction mixture as optically and chemically pure zwitterions (Scheme 12).[110] The product **19** was collected by filtration and carefully washed with cold MeOH to remove unreacted amino acids. In some cases additional recrystallization from hot ethanol was needed to remove small amounts of contamination by dialkylated amino acids. Additional amounts of mono-*N*-methyl amino acids **19** could be obtained from the filtrate after chromatography. This method gives high yields of the monoalkylated product (Table 5), even of amino acids with functionalized side chains such as Glu and Ser. However, this method remained unused for many years since the orthogonal protection of such trifunctional amino acids is cumbersome. This simple method has been applied successfully to side-chain-protected zwitterionic amino acids.[111] The side-chain-protected monoalkylated amino acid could then be easily $N^{\alpha}$-protected (see Section 10.1.1).

*for references see p 265*

**Scheme 12** Reductive Alkylation of Zwitterionic Amino Acids[110]



19

**Table 5** *N*-Alkyl Amino Acids Prepared by the Reductive Alkylation of Zwitterionic Amino Acids[110]

| Amino Acid | Aldehyde | $R^2$ | $R^4$ | Yield (%) of **19** | $[\alpha]_D$ (*c* 1, 2 M HCl) | Ref |
|---|---|---|---|---|---|---|
| Glu[a] | EtCHO | $(CH_2)_2CO_2H$ | Pr | 75 | +22.0 | [110] |
| Glu[a] | iBuCHO | $(CH_2)_2CO_2H$ | $CH_2iBu$ | 77 | +15.6 | [110] |
| Glu[a] | PhCHO | $(CH_2)_2CO_2H$ | Bzl | 55 | +17.5 | [110] |
| Met | MeCHO | $(CH_2)_2SMe$ | Et | 75[b] | +47.5 | [110] |
| Met | EtCHO | $(CH_2)_2SMe$ | Pr | 84 | +40.7 | [110] |
| Met | iBuCHO | $(CH_2)_2SMe$ | $(CH_2)_2iPr$ | 77 | +32.5 | [110] |
| Met | PhCHO | $(CH_2)_2SMe$ | Bzl | 96 | +24.7 | [110] |
| Phe | MeCHO | Bzl | Et | 80[b] | +33.3 | [110] |
| Phe | EtCHO | Bzl | Pr | 80 | +31.2 | [110] |
| Phe | iBuCHO | Bzl | $CH_2iBu$ | 82 | +31.3 | [110] |
| Phe | PhCHO | Bzl | Bzl | 83 | +23.6 | [110] |
| Ser | MeCHO | $CH_2OH$ | Et | 51[b] | +13.6 | [110] |
| Ser | EtCHO | $CH_2OH$ | Pr | 60 | +13.7 | [110] |
| Ser | iBuCHO | $CH_2OH$ | $CH_2iBu$ | 66 | +10.8 | [110] |
| Ser | PhCHO | $CH_2OH$ | Bzl | 73 | +6.8 | [110] |
| Val | MeCHO | iPr | Et | 67 | +28.8 | [110] |
| Val | EtCHO | iPr | Pr | 69 | +24.0 | [110] |
| Val | iBuCHO | iPr | $CH_2iBu$ | 79 | +21.0 | [110] |
| Val | PhCHO | iPr | Bzl | 82 | +14.1 | [110] |

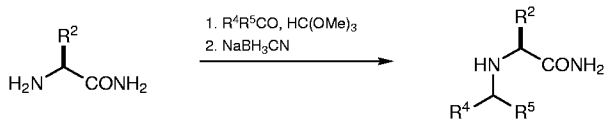
[a] 1.5 equiv of $NaBH_3CN$ was used.
[b] Some dialkylation contaminants were removed by recrystallization (hot EtOH) or chromatography (silica gel, $CHCl_3$/MeOH).

***N*-Alkyl Amino Acids 19 by Reductive Alkylation of Zwitterionic Amino Acids; General Procedure:**[110] To a suspension of zwitterionic amino acid (10 mmol) and $NaBH_3CN$ (7 mmol) in MeOH (15 mL) was added freshly distilled aldehyde (11 mmol) at rt. The mixture was stirred for 18 h. The white precipitate formed was collected by filtration, washed with small amounts of cold MeOH and dried in vacuo to give the mono-*N*-alkyl amino acids. The combined filtrate and washings were taken to dryness in vacuo to give a crude mixture of monoalkylated product and starting material. Chromatography (silica gel, $CHCl_3$/MeOH) afforded additional mono-*N*-alkyl amino acid.

### 10.1.1.3.2 Reductive Alkylation of Amino Acid Esters

The introduction of trimethyl orthoformate[112] as a mild and effective dehydrating solvent for solution and solid-phase imine formation simplified previous attempts to solve the problem of overalkylation (ref[108] and references therein) by conducting the reductive alkylation in two steps: efficient imine formation in trimethyl orthoformate followed by reduction of the imine. Campbell and co-workers[113] were the first to use this one-pot procedure for the preparation of various *N*-alkyl-Phe and *N*-alkyl-Val amides **20** (see Scheme 13) using $NaBH_3CN$ as the reducing agent.

**Scheme 13** Synthesis of *N*-Alkyl Amino Acid Amides by Reductive Alkylation in Trimethyl Orthoformate[113]

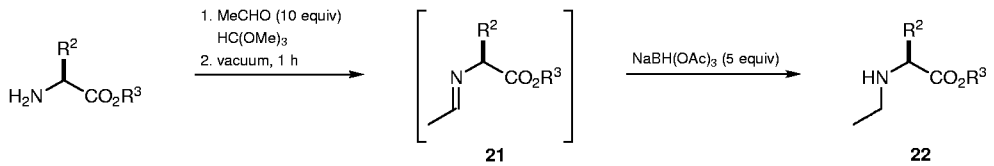



**20**

| Amino Acid | Aldehyde or Ketone | $R^2$ | $R^4$ | $R^5$ | Yield (%) of **20** | Ref |
|---|---|---|---|---|---|---|
| Phe | MeCHO | Bzl | Me | H | 77 | [113] |
| Phe | iPrCHO | Bzl | iPr | H | 69 | [113] |
| Phe | tBuCHO | Bzl | tBu | H | 80 | [113] |
| Phe | PhCHO | Bzl | Ph | H | 74 | [113] |
| Phe | acetone | Bzl | Me | Me | 82 | [113] |
| Phe | cyclohexanone | Bzl | $(CH_2)_5$ | | 83 | [113] |
| Val | tBuCHO | iPr | tBu | H | 82 | [113] |

Acetaldehyde/trimethyl orthoformate and the efficient reducing agent triacetoxyborohydride [$NaBH(OAc)_3$[93]] were used to develop a one-pot synthesis of *N*-ethyl amino acid esters **22**.[114] As can be seen in Scheme 14, excess acetaldehyde was reacted with various amino acid ester in trimethyl orthoformate. The excess acetaldehyde was removed under reduced pressure and the resulting imine **21** was reduced with $NaBH(OAc)_3$ without isolation. The yields of the isolated *N*-ethyl amino acid esters **22** are high.

**Scheme 14** Synthesis of *N*-Ethyl Amino Acid Esters by Reductive Alkylation in Trimethyl Orthoformate[114]




**21** **22**

| Amino Acid | $R^2$ | $R^3$ | Yield (%) | Ref |
|---|---|---|---|---|
| Leu | iBu | tBu | 82 | [114] |
| Lys(Z) | $(CH_2)_2NHZ$ | tBu | 75 | [114] |
| Phe | Bzl | tBu | 78 | [114] |
| Thr | CH(OH)Me | tBu | 75 | [114] |
| Thr | CH(OH)Me | Bzl | 61 | [114] |
| Thr(Bzl) | CH(OBzl)Me | Bzl | 87 | [114] |
| Thr(tBu) | CH(OtBu)Me | tBu | 57 | [114] |
| Val | iPr | tBu | 85 | [114] |
| Val | iPr | Me | 83 | [114] |

***N*-Alkyl-Phe-$NH_2$ Derivatives** 20**; General Procedure:**[113]
To H-Phe-$NH_2$ (0.25 g, 1.52 mmol) dissolved in trimethyl orthoformate (20 mL) under argon was added aldehyde (15.2 mmol) followed by $NaBH_3CN$ (0.478 g, 7.61 mmol) dissolved in trimethyl orthoformate (20 mL). After 1 h at rt, the reaction was cooled to 0 °C, quenched with 2% aq HCl (50 mL), and washed with $Et_2O$ (3 × 25 mL). The $Et_2O$ phase was extracted with 2% aq HCl (2 × 25 mL), and the combined aqueous phases were cooled to 0 °C, treated with concd NaOH until basic, and then extracted with $Et_2O$ (4 × 25 mL). The $Et_2O$ was dried ($MgSO_4$) and filtered. The filtrate was treated with TosOH (1.672 mmol) dissolved in $Et_2O$ (10 mL) and placed in the freezer. The precipitate was collected by filtration, washed with $Et_2O$, and dried under vacuum.

**HCl·H-(Et)Val-OtBu** (22, **$R^2$ = iPr; $R^3$ = tBu); Typical Procedure:**[114]
H-Val-OtBu·HCl (500 mg, 2.4 mmol) and DIPEA (0.42 mL, 2.4 mmol) were dissolved in trimethyl orthoformate (100 mL) under argon. Acetaldehyde (1.34 mL, 24 mmol) was added and the mixture stirred at rt for 30 min. The excess acetaldehyde was removed under vacuum during 1 h followed by the addition of small portions of $NaBH(OAc)_3$ (total 2.5 g, 11.8 mmol). After 15 min the suspension was cooled to

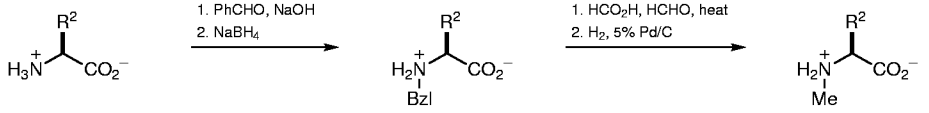
*for references see p 265*

0 °C and quenched with 0.8% aq HCl (60 mL) to pH 4. The organic phase was concentrated in vacuo and the mixture diluted with $H_2O$ (60 mL), and washed with $Et_2O$ (2 × 25 mL). The $Et_2O$ phase was extracted twice with 2% aq HCl and the combined aqueous phases were cooled to 0 °C and adjusted to pH 10 with 2 M NaOH. The milky soln was extracted with $Et_2O$ (4 × 25 mL). The combined ether phases were dried ($Na_2SO_4$) and taken to dryness. The residue was purified by flash chromatography (silica gel, 0.04–0.0633 mm, EtOAc/hexane 2:8). The isolated monoalkylated product was dissolved in dry $Et_2O$ (30 mL) and a soln of HCl in $Et_2O$ was added until it reached pH 3. The mixture was kept in the cold overnight. The precipitate was collected by filtration, washed with dry $Et_2O$ and dried in vacuo to give a colorless solid; yield: 480 mg (85%); $t_R$ 9.32 min (C18 RP-HPLC); mp 183–185 °C; $[\alpha]_D^{25}$ +33.0 (*c* 0.2, MeOH); $^1H$ NMR (400 MHz, $CDCl_3$, δ) (free amino acid): 2.85(d, *J* = 2.85 Hz, 1H), 2.63 (m, 1H), 2.48 (m, 1H), 1.88 (m, 1H) , 1.49 (s, 9H), 1.09 (t, *J* = 7.2 Hz, 3H), 0.92 (dd, *J* = 4.4, 6.8 Hz, 6H); ESI-MS *m/z*: $[M+H^+]$ calcd for $C_{11}H_{23}NO_2$: 201.3; found, 202.6.

### 10.1.1.3.3 Reductive Alkylation of Temporarily *N*-Alkyl Protected Amino Acids

Quitt et al.[115,116] introduced a three-step process to prepare mono-*N*-methyl amino acids **24** without racemization (Scheme 15). The sequence involved the following steps: (1) protection of the α-amino group with the benzyl group using benzaldehyde and $NaBH_4$; (2) reductive methylation of the *N*-benzyl amino acid **23** with formaldehyde/formic acid; and (3) hydrogenation to remove the Bzl group giving **24**. This method was introduced in 1963 before the discovery of $NaBH_3CN$ in 1971 by Borch et al.[109] The use of the benzyl group prevents over alkylation in the second reductive alkylation step even with formaldehyde. The sequence of reactions is performed under mild conditions and without racemization and the overall yields are in the range of 40% for amino acids with nonfunctionalized side chains. This method has been applied to amino acids with unprotected functional side chains, see Scheme 15.[117] The same approach was also used on side-chain-protected amino acids. The original method of Quitt et al.[115] was improved by using $NaBH_3CN$[118] and this method was used to prepare H-(Me)Arg(Tos)-OH and Boc-(Me)Arg(Tos)-OH from H-Arg(Tos)-OH.

**Scheme 15** Synthesis of Mono-*N*-methyl Amino Acids by Temporary Protection with a Benzyl Group[115–117]

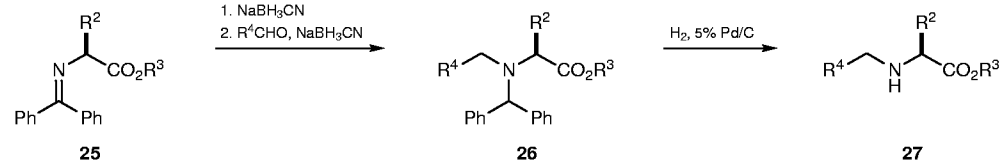



**23** **24**

| Amino Acid | $R^2$ | Yield (%) of **24** | Ref |
|---|---|---|---|
| Arg | $(CH_2)_3NHC(=NH)NH_2$ | 26 | [117] |
| Asp | $CH_2CO_2H$ | 28 | [117] |
| Glu | $(CH_2)_2CO_2H$ | 3 | [117] |
| Tyr | $4\text{-}HOC_6H_4CH_2$ | 34 | [117] |
| Ser | $CH_2OH$ | 27 | [117] |
| Thr | CH(OH)Me | 14 | [117] |

Charuma et al.[119] have improved the sequence of reactions introduced by Quitt et al.[115] Benzophenone[120] was used to form the stable Schiff base **25** (Scheme 16) with amino acid esters, $NaBH_3CN$ was used for both reductive alkylation steps giving *N*-alkyl-*N*-Dpm amino acid esters **26** (alkyl groups other than methyl were also introduced using this method). Catalytic hydrogenation of the *N*-alkyl-*N*-Dpm amino acid esters **26** removed the *N*-Dpm group and, if present, it removed the Dpm ester group simultaneously, to give the product *N*-alkyl amino acids or esters **27**.

**Scheme 16** Synthesis of Mono-*N*-alkyl Amino Acids and Esters by Temporary Protection with Dpm[119]




25 26 27

| Amino Acid | $R^2$ | $R^3$ | $R^4$ | Yield (%) | | Ref |
|---|---|---|---|---|---|---|
| | | | | **26** | **27** | |
| Ala | Me | Dpm | H | 77 | 91[a] | [119] |
| Ala | Me | Me | H | 90 | 90 | [119] |
| Ala | Me | Me | $(CH_2)_5Me$ | 90 | 91 | [119] |
| Ser | $CH_2OH$ | Dpm | H | 71 | 90[a] | [119] |
| Thr | CH(OH)Me | Dpm | H | 63 | 74[a] | [119] |
| Trp | 1*H*-indol-3-ylmethyl | Bzl | H | 53 | 82[a] | [119] |
| Leu | iBu | Et | H | 63 | 92 | [119] |

[a] Catalytic hydrogenation also removed the ester Dpm group hence $R^3$ = H.

**Bzl-Arg(Tos)-OH; Typical Procedure:**[118]
To a soln of H-Arg(Tos)-OH (25 g, 76.1 mmol) and benzaldehyde (8.0 mL, 76.1 mmol) in MeOH (30 mL) was added $NaBH_3CN$ (1.6 g, 25.5 mmol) and the mixture was stirred at rt for 90 min. The pH of the reaction was maintained at 6.5–7.0 by adding the appropriate amount of 1 M HCl. An additional amount of benzaldehyde (8.0 mL) and $NaBH_3CN$ (1.6 g) was added, and stirring continued for 18 h. Bzl-Arg-(Tos)-OH precipitated and was collected by filtration. The solid was triturated with hot MeOH; yield: 10.5 g (33%); mp 160–167 °C dec; FAB-MS *m/z*: 419 $[M+H]^+$; $^1H$ NMR ($CDCl_3$, 90 MHz, δ): 1.9 (m, 4H), 2.4 (s, 3H), 3.3 (m 2H), 4.1 (s 2H), 4.8 (m, 4H), 7.1 (s 5H), 7.5 (d, 2H), 7.9 (d, 2H).

**Bzl(Me)N-Arg(Tos)-OH Modified Procedure of Quitt et al.;**[115] **Typical Procedure:**[118]
To finely powdered Bzl-Arg(Tos)-OH (8.35 g, 20.0 mmol) were added $HCO_2H$ (95–97%) (2.3 mL, 60 mmol) and 37–40% HCHO (1.95 mL, 24 mmol). The mixture was heated on a steam bath for 45 min and then taken to dryness to give a gummy material; $^1H$ NMR ($CDCl_3$, 90 MHz, δ): 1.9 (m, 4H), 2.4 (s, 3H), 2.6 (s 3H), 3.3 (m 2H), 4.1 (s 2H), 4.8 (m, 4H), 7.1 (s 5H), 7.5 (d, 2H), 7.9 (d, 2H); FAB-MS *m/z*: 433 $[M+H]^+$.

**H-(Me)-Arg(Tos)-OH; Typical Procedure**[118]
To a soln of Bzl(Me)N-Arg(Tos)-OH (8.65 g, 20.0 mmol) in glacial AcOH (50 mL) was added a mixture of $H_2O$ (10 mL) and 3 M HCl (5 mL) and then a catalytic amount of 5% Pd/C. The mixture was hydrogenated on a Parr shaker for 6 h for complete uptake of $H_2$. The catalyst was removed by filtration and the filtrate was taken to dryness. The oily residue was dissolved in $H_2O$ and neutralized to pH 7 with dil $NH_4OH$. The white precipitate was collected by filtration and recrystallized (hot $H_2O$, 2 ×) to give pure H-(Me)-Arg(Tos)-OH; yield: 2.5 g (36%); mp 210–216 °C dec; $^1H$ NMR ($D_2O$, DCl, 90MHz, δ): 1.75 (m, 4H), 2.4 (s, 3H), 2.8 (s 3H), 3.3 (t 2H), 4.0 (t 1H), 4.8 (m, 4H), 7.5 (d, 2H), 7.9 (d, 2H). FAB-MS *m/z*: 343 $[M+H]^+$; Anal. calcd for $C_{14}H_{22}N_4O_4S \cdot H_2O$: C, 46.65; H, 6.71; N, 15.55. Found: C, 46.64; H, 6.70; N, 15.02.

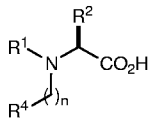
### 10.1.1.4 Synthesis of $N^{\alpha}$-(ω-Functionalized Alkyl) Amino Acids in Solution

The synthesis of amino acids in which a nonfunctionalized alkyl group is covalently connected to the α-nitrogen has been discussed (*vide supra*). In this section the synthesis of N-substituted amino acids in which the alkyl chain bears a protected functional group is discussed. The resulting amino acids have the general formula $N^{\alpha}$-(ω-functionalized alkyl) amino acids **28** (Scheme 17). These amino acids were developed for the solid-phase synthesis of backbone cyclic peptides[70] (see also Vol. E 22b, Section 6.8.3.2.4) [in this case they were called building units (BU)], for the synthesis of PNA[71] and for the synthesis of peptoids.[72] (In this case they were called monomers.)

*for references see p 265*

It is beyond the scope of this review to describe in details the synthesis of the protected ω-functionalized alkyl synthons: **29**, **33**, **37**, **41**, **44** (Schemes 18–22). For these details the reader should consult the original literature.

**Scheme 17** General Structure of $N^{\alpha}$-(ω-Functionalized alkyl) Amino Acids



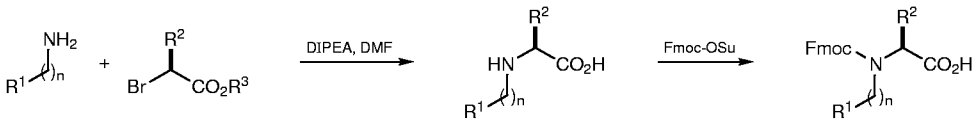
**28**

n = 1–6; $R^1$ = Fmoc, Boc; $R^2$ = H, amino acid side chains, orthogonally protected amino acid side chain
$R^4$ = orthogonally protected functional group, e.g., NHBoc, NHAloc, $CO_2tBu$, $CO_2Al$, STrt, SAcm

### 10.1.1.4.1 Synthesis by Nucleophilic Substitution

The first synthesis[121] of $N^{\alpha}$-(ω-BocHN-alkyl) amino acid building units used nucleophilic substitution of α-halocarboxylic acids or their esters **30** (Scheme 18) with mono-Boc-substituted diamines **29**. The nucleophilic substitution product **31** was reacted with Fmoc-OSu to give the orthogonally protected building unit **32**.

**Scheme 18** Synthesis of Fmoc-Protected $N^{\alpha}$-(ω-BocHN-alkyl) Amino Acids Prepared by Nucleophilic Substitution[121,122]



**29** **30** **31** **32**

| $R^2$ | n | Yield[a] (%) | Ref |
|---|---|---|---|
| Ph | 1 | 92 | [122] |
| iPr | 1 | 94 | [122] |
| SMe | 2 | 86 | [122] |
| NHBoc | 4 | 71 | [122] |
| $CONH_2$ | 2 | 71 | [122] |
| 4-$tBuOC_6H_4$ | 1 | 64 | [122] |

[a] $R^1$ = NHBoc.

This approach gave fair yields of Gly-based building units ($R^2$ = H). A similar approach was used[122] to prepare a variety of monomers (Scheme 18) used in the synthesis of Leu-enkephalin peptoids. In the case of substituted α-halocarboxylic acids or esters bearing $R^2$ side chains the yields were lowered because of racemization and β-elimination resulting in the undesired α,β-dehydrocarboxylic acids in addition to the desired chiral building unit. Another problem was polyalkylation.

This procedure was improved[123] for the synthesis of building units based on amino acids other than Gly, but with nonfunctionalized side chains (Table 6). To suppress the β-elimination and racemization side reactions, triflates of α-hydroxycarboxylic acid esters[124] **34** (L = OTf, Scheme 19) were used as substrates for the nucleophilic substitution. In order to prevent polyalkylation, the nucleophilic amine of ω-BocNH- or ω-$tBuO_2C$-alkylamines **33** were temporarily protected with the benzyl group.[115,116] This protection also improved the yields and purity of Gly-based building units. In this case commercially available benzyl bromoacetate **34** (L = Br) was used as the substrate. In both cases the nucleophilic sub-

stitution resulted in $N^{\alpha}$-Bzl-$N^{\alpha}$-(ω-BocNH-alkyl)- **35** ($R^4$ = NHBoc) or $N^{\alpha}$-Bzl-$N^{\alpha}$-(ω-$tBuO_2C$-alkyl) amino acid esters **35** ($R^4$ = $CO_2tBu$). The last steps were the concomitant removal of both Bzl protecting groups by catalytic hydrogenation to give **36** and the introduction of the Fmoc group to obtain $N^{\alpha}$-Fmoc-$N^{\alpha}$-(ω-BocNH-alkyl)- **32** ($R^4$ = NHBoc) or $N^{\alpha}$-Fmoc-$N^{\alpha}$-(ω-$tBu_2OC$-alkyl) amino acid **32** ($R^4$ = $CO_2tBu$).

**Scheme 19** Synthesis of Fmoc-Protected $N^{\alpha}$-(ω-BocNH-alkyl) and $N^{\alpha}$-(ω-$tBuO_2C$-alkyl) Amino Acids by Nucleophilic Substitution[123]

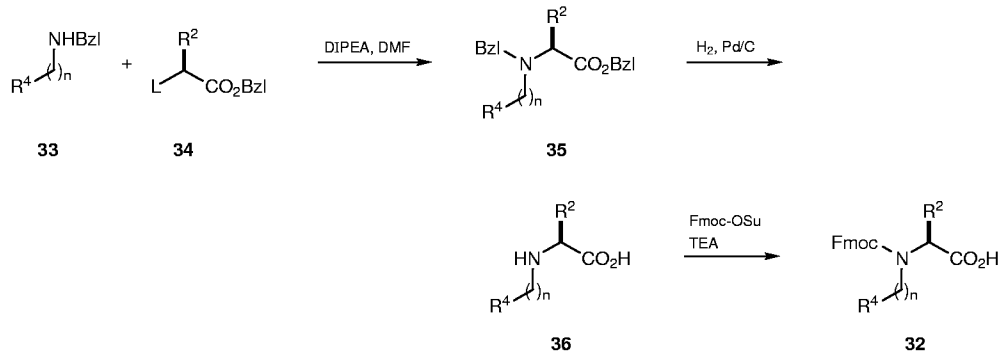



**Table 6** Synthesis of Fmoc-Protected $N^{\alpha}$-(ω-BocNH-alkyl) and $N^{\alpha}$-(ω-$tBuO_2C$-alkyl) Amino Acids by Nucleophilic Substitution[123]

| $R^2$ | Config | $R^4$ | n | L | Yield[a] (%) | Ref |
|---|---|---|---|---|---|---|
| H | | NHBoc | 2 | Br | 80 | [123] |
| H | | NHBoc | 3 | Br | 85 | [123] |
| H | | NHBoc | 4 | Br | 79 | [123] |
| H | | NHBoc | 6 | Br | 81 | [123] |
| H | | $CO_2tBu$ | 1 | Br | 71 | [123] |
| H | | $CO_2tBu$ | 2 | Br | 77 | [123] |
| H | | $CO_2tBu$ | 3 | Br | 64 | [123] |
| H | | $CO_2tBu$ | 4 | Br | 70 | [123] |
| H | | $CO_2tBu$ | 5 | Br | 81 | [123] |
| Me | *S* | NHBoc | 3 | OTf | 75 | [123] |
| Me | *R* | NHBoc | 3 | OTf | 76 | [123] |
| Me | *S* | NHBoc | 6 | OTf | 73 | [123] |
| Bzl | *S* | NHBoc | 3 | OTf | 65 | [123] |
| Bzl | *R* | NHBoc | 3 | OTf | 62 | [123] |
| Bzl | *S* | $CO_2tBu$ | 3 | OTf | 38 | [123] |

[a] Yield of **32**.

An improved and much shorter synthesis of chiral building units suitable for Fmoc SPPS of backbone cyclic peptides has been presented.[125] A variety of Fmoc-protected $N^{\alpha}$-(ω-AlocNH-alkyl) and $N^{\alpha}$-(ω-$AlO_2C$-alkyl) amino acids **32** were synthesized by nucleophilic substitution of ω-AlocHN- and $AlO_2CO$-substituted alkyl bromides **37** (Scheme 20, Table 7) with amino acid *tert*-butyl esters **38** having nonfunctionalized or orthogonally protected functionalized side chains. The resulting $N^{\alpha}$-(ω-AlocNH-alkyl) and $N^{\alpha}$-(ω-$AlO_2C$-alkyl) amino acid *tert*-butyl esters **39** were protected by Fmoc giving **40** and the tBu protecting group removed by TFA to yield the protected building units **32**. The mild reaction conditions using solid $Na_2CO_3$ caused no racemization or β-elimination and did not cause any damage to functionalized side chains (e.g., Asn, Trp) or protecting groups (e.g., Boc, Aloc, OAl). Nevertheless this method is not general and does not give satisfactory yields with many amino acids having functionally protected side chains.

*for references see p 265*

**Scheme 20** Synthesis of Fmoc-Protected $N^{\alpha}$-(ω-AlocNH-alkyl) and $N^{\alpha}$-(ω-$AlO_2C$-alkyl) Amino Acids by Nucleophilic Substitution[125]

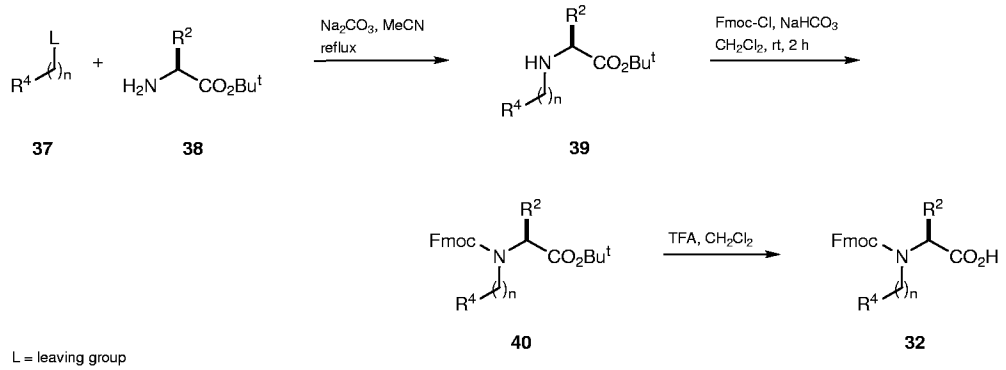



**Table 7** Fmoc-Protected $N^{\alpha}$-(ω-Protected functionalized alkyl) Amino Acids[125]

| Amino Acid | Config | $R^2$ | $R^4$ | n | Yield[a] (%) | Ref |
|---|---|---|---|---|---|---|
| Ala | L | Me | $CO_2Al$ | 1 | 53 | [125] |
| Ala | L | Me | $CO_2Al$ | 3 | 61 | [125] |
| Ala | D | Me | $CO_2Al$ | 3 | 61 | [125] |
| Ala | L | Me | $CO_2Al$ | 4 | 64 | [125] |
| Ala | D | Me | $CO_2Al$ | 4 | 65 | [125] |
| Ala | L | Me | $CO_2Al$ | 5 | 67 | [125] |
| Ala | L | Me | NHAloc | 2 | 60 | [125] |
| Ala | D | Me | NHAloc | 2 | 58 | [125] |
| Ala | L | Me | NHAloc | 3 | 62 | [125] |
| Ala | L | Me | NHAloc | 3 | 70 | [125] |
| Asn | L | $CH_2CONH_2$ | $CO_2Al$ | 4 | 42 | [125] |
| Asn | L | $CH_2CONH_2$ | NHAloc | 2 | 41 | [125] |
| Gly | | H | $CO_2Al$ | 3 | 71 | [125] |
| Gly | | H | NHAloc | 2 | 68 | [125] |
| Gly | | H | NHAloc | 3 | 69 | [125] |
| Ile | L | sBu | $CO_2Al$ | 1 | 58 | [125] |
| Ile | L | sBu | $CO_2Al$ | 3 | 66 | [125] |
| Ile | D | sBu | $CO_2Al$ | 3 | 65 | [125] |
| Ile | L | sBu | $CO_2Al$ | 4 | 67 | [125] |
| Ile | D | sBu | $CO_2Al$ | 4 | 68 | [125] |
| Ile | L | sBu | NHAloc | 2 | 57 | [125] |
| Ile | D | sBu | NHAloc | 2 | 56 | [125] |
| Ile | L | sBu | NHAloc | 3 | 57 | [125] |
| Ile | D | sBu | NHAloc | 3 | 50 | [125] |
| Ile | D | sBu | NHAloc | 4 | 53 | [125] |
| Leu | L | iBu | $CO_2Al$ | 1 | 54 | [125] |
| Leu | D | iBu | $CO_2Al$ | 1 | 51 | [125] |
| Leu | L | iBu | $CO_2Al$ | 3 | 64 | [125] |
| Leu | L | iBu | $CO_2Al$ | 4 | 66 | [125] |
| Leu | D | iBu | $CO_2Al$ | 4 | 69 | [125] |
| Leu | L | iBu | $CO_2Al$ | 5 | 70 | [125] |
| Leu | L | iBu | NHAloc | 2 | 49 | [125] |
| Leu | D | iBu | NHAloc | 2 | 50 | [125] |
| Leu | L | iBu | NHAloc | 3 | 55 | [125] |
| Leu | L | iBu | NHAloc | 4 | 63 | [125] |
| Lys(Boc) | L | $(CH_2)_4NHBoc$ | $CO_2Al$ | 1 | 38 | [125] |
| Lys(Boc) | L | $(CH_2)_4NHBoc$ | $CO_2Al$ | 3 | 41 | [125] |
| Lys(Boc) | D | $(CH_2)_4NHBoc$ | $CO_2Al$ | 3 | 40 | [125] |
| Lys(Boc) | L | $(CH_2)_4NHBoc$ | $CO_2Al$ | 4 | 45 | [125] |
| Lys(Boc) | L | $(CH_2)_4NHBoc$ | $CO_2Al$ | 2 | 44 | [125] |
| Lys(Boc) | D | $(CH_2)_4NHBoc$ | $CO_2Al$ | 2 | 43 | [125] |

**Table 7** (cont.)

| Amino Acid | Config | $R^2$ | $R^4$ | n | Yield[a] (%) | Ref |
|---|---|---|---|---|---|---|
| Lys(Boc) | L | $(CH_2)_4NHBoc$ | NHAloc | 3 | 47 | [125] |
| Phe | L | Bzl | $CO_2Al$ | 1 | 52 | [125] |
| Phe | D | Bzl | $CO_2Al$ | 1 | 59 | [125] |
| Phe | L | Bzl | $CO_2Al$ | 3 | 68 | [125] |
| Phe | D | Bzl | $CO_2Al$ | 3 | 60 | [125] |
| Phe | L | Bzl | $CO_2Al$ | 4 | 70 | [125] |
| Phe | D | Bzl | $CO_2Al$ | 4 | 67 | [125] |
| Phe | D | Bzl | $CO_2Al$ | 5 | 70 | [125] |
| Phe | L | Bzl | NHAloc | 2 | 59 | [125] |
| Phe | D | Bzl | NHAloc | 2 | 60 | [125] |
| Phe | L | Bzl | NHAloc | 3 | 63 | [125] |
| Trp | L | 1*H*-indol-3-ylmethyl | $CO_2Al$ | 1 | 43 | [125] |
| Trp | L | 1*H*-indol-3-ylmethyl | $CO_2Al$ | 3 | 51 | [125] |
| Trp | L | 1*H*-indol-3-ylmethyl | $CO_2Al$ | 5 | 59 | [125] |
| Trp | L | 1*H*-indol-3-ylmethyl | NHAloc | 2 | 48 | [125] |
| Trp | L | 1*H*-indol-3-ylmethyl | NHAloc | 3 | 49 | [125] |
| Val | L | iPr | $CO_2Al$ | 1 | 55 | [125] |
| Val | D | iPr | $CO_2Al$ | 1 | 54 | [125] |
| Val | L | iPr | $CO_2Al$ | 3 | 68 | [125] |
| Val | D | iPr | $CO_2Al$ | 4 | 70 | [125] |

[a] Yield of **32** from **38**.

**Fmoc-Protected $N^{\alpha}$-(ω-BocHN-alkyl) Amino Acids** 32**; General Procedure:**[122]
Ethyl bromoacetate (5.54 mL, 50 mmol) in THF (25 mL) was added dropwise to a cooled (ice bath) soln of amine **29** (110 mmol) in THF (25 mL). After stirring for 2.5 h at rt, the mixture was taken to dryness and the residue resuspended in $Et_2O$. The precipitate was removed by filtration and washed with $Et_2O$. The combined organic phase was taken to dryness and the residue purified by column chromatography (silica gel, $Et_2O$). 4 M NaOH (2.5 mL) was added to the soln of *N*-alkyl-Gly-OEt (10 mmol) in dioxane (35 mL) and MeOH (12.5 mL). After stirring for 30 min at rt, the mixture was taken to dryness. The residue was dissolved in $H_2O$ and the pH adjusted to 9–9.5 with concd HCl. To this mixture a soln of Fmoc-OSu (10 mmol) in MeCN (20 mL) was added in one portion and the pH maintained at 8.5–9.0 with TEA. The MeCN was removed and the residue added to 20% citric acid soln (60 mL). The soln was extracted with EtOAc (3 × 75 mL) and the combined organic layers were washed with $H_2O$ and brine, dried ($Na_2SO_4$), and taken to dryness. The residual oil was recrystallized (EtOAc/hexane).

**$N^{\alpha}$-(ω-BocHN-alkyl) and $N^{\alpha}$-(ω-tBu$O_2$C-alkyl) Amino Acids** 32 **($R^2$ = H); General Procedure:**[123]
A soln of $N^{\alpha}$-Bzl-$N^{\omega}$-(BocHN-alkyl) or $N^{\alpha}$-(ω-tBu$O_2$C-alkyl) amines **33**[123] (32.5 mol) in DMF (10mL) was cooled in an ice bath. Then DIPEA (5.66 mL, 32.5 mol) followed by benzyl bromoacetate (5.15 mL, 32.5 mol) were added. The mixture was stirred at 0 °C for 30 min, and then at rt for 3 h. $Et_2O$ (200 mL) was added, and the white precipitate that formed was removed by filtration. The $Et_2O$ filtrate was washed with $H_2O$ (3 × 100 mL), 1 M HCl (2 × 100 mL), and $H_2O$ (3 × 100 mL), dried ($Na_2SO_4$), and taken to dryness in vacuo. The residue was dissolved in MeOH (60 mL) and 10% Pd/C (1 g) was added. The mixture was hydrogenated for 4 h at rt with $H_2$ (4.1 bar). The catalyst was removed by filtration and washed with MeOH (2 × 40 mL). The combined solns were taken to dryness in vacuo. $Et_2O$ (200 mL) was added to the residual oil, and the thus formed precipitate was collected by filtration, washed with $Et_2O$ (2 × 30 mL), and dried in vacuo. The residue was dissolved in $H_2O$ (40 mL) and TEA (20 mmol) was added. The mixture stirred until a clear soln was obtained. Then a soln of Fmoc-OSu (6.75 g, 20 mmol) in MeCN (80 mL) was added. The mixture was stirred at rt for 4 h. $H_2O$ (150 mL) was added and the mixture was washed with petroleum ether (3 × 100 mL) and then $Et_2O$/petroleum ether (3:7; 2 × 100 mL). The aqueous soln was cooled to 0 °C and acidified with 1 M HCl (40 mL). The product was extracted with EtOAc (4 × 100 mL), the combined organic phases were washed with $H_2O$ (4 × 100 mL), dried ($Na_2SO_4$), and taken to dryness in vacuo. The product was recrystallized ($Et_2O$/petroleum ether 1:10).

*for references see p 265*

### 10.1.1.4.2 Synthesis by Reductive Alkylation

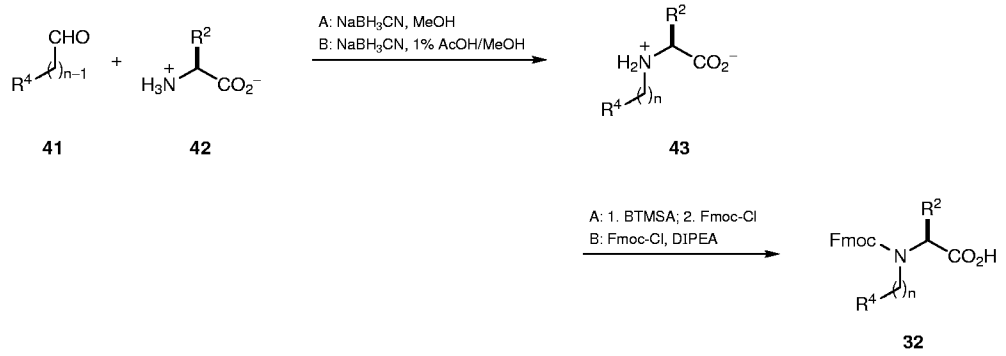
#### 10.1.1.4.2.1 Reductive Alkylation Reaction of Zwitterionic Amino Acids

A convenient synthesis[111] of Fmoc-protected building units is based on the reductive alkylation reaction of zwitterionic amino acids introduced by Ohfune[110] (see Section 10.1.1.3.1 and Scheme 12). Thus, highly concentrated mixtures of ω-protected functionalized aldehydes **41** of the type $R^4$-$(CH_2)_{n-1}$-CHO (Scheme 21, $R^4$ is protected amine, carboxy, or thiol) and chiral zwitterionic amino acids **42** were reacted in methanol with $NaBH_3CN$. Amino acids that have functional groups were protected by Bzl-type protecting groups. The resulting unprotected, optically and chemically pure, $N^{\alpha}$-(ω-protected functionalized alkyl) amino acids **43** precipitated from the reaction mixture as zwitterions. They were collected by filtration and carefully washed with cold MeOH to remove unreacted amino acids (in some cases it was necessary to crystallize the product from hot ethanol). The resulting building units **43** were reacted with Fmoc-Cl to yield the $N^{\alpha}$-Fmoc-protected building units **32**.

An improved version of this procedure has been introduced[125] and used for the preparation of a large variety of Fmoc-protected $N^{\alpha}$-(ω-AlocNH-alkyl)- and $N^{\alpha}$-(ω-$AlO_2C$-alkyl) amino acids **32** that contained sensitive, protected side chains. The ω-protected functionalized aldehydes **41** and the zwitterionic amino acids **42** were dissolved in 1% AcOH/MeOH and then added to the mixture $NaBH_3CN$ in THF. After the completion of the reaction and removal of the solvents, the resulting unprotected $N^{\alpha}$-(ω-protected functionalized alkyl) amino acids **43** were directly reacted with Fmoc-Cl to yield the Fmoc-protected building units **32**.

A variety of monomers for PNA have been prepared.[126] These monomers are based on $N^{\alpha}$-(2-BocNH-ethyl) amino acid esters, which were prepared by reductive alkylation of amino acid benzyl esters with $BocNHCH_2CHO$.

**Scheme 21** Synthesis of Fmoc-Protected $N^{\alpha}$-(ω-Protected functionalized alkyl) Amino Acids by Reductive Alkylation[111,125]




**41** **42** **43** **32**

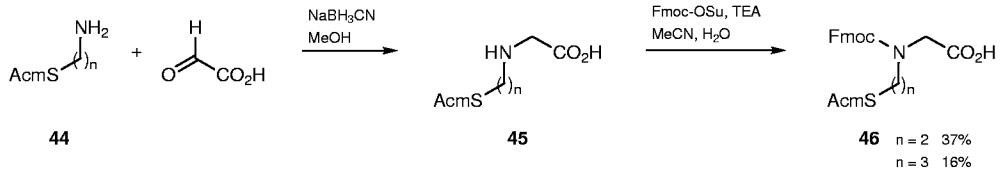
**Table 8** Fmoc-Protected $N^{\alpha}$-(ω-Protected Amino)alkyl Amino Acids Prepared[111,125]

| Amino Acid | Config | $R^2$ | $R^4$ | n | Method[a] | Yield[b] (%) | Ref |
|---|---|---|---|---|---|---|---|
| Arg(Pmc) | L | $(CH_2)_3NHC(=NH)NHPmc$ | NHAloc | 2 | B | 57 | [125] |
| Arg(Pmc) | D | $(CH_2)_3NHC(=NH)NHPmc$ | NHAloc | 2 | B | 53 | [125] |
| Arg(Pmc) | L | $(CH_2)_3NHC(=NH)NHPmc$ | NHAloc | 3 | B | 51 | [125] |
| Asn(Trt) | L | $CH_2CONHTrt$ | NHAloc | 2 | B | 55 | [125] |
| Asn(Trt) | D | $CH_2CONHTrt$ | NHAloc | 2 | B | 60 | [125] |
| Asp(OtBu) | L | $CH_2CO_2tBu$ | NHAloc | 2 | B | 54 | [125] |
| Asp(OtBu) | D | $CH_2CO_2tBu$ | NHAloc | 2 | B | 54 | [125] |
| Cys(Trt) | L | $CH_2STrt$ | NHAloc | 2 | B | 34 | [125] |
| Gln(Trt) | L | $(CH_2)_2CONHTrt$ | NHAloc | 2 | B | 81 | [125] |
| Gln(Trt) | D | $(CH_2)_2CONHTrt$ | NHAloc | 2 | B | 74 | [125] |
| Glu(OtBu) | L | $(CH_2)_2CO_2tBu$ | NHAloc | 2 | B | 51 | [125] |
| Glu(OtBu) | D | $(CH_2)_2CO_2tBu$ | NHAloc | 2 | B | 50 | [125] |
| Gly | | H | NHBoc | 3 | A | 57 | [111] |
| His(Trt) | L | 1-trityl-1*H*-imidazol-5-ylmethyl | NHAloc | 2 | B | 41 | [125] |
| His(Trt) | L | 1-trityl-1*H*-imidazol-5-ylmethyl | NHAloc | 3 | B | 44 | [125] |
| Ile | L | sBu | NHBoc | 3 | A | 50 | [111] |
| Leu | L | iBu | NHBoc | 3 | A | 46 | [111] |
| Met | L | $(CH_2)_2SMe$ | NHBoc | 3 | A | 54 | [111] |
| Phe | L | Bzl | NHBoc | 3 | A | 69 | [111] |
| Ser(Bzl) | L | $CH_2OBzl$ | NHBoc | 2 | A | 29 | [111] |
| Ser(tBu) | L | $CH_2OtBu$ | NHAloc | 2 | B | 43 | [125] |
| Ser(tBu) | D | $CH_2OtBu$ | NHAloc | 2 | B | 43 | [125] |
| Thr(tBu) | L | CH(OtBu)Me | NHAloc | 2 | B | 52 | [125] |
| Thr(tBu) | D | CH(OtBu)Me | NHAloc | 2 | B | 46 | [125] |
| Trp | D | 1*H*-indol-3-ylmethyl | NHAloc | 2 | B | 50 | [125] |
| Trp(Boc) | D | 1-Boc-1*H*-indol-3-ylmethyl | NHAloc | 2 | B | 49 | [125] |
| Tyr(tBu) | L | $4\text{-}tBuOC_6H_4CH_2$ | NHAloc | 2 | B | 73 | [125] |
| Tyr(tBu) | D | $4\text{-}tBuOC_6H_4CH_2$ | NHAloc | 2 | B | 72 | [125] |
| Tyr(tBu) | L | $4\text{-}tBuOC_6H_4CH_2$ | NHAloc | 3 | B | 76 | [125] |
| Val | L | iPr | NHBoc | 3 | A | 59 | [111] |

[a] A and B are defined in Scheme 21.
[b] Yield of **32** from **41** and **42**.

The preparation of Gly based building units containing the $N^{\alpha}$-(sulfanylalkyl) group by the two methods described (*vide supra*) gave low yields, typically 1–23% for $R^4$ = STrt. Building units have been prepared by reacting ω-AcmS-alkylamines **44** (Scheme 22) with glyoxylic acid to give the unprotected $N^{\alpha}$-(ω-AcmS-alkyl)-Gly-OH building units **45**, which were protected with Fmoc to furnish the desired building units **46** in good yields.[127]

**Scheme 22** Synthesis of Fmoc-Protected $N^{\alpha}$-(ω-AcmS-alkyl)-Gly-OH Building Units by Reductive Amination of Glyoxylic Acid[127]




**44** **45** **46** n = 2 37%, n = 3 16%

**Fmoc-Protected $N^{\alpha}$-(ω-Protected functionalized alkyl) Amino Acids** 32 **by Method A; General Procedure:**[111]

A zwitterionic side-chain protected amino acid **42** (5–10 mmol) was dissolved or suspended in MeOH (1.5 mL·mmol$^{-1}$) then the corresponding ω-protected functionalized aldehyde of the type $R_4$-$(CH_2)_{n-1}$-CHO **41**[111] (1.5 equiv) was added followed by $NaBH_3CN$ (1.1 equiv). The mixture was stirred at rt for 18 h. The precipitate product was collected by filtration on a glass sinter, washed with small amount of cold MeOH and dried in vacuo. The product of this reaction **43** (10 mmol) was suspended in $CH_2Cl_2$ (20 mL), and BTMSA (4.33 mL, 17.5 mmol, 1.75 equiv) was added under $N_2$. When the soln was nearly

*for references see p 265*

clear (usually after 5–10 min), Fmoc-Cl (2.72 g, 11.2 mmol, 1.05 equiv) was added and the mixture was stirred for 2 h. MeOH (2 mL) was carefully added and the mixture was stirred for an additional 15 min, diluted with $CH_2Cl_2$ (80 mL), washed successively with 1 M HCl (3 × 50 mL) and sat. aq NaCl (2 × 50 mL), and dried ($MgSO_4$). The solvent was taken to dryness in vacuo and the crude product was crystallized ($Et_2O$/petroleum ether 1:10). In some cases further purification by chromatography was required.

**Fmoc-Protected $N^{\alpha}$-(ω-Protected functionalized alkyl) Amino Acids** 32 **by Method B; General Procedure:**[125]

The amino acid (10 mmol) was added to MeOH/AcOH (99:1, 50–70 mL) until the solid dissolved. Aldehyde **41** (11 mmol) was added in MeOH/AcOH (99:1, 10 mL) and the mixture stirred for 1 h. $NaBH_3CN$ (13 mmol) in THF (3 mL) was added dropwise. The progress of the reaction was controlled by HPLC. The mixture was taken to dryness and the residue taken in $CH_2Cl_2$. The mixture was stirred for 1 h until dissolution, cooled, and DIPEA (30 mL) was added, followed by Fmoc-Cl (10 mmol). After stirring overnight, $CH_2Cl_2$ (100 mL) was added and the organic phase was washed with 1 M HCl (2 ×) and brine, and dried ($Na_2SO_4$). After removal of the $Na_2SO_4$ by filtration, the filtrate was taken to dryness and the residue purified by flash chromatography (silica gel).

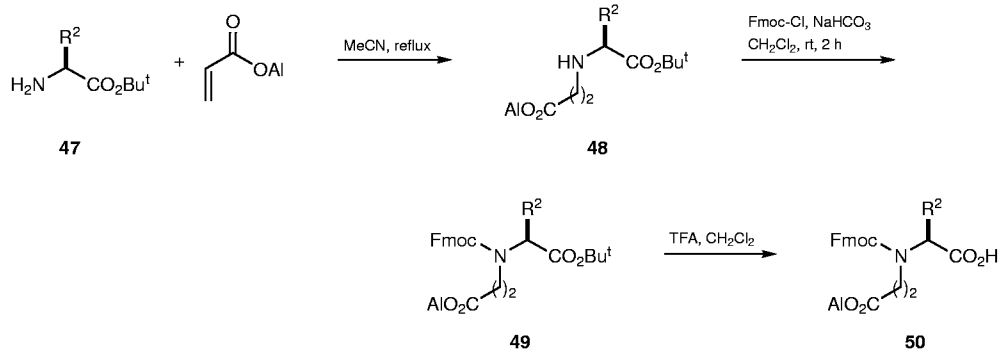
**Fmoc-Protected $N^{\alpha}$-(ω-AcmS-alkyl)-Gly-OH** 46**; General Procedure:**[127]

Acm-protected sulfanylalkylamine **44** (0.1 mol) and $NaBH_3CN$ (6.3 g 0.1 mol) were dissolved in MeOH (200 mL). Glyoxylic acid monohydrate (7.8 g, 0.085 mol) was added and the soln stirred for 19 h. The mixture was taken to dryness and the residual oil dissolved in $H_2O$ (250 mL) and TEA (24 mL, 0.17 mol) was added. To this mixture a soln of Fmoc-OSu (21.5 g, 0.064 mol) dissolved in MeCN (375 mL) was added and the mixture was stirred for 4 h. The soln was washed with petroleum ether (bp 40–60 °C) (3 × 400 mL) followed by $Et_2O$/petroleum ether (7:3; 3 × 400 mL). The soln was brought to pH 3 by 2 M HCl and extracted with EtOAc (4 × 350 mL). The organic phase was washed with 1 M HCl (3 × 250 mL), sat. $KHSO_4$ (3 × 250 mL), dried ($Na_2SO_4$), and taken to dryness.

### 10.1.1.4.3 Synthesis by 1,4-Michael Addition to Amino Acid *tert*-Butyl Esters

The preparation of Fmoc-protected building units that contain the $N^{\alpha}$-[2-(allyloxycarbonyl)ethyl] group by 1,4-Michael addition of commercially available allyl acrylate to various amino acid *tert*-butyl esters has been described.[125] Even though 1,4-Michael addition to primary amines is a well-known and widely used method to create the N–C bond,[128] it is used to N-alkylate amino acids only in special cases.[72] Thus, various chiral amino acid *tert*-butyl esters **47**, including those containing various protected functionalized side chains, were refluxed for 72 hours in acetonitrile with allyl acrylate. The unprotected building units **48** were protected on the α-amino group by Fmoc to furnish **49** (Scheme 23), which provided the desired building units **50** in good yields after *tert*-butyl deprotection. The 1,4-Michael addition to amino acid esters other than *tert*-butyl (such as Me, Et, iPr, or Bzl esters) under various reaction conditions failed, possibly due to the formation of DKP and polyalkylation. However, the use of amino acid *tert*-butyl esters yielded **50** in quantitative yields. It should be mentioned that in the case of Fmoc($AlO_2CCH_2CH_2$)N-Lys(Boc)-OtBu, the Boc protecting group was reintroduced by $Boc_2O$ after *tert*-butyl deprotection with TFA.

**Scheme 23** Synthesis of Fmoc-Protected $N^{\alpha}$-[2-(Allyloxycarbonyl)ethyl] Amino Acids by 1,4-Michael Addition to Amino Acid *tert*-Butyl Esters[125]




47 48 49 50

| Amino Acid | Config | $R^2$ | Yield[a] (%) | Ref |
|---|---|---|---|---|
| Ala | L | Me | 58 | [125] |
| Ala | D | Me | 57 | [125] |
| Asn | L | $CH_2CONH_2$ | 41 | [125] |
| Gly | | H | 73 | [125] |
| Ile | L | sBu | 64 | [125] |
| Ile | D | sBu | 67 | [125] |
| Leu | L | iBu | 53 | [125] |
| Leu | D | iBu | 59 | [125] |
| Lys(Boc) | L | $(CH_2)_4NHBoc$ | 39 | [125] |
| Lys(Boc) | D | $(CH_2)_4NHBoc$ | 40 | [125] |
| Phe | L | Bzl | 59 | [125] |
| Trp | L | 1*H*-indol-3-ylmethyl | 53 | [125] |
| Val | L | iPr | 60 | [125] |
| Val | D | iPr | 64 | [125] |

[a] Yield of **50** from **47**.

**Fmoc($AlO_2CCH_2CH_2$)N-Phe-OH** 50 **($R^2$ = Bzl); Typical Procedure for the Synthesis of Fmoc-Protected, $N^{\alpha}$-[2-(Allyloxycarbonyl)ethyl] Amino Acids by 1,4-Michael Addition:**[125]

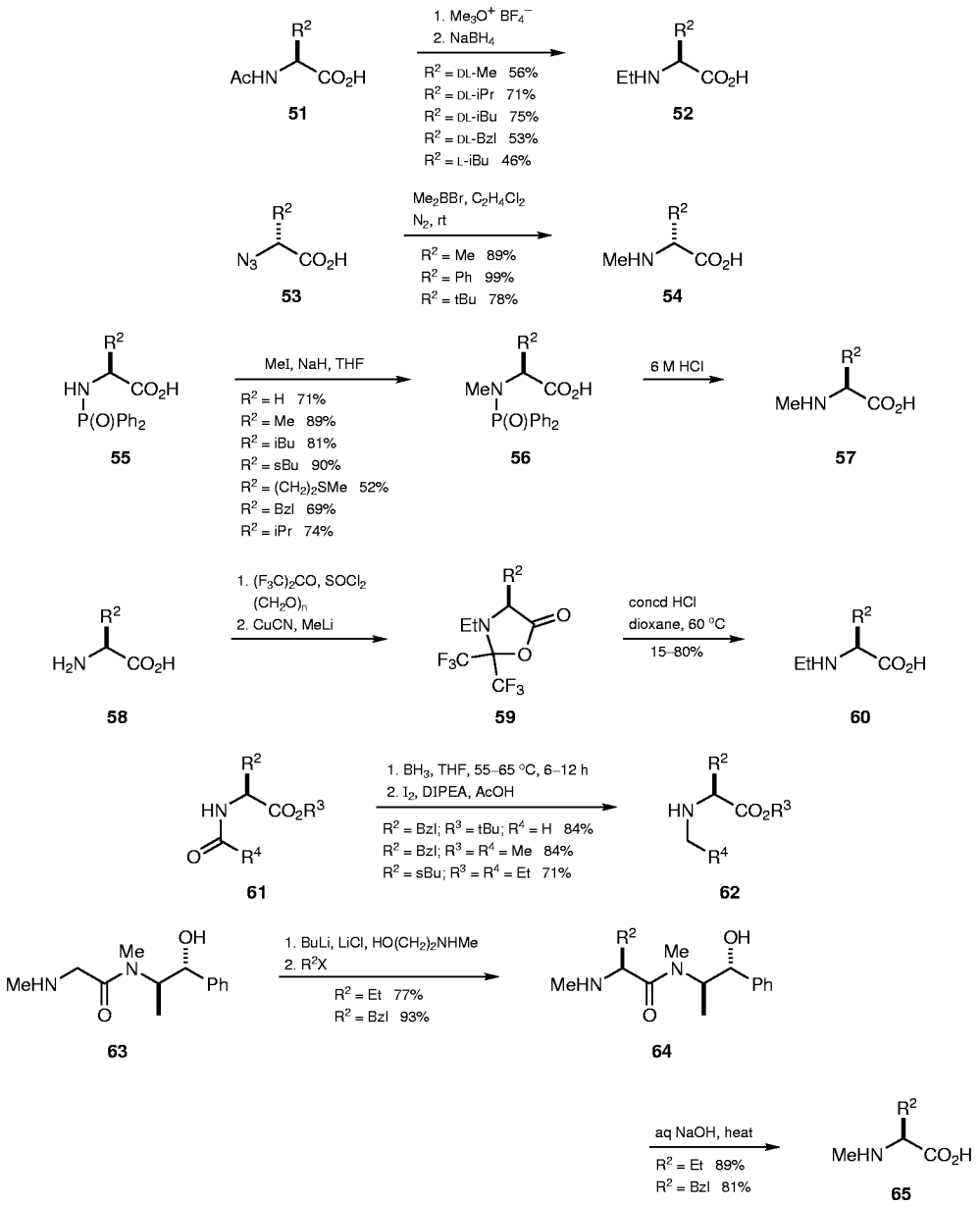
To a mixture of 1 M $Na_2CO_3$ (20 mL) and $Et_2O$ (20 mL) was added HCl·H-Phe-OtBu (1.81 g, 7.6 mmol) with vigorous stirring at 0 °C. The organic phase was separated, dried ($Na_2SO_4$), and taken to dryness to give an oil. To this oil MeCN (20 mL) and allyl acrylate (4.72 g, 42 mmol) were added and the mixture was refluxed for 72 h. The mixture was taken to dryness to give $AlO_2CCH_2CH_2$-Phe-OtBu as a colorless oil; yield: 3.1 g (%); $R_f$ 0.74 (EtOAc); MS *m/z* (%): 333 ($M^+$, 100). The crude oil was protected by Fmoc without further purification. Thus, the oil was dissolved in dry $CH_2Cl_2$ (20 mL) and cooled to 0 °C. Fmoc-Cl (1.96 g, 7.6 mmol) was added in few portions followed by $NaHCO_3$ (4 g). The mixture was stirred at rt for 3 h. TLC monitoring (silica gel, $CHCl_3$) indicated the disappearance of starting material and appearance of new spot ($R_f$ 0.85). $CH_2Cl_2$ (20 mL) was added and the organic phase washed with 1 M $NaHSO_4$ (2 × 20 mL). The organic phase was cooled to 0 °C and a mixture of 95% TFA/$H_2O$ (20 mL) was added. After stirring at rt for 2 h, the mixture was taken to dryness to give a yellow oil, which was redissolved in $CH_2Cl_2$ (30 mL) and washed with 1 M $NaHSO_4$(2 × 20 mL) and brine (20 mL), dried ($Na_2SO_4$), and taken to dryness to give an oil. The oil was triturated with hexane (2 × 20 mL) and purified by flash chromatography (silica gel, 10% MeOH/$CH_2Cl_2$) to give the product; yield: 2.97 g (59%); $R_f$ 0.55 (silica gel, 5% MeOH/$CH_2Cl_2$); MS *m/z* (%): 500 ($M^+$, 100). $^1H$ NMR ($CDCl_3$),$^{13}C$ NMR and elemental analysis conformed with the desired product.

## 10.1.1.5 Miscellaneous Methods for the Synthesis of *N*-Alkyl Amino Acids

Various alternative methods have been described for the preparation of *N*-alkyl amino acids. Most of the methods are applicable only to *N*-alkyl amino acids with nonfunctionalized side chains. The methods are summarized in Scheme 24.

*for references see p 265*

**Scheme 24** Miscellaneous Methods for the Synthesis of *N*-Alkyl Amino Acids[129,131,135,137–139]




In the synthesis of *N*-ethyl amino acids **52**[129] reaction of Meerwein's reagent[130] with *N*-acetyl amino acids **51** gave *N*-alkyl amino ether fluoroborates, which were reduced with $NaBH_4$ to give the product. Reductive alkylation of α-azidocarboxylic acids **53** by bromodimethylborane gives *N*-methyl amino acids **54**.[131] Controlled conditions (temperature and excess regents) were used that suppress epimerization. The α-azidocarboxylic acids **53** are readily accessible by various methods.[132–134] It should be noted that N-methylation of amino acids with bulky side chain (e.g., tBu, Ph) proceeds in high yield without epimerization.

Alkylation[135] of *N*-diphenylphosphoryl amino acids **55** by the method of Benoiton using MeI/NaH in THF[136] (see also Section 10.1.1.1.2) and the subsequent removal of the diphenylphosphoryl activating group by acid from the resulting *N*-diphenylphosphoryl-*N*-

methyl amino acid **56** gives *N*-methyl amino acids **57**. No racemization was observed in this process, however it is not applicable to amino acids with protected functionalized chains owing to the harsh conditions of removal of the diphenylphosphoryl group (6 M HCl). Taking advantage of the reactivity of the 2,2-bis(trifluoromethyl)oxazolidin-5-ones that are formed by the reaction of amino acids **58** with hexafluoroacetone, these oxazolidinones can be N-chloromethylated by a mixture of paraformaldehyde and thionyl chloride, and the chloromethyl group converted into an ethyl group by CuCN and MeLi.[137] Hydrolysis of the resulting 3-ethyl-2,2-bis(trifluoromethyl)oxazolidin-5-ones **59** gives the desired *N*-ethyl amino acids **60** in overall yields of 15–80%. Mild and efficient oxidative cleavage of borane–amine adducts, formed after the reduction of *N*-formyl or *N*-acyl amino acid esters **61** with diborane, gives *N*-alkyl amino acid esters **62**.[138] The method was not employed on amino acid esters with functionalized side chains, but was employed for SPPS. Asymmetric alkylation of pseudoephedrine sarcosinamide **63** gives an alkylation product **64** that is hydrolyzed by heating in aqueous NaOH to give the *N*-methyl amino acids **65**.[139] This reaction sequence proceeds without epimerization, however only two *N*-methyl amino acids with nonfunctionalized side chains were reported.

## 10.1.2 N-Alkylation of Peptides on Solid Supports

The emergence of solid-phase combinatorial chemistry has prompted the conversion of many chemical reactions, which were previously performed in solution, into a version that can be performed on a solid support. In a traditional solid-phase synthesis of N-alkylated peptides, the protected *N*-alkyl amino acids were prepared in solution and incorporated into the peptide as for any other amino acid. There are many advantages in performing the N-alkylation reaction during the peptide synthesis on the free amino group of the peptide resin. A method that allows site-specific quantitative monoalkylation simplifies the synthetic protocol in the sense that the alkylating synthons are easier to prepare than the protected *N*-alkyl amino acids. Moreover, for the preparation of combinatorial libraries in which the alkyl moiety as well as the amino acid being alkylated are diversified, site-specific alkylation has a profound advantage since much smaller numbers of alkylating synthons must be prepared. With the introduction of methods that allow reliable and high yielding monoalkylation of all the protein amino acids on the solid support with various alkylating agents, site-specific on-resin alkylation is gaining momentum. One of the main drawbacks of the on-resin site-specific alkylation is that the monoalkylation reaction must be quantitative since the existence of even small amounts of contaminants (of the nonalkylated amino acids) can have severe effects on the interpretation of biological structure–activity relationship studies, especially in cases where the nonalkylated peptide is several orders of magnitude more active than the N-alkylated peptide. In this sense, the use of premade protected *N*-alkyl amino acids and their incorporation in SPPS has an advantage since these synthons can be rigorously purified and characterized. It should be also mentioned that in many cases where the preparation of the protected alkylated amino acids are relatively simple, their incorporation is preferred by on-resin site-specific alkylation. Few of the methods used for N-alkylation of amino acids in solution (see Section 10.1.1) have been transformed to the solid phase; two of them use temporary protection/activation to ensure monoalkylation. The following guidelines were used in the choice of the temporary protecting group:

(1) There should only be one protecting group. The incorporation of two protecting groups would result in amino groups shielded from methylation. This would result in contamination of the product with non-methylated peptide, which in most cases is difficult to separate from the main product.

*for references see p 265*

(2) Bulky groups would be the best, but they should not be too sterically hindered as this would severely slow down the rate of the reductive alkylation.

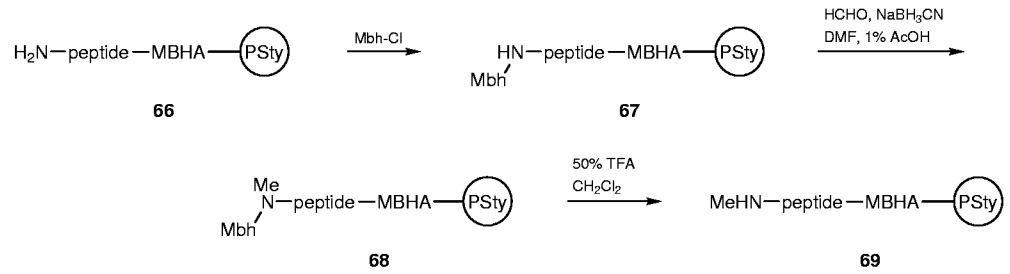
(3) The temporary protecting group should be readily removed to allow peptide elongation.

## 10.1.2.1 Site Specific N-Alkylation of Amino Acids and Peptides on Solid Support

### 10.1.2.1.1 Reductive Methylation of 4,4′-Dimethoxybenzhydryl-Protected Amino Acids

The three-step strategy developed by Quitt[115,116] (see Section 10.1.1.3.3 and Scheme 15) was applied to site-specific monomethylation of amino acids and peptides during SPPS (Scheme 25).[140] The methyl group is introduced by reductive alkylation of the amino groups on the solid phase. To circumvent the problem of polyalkylation of especially sterically unhindered amino acids the primary amino groups must be temporarily monoalkylated. To allow on-resin removal of the temporary protecting group (Quitt used Bzl) the 4,4′-dimethoxybenzhydryl group (Mbh) was used, which is an acid-labile, alkyl-type protecting group, and can be used in SPPS with Boc chemistry. The Mbh group is attached within 20 minutes, in high yield, to the primary amino group of peptide-resin **66**, using Mbh-Cl. The completeness of this reaction is monitored by the quantitative ninhydrin test.[141] If this protection is incomplete, the reductive alkylation that follows will yield a di-N-alkylated, truncated peptide, which is easily removed from the desired product during purification. Reductive methylation of **67** proceeds with a solution of 37% formalin solution diluted in DMF in the presence of $NaBH_3CN$. The completeness of the methylation reaction is controlled by small deprotection (withdrawing a small resin sample, cleaving the Mbh group, and determine the amount of residual primary amine by the quantitative ninhydrin test[141]). Monitoring of the reaction is important, as there is a significant difference in the rate of reductive methylation of different N-terminal Mbh-amino acid residues, trifunctional amino acids, such as Asn, Glu, or Gln, in particular show low activity. Therefore, it is likely, that side-chain interference with hydroxyamine formation and/or the reduction step, rather than steric effects, are the source of the relatively slow rate of reductive methylation. The removal of the Mbh group from **68** with 50% TFA/$CH_2Cl_2$ is rapid and complete within 1 hour giving the desire product **69**. Addition of TFA/$CH_2Cl_2$ produces a strong red color caused by formation of the Mbh cation. The formation of the red color is a convenient method to monitor the removal of the Mbh group.

**Scheme 25** Methylation of N-Terminal Amino Acid Residues on Solid Phase by the Mbh Method[140]



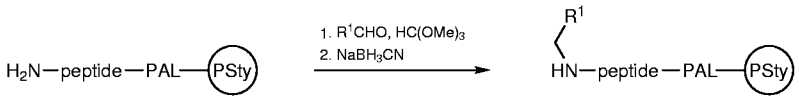

**Alkylation of Amino Groups with 4,4′-Dimethoxybenzhydryl Chloride:**[140]
The peptide-resin (100 mg, 0.5 mmol·g$^{-1}$ primary amino groups) was washed with 10% DIPEA/$CH_2Cl_2$ and $CH_2Cl_2$. The resin was suspended in $CH_2Cl_2$ (3 mL) followed by addition of DIPEA (50 µL, 0.29 mmol) and Mbh-Cl (13 mg, 0.05 mmol). The mixture was shaken for 20 min. The resin was washed with DMF (2 × 4 mL), DIPEA/$CH_2Cl_2$ (1 × 4 mL), and $CH_2Cl_2$ (3 × 4 mL). A small resin sample was tested for primary amino groups with the ninhydrin method.

**Reductive Methylation of 4,4′-Dimethoxybenzhydryl-Protected Amino Groups:**[140]
Formaldehyde soln [3 mL, prepared by diluting 37% formalin (10 mL) with DMF (90 mL) and treatment with anhyd $MgSO_4$ (30 g), after stirring overnight and centrifugation the supernatant was used for methylation] was added to the peptide resin (50 mg) and shaken for 5 min. The solvent was removed by filtration, and an additional portion of formaldehyde soln and AcOH (30 μL) were added. After 5 min $NaBH_3CN$ (30 mg) was added to the mixture and the reaction vessel was shaken for 1 h. The completeness of the methylation reaction was controlled by withdrawing a small resin sample (5 mg), deprotection of the Mbh group and testing for primary amino groups with the ninhydrin method. When the methylation was completed, the Mbh protecting group was removed by treatment with 50% TFA/$CH_2Cl_2$ for 1 h.

### 10.1.2.1.2 On-Resin Reductive Alkylation in Trimethyl Orthoformate

Trimethyl orthoformate has been used as a solvent by Campbell and co-workers for site-selective monoalkylation in SPPS.[113] Trimethyl orthoformate was previously shown to be a highly efficient dehydrating solvent for the formation of imines.[112] Thus, dividing the reductive alkylation step into two separate substeps of (1) imine formation followed by solvent removal and (2) reduction, resulted in on-resin monoalkylation of various amino acids and dipeptides (Scheme 26).

**Scheme 26** On-Resin Reductive Alkylation in Trimethyl Orthoformate[113]




70 71

**Alkylation of H-Phe-Phe-$NH_2$; General Procedure:**[113]
To H-Phe-Phe on PAL resin (170 mg, 0.046 mmol) suspended in trimethyl orthoformate (2.0 mL) under argon was added propanal (74 μL, 1.03 mmol) and the mixture was shaken for 30 min. The resin was drained and $NaBH_3CN$ (64 mg, 1.03 mmol) dissolved in trimethyl orthoformate (1 mL) was added. After 10 min the resin was drained, and either washed with DMF (3 × 2 mL) and the synthesis continued or alternatively the resin was washed with MeOH and $Et_2O$ and dried, and the H-(Pr)Phe-Phe-$NH_2$ removed from the resin with TFA.

### 10.1.2.1.3. On-Resin Alkylation of Nitrophenylsulfonyl-Protected Peptides

Miller and Scanlan described the first general on-resin method for site-selective N-methylation at any position in a given peptide.[142] This method is compatible with Fmoc solid-phase peptide synthesis and like the method of Kaljuste and Unden[140] (Section 10.1.2.1.1) avoids dimethylation of the amino termini and polymethylation of the peptide.

Nitrobenzenesulfonamides were introduced for the alkylation of primary amines (see Section 10.1.1.1.1) and have proved to be applicable to the solid phase. A comparison of three methods for on-resin site-selective alkylation using Nbs is shown in Table 9. The free amine of a peptide-resin **72** can be readily protected with 2- or 4-nitrobenzenesulfonyl chloride (oNbs-Cl or pNbs-Cl) in $CH_2Cl_2$ containing collidine (Scheme 27, step 1). The sulfonamide **73** is selectively deprotonated with the strong, nonionic guanidinium base 1-methyl-1,3,4,6,7,8-hexahydro-2*H*-pyrimido[1,2-*a*]pyrimidine (7-methyl-1,5,7-triazabicyclo-[4.4.0]dec-5-ene, MTBD) and alkylated with methyl 4-nitrobenzenesulfonate in DMF or it was alkylated under Mitsunobu conditions (Scheme 27, step 2). Quantitative cleavage of oNbs and pNbs from **74** was achieved with benzenethiol and DBU in DMF giving the desired product **75** (Scheme 27, step 3). It was found that the oNbs group cleaved more readily than pNbs. 2-Sulfanylethanol was substituted for benzenethiol, since it cleaved the sulfonamide faster and is less toxic and less noxious.[143]

*for references see p 265*

**Scheme 27** On-Resin Alkylation of Nbs-Protected Peptides[85,142,143]

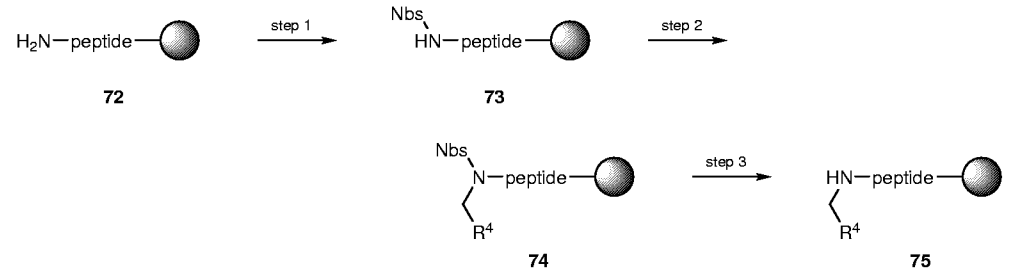



**Table 9** Comparison of Three Methods for On-Resin Site-Selective Alkylation Using the Nbs Method

| | Miller and Scanlan[142] | Reichwein and Liskamp[143] | Wisniewski[144] |
|---|---|---|---|
| Resin | Rink amide MBHA (Fmoc chemistry) | TentaGel-OH (Fmoc chemistry) | Trt resin (Fmoc chemistry) |
| Step 1[a] | ONbs-Cl, collidine, $CH_2Cl_2$, 2 h or coupling of oNbs-Xaa$^1$-OH with HBTU, NMM, DMF, 20 min | ONbs-Cl, DIPEA, DMF 0.5 h | ONbs-Cl, collidine, $CH_2Cl_2$, 2 h |
| Step 2 | MTBD[b] (DMF), pNbs-Me, DMF, 30 min or allyl methyl carbonate, $Pd_2(dba)_3$–$CHCl_3$ adduct, TPP, THF, Ar, 2 h, or $Me_2SO_4$, MTBD[b], DMF, 2 h | Mitsunobu: DEAD, TPP, EtOH, $CH_2Cl_2$, 0.5 h | Mitsunobu: $H_2N$-peptide resin, $R^4CH_2OH$, premade soln of TPP/DIAD, DME, 5 min at 0 °C |
| Step 3 | $HSCH_2CH_2OH$, DBU, DMF, 0.5 h | tBuOK, PhSH, DMF, 0.5 h or $HSCH_2CH_2OH$, DBU, DMF, 0.5 h | $HSCH_2CH_2OH$, DBU, DMF, 1 h |
| Applications | 1. Me-scan of SFLLRN-$NH_2$<br>2. Allyl-scan of LLRN (HATU; 15% yield) | Et-scan of Boc-Y(tBu)GGFL-OMe (Py-BroP/HOAt, DMF; 12–43% yield) | preparation of Fmoc-Gly$\psi$($CH_2NH$)Xaa-OH (conversion 96%) |

[a] See Scheme 27.
[b] MTBD = 1-methyl-1,3,4,6,7,8-hexahydro-2*H*-pyrimido[1,2-*a*]pyrimidine.

#### 10.1.2.1.4 On-Resin Formation of *N*-Alkylglycines

On-resin formation of *N*-alkylglycines can be carried out according to the methods described in Section 10.1.3.4.2.

### 10.1.2.2 Nonspecific N-Alkylation of Peptides

Houghten and co-workers[145] introduced a method for combinatorial synthesis of a per-alkylated peptide library using nonspecific N-alkylation. The peptides were synthesized by SMPS methodology[146] in combination with repetitive amide N-alkylation on the solid support after each coupling step. Peptides were synthesized on MBHA-PSty resin using Fmoc chemistry. After Fmoc deprotection the α-amino group was protected by Trt to prevent $N^{\alpha}$-alkylation and to allow only amide alkylation. The on-resin amide alkylation was achieved by amide proton abstraction using LiOtBu in THF followed by nonfunctionalized alkyl and aryl halides in DMSO.

**On-Resin Alkylation of $N^{\alpha}$-Trt-Xaa$^1$-MBHA-resin:**[145]
All manipulations were carried out under $N_2$ and strictly anhydrous conditions. Trt-Xaa$^1$-MBHA-PSty-resin (250 mg, 11.07 mmol) in polypropylene packet (5 × 5 cm) was dried overnight at 50 mTorr. The packet was placed in a round-bottomed flask. 1 M LiOtBu in THF (22 mL, 22 mmol) and THF (22 mL) were added and the vessels shaken for 15 min. Excess base soln was removed by cannulation, following addition of DMSO (44 mL) and the alkylating agent (66.5 mmol). The mixture was vigorously shaken at

rt for 2 h. The alkylating agent was removed by cannulation and the entire procedure repeated twice. The packet was washed [DMF (3 ×), iPrNH$_2$ (2 ×), $CH_2Cl_2$ (3 ×), MeOH (1 ×); 8 mL each) and dried. Following complete drying overnight at 50 mTorr, the process above was repeated three times for allylation, benzylation, and naphthylmethylation (each alkylation, 2 × 2 h and 1 × 5 h).

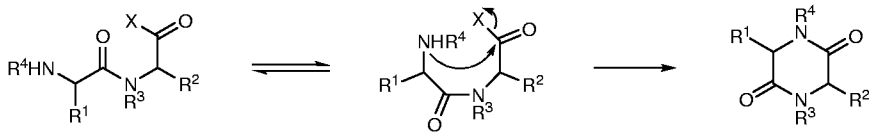
## 10.1.3 Synthesis of N-Alkylated Peptides

The synthesis of peptides containing cyclized *N*-alkyl amino acids such as Pro (or other amino acids that contain $N^{\alpha}$–$C^{\alpha}$ cyclized systems) and acyclic *N*-alkyl amino acids may encounter various problems that lead to side reactions and, therefore, this results in low yields of the desired peptides. In many cases the incorporation of Pro, Sar, alkyl-Gly, or even some *N*-methyl amino acid residues in a peptide is straightforward and the reaction proceeds under normal peptide synthesis conditions. However, in some cases the incorporation of the same residue is not straightforward and leads to undesired reactions and side products. In other cases the incorporation of certain *N*-alkyl amino acids is extremely difficult and even impossible. Some peptides that contain *N*-alkyl amino acids are unstable under various conditions used in peptide synthesis and some decompose spontaneously even under very mild conditions. The difficulties encountered over the years during attempts to synthesize important naturally occurring biologically active peptides or model peptides for proteins containing *N*-alkyl amino acids provides the impetus for the development of new synthetic tools that not only make the preparation of more complicated *N*-alkyl peptides possible but these new synthetic tools have also been applied to the general synthesis of peptides. In this section the problems encountered during the synthesis of peptides containing *N*-alkyl amino acids are listed and the various methods suggested to overcome them are described. For additional literature on this subject see refs[147–152] and references cited therein.

### 10.1.3.1 Preventing Piperazine-2,5-dione Formation

Some peptide derivatives containing *N*-alkyl amino acids, especially Pro or Hyp, have a tendency to undergo cyclization to form piperazine-2,5-diones (diketopiperazines, DKP), or amino acid anhydrides.[149,151] Formation of piperazine-2,5-diones is explained by the occurrence of the *cis*-isomer of the tertiary amide bond (Scheme 28).[19–21]

**Scheme 28** Formation of Piperazine-2,5-diones via a *cis* Peptide Bond Isomer




$R^4$ = H, $R^3$ = alkyl; $R^4$ = alkyl, $R^3$ = H; $R^4$ = $R^3$ = alkyl; $R^4$ = protecting group such as Z
X = $OR^5$ (including unhindered ester linkage to the solid support and active ester), $NH_2$, $NHR^5$

The spontaneous formation of piperazine-2,5-diones occurs mainly during N-deprotection or the acylation step to dipeptide esters (usually unhindered esters such as Me, Et, Bzl, and Pac esters) that contain an *N*-alkyl amino acid especially at the C-terminus.[153] In some cases the formation of piperazine-2,5-diones becomes the major reaction product and thus prevents peptide elongation by the [1+2] or [1+3] segment condensation strategy in solution synthesis or elongation of the peptide from the C-terminus in SPPS. Piperazine-2,5-dione formation

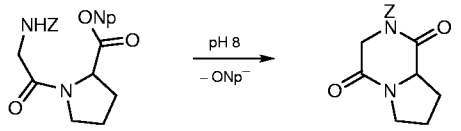
*for references see p 265*

has been noted both during solution and SPPS especially with peptides containing Pro and Hyp[149–151] but also in peptides that contain *N*-methyl amino acids.[154,155]

The extent of piperazine-2,5-dione formation depends on many factors. Among these are: (1) the nucleophilicity of the nitrogen participating in the cyclization step; (2) the stability of the CO–X bond; (3) steric hindrance exerted by the various alkyl groups ($R^1$, $R^2$, $R^3$, $R^4$) and by groups connected to X; (4) solvents and catalysts; and (5) the thermodynamic stability of the piperazine-2,5-dione. Because so many factors influence the rate of piperazine-2,5-dione formation, it is hard to predict when its formation will hamper the synthesis of a given peptide. However, using the vast amount of information that has accumulated over the years, the formation of piperazine-2,5-diones can now be predicted and their formation prevented by using various methods, some of which are described (*vide infra*). For example, the solution synthesis of TRH (pGlu-His-Pro-$NH_2$) could not be performed by the [1+2]-condensation method due to facile cyclization of H-His-Pro-OMe to the corresponding piperazine-2,5-dione *c*[-His-Pro-]. To overcome this side reaction the [2+1]-condensation method was employed. Thus, coupling of pGlu-His-$N_3$ to H-Pro-$NH_2$ gave the desired product.[156] Dipeptide amides having an *N*-alkyl amino acid at the C-terminus are also inclined to form piperazine-2,5-diones. For example, the dipeptide H-Phe(4-$NO_2$)-Pro-$NH_2$ cyclized to the piperazine-2,5-dione upon recrystallization.[157] Another example concerns preparation of the TRH analogue pAad-His-Thz-$NH_2$ (pAad-OH = pyro-α-aminoadipic acid) by the [1+2] method. In this case the attempt to couple pAad-OH to H-His(Dnp)-Thz-$NH_2$ led to *c*[-His(Dnp)-Thz-] as the major product. However, a high yield of the desired product was obtained by preactivation of pAad-OH with DCC/HOBt followed by coupling to H-His(Dnp)-Thz-$NH_2$.[158]

Piperazine-2,5-dione formation can occur not only in the case of dipeptides esters or amides containing a free amino group, but also during the activation of N-blocked or protected *N*-alkyl amino acids containing dipeptides or even tripeptides. Thus, the activated dipeptide Z-Gly-Pro-ONp (**76**) afforded the Z-protected piperazine-2,5-dione **77** when subjected to buffered dioxane solution at pH 8 (Scheme 29).[159]

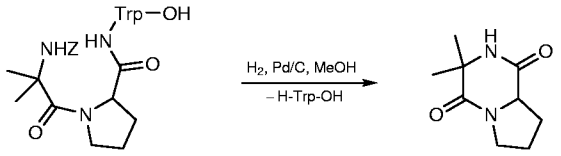
**Scheme 29** Formation of a Z-Piperazine-2,5-dione from a Z-Protected Activated Peptide[159]




**76** **77**

Many examples of peptide chain fragmentation are known when peptides or their derivatives containing *N*-alkyl amino acids are subjected to conditions that promote piperazine-2,5-dione formation. These conditions include high temperatures, solvents, such as pyridine, phenol, and butan-2-ol,[160,161] or hot 50% AcOH for 24 hours.[162] Cyclizations of this type can be minimized by avoiding these harsh conditions. However, in some other cases spontaneous splitting of a peptide bond may occur under extremely mild conditions. Deprotection of the Z group from Z-Aib-Pro-Trp-OH (**78**) by hydrogenolysis in methanol at 0 °C resulted in fast peptide decomposition and formation of *c*[-Aib-Pro-] (**79**) and H-Trp-OH[163] (Scheme 30). This facile peptide fragmentation was interpreted by the ability of the Aib *gem*-dimethyl group to enhance the rate of the cyclization reaction by stereopopulation control.[164]

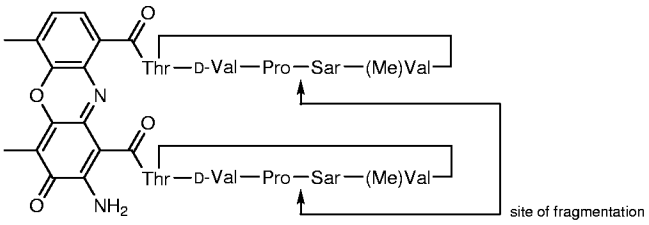
**Scheme 30** Spontaneous Decomposition of Z-Aib-Pro-Trp-OH upon Hydrogenolysis[163]




78 79

Formation of piperazine-2,5-dione is facilitated in dipeptides in which one residue has the L- and the other the D-configuration. The reason for this is the energy destabilization of the L,L- or D,D-piperazine-2,5-dione due to interference of the two side chains $R^1$ and $R^2$ (Scheme 28) being on the same side of the average plane of the ring. In a D,L-DKP the two side chains are on opposite sides of the ring plane which minimizes their interference and thus increases stability. The stability of D,L-piperazine-2,5-diones is a driving force that may lead to spontaneous fragmentation of a peptide chain. For example, during the total synthesis of actinomycin D[165] (Scheme 31), the tripeptide H-D-Val-Pro-Sar-OH underwent spontaneous decomposition to the piperazine-2,5-dione *c*[-D-Val-Pro-] and H-Sar-OH.

**Scheme 31** The Structure of Actinomicin D[165]




This tripeptide was a key intermediate in an attempted preparation of *c*[3-(benzyloxymethyl)-2-nitrobenzoyl-Thr-D-Val-Pro-Sar-(Me)Val-] by the [2+3]-segment condensation strategy peptide and cyclization between Sar and (Me)Val. This side reaction was circumvented by using the dipeptide H-D-Val-Pro-OtBu in a [3+2]-segment condensation strategy. Contrary to its benzyl ester, this dipeptide did not form the piperazine-2,5-dione because of presence of the bulky *tert*-butyl ester.[165,166] Cyclization was performed between Pro and Sar.

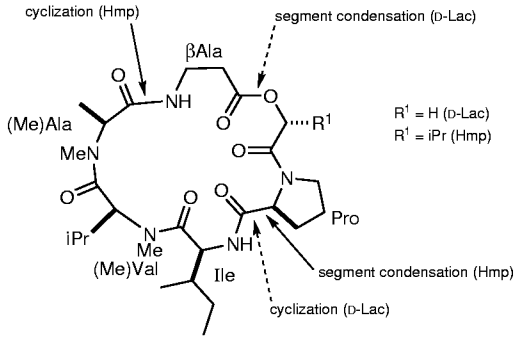
Peptides that contain two or more successive *N*-alkyl amino acids have a high tendency to form piperazine-2,5-diones. Ring closures yielding the piperazine-2,5-dione *c*[-Pro-Pro-] are particularly facile.[166] Also formation of piperazine-2,5-diones upon N-deprotection of the dipeptide H-(Me)Leu-(Me)Val-OBzl was the main obstacle in the synthesis of the 7–11 fragment of cyclosporin, when extending the peptide chain from the C-terminus.[167] The desired peptide, Boc-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-OH, was prepared by using the *tert*-butyl ester instead of the benzyl ester. In this case the steric hindrance exerted by the bulky *tert*-butyl group prevented the formation of the piperazine-2,5-dione upon N-deprotection of the dipeptide H-(Me)Leu-(Me)Val-OtBu and during its acylation. Using the same strategy with different coupling agents,[168] a similar fragment Fmoc-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-OtBu was prepared in higher yield.

Another example may be found in the preparation of destruxin B (Scheme 32). Two syntheses of destruxin B and its D-$Lac^6$ analogue, overcome the high tendency of consecutive *N*-alkyl amino acids to undergo cyclization during peptide synthesis. In both syntheses the choice of strategy was dictated by attempts to minimize piperazine-2,5-dione formation from the sequence -(Me)Val-(Me)Ala-. The synthesis of the D-$Lac^6$ analogue employed a [4+2]-segment condensation and cyclization at the -Pro-Ile- site. The fragment Boc-Ile-(Me)Val-

*for references see p 265*

(Me)Ala-βAla-OH was prepared by stepwise elongation from the C-terminus. The first step in the preparation of this tetrapeptide involved the synthesis of H-(Me)Ala-βAla-OMe. This dipeptide did not cyclize because of the presence of the β-amino acid at the C-terminus. The tetrapeptide fragment was coupled to H-D-Lac-Pro-OtBu. This segment condensation approach involved the formation of the backbone ester bond. Deprotection and cyclization by the DPPA method afforded [D-Lac$^6$]destruxin in 30% yield.[169]

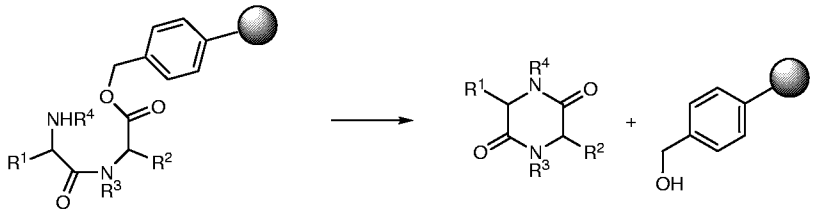
**Scheme 32** Structure of Destruxin and Cyclization Sites within the Peptide Chain




The synthesis of destruxin B [R$^1$ = (*R*)-2-hydroxy-4-methylpentanoic acid (Hmp)] exploited a [3+3]-segment condensation. In this case both segment condensation and cyclization involve peptide bond formation: the first at the -Pro-Ile- site and then at the -(Me)Ala-βAla- site. The key to the success of this synthesis is the finding that the dipeptide H-(Me)Val-(Me)Ala-NHNHBoc undergoes cyclization at a much slower rate ($t_{1/2}$ = 16 h; $CDCl_3$) than the related benzyl ester ($t_{1/2}$ = 30 min; $CDCl_3$). Thus, the fragment H-Ile-(Me)Val-(Me)Ala-NHNHBoc could be prepared in high yield and coupled to the fragment Boc-βAla-Hmp-Pro-OH by the DCC/HOBt method. Final deprotection and cyclization by the azide method afforded destruxin B in 65% yield.[170]

Formation of piperazine-2,5-diones from dipeptide esters containing Pro or other *N*-alkyl amino acids has also been observed during SPPS. The dipeptides most prone to cyclization are those that contain a C-terminal Pro or other *N*-alkyl amino acids connected via an ester bond to a sterically unhindered handle.[171–178] Scheme 33 depicts the formation of piperazine-2,5-diones **81** from dipeptides **80** in which the *N*-alkyl amino acid is connected via a benzyl ester to the solid support.

**Scheme 33** Formation of Piperazine-2,5-diones on SPPS



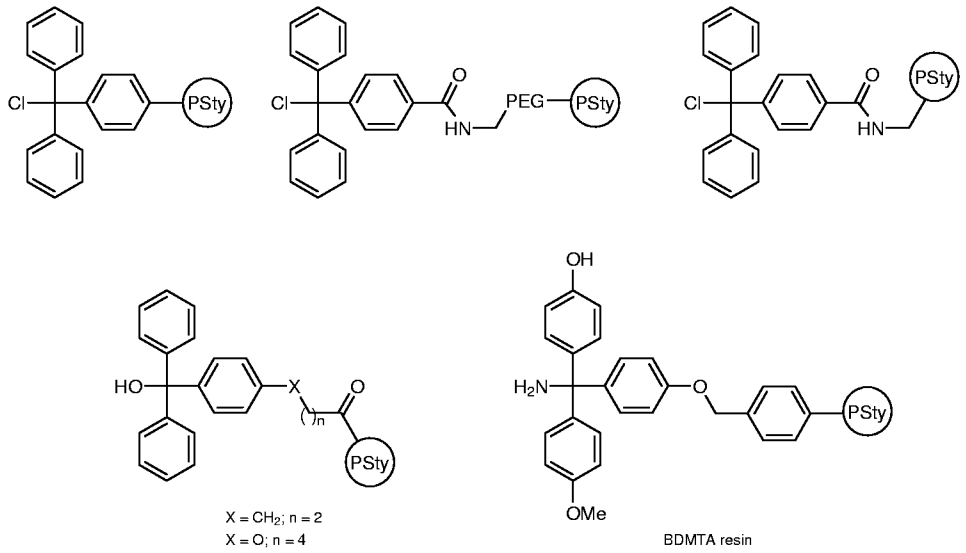

**80** **81**

As in the case of dipeptide esters in solution, the rate of piperazine-2,5-dione formation in SPPS strongly depends on steric and chiral features of the original dipeptide structure.[150] The rate of the cyclization increases upon catalysis by weak carboxylic acids.[176] However, it was also reported that piperazine-2,5-diones are formed under basic conditions, e.g. removal of the Fmoc protecting group. Formation of piperazine-2,5-diones is generally overcome by

the 2+1 strategy, namely, coupling of a premade N-protected dipeptide to the *N*-alkyl amino acid-resin.[179]

Many biologically active peptides (e.g., bradykinin potentiators B and C; hACTH 1–24; h-angiotensin 1–7; calcitonin free acid; fibrinogen fragments used as fibronectin inhibitors, etc.) are characterized by a Pro residue at their C terminus. It is obvious that formation of piperazine-2,5-diones is a serious problem in the synthesis of these peptides and hampers SAR studies. The solution afforded by the 2+1 strategy is useful only in cases when one distinct peptide is to be prepared. However, it is cumbersome and impractical to use in the preparation of combinatorial peptide libraries by the SPPS method since numerous dipeptides must be prepared beforehand. The best solution to the problem of piperazine-2,5-dione formation during SPPS of peptides containing C-terminal Pro or other *N*-alkyl amino acids was furnished by the use of resins bearing the trityl handle. In this case the bulk of the trityl linker prevents cycloaminolysis to piperazine-2,5-diones. The trityl handles were initially introduced to allow SPPS of partially protected peptide fragments by the Fmoc strategy, via their cleavage from the resin as free acids by extremely mild acidic conditions.[180–184] The preparation[180] by SPPS of calcitonin and ACTH fragments bearing Pro free acid at their C-terminus in high yield, led the trityl ester linker (Scheme 35) to become the method of choice for SPPS, by the Fmoc strategy, of peptide acids containing Pro or other *N*-alkyl amino acids at their C-terminus.[185,186] It should be mentioned that the ability to synthesize protected peptides or protein fragments by a method that avoids formation of piperazine-2,5-diones opens additional latitude in the total synthesis of proteins by fragment condensation and in the synthesis of protected fragments as precursors for cyclic peptides.

**Scheme 34** Trityl Anchors Suitable for SPPS of Peptides with C-Terminal *N*-Alkyl Amino Acids[180–187]




Peptides that contain amidated Pro or other *N*-alkyl amino acids at their C-terminus are less prone to formation of piperazine-2,5-diones in SPPS. Apparently, the use of a benzhydrylamine-type resin exerts enough steric hindrance to slow down piperazine-2,5-dione formation. In some cases, e.g., the synthesis of peptide amides that have -D-Pro-Pro-$NH_2$ or -Aib-Pro-$NH_2$ sequences, the use of the 4-benzyloxy-4′-4″-dimethoxytrityl amine resin (BDMTA resin, Scheme 34)[187] is recommended.

*for references see p 265*

**SPPS of Peptides Containing *N*-Alkyl Amino Acids at Their C-Terminus; General Procedure:**[180–184]
*Loading of the chlorotrityl resin:* the Fmoc amino acid derivative (1.5 equiv) was dissolved in dry $CH_2Cl_2$ ($10\,mL \cdot g^{-1}$ resin; loading: 0.8–1 $mmol \cdot g^{-1}$) and DIPEA (3 equiv relative to the Fmoc amino acid derivative) was added. This soln was added directly to the dry resin and the suspension was shaken. After 1 h, MeOH (1 mL/g resin) was added to cap unreacted trityl chloride groups, and the suspension was shaken for a further 20 min. The resin was then washed with $CH_2Cl_2$ (3 × 2 min), DMF (3 × 2 min), and MeOH (3 × 2 min) and dried in vacuo. Substitution of the resin was determined gravimetrically and was found in the range between 0.5 and 0.6 $mmol \cdot g^{-1}$.

*Coupling Protocol:*
Step. reagent, operation time.
1. NMP washing (5 ×), 2 min
2. 20% piperidine/NMP deprotection, 4 min.
3. 20% piperidine/NMP deprotection, 25 min.
4. NMP washing (5 ×), 2 min.
5. Fmoc-Xaa$^1$-OH, HOBt, TBTU, DIPEA, NMP coupling, 30 min.
6. Fmoc-Xaa$^2$-OH, HOBt, TBTU, DIPEA, NMP coupling, 30 min.

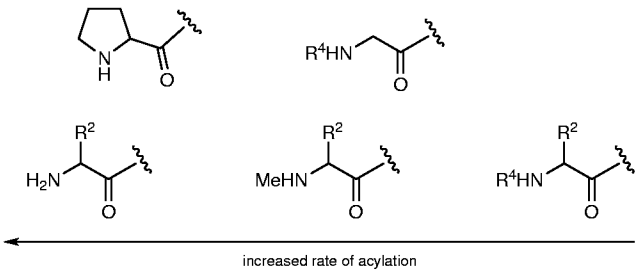
*Cleavage of peptide from the resin (with side-chain deprotection):* peptides were cleaved by a mixture of TFA/$H_2O$/TIS (90:3:7; $10\,mL \cdot g^{-1}$ resin) for 1 h. The resin was filtered off and washed with TFA (3 ×). The collected filtrates were taken to dryness in vacuo and the peptide was precipitated by ice-cold $Et_2O$. The precipitate was collected by centrifugation, resuspended in $Et_2O$ and collected by centrifugation twice. The precipitate was dried and lyophilized.

*Cleavage of peptide from the resin (without side-chain deprotection):* protected peptides were cleaved by a mixture of 20% HFIP/$CH_2Cl_2$ ($10\,mL \cdot g^{-1}$ resin) for 15 min. The procedure was repeated twice and then washed with $CH_2Cl_2$ (3 ×). The collected filtrates were taken to dryness in vacuo and the peptide was dissolved in tBuOH and lyophilized. An alternative more efficient and faster mixture was: $CH_2Cl_2$/HFIP/TFE/TIS (6:2:1:0.5) twice for 5 min.[271]

## 10.1.3.2 Acylation of *N*-Alkyl Amino Acids during Peptide Synthesis

Steric factors play a major role in the acylation of *N*-alkyl amino acids during peptide synthesis. It is expected that the increased nucleophilicity of secondary amines, as compared to primary amines, will increase the rate of acylation. However, the opposite is observed: acylation of *N*-alkyl amino acids (except for Pro) is usually slower than the acylation of primary amines. This is explained by steric hindrance exerted by the *N*-alkyl group, which shields the nucleophilic center. The steric effect increases with the size of the *N*-alkyl group and is enhanced by bulky side chains and, to a lesser extent, by other remote groups. The relative rate of acylation of various types of *N*-alkyl amino acids, as compared to amino acids, is shown in Scheme 35.

**Scheme 35** Relative Rate of Acylation of Various *N*-Alkyl Amino Acids during Peptide Synthesis

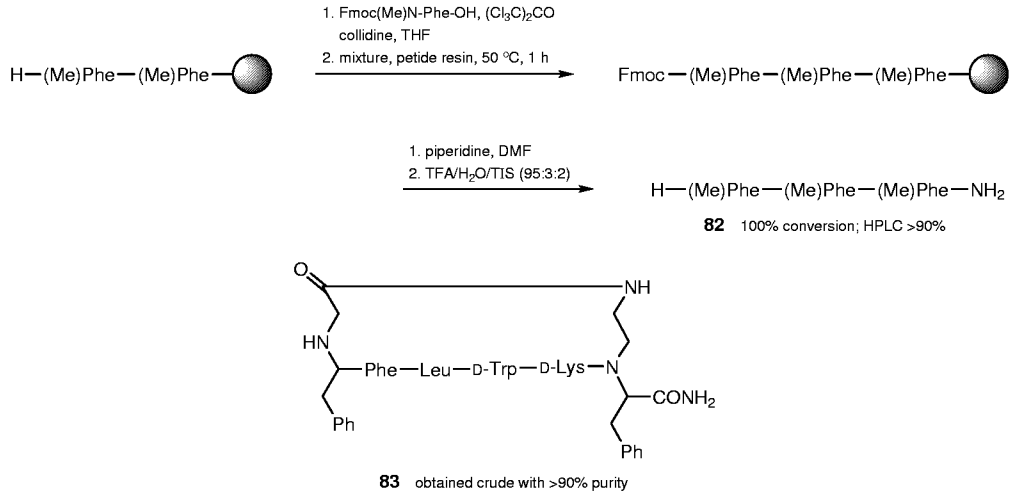



Acylation of the secondary amino group of Pro usually occurs in excellent yields.[150,166] Apparently, the steric effects exerted by the δ-methylene are minimized by the ring and are also compensated by the increase in nucleophilicity of the secondary amino group. In contrast, acylation of acyclic *N*-alkyl amino acids is difficult and standard reagents are often inefficient. Acylation of *N*-methyl amino acids and *N*-alkylglycines ($R^2 = H$; $R^4$ other than

methyl) are easier than acylation of *N*-alkyl amino acids ($R^4$ other than methyl). However, coupling of several *N*-methyl amino acids to each other is enormously difficult (e.g., cyclosporin). To overcome these difficult coupling steps during SPPS, several coupling reagents have been evaluated and recommended. These include: PyBroP,[188] PyCloP, PyAOP, PyCIU (chlorodipyrrolidinocarbenium hexafluorophosphate), HATU, HATU/HOAt,[189,190] TOPPipU,[191] CIP/HOAt, and Mukaiyama's reagent.[192] In addition the active species UNCA[193] and acid fluorides and acid chlorides[194,195] have been also recommended. Among these, HATU was extensively used for the SPPS of cyclosporin analogues and fragments.[196,197]

Bis(trichloromethyl) carbonate has been introduced for the in situ generation of *N*-Fmoc amino acid chlorides and is the method of choice for the preparation of N-alkylated peptides by SPPS.[30] The use of acid chlorides for the generation of a dipeptide was introduced by Fischer.[198] The general use of premade protected amino acid chlorides in SPPS was previously limited mainly due to the fact that chlorides of *N*-Fmoc amino acids bearing acid-labile side-chain protecting groups (e.g., tBu, Boc, Trt) have limited shelf stability. In addition the preparation of chlorides derived from Fmoc-Lys(Boc)-OH, Fmoc-Trp(Boc)-OH, Fmoc-Cys(Trt)-OH, Fmoc-Gln(Trt)-OH, and Fmoc-Arg(Pmc)-OH is problematic because of side reactions and requires special reaction conditions, purification, and storage.[194] These problems hampered the general use of the acid chloride method in the synthesis of peptides and especially in SPPS of N-alkylated peptides, despite its initial success.[194,195] However, in situ generation of *N*-Fmoc amino acid chlorides by using bis(trichloromethyl) carbonate[30] and immediate use of the generated acid chloride is the most efficient method for the acylation of N-alkylated peptides. This method does not cause racemization to any of the 30 *N*-Fmoc amino acids tested (including *N*-Fmoc-*N*-methyl amino acids and building units for backbone cyclization) with the exception of Fmoc-Cys(Trt)-OH (10%) and Fmoc His(Trt)-OH (10%). The coupling was highly efficient with conversions of 100% in most cases [with the exception of couplings involving Fmoc-Asn(Trt)-OH]. The crude peptides obtained were of a high degree of purity. Among these are two peptides **82** and **83** (Scheme 36) that were previously obtained in very poor yield using other coupling methods.

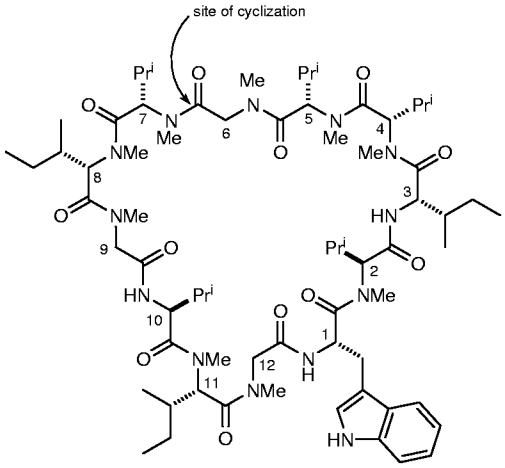
**Scheme 36** SPPS of N-Alkylated Peptides by the Bis(trichloromethyl) Carbonate Method[30]




*for references see p 265*

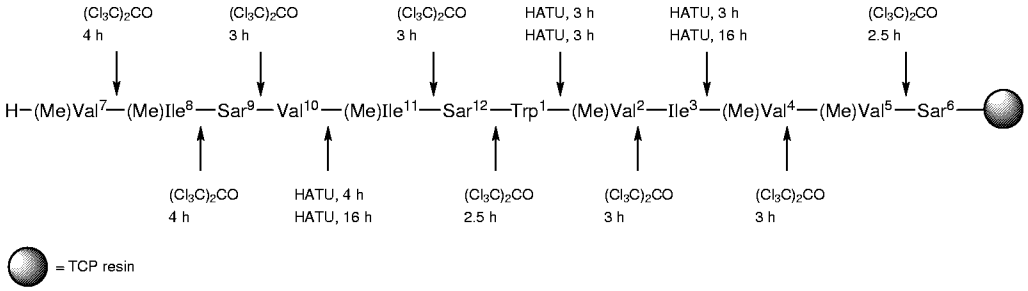
A slightly modified version of the bis(trichloromethyl) carbonate method[30] has been used for the total SPPS of cyclosporin O[29] (Scheme 1) and omphalotin A[40] (Scheme 37). Since in this case an acid sensitive trityl resin was used, excess DIPEA was added to the resin.

**Scheme 37** The Structure of Omphalotin A[40]




An outline of the synthesis of the linear dodecapeptide precursor of omphalotin A is shown in Scheme 38.

**Scheme 38** Synthesis Outline of the Linear Precursor of Omphalotin A[40]




After the reaction times given in Scheme 38, the respective couplings were quantitative according to chloranil and the Kaiser ninhydrin test and/or HPLC of small cleavage. The peak area of the crude Fmoc-deprotected dodecapeptide was 90% (HPLC $\lambda = 214$ nm). The pure cyclic peptide was obtained in an overall yield of 31%. Cyclosporin O was synthesized by a similar procedure.[29]

Tables 10 and 11 summarize the solution and solid-phase synthesis, respectively, of many peptides that contain at least two consecutive N-methyl amino acid residues.

**Table 10** Synthesis of N-Alkylated Peptides in Solution[168–170,199,200]

| Peptide[a] | Reagent[b] | Stoichiometry[c] (equiv) R/N/C/B | Temp[d] (°C) | Time (h) | Yield[e] (%) | Epimer[f] (%) | Ref |
|---|---|---|---|---|---|---|---|
| Z-(Me)Val-(Me)Ala-NHNH-Boc | PyBroP | 1.2:1.1:1:2 | 0 to rt | 18 | 95 (i) | <0.5 | [170] |
| Z-Ile-(Me)Val-(Me)Ala-NHNH-Boc | Bop-Cl | 1.2:1.1:1:4 | 0 to rt | 14 | 84 (i) | 2.3 | [170] |
| Z-(Me)Val-(Me)Val-OMe | PyBroP | 1.5:1.5:1:4 | 0 to rt | 3 | 87 (H) | <0.3 | [170] |
| Fmoc-(Me)Val-(Me)Val-OMe | PyBroP | 1.5:1.5:1:4 | 0 to rt | 3 | 83 (H) | <0.3 | [170] |
| Boc-(Me)Leu-(Me)Val-OMe | PyBroP | 1.5:1.5:1:4 | 0 to rt | 3 | 88 (H) | <0.6 | [170] |
| Z-(Me)Leu-(Me)Val-OMe | CIP/HOAt | 2/1:2:1:6 | 25 | 3 | 91 (i) | <0.5 | [199] |
| Z-(Me)Leu-(Me)Leu-OMe | CIP/HOAt | 2/1:2:1:6 | 25 | 3 | 97 (i) | <1.5 | [199] |
| Z-(Me)Val-Pro-OtBu | PyBroP | 1.5:1:1:3.5 | rt | 1 | 76 (i) | | [200] |
| Z-Val-(Me)Val-Pro-OtBu | PyBroP | 1.5:1:1:3.5 | rt | 1 | 93 (i) | | [200] |
| Boc-Dil-Pro-Hiva-Dpy[g] | PyBOP | 1.5:1:1.3.5 | rt | 1 | 88 (i) | | [209] |
| Boc-Val-Dil-Pro-Hiva-Dpy[g] | PyCloP | 1.5:1:1.3.5 | rt | 24 | 88 (i) | | [209] |
| Fmoc-(Me)Val-Pro-Dap-Doe[g] | BroP | 1.5:1:2:3.5 | rt | | 92 (i) | | [200] |
| Fmoc-Val-(Me)Val-Pro-Dap-Doe[g] | BroP | 1.5:1:2:3.5 | rt | | 80 (i) | | [200] |
| Boc-(Me)Val-(Me)Ala-βAla-OMe[h] | DCC/HOBt | 1.1/1:1:1:1.1 | 0/rt | 1/72 | 60 (i) | | [169] |
| Fmoc-(Me)Val-(Me)Ala-βAla-OMe[h] | BroP | 1.1:1:1.1:4 | 0/rt | 3 | 50 (i) | | [169] |
| Z-(Me)Val-(Me)Ala-βAla-OMe | BroP | 1.1:1:1.1:4 | 0/rt | 14 | 80 (i) | | [169] |
| Boc-Ile-(Me)Val-(Me)Ala-βAla-OMe[h] | DCC/HOBt | 1.1/1:1:1:1.1 | 0/rt | 1/48 | 50 (i) | | [169] |
| Fmoc-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 2 | 90 (i) | <1 | [168] |
| Fmoc-(Me)Leu-D-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 2 | 79 (i) | <1 | [168] |
| Fmoc-D-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 2 | 70 (i) | <1 | [168] |
| Fmoc-(Me)Leu-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 18 | 87 (i) | <1 | [168] |
| Fmoc-D(Me)Leu-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 18 | 85 (i) | <1 | [168] |
| Fmoc-Ala-(Me)Leu-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 18 | 78 (i) | <1 | [168] |
| Fmoc-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-OtBu[h] | BOP-Cl | 1.1:1:1.05:2.2 | 0/rt | 18 | 85 (i) | <1 | [168] |
| Z-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0 | 2 | 93 (i) | <1 | [168] |
| Z-(Me)Leu-(Me)Leu-(Me)Val-OtBu | BOP-Cl | 1.1:1:1.05:2.2 | 0/10 | 24 | 99 (i) | <1 | [168] |

[a] Only peptides that are composed of at least two consecutive N-alkylated amino acids are included.
[b] Solvent: $CH_2Cl_2$: base: DIPEA.
[c] R = coupling reagent: N = N-protected amino acid: C = C-protected amino acid: B = base.
[d] Mixing temp/coupling temp.
[e] (i) = isolated; (H) = determined by HPLC.
[f] Estimated by HPLC.
[g] Doe = dolaproine; Dil = dolaisoleucine; Dpy = (*S*)-dolapyrolidinone; Hiva = 2-hydroxy isovaleric acid.
[h] In comparative studies where the same peptide was synthesized by different coupling agents and different conditions, only the coupling that gave the highest yield is given.

*for references see p 265*

**Table 11** Solid-Phase Synthesis of N-Alkylated Peptides[30,189,196,197,201,202]

| Peptide[a] | Reagent | Resin | Solvent | Base | Stoichiometry[b] (Equiv) R/N/B | Temp[c] (°C) | Time (h) | Yield[d] (%) | Epimer[e] (%) | Ref |
|---|---|---|---|---|---|---|---|---|---|---|
| H-Val-(Me)Val-Ala-$NH_2$ | HATU | MBHA | DMF | DIPEA | 3:3:3 | rt | 2 × 3 | >99 | | [201] |
| H-(Me)Leu-(Me)Leu-Ala-$NH_2$ | HATU | MBHA | DMF | DIPEA | 3:3:3 | rt | 2 × 3 | >99 | | [201] |
| H-(Me)Leu-(Me)Val-Ala-$NH_2$ | HATU | MBHA | DMF | DIPEA | 3:3:3 | rt | 2 × 3 | >99 | | [201] |
| CsA(2–7) | HATU | TentaGel | DMF | DIPEA | 3:3:3 | rt | 2 × 3 | 80 (i) | <4 | [201] |
| H-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-Phe-Val-OH | HATU | Wang | DMF | DIPEA | 4.75:5:10 | 25 | 2 × 2 | 85 | | [189] |
| linear (Me)$Leu^1$-CsA | HATU[f] | HMPB-PEG-PS | DMF | DIPEA | 4.75:5:10 | 25 | 2 × 2 | n.r. | | [196,202] |
| linear [(Me)$Ser^1$]Cs | HATU | HMPB-PEG-PS | DMF[g] | DIPEA | 4.75:5:10 | 25[g] | 2 × 2[h] | 5–16[i] | ~30 | [197] |
| linear [(Me)$Thr^1$]Cs | HATU | HMPB-PEG-PS | DMF[g] | DIPEA | 4.75:5:10 | 25[g] | 2 × 2[h] | 5–16[i] | ~30 | [197] |
| linear [(Me)Leu(3-OH)$^1$]Cs | HATU | HMPB-PEG-PS | DMF[g] | DIPEA | 4.75:5:10 | 25[g] | 2 × 2[h] | 5–16[i] | ~30 | [197] |
| Me-Phe-(Me)Phe-(Me)Phe-$NH_2$ | BTC | MBHA | THF | collidine | 1.65:5:14 | 50 | 1 × 2 | >99 | 0 | [30] |

[a] Only peptides that are composed of at least two consecutive N-alkylated amino acids are included; Cs = cyclosporin; CsA = cyclosporin A.
[b] R = coupling reagent; N = N-protected amino acid; B = base.
[c] Mixing temp/coupling temp.
[d] (i) = isolated; (H) = determined by HPLC.
[e] Estimated by HPLC.
[f] In comparative studies where the same peptide was synthesized by different coupling agents and different conditions, only the coupling that gave the highest yield is given.
[g] The coupling between amino acids in positions 11 and 1 and the following 3 amino acids was performed in NMP at 60 °C.
[h] The coupling between amino acids in positions 11 and 1 and the following 3 amino acids was performed 4 times.
[i] 75% of the heptapeptide resin.

**Bis(trichloromethyl) Carbonate Mediated Coupling; General Procedure:**[30]
The coupling of Fmoc-Xaa$^1$-OH or $N^\alpha$-alkyl amino acids such as Fmoc-(Me)Xaa$^1$-OH to the following: *N*-alkyl amino peptide-resin, *N*-methyl amino acid peptide-resin, building blocks resin or building units peptide-resin was performed as follows: In a well-ventilated hood, Fmoc-Xaa$^1$-OH, Fmoc-(Me)Xaa$^1$-OH, or building units such as $N^\alpha$-Fmoc-$N^\alpha$-(ω-$AlO_2C$-alkyl)-Xaa$^1$-OH or $N^\alpha$-Fmoc-$N^\alpha$-(ω-AlocNH-alkyl)-Xaa$^1$-OH (5 equiv) and bis(trichloromethyl) carbonate (1.65 equiv) were dissolved in dry THF, dry dioxane, dry diglyme, or dry 1,3-dichloropropane (to give a 0.15 M soln) and cooled in an ice bath. Then 2,4,6-collidine (25 equiv) was added to give a white suspension. After 1 min, the suspension was poured into the peptide-resin prewashed with the appropriate solvent and preheated to 50 °C. The mixture was shaken at 50 °C for 1 h, filtered and washed with $CH_2Cl_2$ (10 mL, 5 × 2 min) and DMF (10 mL, 5 × 2 min).

**Bis(trichloromethyl) Carbonate Mediated Coupling; General Procedure:**[40]
Fmoc-peptide-resin was deprotected with 20% piperidine/DMF (2 min + 8 min). After washing, the resin was treated with dry THF (1 mL) for 15 min. Meanwhile, the Fmoc amino acid (3.5 equiv) was added to a 68 mM soln of bis(trichloromethyl) carbonate in dry THF [1.15 mM of bis(trichloromethyl) carbonate]. 2,4,6-Collidine (10 equiv) was added to the clear soln, upon which a precipitate of collidinium chloride formed. DIPEA (8 equiv) was added to the resin, immediately followed by addition of the suspension. The mixture was shaken for the times given in Scheme 39, filtered, and washed.

### 10.1.3.3 Lability of N-Alkylated Peptides to Acids

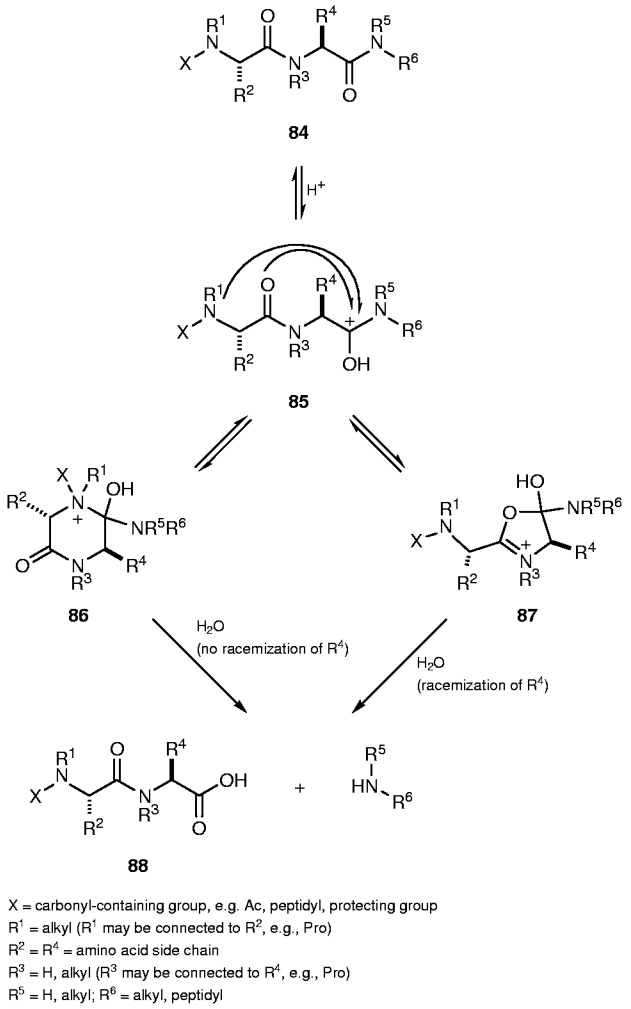
The lability of peptides and proteins to acidic conditions was first reported in 1920 by Dakin,[203] who found that acid hydrolysis of peptides or proteins that contain consecutive *N*-alkyl amino acids leads to the formation of piperazine-2,5-diones (DKP); this side reaction lowered their yield during amino acid analysis. For example, the piperazine-2,5-dione *c*[-Hyp-Pro-] was isolated from the hydrolyzate of gelatine.

Peptides that contain *N*-alkyl amino acids are susceptible to chain fragmentation upon exposure to acids. This fragmentation is troublesome since acids are extensively used in peptide synthesis for the acidolytic removal of protecting groups or for the removal of peptides from the solid support. Chain fragmentation is easily observed by HPLC and leads to the appearance of multiple peaks. The facile cleavage of peptides that contain *N*-alkyl amino acids arises mainly from the relatively favorable formation of *cis* peptide bonds, which brings parts of the *cis*-isomer in close proximity thus leading to tetrahedral cyclic intermediates.

Peptides containing the general structure **84** (Scheme 39) underwent between 50–100% cleavage in 95% TFA/$H_2O$ at room temperature in 60 minutes, that is, under routine conditions for the removal of a peptide from the resin after SPPS. The hydrolysis yielded the *N*-acyl-*N*-methyl peptide fragment **88**.[204] The same peptides were also cleaved in solution but at a much slower rate (13–18 h). The hydrolysis occurs only when a carbonyl group is present at the N-terminal side of the N-methylated residue. The hydrolysis was accompanied by racemization of the N-methylated residue. The suggested mechanism involves the formation of the oxazolonium tetrahedral intermediate **87**. This hydrolysis was circumvented by cleavage of the peptide (that did not contain functional side chains) from the resin with 0.1% NaOH in 50% aq MeOH (rt, 60 min).

*for references see p 265*

**Scheme 39** Proposed Mechanisms for Acid-Catalyzed Hydrolysis of a Peptide Containing *N*-Alkyl Amino Acids[204,205]




X = carbonyl-containing group, e.g. Ac, peptidyl, protecting group
$R^1$ = alkyl ($R^1$ may be connected to $R^2$, e.g., Pro)
$R^2$ = $R^4$ = amino acid side chain
$R^3$ = H, alkyl ($R^3$ may be connected to $R^4$, e.g., Pro)
$R^5$ = H, alkyl; $R^6$ = alkyl, peptidyl

Peptides that contain two or three consecutive *N*-alkyl amino acids (called diad and triad respectively) or two *N*-alkyl amino acid dialkylamides of the type **84** (where X = Boc, Z, peptidyl; $R^1$ connected to $R^2$ as in Pro, or $R^1$ = Me; $R^2$ = Me, Bzl; $R^3$ connected to $R^4$ as in Pro or Pip, or $R^3$ = Me; $R^4$ = Bzl; $R^5$ = $R^6$ = Me or $R^5$ connected to $R^6$ as in Pro-OMe or Pip-OMe; $R^6$ = Ala or Ala-OMe) are extremely labile to acidic conditions which cause peptide fragmentation between the second and third *N*-alkyl amino acid (or dialkylamide).[205] The cleavage can occur within a few minutes to 1 hour in TFA depending on the sequence. This cleavage is facilitated when Pro is the first amino acid in the triad. For example the peptide Z-Pro-(Me)Phe-(Me)Ala-OMe was completely cleaved in TFA to yield Z-Pro-(Me)Phe-OH and Me-Ala-OMe. Triads become resistant to cleavage when Pro or a D-amino acid occupies the second position. In addition two successive *N*-alkyl amino acids followed by Gly are also sensitive to acid hydrolysis. Two possible mechanisms were suggested.[205] Both involve nucleophilic attack on the carbonyl of the amide bond being cleaved with the generation of a relatively stable tetrahedral carbon. In the first mechanism the nucleophile in **85** is the tertiary nitrogen of the first *N*-alkyl amino acid leading to the dioxopiperazinium intermediate **86**. This intermediate has a *cis* amide bond between the first and the second *N*-alkyl

amino acid. The second mechanism involves the formation of the oxazolonium tetrahedral intermediate **87**. Mechanistic studies have suggested that the dioxopiperazinium **86** mechanism is prevalent.

Similar results were obtained during the solution synthesis of cyclosporin fragments. For example, treatment of the peptide Boc-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-(Me)Bmt-Abu-Sar-(Me)Leu-Val-(Me)Leu-Ala-OH [(Me)Bmt = (4*R*)-4-[(*E*)-but-2-enyl]-*N*,4-dimethyl-L-threonine] with a solution of TFA in chloroform at 0 °C to remove Boc gave a complex mixture of products. To avoid the peptide decomposition the Boc deblocking was performed at −20 °C.[28,168] Attempts to synthesize (Me)Leu-CsA by SPPS led to fragmentation during the removal of the peptide from the resin by TFA and scavengers. This was circumvented by synthesizing the peptide on the acid labile HMPB-PEG-PS-resin and its removal by TFA/$CH_2Cl_2$ (1:19, rt, 1 h). The linear peptide also decomposed upon exposure to TFA.[196] Incorporation of (Me)Aib into peptides leads to facile acid-catalyzed cleavage (2% TFA/MeCN) of the peptide bond at the C-terminus of the (Me)Aib. The oxazolonium-type intermediate was proposed for this cleavage.[206]

To overcome the chain fragmentation of sensitive peptides that contain *N*-alkyl amino acids caused by acids as described *vide supra*, the following measures are recommended: (1) in solution synthesis the exploitation of protecting groups removed by acids should be minimized (e.g., apply Fmoc chemistry, use acid-labile side-chain protection as little as possible); (2) if the peptides must be subjected to acid use low temperatures (−20 °C ) for the shortest time possible (monitor the reaction by HPLC ); (3) in SPPS: if the peptide does not contain functionalized side chains, prepare the peptide by Fmoc chemistry on Trt resin and remove the peptide from the resin with HFIP (see Section 10.1.1.2.2). If the peptides have functionalized side chains see point (1).

**Deprotection and Removal from the Resin of N-Alkylated Peptides that Contain Acid-Labile Side-Chain Protection; General Procedure:**[28,168]

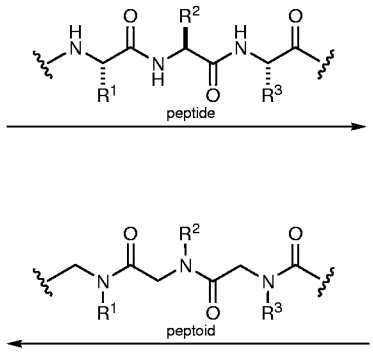
The dry peptide-resin (~1 g) was cooled to −20 °C and pre-cooled (−20 °C) mixture of TFA/TIS (98:2) was added. The mixture was kept under argon at −20 °C for 30 min and at 0 C for 1 h. The resin was removed by filtration and washed with additional cooled TFA. The combined filtrates were taken to dryness by a stream of $N_2$ to give an oil, which solidified upon treatment with cooled (peroxide-free) $Et_2O$ (40 mL). The $Et_2O$ was removed after centrifugation and the solid dried under high vacuum overnight to give the crude peptide. Small amounts of the resin were subjected to fast cleavage to ensure that all the peptide was removed from the resin. If not, the procedure was repeated.

### 10.1.3.4 Peptoids: Peptides Based on N-Alkylated Glycines

Peptoids, oligomers of N-substituted glycines, introduced by Simon et al.[72] represent a new class of N-alkylated peptides.[207] The side chains on the α-carbon atom of the peptide chain are shifted by one position along the peptide backbone to the nitrogen atom to give the N-substituted oligoglycine. A comparison of the backbones of a peptide and a peptoid (Scheme 40) shows that peptoids are achiral and that hydrogen bonds are only possible with hydrogen donors from the side chains. To keep the relative orientation of the carbonyl groups and the side chains the order of the amino acids has to be reversed in the peptoid sequence. This is of importance when the carbonyl group takes part in receptor binding.

*for references see p 265*

**Scheme 40** Comparison of a Peptide and a Peptoid Chain[208]




Peptoids were designed for the generation of chemically diverse libraries[209,210] and offer the following advantages:

(1) Like any N-alkylated peptide, peptoids are distinguished by their metabolic stability. Proteolytic studies showed that peptoids are stable against several known classes of relevant proteases including serine proteases: chymotrypsin, elastase, trypsin; cysteine proteases: papain; acid proteases: pepsin; metallo proteases: carboxypeptidase A.[211] Moreover, in studies in the rat, the tripeptoid [$^3$H]phenyl-acetyl-Nleu-Nglu-Nleu-$NH_2$ was found to be completely stable in blood and was secreted as intact drug.[208]

(2) Due to the simple chemical synthesis a great variability in functional groups is possible which allows convenient synthesis of combinatorial libraries[210,212] (for specific examples see Table 12).

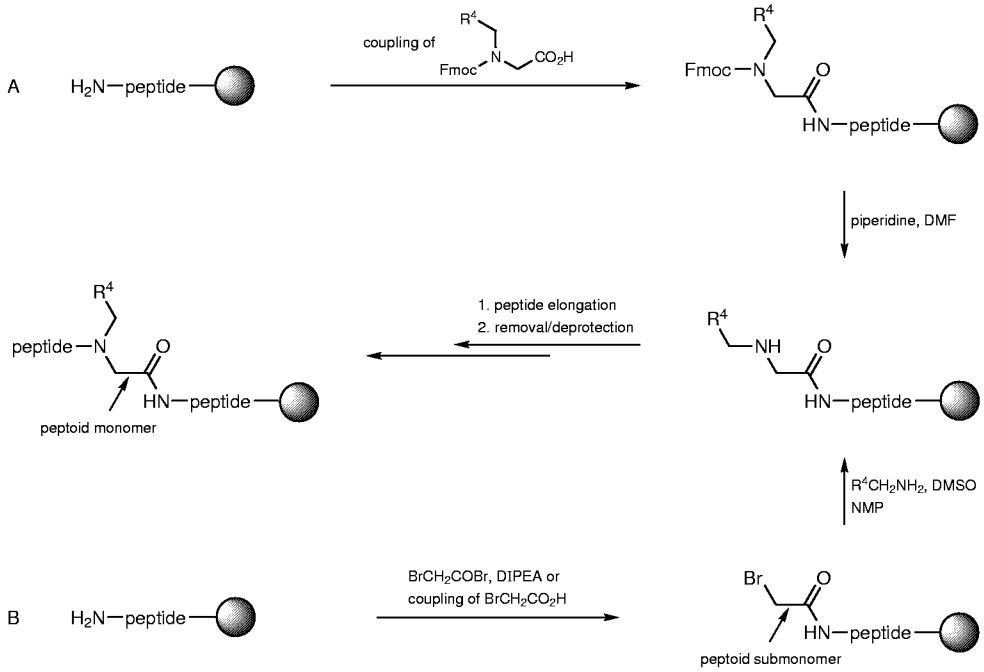
(3) The high flexibility of the backbone and the lack of chirality allows the probing of a large conformational space.

(4) Biologically active small peptoids (di-, tri-, and tetrapeptoids) comply with the Lipinski/Veber rules[213,214] (depends on the structure of the side-chains and end-capping groups) and thus are excellent candidates for orally available drugs. However, a comparative study[208] of the oral availability in rat of labeled tetrapeptide and an end-capped tripeptoid, having the same side chains, revealed that the peptide has a much better oral bioavailability than the peptoid. Nevertheless, some peptoids and peptoid–peptide hybrids are in clinical trials.

The nomenclature of the peptoid monomers that have the same side chain as a corresponding amino acids, consist of an N prefix followed by the three-letter code of the amino acid in small letters. Thus, for example, the peptoid isomer of His is Nhis.

Instead of turning a complete peptide into its peptoid analogue it is also possible to incorporate a single peptoid monomer into a peptide sequence. The resulting peptoid–peptide hybrid is very similar to the original peptide. This allows a peptoid-scan of a peptide sequence to be performed, analogous to an Ala-scan. Peptoids and peptoid–peptide hybrids are synthesized either by solution methods or by SPPS (see Table 12).

**Scheme 41** SPPS of a Peptoid–Peptide Hybrid: (A) The Premade Monomer Method[122,215] (B) The Submonomer Method[216,217]




Two methods are used for the SPPS of peptoids and peptoid–peptide hybrids (Scheme 41). The first method[122,215] (Scheme 41, route A) called the premade monomer method involves the preparation of a Fmoc-protected monomer in solution (see Section 10.1.1.4.1 and Table 8) and its incorporation into the peptoid or the peptoid–peptide hybrid using Fmoc/SPPS. The second method[216,217] (Scheme 41, route B) called the submonomer method involves the formation of the peptoid monomer on the solid support by first forming a bromoacetylated peptide-resin and then substituting the alkyl halide with the appropriate alkyl- or side-chain protected ω-functionalized alkylamine.

Since their introduction in 1992, peptoids and peptoid–peptide hybrids have been extensively used for a large variety of purposes summarized in Table 12.

*for references see p 265*

**Table 12** Examples of Peptoids and Peptoid–Peptide Hybrids[218–262]

| Peptoid/Peptide | Methods | Synthetic Procedure[a] | Ref |
|---|---|---|---|
| polypeptoids with chiral side chain | NMR, CD, Differential Scanning Calorimetry | B | [218] |
| pentapeptoids with chiral side chain | NMR, CD | B | [219] |
| collagen mimetics | NMR, MD | | [220] |
| collagen mimetics | cell binding | | [221] |
| collagen mimetics | host–guest, CD | | [222] |
| polypeptoids with chiral side chain | NMR, CD | B | [223] |
| peptoid oligomers with chiral aromatic side chains | NMR, CD | B | [224,225] |
| membrane-interactive hybrids | CD. MD | A | [226] |
| β-turn formation by tripeptoid | NMR, FTIR | | [227] |
| 15-mer amphiphillic helix | CD | B (library) | [228] |
| metal ion binding | $^{19}F$ NMR | B (library) | [229] |
| HIV-1 Tat analogue transporters | cellular uptake | B | [230,231] |
| 36-mer peptoid | cell delivery of plasmid | B (library, split-and-mix) | [232] |
| lipitoid | cell delivery of plasmid | B (library, split-and-mix) | [233] |
| nonapeptide–peptoid hybrid | suppression of HIV-1 replication | B (library, split-and-mix) | [231,234,235] |
| peptoid–peptide hybrids | MHC class II binding | | [236] |
| tripeptoid | VR1 blockers | B (library) | [237] |
| Nociceptin antagonists | binding to NCR, SAR studies | A | [238] |
| CCK-antagonists | binding to CCK receptors; pharmacokinetics | solution | [239–242] |
| selective CCK-1 agonist | design, binding to CCK receptors. | solution | [243] |
| selective CCK-A agonist and antagonist | design, binding to CCK receptors. | solution | [244] |
| NK1 antagonists | design, synthesis, binding to NK receptors | | [245] |
| NK2 antagonists | design, synthesis, binding to NK receptors | | [246] |
| NK3 antagonists | design, synthesis, binding to NK receptors | solution, library | [247] |
| somatostatin analogues | binding to host receptors. NMR, MD | solution | [248,249] |
| VR1 antagonists | binding to NC receptor | B (library) | [237] |
| screening of peptoid library | MSH, GRP/bombesin receptor ligands | B (library, split-and-mix) | [250] |
| BK linear and cyclic peptoid analogues | binding to BK receptors, NMR, FTIR | B | [251] |
| neuromedin B antagonist | binding SAR studies, in vivo studies | B | [252,253] |
| antimicrobial peptoids | | B (library, split-and-mix) | [254,255] |
| tripeptoid | inhibitors of trypanothione reductase | solution | [256] |
| screening of peptoid library | anti *Neisseria meningitides B* vaccine | B (library, split-and-mix) | [257] |
| phosphorylated peptide–peptoid hybrids | binding to Syk SH2 domains[b] | A | [258] |
| design of selective peptide–peptoid inhibitors | selective binding to SH3 domains, SAR studies | B (library) | [259] |
| peptide–peptoid hybrids | ligands of streptavidin and IR | B (library) | [260] |
| peptoid combinatorial mixture | CE separation | | [261,262] |

[a]The synthetic routes A and B are shown schematically in Scheme 41.
[b]Syk = spleen tyrosine kinase.

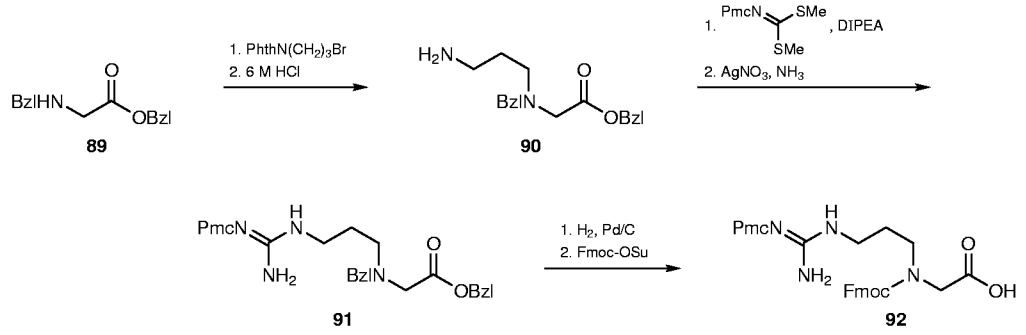
### 10.1.3.4.1 Synthesis of Peptoid Monomers in Solution

The synthesis of peptoid monomers in solutions was performed by the general routes described in Scheme 3. The most applicable synthetic method is route c, namely reductive amination of an alkyl- or ω-protected side-chain amine with glyoxylic acid (see Section 10.1.1.4). The permutation of this method, where a side-chain aldehyde is reacted with glycine methyl ester, is used for the preparation of Nade (*N*-adamantylethylglycine). This method is convenient because stoichiometric amounts of reagents are used and purification is straightforward. An alternative route is alkylation of ω-protected side-chain amines with chloroacetic acid (route b) or, in the case of Ngln, Michael addition of glycine to acrylamide (see Section 10.1.1.4.3 and Scheme 23). Side-chain protection is incorporated prior to formation of the N-substituted glycine monomers. *tert*-Butyl-based groups are used to protect side chains containing amino, carboxy, hydroxy, and indole moieties; guanidine and imidazole moieties are blocked with 2,2,5,7,8-pentamethylchromane-6-sulfonyl (Pmc) and trityl groups, respectively. In the case of hydroxy groups several acid-stable silyl ethers (e.g., TIPS, TBDMS) are used.

These methods allow the synthesis of a large majority of peptoid analogues of the naturally occurring amino acids. Only the synthesis of the arginine analogue is more complicated since the guanidine moiety has to be introduced into a Norn derivative. The first synthesis of Fmoc-Narg(Pmc)-OH was described by Heizmann and Felder.[263] The ornithine analogue Bzl-Norn-OBzl (**90** in Scheme 42) was prepared by the alkylation of *N*-benzylglycine **89** with *N*-3-bromopropylphthalimide followed by deprotection with 6 M hydrochloric acid.

The guanidine moiety was incorporated by dimethyl (2,2,5,7,8-pentamethylchromane-6-sulfonyl N-alkylated amino) dithiocarbonate[264] (Pmc-dithiocarbonate) **91** in two steps. One methylsulfanyl group was replaced by the primary amino group of Bzl-Norn-OBzl in the presence of excess DIPEA and the isothiourea obtained was reacted with silver nitrate under basic conditions to yield the intermediate sulfonylcarbodiimide which reacts immediately with ammonia to yield the guanidine Bzl-Narg(Pmc)-OBzl **91**.

**Scheme 42** Synthesis of Fmoc-Narg(Pmc)-OH[263]

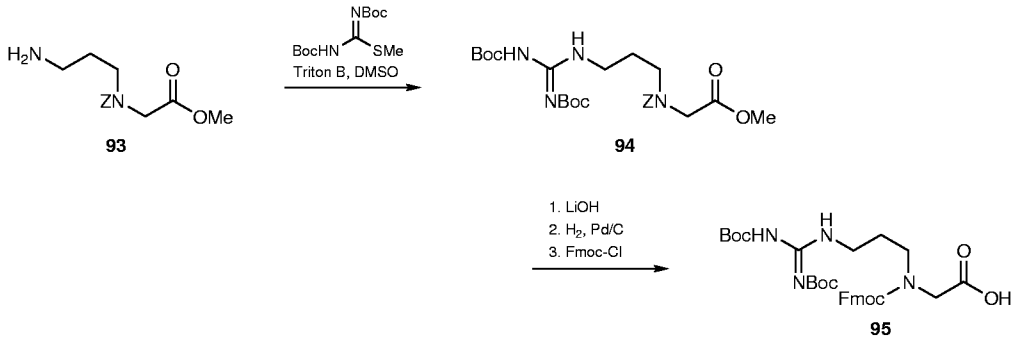



Removal of the *N*-benzyl group of **91** by catalytic hydrogenation and introduction of the Fmoc group proceeded under standard conditions and gave the suitably protected Narg derivative **92** in a total yield of 39%.

A more convenient approach for the preparation of a protected Narg derivative is the synthesis of Fmoc-Narg(Boc)$_2$-OH (**95** in Scheme 43). The key step in this synthesis is the guanylation of Z-Norn-OMe **93** which can be prepared from a monoprotected diaminopropane with bromoacetate or by reductive amination with glyoxylic acid.

*for references see p 265*

**Scheme 43** Synthesis of Fmoc-Narg(Boc)$_2$-OH[265]




The guanylation reagent is *N,N'*-bis(*tert*-butoxycarbonyl)-*S*-methylisothiourea,[265] which is prepared from *S*-methylisothiourea with di-*tert*-butyl dicarbonate and $NaHCO_3$.[266] The guanylation reaction itself is performed in a mixture of DMSO and a solution of 40% Triton-B (benzyltrimethylammonium hydroxide) in methanol. Z-Narg(Boc)$_2$-OMe **94** can then easily be converted into Fmoc-Narg(Boc)$_2$-OH **95** in three steps using standard procedures.

**A: Reductive Amination:**
The amine (5.0 mmol) and glyoxylic acid monohydrate (5.5 mmol) were dissolved in a mixture of MeOH (10 mL) and $H_2O$ (5 mL). The pH of the soln was adjusted to 6 by the addition of 2 M NaOH. After the addition of catalyst [10% Pd/C (0.1 g) with 50% $H_2O$] the mixture was stirred in an autoclave with 50 bar $H_2$ pressure at rt overnight. The catalyst was removed by filtration over silica gel and the filtrate was taken to dryness and dried in vacuo. To remove inorganic impurities, the solid residue was extracted with dry MeOH. The product can be used without further purification.

**B: Direct Alkylation:**
A soln of ethyl bromoacetate (20 mmol) in toluene (7 mL) was added dropwise to a cooled (ice bath) soln of the lipophilic amine (20 mmol) and TEA (20 mmol) in toluene (15 mL) over a period of 4 h. After stirring for 48 h at rt, the mixture was extracted with $H_2O$. The organic phase was dried ($Na_2SO_4$) and taken to dryness in vacuo. Purification of the product was performed by flash chromatography (silica gel); yield: 60–100%.

**C: Michael Addition (Synthesis of H-Ngln-OH):**[267]
KOH (30 mmol) and acrylamide (35 mmol) were added to a soln of glycine (30 mmol) in $H_2O$ (10 mL). The mixture was heated for 25 h at 50–60 °C. After neutralization with AcOH and cooling for 1 d, the precipitated H-Ngln-OH was collected by filtration and washed with EtOH; yield: 91%, mp 210–211 °C.

**Synthesis of Z-Narg(Boc)$_2$-OMe:**[265]
Bis Boc-*S*-methylthiourea (0.63 g, 2.17 mmol) and 40% Triton-B (0.95 mL, 2.17 mmol) in MeOH were added to a soln of Z-Norn-OMe · HCl (0.62 g , 1.96 mmol) in DMSO (7 mL). The mixture was stirred for 4 d at rt. After the addition of 10% $KHSO_4$ and sat. NaCl, the mixture was taken to dryness and the residue was dissolved in EtOAc. The organic phase was washed with 5% $NaHCO_3$, 1 M HCl and sat. NaCl, dried ($Na_2SO_4$), filtered, and taken to dryness and the residue purified by flash chromatography (hexane/EtOAc 3:2, 1:1) to afford Z-Narg(Boc)$_2$-OMe as a colorless oil; yield: 0.64 g (63%); $R_f$ 0.50 (hexane/EtOAc 1:1)

### 10.1.3.4.2 Submonomer SPPS of Peptoids

Zuckermann et al.[216,217] have adapted the method developed by Fischer and Otto[268] for a convenient synthesis of peptoids and peptoid–peptide hybrids by SPPS. The peptoid monomer is built on the solid support by first preparing a bromoacetyl submonomer and then substituting the bromine with an alkyl- or side-chain-protected ω-functional alkylamine (Scheme 41, route B).

The submonomer method has two distinct advantages over the premade monomer method: (1) the formation of the monomer on the resin eliminates the need for $N^{\alpha}$-protection; (2) this method simplifies the synthesis of peptoid libraries, especially those that contain nonfunctionalized N-alkylated side chains, by using commercially available amines. Moreover, the diversity of these libraries is huge.

The first step in the submonomer method involves the acylation of a halogenated acetic acid on the resin under standard conditions. In the second step the side chain of the peptoid is introduced by nucleophilic substitution of the halide with an excess of the primary amine.

The synthesis of peptoids and peptoid–peptide hybrids libraries is described in refs[210,217,254,269] and in references cited in Table 12.

**Synthesis of Individual Peptoids by the Submonomer Method:**[254,269,270]

*Loading to the resin:* To Trt(2-Cl)-PSty-resin (1.0 g) were added bromoacetic acid (0.70 g, 5 mmol) and DIPEA (1.2 mL, 7 mmol) in $CH_2Cl_2$ (15 mL). The mixture was shaken at rt for 1 h and the resin was washed with $CH_2Cl_2$ and $Et_2O$. In general a loading of 0.9–1.2 mmol $\cdot g^{-1}$ was achieved.

*Coupling of bromoacetic acid:* To a stirred soln of bromoacetic acid (10 equiv) in $CH_2Cl_2$ (maximum concentration 0.3 M) were added bromoacetyl bromide (8 equiv) and DIPEA (20 equiv). The mixture was added to amino-peptide-resin and shaken for 10 min with the resin followed by washing with $CH_2Cl_2$.

*Substitution with primary amine:* The resin was shaken for 4 h with a soln of the appropriate amine in NMP (30 equiv, ca. 2 M) followed by washing of the resin with NMP and $CH_2Cl_2$.

*Capping:* the resin was shaken for 10 min with a soln of $Ac_2O$ (10 equiv) and DIPEA in $CH_2Cl_2$ (5 equiv) followed by washing of the resin with $CH_2Cl_2$.

*Cleavage from the resin (with the retention of tBu-type side chain protection):* The resin (1 g) was shaken for 30 min with a mixture of AcOH, TFE, and $CH_2Cl_2$ (1:1:3, 20 mL) and the resin removed by filtration. The procedure was repeated once more and the resin washed with the above mixture ($3 \times 5$ mL) the combined filtrates were taken to dryness in vacuo after addition of toluene (20 mL). The crude product was purified by HPLC.

*Cleavage/deprotection:* Resin (1 g) was shaken for 1 h with TFA/$H_2O$/TIS (95:2.5:2.5, 20 mL) and the resin was removed by filtration. The resin was washed with TFA and the combined filtrates were taken to dryness. The crude product was analyzed by MS and RP-HPLC on C-18 columns (Vydack, 5 µm, 300 A, $4.5 \times 250$ mm$^2$). A linear gradient of 0–80% B in 40 min was used at flow rate of 1 mL $\cdot$ min$^{-1}$ (solvents A = 0.1% TFA/$H_2O$, solvent B = 0.1% TFA/MeCN). The crude peptoid was purified by preparative RP-HPLC on C-4 columns (Vydack, 5 µm, 300 A, $22 \times 250$ mm$^2$). A linear gradient of 5–85% B in 80 min was used at flow rate of 10 mL $\cdot$ min$^{-1}$. Collected fractions were lyophilized, rinsed with AcOH, and relyophilized.


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# 10.2 Synthesis of Depsipeptides

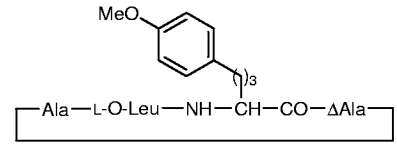
V. T. IVANOV and I. I. MIKHALEVA

The name "depsipeptide" was initially suggested by Shemyakin and Khokhlov[1] at the beginning of the 1950s and was defined as a naturally occurring antibiotic containing amide- and ester-linked amino and hydroxy acid residues. As all natural depsipeptides known at that time had a cyclic structure, Russell and Brown[2] introduced the term "peptolide" for depsipeptides containing "pure" hydroxy acids. In the literature there is also the term "O-peptide" or peptide lactone that refers to the numerous peptide compounds with ester bonds formed with the participation of the side-chain hydroxy groups of Ser or Thr. In accordance with the nomenclature of Schröder and Lübke,[3] peptolides represent heteromeric heterodetic peptides, while O-peptides are homomeric heterodetic peptides. The name depsipeptides is, at present, by far the most frequently used term for compounds containing amino and hydroxy acid residues connected by amide and ester bonds.

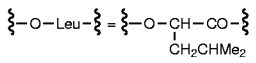
Since the 1950s, the number of natural depsipeptides has expanded by the identification of numerous metabolites of microbes, yeasts, sea hares, and other organisms. The logic of this volume dictates that the main subject of this section is peptidomimetics with CONH → COO modification of the peptide backbone, i.e. depsipeptides whose ester bonds are formed by the hydroxy function of α-hydroxy acids. Representative examples **1**,[4] **2**,[5,6] **3**,[7,8] **4**,[9] **5**,[10] **6**,[11,12] **7**,[13] and **8**[14] of this group of natural depsipeptides are shown in Scheme 1. Most of them have a macrocyclic structure with 12–36 atoms in the ring. The α-hydroxy acid constituents are frequently lactic (Lac), α-hydroxyisovaleric (Hyv), or other residues with aliphatic side chains, of both L- or D-configuration. The remaining part of the depsipeptide molecule comprises coded amino acids or their enantiomers as well as their N-methylated derivatives or some unique units, e.g. aIle (alloisoleucine), occurring only in this particular group of compounds. In contrast to normal peptides, the biosynthesis of natural depsipeptides proceeds through specialized polyenzymatic systems. The vast majority of natural depsipeptides have pronounced biological activity: AM-toxins are known as phytotoxins;[15] aureobasidin A exhibits antifungal activity;[10] enniatins and valinomycin are typical ionophores;[16] and PF1022 A has anthelmintic activity.[17] Some recently discovered depsipeptides exhibit extremely potent cytotoxicity and they are receiving increasing attention as promising potential antineoplastic drugs.[13,14]

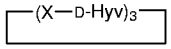
The diversity of the chemical structures of depsipeptides and their biological significance has stimulated the development of appropriate synthetic approaches to their preparation. The isolation of depsipeptides from natural sources and subsequent determination of the chemical structure is generally followed by their total syntheses to finally confirm the proposed structures. Sometimes, as in the case of enniatins and valinomycin, the synthesis has proved instrumental in correcting the originally suggested structures.[5,7,12] The total synthesis of natural depsipeptides also gives the basis for the preparation of analogues, which are required for studies of structure–activity relationships, cellular and molecular targets, and the search for new synthetic derivatives with modified action. The replacement of the amino acid residue by its corresponding hydroxy acid component is used in a technique applicable to the structure–functional studies of any regulatory peptide, regardless of size or the mechanism of action, e.g. depsibradykinin[18] and depsiinsulin.[19] A stereochemical background justifying this approach is the stereoelectronic analogy of the amide and ester moieties predominantly assuming planar *Z* or *E* conformations (Scheme 2).

**Scheme 1** Representative Examples of Depsipeptides Whose Ester Bonds are Formed by the Hydroxy Group of α-Hydroxy Acids[4–14]

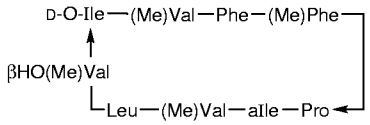



**1** AM-toxin I

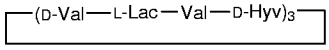


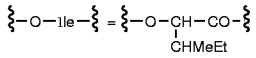

**2** X = (Me)Ile, enniatin A
**3** X = (Me)Val, enniatin B
**4** X = (Me)Phe, beauvericin

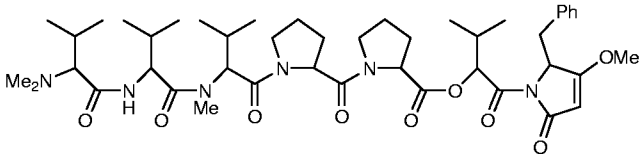



**5** aureobasidin A

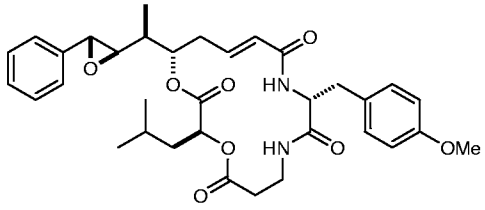



**6** valinomycin



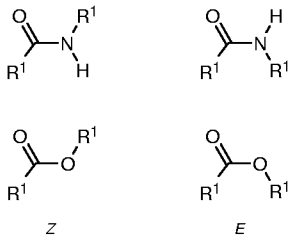




**7** dolastatin 15




**8** arenastatin A

**Scheme 2** *Z* and *E* Conformations of Amide and Ester Bonds



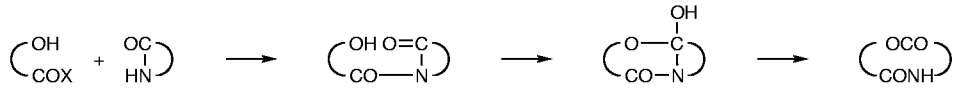
*Z* *E*

A distinct group of synthetic depsipeptides comprises of compounds that do not originate from natural product biodiversity; several artificial substrates of peptidases and esterases belong to this group, as well as polydepsipeptides that are considered as potentially biodegradable polymeric materials. A specific feature of depsipeptide synthesis is the necessity to acylate a hydroxy acid component, which requires stronger activation of the amino acid component in comparison to normal peptide synthesis. Otherwise, the main principles of depsipeptide synthesis are similar to those of peptides. Frequently, formation of the ester

*for references see p 289*

bond takes place at the first stage of the synthesis and standard sets of protecting groups are employed. Consequently, the synthesis of depsipeptides essentially consists of esterification of the amino acid components by the hydroxy acid components. The main goal of this section is to discuss the developments in this field of chemistry. A promising route to cyclic depsipeptides employs hydroxy acyl incorporation (Scheme 3).[20] However, such an approach has never straightforwardly resulted in α-hydroxy acids and therefore will not be discussed.

**Scheme 3** The Hydroxy Acyl Incorporation[20]



Note that we have used the following notation for hydroxy acids in this section, -O-Gly for the glycolic acid residue -$OCH_2C(O)$-, likewise -O-Ile is the isoleucic acid residue -OCH(CHMeEt)C(O)-, -O-Leu is the leucic acid residue -OCH(iBu)C(O)-, -O-Phe is the β-phenyllactic acid residue -OCH(Bzl)C(O)-, and -O-Phg is the phenylglycolic acid residue -OCHPhC(O)-.

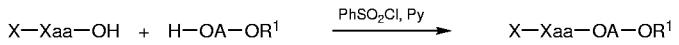
## 10.2.1 Formation of the Ester Bond

Since depsipeptides, in contrast to classical peptides, contain units constructed from amino and hydroxy acid residues, the various methods for their preparation are generally pathways involving the formation of ester bonds. The novel achievements in this area discussed (*vide infra*) are associated with further developments in the mixed anhydride technique, the application of effective catalysts in the carbodiimide procedure, and adaptation of the known Mitsunobu reaction to the depsipeptide case. A number of significant and efficient esterification procedures utilized for the preparation of depsipeptides are considered.

### 10.2.1.1 Mixed Anhydrides

From the beginning of the 1960s, the progress in depsipeptide chemistry was connected with the utilization and further development of the mixed anhydride approach for the initial formation of the ester link between a suitably N-protected amino acid and the hydroxy acid ester. The reagent predominantly applied for the construction of the depsipeptide unit was benzenesulfonyl chloride in pyridine. This reagent, introduced by Shemyakin and co-workers in depsipeptide chemistry, was shown to be an efficient reagent for the formation of the ester bond between a protected amino acid and the hydroxy acid component (Scheme 4).[21,22]

**Scheme 4** General Route to Protected *O*-Aminoacyl Hydroxy Acid Esters by the Benzenesulfonyl Chloride Method




X = Z, Z(NO$_2$); OA = hydroxy acid; R$^1$ = tBu, Bzl

The reaction proceeds through an intermediate mixed anhydride, or the symmetrical anhydride derived from it, and subsequent alcoholysis in the presence of the hydroxy component.[23–25]

This method was used widely and successively in the synthesis of cyclic depsipeptide antibiotics and natural metabolites such as sporidesmolids I–IV,[26–29] angolide,[30] enniatins A

(**2**), B (**3**), and C,[31–33] and valinomycin (**6**).[11,12] The depsipeptide chain was further built up and cyclization was achieved exclusively through amide bond formation by any appropriate method for amide coupling. It is important to note that C-terminal hydroxy acid residues of intermediate depsipeptides show no tendency toward racemization even under such drastic conditions as activation by the acid chloride technique. The most suitable protecting groups used for the $N^{\alpha}$-amino function are the benzyloxycarbonyl, 4-nitrobenzyloxycarbonyl, *tert*-butoxycarbonyl, or the 2-nitrophenylsulfanyl group, and, for the carboxy function of the hydroxy acid, the *tert*-butyl or benzyl ester groups. Using this methodology Shemyakin, Ovchinnikov et al.[26–35] accomplished successful syntheses of a great number of linear and cyclic natural depsipeptides and their analogues. A similar approach was utilized also for the synthesis of enniatins A and B by the Swiss groups of Plattner, Vogler, Quitt et al.[6,8]

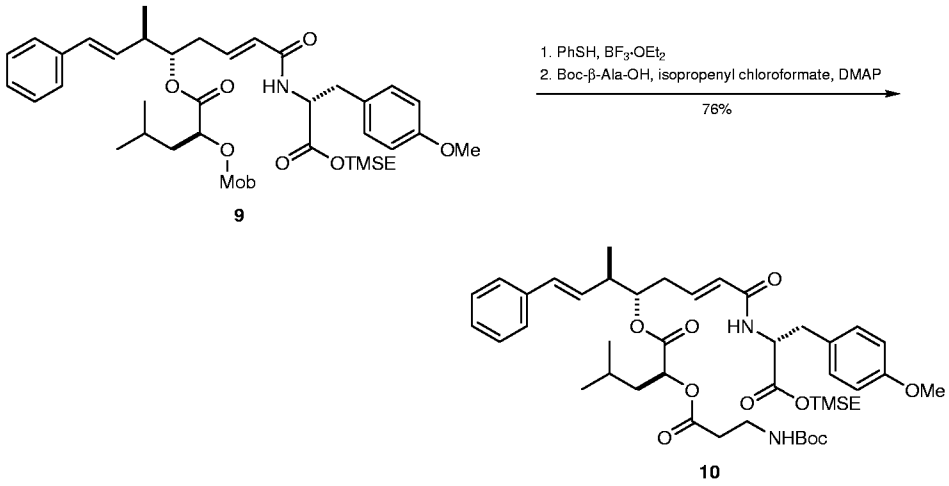
The purity and yield of the desired initial didepsipeptide moieties prepared using benzenesulfonyl chloride in pyridine depends on steric factors and may vary from excellent to low. The nature of the protected amino acid strongly influences the yield and the optical purity of the condensation product. For example, high (65–90%) yields of didepsipeptides derived from the coupling of benzyloxycarbonyl- or 4-nitrobenzyloxycarbonyl derivatives of *N*-methylvaline, *N*-methylleucine, or *N*-methylisoleucine, as well as their nonmethylated derivatives with *tert*-butyl or benzyl esters of α-hydroxyisovaleric or lactic acid, were obtained en route to enniatins, valinomycin, insecticidal cyclodepsipeptide bassianolide,[36] their analogues, and cyclopolymer homologues. However, in some cases, low yields (< 40%) were observed, presumably due to undesirable side reactions of the highly activated mixed anhydrides. There has been no systematic investigation of the factors that usually influence the outcome of these reactions, such as the nature of the amino acid side chain and the protecting group, temperature and activation time, as well as the nature of the nucleophilic hydroxy component. The use of nonurethane amino-protecting phthaloyl or nitroso groups results in complete or partial racemization of the activated amino acid moiety, presumably due to oxazolone formation.[37,38]

Benzenesulfonyl chloride in pyridine remains a powerful and facile method for the efficient synthesis of optically pure depsipeptides. Nevertheless, various new approaches based upon the mixed anhydride methodology have been introduced into depsipeptide chemistry.

In 1987, Jouin et al.[39] reported an efficient one-pot esterification of N-protected amino acids with primary, secondary, or tertiary alcohols by means of isopropenyl chloroformate with 4-(dimethylamino)pyridine (DMAP) for acid activation. In this method the urethane $N^{\alpha}$-protected amino acid is activated in situ as the mixed anhydride, which in turn smoothly reacts in the presence of 4-(dimethylamino)pyridine with the alcohol already present in the reaction mixture giving the corresponding optically pure ester in high yield. This method was applied to the synthesis of the cyclic depsipeptide nordidemnin B as a highly effective and convenient synthetic procedure for depsipeptide mimetics. The desired intermediate tridepsipeptide Boc-D-[*N*,*O*-isopropylidene]Valψ[CHO]Gly-L-Hyv-OBzl was obtained smoothly and quickly as a homogenous product in 92% yield.[40] Advantage was taken of this effective and promising method for the introduction of the ester moiety into the skeleton of cyclic or linear peptides in a number of reported syntheses. Thus, in the total synthesis of the calcium channel blocker cyclodepsipeptide leualacin, McLaren[41] easily prepared two depsipeptides, namely Boc-Leu-L-O-Leu-ODpm, from *N*-(*tert*-butoxycarbonyl)leucine and L-leucic acid diphenylmethyl ester, and Boc-β-Ala-D-O-Leu-ODpm, from *N*-(*tert*-butoxycarbonyl)-β-alanine and D-leucic acid diphenylmethyl ester, in 63 and 96% yield, respectively. Similar results were obtained for depsipeptide **10** starting from the protected hydroxy acid derivatives, such as **9** (Scheme 5), in studies of the natural cytotoxic cyclodepsipeptide arenastatin A (**8**)[42,43] and also for didepsipeptide ester Boc-(Me)Tyr-L-Hyv-OTMSE in the synthesis of the depsipeptide mimic of tendamistat, a microbial inhibitor of α-amylase (Table 1).[44]

*for references see p 289*

**Scheme 5** Synthesis of a Depsipeptide from a Hydroxy Derivative Using Isopropenyl Chloroformate/DMAP[42]




9 10

**Table 1** Yields and Reaction Conditions for Introduction of an Ester Moiety into Peptides[6,8,12,35,36,38,40,44,52,53]

| Starting Materials | | Reaction Conditions[a] | Product | Yield (%) | Ref |
|---|---|---|---|---|---|
| Z-Val-OH | H-L-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (1.2:1:1) | Z-Val-L-Hyv-OtBu | 70 | [38] |
| Z-(Me)Val-OH | H-D-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (2:1:1) | Z-(Me)Val-D-Hyv-OtBu | 50 | [8] |
| Z-Val-OH | H-D-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (1.2:1:1) | Z-Val-D-Hyv-OtBu | 64 | [52] |
| Z-D-Ala-OH | H-L-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (1.2:1:1) | Z-D-Ala-L-Hyv-OtBu | 75 | [53] |
| Z-D-Val-OH | H-L-Lac-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (1.2:1:1) | Z-D-Val-L-Lac-OtBu | 85 | [12] |
| Z-(Me)Leu-OH | H-D-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (2:1:1) | Z-(Me)Leu-D-Hyv-OtBu | 89 | [36] |
| Z-(Me)Ile-OH | H-D-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (2:1:1) | Z-(Me)Ile-D-Hyv-OtBu | 51 | [6] |
| $Z(NO_2)$-D-(Me)Ile-OH | H-L-Hyv-OtBu | pyridine, 0 °C, $PhSO_2Cl$/Xaa/OA (1.2:1:1) | $Z(NO_2)$-D-(Me)Ile-L-Hyv-OtBu | 80 | [35] |
| Boc-D-[*N*,*O*-isopropylidene]-Val ψ[CHO]Gly-OH | H-L-Hyv-OBzl | $CH_2Cl_2$, −5 °C, 30 min, isopropenyl chloroformate/DMAP/Xaa/OA (19.9:3.6:18.1:17.9) | Boc-D-[*N*,*O*-isopropylidene]Val-ψ-(CHO)-Gly-L-Hyv-OBzl | 92 | [40] |
| Boc-(Me)Tyr-OH | H-L-Hyv-OTMSE | $CH_2Cl_2$, −78 °C, 10 min, then 0 °C 1 h, isopropenyl chloroformate/DMAP/Xaa/OA (8:1.4:7.3:8) | Boc-(Me)Tyr-L-Hyv-OTMSE[1] | 65 | [44] |

[a] OA = hydroxy acid.

Some of the versions of the mixed anhydride technique discussed may involve components other than α-hydroxy acids. Therefore, the resultant products cannot be considered as main-chain modified peptidomimetics. However, on many grounds these methods are of general importance in depsipeptide synthesis. For example the classical peptide reagent isobutyl chloroformate appears to be suitable for ester bond formation through the corresponding mixed anhydride with Boc- or Z-protected amino acids and the Thr β-hydroxy group in the synthesis of a number of natural peptide lactones.[45–47]

A rapid and mild procedure utilizing 2,4,6-trichlorobenzoic acid anhydrides in the presence of 4-(dimethylamino)pyridine was utilized for the intramolecular formation of the

ester bond by Yamaguchi et al.[48] This reaction was successfully used for the final lactonization of the linear precursor of jasplakinolide, a cyclic depsipeptide of marine origin, in 82% yield.[49] In this case, the achiral β-tyrosine residue of the precursor was activated during ring closure by treatment with 2,4,6-trichlorobenzoyl chloride in benzene in the presence of triethylamine followed by heating with 4-(dimethylamino)pyridine under reflux. It is interesting to note that this approach allows ester bond formation between the amino and hydroxy acid residues at the final stage of synthesis (cyclization) in contrast to the usual preparation of cyclodepsipeptides based on amide bond cyclization. However, esterification under Yamaguchi conditions may be the method of choice only when the activated peptide moiety is achiral; in other cases this procedure is accompanied by substantial or complete racemization.[50,51]

**_N_-Benzyloxycarbonyl-Substituted Aminoacyl Hydroxy Acid *tert*-Butyl Esters; General Procedure for the Synthesis of Protected Didepsipeptides Using Benzenesulfonyl Chloride:**[35]
To a soln of the N-protected amino acid (120 mmol) in dry pyridine (100 mL) was added dropwise at 0 °C with stirring over 10 min $PhSO_2Cl$ (19.4 g, 110 mmol). A soln of the hydroxy acid *tert*-butyl ester (100 mmol) in pyridine (20 mL) was then added at 0 °C and the mixture was stirred for a further 2 h under cooling and 3 h at rt. The mixture was poured into ice water and extracted with $Et_2O$. The $Et_2O$ soln was washed with 0.5 M aq $H_2SO_4$, sat. aq $NaHCO_3$, and $H_2O$, dried ($MgSO_4$), and then removed in vacuo. The oily residue was purified by column chromatography (silica gel or neutral alumina, benzene/EtOAc) to afford the depsipeptides; yield: 75–90%. For example, analytical data for Z($NO_2$)-D-(Me)Ile-L-Hyv-OtBu: mp 96–97 °C; $[\alpha]_D^{20}$ +86.0 (*c* 0.3, benzene).

**Boc-β-Ala-D-O-Leu-ODpm; General Procedure for Depsipeptide Formation Using Isopropenyl Chloroformate:**[41]
Boc-β-Ala-OH (0.35 g, 1.8 mmol) was added to a soln of H-D-O-Leu-ODpm (0.48 g, 1.6 mmol) in EtOH-free $CH_2Cl_2$ (8 mL) at 0 °C with stirring. The mixture was treated sequentially with TEA (0.55 mL, 0.40 g, 3.9 mmol), DMAP (0.04 g, 0.3 mmol), and isopropenyl chloroformate (0.21 mL, 0.23 g, 1.9 mmol) which resulted in precipitate formation and gas evolution. After 2 h, the mixture was partitioned between $Et_2O$ and 1 M HCl. The organic layer was washed with 5% aq $NaHCO_3$, $H_2O$, and dried ($Na_2SO_4$). The soln was concentrated in vacuo and the residue was purified by MPLC (silica gel 60 column, EtOAc/hexane 1:9) to give the product as a colorless oil; yield: 0.73 g (96%).

### 10.2.1.2 Carbodiimides with or without Additives

The carbodiimide method has been employed in several syntheses of depsipeptides. However, direct application of DCC for the formation of the ester bond between the amino acid and hydroxy acid components under the usual conditions of amide coupling affords the desired depsipeptides in acceptable yields only in the case of unhindered ω-hydroxy units [54] or an active hydroxy group, such as in *N*-benzoyl-α-hydroxyglycine benzyl ester. For example, Ravdel et al.[55] have performed the esterification of various benzyloxycarbonyl- and phthalylamino acids with *N*-benzoyl-α-hydroxyglycine benzyl ester with DCC in 50–65% yield. On the other hand, Shemyakin et al.[21] failed to obtain the expected depsipeptide products on condensation of bulky benzyloxycarbonyl- or phthalylvaline with α-hydroxyisovaleric acid benzyl ester. The main product was acylurea in the first case and phthalylvaline anhydride in the second. Thus, the classical carbodiimide procedure could not be applied in practical depsipeptide preparation.

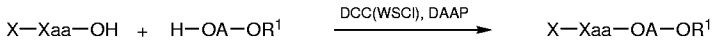
In 1959 Shchukina et al.[56] showed that the addition of pyridine to DCC increases the yield (65–85%) of depsipeptides constructed from Z-protected amino acids and serine, threonine, or salicylic acid derivatives. The *tert*-butyl ester of *N*-(benzyloxycarbonyl)-leucylleucic acid (Table 2) was prepared in 60% yield under the same conditions of addition using a 2 molar excess of pyridine to carbodiimide.[57] The DCC/pyridine technique was successfully utilized in the ring closure between β-alanine and leucic acid residues during the synthesis of the cyclic hexadepsipeptide destruxin B.[58] Under this modified approach the

*for references see p 289*

amount of pyridine in the reaction mixture may vary from 2 equivalents with respect to carbodiimide up to using it as a solvent.

Further developments in the application of the carbodiimide method to the synthesis of depsipeptides originate from the introduction of 4-(dimethylamino)pyridine (DMAP) by Steglich et al.[59–61] as a catalyst to facilitate the esterification of hindered hydroxy moieties. Hassner et al.[62] investigated the efficiency of different types of potent catalytic additives to carbodiimide such as pyridazine, quinoline, and various 4-substituted derivatives of pyridine on the rate of acylation during esterification of hindered alcohols. They found 4-pyrrolidinopyridine (Ppyr), 1,1,3,3-tetramethyl-4-(4-pyridyl)guanidine, and DMAP to be the most effective catalysts. In the case of hindered alcohols where the use of pyridine fails to catalyze the acylation, the use of 0.1 molar equivalent of 4-pyrrolidinopyridine results in complete and smooth esterification. The pronounced effect of these pyridine derivatives in catalyzing the acylation reactions was attributed by the authors to the increased donor ability of the 4-substituent and its stabilizing effect on the intermediate acylpyridinium compounds.[62] These findings significantly improved the efficiency of the carbodiimide approach in the synthesis of depsipeptides (Scheme 6). Thus, the preparation of the sterically hindered Boc-Aib-L-Hyv-OBzl was accomplished in 92% yield by using carbodiimide with an equimolar amount of dialkylaminopyridine, while attempts to obtain this depsipeptide using carbonyldiimidazole, active esters or carbodiimide/pyridine activations were fruitless. The advantages of the method are the mild reaction conditions ($CH_2Cl_2$, 25 °C), short reaction times (6–11 h), and high yields even with sterically hindered compounds.[63]

**Scheme 6** Formation of the Ester Bond by Carbodiimides Catalyzed by 4-(Dialkylamino)pyridine




X = urethane protecting group; OA = hydroxy acid; $R^1$ = alkyl, aryl
DAAP = 4-(dialkylamino)pyridine

Later on, the use of a carbodiimide in the presence of a catalytic additive 4-(dialkylamino)pyridine found wide application in the synthesis of depsipeptide intermediates (see Table 2) in the course of the preparation of some natural depsipeptides and their analogues, such as AM-toxins,[64–67] anthelmintic cyclodepsipeptide PF 1022A,[68,69] aureobasidin A,[70–74] dolastatin 15,[19] and hapalosin.[75] The yields of linear depsipeptide fragments obtained were in the 60–98% range; the lowest yields were obtained in the condensation of Boc-(Me)Leu-OH with various derivatives of β-phenyllactic acid substituted in the *para* position of the phenyl ring. The coupling was performed using EDC with 10% DMAP[69] and the quantity of catalyst used varied from 0.05 to 1 equivalent with respect to the amino acid component. This synthetic procedure proved to be efficient not only in the case of linear depsipeptides, but also for cyclization through ester bond formation. Thus, lactonization of the linear precursor H-L-Hyv-Phe-Ser(Bzl)-Ala-OH into the corresponding AM-toxin II analogue *c*[-L-Hyv-Phe-Ser(Bzl)-Ala-] proceeded in 16% yield in the presence of water-soluble carbodiimide and 4-(dimethylamino)pyridine. Even more successful was a similar experiment related to another AM-toxin II analogue *c*[-Hyv-Phe-Ser(Bzl)-Appa-] (Appa = 2-amino-5-phenylpentanoic acid) (Table 2). However, the authors of this work did not examine the level of possible epimerization under the peptide fragment activation conditions.[64] Jao et al. have observed racemization during the coupling of bulky Boc-β-hydroxy-*N*-methylvaline and (+)-(*R*)-hexahydromandelic acid benzyl ester using DCC and DMAP. As a result, they obtained a difficult to separate diastereomeric mixture of the corresponding didepsipeptides that was further converted into a mixture of two tridepsipeptides elongated by C-terminal addition of a *N*-methylvaline residue (final LDL/DDL ratio 2:1).[71]

**Table 2** Yields and Reaction Conditions for the Preparation of Depsipeptides Using Carbodiimide[57,63,64,66–68,72,74]

| Starting Materials[a] | | Reaction Conditions | Product[a] | Yield (%) | Ref |
|---|---|---|---|---|---|
| Z-Leu-OH | H-L-O-Leu-OtBu | $Et_2O$/pyridine, 4 °C, 20 h, DCC/Xaa/OA/pyridine (1.1:1:1:2) | Z-Leu-L-O-Leu-OtBu | 60 | [57] |
| Boc-D-Val-OH | H-D-Lac-OPac | dry $CH_2Cl_2$, 0 °C, 3 h, then rt, 20 h, DCC/Xaa/OA/DMAP (1.1:1:1:1) | Boc-D-Val-D-Lac-OPac | 83 | [67] |
| Boc-Ala-OH | H-L-Lac-OBzl | dry $CH_2Cl_2$, 0 °C, 24 h, DCC/Xaa/OA/DMAP (1.1:1:1:1) | Boc-Ala-L-Lac-OBzl | 87 | [66] |
| Boc-Ala-OH | H-L-Lac-OBzl | dry $CH_2Cl_2$, 25 °C, DCC/Xaa/OA/DMAP or Ppyr (1.1:1:1:1) | Boc-Ala-L-Lac-OBzl | 98 | [63] |
| H-L-Hyv-Phe-Ser(Bzl)-Appa-OH | | dry $CH_2Cl_2$, 0 °C, 8 h, peptide/EDC/DMAP (1:2:1) | *c*[-Hyv-Phe-Ser(Bzl)-Appa-] | 19 | [64] |
| Boc-Leu-OH | H-D-O-Ile-OPac | dry THF, 0 °C 1 h, rt 20 h, DCC/Xaa/OA/Ppyr (1.1:1:1.1:0.3) | Boc-Leu-D-O-Ile-OPac | 78 | [74] |
| Boc-Ala-OH | H-L-O-Ile-OBzl | dry $CH_2Cl_2$, 0 °C, 24 h, DCC/Xaa/OA/DMAP (1.1:1:1:1) | Boc-Ala-L-O-Ile-OBzl | 90 | [63] |
| Boc-(Me)Leu-OH | H-D-O-Phe-OBzl | dry $CH_2Cl_2$, rt 2 h, DCC/Xaa/OA/DMAP (1:1:1:0.05) | Boc-(Me)Leu-D-O-Phe-OBzl | 95 | [68] |
| Boc-(Me)Leu-OH | H-D-Lac-OBzl | dry $CH_2Cl_2$, rt, 2 h, DCC/Xaa/OA/DMAP (1:1:1:0.05) | Boc-(Me)Leu-D-Lac-OBzl | 76 | [68] |
| Boc-Leu-DL-(β-HO)(Me)Val-OH | H-D-O-Ile-OPac | dry THF, 0 °C 1 h, then rt 18 h, DCC/Xaa/OA/Ppyr (1.1:1:1.1:0.3) | Boc-Leu-DL-(β-HO)(Me)Val-D-O-Ile-OPac | 79 | [72] |

[a] Appa = 2-amino-5-phenylpentanoic acid; O-Ile = 2-hydroxy-3-methylpentanoic acid; O-Leu = 2-hydroxy-4-methylpentanoic acid; O-Phe = β-phenyllactic acid residue, OA = hydroxy acid.

A further development of the carbodiimide/dialkylaminopyridine method is associated with the contribution of Boden and Keck.[76] By analyzing the reasons for the low yields in the course of the preparation of large ring lactones containing ω-hydroxy acid residues, they demonstrated that carbodiimide/dialkylaminopyridine mediated esterification includes a proton transfer step. Therefore, it was concluded that the partial failure might conceivably arise from low effective concentrations of the alcohol and acid partners (both proton sources) under the reaction conditions employed. Bimolecular esterification experiments showed the crucial role of the additive proton-transfer agents in preserving the active species under high dilution. Reactions were re-examined in the presence of such additives (specifically, various amine hydrochlorides) to mediate such a proton-transfer step. The best results were obtained in media where the appropriate salts were completely soluble, 4-(dimethylamino)pyridine hydrochloride being the most efficient and convenient. Such a simple change in the experimental procedure considerably improved the yields of ester bond closure.[76] This improved technique was further successfully applied to the final lactonization step in the total syntheses of such complex natural cyclic depsipeptides of marine origin as deodiamolide A, jasplakinolide, and doliculide.[77–79] There was no risk of racemization in the first two cases where the achiral β-tyrosine residue was activated during the reaction. However, complete racemization of the *N*-methyltyrosine residue was observed under similar conditions in the course of the doliculide synthesis. Although in the aforementioned syntheses of deodiamolide A, doliculide, and jasplakinolide only ω-hydroxy acid moieties were subjected to esterification, the Boden–Keck modification of the carbodiimide coupling method should also be applicable to the synthesis of depsipeptides with classical α-hydroxy acids.

In summary, among the carbodiimide-based methods only the carbodiimide/dialkylaminopyridine technique has been widely used in the synthesis of depsipeptides. It usually affords the desired products in moderate or high yield and has good potential for routine use in the esterification of urethane-protected amino acids or peptide fragments even in the case

*for references see p 289*

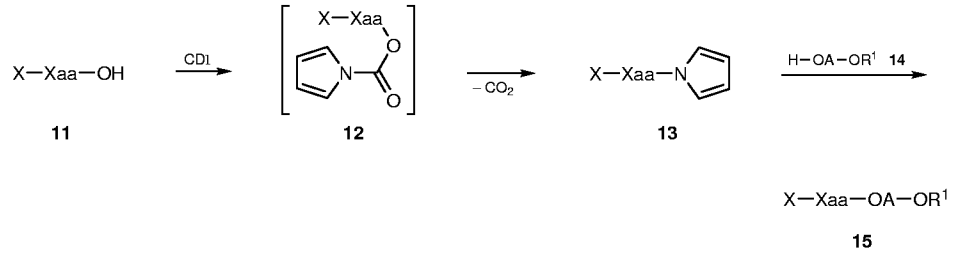
of sterically hindered compounds. A certain risk of racemization obviously exists especially with susceptible residues, requiring the necessity to check the optical purity of the final products.

**Synthesis of Boc-Aminoacyl Hydroxy Acid Benzyl Esters; Typical Procedure:**[63]
To a soln of *N*-Boc-protected-amino acid (Boc-Ala-OH, Boc-D-Val-OH, or Boc-Aib-OH, 1 mmol), an α-hydroxy acid benzyl ester (H-L-Lac-OBzl, or H-L-Hyv-OBzl, 1 mmol) and 4-(dialkylamino)pyridine (DMAP or Ppyr, 1 mmol) in $CH_2Cl_2$ (15 mL), a soln of DCC (1.1 mmol) in $CH_2Cl_2$ (5 mL) was added. The mixture was stirred at 25 °C and followed by TLC (silica gel, $CHCl_3$ or hexane/EtOAc 4:1). The *N,N'*-dicyclohexylurea was filtered off and the filtrate concentrated to dryness. The residue was dissolved in EtOAc, cooled and the *N,N'*-dicyclohexylurea was removed by filtration. The filtrate was washed with 1 M aq $KHSO_4$, $H_2O$, 5% aq $NaHCO_3$, dried ($MgSO_4$), and concentrated to dryness in vacuo. The resulting oil was washed with petroleum ether and dried in vacuo to afford the analytically pure depsipeptide in 90–98% yield. The purity of the products was confirmed by TLC; quantitative analysis; IR and NMR spectra.

### 10.2.1.3 1,1'-Carbonyldiimidazole Method and Related Approaches

1,1'-Carbonyldiimidazole (CDI),[80,81] one of the oldest and most well-known reagents for the esterification of carbonic acids, was initially introduced by Brockmann et al. for the aminoacylation of α-acyl-β-hydroxyamino acids in their studies on synthesis of actinomycin $C_3$.[82,83] Schröder and Lübke[3] performed a systematic investigation of this method on a series of model depsipeptides constructed from D-hydroxyisovaleric acid derivatives and benzyloxycarbonyl- or *tert*-butoxycarbonyl-protected glycine, valine, or related dipeptides. The yields of the corresponding didepsipeptides were high (70–80%) while the yields of the tridepsipeptides were lower (35–60%) or even 0% when the carboxy components were the activated acyldipeptide residues, especially those containing the C-terminal bulky valine. The effect was ascribed to steric constraints.[3] The coupling reaction with CDI proceeds through the highly active intermediate imidazolide **13** of the amino acid component **11** which in turn is presumably formed via the corresponding mixed anhydride **12** (Scheme 7).[84] The imidazolide method was successfully applied to the synthesis of a variety of natural and model depsipeptides in a number of synthetic works from the early 1960s up to the late 1990s. As a rule, amino acid imidazolides **13** with an *N*-urethane protecting group react smoothly with hydroxy acid esters **14** without any notable racemization under cooling in dry dichloromethane or tetrahydrofuran, affording depsipeptides **15** in moderate or high yields, except when especially bulky reacting units are involved.

**Scheme 7** General Route to Protected *O*-Aminoacyl Hydroxy Acid Esters by the 1,1'-Carbonyldiimidazole Method




**11** **12** **13** **15**

X = urethane protecting group; OA = hydroxy acid; $R^1$ = alkyl, aryl

Gisin, Merrifield, and Tosteson in their solid-phase synthesis of valinomycin prepared the two protected didepsipeptides Boc-D-Val-L-Lac-OBzl and Boc-Val-D-Hyv-OBzl (Table 3) in solution using CDI with excellent yields.[85] A quantitative yield of *N*-(*tert*-butoxycarbonyl)-alanyl-hydroxyisovaleric acid benzyl ester (Boc-Ala-L-Hyv-OBzl) was also obtained by Shimohigashi et al.[86] by exploiting the same method during the synthesis of the *O*-methyltyrosine containing analogue of the phytotoxic metabolite AM-toxin. 1,1'-Carbon-

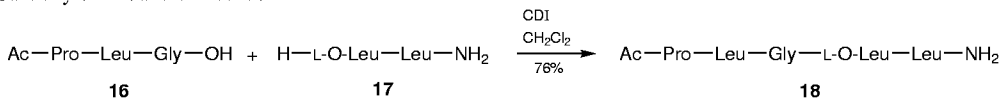
yldiimidazole was also found acceptable for fragment ester bond condensation in the case of the $N^{\alpha}$-blocked tripeptide **16** and $C^{\alpha}$-blocked didepsipeptide **17** that afforded the pentadepsipeptide **18** (Scheme 8), a substrate of mammalian collagenase, in 76% yield.[87]

**Table 3** Yields and Reaction Conditions for the Preparation of Didepsipeptides Using the 1,1′-Carbonyldiimidazole Method[85,86,90–94]

| Starting Material[a] | | Reaction Conditions | Product[a] | Yield (%) | Ref |
|---|---|---|---|---|---|
| Boc-D-Val-OH | H-L-Lac-OBzl | CDI, dry $CH_2Cl_2$, 0 °C, 1.5 h, then rt, 72 h | Boc-D-Val-L-Lac-OBzl | 97 | [85] |
| Boc-Val-OH | H-D-Hyv-OBzl | CDI, dry $CH_2Cl_2$, 0 °C, 1.5 h, then rt, 72 h | Boc-Val-D-Hyv-OBzl | 100 | [85] |
| Boc-Ala-OH | H-L-Hyv-OBzl | CDI, dry THF, 0 °C, 2.5 h, then rt, 12 h | Boc-Ala-L-Hyv-OBzl | 100 | [86] |
| THP-O-Phg-Aib-OH | H-D-O-Phg-Aib-$NMe_2$ | CDI, dry THF, rt, 2 h, then 40 °C, 6 h | THP-O-Phg-Aib-D-O-Phg-Aib-$NMe_2$, | 51 | [90] |
| Boc-Leu-OH | H-D-OCHR[1]-$CO_2$Bzl | CDI, dry THF, 0 °C, 7 h, then rt, 12 h | Boc-Leu-D-O-CHR[1]$CO_2$Bzl | 74 | [91] |
| Boc-Ala-OH | H-L-Lac-OBzl | CDI, dry $CH_2Cl_2$, 0 °C, 2.5 h, then rt, 72 h | Boc-Ala-L-Lac-OBzl | 68 | [92] |
| Z-(Me)Phe-OH | H-D-Hyv-OtBu | CDI, dry $CH_2Cl_2$, 0 °C, 1.5 h, then rt, 72 h | Z-(Me)Phe-D-Hyv-OtBu | 75 | [93] |
| Z-Val-OH | H-D-OCHR[2]-$CO_2$tBu | CDI, dry $CH_2Cl_2$, 0° C, 1.5 h, then rt, 90 h | Z-Val-OCHR[2]-$CO_2$tBu | 68 | [94] |

[a] O-Phg = -O-CH(Ph)-CO-; $R^1 = (CH_2)_{10}Me$; $R^2$ = Cy

**Scheme 8** Synthesis of a Protected Depsipeptide through the Ester Bond Condensation of Peptides by the 1,1′-Carbonyldiimidazole Method[87]




1,1′-Carbonyldiimidazole, in spite of its high moisture sensitivity, is the most popular reagent for the synthesis of depsipeptides among various related *N*-acylimidazoles, such as the thionyl analogue of 1,1′-carbonyldiimidazole or an equimolar mixture of acetylimidazole and acetyl chloride. However, the latter method was more efficient for the lactonization of pentapeptide chains of various actinomycin acids.[88,89]

***N*-(*tert*-Butoxycarbonyl)-D-valyl-L-lactic Acid Benzyl Ester; General Procedure for the Synthesis of Protected Didepsipeptides:**[85]

A soln of Boc-D-Val-OH (8.45g, 39 mmol) in dry $CH_2Cl_2$ (25 mL) was added to stirred 1 M CDI in $CH_2Cl_2$ (39 mL, 39 mmol) at 0 °C during 30 min, followed by H-L-Lac-OBzl (5.4 g, 30 mmol). The mixture was stirred for an additional hour at 0 °C and further kept at rt for 3 d. Then, $H_2O$ (10 mL) was added to destroy the excess of activated acid and the solvent was removed under reduced pressure. The residue was dissolved in $Et_2O$ (100 mL), washed with $H_2O$, aq citric acid, sat. aq $NaHCO_3$, and $H_2O$, dried ($MgSO_4$), and concentrated in vacuo to afford the depsipeptide. The oily product was homogenous by TLC (silica gel); yield: 97%; $[\alpha]_D^{20}$ −9.4 (*c* 3.3, benzene).

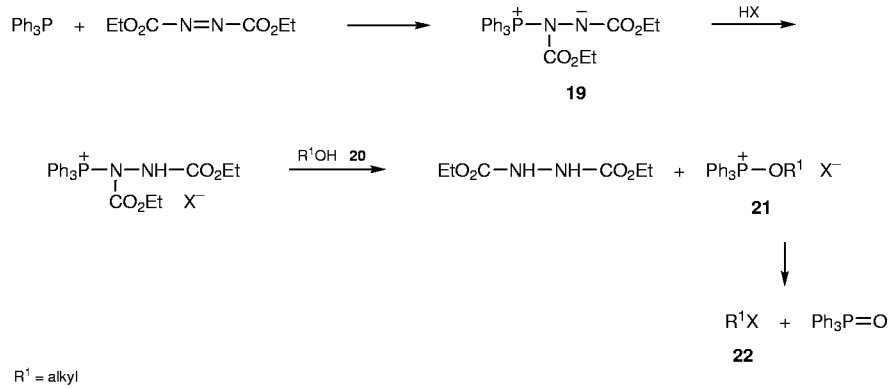
## 10.2.1.4 Activation of the Hydroxy Group

Intermolecular and intramolecular nucleophilic substitution of an alcoholic hydroxy group by the triphenylphosphine/dialkyl azodicarboxylate redox system is widely used in the synthesis and transformation of natural products and is known in organic chemistry as the Mitsunobu reaction.[95] This reaction starts with formation of the zwitterionic phosphonium adduct **19** (Scheme 9) from triphenylphosphine and diethyl (or diisopropyl) azodicarbox-

*for references see p 289*

ylate, followed by protonation of **19** by acid, activation of the hydroxy group of alcohol **20** via the oxyphosphonium salt **21** and subsequent $S_N2$-type substitution of the resulting intermediate **21** leading to the product **22**. The reaction proceeds under mild neutral conditions and exhibits functional selectivity and stereospecificity with complete inversion of the chirality of the hydroxy component.[96,97]

**Scheme 9** The Mitsunobu Reaction




The Mitsunobu reaction has been applied to depsipeptide synthesis in a number of recent works where the ester bond was formed either between two amino acid and hydroxy acid components or intramolecularly through ring closure by lactonization. The latter approach was used by Emmer et al.[98,99] in the semisynthesis of several modified derivatives (see Table 4) of the antifungal cyclic decadepsipeptide **23** (Scheme 10) containing one D-Lac residue in the ring. These authors subjected some of the amino acid residues in the natural compound to chemical modification, opened the cyclic system and performed recyclization under Mitsunobu conditions followed by simultaneous configurational inversion of the Lac residue. The macrolactonization reaction proceeded in toluene for a period of 24–25 hours in 22–67% yield depending on the structure of the linear precursor.[98,99] This reaction was effective for ester linking in the case of hindered and unstable derivatives, for example, in the total synthesis of the antitumor bicyclic S–S bridged depsipeptide FR-901228.[100]

**Scheme 10** The Structure of the Antifungal Cyclic Decadepsipeptide

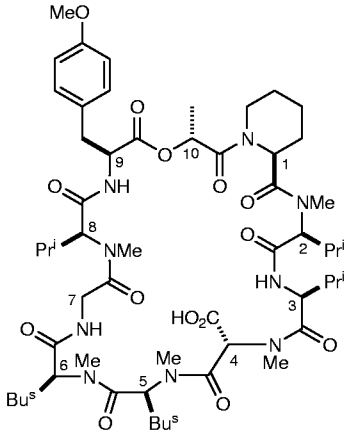


**Table 4** Yields and Reaction Conditions for Depsipeptide Synthesis Using the Mitsunobu Reaction[98,99,101]

| Starting Material | Reaction Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|
| H-D-Lac-Pip-(Me)Val-Val-(Me)Asp(OtBu)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-OH | $Ph_3P$/DEAD (1:1, 3 equiv), dry toluene, rt, 24 h | *c*[-Pip-(Me)Val-Val-(Me)-Asp(OtBu)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-L-Lac-] | 67 | [98] |
| H-D-Lac-Pip-(Me)Val-Val-(Me)Asp(*tert*-butylamide)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-OH | $Ph_3P$/DEAD (1:1, 2 equiv), dry toluene, rt, 5 h (mixing), then 20 h | *c*[-Pip-(Me)Val-Val-(Me)-Asp(*tert*-butylamide)-(Me)Ile-(Me)Ile-Gly-(Me)-Val-Tyr(Me)-L-Lac-] | 38 | [99] |
| H-D-Lac-Pip-(Me)Val-Val-(Me)Asp(Val-OtBu)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-OH | $Ph_3P$/DEAD (1:1, 2 equiv), dry toluene, rt, 5 h (mixing), then 20 h | *c*[-Pip-(Me)Val-Val-(Me)-Asp(Val-OtBu)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-L-Lac-] | 43 | [99] |
| Boc-β-Ala-OH, H-L-O-Leu-OBzl | $Ph_3P$/DEAD, toluene, 0 °C, 20 h | Boc-β-Ala-D-O-Leu-OBzl | 69 | [101] |

Smidt and Langner[101] in their synthesis of the natural calcium blocker leualacin utilized the Mitsunobu reaction for the construction of the linear depsipeptide *N*-(*tert*-butoxycarbonyl)-β-alanyl-*O*-D-leucic acid benzyl ester in 69% yield. Higher yields (94–96%) were reported for two didepsipeptide units *N*-(*tert*-butoxycarbonyl)-*N*-methylleucyl-D-lactic acid diphenylmethyl ester and *N*-(*tert*-butoxycarbonyl)-*N*-methylleucyl-D-phenyllactic acid diphenylmethyl ester in the course of the total synthesis of the anthelmintic cyclooctadepsipeptide PF1022A based on the Mitsunobu reaction.[102]

Mitsunobu activation of a hydroxy group has been successfully utilized for the construction of depsipeptide units in only a limited number of published synthetic works. Nevertheless, this procedure might be considered as an effective and promising approach to depsipeptide synthesis even in complicated cases due to its simplicity, mild conditions, and absence of racemization.

**c[-Pip-(Me)Val-Val-(Me)Asp(OtBu)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-L-Lac-]; General Procedure for Lactonization:**[98]

A soln of DEAD (2.37 mL, 15 mmol) in toluene (500 mL) was added under vigorous stirring to a soln of H-D-Lac-Pip-(Me)Val-Val-(Me)Asp(OtBu)-(Me)Ile-(Me)Ile-Gly-(Me)Val-Tyr(Me)-OH (6.0 g, 5 mmol) and $Ph_3P$ (4.08 g, 15 mmol) over a period of 24 h. The solvent was removed in vacuo, and the crude product was purified by column chromatography (silica gel, hexane/EtOAc) and then freed from traces of $Ph_3PO$ on Sephadex LH 20 ($CH_2Cl_2$) to afford the title compound; yield: 4.02 g (67%); mp 146–148 °C; $[\alpha]_D^{20}$ −221 (*c* 1.0, $CH_2Cl_2$).

## 10.2.1.5 Active Esters

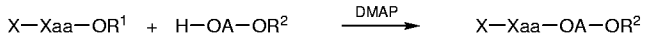
The active ester methodology, which is widely used in peptide chemistry, has found only limited application in depsipeptide synthesis. A more vigorous activation of the carboxy component is apparently required to form an ester bond compared to the peptide analogue. Nevertheless, active esters have been utilized for this purpose in combination with some catalyst additives. The first successful attempt in this direction was described by Mazur.[103] The modification of the 4-nitrophenyl ester procedure included addition of 1–10 equivalents of imidazole to the reaction mixture. This accelerated technique presumably involves formation of the highly reactive intermediate imidazolide. The reaction resulted in the preparation of model benzyloxycarbonyl didepsipeptide esters in good yields within several hours at room temperature from 4-nitrophenyl esters of Z-amino acids and the pentamethylbenzyl ester of glycolic acid, while in the absence of imidazole this reaction failed to give any product.

*for references see p 289*

The above described technique was further extended by Mammi and Goodman[104] who applied 4-(dimethylamino)pyridine to enhance the reactivity of Boc-alanine *N*-hydroxysuccinimide ester in the synthesis of depsipeptide collagen model structures. Homogenous Boc-alanyl-glycolic acid benzyl ester was obtained in 94% yield. A similar approach was also successfully employed for the construction of depsipeptide building blocks in 84–85% yield in the course of the total synthesis of the cyclic natural cytotoxic cryptophycins A, B, C, and D[105] as well as in the synthesis of an esterase-sensitive cyclic prodrug with a lactone structure.[106]

The application of hydroxysuccinimide, 4-nitrophenyl, or other types of active esters together with 4-(dimethylamino)pyridine as a catalyst (Scheme 11) is a possibly complementary procedure to existing popular methods leading to depsipeptides. Nevertheless, this approach, in spite of its simplicity and mildness, has so far found limited application.

**Scheme 11** General Route to Protected *O*-Aminoacyl Hydroxy Acid Esters by Active Esters



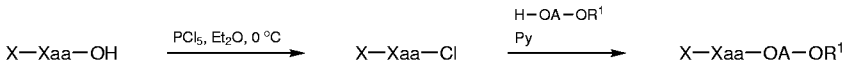

X = urethane protecting group; OA = hydroxy acid
$OR^1$ = OSu, ONp, or other active ester; $R^2$ = alkyl, aryl

**Boc-Ala-O-Gly-OBzl; General Procedure for Depsipeptide Synthesis through Active Esters:**[104]
To a soln of Boc-Ala-OSu (11.8 g, 0.04 mol) in $CH_2Cl_2$ (200 mL), glycolic acid benzyl ester (6.65 g, 0.04 mol) and DMAP (4.89 g, 0.04 mol) were added and the mixture was allowed to stand overnight at rt. The resulting soln was washed with 1 M HCl followed by $H_2O$, and dried ($Na_2SO_4$). The solvent was removed in vacuo to afford the product (homogenous by TLC); yield: 12.6 g (94%).

## 10.2.1.6 Acid Chlorides

Acid chlorides of N-protected amino acids are extremely activated derivatives and have been applied to ester bond formation, together with mixed anhydrides, as the main synthetic approach from the early 1960s. Amino acid chlorides are easily formed by treatment of N-protected amino acid derivatives in dry diethyl ether with phosphorus pentachloride or thionyl chloride at 0 °C and may be isolated in crystalline form (e.g., Phth-Val-Cl), but are more often used as coupling reagents without isolation (Scheme 12). The activated intermediates, even the phthaloyl-containing derivatives with their relatively active $C^{\alpha}$-hydrogen, do not notably racemize in the absence of a base. In the presence of a base, however, amino acid chlorides undergo pronounced racemization, therefore excess base should be avoided. For example, using a strictly controlled amount of pyridine in the reaction mixture, Shemyakin et al.[38] succeeded in preparing phthaloyl-D-valyl-L-lactic acid benzyl ester in 90% yield (Table 5). This approach was later employed by Losse and Bachmann[107] for the synthesis of other phthaloylaminoacyl hydroxy acid benzyl esters (see Table 5) in good yield. Alternatively, optically pure related didepsipeptides were obtained from phthaloyl amino acid chlorides in moderate yields without any organic base when hydroxy components were taken in the form of mixed magnesium alcoholates.[22,38] Quitt, Studer, and Vogler[37] prepared the didepsipeptide NO-(Me)Ile-D-Hyv-OBzl($NO_2$) by coupling *N*-methyl-*N*-nitrosoisoleucine acid chloride with the appropriate hydroxy component under deep cooling in a pyridine-containing solution. The product served as the principal building block in the total synthesis of enniatin A.

**Scheme 12** General Route to Protected *O*-Aminoacyl Hydroxy Acid Esters by Acid Chlorides




X = Phth, NO; OA = hydroxy acid; $R^1$ = alkyl, aryl

Generally, the enhanced risk of racemization of amino acid chlorides, due to their known susceptibility to cyclization into oxazolones, as well as the instability of urethane-protected derivatives easily convertible into the corresponding *N*-carboxy anhydrides,[108] has resulted in very limited application of this technique in depsipeptide chemistry in spite of the successful examples discussed.

**Table 5** Yields and Reaction Conditions for Depsipeptide Synthesis Using Acid Chlorides[38,107]

| Starting Material | | Reaction Conditions[a,b] | Product | Yield (%) | Ref |
|---|---|---|---|---|---|
| Phth-D-Val-Cl | H-L-Lac-OBzl | dry benzene, 0 °C, Xaa/OA/pyridine (1.1:1.2:1) | Phth-D-Val-L-Lac-OBzl | 90 | [38] |
| Phth-DL-Val-Cl | H-DL-Lac-OBzl | dry $Et_2O$, −10 °C, Xaa/OA/pyridine (1:1:1) | Phth-DL-Val-DL-Lac-OBzl | 71 | [107] |
| Phth-Val-Cl | H-L-Lac-OBzl | dry $Et_2O$, −10 °C, Xaa/OA/pyridine (1:1:1) | Phth-Val-L-Lac-OBzl | 91 | [107] |
| Phth-DL-Leu-Cl | H-DL-Lac-OBzl | dry $Et_2O$, −10 °C, Xaa/OA/pyridine (1:1:3) | Phth-DL-Leu-DL-Lac-OBzl | 75 | [107] |
| Phth-Val-Cl | H-D-Hyv-OBzl | dry benzene, 0 °C, Xaa/OA/pyridine (1.1:1.2:1) | Phth-Val-D-Hyv-OBzl | 95 | [38] |

[a] **CAUTION:** *Benzene is a carcinogen!*
[b] OA = hydroxy acid.

***N*-Methyl-*N*-nitrosoisoleucyl-D-hydroxyisovaleric Acid 4-Nitrobenzyl Ester:**[37]
$PCl_5$ (16.7 g, 80 mmol) was added at 0 °C to a soln of NO-(Me)Ile-OH (14 g, 80 mmol) in dry $Et_2O$ (150 mL). The mixture was stirred for 15 min under cooling and then was dried in vacuo at 35 °C. The residue was treated twice with toluene and concentrated in vacuo to remove $POCl_3$, then dissolved in dry $Et_2O$ (50 mL) followed by cooling to −60 °C. The mixture was added under vigorous stirring at −60 °C to H-Hyv-OBzl($NO_2$) (19 g, 75 mmol) in dry $Et_2O$ (100 mL) and then to a mixture of pyridine (15 mL) and $Et_2O$ (15 mL). The temperature was allowed to increase to 0 °C and the soln was stirred for 2 h. The mixture was washed with 1 M HCl, 10% aq $KHSO_4$, and $H_2O$, dried, and concentrated in vacuo. The oily residue was purified by column chromatography (neutral alumina, benzene, and benzene/EtOAc 9:1) to afford the title compound; yield: 16.4 g (53%); $[\alpha]^{20}_D$ −37.5 (*c* 1, benzene).

## 10.2.1.7 Other Synthetic Approaches

In addition to the more or less popular methods of depsipeptide synthesis discussed *vide supra*, there are also a limited number of complementary and effective synthetic procedures that have been described for this purpose. Among these, the well-known method of symmetric anhydrides from N-protected amino acids has to be considered. This method has found successful use in the esterification of hydroxy acids in the presence of some catalyst additives. Initially, the addition of pyridine[109] or 1-hydroxybenzotriazole in pyridine[110] to a symmetric anhydride was utilized for ester bond formation. As an example, Katakai has prepared a number of didepsipeptides in 85–96% yield by means of a 2-nitrophenylsulfenyl *N*-carboxy anhydride with lactic acid derivatives in the presence of pyridine.[109]

This procedure was later modified by the application of 4-(dimethylamino)pyridine as a catalyst in combination with Fmoc amino acid anhydrides in the course of the solid-phase syntheses of several depsipeptide substrates of human hepatitis C virus protease. Thus, Bianchi et al.[111] synthesized the 13-membered depsipeptide Ac-Asp-Glu-Met-Glu-Glu-

*for references see p 289*

αAbu-Lac-Ser-His-Leu-Pro-Tyr-Lys(Ac)-$NH_2$ by solid-phase synthesis using the continuous-flow Fmoc-polyamide technique. The reaction of $(Fmoc\text{-}\alpha Abu)_2O$ with the Lac-containing peptidyl polymer proceeded in the presence of 0.1 molar equivalent of 4-(dimethylamino)-pyridine for 30 minutes at room temperature and was repeated twice to afford a 70% coupling yield. Using this technique the authors also prepared in the same yield another closely related depsipeptide substrate Ac-Asp-Glu-Met-Glu-Glu-Thr-L-Lac-Ser-His-Leu-Pro-Tyr-Lys(Ac)-$NH_2$ containing a Thr residue instead of αAbu. Modest racemization of the Thr residue during esterification was detected and the resulting diastereomers were separated by HPLC.[111] The same Italian group used an alternative route during the synthesis of the fluorogenic depsipeptide substrates **24–26** (Scheme 13) of human hepatitis C virus protease. The synthesis, performed on a derivatized Tentagel resin using the continuous-flow Fmoc-polyamide solid-phase procedure and the same method of ester link forming between αAbu and Lac residues, allowed the authors to achieve the desired depsipeptide polymers in 92% yield after double coupling.[112,113]

**Scheme 13** Fluorogenic Depsipeptide Substrates of Human Hepatitis C Virus Protease

Ac-DED(EDANS)EE-αAbu-Lac-SK(DABCYL) **24**

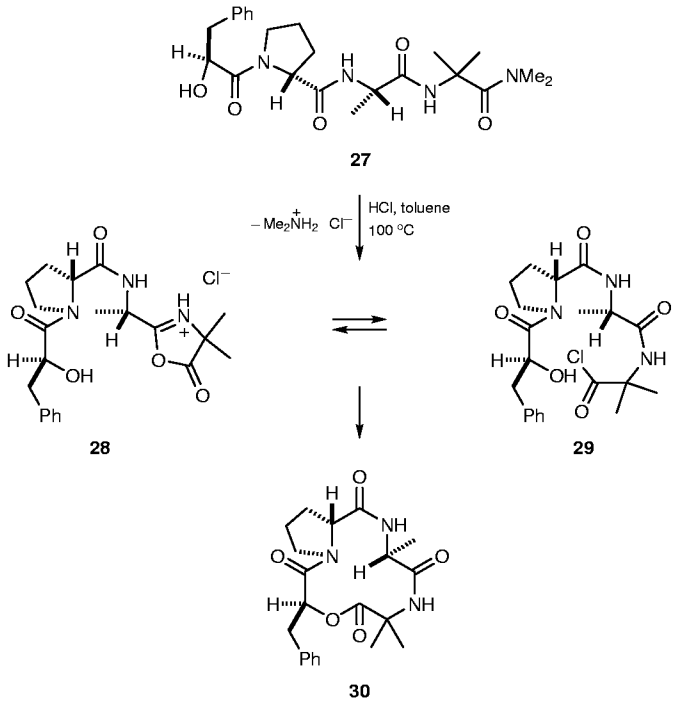
Ac-DD(EDANS)MEE-αAbu-Lac-SK(DABCYL) **25**

Ac-DED(EDANS)EE-αAbu-Lac-SK(DABCYL)GG-$NH_2$ **26**

EDANS = 5-[(2-aminoethyl)amino]naphthalenesulfonic acid
DABCYL = 4-{[4-(dimethylamino)phenyl]azo}benzoic acid

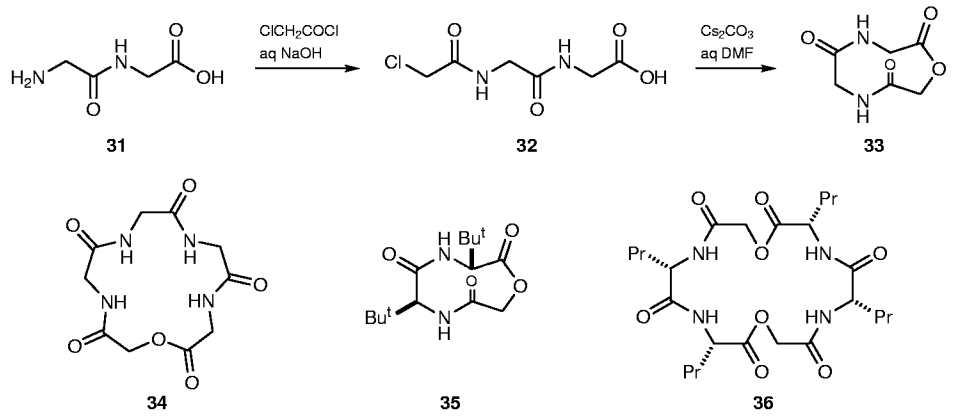
The Swiss authors Obrecht and Heimgartner[114] have described a rather unusual method for the closure of a depsipeptide ring through ester bonding via direct amide cyclization. They have shown that a linear tetradepsipeptide **27** (phenyllactoyl-prolyl-alanyl-α-isobutyric acid dimethylamide) is smoothly converted into the cyclic depsipeptide **30** in 72% yield under the acidic conditions of HCl gas in dry toluene at 100 °C for 15 minutes. The authors have suggested that the cyclization reaction proceeds through formation of intermediate oxazolone **28** and corresponding acyl chloride **29** with release of dimethylamine hydrochloride and subsequent formation of the cyclodepsipeptide (Scheme 14). This reaction proceeds more smoothly in the case of peptides with C-terminal α,α-disubstituted amino acid dimethylamide residues.[114] This technique was further applied by the same authors in the synthesis of two other cyclic tetradepsipeptides *c*[-Aib-O-Phg-]$_2$ and *c*[-Aib-O-Phe-]$_2$ in 88% yield from their linear depsipeptide precursors.[90]

**Scheme 14** Route to Cyclic Depsipeptides via Direct Amide Cyclization[114]




The next synthetic approach that may contribute to the existing arsenal of methods of depsipeptide synthesis consists of the esterification of cesium salts of protected amino acids by halo-containing progenitors of hydroxy acids. This approach originated from the early observation of Fischer and Scheiber[115] who in 1908 reported the reaction of chloroacetyl chloride with α-amino acids followed by treatment with tertiary amines providing 6-membered morpholine-2,4-diones. Based on this observation Kruizinga et al.[116] reported a synthetic route to relatively simple macrocycles containing one ester linkage and a peptide backbone. Schotten–Baumann acylation of an amino acid, short peptide, or depsipeptide (e.g., **31**) with 2-chloroacetyl chloride, followed by cesium carbonate mediated ring closure of the resulting acyl peptide **32** affords the corresponding cyclic depsipeptides **33** or **34–36** in 80–90% yield (Scheme 15).

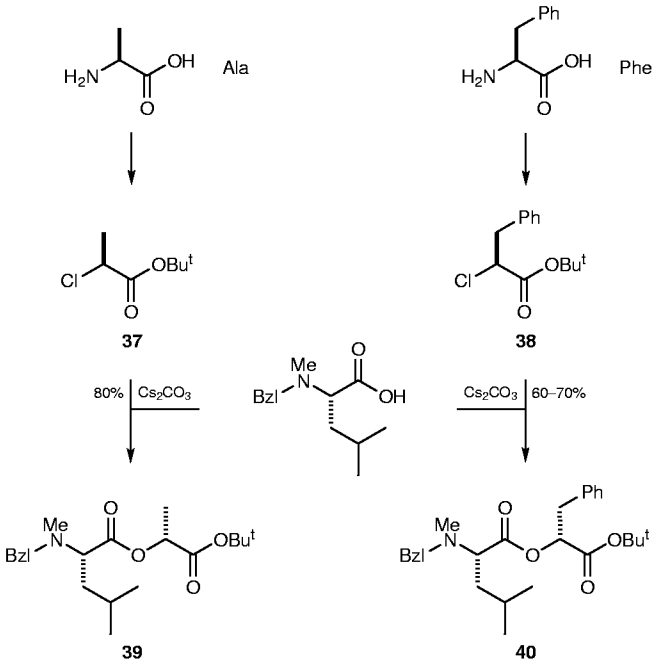
**Scheme 15** Route to Cyclic Depsipeptides through Acyl Peptide Cesium Salt[116]




*for references see p 289*

The same approach, applied in this case to intermolecular ester bond formation between a protected amino acid or a peptide cesium salt and an appropriate haloacyl derivative, was successfully utilized in syntheses of linear and cyclic depsipeptides. Thus, a highly efficient synthesis of the anthelmintic cyclooctadepsipeptide PF1022A was performed by Scherkenbeck et al.[117] This depsipeptide consists of four *N*-methylleucine, D-lactoyl, and D-phenyllactoyl moieties linked via amide and ester bonds. The corresponding linear didepsipeptide fragments **39** and **40** were prepared from *N*-benzyl-*N*-methylleucine cesium salt and the chlorocarboxylic acid derivatives **37** and **38** (Scheme 16), respectively. The latter were obtained in turn from alanine and phenylalanine according to a standard procedure for transformation of amino acids to their halo counterparts with full retention of configuration. Coupling of these derivatives with the amino acid component proceeded in high yield and high degree of inversion. Indeed, no diastereomers were found for **39** and only partial racemization was detected for **40**. The same group has further extended the above described convenient procedure to the synthesis of a number of related cyclic tetra- and hexadepsipeptides suitable for attacking pathogenic endoparasites.[118,119]

**Scheme 16** Route to Protected *O*-Aminoacyl Hydroxy Acid Esters through Amino Acid Cesium Salts and Haloacyl Derivatives[117]




The synthetic procedures mentioned in this section certainly do not cover all the known routes leading to depsipeptides. Some seldom-used or potentially useful but not yet practically employed techniques are inevitably missing. For example, such potential might be attributed to the method of carboxylic acid activation by pyrocarbonates suggested by Pozdnev in 1992.[120] In particular, this author has shown in some model cases that $Boc_2O$ in the presence of pyridine and a catalytic amount of 4-(dimethylamino)pyridine forms an ester link between an amino acid and an α-hydroxy acid in 80–90% yield.

***N*-Benzyl-*N*-methylleucyl-D-lactic Acid *tert*-Butyl Ester 39**:[117]
The dry cesium salt of *N*-benzyl-*N*-methylleucine (77.7 g, 0.212 mol) and *tert*-butyl 2-chloropropanoate (34.76 g, 0.212 mol) were stirred in DMSO (530 mL) for 20 h at rt. The mixture was poured into brine and

extracted with EtOAc. The organic layer was washed with $H_2O$, dried ($Na_2SO_4$), filtered and concentrated. The residue was column chromatographed (silica gel, cyclohexane/EtOAc 60:1) to give **39**; yield: 63.5 g (82%); $[\alpha]_D^{25}$ −35.0 (*c* 2.1, MeOH).


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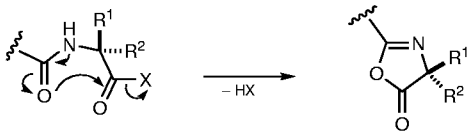
# 10.3 Synthesis of Peptides Based on $C^{\alpha}$-Tetrasubstituted α-Amino Acids

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The increasing number of papers dealing with applications of $C^{\alpha}$-tetrasubstituted α-amino acids ($C^{\alpha,\alpha}$-disubstituted glycines) or giving new synthetic methodologies for their incorporation into peptide sequences is motivated by the following factors: (i) The occurrence of some $C^{\alpha}$-tetrasubstituted α-amino acids in peptaibols,[1] an important family of membrane-active, natural antibiotics.[2,3] (ii) The usefulness of these residues in the synthesis of conformationally constrained, enzyme-resistant agonists and antagonists of bioactive peptides.[4,5] (iii) The high crystallinity of their derivatives which, by means of X-ray diffraction techniques, gives access to the conformational and electronic structure characterization of the N- and C-protecting groups, and C-activating groups (including reactive intermediates) commonly used in peptide synthesis.[6] (iv) The unique stereochemistry of peptides containing these residues[7–9] that, by severely restricting the conformational freedom, allows for the exploitation of such compounds as precise molecular rulers,[10] scaffolding blocks in the de novo design of protein and enzyme mimetics,[11,12] or suitable models for spectroscopic studies.[13,14] As only a few $C^{\alpha}$-tetrasubstituted α-amino acids are commercially available, many synthetic methods have been devised for their preparation.[15]

The main problem a peptide chemist has to tackle with $C^{\alpha}$-tetrasubstituted α-amino acids is their rather poor reactivity in peptide-bond formation due to steric hindrance at the α-carbon. In addition, during the activation process of the carboxylic function of peptides and urethane or amide N-protected derivatives, an intramolecular reaction leading to the oxazol-5(4*H*)-one heterocyclic skeleton is greatly favored by the *gem*-dialkyl effect (Scheme 1).[16,17]

**Scheme 1** Oxazol-5(4*H*)-one Formation[18–22]




Since the pioneering work of Goodman and McGahren,[18,19] oxazol-5(4*H*)-one formation has been recognized as the main pathway leading to racemization (or epimerization) of C-activated derivatives and peptide fragments involving $C^{\alpha}$-trisubstituted protein amino acids.[20,21] This is unlikely to occur with $C^{\alpha}$-tetrasubstituted α-amino acids, since racemization (or epimerization) would require a C–C or C–N bond breakage. Indeed, the shelf-stable oxazol-5(4*H*)-ones from $C^{\alpha}$-tetrasubstituted α-amino acid derivatives and peptides are easy to isolate and can be directly exploited in peptide synthesis. Following ring-opening by the nucleophilic attack of an amino function at the C5 atom, a peptide bond can form. Despite the relatively poor reactivity of the oxazol-5(4*H*)-ones at room temperature, such intermediates were extensively employed in early work and are still a viable choice in peptide-fragment condensations which do not involve racemizable residues (see Section 10.3.2.1.1).

Since the first systematic synthetic effort of Leplawy et al.[22] the preparation of peptides containing the slow-reacting $C^{\alpha}$-tetrasubstituted α-amino acids has been achieved mainly by solution procedures. However, recent improvements in activation methodologies make solid-phase peptide synthesis (SPPS) an attractive tool as well, at least with the less sterically hindered $C^{\alpha}$-tetrasubstituted α-amino acids.

## 10.3.1 Peptides with Non-consecutive $C^{\alpha}$-Tetrasubstituted α-Amino Acids

Compared to couplings between protein amino acids, those involving one $C^{\alpha}$-tetrasubstituted α-amino acid usually give lower yields even with longer reaction times and, where possible, higher temperatures. Long-lasting couplings may lead to racemization of C-activated protein amino acids. In addition, when a peptide sequence bearing a $C^{\alpha}$-tetrasubstituted α-amino acid at the C-terminus is activated, racemization of the penultimate residue may also occur through oxazol-5(4*H*)-one formation.[23,24] Therefore, segment condensation has to be carefully planned and the use of agents to suppress racemization or fast-reacting activation methods are recommended.

### 10.3.1.1 Synthesis in Solution

A large variety of methods have been devised in order to incorporate a $C^{\alpha}$-tetrasubstituted α-amino acid into a peptide sequence.[25] Selected examples are given in Table 1.

**Table 1** Selected Examples of Methods Used To Incorporate a $C^{\alpha}$-Tetrasubstituted Amino Acid into a Peptide Sequence[27–36]

| Amino Acids | Coupling Method | Yield (%) | Ref |
|---|---|---|---|
| Aib + Gly | DCC | 55 | [26] |
| Aib + Gly | BOP | 92 | [27] |
| Aib + Gly | SA | 94 | [28] |
| Aib + Val | oxazolone | 70 | [29] |
| Aib + Pro | BOP/DMAP | 95 | [30] |
| Aib + Pro | acyl chloride | 98 | [30] |
| Pro + Aib | PyBOP | 88 | [27] |
| Pro + Aib | MA | 78 | [31] |
| Phe + Aib | UNCA | 86 | [32] |
| Ile + Aib | azirine | 99 | [33] |
| Val + Aib | CIP | 60 | [34] |
| Val + Aib | CIP/HOAt | 85 | [34] |
| HmS(Ipr) + Gly | HBTU | 88 | [35] |
| $Ac_7c$ + Val | EDC/HOBt | 72 | [36] |

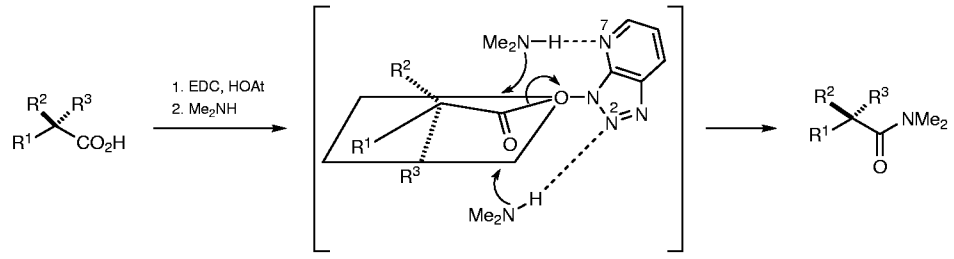
Tetrasubstitution at the α-carbon usually affects the reactivity of the amino more than that of the carboxyl function. In the preparation of the cyclic dipeptide *c*[-His(Trt)-Phe(αMe)-] from the N-terminal-free dipeptide H-(αMe)Phe-His(Trt)-OMe, the yield is only 20% after refluxing in methanol for one week, whereas H-His(Trt)-(αMe)Phe-OMe gives 50% yield of the cyclization product after refluxing for only one day.[37] In general, with any coupling method, yields are higher when the nucleophile is the protein amino acid, the main exception being represented by the C-terminal activation procedures leading to mixed anhydride (MA) formation. In a comparative study of nine different coupling methods, performed on the Dphg-Ala and Ala-Dphg peptide bond, Yamada et al.[38] found that with seven activation procedures, yields are higher in the Dphg-Ala peptide-bond formation. With the remaining two coupling procedures, i.e. the MA (with isobutyl chloroformate) and the EEDQ-mediated MA methods, the results are reversed. In these cases the bulkiness of the $C^{\alpha}$-tetrasubstituted α-amino acid may favor the nucleophilic attack of Ala at the "wrong" carbonyl of the MA, thus dramatically decreasing the yield of the desired product. Analogous observations were made as early as 1965 by studying the reaction of the MA between $N^{\alpha}$-protected (αMe)Phe and pivalic acid with cyclohexylamine.[39] Therefore, the MA methodologies, either of the carboxylic-carbonic or carboxylic-carboxylic type, should be limited to couplings where the $C^{\alpha}$-tetrasubstituted α-amino acid is the nucleophile.

With the above limitations in mind, almost all of the methods described in the literature give acceptable to good results in couplings involving only one $C^{\alpha}$-tetrasubstituted α-amino acid. However, when a very bulky residue is involved, the use of the amino acyl fluoride or other highly effective methods has to be considered (see Section 10.3.2.1). Because of possible racemization (or epimerization), procedures involving DMAP as the acylation catalyst should be avoided when the C-activated residue is a $C^{\alpha}$-trisubstituted amino acid. Taking into account coupling efficiency, ease of the procedure, and racemization (or epimerization) risks, the EDC-mediated active ester formation with HOAt[40] is probably the method of choice for peptide-bond formation involving a $C^{\alpha}$-tetrasubstituted and a $C^{\alpha}$-trisubstituted α-amino acid (see Section 10.3.1.1.1). Similar results can be achieved with the symmetrical anhydride (SA) method (see Section 10.3.1.1.2).

#### 10.3.1.1.1 *N*-[3-(Dimethylamino)propyl]-*N'*-ethylcarbodiimide Hydrochloride/ 7-Aza-1,2,3-benzotriazol-1-ol Activation

In this procedure the acylating molecule is the active ester formed between the amino acyl moiety and the hydroxy group of HOAt.[40] Compared to the earlier methods utilizing HOSu[41] and HOBt,[42] HOAt maintains the known ability to suppress racemization, but is much more effective in catalyzing peptide-bond formation. This efficiency of HOAt may be attributed to the possible assistance of both N2 and N7 to the amine nucleophilic attack (Scheme 2). If the preferred active ester conformation found in the crystalline state (the heterocyclic At plane lies approximately perpendicular to that of the carboxylic ester moiety)[43] is preserved in solution, then assistance to nucleophilic attack on both faces of the ester is likely.

**Scheme 2** Possible Catalytic Pathways in Carbodiimide/HOAt Coupling[40,43]




Active ester formation is conveniently achieved by treating an equimolar amount of HOAt and a compound with a free carboxylic acid moiety with the water-soluble carbodiimide EDC, which greatly facilitates isolation and purification procedures compared to the classic use of DCC. The active ester with HOAt (or HOBt) is also believed to be the acylating species with the CIP/HOAt,[44] BOP,[45] and PyBOP[27] activation procedures. Nevertheless, a carbodiimide seems to be the most practical reagent for use in the active ester formation process. Couplings can be performed with good results, even without the addition of any base.[37] In fact, the increasing acidity, governed by the HOAt molecules liberated during the reaction process, does not seem to have any major impact on coupling yields. This can be advantageous when base-sensitive protecting groups are present. Furthermore, it is possible to add extra amounts of EDC during the course of the reaction in order to reconstitute the partially hydrolyzed acylating active ester. Thus, prolonged reaction times may effectively increase the yields in difficult couplings.

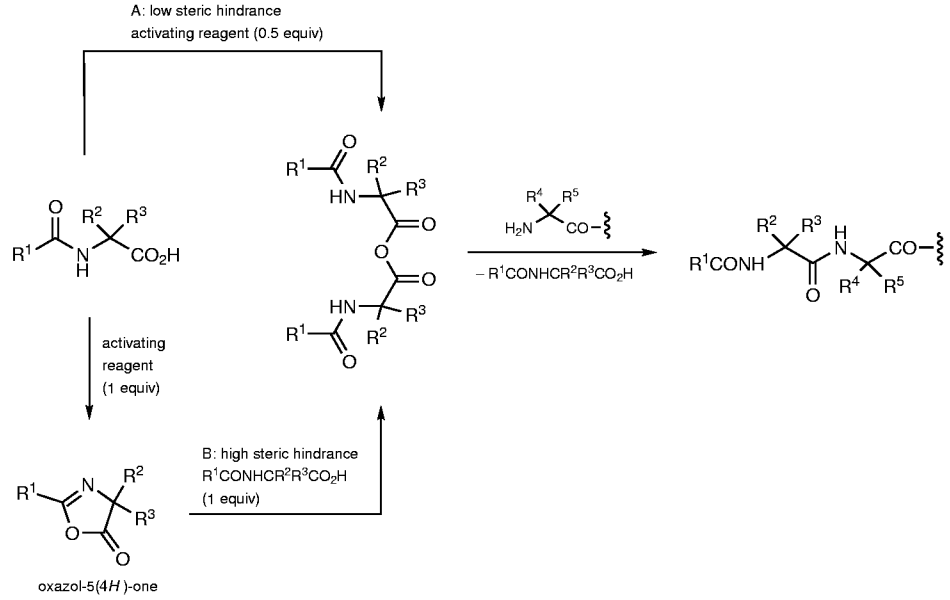
**Z-(αMe)Phg-Ala-Ala-(αMe)Phg-Ala-OMe:**[46]
Z-(αMe)Phg-OH (60 mg, 0.20 mmol) was dissolved in anhyd $CH_2Cl_2$ (2 mL) and cooled to 0 °C. HOAt (29 mg, 0.21 mmol) was added (a suspension was formed) followed by EDC (40 mg, 0.21 mmol). After stirring the mixture for 5 min (after about 1 min the soln became clear, thereby indicating that active ester formation had occurred), H-Ala-Ala-(αMe)Phg-Ala-OMe [0.17 mmol, obtained by Pd-catalyzed hydrogenation of the corresponding Z-protected tetrapeptide ester (90 mg, 0.17 mmol)] dissolved in anhyd $CH_2Cl_2$ (1 mL) was added, followed by NMM (0.025 mL, 0.23 mmol). After stirring the soln at rt for 3 d, the solvent was removed under reduced pressure and the mixture was dissolved in EtOAc. The organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), and evaporated to dryness. The peptide was crystallized from EtOAc by the addition of light petroleum ether; yield: 80 mg (70%); mp 123–124 °C; $[\alpha]_D^{20}$ −19.5 (*c* 0.5, MeOH).

**Z-Trp-Aib-OtBu:**[47]
To a soln of Z-Trp-OH (1.30 g, 3.84 mmol) in anhyd $CH_2Cl_2$ (8 mL) at 0 °C was added HOAt (0.522 g, 3.84 mmol) followed by EDC (0.736 g, 3.84 mmol). After stirring for 5 min (after about 1 min the soln became clear, thereby indicating that active ester formation had occurred), H-Aib-OtBu [3.84 mmol, obtained by Pd-catalyzed hydrogenation of the corresponding Z-protected amino ester (1.13 g, 3.84 mmol)] dissolved in anhyd $CH_2Cl_2$ (4 mL) was added, followed by NMM (0.423 mL, 3.84 mmol). After stirring the soln at rt for 3 d, the mixture was dissolved in EtOAc, and washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), and evaporated to dryness. The product was crystallized from EtOAc by the addition of light petroleum ether; yield: 1.54 g (83%); mp 67–68 °C; $[\alpha]_D^{20}$ −14.9 (*c* 0.5, MeOH).

### 10.3.1.1.2 Symmetrical Anhydride Activation

Symmetrical anhydrides (SAs) of protein amino acids are conveniently prepared in situ, whereas the more stable SAs from $C^{\alpha}$-tetrasubstituted α-amino acids generally give better results in coupling reactions when they are isolated beforehand. Only SAs of the least sterically hindered $C^{\alpha}$-tetrasubstituted α-amino acids, e.g. Aib, can be prepared in good yields upon treatment of the urethane N-protected amino acid with half an equivalent of carbodiimide or thionyl chloride. In most of the other cases a mixture of SA and oxazol-5(4*H*)-one is obtained;[48] however, SAs can be obtained in satisfactory yields by reacting one equivalent of preformed oxazol-5(4*H*)-one with one equivalent of the corresponding free acid (Scheme 3).[49,50]

**Scheme 3** Symmetrical Anhydrides: Preparation and Peptide Bond Formation[48–51]




*for references see p 307*

Peptide bond formation via SAs may proceed with satisfactory results even in the absence of any base, although the N-protected amino acid liberated during anhydride aminolysis is commonly neutralized with a tertiary amine. A drawback of the SA procedure is the need for two equivalents of N-protected amino acid derivative, the excess of which can only be recovered in part during the workup procedure.

**Symmetrical Anhydride Preparation:**

**$(Z\text{-Aib})_2O$ (Pathway A, Scheme 3):**[51]

To a soln of Z-Aib-OH (30 g, 126 mmol) in MeCN/EtOAc (3:1, 160 mL) at 0 °C was added EDC (12.08 g, 63 mmol). The mixture was stirred at 0 °C for 2 h and at rt for 24 h. The solvent was removed under reduced pressure, the residue was dissolved in EtOAc, and washed with 0.5 M citric acid, $H_2O$, 5% $NaHCO_3$, and $H_2O$. The organic layer was dried ($Na_2SO_4$), filtered, and concentrated to dryness. The product was crystallized from EtOAc by the addition of light petroleum ether; yield: 22.01 g (83%); mp 100–101 °C; IR (KBr): 1814, 1742 $cm^{-1}$ (anhydride carbonyl stretchings).

**$(Z\text{-Ac}_9c)_2O$ (Pathway B, Scheme 3):**[50]

To a suspension of Z-$Ac_9c$-OH (1.456 g, 4.56 mmol) in MeCN (4 mL) was added EDC (0.961 g, 5.01 mmol). After stirring at rt for 1 h, the solvent was removed under reduced pressure. The residue was dissolved in EtOAc and washed with $H_2O$. The organic layer was dried ($Na_2SO_4$), filtered, and concentrated to dryness to give the oily oxazol-5(4*H*)-one; yield: 1.333 g (97%); IR (film): 1827 $cm^{-1}$ (C=O stretching), 1684 $cm^{-1}$ (C=N stretching). To this oxazolone (1.333 g, 4.42 mmol) were added MeCN (8 mL) and Z-$Ac_9c$-OH (1.387 g, 4.42 mmol) and the mixture was stirred for 2 d at rt. The solvent was removed under reduced pressure, the residue was dissolved in EtOAc, and washed with 5% $NaHCO_3$ and $H_2O$. The organic layer was dried ($Na_2SO_4$), filtered, and concentrated to dryness. The product was crystallized from EtOAc by the addition of light petroleum ether; yield: 1.975 g (72%); mp 146–147 °C; IR (KBr): 1813, 1749 $cm^{-1}$ (anhydride carbonyl stretchings).

**Coupling Procedures:**

**Z-Aib-Ser(tBu)-Leu-Aib-Ser(tBu)-Ser(tBu)-Leu-Aib-Ser(tBu)-Ile-Leu-OMe:**[52]

To a soln of H-Ser(tBu)-Leu-Aib-Ser(tBu)-Ser(tBu)-Leu-Aib-Ser(tBu)-Ile-Leu-OMe [1.20 mmol, obtained by Pd/C-catalyzed hydrogenation of the Z-protected decapeptide ester (1.63 g, 1.20 mmol)] and NMM (0.130 mL, 1.18 mmol) in anhyd MeCN (4 mL) was added $(Z\text{-Aib})_2O$ (0.538 g, 1.18 mmol). The mixture was stirred at rt overnight, then the solvent was removed in vacuo. The residue was dissolved in EtOAc, and washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$. The organic layer was dried ($Na_2SO_4$), filtered, and concentrated to dryness. The product was crystallized from EtOAc by the addition of light petroleum ether; yield: 1.646 g (96%); mp 185–187 °C; $[\alpha]_D^{20}$ −8.2 (*c* 0.5, MeOH).

**Z-Ala-$Ac_9c$-Ala-OMe:**[53]

To a soln of Z-Ala-OH (0.182 g, 0.82 mmol) in $CH_2Cl_2$ (1 mL) at 0 °C was added EDC (0.078 g, 0.41 mmol). After stirring at rt for 1 h, a soln of H-$Ac_9c$-Ala-OMe [0.37 mmol, obtained by Pd/C-catalyzed hydrogenation of the corresponding Z-protected dipeptide ester (0.15 g, 0.37 mmol)] in $CH_2Cl_2$ (2 mL) was added. The mixture was stirred for 2 d at rt. The solvent was removed under reduced pressure and the residue was dissolved in EtOAc, and washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$. The organic layer was dried ($Na_2SO_4$), filtered, and concentrated to dryness. The product was crystallized from EtOAc by the addition of light petroleum ether; yield: 0.144 g (82%); mp 168–169 °C; $[\alpha]_D^{20}$ −50.0 (*c* 0.5, MeOH).

## 10.3.1.2 Synthesis on Solid Support

For an efficient solid-phase synthesis, almost quantitative yields in peptide bond formation are strictly required. For years, the low reactivity of $C^{\alpha}$-tetrasubstituted α-amino acids has therefore limited extension of the techniques of SPPS to such sterically hindered residues. Moreover, some activation methodologies, widely exploited in solution-phase synthesis, require reaction conditions which are not compatible with SPPS. Only recent improvements, such as the introduction of the HATU[40] and the amino acyl fluoride[54,55] methods, allow for a relatively easy incorporation of $C^{\alpha}$-tetrasubstituted α-amino acids into peptides using SPPS. Nevertheless, a few successful syntheses have been performed using classic activation procedures, employing both Boc and Fmoc strategies. As early as 1972, five angiotensin II analogues, each incorporating a single $Ac_5c$ residue, were prepared with DCC activation.[56]

Related analogues with Aib were synthesized by exploiting the DCC/HOBt procedure.[57] A remarkable achievement using classic activation procedures is the synthesis of alamethicin, a 20-residue peptaibol bearing eight Aib units, two of them contiguous, reported by Gisin and co-workers in 1981.[58] However, a laborious strategy with stepwise and fragment condensations had to be devised in order to limit the occurrence of the deletion peptides generated by poor coupling yields. Similarly, the introduction of dipeptide units was proposed to bypass difficult incorporations of single $C^{\alpha}$-tetrasubstituted residues.[59] The preformed SA has been used to incorporate Aib into a parathyroid hormone agonist and an antagonist,[60] and into a substance P(6–11) analogue.[61] Also, the BOP/HOBt activation method has been successfully applied to syntheses involving Aib or $Ac_6c$.[62,63] Currently, the extensively used uronium salts, which involve active ester intermediates as in the case of the DCC/HOBt or BOP/HOBt activation procedures, appear to be quite efficient with $C^{\alpha}$-tetrasubstituted α-amino acids. In particular, HATU[40] is superior to classic methods and to other uronium salts both in terms of coupling efficiency[64–66] and peptide optical stability.[67,68] Only the acyl fluoride procedure (see Section 10.3.2.2) stands one step ahead of the HATU activation method. Finally, silylation of the amino group, prior to the acylation step, considerably increases its nucleophilicity, thus significantly raising coupling yields,[69,70] and allowing for good results even with the classic DCC/HOBt activation method.[70]

In summary, the use of HATU, possibly combined with silylation, represents a good choice for incorporation of isolated $C^{\alpha}$-tetrasubstituted α-amino acids using SPPS, whereas the amino acyl fluoride method is a must for coupling contiguous, sterically hindered residues (see Section 10.3.2.2). Better results are usually obtained with repetitive couplings, longer reaction times, and capping after every acylation step. As observed in solution synthesis, $C^{\alpha}$-tetrasubstituted amino acids are less reactive at the N-terminus than at the C-terminus. Both Boc and Fmoc strategies can be used and a number of different resins have been employed.

**Boc-(αMe)Phe-Gly-Leu-Met-Resin:**[70]
H-Gly-Leu-Met-resin was treated with a $CH_2Cl_2$ soln of TMSCl (10 equiv per $NH_2$ substitution) for 1 h at 37 °C (at rt comparable results were obtained). After washing with $CH_2Cl_2$, TEA/$CH_2Cl_2$ (1:9), $CH_2Cl_2$, and NMP, a soln of Boc-(αMe)Phe-OH (3 equiv), DCC/HOBt (3 equiv), and DIPEA (3 equiv) in NMP was added. Reaction was allowed to proceed for 3 h at rt; yield: 83±6% (after a single coupling).

## 10.3.2 Peptides with Consecutive $C^{\alpha}$-Tetrasubstituted α-Amino Acids

Efficient peptide bond formation is not an easy achievement when both the carboxy and the amino components are $C^{\alpha}$-tetrasubstituted α-amino acids. Nevertheless, the degree of steric congestion is not the same for different residues, heavily depending on the nature of the side chains. For instance, two Aib residues are easier to couple than two (αMe)Val residues, where the steric hindrance of the β-branched, isopropyl side chains is definitely higher than that of two methyl groups. Peptide bond formation between two Aib residues can be accomplished, although with variable yields, with most of the classic activation methods. Recently proposed activation procedures allow for a relatively easy coupling even when residues with steric hindrance greater than Aib are involved. These developments also make feasible the solid-phase synthesis of peptides with sequences based exclusively on $C^{\alpha}$-tetrasubstituted amino acids.

*for references see p 307*

### 10.3.2.1 Synthesis in Solution

Selected examples of coupling procedures involving two consecutive $C^{\alpha}$-tetrasubstituted α-amino acids are listed in Table 2.

**Table 2** Selected Examples of Coupling Procedures Involving Two Consecutive $C^{\alpha}$-Tetrasubstituted Amino Acids[10,22,27,34,36,71–75]

| Amino Acids | Coupling Method | Conditions | Yield (%) | Ref |
|---|---|---|---|---|
| Aib +Aib | MA | toluene, 60 °C, 3 h; rt, 12 h | 54–89 | [22,31] |
| Aib + Aib | PyBroP | $CH_2Cl_2$, rt, 1 d | 25 | [27] |
| Aib + Aib | PyBroP/DMAP | $CH_2Cl_2$, rt, 1 h | 77 | [27] |
| Aib + Aib | CIP | rt, 1 h | 11 | [34] |
| Aib + Aib | CIP/HOAt | rt, 1 h | 82 | [34] |
| Aib + Aib | DCC/HOBt | DMF, rt, 2 d | 63 | [71] |
| Aib + Aib | azirine | MeCN, rt, 12 h | 88 | [72] |
| Iva + Iva | DCC/HOBt | DMF, rt, 5 d | 63 | [73] |
| $Ac_6c + Ac_6c$ | SA | benzene, rt, overnight | 92 | [74] |
| $Ac_7c + Ac_7c$ | SA | $CH_2Cl_2$, rt, 4 d | 62–98 | [36] |
| Dpg + Dpg | KOBt | $CH_2Cl_2$, rt, 3 h | 80 | [75] |
| (αMe)Val + (αMe)Val | acyl fluoride | $CH_2Cl_2$, rt, 5 d | 73 | [10] |

For decades, oxazol-5(4*H*)-ones of low reactivity from $N^{\alpha}$-protected derivatives or peptides have been used to couple consecutive $C^{\alpha}$-tetrasubstituted α-amino acids, and they are still useful intermediates in the fragment condensation and the N- to C-terminus peptide-chain elongation strategy.[76] Nevertheless, in the step-by-step incorporation of $C^{\alpha}$-tetrasubstituted α-amino acid residues into a peptide chain, other activation methods are more efficient. Among these, the symmetrical anhydride and the carbodiimide/HOAt approaches (see Section 10.3.1.1) give good results. A superior coupling procedure is represented by the amino acyl fluoride method.[77,78] A further improvement results from the combination of acyl chloride activation, stronger than acyl fluoride, with an $N^{\alpha}$-protecting sulfonyl group lacking the carbonyl moiety typical of urethanes and amides.[79] This structural modification prevents the drop in reactivity associated with oxazol-5(4*H*)-one formation. Alternatively, replacement of the tertiary amine that catalyzes oxazol-5(4*H*)-one formation,[78,80] with the potassium salt of HOBt, allows the amino acyl chloride method with urethane protection to be used with promising results.[75,81]

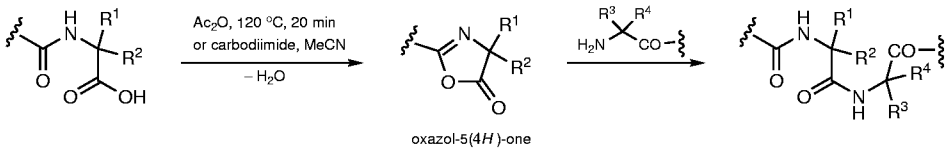
Other coupling procedures are available, although literature applications are limited in number. The azirine approach[82] is fairly efficient, but requires the synthesis of an appropriate three-membered ring heterocyclic compound and only allows for N- to C-terminus peptide chain elongation. In a comparative study[83] dealing with peptide bond formation between two Aib residues, the NCA method with urethane protection[84] gave a better yield than PyBroP activation.[27] Nevertheless, when high steric hindrance is involved, nucleophilic attack of the carbonyl at position 2 of the UNCA ring may become more favorable, yielding an undesired urea derivative.[54] Although not straightforward, a highly effective synthetic approach is represented by the four-component Ugi reaction.[85] Despite a few remarkable achievements, e.g. the synthesis of the -Dipg-Dphg-[86] and $-(Dphg)_3-$[87] sequences, no example of a peptide longer than three residues is as yet available.

#### 10.3.2.1.1 Oxazol-5(4*H*)-one Activation

The oxazol-5(4*H*)-one coupling method (Scheme 4) is uniquely appropriate for $C^{\alpha}$-tetrasubstituted α-amino acids, since racemization or epimerization is unlikely to occur with these compounds. It represents the main methodology exploitable for fragment condensation of peptides based on $C^{\alpha}$-tetrasubstituted α-amino acids. Due to the *gem*-dialkyl effect on the $C^{\alpha}$

side chains,[16,17] cyclization of the desired peptide free acid or the $N^{\alpha}$-protected amino acid to the oxazolone occurs easily, either by heating in acetic anhydride[22] or by treatment with one equivalent of carbodiimide (Scheme 4).[88] The oxazolones, most of which are stable for very long times at low temperatures, are usually isolated before reaction with the amino nucleophile. The low reactivity of oxazolones can be compensated for by heating the reaction mixture and allowing a sufficient time for coupling to be completed. Reactions can also be performed by melting the reagents in the absence of any solvent. It has to be remembered that racemization of proteinogenic amino acids can occur even at the penultimate position of an oxazolone peptide chain.[23,24] Selected examples of the oxazolone coupling procedure are given in Scheme 4.

**Scheme 4** Oxazol-5(4*H*)-ones: Preparation and Peptide Bond Formation[22,29,39,88–91]




| Amino Acids | $R^1$ | $R^2$ | $R^3$ | $R^4$ | Conditions | Yield (%) | Ref |
|---|---|---|---|---|---|---|---|
| Aib + Aib | Me | Me | Me | Me | MeCN, reflux, 10 min–6 h | 60–98 | [22] |
| Iva + Iva | Me | Et | Me | Et | $CH_2Cl_2$, reflux, 1 h | 70–98 | [39] |
| (αMe)Leu + (αMe)Leu | Me | iBu | Me | iBu | MeCN, reflux, 70 h | 75 | [89] |
| (αMe)Phe + (αMe)Phe | Me | Bzl | Me | Bzl | MeCN, reflux, 15–64 h | 52–71 | [88] |
| Aib + (αMe)Val | Me | Me | Me | iPr | MeCN, reflux, 15 h | 85 | [29] |
| Deg + Deg | Et | Et | Et | Et | MeCN, reflux, 2 h | 90 | [90] |
| Dpg + Dpg | Pr | Pr | Pr | Pr | no solvent, 120 °C, 2 h | 82 | [91] |

**General Procedures for Oxazol-5(4*H*)-one Synthesis:**

**Procedure A:**[22]

A soln of the $N^{\alpha}$-acylamino acid or peptide (5 mmol) in $Ac_2O$ (20 mL) was heated at 120 °C for 15 min. The solvent was removed and the last traces of $Ac_2O$ were removed by the addition of toluene (20 mL) and subsequent evaporation.

**Procedure B:**[88]

To a soln of the $N^{\alpha}$-acylamino acid or peptide (1 mmol) in MeCN (or $CH_2Cl_2$ or EtOAc) (4 mL) at 0 °C was added EDC (1.1 mmol). The mixture was stirred for 2 h at rt. The solvent was concentrated, and the residue was taken up in EtOAc and washed with 5% aq $KHSO_4$, $H_2O$, 5% aq $NaHCO_3$, and again $H_2O$. The organic layer was dried ($Na_2SO_4$) and concentrated to dryness. The crude oxazol-5(4*H*)-one was recrystallized (toluene/light petroleum ether). A strong IR carbonyl stretching absorption in the 1840–1820 $cm^{-1}$ region was indicative of the oxazol-5(4*H*)-one ring.

**Coupling Procedure:**

**Z-$(Aib)_{10}$-OtBu:**[76,92]

The oxazol-5(4*H*)-one from Z-$(Aib)_5$-OH (5.38 g, 9.61 mmol) and H-$(Aib)_5$-OtBu (4.80 g, 9.61 mmol) in butyronitrile[76] (150 mL) were heated for 26 h at 100 °C. The solvent was removed, and the residue dissolved in $CHCl_3$ and washed (3 ×) with 5% aq $KHSO_4$, $H_2O$, 5% aq $KHCO_3$, and $H_2O$. The organic layer was dried ($Na_2SO_4$) and removed to dryness. The peptide was crystallized from $CHCl_3$ by the addition of light petroleum ether; yield: 7.74 g (76%).

### 10.3.2.1.2 *N*-[3-(Dimethylamino)propyl]-*N'*-ethylcarbodiimide Hydrochloride/ 7-Aza-1,2,3-benzotriazol-1-ol Activation

**Z-(D-Iva)$_5$-OtBu:**[93]

Z-D-Iva-OH (76 mg, 0.30 mmol) was dissolved in anhyd $CH_2Cl_2$ (2 mL) and cooled to 0 °C. HOAt (41 mg, 0.30 mmol) was added (a suspension was formed) followed by EDC (77 mg, 0.40 mmol). After stirring the mixture for 2 min (after about 1 min the soln became clear, thereby indicating that active ester formation had occurred), H-(D-Iva)$_4$-OtBu [0.30 mmol, obtained by Pd-catalyzed hydrogenation of the

*for references see p 307*

corresponding Z-protected tetrapeptide ester (0.181 g, 0.30 mmol)] dissolved in anhyd $CH_2Cl_2$ (2 mL) was added, followed by NMM (0.033 mL, 0.30 mmol). After stirring the soln at rt for 6 d, the solvent was removed under reduced pressure and the mixture was dissolved in $CHCl_3$. The organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), and concentrated to dryness. The peptide was purified by flash chromatography (EtOAc/light petroleum ether 1:1); yield: 0.141 g (67%); mp 204–205 °C (EtOAc/light petroleum ether).

### 10.3.2.1.3 Symmetrical Anhydride Activation

**Z-[L-(αMe)Val]$_2$-OtBu:**[10]

Z-[L-(αMe)Val]$_2$O (0.269 g, 0.52 mmol) and H-L-(αMe)Val-OtBu [0.57 mmol, obtained by Pd-catalyzed hydrogenation of the corresponding Z-protected ester (0.183 g, 0.57 mmol)] were dissolved in $CH_2Cl_2$ (2 mL). Then, NMM (0.058 mL, 0.52 mmol) was added. After stirring the mixture for 7 d at rt, the solvent was removed under reduced pressure and the residue dissolved in EtOAc. The organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), and concentrated to dryness. The terminally protected dipeptide was purified by flash chromatography (EtOAc/light petroleum ether 1:5); yield: 58%; mp 88–90 °C ($Et_2O$/light petroleum ether).

### 10.3.2.1.4 Amino Acyl Fluoride Activation

The remarkable syntheses of the octapeptide -[(αMe)Val]$_8$-[10] and heptapeptide -Iva-(αMe)Val-Iva-(αMe)Val-(αMe)Phe-(αMe)Val-Iva-[14] sequences clearly demonstrate the efficiency of the amino acyl fluoride method. In a comparative study of the synthesis of Z-[(αMe)Val]$_2$-OtBu,[10] the amino acyl fluoride method was superior to the EDC/HOBt, EDC/HOAt, and SA procedures. Furthermore, longer reaction times (days) and relatively large excesses of the acylating reagent (2–3 equivalents) considerably improved the yields. The effectiveness of the amino acyl fluoride method is attributed to (i) the small size of the leaving group, the fluoride ion, (ii) the high electronegativity of the fluorine which activates the carboxy group, and (iii) the fact that $N^{\alpha}$-protected amino acyl fluorides are less likely than chloride derivatives to be transformed into the oxazol-5(4*H*)-ones of low reactivity.[78,94] Suitably $N^{\alpha}$-protected amino acyl fluorides are conveniently prepared from the corresponding free carboxylic α-amino acids by means of cyanuric fluoride[95] or DAST.[96] They can be stored for days and employed both in solution and solid-phase peptide synthesis. The use of TFFH[97] allows for the in situ preparation of acyl fluorides. Nevertheless, peptide bond formation is most efficiently achieved when a previously isolated amino acyl fluoride is used.

**Z-L-(αMe)Val-F:**[10]

Z-L-(αMe)Val-OH · $H_2O$ (0.680 g, 2.38 mmol) was dissolved in anhyd $CH_2Cl_2$ (50 mL) and dried ($Na_2SO_4$). The salt was filtered off, and the soln was concentrated and cooled to −15 °C. Then, cyanuric fluoride (0.404 mL, 4.79 mmol) and pyridine (0.408 mL, 4.77 mmol) were added, and a precipitate was formed. After stirring the mixture for 1 h at −15 °C and 1 h at rt, the product was purified by flash chromatography (EtOAc/light petroleum ether 1:6) to give an oil; yield: 0.54 g (85%); IR (film): 1836 $cm^{-1}$.

**Z-[L-(αMe)Val]$_8$-OtBu:**[10]

Z-L-(αMe)Val-F (0.30 g, 1.13 mmol) and H-[L-(αMe)Val]$_7$-OtBu [0.37 mmol, obtained by Pd/C-catalyzed hydrogenation of the Z-protected heptapeptide ester (0.372 g, 0.37 mmol)] were dissolved in $CH_2Cl_2$ (5 mL). To this soln was added NMM (0.085 mL, 0.77 mmol) and the mixture was stirred at rt for 5 d. The solvent was removed under reduced pressure and the residue was dissolved in $CHCl_3$. The organic phase was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), and concentrated to dryness. The peptide was crystallized from $CHCl_3$ by the addition of light petroleum ether; yield: 0.301 g (73%); mp 230–232 °C.

### 10.3.2.2 Synthesis on Solid Support

The amino acyl fluoride activation procedure is undoubtedly the best method for SPPS involving consecutive $C^{\alpha}$-tetrasubstituted α-amino acids. Only by means of this method were four Aib residues in a row successfully linked.[54,55] With the same procedure, a few peptaibols have been prepared in high purity and excellent yields.[98,99] Furthermore, difficult sequences bearing unnatural, sterically hindered amino acids have also been prepared.[100,101] The coupling efficiency of such an activation procedure is superior to that of the HATU method, as shown in comparative experiments.[66,94,102] The resulting amino acid fluorides are more stable in nonpolar solvents,[103] while silylation may significantly increase their reactivity.[69,103] In addition, coupling proceeds smoothly even in the absence of any base, the presence of which might cause premature deblocking of the Fmoc protecting group or undesired oxazol-5(4*H*)-one formation.[78,103] An original and extremely efficient method employing α-azido acids is also available for difficult sequences: up to four successive Dphg residues have been linked.[104] So far, this method, requiring the preparation of appropriate α-azido acid precursors, has not found as wide an application as the amino acyl fluoride activation procedure.

**Stepwise Solid-Phase Synthesis of Peptaibols:**[99]
Up to 8 equiv of Fmoc-protected amino acid fluoride, at a concentration of 0.3 M in $CH_2Cl_2$, in the presence of DIPEA (1 equiv) were used. Single couplings of 30 min were performed. Only in the synthesis of trichotoxin A-50 were the introduction of Fmoc-Iva-F and the following Aib residue accomplished using double couplings of 1 h each. The yields of the final crude peptides were in the range 60–84%.

## 10.3.3 Peptides Containing $C^{\alpha}$-Trifluoromethylated α-Amino Acids

$C^{\alpha}$-Trifluoromethylated α-amino acids [(αTfm)Xaa] form a special class of building blocks for peptide synthesis, due to their unique electronic properties generated by the fluorinated substituent. The polarization effects on the neighboring substituents may influence the hydrolytic stability of peptides, conferring on them resistance to peptidase degradation and thus a prolonged activity.[105–109] Furthermore, the high lipophilicity may enhance in vivo absorption, improving the permeability through some barriers in the body.[107] Trifluoromethyl groups are capable of participating in hydrogen bonding, thus offering additional ways of interacting with enzyme or receptor subsites.[107–109] There is still controversy over the steric bulk of a trifluoromethyl group; the trifluoromethyl group seems to be closer in bulk to an isopropyl rather than a methyl group.[107,109] This in turn suggests that $C^{\alpha}$-trifluoromethylated α-amino acids severely restrict the conformational freedom of peptides.[110] An additional interesting property of the trifluoromethyl group is its relatively low toxicity and high stability.[107] Finally, the conformational behavior of derivatives and peptides consisting of fluorinated α-amino acids can be conveniently monitored by means of $^{19}F$ NMR spectrometry.

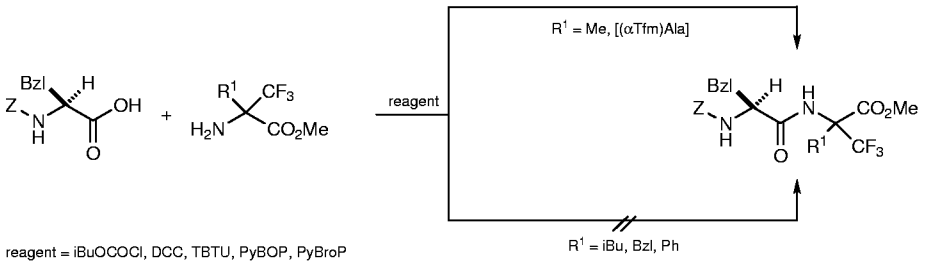
### 10.3.3.1 Synthesis in Solution

The presence of the electron-withdrawing trifluoromethyl substituent in the α-position has a dramatic effect on the reactivity of both the carboxy and the amino groups. The latter, in particular, shows a very marked decrease (~ 4 units) in the $pK_a$ value with respect to the corresponding natural amino acids, whereas the carboxy group experiences a more limited deviation (0.3–1.1 units).[111] As a consequence, activation of the carboxy group of

*for references see p 307*

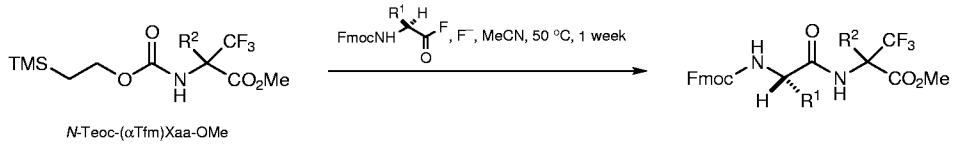
(αTfm)Xaa and subsequent peptide bond formation is easily accomplished with most of the classic methods, but activation of the amino group represents a very difficult task. A number of (αTfm)Xaa have been activated and reacted at their carboxy group. Carbodiimide treatment, the most widely exploited method, produces the stable oxazol-5(4*H*)-ones as the reactive species, even in the presence of HOBt.[106] The same intermediates are formed upon MA activation.[106] (αTfm)Xaa NCAs have also been prepared and used for the synthesis of dipeptides.[112] Polymerization cannot occur because of the poor nucleophilicity of the amino group of (αTfm)Xaa.[106,112] Peptide bond formation involving the amino group of an (αTfm)Xaa can be achieved by means of classic methods only for (αTfm)Ala (Scheme 5).[113,114]

**Scheme 5** Nucleophilicity of (αTfm)Xaa in Peptide Bond Formation[113,114]




Bulkier $C^{\alpha}$-trifluoromethylated α-amino acids do not react or result in substantial racemization (epimerization) of the nonfluorinated amino acids in the sequence.[106,113,114] Therefore, alternative pathways have been developed. The in situ deprotection of *N*-Teoc-(αTfm)Xaa derivatives, in the presence of a catalytic amount of fluoride ions and Fmoc-Xaa-F, allows for the N-acylation of the (αTfm)Xaa with acceptable yields (Scheme 6).[111,113] Very recently, the multicomponent Ugi reaction has also been explored with the aim of incorporating (αTfm)Xaa at the C-terminal position of peptides.[114]

**Scheme 6** *N*-Teoc/Amino Acyl Fluoride Method of Activation for the Amino Group of Xaa(αTfm)[111,113]




### 10.3.3.1.1 $C^{\alpha}$-Tetrasubstituted α-Amino Acids as Acyl Donors

**General Procedure Using Carbodiimide:**[109,115]
To a stirred soln of Z-(αTfm)Xaa-OH (2 mmol) in $CH_2Cl_2$ (10 mL) were added DIC[109] (or DCC,[115] 2.2 mmol) and the N-deprotected peptide or amino acid derivative (2 mmol). The mixture was stirred at rt for 12 h [the progress of the reaction, indicated by the formation and disappearance of the oxazol-5(4*H*)-one, can be monitored by TLC].[115] The precipitate was separated by filtration, the mother liquor was concentrated to dryness, and the residue was purified by flash chromatography.

### 10.3.3.1.2 $C^{\alpha}$-Tetrasubstituted α-Amino Acids as Nucleophiles

**The Mixed Anhydride Method [Scheme 5; Only for (αTfm)Ala]:**[109]
To a stirred soln of Z-Xaa-OH (2 mmol) in EtOAc (20 mL), kept at −15 °C, were added isobutyl chloroformate (2 mmol) and NMM (2 mmol). After 30 min, a soln of H-(αTfm)Xaa-$OR^1$ (2 mmol) in EtOAc (10 mL) was added. The mixture was stirred for 1 h at −15 °C and for 12 h at rt. Then, it was extracted with $H_2O$, aq citric acid, $H_2O$, sat. $NaHCO_3$, and $H_2O$, dried ($MgSO_4$), and concentrated to dryness. Purification was achieved by flash chromatography.

**The *N*-Teoc/Amino Acyl Fluoride Method [Scheme 6; for Most (αTfm)Xaa]:**[111]
A well-stirred soln of *N*-Teoc-(αTfm)Xaa-OMe (2 mmol), Fmoc-Xaa-F (3 mmol) and tetraethylammonium fluoride (cat. amount) in MeCN (3 mL) was heated at 50 °C under an inert gas atmosphere until complete conversion was reached (usually 1–2 weeks, as monitored by $^{19}F$ NMR spectroscopy). The solvent was removed and the product purified by flash chromatography; Fmoc-Gly-(αTfm)Leu-OMe, yield: 77%; Fmoc-Ala-(αTfm)Leu-OMe, yield: 77%; Fmoc-Ala-(αTfm)Phe-OMe, yield: 55%.

### 10.3.3.2 Synthesis on Solid Support

Due to the very poor nucleophilicity of the amino group of $C^{\alpha}$-trifluoromethylated α-amino acids, their incorporation into peptides by means of SPPS still represents a major challenge. To study the influence on proteolytic stability and receptor affinity, the (αTfm)pGlu residue has been introduced at the N-terminal position of the tripeptide TRH[107,116,117] and the decapeptide GnRH.[107,116] The introduction of (αTfm)Ala at an internal position of GnRH was possible only by coupling a tripeptide building block, Fmoc-Tyr(tBu)-(αTfm)Ala-Leu-OH, to the peptide resin.[107] The amino acyl fluoride method, which has been recently exploited in the SPPS of peptaibols,[99] is also a promising procedure for incorporating (αTfm)Ala into peptaibols.[118]

**H-(αTfm)pGlu-His-Pro-$NH_2$:**[117]
The H-His(Trt)-Pro-Knorr-resin was allowed to react for 8 h with a 4–5-fold excess of H-(*S,R*)-(αTfm)pGlu-OH and the DIC/HOBt activating mixture. The resulting peptide was cleaved from the resin using a soln consisting of 95% TFA, 2.5% $H_2O$, and 2.5% ethane-1,2-dithiol. The two diastereomeric products were precipitated with $Et_2O$, desalted by size-exclusion chromatography (Sephadex LH20, MeOH/$H_2O$ 1:9), and purified by RP-HPLC (MeCN/$H_2O$ containing 0.1% TFA); yield: 21%.

## 10.3.4 Peptides Containing TOAC Residues

Among $C^{\alpha}$-tetrasubstituted α-amino acids, TOAC (see Scheme 7 for molecular structure) deserves special attention, in view of its varied and peculiar properties.[119] TOAC is characterized by a saturated heterocyclic structure containing a stable, free-radical nitroxide moiety. The hampered rotation about side-chain bonds dramatically restricts the mobility of the paramagnetic probe, whose movements, strictly reflecting those of the α-carbon atom, give valuable information about the peptide skeleton into which TOAC is inserted. This is a very favorable property of TOAC over other commonly used, side-chain-modified, spin-labeled protein amino acids, where flexible links connect the label to the peptide backbone. Therefore, TOAC represents an extremely useful tool for the exploitation of sensitive ESR techniques in peptide and protein chemistry.[120–132] In addition, TOAC (i) is a strong fluorescence quencher,[133] (ii) is a useful CD probe in the visible spectral region ($\lambda$ = 420–450 nm, weak n $\rightarrow$ $\pi^*$ transition of the nitroxide chromophore) when inserted into a chiral peptide,[125] (iii) is able to undergo a reversible, nitroxide-based redox process which can be monitored by cyclic voltammetry,[125] (iv) does not exhibit any nitroxide vibrational band in its infrared absorption spectrum which may interfere with the amide A, amide I, and amide II absorptions, commonly used to determine peptide secondary structure,[125] (v) gives derivatives and peptides of extremely high crystallinity which has allowed the X-ray diffraction structures of a number of them to be solved, [125,132–136] and (vi) is a strong inducer of turns and helical structures, both in the crystal state and in solution.[13,125–128,133,134] The only relevant limitation of this amino acid to physicochemical investigations is that $^1H$ NMR spectroscopy cannot be routinely performed, because most of the proton resonances of a TOAC-containing peptide are significantly broadened by the paramagnetic effect of the nitroxide free radical.

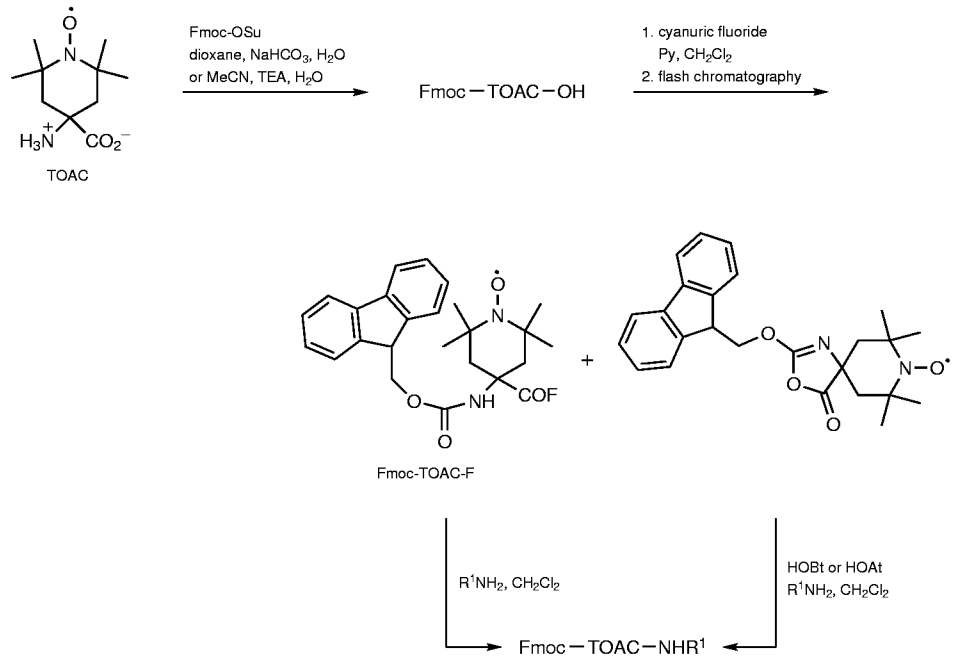
*for references see p 307*

During peptide synthesis, acidic, reducing, or oxidizing conditions should be avoided in order to preserve the chemical integrity of the nitroxide moiety. Therefore, $N^{\alpha}$-protection with the Fmoc group is strongly recommended for the introduction of a TOAC residue in an internal position and subsequent elongation of the peptide chain. The acidic or reducing conditions required to remove the extensively exploited N-protecting groups Boc and Z, respectively, greatly affect the nitroxide group. Nevertheless, acidic treatment with neat trifluoroacetic acid, both in solution[137] and in the solid phase,[127] or with hydrogen fluoride[123,126,131] can be performed, provided loss of TOAC radical character is reversed by treatment with aqueous ammonia or ammonium acetate at pH ~ 9.[123,126,131] Removal of the Z group from a TOAC homodipeptide under reducing conditions has also been reported, but a five-day exposure to atmospheric oxygen only allowed for a partial recovery of the N-deprotected dipeptide with full nitroxide integrity.[138]

### 10.3.4.1 Synthesis in Solution

From the limited information available to date, TOAC appears to be slightly less reactive than the closely related $Ac_6c$. Therefore, the general observations of Sections 10.3.1.1 and 10.3.2.1 can also be extended to TOAC. In particular, when an N-protected TOAC residue is to be linked to a protein amino acid ester or amide, most of the classic activation methods are expected to give reasonable yields. The HOBt or, better, the HOAt carbodiimide-mediated coupling method represents a valuable choice because of the easy synthetic procedure and coupling effectiveness.[13] The SA method is also excellent for this purpose, and can even be effectively exploited in the acylation of the poorly nucleophilic TOAC amino group. The Aib-TOAC peptide bond can also be easily formed using this procedure.[13] The EDC/HOAt and the amino acyl fluoride methods, in turn, give satisfactory results in TOAC-TOAC and TOAC-Aib peptide bond formation.[13] As expected, esters of low reactivity (such as OSu and ONp esters) fail in coupling two consecutive TOAC residues, whereas the MA method with ethyl chloroformate yields mainly the wrong acylation product.[138] In the preparation of the useful derivative Fmoc-TOAC-F (Scheme 7), the corresponding oxazol-5(4*H*)-one,[135] formed as a byproduct, can be isolated by column chromatography and stored for months due to its unusual stability.[135] It can also be used for acylations in combination with one equivalent of HOBt or HOAt, even without any addition of base (Scheme 7).

**Scheme 7** Fmoc-TOAC-F and Related Oxazolone Preparation and Reactivity[13,37,135]




### 10.3.4.1.1 *N*-[3-(Dimethylamino)propyl]-*N'*-ethylcarbodiimide Hydrochloride/ 1,2,3-Benzotriazol-1-ol Activation

**Fmoc-TOAC-(L-Ala)$_3$-OtBu:**[13]
To a suspension of Fmoc-TOAC-OH (1.750 g, 4.0 mmol) and HOBt (0.578 g, 4.0 mmol) in $CH_2Cl_2$ (10 mL) at 0 °C was added EDC (0.770 g, 4.0 mmol). When the suspension had become clear, a soln of H-(L-Ala)$_3$-OtBu (1.150 g, 4.0 mmol) in $CH_2Cl_2$ (5 mL) was added. After stirring overnight at rt, the solvent was removed under reduced pressure. The residue was dissolved in EtOAc and the organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), filtered, and concentrated under reduced pressure. The peptide was precipitated by the addition of light petroleum ether; yield: 2.540 g (90%); mp 161–163 °C; $[\alpha]_D^{20}$ −21.0 (*c* 0.5, MeOH).

### 10.3.4.1.2 Symmetrical Anhydride Activation

**Fmoc-L-Leu-TOAC-Gly-L-Ile-L-Leu-OMe:**[132]
To a soln of Fmoc-L-Leu-OH (28.98 g, 82.0 mmol) in $CH_2Cl_2$ (120 mL) at 0 °C was added EDC (7.86 g, 41.0 mmol). After stirring for 30 min at 0 °C, a soln of H-TOAC-Gly-L-Ile-L-Leu-OMe (21.02 g, 41.0 mmol) and NMM (4.52 mL, 41.0 mmol) in $CH_2Cl_2$ (120 mL) was added. After stirring overnight at rt, the solvent was removed under reduced pressure. The residue was dissolved in EtOAc, and the organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), filtered, and concentrated to dryness. The peptide was crystallized from EtOAc/light petroleum ether; yield: 29.56 g (85%); mp 166–168 °C; $[\alpha]_D^{20}$ −54.7 (*c* 0.5, MeOH).

**Fmoc-Aib-TOAC-(L-Ala)$_2$-OtBu:**[13]
To a stirred soln of H-TOAC-(L-Ala)$_2$-OtBu (0.340 g, 0.82 mmol) in $CH_2Cl_2$ (3 mL) was added the SA of Fmoc-Aib-OH (0.570 g, 0.90 mmol), followed after 30 min by NMM (0.050 mL, 0.045 mmol). After stirring overnight at rt, the solvent was removed under reduced pressure. The residue was dissolved in EtOAc, and the organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), filtered, and concentrated to dryness. The peptide was purified by flash chromatography (silica gel, $CHCl_3$/EtOH 98:2) and crystallized from EtOAc/light petroleum ether; yield: 0.485 g (82%); mp 120–122 °C; $[\alpha]_D^{20}$ −3.5 (*c* 0.5, MeOH).

*for references see p 307*

#### 10.3.4.1.3 *N*-[3-(Dimethylamino)propyl]-*N'*-ethylcarbodiimide Hydrochloride/ 7-Aza-1,2,3-benzotriazol-1-ol Activation Activation

**Boc-(TOAC)$_2$-(L-Ala)$_4$-OtBu:**[13]
To a suspension of Boc-TOAC-OH (0.035 g, 0.11 mmol) and HOAt (0.015 g, 0.11 mmol) in $CH_2Cl_2$ (1 mL) at 0 °C was added EDC (0.021 g, 0.11 mmol). When the suspension had become clear, a soln of H-TOAC-(L-Ala)$_4$-OtBu (0.050 g, 0.09 mmol) in $CH_2Cl_2$ (1 mL) was added. After stirring for 4 d at rt, the solvent was removed under reduced pressure. The peptide was purified by flash chromatography (silica gel, $CHCl_3$/EtOH 95:5) and crystallized from EtOAc/light petroleum ether; yield: 0.027 g (35%); mp 185–186 °C; $[\alpha]_D^{20}$ 39.5 (*c* 0.5, MeOH).

#### 10.3.4.1.4 Amino Acyl Fluoride Activation

**Fmoc-TOAC-F:**[37]
To a soln of Fmoc-TOAC-OH (0.437 g, 1.0 mmol) in $CH_2Cl_2$ at −10 °C were added cyanuric fluoride (0.085 mL, 1 mmol) and pyridine (0.081 mL, 1 mmol) under nitrogen. After stirring for 1 h at 0 °C the mixture was directly applied to a silica gel column and eluted with EtOAc/light petroleum ether (1:5). The product eluted first was identified as the oxazol-5(4*H*)-one[135] (0.110 g, 0.026 mmol; yield: 26%). The fluoride, which was eluted from the column immediately after the oxazolone, was obtained as a thick, orange oil; yield: 0.308 g (70%); IR (film): 1839, 1715 $cm^{-1}$.

**Fmoc-(TOAC)$_2$-(L-Ala)$_4$-OtBu:**[37]
Fmoc-TOAC-F (0.170 g, 0.39 mmol), H-TOAC-(L-Ala)$_4$-OtBu (0.200 g, 0.36 mmol), and NMM (0.043 mL, 0.39 mmol) were dissolved in $CH_2Cl_2$ (3 mL) and stirred for 3 d at rt. The solvent was removed under reduced pressure, the residue was dissolved in EtOAc, and the organic layer was washed with 10% $KHSO_4$, $H_2O$, 5% $NaHCO_3$, and $H_2O$, dried ($Na_2SO_4$), filtered, and concentrated to dryness. The peptide was purified by flash chromatography (silica gel, $CHCl_3$/EtOH 95:5) and crystallized from EtOAc/light petroleum ether; yield: 0.165 g (47%); mp 174–175 °C; $[\alpha]_D^{20}$ 8.0 (*c* 0.5, MeOH).

### 10.3.4.2 Synthesis on Solid Support

The Merrifield method (Boc strategy) was the first to be employed for the SPPS[120,131] of TOAC-containing peptides, but TOAC was only inserted in the last step (as the N-terminal residue) in order to avoid possible nitroxide decomposition subsequent to repetitive acidic deprotection steps. Hydrogen fluoride was used to cleave the peptide from the resin because NO• protonation can be easily reversed by treatment with base.[120,131] Introduction of the use of Fmoc-TOAC-OH[123] has allowed the incorporation of TOAC at an internal position, since Fmoc removal conditions are fully compatible with nitroxide chemical integrity. Nevertheless, in this Fmoc strategy, hydrogen fluoride was also used for the final cleavage from the resin.[123,126] Recently, the exploitation of the less toxic trifluoroacetic acid, believed to cause nitroxide decomposition,[123] for this purpose has been shown to be feasible, provided no sulfur-containing additives are present.[127,137] So far, TOAC has only been linked to protein amino acids using SPPS. The carbodiimide/HOBt activation method has been widely exploited, and the HBTU/HOBt combination has also proven to be valuable.[127] In analogy with most $C^{\alpha}$-tetrasubstituted α-amino acids, steric hindrance affects the reactivity at the TOAC N-terminus more than at the C-terminus.[123] HPLC purification of the products can be performed by employing the common TFA-containing eluants, but treatment with aqueous ammonia or ammonium acetate at pH ~ 9 has to be carried out.

**Fmoc Strategy Procedure for TOAC-Containing Peptides:**[127]
*Activation and Coupling*: Fmoc-TOAC-OH (1 mmol) was dissolved in a DMF soln (2 mL), 0.45 M in both HBTU and HOBt.[139] To this soln was added 2 M DIPEA in DMF (1mL). After 5 min, the activated mixture and NMP (1 mL) were added to the peptide-resin (0.25 mmol). The resulting mixture was shaken for at least 30 min.
*Deprotection*: The Fmoc group was removed by two treatments (3 and 10 min each) with a 20% piperidine soln in NMP.

*Cleavage from the Resin*: The peptide was simultaneously deprotected and cleaved from the resin by a 30-min treatment with neat TFA (sulfur-containing scavengers have to be absent). The resin was filtered and the soln added to cold MTBE. The precipitated peptide was collected and treated with aq $NH_3$ (pH 9.0–9.5) for 3 h at rt in order to restore the chemical integrity of the nitroxide moiety.[120]
*HPLC Purification*: The peptide was purified by RP-HPLC ($MeCN/H_2O$ containing 0.1 % TFA). The above-noted $NH_3$ treatment was repeated immediately after the HPLC elution.


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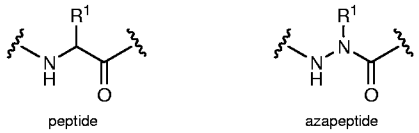
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# 10.4 Synthesis of Azapeptides

J. Gante†, H. Kessler, and C. Gibson

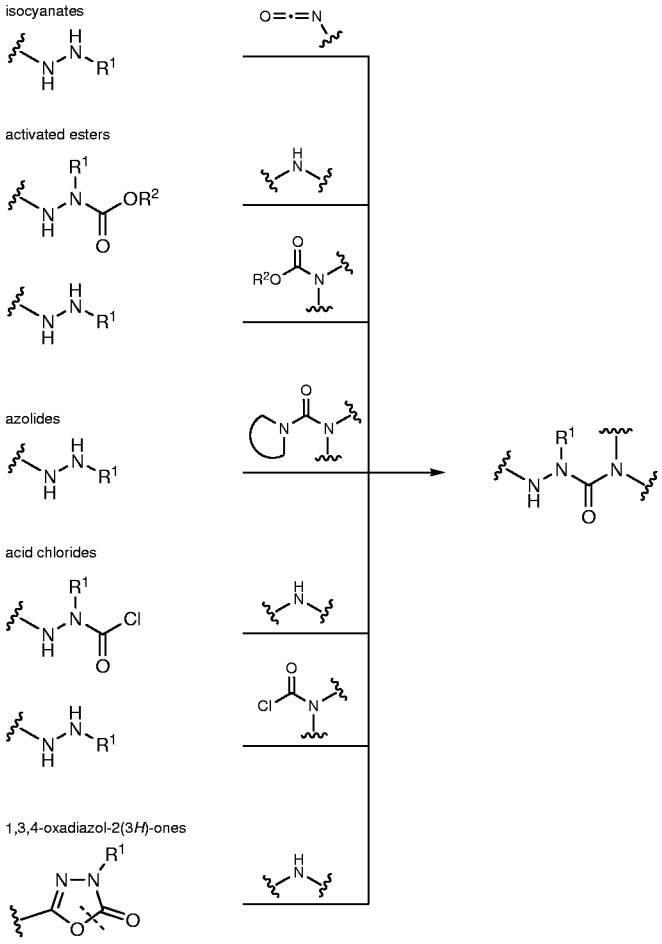
Over the last few decades, there has been increased interest in azapeptides in biological and medicinal chemistry. In these main-chain-modified peptide analogues the α-CH group of one or more amino acid residues in the chain is replaced by nitrogen (Scheme 1).

**Scheme 1** The Structures of Peptides and Azapeptides

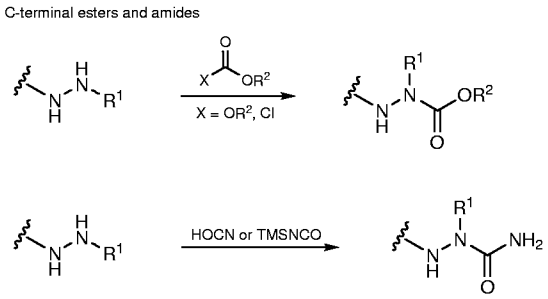



The generation of azaamino acid residues and their incorporation into the peptide chain can be accomplished by different methods that closely resemble the methods used for peptide coupling but are derived from the chemistry of semicarbazides. In principle, the formation of an azaamino acid residue takes place by the reaction of a hydrazine derivative (hydrazine or hydrazide) with "carbonyl donors", different types of activated *N*-carboxylic acid derivatives (Scheme 2).

**Scheme 2** Methods for the Generation of an Azaamino Acid Residue




*for references see p 322*




The preparation, properties, and biological activities of azapeptides have been reviewed,[1–3] and many other publications on this subject have appeared, e.g. polymer-supported syntheses of azapeptides.[4–9]

## 10.4.1 Addition of Hydrazine Derivatives to Isocyanates

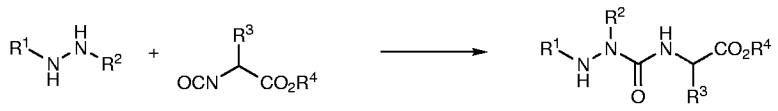
One of the most common methods for the preparation of azapeptides is the addition of hydrazines or hydrazides to α-isocyanato esters of carboxylic acids. In general, this is a very effective reaction affording only a few side products and relatively high yields. The method allows the incorporation of the aza analogues of simple amino acids with an alkyl side chain[3,10,11] or of more complex analogues such as those of tyrosine,[3] tryptophan,[3] asparagine and glutamine,[12–14] ornithine,[15] proline,[3,16] pyroglutamic acid,[3] or aspartic acid.[17] The use of α-isocyanato-α-trifluoromethylcarboxylates, which give the corresponding trifluoromethyl-substituted azapeptides, has also been described.[18]

Use of hydrazine hydrate and alkyl hydrazines results in azapeptides with the corresponding N-terminal unprotected azaamino acid residues.[11] Reaction at the substituted amino group of the hydrazine is favored, affording predominantly the "correct" (α-substituted) product.

Compounds with an N-terminal protected azaamino acid residue can be produced by using acylated hydrazines, those with Boc and Z groups being favored.[3,19] Tosyl-protected compounds can likewise be prepared; however, attempts to remove the tosyl protecting group using sodium in liquid ammonia were unsuccessful.[19]

Reaction of an isocyanate with an amino acid or peptide hydrazide affords products with a "central" azaamino acid residue[3,10,19,20] (Scheme 3).

**Scheme 3** Addition of Hydrazine Derivatives to Isocyanates[3,10–20]



1 2 3

| $R^1$ | $R^2$ | $R^3$ | $R^4$ | Conditions | Yield (%) | Ref |
|---|---|---|---|---|---|---|
| H | H | H | Et | **1**[a]/**2** (1:1), EtOH, 0 °C, 10 min, then 25 °C, 30 min | 44 | [18] |
| H | Bzl | H | Et | **1**/**2** (1:1), $Et_2O$, 0 °C, 24 h | 74 | [18] |
| Boc | Bzl | iBu | Me | **1**/**2** (1:1), THF, rt, 16 h | 92 | [3] |
| Z | Me | iBu | Me | **1**[b]/**2**/TEA (1:1:1), $CHCl_3$, rt, 16 h | 85 | [3] |
| Tos | H | H | Et | **1**/**2** (1:1), without solvent, mixing at rt followed by heating | 86 | [19] |
| Z-Gly | $CH_2CO_2Et$ | H | Et | **1**/**2** (1:1), dioxane, 80 °C, 10 min | 97 | [12] |
| Z-Gly | $CH_2CO_2tBu$ | H | Me | **1**/**2** (1:1), dioxane, <30 °C, 3 h | 63 | [12] |
| Tos-AzGly-Gly | Me | Bzl | Bzl | **1**/**2** (1:1), dioxane, 80 °C, 2 h | 45 | [19] |
| Boc-Gly-Val-Gly | Me | Bzl | Bzl | **1**/**2** (1:1.1), $CHCl_3$, rt, overnight | 77 | [10] |

[a]Hydrate.
[b]Hydrochloride.

**AzPhe-Gly-OEt (3, $R^1 = R^3 = H$; $R^2 = Bzl$; $R^4 = Et$):**[11]
To a chilled and well-stirred soln of benzylhydrazine (2.0 mL, 17.2 mmol) in $Et_2O$ (100 mL) was added ethyl glycinate isocyanate (2.1 mL, 17.9 mmol) within 10 min. After 24 h in the refrigerator the undissolved material was filtered off and the solvent was removed at ambient pressure (traces in vacuo). The product was crystallized (EtOAc/petroleum ether); yield: 3.2 g (74%); mp 79–80 °C.

**Boc-AzPhe-Leu-OMe (3, $R^1$ = Boc; $R^2$ = Bzl; $R^3$ = iBu; $R^4$ = Me):**[3]
A soln of 1-benzyl-2-*tert*-butoxycarbonylhydrazine (11.1 g, 50 mmol) and methyl leucinate isocyanate (8.55 g, 50 mmol) in THF (70 mL) was kept at rt for 16 h. The solvent was removed in vacuo and the residue, dissolved in EtOAc, was washed with aqueous 20% citric acid and $H_2O$. The solid obtained by removal of the solvent was crystallized (EtOAc/petroleum ether); yield: 18 g (92%); mp 91–92 °C; $[\alpha]_D^{25}$ −23.2 (*c* 2.3, MeOH).

**Boc-Gly-Val-Gly-AzAla-Phe-OBzl (3, $R^1$ = Boc-Gly-Val-Gly; $R^2$ = Me; $R^3 = R^4$ = Bzl):**[10]
To a stirred suspension of Boc-Gly-Val-Gly-NHNHMe (0.4 g, 1.1 mmol) in $CHCl_3$ (5 mL) benzyl phenylalaninate isocyanate (0.31 g, 1.2 mmol) was added. A clear soln, obtained in a few min, was left at rt overnight. The solvent was removed and the residue was purified by chromatography (silica gel, $CHCl_3$ and 1–2% MeOH/$CHCl_3$) to give an oil; yield: 0.54 g (77%).

## 10.4.2 Reaction of Activated Esters of *N*-Carboxylic Acids with Amino Groups

The use of activated azaamino or amino acid derivatives of the ester type has rendered the synthetic strategy for azapeptides considerably more independent of the sometimes difficult to access and unstable isocyanates.

The use of activated esters provides a relatively high degree of synthetic flexibility. Thus, it is possible, for example, not only to synthesize any "mixed" peptide analogue required, but also to link azaamino acid residues directly in almost any desired number and sequence, enabling the preparation of "pure" azapeptides.

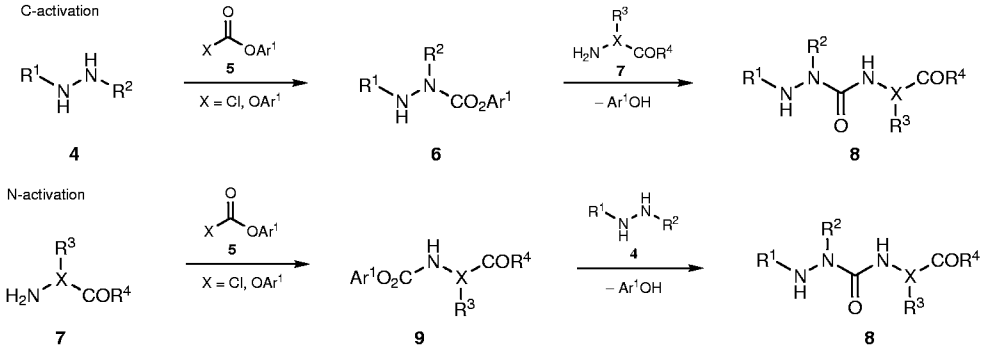
The activation for coupling by this method can be carried out either at the C-terminus (C-activation, for examples see Table 1) or at the N-terminus (N-activation, for examples see Table 2) of the corresponding components. The method of use of these compounds com-

*for references see p 322*

prises the introduction of the corresponding C- or N-terminal ester moiety to **4** or **7** as the first step giving **6** or **9**, respectively, and coupling with the amino component **7** or **4** as the second step giving **8** (Scheme 4, see also Table 3).

C-Activated esters **6** are prepared by reacting hydrazides **4** with aryl chloroformates **5** (X = Cl) or carbonic diesters **5** (X = OAr),[1] whereas N-activated esters **9** are obtained by reaction of the free amino group of N-terminal amino acid **7** or azaamino acid residues with the same reagents.

**Scheme 4** Generation and Coupling To Give Azapeptides via C- and N-Activated Esters[6,8,12–14,21–29]




Solid-phase techniques can be applied to the N-activated esters.[6] In the activated ester method, 4-nitrophenyl,[12,21–27] 2,4-dinitrophenyl,[6,14,28] 2,4,5-trichlorophenyl,[29] and pentafluorophenyl[8] esters can be used, the amide bonds being formed with aminolytic release of the corresponding phenol (Scheme 4). For example, the use of pentafluorophenyl esters in the synthesis of "pure" azapeptides (named "azatides" for the first time) proved to be advantageous.[8]

**Table 1** Preparation of Activated Esters by C-Activation[8,23,25,27,29][a]

| $R^1$ | $R^2$ | $Ar^1$ | X | Conditions | Yield (%) of **6** | Ref |
|---|---|---|---|---|---|---|
| Z-Gly | H | Np | ONp | **4**/**5** (1:1), EtOAc, reflux, 2 h | 71 | [23] |
| Boc | Bzl | Np | Cl | **4**/**5**/NMM (1:1:1), THF, 0 °C, 3 h | 91 | [25,27] |
| Boc | H | 2,4,5-Tcp | Cl | **4**/**5**/TEA (1:1:2), $CHCl_3$, rt, ca. 15 h | quant[b] | [29] |
| Boc | iPr | Pfp | OPfp | **4**/**5**/DMAP (1:1:1), $CH_2Cl_2$, rt,30 min | quant[b] | [8] |

[a] See Scheme 4.
[b] Quantitative, not isolated.

**Table 2** Preparation of Activated Esters by N-Activation[13,24][a]

| X | $R^3$ | $R^4$ | $Ar^1$ | X | Conditions | Yield (%) of **9** | Ref |
|---|---|---|---|---|---|---|---|
| N | H | Gly-OEt | Np | ONp | **7**/**5** (1:1), EtOAc, reflux, 2.5 h | 47 | [24] |
| CH | Bzl | OBzl | 2,4-Dnp | 2,4-Dnp-O | **7**/**5** (1:1), EtOAc, rt, 3 h | 67 | [13] |
| CH | H | OBzl | 2,4-Dnp | 2,4-Dnp-O | **7**/**5** (1:1), EtOAc, rt, 3 h | 68 | [13] |

[a] See Scheme 4.

**Table 3** Examples of Azapeptides Generated by the Activated Ester Method[8,13,23,25,29][a]

| $R^1$ | $R^2$ | X | $R^3$ | $R^4$ | Method[b,c] | Conditions | Yield (%) of **8** | Ref |
|---|---|---|---|---|---|---|---|---|
| Z-Gly | H | N | H | Gly-OEt | C[b] | **6**/**7** (1:1), EtOAc, reflux, 3 h | 24 | [23] |
| Boc | Bzl | $CH_2CH$ | H (βAla) | OBzl | C | **6**/**7** (1:2), THF, 50 °C, 24 h | 61 | [25] |
| Boc | H | CH | Bzl | Leu-OBzl | C | **6**/**7**/HOBt (1:1:1), DMF, rt, overnight | 81 | [29] |
| Boc | iPr | N | iPr | OtBu | C | **6**/**7**/DMAP (1:1:1), $CH_2Cl_2$, rt, 0.5 h | 84 | [8] |
| H | H | N | H | Gly-OEt | N[c] | **4**[d]/**9** (1:0.7), EtOH, 0 °C, 0.5 h then 20 °C, 1.5 h | 89 | [23] |
| Z-AzGly | H | N | H | Gly-OEt | N | **4**/**9** (1:1), EtOAc, reflux, 2 h | 77 | [23] |
| Boc-Ala | $(CH_2)_2$ $CONH_2$ | CH | Bzl | OBzl | N | **4**/**9** (1:1), EtOAc, rt, overnight | 32 | [13] |

[a] See Scheme 4.
[b] C = C-activation.
[c] N = N-activation.
[d] Hydrate.

**Boc-AzPhe-ONp (6, $R^1$ = Boc; $R^2$ = Bzl; $Ar^1$ = Np):**[27]
To a stirred soln of 1-benzyl-2-*tert*-butoxycarbonylhydrazine (25.7 g, 115.8 mmol) in THF (150 mL) were added dropwise at 0 °C during 30 min 4-nitrophenyl chloroformate (23.3 g, 115.8 mmol) in THF (50 mL) and NMM (11.7 g, 115.8 mmol) in THF (50 mL). After 3 h at 0 °C the precipitate was filtered off and the resulting soln concentrated to dryness. The residue was taken up in $CHCl_3$ and the soln washed with aq $KHSO_4$, $Na_2CO_3$ soln, and $H_2O$, dried, and the solvent removed to dryness. The residue was chromatographed (silica gel, $CHCl_3$/hexane 97:3) to give a foam; yield: 40.8 g (91%); IR ($CDCl_3$) $\tilde{\nu}_{max}$: 3410, 1760, 1720 $cm^{-1}$.

**Z-Gly-AzGly-ONp (6, $R^1$ = Z-Gly; $R^2$ = H; $Ar^1$ = Np):**[23]
Z-Gly-$NHNH_2$ (1.62 g, 7.3 mmol) and bis(4-nitrophenyl) carbonate (2.21 g, 7.3 mmol) were refluxed under stirring in EtOAc (30 mL) for 2 h. After cooling to rt, the crystalline product was isolated; yield: 1.98 g (71%); mp 127–128 °C.

**Z-Gly-AzGly-AzGly-Gly-OEt (8, $R^1$ = Z-Gly; $R^2$ = H; X = N; $R^3$ = H; $R^4$ = Gly-OEt):**[23]
A soln of Z-Gly-AzGly-ONp[23] (0.260 g, 0.67 mmol) and H-AzGly-Gly-OEt[11] (0.108 g, 0.67 mmol) in EtOAc (10 mL) was refluxed for 3 h. The soln was filtered, the solvent removed in vacuo, and the residue recrystallized (EtOH/$H_2O$ 1:1.5 mL); yield: 0.065 g (24%); mp 189–190 °C.

***N*-(2,4-Dinitrophenoxycarbonyl)-Phe-OBzl [9, X = CH; $R^3$ = Bzl; $R^4$ = OBzl, $Ar^1$ = 2,4-Dnp]:**[14]
H-Phe-OBzl (TosOH salt) (3.95 g; 8.95 mmol) was dissolved in 0.1 M NaOH (100 mL) and extracted with $Et_2O$ (4 × 50 mL). The combined $Et_2O$ extracts were dried ($MgSO_4$) and the solvent removed in vacuo to give a light brown liquid (0.98 g, 65%). The free amino acid ester was dissolved in EtOAc (20 mL) and bis(2,4-dinitrophenyl) carbonate (2.30 g, 5.8 mmol) in EtOAc (20 mL) was added dropwise with stirring over 1 h. The soln was stirred for a further 2 h and petroleum ether (bp 40–60 °C, 400 mL) was added. An oil precipitated which on vigorous shaking solidified. After cooling for 3 d, the precipitate was collected and recrystallized (EtOAc/petroleum ether bp 40–60 °C) to give a white solid; yield: 1.50 g (67%); mp 104–104.5 °C; MS *m/z*: 466 $[M+H]^+$.

**Boc-Ala-AzGln-Phe-OBzl [8, $R^1$ = Boc-Ala; $R^2$ = $(CH_2)_2CONH_2$; X = CH; $R^3$ = Bzl; $R^4$ = OBzl]:**[14]
Boc-Ala-NHNH-$(CH_2)_2$-$CONH_2$[14] (0.100 g, 0.365 mmol) was dissolved in dry EtOAc (5 mL) and *N*-(2,4-dinitrophenoxycarbonyl)-Phe-OBzl (0.170 g, 0.365 mmol) in dry EtOAc (3 mL) was added and stirred overnight. Then, hexane (25 mL) was added to give a yellow solid on cooling. The solid material was dissolved in $CH_2Cl_2$ (20 mL) and the soln washed with 0.1 M NaOH (3 × 25 mL) and dried ($MgSO_4$). The solvent was removed in vacuo to give a white solid (0.120 g), which was crystallized (THF/hexane); yield: 0.065 g (32%); mp 151–151.5 °C; MS *m/z*: 557 $[M+H]^+$.

*for references see p 322*

**Boc-AzGly-Phe-Leu-OBzl (8, $R^1$ = Boc; $R^2$ = H; X = CH; $R^3$ = Bzl; $R^4$ = Leu-OBzl); Typical One-Pot Procedure:**[29]

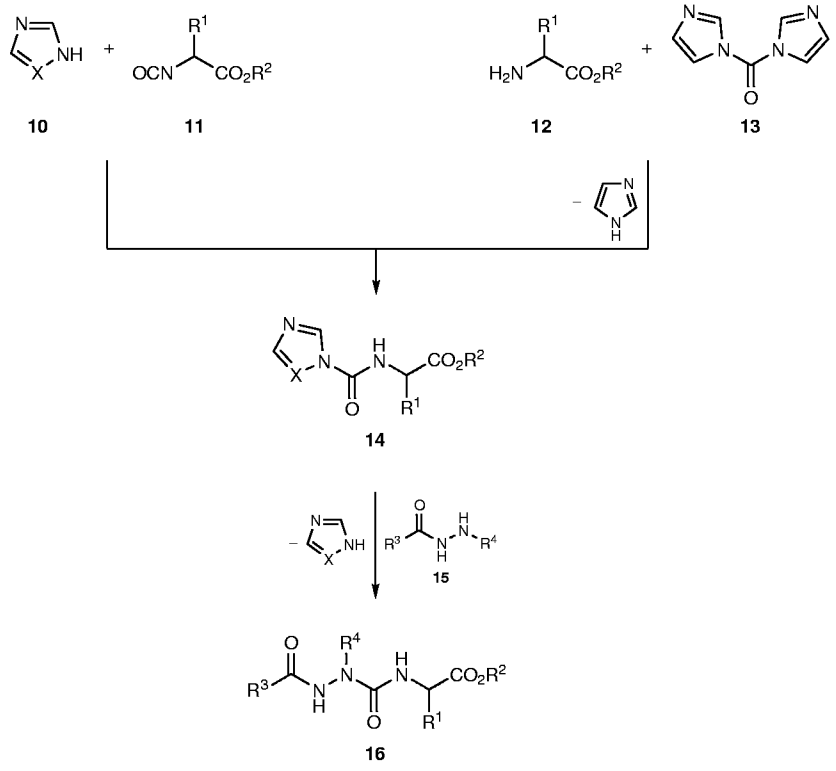
A cold (−10 °C) $CHCl_3$ soln (5 mL) of 2,4,5-trichlorophenyl chloroformate (520 mg, 2 mmol) was added with stirring to a cold $CHCl_3$ soln (10 mL) of Boc-$NHNH_2$ (264 mg, 2 mmol) and TEA (404 mg, 4 mmol). After stirring overnight at 23 °C to assure complete conversion into the active ester, HCl·H-Phe-Leu-OBzl (810 mg, 2 mmol) dissolved in DMF (2 mL) was added, followed in 1 h by HOBt (270 mg, 2 mmol). The mixture was stirred at 23 °C for 15 h, and EtOAc (50 mL) was added. The organic phase was washed with aq 0.5 M $Na_2CO_3$ (3 ×), 10% citric acid, and $H_2O$. The soln was dried ($MgSO_4$), EtOAc was removed in vacuo, and $Et_2O$ (20 mL) was added to the oily residue; yield: 856 mg (81%); mp 156–159 °C; $[\alpha]_D^{23}$ −10.6 (*c* 1, DMF).

## 10.4.3 Reaction of *N*-Azolides with Amino Groups

Like N-activated esters, N-activated azolides can be used in azapeptide synthesis. These include imidazolides[30–33] and 1,2,4-triazolides.[34] These intermediates **14** can be synthesized by addition of the corresponding heterocycle **10** to α-isocyanato carboxylates **11**[30,31] or from the amino component **12** and 1,1′-carbonyldiimidazole (**13**).[32,33]

Their potential in azapeptide synthesis is similar to that of the N-activated esters or isocyanates as they react readily with hydrazides **15** to give the azapeptides **16** with release of imidazole or 1,2,4-triazole as shown in the general Scheme 5 and in an illustrative example (Scheme 6).[32]

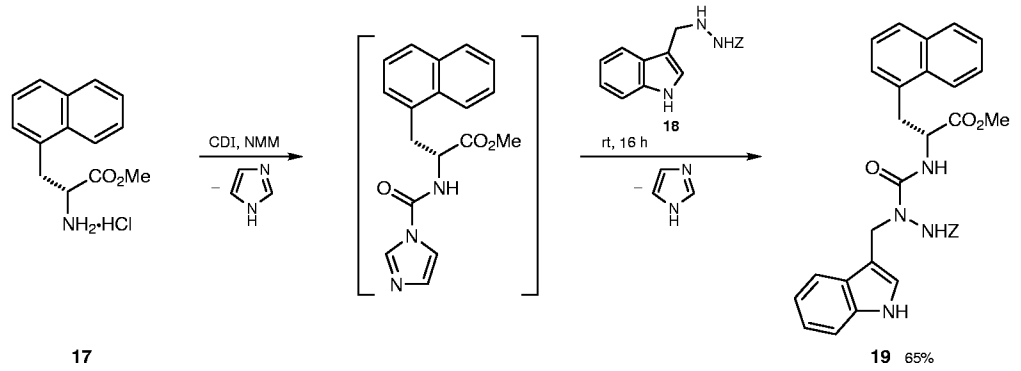
**Scheme 5** Azapeptides via *N*-Azolides



The isocyanate reaction pathway seems to have no major advantage over the other methods with N-activation (*vide supra*), since it implies an additional reaction step and the isocyanates react directly with hydrazides giving azapeptides. Advantages may be the easier handling of the solid and stable *N*-azolides in comparison to the liquid, toxic, and relatively unstable isocyanates.

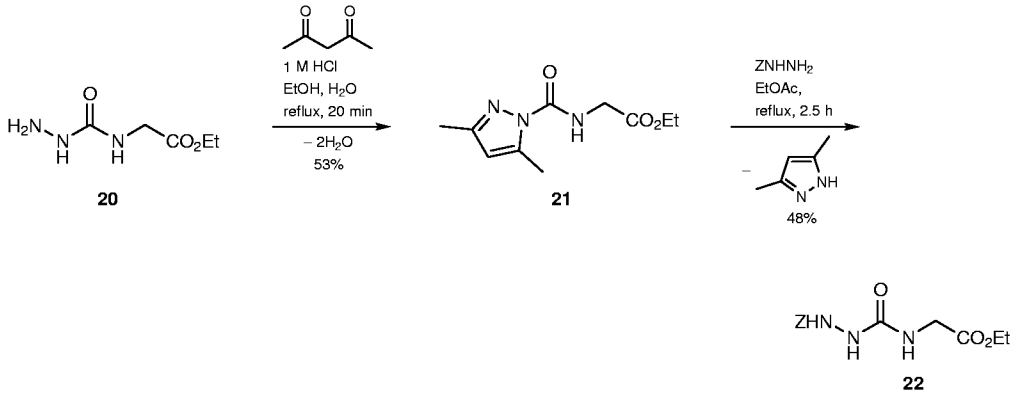
An additional route for the generation and reaction of an activated *N*-azolide is given in Scheme 7, though this is the sole example of its type.[30] It involves the ring closure of the azapeptide H-AzGly-Gly-OEt (**20**) with acetylacetone to give the corresponding activated pyrazole derivative **21** and further reaction with Z-protected hydrazine leading to the azapeptide derivative **22**.

**Scheme 6** Synthesis of an Azapeptide by the 1,1′-Carbonyldiimidazole (CDI) Method[32]




17 19 65%

**Scheme 7** Synthesis of an Azapeptide by the *N*-Pyrazolide Method[30]




20 21 22

**Z-AzTrp-D-(1-naphthyl)Ala-OMe (19):**[32]

To HCl·H-D-(1-naphthyl)Ala-OMe (**17**; 190 mg, 0.715 mmol) in DMF (3 mL) at 0 °C was added NMM (80 µL, 0.73 mmol), followed by CDI (116 mg, 0.715 mmol). After 1 h, 1-(benzyloxycarbonyl)-2-(indol-3-ylmethyl)hydrazine[32] (**18**; 210 mg, 0.711 mmol) was added, and the mixture was warmed to rt and stirred for 16 h. The mixture was poured into EtOAc and then washed sequentially with $H_2O$, sat. $NaHCO_3$, and brine, dried ($Na_2SO_4$), and the solvent removed under reduced pressure. Column chromatography (20% EtOAc/hexane) of the residue afforded a foam: yield: 256 mg (65%).

***N*-(3,5-Dimethylpyrazol-1-ylcarbonyl)-Gly-OEt (21):**[30]

A soln of H-AzGly-Gly-OEt[11] (**20**; 1.073 g, 6.67 mmol), acetylacetone (0.68 mL, 6.67 mmol), and 1 M HCl (1.5 mL) in EtOH/$H_2O$ (1:1, 20 mL) was refluxed for 20 min. Then, $H_2O$ (25 mL) was added; the precipitated oil was crystallized over several hours; yield: 0.80 g (53%); mp 71–72 °C.

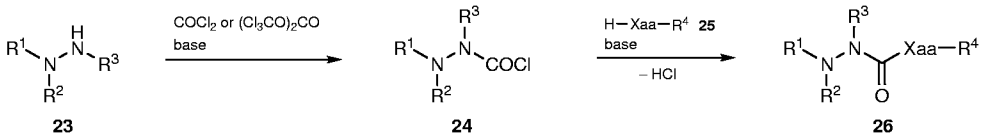
*for references see p 322*

**Z-AzGly-Gly-OEt** (22):
A soln of *N*-(3,5-dimethylpyrazol-1-ylcarbonyl)-Gly-OEt (**21**; 0.286 g, 1.27 mmol) and benzyloxycarbonylhydrazine (0.211 g, 1.27 mmol) in EtOAc (5 mL) was refluxed for 2.5 h. Upon addition of petroleum ether (8 mL) the product was crystallized; yield: 0.18 g (48%); mp 139–141 °C.

## 10.4.4 Reaction of *N*-Carboxylic Acid Chlorides with Amino Groups

The acid chloride method of linking two amino acid residues giving a peptide has its parallel in the azaamino acid series. Though in principle, as in the activated ester series, C- and N-activation is possible, nearly all of the applications of this method deal with C-terminal activation of (substituted) amino acid hydrazides. Reaction of hydrazine **23** with phosgene, or its substitute triphosgene, in the presence of a base gives the *N*-carboxylic acid chlorides **24** which are reacted with the amino component **25** with release of HCl[5,7,35–38] to give the azaamino acid derivative **26** (Scheme 8, Table 4). Both Fmoc-protected azaalanine and azasarcosine are quantitatively converted into the corresponding carbazic acid chlorides by using a solution of phosgene in toluene in the absence of a base. These carbazic acid chlorides were isolated as pure solids and proved to be stable in long-term storage at 4 °C.[9]

**Scheme 8** Coupling of *N*-Carboxylic Acid Chlorides with Amino and Azaamino Acid Derivatives[7,9,35–38]




This method has been applied preferentially to proline derivatives as coupling partners for the acid chlorides. The relatively strong activation of these intermediates may recommend them for the generally difficult coupling with secondary amines, as is the case with amino acid chlorides in the peptide series.

This coupling method seems to be equally suited to the synthesis of "normal" azaamino acid compared to the activated ester method.

In the case of the unsubstituted hydrazide Z-AzGly-Cl the thermal instability of this compound causes polymerization,[35] whereas Fmoc-AzGly-Cl can be prepared at room temperature with phosgene in the absence of a base as a stable solid.[39]

The solid-phase technique has been applied in this field with the amino component linked to a polymer.[7,9] To meet the requirements of combinatorial chemistry the coupling conditions of resin-bound Asp with carbazic acid chloride of Fmoc-AzSar-OH and Fmoc-AzAla-OH were individually optimized. Remarkably, the carbazic acid chloride of Fmoc-AzSar-OH proved to be much more reactive than the activated Fmoc-AzAla-OH.[9]

In the only application of this method via N-activation, use of an *N*-(chloroformyl)-proline ester leads to the AzAla analogue of the powerful antihypertensive agent enalapril with comparable biological activity.[36]

**Table 4** Azapeptides Synthesized by the Acid Chloride Method[7,9,35,38][a]

| $R^1$ | $R^2$ | $R^3$ | Xaa | $R^4$ | Conditions | Yield (%) of **26** | Ref |
|---|---|---|---|---|---|---|---|
| Z | H | H | AzGly | OBzl | **24**/**25** (1:2), $Et_2O$, rt, 15 min | 68 | [35] |
| Boc | H | $CH_2CO_2Et$ | Pro-Ser(Bzl) | NHMe | NMM, $(Cl_3CO)_2CO$, then **23** (1:0.33:1), THF, −10 °C, 45 min; then **25**/NMM (1:1), THF −10 °C, 1 h; rt, 2 h | 60 | [38] |
| Boc | H | Me | Pro | $NMe_2$ | NMM, $(Cl_3CO)_2CO$, then **23** (1:0.33:1), THF, −10 °C, 45 min; then **25**/NMM (1:1), THF −10 °C, 1 h; rt, 2 h | 60 | [7] |
| Z | H | $CH_2CO_2Et$ | Pro | OtBu | NMM, $(Cl_3CO)_2CO$, then **23** (1:0.33:1), THF, −10 °C, 45 min; then **25**/NMM (1:1), THF −10 °C, 1 h then rt, 2 h | 75 | [7] |
| Fmoc | Me | H | Asp(Bu) | $NH_2$ | $COCl_2$/**23** (2.4:1), dioxane, 10 °C, 5 min then rt, 70 min; **24** and resin-bound **25**, then DIPEA (5:1:6), $CH_2Cl_2$, rt, 2 h | –[b] | [9] |
| Fmoc | H | Me | Asp(Bu) | $NH_2$ | $COCl_2$/**23** (2:1), dioxane, 10 °C, 5 min then rt, 90 min; **24** and resin-bound **25**, then DIPEA (5:1:5.5), DMF, rt, 15 h | –[c] | [9] |

[a] See Scheme 8.
[b] >95% purity.
[c] 89% purity.

**Boc-AzAla-Pro-$NMe_2$** (26, **$R^1$ = Boc; $R^2$ = H; $R^3$ = Me; Xaa = Pro; $R^4$ = $NMe_2$**):[7]
A soln of NMM (0.44 mL, 4 mmol) in THF (1 mL) was added dropwise under stirring to a cold soln (−10 °C) of triphosgene (0.39 g, 1.3 mmol) and 1-*tert*-butoxycarbonyl-2-methylhydrazine (0.58 g, 4 mmol) in THF (4 mL). Stirring at −10 °C was maintained for 45 min. Then, H-Pro-$NMe_2$ (0.569 g, 4 mmol) and NMM (0.44 mL, 4 mmol) were added progressively under stirring at −10 °C for 1 h, and then at rt for 2 h. After filtration, THF was removed in vacuo, and the crude material, dissolved in $CH_2Cl_2$, was washed with aq 0.5 M $KHSO_4$ (2 ×) and sat. aq NaCl (2 ×), dried ($Na_2SO_4$), and chromatographed (silica gel, iPrOH/$CH_2Cl_2$ 7:93); yield: 60%; mp 154 °C.

**2-(Chloroformyl)-1-(9-fluorenylmethoxycarbonyl)-1-methylhydrazine** (24, **$R^1$ = Fmoc; $R^2$ = Me; $R^3$ = H**):[9]
1.89 M Phosgene in toluene (4.2 mL, 7.8 mmol) was slowly added at 10 °C to a stirring soln of 1-(9-fluorenylmethoxycarbonyl)-1-methylhydrazine (0.925 g, 3.30 mmol) in anhyd dioxane (20 mL). The temperature was kept at 10 °C for 5 min, after which the mixture was allowed to warm to rt and stirred for 70 min. The mixture was lyophilized; yield: 1.092 g (~100%); mp 146 °C; IR (KBr) $\tilde{\nu}_{max}$: 3250, 1782, 1694 $cm^{-1}$.

**2-(Chloroformyl)-1-(9-fluorenylmethoxycarbonyl)-2-methylhydrazine** (24, **$R^1$ = Fmoc; $R^2$ = H; $R^3$ = Me**):[9]
1.89 M Phosgene in toluene (1.7 mL, 3.2 mmol) was slowly added at 10 °C to a stirring soln of 1-(9-fluorenylmethoxycarbonyl)-2-methylhydrazine (0.449 g, 1.60 mmol) in anhyd dioxane (5 mL). The temperature was kept at 10 °C for 5 min, after which the mixture was allowed to warm to rt and stirred for 90 min. The solvent was removed under reduced pressure; yield: 0.531 g (~100%); mp 62 °C; IR (KBr) $\tilde{\nu}_{max}$: 3298, 1749 $cm^{-1}$.

**Fmoc-AzSar-Asp(Bu)-amide Photolinker TentaGel Resin** (26, **$R^1$ = Fmoc; $R^2$ = Me; $R^3$ = H; Xaa = Asp(Bu); $R^4$ = $NH_2$**):[9]
H-Asp(Bu)-amide photolinker TentaGel resin (0.372 g, 0.18 mmol·$g^{-1}$, 67 μmol) was washed with anhyd $CH_2Cl_2$ (3 × 5 mL) and a soln of 2-(chloroformyl)-1-(9-fluorenylmethoxycarbonyl)-1-methylhydrazine (0.112 g, 0.339 mmol) in anhyd $CH_2Cl_2$ (3 mL) was added. The mixture was shaken at rt for 10 min. A soln of DIPEA (69 μL, 52 mg, 0.40 mmol) in $CH_2Cl_2$ (0.9 mL) was then added to the mixture in 3 equal portions over a period of 1 h. After shaking for 1 h, the resin was washed with $CH_2Cl_2$ (3 × 7 mL), NMP (5 × 7 mL), and $Et_2O$ (4 × 7 mL) and dried in vacuo. A small sample of resin (10.4 mg) was photolyzed in a soln of 20% $H_2O$ in MeCN (1 mL) for 1 h to yield the product (with >95% purity) in soln; HPLC [220 nm, 1 mL·$min^{-1}$, 30 min linear gradient from 10 to 90% MeCN (0.1% TFA) in $H_2O$ (0.1% TFA)] $t_R$ 23.5 min.

*for references see p 322*

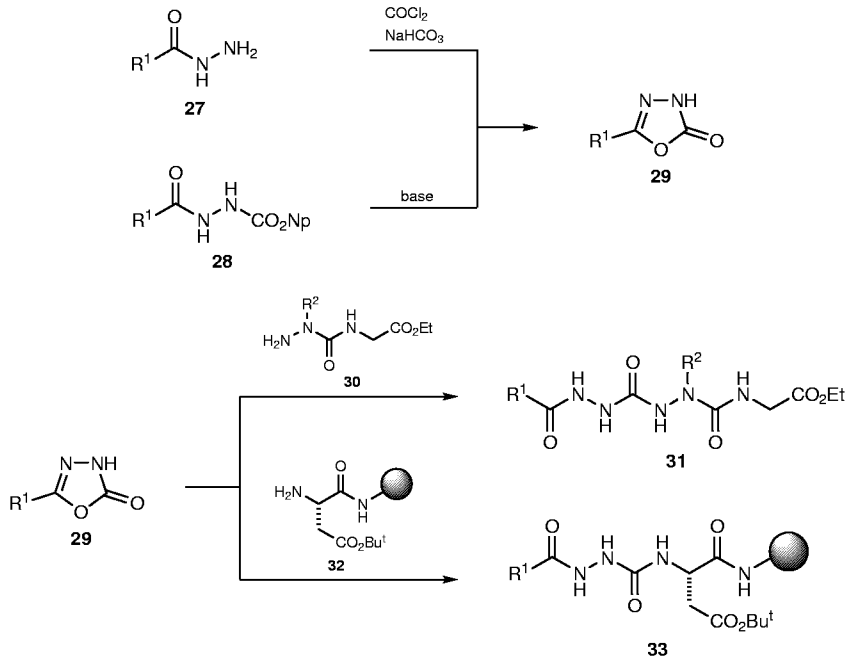
**Fmoc-AzAla-Asp(Bu)-amide Photolinker TentaGel Resin (26, $R^1$ = Fmoc; $R^2$ = H; $R^3$ = Me; Xaa = Asp(Bu); $R^4$ = $NH_2$):**[9]

H-Asp(Bu)-amide photolinker TentaGel resin (0.577 g, 0.17 mmol·g$^{-1}$, 98.1 μmol) was washed with anhyd DMF (3 × 5 mL) and a soln of 2-(chloroformyl)-1-(9-fluorenylmethoxycarbonyl)-2-methylhydrazine (0.164 g, 0.496 mmol) and DIPEA (92 μL, 69 mg, 0.54 mmol) in anhyd DMF (5 mL) was added. The mixture was shaken at rt for 15 h and washed with NMP (5 × 7 mL) and $Et_2O$ (4 × 7 mL) and dried in vacuo. A small sample of resin (9.5 mg) was photolyzed in 20% $H_2O$/MeCN (1 mL) for 1.5 h to yield the product (with 89% purity) in soln; HPLC [220 nm, 1 mL·min$^{-1}$, 30 min linear gradient from 10 to 90% MeCN (0.1% TFA) in $H_2O$ (0.1% TFA)] $t_R$ 20.8 min.

## 10.4.5 Ring Opening of 1,3,4-Oxadiazol-2(3*H*)-ones by Amino Groups

Another method of generating azapeptides is to react 1,3,4-oxadiazol-2(3*H*)-ones **29**, which are alkyl-, aryl-, Z-aminoalkyl, or 9-fluorenylmethoxy-substituted in the 5-position, with various N-protected azadipeptides **30** or resin-bound β-protected Asp (**32**). Ring opening and addition give azatri- and azatetrapeptides **31** with two directly linked azaamino acid residues and polymer-bound Fmoc-protected azapeptide **33**, respectively. The heterocycles **29** can be obtained from the corresponding hydrazides **27** with phosgene or from protected azaamino acid 4-nitrophenyl esters **28** with base[9,11,23] (Scheme 9). They can be regarded as aza analogues of the azlactones which react similarly with amines and give acylamino acid amides and peptides.[40]

**Scheme 9** Synthesis of Azapeptides via 1,3,4-Oxadiazol-2(3*H*)-ones[9,11,18]




| $R^1$ | $R^2$ | Conditions | Yield (%) of **31** or **33** | Ref |
|---|---|---|---|---|
| Me | Me | **29/30** (1:1), without solvent, 130 °C, 4 h | 23 | [18] |
| Ph | Bzl | **29/30** (1:1), without solvent, 125–130 °C, 1 h | 17 | [18] |
| $CH_2NHZ$ | H | **29/30** (1:1), without solvent, 145 °C, 10 min | 71 | [18] |
| $CH_2NHZ$ | Bzl | **29/30** (1:1), without solvent, 130 °C, 1 h | 25 | [18] |
| DL-CH(Me)NHZ | H | **29/30** (1:1), without solvent, 125 °C, 30 min | 57 | [18] |
| L-CH(iBu)NHZ | H | **29/30** (1:1), without solvent, 130 °C, 1 h | 47 | [18] |
| OFm | | **29/32** (3:1), $CH_2Cl_2$, rt, 90 min | –[a] | [9] |

[a] 88% purity.

However, 1,3,4-oxadiazol-2(3*H*)-ones with a methyl, phenyl, or an alkylamino group at C5 appear to be much more stable and thus react more sluggishly than azlactones. This is in accordance, for example, with the relatively high reaction temperature (110–145 °C) required and with the fact that the amino group of an amino acid does not react at all with the heterocycle.[28,41] In addition, a 3-methylated oxadiazolone reacted neither with H-Phe-OMe nor with H-AzAla-Phe-OMe[41] due to further deactivation of the heterocyclic system. In contrast to these compounds, the 5-(9-fluorenylmethoxy)-1,3,4-oxadiazol-2(3*H*)-one (**29**, $R^1$ = OFm) exhibited extremely high reactivity. Thus, the reaction of **29** ($R^1$ = OFm) with resin-bound Asp **32** at room temperature was complete within 50 minutes.[9] It is reasonable to assume that the substitution with an oxygen at C5 of the 1,3,4-oxadiazol-2(3*H*)-one, resulting from the urethane group of 9-fluorenylmethoxycarbonylhydrazine (**27**, $R^1$ = OFm), would be the reason for the observed high reactivity of **29** ($R^1$ = OFm).

**5-(9-Fluorenylmethoxy)-1,3,4-oxadiazol-2(3*H*)-one** (29, **$R^1$ = OFm**):[9]
9-Fluorenylmethoxycarbonylhydrazine (0.995 g, 3.91 mmol) was suspended in $CH_2Cl_2$ (40 mL) and sat. aq $NaHCO_3$ (40 mL). The biphasic mixture was cooled to 0 °C in an ice bath with vigorous stirring for 5 min. Stirring was stopped, the layers were allowed to separate, and 1.89 M phosgene in toluene (4.2 mL, 7.8 mmol) was added in a single portion by syringe to the lower (organic) phase. Stirring was resumed immediately, and the ice-cooled mixture was vigorously stirred for 10 min. Then, the layers were separated, the aqueous phase was extracted with $CH_2Cl_2$ (3 × 40 mL), and the combined organic layers were dried ($Na_2SO_4$), filtered, and concentrated under reduced pressure; yield: 0.958g (87%); mp >260 °C dec; IR (KBr) $\tilde{\nu}_{max}$: 3300, 1780, 1650 $cm^{-1}$.

**Z-Gly-AzGly-AzGly-Gly-OEt** (31, **$R^1$ = $CH_2NHZ$; $R^2$ = H**):[11]
A well-homogenized mixture of 2-{[(benzyloxycarbonyl)amino]methyl}-1,3,4-oxadiazol-2(3*H*)-one[11] (0.33 g, 1.33 mmol) and H-AzGly-Gly-OEt (0.22 g, 1.37 mmol) was heated at 145 °C for 10 min (the initially liquefied mixture solidified at the end of the reaction). After recrystallization (EtOH/$H_2O$ 1:1, 15 mL) the product was obtained; yield: 0.39 g (71%); mp 188–190 °C.

**Fmoc-AzGly-Asp(Bu)-amide Photolinker TentaGel Resin** (33, **$R^1$ = OFm**):[9]
H-Asp(Bu)-amide photolinker TentaGel resin (1.12 g, 0.18 mmol · $g^{-1}$, 0.20 mmol) was washed with anhyd $CH_2Cl_2$ (3 × 7 mL) and a soln of 5-(9-fluorenylmethoxy)-1,3,4-oxadiazol-2(3*H*)-one (**29**, $R^1$ = OFm; 0.176 g, 0.629 mmol) in anhyd $CH_2Cl_2$ (5 mL) was added. The mixture was shaken at rt for 90 min, washed with $CH_2Cl_2$ (3 × 7 mL), NMP (5 × 7 mL), and $Et_2O$ (4 × 7 mL), and dried in vacuo. A small sample of resin (3.1 mg) was photolyzed in a soln of 20% $H_2O$/MeCN (1 mL) for 1.5 h to give the product (with 88% purity) in soln: HPLC [220 nm, 1 mL · $min^{-1}$, 30 min linear gradient from 10 to 90% MeCN (0.1% TFA) in $H_2O$ (0.1% TFA)] $t_R$ 28.2 min.

## 10.4.6 Reaction of Hydrazides with Carbonic Acid Derivatives or Isocyanic Acid

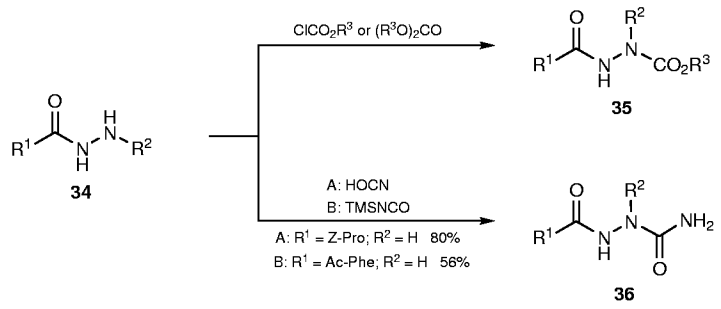
Special azapeptides are those with C-terminal azaamino acid residues that can only be obtained as esters or amides; the corresponding free acids are not available because of their lability that results in decarboxylation to hydrazides.

The esters **35** are obtained by reaction of hydrazides **34** with chloroformate esters or carbonic diesters[3,23,28,42] as previously discussed for the special case of the activated esters (see Section 10.4.2). The amides **36**, on the other hand, are formed from hydrazides **34** with isocyanic acid[3,43,44] or with trimethylsilyl isocyanate[42] (Scheme 10).

Another possibility of generating C-terminal azaglycine amides is given by coupling the C-terminal amino acid of a peptide chain with semicarbazide by the conventional methods of peptide synthesis.[43]

*for references see p 322*

**Scheme 10** Synthesis of C-Terminal Azapeptide Esters and Amide[42–44]




| $R^1$ | $R^2$ | $R^3$ | Yield (%) of **35** | Ref |
|---|---|---|---|---|
| Ac | H | iBu | 84 | [42] |
| Ac | H | $CH_2CH_2Cl$ | 64 | [42] |
| Ac | H | $CH_2CH_2Br$ | 94 | [42] |
| Ac | H | $CH_2CCl_3$ | 76 | [42] |
| Ac | H | Bzl | 54 | [42] |
| Ac | H | Ph | 57 | [42] |
| Ac | H | Np | 26 | [42] |

**Synthesis of C-Terminal Azapeptide Esters** 25**; General Procedure:**[42]
Ac-Phe-$NHNH_2$[42] and NMM (molar ratio 1:1) were dissolved in $CHCl_3$ at −10 °C followed by slow dropwise addition of the chloroformate ester (1 equiv) dissolved in $CHCl_3$ (5 mL). The mixture was stirred at −10 °C for 30 min and rt for additional 6 h. After addition of more $CHCl_3$, the organic phase was extracted with sat. $NaHCO_3$ and then $H_2O$, and dried. Removal of the $CHCl_3$ left a crude product as white paste, from which a pure product was isolated by chromatography and/or recrystallization.

**Z-Pro-AzGly-$NH_2$ (36, $R^1$ = Z-Pro; $R^2$ = H):**[44]
To Z-Pro-$NHNH_2$[44] (0.179 g, 0.41 mmol) dissolved in warm $H_2O$ (2 mL) containing AcOH (0.05 mL) NaOCN (0.031 g, 0.48 mmol) was added. The soln was kept at rt for 45 min and at 4 °C for 18 h. Filtration afforded the product; yield: 0.101 g (80%); mp 187–189 °C. After recrystallization (iPrOH) the mp was 187–189.5 °C.

**Ac-Phe-AzGly-$NH_2$ (36, $R^1$ = Ac-Phe; $R^2$ = H):**[42]
Ac-Phe-$NHNH_2$[42] (0.44 g, 2.0 mmol) in THF/EtOH (1:1, 15 mL) was combined with TMSNCO (0.54 g, 4.0 mmol) and the mixture was heated to reflux for 20 h, after which $H_2O$ was added and the mixture was stirred for 1 h. After removal of the solvent in vacuo, the residue was chromatographed yielding a white solid that was recrystallized (MeOH/$Et_2O$); yield: 56%; mp 123–126 °C; MS (CI, $NH_3$) *m/z*: 265 [M + $H^+$].


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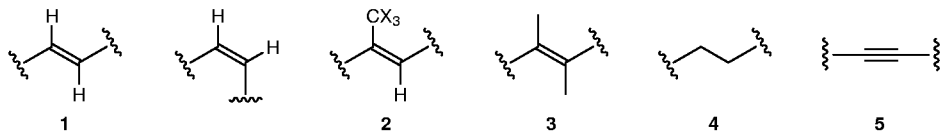
# 10.5 Synthesis of Peptides Containing Carbaamide Bond Replacements

D. Tourwé, K. Iterbeke, and E. Mannekens

The idea of replacing the peptide CO–NH function by an isosteric function originated more than a decade ago in order to investigate the importance of specific peptide bonds as hydrogen donors or acceptors during receptor interaction.[1,2] Since the amide bond of peptides constitutes the major site of enzymatic metabolism, this replacement is also able to stabilize the peptide even when an amide bond adjacent to the cleavage site is replaced.[3] The altered lipophilicity of the carba replacements can affect the transport properties (e.g., blood–brain) of the resulting pseudopeptides.

Because of the preferred *trans* conformation of secondary amide bonds, the replacement by an *E*-configurated C=C bond **1** (Scheme 1 and Table 1) was considered as an obvious isostere, maintaining the rigidity and overall geometry.[4] This similarity has been confirmed by theoretical calculations,[5,6] which, however, also show a wider φ,ψ range in the Ramachandran plots. The resulting increased conformational space accessible to the pseudopeptide can be reduced by introducing substituents on the double bond (**2** and **3**). In order to restore the polarity and the hydrogen bond acceptor character present in the peptide function, a fluorine substitutent was introduced.[7,8] The best mimic of the electronic properties of the amide bond was found to be the (trifluoromethyl)alkene **2** (X = F).[9]

**Scheme 1** Carbaamide Bond Replacements




1 2 3 4 5

The dimethylene replacement **4** is used to investigate the influence of rigidity. The use of 5-aminovaleric acid as a Gly$\psi$[$CH_2$–$CH_2$]Gly replacement has been extended to other ω-amino acids as a means to influence the distance between crucial amino acid residues, and to function as a flexible spacer, replacing parts of the peptide.

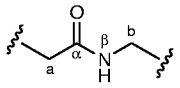
The alkyne replacement **5** has been less exploited and has a distinctly different geometry. It has been proposed as a turn inducer.[10]

Methods to prepare vinylogous ($NH_2$–CHR–CH=CH–$CO_2H$)[11–13], alkynylogous ($NH_2$–CHR–C≡C–$CO_2H$),[14] and $NH_2$–CHR–$CH_2$–$CH_2$–$CO_2H$[15] amino acids have been reported. These are lacking one carbon atom to be isosteric with a dipeptide, and therefore are not treated here.

## 10.5.1 $\psi$[$CH_2$–$CH_2$] Peptides

A comparison of the geometrical, steric, and lipophilic properties of this isostere with those of the parent amide bond, as shown in Table 1, reveals that this replacement maintains the distance between the α-carbons.[16] This distance has been changed by using ω-amino acids $NH_2$–$(CH_2)_n$–$CO_2H$ with varying chain length.[17,18] As expected, the bond angles have changed owing to the modification in hybridization. A systematic molecular mechanics calculation confirms the similarity of the accessible distances and pseudo torsion angles for the dipeptide and the $\psi$[$CH_2$–$CH_2$] isostere.[19] The molar volume of the isostere is slightly increased (Table 1). Lipophilicity is much higher [Table 1 and log $p$($CH_2$–$CH_2$) = +1.54 versus −2.18 for CONH].[20]

**Table 1** Calculated Geometrical and Physicochemical Properties of Carbaamide Bond Replacements in *N*-Methylacetamide




| Compound | Distance between $C_a$ and $C_b$ (Å) | α (°) | β (°) | Mol Vol (Å$^3$) | CLOG P[a] |
|---|---|---|---|---|---|
| MeCONHMe | 3.8 | 119 | 120 | 69.3 | −1.08 |
| ψ(*E*, CH=CH) | 3.9 | 122 | 122 | 68.2 | 2.27 |
| ψ(*Z*, CH=CH) | 3.0 | 125 | −125 | 68.5 | 2.27 |
| ψ($CH_2CH_2$) | 3.9 | 111 | 112 | 76.0 | 2.81 |

[a]Calculated log *p* values

NMR studies of the opioid tetrapeptide H-Tyr-D-Ala-Phe-Gly-$NH_2$ and of its Phe$^3$ψ[$CH_2$–$CH_2$]Gly$^4$ analogue demonstrated little effect on the overall conformational equilibrium, including the characteristic high shielding and temperature dependence of the Ala$^2$ methyl signal.[21] An NMR and molecular modeling study on substance P analogues indicated a shift from predominant α-helix for native substance P to an equilibrium of β-turn and $3_{10}$-helix in the Gly$^9$ψ[$CH_2$–$CH_2$]Leu$^{10}$ analogue, probably owing to the absence of an interaction of the C-terminal tripeptide with the core of the molecule.[22] 5-Aminovaleric acid has been incorporated into a folded helical structure[23] and pseudohexapeptides containing a β-turn/β-sheet conformation.[24] In a linear octapeptide analogue, no significant β-turn formation was observed.[10]

The ψ[$CH_2$–$CH_2$] dipeptides have been incorporated into the sequence of biologically active peptides, as shown in Table 2. The first reported example, described by Szelke et al.[25] in an LHRH analogue, showed that full agonism can be maintained, but with a drop in potency. Later, the substitution in a Met-enkephalin analogue resulted in a dramatic loss in receptor affinity, which is not observed for the ψ[CH=CH] analogue.[4] A similar observation has been reported for CCK-4,[26] enkephalins,[27–29] and RasFPT-ase,[30] indicating the importance of the peptide bond geometry in these peptides. In IL-8, potency was maintained,[31] as was the case in CCK[27,32] and bradykinin analogues, where a large portion of the peptide could be replaced by an ω-amino acid spacer.[33,34] An increased receptor selectivity has also been observed.[21,22]

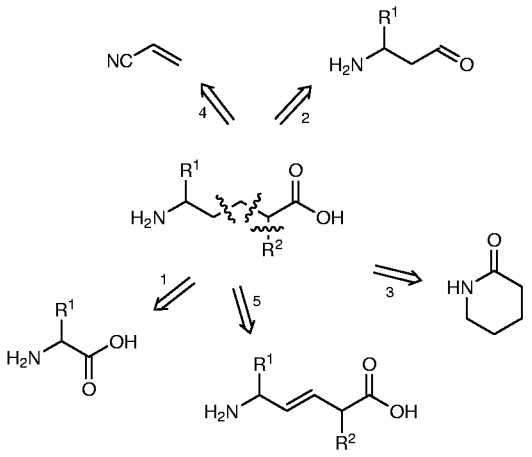
*for references see p 369*

**Table 2** Peptides Containing the ψ[$CH_2$–$CH_2$] Replacement

| ψ[$CH_2$–$CH_2$] Dipeptide | Section | Peptide (Class) | Ref |
|---|---|---|---|
| Glyψ[$CH_2$–$CH_2$]Gly (5-Ava) | | enkephalin | [35] |
| | | CCK | [6] |
| Glyψ[$CH_2$–$CH_2$]Gly (5-Ava) | | substance P | [22] |
| Glyψ[$CH_2$–$CH_2$]Gly (5-Ava) | | enkephalin | [134] |
| Glyψ[$CH_2$–$CH_2$]Gly (5-Ava) | | helical octapeptide | [23] |
| Glyψ[$CH_2$–$CH_2$]Gly (5-Ava) | | cyclic hexapeptide | [24] |
| Glyψ[$CH_2$–$CH_2$]Gly (5-Ava) | | interleukin-8 | [31] |
| $NH_2(CH_2)_5CO_2H$ | | enkephalin | [135] |
| $NH_2(CH_2)_{2\text{-}5}CO_2H$ | | enkephalin | [17] |
| $NH_2(CH_2)_{1\text{-}4}CO_2H$ | | CCK | [20] |
| $NH_2(CH_2)_{1\text{-}5}CO_2H$ | | CCK | [18] |
| $NH_2(CH_2)_{5\text{-}11}CO_2H$ | | bradykinin | [33] |
| Alaψ[$CH_2$–$CH_2$]Gly | 10.5.1.2 | interleukin-8 | [31] |
| D-Alaψ[$CH_2$–$CH_2$]Gly | 10.5.1.5 | enkephalin | [28] |
| Aspψ[$CH_2$–$CH_2$]D,L-Phe | 10.5.1.2 | CCK | [26] |
| Glyψ[$CH_2$–$CH_2$]Ala | 10.5.1.4 | interleukin-8 | [31] |
| Glyψ[$CH_2$–$CH_2$]Leu | 10.5.1.3 | substance P | [22] |
| Glyψ[$CH_2$–$CH_2$]D-Leu | 10.5.1.3 | substance P | [22] |
| Glyψ[$CH_2$–$CH_2$]D,L-Phe | 10.5.1.2 | CCK | [39] |
| Glyψ[$CH_2$–$CH_2$]D,L-Phe | 10.5.1.5 | enkephalin | [28] |
| Glyψ[$CH_2$–$CH_2$]D,L-Phe | 10.5.1.3, 10.5.1.5 | RasFPT-ase | [43] |
| Glyψ[$CH_2$–$CH_2$]D,L-Phe | 10.5.1.3 | DPDPE | [29] |
| Glyψ[$CH_2$–$CH_2$]D,L-Leu | 10.5.1.3 | CCK | [39] |
| Ileψ[$CH_2$–$CH_2$]Nva | 10.5.1.5 | RasFPT-ase | [30] |
| Ile[$CH_2$–$CH_2$]Phe | 10.5.1.5 | RasFPT-ase | [30] |
| Ileψ[$CH_2$–$CH_2$]Val | 10.5.1.5 | RasFPT-ase | [30] |
| Leuψ[$CH_2$–$CH_2$]Leu | 10.5.1.5 | – | [42] |
| Leuψ[$CH_2$–$CH_2$]D,L-Leu | 10.5.1.2 | bombesin | [15] |
| Leuψ[$CH_2$–$CH_2$]D,L-Phe | 10.5.1.2 | – | [40] |
| Nleψ[$CH_2$–$CH_2$]Asp | 10.5.1.2 | CCK | [26] |
| Nleψ[$CH_2$–$CH_2$]Gly | 10.5.1.5 | CCK | [27,32,136] |
| Pheψ[$CH_2$–$CH_2$]D,L-Ala | 10.5.1.2 | – | [38] |
| Pheψ[$CH_2$–$CH_2$]Gly | 10.5.1.5 | dermorphin | [21] |
| Pheψ[$CH_2$–$CH_2$]Gly | 10.5.1.1 | LHRH | 25] |
| Tyrψ[$CH_2$–$CH_2$]Gly | 10.5.1.1, 10.5.1.5 | enkephalin | [4,35] |
| Tyrψ[$CH_2$–$CH_2$]Gly | 10.5.1.1 | enkephalin | [134] |
| Tyrψ[$CH_2$–$CH_2$]Gly | 10.5.1.1 | LHRH | [25] |

Synthetic pathways towards the dimethylene isosteres comprise several disconnections, indicated by the essential bond forming reactions (Scheme 2). Several methods yield racemic dipeptide analogues. These are usually incorporated into the peptide sequence and the resulting epimeric peptides are separated. When either $R^1$ or $R^2 = H$, asymmetric syntheses towards the required enantiomer are available. When both $R^1$ and $R^2 \neq H$, only the reduction of the ψ[CH=CH] precursor yields homochiral compounds. As many ω-amino acids ($R^1 = R^2 = H$) are commercially available, their synthesis needs not be discussed here.

**Scheme 2** Retrosynthetic Pathways towards the ψ[$CH_2$–$CH_2$] Dipeptides

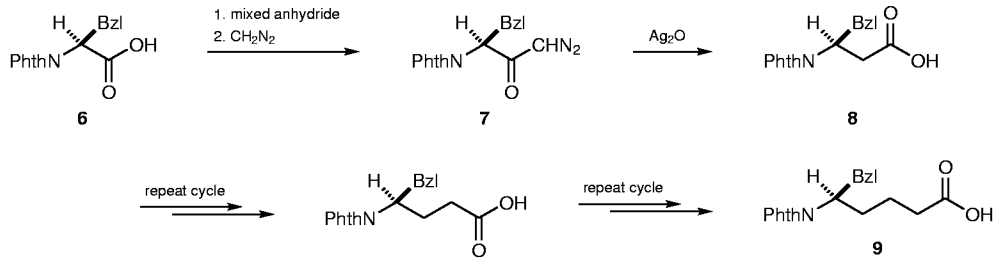


Peptide synthesis with ψ[$CH_2$–$CH_2$] dipeptides has been performed in solution[27,29] as well as on solid-phase supports, using either Boc[22,28] or Fmoc chemistry[33] and various coupling methods (carbodiimide/HOBt, mixed anydride, active ester, BOP). The originally reported slower couplings[25,35] or the side reaction described by Stewart,[17] resulting in lactam formation for 5-aminovaleric acid, has not been reported to be a problem using these coupling methods.

### 10.5.1.1 Arndt–Eistert Homologations

This method was first reported in 1976[25,36] and yields dipeptides of the type L-Xaaψ[$CH_2$–$CH_2$]Gly **9** (Scheme 3). Yields for the three successive homologations were around 60% each. This method has been replaced by more efficient ones.

**Scheme 3** Repetitive Arndt–Eistert Homologations[25,36]



**4-Phenyl-3-phthalimidobutanoic Acid (8):**[36]

**CAUTION:** *Diazomethane is highly toxic and irritating. All operations with diazomethane should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

A soln of Phth-Phe-OH (**6**; 1.60 g, 5.42 mmol) in dry EtOAc (15 mL) and NMM (0.60 mL, 5.42 mmol) was cooled to −15 °C and iBuOCOCl (0.71 mL, 5.42 mmol) was added dropwise. The mixture was stirred for 8 min at −15 °C and the suspension was filtered. To the filtrate was added a precooled soln of $CH_2N_2$ (12 mmol) in dry $Et_2O$ (70 mL). The yellow soln was kept at 4 °C for 5 h and then evaporated, giving the diazo ketone **7** as a yellow solid; IR ($CHCl_3$) $\tilde{\nu}_{max}$: 2115, 1780, 1720 $cm^{-1}$. The diazo ketone was dissolved in dry dioxane (34 mL) and dry Bzl-OH (5.85 g, 54.2 mmol) was added, followed by $Ag_2O$ (0.25 g, 1.08 mmol). The suspension was heated for 0.5 h in an oil bath at 70 °C, cooled to rt, filtered, and the solvent removed to give the crude benzyl ester. The latter was dissolved in MeOH (35 mL) and hy-

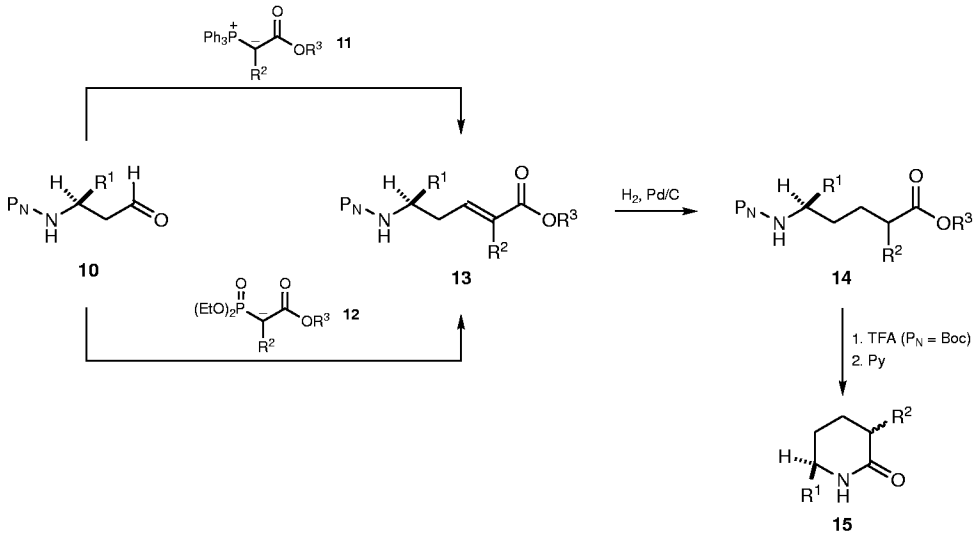
*for references see p 369*

drogenated at atmospheric pressure over 10% Pd/C (0.20 g) at rt for 18 h. Extraction with 1 M $Na_2CO_3$ (3×25 mL) followed by acidification yielded **8** as a white solid which was recrystallized (aq MeOH); yield: 1.0 g (60%); mp 123–124.5 °C.

### 10.5.1.2 Wittig Alkenation of β-Amino Aldehydes

The required β-amino aldehydes **10** (Scheme 4) are obtained from the corresponding β-amino acids, which are converted into the *O,N*-dimethylhydroxamates and reduced with $LiAlH_4$, according to the method of Fehrentz–Castro.[37] The aldehyde is treated with the phosphorus ylide without prior purification. Using the α-unsubstituted ylide ($R^2$ = H) the *E*-isomer is formed.[27] For other Wittig or Wittig–Horner reagents, where $R^2 \neq H$, the *E*/*Z* proportion depends on both the $R^1$ and $R^2$ substituents.[38,39] Catalytic hydrogenation using Pd/C and hydrogen either at atmospheric pressure (overnight)[39] or at 5 atm (1 h)[27] then yields the ψ[$CH_2$–$CH_2$] dipeptide. Because of lack of stereocontrol in the last step, a mixture of epimeric dipeptides is formed. The configuration at position 2 can be determined after cyclization to the lactam **15**, on the basis of specific NOE effects in the NMR spectra.[38,40] These lactams are also easier to separate by silica gel chromatography,[15,38,40] after which acid hydrolysis converts them back into the ψ dipeptide epimers.

**Scheme 4** Wittig Alkenation of β-Amino Aldehydes




$P_N$ = Boc, Z, Fmoc
$R^1$ = H, Me, Bzl, iBu, Bu, $CH_2CO_2tBu$
$R^2$ = H, Me, Bzl, iBu, $CH_2CO_2tBu$
$R^3$ = Me, Et, Bzl

**Ethyl (*Z*)-2-Benzyl-5-[(*tert*-butoxycarbonyl)amino]pent-2-enoate** (13, $P_N$ = Boc; $R^1$ = H; $R^2$ = Bzl; $R^3$ = Et):[39]

To a cold (0 °C) soln of ethyl 2-(diethyloxyphosphoryl)-3-phenylpropanoate (**12**, $R^2$ = Bzl; $R^3$ = Et; 5.26 g, 16.74 mmol) in dry DME (50 mL), NaH (60% in mineral oil; 670 mg, 16.74 mmol) was added at once and the mixture was stirred at rt until absence of gas evolution (ca. 1 h). Boc-β-alaninal (**10**, $P_N$ = Boc; $R^1$ = H; 2.9 g, 16.74 mmol) was then added and the mixture stirred for 15 min at rt. The solvent was concentrated under reduced pressure and the residue dissolved in EtOAc. The soln was then washed with sat. aq $NaHCO_3$, brine, aq 1 M $KHSO_4$, and brine, dried ($Na_2SO_4$), and concentrated in vacuo to give (*Z*)-**13** as an oil, which was purified by chromatography (silica gel, EtOAc/hexane 3:7); yield: 4.2 g (75%); $R_f$ 0.75 (EtOAc/hexane 3:7).

**Ethyl 2-Benzyl-5-[(*tert*-butoxycarbonyl)amino]pentanoate (Boc-Gly$\psi$[$CH_2$–$CH_2$]-D,L-Phe-OEt) (14, $P_N$ = Boc; $R^1$ = H; $R^2$ = Bzl; $R^3$ = Et):**[39]

Compound (*Z*)-**13** ($P_N$ = Boc; $R^1$ = H; $R^2$ = Bzl; $R^3$ = Et; 4.1 g, 12.3 mmol) was hydrogenated overnight in 95% EtOH at rt and atmospheric pressure in the presence of 10% Pd/C catalyst. The catalyst was removed by filtration, and the residue concentrated under reduced pressure to afford the title compound as a colorless oil; yield: 4.1 g (97%); $R_f$ 0.74 (EtOAc/hexane 3:7).

**(2*E*,5*S*)-5-[(*N*-*tert*-Butoxycarbonyl)amino]non-2-enoic Acid (13, $P_N$ = Boc; $R^1$ = Bu; $R^2$ = $R^3$ = H):**[27]

To a cold (0 °C) soln of Boc-β-homo-Nle-N(Me)OMe (5.08 g, 17.62 mmol) in anhyd $Et_2O$ (100 mL), $LiAlH_4$ (2 g, 54 mmol) was added under vigorous stirring, in portions, over a period of 15 min. After an additional 15 min, EtOAc (300 mL) was carefully added, followed by aq 1 M $KHSO_4$ (500 mL), and the mixture was stirred at rt for 30 min. The organic layer was collected, washed with brine, dried ($MgSO_4$), and concentrated in vacuo to give the corresponding aldehyde **10** ($P_N$ = Boc; $R^1$ = Bu), which was quickly used in the next step without any further purification. To a soln of the aldehyde in benzene (80 mL), methyl (triphenylphosphoranylidene)acetate (**11**, $R^2$ = H; $R^3$ = Me; 5.89 g, 17.62 mmol) was added. After 1 h stirring at rt, the solvent was removed in vacuo and the residue applied to a column (silica gel, EtOAc/hexane 2:8) to afford the corresponding methyl ester as a clean oil (4.47 g, 15.66 mmol). The ester was dissolved in MeOH and treated with aq 2 M NaOH (25 mL, 17 mmol). After 2 h at 30 °C, **13** ($P_N$ = Boc; $R^1$ = Bu; $R^2$ = $R^3$ = H) was precipitated upon addition of aq 1 M $KHSO_4$ (300 mL), washed with $H_2O$, and dried in vacuo; yield: 4.04 g (83%); mp 81–83 °C; $R_f$ 0.86 ($CHCl_3$/MeOH/AcOH 85:10:5); $[\alpha]_D^{20}$ –23.5 (*c* 1, DMF).

**5-[(*tert*-Butoxycarbonyl)amino]nonanoic Acid (Boc-Nle$\psi$[$CH_2$–$CH_2$]Gly-OH) (14, $P_N$ = Boc; $R^1$ = Bu; $R^2$ = $R^3$ = H):**[27]

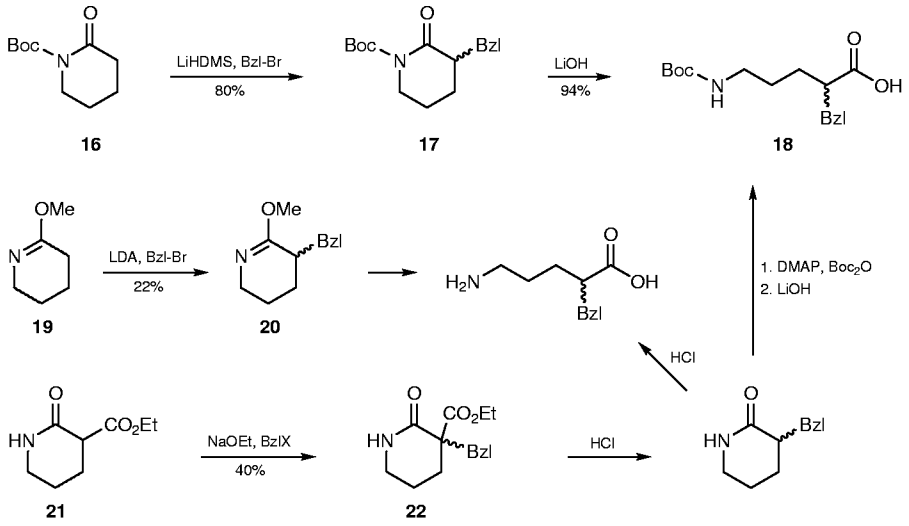
Compound (*E*)-**13** described above (3.73 g, 13.7 mmol) was hydrogenated for 1 h in EtOAc (200 mL) in the presence of 10% Pd/C catalyst under 5 atm and at rt. The catalyst was filtered off, the solvent concentrated under reduced pressure, and the residue crystallized upon trituration with $Et_2O$ to give **14** ($P_N$ = Boc; $R^1$ = Bu; $R^2$ = $R^3$ = H); yield: 2.74 g (73%); mp 72–74 °C; $R_f$ 0.44 (EtOAc/hexane/AcOH 20:80:1); $[\alpha]_D^{20}$ 5.4 (*c* 1, DMF).

## 10.5.1.3 Piperidin-2-one Enolate Alkylation

This method gives rapid access to racemic dipeptides of the type Gly$\psi$[$CH_2$–$CH_2$]-D,L-Xaa (**18**, Scheme 5), using a piperidin-2-one derivative as the starting material and introducing the $R^2$ substitutent.[29] Excellent yields (80%) are obtained for reaction with benzyl bromide, despite the formation of some dialkylated compounds when starting from **16**. A similar alkylation of the piperidin-2-one lactam ether **19** gave only 22% yield of **20**. Direct introduction of the isobutyl substituent on the ethyl 2-oxopiperidine-3-carboxylate (**21**) gave a low yield (40%).[39] The use of 3-chloro-2-methylpropene, followed by a hydrogenation step, did not significantly improve the overall yield. The 3-ethoxycarbonyl lactam **22** can be converted into the free $\psi$[$CH_2$–$CH_2$] dipeptide by acid hydrolysis, or into its Boc-protected form through different pathways. In terms of yield and number of steps, the method starting from the Boc-protected piperidin-2-one **16** is preferable.

*for references see p 369*

**Scheme 5** Piperidin-2-one Enolate Alkylation



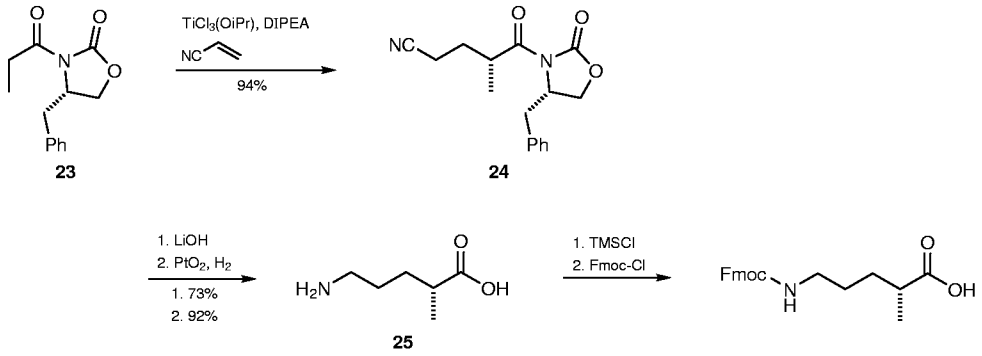

**3-Benzyl-1-(*tert*-butoxycarbonyl)piperidin-2-one (17):**[29]
To a stirred soln of 1 M LiHMDS in pentane (100 mL, 0.1 mol) in THF (500 mL) at −78 °C, 1-Boc-piperidin-2-one (**16**; 18.8 g, 0.095 mol) in THF (50 mL) was added. After 15 min, Bzl-Br (11.2 mL, 0.095 mol) was added rapidly. Stirring was continued for 2 h, and then the temperature was allowed to rise to 0 °C over 2 h. Excess $NH_4Cl$ was added to the mixture, and the layers were separated. The organic phase was washed with half-sat. NaCl, dried, and concentrated. The residue was flash chromatographed [Merck silica gel 60 (230–400 mesh), 15% EtOAc/hexane] to give, in the order of elution, a dialkylated product [$R_f$ 0.51(EtOAc/hexane 3:17)], followed by **17** as a thick liquid; yield: 22 g (80%); $R_f$ 0.39 (EtOAc/hexane 3:17); HRMS (CI) *m/z*: 290.2 $[M+H]^+$, calcd. for $C_{17}H_{23}NO_3$ 290.2.

**2-Benzyl-5-[(*tert*-butoxycarbonyl)amino]pentanoic Acid (Boc-Glyψ[$CH_2$–$CH_2$]D,L-Phe-OH) (18):**[29]
To a stirred soln of **17** (8 g, 27.6 mmol) in THF (125 mL) at rt a 0.5 M soln of LiOH (80 mL) was added. Stirring was continued for 16 h, and the solvent was then removed in vacuo. The residue was taken up in $H_2O$, acidified (20% AcOH), and extracted (EtOAc). The organic phase was dried and concentrated in vacuo to give **18** as a thick gum; yield: 8 g (94%); $R_f$ 0.55 (EtOAc/hexane 1:1); HRMS (CI) *m/z*: 308.2 $[M+H]^+$, calcd. for $C_{17}H_{25}NO_4$ 308.2.

## 10.5.1.4 Chiral Enolate Addition to Acrylonitrile

This method (Scheme 6) gives high yields of dipeptides of the type Glyψ[$CH_2$–$CH_2$]Xaa.[31] The configuration can be chosen using the appropriate configuration of the chiral auxiliary. It can therefore be considered the method of choice for this type of dimethylene dipeptide.

**Scheme 6** Chiral Enolate Addition to Acrylonitrile[31]




**(*R*)-5-Amino-2-methylpentanoic Acid (25):**[31,41]

To a soln of (*S*)-4-benzyl-3-propanoyloxazolidin-2-one (**23**; 1 g, 4.29 mmol) in dry $CH_2Cl_2$ (14 mL) was added DIPEA (0.78 mL) at 0 °C with stirring followed by $TiCl_3$ (OiPr) (0.95 g, 4.48 mmol) (prepared from $TiCl_4$ and iPrOH in $CH_2Cl_2$, and sublimation at 90 °C/0.5 Torr). To the resulting dark red soln, acrylonitrile (1.5 equiv, 0.42 mL) was added and the reaction was stirred at 0 °C for 16 h. The reaction was then quenched with $H_2O$ and extracted ($CH_2Cl_2$). The organic extracts were combined and washed sequentially with $H_2O$ and brine, dried ($Na_2SO_4$), and concentrated. The residue was purified by flash chromatography (EtOAc/hexane 1:1) to afford (4*S*)-4-benzyl-3-[(2*R*)-(4-cyano-2-methylbutanoyl)]oxazolidin-2-one (**24**); yield: 1.153 g (94%; dr 97:3).

Compound **24** (0.5 g, 1.75 mmol) was dissolved in THF/$H_2O$ (3:1, 30 mL). The soln was cooled in ice and LiOH (88 mg, 3.7 mmol) was added. After 1 h, further LiOH (36 mg, 1.5 mmol) was added and the reaction was stirred for a further 20 min. The THF was removed in vacuo and the resultant aqueous residue washed with $CH_2Cl_2$. The aqueous layer was adjusted to pH 2 with 2 M HCl and then extracted ($CH_2Cl_2$). The extracts were combined, dried ($Na_2SO_4$), and concentrated to afford (*R*)-4-cyano-2-methylbutanoic acid; yield: 0.163 mg (73%).

The butanoic acid (163 mg, 1.28 mmol) was then dissolved in MeOH (5 mL) and the soln added to a suspension of $PtO_2$ in $H_2O$ (2 mL). The resulting mixture was stirred under an atmosphere of $H_2$ for 16 h. The catalyst was removed by filtration and the filtrate concentrated in vacuo to afford **25**; yield: 0.155 mg (92%).

## 10.5.1.5 Reduction of ψ[CH=CH] Dipeptides

Catalytic hydrogenation of ψ[CH=CH] dipeptides using hydrogen[21,28,30,42,43] or ammonium formate[44] and a Pd or Pt catalyst, prior to incorporation into the peptide sequence, or in the final peptide, proceeds with high yields. Since a variety of methods is available for the stereoselective preparation of ψ[CH=CH] dipeptides having $R^1$ and $R^2 \neq H$, this method gives access to the corresponding homochiral ψ[$CH_2$–$CH_2$] analogues. Complete hydrogenation of a ψ[C≡C]-containing peptide has also been reported.[10]

**5-[((2*R*)-2-{[2-Amino-3-(4-hydroxyphenyl)propanoyl]amino}propanoyl)amino]-6-phenylhexanamide (H-Tyr-D-Ala-Pheψ[$CH_2$–$CH_2$]Gly-$NH_2$):**[21]

To a soln of H-Tyr-D-Ala-Pheψ[CH=CH]Gly-$NH_2$ (20 mg, 0.046 mmol) in iPrOH (12 mL), 20% $Pd(OH)_2$/C (10 mg) was added and $H_2$ was bubbled through the soln for 5 h. The catalyst was filtered off, washed with iPrOH, and the filtrate was concentrated in vacuo. The peptide was purified by RP-HPLC with a MeCN/0.1% aq TFA eluant system; yield: 13 mg (65%); $R_f$ 0.37 (EtOAc/BuOH/AcOH/$H_2O$ 1:1:1:1); FAB-MS *m/z*: 441 $[M+H]^+$, calcd. for $C_{24}H_{32}N_4O_4$ 440.

*for references see p 369*

## 10.5.2 ψ[CX=CY] Peptides

The *E*-alkene isosteric replacement of a peptide bond preserves the overall geometry almost perfectly (Table 1). Theoretical conformational analyses of this isostere have confirmed this property, but have also shown that the Ramachandran φ/ψ map has an enlarged conformational space available.[5,6] Further molecular mechanics calculations for Tyrψ[*E*, CH=CH]Gly indicated conformational properties which are close to those of the parent peptide.[19] An X-ray structure of Boc-Leuψ[*E*, CH=CH]Leu-Sta-Ala-OMe showed a good similarity of this modified pentapeptide with the structure observed for H-Leu-Leu-Val-Tyr-OMe.[45] An NMR study of dermorphin and its $Phe^3$ψ[*E*, CH=CH]$Gly^4$ analogue indicated very similar overall conformations for both peptides, including the characteristic high-field position of the $Ala^2$ methyl signal. This high-field position is ascribed to a shielding effect by the aromatic rings of $Tyr^1$ and $Phe^3$ in a β-turn conformation. In the case of the $Phe^3$ψ[*E*, CH=CH]$Gly^4$ analogue, this β-turn cannot be stabilized by an intramolecular N–H···O=C hydrogen bond with the $Gly^4$NH, since the latter is absent. Nevertheless, biological activity is maintained. In the cyclic somatostatin analogue *c*(-Pheψ[*E*, CH=CH]Gly-Phe-D-Tyr-Lys-Thr-) the β-turn structure around the central residue D-Trp-Lys, which is seen in the parent peptide, remains present, as indicated by characteristic NOEs and by the high-field position of the Lys-γ($CH_2$) signals.[46,47] Overall, these data present evidence that the introduction of an *E*-alkene isostere does not induce any major change in peptide conformation.

The replacement of a peptide bond in a biologically active peptide is expected to stabilize this compound against enzymatic degradation. Indeed, examples of prolonged action have been reported for bradykinin,[48] oxytocin,[49] and LHRH analogues.[3] This type of replacement also provides a means to investigate the importance of the specific amide bond for receptor interaction or for stabilizing a particular conformation by formation of intramolecular hydrogen bonds. Thus, the immunogenic peptide HEL(52–61), containing a $Tyr^{53}$ψ[*E*, CH=CH]$Gly^{54}$ replacement, is presented by MHC-II molecules and recognized by specific T-cell clones, whereas its ψ[$CH_2$NH] analogue is not.[50] Similarly, the replacement of the $Tyr^1$–$Gly^2$ bond in opioid peptides is tolerated, whereas that of the $Gly^2$–$Gly^3$ or D-$Ala^2$–$Gly^3$ bond is not.[51,52]

Other selected examples are reported in Table 3 (see also Scheme 7), and include applications as enzyme inhibitors. For renin, the *E*-isostere showed a good potency, whereas the *Z*-isostere was inactive.[53] In several cases a dramatic loss in potency is observed, but in others potency is maintained or even increased.[54] It is interesting to note that the introduction of the alkene isostere into the bombesin (6–14) sequence resulted in an antagonist.[55] In neurotensin analogues, a Pro-Tyr isostere showed good in vivo potency, despite a drop in receptor affinity compared to the parent compound. This may reflect an increased stability.[56] The importance of the rigidity of the amide bond can be evaluated by comparing biological data of the alkene isostere with its corresponding ψ[$CH_2CH_2$] analogue (see Section 10.5.1). The alkene function prohibits a *cis/trans* isomerization which occurs mainly in peptides containing a tertiary amide function (e.g., Xaa-Pro). It could therefore be used to confirm that a particular *cis* or *trans* conformation is present during receptor interaction, such as in cyclic somatostatin analogues.[46,47]

**Table 3** ψ[CX=CY] Dipeptides

| ψ[*E*, CH=CH] Dipeptide | Section | Ref |
|---|---|---|
| Ala-Ala | 10.5.2.1.2.4 | [77] |
| Ala-Ala | 10.5.2.1.3.2 | [102,103] |
| Ala-Ala | 10.5.2.1.3.2 | [105] |
| Ala-Ala | 10.5.2.1.4.1 | [114] |
| Ala-Gly | 10.5.2.1.2.1 | [137] |
| Ala-Gly | 10.5.2.1.2.2 | [138] |
| Ala-Gly | 10.5.2.1.2.3 | [73] |
| Ala-Leu | 10.5.2.1.3.2 | [105] |
| Ala-Nle | 10.5.2.1.3.2 | [105] |
| Ala-Phe | 10.5.2.1.2.4 | [82] |
| Ala-D-Ala | 10.5.2.1.3.1 | [66,88] |
| Ala-D-Ala | 10.5.2.1.3.2 | [102,103] |
| Ala-D-Nle | 10.5.2.1.3.2 | [102] |
| Ala-D-Val | 10.5.2.1.3.2 | [102] |
| D-Ala-Ala | 10.5.2.1.2.4 | [77] |
| D-Ala-Ala | 10.5.2.1.3.2 | [104] |
| D-Ala-Ala | nd[a] | [115] |
| D-Ala-D,L-Ala | 10.5.2.1.2.1 | [64] |
| D-Ala-D-Ala | nd | [115] |
| D-Ala-D-Leu | nd | [115] |
| D-Ala-Glu | 10.5.2.1.3.2 | [104] |
| D-Ala-Gly | 10.5.2.1.2.1 | [28] |
| D-Ala-Gly | 10.5.2.1.2.4 | [77] |
| D-Ala-Nle | 10.5.2.1.3.2 | [104] |
| Asn-Val | 10.5.2.1.2.2 | [54,72] |
| Asp-D,L-Phe | nd | [44,69,85,86] |
| Gly-Phe | 10.5.2.1.1 | [48] |
| Gly-Gly | 10.5.2.1.1 | [48,59–61] |
| Gly-D,L-Ala | 10.5.2.1.4.1 | [112,113] |
| Gly-D,L-Phe | 10.5.2.1.4.3 | [29] |
| Gly-D,L-Trp | nd | [44,69,85,86] |
| Ile-Ala | 10.5.2.1.3.1 | [92] |
| Ile-Ala | 10.5.2.1.3.1 | [89] |
| Ile-Ala | 10.5.2.1.3.1 | [30] |
| Ile-Ile | 10.5.2.1.3.1 | [30] |
| Ile-Leu | 10.5.2.1.3.1 | [89] |
| Ile-Nle | 10.5.2.1.3.1 | [30] |
| Ile-Nva | 10.5.2.1.3.1 | [30] |
| Ile-Phe | 10.5.2.1.3.1 | [92] |
| Ile-Phe | 10.5.2.1.3.1 | [30] |
| Ile-Val | 10.5.2.1.3.1 | [89] |
| Ile-Val | 10.5.2.1.3.1 | [89] |
| Ile-Val | 10.5.2.1.3.1 | [30] |
| Ile-Leu | 10.5.2.1.3.1 | [89] |
| Ile-D-Ala | 10.5.2.1.3.1 | [89] |
| Leu-Ala | 10.5.2.1.2.4 | [84] |
| Leu-Asp | 10.5.2.1.2.1 | [68,69] |
| Leu-Asp | 10.5.2.1.2.1 | [68] |
| Leu-Gly | 10.5.2.1.2.1 | [139,140] |
| Leu-Gly | 10.5.2.1.2.1 | [68,69] |
| Leu-Gly | 10.5.2.1.2.1 | [137] |
| Leu-Leu | 10.5.2.1.2.2 | [70,71] |
| Leu-Leu | 10.5.2.1.3.1 | [92] |
| Leu-Leu | 10.5.2.1.3.1 | [89] |
| Leu-Leu | 10.5.2.1.3.2 | [108] |
| Leu-Leu | 10.5.2.1.4.1 | [42] |
| Leu-Leu | 10.5.2.1.4.3 | [117] |
| Leu-Phe | 10.5.2.1.2.2 | [133] |
| Leu-Phe | 10.5.2.1.2.2 | [70,71] |
| Leu-Val | 10.5.2.1.2.4 | [84] |

*for references see p 369*

**Table 3** (cont.)

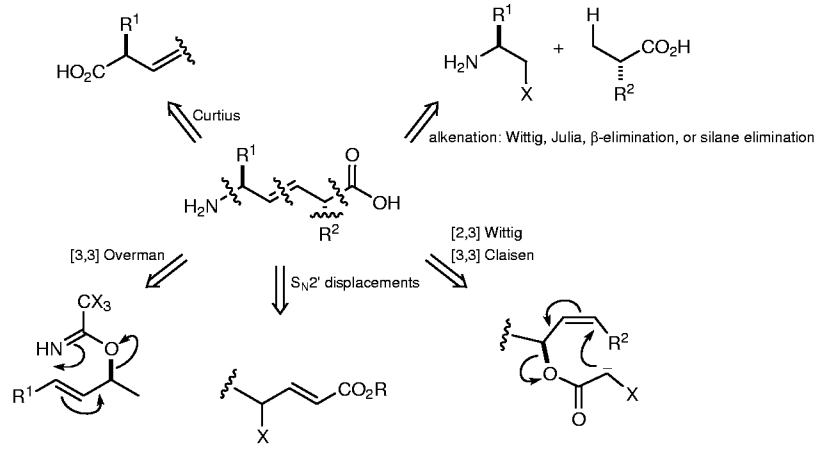
| ψ[*E*, CH=CH] Dipeptide | Section | Ref |
|---|---|---|
| Leu-Val | 10.5.2.1.3.1 | [89] |
| Leu-D,L-Asp | nd | [44,69,85,86] |
| Leu D-Leu | 10.5.2.1.4.1 | [42] |
| 2-Nal-D,L-Nle | nd | [69,85] |
| Nle-Ala | 10.5.2.1.4.3 | [117] |
| Nle-Val | 10.5.2.1.4.3 | [117] |
| Phe-Phe(αBzl) | 10.5.2.1.2.2 | [65,141] |
| Phe-Ala | 10.5.2.1.3.1 | [92] |
| Phe-Ala | 10.5.2.1.3.1 | [66,88] |
| Phe-Asp | 10.5.2.1.4.2 | [116] |
| Phe-Gly | 10.5.2.1.2.1 | [21] |
| Phe-Gly | 10.5.2.1.2.1 | [48] |
| Phe-Gly | 10.5.2.1.2.1 | [142] |
| Phe-Gly | 10.5.2.1.2.1 | [53] |
| Phe-Gly | 10.5.2.1.2.3 | [73] |
| Phe-Gly | 10.5.2.1.3.1 | [90] |
| Phe-Gly | 10.5.2.1.3.1 | [1] |
| Phe-Gly | 10.5.2.1.4.1 | [113] |
| Phe-Leu | 10.5.2.1.3.1 | [66,88] |
| Phe-Leu | 10.5.2.1.3.1 | [1] |
| Phe-Nle | 10.5.2.1.3.2 | [111] |
| Phe-Phe | 10.5.2.1.2.2 | [133] |
| Phe-Phe | 10.5.2.1.2.2 | [70,71] |
| Phe-Phe | 10.5.2.1.2.2 | [65,141] |
| Phe-Phe | 10.5.2.1.1.1 | [60] |
| Phe-Phe | 10.5.2.1.3.1 | [1] |
| Phe-Phg | 10.5.2.1.3.1 | [1] |
| Phe-Val | 10.5.2.1.3.2 | [109] |
| Phe-D-Ala | 10.5.2.1.3.1 | [66,88] |
| Phe-D-Asp | 10.5.2.1.4.2 | [116] |
| Phe-D-Leu | 10.5.2.1.3.1 | [66,88] |
| Phe-D-Leu | 10.5.2.1.3.1 | [1] |
| Phe-D-Nle | 10.5.2.1.3.2 | [111] |
| Phe-D-Phe | 10.5.2.1.3.1 | [1] |
| Phe-D-Val | 10.5.2.1.3.2 | [109] |
| D,L-Phe-Gly | 10.5.2.1.1 | [60] |
| Phg-Ala | 10.5.2.1.3.2 | [110] |
| Phg-D-Ala | 10.5.2.1.3.2 | [103] |
| Pro-Ala | 10.5.2.1.3.1 | [93] |
| Pro-Gly | 10.5.2.1.2.1 | [48] |
| Pro-Gly | 10.5.2.1.2.1 | [64] |
| Pro-Leu | 10.5.2.1.3.1 | [93] |
| Pro-Phe | 10.5.2.1.3.1 | [93] |
| Pro-Tyr | 10.5.2.1.3.1 | [56] |
| Pro-Val | 10.5.2.1.3.1 | [93] |
| Pro-D,L-Gly | 10.5.2.1.2.1 | [48] |
| Pro-D,L-Leu | 10.5.2.1.2.1 | [64] |
| Pro-D,L-Phe | 10.5.2.1.2.1 | [64] |
| Ser-Ala | 10.5.2.1.3.1 | [88,92] |
| Ser-Ala | 10.5.2.1.3.1 | [66,88] |
| Ser-Asn | 10.5.2.1.2.2 | [54,72] |
| Ser-Gly | 10.5.2.1.3.1 | [90] |
| Ser-Ala | 10.5.2.1.3.1 | [93] |
| Ser-Leu | 10.5.2.1.3.1 | [93] |
| Ser-Nva | 10.5.2.1.3.1 | [93] |
| Ser-Val | 10.5.2.1.3.1 | [93] |
| Ser-Leu | 10.5.2.1.3.1 | [66,88] |
| Ser-Leu | 10.5.2.1.3.1 | [89] |
| Ser-Phe | 10.5.2.1.3.1 | [66,88] |
| Ser-Val | 10.5.2.1.3.1 | [89] |

**Table 3** (cont.)

| ψ[*E*, CH=CH] Dipeptide | Section | Ref |
|---|---|---|
| Ser-D-Ala | 10.5.2.1.3.1 | [66,88] |
| Ser-D-Ala | 10.5.2.1.3.1 | [93] |
| Ser-D-Leu | 10.5.2.1.3.1 | [66,88] |
| Ser-D-Leu | 10.5.2.1.3.1 | [93] |
| Ser-D-Nva | 10.5.2.1.3.1 | [93] |
| Ser-D-Phe | 10.5.2.1.3.1 | [66,88] |
| Ser-D-Val | 10.5.2.1.3.1 | [93] |
| Thr-Ala | 10.5.2.1.3.1 | [66,88] |
| Thr-Gly | 10.5.2.1.3.1 | [90] |
| Thr-Nle | 10.5.2.1.3.1 | [93] |
| Thr-Val | 10.5.2.1.3.1 | [93] |
| Tyr-Ala | 10.5.2.1.2.2 | [70,71] |
| Tyr-Gly | 10.5.2.1.2.1 | [4,143] |
| Tyr-Gly | 10.5.2.1.2.1 | [50] |
| Tyr-Gly | 10.5.2.1.1 | [60] |
| Tyr-Gly | 10.5.2.1.3.3 | [97] |
| Tyr-Gly | nd | [71] |
| Tyr-Val | 10.5.2.1.3.1 | [96] |
| Thr-D-Ala | 10.5.2.1.3.1 | [66,88] |
| Thr-D-Nle | 10.5.2.1.3.1 | [93] |
| Thr-D-Val | 10.5.2.1.3.1 | [93] |
| Val-Leu | 10.5.2.1.3.1 | [89] |
| Val-Leu | 10.5.2.1.3.2 | [109] |
| Val-Nle | 10.5.2.1.3.2 | [110] |
| Val-Phe | 10.5.2.1.3.1 | [30] |
| Val-Phe | 10.5.2.1.3.2 | [111] |
| Val-Val | 10.5.2.1.3.1 | [98] |
| Val-Val | 10.5.2.1.3.1 | [89] |
| Val-Val | 10.5.2.1.3.1 | [30] |
| Val-D-Ala | 10.5.2.1.3.1 | [89] |
| Val-D-Phe | 10.5.2.1.3.2 | [111] |
| D-Val-Ala | 10.5.2.1.2.4 | [77] |

[a] nd = method not described in the present work.

**Scheme 7** Retrosynthetic Pathways towards ψ[CH=CH] Dipeptides




Alkene isosteres have also been used in precursors of epoxy peptides as potential irreversible inhibitors of renin[57] or HIV protease.[58]

*for references see p 369*

The synthesis of ψ[CH=CH] peptides containing two chiral centers and an *E* or *Z* double bond geometry presents an important synthetic challenge. Whereas initial efforts focused on pseudodipeptides containing one glycine residue, efficient methods are now available to produce stereodefined dipeptide isosteres, although the number of synthetic steps required is still high. The incorporation of these isosteres into peptides has been performed using many different coupling reagents, both in solution and in solid-phase synthesis. No difficulties have been observed.

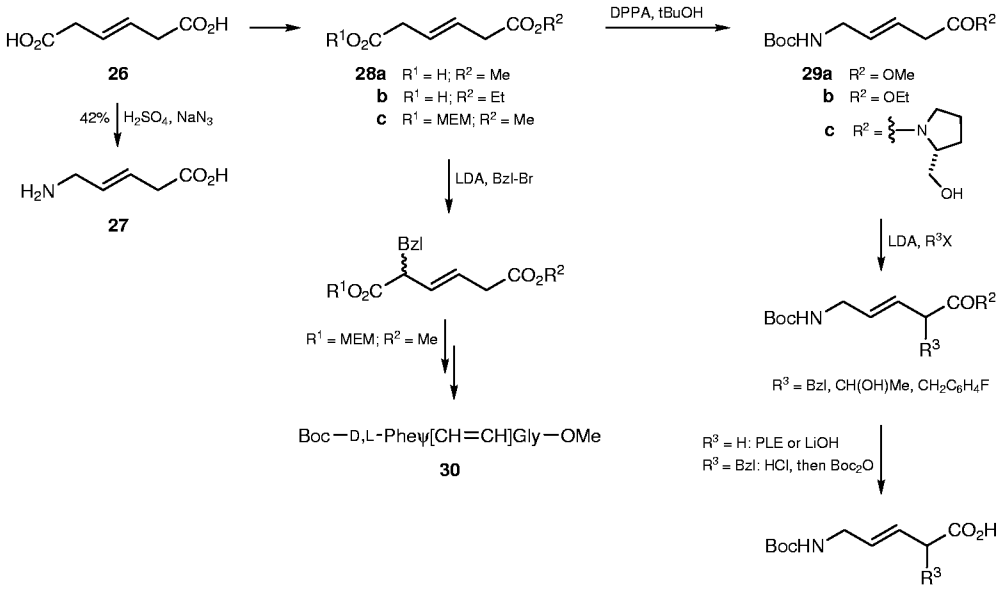
## 10.5.2.1 ψ[*E*, CH=CH] Isosteres

### 10.5.2.1.1 Curtius Rearrangement

Commercial (*E*)-hex-3-enedioic acid (**26**) (Scheme 8) has been submitted directly to a Curtius rearrangement to yield 42% of (*E*)-5-aminopent-3-enoic acid (**27**), the Glyψ[*E*, CH=CH]Gly equivalent.[59] The *Z*-isomer was prepared similarly. Higher yields of Boc-protected compounds (62%[60] or 56–77%[48]) are obtained from the monoesters **28a**,**b** either using diphenylphosphoryl azide or via acid chloride formation and reaction with sodium azide (70–80%),[61] followed by the addition of tBuOH to the intermediate isocyanate. The original method for the preparation of the monomethyl ester is not reproducible. The monoethyl ester **28b** can be isolated from the equilibrium mixture of diacid and mono- and diesters,[48] but a high yield (93%) of **28a** is obtained by pig liver esterase catalyzed hydrolysis of the diester.[61] A regioselective alkylation α to the Mem ester **28c**, followed by selective deprotection and Curtius rearrangement, led to racemic Boc-Pheψ[*E*, CH=CH]Gly-OMe (**30**).[60] Alkylation α to the ester in **29** resulted in Boc-Glyψ[*E*, CH=CH](*R*,*S*)-Xaa type dipeptides. An asymmetric synthesis using (*S*)-prolinol as a chiral auxiliary gave Boc-Glyψ[*E*, CH=CH]Phe-OH in 51% yield but with only 40% de.[28]

Thus, the first method provides a rapid access to the Gly-Gly alkene isostere. For isosteres containing a side chain, this method is not competitive with more recent asymmetric synthesis methods.

**Scheme 8** Curtius Rearrangement



**(*E*)-5-Aminopent-3-enoic Acid (27):**[59]
To a suspension of (*E*)-hex-3-enoic acid (**26**; 1.4 g, 9.7 mmol) in $CHCl_3$ (40 mL) was added concd $H_2SO_4$ (4 mL). Small portions of $NaN_3$ (0.65 g, 10 mmol) were added over 30 min, while the mixture was stirred rapidly. After a further 4 h at 40 °C the $CHCl_3$ layer was decanted from the viscous residue, which was washed again with $CHCl_3$ (30 mL). The acid layer was diluted with $H_2O$ (150 mL), filtered, and added to a column of Dowex 50W ($H^+$) ion-exchange resin (50 mL). The column was washed with $H_2O$ to neutral pH and then the amino acid eluted with aq 1 M pyridine (300 mL). Removal of the solvent afforded an oil (600 mg) which slowly solidified. Recrystallization ($H_2O$/EtOH) gave **27** as white needles; yield: 437 mg (42%); $R_f$ 0.44 (BuOH/AcOH/$H_2O$ 4:1:1); MS *m/z* (relative intensity): 116 $[M+H]^+$, (100%), 99 (86%), 98 (11%), 85 (10%).

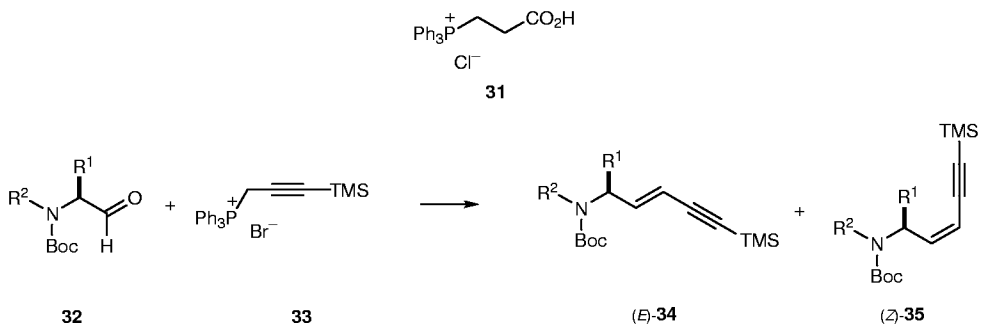
**Ethyl (*E*)-5-[(*tert*-Butoxycarbonyl)amino]pent-3-enoate (29b, $R^2$ = OEt):**[62]
A soln of (*E*)-hex-3-enedioic acid monoethyl ester (**28b**; 5.32 g, 0.31 mol) in dry toluene (28 mL) was cooled in an ice bath, and a soln of TEA (4.3 mL, 0.31 mol) in toluene (5 mL) was added, followed by a soln of DPPA (8.48 g, 0.31 mol) in toluene (5 mL). The mixture was stirred at 0 °C for 1.75 h, and was then heated to 75 °C. At that temperature, dry tBuOH (16 mL) in toluene (2 mL) was added and heating was continued at 90 °C for 18 h. After cooling, benzene (40 mL) was added and the soln was extracted successively with sat. $NaHCO_3$ (3×), 10% citric acid (3×), and $H_2O$. The organic phase was dried ($MgSO_4$), filtered, and the solvent was removed. The residual oil was distilled in a Kugelrohr apparatus at 100 °C/0.05 Torr to give **29b** (a small solid fraction distilling at 80 °C/0.05 Torr was discarded); yield: 4.23–5.04 g (56–71%); $R_f$ 0.53 ($CHCl_3$); MS *m/z* (relative intensity): 244 $[M+H]^+$, (12%), 188 (50%), 144 (78%), 127 (44%), 114 (32%), 57 (100%).

### 10.5.2.1.2 Alkenation Reactions

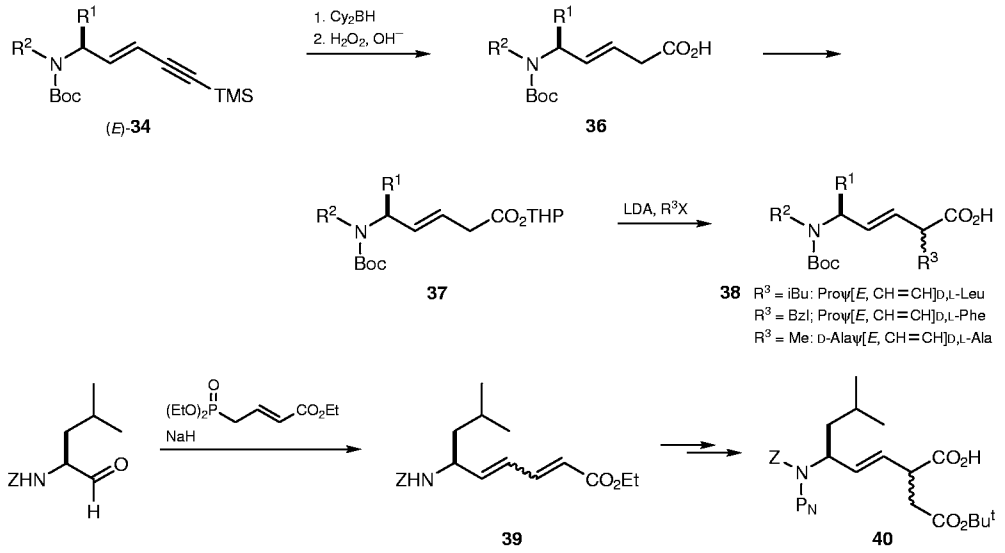
#### 10.5.2.1.2.1 Wittig Type

This type of disconnection is mainly used for the preparation of dipeptides of type Xaaψ[*E*, CH=CH]Gly. It allows control of the stereochemistry of the Xaa residue by starting from chiral α-amino aldehydes. For the construction of the *trans*-β,γ-unsaturated carboxylic acid moiety, the use of the triphenylphosphonium salt **31** (Scheme 9) derived from 3-chloropropanoic acid was not suitable.[4] Instead, the trimethylsilylprop-2-ynyl phosphonium salt **33** serves as a three-carbon unit, which can be converted into the β,γ-unsaturated acid by hydroboration and oxidation. The required Boc-protected α-amino aldehyde **32** can be prepared using virtually racemization-free procedures.[37] However, at the end of the reaction sequence, racemization has been detected, especially for Boc-Pheψ[*E*, CH=CH]Gly-OH, but not for the Ala and Pro analogues.[63] A mixture of *E*- and *Z*-enynes **34** and **35** is formed (8:2 to 9:1), which can be separated by column chromatography.[4,48,50,53,64,65]

**Scheme 9** Wittig Alkenation




*for references see p 369*




A side chain can be introduced at the position α to the carbonyl group, using either the methyl ester[64] or the tetrahydropyranyl ester.[48] Saponification of the methyl ester has been reported,[64] but concomitant shift of the unsaturation to the α,β-position has also been observed under these conditions.[66] This side reaction is avoided by the use of the acid-sensitive tetrahydropyranyl ester.[48] No asymmetric induction occurred during these alkylations, resulting in the diastereomeric dipeptide analogues **38**, which could not be separated by chromatographic means. A method for assigning the configuration of the new chiral center, after reduction and lactam formation, has been described for the Proψ[*E*, CH=CH]D,L-Phe isomer.[67] For the specific introduction of the Asp side chain, a Horner alkenation using ethyl (*E*)-4-(diethoxyphosphoryl)but-2-enoate yields the diene ester **39**, which after reduction to the alcohol was converted, through an orthoester Claisen rearrangement and further oxidation, in the desired Leuψ[*E*, CH=CH]D,L-Asp dipeptide **40**.[68,69] The method requires double nitrogen protection and the overall yield is low; better methods are available.

**(2*S*,3*E*)-4-[(1-*tert*-Butoxycarbonyl)pyrrolidin-2-yl]but-3-enoic Acid (Boc-Proψ[*E*, CH=CH]Gly-OH) [36, $R^1$,$R^2$ = $(CH_2)_3$]:**[64]

**(2*S*,3*E*)-4-[(1-*tert*-Butoxycarbonyl)pyrrolidin-2-yl]-1-(trimethylsilyl)but-3-en-1-yne [34, $R^1$,$R^2$ = $(CH_2)_3$]:** A soln of triphenyl[3-(trimethylsilyl)prop-2-ynyl]phosphonium bromide (**33**; 146 g, 0.32 mol) in THF (1.5 L) was stirred under $N_2$ at −60 °C, whilst 1.55 M BuLi in hexane (255 mL, 0.39 mol) was added over 20 min. The mixture was stirred for a further 30 min at −60 °C before the slow addition of a soln of freshly prepared Boc-Pro-aldehyde **32** (68.2 g, 0.34 mol) in THF (200 mL). The soln was allowed to warm to rt over 30 min and, after a further 1 h, brine (650 mL) was added and the mixture extracted with petroleum ether (3×650 mL). The combined organic extracts were dried and the solvent removed under reduced pressure. The residue was triturated with petroleum ether (100 mL) and the precipitated $Ph_3P$=O removed by filtration. The filtrate afforded a brown oil (79.3 g) which was first filtered through a short column (silica gel, EtOAc/petroleum ether 1:10), followed by chromatography (silica gel 60G, 2 kg, EtOAc/petroleum ether 7:93) using a Jobin-Yvon HPLC apparatus. The major fraction was the *E*-isomer **34**, isolated as a viscous oil; yield: 30.2 g (32%); $R_f$ 0.50 ($CHCl_3$);[48] $[\alpha]_D^{20}$ −122 (*c* 0.07, MeOH). As a minor, more polar, fraction, the corresponding *Z*-isomer **35** was isolated; yield: 3.3 g (3.5%). Separation of the *E*- and *Z*-isomers has also been described using chromatograpy with EtOAc/hexane (4:1).[48]

**(2*S*,3*E*)-4-[(1-*tert*-Butoxycarbonyl)pyrrolidin-2-yl]but-3-enoic Acid (Boc-Proψ[*E*, CH=CH]Gly-OH) [36, $R^1$,$R^2$ = $(CH_2)_3$]:**

A 1 M soln of $BH_3$·THF (210 mL, 210 mmol) in THF (220 mL) at 0 °C under $N_2$ was treated with cyclohexene (37.5 g, 457 mmol), added dropwise, the temperature being maintained below 5 °C. The

mixture was stirred for a further 1 h, when a precipitate of $Cy_2BH$ formed. To the suspension was then added, at 0 °C, a soln of the enyne **34** (20 g, 68 mmol) dissolved in THF (55 mL). The turbidity gradually dissipated over a period of 2 h. MeOH (75 mL) was added followed by aq 1 M KOH (220 mL), and then, with care, a soln of $H_2O_2$ (100 vol, 58 mL), the temperature of the mixture being kept below 20 °C throughout these operations. The mixture was then stirred at rt for 1.5 h, poured into $H_2O$ (900 mL), and extracted with $Et_2O$ (2×350 mL). The aqueous phase was acidified with phosphoric acid and extracted with $Et_2O$ (3×300 mL). The combined organic extracts were dried and concentrated under reduced pressure to yield the title acid as a waxy solid; yield: 13.1 g (75%); mp 68–72 °C; $[\alpha]_D^{20}$ −16.8 (*c* 1, MeOH).

**(2*S*,3*E*)-2-Benzyl-4-[(1-*tert*-butoxycarbonyl)pyrrolidin-2-yl]but-3-enoic Acid (Boc-Pro$\psi$[*E*, CH=CH]D,L-Phe-OH) [38, $R^1$,$R^2$ = $(CH_2)_3$; $R^2$ = Bzl]:**[48]

To a soln of Boc-Pro$\psi$[*E*, CH=CH]Gly-OH [**36**, $R^1$,$R^2$ = $(CH_2)_3$; 5 g, 19.6 mmol] in $CH_2Cl_2$ (150 mL), dihydropyran (3.29 g, 39.2 mmol) and TosOH (0.5 g) were added and the mixture was stirred for 5 h at rt. The solvent was then removed, the residue dissolved in $Et_2O$, extracted ($NaHCO_3$ soln), washed ($H_2O$), dried ($MgSO_4$), and concentrated to give **37** [$R^1$,$R^2$ = $(CH_2)_3$] as a colorless oil; yield: 6.6 g (100%).

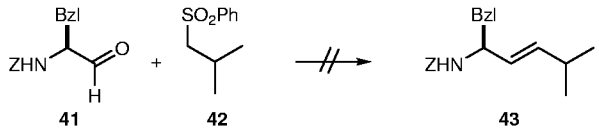
The crude tetrahydropyranyl ester **37** (4.13 g, 12.2 mmol) was dissolved in dry THF (few mL) and added dropwise to a soln of LDA in dry THF (30 mL) at −78 °C [prepared from $iPr_2NH$ (1.26 g) and 1.5 M BuLi in hexane (8.53 mL)]. The soln was stirred at this temperature for 3 h and then Bzl-Br (2.56 g, 15 mmol) was added and stirring was continued at −78 °C for 16 h. Then, AcOH (5 mL) was added, the soln warmed to rt, and the solvent removed. The residue was dissolved in $Et_2O$, extracted (citric acid soln), washed ($H_2O$), and concentrated. The residue was then dissolved in MeOH (100 mL) to which 0.1% aq HCl (100 mL) was added. This soln was stirred for 2 h at rt. The solvent was then removed and the aqueous phase extracted ($Et_2O$). After washing ($H_2O$) and drying ($MgSO_4$), the solvent was removed. The resulting brown oil was purified by flash chromatography (silica gel, $CHCl_3$/EtOAc gradient). A yield of 6.1 g (55%) of a mixture of diastereomers was obtained as a colorless oil. By addition of 1 equiv of (±)-α-methylbenzylamine to an $Et_2O$ soln of the dipeptide isostere, a white precipitate was obtained which could be recrystallized (EtOAc); mp 156–157 °C.

In the same way, a crystalline dicyclohexylamine salt can be obtained; mp 158.5–159.5 °C; $R_f$ 0.35 ($CHCl_3$/EtOAc 4:1+0.5% AcOH); FAB-MS *m/z* (relative intensity): 467 ($M^+$+1), 346 (1%), 290 (5%), 244 (13%), 122 (46%), 114 (24%), 105 (75%), 91 (47%), 57 (100%).

### 10.5.2.1.2.2 Julia Type

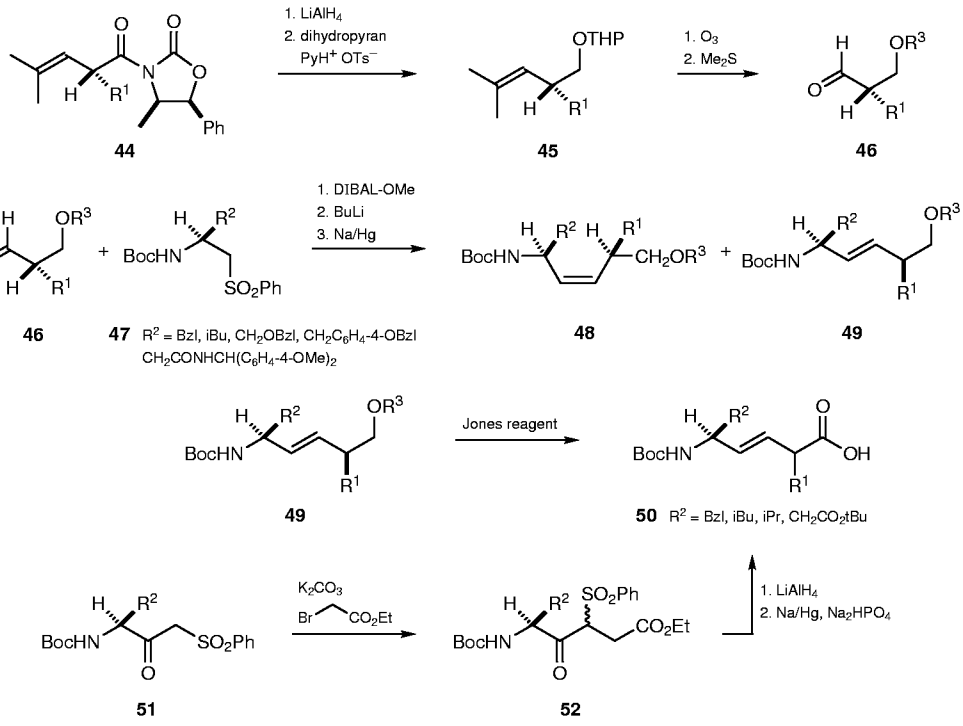
This alkenation method was developed for the preparation of pseudodipeptides which bear nonglycine residues at both termini. A chiral sulfone and a chiral aldehyde are coupled, allowing stereocontrol over both centers in a convergent strategy.[70] In principle, two combinations are possible. Coupling of an N-protected α-amino aldehyde **41** with a sulfone **42** (Scheme 10) resulted in only traces of the required compound **43**.[71] Therefore, the polarity was reversed: the N-protected β-amino sulfones **47** were readily prepared from $N^{\alpha}$-Boc-α-amino acids in a four-step procedure in 60–80% overall yields. Chiral α-hydroxymethyl aldehydes **46** are obtained by ozonolysis of a chiral homoallylic alcohol **45**, which was prepared using Evans chiral alkylation technology on the 3-(4-methylpent-3-enoyl)oxazolidinone **44**. The elimination reaction of the β-hydroxy sulfone is not completely stereoselective: both *Z*- and *E*-isomers **48** and **49** can be formed in up to a 1:1 proportion.[65]

**Scheme 10** Julia Alkenation




**41** **42** **43**

*for references see p 369*




The presence of an NH group is essential, since the Boc-β-amino sulfone derived from Pro could only be coupled with lower yields (~20%). This method is also suitable for the preparation of pseudodipeptides having functionalized side chains such as those of Ser and Asn.[72] The β-oxo sulfone **51**, which was obtained by reacting Boc-βAla-OMe with the dilithium anion of methyl phenyl sulfone, cleanly reacted with ethyl bromoacetate to give the β-oxo sulfone **52** (86%). The latter was reduced using $LiAlH_4$, which on reductive elimination produced the over-reduced Boc-Ala$\psi$($CH_2$–$CH_2$]Gly-OMe isostere (80%).

**(5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-6-phenylhex-3-en-1-ol** (49, **$R^1$ = $R^3$ = H; $R^2$ = Bzl**):[65]
A suspension of sulfone **47** ($R^2$ = Bzl; 6.7 g, 18 mmol) in THF (300 mL) was refluxed until a clear soln was obtained. The soln was cooled to −78 °C and 1.6 M BuLi in hexane (25 mL) was added dropwise. The soln was then stirred for 30 min at −78 °C. In a separate flask, a soln of **46** ($R^1$ = H; $R^3$ = TBDMS; 6.7 g, 35 mmol) in THF (5 mL) was treated with DIBAL-OMe [prepared by the addition of MeOH (1.7 mL, 42 mmol) and THF (5 mL) to 20% DIBAL-H in toluene (30 mL, 42 mmol) at −78 °C]. The soln of the aluminum complex was transferred via cannula to the soln of the sulfone dianion, and the mixture was stirred for 30 min at −78 °C. The reaction was quenched at −78 °C with sat. aq $NH_4Cl$ and extracted ($CH_2Cl_2$). The organic layer was dried ($MgSO_4$) and concentrated. MeOH (100 mL) was added, and undissolved **47** was filtered off. The MeOH soln was cooled to 0 °C and treated with $Na_2HPO_4$ (14 g, 37 mmol) and 6% Na/Hg (140 g). The mixture was stirred overnight at 0 °C, diluted ($H_2O$), and extracted ($CH_2Cl_2$). The organic phase was dried ($MgSO_4$) and concentrated. Flash chromatography (pentane/$Et_2O$ 9:1) afforded 5.8 g of an *E*/*Z* isomeric mixture of (*S*)-1-benzyl-*N*-(*tert*-butoxycarbonyl)-5-(*tert*-butyldimethylsiloxy)pent-2-enylamine (**48**) and **49** ($R^1$ = H, $R^2$ = Bzl, $R^3$ = TBDMS) as a colorless oil, which was used in the next step without further purification.
The crude product (5.8 g, 14 mmol) was dissolved in MeCN (100 mL) containing 2% HF (5 mL of a 40% aq HF soln). The mixture was stirred at rt until the deprotection was complete according to TLC. Then, $CH_2Cl_2$ and $H_2O$ were added. The organic phase was separated, dried ($MgSO_4$), and concentrated. Purification by flash chromatography (hexane/$CHCl_3$/MeOH 5:4:1) yielded 3.55 g (68% yield calculated from sulfone **47**) of a mixture of the *Z*- and *E*-isomers **48** and **49** ($R^1$ = $R^3$ = H, $R^2$ = Bzl) as a colorless oil which solidified upon standing at rt. According to analytical HPLC the *E*/*Z* ratio was 86:14. Double recrystallization ($CHCl_3$/hexane) afforded pure **49**. Isomer **48** was purified by preparative HPLC (1.5% EtOH/hexane, $t_R$ **49** 67 min, **48** 94 min). For isomer **49**: HPLC (LiChrosphere Si60, 50% EtOAc/hexane), 1.5 mL·min$^{-1}$, $t_R$ 6.97 min; mp 80–81 °C; $[\alpha]_D$ +12.7 (*c* 1, $CHCl_3$).

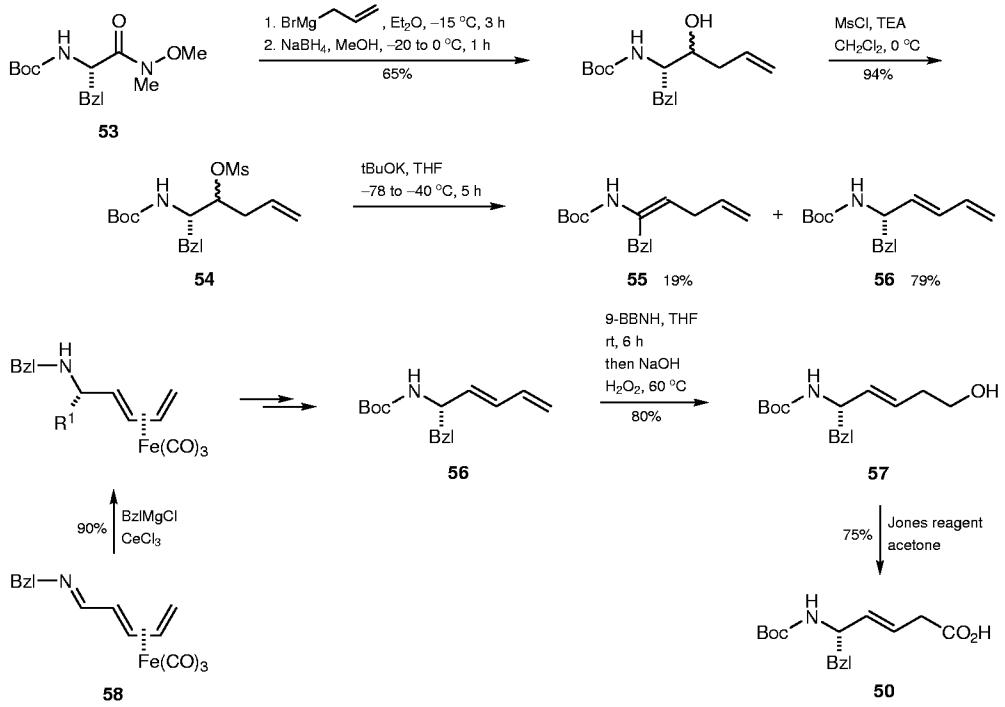
**(3*E*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-6-phenylhex-3-enoic Acid, Boc-Phe$\psi$[*E*, CH=CH]Gly-OH (50, $R^1$ = H; $R^2$ = Bzl):**[65]

A soln of **49** ($R^1$ = $R^3$ = H; $R^2$ = Bzl; 1.05g, 3.6mmol) in acetone (30mL) was treated with 0.67M Jones reagent (13mL, 9.0mmol) at 0°C and stirred for 30min. Workup by acid/base extractions afforded 1.0g (91%) of crude **50**, which was purified by recrystallization ($Et_2O$/hexane); $R_f$ 0.34 ($CH_2Cl_2$/MeOH 9:1); mp 91–93°C: $[\alpha]_D$ −11.8 (*c* 1, MeOH).

### 10.5.2.1.2.3 β-Elimination

Mesylate **54** (Scheme 11) was prepared by addition of allylmagnesium bromide to Boc-Phe dimethylhydroxamate **53**, followed by reduction of the resulting ketone,[73] a method which requires careful optimization.[74] β-Elimination of the mesyloxy moiety of **54** using tBuOK resulted in an isomeric mixture of dienes **55** and **56** (19:79) which was separated by flash chromatography. Diene **56** was hydroborated regioselectively to homoallyl alcohol **57**, which was further oxidized to Boc-Phe$\psi$[*E*, CH=CH]Gly-OH (**50**, $R^2$ = Bzl, iBu, iPr, $CH_2CO_2tBu$) in 35% overall yield. Similarly, Boc-Ala$\psi$[*E*, CH=CH]Gly-OH and *Z*-Phe[*E*, CH=CH]Gly-OH were obtained in 15 and 8% overall yields, respectively. The main limitation appears to be the separation of dienes **55** and **56**, which requires careful silica gel flash chromatography. Chiral α-amino-substituted dienes of type **56** are also available from diastereoselective nucleophilic addition to chiral tricarbonyl(iminobutadiene)iron complexes **58**, which broadens the scope of this method.[75]

**Scheme 11** β-Elimination




**(3*E*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-6-phenylhexa-1,3-diene (56):**[76]

A soln of (5*S*)-5-(*tert*-butoxycarbonylamino)-4-mesyloxy-6-phenylhex-1-ene (**54**; 1.85g, 5mmol) in dry THF (50mL) was cooled to −78°C, and 1M tBuOK in dry THF (25mL) was added slowly by syringe. The soln was slowly allowed to warm to −40°C and, after 5h, 2M HCl was carefully added. The solvent was removed under vacuum and the residue dissolved in EtOAc (50mL) and washed three times with sat. solns of $KHSO_4$, $NaHCO_3$, and NaCl (30mL). After drying ($MgSO_4$) and removal of the solvent, the

*for references see p 369*

residue was flash chromatographed (EtOAc/hexane); yield: 1.01 g (79%) of **56** as a yellow oil and 0.26 g (19%) of its isomer **55**. Isomer **56**: $[\alpha]_D^{20}$ +10.5 (*c* 1.28, $CH_2Cl_2$); MS *m/z*: 273 (M), 217, 182, 162.

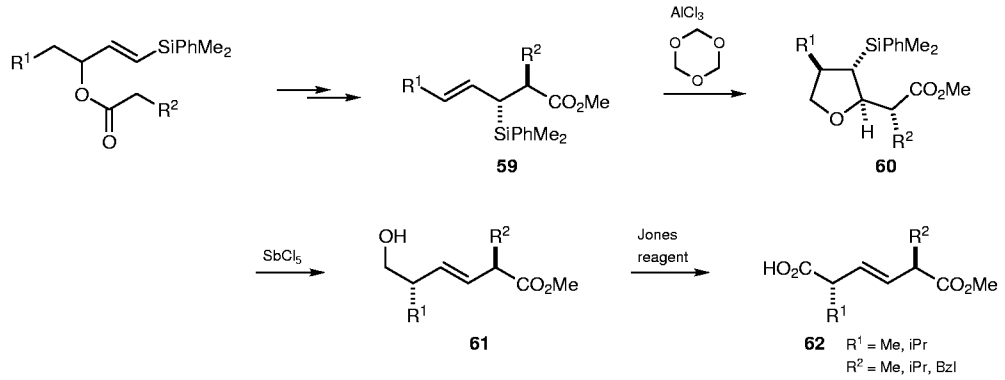
**(3*E*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-6-phenylhex-3-en-1-ol (57):**[76]
To a soln of diene **56** (2.65 g, 2 mmol) in dry THF (15 mL), 0.5 M 9-BBNH in THF (1.3 equiv) was added by syringe. The soln was stirred at rt, and after 6 h a 3 M NaOH soln (1.3 equiv) was added, followed by 30% $H_2O_2$ (0.85 mL). The mixture was warmed to 70 °C (oil bath temperature) for 1 h, and then partitioned between $Et_2O$ and a $K_2CO_3$ soln. The aqueous phase was extracted ($Et_2O$ and then EtOAc), the combined organic phases were washed (3×) with sat. NaCl, dried ($MgSO_4$), and concentrated. The residue was flash chromatographed; yield: 0.47 g (80%) of **57**; mp 67–70 °C; $[\alpha]_D^{20}$ –4.1 (*c* 1, MeOH); MS *m/z*: 292 ($M^+$ + 1), 236.

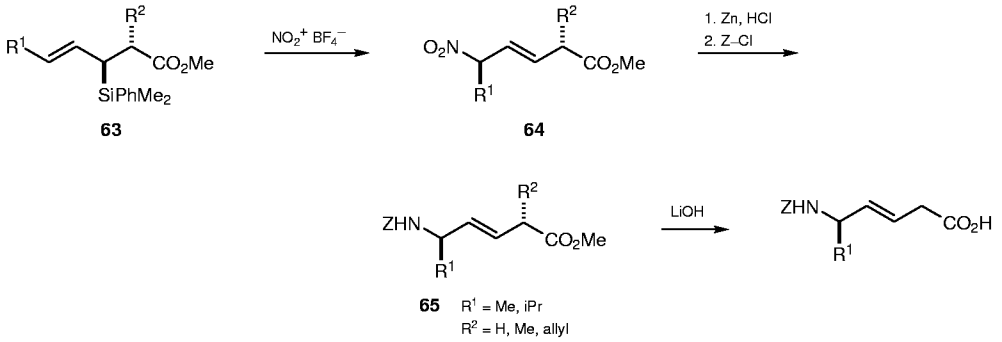
**(3*E*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-6-phenylhexa-3-enoic Acid, Boc-Phe$\psi$[*E*, CH=CH]Gly-OH (50, $R^2$ = Bzl, iBu, iPr, $CH_2CO_2tBu$):**[76]
A 2 M Jones soln was prepared from $CrO_3$ (4.98 g) and concd $H_2SO_4$ (5.5 mL), to which $H_2O$ (25 mL) was added. This reagent (7.5 mL, 15 mmol) was slowly added to a soln of alcohol **57** in acetone (15 mL) at 0 °C. After 1 h, $H_2O$ (20 mL) was added and the aqueous phase extracted (5% NaOH soln, 5×15 mL). The pH of the combined aqueous phases was adjusted to 2 using 4 M HCl, and then extracted ($Et_2O$). After drying ($MgSO_4$), the solvent was removed to dryness to give the product; yield: 0.23 g (75%); FAB-MS *m/z*: 351 ($M^+$+2Na), 328 ($M^+$+Na), 250 ($M^+$+H–$C_4H_8$).

### 10.5.2.1.2.4 Silane Elimination: $S_{E'}$ Reaction

Chiral *E*-crotylsilanes **59** (Scheme 12) function as carbon nucleophiles in an asymmetric condensation reaction with aldehydes. Thus reaction of **59** with *s*-trioxane in an anti-$S_{E'}$ addition afforded the tetrahydrofuran **60**, which was converted to homoallyl alcohol **61** by treatment with a strong Lewis acid.[17] After Jones oxidation, the monoester of hex-3-enedioic acid **62** was obtained. The latter was not converted into a dipeptide isostere (see Section 10.5.2.1.1), however. Rather, the (*E*)-crotylsilanes can be reacted with the nitronium cation $NO^{2+}$ to give unstable allylic nitro compounds **64**, which can be reduced using Zn/HCl in MeOH at 0 °C (95%), and immediately acylated with benzyloxycarbonyl chloride to give the isosteres **65** in 18–36% overall yield.[77] The methyl ester group in **65** was saponified in quantitative yield using lithium hydroxide. Neither $R^1$ nor $R^2$ can be benzylic groups because of competing nitration of the aromatic rings. The required homochiral *syn*- and *anti*-α-alkyl-β-silyl-(*E*)-hex-4-enoates **59** and **63** are obtained in good yields and with a high level of diastereoselection (16 to >25:1) through an Ireland–Claisen ester rearrangement (to give mostly *syn* compounds) of enantiomerically pure C3-(acyloxy)-(*E*)-vinylsilanes ($R^2$ = Me) (Scheme 12),[78–81] or by alkylation ($R^2$ = higher alkyl) of the enolate of **77** ($R^2$ = H, mostly *anti*, 92%) (see Scheme 14).

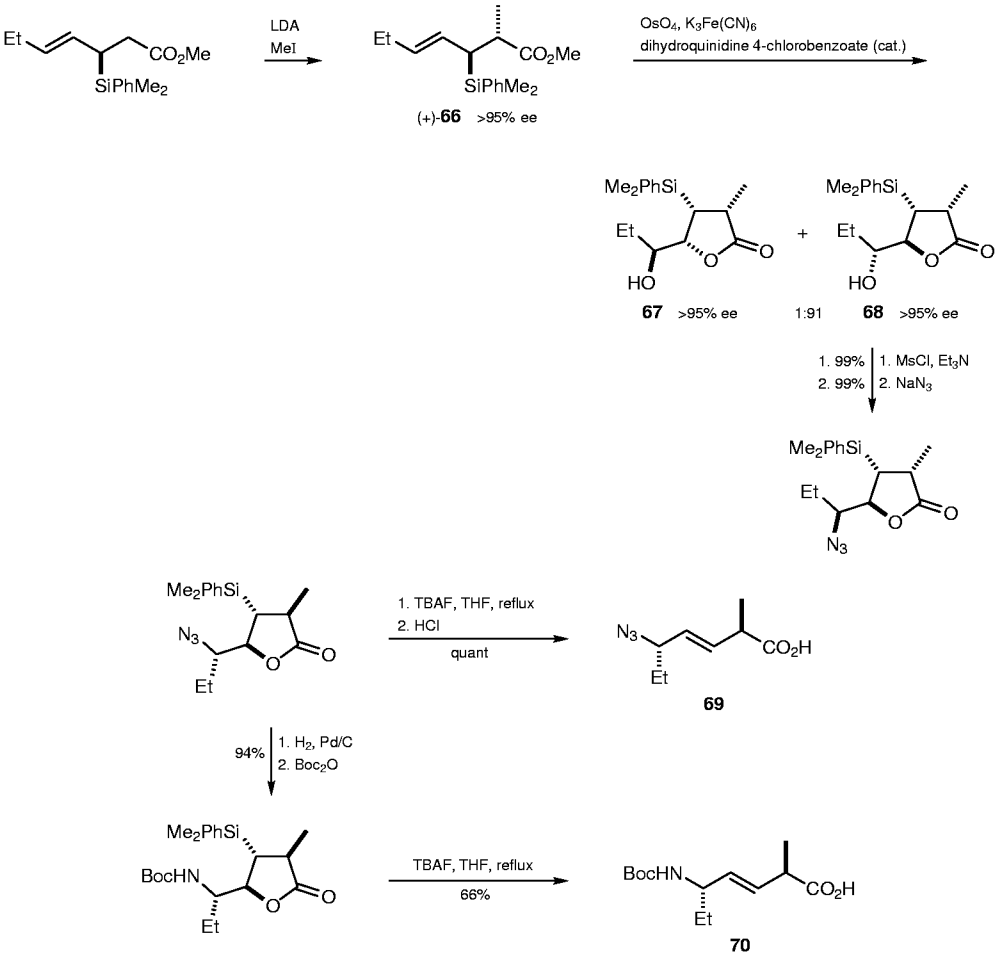
**Scheme 12** Silanes as Carbon Nucleophiles[78–81]






The same β-silyl-(*E*)-hex-4-enoates have been transformed into γ-lactones **67** and **68** by asymmetric dihydroxylation (Scheme 13).[62,82] After mesylation the remaining hydroxyl was substituted by an azide with inversion of configuration. When the silane group was first eliminated to give the allyl azide **69**, reduction of the latter could not be achieved. Therefore, reduction and Boc protection of the resulting amine was performed before silane elimination, to give the Boc-protected dipeptide isostere **70**.

**Scheme 13** Dihydroxylation of Allylsilanes[62,82]

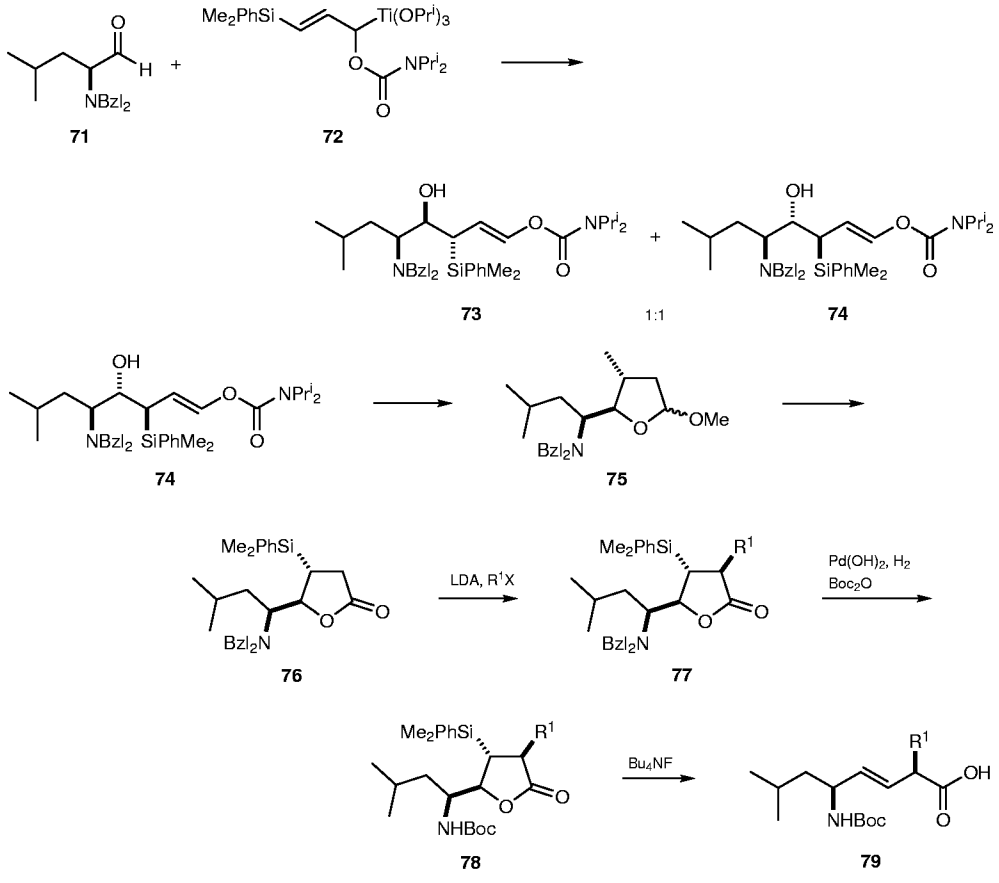


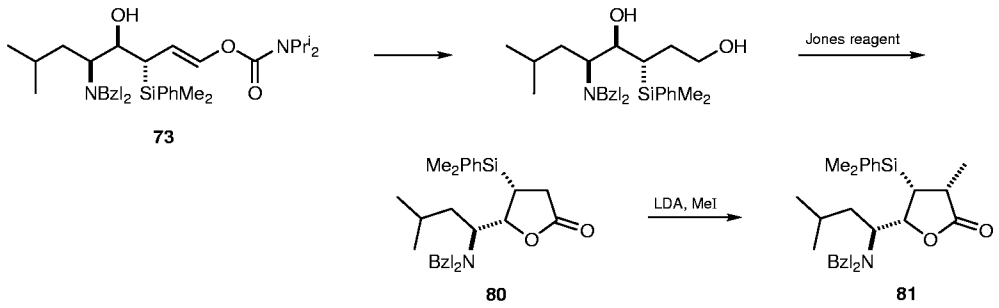

*for references see p 369*

In both strategies (Schemes 12 and 13), the $R^2$ side chain is introduced in the linear crotylsilanes **59**, **63**, and **66**. High levels of stereoselection are achieved when the $R^2$ group is introduced at the γ-lactone stage.[82] However, no experimental details have been published for these procedures.

A more convergent approach towards these lactones, which already contain the protected nitrogen substituent, was developed by Rehders and Hoppe (Scheme 14).[83,84] Reaction of (*S*)-*N*,*N*-dibenzylleucinal **71** with the homoenolate reagent **72** resulted in equal amounts of two diastereomers **73** and **74**, which were separated by chromatography and converted into the γ-lactones **76** and **80**. Alkylation of the enolates of **77** and **81** was completely stereoselective and gave only one isomer **77** or **81**. After conversion of the dibenzyl N-protection into Boc protection, treatment of **78** with tetrabutylammonium fluoride gave the diastereomerically pure dipeptide isostere **79**. This method provides a rapid access to all stereoisomers of ψ[*E*, CH=CH] dipeptides, with good overall yields. Almost all intermediate compounds were purified by column chromatography.

**Scheme 14** Diastereoselective Introduction of $R^2$[83,84]






**(1*Z*,3*R*,4*S*,5*S*)- and (1*Z*,3*S*,4*R*,5*S*)-5-(Dibenzylamino)-3-(dimethylphenylsilyl)-4-hydroxy-7-methyloct-1-enyl *N*,*N*-Diisopropylcarbamate (73 and 74):**[83]

To a soln of (*Z*)-3-(dimethylphenyl)silylprop-2-enyl *N*,*N*-diisopropylcarbamate (3.50 g, 11.0 mmol) and TMEDA (1.28 g, 11.0 mmol) in $Et_2O$ (20 mL) under an argon layer, 1.6 M BuLi in hexane (6.9 mL, 11.0 mmol) was added dropwise with magnetic stirring at below −70 °C. After 1 h, to the red-colored soln was added $Ti(OiPr)_4$ (15.6 g, 55 mmol) and stirring was continued for 1 h. (*S*)-*N*,*N*-Dibenzylleucinal (**71**; 2.95 g, 10.0 mmol) in $Et_2O$ (15 mL) was added dropwise and the mixture stirred at −78 °C for 17 h. The mixture was allowed to warm to 0 °C, then aq 2 M HCl (100 mL) and $Et_2O$ (150 mL) were added. The aqueous soln was extracted ($Et_2O$, 3×20mL) and the combined ethereal soln washed (sat. aq $NaHCO_3$, 50 mL) and dried ($MgSO_4$). After removal of the solvent in vacuo, the residue was chromatographed (silica gel, 550 g, $Et_2O$/pentane 1:8 to 1:4), affording **73**; yield: 2.83 g (46%) and **74**; yield: 2.83 g (46%).

**(2*S*,3*R*,5*S*)- and (2*S*,3*R*,5*R*)-2-[(*S*)-1-(Dibenzylamino)-3-methylbutyl]-3-(dimethylphenylsilyl)-5-methoxytetrahydrofuran (75):**[83]

A soln of **74** (1.23 g, 2.00 mmol) and $Hg(OAc)_2$ (160 mg, 25 mol%) in MeOH (12 mL) was stirred at rt. After 1 h the soln was cooled to 0 °C and MsOH (3.25 mL, 50 mmol) was added. The mixture was allowed to warm to rt and stirred for 14 h. $Et_2O$ (30 mL) and $H_2O$ (15 mL) were added and the soln neutralized with solid $KHCO_3$. The aqueous soln was extracted ($Et_2O$, 3×20 mL) and the combined ethereal soln dried ($Na_2SO_4$). After removal of the solvent in vacuo the residue was chromatographed (silica gel, 140 g, $Et_2O$/pentane 1:8) to afford **75** as a mixture of the two diastereomers (3.3:1); yield: 858 mg (86%).

**(3*R*,4*S*,5*S*)-5-(Dibenzylamino)-3-(dimethylphenylsilyl)-7-methyloctan-4-olide (76):**[83]

A soln of **75** (903 mg, 1.8 mmol) and MsOH (0.12 mL, 1.8 mmol) in $CH_2Cl_2$ (16 mL) was stirred at rt and, after 1 h, MCPBA (617 mg of 80% purity, 3.58 mmol) and MsOH (0.12 mL, 1.8 mmol) were added and the mixture stirred for 24 h at rt. The soln was reduced with $Me_2S$ (0.05 mL, 0.68 mmol), and then $Et_2O$ (20 mL) and sat. aq $NaHCO_3$ (15 mL) were added. The aqueous soln was extracted ($Et_2O$, 3×15 mL) and the combined ethereal soln dried ($Na_2SO_4$). After removal of the solvent, the residue was chromatographed (silica gel, 105 g, $Et_2O$/pentane 1:8) to afford **76**; yield: 709 mg (81%).

**(2*S*,3*R*,4*S*,5*S*)-2-Benzyl-5-(dibenzylamino)-3-(dimethylphenylsilyl)-7-methyloctan-4-olide (77, $R^1$ = Bzl):**[83]

A soln of **76** (242 mg, 0.5 mmol) and LDA (1.0 mmol) in THF (2 mL) was stirred at −78 °C for 0.5 h and Bzl-Br (0.18 mL, 0.75 mmol) was added. After 20 h stirring at −78 °C the mixture was allowed to warm to 0 °C, and $Et_2O$ (20 mL) and $H_2O$ (15 mL) were added. The aqueous soln was extracted ($Et_2O$, 3×10 mL) and the combined ethereal solns were dried ($Na_2SO_4$). After removal of the solvent in vacuo the residue was chromatographed (silica gel, 45 g, $Et_2O$/pentane 1:4) to afford **77** ($R^2$ = Bzl); yield: 229 mg (80%).

**(2*S*,3*R*,4*S*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino])-2,7-dimethyl-3-(dimethylphenylsilyl)octan-4-olide (78, $R^1$ = Me):**[84]

To a soln of **77** ($R^2$ = Me; 283 mg, 0.57 mmol) in MeOH (3 mL), 20% $Pd(OH)_2$/C (213 mg) was added and the suspension stirred in a $H_2$ atmosphere for 24 h. The mixture was filtered (silica gel) and the residue dissolved in THF (3 mL). Then, 80% NaH in mineral oil (20 mg, 0.68 mmol) was added and the mixture stirred for 1 h. $Boc_2O$ (122 mg, 0.56 mmol) in THF (1 mL) was added dropwise and stirring was continued for 14 h. $Et_2O$ (15 mL) and $H_2O$ (10 mL) were added, the aqueous soln was extracted with $Et_2O$ (3×10 mL), and the combined ethereal solns dried ($Na_2SO_4$). After removal of the solvent in vacuo the residue was chromatographed (silica gel, 45 g, EtOAc/hexane 1:10) to afford **78** as an oil; yield: 142 mg (61%).

*for references see p 369*

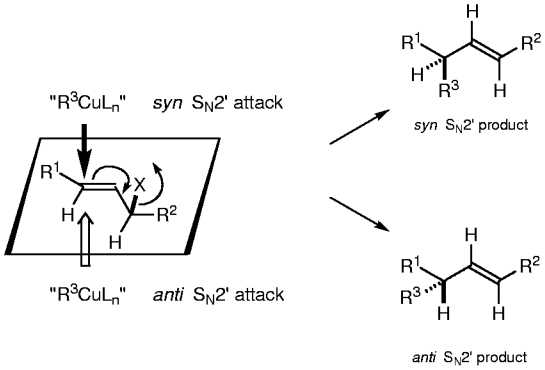
**(2*R*,3*E*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-2,7-dimethyloct-3-enoic Acid (79, $R^1$ = Me):**[84]
To a soln of **78** (244 mg, 0.58 mmol) in THF (5 mL) a soln of 1 M TBAF in THF (0.7 mL, 0.7 mmol) was added and the mixture stirred at 70 °C for 6 h. $Et_2O$ (20 mL) and sat. aq $NaHCO_3$ were added and the aqueous soln extracted ($Et_2O$, 3×10 mL). The ethereal solns were dried ($Na_2SO_4$) and the solvent was removed in vacuo. The residue was chromatographed (silica gel, 45 g, $Et_2O$/pentane 1:1) to afford **79** as an oil; yield: 141 mg (85%); $R_f$ 0.23.

### 10.5.2.1.3 $S_N2'$ Displacements

#### 10.5.2.1.3.1 Allylic Mesylates

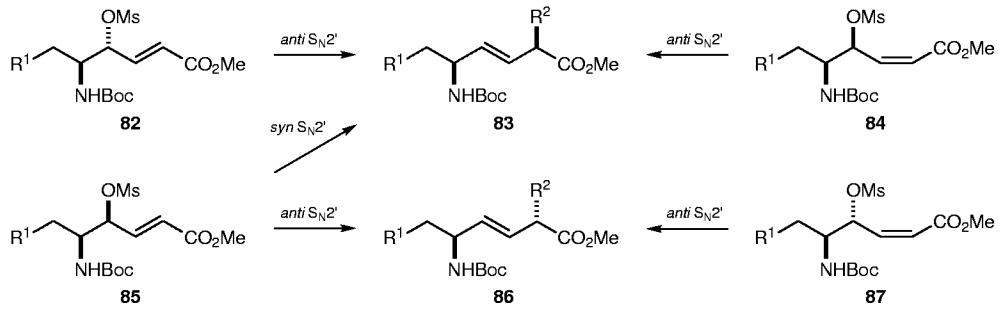
The concept (see Scheme 15) was introduced by Shue et al.,[44,85,86] who used a zinc-induced reductive isomerization of diasteromeric allylic bromides which already contained the $R^2$ side chain. Because of the lack of stereocontrol at the α-carbon, and of difficulties in obtaining the allylic bromide, this method has not been further developed; instead, homochiral allylic acetates or mesylates have been used.

**Scheme 15** General Concept of 1,3-Chirality Transfer in an anti-$S_N2'$ Reaction[44,85,86]




Starting from the homochiral γ-mesyloxy-α,β-enoates **82**, **84**, **85**, and **87** (Scheme 16), an *anti*-$S_N2'$ substitution has been developed by Kempf et al.[87] and Ibuka et al.[88] No such reaction was obtained for the corresponding γ-acetoxy compounds.[66] Whereas Kempf et al.[87] used Grignard reagents with catalytic copper(I) cyanide, Ibuka et al.[88] reported that the presence of the Lewis acid $BF_3$ is essential for clean 1,3-chirality transfer.[66] Lower-order organocyanocopper $BF_3$ complexes, $RCu(CN)Li \cdot BF_3$ (R = primary, secondary, or tertiary) and $RCu(CN)MgX \cdot BF_3$ (R= primary, secondary, or tertiary), could be used, giving clean anti-$S_N2'$ substitution irrespective of the steric bulk of the reagents. In the absence of $BF_3 \cdot OEt_2$, substantial amounts of reductive elimination products are formed. This reductive elimination product **83** ($R^2$ = H) was also observed by Kempf et al.[87] from the *anti*-stereoisomer **82**, whereas the *syn*-isomer **85** gave cleanly the substitution product. Kim and Ahn[89] reported the formation of a reductive elimination product as a side reaction (6–15%) for both *syn*- and *anti*-mesylates. This reductive elimination reaction has been exploited to prepare a range of isosteres of the type Xaaψ[*E*, CH=CH]Gly.[90] Whereas alkylCu(CN)M·$BF_3$ (M = Li or Mg) readily reacted with the mesylates **82**, **84**, **85**, and **87** via an *anti*-$S_N2'$ pathway, alkenylcoppers such as alkenylCu(CN)MgX (lower-order reagents) or $(alkenyl)_2Cu(CN)MgX_2$ (higher-order reagents) and $(alkenyl)_2CuMgX$ (Gilman-type reagents) afforded the elimination products **83** ($R^2$ = H) in high yields. Under these conditions the γ-acetoxy enoates also react to give the corresponding ψ[*E*, CH=CH]Gly isosteres.[90] A vinyl substituent as $R^2$ can nevertheless be introduced via the *trans*-$S_N2'$ substitution reaction using higher-order zinc cuprate reagents.[91] A major side reaction under these conditions is aziridine formation.

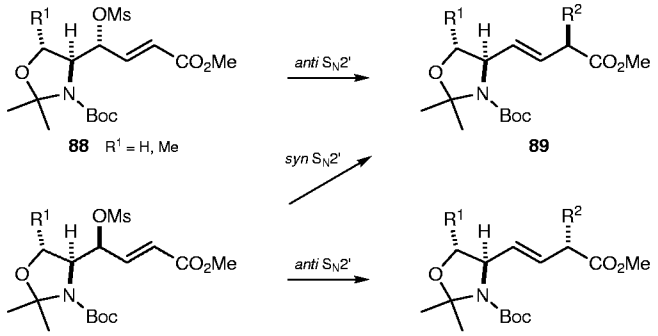
**Scheme 16** $S_N2'$ Displacement of Allylic Mesylates




Organozinc reagents with copper catalysis were also shown to give excellent yields with high diastereoselection in the substitution process. These conditions have some advantages because of the easier preparation of organozinc halides from Grignard reagents in comparison with organolithiums (e.g., benzyllithium).[92]

For δ-NH-Boc-γ-mesyloxy-α,β-enoates, only *anti*-$S_N2'$ substitution has been observed. However, when the NH is absent, such as in Pro-derived enoates, or Ser and Thr acetonide-derived enoates **88** or **89** (Scheme 17), the *syn*-$S_N2'$ stereochemistry has also been observed.[93]

**Scheme 17** *Syn* $S_N2'$ Displacements




Whereas the *anti*-mesylates exclusively react in an *anti* fashion, the *syn*-mesylates afford a mixture of *syn* and *anti* substitution. In this case the bulk of the alkyl group in organocyanocopper–boron trifluoride reagents strongly influences the level of diastereoselection. Using iPrCu(CN)MgCl·$BF_3$, or the *tert*-butyl analogue, the *syn*-$S_N2'$ substitution became strongly favored (*syn*/*anti* 91–98/9–2). Changing Boc N-protection to Moc N-protection decreases the extent of *syn* substitution.

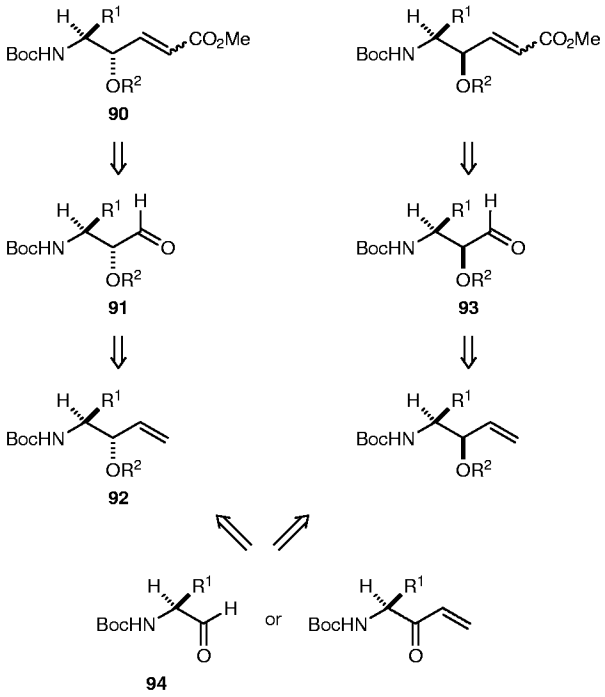
The stereochemistry of the dipeptide isosteres has been determined by X-ray crystallography,[87,93] by chemical conversion,[66] by circular dichroism,[93,94] and by enantiospecific synthesis.[43]

Crucial to the success of this methodology is the preparation of diastereomerically pure γ-mesyloxy-α,β-enoates. These are obtained (Scheme 18) through a Wittig reaction using (methoxycarbonylmethylene)triphenylphosphorane[66,95] or methyl (dimethoxyphosphoryl)acetate[87] on α-acetoxy (**91**, **93**, $R^2$ =Ac), α-tetrahydroxypyranyloxy (**91**, **93**, $R^2$ = THP), or α-*tert*-butyldimethylsilyl aldehydes (**91**, **93**, $R^2$ = TBDMS).[43] The formation of pure *E*-enoates,[66,95] as well as of *E*/*Z* mixtures,[96] has been observed. The α-hydroxy aldehydes were obtained in a three-step procedure, starting with the addition of vinylmagnesium chloride[66] to Boc-protected α-amino aldehydes **94**. This reaction resulted in a mixture of

*for references see p 369*

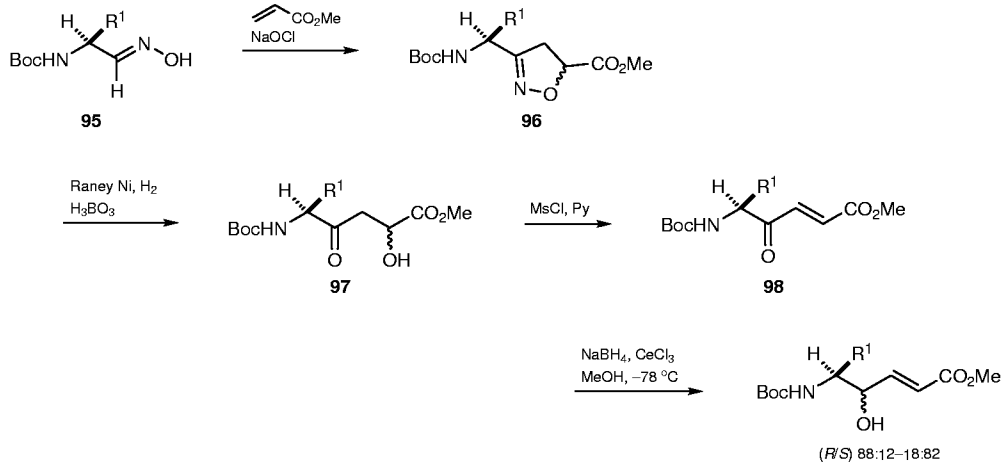
*syn*- and *anti*-vinyl alcohols (7:3 for Boc-Ala aldehyde). The *syn* selectivity could be increased (>15:1) by reducing Boc-Ala-OMe with DIBAL-H, followed by treatment with vinylmagnesium chloride in a one-pot reaction.[66] Enhanced *anti* selectivity (typically 8:1) was reported using vinylmagnesium bromide in the presence of stoichiometric ethereal zinc chloride.[97] The vinyl alcohols have to be separated by HPLC[95] or by flash chromatography of the *tert*-butyldimethylsilyl ether,[66] a procedure which limits large-scale application.

**Scheme 18** Preparation of γ-Mesyloxy-α,β-enoates



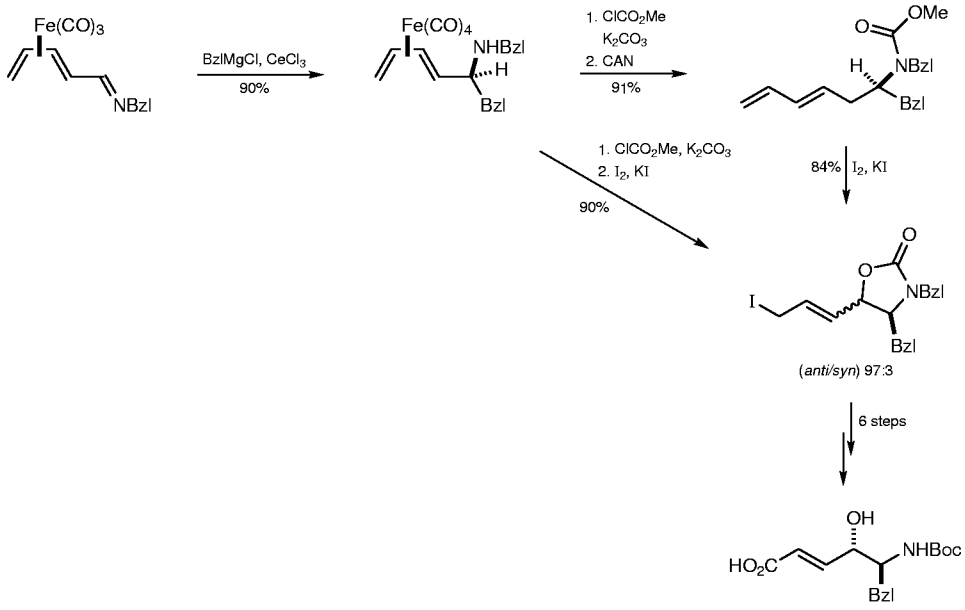

An alternative preparation of the γ-hydroxy-α,β-enoates has been developed by Kim and coworkers (Scheme 19).[98] α-Amino nitrile oxides, which are generated in situ from the reaction of the corresponding α-amino aldoximes **95** and sodium hypochlorite, undergo 1,3-dipolar cycloadditions with methyl acrylate smoothly to give a diastereomeric mixture of 4,5-dihydroxyisoxazoles **96** (44–76%).[99] These are subjected to reductive cleavage conditions [Raney Ni, $H_2$, $B(OH)_3$, MeOH/$H_2O$ 5:1][100] to give the diastereomeric α-hydroxyoxomethylene isosteres **97** (44–94%), which were dehydrated (MsCl, pyridine) to the oxovinyl compound **98**.[101] Reduction of **98** using various reducing agents produced the required γ-hydroxy-(*E*)-α,β-enoates.[89] The stereoselectivities of the reductions were moderate (88:12–18:82) and could be reversed by using additives ($CeCl_3$, $SmCl_3$), but a separation step is still needed.

**Scheme 19** Preparation of γ-Mesyloxy-α,β-enoates by 1,3-Dipolar Addition[98]




An alternative preparation giving access to *anti*-hydroxyvinyl compounds uses diastereoselective nucleophilic addition to a chiral diene–iron tricarbonyl complex, followed by iodocarbamation and ring opening (Scheme 20).[75]

**Scheme 20** Preparation of γ-Mesyloxy-α,β-enoates from Chiral Diene–Iron Tricarbonyl Complexes[75]




**Methyl (2*E*,4*S*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-4-hydroxyhex-2-enoate (90, R¹ = Me; R² = H):**[66]
To a soln of acetate **92** (R¹ = Me; R² = Ac; 4.02 g, 20 mmol) in $CH_2Cl_2$ (30 mL), ozone was bubbled at −78 °C until a blue color persisted. The soln was warmed to 0 °C and stirred for 10 min. To the above soln, $Ph_3P$ (5.24 g, 20 mmol) and (methoxycarbonylmethylene)triphenylphosphorane (13.36 g, 40 mmol) were added at −78 °C, and the mixture was stirred for 2 h at 0 °C. The mixture was concentrated under reduced pressure to an oil, which was flash chromatographed (silica gel, hexane/EtOAc 3:1) to give 4.6 g of a colorless oil. Powdered $Na_2CO_3$ (8.1 g) was added to a soln of the above oil (4.6 g) in MeOH (38 mL) under stirring at rt and the mixture was stirred for 18 h. The mixture was filtered and the filtrate was concentrated under reduced pressure to leave a slightly yellow oil, which was flash chromatographed (silica gel, hexane/EtOAc 3:1) to give **90** (R¹ = Me; R² = H) as a colorless oil; yield: 1.75 g (34%); $[\alpha]_D^{30}$

*for references see p 369*

−54.5 (*c* 0.955, $CHCl_3$); IR ($CHCl_3$) ṽ: 3450, 2980, 2960, 1709, 1504, 1453, 1440, 1397, 1372, 1316, 1163, 1052, 983, 931, 864 $cm^{-1}$; MS *m/z*: 260 $[M+H]^+$, 244, 204, 186, 172, 144, 116, 88, 57, 44 (base peak); exact mass *m/z*: 260.1499, calcd. for $C_{12}H_{21}NO_5H^+$ 260.1498.

**Methyl (2*E*,4*S*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-4-(methylsulfonyloxy)hex-2-enoate (85, $R^1$ = H):**[66]
To a stirred soln of alcohol **90** (1.57 g, 6.06 mmol) in a mixture of pyridine (6 mL) and $CH_2Cl_2$ (10 mL) at −78 °C, MsCl (2 mL) was added dropwise and the mixture stirred for 18 h with warming to 0 °C. The mixture was poured into a cold soln of 5% $NaHCO_3$ (25 mL) and extracted ($Et_2O/CH_2Cl_2$ 4:1). The extract was washed successively with 5% citric acid, 5% $NaHCO_3$, and $H_2O$, and dried ($MgSO_4$). Concentration under reduced pressure below 25 °C yielded an oily residue, which was purified by flash chromatography (silica gel, hexane/EtOAc 3:1) to give a crystalline residue. Recrystallization (hexane/$Et_2O$ 1:1) gave **85** ($R^1$ = H) as a colorless oil; yield: 2.025 g (99%); $[\alpha]_D^{18}$ −25.3 (*c* 0.978, $CHCl_3$); IR ($CHCl_3$) ṽ: 3440, 2980, 1715, 1501, 1453, 1440, 1369, 1315, 1285, 1175, 1095, 974, 954, 926 $cm^{-1}$; MS *m/z*: 337 ($M^+$), 322, 282, 264, 206, 194, 186, 168, 144, 115, 110, 88, 83, 57, 44 (base peak), exact mass *m/z*: 337.1188, calcd. for $C_{13}H_{23}NO_7S$ 337.1195.

**Use of RCu(CN)Li·$BF_3$ (R = Me, iBu, tBu); General Procedure:**[66]
The following procedure is representative for all reactions of δ-aminated γ-mesyloxy-(*E*)-α,β-enoates with MeCu(CN)Li·$BF_3$, iBuCu(CN)Li·$BF_3$, or tBuCu(CN)Li·$BF_3$.

**Methyl (2*S*,3*E*,5*S*)-5-[(*tert*-Butyloxycarbonyl)amino]-2-methylhex-3-enoate (86, $R^1$ = H; $R^2$ = Me):**[66]

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

To a stirred slurry of CuCN (72 mg, 0.8 mmol) in dry THF (5 mL) at −78 °C, 1.5 M MeLi/LiBr in $Et_2O$ (0.533 mL, 0.8 mmol) was added by syringe and the mixture was allowed to warm to −20 °C and to stir at this temperature for 10 min. $BF_3 \cdot OEt_2$ (0.1 mL, 0.8 mmol) was added to the above mixture at −78 °C and the mixture was stirred for 5 min. A soln of α,β-enoate **85** (67.4 mg, 0.2 mmol) in dry THF (2 mL) was added dropwise to the above reagent at −78 °C with stirring; the stirring was continued for 30 min, followed by quenching with a sat. $NH_4Cl$/28% $NH_4OH$ soln (2:1, 3 mL). The mixture was extracted ($Et_2O$) and the extract washed successively with 5% citric acid, 5% $NaHCO_3$, and $H_2O$, and dried ($MgSO_4$). Concentration under reduced pressure gave an oily residue, which was purified by flash chromatography (silica gel, hexane/EtOAc 4:1) to give **86** ($R^1$ = H, $R^2$ = Me) as a colorless syrup of better than 99% optical purity (capillary GC and $^1$H NMR); yield: 47 mg (93%); Kugelrohr distillation, 125 °C/1 Torr; $[\alpha]_D^{15}$ +2.74 (*c* 0.73, $CHCl_3$); IR ($CHCl_3$) ṽ: 3440, 2980, 2940, 2880, 1722, 1710, 1494, 1453, 1438, 1395, 1378, 1369, 1168, 1070, 1049, 969 $cm^{-1}$; MS *m/z*: 257 ($M^+$), 201, 186, 142, 141, 114, 70 (base peak), 57; exact mass *m/z*: 257.1630, calcd. for $C_{13}H_{23}NO_4$ 257.1626.

**Reductive Elimination of Mesylates; General Procedure:**

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

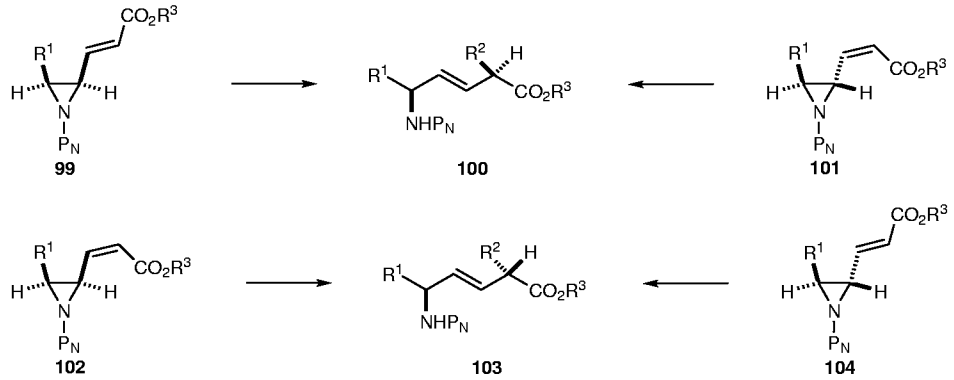
To a stirred suspension of CuCN (90 mg, 1 mmol) in dry THF (4 mL) under argon at −78 °C, freshly prepared 2.2 M $H_2C$=CHMgCl in THF (0.91 mL, 2 mmol) was added by syringe. The mixture was allowed to warm to 0 °C and then stirred at this temperature for 10 min. A soln of mesylate **82**, **84**, **85**, or **87** (120 mg, 0.25 mmol) in dry THF (2 mL) was added to the above reagent at −78 °C with stirring. The stirring was continued for 30 min followed by quenching with a sat. $NH_4Cl$/28% $NH_4OH$ soln (2:1, 3 mL). After the usual workup, the product was purified by flash chromatography (silica gel, hexane/EtOAc 4:1) to give **83** ($R^2$ = H) as a colorless oil of better than 99% purity (capillary GC and $^1$H NMR); yield: 86 mg (89%); Kugelrohr distillation, 160 °C/1 Torr; $[\alpha]_D^{18}$ +7.44 (*c* 0.941, $CHCl_3$). The synthesized protected isostere **83** ($R^2$ = H) exhibited $^1$H NMR (in $CDCl_3$) and IR (in $CHCl_3$) spectra consistent with the assigned structure.

### 10.5.2.1.3.2 Allylic Aziridines

The β-(*N*-tosylaziridinyl)-α,β-enoates **99**, **101**, **102**, and **104** ($P_N$ = Tos, Scheme 21) have the experimental advantage that they are readily purified by recrystallization and are more stable than the γ-mesyloxy-α,β-enoates, whose application is hindered by a rather low stability in some cases.[102,103] When activated by the electron-withdrawing tosyl group as

nitrogen protection, it was shown that, like their γ-mesyloxy analogues, these aziridines underwent *anti*-$S_N2'$ substitution in organocopper-mediated reactions. Whereas higher-order cuprates or Gilman reagents only resulted in reduction to (*E*)-β,γ-unsaturated esters **100** or **103** ($R^2$ = H), the use of lower-order methylcyanocuprate [MeCu(CN)Li·LiBr] or dimethylzinc [$Me_2Zn$·2LiCl·2LiBr] in the presence of 20–30 mol% CuCN gave predominantly the *anti*-$S_N2'$ reaction. However, a small amount (3–6%) of the $S_N2$ reaction was also observed under these conditions.[103] It was found that addition of LiCl to the mixture of dialkylzinc and 20 mol% CuCN was essential for optimizing reaction rates and chemical yields. Both *E*- and *Z*-enoates gave the *E*-alkene isostere.

**Scheme 21** $S_N2'$ Displacements of Allylic Aziridines



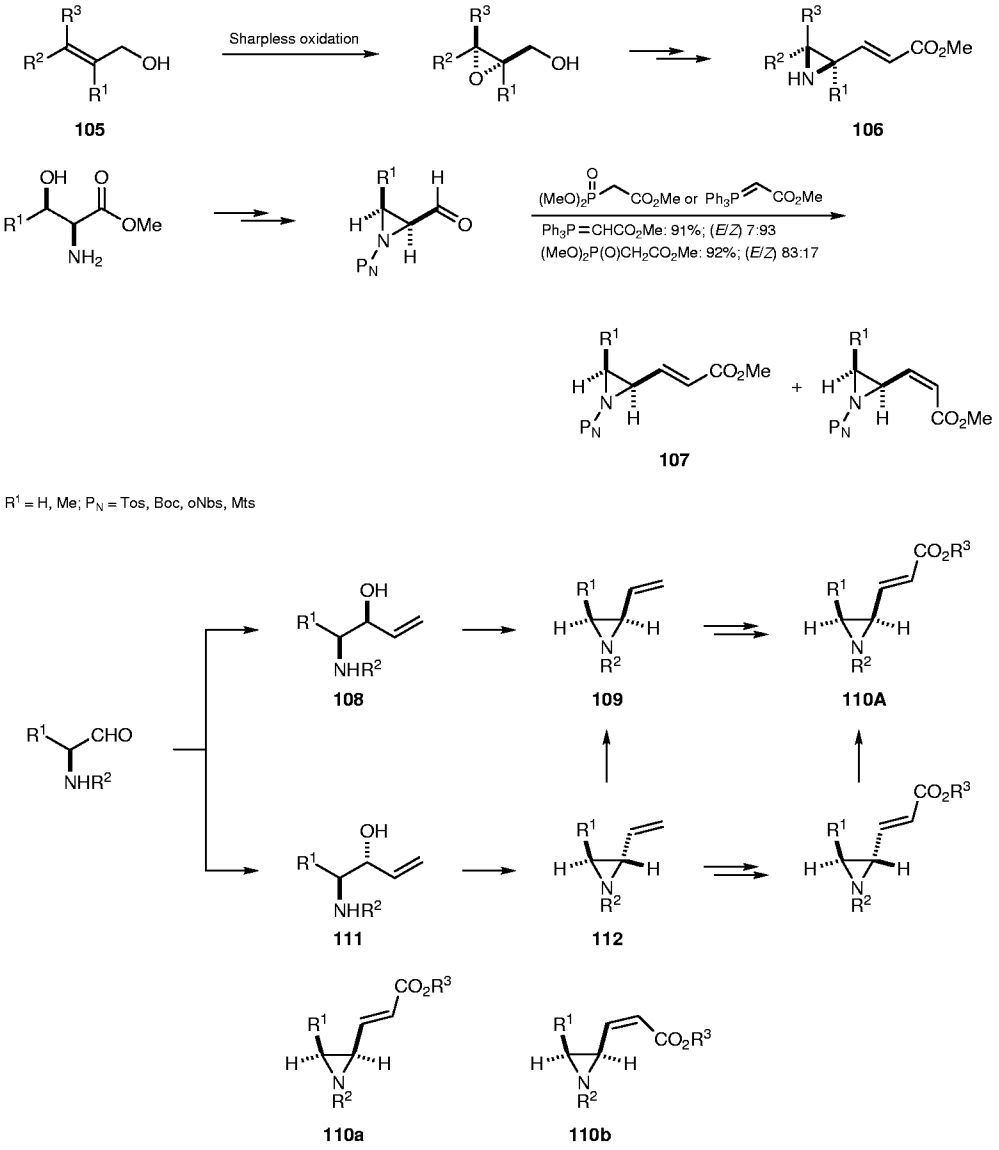

In a parallel study, Wipf and Fritch[104] have shown that also urethane-protected (Boc), and even amino acid segments, are tolerated as acyl compounds on the aziridine nitrogen. The best results were obtained with alkylcopper reagents derived from CuCN and an alkyllithium in the presence of boron trifluoride–diethyl ether complex. Some δ-alkylated compounds (11–15%) were isolated as well. This work was extended to a solid-phase procedure that resulted in resin-bound alkene isosteres that could immediately be used in further peptide synthesis.[105] For this purpose, the 2-nitrophenylsulfonyl (oNbs) group was used for nitrogen protection and aziridine activation. It could be readily cleaved with benzenethiolate, which was compatible with the acid-sensitive Wang linker used.

The starting γ,δ-epimino-α,β-enoates **99**, **101**, **102**, and **104** could be obtained through an asymmetric Sharpless oxidation of substituted allylic alcohols **105** (Scheme 22), followed by oxirane to aziridine transformation under modified Staudinger conditions, and Wittig chain extension. It allows the presence of a variety of substituents ($R^1$, $R^2$, $R^3$ in **106**), but is rather lengthy.[104] Unsubstituted (**107**, $R^1$ = H) or methyl-substituted (**107**, $R^1$ = Me) compounds were readily available from the β-hydroxyamino acids Ser and Thr.[9,102–104] By changing the phosphorane to the phosphonoacetate reagent, the *E* to *Z* selectivity in the alkenation reaction could be reversed.[103] An extensive experimental description of aziridine formation from Thr via asymmetric epoxidation, and the reaction of the alkenylaziridines with organocuprates leading to the Ala-Ala and Ala-Nle alkene isosteres, has been published.[106] A more general synthesis consists of aziridine formation from the chiral *syn*- and *anti*-amino alcohols **108** and **111**. However, as reported in Section 10.5.2.1.3.1, the addition of vinylmagnesium bromide or chloride to chiral N-protected amino aldehydes results in a mixture of *syn*- and *anti*-amino alcohols, which have to be separated by chromatography. Moreover, after oxidative transformation of the vinyl group to the aldehyde, subsequent Wittig or Horner–Emmons reaction results in mixtures of the *E*- and *Z*-enoates. It was shown that the vinylaziridines **109** and **112** can be equilibrated using Pd(0) catalysis, with a strong preference for the *syn*-isomer.[107,108] Conditions were elaborated that allowed the

*for references see p 369*

isomerization of the four stereoisomeric γ,δ-epimino-α,β-enoates to the more stable γ,δ-*syn*-epimino-(2*E*)-isomer **110A** using palladium catalysis. This procedure allows a ready access to the isomer **110**, which results in an *E*-alkene isostere having a stereochemistry equivalent to a L,L-dipeptide.[109,110] It should be noted that also the Pmc and PNBS nitrogen protecting groups can be used, which are easier to deprotect.[108]

**Scheme 22** Preparation of γ,δ-Epimino-α,β-enoates




The ready availability of γ,δ-epimino-*syn*-(2*E*)-enoates **99** and **110A** was further exploited to prepare L,L-type **100** (or D,D-type) isosteres as well as L,D-type **103** (or D,L-type) isosteres from the same precursor.[111] Direct treatement of the *N*-(arylsulfonyl)aziridines **99** and **110** with $R^2Cu(CN)MgCl \cdot 2LiCl$ gives rise to an $S_N2'$ substitution, leading to the L,L-compound **100** (75%). The same aziridines **99** and **110** could be treated with methanesulfonic acid to give a *syn*-mesylate.[93] A subsequent *anti*-$S_N2'$ reaction using $R^2Cu(CN)MgCl \cdot BF_3$ led to the L,D-type isostere **103** (94%).

**(*E*)- and (*Z*)-4,5-Epimino-2-enoates:**
***tert*-Butyl (4*R*,5*S*,2*E*)-4,5-Epimino-6-methyl-*N*-[(2,4,6-trimethylphenyl)sulfonyl]hept-2-enoate** (110A, **$R^1$ = iPr; $R^2$ = Mts; $R^3$ = tBu) and *tert*-Butyl (4*R*,5*S*,2*Z*)-5,5-Epimino-6-methyl-*N*-[(2,4,6-trimethylphenyl)sulfonyl]hept-2-enoate** (110B, **$R^1$ = iPr; $R^2$ = Mts; $R^3$ = tBu):**[109]
Ozone was bubbled through a soln of the vinylaziridine **109** ($R^2$ = Mts; 1.3 g, 4.436 mmol) in EtOAc (15 mL) at −78 °C until a blue color persisted. Zn powder (0.7 g) was added to the soln, and the mixture was stirred for 30 min, during which time it was allowed to warm to 0 °C. To the mixture at 0 °C, (*tert*-butoxycarbonylmethylene)triphenylphosphorane (3.3 g, 8.87 mmol, 2.0 equiv) was added, and the mixture was stirred for 1 h. The mixture was concentrated under reduced pressure to leave a semisolid, which was purified by flash chromatography (silica gel, hexane/EtOAc 4:1) to give the *Z*-enoate; yield: 760 mg (44%). Continued elution gave the *E*-enoate; yield: 950 mg (55%). *Z*-enoate: colorless crystals; mp 118 °C (hexane/$Et_2O$ 2:1); $[\alpha]_D^{20}$ −54.7 (*c* 1.64, $CHCl_3$). *E*-enoate: colorless oil; $[\alpha]_D^{20}$ −54.2 (*c* 1.84, $CHCl_3$ ); LRMS (FAB) *m/z*: 394 ($MH^+$), 338, 320, 254 (base peak), 154, 119; HRMS (FAB) *m/z*: 394.2057, calcd. for $C_{21}H_{32}NO_4S$ ($MH^+$) 394.2052.

**Isomerization of (*Z*)-4,5-Epimino-2-enoates:**
***tert*-Butyl (4*R*,5*R*,2*E*)-4,5-Epimino-6-methyl-*N*-[(2,4,6-trimethylphenyl)sulfonyl]hept-2-enoate** (110A, **$R^1$ = iPr; $R^2$ = Mts; $R^3$ = tBu):**[109]
To a stirred soln of the *Z*-enoate **110b** (1.1 g, 2.8 mmol) in dry THF (7 mL) at 0 °C under argon, a soln of $Pd(PPh_3)_4$ (64.6 mg, 0.056 mmol, 2 mol%) in dry THF (3 mL) was added by syringe, and the mixture was stirred at 15 °C for 15 h. Concentration under reduced pressure at 0 °C followed by flash chromatography (silica gel, hexane/EtOAc 4:1) gave 1.08 g of a crystalline mass. Recrystallization (hexane) gave the 2*E*-isomer as colorless crystals; yield: 690 mg (63%). The mother liquor was concentrated under reduced pressure to leave 410 mg of a colorless semisolid; this was treated with $Pd(PPh_3)_4$ (24 mg) in dry THF (5 mL) for 16 h at 15 °C. Usual workup followed by flash chromathgraphy (silica gel, hexane/EtOAc 5:1) and recrystallization (hexane) gave the *E*-enoate **110A**; yield: 230 mg (21%). The total product thus obtained amounts to 920 mg (84%); mp 118 °C (hexane); $[\alpha]_D^{26}$ −54.8 (*c* 0.945, $CHCl_3$).

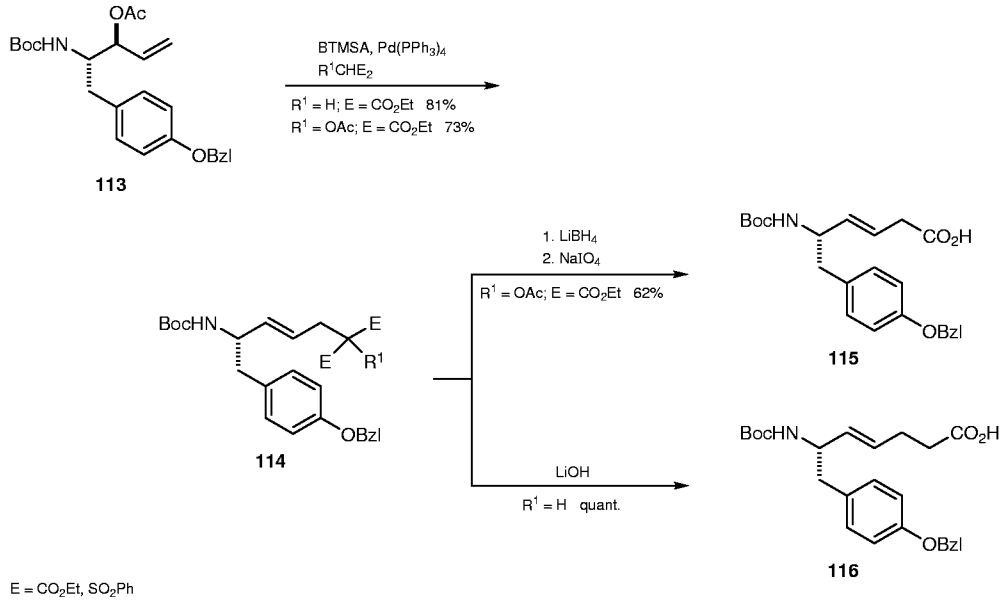
***tert*-Butyl (2*R*,3*E*,5*S*)-2-Isobutyl-6-methyl-5-[[(2,4,6-trimethylphenyl)sulfonyl]amino]hept-3-enoate (Mts-Valψ[*E*, CH=CH]Leu-OH)** (100, **$R^1$ = iPr; $R^2$ = iBu; $R^3$ = tBu; $P_N$ = Mts):**[109]
To a stirred soln of CuCN (916 mg, 10.2 mmol) and LiCl (860 mg, 20.4 mmol) in dry THF (20 mL) under argon, 1.1 M iBuMgCl in THF (9.27 mL, 10.2 mmol) was added by syringe at −78 °C, and the mixture was allowed to warm to 0 °C and stirred at this temperature for 10 min. The enoate **99** ($R^1$ = iPr; $P_N$ = Mts; $R^3$ = tBu; 1.0 g, 2.544 mmol) in dry THF (10 mL) was added dropwise to the above reagent at −78 °C with stirring, and the stirring was continued for 30 min followed by quenching with a sat. $NH_4Cl$/28% $NH_4OH$ soln (1:1, 20 mL). The mixture was extracted ($Et_2O$) and the extract was washed ($H_2O$) and dried ($MgSO_4$). Concentration under reduced pressure gave an oily residue, which was flash chromatographed (silica gel, hexane/EtOAc 4:1) to give the title compound as a crystalline mass; yield: 1.091 g (95%). Recrystallization (hexane) gave colorless crystals; mp 108 °C; $[\alpha]_D^{20}$ −42.1 (*c* 1.35, $CHCl_3$).

### 10.5.2.1.3.3 Trost Alkylation of Allylic Acetates

Palladium-catalyzed coupling of allylic acetate **113** (Scheme 23) with diethyl malonate, or with acetylated diethyl tartronate, yielded the Boc-protected allylic amines **114** in 99:1 *E* stereoselectivity.[97] Saponification of **114** ($R^1$ = H) and decarboxylation gave the Tyr-βAla alkene isostere **116** in quantitative yield, whereas **114** ($R^1$ = OAc) was converted into the Tyr-Gly alkene isostere **115** in a two-step procedure in 62% overall yield. Considering the number of steps and the overall yield, this procedure is among the most efficient to prepare Xaaψ[*E*, CH=CH]Gly dipeptide isosteres.

*for references see p 369*

**Scheme 23** Trost Alkylation of Allylic Acetates[97]




**(1*S*)-1-{(1*S*)-[2-(4-Benzyloxyphenyl)-1-[(*tert*-butoxycarbonyl)amino]ethyl]}prop-2-enyl Acetate (113):**[97]

To a stirred soln of 1 M $ZnCl_2$ in $Et_2O$ (160 mL), THF (400 mL) was added. The soln was cooled to 0 °C (internal temperature) and 1 M $H_2C{=}CHMgBr$ in THF (240 mL) was added. After 30 min the suspension was cooled to −30 °C and a soln of *N*-*tert*-butoxycarbonyl-*O*-benzyl-L-tyrosinal (20 g, 56 mmol) in THF (300 mL) was added over 20 min, keeping the internal temperature at −30 ± 5 °C. The mixture was allowed to warm to 0 °C over 1.5 h, then cooled to −30 °C, cautiously quenched with 5% aq citric acid (1 L), and diluted with $Et_2O$ (2 L). The organic layer was washed with $H_2O$ (600 mL) and sat. $NaHCO_3$ (600 mL), and dried ($MgSO_4$). Purification (silica gel, $CH_2Cl_2$/EtOAc 8:2) gave the *threo*-allylic alcohol as a white solid; yield: 13.7 g (63%). Acetylation with $Ac_2O$ and TEA (1.1 equiv of each) in the presence of DMAP (0.1 equiv) in $CH_2Cl_2$ gave acetate **113** in quantitative yield; mp 73–74 °C.

**(5*E*,7*S*)-8-(4-Benzyloxyphenyl)-7-[(*tert*-butoxycarbonyl)amino]-1-(*tert*-butyldimethylsilyloxy)-3,3-bis(phenylsulfonyl)oct-5-ene (114, E = $SO_2Ph$; $R^1 = CH_2CH_2OTBDMS$):**[97]

A soln of allylic acetate **113** (150 mg, 0.37 mmol), $Pd(PPh_3)$ (21 mL, 0.02 mmol), and $Ph_3P$ (10 mg, 0.04 mmol) in THF (5 mL) was added to a mixture of 1-(*tert*-butyldimethylsiloxy)-3,3-bis(phenylsulfonyl)propane (255 mg, 0.56 mmol)[97] and BTMSA (115 mg, 0.57 mmol) [or 60% NaH (22 mg, 0.56 mmol)] in THF (4 mL). The resulting mixture was stirred under reflux for 1.5 h (until reaction completion, monitored by TLC), then concentrated to dryness and purified by chromatography (silica gel, hexane/EtOAc 3:1). After drying, the product was obtained as a colorless solid; yield: 230 mg (77%); mp 119–120.5 °C.

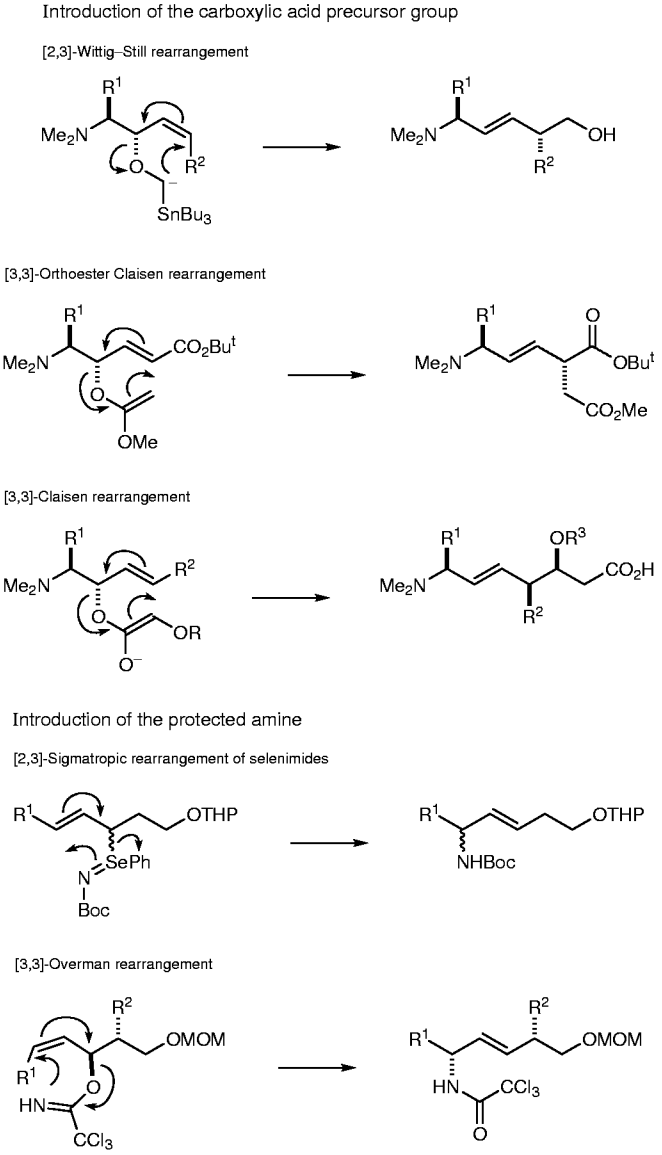
**(3*E*,5*S*)-6-(4-Benzyloxyphenyl)-5-[(*tert*-butoxycarbonyl)amino]hex-3-enoic Acid (Boc-Tyr(OBn)ψ[*E*, CH=CH]Gly-OH) (115):**[97]

A soln of the triester **114** (E = $CO_2Et$; $R^1$ = OAc; 0.5 g) in DME (10 mL) was added to commercial 2 M $LiBH_4$ in THF (5 mL). After 20 h the reaction was quenched with 5% aq citric acid and the product isolated by extraction (EtOAc). The crude product was allowed to react with $NaIO_4$ (0.8 g) in dioxane/$H_2O$ (1:1, 30 mL) for 1.5 h. The product **115** was isolated by extraction (EtOAc) and purified by chromatography (silica gel, 2% MeOH/$CHCl_3$); yield: 220 mg (94%); mp 129–130.5 °C; $[\alpha]_D^{25}$ −4.5 (*c* 0.5, MeOH).

### 10.5.2.1.4 Sigmatropic Rearrangements

The [2,3]- or [3,3]-sigmatropic rearrangements (Scheme 24) provide a means to introduce either the protected amine or the carbon atom which will become the carboxylic acid, while also positioning the double bond in the correct position for the alkene isosteres. Moreover, when starting from homochiral allyl alcohols, a very effective chirality transfer assures the stereospecific construction of the $R^1$ and $R^2$ side-chain stereochemistries.

**Scheme 24** Concept of Sigmatropic Rearrangement for the Stereospecific Synthesis of ψ[CH=CH] Isosteres



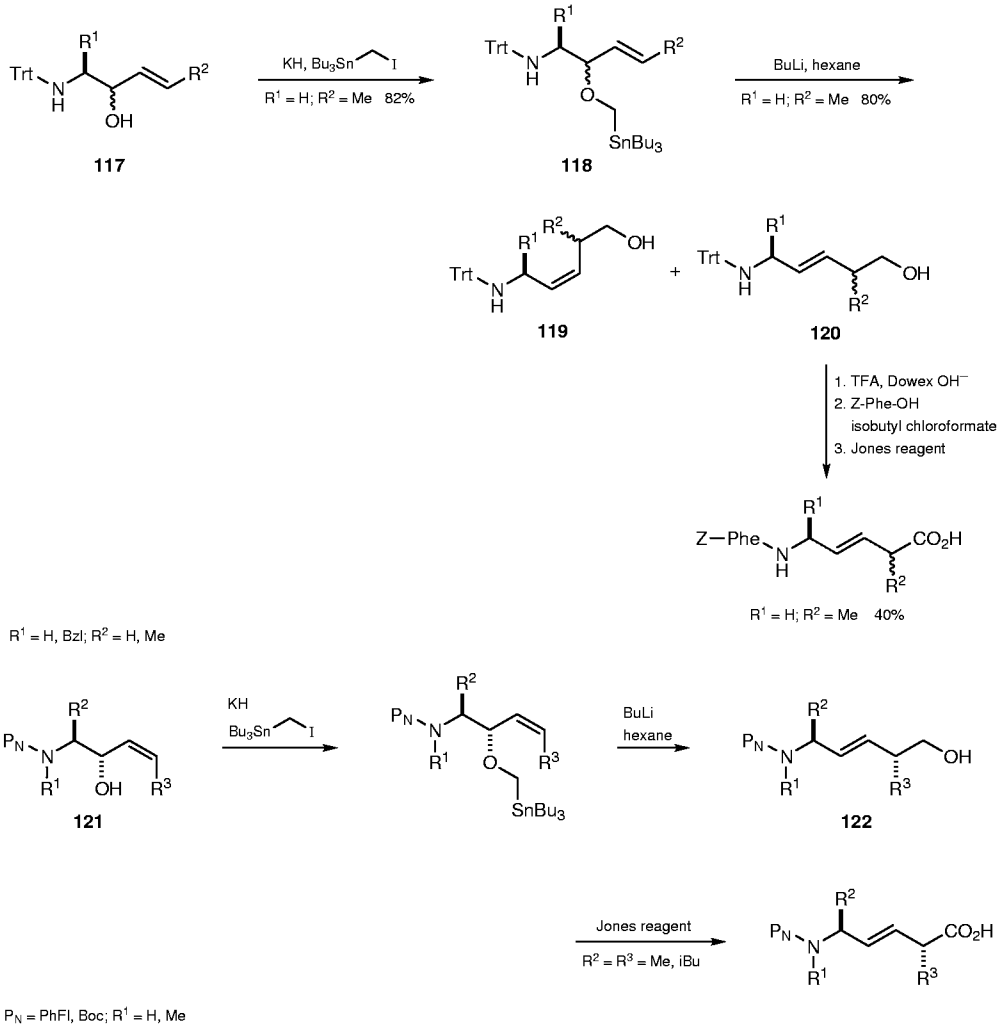

#### 10.5.2.1.4.1 [2,3]-Wittig–Still Rearrangement

Bol and Liskamp[112] introduced the [2,3]-Wittig rearangement in order to install the regiochemistry of the alkene and the oxygenation of the incipient C-terminus in one operation. A racemic β-*N*-trityl-α-hydroxy-(*E*)-alkene **117** ($R^1$ = H; $R^2$ = Me) was prepared from cro-

*for references see p 369*

tonaldehyde (Scheme 25). Subsequently, the Wittig precursor **118** was prepared and the rearrangement was performed using BuLi. The use of THF as solvent led mainly to a [1,2]-Wittig rearrangement, while in hexane, 80% of the [2,3]-Wittig rearrangement was observed, leading to an *E*/*Z* mixture (1.5:1) of alkenes **119** and **120**. The choice of the N-protection was crucial: Boc, 1,3-dimethylhexahydro-1,3,5-triazine-2-one, or Tos groups could not be used. Elaboration of this reaction sequence on the diastereomeric mixture of the allylic alcohols **117** ($R^1$ = Bzl; $R^2$ = H), obtained from the Trt-Phe aldehyde, gave only the *E*-alkene **120** ($R^1$ = Bzl; $R^2$ = H) (80%), which was converted into the Phe$\psi$[*E*, CH=CH]Gly-OH isostere in excellent overall yield.[113]

**Scheme 25** [2,3]-Wittig–Still Rearrangement




Enantioselective variants have been developed since, allowing chirality control at both chiral centers in the alkene dipeptide isosteres.[42,114] *Z*-Alkene precursors **121** were used to provide good stereocontrol during the rearrangement. Indeed, only the (*E*)-β,γ-alkene with *R* stereochemistry at the new stereogenic center in **122** was observed.[114] In this case the bulky 9-phenylfluoren-9-yl (PhFl) group[114] was used (or a Boc group).[42] The required *anti*-vinyl alcohols were obtained either by addition of (*Z*)-1-lithiopropene to the α-amino aldehyde

(7:1 *anti/syn*, 61% yield)[114] or by *Z* selective hydrogenation of the acetylene precursor using $Pd/BuSO_4$ in pyridine.[42]

**(*E*)-*N*-Trityl-2-[(tributylstannyl)methoxy]pent-3-en-1-amine** (118, **$R^1$ = H; $R^2$ = Me**):[113]
To a soln of (*E*)-1-aminopent-3-en-2-ol (1.11 g, 11.0 mmol) in $CH_2Cl_2$ (30 mL), Trt-Cl (3.07 g, 11 mmol) and TEA (8 mL, 22 mmol) were added. The mixture was stirred for 1 h at rt, followed by concentration under reduced pressure. The residue was suspended in EtOAc and filtered to remove the TEA·HCl. After column chromatography (EtOAc/hexanes/TEA 1:9:0.05), **117** ($R^1$ = H; $R^2$ = Me) was obtained as a colorless oil; yield: 79%; $R_f$ 0.48 (EtOAc/hexanes 1:1). After coevaporation twice with dry THF, **117** ($R^1$ = H; $R^2$ = Me; 0.86 g, 2.5 mmol) was dissolved in THF (15 mL). In a separate flask, a KH suspension (1.00 g, 20–25%, 5 mmol) was washed under argon with pentane (2×4 mL), followed by suspension in THF (5 mL) and addition to the soln of **117**. After stirring under argon for 15 min at rt, $Bu_3SnCH_2I$ (1.62 g, 3.75 mmol) in THF (10 mL) was added. Stirring was continued for 1 h, after which the mixture was poured into a sat. $NH_4Cl$ soln containing crushed ice. Organic and aqueous layers were separated and the aqueous layer was extracted ($Et_2O$, 4×). The combined organic layers were dried ($MgSO_4$), concentrated under reduced pressure, and the resulting yellow oil was chromatographed (silica gel, EtOAc/hexanes/TEA 1:9:0.005) to give compound **118** as a colorless oil; yield: 1.31 g (82%); $R_f$ 0.64 (EtOAc/hexanes 1:1).

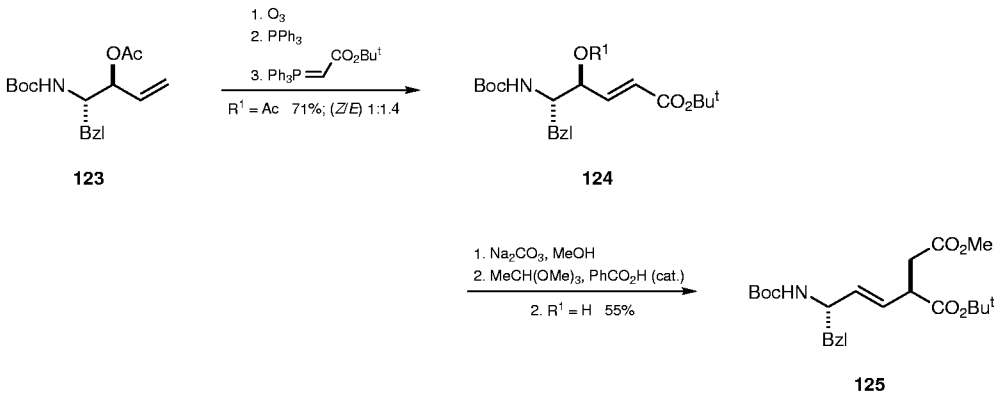
**(*E*)-5-Tritylamino-2-methylpent-3-en-1-ol** (120, **$R^1$ = H; $R^2$ = Me**) **and its *Z*-Isomer (119)**:[113]
The tin derivative **118** ($R^1$ = H; $R^2$ = Me; 0.65 g, 1 mmol) was coevaporated with dry THF (3×10 mL) and dissolved in dry hexanes (13 mL). Then, 1.6 M BuLi (2 mmol, 1.25 mL) was added at −100 °C, under argon. The yellow- or red-colored mixture was allowed to reach rt in the cooling bath and stirring was continued for 1 h. The mixture was quenched with sat. $NH_4Cl$ soln (20 mL) containing crushed ice. The aqueous layer was extracted with $Et_2O$ (4×), dried ($MgSO_4$), and concentrated under reduced pressure. The resulting yellow oil was chromatographed ($Et_2O$/petroleum ether gradient 0:1 to 1:1) to give the [2,3]-*trans*-product **120** as a white solid; yield: 0.161 g (45%) and the [2,3]-*cis*-product **119** as an oil; yield: 0.090 g (25%). In addition, 6% of the [1,2]-rearranged product was isolated. Isomer **120**: $R_f$ 0.60 (EtOAc/hexanes 1:1); MS *m/z*: 357.3; isomer **119**: $R_f$ 0.65 (EtOAc/hexanes 1:1); MS *m/z*: 357.2.

### 10.5.2.1.4.2 Orthoester Claisen Rearrangement

Although an elegant synthetic route using the [3,3]-Claisen rearrangement to prepare Alaψ[*E*, CH=CH]Ala and Alaψ[*E*, CH=CH]Leu isosteres with full stereocontrol has been reported, no experimental data were published.[115] The orthoester Claisen rearrangement specifically provides access to isosteres of the type Xaaψ[*E*, CH=CH]Asp. Allylic acetate **123** was synthesized using the previously described route (Section 10.5.2.1.3.3.) from Boc-L-Phe-OMe.[66,95] The Wittig alkenation produced (Scheme 26) a mixture of *E*- and *Z*-enoates that were separated by flash chromatography. The orthoester Claisen rearrangement produced the Pheψ[*E*, CH=CH]Asp(OMe)-OtBu isostere (**125**) with differentially protected carboxylic groups.[116] Starting from the *syn*-epimer of acetate **123**, the corresponding L,D-isomer has been obtained. The use of the Wittig–Still rearrangement was suggested to provide Xaaψ[*E*, CH=CH]Ser isosteres.

*for references see p 369*

**Scheme 26** Orthoester Claisen Rearrangement




***tert*-Butyl (2*R*,3*E*,5*S*)-5-[(*tert*-Butoxycarbonyl)amino]-2-(methoxycarbonylmethyl)-6-phenylhex-3-enoate Pheψ[*E*, CH=CH]Asp(OMe)-OtBu** (125):[116]

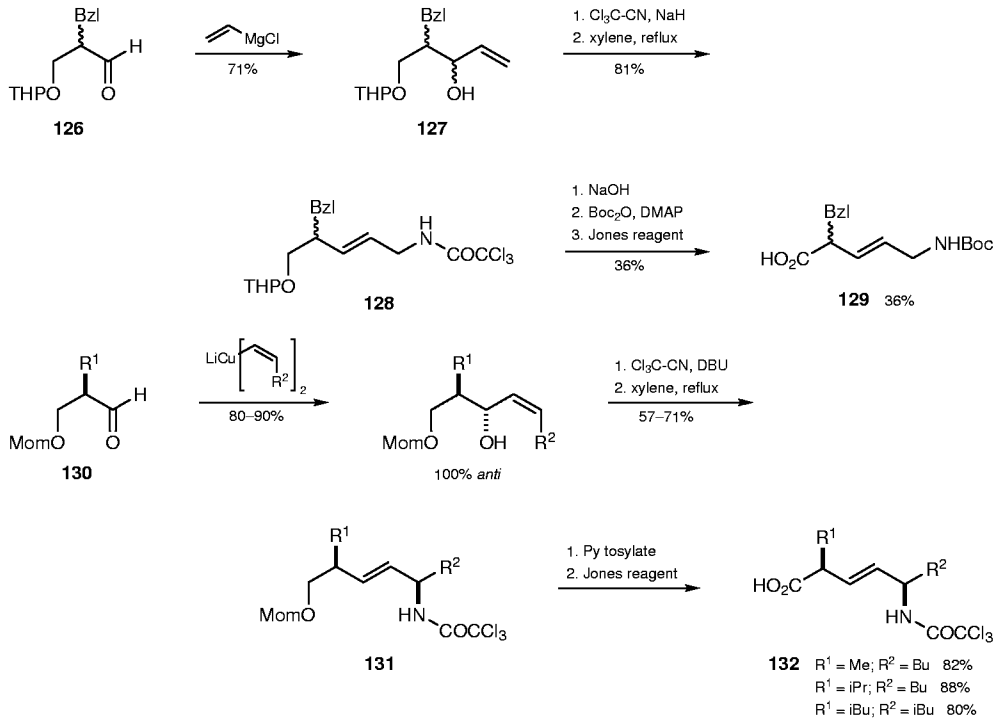
Allylic alcohol **124** ($R^1$ = H; 113 mg, 0.34 mmol), trimethyl orthoacetate (0.21 mL, 1.7 mmol), and benzoic acid (4 mg, 0.033 mmol) were dissolved in *o*-xylene (5 mL) and refluxed for 4 h, when a small amount of starting material was still present by TLC. Additional trimethyl orthoacetate (0.21 mL, 1.6 mmol) and benzoic acid (4 mg, 0.033 mmol) were added and the solvents removed. The residue was flash chromatographed (silica gel, hexanes/EtOAc 5:1) to provide the *trans*-alkene **125**; yield: 81 mg (55%).

### 10.5.2.1.4.3 Overman Rearrangement

Among the first reported synthetic methods for alkene isosteres, a sigmatropic rearrangement of oxidatively activated allylic selenides to provide Boc-protected allylic amines was used for the synthesis of the D,L-Tyrψ[*E*, CH=CH]Gly isostere.[71] The method resulted in a racemic dipeptide isostere, and only a Gly residue at the C-terminus is possible. It is no longer competitive compared with more recent methods using rearrangement of allylic trichloroacetimidates.

A synthesis of racemic Boc-Glyψ[*E*, CH=CH]D,L-Phe-OH (**129**) using the Overman rearrangement has been reported (Scheme 27).[29] However, the full stereochemical potential has been exploited by setting up the required stereochemistry of the [3,3]-sigmatropic rearrangement of trichloroacetimidates in the C-terminal part. *Z*-Vinylic cuprates were found to react with enantiopure β-alkoxy aldehydes having a strongly chelating O-protecting group (Mom) with 100% *anti* selectivity.[117] Conversion to the acetimidates and subsequent sigmatropic rearrangement gave single diasteromerically pure *E*-alkenes **131**, which were further transformed into the β,γ-unsaturated acids **132** in high yield. The required 2-substituted 3-hydroxypropanals **130** are readily accessible by $BF_3$-promoted ring opening of homochiral styrene oxides by lithium cyanocuprates, followed by oxidative cleavage of the aromatic ring.[52]

**Scheme 27** Allylic Trichloroacetimidate Rearrangement[29]




**4-Benzyl-5-(tetrahydro-2*H*-pyran-2-yloxy)pent-1-en-3-ol (127):**[29]

To a stirred soln of **126** (10 g, 34.2 mmol) in $Et_2O$ (250 mL) at −78 °C, a soln of 1 M DIBAL-H in toluene (35 mL, 35 mmol) was added using a syringe pump at the rate of 1.6 mL $min^{-1}$. Stirring was continued for another 45 min. Then, 1 M $H_2C$=CHMgBr in THF (40 mL, 40 mmol) was added rapidly to the mixture, which was warmed to 0 °C over 30 min. Then excess sat. $NH_4Cl$ was added. The mixture was filtered through Celite and the organic phase dried and concentrated. The residue was chromatographed (25% EtOAc/hexane) to give **127** as a thick liquid; yield: 6.7 g (71%). Though the TLC of this material seemed to indicate a homogeneous substance, it was a mixture of stereoisomers as shown by NMR; $R_f$ 0.26 (EtOAc/hexane 3:17), $R_f$ 0.53 (EtOAc/hexane 3:7); HRMS (CI) *m/z*: 277.1791 $[M+H]^+$, calcd. for $C_{17}H_{24}O_3$ 277.1803.

***N*-[(2*E*)-4-Benzyl-5-(tetrahydro-2*H*-pyran-2-yloxy)pent-2-enyl]-2,2,2-trichloroacetamide (128):**[29]

A 60% mineral oil suspension of NaH (126 mg, 5.25 mmol) was washed with hexane and suspended in $Et_2O$ (100 mL) at 0 °C. To this stirred suspension, **127** (6.69 g, 24 mmol) dissolved in $Et_2O$ (30 mL) was added. After 30 min, trichloroacetonitrile (2.5 mL, 25 mmol) was added and the cooling bath removed. After 1 h the solvents were removed in vacuo. The residue was shaken with hexane (100 mL) containing MeOH (125 mL) and filtered (gravity). The filtrate was concentrated in vacuo to give the imidate which was found to be pure by $^1H$ NMR analysis. This material was taken up in xylene (400 mL) and refluxed for 2 h. The solvent was removed in vacuo. A soln of the residue in 20% EtOAc/hexane was filtered through a short column of silica gel. The filtrate was concentrated to give **128** as a thick gum; yield: 8.3 g (81%); $R_f$ 0.33 (EtOAc/hexane 3:17), $R_f$ 0.54 (EtOAc/hexane 3:7).

**(3*E*)-2-Benzyl-5-[(*tert*-butoxycarbonyl)amino]pent-3-enoic Acid (Boc-Gly[*E*-CH=CH]Phe-OH) (129):**[29]

A mixture of **128** (12.3 g, 29 mmol) and 6 M NaOH (148 mL) in 95% EtOH (155 mL) was stirred at rt for 48 h. The mixture was concentrated in vacuo to remove most of the EtOH and the residue was extracted ($Et_2O$). The $Et_2O$ extract was dried and concentrated in vacuo. The residue was chromatographed (EtOAc/hexane containing 2% TEA 1:1) to give (2*E*)-4-benzyl-5-(tetrahydro-2*H*-pyran-2-yloxy)pent-2-enamine as a thick pale yellow oil; yield: 5.3 g (66%); $R_f$ 0.45 (2% TEA in EtOAc/hexane 1:1).

A soln of this amine (5.22 g, 19 mmol) and $(Boc)_2O$ (4.2 g, 19 mmol) in $CHCl_3$ (100 mL) containing a trace of DMAP was allowed to stand at rt for 16 h. The solvent was removed in vacuo and the residue was taken up in $Et_2O$. The ether extract was washed ($H_2O$), dried, and concentrated. An $Et_2O$ soln of the

*for references see p 369*

residue was filtered through a short column of silica gel and the filtrate concentrated to give *tert*-butyl (2*E*)-[4-benzyl-5-(tetrahydro-2*H*-pyran-2-yloxy)pent-2-enyl]carbamate as a yellowish gum; yield: 6.56 g (92%); $R_f$ 0.59 (EtOAc/hexane 3:17).

To a cold (5 °C) soln of 5.8 g of $CrO_3$ in $H_2O$ (30 mL), concd $H_2SO_4$ (5 mL) was added. The thus-prepared Jones reagent was cooled to 0 °C and added to a stirred soln of the carbamate described above (6.52 g) in acetone (350 mL) at 0 °C. The mixture was stirred for 3 h and added to excess $H_2O$ and $Et_2O$. The aqueous phase was separated and extracted with $Et_2O$ (5×). The combined organic extract was concentrated in vacuo to a volume of 100 mL and extracted with 10% NaOH (200 mL, 5×). The organic extract was saved (extract A). The combined basic extract was neutralized with 1 M HCl and extracted with $Et_2O$. This extract was dried and concentrated to give acid **129** as a pale yellow oil; yield: 1.58 g (60%) [the original neutral extract (extract A) was concentrated and the residue chromatographed to give 3.25 g of starting material]; $R_f$ 0.47 (EtOAc/hexane 1:1), $R_f$ 0.53 (EtOAc/hexane 3:2).

## 10.5.2.2 ψ[*Z*, CH=CH] Isosteres

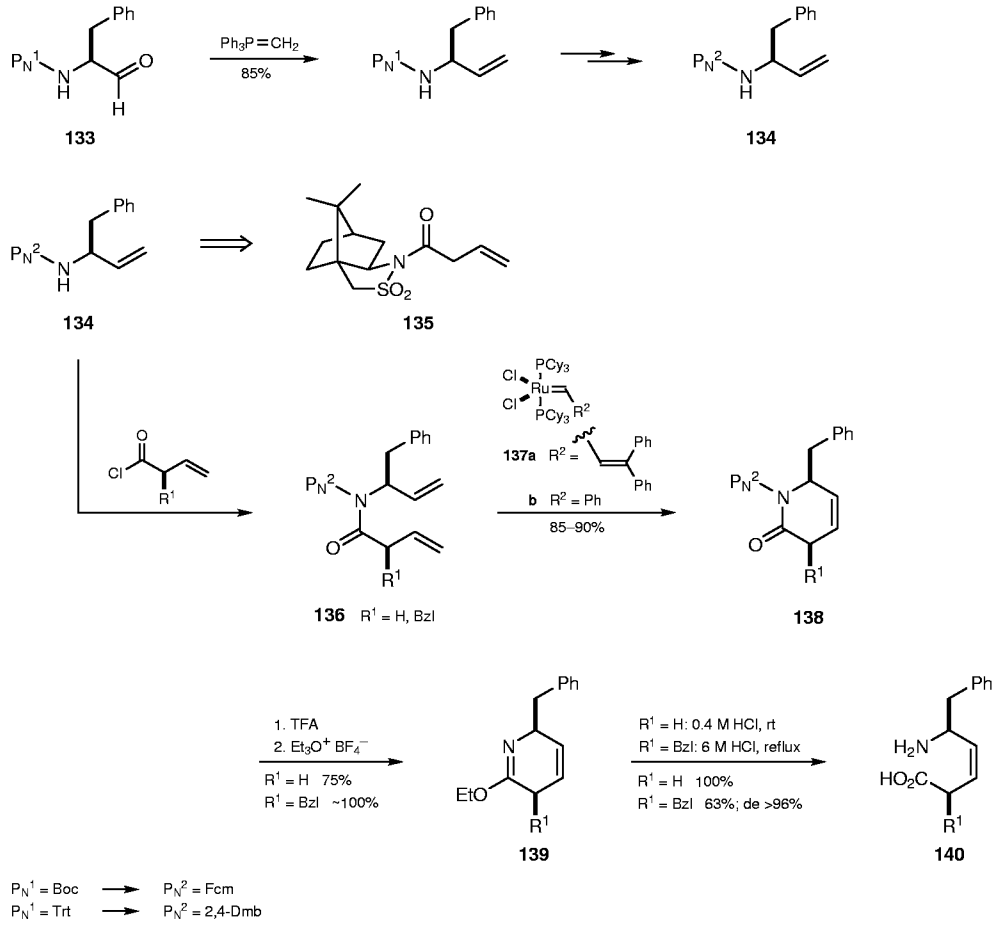
The *Z*-alkene isostere mimics the *cis*-amide bond conformation. Its molecular volume and log *p* values are almost identical to those of the *E*-isostere (Table 1). Its use in bioactive peptides is very limited, probably because of the synthetic challenge involved. One study reports the migration of the double bond to the α,β-position in an enkephalin analogue.[4] The *Z*-alkene in the Alaψ[*Z*, CH=CH]Pro dipeptide isostere, however, was reported to be stable towards isomerization.[118] *ortho*-Substituted aromatic or tetrazole rings have been used more frequently as *cis*-amide bond mimics.

Although *Z*-alkene isosteres have been obtained from Wittig alkenation reactions of α-amino aldehydes using triphenyl[3-(trimethylsilyl)prop-2-ynylidene]phosphorane (Section 10.5.2.1.2.1), this stereoisomer was always obtained in minor amounts. *Z*-Selective alkenations were obtained using ylides containing dioxolane-protected aldehydes[42] or orthoester-protected carboxylic acid functions.[58] No experimental data were published, however.

### 10.5.2.2.1 Ring-Closing Metathesis

A convergent approach towards *Z*-ethylenic isosteres has been proposed, based on a metathesis reaction using Grubbs ruthenium catalyst. The bulky ferrocenylmethyl (Fcm) group was used for N-protection to induce a maximum of *cis* configuration of the peptide bond in **136** (Scheme 28). Fcm-allylic amine **134** ($P_N^2$ = Fcm) was prepared from Boc-L-Phe aldehyde **133** using Wittig alkenation, followed by a change of N-protective group. Acylation of **134** with but-3-enoyl chloride gave **136** ($R^1$ = H) in 95% yield, which was submitted to the metathesis reaction to give lactam **138** ($R^1$ = H) in 85–92% yield. After removal of the Fcm group the lactam was hydrolyzed by conversion into the lactim ether, followed by treatment under mild acidic conditions that also resulted in hydrolysis of the ester. An overall yield of the *Z*-alkene isostere **140** of 35% was obtained. Unfortunately, during the Boc to Fcm transprotection step, significant racemization occurred, lowering the ee to 62%. For that reason the Fcm protection was changed for the 2,4-dimethoxybenzyl (2,4-Dmb) group.[119] Compound **134** ($P_N^2$ = 2,4-Dmb) was reacted with (*R*)-2-benzylbut-3-enoyl chloride, which was prepared by regio- and diastereoselective (de >99%) alkylation of (2*R*)-bornana-10,2-sultam **135**. Ring-closing metathesis of the disubstituted amide **136** ($P_N^2$ = 2,4-Dmb; $R^1$ = Bzl) required more drastic reaction conditions (68 °C, 12 h vs. rt, 6.5 h for $R^2$ = H) and another ruthenium catalyst **137b** ($R^2$ = Ph), and led to 100% conversion into one single stereoisomer. After removal of the Dmb protection (neat TFA, reflux, 5 min), and treatment with triethyloxonium tetrafluoroborate, **139** ($R^1$ = Bzl) was obtained in quantitative yield. Its hydrolysis to the Phe[*Z*, CH=CH]Phe isostere **140** also needed more drastic conditions.[120]

**Scheme 28** Ring-Closing Metathesis




**Metathesis Reaction:**[121,122]

**(*S*)-6-Benzyl-1-(ferrocenylmethyl)-3,6-dihydropyridin-2(1*H*)-one** (138, $P_N^2$ = Fcm; $R^1$ = H):

In a Schlenk tube in an argon atmosphere, the substrate **136** ($P_N^2$ = Fcm; $R^1$ = H), 10% of Grubbs ruthenium catalyst **137a** ($R^2$ = CH=CPh$_2$; 0.496 g, 0.12 mmol), and benzene (10 mL) were added successively. The reaction was allowed to proceed at rt for 6.5 h. Then the solvent was removed in vacuo and the residue was directly purified by chromatography (silica gel, $Et_2O$/pentane 4:6) to give **138** ($P_N^2$ = Fcm; $R^1$ = H) as an orange solid; yield: 90%; mp 68–69 °C; $[\alpha]_D$ +238 (*c* 7, $CH_2Cl_2$); MS *m/z* (relative intensity): 385 ($M^+$, 51%), 199 (100%), 121 (30%).

**(3*Z*,5*S*)-5-Amino-6-phenylhex-3-enoic Acid** (140, $R^1$ = H):[121,122]

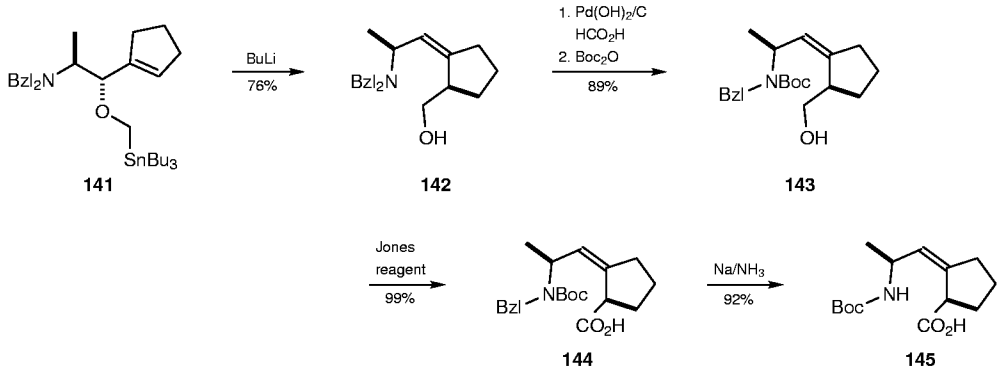
To a soln of dihydropyridone **138** ($R^1$ = H; $P_N^2$ = H; 0.3 g, 1.60 mmol) in $CH_2Cl_2$ (6 mL), $Et_3O^+BF_4^-$ (0.38 g, 2 mmol) was added in an argon atmosphere. The mixture was stirred for 30 min at rt. Then the temperature was decreased to 5 °C and the soln washed successively with a phosphate buffer (6 mL of a soln of 0.3 mol $NaH_2PO_4$ and 0.8 mol of $Na_2HPO_4$ in 800 mL $H_2O$) and with $H_2O$ (2×6 mL). $CH_2Cl_2$ was removed in vacuo. A mixture of lactim ether **139** ($R^1$ = H, 75%) and starting material (25%) was obtained (conversion was 100% but partial hydrolysis occurred in the phosphate buffer). It was used immediately in the hydrolysis reaction by adding 0.38 M HCl (2 equiv) and stirring at rt for 8 h. The mixture was then purified by chromatography (Dowex-50-X8 column, 24 mL resin, 1 M NaOH). The solid product was washed several times with $CH_2Cl_2$ to give **140** ($R^1$ = H); yield: 0.245 g (100%); $[\alpha]_D$ −10 (*c* 1, 0.1 M HCl); MS *m/z*: 206 ($M^+$+1).

*for references see p 369*

### 10.5.2.2.2 Wittig–Still Sigmatropic Rearrangement

The [2,3]-sigmatropic rearrangement of the homochiral cyclopentenyl tributyltin compound **141** gave the *Z*-alkene **142** as a single isomer (Scheme 29),[118] which is in contrast to the corresponding open-chain compounds which gave predominantly the *E*-alkenes (e.g., see **118**, Scheme 25). The removal of the benzyl groups in **142** could not be accomplished in a single step, but required a double-step procedure. An overall yield of 49% was obtained for the eight-step synthesis starting from dibenzyl Ala-*O,N*-dimethylhydroxamate.

**Scheme 29** Wittig–Still Sigmatropic Rearrangement[118]




**(1*R*,2*Z*)-2-[(2*S*)-2-(Dibenzylamino)propylidene]cyclopentanemethanol** (142):[118]
Stannane **141** (1.74 g, 2.8 mmol) (obtained from the corresponding allylic alcohol using $Bu_3SnCH_2I$, KH, and 18-crown-6, using the procedure described in Section 10.5.2.1.4.1) was dissolved in THF (50 mL). BuLi (2.6 M, 1.34 mL, 3.34 mmol) was added slowly and stirred for 11 h. The mixture was quenched with MeOH (10 mL), diluted with EtOAc (100 mL), washed with sat. $NH_4Cl$ (3×25 mL) and brine (25 mL), dried ($MgSO_4$), and concentrated. Chromatography silica, (10% EtOAc/petroleum ether) gave **142** as an off-white glassy solid; yield: 709 mg (76%); CI-MS *m/z*: 336 ($M^+ + 1$); $[\alpha]_D^{20}$ −4.3 (*c* 1, MeOH).

**(1*R*,2*Z*)-2-[(2*S*)-2-[Benzyl(*tert*-butoxycarbonyl)amino]propylidene]cyclopentanemethanol** (143):[118]
MeOH (40 mL) was added under $N_2$ to a flask containing dibenzylamine **142** (770 mg, 2.30 mmol) and 20% $Pd(OH)_2/C$ (200 mg). Formic acid (88% aq, 10 mL) was added and the reaction was stirred for 30 min before filtering through Celite and concentration. The residue was dissolved in $H_2O$ (150 mL) and treated with sat. $NaHCO_3$ until gas evolution ceased. The mixture was extracted ($CH_2Cl_2$, 10×25 mL), dried ($MgSO_4$), and concentrated to yield a clear oil; yield: 500 mg (89%); MS (CI) *m/z*: 246 ($M^+ + 1$).
This monobenzylamine (50 mg, 0.2 mmol) was dissolved in $CH_2Cl_2$ (20 mL) and $(Boc)_2O$ (111 mg, 0.51 mmol) was added and stirred for 12 h. The mixture was concentrated. Chromatography (silica gel, 15% EtOAc/petroleum ether) yielded **143** as a clear oil; yield: 67 mg (99%); CI-MS *m/z*: 346 ($M^+$+1); $[\alpha]_D^{20}$ −23.1 (*c* 1, MeOH).

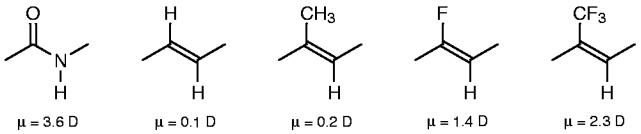
**(1*R*,2*Z*)-2-[(2*S*)-2-[Benzyl(*tert*-butoxycarbonyl)amino]propylidene]cyclopentanecarboxylic Acid** (144):[118]
Alcohol **143** (360 mg, 1.04 mmol) was dissolved in acetone (20 mL) and cooled in an ice bath. Jones reagent (2.7 M $CrO_3/H_2SO_4$ in acetone/$H_2O$, 970 µL, 2.61 mmol) was added and the reaction was stirred for 30 min. iPrOH (5 mL) was added to quench the Jones reagent and allowed to stir for 5 min. After diluting with $H_2O$ (30 mL) the mixture was extracted (EtOAc, 5×5 mL). The organic layer was washed with $H_2O$ (25 mL), dried ($MgSO_4$), and concentrated to yield **144** as a colorless oil; yield: 339 mg (99%); CI-MS *m/z*: 360 ($M^+ + 1$); $[\alpha]_D^{20}$ 60.8 (*c* 1, MeOH).

**(1*R*,2*Z*)-2-[(2*S*)-2-(*tert*-Butoxycarbonylamino)propylidene]cyclopentanecarboxylic Acid** (145):[118]
Boc-benzylamine **144** (230 mg, 0.64 mmol) was dissolved in THF (10 mL) and cooled to −78 °C. $NH_3$ (20 mL) was distilled into the soln and allowed to reflux for 40 min. The reaction was quenched with sat. $NH_4Cl$ (5 mL) and allowed to warm to rt while the $NH_3$ slowly evaporated. After the mixture reached rt, it was added to $H_2O/CH_2Cl_2$ (1:1, 50 mL) and treated with 1 M HCl until the aqueous layer was <pH 2. The mixture was extracted ($CH_2Cl_2$, 5×25 mL), dried ($MgSO_4$), and concentrated. Chromatography (silica, 30% EtOAc/petroleum ether) yielded **145** as a colorless oil; yield: 159 mg (92%); CI-MS *m/z*: 270 ($M_+$+1); $[\alpha]_D^{20}$ −106.3 (*c* 1, MeOH).

### 10.5.2.3 ψ[*E*, CX=CY] Isosteres

Semi-empirical energy calculations of model alkene dipeptide isosteres have indicated an increased flexibility compared to the parent peptide.[5,6] This flexibility resulted from the replacement of the oxygen by the smaller hydrogen. Introduction of a methyl substituent at this position was predicted to result in a strong type-I β-turn tendency.[5] The presence of two methyl substituents on the alkene was expected to disfavor extended conformations.[123] The incorporation of the alkene moiety into a cyclohexene ring has been proposed to lead to type-I and type-II β-turn mimetics.[124] A comparison of the NMR parameters of the tetrapeptide isostere Ac-Ala-D,L-Pheψ[*E*, CX=CH]Gly-Ala-$NH_2$ with (X = Me) and without (X = H) methyl substitution did not indicate any difference in conformational rigidity.[125] On the other hand, an IR absorption study on iBu-Gly[*E*, CX=CY]Gly-NHiPr did show a β-turn conformation for the tetrasubstituted (X = Y = Me) alkene, but not for the disubstituted (X = Y = H) one.[123] X-ray structural analysis of Boc-Alaψ[*E*, CX=CH]D-Ala-NHMe revealed an opened bend structure for the disubstituted analogue (X = H) and an ideal type-II′ β-turn for the methyl-substituted analogue (X = Me).[9] Modeling data indicate an even greater tendency for β-turn formation for the (trifluoromethyl)alkene derivative (X = $CF_3$). The latter represents a better match of the amide bond electrostatic properties than any other common alkene isostere, including the fluorine-substituted alkene (Scheme 30).[126–128]

**Scheme 30** AM1-Calculated Dipole Moments[126–128]



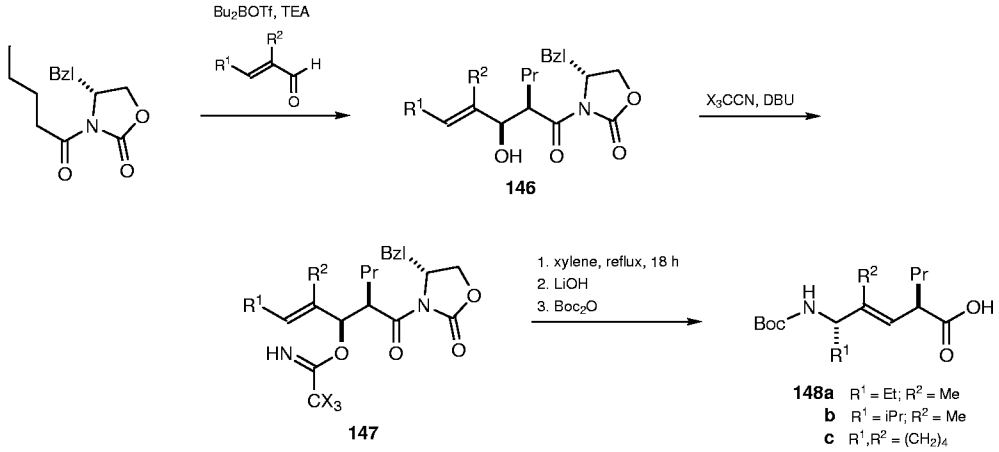

For the construction of the methyl-substituted alkene, the Wittig alkenation method described in Section 10.5.2.1.2.1 has been used, starting from the Boc-Phe-methyl ketone.[125] The more nucleophilic diethyl [3-(trimethylsilyl)prop-2-ynyl]phosphonate gave the best yield (57%) of a mixture of *Z*- and *E*-enynes (81:19). A major limitation, however, was extensive racemization, resulting in an ee of only 48% of the final dipeptide isostere. The methods described below are therefore preferable.

#### 10.5.2.3.1 Overman Rearrangement

Evans asymmetric aldol reaction was used to set up the absolute stereochemistry of the two chiral centers in **146** (Scheme 31).[129] The stereochemistry at the β-carbon is then transposed to the δ-carbon during the [3,3]-sigmatropic rearrangement. The Boc-protected dipeptide isosteres **148** were isolated in 55–67% yield in eight steps without any detectable level of racemization at the α-carbon. For the more sterically hindered **148b**, the use of trifluoroacetonitrile was necessary to allow the hydrolysis of the trifluoroacetamide formed. The ring-constrained analogue **148c** was also prepared this way.

*for references see p 369*

**Scheme 31** Overman Rearrangement[129]



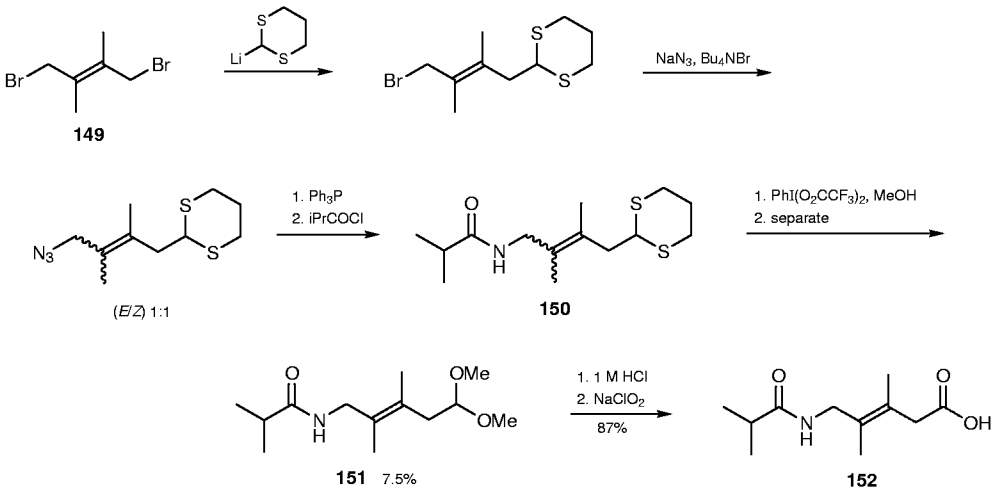

**(1*R*,2*E*)-1-((1*R*)-1-{[(4*R*)-4-Benzyl-2-oxooxazolidin-3-yl]carbonyl}butyl)-2,4-dimethylpent-2-enyl Trifluoroacetimidate** (147, $R^1$ = iPr; $R^2$ = Me; X = F):[129]

In a cold (liquid $N_2$ bath), dry, three-necked round-bottomed flask (50 mL) equipped with a condenser (liquid $N_2$), trifluoroacetonitrile (~20 mL, bp −64 °C) was condensed and solidified onto the bottom of the flask. A soln of alcohol **146** ($R^1$ = iPr; $R^2$ = Me; 2.0 g, 5.35 mmol) and DBU (0.9 mL, 6.02 mmol) in $CHCl_3$ (5 mL) was added dropwise. After addition was complete, the liquid $N_2$ bath was removed and the condenser was charged with dry ice/acetone. The reacting clear bilayer was stirred vigorously under reflux for 6 h. The resultant mixture was concentrated (silica gel, 25% EtOAc/hexane) to provide the product; yield: 1.9 g (85% based on recovery of 0.2 g of starting alcohol **146**).

### 10.5.2.3.2 Substitution of a 1,4-Dibromobut-2-ene

The dibromobutene **149** could be monosubstituted using 2-lithio-1,3-dithiane as a carboxylic acid precursor, followed by substitution of the remaining bromide with sodium azide as the amine precursor (Scheme 32). The azide was reduce under Staudinger conditions and acylated. After separation of the *E*/*Z* isomers, further functional group transformations gave the dimethyl-substituted Gly-Gly alkene isostere **152**.[123]

**Scheme 32** Substitution of a 1,4-Dibromo-2,3-dimethylbut-2-ene



***N*-[(2*E*)-5,5-Dimethoxy-2,3-dimethylpent-2-enyl]-2-methylpropanamide (151):**[123]
A soln of 1,3-dithiane (7.08 g, 59 mmol) in anhyd THF (130 mL) was cooled to −30 °C under $N_2$. A soln of 2.6 M BuLi in hexane (22.5 mL, 59 mmol) was added dropwise over 40 min. After stirring for an additional 80 min the soln was added dropwise over 2 h to a soln of (3*E*)-1,4-dibromo-2,3-dimethylbut-2-ene (**149**, 14.4 g, 59.5 mmol) in anhyd THF (100 mL) at −78 °C. The addition was performed using a cannula. The resulting yellow soln was stirred for 6 h while warming to rt. The mixture was quenched with $H_2O$ (50 mL) and then concentrated to a total volume of ca. 100 mL. The organic layer was separated and the aqueous layer was extracted ($Et_2O$, 2×60 mL). The combined organic layers were concentrated in vacuo and the semisolid residue was taken up in $H_2O$ (50 mL). This mixture was treated with $NaN_3$ (4.36 g, 67 mmol) and TBAB (1.9 g, 5.9 mmol), and the two-phase mixture was stirred at 80 °C for 10 h. The mixture was cooled and the organic phase was separated. The aqueous layer was extracted ($Et_2O$, 50 mL) and the combined organic layers were evaporated to yield a brown viscous liquid. This material was taken up in THF (170 mL) and $H_2O$ (30 mL), and then was treated with $Ph_3P$ (23.2 g, 88.5 mmol) and stirred at rt for 30 h. THF was removed in vacuo and the resulting material was mixed with $CHCl_3$. This soln was extracted (1 M HCl, 4×50 mL) and the combined acid extracts were carefully treated with 10 M NaOH until the pH of the soln was >10, which resulted in the formation of a dark organic layer at the top of the mixture. The two-phase mixture was extracted ($Et_2O$, 5×50 mL) and the combined organic layers were dried ($Na_2SO_4$) to yield the amine as a mixture of isomers; yield: 4.32 g (34% crude yield from the dibromide). The *E*/*Z* isomer ratio was ~1:1. The isomeric mixture was carried on to the next reaction without further purification or characterization.
A stirred soln of isomeric amines (4.32 g, 19.9 mmol) in anhyd THF (120 mL) under $N_2$ at 0 °C was treated with TEA (4.2 mL, 29.9 mmol) and isobutyryl chloride (3.11 mL, 29.9 mmol). A white solid precipitated upon addition of the isobutyryl chloride. After 30 min, the reaction mixture was suction filtered, and the filtrate was concentrated to a solid in vacuo, yielding crude amide **150**; yield: 6.34 g. The material contained the *E*- and *Z*-alkene isomers (1:1) and was carried on to the next reaction without purification or characterization.
A soln of crude **150** (6.34 g, 22.1 mmol) in anhyd MeOH (50 mL) was treated with small portions of [bis(trifluoroacetoxy)]iodobenzene (14.25 g, 33.1 mmol). All solids dissolved quickly and a slight exothermic reaction resulted upon the addition. After 30 min stirring under $N_2$ the reaction was quenched with 1 M $NaHCO_3$ and then extracted ($Et_2O$, 3×80 mL). The combined organic layers were dried ($Na_2SO_4$) and then concentrated. The crude product was purified by chromatography (silica gel, 15% EtOAc/hexanes). The second isomer to elute was the *E*-isomer **151** as a white solid; yield: 1.07 g (7.5% from dibromide); mp 60–61.5 °C; IR (1 mM in $CH_2Cl_2$) $\tilde{\nu}$: 3442 (w), 2930 (m), 2873 (w), 2856 (w), 1668 (s), 1511 (s), 1447 (m), 1368 (m), 1223 (w), 1200 (w), 1119 (s), 1073 (w), 1014 $cm^{-1}$ (w); EI-MS *m*/*z*: 243.1834, calcd. for $C_{13}H_{25}NO_3$ 243.1884.

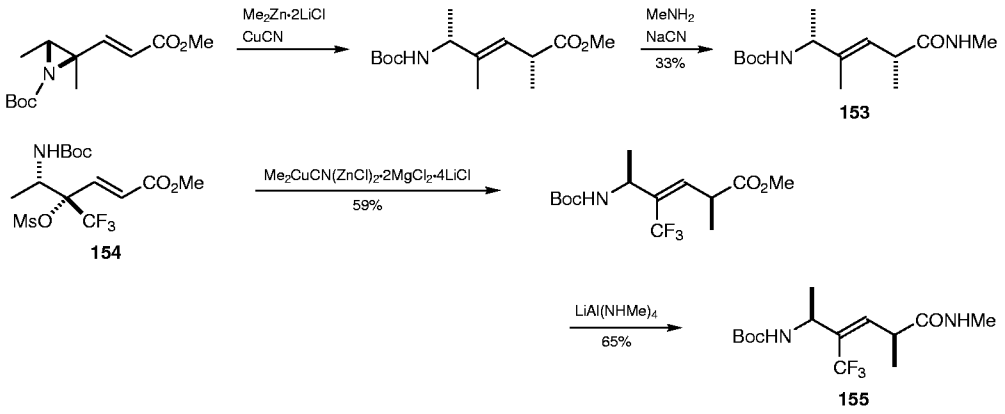
**(3*E*)-5-(Isobutyrylamino)-3,4-dimethylpent-3-enoic Acid (iPrCO-Gly$\psi$[*E*, C(Me)=C(Me)]Gly-OH) (152):**[123]
A soln of crude acetal **151** (97 mg, 0.4 mmol) in THF (1.4 mL) was treated with 1 M HCl (1.4 mL, 1.4 mmol). After stirring for 30 min the mixture was extracted with $CHCl_3$ (5×5 mL). The combined organic layers were dried ($Na_2SO_4$) and yielded 76.3 mg of aldehyde (79% crude yield) as a solid. This material was used without further purification or characterization.
A soln of the crude aldehyde (76.3 mg, 0.38 mmol) in tBuOH (2 mL) and phosphate buffer (2 mL, 1 M, pH 4) was treated with 2-methylbut-2-ene (0.1 mL, 0.94 mmol) and $NaClO_2$ (36 mg, 0.40 mmol), and was stirred at rt for 30 min. The mixture was extracted ($CHCl_3$, 10×5 mL) and the combined organic layers were dried ($Na_2SO_4$) to yield crude **152** as a yellow viscous liquid; yield: 74 mg (87%).

### 10.5.2.3.3 $S_N2'$ Substitutions

The *anti*-$S_N2'$ substitution of allylic aziridines (see Section 10.5.2.1.3.2) has been extended to the synthesis of the methyl-substituted D-Ala-Ala alkene isostere **153** (Scheme 33).[9] For the corresponding (trifluoromethyl)alkene isostere, the aziridine precursor could not be formed. Instead, allylic $S_N2'$ displacement (see Section 10.5.2.1.3.1) of the mesylate **154** occurred readily to provide the Ala-D-Ala isostere **155**.

*for references see p 369*

**Scheme 33** $S_N2'$ Substitutions[9]




**(2*R*,3*E*,5*R*)-5-[(*tert*-Butoxycarbonyl)amino]-*N*,2,4-trimethylhex-3-enamide (Boc-D-Alaψ[*E*, C(Me)=CH]Ala-NHMe) (153):**[9]

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

An aliquot of a soln of 1.37 M $ZnCl_2$ in THF (1.3 mL, 1.78 mmol) was diluted with THF (1.5 mL) and cooled to −78 °C. After addition of a soln of 0.9 M MeLi in $Et_2O$ (4.2 mL, 3.78 mmol), the mixture was warmed to 0 °C for 10 min and then cooled to −78 °C. To this mixture, CuCN (32 mg, 0.36 mmol) was added quickly, and after 5 min a soln of the *N*-Boc-aziridine (120 mg, 0.47 mmol) in THF (2.5 mL) was added. The resulting mixture was stirred at −78 °C for 15 min and at −20 °C for 45 min. The mixture was quenched with sat. aq $NH_4Cl/NH_4OH$ soln (1:1, 1.5 mL) and extracted ($Et_2O/CH_2Cl_2$ 4:1, 3×15 mL). The combined extracts were washed with $H_2O$ (20 mL) and brine (20 mL), dried ($MgSO_4$), and concentrated to give a ~3:1 mixture of C4-methylated and reduced products as a colorless oil; yield: 106 mg. The crude product mixture was dissolved in a soln of 3.2 M $MeNH_2$ in MeOH (7 mL), NaCN (10 mg, 0.20 mmol) was added, and the soln was stirred for 48 h. Concentration gave a residue which was taken up in $H_2O$ (20 mL) and extracted ($CH_2Cl_2$, 1×20 mL, 2×10 mL). The combined organic extracts were washed with brine (20 mL), dried ($MgSO_4$), and evaporated to yield an oil. Purification by chromatography (silica gel, 50–75% EtOAc/hexanes) yielded 14 mg (12%) of the C4-reduced compound and 42 mg (33%) of **153**; $[\alpha]_D^{23}$ −10.4 (*c* 0.2, $CHCl_3$); mp 112–114 °C.

**(3*Z*)-5-[(*tert*-Butoxycarbonyl)amino]-*N*,2-dimethyl-4-(trifluoromethyl)hex-3-enamide (Boc-Alaψ[*E*, C($CF_3$)=CH]D-Ala-NHMe):**[9]

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

To a soln of LiCl (76 mg, 1.79 mmol) in THF (3 mL) at 0 °C, a 0.5 M $ZnCl_2$ soln in THF (1.8 mL, 0.90 mmol) was added followed by a soln of 3.0 M MeMgCl in THF (0.30 mL, 0.90 mmol). The resulting clear, colorless soln was stirred at 0 °C for 30 min, CuCN (40 mg, 0.45 mmol) was added, and stirring was continued for 30 min at 0 °C. To the resulting colorless soln, mesylate **154** (51 mg, 0.13 mmol) in THF (2 mL) was added. After 30 min at 0 °C, the soln was cooled to −78 °C and quenched with sat. $NH_4OH$ (1.5 mL). After allowing the mixture to warm to rt, it was extracted ($Et_2O$, 3×10 mL) and the combined extracts were washed with $H_2O$ (10 mL) and brine (10 mL), dried ($MgSO_4$), and concentrated. Purification by chromatography (silica gel, hexanes/$Et_2O$ 2:1) provided the $S_N2'$ product as a colorless oil; yield: 24 mg (59%); $[\alpha]_D^{24}$ −8.2 (*c* 1.2, $CHCl_3$).

## 10.5.3 ψ[C≡C] Peptides

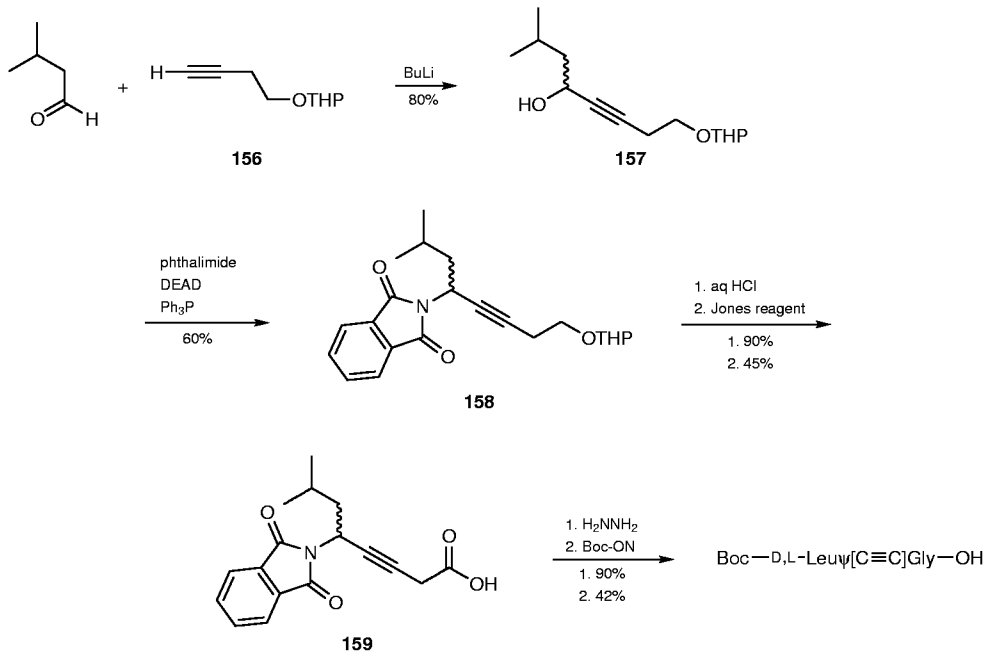
The linearity of the alkyne function results in a substantial different local geometry from that in the normal peptide bond. Nevertheless, computer modeling had indicated that an acetylene pseudodipeptide provides the correct $C^{\alpha}$ to $C^{\alpha}$ distance of 4.5–5.3 Å typical for β-turns. NMR and CD studies on an octapeptide containing the Glyψ[C≡C]Gly pseudodipeptide did indicate that a β-turn conformation was not significantly populated.[10] Since turns containing Gly residues at both the $i+1$ and $i+2$ positions are relatively rare, it is expected that 5-aminopent-3-ynoic analogues having C2 and C5 substituents will have higher tendencies to form β-turns. The possibility for introducing side chains in the reverse turn analogue is a feature that is not available to other turn inducers containing aromatic moieties.

The but-3-ynoic acid moiety which is present in this isostere may constitute the reactive part in enzyme-activated irreversible inhibitors.[130]

### 10.5.3.1 Mitsunobu Reaction

Addition of the lithium acetylide of tetrahydropyranyl-protected but-3-yn-1-ol **156** provided the racemic alcohol **157** (Scheme 34). The nitrogen was introduced through a Mitsunobu reaction, followed by oxidation of the primary alcohol to the carboxylic acid and a change of the phthaloyl protecting group to Boc protection. The latter reaction was necessary because hydrazinolysis of the C-terminal amide analogue of **159** did result in deeply red-colored mixtures, indicating that phthaloyl removal by this method should occur prior to peptide synthesis.[131]

**Scheme 34** Mitsunobu Reaction[131]




**7-Methyl-5-phthalimido-1-(tetrahydro-2*H*-pyranyloxyoct-3-yn-1-ol** (158)**:**[131]
A soln of **157** (16.8 g, 70 mmol), phthalimide (10.73, 73 mmol), and $Ph_3P$ (20.29 g, 73 mmol) was heated to reflux in dry THF (150 mL), and DEAD (12.7 g, 73 mmol) was added dropwise. A very exothermic reaction occurred and reflux was continued for 16 h. The THF was removed and dry $Et_2O$ (250 mL) was

*for references see p 369*

added. Any residue of $Ph_3PO$ was removed by filtration. Alumina was added to the filtrate and stirring was continued for 1 h to decompose residual DEAD. The suspension was filtered again, the $Et_2O$ was concentrated to half its volume, and cooled in the refrigerator. The precipitated $Ph_3PO$ was filtered off. By adding petroleum ether (bp 40–60 °C) the remaining $Ph_3PO$ could be precipitated. The solvent was then removed and the mixture chromatographed (silica gel, benzene/$CHCl_3$ gradient). After a small $Ph_3P$ fraction, **158** was collected as a colorless oil; yield: 15.5 g (60%); $R_f$ 0.30 ($CHCl_3$); FAB-MS *m/z* (relative intensity): 369 ($M^+$, not observed), 268 (0.05%), 174 (1%), 160 (2%), 130 (1.5%).

**7-Methyl-5-phthalimidooct-3-ynoic Acid** (159):[131]

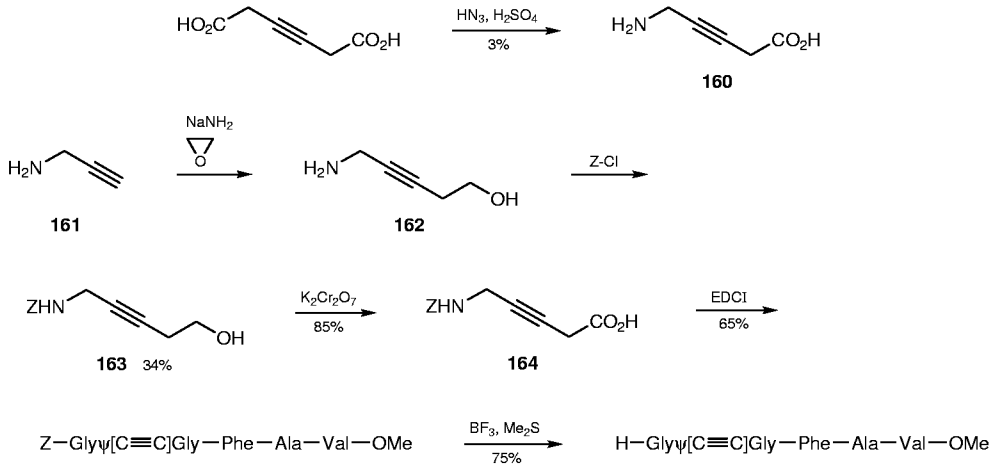
A soln of **158** (36.9 g) in MeOH (330 mL) was stirred overnight with 10% aq HCl (110 mL). The MeOH was removed and the residue was extracted ($Et_2O$). The organic phase was washed with sat. $NaHCO_3$ and $H_2O$, dried ($MgSO_4$), and concentrated to yield the alcohol as an oil; yield: 27.10 g (95%); $R_f$ 0.33 ($CHCl_3$/15% EtOAc); MS (FAB) *m/z* = 286 (13%, $M^+ + 1$).

The alcohol (5.96 g, 21 mmol) was dissolved in acetone (100 mL), Jones reagent [1.50 equiv, prepared from $CrO_3$ (3.15 g), $H_2SO_4$ (5.04 g), and $H_2O$ (15 mL)] was added while maintaining the reaction temperature at −10 °C in an ice/salt bath. The cooling bath was removed, and the reaction was carried on for 2 h at rt. The soln was then diluted with $H_2O$, the acetone was removed, and the residue was extracted ($Et_2O$). The combined $Et_2O$ phases were then extracted (sat. $NaHCO_3$ soln). The $NaHCO_3$ phases were acidified with 20% aq HCl and extracted ($Et_2O$). After drying ($MgSO_4$) and removal of the solvent, a crude yield of 3.43 g (55%) of **159** was obtained. Crystallization ($Et_2O$/petroleum ether) yielded pure **159**; yield: 2.79 g (45%); mp 102–103 °C; $R_f$ 0.25 ($CHCl_3$/EtOAc 8:2); MS (FAB) *m/z*: 300 (20%, $M^+ + 1$).

## 10.5.3.2 Acetylide Addition to Oxirane

The synthesis of the Gly-Gly pseudodipeptide 5-aminopent-3-ynoic acid (**160**; Scheme 35) was first reported by Gani and co-workers[10] using a Schmidt reaction. The yield was extremely low, however. The acetylide of unprotected propargylamine with ethylene oxide in liquid ammonia gave amino alcohol **162**, which after Z-protection was oxidized to the Z-protected acetylene dipeptide analogue **164**.[10] Coupling of this dipeptide analogue to the tetrapeptide H-Phe-Ala-Val-OMe using mixed isobutyl carbonic anhydride resulted in rearrangement of the acetylenic moiety. Using EDC, the required peptide was obtained in good yield. Also the Z-deprotection required careful optimization to avoid destruction of the acetylenic moiety. This method could provide an entry to substituted dipeptide acetylene isosteres, for instance by starting from chiral propargylic amines for which good preparations are available.[14,132,133]

**Scheme 35** Acetylide Addition to Oxirane[10]



**5-[(Benzyloxycarbonyl)amino]pent-3-yn-1-ol** (163):[10]
To a stirred soln of liq $NH_3$ (250 mL) at −78 °C under $N_2$, Na metal (1.84 g, 80 mmol) was added, followed by a catalytic amount of $FeCl_3 \cdot 6H_2O$ (40 mg). The cold bath and $N_2$ supply were removed and the mixture was allowed to warm to −25 °C and stirred under reflux for a further 3 h. The soln was then cooled to −78 °C while flushing with $N_2$. Propargylamine (**161**; 4.00 g, 72.60 mmol) was added via a glass syringe to the deep blue-colored soln of $NaNH_2$ at −78 °C and the mixture was again stirred under reflux for a further 2 h at −35 °C. The mixture was cooled to −78 °C and ethylene oxide (6.40 g, 145 mmol) was added and the mixture was stirred under reflux for a further 12 h. The excess $NH_3$ was gradually removed at rt for 3 h. The resulting residue was dissolved in aq 1 M $(NH_4)_2SO_4$ (100 mL) and the solvents were removed under reduced pressure. The residue was dissolved in EtOH (50 mL), filtered, and the solvent was removed under reduced pressure to afford **162** as a crude brown oil which resisted further purification. To a stirred soln of the crude brown oil in aq $Na_2CO_3$ (50 mL, 15.40 g, 0.145 mol) was added Z-Cl (14.82 g, 87.12 mmol) and the mixture was stirred for 15 h at rt. The aqueous phase was washed (EtOAc, 2×30 mL), carefully acidified with aq 6 M HCl, and then extracted (EtOAc, 3×50 ml). The combined organic layers were washed with brine (50 mL) and then dried ($MgSO_4$) and concentrated under reduced pressure to give a brown oil. Purification by flash chromatography (silica gel, 40% EtOAc/hexane) afforded the alcohol **163** as a yellow oil; yield: 8.1 g (48%); HRMS found 234.1128 $[M+H]^+$, calcd. for $C_{13}H_{16}NO_3$ 234.1130; CI-MS *m/z* (relative intensity): 234 (31%, $[M^+ + H]$), 152 (59%, $[PhCH_2CO_2NH_2 + H]^+$) and 91 (100%, $PhCH_2^+$).

**5-[(Benzyloxycarbonyl)amino]pent-3-ynoic Acid** (164):[10]
To a stirred soln of alcohol **163** (0.51 g, 2.19 mmol) in acetone (20 mL) at 0 °C an excess of Jones reagent (1.2 equiv, 3.49 mL) was added. The soln was allowed to warm to rt, stirred for a further 30 min, and then iPrOH (5 mL) was added to the soln. The solvents were removed under reduced pressure and the resulting residue was dissolved in aq 1 M, HCl (5 mL). The aqueous layer was extracted (EtOAc, 3×15 ml) and the combined organic layers were dried ($MgSO_4$) and concentrated under reduced pressure to give a colorless oil. Crystallization (EtOAc/hexane) afforded the pure acid **164** as a white crystalline solid; yield: 0.45 g (83%); mp 85–86 °C; HRMS found 248.0928 $[M^+ + H]$, calcd. for $C_{13}H_{14}NO_4$ 248.0923; CI-MS *m/z* (relative intensity): 248 (100%, $[M^+ + H]$).


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# 10.6 Synthesis of Peptides Containing C/O Amide Bond Replacements

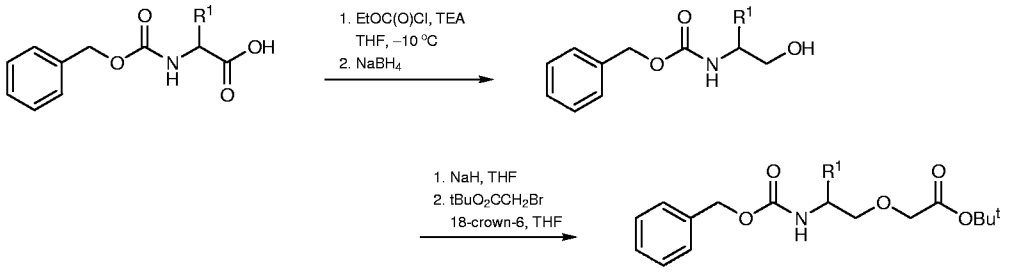
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The C/O amide bond replacements include methyleneoxy, 1-hydroxyethylene, 2-hydroxyethylene, 1,2-dihydroxyethylene, 1-oxoethylene, and epoxide isosteres. This chapter focuses on the synthetic methods of dipeptide building blocks containing these isosteres.

## 10.6.1 ψ[$CH_2O$] (Methyleneoxy) Peptides

One of the earliest syntheses of ψ[$CH_2O$](methyleneoxy) peptide isosteres was reported by Rubini et al.[1,2] These authors used an intermolecular Williamsons reaction between a urethane-protected amino alcohol and *tert*-butyl bromoacetate (Scheme 1).[1,2] However, this method resulted in low yields and was not applicable to bulky amino acids. Thus, only isosteres containing glycine as the C-terminal component could be synthesized by this method.

**Scheme 1** Synthesis of Methyleneoxy Peptides using an Intermolecular Williamson Reaction[1,2]

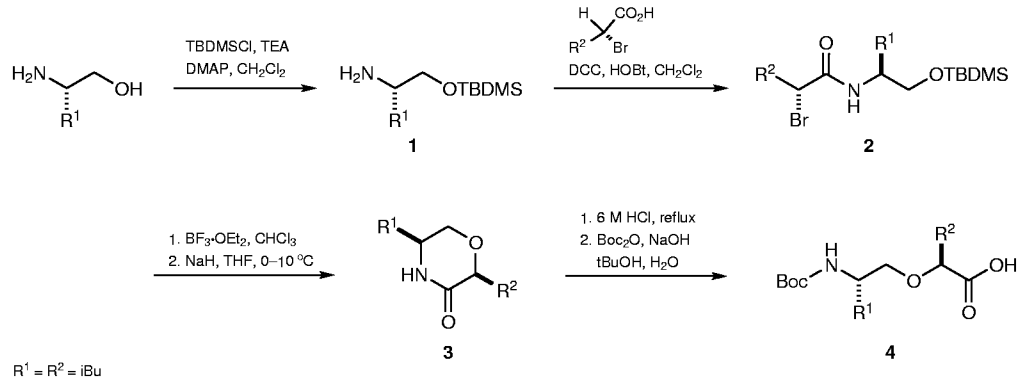



A more efficient synthesis has been described by Nicolaides et al.[3] and TenBrink[4] (Scheme 2). The key step of the synthesis is an intramolecular Williamson's reaction between a bromide and an alkoxide to provide 1,4-oxazin-5-ones **3**. The cyclic product is subsequently hydrolyzed to the desired methyleneoxy dipeptide. This method provides better yields and allows for the preparation of dipeptide surrogates containing bulky side chains.

This method has been exploited to synthesize various peptide analogues.[3] As summarized in Scheme 2,[4] the synthesis is initiated with the protection of an amino alcohol using the *tert*-butyldimethylsilyl group. Then, the resulting compound **1** is coupled to a 2-bromoalkyl carboxylic acid. After removal of the silyl protecting group by boron trifluoride–diethyl ether complex, an alkoxide is generated with sodium hydride and spontaneous ring formation with inversion of stereochemistry takes place at 0–10 °C. Hydrolysis of the resulting lactam **3** with 6 M HCl and protection of the amino group with the *tert*-butoxycarbonyl group provides the desired product.

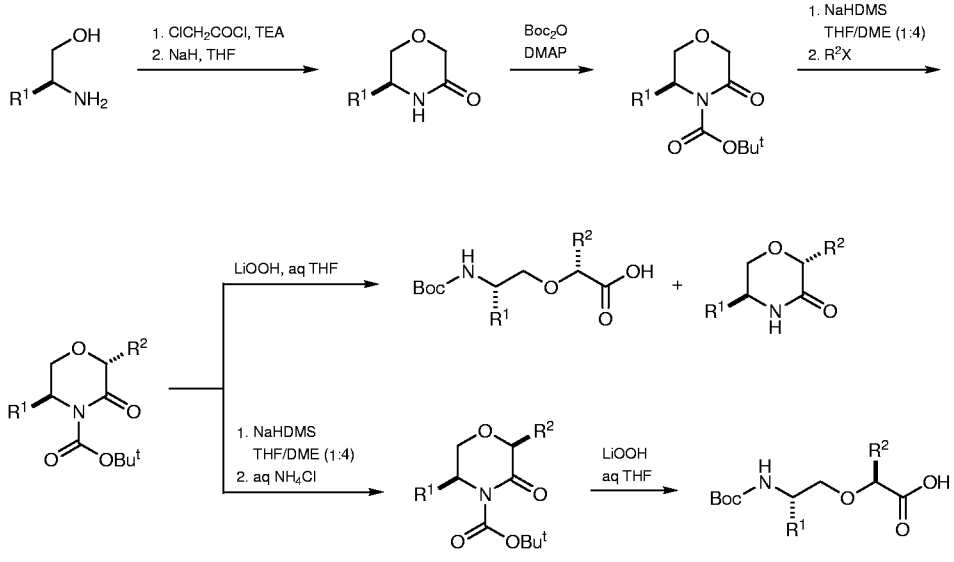
*for references see p 397*

**Scheme 2** Synthesis of Methyleneoxy Peptides using an Intramolecular Williamson Reaction via 1,4-Oxazin-5-ones[3,4]

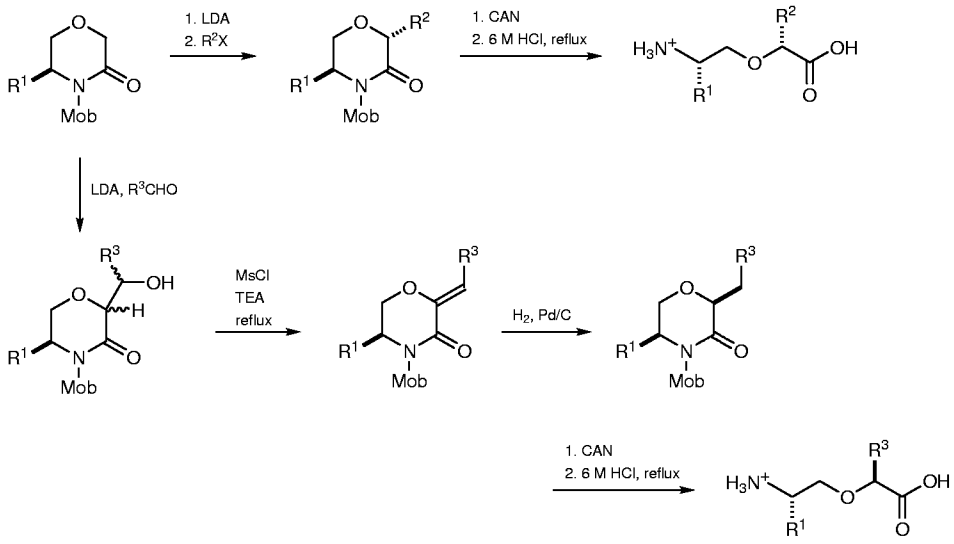



Mayer and co-workers improved the amide bond formation of TenBrink's method using unprotected amino alcohols and active esters of bromoalkyl carboxylic acids.[5] More recently, Anthony et al.,[6] and Norman and Kroin[7] reported stereospecific alkylation of acylmorpholinone (Schemes 3 and 4) using sodium hexamethyldisilazanide and lithium diisopropylamide, respectively.

**Scheme 3** Synthesis of Methyleneoxy Peptides using an Intramolecular Williamson Reaction and Sodium Hexamethyldisilazanide[6]



**Scheme 4** Synthesis of Methyleneoxy Peptides using an Intramolecular Williamson Reaction and Lithium Diisopropylamide[7]




**(2*S*)-2-{[(2*S*)-2-(*tert*-Butoxycarbonylamino)-4-methylpentyl]oxy}-4-methylpentanoic Acid (4, $R^1 = R^2 =$ iBu):**[4]

***O*-(*tert*-Butyldimethylsilyl)-(*S*)-leucinol (1, $R^1 =$ iBu):**
To an ice-cooled soln of (*S*)-leucinol (3.67 g, 0.0313 mol), $Et_3N$ (3.80 g, 0.0375 mol), and DMAP (0.15 g, 0.0012 mol) in $CH_2Cl_2$ (30 mL), TBDMSCl (5.19 g, 0.0344 mol) was added. After stirring overnight, the precipitated $Et_3N \cdot HCl$ was filtered off and the filtrate was extracted with $H_2O$ and aq $NH_4Cl$. The organic layers were filtered through $Na_2SO_4$ and concentrated. The crude product was purified on silica gel (4% MeOH/$CH_2Cl_2$) to provide **1** ($R^1 =$ iBu) as a clear liquid; yield: 6.76 g (93%); $[\alpha]_D$ +1.9 (*c* 0.914, $CH_2Cl_2$); CI-MS *m/z*: 249 [$NH_3$].

***N*-(2*R*)-2-Bromo-4-methylpentanoyl]-*O*-(*tert*-butyldimethylsilyl)-(*S*)-leucinol (2, $R^1 = R^2 =$ iBu):**
A mixture of (2*R*)-2-bromo-4-methylpentanoic acid (2.86 g, 0.01466 mol) and HOBt (2.24 g, 0.01466 mol) was stirred in $CH_2Cl_2$ at 0 °C. To this mixture DCC (3.02 g, 0.01466 mol) was added. Stirring was continued at 0 °C for 11 h, at which time the mixture was added to **1** ($R^1 =$ iBu; 3.23 g, 0.014 mol) in $CH_2Cl_2$ and cooled to 0 °C. After 2 h the mixture was concentrated, and DCU was filtered off and discarded. The filtrate was extracted with aq $NaHCO_3$ (2×), 0.5 M citric acid (1×), and brine. The organic layers were filtered through $Na_2SO_4$ and concentrated. EtOAc was added to the residue, and DCU was again filtered off. The crude product was purified on silica gel (5% EtOAc/hexane) to give **2** ($R^1 = R^2 =$ iBu); yield: 3.26 g (57%); mp 74.5–75.5 °C; $[\alpha]_D$ −12.2 (*c* 0.9445, $CH_2Cl_2$); MS *m/z*: 408.

**(2*S*,5*S*)-2,5-Diisobutyl-1,4-oxazin-3-one (3, $R^1 = R^2 =$ iBu):**
A soln of **2** ($R^1 = R^2 =$ iBu; 1.03 g, 0.00252 mol) in $CHCl_3$ (20 mL) was cooled to 0 °C and $BF_3 \cdot OEt_2$ (0.34 mL, 0.00277 mol) was added. After 5 min the ice bath was removed and the soln was allowed to warm to rt. After 90 min additional $BF_3 \cdot OEt_2$ (0.34 mL) was added. The mixture was stirred for 1 h, after which it was extracted with $H_2O$ and sat. aq $NaHCO_3$ (2×). The organic layers were filtered through $Na_2SO_4$, and the solvent was removed in vacuo. The crude product was purified on a silica gel column (4% MeOH/$CH_2Cl_2$) to provide *N*-[(2*R*)-2-bromo-4-methylpentanoyl]-(*S*)-leucinol; yield: 0.66 g (89%); mp 57–58 °C; $[\alpha]_D$ +8.7 (*c* 0.859, $CH_2Cl_2$); MS *m/z*: 262.

NaH (0.34 g, 0.00709 mol, 50% in oil) was washed with hexane (3×) to remove the oil. To the NaH THF (5 mL) was added and the suspension was cooled to −25 °C. To the NaH suspension precooled (to −25 °C) *N*-[(2*R*)-2-bromo-4-methylpentanoyl]-(*S*)-leucinol (1.98 g, 0.006755 mol) in THF (25 mL) was added. To this further THF (30 mL) was slowly added. After 10 min the MeOH–ice bath was replaced by an $H_2O$–ice bath. After stirring for 1 h (allowing to warm slowly) the mixture was extracted with water and brine. The organic layers were filtered through $Na_2SO_4$ and taken to dryness. The crude product was filtered through a short silica gel column (4% MeOH/$CH_2Cl_2$) and then recrystallized ($Et_2O$/hexane) to provide **3** ($R^1 = R^2 =$ iBu); yield: 1.28 g (89%); mp 108.5–110.0 °C; $[\alpha]_D$ −78.0 (*c* 0.924, $CH_2Cl_2$); MS *m/z*: 213.

*for references see p 397*

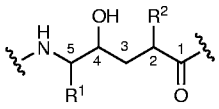
**(2*S*)-2-{[(2*S*)-2-(*tert*-Butoxycarbonylamino)-4-methylpentyl]oxy}-4-methylpentanoic Acid (4, $R^1 = R^2 = iBu$):**

A soln of **3** ($R^1 = R^2 = iBu$; 1.0 g, 0.00469 mol) and 6 M HCl (25 mL) were heated at 100 °C for 19 h and then cooled and concentrated. $Et_2O$ was added to the wet solid, and the solid was collected (0.82 g) and washed ($Et_2O$). The solid was dissolved in $H_2O$ (2 mL) and the pH was adjusted to 8–10 with 1 M NaOH. To this soln *t*-BuOH (10 mL) and $Boc_2O$ (0.63 g) were added. The pH was kept at 8–9 by the addition of 1 M NaOH as needed. In the same manner, the $Et_2O/H_2O$ filtrate was treated with aq NaOH, *t*-BuOH, and $Boc_2O$ (0.50 g). After the mixture had been stirred overnight, *t*-BuOH was removed from the mixture in vacuo, and the aqueous residue was washed with pentane (1×). The aqueous layers were acidified (3 M HCl) and extracted ($CH_2Cl_2$). The organic layers were filtered through $Na_2SO_4$ and taken to dryness. The crude product was purified on silica gel by elution with a gradient of 100% $CH_2Cl_2$ to 4% $MeOH/CH_2Cl_2$ to provide **4** ($R^1 = R^2 = iBu$); yield: 0.93 g (60%); $[\alpha]_D$ −62.6 (*c* 0.7695, EtOH); MS *m/z*: 332.

## 10.6.2 ψ[CH(OH)$CH_2$] (1-Hydroxyethylene) Peptides

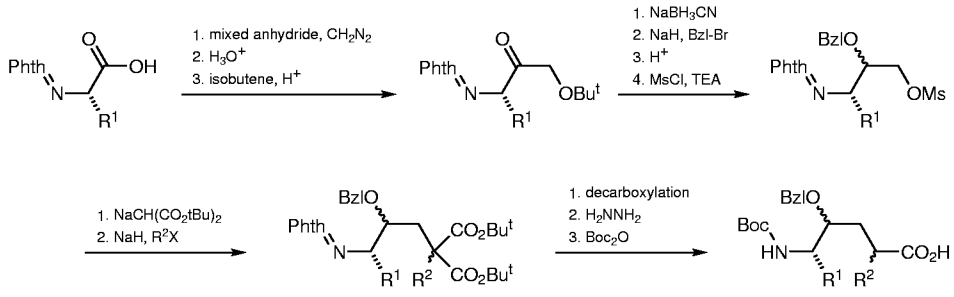
1-Hydroxyethylene (2-alkyl-4-hydroxy-5-amino acid; Scheme 5) is one of the most extensively studied amide isosteres.[8] It is a common structural motif in aspartyl protease inhibitors. The hydroxyl group of this moiety serves as a mimic for the tetrahedral intermediate of proteolysis. There are three stereogenic centers in the 1-hydroxyethylene dipeptides (Scheme 5). For the design of aspartyl protease inhibitors, the *S* configuration is generally preferred at the C4 and C5 atoms. Both *R* and *S* configurations are allowed at the C2 atom, although the *R* configuration would be the same stereochemistry as the side chain of the natural peptide.

**Scheme 5** Structure of 1-Hydroxyethylene Peptides



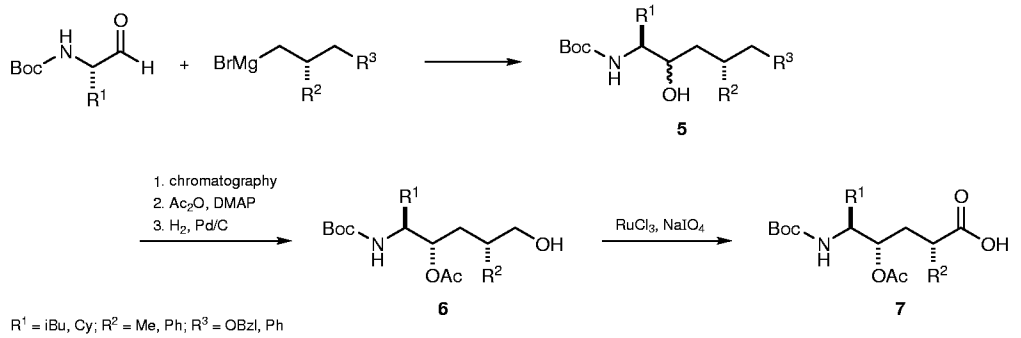

The first synthesis of this amide isostere was reported by Szelke et al.[9,10] As summarized in Scheme 6, the synthesis is initiated by the conversion of a N-protected amino acid to an alkoxy ketone. Reduction of the ketone gives the C4 hydroxyl group of the isostere. Then, the carboxylic acid moiety is introduced in the form of malonate by displacement of the terminal hydroxyl group and the side chain of the C2 atom is incorporated by α-alkylation. Finally, decarboxylation followed by change of the N-protecting group leads to the desired isostere. Although this method is synthetically straightforward, among the three stereogenic centers only the stereochemistry of C5 can be rigorously defined.

**Scheme 6** Synthesis of 1-Hydroxyethylene Peptides[9,10]



Rich and co-workers[11–13] reported a stereospecific synthesis of this amide isostere initiated by the addition of a Grignard reagent to an α-amino aldehyde (Scheme 7). In this synthesis, the stereochemistries of C2 and C5 were established from the use of an optically active Grignard reagent and α-amino aldehyde, respectively (see Section 10.6.2.1 for further details). Although this reaction produces an epimeric mixture of C4 alcohols, the 4*S*-diastereomer prevails. To complete the synthesis, the carboxylic acid group is obtained by oxidation of the terminal hydroxy group. Rasetti et al.[14] used a similar approach to synthesize highly potent renin inhibitors.

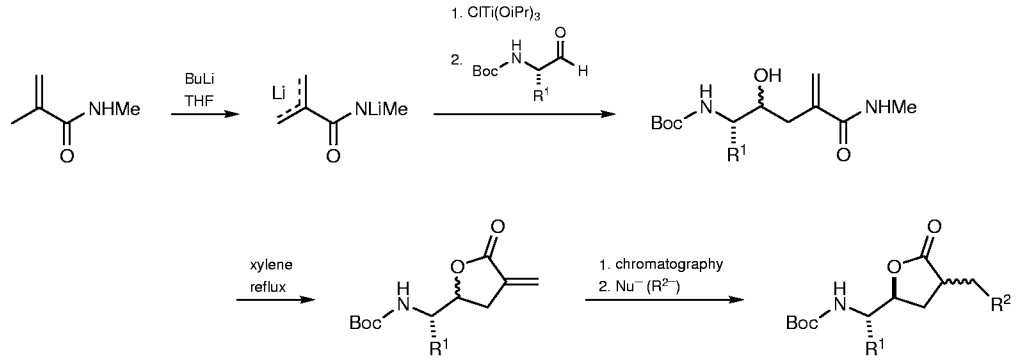
**Scheme 7** Stereospecific Synthesis of 1-Hydroxyethylene Peptides[11–13]




Jones et al.[15] modified this method using a phenyl ring as the synthon for the terminal carboxylic acid. They employed (2*S*)-1-bromo-2-phenylpropylmagnesium bromide as the optically active Grignard reagent. From such a modification, high stereoselectivity (4*S*:4*R* = 6:1) was obtained and the reaction could be carried out on a large scale. Oxidation of the phenyl ring by ruthenium(IV) oxide was used to complete the synthesis. In an additional variation, Diederich and Ryckman[16] coupled a Weinreb amide of an amino acid to a Grignard reagent.

In place of a Grignard reagent, several homoenolate equivalents have also been employed. Kempf[17] reported the titanium-mediated addition of *N*-alkylmethylacrylamide dianions to N-protected α-amino aldehydes (Scheme 8). Pyrolytic cyclization affords a 3-methylenetetrahydrofuran-2-one and the side chain of C3 is appended via conjugate addition. The resulting lactone can be converted into the 1-hydroxyethylene dipeptide by hydrolysis. The stereochemistry of the C6 atom is the same as that of the α-amino aldehyde. However, the stereoselectivities of the reactions regarding the C3 and C5 atoms are unsatisfactory.

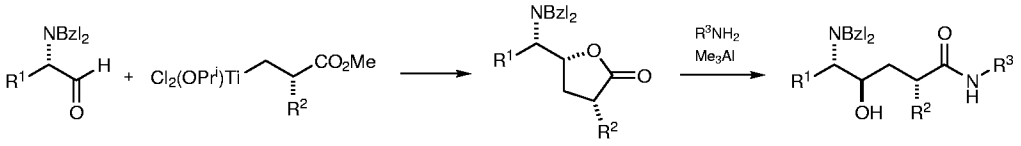
**Scheme 8** Synthesis of 1-Hydroxyethylene Peptides by Addition of a Titanium Homoenolate to an α-Amino Aldehyde[17]



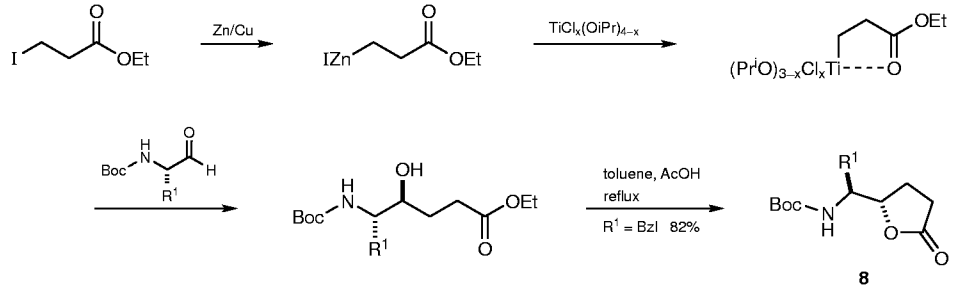

*for references see p 397*

Stereoselective routes to the lactone precursors of the 1-hydroxyethylene isosteres using a titanium-mediated condensation has been reported by both Shibuya and co-workers[18] and DeCamp et al.[19] The former method (Scheme 9) features the reaction between (2*S*)-2-dibenzylamino aldehydes and optically active dichloroisopropoxytitanium ester homoenolates to provide the desired lactone with high *erythro* selectivity. Then, the lactone is opened by treatment with an amine and trimethylaluminum to give the 1-hydroxyethylene isostere. The experimental details of DeCamp et al.'s high yielding and stereoselective synthesis[19] of lactone **8** (Scheme 10) is discussed in Section 10.6.2.2.

**Scheme 9** Synthesis of 1-Hydroxyethylene Peptides by Addition of a Titanium Homoenolate to an α-Amino Aldehyde[18]

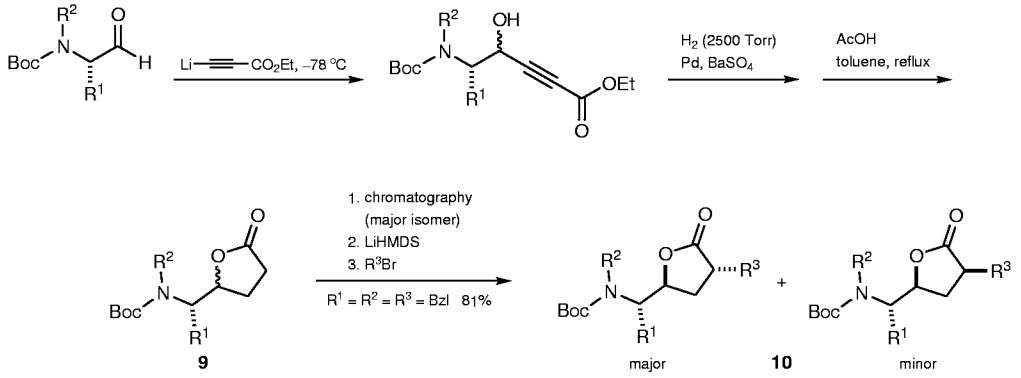


**Scheme 10** Synthesis of 1-Hydroxyethylene Peptides by Addition of a Titanium Homoenolate to an α-Amino Aldehyde[19]



Kleinman and co-workers[20] synthesized a lactone precursor to the (2*R*,4*S*,5*S*)-1-hydroxyethylene dipeptide stereoselectively in four steps using the lithium salt of ethyl propiolate as a homoenolate equivalent. As summarized in Scheme 11, addition of ethyl lithiopropiolate to a protected α-amino aldehyde affords hydroxy acetylenic esters as a mixture of diastereomers. Reduction of the acetylene group and subsequent lactonization gives a readily separable (4*S*)-lactone-enriched mixture. Direct alkylation with alkyl halide and lithium hexamethyldisilanazide yields the *trans*-lactone as the major stereoisomer.

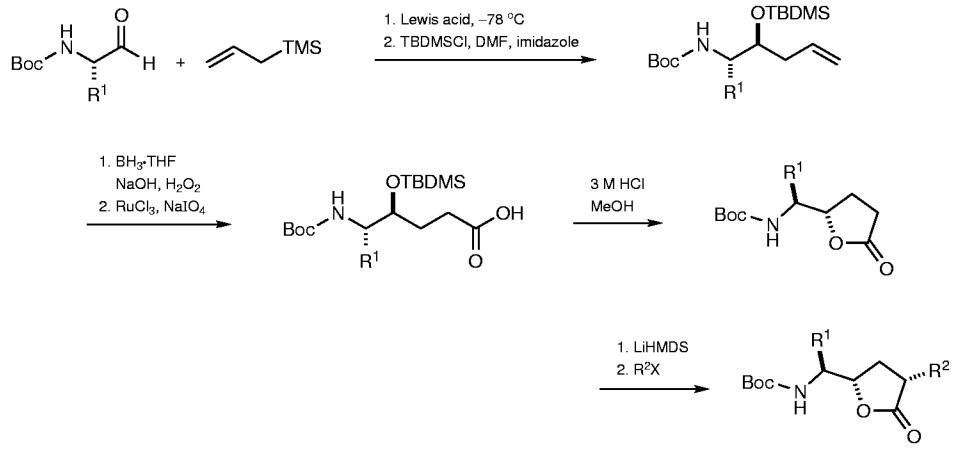
**Scheme 11** Synthesis of 1-Hydroxyethylene Peptides by Addition of Ethyl Lithiopropiolate to an α-Amino Aldehyde[20]



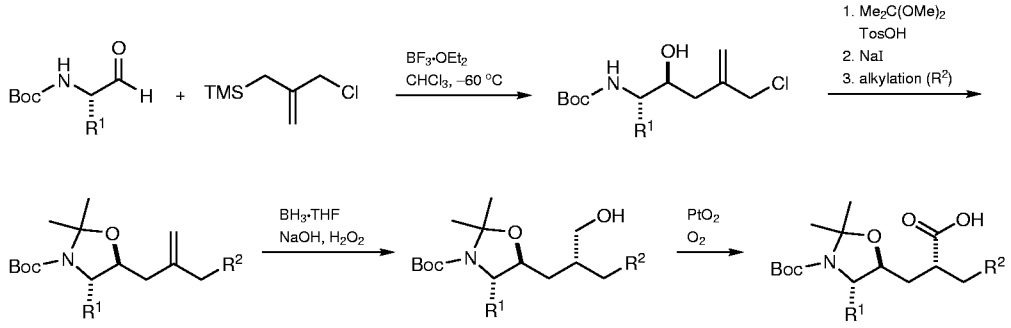
In addition, Cushman and co-workers[21] reported the synthesis of a 1-hydroxyethylene dipeptide with a Pro moiety at the C-terminus using the reaction between an α-amino aldehyde and a lithium cyclopentanone enolate. Matternich and Lüdi[22] described the synthesis of a γ-(aminoalkyl)-α-hydroxy-γ-lactone starting from the addition of an α-amino aldehyde to a pyruvate enolate.

The addition of an allyl metal to α-amino aldehydes has been used by Vara Prasad and Rich (Scheme 12).[23] After the addition step, the allyl group is oxidized to a carboxylic acid and lactonized. Then, the α-carbon of the lactone is alkylated stereoselectively. These investigators also systematically examined the addition reaction to determine its diastereoselectivity. The highest diastereoselectivity was obtained when allyltrimethylsilane was used in the presence of tin(IV) chloride. An increase in the steric bulk of the protecting group and of the side chain also resulted in a better diastereoselection. Alternatively, Taddei and coworkers[24–26] used a 2-(halomethyl)allylsilane and the side chains were introduced by nucleophilic substitution of the halogen (Scheme 13).

**Scheme 12** Synthesis of 1-Hydroxyethylene Peptides by Addition of an Allyl Metal to an α-Amino Aldehyde[23]




**Scheme 13** Synthesis of 1-Hydroxyethylene Peptides by Addition of an Allyl Metal to an α-Amino Aldehyde[24–26]

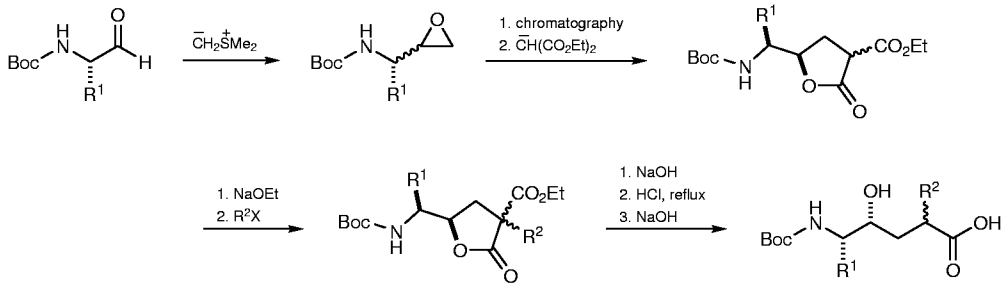



In several of the previously described methods the anion of the three-carbon fragments was added to an α-amino aldehyde to construct a C–C bond between the C3 and C4 atoms of the 1-hydroxyethylene isostere. However, Sakurai et al.[27] used an acid chloride instead of an α-amino aldehyde. They synthesized δ-phthalimido-γ-oxo esters by a palladium-catalyzed reaction between acid chlorides and organozinc reagents derived from β-iodo esters. Then, the oxo esters were converted into the γ-lactone precursors.

*for references see p 397*

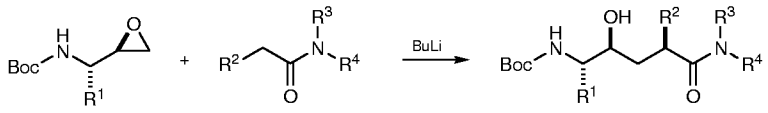
The construction of the C2–C3 bond of the 1-hydroxyethylene moiety by the addition of a two-carbon fragment to aminoalkyl epoxides, amino ketones, β-amino-α-hydroxy aldehydes, or 4-amino-3-oxo phosphonates was also studied. As summarized in Scheme 14, Evans et al.[28] added a malonate to an aminoalkyl epoxide and the product spontaneously cyclized, leading to a lactone.[29] Then, the second side chain was incorporated into the α-carbon of the lactone, providing a mixture of C2 epimers. The final compound was obtained by hydrolysis and decarboxylation. The lack of diastereoselectivity of this method is offset by the small number of steps and the accessibility to all eight possible stereoisomers. Bühlmayer et al.[30] also used a similar method, but they transformed the epoxide into the apparently more reactive iodide. Then, the iodide compound was treated with an enolate.

**Scheme 14** Synthesis of 1-Hydroxyethylene Peptides using an Aminoalkyl Epoxide[28]

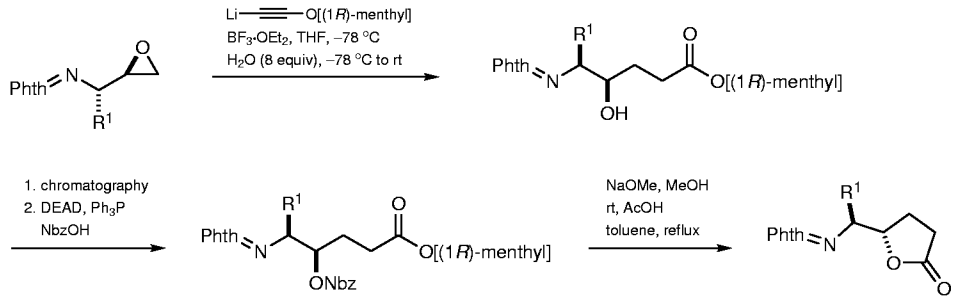



The stereoselectivity of the method using an aminoalkyl epoxide was improved by Askin et al. (Scheme 15).[31] After having prepared an aminoalkyl epoxide in the *R* configuration from an aminoalkene through a carbamate-derived epoxidation, the epoxide was directly coupled to a chiral amide enolate. The reaction led to the product with the 2*R*,4*S*,5*S* configuration. If the electrophilicity of the epoxide is enhanced by pre-lithiation of the carbamate moiety using an additional equivalent of butyllithium, the reaction proceeds with both high yield and diastereoselectivity. Aguilar et al.[32] treated a chiral epoxide with lithium (1*R*)-menthyloxyacetylide in the presence of boron trifluoride diethyl ether complex (Scheme 16). Then, the reaction is followed by in situ hydrolysis of the ynol ether, leading to a (4*S*,5*S*)-5-phthalimido-4-hydroxy ester. A Mitsunobu reaction with 4-nitrobenzoic acid and subsequent selective saponification of the benzoate and cyclization of the inverted hydroxy ester affords the target N-protected (5*S*)-5[(1*S*)-1-aminoalkyl]-tetrahydrofuran-2-one. Similarly, Pégorier and Larchevêque[33] have reported a synthetic method using the stereocontrolled hydroboration of homoallylic alcohols derived from *syn*-protected amino epoxides followed by chemoselective oxidation of the resulting primary alcohols.

**Scheme 15** Synthesis of 1-Hydroxyethylene Peptides using an Aminoalkyl Epoxide[31]

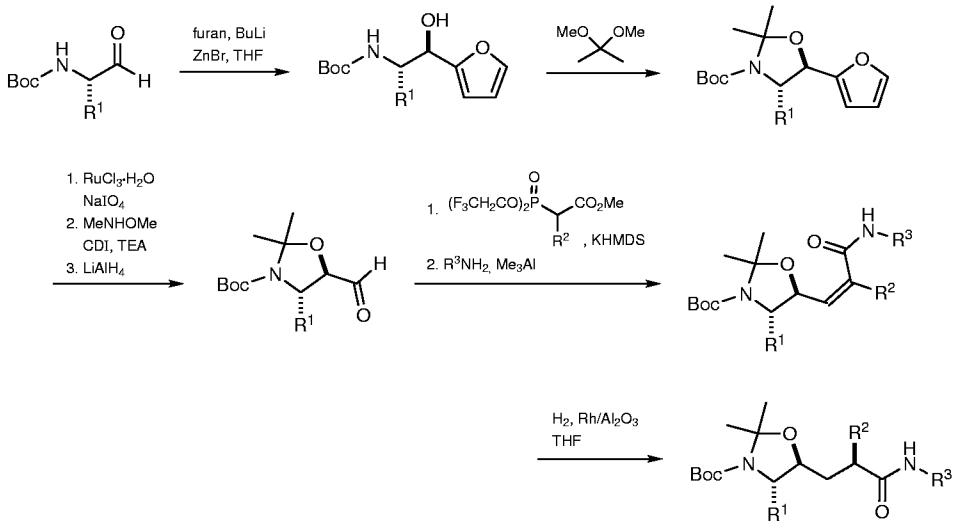


**Scheme 16** Synthesis of 1-Hydroxyethylene Peptides using an Aminoalkyl Epoxide[32]




To connect the C2 and C3 atoms of the 1-hydroxyethylene moiety, Wuts et al.[34] and Poss and Reid[35] used the Wadsworth–Emmons reaction between a β-amino-α-hydroxy-aldehyde and a phosphonate (two-carbon fragment) to give the desired isostere after reduction of the alkene. In the synthesis reported by Poss and Reid (Scheme 17) the stereochemistry of the hydroxyl group at C4 is established by a highly diastereoselective aldol addition to furan. A slight modification was employed by Chakravarty et al.[36] and Plata et al.[37] where 4-amino-3-oxo phosphonates were reacted with an aldehyde or ketone form of the two-carbon fragment.

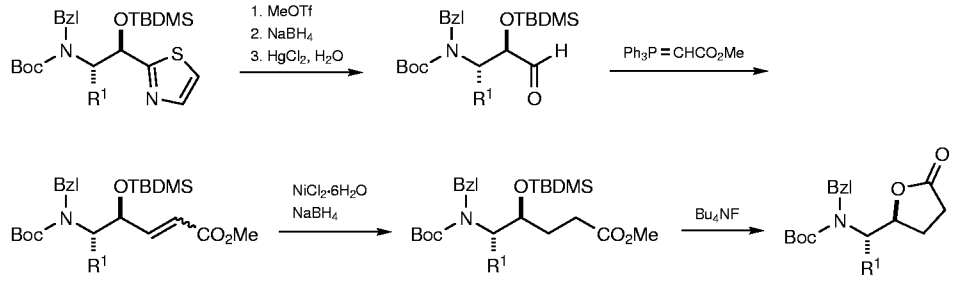
**Scheme 17** Synthesis of 1-Hydroxyethylene Peptides using the Wadsworth–Emmons Reaction[35]




Dondoni et al. prepared a 1-hydroxyethylene peptide using a thiazole–aldehyde synthesis starting from an amino acid.[38] The aldehyde is converted into an alkanoate by Wittig alkenation and reduction of the double bond (Scheme 18). Then, removal of the *tert*-butyldimethylsilyl group gives the unsubstituted lactone. In the last step, the lactone is alkylated using the method reported by Kleinman and co-workers.[20]

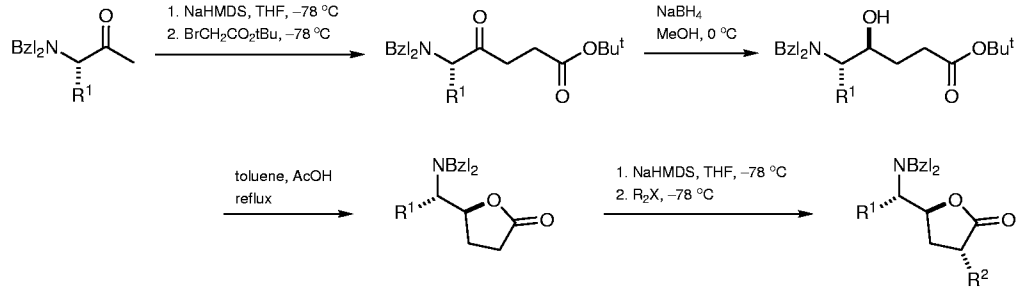
*for references see p 397*

**Scheme 18** Formation of 1-Hydroxyethylene Peptides using a Thiazole–aldehyde Synthesis[38]



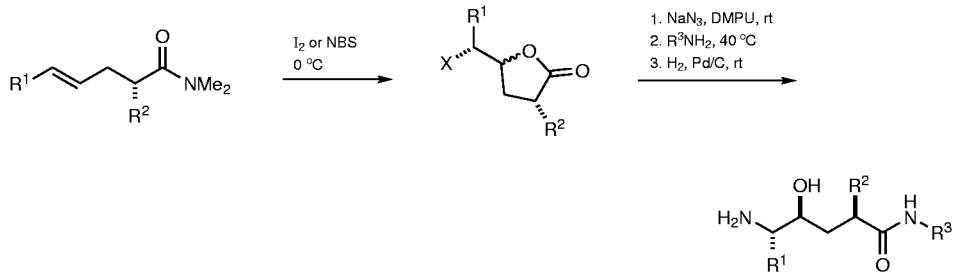

Radunz et al.[39] and Lagu and Liotta[40] initiated the synthesis of the 1-hydroxyethylene isostere with an α-amino ketone, as summarized in Scheme 19. The α-amino ketone is converted into a γ-keto ester, and then the keto group is reduced. When the amino group is protected with two benzyl groups, the keto ester is reduced to the corresponding vicinal amino alcohol possessing *S,S* stereochemistry with excellent diastereoselectivity. The resulting amino alcohol is lactonized, and the side chain is introduced by stereoselective alkylation. An important feature of this method is that the introduction of an alkyl group at the C3 atom is possible.

**Scheme 19** Synthesis of 1-Hydroxyethylene Peptides using an α-Amino Ketone[39,40]



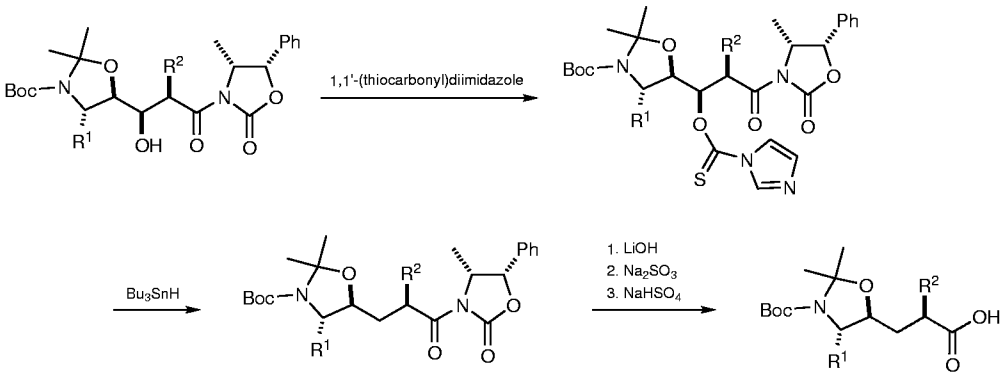

Haster et al.[41] reported a method using a 2,5-disubstituted pent-4-enylamide. The alkene is treated with osmium(VIII) oxide and the resulting dihydroxy product is lactonized under acidic conditions. The remaining hydroxy group is converted into a protected amine and the lactone is hydrolyzed. Herold et al.[42] and Bradbury et al.[43] lactonized the 2,5-disubstituted pentenoic amide using the halolactonization reaction (Scheme 20), while Wuts et al.[44] lactonized via epoxidation of an alkene and subsequent acidolytic ring opening of the epoxide. In each case the lactonization reaction leads to the formation of the thermodynamically more stable *trans* product as the major stereoisomer. For example, Herold et al.[42] used *N*-bromosuccinimide for halolactonization and obtained a 20:1 ratio of the *trans*- to *cis*-lactone.

**Scheme 20** Synthesis of 1-Hydroxyethylene Peptides using a 2,5-Disubstituted Pent-4-enylamide[42]

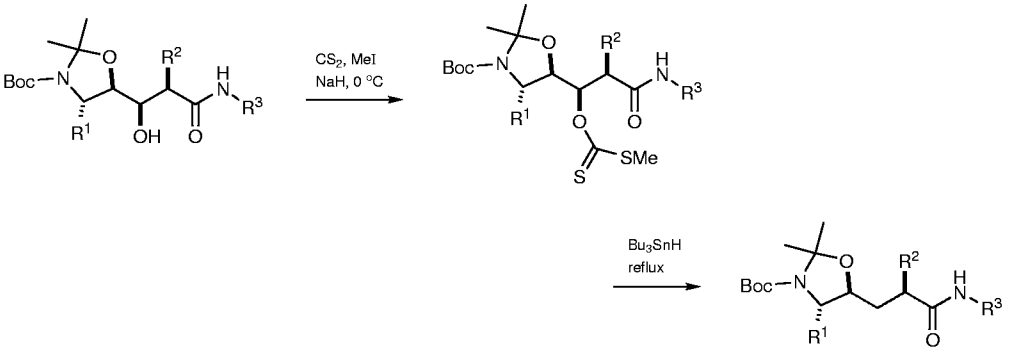



Modifications of other amide isosteres have also been reported. Boyd et al.[45] attempted two deoxygenation methods of the 1,2-dihydroxyethylene isostere. One method proceeds by Barton–McCombie deoxygenation using a thiocarbamate (Scheme 21). Another method exploits carbon disulfide, methyl iodide, and sodium hydride to form a xanthate (Scheme 22).[45] Subsequent reduction by tributyltin hydride in refluxing toluene gives the identical compound prepared by the previous method. Reduction of an 1-oxoethylene isostere by sodium borohydride was carried out by Nishi et al.,[46] Sakurai et al.,[27] and Tao and Hoffman[47] (Scheme 23). When a *N,N*-dibenzyl-protected 1-oxoethylene is used, the reduction proceeds with excellent diastereoselectivity.

**Scheme 21** Synthesis of 1-Hydroxyethylene Peptides using Deoxygenation of the 1,2-Dihydroxyethylene Isostere[45]

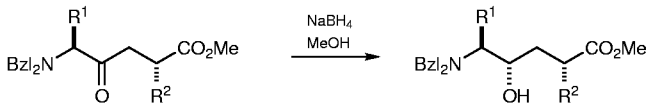



**Scheme 22** Synthesis of 1-Hydroxyethylene Peptides using Deoxygenation of the 1,2-Dihydroxyethylene Isostere[45]



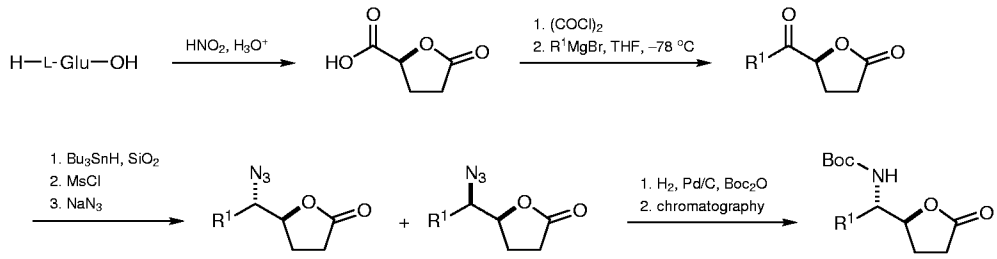

*for references see p 397*

**Scheme 23** Synthesis of 1-Hydroxyethylene Peptides by Reduction of an 1-Oxoethylene Isostere[27,46,47]




Peyrat et al.[48] carried out a stereocontrolled synthesis starting from the readily available L-Glu (Scheme 24). This method allows access to further analogues in which the $R^1$ substituent can be easily varied through an acylation reaction of the corresponding Grignard reagent with the carboxylic acid chloride. The key step is the reduction of the carbonyl group of a ketone with tributyltin hydride in the presence of silica gel. The reaction affords a mixture of the *syn* (4*S*,5*S*) and *anti* (4*S*,5*R*) alcohols enriched in the *anti* isomer. Then, the hydroxy group is converted into azide, which is reduced and treated with $Boc_2O$ to give the protected amino group. Since the $S_N2$ mechanism is involved in the azide formation, the 4*S*,5*S* isomer is obtained as the major stereoisomer.

**Scheme 24** Synthesis of 1-Hydroxyethylene Peptides Starting from Glutamic Acid[48]




Carbohydrates and their derivatives have been exploited as starting materials to unambiguously control the configuration at the C4 and C5 atoms. Shiozaki and Kobayashi[49,50] reported the synthesis of the 1-hydroxyethylene isostere from D-glucal derivatives via a lactone precursor. This route allows the incorporation of various substituents at the C2 and C5 atoms. Ghosh et al.[51] reported the synthesis of lactones with an absolute stereocontrol at the C4 and C5 positions starting from D-mannose. Additionally, Chakraborty et al.[52,53] and Kotsuki et al.[54] used D-glucose and D-ribose, respectively. However, the syntheses using carbohydrates are often long and inefficient.

Statine was identified as a starting material that contains predetermined stereochemistry at the C4 and C5 atoms. Nakano and co-workers[55,56] degraded statine to an aldehyde, (4*S*,5*R*)-4-alkyl-3-benzyloxycarbonyl-5-formyl-2,2-dimethyloxazolidine, which is an intermediate of the method described in Scheme 17. Then, the elongation of this aldehyde was carried out by the Wadsworth–Emmons reaction with diethyl phosphonate.

### 10.6.2.1 Synthesis by Addition of a Grignard Reagent

As mentioned in Section 10.6.2, synthesis of 1-hydroxyethylene peptides can be initiated by adding a *tert*-butoxycarbonyl N-protected α-amino aldehyde to an optically active Grignard reagent (Scheme 7).[11–13] This reaction affords a diastereomeric mixture of the C4 epimers of the hydroxy ether in good yields. In most cases the mixture is enriched in the 4*S*-epimer and the epimers are readily separable. The yields and the ratios of the resulting 4*S*- and 4*R*-epimers obtained from several examples of this reaction are summarized in Table 1. When this reaction was attempted with the aldehyde prepared from $N^{\alpha}$,$N^{\varepsilon}$-bis-*tert*-butoxycarbonyl-protected Lys, the desired product was not obtained. The anion of the Lys $N^{\varepsilon}$-*tert*-butoxycarbonylamino group probably reacts with the aldehyde to form a cyclic aminol that does not

undergo the desired addition of the Grignard reagent. The reaction with $N^{\alpha}$-benzyloxycarbonyl derivatives was also found to be incompatible owing to the intramolecular formation of a cyclic carbamate. The Grignard addition was also found to be incompatible with any side chain or protecting groups that would be sensitive to the strongly basic, nucleophilic Grignard reagent.

**Table 1** Addition of an α-Amino Aldehyde to a Grignard Reagent[11–15]

| $R^1$ | $R^2$ | $R^3$ | Yield of **7** (%) | 4*S*/4*R* | Ref |
|---|---|---|---|---|---|
| iBu | Me | OBzl | 66 | 4:1 | [11] |
| iBu | Me | OBzl | 52.5 | – | [13] |
| iBu | Ph | OBzl | 30 | 3:2 | [12] |
| iBu | 4-[3-(2,2-dimethylpentyl)phenyl]benzyl | OBzl | 61 | 3:1 | [14] |
| iBu | 2,2-dimethyl-1-naphthylpropyl | OBzl | 80 | 5:1 | [14] |
| MeCy | Me | Ph | 66[a] | 6:1 | [15] |

[a] *S*-isomer only (*R*-isomer not included).

The remaining three steps are accomplished without purification of the intermediate products. The secondary hydroxy group is protected by acetylation and the benzyl ether is removed by hydrogenolysis to provide a primary alcohol. The alcohol is oxidized to a carboxylic acid by ruthenium(III) chloride or pyridinium dichromate. This method has been applied to the synthesis of various enzyme inhibitors containing the 1-hydroxyethylene isostere.

**(2*R*,4*RS*,5*S*)-1-(Benzyloxy)-5-[(*tert*-butoxycarbonyl)amino]-2,7-dimethyloctan-4-ol (5, $R^1$ = iBu; $R^2$ = Me; $R^3$ = OBzl):**[13]

Mg powder (0.90 g, 37 mmol) was placed in a dry 50-mL two-necked round-bottomed flask under an argon atmosphere. The Mg was suspended in dry THF (10 mL) and an iodine crystal was added. 3-Benzyloxy-2-methylpropyl bromide (6.0 g, 24.7 mmol) was placed in a 10-mL addition funnel, and two drops were added to the Mg mixture. The reaction was started by gently warming and then maintained at a gentle reflux by dropwise addition of the bromide. The addition was complete in 20 min, and the mixture was stirred for 2 h at 25 °C. The resulting Grignard reagent was chilled to 5 °C before adding a soln of Boc-L-leucinal (2.42 g, 11.2 mmol) in THF (10 mL) over a 2 h period. The reaction was stirred for 6 h before quenching with aq $NH_4Cl$. The reaction was poured into $H_2O$ (150 mL) and extracted with $Et_2O$ (3×40 mL). The organic phase was dried ($MgSO_4$), suction filtered, and concentrated in vacuo to a yellow oil (5.5 g). The product was purified by MPLC to provide the separated diastereomers and a mixture of diastereomers; total yield: 2.24 g (5.9 mmol, 52.5%); yield of 2*R*,4*S*,5*S*-isomer: 0.927 g (2.4 mmol, 22%); $[\alpha]_D$ −21.4 (*c* 2.8, $CHCl_3$); MS *m/z*: 380 [$M^+$+1], 324, 280, 130, 91; yield of 2*R*,4*R*,5*S*-isomer: 0.41 g (1.1 mmol, 9.6%); $[\alpha]_D$ −10.6 (*c* 2.89, $CHCl_3$); MS *m/z*: 380 [$M^+$+1], 324, 280, 130, 91; yield of mixture: 0.907 (2.4 mmol, 21.3%).

**(2*R*,4*S*,5*S*)-4-Acetoxy-5-[(*tert*-butoxycarbonyl)amino]-2,7-dimethyloctanol (6, $R^1$ = iBu; $R^2$ = Me):**[13]

The 2*R*,4*S*,5*S*-isomer of **5** ($R^1$ = iBu; $R^2$ = Me; 0.927 g, 2.44 mmol) was dissolved in EtOAc (3 mL). To this soln NMM (0.28 mL, 0.26 g, 0.24 mmol) and DMAP (0.03 g, 0.24 mmol) were added, followed by $Ac_2O$ (0.70 mL, 0.76 g, 7.4 mmol). The mixture was then stirred for 24 h at 25 °C. Sat. $NaHCO_3$ (3 mL) was added and the mixture stirred for 30 min. This mixture was then poured into EtOAc (80 mL), and the organic phase was washed with sat. $NaHCO_3$ (3×30 mL) and cold 0.5 M HCl soln (2×30 mL), dried ($MgSO_4$), filtered, and concentrated in vacuo to give the desired product as an oil; yield: 0.984 g (2.34 mmol, 96%); MS *m/z*: 422 [$M^+$+1], 366, 336, 322, 306, 262, 186, 130, 91.

The benzyl ether was then removed by catalytic transfer hydrogenation. The acetylated product (0.984 g, 2.34 mmol) was dissolved in iPrOH/AcOH (1:1, 30 mL), and this soln was flushed with argon before adding ammonium formate (1.5 g, 23.8 mmol) and 10% Pd/C (0.60 g). This mixture was stirred for 6 h at 25 °C and then filtered through a Celite pad (EtOAc wash). The filtrate was diluted with EtOAc (200 mL) and washed with $H_2O$ (3×100 mL) and sat. $NaHCO_3$ soln (3×100 mL). The organic phase was then dried ($MgSO_4$), filtered, and concentrated in vacuo to give the desired alcohol as an oil, which was used without further purification; yield: 0.712 g (2.15 mmol, 92%); MS *m/z*: 332 [$M^+$+1], 286, 232, 216, 186, 157, 130, 91.

*for references see p 397*

**(2*R*,4*S*,5*S*)-4-Acetoxy-5-[(*tert*-butoxycarbonyl)amino]-2,7-dimethyloctanoic Acid (7, $R^1$ = iBu; $R^2$ = Me):**[13]

The alcohol **6** ($R^1$ = iBu; $R^2$ = Me; 0.38 g, 1.15 mmol) was oxidized to the carboxylic acid **7** ($R^1$ = iBu; $R^2$ = Me) by dissolving it in a $CCl_4$/MeCN (1:1, 4.5 mL) solvent mixture and then adding $H_2O$ (3.4 mL), $NaIO_4$ (0.724 g, 3.4 mmol), and $RuCl_3$ (4.5 mg, 0.02 mmol). The mixture was stirred vigorously for 2.5 h and then diluted into a cold 0.5 M HCl soln (100 mL). The aqueous phase was extracted with $Et_2O$ (3×40 mL) and the extract dried ($MgSO_4$), suction filtered, and concentrated in vacuo to an oil which was purified by chromatography (20 g, 2% MeOH/0.5% AcOH/$CH_2Cl_2$ as eluant) to provide **7** ($R^1$ = iBu; $R^2$ = Me) as a glassy foam; yield: 0.264 g (0.76 mmol, 66%); MS *m/z*: 346 [$M^+$+1], 246, 230, 212, 186, 142, 130, 86.

## 10.6.2.2 Synthesis and Alkylation of the Lactone Intermediate

DeCamp et al.[19] synthesized the lactone intermediate of the 1-hydroxyethylene isostere with high yields and stereoselectivity. As summarized in Scheme 10 (Section 10.6.2), the titanium homoenolate is prepared from ethyl 3-iodopropionate. The iodide is metalated with zinc/copper couple to give the iodozinc homoenolate species. The alkyltitanium homoenolate is then generated by transmetalation of the iodozinc precursor with one of the several chlorotitanium isopropoxide species. The resulting titanium homoenolate reacts with a N-protected α-amino aldehyde, leading to a mixture of 4*S*-diastereomers. In the last step, the product is lactonized.

The stereoselectivity of the addition reaction is strongly dependent on the titanium ligand. For example, the 4*S*/4*R* ratio of the product increases as the number of chlorines on the titanium homoenolate anion increases. The ratio is 2.8:1, 9:1, and 16:1 over the series of homoenolates [Ti($CH_2CH_2CO_2Et$)(OiPr)$_3$], [Ti($CH_2CH_2CO_2Et$)Cl(OiPr)$_2$], and [Ti($CH_2CH_2CO_2Et$)$Cl_2$(OiPr)]. The highest 4*S* selectivity is obtained upon transmetalation with trichlorotitanium isopropoxide to give the dichloro(isopropoxy)titanium homoenolate. The stereoselectivity decreases as the stoichiometry of the homoenolate anion is increased from two to six equivalents (fewer than two equivalents result in an incomplete consumption of the aldehyde). In addition, the ratio is dependent upon the N-protecting group of the α-amino aldehyde. When dibenzyl was used as the N-protecting group the 4*S*/4*R* ratio was 1:20.

The reactions described here do not cause any racemization of the readily epimerizable aldehyde starting material. No 5*R*-isomers were detected by chiral HPLC of the final lactone product. In addition, these conditions are suitable for large-scale reactions. DeCamp et al.[19] mentioned that they converted 600 g of α-amino aldehyde derivatives into the desired lactone.

Most stereoselective alkylations to the lactone intermediates are accomplished using the method reported by Kleinman and co-workers.[20] The dianion form is prepared by treating the lactone with lithium or sodium hexamethyldisilanazide. Then, the reaction between the dianion and an alkyl halide produces the 2-alkylated lactone (Scheme 11, Section 10.6.2). Kleinman and co-workers reported that the ratio of the *trans*- and *cis*-lactone is 47:3. When a doubly protected α-amino aldehyde [e.g., dibenzylamino aldehyde, (*tert*-butoxycarbonyl)-benzylamino aldehyde] is used, the *cis*-lactone is not obtained. As an example, the alkylation of (5*S*,1′*S*)-5-[1′-[*N*-(*tert*-butoxycarbonyl)benzylamino]-2′-phenylethyl]dihydrofuran-2(3*H*)-one to produce (3*R*,5*S*,1′*S*)-3-benzyl-5-[1′-[*N*-(*tert*-butoxycarbonyl)benzylamino]-2′-phenylethyl]dihydrofuran-2(3*H*)-one is described in this section.

**(5*S*)-5-[(1*S*)-1-(*tert*-Butoxycarbonyl)amino)-2-phenylethyl]dihydrofuran-2(3*H*)-one (8, $R^1$ = Bzl):**[19]

The iodozinc homoenolate was prepared by stirring ethyl 3-iodopropionate (1.06 g, 4.66 mmol), freshly prepared Zn/Cu couple (0.49 g, 7.53 mmol), anhyd toluene (7.9 mL), and DMA (0.79 mL) under $N_2$ for 1 h at rt and 4 h at 80 °C. The mixture was cooled to 22 °C and the excess of Zn/Cu couple allowed to settle. To a flask equipped with a mechanical stirrer, dry toluene (1.0 mL), dry $CH_2Cl_2$ (5.6 mL), and Ti(OiPr)$_4$ (0.34 mL, 1.1 mmol) were added. The soln was cooled to 17 °C and $TiCl_4$ (0.36 mL, 3.3 mmol) was added dropwise (<25 °C). The pale yellow soln was stirred at rt for 15 min, and then cooled to −40 °C (a thin white slurry formed at the lower temperature). The iodozinc homoenolate supernatant was added by cannula (leaving metal solids behind) to the [Ti(OiPr)$Cl_3$] mixture (<-20 °C; a dark red

soln formed). The resulting dichloro(isopropoxy)titanium homoenolate soln was stirred at −25 °C for 5 min and then cooled to −40 °C. A soln of Boc-(*S*)-phenylalaninal (0.55 g, 2.2 mmol) in $CH_2Cl_2$ (2 mL) was added by cannula to the homoenolate soln (<− 35 °C). The mixture was stirred vigorously at −20 °C for 18–24 h. The reaction was then quenched by addition to a stirred mixture of $H_2O$ (25 mL) and tBuOMe (35 mL) at 0 °C (using $CH_2Cl_2$ to rinse the reaction flask of any thick residue). The mixture was stirred for 7 min (until a colorless two-phase mixture was observed). The aqueous phase was extracted with tBuOMe (25 mL), and the combined organic phases washed with $H_2O$, sat. $NaHCO_3$ soln, $H_2O$, and brine. The organic phase was dried ($MgSO_4$) and concentrated in vacuo to give the hydroxy esters (the 4*S*,5*S*-diastereomer predominates) as a white solid; yield: (90%). The crude hydroxy ester mixture was dissolved in toluene (18 mL) and glacial AcOH (0.55 mL) then heated at reflux for 2.5 h. The mixture was cooled to ambient temperature, diluted with iPrOAc, and washed with 25 mL portions of $H_2O$, sat. $NaHCO_3$ soln, $H_2O$, and brine. The organic phase was dried ($MgSO_4$) and concentrated in vacuo to give an oil which was crystallized (hexanes) to give the desired lactone as a cream-colored solid; yield: 82% (from hydroxy ester). The ratio of 4*S*/4*R*-diastereomers was determined by HPLC to be 16:1.

**(3*R*,5*S*,1′*S*)-3-Benzyl-5-[1′-[*N*-(*tert*-butoxycarbonyl)benzylamino]-2′-phenylethyl]dihydrofuran-2(3*H*)-one (10, $R^1 = R^2 = R^3 =$ Bzl):**[20]

To a cold (−78 °C) soln of LiHMDS (1.52 mmol) in THF (7.5 mL), a soln of the lactone **9** ($R^1 = R^2 =$ Bn; 0.30 g, 0.76 mmol) in THF (2 mL) was added dropwise. The mixture was stirred at −78 °C for 20 min, and then freshly distilled BnI (0.18 g, 0.84 mmol) in THF (1 mL) was added over 2 min. The resulting mixture was stirred at −78 °C for 30 min and then quenched with aq phosphate buffer (pH 7, 10 mL) and allowed to warm to rt over 15 min. The mixture was treated with EtOAc (9 mL) and the phases separated. The aq phase was extracted with EtOAc (3×9mL), and the combined organic phases were dried ($Na_2SO_4$) and concentrated. Flash chromatography of the residue (silica gel, hexane/$Et_2O$ 7:3) afforded **10** ($R^1 = R^2 = R^3 =$ Bzl) as a white solid; yield: 0.3 g (81%); mp 123–125 °C; $[\alpha]_D$ −23.3 (*c* 1.3, $CHCl_3$).

## 10.6.3 ψ[$CH_2CH(OH)$] (2-Hydroxyethylene) Peptides

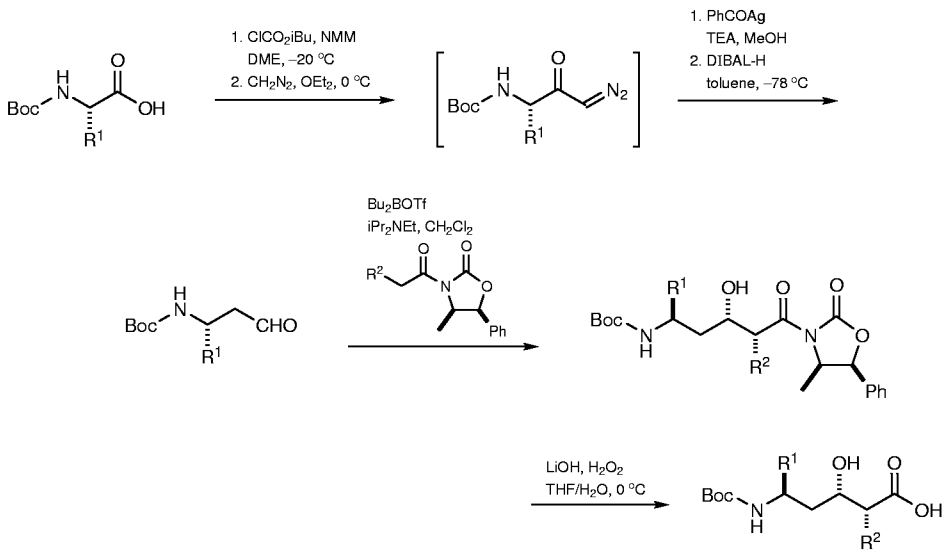
The 2-hydroxyethylene isostere is different from the 1-hydroxyethylene isostere only in the position of the hydroxy goup. In the 2-hydroxyethylene isostere the hydroxy group is attached to the C2 of ethylene, while it is on C1 in the 1-hydroxyethylene isostere. The 2-hydroxyethylene isostere has been applied to the design HIV-1 protease inhibitors.[51] However, the activity of the resulting compounds is lower than that of inhibitors in which the isostere is replaced with the 1-hydroxyethylene group.

### 10.6.3.1 Synthesis of 2-Hydroxyethylene Peptides

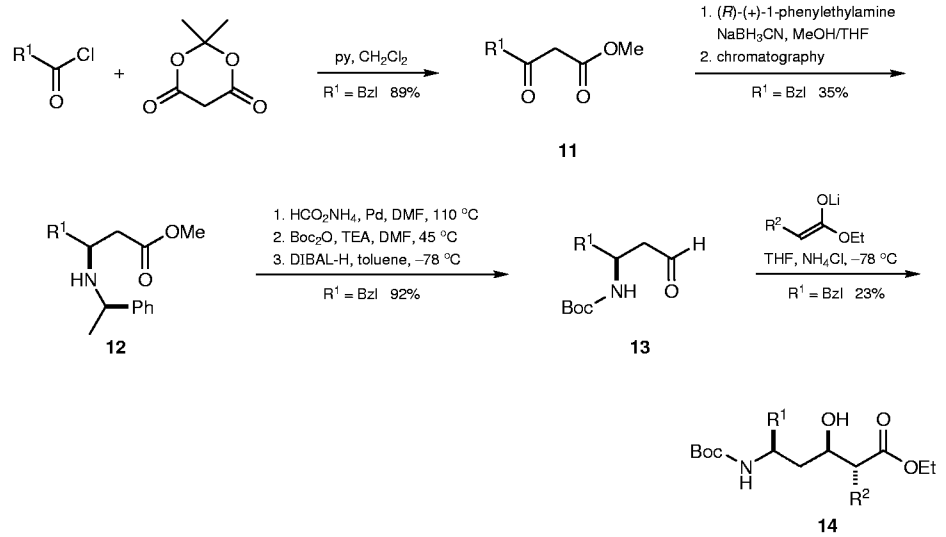
Chen et al.[57] reported the synthesis of the 2-hydroxyethylene isostere using two methods. The first method is initiated with readily available α-amino acids and utilizes the Evans chiral aldol condensation to control the stereochemistry (Scheme 25). The second method does not start with α-amino acids, and thus allows for the synthesis of isosteres having side chains other than those obtained from the available α-amino acids (Scheme 26). Thus, this synthesis relies on an anti-aldol product for the 2-hydroxyethylene isosteres via an *E*-selective ethyl hydrocinnamate enolization.

*for references see p 397*

**Scheme 25** Synthesis of 2-Hydroxyethylene Peptides Utilizing the Evans Chiral Aldol Condensation[57]




**Scheme 26** Alternative Synthesis of 2-Hydroxyethylene Peptides[57]




As described in Scheme 25, the first method starts with the conversion of an α-amino acid into a β-amino-β-alkyl aldehyde. Then, the chiral aldol condensation between the resulting aldehyde and an oxazolidinone derivative is carried out. In the last step, removal of the oxazolidinone provides the desired product.

In the second method (Scheme 26) a β-oxo ester is synthesized from an acid chloride and 2,2-dimethyl-1,3-dioxane-4,6-dione. Then, reductive amination of the β-oxo ester with α-methyl(phenyl)methylamine provides a β-amino-β-alkyl propionic acid ester. This compound is then converted into the corresponding aldehyde, which is condensed with an enolate to afford the final product. A representative synthetic procedure of this method is given in detail.

**(2*R*,3*R*,5*S*)-Ethyl 2-Benzyl-5-(*N*-*tert*-butoxycarbonyl)amino-3-hydroxy-6-phenylhexanoate (14, $R^1 = R^2 = Bzl$):**[57]

**Methyl 3-Oxo-4-phenylbutyrate (11, $R^1 = Bzl$):**[57]

A 50 mL, two-necked flask, equipped with a magnetic stirrer, was flame dried and then cooled in a $N_2$ atmosphere. The flask was charged with 2,2-dimethyl-1,3-dioxane-4,6-dione (Meldrum's acid; 1.69 g, 11.76 mmol) dissolved in $CH_2Cl_2$ (20 mL) and cooled to 0 °C. The soln was treated with pyridine (2.09 mL, 25.8 mmol) followed by phenylacetyl chloride (1.7 mL, 12.9 mol). The mixture was stirred at 0 °C for 1 h and at rt for 1 h. The mixture was then partitioned between $CHCl_3$ and $H_2O$. The organic layer was separated, washed with brine, dried ($Na_2SO_4$), and concentrated in vacuo. The resulting red residue was dissolved in MeOH and refluxed for 2 h. The soln was then concentrated in vacuo. The crude product (3.1 g) was purified by flash chromatography (silica gel, $CHCl_3$) to give **11** ($R^1 = Bzl$); yield: 2.01 g (89%); MS *m/z*: 192 [$M^+$].

**(3*R*)-Methyl 3-[(1′-Phenyl-(1′*R*)-ethyl)amino]-4-phenylbutyrate and (3*S*)-Methyl 3-[1′-Phenyl-(1′*R*)-ethylamino]-4-phenylbutyrate (12, $R^1 = Bzl$):**[57]

To a soln of **11** ($R^1 = Bzl$; 2.0 g, 10.4 mmol) in MeOH/THF (1:1, 40 mL) (*R*)-(+)-1-phenylethylamine (6.7 mL, 52 mmol) was added followed by $NaBH_3CN$ (1.31 g, 20.8 mmol) and the mixture was stirred at rt for 18 h. The pH was adjusted to 11 with 1.0 M NaOH and the slurry was concentrated in vacuo to half the original volume. The resulting aqueous soln was extracted with $CHCl_3$ (3×). The combined organic layers were washed with brine, dried ($Na_2SO_4$), and concentrated in vacuo. The crude product (6.7 g) was then chromatographed (silica gel, hexane/EtOAc 4:1). The product eluted as two peaks and fractions were collected based on TLC homogeneity to give pure 3*R*,2′*R*-diastereomer [yield: 913 mg (31%)], and pure 3*S*,2′*R*-diastereomer [yield: 1.03 g (35%)]. (3*R*)-Methyl 3-[1′-phenyl-(1′*R*)-ethylamino]-4-phenylbutyrate: MS *m/z*: 282 [$M - 15]^+$; $[\alpha]_D = 25$ (*c* 1.0, $CHCl_3$). (3*S*)-Methyl 3-[1′-phenyl-(1′*R*)-ethylamino]-4-phenylbutyrate (**12**, $R^1 = Bzl$): MS *m/z*: 282 [$M - 15]^+$; $[\alpha]_D = 23$ (*c* 1.0, $CHCl_3$).

**(3*S*)-3-[(*tert*-Butoxycarbonyl)amino]-4-phenylbutanal (13, $R^1 = Bzl$):**[57]

To a soln of **12** ($R^1 = Bzl$; 942 mg, 3.17 mmol) in dry DMF (45 mL), ammonium formate (1.0 g, 15.85 mmol) was added followed by Pd black (942 mg). The mixture was stirred at 110 °C for 1.5 h and then filtered through a medium-sintered glass funnel. The solids were washed several times with fresh DMF and the filtrate was concentrated, affording pure (3*S*)-methyl 3-amino-4-phenylbutyrate; yield: 613 mg (100%); MS *m/z*: 194 [$M+H]^+$.

To a soln of (3*S*)-methyl 3-amino-4-phenylbutyrate (613 mg, 3.17 mmol) in dry DMF (40 mL), TEA (1.85 mL, 13.3 mmol) was added followed by $Boc_2O$ (1.38 g, 6.34 mmol). After stirring at 45 °C for 1 h, the mixture was concentrated in vacuo. The crude product (1.38 g) was chromatographed (silica gel, $CHCl_3$/acetone 98:2) to yield pure (3*S*)-methyl 3-[(*tert*-butoxycarbonyl)amino]-4-phenylbutyrate; yield: 573 mg (62%); MS *m/z*: 293 [$M^+$]; $[\alpha]_D$ −8 (*c* 1.0, $CHCl_3$).

To a soln of (3*S*)-methyl 3-[(*tert*-butoxycarbonyl)amino]-4-phenylbutyrate (2.93 g, 10 mmol) in dry toluene (refluxed over $CaH_2$ overnight and freshly distilled, 20 mL), precooled to −78 °C and under an argon atmosphere, 1 M DIBAL-H in toluene (20 mL, 20 mmol) was added at such a rate so as to keep the inner temperature below −75 °C for 1 h. The mixture was stirred at −78 °C for another 2 h and quenched with MeOH (10 mL), followed by Rochelle salt (sat. potassium sodium tartrate) (20 mL). The mixture was stirred at 0 °C for 30 min, and then diluted with EtOAc/$Et_2O$ (1:1, 200 mL). The aqueous phase was separated and extracted with the above solvent system. The combined organic soln was washed with 1 M potassium bisulfate, sat. $Na_2CO_3$, brine, and dried ($MgSO_4$). The concentrated oil was flash chromatographed (hexane/EtOAc 4:1) to give **13** ($R^1 = Bzl$) as a white solid; yield: 2.40 g (92%); IR (oil mull) $\tilde{\nu}_{max}$ 3359, 1729, 1692 $cm^{-1}$; FAB-MS *m/z*: 263 [$M^+$]; $[\alpha]_D$ −20 (*c* 1.01, $CHCl_3$).

**(2*R*,3*R*,5*S*)-Ethyl 2-Benzyl-5-(*N*-*tert*-butoxycarbonyl)amino-3-hydroxy-6-phenylhexanoate (14, $R^1 = R^2 = Bzl$):**[57]

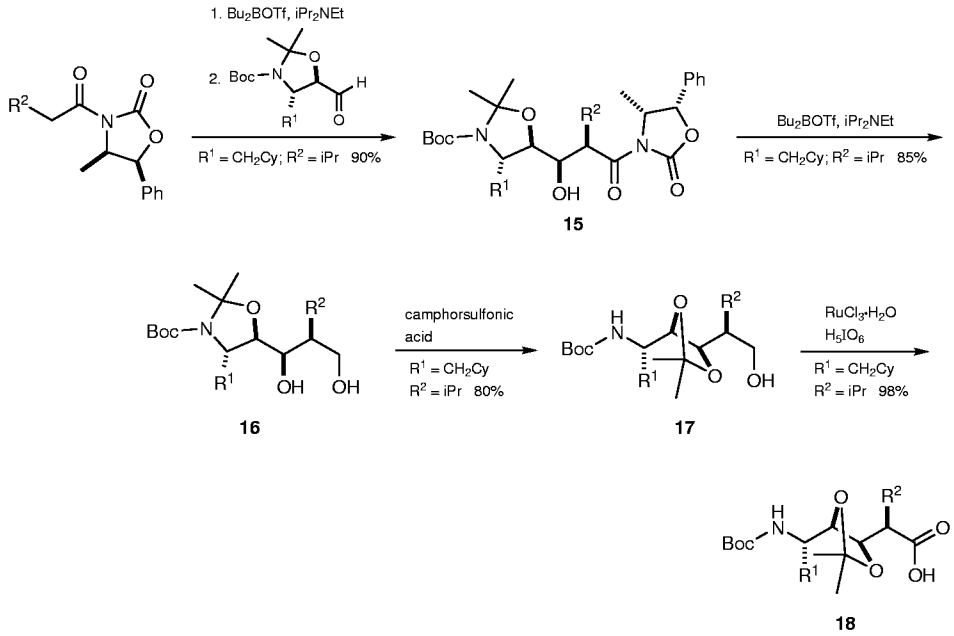
To a stirred slurry of 2,2,6,6-tetramethylpiperidine hydrobromide (105 mg, 0.5 mmol) in freshly distilled THF (6 mL) cooled to 0 °C under anhyd $N_2$, 1.6 M BuLi in hexane (0.58 mL, 0.92 mmol) was added. The mixture was stirred at 0 °C for 15 min and then cooled to −78 °C. A cold soln containing of ethyl hydrocinnamate (67 µL, 0.39 mmol) in freshly distilled THF (1 mL) was added at such a rate as to maintain a −75 °C reaction temperature. After stirring at −78 °C for 30 min, **13** ($R^1 = Bzl$; 100 mg, 0.38 mmol) dissolved in freshly distilled THF (1.5 mL) was added at such a rate as to maintain a −75 °C reaction temperature. After stirring at −78 °C for 45 min, the cooling bath was removed and sat. $NH_4Cl$ (3 mL) was immediately added. The mixture was allowed to warm to rt and then concentrated in vacuo. The resulting residue was partitioned between $CHCl_3$ and $H_2O$. The organic layer was separated and the aqueous layer was re-extracted with $CHCl_3$. The combined organic layers were washed with brine, dried ($Na_2SO_4$), and concentrated in vacuo. The crude product (153 mg) was chromatographed on silica gel (60 g, 230–400 mesh, hexane/EtOAc 2:1) to afford pure **14** ($R^1 = Bzl$); yield: 18 mg (23%); MS *m/z*: 442 [$M+H]^+$.

*for references see p 397*

## 10.6.4 ψ[CH(OH)CH(OH)] (1,2-Dihydroxyethylene) Peptides

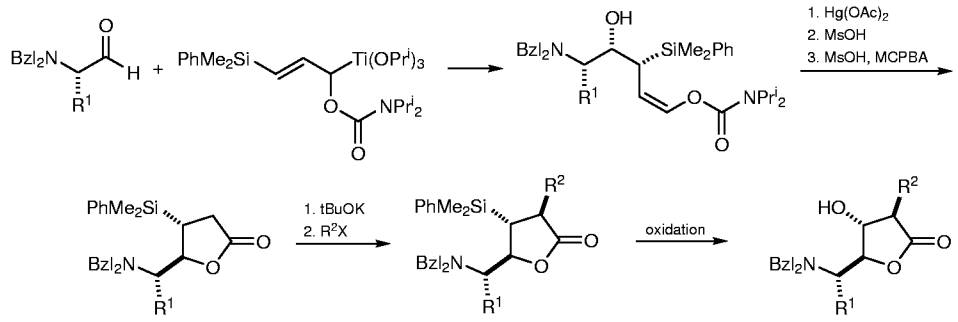
The first and widely applied synthesis of ψ[CH(OH)CH(OH)] isosteres was reported by Thaisrivongs et al.[58–60] This synthesis starts with a diastereoselective aldol addition between a chiral aldehyde and an oxazolidinone (Scheme 27). Then, the chiral auxiliary is removed to provide the diol. The two hydroxy groups can be differentiated by an acid-catalyzed migration of the isopropylidene group from the *N,O*-acetal to the *O,O*-acetal, the structures of which contain *trans* substitution in the five-membered rings. The remaining primary alcohol is oxidized using a ruthenium(III) chloride/periodic acid mixture to yield the desired carboxylic acid.

**Scheme 27** Synthesis of 1,2-Dihydroxyethylene Peptides[58–60]



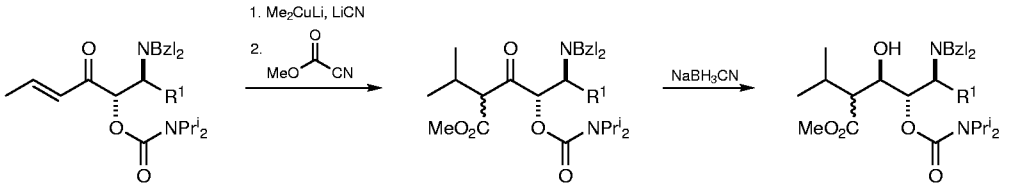

Rehders and Hoppe[61,62] reported a method for the synthesis of 1,2-dihydroxyethylene peptides starting from condensation between an α-amino aldehyde and a homoenolate (Scheme 28). The resulting adduct is converted into a lactone and the second side chain is introduced using the enolate methodology. In this reaction the dimethylphenylsilyl group is used as a directing device for diastereofacial differentiation. In the last step a stereospecific oxidative desilylation of the alkylated lactone by Fleming oxidation yields the desired hydroxylated lactone.

**Scheme 28** Synthesis of 1,2-Dihydroxyethylene Peptides via Condensation Reaction[61,62]




More recently, Hoppe et al.[63] also reported a stereoselective synthesis of a β′-amino-α′-hydroxy enone and its transformation to a 1,2-dihydroxyethylene isostere (Scheme 29). The addition of dilithium dimethylcyanocuprate to the conjugated C=C bond proceeds smoothly to produce a *Z*-enolate. The C-methoxycarbonylation with methyl cyanoformate forms the epimeric mixture of the β-oxo ester. The product is then reduced with sodium cyanoborohyde to provide a protected form of the 1,2-dihydroxyethylene isostere.

**Scheme 29** Alternative Synthesis of 1,2-Dihydroxyethylene Peptides[63]




Additionally, 1,2-dihydroxyethylene dipeptide analogues without the C-terminal carboxylic acid have been used to obtain aspartyl proteases inhibitors.[64] These efforts include stereoselective alkylation of imines, one-pot reductive amination of epoxy ketones, ring opening of epoxides with sodium azide, diastereoselective dihydroxylation of allylic amines, and enzymatic resolution and stereocontrolled intramolecular amidation.

**(2*R*)-2-[(4*R*,5*R*)-4-{(1*S*)-1-[(*tert*-Butoxycarbonyl)amino]cyclohexylethyl}-2,2-dimethyl-1,3-dioxolan-5-yl]-3-methylbutanoic Acid** (18, $R^1 = CH_2Cy$; $R^2 = iPr$):[58–60]

**(4*R*,5*S*)-3-{(2*R*,3*R*)-[(4*S*,5*R*)-3-(*tert*-Butoxycarbonyl)-4-(cyclohexylmethyl)-2,2-dimethyloxazolidin-5-yl]-3-hydroxy-2-isopropyl-1-oxopropyl}-4-methyl-5-phenyloxazolidin-2-one** (15, $R^1 = CH_2Cy$; $R^2 = iPr$); **Typical Procedure:**[58,60]

To a stirred soln of (4*R*,5*S*)-4-methyl-3-(1-oxo-3-methylbutyl)-5-phenyloxazolidin-2-one (460 mg, 1.75 mmol) in $CH_2Cl_2$ (1.8 mL) at 0 °C under argon, dibutylboron triflate (0.49 mL, 1.92 mmol) was added dropwise followed by $iPr_2NEt$ (0.36 mL, 2.1 mmol). After 30 min the mixture was cooled to −78 °C and (4*S*,5*R*)-3-(*tert*-butoxycarbonyl)-4-(cyclohexylmethyl)-2,2-dimethyl-5-formyloxazolidine (520 mg, 1.59 mmol) in $CH_2Cl_2$ (1.5 mL) was added dropwise. After 30 min the mixture was allowed to warm to rt. After an additional 90 min it was cooled in an ice bath and treated with MeOH (1.3 mL) and 30% $H_2O_2$ (1.3 mL) in MeOH (2.6 mL). After 1 h the mixture was partitioned between $CH_2Cl_2$ and aq phosphate (pH 7) buffer. The organic phase was dried ($MgSO_4$) and then concentrated. The residue was chromatographed (silica gel, 15% EtOAc/hexane) to give **15** ($R^1 = CH_2Cy$; $R^2 = iPr$); yield: 860 mg (1.46 mmol, 90%); IR (oil mull) $\tilde{\nu}_{max}$ 3470, 2868, 2853, 1789, 1692 $cm^{-1}$; $[\alpha]_D$ +18 (*c* 1.0, $CHCl_3$); FAB-HRMS *m/z*: 587.3706 $[M+H]^+$.

**(1*R*,2*S*)-1-[(4*S*,5*R*)-3-(*tert*-Butoxycarbonyl)-4-(cyclohexylmethyl)-2,2-dimethyloxazolidin-5-yl]-2-isopropylpropane-1,3-diol** (16, $R^1 = CH_2Cy$; $R^2 = iPr$):[58–60]

To a stirred soln of **15** ($R^1 = CH_2Cy$; $R^2 = iPr$; 1.6 g, 2.72 mmol) in THF (5.4 mL) at 0 °C under argon, $iPr_2NEt$ (0.59 mL, 3.4 mmol) was added dropwise followed by dibutylboron triflate (0.76 mL, 3.0 mmol).

*for references see p 397*

After 30 min, 2 M $LiBH_4$ in THF (3.4 mL, 6.8 mmol) was added dropwise. After 90 min the mixture was allowed to warm to rt. After a further 30 min it was cooled in an ice bath and treated with a soln containing MeOH (5.4 mL), aq phosphate (pH 7) buffer (5.4 mL), and 30% $H_2O_2$ soln (2.7 mL). After stirring at rt for 14 h the mixture was partitioned between $CH_2Cl_2$ and aq phosphate (pH 7) buffer. The organic phase was dried ($MgSO_4$) and then concentrated. The residue was chromatographed (silica gel, 20% EtOAc/hexane) to give **16** ($R^1 = CH_2Cy$; $R^2 = iPr$); yield: 955 mg (2.31 mmol, 85%); IR (neat) $\tilde{\nu}_{max}$ 3470, 2926, 1700, 1680 $cm^{-1}$; $[\alpha]_D$ +11 (*c* 0.32, $CHCl_3$); FAB-HRMS *m/z*: 414 $[M+H]^+$.

**(2*S*)-2-[(4*R*,5*R*)-4-{(1*S*)-1-[(*tert*-Butoxycarbonyl)amino]-2-cyclohexylethyl}-2,2-dimethyldioxolan-5-yl]-3-methylbutanol** (17, $R^1 = CH_2Cy$; $R^2 = iPr$):[58–60]

A soln of **16** ($R^1 = CH_2Cy$; $R^2 = iPr$; 562 mg, 1.36 mmol) in 0.1 M camphorsulfonic acid in $CH_2Cl_2$ (5.5 mL) was allowed to stir at rt for 45 min, and then acetone (7.5 mL) was added. After 14 h the mixture was treated with excess solid $NaHCO_3$. After 40 min it was filtered through Celite and the filtrate concentrated. The residue was chromatographed (silica gel, 10% EtOAc/hexane) to give **17** ($R^1 = CH_2Cy$; $R^2 = iPr$); yield: 450 mg (1.08 mmol, 80%); IR (oil mull) $\tilde{\nu}_{max}$ 3397, 3267, 2954, 1671 $cm^{-1}$; $[\alpha]_D$ −26 (*c* 0.89, $CHCl_3$); FAB-HRMS *m/z*: 414.3207 $[M+H]^+$.

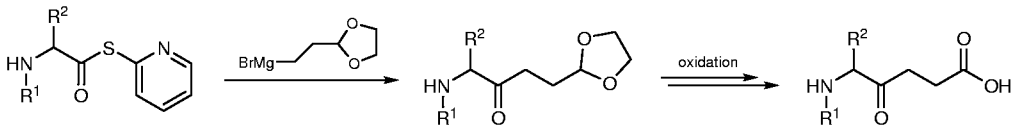
**(2*R*)-2-[(4*R*),(5*R*)-4-{(1*S*)-1-[(*tert*-Butoxycarbonyl)amino]cyclohexylethyl}-2,2-dimethyl-1,3-dioxolan-5-yl]-3-methylbutanoic Acid** (18, $R^1 = CH_2Cy$; $R^1 = iPr$):[58–60]

To a stirred soln of of **17** ($R^1 = CH_2Cy$; $R^2 = iPr$; 450 mg, 1.08 mmol) in MeCN (2 mL), $CCl_4$ (2 mL), and $H_2O$ (3 mL), periodic acid (993 mg, 4.35 mmol) and $RuCl_3{\cdot}3H_2O$ (5.7 mg, 0.02 mmol) were added. The resulting mixture was vigorously stirred for 2 h and then partitioned between $CH_2Cl_2$ and $H_2O$. The organic phase was dried ($MgSO_4$) and then concentrated. The residue was chromatographed (silica gel, 30% EtOAc/hexane) to give **18** ($R^1 = CH_2Cy$; $R^2 = iPr$); yield: 455 mg (1.06 mmol, 98%); IR (oil mull) $\tilde{\nu}_{max}$ 3333, 2934, 1710, 1649 $cm^{-1}$; FAB-HRMS *m/z*: 428.2007 $[M+H]^+$.

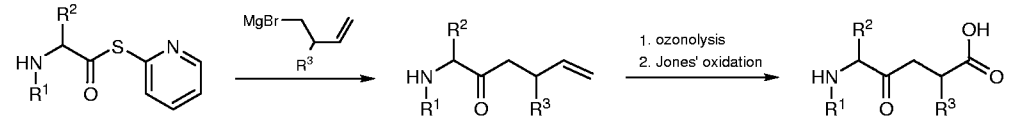
## 10.6.5 ψ[C(=O)$CH_2$] (1-Oxoethylene) Peptides

Several approaches towards the synthesis of 1-oxoethylene isosteres have been reported. Almquist et al.[65] synthesized 1-oxoethylene Gly-containing peptides starting with the reaction between a 2-pyridyl thioester of an $N^{\alpha}$-protected α-amino acid and a Grignard reagent, followed by oxidizing the resulting ketal (Scheme 30). To facilitate the introduction of a side chain into the 1-oxoethylene residue, Almquist and co-workers[66,67] employed homoallylic Grignard reagents, and the terminal carboxylic acid was obtained by ozonolysis of the allyl group followed by Jones' oxidation (Scheme 31).[66,67] Using this method, DeGraw et al.[68,69] synthesized various 1-oxoethylene dipeptides and employed the resulting dipeptides as building blocks for the solid-phase synthesis of thymopentin analogues. In addition, Almquist et al.[70] synthesized an enkephalin analogue containing two consecutive 1-oxoethylene isosteres using an amino acid chloride instead of an amino acid 2-pyridyl thioester.

**Scheme 30** Synthesis of 1-Oxoethylene Peptides using an α-Amino Acid Thioester and a Grignard Reagent[65]

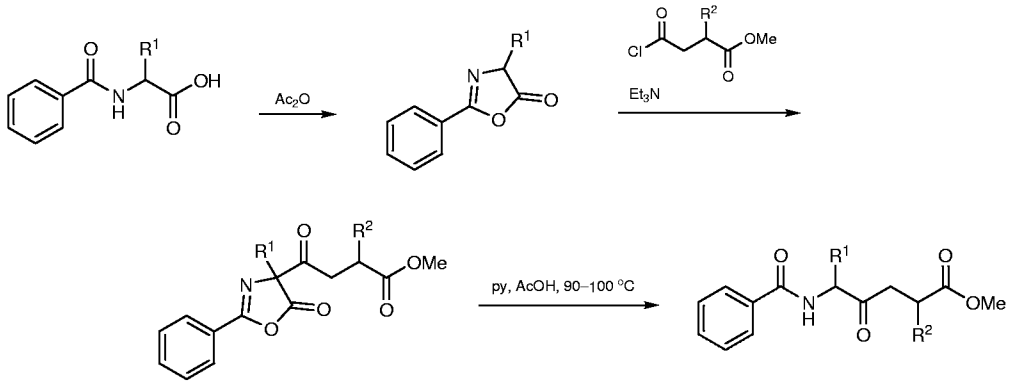



**Scheme 31** Synthesis of 1-Oxoethylene Peptides using an α-Amino Acid Thioester and a Homoallylic Grignard Reagent[66,67]



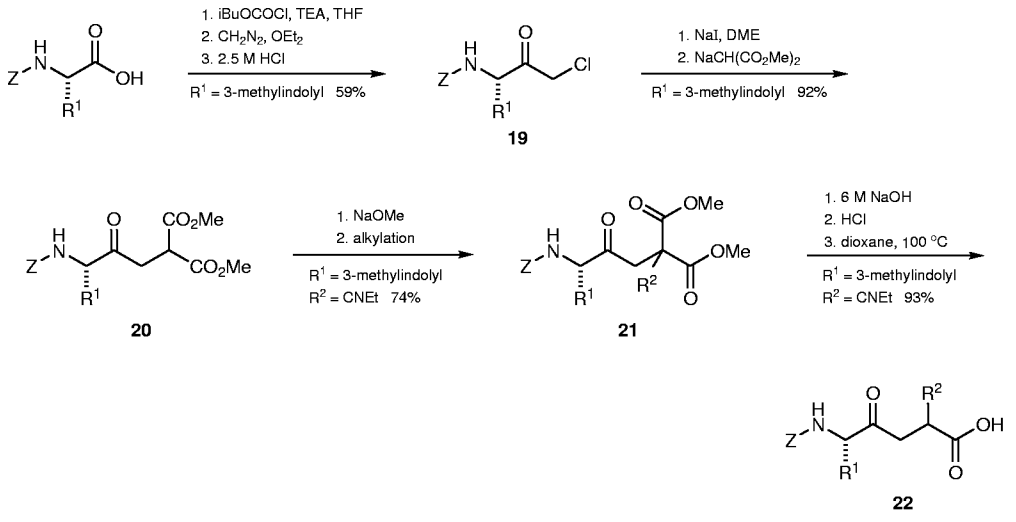
A modified form of the Dakin–West reaction[71] was also employed to synthesize an 1-oxoethylene isostere by Meyer et al.[72] and Almquist et al.[73] (Scheme 32). To obtain the oxazol-5(4*H*)-one an $N^{\alpha}$-acylated α-amino acid is treated with acetyl anhydride. As for the acyl group of the starting amino acid, the benzoyl group is favored because the benzene ring can activate the oxazolone intermediate. The oxazolone can also be synthesized by treatment with DCC[74] or by catalytic hydrogenation of the corresponding unsaturated oxazolone obtained from the reaction of aldehydes and hippuric acid (Erlenmeyer reaction).[75] The resulting oxazolone is reacted with an acid chloride to yield an N-blocked 1-oxoethylene dipeptide ester. Using similar methods, Cheng et al.[76] and Casimir et al.[77] accomplished the direct incorporation of 1-oxoethylene Gly at the C-terminus of urethane Boc-, Z-, or Fmoc-protected dipeptides or tripeptides.

**Scheme 32** Synthesis of 1-Oxoethylene Peptides using a Modified Dakin–West Reaction[72,73]




Umezawa et al.[78] and García-Lopez et al.[79,80] reported the synthesis of Z-protected 1-oxoethylene dipeptides starting with a halomethyl ketone and a malonate (Scheme 33). Various side chains can be introduced into the 1-oxoethylene residue by treating the anion of the oxo diester with appropriate alkylating agents. Using similar methods, Boc-protected 1-oxoethylene dipeptides were also synthesized by Gonzalez-Muniz et al.[81] In both cases the α-carbon of the resulting 1-oxoethylene residue is fully racemic while the configuration of the asymmetric center of the starting amino acid derivatives is maintained.

**Scheme 33** Synthesis of 1-Oxoethylene Peptides using a Halomethyl Ketone and a Malonate[78–80]



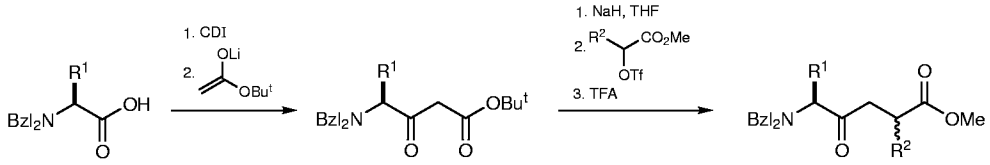

*for references see p 397*

In addition, Holladay and Rich[11] synthesized 1-oxoethylene moieties by oxidation of 1-hydroxyethylene isosteres with pyridinium dichromate (Scheme 34). Since the 1-hydroxyethylene isostere can be synthesized asymmetrically, the stereochemistry of the α-carbon of the resulting 1-oxoethylene dipeptide is defined. Hoffman and co-workers[47,82] synthesized protected 1-oxoethylene dipeptide isosteres by constructing the central bond of a γ-keto ester via alkylation of a *tert*-butyl β-oxo ester with an α-hydroxy ester (Scheme 35). Déziel et al.[83] synthesized 1-oxoethylene dipeptides containing an Asp residue, Boc-AAψ[C(=O)$CH_2$]Asp(OtBu)OBzl using a diastereoselective Michael addition as the key reaction (Scheme 36). In addition, modified 1-oxoethylene isosteres such as dehydrooxoethylene, fluorooxoethylene, and α-hydroxyoxoethylene were synthesized by Ewenson et al.,[84] Garrett et al.,[85] and Kim et al.,[86,87] respectively.

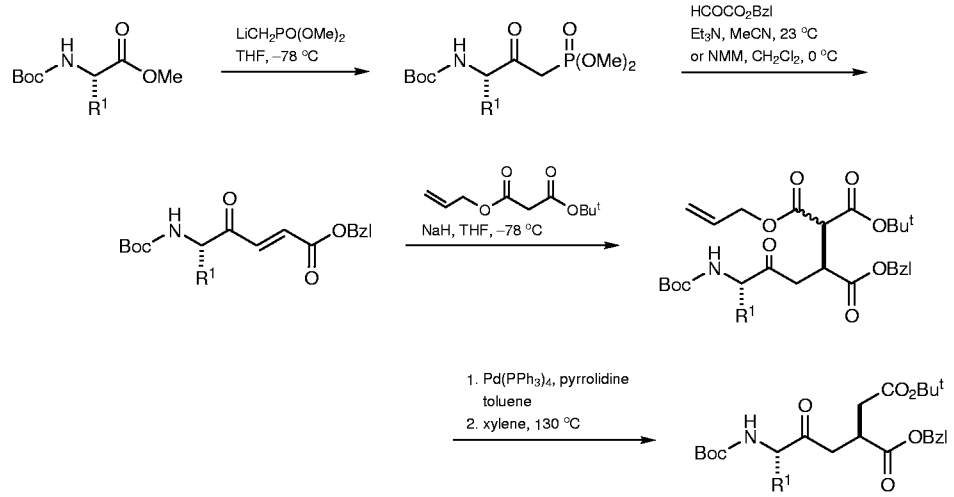
**Scheme 34** Synthesis of 1-Oxoethylene Peptides by Oxidation of a 1-Hydroxyethylene Isostere[11]




**Scheme 35** Synthesis of 1-Oxoethylene Peptides by Alkylation with an α-Hydroxy Ester[47]




**Scheme 36** Synthesis of 1-Oxoethylene Peptides by Diastereoselective Michael Addition[83]




## 10.6.5.1 Synthesis by Reaction of a Halomethyl Ketone

Among the methods described in Section 10.6.5, the syntheses reported by Umezawa et al.[78] and García-Lopez et al.[79,80] have been most widely used. As summarized in Scheme 33, the synthesis is initiated with the preparation of a diazo ketone through the reaction between a N-protected α-amino acid and isobutyl chloroformate followed by treatment with diazomethane. The chloromethyl ketone is prepared by adding 2.5 M hydrochloric acid to the diazo ketone. Transhalogenation is exploited to obtain the iodomethyl ketone. Through in situ reaction with the sodium derivative of dimethyl malonate, the 4-oxo diester is obtained.

Saponification and acidification with concentrated hydrochloric acid, followed by decarboxylation, lead to the N-protected 1-oxoethylene Gly dipeptide.

To introduce a side chain into the 1-oxoethylene residue, the sodium derivative of the oxo diester is treated with the appropriate alkylating agent. These alkylations provide reasonable yields: iodomethane, 93%; methyl bromoacetate, 89%; benzyl bromide, 87%; gramine methiodide, 72%; iodobutanenitrile, 67%; iodopropanenitrile, 40%. When acrylonitrile instead of iodopropanenitrile is used, the desired product (Orn residue) is obtained in 80% yield. Hydrolysis and decarboxylation of the resulting the oxo ester derivatives lead to the desired Z-protected 1-oxoethylene dipeptides containing Ala, Asp, Phe, Trp, Lys, and Orn residues. The 1-oxoethylene Arg residue can be obtained by amidination of the Orn residue using 1-amidino-3,5-dimethylpyrazole nitrate.

**(2*R/S*,5*S*)-2-(2-Aminopropyl)-5-[(benzyloxycarbonyl)amino]-6-(indol-3-yl)-4-oxohexanoic Acid Hydrochloride Salt (22, $R^1$ = 3-methylindolyl; $R^2$ = 2-$NH_2$Pr) and (2*R/S*,5*S*)-5-[(Benzyloxycarbonyl)amino]-2-(2-guanidylpropyl)-6-(indol-3-yl)-4-oxohexanoic Acid Hydrochloride Salt (22, $R^1$ = 3-methylindolyl; $R^2$ = 2-guanidinopropyl):**

**(3*S*)-3-[(Benzyloxycarbonyl)amino]-4-(indol-3-yl)-2-oxobutyl Chloride (19, $R^1$ = 3-Methylindolyl):**
To a cooled soln (−20 °C) of Z-(*S*)-Trp-OH (20 g, 59 mmol) in dry THF (100 mL), NMM (6.5 mL, 59 mmol) and isobutyl chloroformate (8.1 mL, 70 mmol) were added. The mixture was stirred at that temperature for 30 min and then filtered. An $Et_2O$ soln of diazomethane, prepared from nitrosomethylurea (7.2 g, 70 mmol), was added to the filtrate. The mixture was stirred for 15 min at 0 °C and concentrated to a small volume. Then, 2.5 M ethanolic HCl was added at rt until $N_2$ evolution ceased. Solvents were removed and the residue purified by column chromatography (EtOAc/hexane 1:3). The product was recrystallized (EtOAc/hexane) to give **19** ($R^1$ = 3-methylindolyl); yield: 12 g (59%); mp 134–136 °C.

**Methyl (5*S*)-5-[(Benzyloxycarbonyl)amino]-6-(indol-3-yl)-2-methoxycarbonyl-4-oxohexanoate (20, $R^1$ = 3-methylindolyl):**
A mixture of **19** ($R^1$ = 3-methylindolyl; 7.4 g, 20 mmol) and NaI (3 g, 20 mmol) in DME (60 mL) was stirred at rt for 15 min, and the Na salt of dimethyl malonate (3.43 g, 22 mmol), which was freshly prepared from the corresponding diester and NaOMe in DME (20 mL), was added. Stirring was continued for 1 h and the solvents were removed. The residue was extracted with $CHCl_3$ (100 mL) and washed with $H_2O$ (100 mL). The organic extract was dried ($Na_2SO_4$) and concentrated. The crude product was purified on a column (silica gel, EtOAc/hexane 1:2) to provide **20** ($R^1$ = 3-methylindolyl) as a syrup; yield: 8.6 g (92%).

**Methyl (5*S*)-5-[(Benzyloxycarbonyl)amino]-2-(2-cyanoethyl)-6-(indol-3-yl)-2-methoxycarbonyl-4-oxohexanoate (21, $R^1$ = 3-methylindolyl; $R^2$ = CNEt):**
A stirred soln of **20** ($R^1$ = 3-methylindolyl; 5.3 mmol) and freshly prepared NaOMe (5.8 mmol) in MeOH (40 mL) was treated with acrylonitrile (5.8 mmol). After 3 h of stirring at rt the mixture was neutralized with AcOH and the solvents were removed. The residue was purified on a column (silica gel, EtOAc/hexane 1:2) to give **21** ($R^1$ = 3-methylindolyl; $R^2$ = CNEt) as a homogeneous syrup; yield: 74%.

**(2*R/S*,5*S*)-5-[(Benzyloxycarbonyl)amino]-2-(2-cyanoethyl)-6-(indol-3-yl)-4-oxohexanoic Acid (22, $R^1$ = 3-methylindolyl; $R^2$ = CNEt):**
A soln of the 2-substituted diester **21** ($R^1$ = 3-methylindolyl; $R^2$ = CNEt; 4 mmol) in MeOH (40 mL) was treated with 6 M NaOH (2 mL) and the mixture was stirred at rt for 3 h. After removal of the MeOH, the remaining aqueous mixture was acidified with conc HCl to pH 3, and extracted with EtOAc (100 mL). The extract was dried ($Na_2SO_4$) and concentrated. The residue was dissolved in dioxane (30 mL) and heated under reflux for 4 h. After removal of the solvent the remaining syrup was purified on a column (silica gel, $CHCl_3$/MeOH 15:1) to give **22** ($R^1$ = 3-methylindolyl; $R^2$ = CNEt); yield: 93%.

**(2*R/S*,5*S*)-5-[(Benzyloxycarbonyl)amino]-2-(2-aminopropyl)-6-(indol-3-yl)-4-oxohexanoic Acid Hydrochloride Salt (22, $R^1$ = 3-methylindolyl; $R^2$ = 2-$NH_2$Pr):**
A soln of **22** ($R^1$ = 3-methylindolyl; $R^2$ = CNEt; 2.9 mmol) in sat. methanolic $NH_3$ (50 mL) was hydrogenated in a Parr apparatus in the presence of wet Raney Ni (0.94 g) at rt and 2000 Torr for 7 h. The insoluble material was discarded by filtration and the filtrate was concentrated. The resulting residue was dissolved in 2 M HCl (30 mL) and washed (EtOAc). The aqueous layer was concentrated to a gum. The residue was purified on a column (silica gel, $CHCl_3$/MeOH 6:1) to give **22** ($R^1$ = 3-methylindolyl; $R^2$ = $NH_2$Pr); yield: 70%.

*for references see p 397*

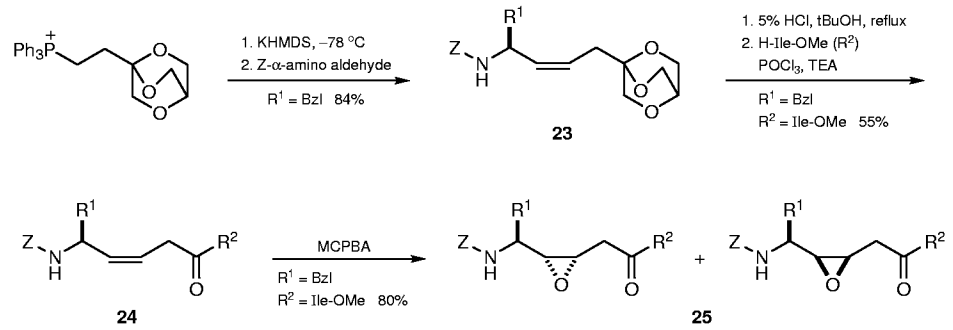
**(2*R*/*S*,5*S*)-5-[(Benzyloxycarbonyl)amino]-2-(2-guanidylpropyl)-6-(indol-3-yl)-4-oxohexanoic Acid Hydrochloride Salt (22, $R^1$ = 3-methylindolyl; $R^2$ = 2-guandinopropyl):**

A soln of **22** ($R^1$ = 3-methylindolyl; $R^2$ = 2-$NH_2$Pr; 1.8 g, 3.6 mmol) in dry DMF (10 mL) was added to a soln of 1-amidino-3,5-dimethylpyrazole nitrate (0.72 g, 3.6 mmol) in dry DMF (10 mL) and adjusted to pH 8–9 with TEA (0.5 mL). The pH of the mixture was maintained at pH 8–9 with more TEA (1.0 mL). After 7 days at rt the solvent was removed, the residue was dissolved in 2 M HCl (20 mL), and washed (EtOAc). The aqueous phase was concentrated to dryness and the residue was purified on a column (silica gel, $CHCl_3$/MeOH 3:1) to give **22** ($R^1$ = 3-methylindolyl; $R^2$ = 2-guanidinopropylamino); yield: 1.15 g (63.4%).

## 10.6.6 ψ[CH(O)CH] (Oxirane) Peptides

Kaltenbronn et al.[88] and Kim et al.[89–93] synthesized oxirane (epoxide) isosteres by epoxidation of *cis*- and *trans*-alkene isosteres (Scheme 37). Kim et al.[89–93] applied these methods to the design of HIV-1 protease inhibitors. The resulting oxirane isostere can react with the Asp side chains of HIV-1 protease to form ester bonds. Thus, the compounds are acting as irreversible inhibitors.

**Scheme 37** Synthesis of Oxirane Peptides[88–93]




The method of Kim et al.[89–93] starts from the synthesis of the three-carbon phosphonium salt according to the modified method of Corey et al.[94,95] The Wittig reaction of the phosphonium salt with a Z-protected α-amino aldehyde using potassium hexamethyldisilazanide provides the *cis*-alkene without racemization. Efficient hydrolysis of the orthoester without double bond migration is achieved by acidolytic hydrolysis with aqueous hydrochloric acid in *tert*-butyl alcohol under reflux conditions. Then, an α-amino acid methyl ester is coupled. The desired epoxide product is obtained by treatment with 3-chloroperoxybenzoic acid. The epoxidation reaction is stereoselective and predominantly provides one isomer (*R*,*S*:*S*,*R* = 4–10:1). The *trans*-epoxide can also be prepared using a *trans*-alkene-containing peptide. A representative synthetic procedure to obtain the *cis*-epoxide isostere is detailed below.

**Methyl (2*S*,3*S*,8*S*,9*R*,10*S*)-2-Amino-10-[(benzyloxycarbonyl)amino]-8,9-epoxy-6-oxo-11-phenylundecanoate (25, $R^1$ = Bzl; $R^2$ = Ile-OMe):**

**(2*Z*,2*S*)-*N*-(Benzyloxycarbonyl)-1-phenyl-4-(2,5,7-trioxabicyclo[2.2.2]oct-1-yl)but-2-en-2-amine (23, $R^1$ = Bzl):**

The triphenyl[2-(2,5,7-trioxabicyclo[2.2.2]oct-1-yl)ethyl]phosphonium bromide (60.8 g, 0.12 mol) was dissolved in THF (400 mL) and the mixture was stirred at –78 °C for 10 min. Then, KHMDS (0.5 M, 220 mL, 0.11 mol) was added and the whole mixture was stirred at –78 °C for 1 h. To the mixture (*N*-benzyloxycarbonyl)-*S*-phenylalaninal (30 g, 0.106 mol) dissolved in THF (150 mL) was added and the

soln was cooled to −78 °C slowly over 40 min. The mixture was stirred at −78 °C for 1 h and subsequently at rt for 1 h. Then, the reaction was quenched with $H_2O$. The solvent was removed from the mixture and the residue was dissolved in $CH_2Cl_2$ (500 mL) and washed with sat. $NaHCO_3$ and $H_2O$. The organic layer was dried ($Na_2SO_4$) and the residue was purified by column chromatography (hexane/EtOAc/TEA 70:25:5) to give **23** ($R^1$ = Bzl); yield: 36.5 g (84%).

**Methyl (2*S*,3*S*)-2-[{(3*Z*,5*S*)-5-[(Benzyloxycarbonyl)amino]-6-phenylhex-3-enoyl}amino]-3-methylpentanoate (24, $R^1$ = Bzl; $R^2$ = Ile-OMe):**

Compound **23** ($R^1$ = Bzl; 2.5 g, 6 mmol) was dissolved in a mixture of $H_2O$ (12 mL) and *t*-BuOH (60 mL) containing less than 5% of concd HCl soln. The resulting soln was refluxed for 20 h. The solvent was removed under reduced pressure and the residue was adjusted to pH 10 with sat. $K_2CO_3$ and then washed with $CH_2Cl_2$ (3×300 mL). The organic layer was removed and the aqueous layer was adjusted to pH 2, extracted (EtOAc), and dried ($MgSO_4$). Concentration of the organic solvent under reduced pressure gave (3*Z*,5*S*)-5-[(benzyloxycarbonyl)amino]-6-phenylhex-3-ene-1-carboxylic acid; yield: 1.62 g (80%); FAB-MS *m/z*: 340 [M+1]; $[\alpha]_D$ +25.2 (*c* 0.50, MeOH).

The above acid (339 mg, 1 mmol) was dissolved in dry $CH_2Cl_2$ and (*S*)-H-Ile-OMe (1 equiv) and TEA (4 equiv) were added. The mixture was cooled to 0 °C and $POCl_3$ (1.5 equiv) was added dropwise to the mixture. After 3 h the resulting soln was diluted by adding $CH_2Cl_2$ and washed (basic $H_2O$). The organic layer was dried ($MgSO_4$) and distilled under reduced pressure to remove the solvent. Then the residue was purified by column chromatography (hexane/EtOAc 7:3) to give **24** ($R^1$ = Bzl; $R^2$ = Ile-OMe); yield: 256 mg (55%).

**Methyl (2*S*,3*S*,8*S*,9*R*,10*S*)-2-Amino-10-[(benzyloxycarbonyl)amino]-8,9-epoxy-6-oxo-11-phenylundecanoate (25, $R^1$ = Bzl; $R^2$ = Ile-OMe):**

To a $CH_2Cl_2$ soln (20 mL) of **24** ($R^1$ = Bzl; $R^2$ = Ile-OMe; 256 mg, 5.5 mmol), 50% MCPBA (2 equiv) was added and the mixture was stirred for 20 h. The mixture was washed successively with $Na_2S_2O_3$ soln (30 mL) and sat. $NaHCO_3$ soln (30 mL). The organic layer was dried ($MgSO_4$) and the organic solvent was removed by distillation under reduced pressure. The residue was purified by column chromatography (hexane/EtOAc 1:1) to give **25** ($R^1$ = Bzl; $R^2$ = Ile-OMe); yield: 212 mg (80%); FAB-MS *m/z*: 483 [M+1].


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# 10.7 Synthesis of Peptides Containing C/N Amide Bond Replacements

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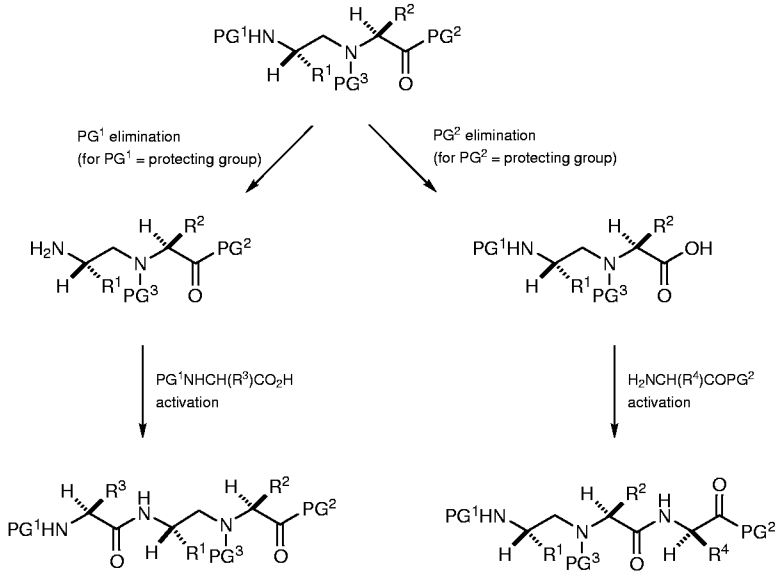
*Dedicated to the memory of Dr. Marc Rodriguez*

## 10.7.1 ψ[$CH_2NH$] (Methyleneamino) (Reduced Amide) Peptides

The isosteric replacement of the amide bond by a reduced peptide bond ψ[$CH_2NH$] was first introduced by Atherton et al. as early as 1971.[1] Later, an analogue of renin containing a reduced peptide bond was found to be a potent renin inhibitor.[2] This analogue represented an interesting approach to the design of bioactive enzyme inhibitors. Indeed, among the various pseudopeptide bonds introduced in peptide chemistry, the reduced peptide bond is one of the more widely used. In addition to its potential to limit enzymatic degradation and to increase hydrophilicity, the reduced amide bond, when introduced into peptide hormones, has also been shown to lead to antagonists.[3,4] The procedure for introducing a reduced amide bond is well established and relatively easy both in solution and solid-phase synthesis. It has been shown that the -$CH_2NH$- moiety is isosteric with the tetrahedral transition state formed during amide bond hydrolysis and it is therefore bound to the enzyme more tightly than the substrate.[5] The introduction of an amino group that could be protonated under physiological conditions into a peptide can also influence receptor interaction and increase solubility in aqueous systems. $^1H$ and $^{13}C$ NMR studies have shown that the modifications produced by the replacement of -CONH- by -$CH_2NH$- are localized around this moiety.[6] However, as a peptide bond induces some rigidity to the molecule, its replacement by a reduced bond increases flexibility.

Elongation of the peptide bearing a reduced bond can be performed stepwise or by fragment condensation. In the latter case, the carbonyl of the amino acid involved in the reduced peptide bond can be activated (Scheme 1) as no racemization by the oxazol-5(4*H*)-one mechanism can occur.

**Scheme 1** Elongation of a Reduced Pseudopeptide




$PG^1$ = N-terminal PG, peptide sequence
$PG^2$ = OH, C-terminal PG, peptide sequence, solid support
$PG^3$ = H, PG

As one of the characteristics of ψ[$CH_2NH$] replacement is the introduction of a new basic site, protection of the secondary amine can be required during peptide elongation. However, it has been shown by Coy and co-workers that the ψ[$CH_2NH$] surrogate is generally resistant to further acylation, except for the incorporation of the Gly residue, probably due to a steric factor.[7,8] In the case of pseudopeptide libraries, protection of the secondary amine seems to be important for affording products in better yields and purity.[9]

### 10.7.1.1 Condensation of a $N^{\alpha}$-Urethane Protected α-Amino Aldehyde with a Free Amino Component

α-Amino aldehydes are important compounds widely used in peptide and organic syntheses. They also provide an interesting access to numerous unusual amino acids of biological interest, such as statine and norstatine, and they are key intermediates for the synthesis of pseudopeptide bonds (reduced, carba-, hydroxyethylene peptide bonds, etc.). Due to the possible use of these molecules, which mimic the transition state of the substrate during hydrolysis, in the synthesis of pseudopeptide bonds and in stereocontrolled organic syntheses, efficient methods for their preparation have been developed. α-Amino aldehydes are mainly obtained by direct reduction of $N^{\alpha}$-protected α-amino acid derivatives. A second approach is based on oxidation of N-protected β-amino alcohols obtained from α-amino acids. In this section the focus will be on different methods used for the preparation of $N^{\alpha}$-protected α-amino aldehydes.

$N^{\alpha}$-Protected α-amino aldehydes and peptide aldehydes are relatively unstable both chemically and configurationally under most reaction conditions. A study of the optical stability of Boc-leucinal during storage at various temperatures was carried out by Rittle et al.[10] They demonstrated that Boc-leucinal essentially retains its chirality at low temperature but it epimerizes at room temperature (Table 1). Furthermore, Ito et al.[11] have shown that purification by chromatography always results in some racemization through keto–enol tautomerization (Scheme 2) and that the percentage of racemization of Z-α-amino aldehydes during purification on silica gel depends on the time of adsorption on silica gel and on the amino acid side chain (Table 2). They observed little racemization of $N^{\alpha}$-Z-$N^{\omega}$-nitroargininal **1**, which seems to be related to its cyclic carbinolamine structure **2** (Scheme 3).

**Scheme 2** Racemization Mechanism of Aldehydes through Keto–Enol Tautomerism[11]



**Scheme 3** Cyclized Carbinol Amine Form of Z-Arg($NO_2$)-H[11]

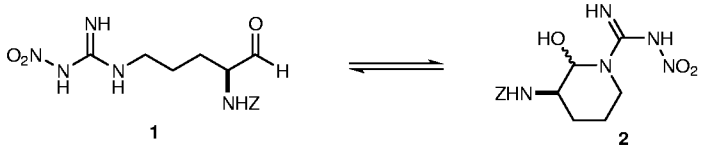


**Table 1** Optical Stability of Boc-leucinal During Storage[10]

| Storage Time (days) | Storage Temp (°C) | $[\alpha]_D^{24}$ (c 1, MeOH) | % Epimerization |
|---|---|---|---|
| 0 | | +18.2 | 0 |
| 1 | –30 | +17.9 | <5 |
| 9 | –30 | +17.4 | <5 |
| 9 | 24 | +6.9 | 60 |

*for references see p 420*

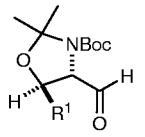
**Table 2** Optical Stability of Z-α-Amino Aldehydes and their Semicarbazone Derivatives During Silica Gel Purification[11]

| α-Amino Aldehyde | Degree of Racemization[a] (%) Exposure Time | | | Degree of Racemization[b] (%) |
|---|---|---|---|---|
| | 0 h | 6 h | 22 h | |
| Z-$N^{\omega}$-nitroargininal | 0 | 5 | 9 | 0 |
| Z-leucinal | 0 | 32 | 65 | 0 |
| Z-phenylalaninal | 0 | 53 | 85 | 0 |
| Z-*S*-Bzl-cysteinal | 7 | 99 | 100 | 5 |

[a] After purification on silica gel chromatography.
[b] After purification as their semicarbazones.

Therefore, it is recommended that aldehydes are used immediately after they have been prepared or that they are derivatized into intermediates which retain chiral integrity during purification or storage. Several studies on the stability of $N^{\alpha}$-protected α-amino aldehydes or derivatives have been reported. The use of 9(9-phenylfluorenyl) as the $N^{\alpha}$-terminal protecting group prevents racemization during preparation, purification, and subsequent reactions.[12] Formation of a more stable semicarbazone (e.g., $ZHN{-}CH(R^1){-}CH{=}NNHCONH_2$), which can be purified and treated with hydrochloric acid to regenerate the pure aldehyde, has been reported[11] (Table 2). Garner and Park[13] have described the synthesis of N,O-diprotected L-serinal **3** ($R^1 = H$) or L-threoninal **3** ($R^1 = Me$) (Scheme 4) and they showed that no racemization occurred after purification of these synthetic intermediates by flash chromatography.

**Scheme 4** N,O-Diprotected L-Serinal and L-Threoninal[13]



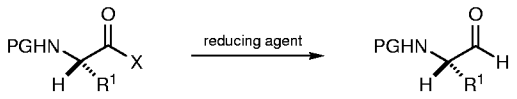

**3** $R^1 = H$
$R^1 = Me$

More recently, a method of purification by flash chromatography on silica gel using 0.1% pyridine in the eluent solvent system[14] has been reported to produce enantiomerically pure aldehyde derivatives.

### 10.7.1.1.1 Preparation of a $N^{\alpha}$-Urethane Protected α-Amino Aldehyde by Reductive Methods

$N^{\alpha}$-Protected α-amino aldehydes **4** are mainly obtained from their corresponding α-amino acid derivatives. Generally the synthetic route proceeds via acid halides, esters, or active amides of α-amino acids that are then reduced. The reduction of $N^{\alpha}$-protected acid halides and esters is often accompanied by some overreduction to the respective alcohols. However, reduction of active amides is apparently free from overreduction. The different procedures described in this review are listed in Table 3.

**Table 3** Preparation of $N^{\alpha}$-Protected Amino Aldehydes by Reduction of Ester and Amide Derivatives




4

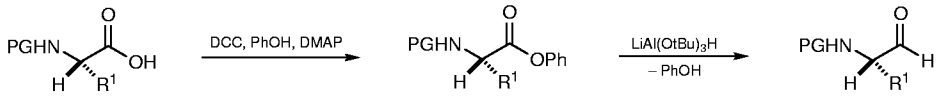
| X | PG | Reductive Agent | Temp (°C) | Time (min) | Yield (%) | Ref |
|---|---|---|---|---|---|---|
| OMe | Z | DIBAL-H | −50 | 50 | 33–68 | [11] |
| OEt | Boc | DIBAL-H | −65 | 45 | 80 | [19] |
| | Boc | | −78 | 6 | 85–97 | [18,20] |
| OPh | Boc | $LiAl(OtBu)_3H$ | 0 | 60 | 71–85 | [21] |
| SBzl | Boc | $Et_3SiH$, 10% Pd/C | 24 | 30–60 | 80–97 | [22] |
| | Fmoc | | 24 | 240 | 67–77 | [14] |
| N(OMe)Me | Boc | $LiAlH_4$ | 0 | 20 | 86–96 | [26] |
| | Fmoc | | 0 | 60 | 87 | [28] |
| DMP | Z, Boc | $LiAlH_4$ | −20 | 120 | 85–95 | [25] |

### 10.7.1.1.1.1 Reduction of α-Amino Acid Halides and Esters

A classical method for the preparation of α-amino aldehydes is the conversion of $N^{\alpha}$-protected α-amino acid chloride derivatives to aldehydes by hydrogenation in the presence of palladium on charcoal[15] or barium sulfate.[16] This method is especially used for the conversion of phthalyl-α-amino acid chlorides into the corresponding aldehydes. $N^{\alpha}$-Protected amino acid chlorides or fluorides are also readily reduced by lithium tri-*tert*-butoxyaluminum hydride[17] at low temperature without epimerization, but this method is of limited application because overreduction into the respective alcohols also occurs.

Several synthetic methods for the reduction of α-amino esters have also been reported. The reduction of methyl or ethyl esters by diisobutylaluminum hydride (DIBAL-H) at low temperature (−78 °C) has been described as useful for the preparation of α-amino aldehydes.[11,18–20] Again, this method suffers from overreduction. Reductive methods involving mild reductive agents have been described. The reduction of phenyl esters **6**[21] (readily prepared from the corresponding amino acid **5**) with lithium tri-*tert*-butoxyaluminum hydride is efficient for the preparation of various Boc-α-amino aldehydes including $N^{\alpha}$-Boc-$N^{\omega}$-nitroargininal and $N^{\alpha}$-Boc-$N^{\omega}$-Z-argininal (Scheme 5).

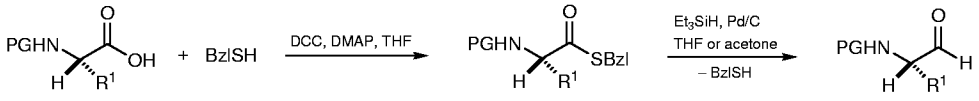
**Scheme 5** Synthesis of an α-Amino Aldehyde by Reduction of an α-Amino Acid Phenyl Ester[21]




5 6 4

A highly efficient reduction of Fmoc-α-amino *S*-ethyl or *S*-benzyl **7** thioesters to aldehydes with triethylsilane and a catalytic amount of palladium on charcoal has been described[14,22] (Scheme 6).

**Scheme 6** Preparation of an α-Amino Aldehyde by Reduction of a *S*-Benzyl α-Amino Thioester[14]




5 7 4

α-Amino aldehydes obtained by this reductive method undergo little epimerization. These reaction conditions can also be applied to Fmoc-protected α-amino aldehydes.[14] It is worth noting that the reaction conditions are sufficiently mild for the selective reduction of a $N^{\alpha}$-protected *S*-benzyl thioester in the presence of other esters.

*for references see p 420*

**Preparation of a Boc-Protected α-Amino Aldehyde via Reduction of the Corresponding Phenyl Ester; General Procedure:**[21]

To a soln of a Boc-protected α-amino acid (10 mmol) in dry THF/MeCN (1:1, 20 mL), freshly distilled phenol (0.94 g, 10 mmol) was added and the mixture cooled under stirring to −10 °C. Then, DCC (2.06 g, 10 mmol) was added followed by DMAP (50 mg). The mixture was stirred overnight at rt. DCU was filtered off and washed with two small portions of dry THF. The filtrate was transferred to a three-necked flask and cooled to 0 °C. The soln of $LiAl(OtBu)_3H$ (2.6 g, 11 mmol) in dry THF (25 mL) was added dropwise under argon at 0 °C. Then, the mixture was stirred for 1 h at 0 °C, poured into 20% citric acid (30 mL) and extracted with EtOAc (× 4). The organic layer was dried ($Na_2SO_4$) and concentrated to a small volume. Then, $iPr_2O$ (25 mL) was added. A precipitate (or an oil) which separated was dissolved in EtOAc (10 mL). Addition of $iPr_2O$ separated the Boc-α-amino aldehyde as an amorphous powder or oil. Aldehyde derivatives were purified by flash chromatography (silica gel, column 5 × 40 cm, $Et_2O$/hexane 2:9, then $Et_2O$/MeOH 10:1).

Boc-L-Ala-H; yield: 76%; $[\alpha]_D^{20}$ −25.5 (*c* 1.0, MeOH).

**Preparation of a $N^{\alpha}$-Protected α-Amino Aldehyde via Reduction of the Corresponding *S*-Benzyl Thioester; General Procedure:**[14]

To a stirred soln of a Fmoc-protected α-amino acid (32 mmol) and BzlSH (64 mmol) in anhyd THF (300 mL) at rt under argon were added DMAP (3.2 mmol) and DCC (35 mmol). The mixture was stirred overnight. The solvent was removed under reduced pressure, the residue was dissolved in EtOAc (400 mL) and washed with 1 M HCl (2 × 300 mL). The organic soln was dried (anhyd $Na_2SO_4$), and filtration and removal of the solvent gave the crude product which was crystallized ($Et_2O$) to yield the pure *S*-benzyl Fmoc-α-amino thioesters.

Fmoc-L-Asp(OtBu)-SBzl; yield: 85%; mp 126–127 °C; $[\alpha]_D^{20}$ +1.1 (*c* 1.0, $CHCl_3$).

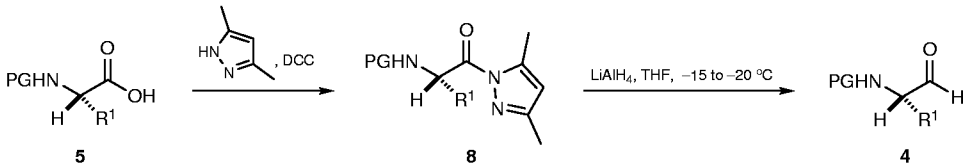
To a stirred soln of *S*-benzyl Fmoc-α-amino thioester (11.5 mmol) and 10% Pd/C (25% by weight) in anhyd acetone (150 mL) or THF (100 mL) at rt under argon was slowly added a soln of $Et_3SiH$ (34.5 mmol) in anhyd acetone (5.5 mL). After 2 h stirring, if the reaction was not completed, a second portion of $Et_3SiH$ (34.5 mmol) was added. The mixture was stirred for additional 2 h and filtered on Celite. Removal of solvent under reduced pressure yielded the crude product which was purified by flash chromatography (silica gel, EtOAc/petroleum ether 1:5 containing 0.1% pyridine). Crystallization ($Et_2O$/petroleum ether) gave the pure Fmoc-α-amino aldehyde.

Fmoc-L-Asp(OtBu)-H; yield: 75%; mp 61–62 °C; $[\alpha]_D^{20}$ −1.4 (*c* 1.0, $CHCl_3$).

### 10.7.1.1.1.2 Reduction of 3,5-Dimethylpyrazole Derivatives or Weinreb Amides

The two-step procedure, involving the transformation of $N^{\alpha}$-protected α-amino acids to suitable amides followed by $LiAlH_4$ reduction, is a reliable methodology for the preparation of $N^{\alpha}$-protected α-amino aldehydes. $N^{\alpha}$-Protected α-amino acid imidazole derivatives are readily converted into the corresponding aldehydes by in situ reduction with $LiAlH_4$, but a high percentage of alcohol is also obtained.[23,24] For example, reduction of 1-(Z-phenylalanyl)imidazole produces Z-phenylalaninal (50%) and Z-phenylalaninol (23%). However, efficient methods of $LiAlH_4$ amide reduction free from racemization and overreduction have been described.[25,26] One of these is based on the reduction of 3,5-dimethylpyrazole derivatives **8** (Scheme 7) of $N^{\alpha}$-protected α-amino acid obtained by coupling 3,5-dimethylpyrazole to $N^{\alpha}$-protected α-amino acids in the presence of DCC.[25] The reduction was carried out with $LiAlH_4$ at −15 °C for one hour.

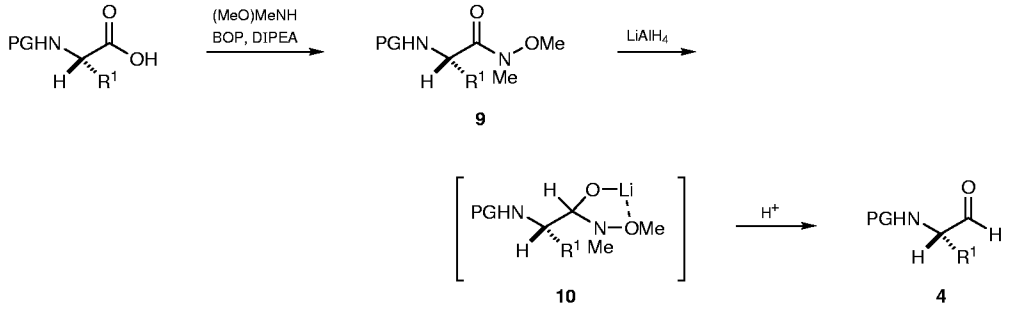
**Scheme 7** Preparation of an α-Amino Aldehyde by Reduction of the Corresponding 3,5-Dimethylpyrazole Derivatives[25]




5 8 4

The other method, proposed by Fehrentz and Castro,[26] described the preparation of very pure Boc-protected α-amino aldehydes via $LiAlH_4$ reduction of the corresponding Weinreb amides **9** (Scheme 8).[27] The possibility of obtaining an aldehyde from a Fmoc-protected α-amino acid was investigated and found to proceed without alteration of the Fmoc group.[28] The reduction proceeds through a stable lithium chelated intermediate **10** (Scheme 8) and the aldehyde is obtained upon acid hydrolysis, thus avoiding the problem of the over-reduction of aldehydes to the corresponding alcohols.

**Scheme 8** Preparation of an α-Amino Aldehyde by Reduction of the Corresponding Weinreb Amide[26]




The reduction of Weinreb amide derivatives by $LiAlH_4$ is not suitable for the synthesis of aspartyl or glutamyl aldehydes. Recently, Paris et al.[29] described the synthesis of various N- and side-chain protected aspartyl or glutamyl aldehydes from the corresponding Weinreb amides by using bulky hydrides such as lithium tri-*tert*-butoxyaluminum hydride.

**Preparation of a $N^{\alpha}$-Protected α-Amino Aldehyde via Reduction of the Corresponding $N^{\alpha}$-Protected α-Amino Acid 3,5-Dimethylpyrazole Derivative; General Procedure:**[25]

A Z-protected α-amino acid (10 mmol) and 3,5-dimethylpyrazole (12 mmol) were treated with DCC (10 mmol) in $CHCl_3$ (200 mL) at −10 °C for 1 h and then left overnight at rt. After removal of DCU the solvent was removed. The solid residue was dissolved in EtOAc and washed with 1 M HCl and $H_2O$. The organic layer was dried ($MgSO_4$) and the solvent removed to give the title compound as an oil or a solid which could be used without further purification.

Z-Phe-3,5-dimethypyrazole;[25] yield: 93%; mp 142 °C; $[\alpha]_D^{20-25}$ +73.9 (*c* 1.0, AcOH).

To a suspension of $LiAlH_4$ (20 mmol) in THF (200 mL) was added a soln of the $N^{\alpha}$-protected α-amino acid 3,5-dimethylpyrazole derivative (10 mmol) in THF (200 mL) over a period of 1 h keeping the temperature between −15 °C and −20 °C under argon. After standing a further 1 h at the same temperature, 2 M HCl (12 mL) was added slowly at a temperature below −20 °C. After removal of $Al(OH)_3$ by filtration, the solvent was concentrated. The residue was dissolved in $Et_2O$, washed with $H_2O$, and then the solvent removed to give the crude aldehyde, which was used without purification in the subsequent reaction.

**Preparation of a Boc-Protected α-Amino Aldehyde via Reduction of the Corresponding Boc-Protected α-Amino Acid *N*-Methoxy-*N*-methylamide; General Procedure:**[26]

TEA (1.012 g, 10 mmol) was added to a stirred soln of the Boc-protected α-amino acid (10 mmol) in $CH_2Cl_2$. Then, BOP (3.483 g, 10 mmol) was added, followed after a few min by (MeO)MeNH·HCl (1.12 g, 11 mmol) and TEA (1.11 g, 11 mmol). The reaction, monitored by TLC, was complete within 30–60 min depending on the substrate. The mixture was then diluted with $CH_2Cl_2$ (250 mL) and washed successively with 3 M HCl (3 × 30 mL), sat. $NaHCO_3$ (3 × 30 mL), and sat. brine (3 × 30 mL). The organic soln was dried ($MgSO_4$) and the solvent removed. The residual crude product was purified by chromatography (silica gel) or by crystallization.

Boc-L-Ala-*N*-methoxy-*N*-methylamide; yield: 85%; mp 150 °C.

$LiAlH_4$ (95 mg, 2.5 mmol) was added to a stirred soln of the Boc-protected α-amino acid *N*-methoxy-*N*-methylamide (2 mmol) in $Et_2O$ or THF (20 mL) at 0 °C. Reduction was complete within 15–20 min. The mixture was hydrolyzed with sat. $KHSO_4$ (10 mL), then $Et_2O$ (50 mL) was added and the aqueous phase was separated and extracted with $Et_2O$ (3 × 50 mL). The organic phases were combined, washed with 1 M HCl (3 × 20 mL), sat. $NaHCO_3$ (3 × 20 mL), and sat. brine (3 × 20 mL), and dried ($MgSO_4$). The solvent was removed to yield the aldehyde.

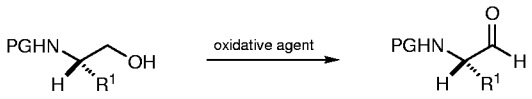
Boc-L-Ala-H; yield: 88%; mp 88–89 °C; $[\alpha]_D^{20}$ +36.7 (*c* 1, $CH_2Cl_2$).

*for references see p 420*

### 10.7.1.1.2 Preparation of a $N^{\alpha}$-Urethane Protected α-Amino Aldehyde by Oxidative Methods

The synthesis of chiral $N^{\alpha}$-protected α-amino aldehydes by a reductive–oxidative sequence from the corresponding α-amino acid derivatives involves the preparation of the corresponding β-amino alcohols **11** (Table 4). They can be synthesized by direct reduction of the $N^{\alpha}$-protected α-amino acids by a complex of borane–tetrahydrofuran[30] or by reduction of the corresponding methyl or ethyl esters by $NaBH_4$ in presence of LiCl[31] or $CaCl_2$.[32] They are also readily and rapidly obtained by $NaBH_4$ reduction of active ester of α-amino acids,[33] mixed anhydrides of $N^{\alpha}$-protected α-amino acids[34,35] and $N^{\alpha}$-urethane protected α-amino acid *N*-carboxyanhydrides (UNCAs).[36] Collin's reagent ($CrO_3$–pyridine) was the first oxidizing reagent shown to be efficient for the preparation of Boc-protected α-amino aldehydes.[10] Oxidation of the corresponding β-amino alcohol with $CrO_3$–pyridine at −10 °C for 30 min results in a rapid reaction to give the aldehyde but the yields are fairly low. Pyridinium dichromate (PDC)[37] has been described as providing highly pure and optically active $N^{\alpha}$-protected α-amino aldehydes in good yield. However, according to Rittle et al.,[10] this method causes significant racemization. Oxoammonium oxidation using a catalytic amount of nitroxyl radical species (2,2,6,6-tetramethyl-1-piperidinyloxy, TEMPO)[38] with commercial bleach as the net oxidant produces $N^{\alpha}$-protected α-amino aldehydes (α-amino acids with hydrophobic side chains and Z and Boc $N^{\alpha}$-protections have been used) in good yield and high optical purity. However, in the case of substrates possessing less hydrophobic side chains, bis-protection of the amino group is recommended as giving aldehydes in good yield. To avoid the problem of overoxidation into the corresponding carboxylic acid, it is necessary to use vigorous stirring of the reaction medium. Further methods involve activated forms of dimethyl sulfoxide, such as DMSO–$SO_3$–pyridine,[39] DMSO–oxalyl chloride,[32,40] DMSO–trifluoroacetic anhydride,[41] or DMSO–DCC,[42] as oxidizing agents. These methods are nonracemizing and have been used for the oxidation of sensitive amino acids such as Trp and Met. They are effective for the preparation of $N^{\alpha}$-protected α-amino aldehydes and also C-terminal peptide aldehydes.[39,43] However, it has been reported[38] that these methods require the use of low temperature and strict adherence to the documented reaction conditions in order to avoid racemization. Methods of preparation of $N^{\alpha}$-protected α-amino aldehydes from their corresponding N-protected β-amino alcohols are summarized in Table 4.

**Table 4** Preparation of $N^{\alpha}$-Protected α-Amino Aldehydes by Oxidation of β-Amino Alcohols[10,32,37–42]




**11** **4**

| Oxidative Agent | PG | Temp (°C) | Time (h) | Yield (%) | Ref |
|---|---|---|---|---|---|
| $CrO_3$–pyridine | Boc | 0 | 0.5 | 67 | [10] |
| PDC | Boc | 20 | 24 | 75–95 | [37] |
| TEMPO | Boc, Z | 0 | 1 | 51–96 | [38] |
| DMSO–$SO_3$–pyridine | Boc | 20 | 0.25 | 66–96 | [39] |
| DMSO–$(COCl)_2$ | Boc | −60 | 0.5 | 89–98 | [32,40] |
| DMSO–$(Tfa)_2O$ | Boc | −78 | 1 | 88 | [41] |
| DMSO–DCC | Boc | 20 | 48 | 43 | [42] |

**Preparation of a $N^{\alpha}$-Protected α-Amino Aldehyde via Oxidation of the Corresponding β-Amino Alcohol: Preparation of a β-Amino Alcohol by Reduction of the Mixed Anhydride of a $N^{\alpha}$-Protected α-Amino Acid:**[35]

To a cold soln (−15 °C) of the $N^{\alpha}$-protected α-amino acid (10 mmol) in DME (10 mL) were successively added NMM (1.11 mL, 10 mmol) and isobutyl chloroformate (1.36 mL, 10 mmol). After 1 min the precipitated NMM·HCl was removed by filtration and washed with DME (5 × 2 mL). The filtrate and

washings were combined in a large flask in an ice-salt bath. A soln of $NaBH_4$ (570 mg, 15 mmol) in $H_2O$ (5 mL) was added at once, producing a strong evolution of gas, and this was followed by $H_2O$ (250 mL) immediately afterwards. If the alcohol precipitated, it was collected and washed thoroughly with $H_2O$ and hexane. If the alcohol did not precipitate, it was extracted with the appropriate solvent and purified by the classical aqueous workup.

Boc-L-Phe-ol; yield: 81%; mp 93–95 °C; $[\alpha]_D^{20}$ −26 (*c* 1.05, MeOH).

**Preparation of a β-Amino Alcohol by Reduction of a $N^{\alpha}$-Protected α-Amino Acid; General Procedure:**[30]

A soln of the $N^{\alpha}$-protected α-amino acid (10 mmol) in THF (10 mL) was added dropwise to a 0 °C soln of 1 M $BH_3$·THF (20 mL, 20 mmol) under $N_2$. The addition took place over a 30 min period, and the reaction was allowed to proceed for 1–2 h at 0 °C. The reaction was quenched using 10% AcOH in MeOH (40 mL). After solvent evaporation the crude product was dissolved in EtOAc and washed with 1 M HCl, $H_2O$, and 1 M $NH_4HCO_3$. The organic layer was dried ($MgSO_4$) and the product was recovered by removal of the solvent.

Boc-L-Ala-ol;[30] yield: 95%; $[\alpha]_D^{20}$ −1.0 (*c* 1.3, $CHCl_3$).

Boc-L-Leu-ol;[10] yield: 81%; $[\alpha]_D^{24}$ −27.4 (*c* 2.03, MeOH).

**Preparation of a $N^{\alpha}$-Protected α-Amino Aldehyde; General Procedure:**[10]

**CAUTION:** *With the DMSO methods, dimethyl sulfide and toxic gaseous side products are produced. With the TEMPO method, care should be taken when handling highly toxic nitroxyl radicals.*

To a soln of dry pyridine (362 g, 4.59 mol) in $CH_2Cl_2$ (5 L) stirred at 0 °C under a $N_2$ was added $Cr_2O_3$ (229 g, 2.29 mol) portionwise while the temperature was maintained below 5 °C. Celite (229 g) was added and the suspension cooled to −10 °C. The mixture was stirred vigorously and added to a soln of Boc-leucinol (50 g, 0.23 mol) in $CH_2Cl_2$ (200 mL) over a 5 min period. The mixture was stirred for 30 min at −10 °C and then the $CH_2Cl_2$ phase was decanted and the residue extracted with fresh $CH_2Cl_2$ (2 × 200 mL). The organic extracts were combined and the solvent was removed under reduced pressure. The brown oily residue was treated with $Et_2O$ (1250 mL) and filtered through a pad of Celite. The filtrate was concentrated under reduced pressure and dried under high vacuum without heat to yield the aldehyde as a light purple oil (46.6 g). Flash chromatography (silica gel, EtOAc) produced the pure aldehyde as a colorless oil.

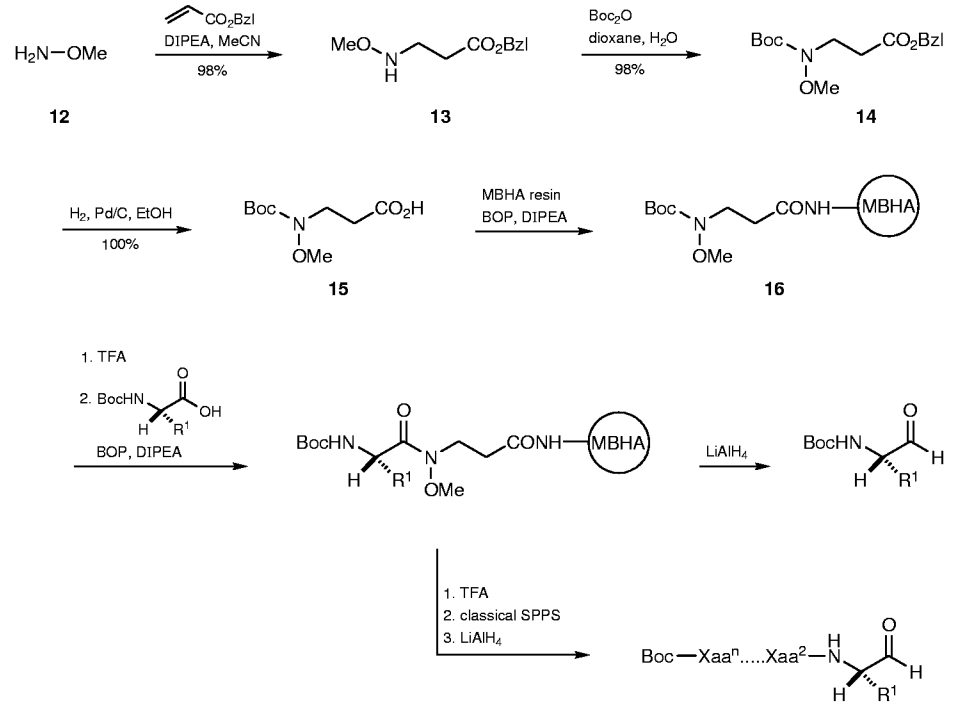
Boc-L-Leu-H; yield: 33 g (67%); $[\alpha]_D^{24}$ −0.71 (*c* 16.8, MeOH).

### 10.7.1.1.3 Preparation of a $N^{\alpha}$-Urethane Protected α-Amino or Peptide Aldehyde on a Solid Support

Several methods for the solid-phase synthesis of peptide aldehydes have been reported. Murphy et al.[44] described a linker based on semicarbazone derivatives. An improvement of this linker has been recently described.[45] Linkers based on the Weinreb amide (Scheme 9) and phenolic esters[46] were developed for the preparation of $N^{\alpha}$-protected α-amino aldehydes and peptide aldehydes. In these two methods peptide aldehydes were released from the resin by $LiAlH_4$ and $LiAl(OtBu)_3H$ reduction, respectively. However, the Weinreb amide approach was preferred due to an uncontrolled overreduction of phenyl esters. Recently, the use of an α,β-unsaturated γ-amino acid as a linker was investigated for the generation of peptide aldehydes from the solid support by ozonolysis. This strategy allowed the preparation of a supported Wittig reagent, which can react with various α-amino aldehydes.[47] Other reports mentioned the use of thiazolidines.[48] The release of the peptide aldehyde was carried out under copper salt mediated neutral hydrolysis, but each α-amino acid has to be derivatized as its thiazolidine by a multistep reaction. Oxazolidines[49] as aldehyde precursors were investigated and used in particular in conjunction with multipin technology. A mild method for the generation of peptide aldehydes from thioester linkers has been developed by Tam and co-workers.[50]

*for references see p 420*

**Scheme 9** Solid-Phase Synthesis of an α-Amino or Peptide Aldehyde by Reduction of the Corresponding Weinreb Amide Linker[46]




**Preparation of a $N^{\alpha}$-Protected α-Amino (and Peptide) Aldehyde on Solid Support via the Weinreb Amide Linker; General Procedure:**[46]

Benzyl acrylate (9.65 g, 60 mmol) was added to a soln containing methoxyamine hydrochloride (**12**) (10 g, 120 mmol) in MeCN (50 mL) in the presence of DIPEA (20 mL). After 24 h reflux, the oily residue was dissolved with EtOAc, washed with sat. $NaHCO_3$, and sat. brine. After drying of the organic layer ($Na_2SO_4$), the soln was concentrated under reduced pressure to yield compound **13** as an oil; yield: 12 g (98%). $Boc_2O$ (12.4 g, 57 mmol) were added to a soln of dioxane/$H_2O$ (2:1) containing compound **13** (12 g, 57 mmol) in the presence of 1 M NaOH in order to maintain the pH between 12 and 13. After 3 h, the solvent was concentrated under reduced pressure, the mixture dissolved in EtOAc and washed with a 1 M $KHSO_4$. After drying and concentration of the organic solvent under reduced pressure, an oily residue **14** was obtained; yield: 17 g (98%).

Compound **14** was hydrogenated with Pd/C as catalyst in 95% EtOH for 3 h. After filtration of the catalyst and concentration of the solvent under reduced pressure, compound **15** was quantitatively obtained as an oil; yield: 12.2 g (100%).

The MBHA resin (substitution 0.63 mmol·$g^{-1}$) was acylated in $CH_2Cl_2$ with the Boc N-protected linker **15** (2 equiv) in the presence of BOP as coupling agent and DIPEA as base. After 4 h and classical washings, the coupling was controlled by Kaiser's test. The protected resin **16** was ready for classical solid-phase synthesis.

The temporary *N*-Boc protection of **16** was removed by treatment with a mixture of TFA/$CH_2Cl_2$ (1:1) for 1 h. Then, successive couplings and deprotection steps were carried out according to the general procedure used for SPPS (Boc-protected α-amino acids and BOP were used). The resin was suspended in anhyd THF and placed in an ice bath. $LiAlH_4$ (5 molar equiv) was added and the mixture was stirred for 30 min. The mixture was then hydrolyzed with a 1 M $KHSO_4$. The resulting mixture was filtered, the resin was washed with $CH_2Cl_2$ (× 2). The solvents were recovered, diluted with $CH_2Cl_2$ and washed with a 1 M $KHSO_4$, sat. $NAHCO_3$ and sat. brine. After drying and concentration of the solvent under reduced pressure, $N^{\alpha}$-protected α-amino (or peptide) aldehyde was obtained.

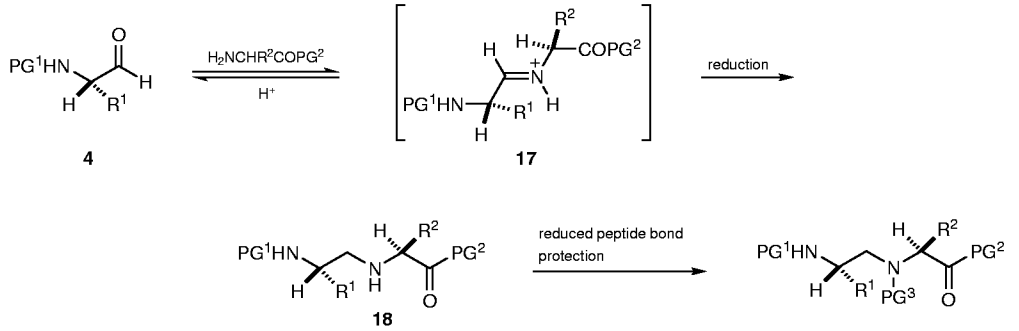
Boc-L-Phe-H; yield: 40%; $[\alpha]_D^{20}$ −46 (*c* 1, MeOH).

Boc-L-Phe-L-Val-L-Ala-H; yield: 45% [after purification by chromatography (silica gel, isocratic soln of EtOAc/hexane 7:3 containing 0.1% pyridine)] or yield: 40% {by RP-HPLC [gradient mode from 100% A to 100% B in 50 min of solns A ($H_2O$ with TFA 0.1%) and B (MeCN with TFA 0.1%)]; $[\alpha]_D^{20}$ −21 (*c* 1, MeOH).

### 10.7.1.1.4 Reductive Amination

A reduced peptide bond can be obtained either in soln or by solid-phase synthesis. The aldehyde of the $N^{\alpha}$-protected α-amino acid reacts with the α-amino function of the C-protected α-amino acid or peptide, to yield a Schiff base **17** which is reduced to give the pseudopeptide **18** (Scheme 10).

**Scheme 10** Solution or Solid-Phase Synthesis of a Reduced Peptide Bond




$PG^1$ = N-terminal PG, peptide sequence
$PG^2$ = OH, C-terminal PG, peptide sequence, solid support
$PG^3$ = PG
reduction: $NaBH_3CN$, MeOH, 1% AcOH or DMF or $H_2$, Pd/C or $PtO_2$, AcOH

#### 10.7.1.1.4.1 Reductive Amination in Solution

In solution, the protonated Schiff base **17** is reduced in the presence of sodium cyanoborohydride ($NaBH_3CN$) as the reductive reagent in a MeOH/AcOH mixture[2,51,52] or by catalytic hydrogenolysis over 10% palladium on charcoal[53] or with $PtO_2$/AcOH[51,54] (Scheme 10). The Schiff base is not isolated, but it is generated in the presence of reducing agents to avoid the problem of racemization.[53,55] It has been reported that in both the $NaBH_3CN$ and the catalytic hydrogenation procedures, the presence of molecular sieves is required for high yields.[53] The reductive amination method with $NaBH_3CN$ is the most commonly used often without adding molecular sieves. In the reductive amination of aldehydes with an unhindered primary amine, particularly with Gly, formation of dialkylated amines has been observed.[56]

**Preparation of Z-L-Nleψ[$CH_2$NH]Gly-OMe in Solution:**[52]
Z-L-Nle-H (6.5 mmol) was dissolved in a mixture of MeOH/AcOH (99:1, 30 mL), containing HCl·H-Gly-OMe (0.82 g, 6.5 mmol). $NaBH_3CN$ (1.23 g, 19.5 mmol) was added portionwise over a period of 45 min. After 1 h, no more H-Gly-OMe could be detected by TLC. The mixture was cooled in an ice water bath, and sat. $NaHCO_3$ (100 mL) was added under stirring, followed by EtOAc (150 mL). The organic layer was collected, washed with $H_2O$ (1 × 20 mL), dried ($Na_2SO_4$), and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel) to give Z-L-Nleψ[$CH_2$NH]Gly-OMe as a pure oily compound; yield: 0.71 g (34%); $R_f$ 0.41 (EtOAc/hexane 7:3).

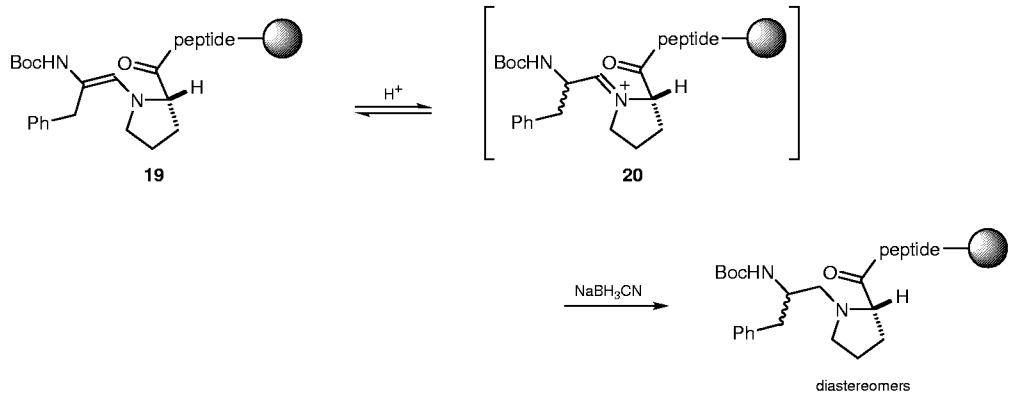
#### 10.7.1.1.4.2 Reductive Amination on Solid Support

In solid-phase synthesis the standard conditions used in solution resulted in an incomplete reaction probably due to a swelling problem of the resin in MeOH. In this case the use of DMF in place of MeOH results in an approximately complete reaction.[7] It has been shown that the introduction of a reduced amide linkage between Phe and Pro residues, by reductive alkylation of the Pro nitrogen with Boc-Phe-H in the presence of $NaBH_3CN$ under acidic conditions leads to epimerization of the Phe residue.[57] This problem of racemization is

*for references see p 420*

probably due to the fact that the Pro residue, being a secondary amine, forms an enamine intermediate **19** which results in loss of the Phe chirality (Scheme 11). Subsequent reduction of the iminium ion **20** results in a diastereomeric mixture of resin-bound peptides.

**Scheme 11** Racemization of the Phenylalanine Residue During the Preparation of Boc-Phe$\psi$($CH_2N$)Pro-Peptide-Resin[57]



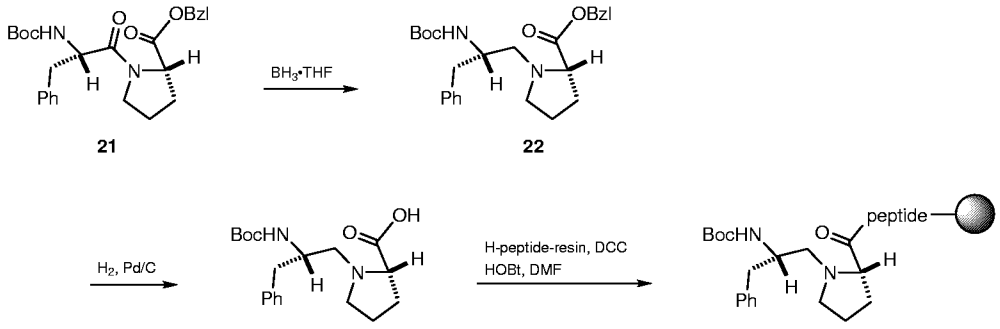

To avoid the production of diastereomers during reductive alkylation of Pro residues, the corresponding dipeptide should be first synthesized and the central amide bond subsequently selectively reduced with diborane (see Section 10.7.1.2.1).

**Solid-Phase Synthesis of a Reduced Peptide Bond; General Procedure:**[8]
Solid-phase synthesis was carried out on a 1% cross-linked benzhydrylamine resin hydrochloride (0.3 equiv·$g^{-1}$) using a manual apparatus. Boc-protected α-amino acids and DCC were used. After removal of the Boc $N^{\alpha}$-protecting group, the trifluoroacetate peptide amide resin was washed with DMF (× 3) and the Boc-protected α-amino aldehyde (2.5 equiv) in DMF containing 1% AcOH was added, followed by the portionwise addition of $NaBH_3CN$ (2.5 equiv) over 40–60 min. The resin was checked by the Kaiser ninhydrin test,[58] and if necessary, the reaction was repeated with 0.5 equiv amounts of alkylating reagents. According to the preliminary study,[7] the secondary amine function of the -$CH_2NH$- bond has poor reactivity toward the DCC-mediated acylation with various amino acids except for Boc-Gly-OH. Thus, the resulting secondary amine function was not protected during the subsequent DCC couplings.

## 10.7.1.2 Reduction of a Peptide or Thioamide Bond

Direct reduction of a peptide bond with diborane[59] or a thioamide bond with several reductive procedures[60] is an alternative route for the production of a reduced peptide bond in a peptide. In some cases the reductive amination does not give satisfactory results. As described earlier, preparation of Boc-Phe$\psi$[$CH_2N$]Pro-OH by reductive amination yields two diastereomers (Scheme 11).[57] In this case treatment of Boc-Phe-Pro-OBzl by diborane yielded the reduced pseudodipeptide Boc-Phe$\psi$[$CH_2N$]Pro-OBzl without epimerization (Scheme 12). However, in some cases diborane is not entirely selective for the amide bond and can reduce ester functions when they are present. Another procedure is to prepare endothiopeptides directly from protected dipeptides[61–66] followed by their selective reduction.[60]

**Scheme 12** Preparation of Racemization-free Boc-Phe$\psi$[$CH_2N$]Pro-Peptide-Resin[57]




### 10.7.1.2.1 Amide Bond Reduction

The facile reduction of amides by borane[67–69] was used for the reduction of the carbonyl group of the peptide bond.[59] This reduction procedure is compatible with Boc, Z, OMe, and OBzl protecting groups. Generally, yields are relatively low. The following experimental procedure describing a dipeptide reduction, is the one which produces the best results.[57]

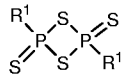
**Boc-Phe$\psi$[$CH_2N$]Pro-OBzl** (22)**:**[57]
The dipeptide Boc-Phe-Pro-OBzl (**21**) (3.53 g, 7.81 mmol) was added to a soln of 1 M $BH_3$·THF in THF (15.6 mL, 15.6 mmol) in THF (52 mL) at 0 °C under $N_2$ over a period of 5 min. When the addition was complete, the resulting mixture was stirred at rt for 24 h. The reaction flask was cooled to 0 °C and sat. $KHSO_4$ (50 mL) was added slowly. THF was removed under reduced pressure. $Na_2CO_3$ (50 mL) was added and the aqueous layer was extracted with EtOAc (× 3, total: 450 mL). The organic soln was dried and concentrated under reduced pressure to afford the reduced peptide as a solid. The product was partially purified by a flash chromatography (silica gel, 96 g, 4.5 × 12 cm, EtOAc/hexane 1:8) and further purified by crystallization (EtOAc/hexane 1:5) to give Boc-Phe$\psi$[$CH_2N$]Pro-OBzl (**22**); yield: 1.81 g (53%); mp 78–79 °C. The C-terminal benzyl ester group was removed by catalytic hydrogenation in MeOH with 10% Pd/C catalyst over 1 h. The C-terminal free pseudopeptide was used in the subsequent step without further purification.

### 10.7.1.2.2 Thioamide Bond Formation and Reduction

Two types of amide thiolation reagents (Scheme 13) are generally used: the traditional Lawesson's reagent **23**,[70] [2,4-bis(4-methoxyphenyl)-1,2,3$\lambda^5$,4$\lambda^5$-dithiaphosphetane-2,4-dithione] and less frequently, Davy's reagents **24** and **25**,[71,72] [2,4-bis(methylthio)-1,2,3$\lambda^5$,4$\lambda^5$-dithiaphosphetane-2,4-dithiane and 2,4-bis(4-methylphenylthio)-1,2,3$\lambda^5$,4$\lambda^5$-dithiaphosphetane-2,4-dithiane, respectively]. Davy's reagents are less bulky than the Lawesson's reagent and seems to be more efficient in the thionation of several peptide bonds.[71–73] For the endothiopeptide reduction at least four methods have been described: (1) direct reduction using Raney nickel; (2) alkylation with triethyloxonium tetrafluoroborate followed by $NaBH_4$ reduction; (3) direct reduction with nickel boride;[60] and (4) reduction with diborane.[59] The nickel boride procedure is described as producing better yields and reproducibility, and requiring shorter times for reaction and purifications.[59,60] This method was recently used to prepare thionated and reduced biologically active peptide analogues such as *Rubia akane* derivatives,[74] hPTH(1–36) analogues,[75] and cyclic RGD peptides containing a reduced bond.[76]

*for references see p 420*

**Scheme 13** Lawesson and Davy's Reagents



**23** $R^1$ = 4-$MeOC_6H_4$
**24** $R^1$ = SMe
**25** $R^1$ = S-4-Tol

**Boc-D-Phe$\psi$[$CH_2$N]Pro-OBzl:**[59]
Boc-D-Phe-Pro-OBzl (7.1 g, 15.6 mmol) was reacted with Lawesson's reagent (4.5 g, 11 mmol, 0.7 equiv) in refluxing THF (50 mL) for 2 h. The mixture was dissolved in EtOAc and washed with sat. $NaHCO_3$. Crude Boc-D-Phe$\psi$[C(=S)N]Pro-OBzl was obtained in nearly quantitative yield. The reaction was monitored by $^1$H NMR [(DMSO-$d_6$, $\delta$): 10 (CSNH)]. The endothiopeptide (7 g, 15 mmol) and $NiCl_2 \cdot 6H_2O$ (29.7 g, 125 mmol) were dissolved in THF/MeOH (1:1 100 mL), the mixture was cooled to 0 °C, and $NaBH_4$ (14.5 g, 374 mmol) was added in small portions. After 30 min the mixture was filtered over Celite and the solvent was removed. The dry gray mass was suspended in EtOAc and the peptide fraction extracted from the solid. After flash chromatography (silica gel, EtOAc/hexane) Boc-D-Phe$\psi$[$CH_2$N]Pro-OBzl crystallized; yield: 3.5 g (53%); mp 77–79 °C; $[\alpha]_D^{20}$ −7.4 (*c* 0.5, MeOH). The C-terminal benzyl ester group was removed by catalytic hydrogenation in MeOH with 10% Pd/C catalyst. The C-terminal free pseudopeptide was used in peptide synthesis without further purification.

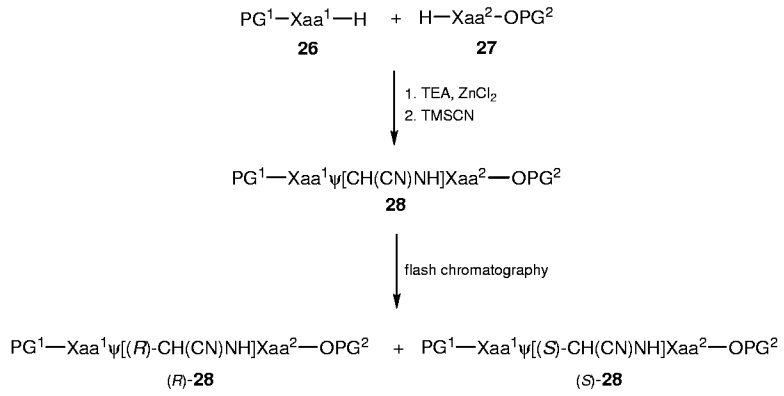
## 10.7.2 $\psi$[CH(CN)NH] [(Cyano)methyleneamino] Peptides

The first synthesis of $\psi$[CH(CN)NH] as a new peptide bond surrogate was described in 1991.[77,78] Semiempirical quantum mechanic calculations have shown that this isosteric peptide bond is similar to the peptide bond from the electronic point of view. In [(cyano)-methyleneamino] pseudopeptides, the cyano group keeps some H-bonding acceptor properties and the new asymmetric center can impart higher backbone rigidity than the reduced peptide bond $\psi$[$CH_2$NH] which is widely and successfully used in the design of receptor agonists and antagonists or enzyme inhibitors (see Section 10.7.1). The synthesis of [(cyano)methyleneamino] pseudopeptides has been shown to be suitable for large peptides.[79] This was demonstrated by incorporation of $\psi$[CH(CN)NH] in pseudopeptide analogues of the C-terminal hexapeptide of neurotensin[80,81] and CCK-4.[82] Furthermore, the versatility of the cyano group makes [(cyano)methyleneamino] peptides suitable starting materials for the preparation of further analogues. For example, a general method of preparation of $\psi$[CH($CONH_2$)NH] pseudopeptides[83] was described during the synthesis of angiotensin II octapeptide analogues,[77] and transformation to $\psi$[CH($CH_2NH_2$)NH] was used to prepare branched peptides.[84] The synthesis of $\psi$[CH(CN)NH] pseudopeptides was performed in solution and then adapted to solid support.[80]

### 10.7.2.1 Synthesis of $\psi$[CH(CN)NH] Pseudopeptides in Solution

As indicated in Scheme 14, introduction of the $\psi$[CH(CN)NH] peptide bond surrogate into pseudopeptides **28** is carried out via a modified asymmetric Strecker synthesis by the Lewis acid catalyzed reaction of $N^{\alpha}$-protected α-amino aldehydes **26** with α-amino esters **27**.[79]

**Scheme 14** Synthesis of Protected Xaa$^1$ψ[CH(CN)NH]Xaa$^2$ Pseudodipeptides[79]




**Protected Xaa$^1$ψ[CH(CN)NH]Xaa$^2$ Pseudodipeptides; General Procedure:**[79]

**CAUTION:** *Trimethylsilyl cyanide (TMSCN) is toxic by inhalation, if brought into contact with the skin, and if swallowed. Keep container in a well-ventilated place and wear suitable protective clothing, gloves, and eye/face protection. Do not empty into drain.*

TEA (0.55 mL, 4 mmol) was added to a stirred soln of the corresponding α-amino acid methyl ester hydrochloride **27**·HCl (PG$^2$ = Me) (4 mmol) in MeOH (25 mL). After 15 min at rt, the mixture was cooled to −20 °C. $ZnCl_2$ (0.27 g, 2 mmol) and the appropriate freshly prepared N$^\alpha$-protected α-amino aldehyde **26** (2 mmol) were added and stirring was continued for 1 h at this temperature. TMSCN (0.30 mL, 2.4 mmol) was added and the mixture was stirred at 0 °C for 24 h. The solvent was removed under reduced pressure, the residue dissolved in EtOAc (25 mL), and the resulting soln was washed with $H_2O$ (20 mL), brine (20 mL), and then dried ($Na_2SO_4$). After removal of the solvent and flash chromatography (EtOAc/hexane), the protected pseudodipeptide **28** was obtained as a mixture of two diastereomers (*R*)-**28** and (*S*)-**28** which could be separated in some cases (**28e**, **28j**). The results are summarized in Table 5. Analytical and spectroscopic data for compounds (*R*)-**28e**, (*S*)-**28e**, (*R*)-**28j**, and (*S*)-**28j** are reported in Table 6.

**Table 5** Yields and Stereoisomeric Ratio of Pseudodipeptides PG$^1$-Xaa$^1$ψ[CH(CN)NH]Xaa$^2$-OMe (PG$^2$ = Me)[79]

| Compound | PG$^1$Xaa$^1$ | Xaa$^2$ | ψ[CH(CN)NH] | | |
|---|---|---|---|---|---|
| | | | Ratio[a] [(*R*)-**28**/(*S*)-**28**] | Yield (%)[b] | $t_R$ (*R*/*S*) (Solvent A/B)[c] |
| **28a** | Boc-L-Phe | L-Leu | 1:1 | 95 | 16.45/15.10 (40:60) |
| **28b** | Z-L-Phe | L-Leu | 1:1 | 90 | 17.60/16.35 (40:60) |
| **28c** | Boc-L-Phe | D-Leu | 1:3[d] | 92 | 15.66 (40:60) |
| **28d** | Boc-L-Phe | L-Val | 2:1 | 80 | 20.48/19.43 (20:80) |
| **28e** | Boc-L-Phe | L-Phe | 1:1 | 82 | 14.81/13.51 (40:60) |
| **28f** | Boc-L-Phe | D-Phe | 1:4[e] | 85 | 14.51 (40:60) |
| **28g** | Boc-L-Phe | L-Lys(Z) | 1:1 | 76 | 11.65/11.36 (20:80) |
| **28h** | Boc-L-Phe | L-Glu(OMe) | 1:1 | 88 | 14.51/14.11 (50:50) |
| **28i** | Z-L-Leu | L-Leu | 1:1 | 85 | 11.31/10.50 (40:60) |
| **28j** | Z-L-Trp | L-Phe | 1:1.5 | 75 | 16.33/15.66 (25:45) |

[a] Determined by RP-HPLC analysis, except for **28c** and **28f**.
[b] Overall yield of (*R*)- and (*S*)-epimers.
[c] A = 0.05% TFA in $H_2O$; B = MeCN.
[d] Determined by measuring the respective OMe, 4-H and 4-H′ integrals for (*R*)-**28c** and (*S*)-**28c** in the $^1$H NMR spectrum of the mixture.
[e] Determined by measuring the respective OMe and α-H integrals for (*R*)-**28f** and (*S*)-**28f** in the $^1$H NMR spectrum of the mixture.

*for references see p 420*

**Table 6** Analytical and Relevant Spectroscopic Data for Protected Pseudopeptides PG$^1$-Xaa$^1$ψ[CH(CN)NH]Xaa$^2$-OMe (PG$^2$ = Me)[79]

| Compound | Formula[a] | UV $\lambda_{max}$ (nm) (MeOH, ε) | $^{13}$C NMR δ[b] | | $^1$H NMR δ[b] | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Xaa$^1$ | | | | Xaa$^2$ | | |
| | | | C2 (Xaa$^1$) | CN | 2-H | 3-H | 4-H | 4-H | NH | α-H | OMe |
| (*R*)-**28e** | $C_{25}H_{31}N_3O_4$ | 214 (6835) | 32.31 | 118.38 | 3.55 | 3.99 | 2.86 | 2.05 | 3.55 | 3.72 | |
| (*S*)-**28e** | $C_{25}H_{31}N_3O_4$ | 213 (6940) | 52.87 | 117.61 | 3.70 | 4.10 | 2.79 | 2.01 | 3.70 | 3.62 | |
| (*R*)-**28j** | $C_{30}H_{30}N_4O_4$ | 224 (12365), 280 (4605) | 52.47 | 118.53 | 3.48 | 4.21 | 2.96 | 3.10 | 1.80 | 3.48 | 3.72 |
| (*S*)-**28j** | $C_{30}H_{30}N_4O_4$ | 224 (11915), 280 (5811) | 52.70 | 117.66 | 3.79 | 4.30 | 2.99 | 1.95 | 3.75 | 3.56 | |

[a] Satisfactory C, H, N analyses are reported.
[b] $CDCl_3$.

## 10.7.2.2 Solid Phase Synthesis of ψ[CH(CN)NH] Pseudopeptides

Solid-phase peptide synthesis of ψ[CH(CN)NH] containing pseudopeptides has been performed using Boc and Fmoc strategies.[80] The Fmoc strategy was successful, while in the Boc chemistry the resulting pseudopeptides were unstable under the standard conditions used for HF cleavage from the resin. The introduction of the ψ[CH(CN)NH] peptide bond surrogate into pseudopeptides was carried out on solid support under conditions similar to those described in Section 10.7.2.1 for solution synthesis. This method was applied to the synthesis of neurotensin [NT(8–13)] analogues indicating that ψ[CH(CN)NH] pseudopeptides can be readily obtained in good yield using Fmoc chemistry on solid support. However, the pseudopeptides bearing the ψ[CH(CN)NH] peptide bond surrogate in the C-terminal position were not stable.

**PG$^1$-Xaa$^1$ψ[CH(CN)NH]Xaa$^2$-Resin Pseudopeptide; General Procedure:**[80]

**CAUTION:** *Trimethylsilyl cyanide (TMSCN) is toxic by inhalation, if brought into contact with the skin, and if swallowed. Keep container in a well-ventilated place and wear protective clothing, gloves, and eye/face protection. Do not empty into drain..*

Typically, dry $ZnCl_2$ (27 mg, 0.2 mmol) and the appropriate N$^\alpha$-protected α-amino aldehyde (0.4 mmol) were added to a suspension of H-Leu-resin or H-Ile-Leu-resin (0.2 mmol) in DMF (8 mL) cooled to −20 °C. After 1 h stirring, TMSCN (0.11 mL, 0.8 mmol) was added and the mixture was left overnight with stirring at 0 °C. A quantitative ninhydrin analysis was performed on a resin sample and the reaction was repeated if necessary. The reaction yields are listed in Table 7.

**Table 7** Reaction Yields in the Synthesis of the PG$^1$-Xaa$^1$ψ[CH(CN)NH]-Xaa$^2$-Resin Pseudopeptide According to the Quantitative Ninhydrin Analysis[80]

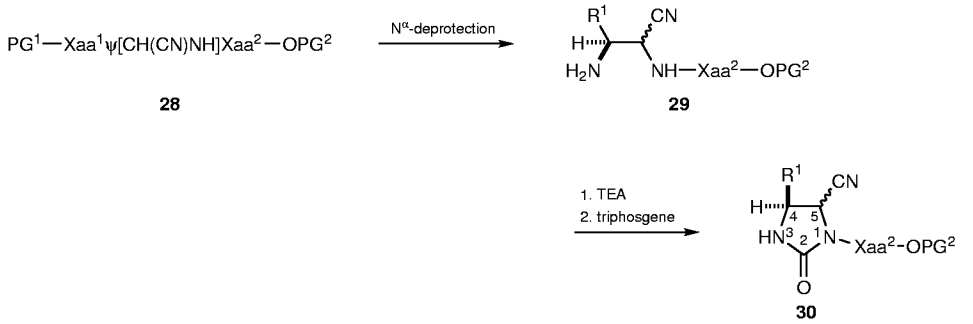
| Aldehyde | Resin | % 1st Coupling | % 2nd Coupling |
|---|---|---|---|
| Boc-Ile-H | H-Leu-resin[a] | 92 | not reported |
| Boc-Tyr[Z(2-Br)]-H | H-Ile-Leu-resin[a] | 64 | 87 |
| Fmoc-Ile-H | H-Leu-resin[b] | 86 | 93 |
| Fmoc-Tyr(tBu)-H | H-Ile-Leu-resin[b] | 86 | 91 |

[a] 4-(Oxymethyl)phenylacetamido-polystyrene resin.
[b] 4-Alkoxybenzyl-polystyrene resin.

### 10.7.2.3 Assignment of Configuration in ψ[CH(CN)NH] Pseudopeptides

The configuration of the new chiral center was established by examining the $^1$H NMR spectra of the imidazolidin-2-ones **30**, obtained by reacting the N$^{\alpha}$-deprotected ψ[CH(CN)NH] pseudodipeptide **29** with bis(trichloromethyl) carbonate (triphosgene) according to Scheme 15.[77–80] The assignment of the absolute configuration of C2 in Xaa$^1$ in the separated diastereomers (*R*)- and (*S*)-**28e** and (*R*)- and (*S*)-**28j** was established by the $J_{4,5}$ coupling constant value in the $^1$H NMR spectra of the corresponding imidazolidin-2-ones (*R*)- and (*S*)-**30e** and (*R*)-**30j**. The *threo*-imidazolidin-2-ones (*R*)-**30e** and (*R*)-**30j** showed a $J_{4,5}$ value of 3–4 Hz, consistent with a $H_4$-$H_5$ *trans* disposition, while in the *erythro*-isomer (*S*)-**30e** and (*S*)-**30j** the value of this coupling constant is 8 Hz.

**Scheme 15** Synthesis of the Imidazolidin-2-ones **30**[79]




**Preparation of Imidazolidin-2-ones 30; General Procedure:**[79]
TEA (0.28 mL, 3 mmol) was added to a suspension of the corresponding N$^{\alpha}$-deprotected pseudopeptide **29** (1 mmol) in dry $CH_2Cl_2$ (10 mL), and the mixture was stirred at rt for 15 min. Then triphosgene (119 mg, 0.4 mmol) and TEA (0.34 mL, 2.4 mmol) were added at 0 °C. The mixture was stirred for 5 h. The mixture was then diluted with $CH_2Cl_2$ (25 mL), washed with $H_2O$ (20 mL) and brine (20 mL), and dried ($Na_2SO_4$). After removal of the solvent a flash chromatography purification of the residue (hexane/EtOAc mixtures) yielded the imidazolidin-2-ones **30** as a foam. In this chromatographic procedure the diastereomeric (5*R*)- and (5*S*)-imidazolidin-2-ones were separated. Analytical and relevant spectroscopic data of these compounds are summarized in Table 8 .

**Table 8** Analytical and Relevant Spectroscopic Data for Imidazolidin-2-ones **30**[79]

| Compound | Yield (%) | Formula | $^1$H NMR δ, *J* (Hz) | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Imidazolidin-2-one | | | | | Xaa$^2$ | |
| | | | 4-H | 5-H | $J_{4,5}$ | 4-$CH_2$ | NH | α-H | OMe |
| (*R*)-**30e** | 80 | $C_{21}H_{21}N_3O_3$ | 3.99 | 4.29 | 4.1 | 2.74 | 4.98 | 4.69 | 3.76 |
| (*S*)-**30e** | 78 | $C_{21}H_{21}N_3O_3$ | 3.65 | 4.46 | 7.9 | 2.72, 2.93 | 4.63 | 4.62 | 3.76 |
| (*R*)-**30j** | 40 | $C_{23}H_{22}N_4O_3$ | 4.22 | 4.41 | 3.2 | 2.86, 3.06 | 5.54 | 4.79 | 3.76 |

## 10.7.3 ψ[NHCH$_2$] (Aminomethylene) (Reduced Retro-amide) Peptides

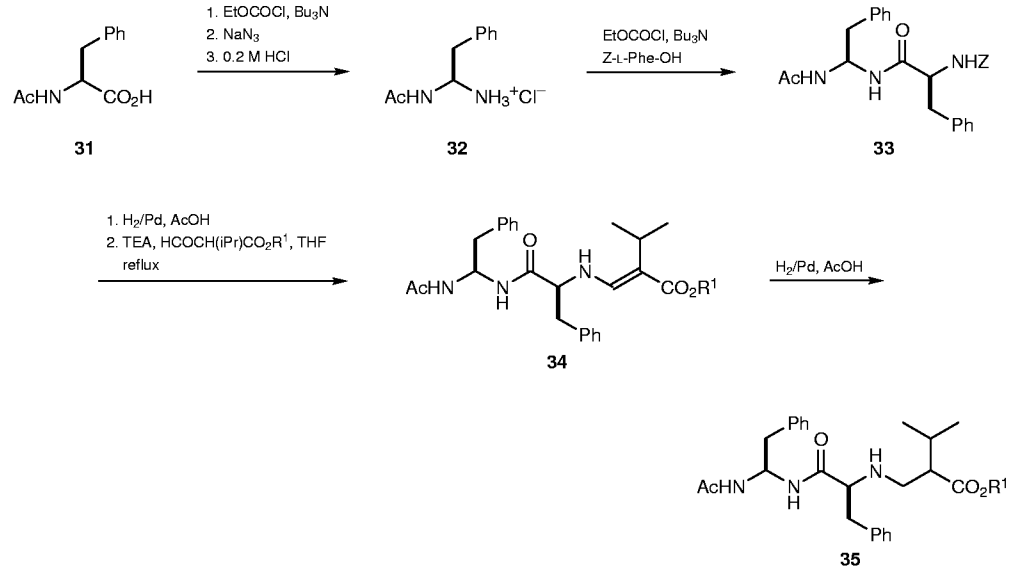
The first synthesis of ψ[NHCH$_2$] containing pseudopeptides was described in 1989.[85] This reduced retro-amide bond isostere was presumed by molecular graphic analysis to provide potential protease inhibitors. A complete study of the conformational effects by modifying the amide link with ψ[NHCH$_2$] was reported.[86,87]

*for references see p 420*

### 10.7.3.1 Synthesis of ψ[NHCH$_2$] Pseudopeptides

Preparation of ψ[NHCH$_2$] pseudodipeptides from a *N*-acetyl *gem*-diamine[88] and an α-aldehydo ester[89] was unsuccessful.[85] The desired pseudodipeptide was unstable and eliminated the acetamide moiety. To overcome this instability a pseudotripeptide was prepared that incorporates the reduced retro isostere from a retro-inverso pseudodipeptide;[85] the synthesis of this compound is shown in Scheme 16. Coupling of Z-L-Phe-OH with **32** using the mixed anhydride method afforded the retro-inverse pseudodipeptide **33**. Reaction of **33** after hydrogenation with an α-aldehydo ester provides the reduced-retro pseudotripeptide **35** as a pair of diastereomers.

**Scheme 16** Synthesis of a Protected ψ[NHCH$_2$] Pseudotripeptide[85]




**Reduced-Retro Pseudotripeptide** 35:[85]
Curtius rearrangement of the acyl azide of Ac-Phe-OH **31**, at rt in dilute HCl, provided Ac-Phe-NH$_2$·HCl **32** in up to 65% yield. Coupling of Z-Phe-OH with **32** using the mixed anhydride method afforded Ac-Pheψ[NHCO]Phe-Z **33** as a single diastereomer in 55% yield. Reaction of **33** after hydrogenation with HCOCH(iPr)CO$_2$Me provides **34**. Compound **34** was hydrogenated to afford the reduced-retro pseudotripeptide **35** in 65% yield for the two steps.

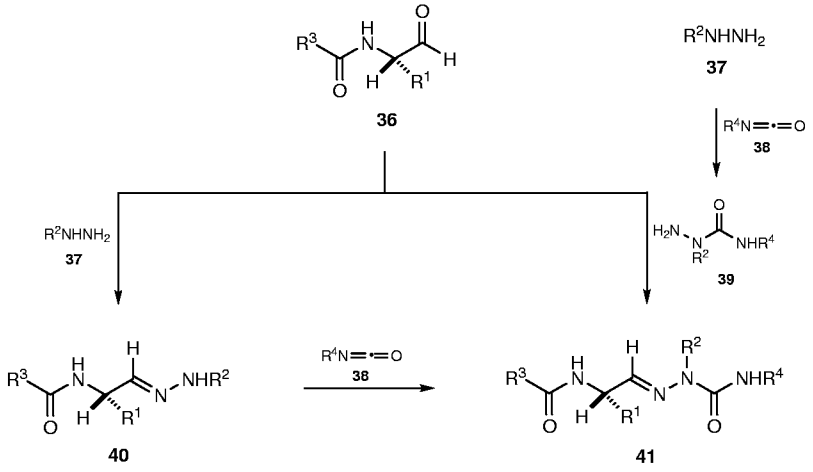
## 10.7.4 ψ[CH=NNH] (Iminoaza) Peptides

Several peptide isosteres have been proposed for the design of less biodegradable and geometrically constrained peptide analogues. However, the modification of the peptide backbone can affect its geometry and electronic properties and can result in various possible hydrogen bonding interactions of the amide surrogate. It is also important to take into account the possibility of protonation of the pseudopeptide bond at physiological pH since it has been shown that protonation of a reduced amide bond changes its conformational properties. As the semicarbazide moiety is not protonated at physiological pH,[90] Marraud and co-workers[91] have investigated the preparation and the structural properties of iminoaza peptides, namely peptides containing a semicarbazone fragment. Studies of various iminoaza dipeptides performed in solution by $^1$H NMR and IR absorption spectroscopy and in the solid state by X-ray diffraction show that the iminoaza fragment can be considered as a *cis* planar structure with a NH to N hydrogen bond closing a 5-membered pseudocycle.[91,92] An iminoaza tetrapeptide analogue of a major immunogenic region of the *Torpedo californica* acetylcholine receptor[93] has been prepared.[94]

### 10.7.4.1 Synthesis of Iminoaza Peptides

Iminoaza peptides of the general formula $R^3CO–Xaa^1\psi[CH=NNH]Xaa^2–NHR^4$ **41** ($R^3$ = alkoxy or amino acid sequence; $R^4$ = H or alkyl group) can be obtained in solution by coupling aldehyde **36** with semicarbazide **39** or by condensing aldehyde **36** with monosubstituted hydrazine **37** giving hydrazone **40**, which in turn is reacted with isocyanate **38** (Scheme 17).

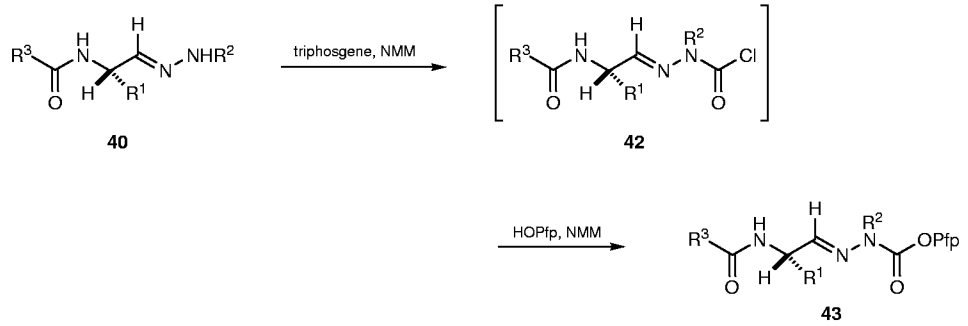
**Scheme 17** Synthesis of Protected Iminoaza Peptides




$R^1$, $R^2$ = amino acid side chain; $R^3$ = alkoxy or amino acid sequence; $R^4$ = H or alkyl group

For incorporating an iminoaza fragment into a peptide chain the hydrazone **40** might be converted into the synthon **43** (Scheme 18). Action of triphosgene on compound **40** produces an intermediate **42**, which is not isolated. Addition of pentafluorophenol to **42** yields the stable activated synthon **43** (Scheme 18).

**Scheme 18** Synthesis of the Activated Synthon **43**

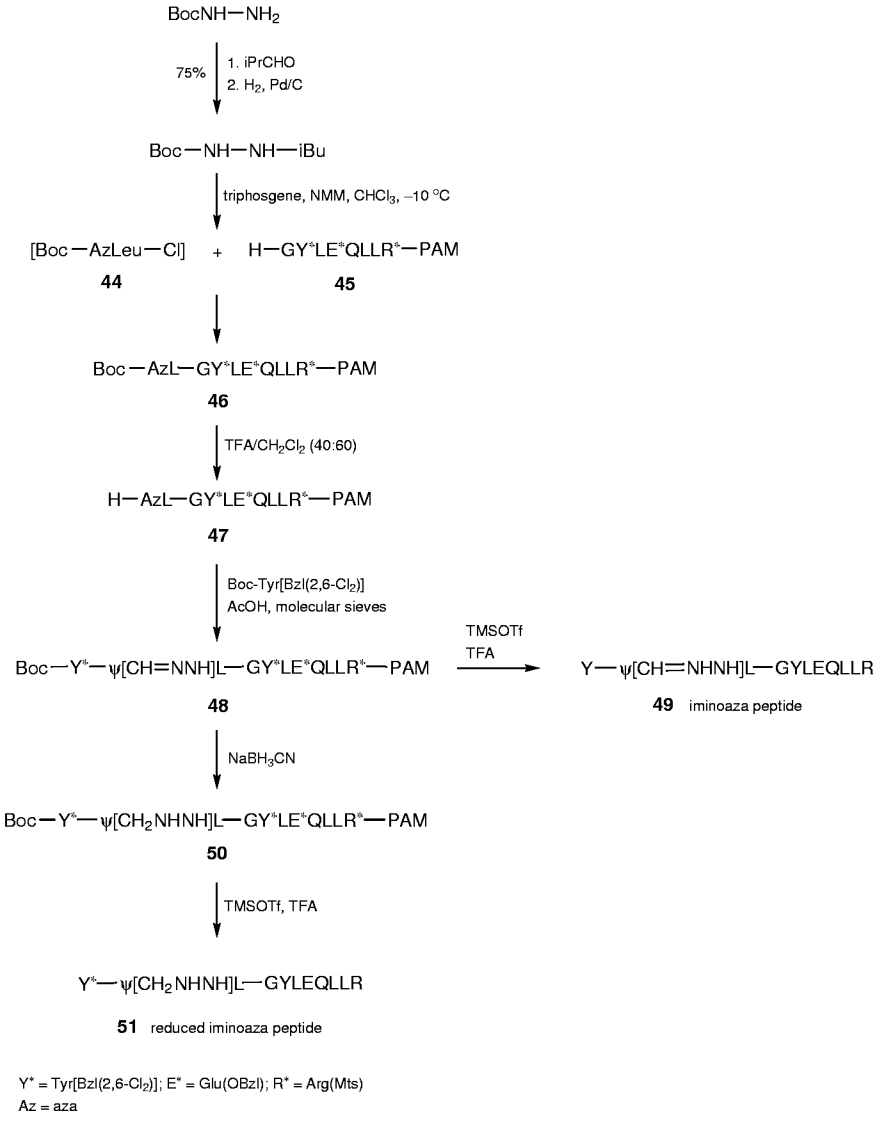



$R^1$, $R^2$ = amino acid side chain; $R^3$ = alkoxy or amino acid sequence

Iminoaza peptides can also be prepared by solid-phase synthesis. In this case, the $N^{\alpha}$-protected aza-amino acid chloride **44** is coupled to the amino group of the growing peptide chain **45**. The resulting peptide-PAM **46** is deprotected to give compound **47** which is then converted into the iminoaza peptide **48** by condensation with a $N^{\alpha}$-protected α-amino aldehyde. The synthesis of the casein iminoaza decapeptide fragment **49** has been reported using a Boc strategy and a stepwise synthesis on the PAM solid support (Scheme 19).[95] Although the affinity of this iminoaza decapeptide for the $GABA_A$ receptor is lower than that of the benzodiazepines, its recognition by the same site of receptor is eightfold better than that of the native peptide.

*for references see p 420*

**Scheme 19** Solid-Phase Synthesis of H-Tyrψ[CH=NNH]Leu-Gly-Tyr-Leu-Glu-Gln-Leu-Leu-Arg-OH and H-Tyrψ[$CH_2$NHNH]Leu-Gly-Tyr-Leu-Glu-Gln-Leu-Leu-Arg-OH[95]




**Synthesis of Boc-Proψ[CH=NNH]Ala-Asp(OBzl)-NHMe:**[94,96]

Boc-Pro-H (169 mg, 8.5 mmol) was slowly added to a soln of $MeNHNH_2$ (0.54 mL, 10.2 mmol) in EtOH (20 mL). The mixture was stirred for 5 min at rt, then for 3 h at 60 °C. After removal of the solvent, Boc-Proψ[CH=NNH]–NHMe was purified by flash chromatography to give a colorless oil; yield: 52%; $R_f$ 0.36 (EtOAc/petroleum ether 1:1).

To a soln of Boc-Proψ[CH=NNH]–NHMe (4.4 mmol) in THF (10 mL) was added dropwise at –15 °C triphosgene (1.46 mmol) and NMM (5.28 mmol). After 1 h, PfpOH (4.4 mmol) and NMM (4.4 mmol) were added and the mixture was stirred for 2 h. Boc-Proψ[CH=NNH]Ala-OPfp was purified by flash chromatography to give a colorless oil; yield: 71%; $R_f$ 0.61 (EtOAc/petroleum ether 1:1).

Boc-Proψ[CH=NNH]Ala-OPfp (3 mmol) was reacted with H-Asp(OBzl)-NHMe trifluoroacetate (3.6 mmol) in the presence of DIPEA (3 mmol) in dioxane (10 mL) at 80 °C for 2 h. The crude product was washed (5% aq $NaHCO_3$, 5% aq $KHSO_4$, and brine) and the solvent removed. The resulting iminoaza tripeptide was purified by flash chromatography affording a white solid; yield: 80%; mp 55 °C; $R_f$ 0.48 (EtOAc/petroleum ether 1:1).

**Solid-Phase Synthesis of H-Tyr-ψ[CH=NH]Leu-Gly-Tyr-Leu-Glu-Gln-Leu-Leu-Arg-OH:**[95]
iPrCHO (0.91 mL, 10 mmol) was added at 20–25 °C over 10 min to $BocNHNH_2$ (1.32 g, 10 mmol) in THF (10 mL). After 3 h the hydrazone was filtered off and hydrogenated over 10% Pd/C in iPrOH. Chromatography (silica gel, EtOAc/petroleum ether 1:1) gave BocNHNHiBu as a colorless oil: yield: 75%. To a cold soln (−10 °C) of BocNHNHiBu (0.94 g, 5 mmol) and triphosgene (0.49 g, 1.67 mmol) in $CHCl_3$ (5 mL) was added dropwise under a $N_2$ stream with vigorous stirring NMM (0.55 mL, 5 mmol), and the mixture was maintained at −10 °C for 1 h. The product Boc-AzLeu-Cl (**44**) prepared just before use, was not isolated.

H-Gly-Tyr[Bzl(2,6-$Cl_2$)]-Leu-Glu(OBzl)-Gln-Leu-Leu-Arg(Mts)-PAM (**45**) (0.2 mmol) was introduced at −10 °C under a $N_2$ stream into a moderately stirred soln of Boc-AzLeu-Cl (5 mmol) in $CHCl_3$. NMM (5 mmol) was added dropwise. The mixture was kept at −10 °C for 1 h and then allowed to reach rt overnight. The Boc group was cleaved from **46** using 40% TFA in $CH_2Cl_2$ to give the AzLeu-peptide-PAM **47**.

To the AzLeu-peptide-PAM **47** (0.1 mmol) in DMF/$CH_2Cl_2$ (1:3) was added Boc-Tyr[Bzl(2,6-$Cl_2$)]-H (2.12 g, 0.5 mmol) in the presence of NaOAc (0.164 g, 2 mmol) and molecular sieves. The reaction was stirred for 5 h at rt and the coupling step was repeated to obtain the protected iminoaza peptide-PAM **48**. A standard deprotection and cleavage procedure (TMSOTf/TFA), with thioanisole and ethane-1,2-dithiol as scavengers afforded the crude deprotected iminoaza peptide **49**. The yield was estimated at about 45%. Fractions determined to be pure by analytical RP-HPLC are pooled, concentrated under reduced pressure and lyophilized to yield the iminoaza peptide **49** as a white powder (>95% pure), which was checked by TOF-MS: calcd, 1253.48 $[M+1]^+$; found, 1252.60.

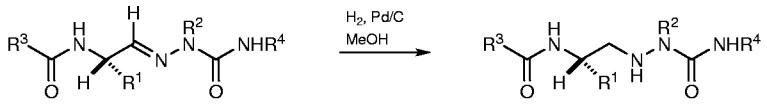
## 10.7.5 ψ[$CH_2NHNH$] (Reduced Iminoaza) Peptides

An iminoaza peptide has a rigid *cis*-planar configuration. In order to restore some conformational flexibility, Marraud and co-workers have investigated the preparation and structural properties of a reduced iminoaza peptide.[97] Structural studies have shown that the reduced iminoaza peptide exhibits properties similar to those of the neutral reduced peptide homologue but differs by the absence of protonation at physiological pH, which is known to considerably affect the conformational properties of the reduced amide link.[98–100] Because the electronic state of a reduced iminoaza peptide is probably independent from the nature of the environment, this pseudopeptide bond seems to be a good candidate for the design of a neutral mimic of the amide hydrolysis transient state.

### 10.7.5.1 Synthesis of Reduced Iminoaza Peptides

In the liquid-phase procedure the reduced iminoaza peptide **52** is readily obtained by catalytic hydrogenation of the -CH=N- bond of the corresponding iminoaza peptide **41** (Scheme 20). In solid-phase synthesis the reduction of the iminoaza peptide is performed by $NaBH_3CN$. As shown in Scheme 19 reduction by $NaBH_3CN$ of compound **48** followed by acidolytic treatment gives the reduced iminoaza peptide **51**.

**Scheme 20** Solution Synthesis of a Reduced Iminoaza Peptide




41 52

$R^1$, $R^2$ = amino acid side chain; $R^3$ = alkoxy or amino acid sequence; $R^4$ = H or alkyl group

**Piv-Pro-ψ[$CH_2NHNH$]Ala-NHiPr by Solution Synthesis:**[94]
Catalytic hydrogenation (96 h) on 10% Pd/C (10 mg) under atmospheric pressure of Piv-Pro-ψ[CH=NNH]Ala-NHiPr (40 mg, 0.14 mmol) in MeOH (10 mL) and AcOH (0.2 mL, 3 mmol) gave

*for references see p 420*

Piv-Pro-ψ[$CH_2NHNH$]Ala-NHiPr which was crystallized (EtOAc); yield: 90%; mp 78°C, $R_f$ 0.40 (iPrOH/$CH_2Cl_2$ 5:95).

**Y-ψ[$CH_2NHNH$]L-GYLEQLLR 51 by Solid-Phase Synthesis:**[95]
On-resin reduction of the imino bond of the Boc-Tyr[Bzl(2,6-$Cl_2$)]ψ[CH=NNH]Leu-peptide-PAM **48** (300 mg, 0.204 mmol) was carried out for 12 h in a DMF/$CH_2Cl_2$ (1:3) by $NaBH_3CN$ following the procedure used for the reductive amination to afford **50**.[7] The same procedures for deprotection and cleavage from the resin, as described for the iminoaza peptide, were applied. The yield was estimated at about 45%. Fractions determined to be pure by analytical RP-HPLC are pooled, concentrated under reduced pressure and lyophilized to yield the expected reduced iminoaza peptide **51** (white powder >95% pure) which was checked by TOF-MS: calcd, 1255.49 $[M+1]^+$; found, 1255.45.


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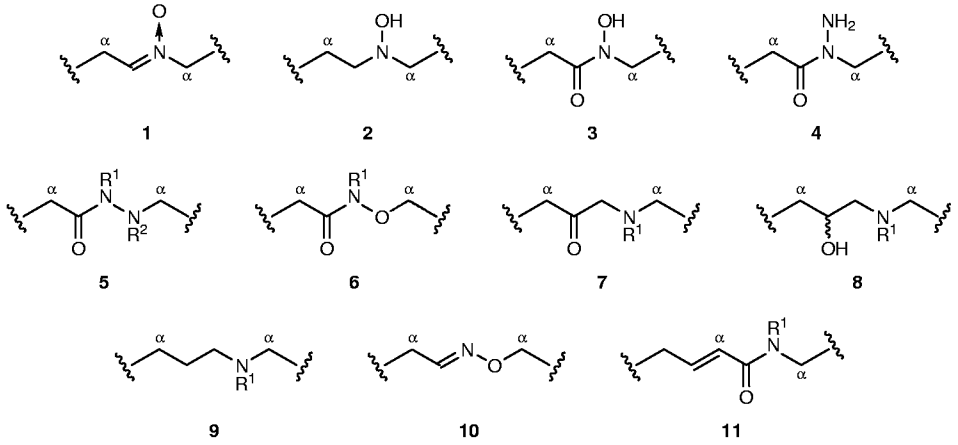
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# 10.8 Peptides Containing C/N/O Amide Bond Replacements

M. Marraud and R. Vanderesse

In this section we consider peptide analogues containing the amide surrogates **1** to **11** (Scheme 1). These can be isosteric with the amide group in the sense that consecutive α-carbons are separated by three bonds, as in link **1**, the (nitrono) peptides, and link **2**, the [methyleneamino(hydroxy)] or (*N*-hydroxy reduced amide) peptides. They also can be an N-modified amide, as in link **3**, the (*N*-hydroxyamide) peptides, and link **4**, the (*N*-aminoamide) peptides. Elongation of the peptide unit by one covalent bond has been realized by the introduction of a heteroatom or a methylene into the backbone, as in link **5**, the (hydrazide) peptides, link **6**, the (amidoxy) peptides, link **7**, the (oxomethyleneamino) peptides, link **8**, the [(hydroxy)ethyleneamino] peptides, link **9**, the (ethyleneamino) peptides, and link **10** the (oxime) peptides. Finally, insertion of an ethylenic bond (two covalent bonds) between the α-carbon and the carbonyl gives rise to link **11**, the (but-2-enamide) or (vinylogous amide) peptides.

**Scheme 1** The Chemical Structures of Amide Surrogates Used in Peptide Analogues

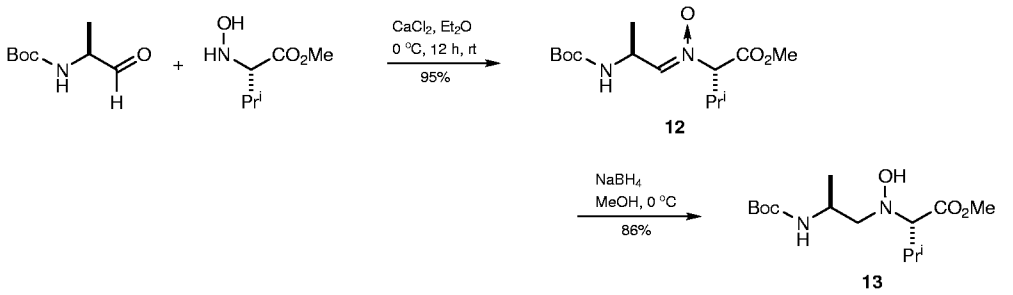



## 10.8.1 (Nitrono) Peptides

(Nitrono) or ψ[CH=N⁺(O⁻)] **1** peptides were introduced in 1987 by Grundke et al.[1] This pioneering work on simple model (nitrono) peptides has not been continued, and no (nitrono) analogue of a bioactive peptide has been reported so far. However, model (nitrono) dipeptides (e.g., **12**) have been prepared in good yield (Scheme 2) by coupling an N-protected α-amino aldehyde[2] with the hydroxyamide terminus of an α-hydroxyamino acid derivative,[3] the synthesis of which is presented in Section 10.8.3.1.

*for references see p 453*

**Scheme 2** Synthesis of a (Nitrono) Dipeptide[1] and a (*N*-Hydroxy reduced amide) Dipeptide[4]

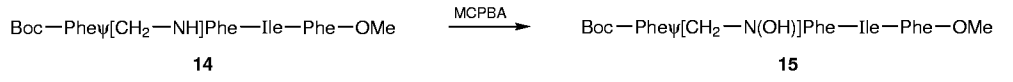



**Boc-L-Alaψ[CH=N⁺(O⁻)]L-Val-OMe (12):**[1]
Boc-L-Ala-H[2] (0.35 g, 2 mmol) in $Et_2O$ (10 mL) at 0 °C was added with stirring to H-L-(OH)Val-OMe[3] (0.29 g, 2 mmol) in $Et_2O$ (10 mL) containing $CaCl_2$ (0.3 g; for trapping $H_2O$). Stirring was continued for 12 h at rt, and reaction progress was monitored by TLC. $CaCl_2$ was filtered off and washed with $CH_2Cl_2$, and the filtrate was concentrated under reduced pressure. Compound **12** was recrystallized ($Et_2O$/petroleum ether); yield: 0.57 g (95%); mp 138–139 °C; $[\alpha]_D^{20}$ −52.3 (*c* 1, $CH_2Cl_2$).

## 10.8.2 [Methyleneamino(hydroxy)] or (*N*-Hydroxy reduced amide) Peptides

The methyleneamino(hydroxy) or ψ[$CH_2$-N(OH)] link **2**, which may be also called an *N*-hydroxy reduced amide bond, was introduced by Grundke et al. in 1987.[4] Compound **13** can be obtained in good yield by reduction of the homologous (nitrono) peptide **12** (Scheme 2). Alternatively, Urban et al.[5] have prepared the (*N*-hydroxy reduced amide) peptide **15**, a potent HIV protease inhibitor, by direct oxidation of the amine nitrogen in the homologous reduced peptide **14**, obtained by classical reductive amination (Scheme 3).

**Scheme 3** Synthesis of an (*N*-Hydroxy reduced amide) Peptide by N-Oxidation of a Reduced Peptide[5]




14 15

**Boc-L-Alaψ[$CH_2$-N(OH)]L-Val-OMe (13):**[4]

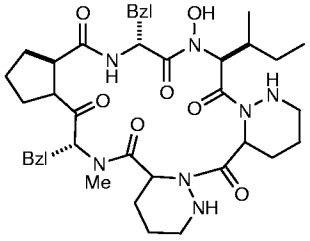
**CAUTION:** *$H_2$ is evolved! This reaction should be performed in a well-ventilated fume cupboard.*

To the (nitrono) dipeptide **12** (0.29 g, 1 mmol), dissolved in abs MeOH (10 mL) at 0 °C, was added $NaBH_4$ (76 mg, 2 mmol). After 10 min, $H_2$ emission stopped and reaction progress was monitored by TLC. MeOH was removed under reduced pressure. The residue was taken up with $Et_2O$ and filtered. The product **13** was purified by chromatography (silica gel, $Et_2O$); yield: 0.27 g (86%); mp 102–103 °C; $[\alpha]_D^{20}$ −10.4 (*c* 4, $CH_2Cl_2$).

## 10.8.3 (*N*-Hydroxyamide) Peptides

The *N*-hydroxyamide or [CO-N(OH)] link **3** is an oxidized form of the amide group present in a great number of metabolites from microorganisms having a large spectrum of antibiotic, antitumoral, and ionophoric properties.[6] However, in most cases, the *N*-hydroxyamide group is found in side chains, rather than in the backbone itself. In 1988 Chimiak and Milewska[6] pointed out that only two natural (*N*-hydroxyamide) peptides had been isolated: H-Asnψ[CO-N(OH)]Asn-OH, which is capable of complexing the Fe(III) cation,[7] and H-Aspψ[CO-N(OH)]Asp-D-*c*Ser-$NH_2$, which has no clear biological properties.[8] More recently, the (*N*-hydroxyamide) cyclohexapeptide *c*{L-Pro-D-Pheψ[CO-N(OH)]L-Ile-D-Piz-L-Piz-D-(Me)Phe} (**16**) (Scheme 4), isolated from *Streptomyces silvensis*, has been found to be a potent selective oxytocin antagonist.[9] From the chemical point of view, no example of amide oxidation has been reported. Thus, N-acylation of the α-hydroxyamino acids remains the only way of preparing (*N*-hydroxyamide) peptides.

**Scheme 4** Chemical Structure of (*N*-Hydroxyamide) Cyclohexapeptide *c*{L-Pro-D-Pheψ[CO-N(OH)]L-Ile-D-Piz-L-Piz-D-(Me)Phe}[9]




16

### 10.8.3.1 Synthesis of α-Hydroxyamino Acids and Derivatives

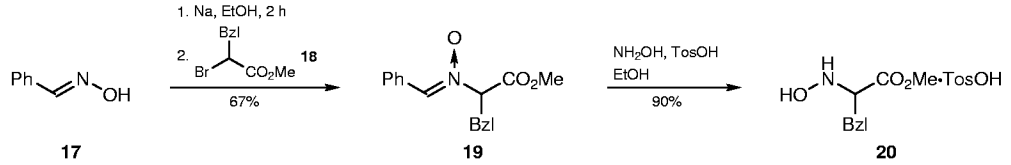
Due to the disproportion of hydroxylamines, which makes them rather unstable products on heating,[10] *N*-hydroxyamino acids have never been isolated as such from living organisms. Not surprisingly, extensive studies on the synthesis of *N*-hydroxyamino acids began only in the early 1970s. The procedures for obtaining *N*-hydroxyamino acids have been reviewed by Chimiak and Milewska[6] and by Ottenheijm and Herscheid.[11] Two additional procedures have appeared since then.[12,13]

An adapted Strecker reaction using NaCN and $NH_2OH$ was the first procedure used for obtaining racemic *N*-hydroxyamino acids.[14] Although it has been modified repeatedly,[15–19] the yields are rather low. Racemic *N*-hydroxyamino acids were also prepared as early as 1894 by addition of nitrogen oxide to the sodium salt of diethyl *C*-alkylmalonate, followed by acidolysis with concentrated hydrochloric acid.[20,21] The procedure has been reinvestigated later with a 45–65% overall yield.[19]

*N*-Hydroxyglycine was obtained in 1896 by acidolysis of the nitrone resulting from the condensation of *anti*-benzaldoxime **17** with chloroacetic acid.[22] N-Alkylation of **17** with α-bromoesters was extended much later to the synthesis of various *N*-hydroxyamino acids.[23,24] It is notable that *syn*-benzaldoxime is O-alkylated under the same conditions.[25] As substitution of α-bromoacids proceeds with inversion of configuration, Liberek and Palacz[26] prepared H-D-(OH)Phe-OH from (*S*)-α-bromo-β-phenylpropanoic acid, obtained by $HNO_2$ deamination of L-Phe in the presence of KBr. Treatment of nitrone **19** with an *N*-hydroxylamine salt yields the corresponding *N*-hydroxyamino ester **20** (Scheme 5).[24,27]

*for references see p 453*

**Scheme 5** Synthesis of an *N*-Hydroxyamino Ester by Nitrone Hydrolysis[27]



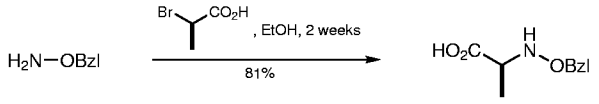

***N*-Hydroxy-D,L-phenylalanine Methyl Ester 4-Toluenesulfonate (20):**[27]
Na (0.92 g, 40 mmol) was dissolved in EtOH (40 mL). To this soln were added *anti*-benzaldoxime **17** (4.84 g, 40 mmol) and methyl (*R*,*S*)-2-bromo-3-phenylpropanoate (**18**; 10.7 g, 44 mmol). The mixture was allowed to stand until it reached pH 7 (2 h), at which point it was concentrated to dryness. $H_2O$ (30 mL) was added to the residue and the aqueous phase was extracted with $CHCl_3$ (40 mL). The organic layer was dried ($MgSO_4$) and concentrated to dryness. Nitrone **19** was crystallized (EtOH/$Et_2O$); yield: 7.58 g (67%); mp 142 °C.

Nitrone **19** (1.42 g, 5 mmol) and hydroxylamine 4-toluenesulfonate (1.02 g, 5 mmol) were dissolved in EtOH (10 mL) with heating, then the solvent was removed. The residue was dissolved in $CHCl_3$, filtered, and the filtrate concentrated. The title product **20** was crystallized (EtOH/$Et_2O$); yield: 1.66 g (90%); mp 107–109 °C.

### 10.8.3.1.1 Halogen Substitution

Halogen displacement from α-halo acids by *N*-hydroxylamine was one of the earliest procedures used to prepare *N*-hydroxyamino acids. It results in very moderate yields due to their disproportion.[19] However, by taking advantage of the inversion of configuration with bromine substitution, La Noce et al.[28] obtain enantiomerically pure H-L- and H-D-(HO)Leu-OH. The disproportion reaction can be eliminated by using O-substituted *N*-hydroxylamine,[29] in particular *O*-benzyl-*N*-hydroxylamine (**21**), although its reduced nucleophilicity requires a longer reaction time (up to 10–20 days) (Scheme 6).[30] The *O*-benzyl group may then be easily eliminated with boron trifluoroacetate to generate the *N*-hydroxyamino acid.[30]

**Scheme 6** Preparation of an *N*-Hydroxyamino Acid by Halogen Displacement of an α-Halo Acid[29,30]



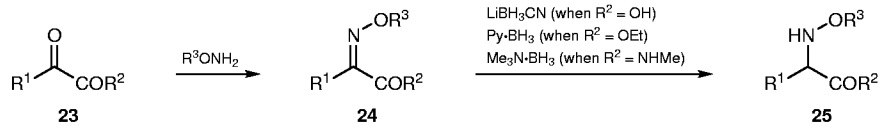

***N*-Benzyloxy-L-alanine (22):**[30]
(*R*)-2-Bromopropanoic acid (15.3 g, 100 mmol) was added to a soln of *O*-benzyl-*N*-hydroxylamine **21** (4.92 g, 400 mmol) in anhyd EtOH (200 mL). The mixture was allowed to stand at rt for 2 weeks. The soln was acidified to pH 4 using ethereal HCl and concentrated to dryness under reduced pressure; then anhyd $Et_2O$ (100 mL) was added. The unreacted hydrochloride of **21** was filtered off and washed with dry $Et_2O$. The combined $Et_2O$ phases were concentrated and unreacted α-bromo acid was removed by several extractions with petroleum ether. Compound **22** was crystallized ($Et_2O$/petroleum ether); yield: 15.8 g (81%); mp 113–114 °C; $[\alpha]_D^{20}$ –25.1 (*c* 2, EtOH).

### 10.8.3.1.2 Oxime Reduction

Oxime reduction is a very efficient procedure for obtaining racemic *N*-hydroxyamino acids in good yields.[19,31–36] *N*-Hydroxylamine and O-protected *N*-hydroxylamines react smoothly and rapidly with α-oxo acid derivatives **23** to give the corresponding oximes **24**. Acids **24** ($R^2$ = OH) can be reduced with $NaBH_3CN$ or $LiBH_3CN$ (Scheme 7) to give *N*-hydroxyamino acids **25** ($R^2$ = OH),[31] but reaction yields are unsatisfactory for esters **24** (e.g., $R^2$ = OEt) or

amides **24** (e.g., $R^2$ = NHMe). Much better results for esters and amides **24** are obtained with pyridine–borane[32,33] or trimethylamine–borane[33] as the reducing agent. Addition of an alkyllithium reagent to an oxime derived from an α-oxo acid, particularly a glyoxalate oxime, is also a direct route to mono-$C^{\alpha}$-alkylated *N*-hydroxyamino acid derivatives.[37,38] Using a glyoxalate oxime with a chiral O-substituent or C-terminal group results in a stereoselective addition with 20–40% diastereomeric excess.[38]

**Scheme 7** Synthesis of *N*-Hydroxyamino Acid Derivatives by Oxime Reduction[31–33]



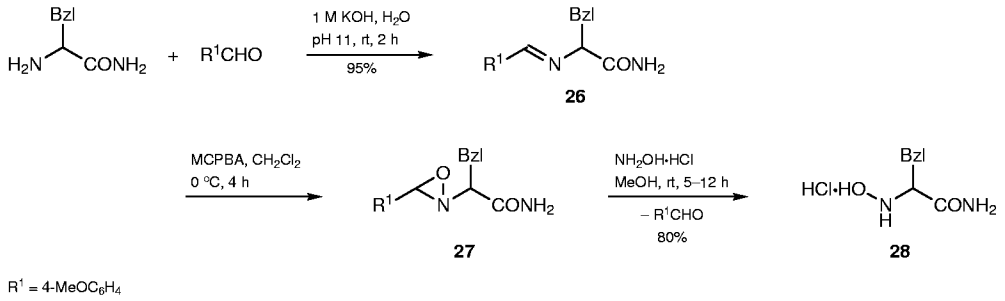

**23** **24** **25**

***N*-Benzyloxy-D,L-alanine Methylamide (25, $R^1$ = Me; $R^2$ = NHMe; $R^3$ = Bzl):**[33]
Ethanolic 1 M HCl (15 mL) was added in one portion to a stirred mixture of the *O*-benzyloxime of *N*-methyl pyruvamide (**24**, $R^1$ = Me; $R^2$ = NHMe; $R^3$ = Bzl; 0.41 g, 2 mmol) and the $Me_3N \cdot BH_3$ complex (0.3 g, 4 mmol) at rt (the temperature of the soln increased only slightly). Stirring was continued at rt for 16 h, after which the solvent was removed. $CH_2Cl_2$ (25 mL) was added, together with solid $Na_2CO_3$ (1 g). After stirring for several hours the suspension was filtered and the solvent removed. The residue was chromatographed (preparative HPLC, $CH_2Cl_2$/MeOH) to give the title compound; yield: 0.37 g (89%); mp 32–33 °C; MS *m/z*: 209.12 $[M+H]^+$.

### 10.8.3.1.3 Oxaziridine or Imidazolidine-4-one Hydrolysis

Oxaziridines are obtained by oxidation of the Schiff base resulting from the addition of a 4-substituted benzaldehyde to an α-amino ester. Hydrolysis of the oxaziridines by concentrated HCl gives the *N*-hydroxyamino acid, while *N*-hydroxylamine 4-toluenesulfonate affords the corresponding *N*-hydroxyamino acid ester 4-toluenesulfonate.[39] Because the Schiff base racemizes only very slowly, the reaction essentially proceeds with retention of configuration of the starting α-amino ester.[3] This procedure has been reinvestigated by Feenstra et al.[12] with α-amino acid amides as chiral starting materials (Scheme 8).

**Scheme 8** Synthesis of an *N*-Hydroxyaminoamide by 3-Aryloxaziridine Hydrolysis[12]



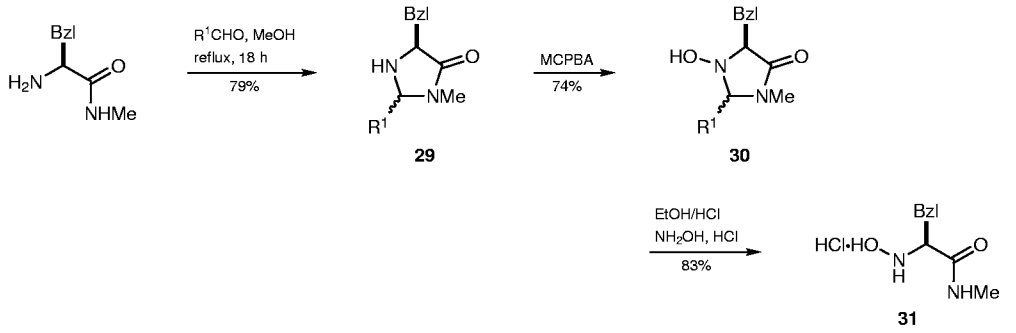

**26** **27** **28**

$R^1$ = 4-$MeOC_6H_4$

The imidazolidine-4-one **29**, obtained by heating the methanol solution of an α-aminoamide and a 4-substituted benzaldehyde, is a mixture of two diastereomers which can be separated by chromatography. They are oxidized by MCPBA, separately or as a mixture, into the two diastereomeric 1-hydroxyimidazolidine-4-ones **30**. Hydrolysis of **30** by ethanolic HCl and hydroxylamine hydrochloride gives the optically pure *N*-hydroxyamino acid amide **31** (Scheme 9).[12]

*for references see p 453*

**Scheme 9** Synthesis of an *N*-Hydroxyaminoamide by Imidazoline-4-one Hydrolysis[12]



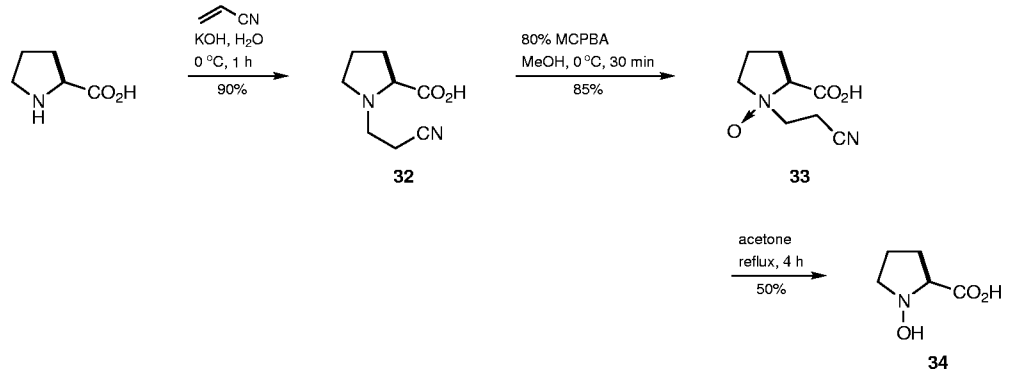

$R^1$ = 4-$MeOC_6H_4$

***N*-Hydroxy-D-phenylalanine Amide Hydrochloride (28):**[12]
H-D-Phe-$NH_2$ (1.63g, 10 mmol) was dissolved in $H_2O$ at 40 °C and the pH was adjusted to 11 by adding 1 M KOH. 4-Methoxybenzaldehyde (1.48 g, 11 mmol) was added over 15 min. After stirring at rt for 2 h, the crystallized Schiff base **26** was isolated by filtration, washed with $H_2O$, and dried; yield: 2.74 g (95%); mp 130–132 °C. Compound **26** was dissolved in dry $CH_2Cl_2$ (10–15 mL) and the soln was cooled to 0–5 °C. 85% MCPBA (1.9 g, 11 mmol) was added in portions, and stirring was continued for 4 h at rt, after which 3-chlorobenzoic acid was filtered off. The residue was washed several times with $CH_2Cl_2$ and the combined organic layers concentrated under reduced pressure (temperature < 30 °C). The solid oxaziridine **27** was not purified due to its instability; rather, it was directly dissolved in MeOH (15 mL) and hydroxylamine hydrochloride (0.77 g, 11 mmol) was added. After stirring at rt for 5–12 h, the soln was triturated with dry $Et_2O$ (100 mL), and the precipitated **28** isolated by filtration; yield: 1.72 g (80%); mp 145–146 °C; $[\alpha]_D^{20}$ +4.3 (*c* 1, MeOH).

### 10.8.3.1.4 α-Amino Acid Oxidation

Tertiary amine oxides can be converted into *N*-hydroxy secondary amines provided that one of the *N*-substituents can be selectively eliminated. This procedure has been applied to the synthesis of secondary *N*-hydroxy-α-amino acids **34** from the corresponding secondary α-amino acids using the *N*-cyanoethyl group for transient protection of the secondary amine (Scheme 10).[40] More recently, direct oxidation with 2,2-dimethyldioxirane of a primary amine has been described for H-L-Val-OMe (82% yield) and H-L-Phe-OMe (54% yield).[13] The reaction proceeds smoothly without epimerization, but no experimental details have been reported.

**Scheme 10** Synthesis of *N*-Hydroxy-L-proline by Amine Oxidation[40]



***N*-Hydroxy-L-proline (34):**[40]

**CAUTION:** *Acrylonitrile is an eye, skin, and respiratory tract irritant.*

Acrylonitrile (14.6 g, 0.275 mol) was added dropwise over 30 min to a cooled (0 °C) stirred soln of H-L-Pro-OH (28.8 g, 0.25 mol) and KOH (16.2 g, 0.25 mol) in $H_2O$ (75 mL). Stirring was continued for 1 h at 0 °C, and 1 h without cooling. The pH was then adjusted to 6.6 with 6 M HCl and the mixture taken to incipient dryness under reduced pressure (keeping the temperature below 70 °C). The residue was extracted continuously on Soxhlet apparatus for 96 h with acetone (1.5 L). *N*-Cyanoethyl-L-proline **32** crystallized in two crops on cooling in the refrigerator; yield: 37.8 g (90%); mp 155–156 °C; $[\alpha]_D^{28}$ −63.0 (*c* 2.28, $H_2O$).

To a cooled, stirred suspension of **32** (16.8 g, 0.1 mol) in MeOH (300 mL) was added dropwise over 30 min a filtered soln of 80% MCPBA (21.6 g, 0.1 mol) in MeOH (100 mL). After 1 h of additional stirring at 0–2 °C, the mixture was diluted with precooled anhyd $Et_2O$ (1.5 L). *N*-Cyanoethyl-L-proline oxide **33**, which crystallized in the refrigerator, was collected by filtration, washed with $Et_2O$, dried, and stored at −5 °C (as it undergoes spontaneous decomposition at rt); yield: 15.6 g (85%); mp 109–111 °C (dec); $[\alpha]_D^{28}$ +25.9 (*c* 2, $H_2O$).

A suspension of **33** (10.1 g, 55 mmol) in acetone (1 L) was refluxed for 4 h, and the acrylonitrile was entrained when formed by distillation of the solvent while fresh acetone was added in order to maintain the initial volume. *N*-Hydroxy-L-proline **34** crystallized upon cooling; yield: 3.6 g (50%); mp 157–160 °C (dec); $[\alpha]_D^{28}$ −79.6 (*c* 2.18, $H_2O$). Unreacted **33** recovered in the mother liquor could be recycled.

## 10.8.3.2 (*N*-Hydroxyamide) Peptide Synthesis

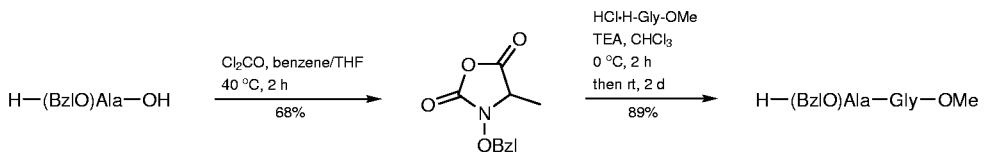
The first synthetic (*N*-hydroxyamide) dipeptide H-Gly$\psi$[CO-N(OH)]Gly-OH was obtained in 14% overall yield from Phth-Gly-Cl and the sodium salt of H-(HO)Gly-OH, with further *N*-hydroxylaminolysis of the Phth protecting group.[41] The results have been improved to some extent by using H-(HO)Gly-OEt or H-(HO)Gly-OtBu, but more conventional coupling agents give lower yields.[42] Moreover, the activated 4-nitrophenyl or *N*-hydroxyphthaliminyl esters result essentially in O-acylation of *N*-hydroxyglycine.[42] Zvilichovsky and Heller[43] prepared the H-Xaa$\psi$[CO-N(OH)]Gly-OH (*N*-hydroxyamide) dipeptides from H-(HO)Gly-OH and various NCAs derived from Gly, Ala, Phe, and Glu(OBzl) with overall yields ranging from 38 to 67%. Nps-NCAs derived from Ala, Leu, and Phe are coupled in good yields with *N*-hydroxyamino acid esters without O-protection; no O-acylation was observed in this case.[42]

Indeed, O-benzylation increases N-coupling yields of the hydroxyamine terminus and has three other advantages: (i) it prevents undesirable O-acylation in further coupling steps, (ii) it decreases the polarity of the (*N*-hydroxy) peptide formed, hence facilitating its purification, and (iii) it is easily cleaved by catalytic hydrogenolysis or treatment with $B(TFA)_3$.[44]

### 10.8.3.2.1 C-Coupling of *N*-Benzyloxy α-Amino Acids

Due to the relatively weak nucleophilicity of the hydroxyamide nitrogen, *N*-benzyloxy α-amino acids do not require N-protection during C-coupling. Two activation methods of the carboxy group have been used to introduce the *N*-benzyloxy α-amino acid NCAs (Scheme 11)[45] and the OSu esters[46] as smooth and efficient acylating reagents giving satisfactory C-coupling yields in the synthesis of the (*N*-hydroxyamide) dipeptides.

**Scheme 11** (*N*-Hydroxyamide) Peptide Synthesis by C-Coupling of *N*-Benzyloxy α-Amino Acids[45]




**35** **36**

*for references see p 453*

**H-D,L-(BzlO)Ala-Gly-OMe** (36):[45]

**CAUTION:** *Benzene is a carcinogen!*

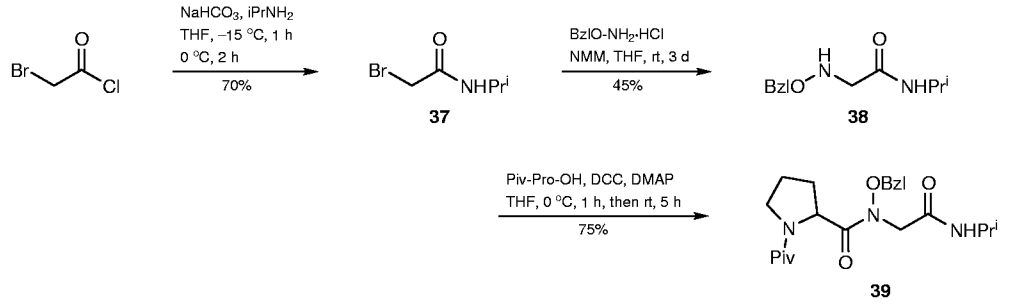
**CAUTION:** *Phosgene ($COCl_2$) is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a suspension of H-D,L-(BzlO)Ala-OH (9.75 g, 50 mmol) in benzene/THF (120 mL), was added phosgene [generated from trichloromethyl chloroformate (37.1 g, 125 mmol)] and the mixture was stirred at 40 °C for 2 h. Solvents were removed to give *N*-benzyloxy-D,L-alanine-NCA (**35**), which was recrystallized from hexanes/THF; yield: 7.58 g (68%); mp 71–72 °C. Compound **35** (2.23 g, 10 mmol) in $CHCl_3$ (20 mL) was added to a cooled (0 °C) stirred mixture of HCl·H-Gly-OMe (1.26 g, 10 mmol) and $Et_3N$ (1.4 mL, 10 mmol) in $CHCl_3$ (50 mL). The stirred mixture was kept at 0 °C for 2 h and then at rt for 2 d. After the usual work-up, the title compound was obtained as a HCl salt with HCl/dioxane, and was recrystallized (MeOH/$Et_2O$); yield: 2.37 g (89%); mp 165–166 °C.

### 10.8.3.2.2 N-Coupling of *N*-Benzyloxy α-Amino Acid Derivatives

Kolasa and Chimiak[44] and, more recently, Shimizu et al.[46] conclude that the best yield in the N-coupling of *N*-benzyloxy α-amino acids is obtained with the acid chloride or the mixed anhydride of the C-component, whereas activated esters completely fail. DCC associated with DMAP as a catalyst is used with success (75% yield) in the coupling of Piv-Pro-OH with H-(BzlO)Gly-NHiPr (**38**) (Scheme 12).[47]

**Scheme 12** (*N*-Hydroxyamide) Peptide Synthesis by N-Coupling of *N*-Benzyloxy α-Amino Acid Derivatives[47]




**Piv-Proψ[CO-N(OBzl)]Gly-NHiPr** (39):[47]

**CAUTION:** *Dicyclohexylcarbodiimide (DCC) is a severe eye, skin, and respiratory tract irritant and a skin sensitizer.*

To a cooled (−15 °C) suspension of $NaHCO_3$ (0.17 g, 2 mmol) in THF (40 mL) containing bromoacetyl chloride (3.1 mL, 20 mmol) was added over 30 min a cooled soln of $iPrNH_2$ (1.18 mL, 20 mmol) in THF (5 mL). The mixture was stirred for 1 h at −15 °C, then 2 h at 0 °C. The salts were filtered off and the solvent was removed. Oily 2-bromo-*N*-isopropylacetamide (**37**) was crystallized from $CH_2Cl_2$/hexanes; yield: 2.52 g (70%); mp 62–64 °C. Compound **37** (1.63 g, 9 mmol) in THF (10 mL) was added to *O*-benzylhydroxylamine hydrochloride (4.33 g, 27 mmol) and NMM (3 mL, 27 mmol) in THF (20 mL) at 0 °C. Stirring was maintained for 3 d at rt. H-(BzlO)Gly-NHiPr (**38**) was obtained as a colorless oil after filtration of the salts and flash chromatography with EtOAc as eluent; yield: 0.78 g (45%). To compound **38** (0.96 g, 5 mmol) and Piv-Pro-OH (1.0 g, 5 mmol) in THF (20 mL) at 0 °C were added DCC (1.03 g, 5 mmol) and DMAP (0.06 g, 0.5 mmol). The mixture was stirred for 1 h at 0 °C and 5 h at rt. After the usual workup, the title compound **39** was purified by chromatography (silica gel, 4% iPrOH/$CH_2Cl_2$); yield: 1.51 g (75%); mp 149 °C.

### 10.8.3.2.3 Cleavage of the *O*-Benzyl Group

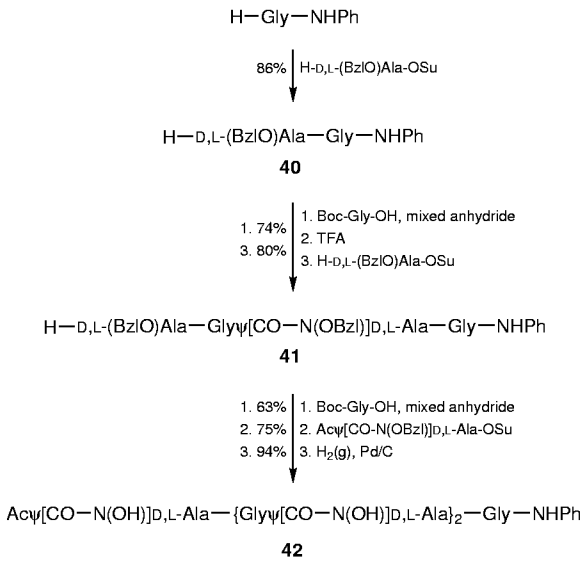
In most cases, the *O*-benzyl group is cleaved by hydrogenolysis under atmospheric pressure over 10% Pd/C,[44,46] 5% Pd/C,[47] or $Pd(AcO)_2$[48] with yields ranging from 40 to 100%. Another procedure with $B(TFA)_3$ in TFA solution has the advantage of giving excellent yields and cleaving the Boc and OtBu protecting groups in the same step.[30,44]

**Phth-Glyψ[CO-N(OH)]D,L-Ala-OMe:**[44]
A soln of Phth-Glyψ[CO-N(OBzl)]D,L-Ala-OMe (0.15 g, 0.4 mmol) in TFA (2 mL) was combined with a soln of 1 M $B(TFA)_3$ in TFA (2 mL, 2 mmol). The mixture was kept for 2 h at rt. After removal of the TFA and addition of aq $Na_2CO_3$, the title compound was extracted with EtOAc, the solvent was removed and the product recrystallized (EtOAc/petroleum ether); yield: 0.1 g (83%); mp 162–165 °C.

### 10.8.3.2.4 Methodologies for (*N*-Hydroxyamide) Peptide Synthesis

A relatively small number of (*N*-hydroxyamide) peptides have been prepared using the step-by-step or the fragment coupling strategy in solution,[45,46,48–50] or the solid-phase synthesis approach.[51] The (*N*-hydroxyamide) tetrapeptide H-D,L-(BzlO)Ala-Glyψ[CO-N(OBzl)]D,L-Ala-Gly-NHPh **41** has been prepared by a step-by-step procedure,[45] which has been extended to the synthesis of the terminally blocked (*N*-hydroxyamide) hexapeptide Acψ[CO-N(OH)]Ala-{Glyψ[CO-N(OH)]Ala}$_2$-Gly-NHPh **42** having alternate *N*-hydroxyamide/amide links (Scheme 13).[46]

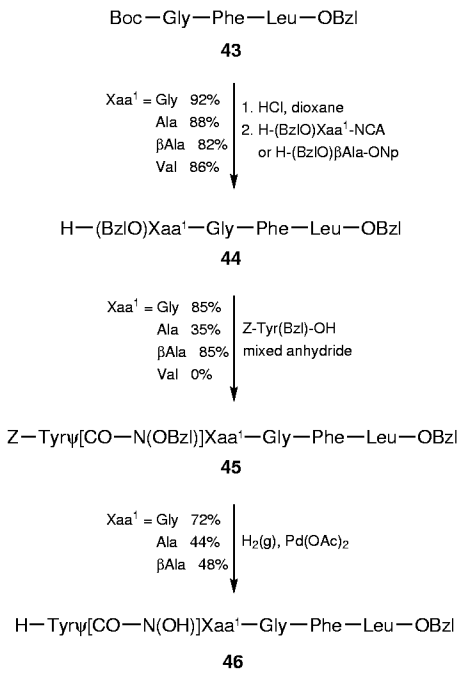
**Scheme 13** Step-by Step Synthesis of (*N*-Hydroxyamide) Peptides[46]




Three (*N*-hydroxyamide) analogues of Leu-enkephalin **46**, in which $Gly^2$ has been replaced with (HO)Gly, (HO)Ala, and (HO)βAla, have been prepared as indicated in Scheme 14.[48] Z-Tyr(Bzl)-OH is coupled to the *N*-benzyloxyamino terminus of **44** using the mixed anhydride method to give **45**. The coupling gives excellent yields (85%) for (HO)Gly and (HO)βAla, decreases to 35% for (HO)Ala, and fails with the sterically hindered (HO)Val, so that this analogue of **46** was not obtained.

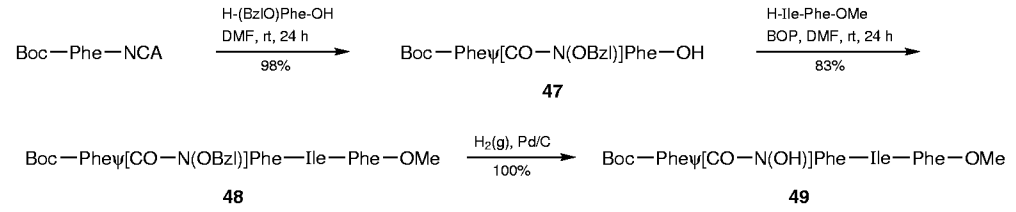
*for references see p 453*

**Scheme 14** Synthesis of (*N*-Hydroxyamide) Analogues of Leu-Enkephalin[48]



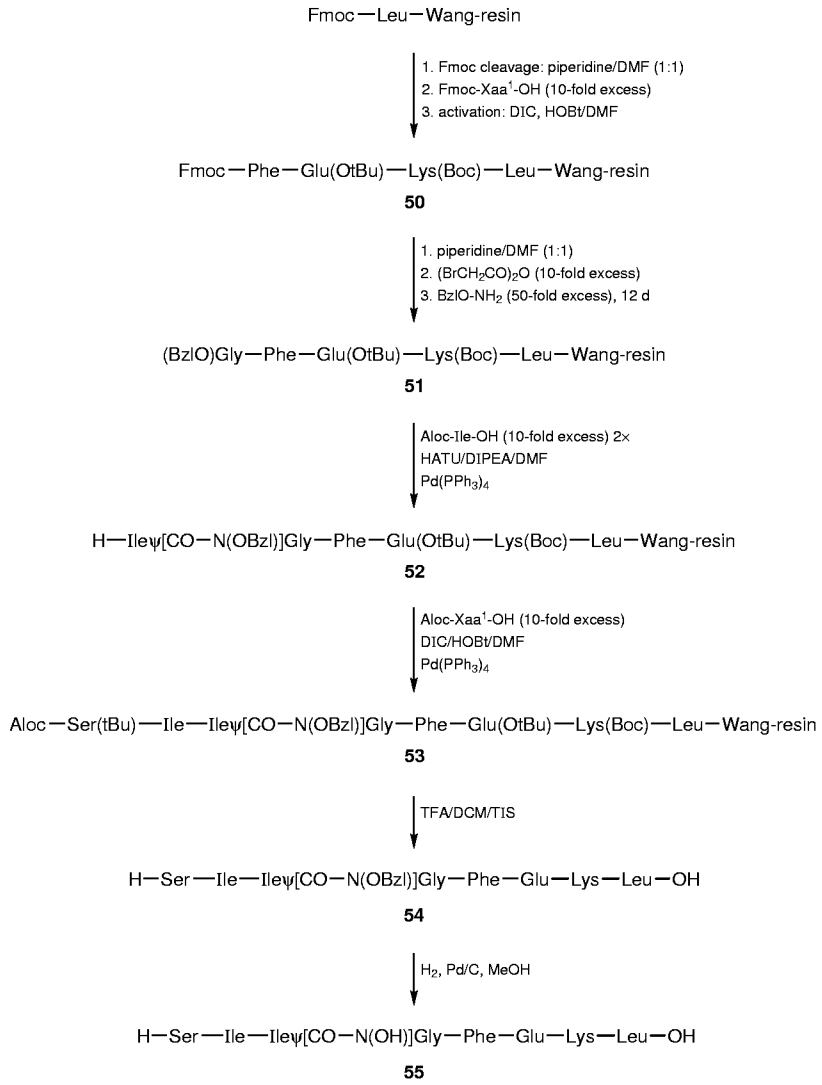

Recently, two moderately active (*N*-hydroxyamide) analogues of the HIV protease substrate have been prepared in two different ways, with *O*-benzyl protection of the *N*-hydroxy-α-amino acid residue.[50] Boc-Pheψ[CO-N(OH)]Gly-Ile-Phe-OMe is constructed by the mixed anhydride procedure in a step-by-step elongation of the peptide backbone. In contrast, Boc-Pheψ[CO-N(OH)]Phe-Ile-Phe-OMe **49** is prepared by fragment condensation (Scheme 15).

**Scheme 15** Synthesis of an (*N*-Hydroxyamide) Analogue of a HIV Protease Substrate by Fragment Condensation[50]




Various (*N*-hydroxy) peptide analogues of the SIINFEKL-epitope, where (HO)Gly has been substituted for each residue, have been obtained by the solid-phase synthesis procedure on Wang resin (Scheme 16).[51] The (BzlO)Gly residue is introduced by successive treatments of the N-terminal peptide amino group with bromoacetic acid and *O*-benzylhydroxylamine in very great excess, and the *N*-benzyloxyamide link is formed with the Aloc-protected amino acid using the strong HATU/HOBt acylating system. The peptide N-terminus is then elongated using either step-by-step Aloc-protected amino acid or fragment condensation. The *O*-benzyl group is removed by catalytic hydrogenolysis after cleavage of the (*N*-benzyloxyamide) peptide from the resin. Peptides containing a hydroxamic acid in their C-terminal position[52] or as a side function[53] have also been obtained by using an *N*-Fmoc-aminoxy-2-chlorotrityl polystyrene resin or a 4-methylbenzhydrylamine (pMeBHA) resin, respectively.

**Scheme 16** Liquid Phase Synthesis of an (HO)Gly-Containing (*N*-Hydroxyamide) Peptide[51]




**H-Ser-Ile-Ileψ[CO-N(OH)]Gly-Phe-Glu-Lys-Leu-OH** (55):[51]

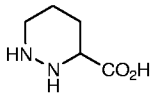
The Fmoc-Phe-Glu(OtBu)-Lys(Boc)-Leu-Wang conjugate **50** was prepared using the Fmoc/tBu strategy with DIC/HOBt activation in DMF and tenfold excess of the Fmoc amino acid. After cleavage of the *N*-terminal Fmoc group with 1:1 piperidine/DMF, a tenfold excess of the symmetric anhydride $(BrCH_2CO)_2O$, obtained by treating bromoacetic acid with DIC in DMF, was introduced; this step was carried out twice. *O*-Benzylhydroxylamine was generated from its hydrochloride with DIPEA in DMF and was added to the resin in 50-fold excess. The mixture was shaken at rt for 12 d to give **51**. Aloc-Ile-OH was activated with HATU/DIPEA in DMF and coupled (twice) in tenfold excess for 20 h to create the *N*-benzyloxyamide link in **52**. After Aloc cleavage with $Pd(PPh_3)_4$ (1.5 equiv) in $CHCl_3$/AcOH/NMM for 2 h, the (*N*-benzyloxy) peptide was step-by-step elongated using *N*-Aloc-protected amino acids activated with DIC/HOBt in DMF and coupled in tenfold excess for 90 min to give **53**. The (*N*-benzyloxy) peptide **54** was cleaved from the resin with a 9:2:1 mixture of TFA/$CH_2Cl_2$/TIS within 3 h, and the *O*-benzyl group was removed by catalytic hydrogenolysis with Pd/C in MeOH. After precipitation, washing with cold $Et_2O$, and lyophilization from tBuOH/$H_2O$ (4:1), the crude (*N*-hydroxy) peptide **55** was purified and analyzed by RP-HPLC using a Nucleosil 300 C18 column; yield: 54%; MS *m/z*: 922.5 $[M+H]^+$.

*for references see p 453*

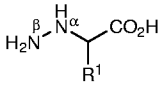
## 10.8.4 (*N*-Aminoamide) Peptides

To our knowledge, no example of a natural peptide having a hanging *N*-$NH_2$ group, as in link **4** (see Scheme 1), has been described, but *c*{L-Pro-D-Phe$\psi$[CO-N(OH)]L-Ile-D-Piz-L-Piz-D-(Me)Phe} **16** (Scheme 4), a potent selective oxytocin antagonist isolated from *Streptomyces silvensis*,[9] contains two piperazic acid (Piz) residues **56** (Scheme 17) and can be considered as a modified (*N*-aminoamide) peptide. There is no N-aminating reagent of the amide group compatible with the chemical and stereochemical stability of the peptides. Thus, the only practical synthetic route is acylation of the α-nitrogen in the α-hydrazino acids **57**. The α-hydrazino acids are found in various natural products[54] and present interesting biological activities, such as enzyme inhibition.[55] There is, therefore, a growing interest in the synthesis of α-hydrazino acids, which are also convenient intermediates in the synthesis of optically active α-amino acids by catalytic hydrogenolysis of the N–N bond.[56]

**Scheme 17** Chemical Structures of a Piperazic Acid (Piz) Residue and an α-Hydrazino Acid




**56**




**57**

### 10.8.4.1 Synthesis of α-Hydrazino Acids

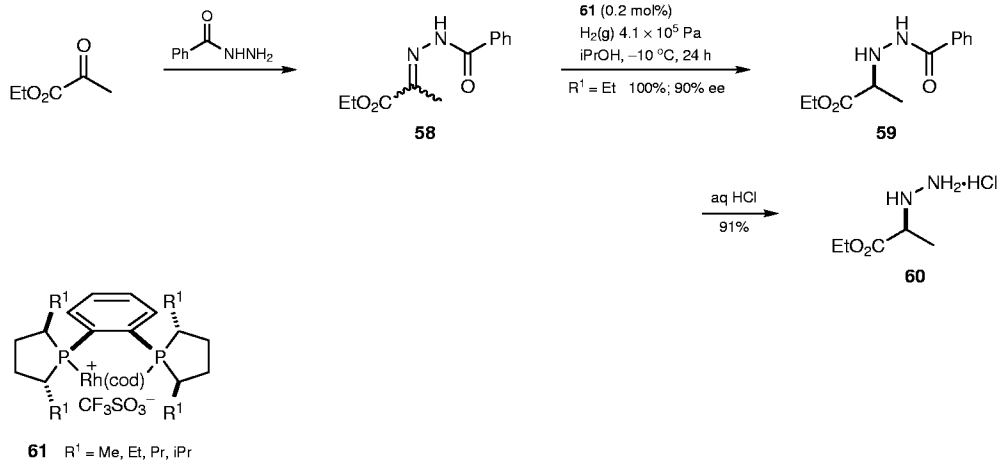
In 1896 Traube and Longinescu[57] reported the first synthesis of α-hydrazino acids by reduction of the α-nitrosohydroxylamines derived from α-amino acids. This work was later reinvestigated and extended by Darapsky.[58] The $S_N2$ reaction of hydrazine with an optically active acid derivative having an appropriate electron-attracting group (halide or sulfonate) in the α-position has been reported to give an α-hydrazino acid with inverted configuration in most cases.[59–61] However, partial or total epimerization has been reported, especially in cases where there is a bulky side chain.[62] Recently the $S_N2$ reaction using Boc-$NHNH_2$ as a nucleophile has been applied to triflates of α-hydroxyacid esters with high enantiomeric excess.[63]

Various synthetic routes to α-hydrazino acids have been proposed via reduction of carboxylic acid derivatives bearing an N–N (diazo, nitrosoamine, hydrazone) moiety in the α-position. For example, the α-diazoesters of α-amino acids are reduced in moderate yield into the corresponding racemic α-hydrazino acids by sodium amalgam.[64] Nitrosation of a secondary amine followed by treatment with zinc dust in aqueous acetic acid has been applied with success to N-amination of proline.[65] This procedure has been extended to α-amino acids having a primary amino group, provided that a temporary *N*-benzyl protection, which is subsequently eliminated by catalytic hydrogenolysis, had been previously introduced.[66] The overall yield of this multistep procedure is rather low, but the configuration of the starting α-amino acid is retained.

#### 10.8.4.1.1 Asymmetric Synthesis

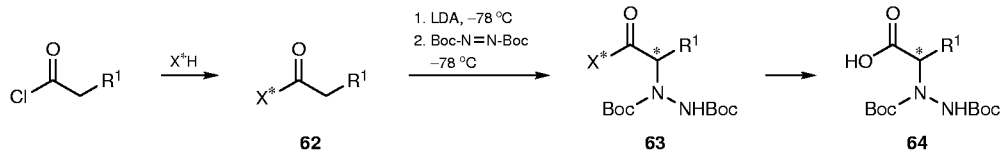
Hydrazones, obtained by reaction of hydrazine on α-oxo acids, have been hydrogenated over Pd/C to give racemic α-hydrazino acids.[58] Using a chiral rhodium complex **61**, *N*-acyl-hydrazones **58** have been hydrogenated into the $N^{\beta}$-acyl-α-hydrazino acid esters **59** with a high enantiomeric excess depending on the experimental conditions, the catalyst, and the nature of the molecule. The best results (up to 95% ee) are obtained at low temperature (−10 °C) in iPrOH with the benzoyl N-blocking group and catalyst **61** ($R^1$ = Et) (Scheme 18).[67] Interestingly, catalytic hydrogenation allows the introduction of deuterium on the α-carbon.[67]

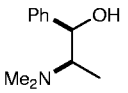
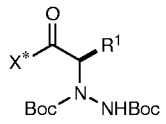
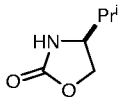
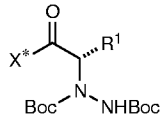
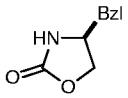
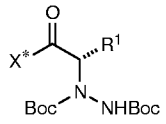
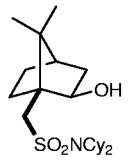
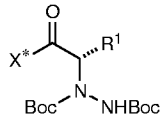
**Scheme 18** Synthesis of an α-Hydrazino Ester by Asymmetric Hydrogenation of an *N*-Benzoylhydrazone[67]




In the late 1980s various asymmetric procedures for the synthesis α-hydrazino acids, based on the electrophilic amination of an enolate bearing a chiral auxiliary by a dialkylazodicarboxylate, have appeared in the literature (Table 1).[55,68–72] The α-hydrazino acid is recovered in high yield after cleavage of the chiral auxiliary and acidolysis of the Boc groups in **63**. The D-enantiomer is obtained with **65**,[68] and the L-enantiomer with **66**,[55] **67**,[69,71] and **68**.[70] A chiral auxiliary is not required for hydrazinating chiral β-hydroxyesters.[72]

**Table 1** Asymmetric Synthesis of α-Hydrazino Acid Derivatives Using a Chiral Auxiliary[55,68–71]




| Chiral Auxiliary X*H | Stereochemistry of **63** | $R^1$ | ee (%) in **63** | Yield (%) of **63** | Ref |
|---|---|---|---|---|---|
|  **65** |  | Me<br>Bzl | 91<br>91 | 70[a]<br>45[a] | [68]<br>[68] |
|  **66** |  | Me<br>Bzl | 72<br>86 | 92[a]<br>88 | [55]<br>[55] |
|  **67** |  | Me<br>Bzl | 96<br>94 | 92<br>91 | [69,71]<br>[69,71] |
|  **68** |  | Me<br>Bzl | 94<br>96 | 81<br>76 | [70]<br>[70] |

[a] Calculated from the resulting α-hydrazino acid.

*for references see p 453*

***N*-Amino-L-alanine Ethyl Ester (60):**[67]

**CAUTION:** *Hydrogen is extremely flammable. Use appropriate safety precautions.*

Benzoylhydrazone **58** was prepared classically by treatment of benzoylhydrazine in THF with ethyl pyruvate in the presence of cat. concd HCl (3 drops). From the mixture, stirred for 12 h at rt, the hydrazone **58** precipitated and was filtered off. NMR analysis showed the presence of both *Z*- and *E*-isomers in approximately 1:5 ratio. In a $N_2$-filled dry box a 100-mL Fischer–Porter tube was charged with a stirrer bar and **58** (0.5 mmol), followed by degassed iPrOH (10 mL) and catalyst **61** ($R^1$ = Et). After six vacuum/$H_2$ cycles to purge the lines of air and two vacuum/$H_2$ cycles on the mixture, the tube was pressurized to an initial pressure of (1.0–4.1) × $10^5$ Pa $H_2$. The reaction was stirred at temperatures ranging from −15 to 50 °C until no $H_2$ uptake was observed. The mixture was concentrated and the residue passed through a short silica column with EtOAc/hexanes as eluent to give Bzψ[CO-NH-NH]L-Ala-OEt (**59**). The enantiomeric excess was estimated by HPLC (89% ee). The ester **59** (1.2 g, 5.08 mmol) was refluxed in 3 M HCl (10 mL) for 3 h. Upon cooling, benzoic acid precipitated. The aqueous soln was extracted with $Et_2O$ and concentrated to afford **60** as the hydrochloride salt; yield: 0.653 g (91%); white powder decomposed on heating; $[\alpha]_D^{25}$ +17.0 (*c* 1, $H_2O$).

**(2*S*)-2-[1,2-Bis(*tert*-butoxycarbonyl)hydrazino]-3-phenylpropanoic Acid (64; $R^1$ = Bzl):**[71]
A soln of the chiral auxiliary **67** (22.55 g, 127.3 mmol) and $Ph_3CH$ (50 mg, indicator) in dry THF (250 mL), stirred at −78 °C under $N_2$, was treated dropwise with 1.55 M BuLi in hexanes until an orange color persisted. The mixture, maintained at −78 °C, was treated with 3-phenylpropanoyl chloride (21 mL, 141 mmol) over 2 min, warmed to 0 °C, treated with sat. aq $NaHCO_3$ (100 mL), and stirred at 20 °C for 30 min. The mixture was extracted with $CH_2Cl_2$ (× 3), and the combined extracts were successively partitioned with 5% aq $Na_2CO_3$ and sat. NaCl, dried ($MgSO_4$), and concentrated to dryness under reduced pressure. The imide **62** (X* = **67**; $R^1$ = Bzl) was recrystallized (hexanes/EtOAc, 2:1); yield: 36.4 g (92%); [α] +78.6 (*c* 1, $CH_2Cl_2$).

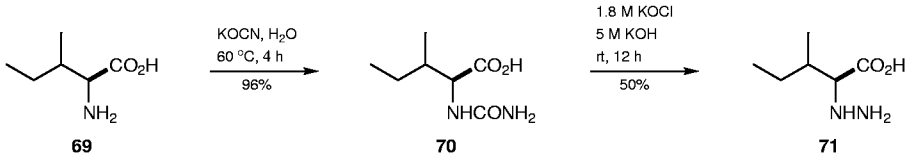
A soln of **62** (X* = **67**; $R^1$ = Bzl; 6.19 g, 20 mmol) in THF (60 mL) was added via cannula over 50 min to a stirred soln of LDA in THF (20 mL) and hexanes (10 mL) at −78 °C to give the enolate. The cold soln of the enolate was added via cannula to a stirred, cold (−78 °C) soln of Boc-N=N-Boc (5.3 g, 23 mmol) in $CH_2Cl_2$ (120 mL) over 9 min. The reaction was quenched with glacial AcOH (3 mL, 52 mmol). The mixture was repeatedly partitioned between $CH_2Cl_2$ and phosphate buffer (pH 7), the $CH_2Cl_2$ phases were combined, washed with sat. aq $NaHCO_3$, dried ($MgSO_4$), and concentrated under reduced pressure to give the hydrazine derivative **63** (X* = **67**; $R^1$ = Bzl); yield: 9.83 g (91%); 94% ee. The major (2*S*)-isomer of **63** (X* = **67**; $R^1$ = Bzl), was purified by chromatography; $[\alpha]_D^{26}$ +84.4 (*c* 1, $CHCl_3$).

An ice-cooled soln of (2*S*)-**63** (X* = **67**; $R^1$ = Bzl; 0.27 g, 0.5 mmol) in THF (2 mL) was treated in one portion with a cold (0 °C) soln of LiOH (28 mg, 1.2 mmol) in $H_2O$ (1 mL). After 3 h, the mixture was repeatedly partitioned between 1 M aq $NaHSO_4$ and $CH_2Cl_2$, the combined organic phases were dried ($Na_2SO_4$) and concentrated in vacuo to give **64** ($R^1$ = Bzl); yield: 0.16 g (82%).

### 10.8.4.1.2 Shestakov Rearrangement

The conversion of urea to hydrazine using hypochlorite is known as Shestakov rearrangement.[73] It has been used to obtain α-hydrazino acids from α-amino acid derived hydantoic acids, with chiral retention but in a moderate yield (about 20–40%), using NaOCl.[74,75] The use of KOCl, instead of NaOCl, greatly increases the yield of Shestakov rearrangement (Scheme 19).[76–78]

**Scheme 19** Synthesis of an α-Hydrazino Acid by Shestakov Rearrangement[78]




**69** **70** **71**

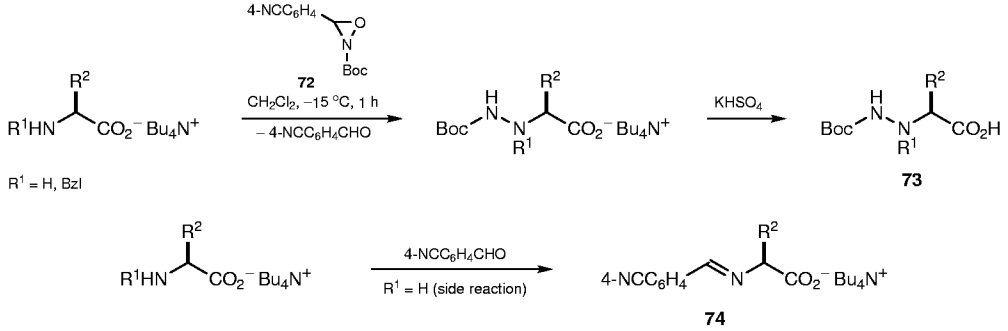
***N*-Amino-L-isoleucine (71):**[78]
KOCN (72 g, 7 equiv) was added to a suspension of H-L-Ile-OH (**69**; 17 g, 0.13 mol) in $H_2O$ (250 mL). After 4 h at 60 °C under stirring, the mixture was cooled to 0 °C. The Ile-derived hydantoic acid **70** was

precipitated by slow addition of 10 M HCl (120 mL), and collected; yield: 21.7 g (96%); $[\alpha]_D^{23}$ +160 (*c* 0.9, 1 M NaOH). To **70** (4.35 g, 25 mmol) in 5 M KOH (15 mL) at 0 °C 1.88 M KOCl (16 mL) was added dropwise, and the mixture was kept in the dark for 19 h at rt. A gummy material, precipitated on acidification (0 °C, 1 M HCl), was extracted by $Et_2O$. The aqueous phase was percolated through a Dowex 1X2–100 ion-exchange resin ($OH^-$ form). The resin was eluted with $H_2O$ until the eluent was neutral, then H-L-($H_2N$)Ile-OH (**71**) was recovered by elution with 1 M AcOH. Recrystallization (95% EtOH) gave a white powder which decomposed on heating; yield: 1.81 g (50%); $[\alpha]_D^{23}$ +2.7 (*c* 0.7, 5 M HCl).

### 10.8.4.1.3 *N*-Amination of α-Amino Acids

Sodium hydroxylamine-*O*-sulfonate and chloramine ($NH_2Cl$) have been used with success for the synthesis of DOPA hydrazino-analogues.[74] Oxaziridine and its derivatives are potential amine N-aminating reagents,[79–81] and *N*-alkyloxycarbonyl-3-aryloxaziridines, such as **72**, have been shown to transfer with appreciable yields the *N*-alkyloxycarbonylamino group to the amine nitrogen of α-amino acids to give **73** (Scheme 20).[82–85] The best yield is obtained with a secondary α-amino acid, such as Pro. With primary amines a side reaction with the liberated arylaldehyde gives the Schiff base **74**.[82] The yield is increased by prior N-benzylation of the primary amine, to give the orthogonally protected *N*-(Boc-amino)-*N*-Bzl-α-amino acid (**73**, $R^1$ = Bzl).[82–84]

**Scheme 20** Electrophilic N-Amination of α-Amino Acids Using *N*-Boc-(3-cyanophenyl)oxaziridine[82]




**1-[(*tert*-Butoxycarbonyl)amino]proline** [73, $R^1$, $R^2 = (CH_2)_3$]:[82]
A suspension of H-Pro-OH (0.575 g, 5 mmol) in MeOH (2.5 mL) was treated with 2.15 M $Bu_4N^+ \cdot OH^-$ in MeOH (2.3 mL) for 30 min. The MeOH was concentrated to dryness and replaced by $CHCl_3$ (5 mL). The soln was cooled to −30 °C then oxaziridine **72** (1.3 g, 5.25 mmol) was slowly added, and stirring was maintained for 1 h at −30 °C. The $CHCl_3$ was removed under reduced pressure, $H_2O$ (20 mL) was added, and the precipitate of 4-$CNC_6H_4CHO$ was filtered off. The basic aqueous phase was extracted with $Et_2O$ (10 mL) and acidified to pH 3 with solid $KHSO_4$. The title product was extracted from the acidified aqueous phase with EtOAc, which was then dried ($Na_2SO_4$) and concentrated; yield: 1.09 g (95%); mp 128 °C; $[\alpha]_D^{25}$ −41.2 (*c* 1, 95% EtOH).

## 10.8.4.2 (*N*-Aminoamide) Peptide Synthesis

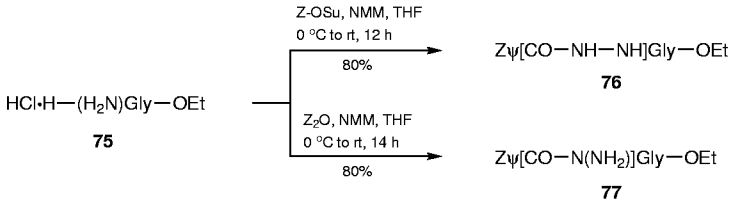
α-Hydrazino acids are monosubstituted hydrazines in which the preferential reactivity of the α-nitrogen may be counterbalanced by the steric effect of the side chain and the electronic influence of the carboxylic group. Tentative coupling of an N-protected α-amino acid with HCl·H-($H_2N$)Xaa-OMe (Xaa = Gly, Ala, Phe) is reported to give exclusively β-acylation whatever the coupling mode.[86–91] $N^\beta$-Protection of the α-hydrazino acids is recommended before $N^\alpha$-coupling. This pioneering work has been reinvestigated by Viret[92] and Lecoq et al.[93,94]

*for references see p 453*

### 10.8.4.2.1 Regioselective N-Protection of α-Hydrazino Acid Derivatives

*N*-Aminoglycine behaves differently from the $C^{\alpha}$-alkyl substituted α-hydrazino acids. The α- and β-protections of the commercially available HCl·H-($H_2N$)Gly-OEt **75** by the usual Boc and Z groups depend both on the acylating agent and the experimental conditions. α-Acylation generally predominates, but occurrence of β-acylation, and even α,β-diacylation, is not negligible. N-Acylation by the Boc group, except with $Boc_2O$ (80% α-acylation),[93] is generally less regioselective than Z acylation. In this latter case regioselectivity is practically inverted when using $Z_2O$ (80% α-acylation, giving **77**) and Z-OSu (80% β-acylation, giving **76**) (Scheme 21).[93] Niedrich[88] has shown the occurrence of the spontaneous partial rearrangement of **76** into **77**. The same phenomenon is observed for Zψ[CO-NH-NH]Gly-OEt, which slowly turns into Zψ[CO-N($NH_2$)]Gly-OEt above −10 °C.[93]

**Scheme 21** α- and β-Acylation of an α-Hydrazino Acid Derivative[93]




The regioselective N-protection of the $C^{\alpha}$-alkyl substituted α-hydrazino acids has been much less investigated. The bulky benzyl side group in H-($H_2N$)Phe-OH results exclusively in $N^{\beta}$-protection, but the α-nitrogen can be further acylated by using a strong acylating agent, such as an acyl chloride, thus giving access to orthogonal $N^{\alpha}$ and $N^{\beta}$-protections, which can subsequently be selectively eliminated.[91,95] The case of H-($H_2N$)Ala-OH is slightly more complicated, as a mixture of the $N^{\alpha}$-monoprotected, $N^{\beta}$-monoprotected, and $N^{\alpha}$,$N^{\beta}$-diprotected derivatives is obtained, although $N^{\beta}$-protection predominates.[90,96]

**Zψ[CO-NH-NH]Gly-OEt (76):**[93]
NMM (2.1 mL, 20 mmol) in THF (5 mL) was added dropwise to a cooled (0 °C) suspension of HCl·H-($H_2N$)Gly-OEt (75; 3 g, 20 mmol) and Z-OSu (5.98 g, 24 mmol) in THF (15 mL). Stirring was maintained for 12 h while the mixture was allowed to reach rt. The salts were filtered off and the solvent was removed. The residue was taken up in $CH_2Cl_2$ (70 mL) and the soln was washed with sat. aq NaCl and dried ($Na_2SO_4$). The $CH_2Cl_2$ was removed and **76** crystallized from EtOAc; yield: 3.89 g (80%); mp 98–100 °C.

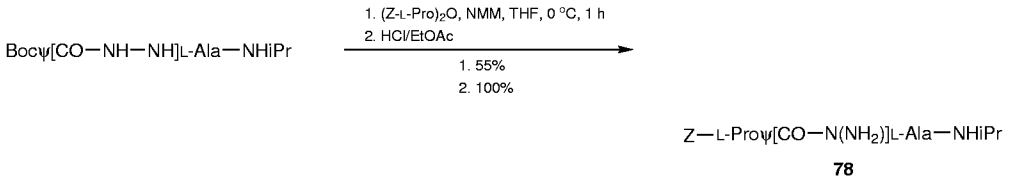
**Zψ[CO-N($NH_2$)]Gly-OEt (77):**[93]
$Z_2O$ (5.72 g, 20 mmol) in THF (10 mL) was added dropwise to a cooled (0 °C) suspension of HCl·H-($H_2N$)Gly-OEt (**75**; 3 g, 20 mmol) and NMM (2.1 mL, 20 mmol) in THF (10 mL). Stirring was maintained for 2 h at 0 °C, and for 12 h at rt. The salts were filtered off and the solvent was removed. The residue was taken up in the minimum quantity of $CH_2Cl_2$, and Zψ[CO-N($NH_2$)]Gly-OEt (**77**) and Zψ[CO-N(NHZ)]Gly-OEt were precipitated by adding $iPr_2O$/hexanes (2:1) at 0 °C. Compound **77** was purified by flash chromatography $R_f$ 0.40 (silica gel, 3% iPrOH/$CH_2Cl_2$) to give a colorless oil; yield: 4.0 g (80%).

### 10.8.4.2.2 C- and $N^{\alpha}$-Coupling of $N^{\beta}$-Protected α-Hydrazino Acid Derivatives

C-Coupling of α-hydrazino acids can be carried out by the standard procedures. $C^{\alpha}$-Alkyl substituted α-hydrazino acids require only $N^{\beta}$-protection, whereas protection of both α- and β-nitrogens is recommended for H-($H_2N$)Gly-OH to prevent undesirable reactions. Selective cleavage of $N^{\alpha}$-protection gives access to the *N*-aminoamide link **4**, which allows in turn for further branching of the main chain.

The poor accessibility of the α-nitrogen and its reduced nucleophilicity due to $N^{\beta}$-protection requires strong $N^{\alpha}$-acylating conditions. Coupling assays of Boc-Pro-OH or Fmoc-Pro-OH with H-(ZNH)Gly-OEt under various conditions show that the best yield is obtained with a symmetric anhydride or an acyl chloride.[93] The symmetric anhydride method affords Z-Proψ[CO-N($NH_2$)]Ala-NHiPr (**78**) in 55% yield (Scheme 22).[97]

**Scheme 22** $N^{\alpha}$-Coupling of $N^{\beta}$-Protected α-Hydrazino Acid Derivatives[97]




An alternative approach is to enhance the nucleophilicity of the α-nitrogen by silylation. Treatment of the sterically hindered H-(ZNH)Aib-OEt derivative with bis(trimethylsilyl)-acetamide results in 35–55% $N^{\alpha}$-coupling yield with Boc-Asp(OBzl)-F or Boc-Glu(OBzl)-F.[98]

**Z-L-Proψ[CO-N($NH_2$)]L-Ala-NHiPr** (78):[97]

**CAUTION:** *Dicyclohexylcarbodiimide (DCC) is a severe eye, skin and respiratory tract irritant and a skin sensitizer.*

To a cooled (0 °C) soln of Z-L-Pro-OH (1.99 g, 8 mmol) in THF (10 mL), DCC (0.82 g, 4 mmol) was added in one portion. DCHU was filtered off after 1 h. The filtrate, containing the symmetric anhydride $(Z\text{-L-Pro})_2O$, was added dropwise with stirring to Bocψ[CO-NH-NH]L-Ala-NHiPr (0.98 g, 4 mmol) in a cold (0 °C) mixture of THF (10 mL) and NMM (0.44 mL, 4 mmol), and the soln was allowed to reach rt for 1 h. After addition of $CH_2Cl_2$ (50 mL), the organic phase was successively washed with 10% citric acid (3 × 10 mL) and 5% aq $NaHCO_3$. The organic phase was dried ($Na_2SO_4$) and concentrated under reduced pressure to give Z-L-Proψ[CO-N(NHBoc)]L-Ala-NHiPr; yield: 1.05 g (55%).

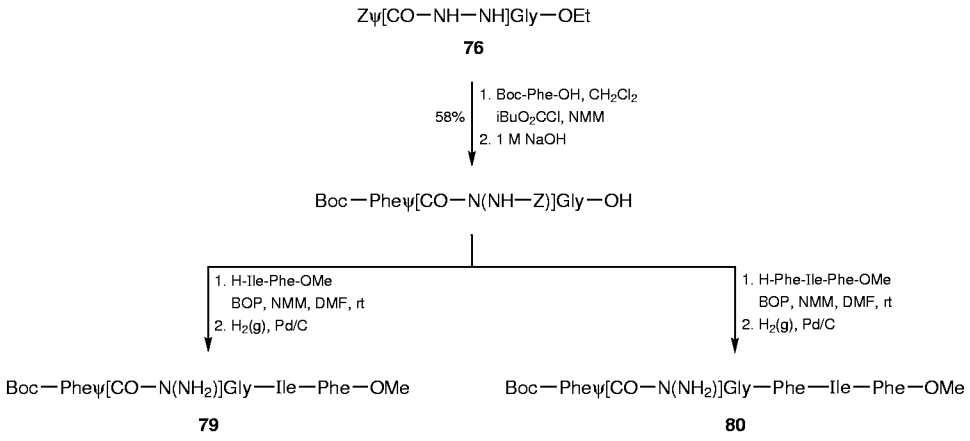
After acidolysis of the Boc group with 3 M HCl in EtOAc, the title compound **78** was purified by chromatography $R_f$ 0.45 (silica gel, 5% iPrOH/$CH_2Cl_2$); yield: 0.83 g (100%); mp 136 °C.

### 10.8.4.2.3 Strategies for (*N*-Aminoamide) Peptide Synthesis

The syntheses of only a very few (*N*-aminoamide) peptides, with no examples of solid-phase synthesis, have been reported in the literature. The low yield in $N^{\alpha}$-coupling calls for the fragment condensation methodology, which has been applied to the synthesis of the HIV protease substrate analogues **79** and **80** (Scheme 23).[50]

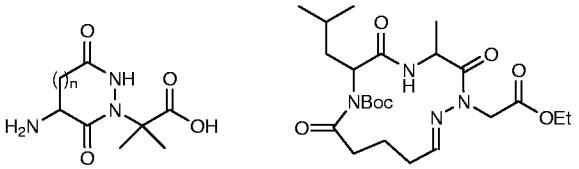
*for references see p 453*

**Scheme 23** (*N*-Aminoamide) Peptide Synthesis by Fragment Condensation[50]




The (*N*-aminoamide) link **4** is a convenient intermediate to cyclization of the peptide chain as in the cyclic initiators **81** (n = 1, 2) of an α-helix (Scheme 24).[98] Replacement of the N–H···O=C hydrogen bond by the N–N=CH–C hydrazone moiety gives another α-helix initiator **82**, synthesized by coupling the *N*-amino group with an aldehyde carbonyl.[99]

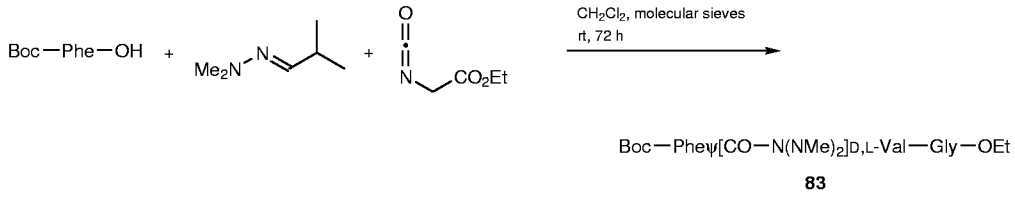
**Scheme 24** Chemical Structures of the Cylic Initiators of the α-Helix[98,99]




## 10.8.4.3 (*N*-Aminoamide) Peptide Synthesis by Ugi's Reaction

In 1973 the (*N*,*N*-dimethylaminoamide) hexapeptide Boc-Ala-Pheψ[CO-N($NMe_2$)]Val-Gly-Leu-Met-$NH_2$ was prepared by fragment condensation methodology from the (*N*,*N*-dimethylaminoamide) tripeptide **83**, which in turn had been obtained by adapting Ugi's reaction (Scheme 25).[100] The Ugi's reaction gave access to **83** containing a racemic $N^{\beta}$,$N^{\beta}$-dialkylhydrazino acid unit, the corresponding diastereomers of which could be separated, but not assigned, by chromatography.

**Scheme 25** Synthesis of an (*N*,*N*-Dimethylaminoamide) Tripeptide by Modified Ugi's Condensation[100]



**Boc-Pheψ[CO-N(NMe$_2$)]L-(or D)-Val-Gly-OEt (83):**[100]

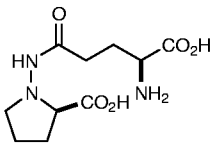
| **CAUTION:** *Benzene is a carcinogen!* |
|---|

2-Methylpropanal dimethylhydrazone (6 g, 0.06 mol) and ethyl isocyanatoacetate (7.8 g, 0.06 mol) in $CH_2Cl_2$ (25 mL) were stirred at 0 °C with 4-Å molecular sieves (10 g). Boc-L-Phe-OH (13.6 g, 0.05 mol) in $CH_2Cl_2$ (50 mL) was added at 0 °C, and the mixture was allowed to reach rt with stirring. The molecular sieves were filtered off and the organic soln was successively washed with aq $NH_3$ and sat. aq NaCl. The organic solvent was removed under reduced pressure to give the crude product; yield 8.8 g (36%). The two oily diastereomers were separated by chromatography (silica gel, benzene/EtOAc 5:2) $[\alpha]_D^{25}$ −66.9 (*c* 1, DMF); MS *m/z*: 493 $[M+H]^+$; and $[\alpha]_D^{25}$ +40.0 (*c* 1, DMF); MS *m/z*: 493 $[M+H]^+$.

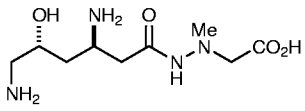
## 10.8.5 (Hydrazide) Peptides

Two natural (hydrazide) or ψ[CO-NR$^1$-NR$^2$] **5** peptides with antibacterial properties have been isolated so far: linatine (**84**) from flaxseed,[65] and negamycin (**85**) from *Streptomyces purpeofuscus*.[101] The first synthetic (hydrazide) peptides, reported in the 1960s, were analogues of the C-terminal octapeptide of eledoisin (**86**),[86–91,102–104] oxytocin (**87**),[105] and norophthalmic acid (**88**) (Scheme 26).[106] Due to the difficulties in obtaining optically pure α-hydrazino acids, many of these analogues are diastereomeric mixtures. The α-hydrazino acid unit is shown to give a folded structure,[107–109] and could induce local geometrical constraints in the (hydrazide) peptides.

**Scheme 26** Chemical Structures of Some (Hydrazide) Peptides



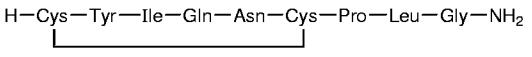

84



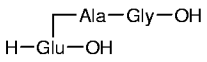

85

pGlu—Pro—Ser—Lys—Asp—Ala—Phe—Ile—Gly—Leu—Met—$NH_2$

86




87




88

### 10.8.5.1 (Hydrazide) Peptide Synthesis in Solution

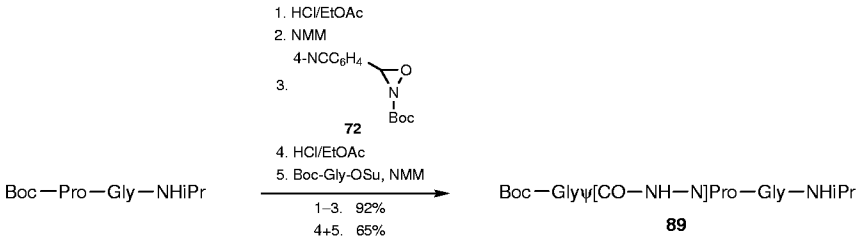
In the pioneering work of Niedrich's group[86–91,102–105] the α-hydrazino acid residues are usually C-coupled by exploiting activated esters, without protection of the α-nitrogen. The synthesis of (hydrazide) peptides was later reinvestigated by Viret,[92] Dubar,[96] and Guy et al.[84] They observed that the mixed anhydride method might give quite poor yields for C-coupling of a N$^\beta$-protected α-hydrazino acid, because the alkyloxycarbonyl group is transferred to the unprotected α-nitrogen.[84] In spite of this, the DCC/HOBt or DCC/DMAP method works satisfactorily. Bentley and Morley[106] pointed out that a C-terminal α-hydrazino acid residue with unprotected α-nitrogen might give a hydantoin, while Guy et al.[84] observed the formation of a piperazine-2,5-dione (diketopiperazine, DKP) and a mixture of related oligomers, thus preventing activation of this unit for fragment condensation.

*for references see p 453*

Z-Ala-Ala-Pro-Valψ[CO-NH-NH]Ile-Leu-OMe, a (hydrazide) hexapeptide analogue of a human leukocyte elastase (HLE) substrate, has been prepared in solution.[78,84] First, Bocψ[CO-NH-NH]Ile-OH is coupled to H-Leu-OMe with DCC/HOBt in 67% yield. Subsequently, the resulting (hydrazide) dipeptide is elongated by the mixed anhydride method to the H-Valψ[CO-NH-NH]Ile-Leu-OMe (hydrazide) tripeptide, which in turn is coupled to the Z-Ala-Ala-Pro tripeptide by the azide[78] or mixed anhydride method.[84] C-Coupling of an $N^{\beta}$-protected α-hydrazino acid is improved by $N^{\alpha}$-benzylation, and the best result is obtained with PyBOP as coupling agent.[84] However, further $N^{\beta}$-acylation of the hydrazide terminus requires strong acylating conditions, such as NCAs. This procedure has been applied to the synthesis of various (hydrazide) analogues of the HLE substrate Z-Ala-Ala-Pro-Val-Ala-Ala-NHiPr.[84]

The easy N-amination of Pro by the oxaziridine reagent **72**[82] allows for a step-by-step synthesis of **89** where N-amination is performed on the growing peptide and $N^{\beta}$-coupling of the *N*-amino proline residue is carried out classically with the activated ester procedure (Scheme 27).[109]

**Scheme 27** (Hydrazide) Peptide Synthesis in Solution[109]



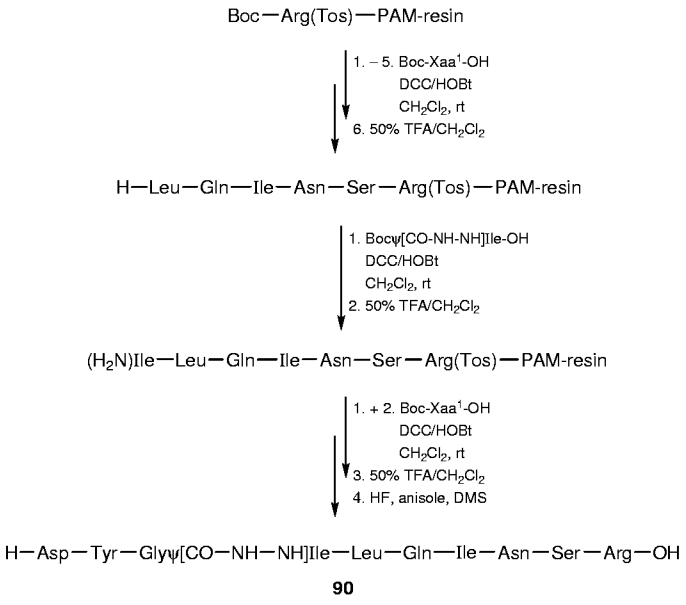

**Boc-Glyψ[CO-NH-N]Pro-Gly-NHiPr (89):**[109]
Boc-Pro-Gly-NHiPr (1.57 g, 5 mmol) was dissolved in 3 M HCl/EtOAc (10 mL) with stirring for 1 h to give HCl·H-Pro-Gly-NHiPr. The solvent and excess of HCl were eliminated under reduced pressure and the amine regenerated with NMM (0.55 mL, 5 mmol) in $CH_2Cl_2$ (5 mL). The HCl·NMM salt was filtered off. The soln was cooled to −30 °C, oxaziridine **72** (1.3 g, 5.25 mmol)[82] was slowly added, and stirring was maintained for 1 h at −30 °C and then 1 h at rt. Bocψ[CO-NH-N]Pro-Gly-NHiPr was purified by chromatography $R_f$ 0.35 (silica gel, 8% MeOH/$Et_2O$); yield: 1.51 g (92%); mp 70 °C.

After removal of the Boc group with 3 M HCl/EtOAc (10 mL), the resulting peptide hydrochloride (1.39 g, 5 mmol) was treated with Boc-Gly-OSu (1.32 g, 5 mmol) and NMM (0.55 mL, 5 mmol) in $CH_2Cl_2$ (10 mL). After the usual work-up procedures, the title compound **89** was purified by chromatography $R_f$ 0.40 (silica gel, 15% iPrOH/$CH_2Cl_2$); yield: 1.25 g (65%); mp 128 °C.

### 10.8.5.2 Solid-Phase (Hydrazide) Peptide Synthesis

Only the H-Asp-Tyr-Glyψ[CO-NH-NH]Ile-Leu-Gln-Ile-Asn-Ser-Arg-OH (hydrazide) decapeptide **90**, an analogue of a HEL T-epitope (52–61 segment),[110] has been prepared by the solid-phase procedure, using the step-by-step Boc/TFA/HF strategy (Scheme 28).[96] Bocψ[CO-NH-NH]Ile-OH is incorporated using DCC/HOBt, and the coupling step is repeated 3 times with 3 equivalents each time. It is noteworthy that, due to the presence of the unprotected α-nitrogen in the growing (hydrazide) peptide, capping with $Ac_2O$ is precluded. The Kaiser ninhydrin test gave a dark red color with the N-terminal hydrazine, and turned to an orange color after formation of the hydrazide linkage.

**Scheme 28** Solid-Phase Synthesis of a (Hydrazide) Decapeptide, an Analogue of a HEL T-Epitope Segment[96]

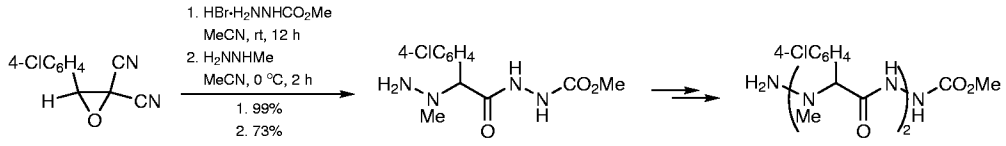



**90**

## 10.8.5.3 (Hydrazide) Polypeptide Synthesis

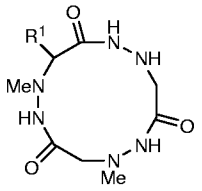
The treatment of a N-substituted hydrazine hydrobromide successively with the α,α-dicyano-epoxide **91** and with *N*-methylhydrazine results in the racemic α-hydrazino hydrazide **92** (Scheme 29).[111] Repeating the above two reactions allows the elongation of **92** into the racemic (hydrazide) peptide **93**.[112] A derived procedure, with $ClCH_2COCl$ instead of **91**, is applied to the synthesis of the first (hydrazide) cyclotripeptide **94** ($R^1 = 4\text{-}ClC_6H_4$ or $4\text{-}MeC_6H_4$) (Scheme 30), by coupling of the N-terminal chloroacetyl and C-terminal hydrazine groups.[113] It has been later extended to the synthesis of linear (hydrazide) polypeptides.[114] It is noteworthy that the α-carbon is racemic in all of the above cases.

**Scheme 29** Synthesis of (Hydrazide) Polypeptides[111,112]




**91** **92** **93**

**Scheme 30** Structure of the First Synthesized (Hydrazide) Cyclotripeptide[113]




**94**

*for references see p 453*

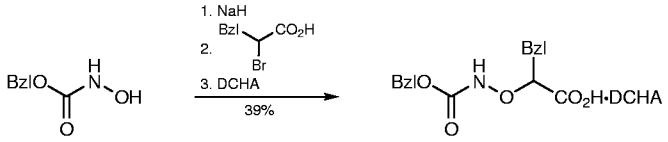
## 10.8.6 (Amidoxy) Peptides

The amidoxy, or ψ[CO-NH-O], link **6** ($R^1 = H$) results from N-acylation of an α-aminoxy acid. α-Aminoxy acids have been isolated from natural sources and some of them present antibacterial properties as such, or when coupled to an α-amino acid, as in malioxamycin H-L-Valψ[CO-NH-O]D-Asp-OH.[115–117] The α-aminoxy acids are homologous to the α-hydrazino acids with the advantage that there is no problem of regioselectivity for N-acylation. Despite the easy preparation of α-aminoxy acids,[118–120] the number of (amidoxy) peptides reported in the literature is rather limited, with no examples of solid-phase synthesis. Recently, it has been observed that the α-aminoxy and α-hydrazino acid units induce quite similar local folded structures.[121–123]

### 10.8.6.1 Synthesis of α-Aminoxy Acid Derivatives

The synthesis of an α-aminoxy acid was reported as early as 1893,[124,125] and was later reinvestigated in the 1960s.[126–130] More recently, various syntheses of N-protected α-aminoxy acid derivatives, based on the same procedure (Scheme 31), have been proposed. An α-halo acid, obtained via $HNO_2$ deamination of the corresponding α-amino acid , is treated with the sodium salt of an N-protected hydroxylamine. The reaction is described for the Boc,[118,119] Z,[118,120] and Bz[117] protections, which may be subsequently eliminated by acidolysis. Owing to this synthetic procedure, the chirality of the resulting α-aminoxy acid is opposed to that of the starting α-amino acid.

**Scheme 31** Synthesis of an α-Aminoxy Acid Derivative[120]




**95**·DCHA

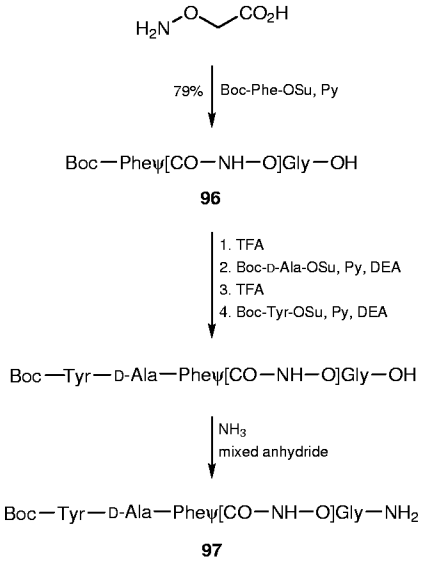
**Zψ[CO-NH-O]L-Phe-OH (95):**[120]
Z-NH-OH (16.7 g, 0.1 mol) in freshly distilled DMF (20 mL) was stirred at 0–5 °C and 80% NaH oil dispersion (6.0 g, 0.2 mol) was added portionwise over 30 min at 0–5 °C. Stirring was continued at 0–5 °C, then a soln of (*R*)-2-bromo-3-phenylpropanoic acid (22.9 g, 0.1 mol) in DMF (20 mL) was added dropwise over 15 min at 0–5 °C. The mixture was left at 20–25 °C overnight, then diluted with ice water and extracted with $Et_2O$ (2 × 150 mL) (backwashing with $H_2O$). The aqueous phase and $H_2O$ backwashings were combined, acidified with concd aq HCl and ice, then extracted with $Et_2O$ (3 × 200 mL). The combined extracts were washed with $H_2O$ (× 2), dried ($MgSO_4$), and removed to ca. 200 mL. DCHA (20 mL) was added and the solid DCHA salt of **95** was collected and washed with $Et_2O$. The pure L-enantiomer was recrystallized (EtOAc); yield: 19.9 g (39%); mp 161–162 °C.

### 10.8.6.2 (Amidoxy) Peptide Synthesis

Various (amidoxy) analogues of pentagastrin,[120,131,132] Leu-enkephalin,[133] dermorphin,[134] and *N*-formyl chemotactic tripeptide[135] have been prepared by solution procedures, with classical C-coupling, N-deprotection, and N-coupling steps. The $pK_a$ of the carboxylic group in the α-aminoxy acids is similar to that in the α-amino acids, while the amino group has lower basicity.[118] As a consequence, C-coupling of an N-protected α-aminoxy acid may be carried out classically by using the $OC_6Cl_5$[131,132] or OSu[133,134] activated ester, the mixed anhydride with isobutyl chloroformate,[132,133] and the DCC–HOBt[135] or TBTU[121] procedure. As catalytic hydrogenation is reported to cleave the N–O bond,[118] the Z protection

must be removed with HBr in AcOH,[120] and the Boc protection with TFA[131,132,134] or, better, with $B(TFA)_3$.[133] N-Coupling of the N-terminal aminoxy group works satisfactorily by using the mixed anhydride,[132] the DCC–HOBt procedure,[120,135] and the OSu activated ester.[117,133,134] It is remarkable that N-acylation with an activated ester, and further elongation of the N-terminus, may be carried out in pyridine without protection of the α-aminoxy acid carboxy group. Then, activation of the carboxy group allows for the exploitation of the segment condensation procedure.[133–135] This strategy has been applied to the synthesis of the (amidoxy) analogue **97** of dermorphin (Scheme 32).[134]

**Scheme 32** Synthesis of a Dermorphin Amidoxy Analogue[134]




**Boc-Pheψ[CO-NH-O]Gly-OH (96):**[134]
Boc-Phe-OSu (6.84 g, 19 mmol) was added to a soln of bis(aminoxyacetic acid) hydrochloride (4.36 g, 20 mmol) in pyridine (30 mL) containing TEA (5.6 mL, 40 mmol). The mixture was stirred overnight at rt, the solvent was removed, and the residue dissolved in $H_2O$ (30 mL). The soln was acidified with solid citric acid to pH 4–5, saturated with NaCl and extracted with EtOAc (3 × 30 mL); the combined organic layers were washed with brine, dried, and concentrated to dryness. The title product **96** was crystallized ($Et_2O$); yield: 5.1 g (79%); mp 135–136 °C; $[\alpha]_D^{20}$ −15.2 (*c* 1, MeOH).

## 10.8.7 (Oxomethyleneamino) Peptides

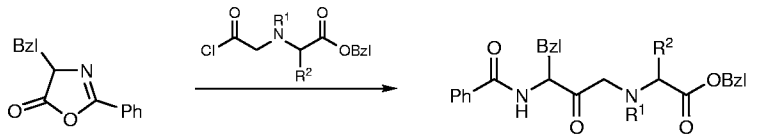
The oxomethyleneamino or ψ[CO-$CH_2$-NH] link **7** has been introduced by Meyer et al.[136] for the design of ACE inhibitors. It has been later used as such or in various N-alkylated forms to give ACE,[137–139] furin,[140] and HIV protease inhibitors.[141,142] It has been constructed in two different ways, either by using a modified Dakin–West procedure,[136–139] or by condensing a chloromethyl ketone with a peptide N-terminal amino group.[137–142] Link **7** can be considered as a precursor of the (hydroxy)ethyleneamino link **8** (see Section 10.8.8). Spontaneous epimerization of the α-carbon preceding the ketone carbonyl during silica gel chromatography has been reported, so that the separation of the L- and D-epimers is usually not possible.[142]

*for references see p 453*

### 10.8.7.1 Modified Dakin–West Reaction

Treatment of an α-amino acid with an anhydride in the presence of pyridine, known as the Dakin–West reaction,[143] gives access to a racemic α-amido ketone.[144,145] The mechanism involves formation of an oxazolinone (**98**).[146] In the modified Dakin–West reaction, the oxazolinone is reacted with an acid chloride (Scheme 33).[136–139] Due to the rapid epimerization of the oxazolinones, a diastereomeric mixture of product **99** is obtained. No detailed protocol has been published so far for the modified Dakin–West synthesis applied to (oxomethyleneamino) peptides, but the procedure is adapted from that already reported for (oxomethylene) peptides containing the ψ[CO-$CH_2$] link.[147]

**Scheme 33** (Oxomethyleneamino) Peptide Synthesis by the Modified Dakin–West Reaction[139]

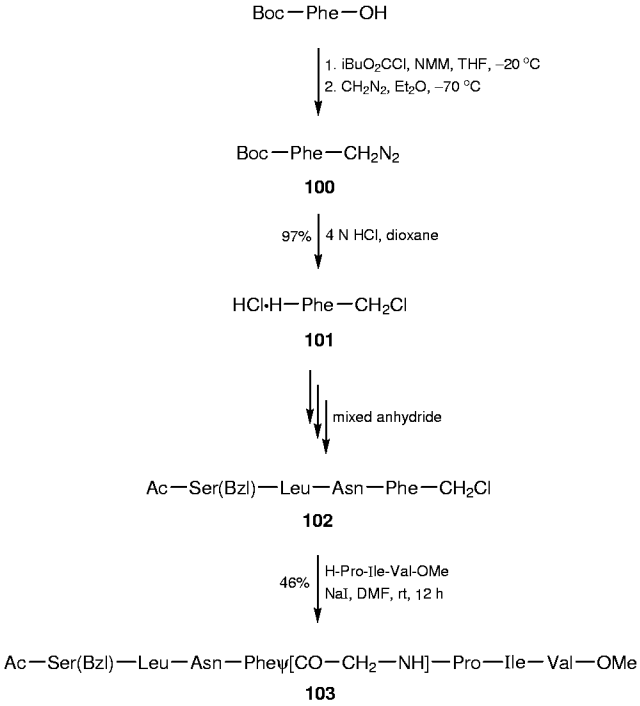



**98** **99**

### 10.8.7.2 From Chloromethyl Ketones

An efficient synthesis of the α-amino chloromethyl ketones **101** (although some epimerization may occur) consists in the acidolysis of the diazomethyl ketone **100** resulting from the action of diazomethane on a N-protected α-amino acid mixed anhydride (Scheme 34).[140,141] Elongation of the N-terminus can be carried out either on the intermediate diazomethyl ketone[140] or on the chloromethyl ketone **101**.[142] The elongated chloromethyl ketone **102** is coupled to the N-terminal amino group of a peptide to create link **7** in **103**.

**Scheme 34** Synthesis of an (Oxomethyleneamino) Peptide by Segment Condensation using a Chloromethyl Ketone[142]



**Preparation of an N-Protected Amino Acid Diazomethyl Ketone (e.g., 100); General Procedure:**[142]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate safety precautions should be taken when using this reagent.*

The N-protected α-amino acid (50 mmol) was dissolved in dry THF (100 mL) and cooled to −25 °C. NMM (50 mmol) was added, followed by $iBuO_2CCl$ (50 mmol), with the temperature maintained at −25 °C during addition. After stirring for 5 min, anhyd $Et_2O$ (100 mL) was added while the soln was simultaneously cooled to −70 °C. The cooled soln was quickly filtered under a blanket of $N_2$ and the cooled filtrate treated with a soln of $CH_2N_2$ (ca. 70 mmol) in $Et_2O$ (250 mL). The reaction was allowed to gradually warm to rt and the excess of $CH_2N_2$ was removed by purging the soln with a stream of $N_2$ for 15–30 min. The soln was concentrated to dryness and the residue taken up in $Et_2O$ (200 mL), washed with sat. $NaHCO_3$ and brine, and dried ($MgSO_4$). The title product was crystallized from $Et_2O$/hexanes or hexanes.

**Preparation of an Amino Acid Chloromethyl Ketone (e.g., 101); General Procedure:**[142]

A soln of 4 M HCl in dioxane was added to the N-protected amino acid diazomethyl ketone dissolved in anhyd dioxane or $Et_2O$ (10 mL·g$^{-1}$) cooled to 0 °C. After slow addition of the first equivalent of HCl in dioxane was complete with rapid evolution of $N_2$, a further 6–10 equivalents were added and the cooling bath removed. The mixture was stirred at rt for 1 h and the aminoacid chloromethylketone was precipitated by addition of $Et_2O$, filtered, washed with copious amounts of $Et_2O$, and dried in vacuo (on NaOH pellets). Prolonged storage resulted in significant coloration and decomposition.

**Preparation of an (Oxomethyleneamino) Peptide (e.g., 103); General Procedure:**[142]

The amino acid chloromethyl ketone (1.0 mmol) was dissolved in DMF (4–7 mL·mmol$^{-1}$) and NaI (1.1 mmol) was added. The mixture was stirred for 15 min. The peptide ester salt (1.1 mmol) in DMF (2–5 mL·mmol$^{-1}$) was added, followed by $NaHCO_3$ (2.1 mmol), and the reaction left to stir for 12–18 h at rt. The soln was then diluted with EtOAc (50–70 mL) and washed with $H_2O$. The aqueous layer was extracted (× 2) with EtOAc and the combined extracts were washed with $H_2O$ and brine before drying ($Na_2SO_4$). The resulting (oxomethyleneamino) peptide was of good purity; silica gel chromatography, carried out to remove trace impurities, may contribute to epimerization at the chiral center α to the ketone carbonyl.

## 10.8.8 [(Hydroxy)ethyleneamino] Peptides

Reduction of the carbonyl in the ψ[CO-$CH_2$-NH] link **7** ($R^1$ = H) results in the (hydroxy)ethyleneamino or ψ[CH(OH)-$CH_2$-NH] link **8** ($R^1$ = H), which has proved to be a very potent analogue of the tetrahedral hydrated intermediate of the scissile amide bond. It has been widely used for the design of various inhibitors of HIV protease[141,142,148–154] and ACE,[155–157] and to synthesize angiotensin II, III, and IV analogues.[158,159] Indeed, the chirality of the hydroxylated carbon is critical for HIV protease inhibition, but separation of the epimers may be difficult. Therefore, the stereoselective synthesis from epoxides has been actively investigated. An example of a C-methylated ψ[CMe(OH)-$CH_2$-NH] link, obtained from an epoxide with chromatographic separation of the epimers, has also been described.[157] Most of the [(hydroxy)ethyleneamino] peptides have been prepared by solution procedures, but two examples of solid-phase synthesis have been reported. A theoretical study of the (hydroxy)ethyleneamino replacement for the amide bond has been carried out on a HIV protease inhibitor.[160]

### 10.8.8.1 From (Oxomethyleneamino) Peptides

Compounds containing the (hydroxy)ethyleneamino link **8** were originally prepared in solution by reduction with $NaBH_4$ of the carbonyl in the oxomethyleneamino link **7**.[141,155,156] This reaction was later adapted to solid-phase conditions.[150,159] Two diastereomers, differing in the chirality of the hydroxylated carbon, have been characterized and their absolute configuration assigned by X-ray diffraction in some cases,[149,156] but their separation is usually not possible.[142]

*for references see p 453*

**Preparation of a [(Hydroxy)ethyleneamino] Peptide; General Procedure:**[142]

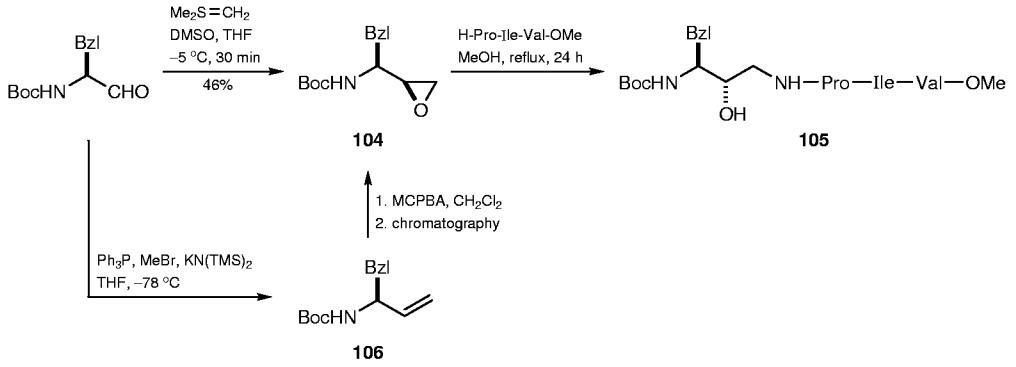
| **CAUTION:** *Hydrogen is extremely flammable. Use appropriate safety precautions.* |
|---|

The crude (oxomethyleneamino) peptide (see Section 10.8.7.2) (10 mmol) was dissolved in anhyd MeOH (10 mL), cooled to 0 °C, and treated with an excess of $NaBH_4$ in a single portion. The course of reaction was monitored by TLC, and addition of extra $NaBH_4$ was occasionally necessary. Upon completion of reduction, the soln was treated with EtOAc (50 mL) and $H_2O$ (50 mL). The aqueous phase was extracted with EtOAc (2 ×), and the combined extracts were washed with $H_2O$ and brine, and dried ($Na_2SO_4$). The [(hydroxy)ethylamino] peptide was purified by chromatography (silica gel gradient of MeOH in $CHCl_3$ for optimal separation). Diastereomer separation is usually not possible.

## 10.8.8.2 Stereoselective Opening of Epoxides

Very different potencies, depending on the chirality of the hydroxylated carbon, are reported for some [(hydroxy)ethyleneamino] peptide inhibitors of HIV protease.[148,150,152] As the diastereomers obtained by reduction of the (oxomethyleneamino) intermediate cannot be usually separated, stereoselective opening of an optically pure N-protected α-amino epoxide **104** is the preferred method of preparation (Scheme 35).[142,151,152] The epoxide **104** is obtained from the α-amino aldehyde either by using the Corey ylide[161] or via the α-amino ethylene **106**, which is subsequently epoxidized with MCPBA.[142,162] The two diastereomeric epoxides can be separated by fractional crystallization and silica gel chromatography, and their stereochemistry determined by NMR.[142,162] Due to the chemical interest in N-protected α-amino epoxides, scale-up productions from α-amino chloromethyl ketones, which can be separated by recrystallization, have also been reported.[151,163] The epoxide **104** is stereospecifically opened by $S_N2$ nucleophilic attack of a peptide N-terminal amino group to give link **8**. This reaction has also been carried out in the solid phase.[152]

**Scheme 35** Synthesis of a [(Hydroxy)ethyleneamino] Peptide by Stereoselective Opening of an Epoxide[142,161,162]




**(1*S*)-*N*-(*tert*-Butoxycarbonyl)-1-[(2*R*)-oxiranyl]-2-phenylethanamine (104):**[161]
Dimethylsulfonium methylide (0.21 mol) in THF[164] was stirred for 1 min at −5 °C, and added fairly rapidly to Boc-Phe-H (0.1 mol) dissolved in THF (135 mL). The resulting mixture was stirred in the cold for 30 min and then quenched in cold $H_2O$ (3 L). The suspension was extracted with $Et_2O$ (3 × 100 mL) and the combined organic layers were washed with $H_2O$ (2 × 100 mL), dried over $Na_2SO_4$, filtered, and the filtrate concentrated to dryness in vacuo. The residue was purified by chromatography (silica gel, hexanes/EtOAc 3:1). Concentration of the product fractions in vacuo gave the solid epoxide as a ca. 1:1 diastereomeric mixture; yield: 46%. The epoxide **104** was crystallized from $Et_2O$ by addition of petroleum ether (× 10), and then recrystallized (hexanes); mp 121.5–123.5 °C.

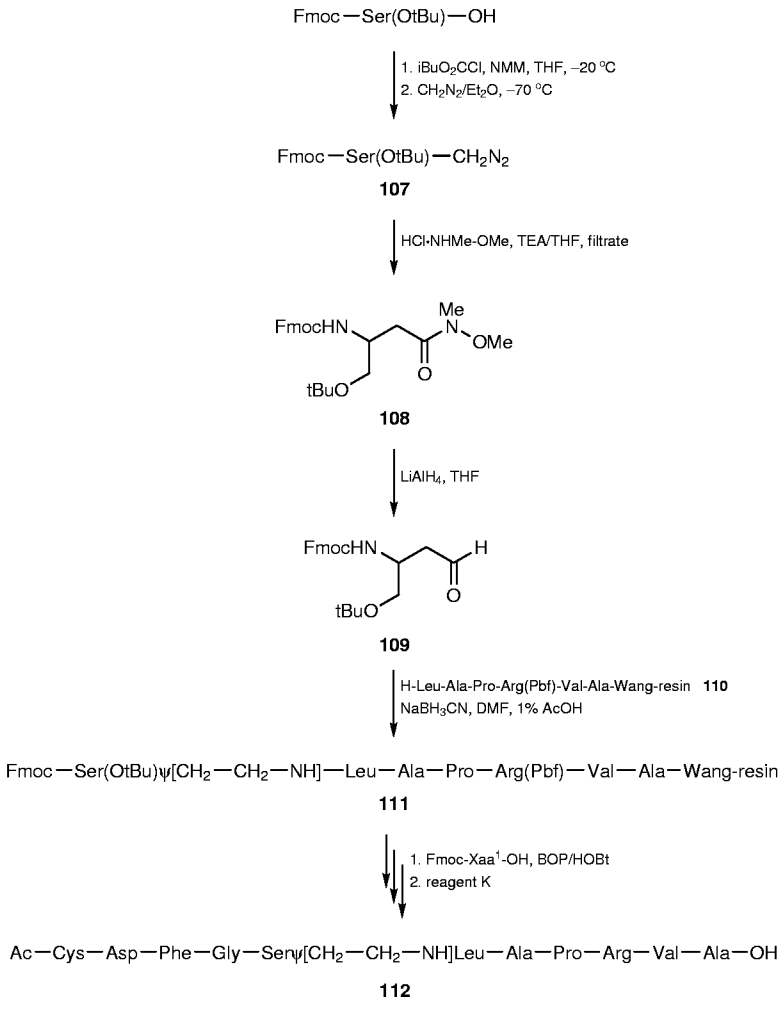
**Boc-Phe$\psi$[(*S*)CH(OH)-$CH_2$-NH]Pro-Ile-Val-OMe (105):**[142]
To a stirred soln of the epoxide **104** (0.29 g, 1.10 mmol) in dry MeOH (10 mL) HCl·H-Pro-Ile-Val-OMe (0.50 g, 1.34 mmol) and TEA (1.34 mmol) were added. The resulting soln was refluxed for 24 h and then

concentrated to give a residue, which was purified by chromatography (silica gel, 3–4% MeOH in $CHCl_3$) to give **105**; yield: 0.58 g, (87%); $[\alpha]_D^{25}$ −87.1 (*c* 1, MeOH); MS *m/z*: 605.39 $[M+H]^+$.

## 10.8.9 (Ethyleneamino) Peptides

The ethyleneamino or ψ[$CH_2$-$CH_2$-NH] link **9** ($R^1$ = H) may be considered as a reduced amide bond between the carboxy group of a $H_2N$-$CHR^1$-$CH_2$-$CO_2H$ β-amino acid and the amino group of an α-amino acid. It has been introduced, using a solid-phase procedure (Scheme 36), in the major antigenic site of the foot-and-mouth disease virus, corresponding to the region 147–156 (**112**) of the viral protein.[165,166] The ethyleneamino link **9** is created by condensing the amino group of a resin-bound peptide (**110**) with a Fmoc-protected β-amino aldehyde (**109**), and followed by reduction with $NaBH_3CN$. The N-protected β-amino aldehyde **109** is obtained by treatment of the corresponding diazoketone **107** (see Section 10.8.7.2) with NHMe-OMe to give **108**, and subsequent reduction of **108** with $LiAlH_4$.

**Scheme 36** Synthesis of an (Ethyleneamino) Peptide[165,166]




*for references see p 453*

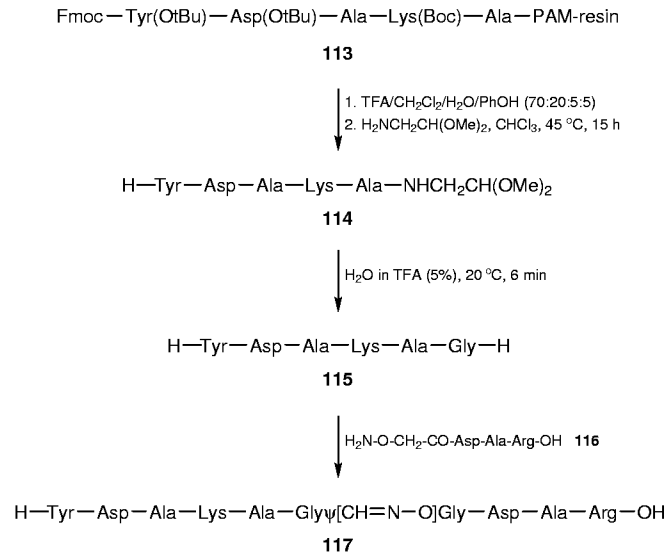
**3-(9-Fluorenylmethoxycarbonylamino)-4-*tert*-butoxy-*N*-methoxy-*N*-methylbutanamide** (108):[166]
Fmoc-Ser(OtBu)-$CH_2N_2$ (**107**) was prepared via the mixed anhydride of Fmoc-Ser(OtBu)-OH with $iBuO_2CCl$ (see Section 10.8.7.2). In the presence of silver benzoate (catalytic amount), compound **107** (10.17 g, 25 mmol) was treated with NHMe-OMe, regenerated from HCl·NHMe-OMe with TEA, in THF overnight. The precipitated HCl·TEA was removed by filtration. The resulting pure **108** was obtained as an oil by chromatography (silica gel, EtOAc/hexanes 1:2); yield: 10.12 g (92%); $R_f$ 0.41 (EtOAc/hexanes 1:1).

**Ac-Cys-Asp-Phe-Gly-Serψ[$CH_2$-$CH_2$-NH]Leu-Ala-Pro-Arg-Val-Ala-OH** (112):[166]
The conjugate **110** was constructed step-by-step by the Fmoc protocol with BOP/HOBt activation and piperidine deprotection. A threefold molar excess of the activated amino acid in DMF was used for coupling. The aldehyde **109**, which was prepared from **108** with $LiAlH_4$ just before use, was introduced (2.5 equiv in DMF with 1% AcOH) with $NaBH_3CN$ (6 equiv) to give **111**. The time course of the reaction was monitored by the Kaiser ninhydrin test. After completion of the reaction, the resin was alternatively washed with DMF and MeOH. The Fmoc group was removed and the $N^{\alpha}$-protected amino acids Fmoc-Xaa$^1$-OH [Xaa$^1$ = Gly, Phe, Asp(OtBu), Cys(Trt)] were successively coupled using the same procedure as indicated above. The (ethyleneamino) peptide **112** was cleaved from the Wang resin with reagent K,[167] and purified by RP-HPLC; overall yield: 22%; MS *m/z*: 1178.56 $[M+H]^+$.

## 10.8.10 (Oxime) Peptides

Immunogenicity enhancement of peptide epitopes is obtained by ligation to carriers, and the oxime ψ[CH=N-O] link **10** is of common use in these cases.[168–171] Interestingly, link **10** is easily created by simply mixing the carrier and the deprotected peptides in water. The 3D-structural influence of link **10** had not been investigated until very recently, when the short model dipeptide Boc-Proψ[CH=N-O]Xaa-NHiPr (Xaa = Gly or Ala) was synthesized by condensation of the corresponding aldehyde and aminoxy derivatives, and analyzed in solution.[123] Larger peptide fragments may be also connected by link **10** by combining solid-phase synthesis of a peptide aldehyde and segment coupling in aqueous solution (Scheme 37).[172]

**Scheme 37** Synthesis of an (Oxime) Peptide[172]

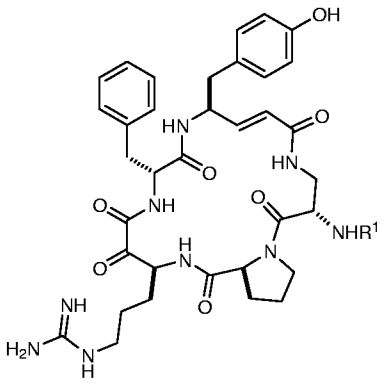


**H-Tyr-Asp-Ala-Lys-Ala-Glyψ[CH=N-O]Gly-Asp-Ala-Arg-Ala-OH** (117):[172]
The conjugate **113** was constructed step-by-step by the Fmoc/tBu chemistry with DCC/HOBt activation. A tenfold molar excess of activated amino acids was used for coupling. After deprotecting the Fmoc group, the side-chain protections were removed by a mixture of TFA/$CH_2Cl_2$/$H_2O$/PhOH (70:20:5:5) for 30 min. Cleavage from the resin by aminolysis with 2,2-dimethoxyethanamine in $CHCl_3$ (2:3) at 45 °C for 15 h gave the peptide aldehyde dimethylacetal **114** which was purified by RP-HPLC and characterized by electrospray MS (*m/z*: 653.39 $[M+H]^+$). The peptide aldehyde **115** was obtained just before use from **114** by treatment with TFA containing 5% of water at 20 °C for 6 min. TFA was immediately removed and **115** was mixed with the deprotected peptide **116** in 0.1 M NaOAc aq soln at pH 4.6. The time course of the ligation reaction was monitored by analytical HPLC, which showed that the reaction was completed in 45 min. The (oxime) peptide **117** was purified by RP-HPLC. Two proximal peaks giving the same molecular weight by electrospray MS (*m/z*: 1095.29 $[M+H]^+$), probably correspond to the *syn* and *anti* forms of the CH=N bond in **117**.

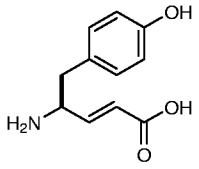
## 10.8.11 (But-2-enamide) Peptides [(Vinylogous amide) Peptides]

This peptide modification was originally mentioned in 1984 as the basis for a new class of potential inactivators of thiol proteases.[173] The first syntheses of 4-aminobut-2-enoic acid (vinylogous α-amino acid) derivatives, containing the ψ[CH=CH-CO-NH] link **11**, were reported in the next few years.[174–177] The natural antithrombin cyclotheonamides A **118** ($R^1$ = CHO) and B **118** ($R^1$ = Ac), isolated from the *Theonella* sponge, were shown to contain a (2*E*)-4-amino-3-(4-hydroxybenzyl)but-2-enoic acid (**119**) unit (*E*-vinylogous Tyr) (Scheme 38).[178,179] With much interest in its biological properties,[180] various total syntheses of **118** have been reported, and hence also of **119**.[179–183] Another reason for interest in 4-aminobut-2-enoic acids is the presence of the rigid ethylenic moiety which may induce specific local 3D-structures according to its *E*- or *Z*-configuration.[184–186]

**Scheme 38** Chemical Structures of Cyclotheonamides A and B and (2*E*)-4-Amino-3-(4-hydroxybenzyl)but-2-enoic Acid (*E*-Vinylogous Tyr)[178,179]




**118** $R^1$ = CHO, Ac




**119**

The synthesis of a (but-2-enamide) peptide [(vinylogous amide) peptide] calls for that of the constitutive 4-aminobut-2-enoic acids with the correct configurations at the α-carbon and double bond. In all of the cases, the 4-aminobut-2-enoic acids are obtained by applying Wittig or Wittig–Horner reactions[187] to the corresponding α-aminoaldehyde (Scheme 39). As the α-aminoaldehydes are available from α-amino acids with high chemical and stereochemical purity,[188] they are easily applicable to the synthesis of 4-aminobut-2-enoic acids. Using the $Ph_3P{=}C(R^1)CO_2R^2$ ylides, the Wittig reaction is known to give almost

*for references see p 453*

exclusively *E*-isomers. However, using phosphonates of the general formula $(R^1O)_2P(O)CH(R^2)CO_2R^3$, it may give in high stereoselective excess the *E*- or the *Z*-isomer derivatives,[189,190] where the *E*/*Z* ratio strongly depends on the experimental conditions (Table 2).

**Scheme 39** Stereoselective Synthesis of 4-Aminobut-2-enoate Esters[190]

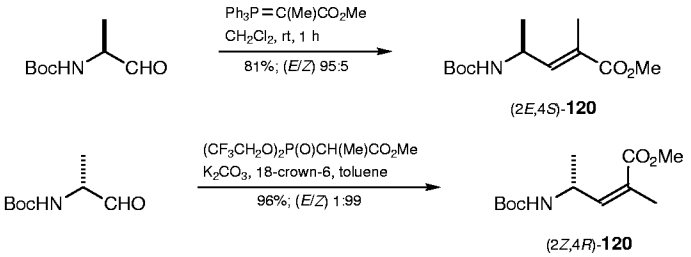
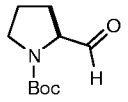
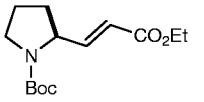
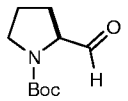
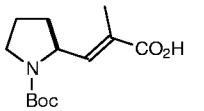
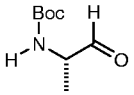
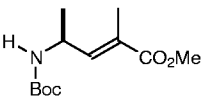
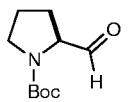
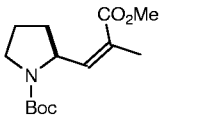
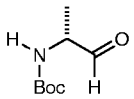
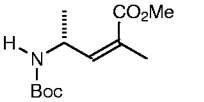
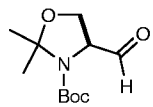
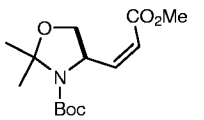



**Table 2** Wittig and Wittig–Horner Methods for the Preparation of 4-Aminobut-2-enoic Acids or Esters

| Aldehyde | Product | Reagents and Conditions | Yield (%) | *E*/*Z* | Ref |
|---|---|---|---|---|---|
|  |  | $(iPrO)_2P(O)CH_2CO_2Et$, NaH, THF, 0 °C | 70 | 99:1 | [175] |
|  |  | $(EtO)_2P(O)CH(Me)CO_2H$, BuLi, THF, −60 °C | 80 | 80:20 | [189] |
|  |  | $Ph_3P{=}C(Me)CO_2Me$, $CH_2Cl_2$, rt | 81 | 95:5 | [190] |
|  |  | $(CF_3CH_2O)_2P(O)CH(Me)CO_2Et$, KH, THF, −78 °C | 79 | 25:75 | [189] |
|  |  | $(CF_3CH_2O)_2P(O)CH(Me)CO_2Me$, $K_2CO_3$, 18-crown-6, toluene, −12 °C | 96 | 1:99 | [190] |
|  |  | $(CF_3CH_2O)_2P(O)CH_2CO_2Me$, NaH, 18-crown-6, THF, −78 °C | 82 | 10:90 | [193] |

## 10.8.11.1 (*E*)-4-Aminobut-2-enoic Acids

Wittig and Wittig–Horner reactions under thermodynamic conditions give the *E*-isomer analogues of Ala,[176,184,189,190] Pro,[175,185,189,191] Leu,[176,189] and Phe,[176,184,192] and 2,2-dimethyl-4-carboxy-1,3-oxazolidine[193] (the Ser pseudoproline analogue)[194] with double bonds CH=CH or CH=CMe (Table 2). As for the functionalized α-amino acids, only the *E*-analogues of Ser,[193,195,196] Tyr,[180–182] Met,[197] and Gln[197] have been prepared to date. Wittig reaction applied to the dipeptide aldehyde Z-Xaa-Ser-H gives the *E*-isomer dipeptide analogues Z-Xaa-NH-CH($CH_2OH$)-CH=CH-$CO_2$tBu (Xaa = Ala, Val, Leu, and Phe).[198]

**Methyl (2*E*,4*S*)-4-(*tert*-Butoxycarbonylamino)-2-methylpent-2-enoate [(2*E*,4*S*)-120]:**[190]
Boc-L-Ala-H (0.87 g, 5.0 mmol) dissolved in $CH_2Cl_2$ (7 mL) was added to a soln of $Ph_3P{=}C(Me)CO_2Me$ (1.92 g, 5.5 mmol) in $CH_2Cl_2$ (10 mL) keeping the temperature at 20 °C (an exothermic reaction took place). After stirring for 1 h, the mixture was concentrated and then triturated with EtOAc/hexanes (30 mL, 1:5) to induce crystallization. The precipitate was filtered and rinsed with EtOAc/hexanes (1:5), the combined filtrate and rinses were concentrated, and the residue was purified by flash chromatography (EtOAc/hexanes 17–33%) to afford the crude product. Recrystallization from hexanes gave the title compound; yield: 0.985 g (81%); mp 79 °C.

## 10.8.11.2 (*Z*)-4-Aminobut-2-enoic Acids

The synthesis of (*Z*)-4-aminobut-2-enoic acids requires Wittig–Horner reaction under kinetic conditions. The yield is noticeably increased by using the Still–Gennari[199] reagent $(CF_3CH_2O)_2P(O)CH(R^1)CO_2R^2$ and strongly dissociated base systems with 18-crown-6 for cation complexation.[193] Only the *Z*-isomer analogues of Pro,[185,189] Ala,[189,190] Phe,[189,192] and Ser[177,193] with double bonds CH=CH or CH=CMe (Table 2) have been reported so far.

**Methyl (2*Z*,4*R*)-4-(*tert*-Butoxycarbonylamino)-2-methylpent-2-enoate [(2*Z*,4*R*)-120]:**[190]
$K_2CO_3$ (29.59 g, 0.214 mol) and 18-crown-6 (18.86 g, 71.4 mmol) were stirred at rt in toluene (100 mL) and MeCN (10 mL) for 1 h, and the mixture was cooled to −12 °C. To this soln Boc-D-Ala-H (6.18 g, 35.7 mmol) and $(CF_3CH_2O)_2P(O)CH(Me)CO_2Me$ (11.85 g, 35.7 mmol) in toluene (45 mL) were added. The mixture was stirred at −12 °C for 5 h and then quenched with 25% aq citric acid. The mixture was extracted with EtOAc (3 × 150 mL), the combined extracts were washed with brine (100 mL), dried ($MgSO_4$), and concentrated to afford the crude product as an oil. This was purified by flash chromatography (18% EtOAc/hexanes) to afford the title compound as a crystalline solid; yield: 8.37 g (96%); mp 49–50 °C; $[\alpha]_D^{25}$ −66.9 (*c* 1, MeOH).

## 10.8.11.3 (But-2-enamide) Peptide Synthesis

Incorporation of 4-aminobut-2-enoic acids in a peptide relies on the classical procedures for peptide synthesis. A critical point might be their possible racemization under basic conditions, due to the higher acidic character of the allylic α-proton compared with the homologous α-amino acids. Substantial epimerization has been observed when incorporating (*E*)-but-2-enamide analogues of Met and Gln, but the step responsible for epimerization is not specified.[197]

The first synthetic (but-2-enamide) peptides were cyclotheonamides A[180–182] and B[179,182,183] (**118**, $R^1$ = CHO, Ac, respectively) from **119**. The different routes for these syntheses have been reviewed by Wipf.[200] N-Coupling of (*E*)-**119** is carried out with BOP[179,180,182] or the OPfp activated ester,[181] while C-coupling is performed with OPfp ester/DMAP,[179] EDC/DMAP,[180,182] or $Ph_2P(O)Cl$ methods.[181] The (*E*)-but-2-enamide analogues of Ala, Val, and Phe are introduced with DCC/HOBt or BOP in various model peptides in order to determine their influence on the extended, helical, and folded structures.[184] Recently, stereoisomers containing the (*E*)- and (*Z*)-but-2-enamide analogues of Pro and Ala have been prepared with BOP,[186] and it has been shown that the *Z*-isomer induces a very stable β-turn-like conformation.[185]


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# 10.9 Synthesis of Peptides with Sulfur-Containing Amide Bond Replacements

A. F. Spatola, K. Darlak, and P. Romanovskis

Due to the good nucleophilic nature of sulfur, formation of new C–S bonds is relatively easy. This property has been used for intramolecular peptide cyclization[1] and was exploited for preparation of methylenethio ether surrogates, ψ[$CH_2$–S], initially as "peptide gap inhibitors" for blocking collagenase action.[2] These surrogates were later converted into sulfoxide and sulfone derivatives. This expansion of amide replacements compelled a more generalized nomenclature system for pseudopeptides (amides with one or more amide bond surrogates) and led directly to the "psi-bracket" convention.[3]

Numerous peptide and hormone analogues have been prepared as enzyme-resistant derivatives. More recently, the activating character of sulfur in the form of thioesters has been exploited to mediate ligation reactions for large protein synthesis[4,5] (see Vol. E 22a, Section 4.1.5). Additionally, dithioesters have proven useful to monitor enzyme–substrate interactions.[6] Other sulfur-containing derivatives such as the sulfinamides and sulfonamides are intriguing surrogates because of their apparent resemblance to the parent amide. But other, more stable versions, such as the methylene sulfonamides, lose the isosteric character usually prized in such replacements.[7] While there is still disagreement about whether the peptide backbone plays a functional role in enzyme or receptor interactions or whether it merely positions the side chains as the important interaction elements, there continues to be great interest in devising improved methods for their synthesis. In this chapter both solution and solid-phase based methods for the synthesis of sulfur-containing amide bond replacements are surveyed.

## 10.9.1 ψ[C(=S)–NH] (Thioamide) Peptides

Replacement of the amide bond (CONH) in the backbone of peptides is a common strategy to enhance stability or to affect the physical/chemical properties of the analogues. In principle, sulfur-containing amide surrogates that retain the adjacent amino acid α-carbon residues of the parent peptide could involve one atom (ψ[C(=S)–NH]), or more than one atom replacement (ψ[$CH_2$–S], ψ[$SO_2$–NH]), or a variety of longer surrogates (e.g., ψ[$CH_2$–S–$CH_2$]; ψ[$CH_2$–S–S–$CH_2$], etc.). However, this section will emphasize those replacements that tend to be isosteric with the original amide bond. The most subtle sulfur-containing peptide bond replacement involves the conversion of the amide C=O into a C=S.

While not necessarily protease resistant,[8] the thioamide replacement for the normal amide linkage is a synthetically accessible and conformationally conservative type of backbone substitution.

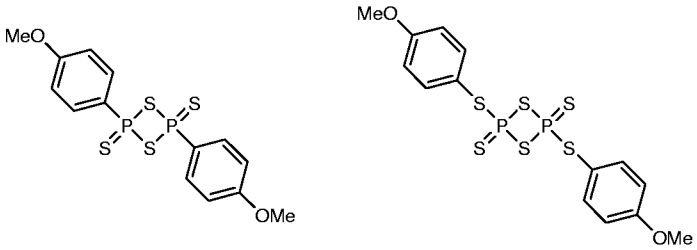
The steric bulk of the thioamide sulfur results in conformational changes that have been documented in cyclopentapeptide and cyclohexapeptide model systems (for more details see Vol. E 22b, Section 6.8.5.2.1). For example, in *c*[-Proψ[C(=S)–NH]Gly-Pro-Gly-D-Phe-], the first synthetic cyclic thiopeptide, an intramolecular γ-turn seen in the all-amide parent compound by NMR methods was perturbed by the putative interaction of sulfur with the adjacent Pro's β-protons.[9] On the other hand, in a cyclic hexapeptide, the enhanced H-bond donor capacity of the thioamide NH led to the formation of a relatively strong intramolecular H-bond stabilized β-turn, which was frame shifted compared to that found in its all-amide parent peptide.[10]

The more lipophilic thioamide unit is also characterized by the poor H-bond acceptor nature of the sulfur and by its larger covalent radius (1.04 vs 0.74 Å for O). The thioamide unit mainly adopts a *Z*-planar conformation, with a rotation energy barrier averaging about 23 vs 18 kcal·mol$^{-1}$ for amides.[11] A recent ab initio computational study[12] suggests that conformational perturbation in linear thioamides is more likely effected in the C-terminal side.

### 10.9.1.1 Selective Thionylation

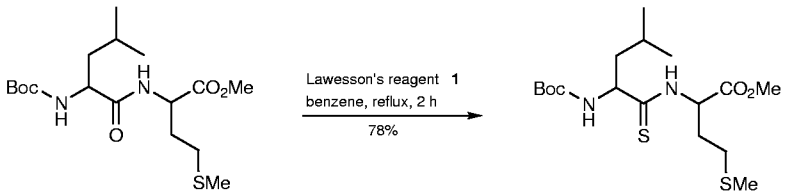
The use of Lawesson's reagent **1** (Scheme 1)[13,14] or several alternative thionylation reagents such as Yokoyama's reagent **2**[15] allows the selective replacement of oxygen by sulfur in amides even in the presence of ester, urethane, or ketone carbonyl functions; the latter are usually transformed to their sulfur counterparts only under more stringent (e.g., higher temperature) conditions.[14] An example of selective thionylation is given in Scheme 2 using Lawesson's reagent **1** and illustrates the selective conversion of a peptide amide into a thioamide in the presence of ester and urethane carbonyls.[16]

**Scheme 1** Structures of Lawesson's Reagent[14] and Yokoyama's Reagent[15]




**1** Lawesson's reagent **2** Yokoyama's reagent

**Scheme 2** Selective Thionylation of a Protected Dipeptide[16]




Side-chain amides (Asn, Gln) are reactive towards thionylating reagents, but in their absence appropriately protected peptides can be converted into thioamide peptides. Examples of short, singly thionylated peptides are given in Table 1.

Phosphorus pentasulfide ($P_2S_5$) is a powerful thionylation reagent that is unfortunately quite nonselective when applied to a peptide backbone.

**Table 1** Examples of Thioamide Pseudopeptides and Their Physical Properties[9,17]

| Protected Peptide | mp (°C) | $[\alpha]_D$ | Ref |
|---|---|---|---|
| Boc-Phe$\psi$[C(=S)–NH]Leu-OH | oil | –14.1 (*c* 1, MeOH) | [17] |
| H-Phe$\psi$[C(=S)–NH]Leu-OH·HCl | 166–168 | +6.6 (*c* 1, MeOH) | [17] |
| Boc-Tyr(Bzl)$\psi$[C(=S)–NH]Gly-Gly-Phe-Leu-OBzl | 184–188 | +12.2 (*c* 1, MeOH) | [17] |
| Boc-Tyr(Bzl)-Gly$\psi$[C(=S)–NH]Gly-Phe-Leu-OBzl | 164–167 (dec) | –9.3 | [17] |
| Boc-Tyr(Bzl)-Gly-Gly$\psi$[C(=S)–NH]Phe-Leu-OBzl | 121–123 | +3.5 | [17] |
| Boc-Tyr(Bzl)-Gly-Gly-Phe$\psi$[C(=S)–NH]Leu-OBzl | 172–174 (dec) | –4.5 (*c* 1, MeOH) | [17] |
| Boc-Pro$\psi$[C(=S)–NH]Gly-OEt | 105–106 | | [9] |
| Boc-Pro$\psi$[C(=S)–NH]Gly-OH | 133–134 | | [9] |
| Boc-Pro$\psi$[C(=S)–NH]Gly-Pro-Gly-OEt | 64–65 | | [9] |
| Pro$\psi$[C(=S)–NH]Gly-Pro-Gly-OEt·HCl | foam | | [9] |

Peptides with an internal $\psi$[C(=S)–NH] replacement, termed endothiopeptides, can be extended at the N-terminus rather easily, while C-terminal extension is more difficult due to the formation of the relatively unreactive thiazol-5(4*H*)-one that is prone to racemization.[18,19] However, with a C-terminal, achiral Gly[9] or Aib[20] extensions were successful.

*for references see p 489*

When more than one amide is present, such as in the tripeptide Boc-Gly-Phe-Leu-OEt, a mild reagent will convert the less-hindered Gly-Phe bond into the thioamide while leaving the Phe-Leu bond, as well as the ester and urethane carbonyls, intact at room temperature to furnish Boc-Gly$\psi$[C(=S)–NH]Phe-Leu-OEt. However, the attempted conversion of *c*-[-D-Lys-Gly-Phe-Leu-] into thioamide product failed using **1**.[9] Only with $P_2S_5$ did thioamide conversion take place, and multiple thionylation products were the result.

On the other hand, **1** was successfully used for the thionylation of the cyclic undecapeptide, cyclosporin A. By using 3–5 equivalents of **1** in the polar solvent DMPU at room temperature for 2–4 days four major products as well as the unreacted starting material were isolated.[21] Three monothio products and one dithionylated product were obtained, with regioselectivity favoring the sterically less hindered non-N-methylated positions. Greater selectivity was seen when the thionylation was carried out in the nonpolar solvent THF, where positions 4–5 and 7–8 were the only ones modified. Application of **1** or alternative reagents, such as the milder Yokoyama equivalent **2**[15] to longer sequences[22] or some cyclic peptides[23] can predictably lead to multiple thionylation products.

An alternative solid-phase synthesis of peptide thioamides that uses methyl dithioesters to introduce the sulfur involves addition of DMAP and cesium trifluoroacetate.[24] These additives greatly accelerate the thioamide-forming reaction and give low levels of racemization according to Marfey's test analysis[25] of an Ala$\psi$[C(=S)–NH]Val product.

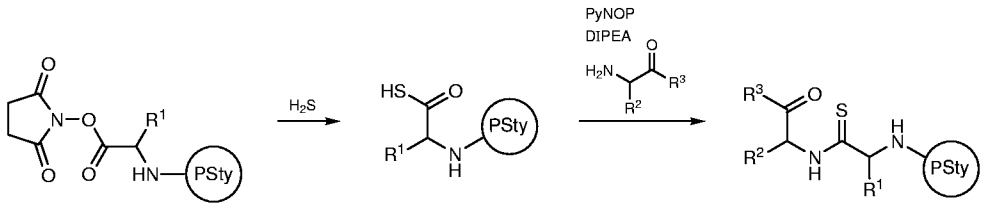
**Synthesis of Boc-Leu$\psi$[C(=S)–NH]Met-OMe:**[16]
A suspension of Boc-Leu-Met-OMe (150 mg, 0.398 mmol) and **1** (88.7 mg, 0.219 mmol) in dry benzene (1 mL) (**CAUTION:** *carcinogen!*) was refluxed under $N_2$ for 2 h. The solvent was removed and the residue was purified by chromatography (silica gel, 15% EtOH/hexane) to give the product as a yellow oil; yield: 122 mg (78%).

**Thionation of Cyclosporin A:**[21]
To a soln of cyclosporin A (6.0 g, 5.0 mmol) in DMPU (200 mL), **1** (6.0 g, 14.8 mmol) was added. The resulting soln was stirred at rt for 4 d. The mixture was diluted with $Et_2O$ (1200 mL), washed with $H_2O$ (4 × 250 mL), brine (2 × 150 mL), and the combined aqueous phase extracted with $Et_2O$ (2 × 100 mL). The combined organic phase was dried ($Na_2SO_4$), the solvent removed, and the crude product was purified by chromatography (silica gel, 1% MeOH/$Et_2O$). The partially purified fractions were rechromatographed (silica gel, 33% $H_2O$-sat. EtOAc/$Et_2O$), then subjected to repeated HPLC (RP8 column, 15–19% $H_2O$/MeOH), and then to a final chromatography (silica gel, MeOH/$Et_2O$); yield: 106 mg (1.7%) of [$^1\psi^2$, C(=S)–NH; $^4\psi^5$, C(=S)–NH]-CsA, 733 mg (12%) of [$^1\psi^2$, C(=S)–NH]-CsA, 135 mg (2.2%) of [$^7\psi^8$, C(=S)–NH]-CsA, 170 mg (2.8%) of [$^4\psi^5$, C(=S)–NH]-CsA, and 2.814 g (46.9%) of unreacted CsA were obtained. The products were analytically characterized.

### 10.9.1.2 Thioacylation with N-Protected Amino Monothioacids

An alternative approach that is applicable to the solid phase involves the preparation of an N-protected amino monothioacid with rapid coupling to a solution or resin-bound amino group (Scheme 3).[22] This procedure is compatible with Fmoc protection. A significant amount of carboxamide byproduct can be separated chromatographically from the less polar thioamide product. Thioamide peptides have been prepared using alkyl dithioesters, but these are prone to racemization[16] when used for peptide elongations. The synthesis of methyl dithioesters from $H_2S$ treatment of a precursor amino acid piperidide is described in Section 10.9.6. These derivatives have been used in the synthesis of analogues of Leu-enkephalin.[26]

**Scheme 3** Thioacylation Approach to Thioamide Pseudopeptides[22]




**Synthesis of a ψ[C(=S)–NH] Tripeptide Analogue of Thyrotropin Releasing Hormone {pGluψ[C(=S)-NH]His-Pro-$NH_2$}:**[22]

**pGlu-SH·DCHA:**[22]

DCHA (1.99 g, 10.0 mmol) in THF (40 mL) was saturated with $H_2S$ gas at 0 °C. A precipitate was observed (DCHA·$H_2S$). pGlu-OPfp (2.95 g, 10.0 mmol) was then added to the mixture, resulting in immediate dissolution of the precipitate. However, within 2 min a new precipitate was formed (pGlu-SH·DCHA). The solvent was removed and $Et_2O$ (15 mL) was added to the residue. The resulting solid was collected and the product washed with more $Et_2O$ (2 × 15 mL); yield: 3.01 g (92%); mp 164–168 °C; FAB-MS *m/z*: 144 $[M+H]^+$.

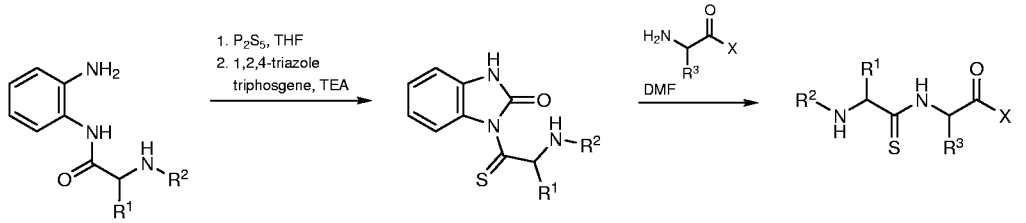
**pGluψ[C(=S)–NH]His-Pro-$NH_2$·TFA:**[22]

Tentagel resin (2.0 g, 0.48 mmol $NH_2$) in a plastic syringe fitted with a Teflon filter was covered with DMF and treated with the Rink linker (777 mg, 3 equiv), TBTU (462 mg, 3 equiv), and DIPEA (493 µL, 6 equiv). After 2 h the resin was washed with DMF (5 ×) and treated with 20% piperidine in DMF (3 + 7 min). The resin was washed with more DMF (5 ×), then mixed with Fmoc-Pro-OH (648 mg, 4 equiv), TBTU (617 mg, 4 equiv), and DIPEA (575 µL, 7 equiv) dissolved in DMF. The peptide-resin was stirred for 4 h and then deprotected as described above. The peptide-resin was then coupled with Fmoc-His(Trt)-OH (1.190 g, 4 equiv) by TBTU/DIPEA activation (same amounts as above). After 4 h the resin was deprotected as above, then washed thoroughly with DMF (5 ×) and $CHCl_3$ (3 ×), and purged with $N_2$. pGlu-SH·DCHA (157 mg, 1 equiv) and PyNOP (327 mg, 1.2 equiv) were added to the peptide-resin in solid form, to which DIPEA (82 µL, 1 equiv) in $CHCl_3$ (enough to cover the resin) was then added. The peptide-resin was left in the dark overnight and then washed with $CHCl_3$ (3 ×), after which the thioacylation reaction was repeated. Again, the peptide-resin was left overnight then washed with DMF (3 ×), MeOH (3 ×), $Et_2O$ (3 ×), and dried in vacuo. Finally, the peptide-resin was treated with a TFA/$H_2O$/EDT mixture (90:5:5, 12 mL) for 30 min. The volatiles were removed in vacuo and the resulting material was dissolved in $H_2O$ and washed with $Et_2O$ (4 ×). The aqueous soln was lyophilized to give the title compound; yield: 106 mg (61%, based on thiopeptide mass); HPLC [linear gradient from 100% of 0.1% TFA in $H_2O$ (soln A) to 70% of 0.1% TFA/9.9% $H_2O$ in MeCN (soln B), in 30 min] showed two major peaks, at 10.24 and 12.86 min, respectively. Comparison with a standard showed the first peak to be the carboxamide tripeptide. The second peak was identified as a thiopeptide on the basis of its UV spectrum ($\lambda_{max}$ 265 nm). Both compounds were collected by preparative HPLC (100% soln A for 3 min, then to 15% soln B in 57 min); MS (thiopeptide) *m/z*: 379 $[M+H]^+$.

## 10.9.1.3 Thioacylation with a Benzimidazolone Derivative

The use of benzimidazolone derivatives **4**, prepared from the anilides **3**, for the thioacylation of peptides (Scheme 4) have also been reported,[27] and three thioamide analogues of tuftsin were prepared. Due to the acid lability of thioamides, Boc deprotection with $SnCl_4$ represents a mild alternative for N-terminal extension.[28]

**Scheme 4** Synthesis of a Thioamide Dipeptide Using a Thiobenzimidazolone[27]




**3** **4**

*for references see p 489*

Another promising approach with an even more reactive thioacylating agent was recently described using the 6-nitrobenzotriazole group.[29] A thioamide derivative of an amino acid 2-amino-5-nitroanilide was prepared using $P_4S_{10}$, followed by intramolecular diazonium cyclization to yield the activated benzimidazolone ester, suitable for acylation. This method was reported to give high yields in most cases and to provide enantiomerically pure products, even with hindered amino acids such as valine.

The solution and solid-phase synthesis of thioamide peptides via thioacylation has been reviewed.[30] Among biologically active thioamide peptides that have been prepared are cyclic enkephalins,[21] analogues of the tripeptide thyrotropin releasing hormone (pGlu-His-Pro-$NH_2$TRH),[23,31] a chemotactic peptide (HCO-Met-Leu-Phe-OR),[32] bombesin,[16] tuftsin,[27] thymopentin,[33] three carboxypeptidase dipeptide substrates,[8] and an oxytocin analogue.[33] A Valψ[C(=S)–NH]Aib-containing decapeptide was prepared by solid-phase methods and shown to be helical by X-ray crystallography.[34] Comparative studies revealed that a cadmium version of carboxypeptidase cleaved a thioamide bond more rapidly than its amide counterpart, revealing the role of the thiophilic metal in thioamide bond hydrolysis.[8]

**Boc-Ala-2-aminoanilide:**[27]
To a mixture of Boc-Ala-OH (2.08 g, 11 mmol), 1,2-phenylenediamine (1 g, 9.2 mmol), and TEA (1.5 mL, 11 mmol) in $CH_2Cl_2$ (50 mL) was added EDC · HCl at 0 °C. After stirring at this temperature for 2 h and then at rt overnight, 5% HCl (25 mL) was added and the mixture was washed with brine, 5% $NaHCO_3$, and brine, and dried ($Na_2SO_4$). Removal of the solvent and recrystallization ($CH_2Cl_2$/hexane) of the residue afforded pure product; yield: 2.4 g (94%); mp 122–124 °C.

**Boc-Ala-2-aminothioanilide:**[27]
Under a flow of argon, $P_2S_5$ (0.75 g, 1.68 mmol) was mixed with $Na_2CO_3$ (0.18 g, 1.7 mmol) in dry THF (100 mL). The mixture was stirred for 1 h at 25 °C and then cooled to 0 °C. To the clear soln was added Boc-Ala-2-aminoanilide (0.5 g, 1.79 mmol), and the reaction was kept at 0 °C overnight. An aqueous soln of sodium tribasic phosphate was then added (12%, 7 mL), followed by EtOAc (20 mL) and heptane (20 mL). The organic layer was separated and washed with brine (2 × 30 mL). The solvent was then removed and the residue was purified by chromatography (silica gel, EtOAc/hexane 1:2) to give pure Boc-Ala-2-aminothioanilide as a pale yellow solid; yield: 0.28 g (53%); mp 124–126 °C.

**Boc-Ala-1*H*-benzimidazol-2(3*H*)-one:**[27]
A soln of 1,2,4-triazole (0.25 g, 3.62 mmol) in dry THF (40 mL) was mixed with triphosgene (0.18 g, 0.61 mmol) and TEA (0.52 mL, 3.65 mmol) under a flow of argon. The mixture was refluxed for 2 h and then cooled to rt. To this soln was added Boc-Ala-2-aminoanilide (0.50 g, 1.81 mmol) and the reaction was stirred overnight at 25 °C. Insoluble products were removed by filtration and the solvent was removed under reduced pressure. The residue was chromatographed (silica gel, EtOAc/hexane 1:1) to give the title compound; yield: 0.34 g (62%).

**Boc-Alaψ[C(=S)–NH]-1*H*-benzimidazol-2(3*H*)-one:**[27]
This compound was prepared according to the procedure described above for the 1*H*-benzimidazol-2(3*H*)-one using Boc-Ala-2-aminothioanilide (0.30 g, 0.99 mmol); yield: 0.16 g (51%); mp 101–103 °C.

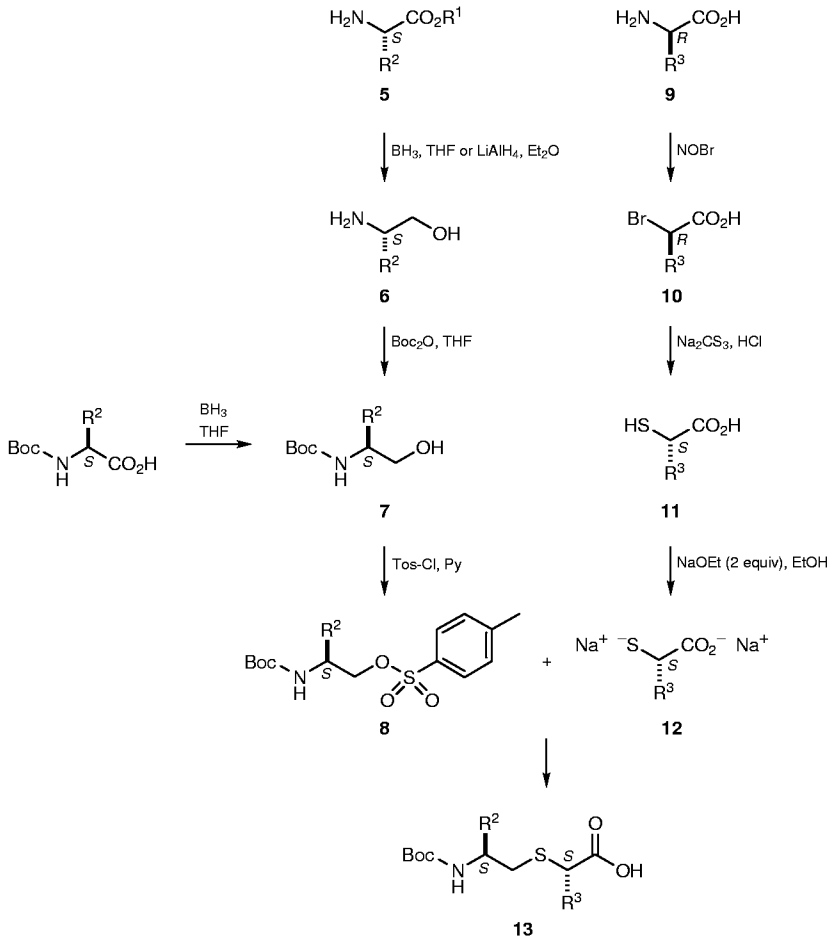
**Coupling of Thioacylating Reagents with Amino Acids or Peptides:**
Free amino acid benzyl ester (2 mmol) was dissolved in dry DMF (0.5 mL) at 0 °C under a stream of $N_2$. To this soln was added thioacylating reagent (2.2 mmol) in portions over a period of 20 min. The resulting mixture was stirred at 0 °C for 2 h and then at rt overnight. Insoluble products were removed by filtration and the solvent was removed under reduced pressure. The remaining residue was dissolved in EtOAc (20 mL) and washed successively with 5% $NaHCO_3$ (25 mL), brine (25 mL), 5% citric acid (25 mL), and brine (25 mL). The organic layer was dried ($MgSO_4$), concentrated and the residue chromatographed (silica gel, EtOAc/hexane 2:3) for di- and tripeptides or (alumina, EtOAc or 6% MeOH/$CHCl_3$) for tetra- and pentapeptides. This procedure gives pure thioamide derivatives in 72–78% yield.

## 10.9.2 ψ[$CH_2$–S] (Methylenethio) Peptides

The formal replacement of the amide by the -$CH_2$–S- group to produce a methylenethio surrogate was one of the first applications of amide bond replacements designed for enzyme inhibition[2] and later as enzyme-resistant peptide analogues.[35]

In principle, the amide group can be replaced by either -$CH_2$–S- or -S–$CH_2$- (thiomethylene). However, since the stability of α-amino thiols is suspect, most researchers have investigated analogues with the -$CH_2$–S- group. Compounds with ψ[$CH_2$–S] replacement can be conveniently prepared by a convergent synthesis from chiral amino acids as shown in Scheme 5. Borane reduction converts an amino acid **5**, or the corresponding methyl ester, into an amino alcohol **6** which is then protected as its Boc derivative **7** using $Boc_2O$. The alcoholic group of **7** is transformed into the displaceable tosylate by reaction with freshly recrystallized Tos-Cl and pyridine to give **8**. The C-terminal component **12** is obtained in three steps from the D-amino acid **9**. Nitrosyl bromide treatment of **9** yields the bromo acid **10** with inverted stereochemistry which is then transformed to the thioacid **11** and then to the disodium salt **12**. Reaction of **8** with **12** yields the *N*-Boc pseudodipeptides **13**. Pseudodipeptides and Boc-protected pseudodipeptides are usually obtained as high melting solids (Table 2).[36]

**Scheme 5** Synthesis of a ψ[$CH_2$–S] Pseudopeptide with the Natural L-L-Amino Acid Configuration Assuming the Stereochemical Transformations as Indicated[36]




$R^1$ = H, Me; $R^2$ = $R^3$ = side chains

*for references see p 489*

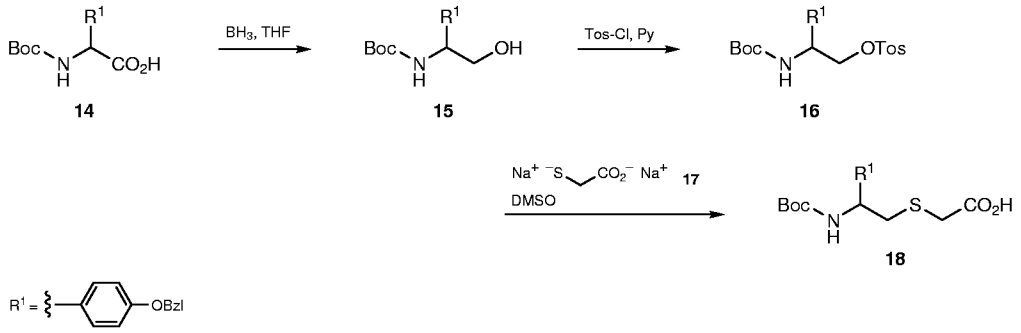
**Table 2** Methylenesulfanyl Peptides[2,36–38]

| | Yield (%) | mp (°C) | $[\alpha]^{20}_D$ | Ref |
|---|---|---|---|---|
| H-Gly$\psi$[$CH_2$–S]Leu-OH | 43 | 207–209 | –29.3 (*c* 1, $H_2O$) | [2] |
| H-Gly$\psi$[$CH_2$–S]D-Leu-OH | 45 | 207–211 | +22.9 (*c* 1, $H_2O$) | [2] |
| H-Phe$\psi$[$CH_2$–S]Leu-OH | – | 87–89 | –4.26 (*c* 2, MeOH) | [36] |
| H-Phe$\psi$[$CH_2$–S]D-Leu-OH | – | 196–198 | +46.7 (*c* 4, MeOH) | [36] |
| Boc-Phe$\psi$[$CH_2$–S]Leu-OH | 30 | 110–111 | –52.5 (*c* 0.88, MeOH) | [37] |
| Boc-Phe$\psi$[$CH_2$–S]D-Leu-OH | 48.5 | 102–103 | +69.1 (*c* 1, DMF) | [37] |
| H-Ser$\psi$[$CH_2$–S]Leu-OH | – | 180–181 | –5.4 (*c* 1.5, $H_2O$) | [38] |
| H-D-Ser$\psi$[$CH_2$–S]Leu-OH | 70 | 179–180 | –53.7 (*c* 2.5, $H_2O$) | [38] |
| H-D-Ser$\psi$[$CH_2$–S]D-Phe-OH | – | 173–174 | –13.3 (*c* 2, $H_2O$) | [38] |
| Boc-D-Ser$\psi$[$CH_2$–S]Leu-OH | – | 149–150 | –88.6 (*c* 2, MeOH) | [38] |
| Boc-D-Ser$\psi$[$CH_2$–S]D-Phe-OH | – | 120–121 | +39.9 (*c* 2, MeOH) | [38] |

## 10.9.2.1 Synthesis of $\psi$[$CH_2$–S]Gly Peptides

If the C-terminal component of a $\psi$[$CH_2$–S] pseudodipeptide derives from Gly, such compounds can be prepared by the simpler approach shown in Scheme 6. Borane treatment reduces the amino acid **14** to the amino alcohol **15** which is then further converted into the tosylate **16** with retention of the chirality in the N-terminal component[36] as described previously. Reaction of **16** with the disodium salt **17** of the commercially available sulfenylacetic acid yields the pseudodipeptide **18**.

**Scheme 6** Synthetic Route to the Pseudodipeptide Boc-Tyr(Bzl)$\psi$[$CH_2$–S]Gly-OH[39,40]




**Synthesis of a $\psi$[$CH_2$–S] Pseudodipeptide with a C-Terminal Gly (18):**[39,40]

***O*-Benzyl-(*N*-*tert*-butoxycarbonyl)tyrosinol (15):**[40]

$BH_3$/THF (30 mL, 30 mmol) was transferred from a sealed bottle to a round-bottomed flask maintained in an acetone–dry ice bath. To the soln was added Boc-Tyr(Bzl)-OH (**14**; 3.71 g, 10 mmol) in THF dropwise for 30 min. The mixture was stirred at –40 °C for 1 h, then at rt for 2 h. The reaction was quenched by the slow addition of ice-cooled water. The soln was concentrated and the residue extracted with EtOAc. The combined organic extracts were washed with $H_2O$, sat. $NaHCO_3$, and sat. NaCl. The EtOAc layer was dried ($MgSO_4$), and the solvent removed under reduced pressure to afford the title compound **15**; yield: 2.96 g (82.8%); mp 100–103 °C.

***O*-Benzyl-(*N*-*tert*-butoxycarbonyl)tyrosinyl Tosyl Ether (16):**[40]

To a soln of **15** (2.9 g, 8.1 mmol) in $CHCl_3$ (20 mL) and pyridine (6.56 mL, 81 mmol) maintained at 0 °C was added Tos-Cl (4.63 g, 24.3 mmol) in $CHCl_3$ over 15 min. On addition of the Tos-Cl soln the color of the mixture changed to yellow, and eventually to red. The mixture was stirred at rt for 1.5 h. The soln was concentrated to dryness under reduced pressure and the residue was extracted with a mixture of EtOAc and $H_2O$. The combined organic extracts were washed with cold $H_2O$, 1 M HCl, $H_2O$, 5% $NaHCO_3$, $H_2O$, and sat. NaCl. The EtOAc layer was dried ($MgSO_4$) and concentrated to afford the crude product (4.15 g) which on column chromatography (silica gel, 100% hexane to 50% hexane/acetone) gave **16**; yield: 3.10 g (74.8%); mp 155–158 °C.

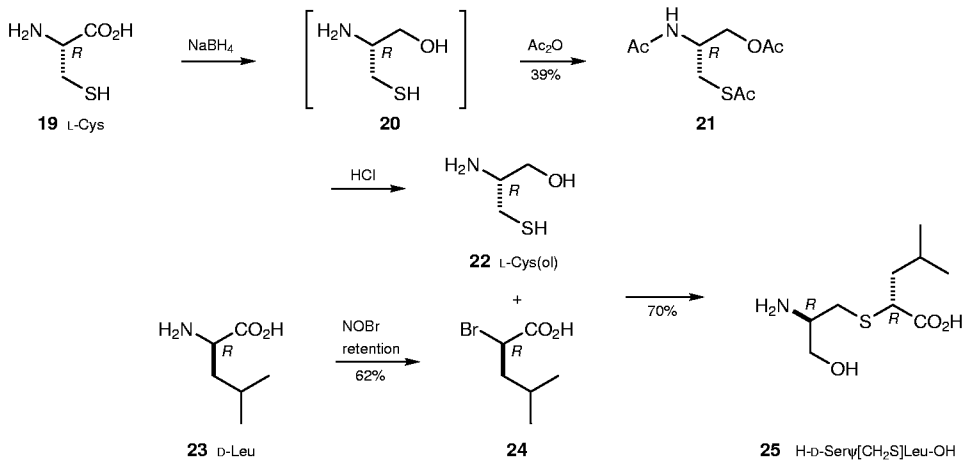
**Boc-Tyr(Bzl)ψ[$CH_2$–S]Gly-OH (18):**[40]
Freshly shaved Na metal (207 mg, 9 mmol) was dissolved in anhyd EtOH (10 mL). To this soln was added freshly distilled $HSCH_2CO_2H$ (0.313 mL, 4.5 mmol). A white solid immediately precipitated out of soln. The suspension was stirred for 10 min, then concentrated under reduced pressure. The residue was dissolved in benzene (**CAUTION:** *carcinogen!*) and the solvent removed under reduced pressure. This process was repeated several times until a white solid was obtained. The solid **17** was further dried and dissolved in DMSO (2–3 mL) under $N_2$. To this soln was added **16** (2.04 g, 4 mmol). The mixture was stirred at rt under $N_2$ for 4 h then cold $H_2O$ (15 mL) was added. The aqueous soln was washed with $Et_2O$, brought to pH 3 by addition of 1 M HCl and extracted with EtOAc (3 × 50 mL). The combined organic extracts were washed with 1 M HCl, $H_2O$, and sat. NaCl. The EtOAc layer was dried ($MgSO_4$), and concentrated to afford of the title product **18**; yield: 1.43 g (83%); mp 54–56 °C.

## 10.9.2.2 Synthesis of ψ[$CH_2$–S]Xaa (Xaa ≠ Gly) Peptides

For more complex C-terminal components, chirality can often be controlled in a predictable fashion. Thus, nitrosyl bromide treatment of an amino acid yields the α-bromo acid with retention of chirality in many cases.[41] This is due to the carboxylate group and thus can be expected to be quite dependent on specific reaction conditions and also be affected by the nature of the amino acid side chain; for example, complete racemization is observed with histidine. The second step, replacement of the bromine by thiol, is carried out using sodium trithiocarbonate with concomitant decomposition of the thiocarbonate intermediate by the accompanying acid.[42] Since this step involves classic configurational inversion, the overall stereochemistry for the C-terminal component usually results in inversion of the initial chirality (Scheme 5). This outcome can be monitored in the subsequent reaction step where the tosylate is displaced by the thiol of the C-terminal component. If both components contain a chiral center, then the resulting molecule is a diastereomer. Replacement of either component with its enantiomer leads to formation of a second diastereomer with the possibility of quantifying loss of chirality throughout the transformations. The resulting Boc-protected pseudodipeptides (δ-amino acid derivatives) can be used to prepare pseudopeptides by appropriate solution or solid-phase methods.

In some cases the N-terminal component can be converted into a thiol [$NH_2$–CH(R)–$CH_2SH$] which is then used to displace the halogen of an α-bromo acid. This approach was used to prepare Boc-Ser ψ[$CH_2$–S]Leu analogues by a role-reversal of the side-chain/acid group of Cys, once the amino acid was converted into an amino alcohol (Scheme 7).[38]

**Scheme 7** Synthetic Route to the Pseudodipeptide H-D-Serψ[$CH_2$–S]-Leu-OH[38]




*for references see p 489*

In this approach cysteine (**19**) was reduced to Cys(ol) (**20**) which was isolated as the triacetyl derivative **21**. This compound was hydrolyzed with hydrochloric acid, the resulting hydrochloride **22** treated with 3 equivalents of NaOEt followed by addition of the α-bromo acid **24** derived from D-leucine (**23**). The resulting product **25** can then be *N*-Boc-protected to yield the pseudodipeptide in a form suitable for solid-phase chain elongation.

**Synthesis of H-D-Serψ[$CH_2$-S]Leu-OH (25) from Cys:**[38]

**(2*R*)-2-Bromo-4-methylpentanoic Acid (24):**[38]
Treatment of D-Leu (16.4 g, 125 mmol) with nitrosyl bromide,[40] followed by distillation of the crude product, gave the title compound **24**; yield: 15.1 g (62%); bp 97–98 °C/0.22 Torr; $[\alpha]_D$ +38.2 (*c* 2, MeOH).

***N,O,S*-Triacetyl-Cys(ol) (21):**[38]
Cys (10.0 g, 83 mmol) was reduced by the procedure of Anhoury et al.[43] The crude Cys(ol) (**20**) obtained was acetylated with $Ac_2O$/NaAc to give the title compound **21**; yield: 7.51 g (39%); mp 95–97 °C; $[\alpha]^{25}{}_D$ −42.6 (*c* 2, $H_2O$).

**H-D-Serψ[$CH_2$–S]Leu-OH (25):**[38]
*N,O,S*-Triacetyl-Cys(ol) (**21**; 1.50 g, 6.44 mmol) was hydrolyzed with 1.7 M HCl (60 mL) at 90 °C under an atmosphere of $N_2$ for 72 h. The solvent was removed under reduced pressure and residual $H_2O$ was removed by azeotropic distillation with absolute EtOH. The crude H-L-Cys(ol)·HCl was dissolved in absolute EtOH (20 mL) containing NaOEt (17.34 mmol). To this soln was added (*R*)-2-bromo-4-methylpentanoic acid (**24**; 0.38 g, 4.46 mmol) in absolute EtOH (5 mL). The mixture was stirred for 72 h at rt followed by removal of the solvent. The residue was dissolved in $H_2O$, acidified with 6 M HCl, and extracted with $Et_2O$. The aqueous portion was neutralized with 2 M NaOH and diluted to 150 mL with deionized $H_2O$. This soln was desalted on a column of Dowex 1 X 8 (2 × 20 cm), according to the procedure of Dréze et al.[44] The ninhydrin-positive fractions were pooled and the solvent was removed under reduced pressure. The residue was recrystallized (47.5% EtOH) to give **25**; yield: 0.70 g (70%); mp 179–180 °C; $[\alpha]^{25}{}_D$ −53.7 (*c* 2.5, $H_2O$).

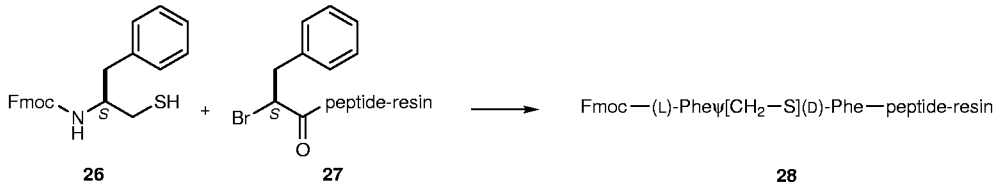
**Boc-Pheψ[$CH_2$–S]Leu-OH:**[37]
2(*S*)-4-Methyl-2-sulfanylpentanoic acid [bp 92–93 °C/0.75 Torr; $[\alpha]_D^{25}$ −23.2 (*c* 1, MeOH)], was prepared from the α-bromo acid according to the literature procedure.[44] The α-bromo acid was in turn synthesized from D-Leu according to the procedure of Levene et al.[45] 2(*S*)-4-Methyl-2-sulfanylpentanoic acid (0.799 g, 5.40 mmol) was added to an anhyd EtOH soln (30 mL) containing freshly shaved Na metal (0.11 g). This soln was concentrated under reduced pressure and repeatedly treated with benzene (**CAUTION:** *carcinogen!*) and concentrated under reduced pressure until dry. The dried disodium salt was immediately dissolved in DMSO (1 mL, dried over molecular sieves) under argon. To this soln was added a soln of 2-(*tert*-butoxycarbonyl)amino-3-phenylpropanyl-4-toluenesulfonate (2.43 g, 5.97 mmol) in DMSO (2 mL). The mixture was allowed to stir for 2 h and then distilled $H_2O$ (150 mL) was added. The aqueous soln was acidified to pH 2 with 1 M $KHSO_4$ on an ice bath with NaCl and EtOAc added. The acidified soln was extracted with EtOAc (3 × 40 mL), the organic extracts were pooled and washed with 1 M HCl (2 × 40 mL), $H_2O$, and sat. NaCl (2 × 40 mL). The EtOAc extract was dried ($Na_2SO_4$), filtered, and concentrated to an oil that, following overnight drying, weighed 950 mg. The DCHA salt of this material was prepared in EtOAc using 1.2 equivalents of DCHA in EtOAc (2 mL). The precipitated solid (950 mg) was collected and dried. The *N*-Boc-protected pseudopeptide was freed from the salt by stirring with 0.2 M $H_2SO_4$ with $Et_2O$ on an ice bath for 30 min. The $Et_2O$ extracts were washed with $H_2O$, sat. NaCl (2 × 40 mL), dried ($Na_2SO_4$), and concentrated yielding a solid which was recrystallized (EtOAc/hexane) and dried in vacuo (over $P_2O_5$) to afford the title compound; yield: 600 mg (30% ); mp 110–111 °C; $[\alpha]^{25}{}_D$ −52.5 (*c* 0.9, MeOH).

**Boc-Pheψ[$CH_2$–S]D-Leu-OH:**[37]
Equivalent aliquots of NaOEt and 2(*S*)-4-methyl-2-sulfanylpentanoic acid (prepared from D-Leu) were combined as described for Boc-Pheψ[$CH_2$-S]Leu-OH to give the disodium salt of 2(*S*)-4-methyl-2-sulfanylpentanoic acid (1.04 g, 5.36 mmol) after drying by repeated removal of benzene (**CAUTION:** *carcinogen!*) under reduced pressure. The dried product was dissolved in dry DMSO (1 mL) and allowed to react with *N*-Boc-phenylalaninyl tosyl ether (2.4 g, 5.9 mmol). The soln was stirred under argon overnight then distilled $H_2O$ (100 mL) was added to the mixture. The aqueous mixture was back-extracted with $Et_2O$ (2 × 30 mL) to remove excess *N*-Boc-phenylalaninyl tosyl ether. The aqueous extract was then acidified to pH 2 with 1 M $KHSO_4$ on an ice bath with NaCl and EtOAc added. The acidified soln was extracted with EtOAc (6 × 50 mL), the combined organic extracts washed with 1 M HCl (2 × 40 mL), $H_2O$ (2 × 40 mL) and sat. NaCl (2 × 40 mL). The EtOAc extract was dried ($Na_2SO_4$), concentrated to an oil that after storage overnight in vacuo afforded the title compound; yield: 1.0 g (48.5%); mp 102–103 °C; $[\alpha]^{25}{}_D$ +69.1 (*c* 1, DMF).

### 10.9.2.3 Solid-Phase Synthesis of ψ[$CH_2$–S] Peptides

A report involving the solid-phase preparation of Fmoc-protected ψ[$CH_2$–S] pseudodipeptides revealed that the expected stereochemical patterns did not hold.[47] It is known that the conversion of an α-bromo acid into the α-thioacid does not always occur with quantitative inversion of chirality, since it is side-chain dependent.[46] With side chains such as benzyl (Phe) or butyl (Leu, Ile), exceptions are not expected. But when the Fmoc-protected amino thiol **26** was condensed to a support-bound bromo acid **27**, as shown in Scheme 8, the (*S*,*R*)-product **28** was obtained instead of the expected (*S*,*S*)-isomer.

**Scheme 8** Unexpected Stereochemical Outcome on Condensation of an Fmoc-Protected Aminothiol with a Resin-Bound α-Bromo Acid[47]




**26** **27** **28**

Another potential problem involving stereochemistry concerns the epimerization of the C-terminal chiral center of the pseudodipeptide unit, which is observed during the resin attachment step when using the cesium salt derivative.[36]

### Biological Activities

While degradation studies have confirmed the improved stability of ψ[$CH_2$–S] pseudopeptides,[48] linear analogues with this modification have generally proven less active than their parent compounds. This property is attributable to the increased flexibility of the unit and its tendency to induce bent structures as seen in an X-ray structure of the dipeptide Boc-Alaψ[$CH_2$–S]Phe-OH.[49]

Peptide cyclization significantly reduces this flexibility, as shown by model studies with cyclic pseudodipeptides. When incorporated into biologically active cyclic peptides, potencies are generally retained or even enhanced. But flexibility is still a feature since pseudopeptide analogues of the μ-receptor-selective opioid parent peptide, Tyr-*c*[-D-Lys-Gly-Phe-Leu-], were potent but nonselective with respect to activity toward μ- and δ-receptors.[50]

Other ψ[$CH_2$–S] peptides that have been reported include both agonists and antagonists of LHRH,[35,51] a series of linear enkephalins,[39] a cyclic mini-somatostatin,[52] and cyclic RGD cell-adhesion antagonists.[53]

In some cases, the ψ[$CH_2$–S] segment has contributed to or was compatible with antagonist activity, as seen in the opioid δ-receptor-selective linear pseudopeptide antagonist H-Tyr(Al)-Glyψ[$CH_2$–S]Gly-Phe-Leu-OAl[54] and in a linear bombesin antagonist.[55]

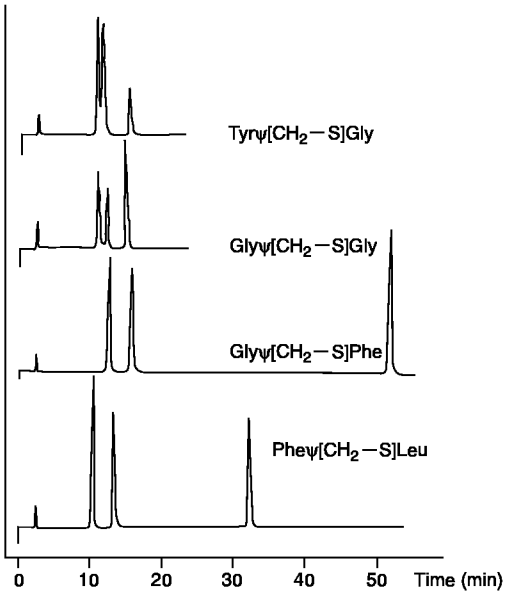
### Characterization

Each methylenethio unit adds three daltons to the peptide weight compared to the parent amide. Mass spectrometry can both confirm the expected mass as well as demonstrate preferred cleavages at both C–S bonds. In a linear pseudopeptide N-terminal or C-terminal fragments allow partial characterization of the location and sequence of the replacement by identification of the fragments using positive and negative ionization, respectively.[56]

*for references see p 489*

As shown in Figure 1 by RP-HPLC, the ψ[$CH_2$–S] unit is considerably more hydrophobic using oxidation to the ψ[$CH_2$–SO] diastereomeric analogues generally renders the products more hydrophilic than the amide bond. Since sulfoxides are configurationally stable and since the characteristic oxidation products are often HPLC-resolvable doublets, these diastereomeric products serve as a facile method to confirm the presence of a thioether in a chiral peptide. Among methods useful for reduction of sulfoxides back to the sulfide moiety, *N*-methylmercaptoacetamide[57] and $NH_4I$/TFA[58] are reported as most effective. The $H_2O_2$ oxidation also aided the characterization of a complex set of cyclic ψ[$CH_2$–S] pseudopeptide libraries, the characteristic ES-MS pattern being shifted 16 units higher following $H_2O_2$ treatment.[40]

**Figure 1** HPLC Profiles of Each of the Four ψ[$CH_2$–S]-Enkephalin Pseudopeptides Following Partial Oxidation to Their Diastereoisomeric ψ[$CH_2SO$] Derivatives.[39][a]




[a] In nearly all cases, the two sulfoxide isomers are baseline resolved from one another and all are characterized by shorter retention times (more polar) than their ψ[$CH_2$–S] precursors in the C-18 $NH_4OAc$/MeOH (20–40%) gradient reversed-phase chromatographic system. The y-axis represents the UV adsorbance at 254 nm. Retention times for the four isomeric methylenethio ether enkephalin analogues are 14.9, 14.6 52.2 and 31.8 min, respectively.

**Synthesis of Fmoc-Tyr(tBu)-D-Lys(Z)-Glyψ[$CH_2$–S]Phe-Leu Polymer:**[59]

(*R*)-BrCH(Bzl)$CO_2H$ (2.5 equiv) was coupled to Boc-Leu-polystyrene-resin (0.50 g, 0.50 mmol) using DCC (2.5 equiv) in DMF/$CH_2Cl_2$ and a standard SPPS cycle. A double coupling procedure was used. Freshly prepared Boc-NH-$CH_2$-$CH_2$-SH (obtained by reduction of Boc-cystamine, 2 mmol) was dissolved in DMF (10 mL) containing TEA (0.21 mL, 1.5 mmol) then added to the resin which was then shaken for 48 h. After washing the resin with DMF (3 × 2 min) the coupling with Boc-NH-$CH_2$-$CH_2$-SH (2 mmol) was repeated. The progress of $CH_2$–S bond formation was determined by quantitative amino acid analysis. Boc-D-Lys(Z)-OH and Fmoc-Tyr(tBu)-OH were then coupled to the peptide resin which was then washed with EtOH (3 × 20 mL), $Et_2O$ (3 × 20 mL), and dried; yield: 0.82 g (79%).

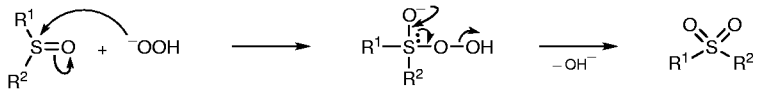
## 10.9.3 ψ[$CH_2$–SO] Sulfinyl Peptides

Sulfoxides are configurationally stable. Oxidation of methylenethio ethers with $H_2O_2$ gives rise to pairs of pseudopeptides of the ψ[$CH_2SO$] type, which are often separable by reversed-phase HPLC. Thus, ψ[$CH_2SO$] diastereomers were observed with each of the four analogues of Leu-enkephalin (Figure 1).[39] Similar separations were observed with peptides as long as decapeptides, e.g. the ψ[$CH_2SO$] derivatives of LHRH.[51] Determination of the absolute configuration of sulfoxides is difficult. In the case of a pair of cyclic ψ[$CH_2SO$] sulfoxides prepared as model compounds for NMR studies, the two analogues of *c*-[-Gly-Proψ[$CH_2$-(*R*,*S*)-SO]Gly-D-Phe-Pro-] were separated by RP-HPLC. An X-ray structure of the more polar component (shorter retention time) established its structure as the (*S*)-sulfoxide isomer, with the polar oxygen oriented outside the five-residue ring.[60]

## 10.9.4 ψ[$CH_2$–$SO_2$] and ψ[$SO_2$–$CH_2$] (Sulfonyl and Retro-sulfonyl) Peptides

Oxidation of a ψ[$CH_2$–S] bond to the (*R*,*S*)-sulfoxide pairs with $H_2O_2$ is more facile than oxidation to the sulfone ψ[$CH_2$–$SO_2$]. However, with excess $H_2O_2$, or other oxidizing agents,[61] the sulfoxides are converted into the sulfone under acid or basic conditions (Scheme 9).

**Scheme 9** Mechanism of Peroxide Oxidation of Sulfoxides to Sulfones[61]



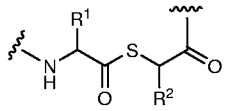
On reversed-phase HPLC, the sulfones usually appear in an intermediate position between the more hydrophobic ψ[$CH_2$–S] peptide and the more polar ψ[$CH_2$–SO] sulfoxides, much as seen in the amino acid analysis by oxidation of Met.[62] Unlike sulfoxides, once formed, sulfones are resistant to reduction to sulfides or sulfoxides.[57] Incorporation of the ψ[$CH_2$–S] element and its oxidized counterparts into litorin (positions 8–9), a bombesin-like peptide, gives rise to receptor antagonists that are more potent when in the sulfoxide forms than in the sulfone form.

## 10.9.5 ψ[C(=O)–S] (Thioester) Peptides

The thioester group (ψ[C(=O)–S]) (Scheme 10) can be distinguished from the thionoester (ψ[C(=S)–O]) by the carbonyl infrared stretching frequency (~1700 $cm^{-1}$) and by the absence of the C=S bond (1200–1050 $cm^{-1}$). In general, thioesters are both more readily synthesized and are more vulnerable to nucleophilic cleavage by virtue of a sulfur leaving group.

*for references see p 489*

**Scheme 10** Structure of a Thioester ψ[C(=O)–S] in a Pseudodipeptide Unit



Thioesters can be prepared either in solution or by solid-phase methods. As with their all-oxygen counterparts, they can be prepared by condensing acids with a thiol using an appropriate condensing agent. Alternatively, a precursor thioacid can be alkylated, usually with a bromoalkyl component. The thioester is stable to acid (TFA, HBr/HOAc, and HBr/TFA). This property makes the group compatible with conditions often used to remove Boc and Z N-protecting groups. However, its lability to nucleophiles makes it incompatible with the base-labile Fmoc group.[63]

Replacement of the peptide bond by a thioester group has been extensively applied in synthetic peptides used as enzyme substrates. Several classes of enzymes such as serine peptidases and metallo-endopeptidases are able to cleave efficiently thioester-modified amino acids and peptides.[6]

Enzymatic cleavage of the thioester bond (located either in the middle or at the C-terminus of the peptide) produces a free thiol compound that can be easily monitored by very sensitive assays (e.g., Ellman's reagent).[64]

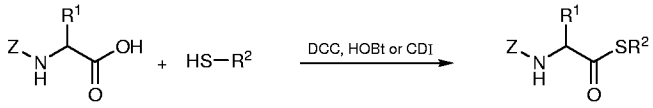
Thioacid- and thioester-modified peptides have been used in the silver-mediated condensations of peptide fragments.[65–70] Thioester-functionalized peptides have also been utilized as the starting precursors for the formation of cyclic peptides[71,72] as well as alkyl and aryl substituted amides.[73]

Other very important applications of C-terminal thioester-functionalized peptides include their usage in the condensation of large unprotected peptide fragments (ligation; see Vol. E 22 a, Section 4.1.5).[4,5,74] In this process the thioester-modified unprotected peptide reacts with the N-terminal cysteine of a second unprotected peptide giving a thioester intermediate. This step is followed by a rapid intramolecular S→N shift with formation of the thermodynamically favored amide bond at the ligation site.

### 10.9.5.1 Solution Synthesis of Thioester Peptides

In general thioesters can be prepared in solution by simple coupling of the N-protected amino acid or peptide with a thiol or a *S*-acylated peptide in the presence of the following activating systems, DCC/HOBt, EDC/HOBt, or CDI (Scheme 11).[6,75,76]

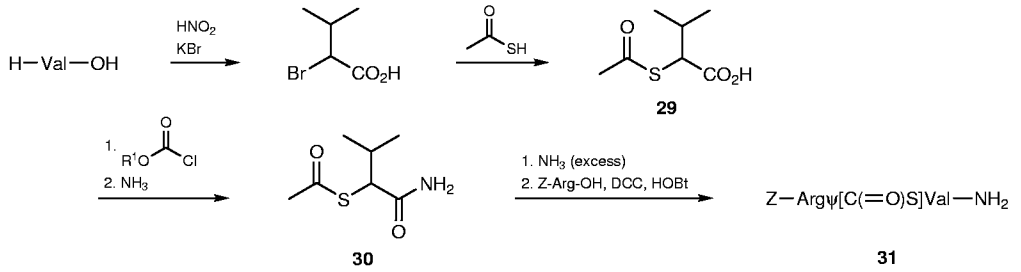
**Scheme 11** Preparation of a Thioester in Solution Using DCC/HOBt or CDI-Mediated Condensation[6]




Formation of the internal thioester bond can be also accomplished by mixed anhydride assisted coupling of the N-protected peptide with a mercaptoacetylated C-terminal peptide fragment.[6] Sulfanylacids used for the preparation of peptide thioesters can be obtained from the common amino acids by the method of Pfister et al.[41] further described by Yankeelov et al.[2]

Another alternative was introduced by McRae et al.[75] in which an S-acetylated mercaptoacid was used for the synthesis of Z-Argψ[C(=O)–S]Val-$NH_2$ (Scheme 12).

**Scheme 12** Preparation of a Thioester Dipeptide with Temporary *S*-Acetyl Protection[75]




Following preparation of the protected thiol derivative **29** (overall retention of configuration is observed when using Val), a mixed anhydride was prepared and converted into the amide **30** with ammonia. The acetyl group of **30** was then removed and the resulting thiol compound condensed with Z-Arg-OH to give the thioester dipeptide **31**.

**Boc-Arg-SBzl·HCl:**[75]
Boc-Arg-OH·$H_2O$·HCl (3.29 g, 10 mmol) and HOBt (1.4 g, 10 mmol) were dissolved in DMF and cooled to 0 °C with stirring. To this soln was added Bzl-SH (2.48 g, 20 mmol) followed by DCC (2.46 g, 12 mmol). After the soln was stirred overnight at 0 °C, DCC (0.834 g, 4 mmol) was added and the reaction was allowed to proceed for another 24 h at 0 °C. The DCU was filtered off and the solvent and excess thiol were removed in vacuo. The remaining oil was chromatographed (silica gel, $CHCl_3$/MeOH) and the product was eluted with $CHCl_3$/MeOH (95/5). Removal of the solvent yielded the title compound as a white foam; yield: 1.86 g (49.3%).

**Boc-Nva-SBzl:**[77]
Boc-Nva-OH (2.17 g, 10 mmol) was dissolved in dry THF (10 mL), CDI (1.62 g, 10 mmol) added and the mixture stirred at 0 °C for 45 min. Bzl-SH (1.16 mL, 10 mmol) was added and the mixture was allowed to warm to rt overnight. After the solvent was removed in vacuo, EtOAc (20 mL) was added and the product washed with 10% citric acid, 4% $NaHCO_3$, and sat. NaCl. The EtOAc soln was dried ($MgSO_4$), the solvent was removed and the solid residue was collected from hexane; yield: 2.58 g (80%).

**L-HS-CH[$CH_2CH(CH_3)_2$]-CO-Leu-Gly-OEt:**[76]
To a soln of *S*-4-methyl-2-sulfanylpentanoic acid (1 g, 6.8 mmol) and H-Leu-Gly-OEt·HCl (2.0 g, 7.9 mmol) in DMF (40 mL) was added HOBt (1.1 g, 7.1 mmol) followed by TEA (1.1 mL, 8 mmol) and EDC (1.6 g, 8.3 mmol). The soln was stirred at rt overnight. TEA·HCl was removed by filtration and the filtrate concentrated under reduced pressure. The residue was partitioned between EtOAc and $H_2O$, the organic layer was washed with dil HCl, $H_2O$, and a $NaHCO_3$ soln adjusted to pH 8. The EtOAc soln was dried ($MgSO_4$) and concentrated in vacuo to afford crude product as a syrup; yield: 0.86 g (59%). This material was used in the next step without further purification.

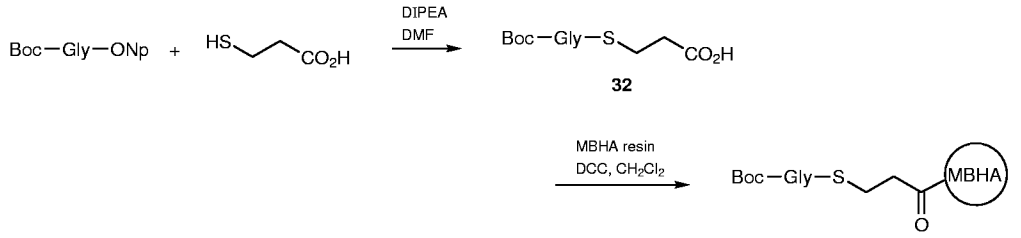
**Ac-Pro-Leu-Glyψ[C(=O)S]Leu-Leu-Gly-OEt:**[76]
To a soln of Ac-Pro-Leu-Gly-OH (0.7 g, 2.1 mmol), HS-CH[$CH_2CH(CH_3)_2$]CO-Leu-Gly-OEt (0.6 g, 1.7 mmol), and HOBt (0.26 g, 1.7 mmol) in DMF (15 mL) was added EDC (0.4 g, 2.1 mmol). The soln was stirred at rt overnight. The solvent was removed in vacuo and the residue partitioned between sat. NaCl and EtOAc. The EtOAc layer was washed with dil $NaHCO_3$, dil HCl, sat. NaCl, 1 mM $CuSO_4$ (to remove any thiol products), and finally sat. NaCl. The EtOAc layer was dried ($MgSO_4$) and concentrated in vacuo to give a glassy residue; yield: 0.8 g (73%). A sample (100 mg) of this crude product was purified by low-pressure reversed-phase (C18) column chromatography. The purified thioester peptide had an acceptable amino acid analysis, consistent mass spectrum, and single peak on HPLC analysis.

## 10.9.5.2 Solid-Phase Synthesis of Thioester Peptides

Preparation of the C-terminal thioacids and thioesters of peptides can be also achieved by solid-phase synthesis. A key step of the solid-phase synthesis is the preparation of the thioester-modified N-protected amino acid in the linkable form **32** so it can be immobilized on the resin as shown in Scheme 13.[67]

*for references see p 489*

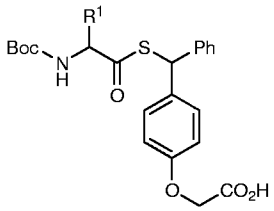
**Scheme 13** Solid-Phase Preparation of Boc-Glyψ[C(=O)–S]$CH_2CH_2CO$-MBHA-Resin[67]




After preparation of the desired resin-bound peptide via SPPS, the thioester-modified product was cleaved by HF[78] from the MBHA resin[79] as the amide.

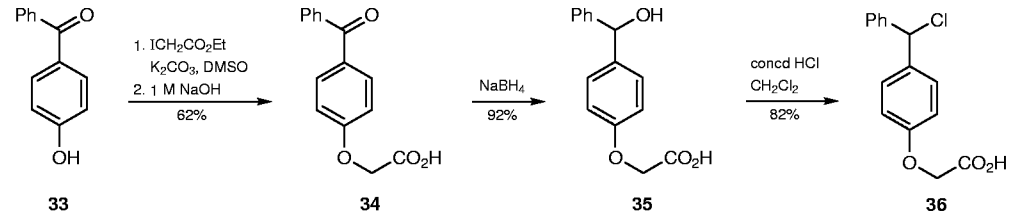
A more versatile route to synthesize peptide thioesters is by the preparation of a peptide thioacid via the solid-phase method and then reaction with a halo derivative to form the thioester. This route can be achieved via the preparation of thioester linkers that can be applied in stepwise solid-phase peptide synthesis. The general structure of the N-protected amino thioacid attached to the linker is shown in Scheme 14.[65,80]

**Scheme 14** Structure of a Boc-Protected Amino Thioacid Attached to a Linker




The linking reagent **36** was synthesized in four steps starting with 4-hydroxybenzophenone (**33**) as outlined in Scheme 15.

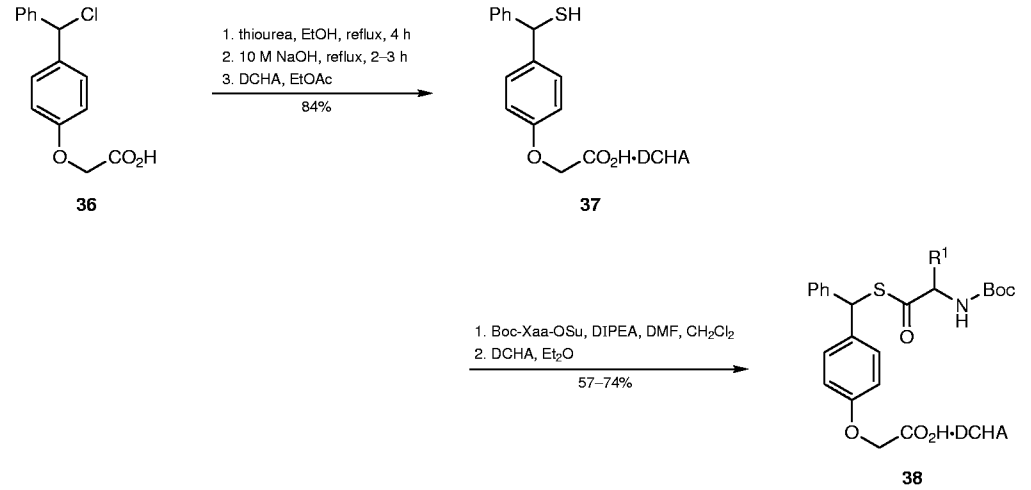
**Scheme 15** Preparation of 4-(α-Chlorobenzyl)phenoxyacetic Acid **6**[80]




The attachment of the first amino acid to the linker **36** was accomplished by the conversion of the amino acid into the thioacid by $H_2S$ treatment of the *N*-hydroxysuccinimide ester followed by formation of the triethyl or *N,N'*-dicyclohexyl ammonium salt and then reaction with the linker.[65,80]

This method was modified by the preparation of the new linking reagent **37** which eliminated the necessity of synthesizing the amino thioacid needed for attachment to **36**. This improved method was described by Canne et al.[81] and is outlined in Scheme 16.

**Scheme 16** Attachment of a Boc-Protected Amino Acid to the Thiol Linker[81]




The linker **37** with the first amino acid attached, compound **38**, can be applied to stepwise solid-phase peptide synthesis. At the end of the synthesis, when the desired peptide sequence is completed, the thioacid-modified peptide fragment is cleaved from the solid support by HF and further S-alkylated with a N-bromoacetylated peptide to form an endothioester bond. The cleaved thioacid can also be reacted with an alkyl bromide to form the corresponding thioester.

Preparation of the thioacids and their esters via SPPS is mainly restricted to the Boc methodology. The popular Fmoc approach is limited by aminolysis of the thioester bond during removal of the Fmoc group by piperidine. However, a modified Fmoc-deprotecting mixture (1-methyl pyrrolidine/hexamethyleneimine/HOBt/DMSO/NMP 25:2:2:35.5:35.5) gave the final desired peptide ester with 24% yield.[24]

**4-Benzoylphenoxyacetic Acid (34):**[80]

A mixture of 4-hydroxybenzophenone (**33**; 23.1 g, 117 mmol), DMSO (100 mL) (dried over 4-Å molecular sieves), ethyl iodoacetate (25 g, 117 mmol), and powdered anhyd $K_2CO_3$ (25 g, 181 mmol) (dried at 100 °C) was stirred overnight at 24 °C. The mixture was cooled, $H_2O$ (920 mL) containing ice added followed by slow addition of 1 M NaOH (200 mL). The mixture was stirred at rt for 3 h, cooled and slowly treated with cold 6 M HCl (96 mL) to decrease the pH of the mixture to below 3. During this step, EtOAc was added to suppress frothing. The mixture was extracted with EtOAc (2 × 100 mL). The combined EtOAc extracts were washed with 1 M HCl (2 × 100 mL) which removed some insoluble material, followed by $H_2O$ (2 × 50 mL). The EtOAc soln was treated with anhyd $MgSO_4$ and Norit A, then the soln was concentrated to about one-fourth its volume in vacuo, at which point the product began to crystallize. After addition of petroleum ether (50 mL) the mixture was kept at 4 °C overnight to give the title product: yield; 18.5 g (62%); mp 155–157 °C.

**4-(α-Hydroxybenzyl)phenoxyacetic Acid (35):**[80]

A soln of 4-benzoylphenoxyacetic acid (**34**; 18.5 g, 72.2 mmol) in EtOH (180 mL), iPrOH (180 mL), and $H_2O$ (73 mL) was treated at 0 °C with $NaBH_4$ (12 g, 317 mmol) over 1 h and stirred to rt for 1 h. The mixture was cooled, treated with glacial AcOH until visible reaction had ceased, and acidified to pH 2.5 with 1 M $H_2SO_4$. The soln was concentrated in vacuo to about one-third volume and then mixed with $H_2O$ (250 mL) and EtOAc (100 mL). The EtOAc layer was separated and the aqueous layer extracted with additional EtOAc (2 × 100 mL). The combined EtOAc layers were washed with $H_2O$ (50 mL), 1 M HCl (50 mL), $H_2O$ (2 × 50 mL), and 50 mL sat. NaCl. After drying ($MgSO_4$), removal of the solvent gave an oil (22 g). This was dissolved in EtOAc (100 mL) and mixed with DCHA (15 mL) with cooling. The DCHA salt rapidly crystallized. The mixture was diluted with anhyd $Et_2O$ (100 mL), stored at 4 °C overnight to afford after filtration and drying the DCHA salt of **35**; yield: 29 g (92%); mp 172–174 °C.

The free acid **35** was released from the above salt by treatment with aq citric acid to give an oil (20 g) which was recrystallized (EtOAc/petroleum ether 3:5) at 4 °C; yield: 14.0 g (82% based on salt); mp 101–103 °C.

*for references see p 489*

**4-(α-Chlorobenzyl)phenoxyacetic Acid (36):**[80]
4-(α-Hydroxybenzyl)phenoxyacetic acid, prepared from the DCHA salt (16.0 g) described above, was dissolved in $CH_2Cl_2$ (220 mL), concd HCl (15 mL) was added and the soln was stirred vigorously for 10–15 min. The lower layer was separated, dried (anhyd $CaCl_2$, 75 g), and the soln concentrated in vacuo to about 60 mL. The product **36** crystallized on addition of petroleum ether (60 mL) and storage at 4 °C overnight; yield: 8.0 g (80%); mp 97–100 °C.

**4-(α-Sulfanylbenzyl)phenoxyacetic Acid·DCHA Salt (37):**[81]
A mixture of 4-(α-chlorobenzyl)phenoxyacetic acid (**36**; 7.5 g, 27 mmol), thiourea (2.3 g, 30 mmol), and EtOH (100 mL) were heated to reflux. After 4 h, conversion into the thiouronium salt was essentially complete as shown by TLC with $CHCl_3$/MeOH/AcOH (90:5:5) as eluant. Then 10 M NaOH (30 mL) was added and the heating under reflux continued for 2–3 h. The mixture was cooled to rt, concentrated in vacuo to approximately half the original volume, acidified with concd HCl to pH 2.0, and extracted with EtOAc (4 × 30 mL). The combined EtOAc extracts were washed with sat. NaCl (1 × 30 mL), dried ($MgSO_4$), and the volatile materials were removed in vacuo. The resulting oil was dissolved in EtOAc (100 mL) and any insoluble material filtered. DCHA (6.0 mL, 30 mmol) was added to the filtrate with stirring. Within a few minutes, a white solid began to precipitate. $Et_2O$ (150 mL) was added and the suspension cooled at −20 °C for several hours. The resulting solid was filtered off, washed with $Et_2O$, and dried in vacuo to give the title product **37**; yield: 10.3 g (84%).

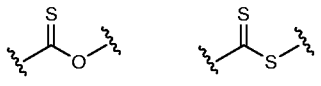
**General Procedure for the Synthesis of the Boc-Protected Amino Thioester Linker DCHA salt (38):**[81]
A mixture of 4-(α-sulfanylbenzyl)phenoxyacetic acid DCHA salt (**37**; 3.67 mmol), Boc-Xaa-OSu (3.68 mmol), DIPEA (5.74 mmol), DMF (35 mL), and $CH_2Cl_2$ (4 mL) was stirred at rt. After several hours the initial white suspension completely dissolved to give a clear, colorless soln. After 24 h the mixture was poured into 1 M HCl (150 mL) and extracted with EtOAc (4 × 35 mL). The combined EtOAc extracts were washed with 1 M HCl (2 × 30 mL), $H_2O$(1 × 30 mL), sat. NaCl (1 × 30 mL), dried ($MgSO_4$), and concentrated. The resulting oil was purified by flash chromatography ($CHCl_3$/MeOH/AcOH 925:50:25) to give an oil contaminated with AcOH. To remove residual AcOH, the oil was dissolved in $CHCl_3$ (40 mL) and washed with 0.1 M HCl (7 × 10 mL), sat. NaCl (1 × 10 mL), and dried ($MgSO_4$). Volatiles were removed in vacuo to give an oil, which was dissolved in $Et_2O$ (10 mL) and DCHA (1 equiv) was added. Dilution with hexane (100 mL) with stirring separated the DCHA salt as a thick oil from any unreacted DCHA. Solvents were decanted from the oil which was dissolved in $CH_2Cl_2$ (30–40 mL). The resulting soln was concentrated in vacuo to give the DCHA salt of the product **38** as a foamy solid.

## 10.9.6 ψ[C(=S)–S] (Dithioester) Peptides

Extensive studies of enzyme–substrate complexes by resonance Raman spectroscopy (RR) have prompted the synthesis of new peptide bond modifications such as thionoesters and dithioesters (Scheme 17)[82,83] within simple model substrates. The resulting acyl–enzyme complexes are especially amenable to RR analysis with cysteine proteases such as papain due to formation of the transient dithioester intermediates.

Several methods for the synthesis of dithiocarboxylic acids and esters have been published.[84–86] They have been summarized by Ramadas et al.[87]

**Scheme 17** Structure of Thionoesters and Dithioesters[82,83]

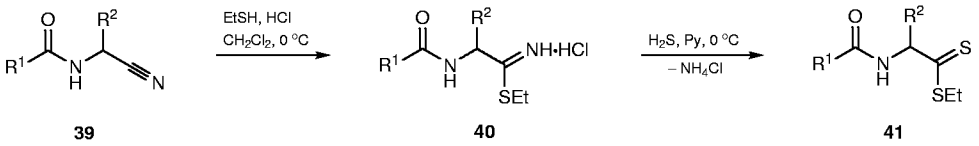



ψ[C(=S)–O] thionoester ψ[C(=S)–S] dithioester

### 10.9.6.1 Preparation of Dithioesters via the Pinner Reaction

The first synthesis of acylglycine dithioester derivatives was described by Storer et al.[86] Their synthesis begins with the Pinner reaction[86,88,89] of the corresponding aminoacyl nitrile **39** in the presence of ethanethiol and HCl gas to produce a thioimidoester hydrochloride **40**. Use of an alcohol instead of a thiol gives the thionoester.[90] The thioimidoester **40** was then treated with $H_2S$ gas in the presence of pyridine to yield the dithioester **41** (Scheme 18).[86]

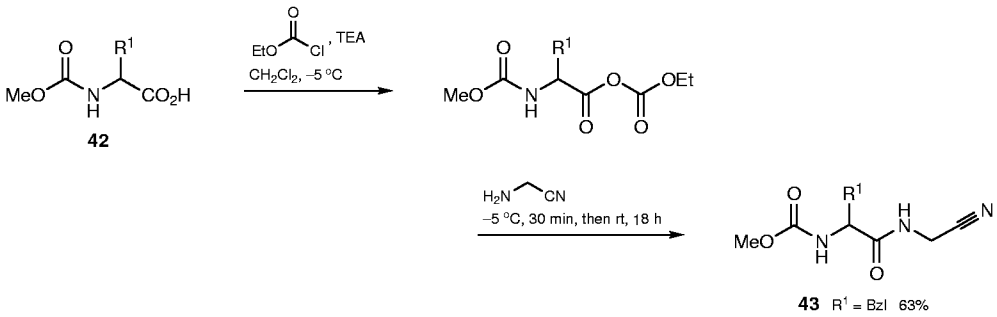
**Scheme 18** Preparation of a Dithioester Modified Peptide[86]




**39** **40** **41**

The synthesis of the starting aminoacyl nitrile **39** was carried out by the modified method of Lowe and Yuthavong[91,92] as shown in Scheme 19. The dipeptide derivative **43** was produced by simple chloroformate-mediated condensation of **42** with aminoacetonitrile formally derived from Gly. This example uses the *N*-methoxycarbonyl protection group, presumably to avoid steric interference in the target enzyme active site.

**Scheme 19** Preparation of the Aminoacyl Nitrile Precursor[91]



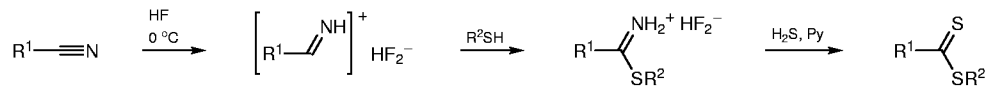

**42**

**43** $R^1$ = Bzl 63%

The above method is applicable only for the preparation of C-terminal dithioesters, and provides a diastereomeric mixture due to the D,L-residue at the C-terminus.

A further modification of this method[86] was described by Neugebauer et al.[93] In this version, the nitrile was dissolved in dry liquid HF at a temperature below 0 °C, instead of HCl saturated $CH_2Cl_2$, to give the adduct $[R^1CNH]^+HF_2^-$ **44** (Scheme 20). This compound was reacted with a thiol to form a thioimidate hydrofluoride **45**, which upon treatment with $H_2S$/pyridine solution provided the dithioester derivative **46**.

**Scheme 20** Preparation of Dithioester-Modified Peptides via the HF Method[93]




**44** **45** **46**

This approach eliminates the problem of the low solubility of *N*-acylaminonitriles in HCl saturated $CH_2Cl_2$ used in the classical Pinner reaction. Overall yields were greatly improved with more reactions in the 60–70% range. The only limitation of this method is the incompatibility of some N-protecting groups with HF.

*for references see p 489*

**Moc-Phe-aminoacetonitrile:**[91]
To a soln of Moc-Phe-OH (10.6 g, 45.5 mmol) and TEA (5.0 g, 49.5 mmol) in $CH_2Cl_2$ (100 mL) at 5 °C was added a soln of ethyl chloroformate (5.4 g, 49.5 mmol) in $CH_2Cl_2$ (20 mL) over 15 min. After stirring for another 15 min, a soln of aminoacetonitrile bisulfate (7.7 g, 50 mmol), adjusted to pH 7 with 2 M NaOH, was added over 30 min while the temperature was maintained at 5 °C. The mixture was stirred for 18 h and allowed to warm gradually to rt. The nitrile was filtered off and recrystallized from $CH_2Cl_2$; yield: 7.5 g (63%); mp 174–175 °C; $[\alpha]_{365}^{20}$ +19.3 (*c* 0.5, EtOH).

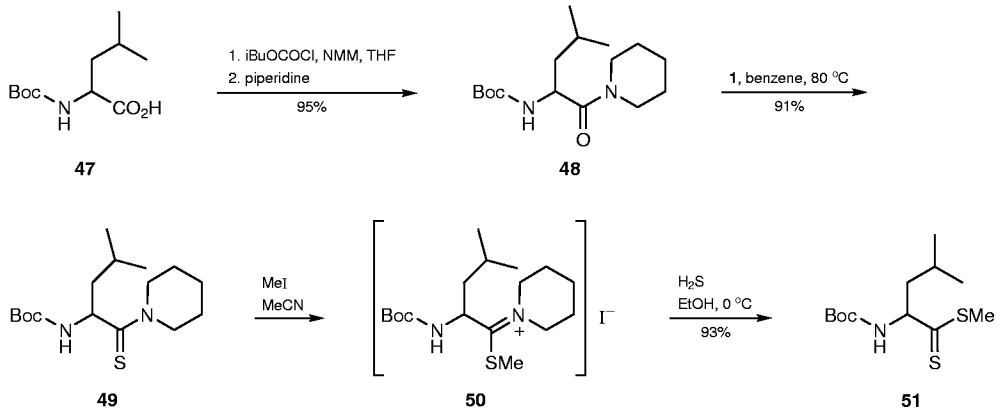
**Moc-Phe-Glyψ[C(=S)S]Et:**[92]
Moc-Phe-aminoacetonitrile (2.61 g, 10 mmol), was added to a 250-mL round-bottomed flask with $CH_2Cl_2$ (50 mL) and ethanethiol (0.52 mL) and cooled to 0 °C. HCl gas was bubbled into the mixture for 20 min. The resultant cloudy soln was treated with an excess of $Et_2O$ (100–150 mL) which produced a sticky precipitate. The $Et_2O$ fraction was decanted and the residue was dried in vacuo for 1 h. The dried solids were then dissolved in ice-cold pyridine (18 mL) and treated with $H_2S$ gas for 10 min at 0 °C. The mixture (a solid formed during $H_2S$ treatment) was stoppered for 5 min then titrated to pH 1–2 with 2 M HCl. The aqueous mixture was extracted with $CH_2Cl_2$ and the organic soln was washed with 1 M HCl, sat. NaCl, and dried ($Na_2SO_4$). The solvent was removed and the solid residue was purified by chromatography (silica gel, $CH_2Cl_2$/acetone 9:1) to give the title compound; yield: 2.5 g (74%).

## 10.9.6.2 Preparation of Dithioesters Using Lawesson's Reagent

An attractive method of preparation of the dithioesters of N-protected amino acids was described by Thorsen et al.[94] The methyl dithioesters were used in an activated form of amino acids for the synthesis of thioamide-modified peptides (see Section 10.9.1).

The preparation starts with the conversion of an N-protected amino acid **47** into the piperidide **48** using a mixed anhydride followed by thionylation with **1**[14] (Scheme 21). The resulting thiopiperide **49** was further methylated with MeI to give **50** which was then treated with $H_2S$ to form the methyl dithioester **51**.

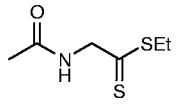
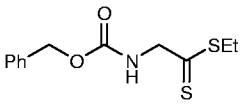
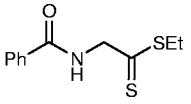
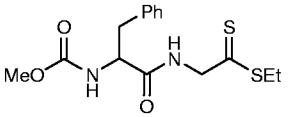
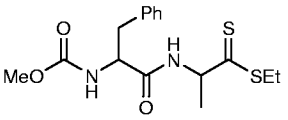
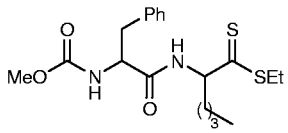
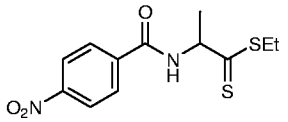
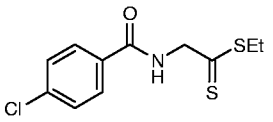
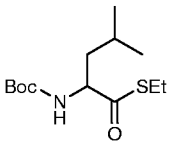
**Scheme 21** Preparation of a Dithioester-Modified Peptide via Lawesson's Reagent[16]




47 48 49 50 51

The advantage of this method is that it can be used to synthesize dithioesters from optically pure amino acids. There are a few compelling examples that this route provides racemization-free dithioesters.[16] Other examples of structures with dithiopeptide modifications are given in Table 3.

**Table 3** Examples of Dithioesters Incorporated in Peptide Structures[16,86,92,93,95–98]

| Compound | Abbreviation | Ref. |
|---|---|---|
|  | Ac-Glyψ[C(=S)–S]Et | [86] |
|  | Z-Glyψ[C(=S)–S]Et | [86] |
|  | Bz-Glyψ[C(=S)–S]Et | [95] |
|  | Moc-Phe-Glyψ[C(=S)–S]Et | [92] |
|  D,L | Moc-Phe-D,L-Alaψ[C(=S)–S]Et | [96] |
|  D,L | Moc-Phe-D,L-Nleψ[C(=S)–S]Et | [97] |
|  D,L | Bz(4-$NO_2$)-D,L-Alaψ[C(=S)–S]Et | [98] |
|  | Bz(4-Cl)-Glyψ[C(=S)–S]Et | [93] |
|  | Boc-L-Leuψ[C(=S)–S]Et | [16] |

**Boc-Leu-Piperidide** (48):[16]

Under an atmosphere of $N_2$, a soln of Boc-Leu-OH·$H_2O$ (**47**; 1.05 g, 4.224 mmol) in anhyd THF (10 mL) was cooled to −10 °C in an ice salt bath. Dry NMM (0.465 mL, 4.224 mmol) was added followed by isobutyl chloroformate (0.548 mL, 4.224 mmol). The mixture was stirred for 15 min, then piperidine (0.418 mL, 4.224 mmol) was added. The mixture was allowed to warm to 0 °C over a period of 2 h and then stirred at rt for 3h. The precipitated salts were filtered off and the soln washed with 1.5 M HCl (2 × 20 mL), 5% $NaHCO_3$ (2 × 20 mL), sat. NaCl (20 mL), and dried ($MgSO_4$). The soln was concentrated in vacuo to give the title product as a colorless oil; yield: 1.195 g (95%); $[\alpha]_D$ −26 (*c* 0.9, MeOH).

*for references see p 489*

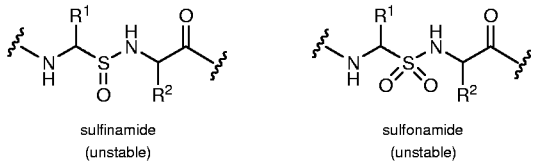
**Boc-Leu-thiopiperidide (49):**[16]
A suspension of Boc-Leu-piperidide (**48**; 119 mg, 0.398 mmol) and **1** (88.7 mg, 0.219 mmol) in dry benzene (**CAUTION:** *carcinogen!*) (1 mL) under $N_2$ was heated under reflux for 2 h. The solvent was removed in vacuo and the residue was purified by flash chromatography (silica gel, 230–400 mesh, EtOAc/hexane 1:9) to give the title compound as an oil; yield: 113 mg (91%).

**Boc-Leuψ[C(=S)–S]Et (51):**[16]
MeI (3.15 mL, 50.6 mmol) was added to a soln of Boc-Leu-thiopiperidide (**49**; 1.59 g, 5.06 mmol) in dry MeCN (5 mL) and the mixture was stirred under $N_2$ for 48 h. The volatiles were removed under reduced pressure with protection from moisture and the residue further dried under high vacuum. The residue was dissolved in abs EtOH (12 mL), cooled to 0 °C and a slow stream of $H_2S$ was bubbled through the soln for 1 h. The mixture was stirred at 0 °C overnight and the solvent was removed in vacuo. The residue was triturated with $Et_2O$ (50 mL), filtered and the ethereal soln concentrated to give a yellow oil which was purified by chromatography (silica gel, hexane/EtOAc 96:4) to give the product as a yellow solid; yield: 1.3 g (93%); mp 55–56 °C; $[\alpha]_D$ −95.5 (*c* 1, MeOH).

## 10.9.7 ψ[SO–NH] and ψ[$CH_2$–SO–NH] (Sulfinamide and Methylene Sulfinamide) Peptides, and ψ[$SO_2$–NH] and ψ[$CH_2$–$SO_2$–NH] (Sulfonamide and Methylene Sulfonamide) Peptides

Replacing an amide group with a sulfonamide bond (Scheme 22) is an attractive goal, because such a substitution creates an amide bond surrogate with significant changes in polarity, hydrogen bonding capability, and acid–base character ($RSO_2$–NHR′, pKa 10–11).[99,100]

**Scheme 22** Structures of Sulfinamide and Sulfonamide Dipeptides




Furthermore, the sulfonamide bond is expected to possess enhanced metabolic stability with structural similarities to the tetrahedral transition state involved in amide bond enzymatic hydrolysis, thus making sulfonamide peptides interesting candidates in the development of protease inhibitors and new drugs. The oligomers and polymers should also be interesting molecular scaffolds, with specific secondary structures enforced by hydrogen bonding.[100,101]

### 10.9.7.1 Preparation of an α-Aminosulfonic Acid

α-Aminoalkanesulfonic acids have been known for more than 95 years.[102,103] Their synthesis proceeds through the action of aqueous ammonia on aldehyde bisulfite compounds. For example, α-aminomethane-, ethane-, isobutane-, isopentane-, and phenylmethanesulfonic acids correspond to Gly, Ala, Val, Leu, and Phg.[104] Some of the α-aminosulfonic acids are known to be limiting factors for growth of some pathogenic organisms. α-Aminobutane sulfonic acid can be prepared from butyraldehyde and sodium metabisulfite.

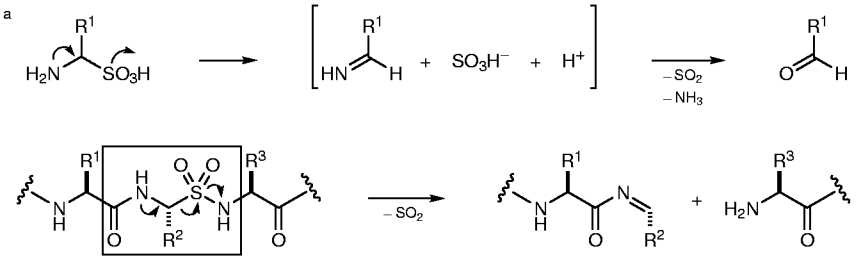
**α-Aminobutane Sulfonic Acid:**[105]
To $Na_2S_2O_5$ (24 g, 126 mmol) in $H_2O$ (40 mL), butyraldehyde (23 mL, 255 mmol) was added with stirring. The soln was cooled to 0 °C and concd $NH_3$ (20 mL) was slowly added with continued stirring at rt for

1 h. The mixture was cooled to 0 °C and acidified with concd HCl to pH 4. The precipitated crystals were filtered off, washed with a small amount of cold $H_2O$, 50% EtOH, and finally 96% EtOH. The air-dried product was recrystallized from $H_2O$ (15 mL, 60 °C) to give the title compound; yield: 2 g (16%); mp 120–121 °C.

### 10.9.7.2 Synthesis of Protected α-Aminosulfonamides

The synthesis of α-sulfonamidopeptides, i.e. the direct substitution of the carboxylic group of α-amino acids with a sulfonic acid group, remains an elusive goal because, unlike the α-amino acids, α-aminoalkyl sulfonic acids are unstable in solution and decompose to give an aldehyde, sulfur dioxide, and ammonia (Scheme 23).[105]

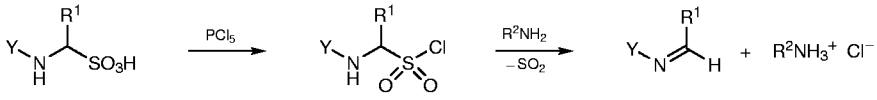
**Scheme 23** Decomposition Pathways of (a) an α-Aminosulfonic Acid and (b) an α-Aminosulfonamide[105]



The α-aminosulfonamides are also prone to decompose in solution with fragmentation.[106] It was suggested that in these compounds the lability of the sulfonic acid group is further increased by the presence of the amino group. The free electron pair on the nitrogen atom is capable of undergoing an electromeric shift to form the unstable imine with rupture of the C–S bond and loss of a proton. Acylation of the amino group greatly enhances the stability of these compounds. However, coupling of the α-aminoalkyl sulfonic acids with a second amino component to form compounds analogous to peptides but containing the sulfonamide group has been unsuccessful by the known methods of peptide synthesis. It has even been concluded that the α-amino sulfonic acids do not form sulfonamides.[105]

Attempts to convert the *N*-acyl derivatives of the moderately stable α-amino sulfonic acids by turning them into sulfonyl chlorides and then directly to sulfonamides have failed; it appears that the critical requirement for C–S bond rupture is related to a sufficient activation of the sulfonate.[106] Any synthetic approach to the desired sulfonamide derivatives which utilizes N-protection of aminomethane sulfonic acid as the initial step seems to offer little likelihood of success. In the context of the overactivation of the sulfonyl group of N-protected aminoalkyl sulfonic acid, β-elimination of the $SO_2Cl$ is favored over the corresponding sulfonamide, with amines acting merely as bases, not as nucleophiles (Scheme 24).[107]

**Scheme 24** β-Elimination on Activation of an N-Protected α-Aminoalkyl Sulfonic Acid[107]

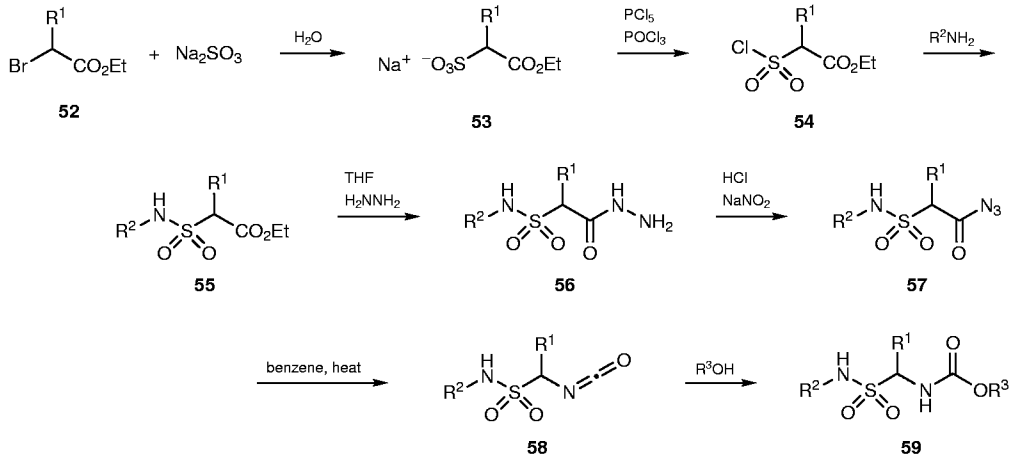


The Curtius rearrangement provides a route to carbamates of α-amino sulfonamides (Scheme 25).[108] Reaction of the ethyl ester of an α-bromo acid **52** with $Na_2SO_3$ yielded the sodium salt of the corresponding sulfonic acid **53**. Treatment of **53** with $PCl_5$ afforded the sulfonyl chloride **54** which on reaction with an amine gave the sulfonamide **55**. The latter

*for references see p 489*

compound was converted into the hydrazide **56** with hydrazine. Diazotization of the hydrazide **56** yielded the azide **57** which on Curtius rearrangement gave the corresponding isocyanate **58**. Treatment of **58** with *tert*-butanol or benzyl alcohol yielded the corresponding carbamate **59**.[109] However, attempts to convert the carbamates into free α-amino alkyl sulfonamides have not been successful.

**Scheme 25** Formation of Carbamates of α-Amino Sulfonamides via Curtius Rearrangement[109]



Several α-aminoalkylsulfo analogues of peptides, such as the α-sulfono analogue of acyl-D-Ala-D-Ala, have been synthesized using the Curtius rearrangement[110] and examined in aprotic solvents. It was hoped that Ac-D-Ala-D-Ala-based sulfonopeptides would inhibit the transpeptidase involved in bacterial cell-wall synthesis by acting on the enzyme like penicillin, namely as a structural analogue of the Ac-D-Ala-D-Ala terminus of nascent peptidoglycan strands. However, these compounds, in accord with what was learned previously, proved to be extremely labile in aqueous and protic media.[110]

The most closely related sulfur-containing transition-state analogue of an α-amino acid incorporated in a peptide is an α-amino alkyl sulfonamide. However these systems tend to fragment as depicted in Scheme 23.

Although the unprotected α-aminosulfinic acids are known,[107] the corresponding α-amino sulfinamide derivatives are expected to be more stable and less prone to fragmentation.[111,112]

A promising approach in which the S–N bond was first generated as a sulfenamide followed by oxidation to sulfinamide has been described.[113] Reaction of bromomethylphthalimide with sodium disulfide produces the corresponding disulfide that, when assisted by silver ions, forms a sulfenamide bond with H-Phe-OMe. Selective oxidation of the latter with 1 equivalent of 3-chloroperbenzoic acid gives two diastereomeric sulfinamides that can be separated by silica gel chromatography. Further oxidation of either sulfinamide with an equivalent of peracid gave the less polar sulfonamide. However any attempt to remove the protecting phthalyl group was unsuccessful due to the apparent instability of α-sulfinamides.[113]

Azasulfonamidopeptides, in which an α-amino acid residue is replaced by the -NH–NR–$SO_2$- group, also appear to be of interest as models for the transition state of peptide-bond hydrolysis. As they are potential proteinase inhibitors, the routes for their synthesis have been described.[114]

**Bis(phthalimidomethyl) Disulfide:**[113]
Precipitated sulfur (0.416 g, 12.5 mmol) was dissolved in aq $Na_2S$ (30 mL, 3.125 g, 12.5 mmol) with heating. The soln was cooled to rt before being added to a soln of *N*-bromomethylphthalimide (5 g, 20 mmol) in acetone (80 mL) over 5 min. The soln turned yellow and then cleared before formation of a colorless precipitate. After 1 h the precipitate was collected and dried to give the title compound; yield: 4.2 g (86%); mp 188–191 °C.

***N*-(Phthalimidomethylsulfenyl)-Phe-OMe:**[113]
A suspension of the above disulfide (20 g, 0.052 mol) and $AcO^-$ $Ag^+$ (16.39 g, 0.104 mol) in EtOAc (200 mL) was stirred at rt in the dark. TEA (15.75 g, 0.156 mol) was then added and a soln of H-Phe-OMe (3 equiv) in EtOAc (50 mL). The mixture was stirred at rt under $N_2$ overnight and the black precipitate that formed was filtered off. The filtrate was concentrated under reduced pressure and the residue was extracted with $Et_2O$. The $Et_2O$ soln was washed with $H_2O$, dried, filtered, and concentrated. The residue was chromatographed (silica gel, EtOAc/light petroleum ether 1:1) to give the title compound as a pale yellow oil; yield: 8.7 g (45%).

***N*-(Phthalimidomethylsulfinyl)-Phe-OMe:**[113]
The above sulfenamide derivative (9.88 g, 26.7 mmol) in $CH_2Cl_2$ (50 mL) was oxidized with a soln of MCPBA (5.41 g, 26.7 mmol) in $CH_2Cl_2$ (50 mL) at 0 °C for 40 min then the soln was washed with 5% $NaHCO_3$. The organic layer was separated, dried, and the solvent removed to give an oil (7.438 g, 73%). The sulfinamide diastereoisomers were separated by column chromatography (silica gel, EtOAc/light petroleum ether 1:1).

The less polar isomer had the following properties; mp 129–131 °C; $[\alpha]_D$ +42 (*c* 1.0, $CHCl_3$). This isomer was subjected to X-ray diffraction analysis and was shown to have the *R*-stereochemistry at the sulfur atom.

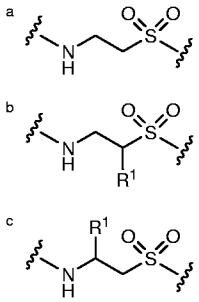
The more polar isomer had the following properties; mp 96–98 °C; $[\alpha]_D$ +26.8.

***N*-(Phthalimidomethylsulfonyl)-Phe-OMe:**
The diastereomeric sulfinamides (200 mg, 0.49 mmol) were dissolved in $CH_2Cl_2$ (5 mL) and stirred at rt with MCPBA (134 mg, 0.77 mmol) for 30 min. Then the reaction was quenched by pouring into 5% $NaHCO_3$. The organic layer was separated, dried, the solvent was removed and the residue chromatographed (silica gel, EtOAc/light petroleum ether 1:1) to afford the title sulfonamide; yield: 170 mg (81%); mp 46–49 °C (decomp.); $[\alpha]_D$ +32.5 (*c* 0.12, $CHCl_3$).

## 10.9.7.3 Synthesis of Taurine-Containing Analogues

In contrast to α-amino sulfinamides or sulfonamides the corresponding β-aminoethane derivatives (methylene sulfinamides and methylene sulfonamides) are stable (Scheme 26).

**Scheme 26** β-Aminoethane Sulfonyl (a) α-Alkyl-β-aminoethane Sulfonyl (b) β-Alkyl-β-aminoethane Sulfonyl (c) Groups Providing Stable Peptide Analogues




A considerable amount of work on the synthesis and crystal structure of peptides containing a β-aminoethane sulfonamide moiety, i.e. taurine (Tau), has been described by Calcagni et al.[7,115–117] and Diaz et al.[118] and a β-aminoethane sulfinamide (hypotaurine) moiety by Moree et al.[111,112,119,120]

*for references see p 489*

Two methods for the synthesis of methylene sulfonamide peptides are available. According to the first, the taurine residue is introduced via its sulfonyl chloride.[121,122] Sulfonyl chlorides can be prepared either by oxidation of a thioester or a disulfide with $Cl_2/Ac_2O$,[123] by the action of $PCl_5$ on a sulfonic acid or its salt in benzene,[121,122] or by reaction of a sulfonate salt with $Ph_3P/SO_2Cl_2$ in $CH_2Cl_2$ (this method is compatible with both acid- and base-sensitive functionalities).[124] An alternative method involves reaction of a sulfonate salt with triphosgene under Vilsmeier–Haack conditions[125] (in $CH_2Cl_2$ in the presence of catalytic amounts of DMF).[126] According to de Bont et al.[127] the latter method is preferable as the former results in a mixture of sulfonyl and sulfinyl chlorides. The other method is a soft base-mediated sulfinyl chloride coupling followed by oxidation of the resulting sulfinamides to sulfonamides.[112]

Introducing the Tau residue into a peptide according to the first approach demands protection of the amino group, usually in the form of a Z-derivative and turning the sulfonic acid into sulfonyl chloride. Synthesis of β-sulfonamidopeptides via an iterative process, both in solution and in the solid phase, has been described.[120] Chiral methylene sulfinamide peptides can be synthesized both in solution and in the solid phase using the sulfonyl chlorides derived from enantiomerically pure 2-substituted taurines under mild coupling conditions (DMAP catalysis and excess methyl trimethylsilyl dimethylketene acetal as a proton trap).[126]

While searching for cell adhesion inhibitors, analogues of the recognition motif-based tripeptide Ac-Leu-Asp-Thr-$NH_2$ were synthesized; the central Asp residue was replaced by the corresponding methylene sulfonamide analogue.[128] The related peptide Ac-Leu-Asp-ψ[$CH_2$–$SO_2$–NH]Val-$NH_2$ turned out to be only a slightly weaker cell adhesion inhibitor than its peptide analogue. The SPPS methodology was extended to Leu-enkephalin-derived methylene sulfonamide peptides: specific amide bonds were replaced one by one by -$CH_2$–$SO_2$–NH- moieties as a positional scan to determine the relative biological effect of each replacement.[127] The binding activities of these methylene sulfonamide peptide hybrids with anti-β-endorphin monoclonal antibody showed that replacement of the C-terminal amide resulted in a nearly equipotent compound, but replacement of other amide bonds in the direction of the N-terminus resulted in decreased and ultimately complete loss of biological activity.

The concept of tweezer-like two-armed molecules was used to construct synthetic receptor molecules based on methylene sulfonamide peptidomimetics.[129] When screened for binding against an encoded tripeptide library, one of the synthetic receptors demonstrated a remarkable binding selectivity;[129] the synthetic receptor affinity could be further increased by varying the hinge part of the peptidomimetic molecule.[130]

When the conformational preferences of methylene sulfonamide peptides were studied in $CHCl_3$ by $^1H$ NMR and FTIR spectroscopy and the experimental data correlated with the hydrogen bond strength, it was found that the amide is a stronger hydrogen-bond acceptor than the sulfonamide.[101]

The presence of the additional β-carbon atom in Tau analogues offers the opportunity to study the influence of functionalization at this β-position as compared to functionalization at the α-position. Naturally occurring α-amino acids can be used as starting materials for the synthesis of homochiral β-substituted methylene sulfinamide and sulfonamide transition-state isosteres incorporated in peptides.[120]

**Z-Tau-ONa:**[121]

Taurine (8 g, 64 mmol) was dissolved in warm $H_2O$ (40 mL). The soln was stirred with ice cooling, $NaHCO_3$ (11.2 g, 130 mmol)) was added followed by the addition of Z-Cl (14.3 g, 84 mmol) over 10 min. Stirring was continued at rt for 4 h. The soln was extracted with $Et_2O$ (3 ×), the aq layer acidified with HCl and concentrated under reduced pressure to about 60 mL when crystals began to form. The soln was warmed, filtered from NaCl, and the product separated on cooling; yield: 16 g (89%). The analytical sample was recrystallized from aq EtOH.

**Z-Tau-Cl:**[122]
Z-Tau-ONa, obtained from 40 g Tau, was finely ground and dried in a desiccator over $P_2O_5$ then suspended in anhyd $Et_2O$. With stirring, finely pulverized $PCl_5$ (70 g) was added in small portions. (The reaction is exothermic and if the addition is too rapid it may become necessary to cool the mixture.) After the addition was complete, the reaction was left for several hours at rt and then the mixture was concentrated in vacuo. On trituration of the residue with ice water, rapid crystallization of the product occurred. The mixture was left in the refrigerator overnight then the crystals were filtered off, washed on the filter with plenty of ice water, and dried in vacuo over $P_2O_5$; yield: 78.5 g (88.5%). An analytical sample was obtained on recrystallization from benzene/petroleum ether; mp 50–51 °C.

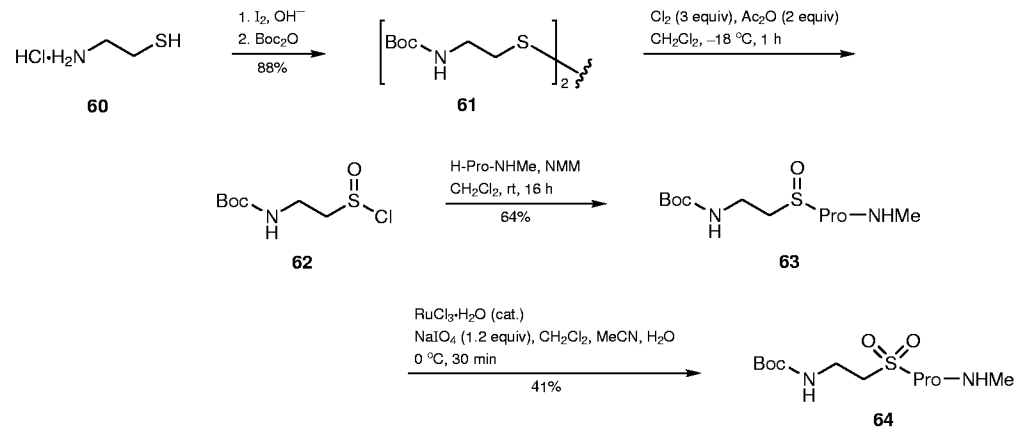
**Z-Tau-Leu-OMe:**[117]
To a stirred soln of H-Leu-OMe·HCl (2.9 g, 16 mmol) in dry THF (30 mL), NMM (4.9 g, 48 mmol) was added. After 10 min Z-Tau-Cl (4.4 g, 16.0 mmol) in dry THF (20 mL) was added at 0 °C and the mixture was stirred at rt for 16 h. Concentration of the mixture under reduced pressure gave a residue which was taken up in EtOAc. The EtOAc soln was washed successively with 1 M $KHSO_4$, sat. NaCl, 5% $NaHCO_3$, sat. NaCl, and dried. Removal of the solvent and chromatography (silica gel, $CH_2Cl_2/Et_2O$ 95:5) of the residue gave the title compound; yield: 3.7 g (60%); mp 45–46 °C ($iPr_2O$/petroleum ether); $[\alpha]_D$ −5.0 (*c* 4, $CHCl_3$).

## 10.9.7.4 Synthesis of Methylene Sulfonamides Through Methylene Sulfinyl Derivatives

In a further approach to the synthesis of methylene sulfonamides the S–N bond is first generated as a sulfinamide followed by oxidation to the sulfonamide by $RuCl_3/NaIO_4$ as illustrated in Scheme 27.[111,112]

Reaction of *N,N'*-bis(Boc)-cystamine (**61**), readily obtained from 2-sulfanylethylamine hydrochloride (**60**), with chlorine in the presence of 2 equivalents of $Ac_2O$, forms *N*-Boc-aminoethanesulfinyl chloride (**62**), the oxygen atom source being the $Ac_2O$. If the same reaction was carried out in the presence of 4 equivalents of $Ac_2O$ the corresponding sulfonyl chloride was generated; however in the latter case the reaction was not unambiguous and there was always some sulfinyl chloride present in the product.[112]

**Scheme 27** Synthesis of a Boc-β-Aminoethane Sulfonyl Dipeptide Using Boc-Aminoethane Sulfinyl Chloride[112]



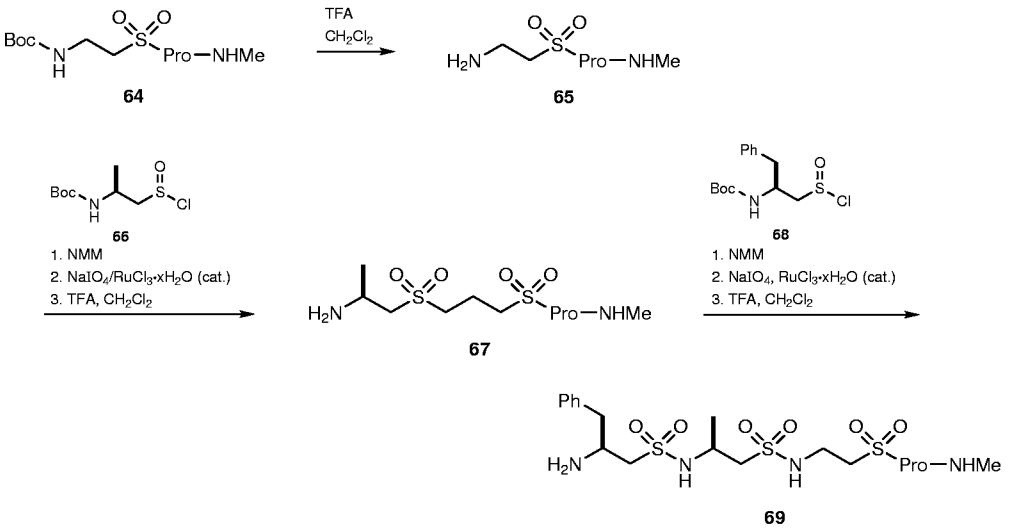

The sulfinyl chlorides are used as building blocks for the preparation of homochiral α- or β-substituted sulfinamide and sulfonamide transition-state isosteres. The corresponding α- or β-substituted *N*-Boc-aminoethanols were converted by the Mitsunobu reaction into thioacetates according to the general procedure[131] and then on treatment with $Cl_2$ or $SO_2Cl_2$ in the presence of 2 equivalents of $Ac_2O$ into sulfinyl chlorides.[119] This methodology has been

*for references see p 489*

applied to the synthesis of methylenesulfonamide peptidomimetics providing, for example, haptens useful for the generation of antibodies and potential HIV protease inhibitors. In addition, the β-substituted aminoethanesulfinyl chlorides have been used as building blocks for the preparation of a tetrapeptido-methylenesulfonamide, which can be considered as a biopolymer mimetic. This approach used a repetition of a cycle of three reactions, coupling of the sulfinyl chloride to the N-terminus of the growing methylenesulfonamide peptide, oxidation to the sulfonamide, and deprotection of the N-terminus.

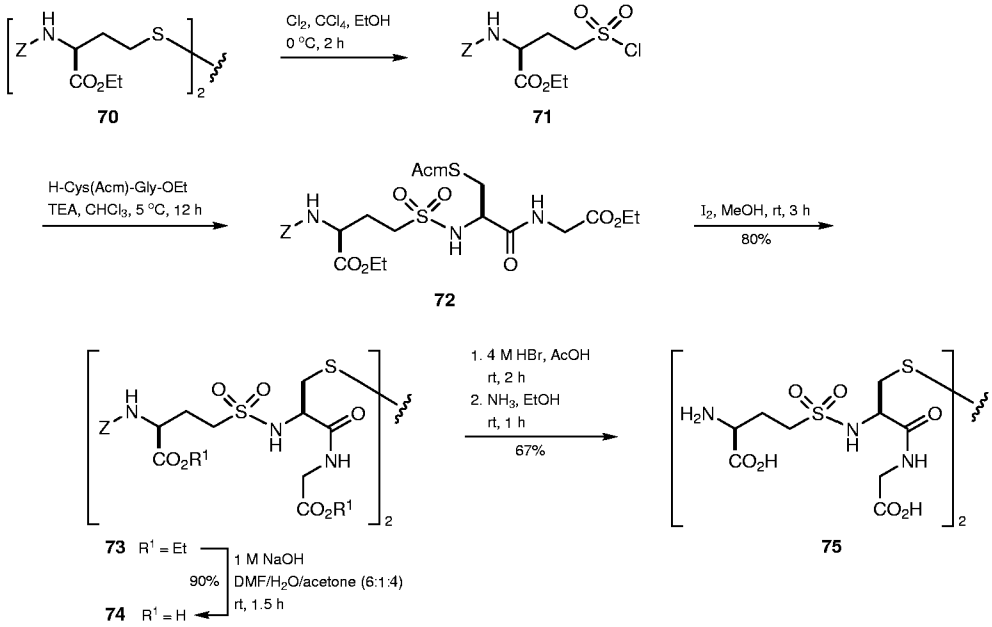
The β-methylene group is necessary to obtain stable sulfonamides. Since all kinds of α-amino acids can be used as starting materials for the preparation of homochiral α- and β-substituted β-aminoethane sulfonamide systems, it is almost imperative to use the sulfinyl chloride synthons as building blocks for the methylenesulfonamide peptidomimetics in which several amide bonds are replaced by a sulfonamide moiety.[120] These unnatural biopolymers or biopolymer mimetics hold considerable promise. This is illustrated by the synthesis of the tetrapeptidosulfonamide **69** (Scheme 28). Starting from the sulfinyl chloride **62**, coupling with H-Pro-NHMe and subsequent oxidation gave Boc-Tau-Pro-NHMe (**64**). The Boc group was then removed from **64** using TFA in $CH_2Cl_2$, and after liberation from its salt, the amine **65** was coupled to the second sulfinyl chloride **66**. The resulting sulfinamide, isolated in 70% yield, was oxidized to the sulfonamide **67** using $NaIO_4/RuCl_3$ in 91% yield. Repetition of the deprotection and coupling procedure with the third sulfinyl chloride **68** gave the sulfinamide in 64% yield, which was oxidized to the corresponding tetrapeptidosulfonamide **69** in 90% yield.[119]

**Scheme 28** Synthesis of a Sulfonamide ψ[$CH_2SO_2NH$] Tetrapeptide Analogue[119]




A glutathione disulfide analogue **75** that is characterized by the presence of the $-CH_2-SO_2-NH-$ junction replacing the γ-Glu-Cys peptide bond was obtained according to Scheme 29.[132] Treating (Z-Cys-OEt)$_2$ (**70**) with chlorine provided ethyl (*S*)-2-benzyloxycarbonyl-amino-4-(chlorosulfonyl)butanoate (**71**) which was used to acylate H-Cys(Acm)-Gly-OEt to yield **72**. Removal of the Acm-protecting group with $I_2$ in MeOH converted the product directly into the symmetrical disulfide **73**. Alkaline hydrolysis of the symmetrical disulfide was achieved with 1 M NaOH in a DMF/$H_2O$/acetone solvent mixture to provide **74**. Removal of the N-terminal Z group with HBr/AcOH, followed by treatment of the resulting bis-hydrobromide with $NH_3$ in EtOH, provided the desired pseudopeptide **75**.

**Scheme 29** Synthesis of a ψ[$CH_2$–$SO_2$–NH] Analogue[132]




***N,N'*-Bis(Boc)-cystamine:**[112]
Cysteamine hydrochloride (5.68 g, 50 mmol), dissolved in 2 M NaOH (25 mL), was oxidized by titration with a soln of $I_2$ (15.23 g, 60 mmol) in dioxane (25 mL). The resulting cystamine was neutralized with 2 M NaOH (25 mL) and treated with $Boc_2O$ (12.0 g, 55 mmol) in dioxane at 0 °C. After stirring for 2.5 h the mixture was concentrated in vacuo and partitioned between EtOAc and $H_2O$. The organic layer was washed with 10% $Na_2S_2O_3$, 1 M $KHSO_4$, and brine, dried ($MgSO_4$) and concentrated in vacuo. Crystallization (EtOAc) afforded the title compound; yield: 88%; mp 107.5–108 °C.

***N*-Boc-aminoethanesulfinyl Chloride (62):**[112]
To a stirred and cooled (−18 °C) soln of *N,N'*-bis(Boc)-cystamine (0.71 g, 2.01 mmol) and $Ac_2O$ (0.38 mL, 4.02 mmol) in $CH_2Cl_2$ (15 mL), a cooled (−10 °C) soln of $Cl_2$ (ca. 0.6 g, 7.9 mmol, dried over concd $H_2SO_4$) in $CH_2Cl_2$ (15 mL) was added via a glass connecting tube. Stirring was continued for 1 h at −18 °C. Concentration and removal of residual solvent and acetyl chloride using an oil vacuum pump gave the sulfinylchloride **62**, which was used without further purification.

***N*-Boc-aminoethanesulfonyl Chloride:**[112]
To a stirred and cooled (−10 °C) soln of *N,N'*-bis(Boc)-cystamine (0.87 g, 2.46 mmol) and $Ac_2O$ (0.93 mL, 9.82 mmol) in $CH_2Cl_2$ (20 mL) a cooled (−10 °C) soln of $Cl_2$ (ca. 0.9 g, 12.3 mmol, dried over concd $H_4SO_4$) in $CH_2Cl_2$ (15 mL) was added via a glass connecting tube and stirring was continued for 4 h at 0 °C. Concentration and removal of residual solvent and AcCl using an oil vacuum pump gave the sulfonyl chloride, which was used without further purification.

**Boc-Glyψ[$CH_2$–SO–NH]Pro-NHMe(63):**[112]
A soln of H-Pro-NHMe (0.539 g, 4.20 mmol) in $CH_2Cl_2$ (10 mL) and NMM (0.44 mL, 4.02 mmol) was added simultaneously to a soln of *N*-Boc-aminoethane sulfonyl chloride (4.02 mmol) in $CH_2Cl_2$ (7 mL) at 0 °C under argon. The mixture was stirred overnight at rt, concentrated in vacuo and purified by column chromatography (silica gel 50 g, $CH_2Cl_2$/MeOH 96:4) to afford the sulfinamide as an oil; yield: 64%.

**Boc-Glyψ[$CH_2$–$SO_2$–NH]Pro-NHMe (64):**[112]
*Method A*: A soln of H-Pro-NHMe (1.26 g, 9.83 mmol) in $CH_2Cl_2$ (10 mL) was added to a stirred, cooled (0 °C) suspension of *N*-Boc-aminoethane sulfonyl chloride (4.92 mmol) in dry $CH_2Cl_2$ (10 mL). After stirring the mixture overnight at rt under argon, the H-Pro-NHMe · HCl precipitate was filtered off and the filtrate concentrated in vacuo. Column chromatography (silica gel, $CH_2Cl_2$/MeOH 95:5) of the residue afforded the sulfonamide **64** (yield: 41%) and the sulfinamide (yield: 19%).

*for references see p 489*

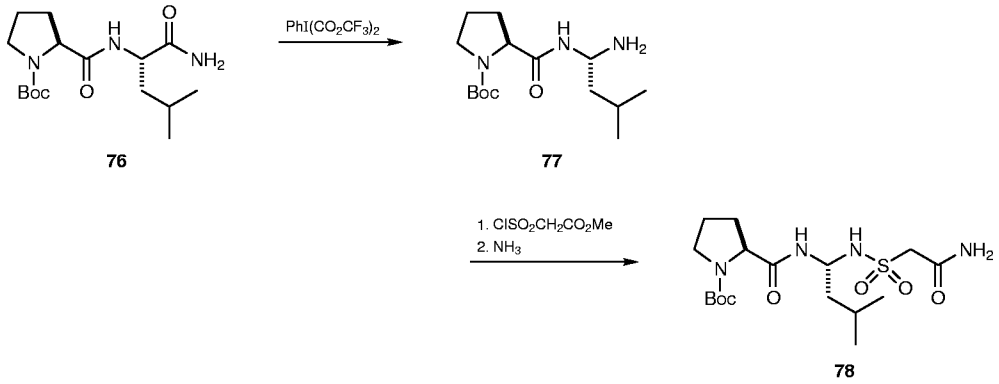
*Method B:* $NaIO_4$ (0.32 g, 1.48 mmol) and a catalytic amount of $RuCl_3 \cdot H_2O$ (2 mg, 8.9 μM) were added to a cooled (0 °C) soln of sulfinamide (0.39 g, 1.23 mmol) in $CH_2Cl_2$ (3.5 mL), MeCN (3.5 mL), and $H_2O$ (5.5 mL). After stirring for 20 min $CH_2Cl_2$ (40 mL) was added and the separated aqueous layer extracted with $CH_2Cl_2$. The combined organic layers were washed with brine, dried ($Na_2SO_4$), and concentrated in vacuo. Column chromatography (silica gel, $CH_2Cl_2$/MeOH 95:5) of the residue afforded the sulfonamide **64** as a solid; yield: 92%.

## 10.9.8 ψ[NH–$SO_2$] (Retro-sulfonamide) Peptides

The retro-inverso concept[133] was adopted to examine the synthesis and conformational properties by X-ray diffraction analysis of retro-sulfonamide ψ[NH–$SO_2$] peptide analogues.[134]

Access to this type of compound is illustrated in Scheme 30 by the preparation of retro-sulfonamide tripeptide Boc-Pro-Leuψ[NH–$SO_2$]Gly-$NH_2$ (**78**). The two "false" termini used are the prochiral *gem*-diamino analogue of Leu and sulfoacetic acid. Amide to amine conversion according to Hofmann, carried out on the dipeptide Boc-Pro-Leu-$NH_2$ (**76**) with iodobenzene 1,1-bis(trifluoroacetate) gave the *gem*-diamino derivative **77**. Coupling of the resulting *gem*-diamino derivative with methyl (chlorosulfonyl)acetate, followed by amidation of the intermediate methyl ester, afforded the desired pseudopeptide **78**.[134] Full experimental details have not yet been reported.

**Scheme 30** Synthesis of the Retro-sulfonamido Peptide Boc-Pro-Leuψ[NH–$SO_2$]Gly-$NH_2$[134]




**76** **77** **78**

**Boc-Pro-Leuψ[NH–$SO_2$]Gly-$NH_2$ (78):**[134]

The title compound was synthesized according to the steps outlined in Scheme 30. The title compound and its immediate precursor were characterized as follows:

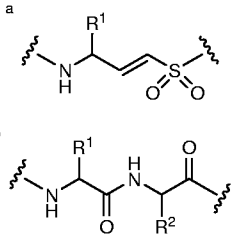
*gem*-Diamino derivative: mp 94–95 °C; $[\alpha]_D$ –110 (c 0.5, $CHCl_3$); IR (KBr): ṽ 3386, 3301, 2958, 1660, 1415 $cm^{-1}$.

ψ[NH-$SO_2$] pseudotripeptide **78**: mp 197–199 °C; $[\alpha]_D$ –56 (c 0.5, DMF); IR (KBr): ṽ 3415, 3282, 3207, 1696, 1677, 1662, 1368, 1167 $cm^{-1}$.

## 10.9.9 ψ[C=C–$SO_2$NH] (Vinylogous Sulfonamide) Peptides

Vinylogous sulfonamide peptides may be considered as pseudopeptides, dipeptide mimetics, or pseudopeptides with five atoms in the backbone, instead of the usual six atoms in proteinogenic dipeptides (Scheme 31).

**Scheme 31** Comparison of (a) a Vinylogous Sulfonamide Peptide with (b) a Dipeptide

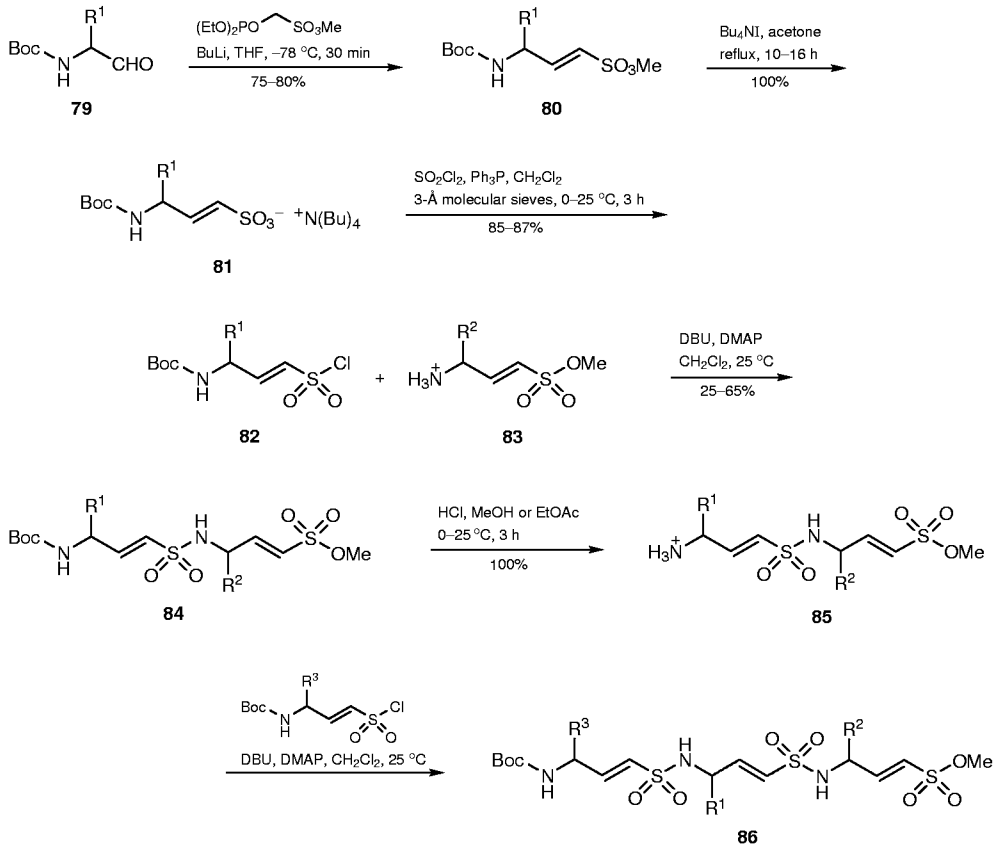



Intramolecular dipolar attraction, including hydrogen bonding, was shown to be a principal driving force for folding in these systems.[135] The precursor vinylogous sulfonyl amino acid residues have been described by adding the "vs" prefix to the three letter code of the corresponding amino acid, e.g. vsAla is vinylogous sulfonyl Ala.[100]

Chiral vinylogous amino sulfonic acids (vs-amino acids) are synthesized starting from either L- or D-α-amino acids via *N*-Boc-α-amino aldehydes **79** (Scheme 32). The Wittig–Horner reaction of an *N*-Boc-α-amino aldehyde with methyl (or ethyl) diethylphosphoryl methanesulfonate and BuLi gave the corresponding γ-amino-α,β-unsaturated sulfonate **80** in high yield and complete *E*-stereoselectivity. Cleavage of the alkyl (methyl or ethyl) ester was effected by treatment of the sulfonate with $Bu_4NI$ in refluxing acetone to give **81**. Treatment of the $Bu_4N^+$ sulfonate salt **81** derived from Ala, Phe, Val, Leu, or Pro with $SO_2Cl_2/PPh_3/CH_2Cl_2$ gave the corresponding sulfonyl chlorides **82** as stable chromatographable compounds. The synthetic sequence was also successful with functionalized α-amino acids (Ser, Tyr, Gln) provided that the side chains were suitably protected. The *N*-Boc-protected, γ-amino-α,β-unsaturated sulfonyl chlorides **82** were coupled with other, γ-amino-α,β-unsaturated sulfonates **83** to give ψ[CH=CH–$SO_2$–NH]-dipeptides **84**. Amine hydrochlorides **85** were prepared from *N*-Boc derivatives **84** by treatment with HCl in methanol or EtOAc. The process was further iterated to give vs-tripeptides and vs-tetrapeptides and it was adapted for the solid-phase synthesis of vinylogous sulfonamide peptides.[136] The above procedure was also used to synthesize mixed peptides, which incorporate both proteinogenic α-amino acids and vs-amino acids. Proteinogenic α-amino acids were incorporated at both the C-terminal and N-terminal positions.[100]

*for references see p 489*

**Scheme 32** Synthesis of Vinylogous Sulfonamide Peptides[100]




**Boc-vsAla-OEt:**[100]

A soln of ethyl (diethylphosphoryl)methanesulfonate (5.0 g, 19.2 mmol) in THF (72 mL) was treated at −78 °C, under $N_2$, with 1.6 M BuLi in hexane (13.2 mL, 21.1 mmol). After 20 min stirring at −78 °C, a soln of Boc-Ala-H (3.3 g, 19.2 mmol) in THF (5.0 mL) was added. After 30 min, a pH 7 phosphate buffer was added and the aqueous phase extracted with $Et_2O$. The combined organic extracts were dried, and concentrated in vacuo. The crude residue was purified by flash chromatography (hexane/EtOAc 7:3) and crystallized (hexane/EtOAc 8:2) to give the title sulfonate; yield 4.18 g (78%); mp 69–71 °C; $[\alpha]_D$ −18.1 (*c* 0.98, $CHCl_3$).

**H-vsAla-OEt · HCl:**[100]

Boc-vsAla-OEt (250 mg, 0.89 mmol) was treated with 3 M HCl in MeOH (5 mL) at 0 °C under $N_2$. The mixture was stirred at rt for 3 h, monitoring the disappearance of the starting material by TLC (hexane/EtOAc 6:4). The solvent was concentrated under reduced pressure and the product dried in vacuo; yield: 192 mg (quant). The hydrochloride salt was used in the subsequent reaction without further purification.

**Boc-vsAlaON(Bu)$_4$:**[100]

A soln of Boc-vsAla-OEt (3.6 mmol) in acetone (20 mL) was treated with $Bu_4NI$ (recrystallized from EtOAc/MeOH 95:5; 1.33 g, 3.6 mmol), under $N_2$ with stirring. The mixture was refluxed for 16–20 h monitoring the disappearance of the starting material by TLC (hexane/EtOAc 6:4). Removal of the solvent under reduced pressure gave quantitatively the tetrabutylammonium sulfonate.

**Boc-vsAla-Cl:**[100]

$SO_2Cl_2$ (180 mg, 0.107 mL, 1.33 mmol) was added to a soln of $Ph_3P$ (320 mg, 1.224 mmol) in $CH_2Cl_2$ (1.5 mL) at 0 °C under $N_2$, in the presence of 3-Å molecular sieves. A soln of Boc-vsAla-ONBu$_4$ (302 mg, 0.611 mmol) in $CH_2Cl_2$ (2.0 mL) was then added at rt under $N_2$ with stirring. The mixture was stirred at rt for 2.5 h, and the solvent was removed in vacuo. The crude product was purified by flash chromatography (hexane/EtOAc 6:4) to give the title compound; yield: 142 mg (85%).

**Boc-vsAla-vsAla-OEt:**[100]
Boc-vsAla-Cl (189 mg, 0.70 mmol) was dissolved in $CH_2Cl_2$ (7.0 mL) at rt under $N_2$, and a soln of H-vsAla-OEt·HCl (74.6 mg, 0.35 mmol) in $CH_2Cl_2$ (2.0 mL), containing DBU (0.104 mL, 0.70 mmol) and DMAP (43 mg, 0.35 mmol) was added at once. The mixture was stirred for 5 h at rt, then diluted with $CH_2Cl_2$ and treated with 10% aq citric acid (2.0 mL). The aq phase was extracted with $CH_2Cl_2$ and the combined organic extracts were dried and concentrated. The crude product was purified by flash chromatography (hexane/EtOAc 1:1) to give the title compound; yield: (65%).


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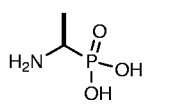
# 10.10 Synthesis of Peptides with Phosphorus-Containing Amide Bond Replacements

P. A. Bartlett, J. E. Hanson, B. P. Morgan, and B. A. Ellsworth

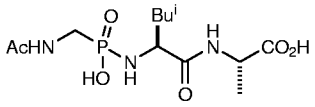
## 10.10.1 Phosphorus Analogues of α-Amino Acids

The methods used for assembly of phosphono-peptide analogues parallel those for conventional peptide synthesis, although different reagents and different coupling and protection strategies are required. Moreover, few α-aminoalkylphosphonic or -phosphinic acid analogues are available commercially, so a key step is preparation of the subunits themselves. Considerable advances have been made since this field was reviewed by Redmore in 1976.[1] This section highlights the most versatile methods currently employed for synthesis of these subunits, for incorporating them into peptide analogues with P–N, P–O, and P–C linkages in place of CO–NH (Scheme 1), and for deprotecting the product selectively. For convenience, the phosphorus analogues are referred to with a superscript P (as in $Ala^P$) for the amino acids, and with explicit indication of the linking atom to the adjacent residue in an oligopeptide, if other than nitrogen (Scheme 1). Conventional nomenclature is followed for the functional groups themselves.[2] Amino acid analogues in which the phosphorus moiety is contained in the side chain are not addressed in this section.

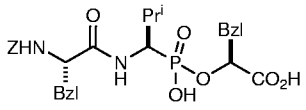
**Scheme 1** Phosphorus Analogues of Amino Acids and Peptides



(*R*)-1-aminoethylphosphonic acid
L-$Ala^P$



Ac-$Gly^P$-Leu-Ala-OH



Z-Phe-$Val^P$-(O)Phe-OH

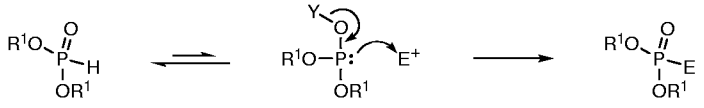
α-Aminoalkyl phosphorus compounds $[H_2NCH(R^1)P(O)(X)(OR^2)]$ are generally prepared in either the phosphonate ($X = OR^3$) or phosphinate (X = H, alkyl) oxidation states, with the choice dictated by their eventual application. Conversion of the hydrogen phosphinates (X = H) into the alkylphosphinates (X = alkyl) by alkylation or into the phosphonates by oxidation is relatively straightforward. Dealkylation of an alkylphosphinate is possible under special circumstances [e.g., $X = CH(OR^3)_2$], while reduction from the phosphonate to the phosphinate oxidation level is generally not possible.

The biological properties of phosphorus amino acid analogues (and their derivatives) depend upon their stereochemistry. Consequently, numerous methods for obtaining these compounds in stereochemically pure form have been developed. Two excellent review articles summarize the work performed prior to 1993.[3,4] Resolution of racemates continues to be a useful approach for obtaining optically pure aminoalkylphosphonic and -phosphinic acid derivatives (*vide infra*), but most of the newer literature describes asymmetric syntheses of these compounds.[5–17] Two methods for resolution and one for asymmetric synthesis are described (*vide infra*); they have been selected since they are relatively easy to perform, work with a variety of side chains, can be carried out on a reasonable scale with readily available starting materials, and produce products of high stereopurity. However, just as in traditional amino acid chemistry, each side chain introduces its own complications, and in many cases, especially for more complex analogues, other methods may be preferred.

### 10.10.1.1 Synthesis of Phosphonate Amino Acid Analogues

Most syntheses of phosphorus amino acid analogues involve the conversion of a nucleophilic, trivalent phosphorus species into a pentavalent adduct. The classical Arbuzov–Michaelis reaction is a well-known transformation that demonstrates this principle.[18] In the case of a phosphite diester, the stable form of the reagent is the P–H derivative, depicted in Scheme 2;[19] this tautomer is converted into a nucleophilic form by deprotonation or by silylation, which favors the trivalent P–O–Si isomer.

**Scheme 2** Tautomeric States of Hydrogen Phosphites and Reactions with Electrophiles[19]



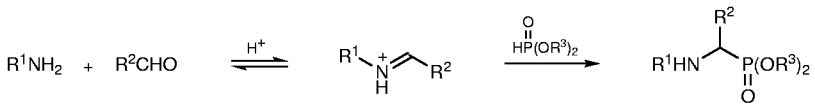
Y = H, Li, Na, TMS; E = electrophile

Nucleophilic phosphorus species are employed in the synthesis of amino acid analogues by condensation with imine derivatives, in parallel with the classical Strecker reaction. A variety of methods are available, depending on the selection of the phosphorus reagents and the imine precursors.

#### 10.10.1.1.1 Strecker-Type Synthesis of α-Aminoalkylphosphonic Acids

Many syntheses of α-aminoalkylphosphonic acids **2** employ the condensation of phosphite esters with aldimines **1** formed in situ.[20] A number of alkylamines have been employed in such condensations,[21] although the most useful procedures lead directly to unsubstituted amines or employ *N*-alkyl groups that are easily removed (Scheme 3). The amino-protecting group is avoided entirely in the condensation with ammonium acetate;[22] although the yields for this transformation are generally modest (30–50%), the reagents are inexpensive and the reaction can be carried out conveniently on large scale. An alternative procedure employs tritylamine as an alkyl group that is readily removed with acid. With most aldehydes, tritylamine forms isolable imines that give high yields (65–95%) of N-protected α-aminoalkylphosphonates on condensation with dimethyl phosphite.[23]

**Scheme 3** Strecker-Type Synthesis[22,23]



**1** **2**

| Amine $R^1NH_2$ | Aldehyde $R^2CHO$ | $R^3$ | Yield (%) | mp[a] (°C) | Ref |
|---|---|---|---|---|---|
| $TrtNH_2$ | HCHO | Me | 95 | 210–211 | [23] |
| $TrtNH_2$ | MeCHO | Me | 75 | 191–195 | [23] |
| $TrtNH_2$ | EtCHO | Me | 70 | 143–145 | [23] |
| $NH_3$ | iPrCHO | Et | 48 | n.r. | [22] |
| $TrtNH_2$ | iPrCHO | Me | 65 | 132–134 | [23] |
| $NH_3$ | iBuCHO | Et | 32 | n.r. | [22] |
| $TrtNH_2$ | iBuCHO | Me | 75 | 91–95 | [23] |
| $NH_3$ | PhCHO | Et | 59 | 159–160[b] | [22] |
| $NH_3$ | 4-$MeOC_6H_4CHO$ | Et | 58 | n.r. | [22] |

[a] n.r. = not reported.
[b] HCl salt.

*for references see p 525*

**Diethyl 1-[Amino(phenyl)methyl]phosphonate Hydrochloride Salt (2·HCl, $R^1$ = H; $R^2$ = Ph; $R^3$ = Et); Typical Procedure:**[22]

To a soln of $NH_4OAc$ (7.7 g, 0.1 mol) in EtOH (200 mL) were added 3-Å molecular sieves (2.0 g), benzaldehyde (10.6 g, 0.1 mol), and diethyl phosphite (13.8 g, 0.1 mol) at rt. The mixture was stirred at 60 °C for 44 h, cooled, then acidified to pH 1 with aq HCl. The soln was washed with $Et_2O$ (×2) to remove neutral materials, the aqueous phase was made alkaline (ca. pH 11) with aq NaOH, and the product was extracted with three portions of $CH_2Cl_2$. The organic phases were combined, dried ($Na_2SO_4$), and concentrated in vacuo to give almost pure product as a pale yellow oil; yield: 14.4 g (59%). The product was treated with HCl in EtOH/$Et_2O$ (1:1, 20 mL) to give the crystalline hydrochloride salt; yield: 14.9 g (53%); mp 159–160 °C.

**1-[(Tritylamino)alkyl]phosphonates** 2 **($R^1$ = Trt); General Procedures:**[23]

***N*-Tritylalkanimines** 1 **($R^1$ = Trt):**

A mixture of tritylamine (13 g, 0.050 mol) and a freshly distilled aldehyde (0.055 mol) in benzene (50 mL) was heated to reflux in a Dean–Stark distillation device with a calibrated receiver. When the calculated amount of $H_2O$ (0.9 mL) had separated, the mixture was concentrated to give the product, which was recrystallized. For acetaldehyde imine **1** ($R^1$ = Trt; $R^2$ = Me); yield: 100%; mp 144–147 °C (dec).

**Dimethyl 1-(Tritylamino)alkylphosphonates** 2 **($R^1$ = Trt):**

A mixture of *N*-tritylalkanimine (**1**, $R^1$ = Trt; 0.010 mol) and dimethyl phosphite (5.5 g, 0.05 mol) was heated at 100 °C (steam bath) for 0.5–4 h. Excess dimethyl phosphite was removed in vacuo, and the residue was crystallized from an appropriate solvent. The product was isolated by filtration, washed with the same solvent, and dried in vacuo at rt. For $Ala^P$ analogue **2** ($R^1$ = Trt; $R^2$ = $R^3$ = Me); yield: 75%; mp 191–195 °C (dec).

### 10.10.1.1.2 Phosphonic Amino Acids and Their Derivatives via *N*-Acyliminium Ion Intermediates

*N*-Acyliminium ions are more reactive partners for condensation reactions than the standard Strecker intermediate;[24] they are not generally isolable, but must be generated in situ and trapped to give the stable product. However, with proper choice of acyl group, they provide α-amidophosphorus analogues that are appropriately protected for peptide synthesis.

#### 10.10.1.1.2.1 Condensation with Triaryl Phosphites

The condensation of an aldehyde, benzyl carbamate, and triphenyl phosphite, first described by Oleksyszyn et al.,[25,26] affords a direct route to α-aminoalkylphosphonates **4** that are conveniently protected for subsequent reactions (Scheme 4). Since dealkylation of the quaternary phosphonium intermediate **3** is not possible in this case, formation of the pentavalent product **4** presumably involves activation of the solvent and formation of phenyl acetate. This method is useful for the synthesis of aliphatic and aromatic amino acid analogues. However, monomers with more elaborate side chains are often incompatible with the reaction conditions. The free amine can be liberated by treatment with HBr/AcOH or by hydrogenolysis after removal of the phenyl esters. The phosphonate moiety can be manipulated by ready exchange of the phenyl esters in alkaline MeOH and activation as described in Section 10.10.2.1.1. Related condensations with other trivalent phosphite derivatives have been reported.[27–30]

**Scheme 4** Synthesis of α-Carboxamidoalkylphosphonates from Triphenyl Phosphite[25–30]

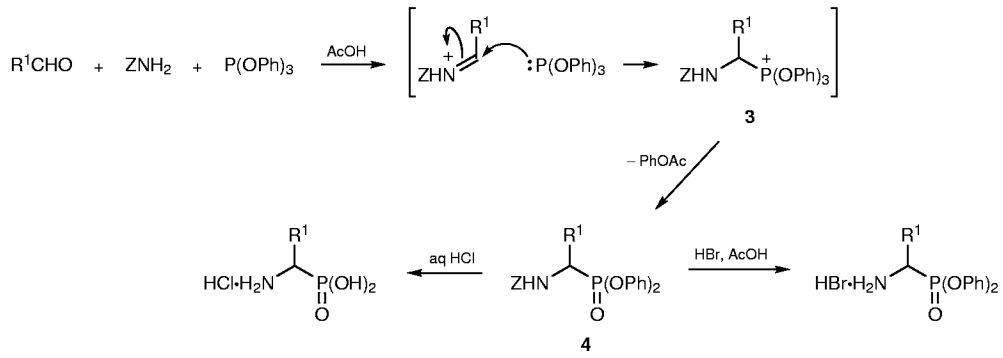
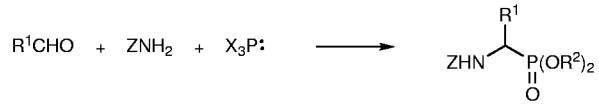



**Table 1** Synthesis of Diphenyl [1-(Benzyloxycarbonylamino)alkyl]phosphonates




| Aldehyde $R^1CHO$ | Reagent | $R^2$ | Yield (%) | mp (°C) | Ref |
|---|---|---|---|---|---|
| MeCHO | $P(OPh)_3$ | Ph | 46 | 115–117 | [26] |
| MeCHO | 1. $PCl_3$; 2. MeOH | Me | 22 | 118–120 | [28] |
| iPrCHO | $P(OPh)_3$ | Ph | 52 | 104–105 | [26] |
| iPrCHO | 1. $PCl_3$; 2. MeOH | Me | 61 | 103–104 | [28] |
| PhthN$(CH_2)_3$CHO | $P(OPh)_3$ | Ph | 50 | 72–74 | [31] |
| PhthN$(CH_2)_4$CHO | $P(OPh)_3$ | Ph | 50 | 112–114 | [31] |
| iBuCHO | $P(OPh)_3$ | Ph | 54 | 122–123 | [26] |
| iBuCHO | $(MeO)_2P(O)H$ | Me | 66 | oil | [27] |
| BzlCHO | $P(OPh)_3$ | Ph | 35 | 119–120 | [26] |
| PhCHO | $P(OPh)_3$ | Ph | 51 | 138–140 | [26] |
| PhCHO | $(MeO)_2P(O)H$ | Me | 88 | 120–121 | [27] |

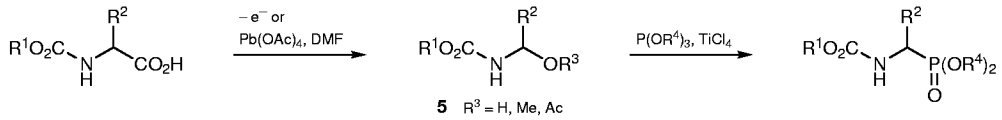
**Diphenyl [1-(Benzyloxycarbonylamino)alkyl]phosphonates 4; General Procedure:**[26]
A mixture of triphenyl phosphite (0.1 mol), aldehyde (0.15 mol), $ZNH_2$ (0.1 mol), and glacial AcOH (15 mL) was stirred for ca. 1 h until the exothermic reaction subsided. The mixture was heated at 80–85 °C for 1 h and volatile products were removed in vacuo. The oily residue was dissolved in MeOH (180 mL) and left for crystallization at −10 °C. After 1–3 h, the crystalline ester was collected by filtration and recrystallized by dissolution in the minimum amount of hot $CHCl_3$ (30–40 mL) and addition of a fourfold volume of MeOH.

### 10.10.1.1.2.2 Condensation with Aminal Derivatives

Other equivalents of, or precursors to, imine derivatives have also been employed in the synthesis of α-aminoalkylphosphonates. *N*-(Hydroxymethyl)phthalimide is converted into the corresponding bromide with $PBr_3$ which is converted via Arbuzov displacement into the phosphonate analogue of Gly.[32] A modification using a silylated phosphonite [$(TMSO)_2PH$] in the Arbuzov reaction leads to the phosphinate analogue of Gly[33] (amino acids of this type will be discussed in Section 10.10.1.2.1). This alkylation approach is popular for the synthesis of Gly analogues since formaldehyde is difficult to use under anhydrous conditions. Conversion of the natural amino acids into their phosphonic acid analogues can be accomplished by oxidative decarboxylation, either electrochemically[34] or with lead(IV) acetate,[35] to give *N*-acyl-*N*,*O*-acetals **5**, which react with phosphite esters via the acylimine (Scheme 5). This method is particularly useful for synthesis of $Ser^P$, $Thr^P$, and $Asp^P$, since the aldehyde precursors are otherwise hard to obtain,[35] and for replacement of the C-terminal carboxylate directly on a more complex peptide derivative.[34]

*for references see p 525*

**Scheme 5** Conversion of α-Amino Acids into α-Aminoalkylphosphonates[34,35]




**5** $R^3$ = H, Me, Ac

**Methyl *N*-[1-Acetoxy-2-(*tert*-butyldimethylsiloxy)propyl]carbamate; Typical Procedure:**[35]
A soln of Moc-Thr(TBDMS)-OH (0.103 g, 0.35 mmol) in dry DMF (0.4 mL) was added to a cold, stirred suspension of $Pb(OAc)_4$ (0.181 g, 0.41 mmol) in dry DMF (0.4 mL). The cooling bath was removed after 30 min, and the suspension was stirred for an additional 3.5 h; sat. aq $NaHCO_3$ was added (4 mL) and the mixture was immediately extracted with EtOAc (4 × 3 mL). The combined organic extracts were washed with sat. aq $NaHCO_3$ and brine, then dried ($Na_2SO_4$), filtered, and concentrated at rt to yield the product; yield: 75.6 mg (70%). [This material was treated with trimethyl phosphite and $TiCl_4$ under the reaction conditions described below for the *O*-methoxy-*N*,*O*-acetal derivative, to give the phosphonic acid analogue of Thr in 98% yield.]

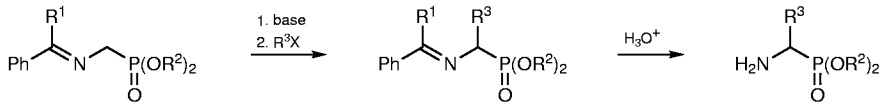
**Diphenyl [(1*S*,2*R*)- and (1*R*,2*R*)-2-(*tert*-Butyldimethylsiloxy)-1-(methoxycarbonylamino])propyl]phosphonate; Typical Procedure:**[34]
A soln of methyl *N*-[(2*R*)-2-(*tert*-butyldimethylsiloxy)-1-methoxypropyl]carbamate (9.26 g, 33 mmol), prepared from the amino acid derivative by electrolysis, and triphenyl phosphite (10.5 mL, 40 mmol) in $CH_2Cl_2$ (100 mL) was cooled to −78 °C and 2 M $TiCl_4$ in $CH_2Cl_2$ (18.5 mL, 37 mmol) was added over 10 min. The soln was allowed to warm to rt over 10 h, added to a 1.25-fold volume of a suspension of $Na_2CO_3 \cdot 10\ H_2O$ (1.75 equiv) in $CH_2Cl_2$, and the mixture was stirred for 30 min at rt. After filtration and removal of the solvent, crude product (19.5 g) was obtained, as a 1:2 ratio of diastereomers. Flash chromatography ($Et_2O$/pentane 1:5) gave the (*R*,*R*)-diastereomer (6.0 g, 38% yield), the (*S*,*R*)-diastereomer (3.30 g, 20%), and a mixture (1.9 g, 12%). The diastereomers were crystallized ($Et_2O$/pentane) to provide analytical samples; (*S*,*R*)-diastereomer: mp 105–106 °C; $[\alpha]_D^{rt}$ −0.2 (*c* 2, $CHCl_3$); (*R*,*R*)-diastereomer: mp 65–66 °C; $[\alpha]_D^{rt}$ +17.8 (*c* 1.4, $CHCl_3$).

**[(1*S*,2*R*)- and (1*R*,2*R*)-1-Amino-2-hydroxypropyl]phosphonic Acid; Typical Procedure:**[34]
The Thr phosphonate diastereomers from above were hydrolyzed, individually, in refluxing 6 M HCl (20 mL/g) for 6 h. The solns were washed with $CH_2Cl_2$ and concentrated, and the residues were purified by ion-exchange chromatography (Dowex 50W X8, $H_2O$) and by recrystallization. (*S*,*R*)-Diastereomer (EtOH/$H_2O$); yield: 250 mg (77%); mp 220–221 °C; $[\alpha]_D^{rt}$ +9.6 (*c* 1.5, $H_2O$); $^{31}P$ NMR ($D_2O$, δ): 1.12. (*R*,*R*)-Diastereomer (MeOH/$H_2O$): yield: 260 mg (80%); mp 234–236 °C; $[\alpha]_D^{rt}$ −9.4 (*c* 1.8, $H_2O$); $^{31}P$ NMR ($H_2O$, δ): 1.52.

### 10.10.1.1.3 Alkylation of Protected Aminomethylphosphonic Acids

Alkylation and deprotection of N-protected aminomethylphosphonate esters **6** are shown in Scheme 6. The nitrogen is protected as the imine derived from benzophenone or a benzaldehyde, and a variety of conditions are used for deprotonation and alkylation (Table 2). The benzaldehyde imine of aminomethylphosphonate can be deprotonated with LDA and alkylated with electrophilic halides (entries 1 and 2). For the best yields, saturated alkyl bromides require an equivalent of HMPA as an additive.[36] Allylic esters can be added to the carbanion with palladium catalysis (entries 3–7).[37,38] For large-scale production, phase-transfer catalysis appears to be effective and inexpensive (entries 8–12).[39,40]

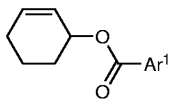
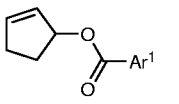
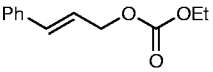
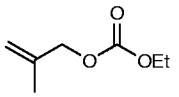
**Scheme 6** Alkylation of Protected Aminomethylphosphonates[36–40]




**6** **7**

$R^1$ = H, Ph

**Table 2** Alkylation of Protected Aminomethylphosphonates

| Entry | Imine **6** | | Conditions | $R^3X$ | Product **7** | | Ref |
|---|---|---|---|---|---|---|---|
| | $Ph(R^1)C=$ | $R^2$ | | | Yield (%) | mp[a] (°C) | |
| 1 | PhCH= | Et | LDA, THF, −78 °C to rt | MeI | 70 | 279 | [36] |
| 2 | PhCH= | Et | LDA, THF, −78 °C to rt | BzlBr | 92 | 265–6 | [36] |
| 3 | $Ph_2C=$ | Et | 1. LDA; 2. $Pd(dppe)_2$,[b] DME, −10 °C | $CH_2=CHCH_2OAc$ | 70 | n.r. | [37] |
| 4 | $4\text{-}ClC_6H_4CH=$ | Et | 1. NaH; 2. $Pd(dppe)_2$,[b] DME, 20 °C |  | 80[c] | n.r. | [37] |
| 5 | $Ph_2C=$ | Et | 1. NaH; 2. $Pd(dppe)_2$,[b] DME, 20 °C |  | 78[c] | n.r. | [37] |
| 6 | $Ph_2C=$ | Et | $Pd(dppe)_2$,[b] DME, 70–80 °C |  | 70 | n.r. | [37] |
| 7 | $Ph_2C=$ | Et | $Pd(dppe)_2$,[b] DME, 20 °C |  | 65 | n.r. | [38] |
| 8 | $Ph_2C=$ | Et | KOH, TBAB, $CH_2Cl_2$ | $HC\equiv CCH_2Br$ | 84 | 87–88 | [39] |
| 9 | $Ph_2C=$ | Et | KOH, Aliquat[d] | BuBr | 70 | 71–72 | [39] |
| 10 | $Ph_2C=$ | tBu | KOH, Aliquat[d] | $CH_2=CBrCH_2Br$ | 69 | n.r. | [40] |
| 11 | $Ph_2C=$ | tBu | KOH, Aliquat[d] | $H_2C=CHCO_2Me$ | 70 | n.r. | [40] |
| 12 | $Ph_2C=$ | tBu | KOH, Aliquat[d] | $H_2C=CHCN$ | 96 | n.r. | [40] |

[a] n.r. = not reported.
[b] $Pd(dppe)_2$ formed in situ with either $Pd(dba)_2$ or $Pd(OAc)_2$ and 1,2-bis(diphenylphosphanyl)ethane.
[c] $Ar^1 = 2,4\text{-}Cl_2C_6H_4$.
[d] No additional solvent used.

When ethyl phosphonates are used, the imine can be selectively deprotected after alkylation by treatment with dilute mineral acid (e.g., 10% HCl),[36–40] and the esters removed subsequently (see Section 10.10.3). When *tert*-butyl phosphonates are used, imine and ester hydrolysis is accomplished simultaneously under these conditions. These mild hydrolysis conditions are preferable when synthesizing amino acids with functionalized side chains.[40]

**Alkylation of Protected Aminomethylphosphonate Using Palladium Catalysis; General Procedure:**[38]
To a stirring suspension of NaH (40 mg, 1.0 mmol, 50% in mineral oil) in dry THF under argon was added diethyl [(diphenylmethylene)aminomethyl]phosphonate (1 mmol) in THF (1 mL) and the resultant mixture was stirred for 1 h. A soln of $Pd(dppe)_2$ (0.5 equiv) and an allylic acetate or benzoate (1.1 mmol) in THF was added at temperatures between −60 °C and 25 °C under argon. The progress of the reaction was monitored by TLC (silica gel). Sat. $NH_4Cl$ (2 mL) was added, and the mixture was extracted with $Et_2O$. The organic layer was washed with brine and concentrated in vacuo, and the crude product was purified by flash chromatography (silica gel).

**Alkylation of Protected Aminomethylphosphonate Using Phase-Transfer Conditions; General Procedure:**[39]
A mixture of diethyl [(diphenylmethylene)aminomethyl]phosphonate (1 mmol), Aliquat 336 (20 mg, 0.05 mmol), the appropriate halide (1.5 mmol), and finely ground KOH (300 mg, 5 mmol) was stirred at rt, and the reaction progress was monitored by TLC (silica gel). The mixture was worked up by adding $CH_2Cl_2$ (10 mL) and slurrying with silica gel (50 mg). After filtration, the solvent was removed in vacuo, and the residue was purified by flash chromatography (silica gel).

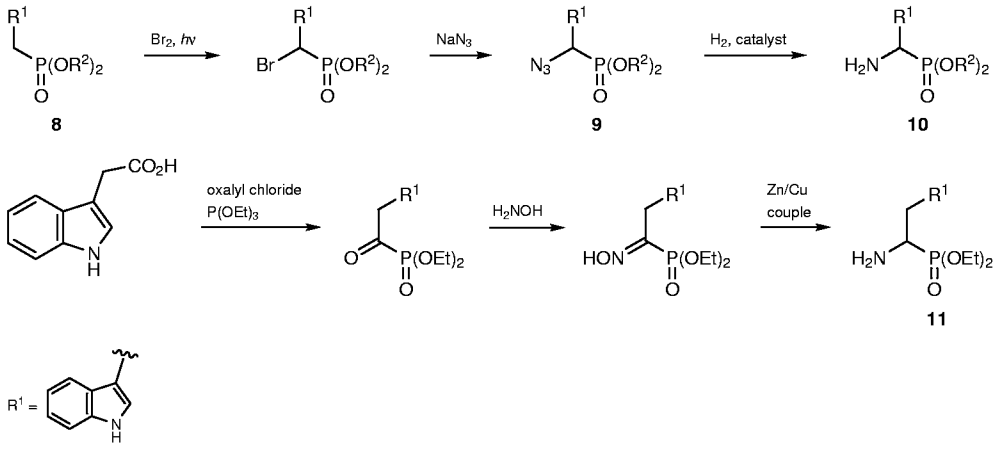
*for references see p 525*

### 10.10.1.1.4 Amination of Alkyl- and Acylphosphonates

Several syntheses of α-aminoalkylphosphonates involve initial P–C bond formation followed by amination of the α-carbon atom. For example, α-azidophosphonates **9** can be generated by radical halogenation of alkyl phosphonates **8** followed by displacement with sodium azide. Hydrogenation of the azide provides the free amino phosphonate **10** (Scheme 7).[41]

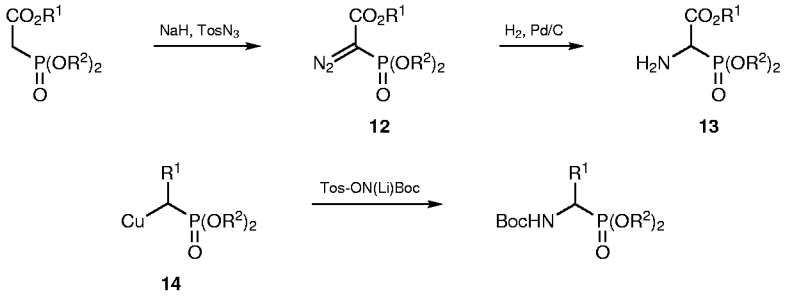
Acylphosphonates, which are accessible via phosphite acylation,[42] can be reductively aminated to give the free amino phosphonates directly (Scheme 7).[43] The phosphonate analogue of Trp **11**, which is difficult to form via other methods, was generated in this manner.[43] Yet another approach is via the α-diazoalkylphosphonate, which can be aminated in the presence of rhodium catalysts.[44,45] These methods are useful when more complex amine components are required; otherwise, the standard Strecker-type syntheses are generally more cost-effective.

**Scheme 7** Amination of Alkyl-[41] and Acylphosphonates[42,43]




α-Aminophosphonoacetates **13**, which are key intermediates for cephalosporin synthesis, have been synthesized by amination of the enolate of α-phosphonoacetates (Scheme 8). A variety of esters have been employed, including methyl, ethyl, *tert*-butyl, 2,2,2-trichloroethyl, or benzyl. Several amination reagents have been used, including *O*-(diphenylphosphoryl)-hydroxylamine,[46] *O*-mesitylenesulfonylhydroxylamine,[47] chloramine, or diazo transfer with 4-toluenesulfonyl azide followed by catalytic hydrogenation.[48] The direct transfer of an electrophilic $BocNH^+$ equivalent to α-cuprophosphonates **14** has been accomplished with lithiated *tert*-butyl *N*-tosyloxycarbamate.[49]

**Scheme 8** Electrophilic Amination of Alkylphosphonates[46–49]



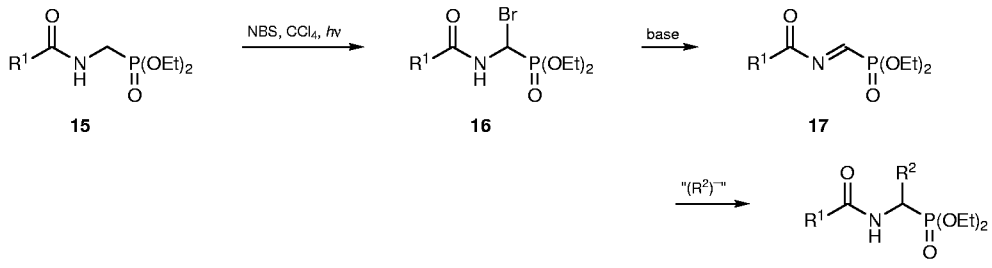
***tert*-Butyl Diazo(diethoxyphosphoryl)acetate (12, $R^1$ = tBu; $R^2$ = Et); Typical Procedure:**[48]
To a stirred suspension of NaH (1.056 g, 40 mmol) in dry DME (40 mL) at 0 °C was added *tert*-butyl diethoxyphosphorylacetate (10.08 g, 40 mmol) dropwise over 20 min, and the mixture was stirred at 0 °C. After 1 h, a soln of $TosN_3$ (7.88 g, 40 mmol) in DME (10 mL) was added over 20 min, and stirring was continued at rt for 3 h. The mixture was filtered, the solvent was removed, and the residue was dissolved in $Et_2O$. The $Et_2O$ soln was washed with 0.5 M NaOH, $H_2O$, and brine, dried ($Na_2SO_4$), filtered, and concentrated to afford the product as a red-brown oil; yield: 9.0 g (81%).

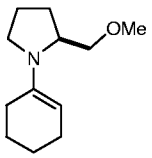
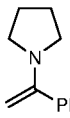
***tert*-Butyl Amino(diethoxyphosphoryl)acetate (13, $R^1$ = tBu; $R^2$ = Et); Typical Procedure:**[48]
A suspension of *tert*-butyl diazo(diethoxyphosphoryl)actetate (**12**, $R^1$ = tBu; $R^2$ = Et; 9.0 g, 32 mmol), 10% Pd/C (4 g), and TosOH (6.18 g, 32 mmol) in MeOH (150 mL) was hydrogenated at 4.1 atm $H_2$ pressure in a Parr apparatus at rt for 2 h. After filtration, the solvent was removed, and the residue was dissolved in 1 M HCl (250 mL). The aqueous soln was washed with EtOAc, neutralized with $Na_2CO_3$, and extracted with $CHCl_3$. The organic soln was washed with brine, dried ($Na_2SO_4$), and concentrated to afford the product as a pale yellow oil; yield: 6.58 g (76%).

### 10.10.1.1.5 Reaction of α-Iminoalkylphosphonates with Nucleophiles

Acylaminomethylphosphonates **15** react with *N*-bromosuccinimide to form the α-bromo derivatives **16** (Scheme 9). In the presence of base, HBr is eliminated to form the acylimi-nomethylphosphonates **17**, which react with nucleophiles such as enamines or organocuprates to afford the alkylated products **18**. DIPEA is generally used when an enamine is the nucleophile; alternatively, 2 equivalents of an organocuprate serve as both the base and nucleophile.[50,51]

**Scheme 9** Nucleophilic Addition to α-Iminoalkylphosphonates[50,51]




| $R^1$ | Base | Nucleophile | $R^2$ | Yield (%) | mp (°C) | Ref |
|---|---|---|---|---|---|---|
| Ph | DIPEA |  | 2-oxocyclohexyl | 43 | 91–92 | [50] |
| $OCH_2CCl_3$ | DIPEA |  | $CH_2COPh$ | 60 | 115 | [50] |
| Ph | $Li_2CuPh(CN)$ | | Ph | 90 | 68 | [51] |
| Ph | $Li_2CuBu(CN)$ | | Bu | 58 | 61 | [51] |

**Diethyl [(Acylamino)bromomethyl]phosphonates 16; General Procedure:**[50]
A soln of diethyl acylaminomethylphosphonate **15** (50 mmol) and NBS (50 mmol) in anhyd $CCl_4$ (150 mL) at 13 °C was photolyzed under a 200-W UV lamp for 1 h. The mixture was filtered, the soln was concentrated, and the crude product was crystallized (EtOAc/hexane) to give the title compounds in 70–95% yield. The bromides are air- and moisture-sensitive and are used directly in the following reaction; alternatively, they can be stored at −40 °C under argon for several weeks.

*for references see p 525*

**Diethyl 1-(Acylamino)alkylphosphonates** 18 **by Addition of an Enamine to Diethyl Acyliminomethylphosphonate** 17**; General Procedure:**[50]

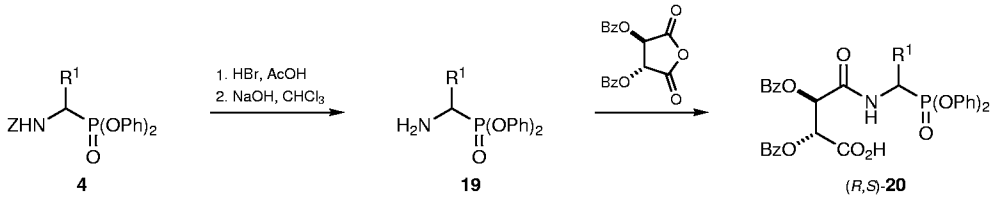
DIPEA (12.5 mmol) was added to a soln of diethyl [(acylamino)bromomethyl]phosphonate **16** (10 mmol) in dry THF (20 mL) at −78 °C. The resultant suspension was warmed to rt for 1 h, then cooled again to −78 °C. The mixture containing the acylimine intermediate **17** was filtered under argon using Schlenk glassware, and the enamine (11 mmol) in THF (20 mL) was added at −78 °C under argon. The mixture was warmed to rt and stirred at 40 °C for 1 h, before 20% aq citric acid was added (10 mL). The mixture was extracted with $Et_2O$ and the organic layer was washed with sat. $NaHCO_3$ and $H_2O$, dried ($MgSO_4$), filtered, and concentrated. Recrystallization of the crude product (EtOAc/hexane) afforded the alkylated product.

### 10.10.1.1.6 Preparation of Optically Active α-Aminoalkylphosphonic Acids via Separation of Covalent Diastereomers

Covalently linking an optically active group (such as an amino acid or peptide) to the amino group of a racemic phosphorus amino acid analogue results in a mixture of two diastereomers that are often separable, either by crystallization or chromatography. This strategy for resolving the racemate is particularly convenient in cases where the ultimate synthetic target has the phosphorus analogue incorporated as the "carboxy" terminal residue of a peptide chain.[52,53] In contrast, covalently linking an optically active group (such as a peptide) to the phosphonyl group of the racemic phosphorus analogue is usually less convenient, since a mixture of four diastereomers is produced due to creation of a new chiral center at phosphorus.

One of the most useful resolution protocols involves reaction of the appropriate diphenyl α-aminoalkylphosphonate **19** (prepared as described in Section 10.10.1.1.2.1) with dibenzoyl L-tartaric anhydride (Scheme 10).[54] The resulting diastereomeric amides **20** have quite different physical and spectral properties (Table 3) and are readily separable by fractional crystallization, with the yields for formation and separation of the diastereomeric amides typically exceeding 70%. The melting point of the diastereomer with the *S* configuration α to phosphorus is dramatically higher than that with the *R*, and the *S*-isomer has a higher $^{31}P$ NMR chemical shift. The diastereomerically pure amides can be converted in good yields (normally >70%) into the enantiomerically pure α-aminoalkylphosphonic acids **21** by treatment with HBr. An attempt to separate the diastereomeric amides resulting from reaction with diphenyl 2-acetoxy-1-aminoethylphosphonate using this approach failed; however separation of the diastereomers was achieved by conversion into the corresponding imide.[55]

**Scheme 10** Resolution of Diphenyl α-Aminoalkylphosphonates Using Dibenzoyl L-Tartaric Anhydride[54]




**4** **19** (*R,S*)-**20**

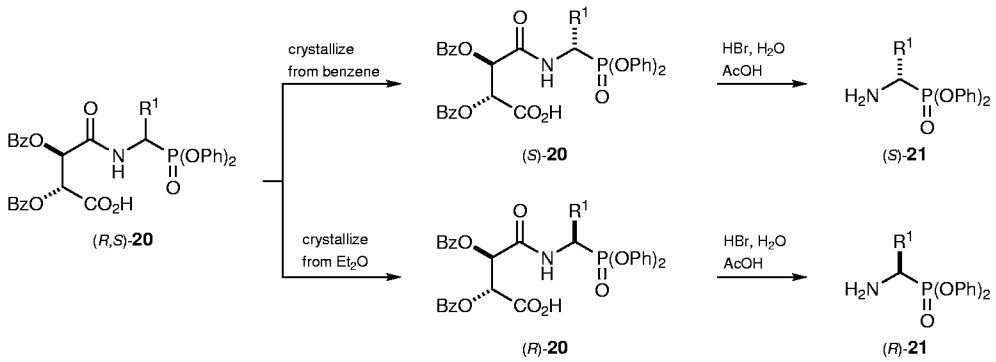



**Table 3** Resolution of α-Aminoalkylphosphonic Acids Using Dibenzoyl L-Tartaric Anhydride[54]

| $R^1$ | Config | **20** | | **21** |
|---|---|---|---|---|
| | | mp (°C) | $^{31}$P NMR (DMSO) δ | $[\alpha]_{578}^{20}$ |
| Me | *S* | 176.5–177 | 14.63 | +17° |
| Me | *R* | 81–83 | 14.45 | −16° |
| iPr | *S* | 186–186.5 | 18.01 | −0.6° |
| iPr | *R* | 82–85 | 17.97 | +0.6° |
| iBu | *S* | 181–182 | 18.69 | +25° |
| iBu | *R* | 89–91 | 18.53 | −24° |
| Ph | *S* | 146–148 | 13.40 | −20° |
| Ph | *R* | 96–98 | 13.09 | +19° |
| Bzl | *S* | 160–161.5 | 17.46 | +52° |
| Bzl | *R* | 79–80 | 17.27 | −49° |

**Diphenyl 1-Aminoalkylphosphonate 19; General Procedure:**[26,54]
A soln of the diphenyl [1-(benzyloxycarbonylamino)alkyl]phosphonate **4** in 45% HBr/AcOH (150 mL per 0.1 mol) was kept at rt for 1 h. Removal of the solvent and volatile products in vacuo yielded the oily hydrobromide, which crystallized upon addition of anhyd $Et_2O$. The free amine was obtained by suspending the hydrobromide in $CHCl_3$ and shaking in a separatory funnel with an equal volume of 2 M NaOH until all the solid dissolved. The $CHCl_3$ layer was separated, washed with brine, dried ($MgSO_4$), and concentrated to yield **19** in >90% yield: mp ($R^1$ = Me) 36–39 °C; mp ($R^1$ = iPr) 60–63 °C; mp ($R^1$ = iBu) 57–58 °C; mp ($R^1$ = Ph) 63–65 °C; mp ($R^1$ = Bzl) 65–67 °C.

**Preparation of Dibenzoyltartramides 20 and Separation of Diastereomers; General Procedure:**[54]

**CAUTION:** *Benzene is a carcinogen.*

Diphenyl 1-aminoalkylphosphonate **19** (0.05 mol) was dissolved in dioxane (200 mL) and dibenzoyl-L-tartaric anhydride (17.0 g, 0.05 mol) was added. The mixture was shaken until all the anhydride dissolved, then left for 3 days at rt. The solvent was concentrated under reduced pressure, benzene (100 mL) was added (with warming until all the oil dissolved) and the mixture was left in the refrigerator to crystallize overnight. The product (*S*)-**20** was collected by filtration and a second crop was obtained by reducing the volume of benzene to 50 mL. This initial diastereomer was further purified by recrystallization from benzene or toluene (75 mL). To obtain the other diastereomer, the benzene was removed from the mother liquor and the resulting oil was dissolved in $Et_2O$ (50 mL). The soln was left overnight in the refrigerator and the precipitate was collected by filtration. A second crop of (*R*)-**20** was obtained by reducing the filtrate volume to 25 mL.

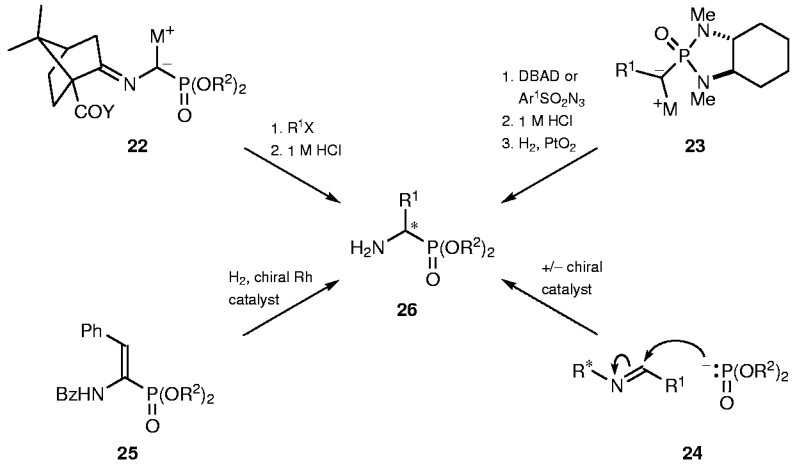
**Optically Active 1-Aminoalkylphosphonic Acids 21; General Procedure:**[54]
The amide **20** (0.1 mol) was dissolved in AcOH (60 mL), 40% HBr/$H_2O$ (60 mL) was added, and the resulting mixture was refluxed for 20 h. After cooling to rt, the solvents were removed in vacuo, the residue was mixed with $H_2O$ (50 mL), benzoic acid was removed by filtration, and the soln was clarified with charcoal and loaded onto a Dowex 50W X8 50–100 mesh column (15 × 300 mm). The column was washed with $H_2O$ and ninhydrin-positive fractions were collected. Concentration yielded pure, crystalline enantiomers of **21** (Table 3).

*for references see p 525*

### 10.10.1.1.7 Asymmetric Synthesis of α-Aminoalkylphosphonic Acids

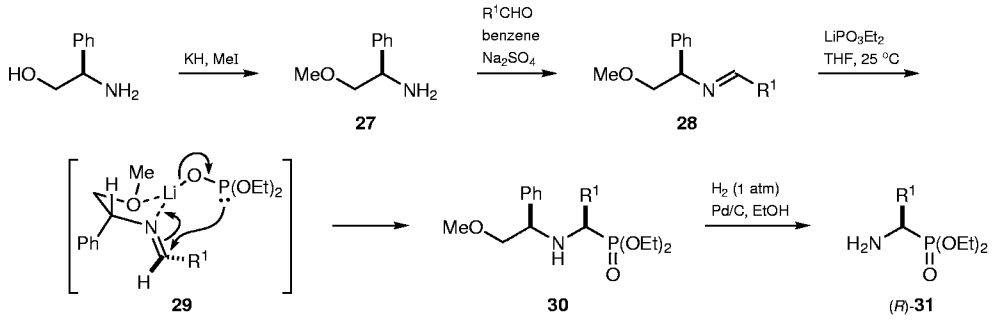
As described in a recent review,[4] formation of each bond to the chiral carbon (C–P, C–N, C–H, and C–C) has served as the basis for stereoselective synthesis of α-aminoalkylphosphonic acid derivatives **26**. New methods illustrating each of these approaches continue to appear (Scheme 11).[5–17] For example, on the basis of molecular modeling studies, some chiral phosphonoglycines **22** were prepared and shown to give improved stereoselectivities in alkylation with small electrophiles.[5] Diastereoselectivities of 64–80% were observed in the electrophilic amination and azidation of the chiral phosphonamidates **23**.[9] Although most stereoselective syntheses of aminoalkylphosphonic acid derivatives **26** use chiral auxiliaries in stoichiometric amounts, a recent report shows that asymmetric hydrogenation of **25** using catalytic amounts of chiral rhodium complexes proceeds with up to 96% ee.[15] Another successful catalytic asymmetric synthesis of α-aminoalkylphosphonates **26** involves formation of the C–P bond by reaction of dimethyl phosphite with an appropriate imine **24** in the presence of 5–20 mol% of a lanthanoid–potassium–BINOL complex.[14]

**Scheme 11** Recently Described Methods for Stereoselective Synthesis of α-Aminoalkylphosphonic Acid Derivatives[5–17]




The most commonly used strategy for the stereoselective formation of the C–P bond is addition of phosphites to chiral imines **24** (or other C=N compounds).[4,6,12,16] Many chiral amines are available and from them the desired imines can be generated by reaction with an appropriate aldehyde. A particularly effective example of this approach was developed by Smith et al. (Scheme 12).[16] They found that aldehyde imines **28** formed from (–)-(*R*)-2-methoxy-1-phenylethanamine (**27**) react with lithium diethylphosphite to give the aminoalkylphosphonic acid derivatives **30** in good yields and with high stereoselectivity (Table 4), and with the new chiral center α to the phosphorus having predominantly the *R* configuration. This high diastereoselectivity was attributed to a chelated intermediate **29** in which the *anti* orientation of the phenyl and phosphite groups directs addition to the *re* face of the imine double bond.

**Scheme 12** Stereoselective Synthesis of α-Aminoalkylphosphonates Utilizing a Chiral Chelating Imine[16]




This approach has a number of advantages: (1) either enantiomer of the necessary chiral auxiliary **27** is readily available by alkylation of commercially available (*R*)- or (*S*)-2-phenylglycinol; (2) high de's (normally greater than 95%) are obtained in reactions of imines formed from a wide variety of aldehydes; (3) the chiral auxiliary is readily removed by hydrogenolysis to produce diethyl α-aminoalkylphosphonates **31**, which are convenient intermediates in the synthesis of phosphonate-containing peptides. Some limitations in the method were observed. Hindered imines, such as those produced from reaction with pivaldehyde or 4,4-dimethylpentan-2-one, react poorly or not at all with lithium diethylphosphite. The protected phosphonate analogue of Phg (**30**, $R^1$ = Ph) was produced in relatively poor stereopurity (76% de), probably due to base-induced epimerization. The phosphonate analogue of Phe could not be produced since the necessary imine apparently isomerizes to the enamine during reaction with lithium diethylphosphite.

**Table 4** Asymmetric Synthesis of Diethyl α-Aminoalkylphosphonates[16]

| $R^1$ | **28** Yield (%) | **30** | | **31** | | |
|---|---|---|---|---|---|---|
| | | de (%) | Yield (%)[a] | ee (%)[b] | Yield (%) | $[\alpha]_D^{25}$ (*c*, $CHCl_3$) |
| Me | 90 | 95.2 | 77 | ≤99 | 99 | −5.4° (1.8) |
| $(CH_2)_5Me$ | 95 | 96.0 | 78 | ≤98 | 98 | −12.2° (1.1) |
| iPr | 82 | 96.5 | 82 | 97 | 86 | −0.8° (1.3) |
| iBu | 84 | >98.3 | 81 | ≤99 | 89 | −20.8° (1.6) |
| Cy | 97 | 96.0 | 80 | 96 | 94 | −52.2° (1.2, acetone) |
| $CH_2Cy$ | 89 | 98.3 | 70 | ≤99 | 87 | −20.9° (0.6) |
| Ph | 82 | 76.0 | 90 | 71 | 88 | −13.3° (1.9) |
| $CH_2CH_2SMe$ | 84 | 96.5 | 69 | 75 | 89[c,d] | −21.6° (0.6) |
| $CH_2OBzl$ | 92 | 96.0 | 38 | ≤98 | 100[e] | −10.6° (1.5) |
| $CH_2CH_2CO_2tBu$ | 95 | 96.0 | 37 | 96 | 83 | −14.3° (0.7) |

[a] Based on BuLi as the limiting reagent; typically 0.8–0.9 equiv of BuLi was used relative to imine **28**.
[b] Determined by analysis of (*S*)-Mosher amides.
[c] Yield based on recovered starting material.
[d] Palladium black in AcOH used instead of 20% $Pd(OH)_2/C$ in EtOH.
[e] The benzyl ether was also removed in this step.

**(−)-(*R*)-2-Methoxy-1-phenylethanamine (27):**[16]
To a stirred suspension of KH (7.82 g, 35% in mineral oil, washed with pentane, 195 mmol) in THF (15 mL) in an oven-dried 2-L flask at 25 °C was added a soln of (−)-(*R*)-2-phenylglycinol (25.0 g, 182.2 mmol) in THF (370 mL) over 2.5 h via an oven-dried, 500-mL pressure-equalizing addition funnel. The resultant pale yellow mixture was stirred overnight, then treated dropwise over 2 h with a soln of MeI (25.2 g, 177.6 mmol) in THF (220 mL). The mixture was stirred for an additional 3 h, poured into cold brine (1.5 L), and extracted with $Et_2O$ (4 × 250 mL), and the combined organic solns were dried ($Na_2SO_4$), filtered, and concentrated. Distillation afforded **27** as a colorless oil; yield: 25.3 g (94%); bp 47–50 °C/0.2 Torr; $[\alpha]_D^{25}$ −49 (*c* 6.3, benzene).

*for references see p 525*

**Asymmetric Synthesis of Diethyl (*R*)-Cyclohexyl[(*R*)-(2-methoxy-1-phenylethyl)amino]methylphosphonate (30, $R^1$ = Cy); Typical Procedure:**[16]

**CAUTION:** *Benzene is a carcinogen.*

A soln of amine **27** (1.80 g, 11.9 mmol) in dry benzene (25 mL) was cooled to 0 °C and treated with cyclohexanecarbaldehyde (1.33 g, 11.9 mmol). The soln was warmed to rt, anhyd $Na_2SO_4$ (12 g) was added, and the mixture was stirred for 1.5 h. Filtration and concentration afforded the crude imine **28** ($R^1$ = Cy; 2.83 g, 97%) as a colorless oil. A soln of freshly distilled diethyl phosphite (3.18 g, 23.1 mmol) in THF (15 mL) was cooled to 0 °C, 1.59 M BuLi in hexanes (6.90 mL, 11.0 mmol) was added slowly, and the mixture was stirred for 30 min and brought to rt. The soln of lithium diethylphosphite was added to the imine soln via cannula, and the mixture was stirred for 20 h and then diluted with $H_2O$ (20 mL). Most of the THF was removed in vacuo, and the aqueous layer was saturated with NaCl and extracted with EtOAc (4 × 50 mL). The combined extracts were dried ($Na_2SO_4$), filtered, and concentrated. Flash chromatography of the crude product (gradient elution, 50–67% EtOAc/hexanes) gave (–)-**30** ($R^1$ = Cy) as a colorless oil; yield: 3.38 g (80%); 96% de; $[\alpha]_D^{25}$ –77.5 (*c* 1.15, $CHCl_3$).

**Diethyl (*R*)-Amino(cyclohexyl)methylphosphonate [(*R*)-31, $R^1$ = Cy]; Typical Procedure:**[16]
Phosphite adduct (–)-**30** ($R^1$ = Cy; 19.12 g, 49.83 mmol) was dissolved in abs EtOH (400 mL), 20% $Pd(OH)_2/C$ (21.0 g) was added, and the flask was connected to a hydrogenation apparatus equipped with a graduated buret containing $H_2O$ to monitor the uptake of $H_2$. The mixture was thoroughly degassed at aspirator pressure and back-filled with hydrogen (× 3). The mixture was stirred under $H_2$ (1 atm) at rt for 15 h, then degassed at aspirator pressure. After filtration of the mixture through Celite, washing with EtOH (3 × 50 mL), and concentration, flash chromatography of the residue (gradient elution, 5–10% $MeOH/CH_2Cl_2$) afforded (*R*)-**31** ($R^1$ = Cy) as a colorless oil; yield: 11.97 g (94%) $[\alpha]_D^{25}$ –52 (*c* 1.3, acetone).

## 10.10.1.2 Synthesis of Hydrogen Phosphinate Amino Acid Analogues

### 10.10.1.2.1 Strecker-Type Condensation

Hydrogen aminoalkylphosphinates, building blocks containing a P–H bond, are useful in the synthesis of peptides with P–C bonds, since the P–C linkage is not readily accessible from the phosphonate oxidation level. In addition, H-phosphinates are useful for the synthesis of P–N and P–O bonds through oxidative activation. Baylis et al.[56] reported the synthesis of a large number of amino acid analogues via the condensation of phosphinic acid [$H_2P(O)OH$] with an imine containing the corresponding amino acid side chain (Scheme 13, Table 5). The yields for this condensation are generally modest (averaging 30–40%), but the starting materials are inexpensive, the chemistry can be performed on large scale, and the N-protected phosphinates **32** are readily isolated via crystallization. The H-phosphinates are identifiable by the characteristic NMR signal for the phosphorus-bound hydrogen, which appears as a doublet centered around δ 7.1 with a very large (ca. 530 Hz) coupling constant. The $^{13}C$ NMR signals also show splitting up to three bonds from phosphorus. Hydrogen phosphinates are prone to oxidation, so care must be taken during their synthesis. However, this tendency to oxidize is quite useful in the elaboration to phosphono-peptides.

**Scheme 13** Synthesis of α-Aminoalkylphosphinates from Phosphinic Acid[56–59]

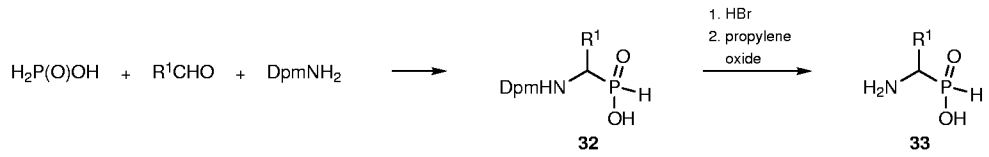


**Table 5** Synthesis of Hydrogen α-Aminoalkylphosphinic Acids[56–59]

| $R^1$ | $Xaa^P$ | Yield (%) of **33** | mp (°C) | Ref |
|---|---|---|---|---|
| H | Gly | 6 | 234–236 | [56] |
| Me | Ala | 68 | 223–224 | [56] |
| iPr | Val | 58 | 201–202 | [56] |
| iBu | Leu | 84 | 222–223 | [56] |
| sBu | Ile | 63 | 203 | [56] |
| Bzl | Phe | 22 | 227–228 | [56] |
| $4\text{-}HOC_6H_4CH_2$ | Tyr | 13 | 235 | [56] |
| indol-3-ylmethyl | Trp | 5 | 253–254 | [56] |
| $CH_2OH$ | Ser | 44 | 210 | [56] |
| CH(OH)Me | Thr | 24 | 218–220 | [56] |
| $CH_2CH_2SMe$ | Met | 13 | 231 | [56] |
| $CH_2SH$ | Cys | 31 | 203–204 | [56] |
| $CH_2CH_2CO_2H$ | Glu | 29 | 154 | [56] |
| $(CH_2)_4NH_2$[a] | Lys | 17 | 217 | [56,57] |
| $(CH_2)_3NH_2$[a] | Orn | 23 | 170 | [56] |
| $3,5\text{-}(HO)_2C_6H_3$ | DHPG[b] | 90 | 230–232 | [58] |
| $(CH_2)_3$[c] | Pro | 90 | –[d] | [59] |

[a] Prepared as the $N^{\omega}$-phthaloyl derivative and hydrolyzed.
[b] DHPG = α-(dihydroxy)phenylglycine.
[c] Prepared from pyrroline trimer.
[d] mp not reported.

**[1-(Benzhydrylamino)alkyl]phosphinic Acid** 32**; General Procedure:**[56]
An aldehyde (1 mmol), dissolved in an equal volume of dry dioxane, was added under $N_2$ to a suspension of an equimolar amount of the phosphinic acid salt of $DpmNH_2$ in dry dioxane (10–20 mL·g$^{-1}$) at 100 °C under $N_2$, at such a rate that the temperature at reflux remained at ca. 100 °C. $H_2O$ was removed azeotropically with dioxane. When two thirds of the dioxane had been removed by distillation, the mixture was cooled to rt and diluted with an equal volume of abs EtOH. The hydrogen phosphinic acid **32** crystallized slowly and was collected by filtration, washed successively with EtOH and $Et_2O$, and dried.

**1-Aminoalkylphosphinic Acid** 33**; General Procedure:**[56]
The [1-(benzhydrylamino)alkyl]phosphinic acid **32** was heated with an excess of 48% HBr (5 times weight) at 100 °C for 1–2 h until two distinct phases separated. (Note: the 48% HBr is pretreated by extraction three times with an equal volume of $CH_2Cl_2$ to remove any bromine, which oxidizes the phosphinic acid.) The mixture was concentrated to dryness under reduced pressure and the residue was taken up in $H_2O$. The aqueous soln was washed several times with $Et_2O$ to remove DpmBr and concentrated to a thick oil. (Note: complete removal of the solvent is important at this point; otherwise, residual HBr interferes in the next step.) The oily residue of 1-aminoalkylphosphinic acid hydrobromide was dissolved in EtOH (10 mL·g$^{-1}$) and propylene oxide was added dropwise until precipitation began. The mixture was allowed to stand until precipitation was complete, and the solid **33** was collected by filtration, washed successively with EtOH and $Et_2O$, and dried. For $Ala^P$ analogue: mp 223–224 °C; $^{31}P$ NMR ($D_2O$, δ): 22.2.

### 10.10.1.2.2 Alkylation of Protected Aminomethylphosphinic Acid

With suitable protection, the phosphinic acid analogue of Gly can be deprotonated at the α-position and alkylated, by analogy to the chemistry of the phosphonate analogue (see Section 10.10.1.1.3). Formation of the benzaldehyde or benzophenone imine is also used in these cases to protect the nitrogen and to make the α-hydrogen more acidic. In the case of the phosphinate analogues, protection of the P–H moiety is also required. A versatile P–H protecting group is the diethoxymethyl moiety, since it is readily introduced (Scheme 14), stable to the strongly basic alkylation conditions, and cleanly removed by dilute mineral acid. Alkyl (dialkoxymethyl)phosphinates such as **34** are formed by the condensation of anhydrous phosphinic acid with excess trialkyl orthoformate and catalytic TosOH.[60]

*for references see p 525*

Conversion of the (dialkoxymethyl)phosphinate **34** into the Gly analogue **35** is accomplished by a Strecker-type condensation with the formaldehyde–benzhydrylamine trimer. Exchange of the N-protecting group in **35** for the benzophenone imine, via hydrogenation to give **36** (acid hydrolysis of which gives **37**) and reaction with benzophenone, then affords intermediate **38**.[61] Acid-catalyzed hydrolysis of the alkylated intermediate **39** results in loss of the dialkoxymethyl group and affords the α-aminoalkylphosphinic acids **33**. Higher yields are obtained with more reactive electrophiles (compare entries 4 and 5 with entry 7 in Table 6). This method is most useful for a range of substituted Phe or Tyr analogues, but with appropriate protection, derivatives with side-chain functionality, such as Asp or Glu (entries 8 and 9), can also be prepared. However, attempts to synthesize Trp analogues using this process have been unsuccessful.[61]

The preparation of α,α-disubstituted aminophosphinic acids **41** was accomplished by further alkylation of the intermediate **39** to give **40** followed by acid hydrolysis. The second alkylation is quite sluggish because of steric congestion, and the overall yield was improved by isolation of **39**, rather than carrying the sequential procedure through in one pot.

**Scheme 14** Synthesis and Alkylation of Protected Aminomethylphosphinates[61]

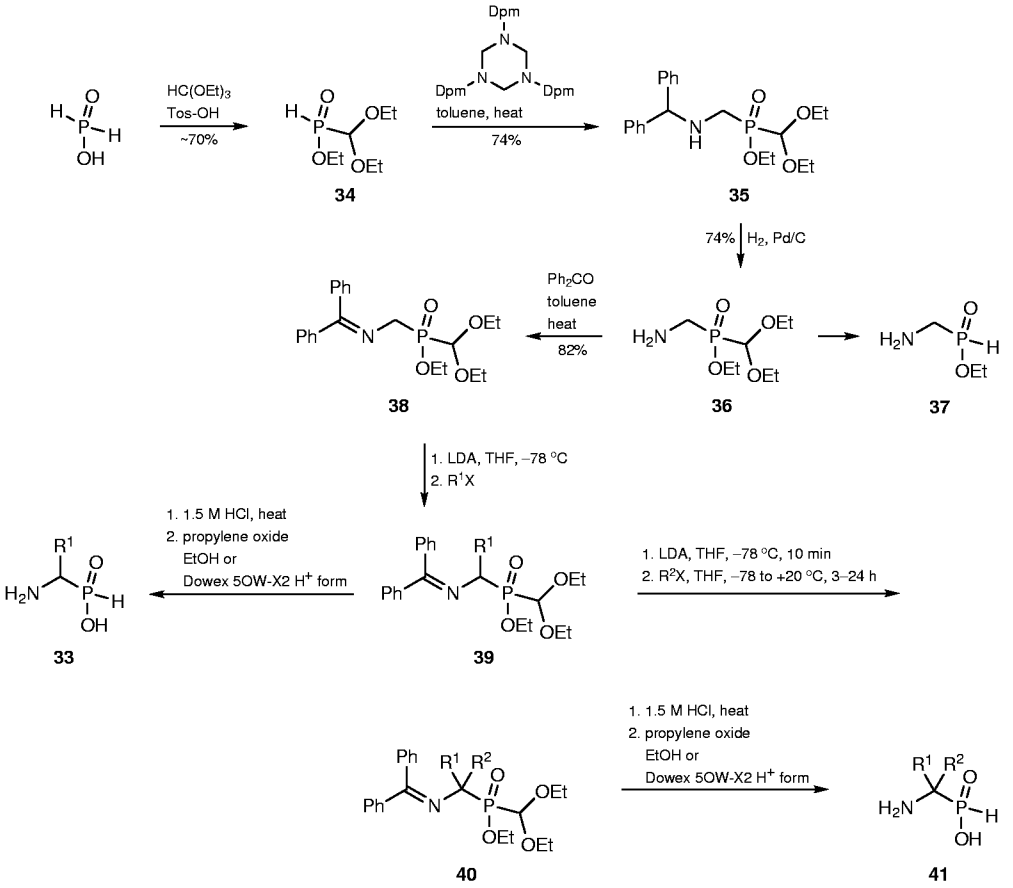


**Table 6** Alkylation of Protected Aminomethylphosphinate[61]

| Entry | $R^1X$ | Yield (%) **36** → **39** | mp (°C) |
|---|---|---|---|
| 1 | MeI | 69 | 223–224 |
| 2 | EtOTs | 77 | 229–230 |
| 3 | $Ph(CH_2)_2Br$ | 55 | 244–246 |
| 4 | BzlBr | 86 | 227–228 |
| 5 | $CH_2{=}CHCH_2Br$ | 77 | 235 |
| 6 | iBuBr | 59 | 222–223 |
| 7 | cPeBr | 28 | 235–237 |
| 8 | $EtO_2CCH_2Br$ | 56 | 209–212 |
| 9 | $EtO_2CCH_2{=}CH_2$ | 30 | ca. 122 |

**Ethyl (Diethoxymethyl)phosphinate (34); Typical Procedure:**[60]
A mixture of anhyd phosphinic acid (1 mol), TosOH (0.2 mol), and triethyl orthoformate (2.2 mol) was allowed to stand at rt for 1.5 h, then concentrated in vacuo (bath temperature <30 °C). A soln of the crude product in $CHCl_3$ was washed with sat. $NaHCO_3$, dried, and concentrated in vacuo (bath temperature <30 °C) to afford **34** as an oil (ca. 70% yield), which can be used directly or purified by carefully controlled distillation (bath temperature <100 °C); bp 70–75 °C/0.3 Torr; $^{31}P$ NMR (δ): 26.7 ($J_{PH}$ = 550 Hz, $J_{PCH}$ = $J_{POCH}$ = 8 Hz); IR (neat) $\tilde{\nu}_{max}$: 2400 (PH), 1233 $cm^{-1}$ (P=O).

**Ethyl [(Benzhydrylamino)methyl](diethoxymethyl)phosphinate (35); Typical Procedure:**[62]
A mixture of 2,4,5-tris(benzhydryl)hexahydro-1,3,5-triazine (9.75 g, 16.6 mmol) and phosphinate **34** (10 g, 50.9 mmol) in toluene (100 mL) was heated to reflux under $N_2$ for 2 h. The mixture was cooled to rt, the solvent was removed in vacuo, and the residue was dissolved in $Et_2O$ (100 mL). A small quantity of solid was removed by filtration, and the soln was concentrated in vacuo to afford **35** as a viscous oil; yield: 14.7 g (74%); $^{31}P$ NMR (δ): 40.89.

**Ethyl Aminomethyl(diethoxymethyl)phosphinate (36); Typical Procedure:**[61]
A mixture of the benzhydryl derivative **35** (28.13 g, 72 mmol) and 5% Pd/C (3.0 g) in abs EtOH (400 mL) was hydrogenated at 50–60 °C and 50–60 atm $H_2$ pressure until the uptake of hydrogen had ceased (20 h). After removal of the catalyst by filtration, and concentration, the crude product was purified by chromatography (silica gel, $CHCl_3$/EtOH 19:1) followed by distillation in vacuo (160–162 °C/0.2 Torr) to afford **36** as a colorless oil; yield: 12 g (74%); $^{31}P$ NMR (δ): 40.95.

**Ethyl (Diethoxymethyl)[(diphenylmethylidene)aminomethyl]phosphinate (38); Typical Procedure:**[61]
A mixture of amine **36** (17.17 g, 76.23 mmol) and benzophenone (15.3 g, 83.96 mmol) in toluene (200 mL) was heated to reflux for 4 h with removal of $H_2O$ (Dean–Stark trap). The mixture was concentrated in vacuo and the residue was first purified by chromatography then by bulb-to-bulb distillation in vacuo (205–210 °C/0.02 Torr) to afford the imine **38** as a viscous oil; yield: 24.4 g (82%); $^{31}P$ NMR (δ): 38.69.

**Ethyl (Diethoxymethyl)[1-(diphenylmethylideneamino)alkyl]phosphinates (39) by Alkylation: General Procedure:**[61]
To a soln of $iPr_2NH$ (1.21 g, 12 mmol) in THF (12 mL) at −78 °C under $N_2$ was added BuLi in hexane (7.5 mL, 12 mmol), and the soln was stirred for 10 min to form LDA. To this soln was added the imine **38** (3.89 g, 10.0 mmol) in THF (10 mL), followed 10 min later by a soln of the appropriate alkylating reagent (24 mmol). The mixture was allowed to warm to rt for 4–16 h, diluted with sat. $NH_4Cl$, and extracted with $Et_2O$. The organic layer was dried ($MgSO_4$) and concentrated to an oil, which was purified by chromatography (silica gel) to afford the alkylated imine **39** as a mixture of diastereomers.

**1-Aminoalkylphosphinic Acids 33 from Hydrolysis of 39; General Procedure**[61]
A mixture of the alkylated adducts **39** and 1.5 M HCl (15–20 mL·mmol$^{-1}$) was heated to reflux for 2–8 h until hydrolysis was complete, as determined by $^{31}P$ NMR. The mixture was concentrated in vacuo, the residue was dissolved in $H_2O$, and the soln was washed with $Et_2O$ and reconcentrated in vacuo. The product was finally purified by ion-exchange chromatography or by dissolution in hot EtOH (twice the minimum volume required) followed by cooling to rt and precipitation of the zwitterion by the addition of propylene oxide. The 1-aminoalkylphosphinic acids **33** were isolated as white solids.

*for references see p 525*

### 10.10.1.2.3 Resolution of α-Aminoalkylphosphinic Acids via Crystallization of Diastereomeric Salts

A number of phosphorus analogues of amino acids have been resolved by crystallization of appropriately protected derivatives with optically active amines or acids.[3] One of the most generally useful approaches is the resolution of Z- and Boc-protected aminoalkylphosphinic acids (Scheme 15 and Table 7).[56,63] In reactions analogous to those for normal amino acids, the aminoalkylphosphinic acids **33** (for preparation see Section 10.10.1.2) are N-protected using Z-Cl or Boc-ON to produce the corresponding Z- and Boc-derivatives (**42** and **43**, respectively). The derivatives can be resolved by recrystallization of the 1-phenylethanamine salts, **44** and **45**, respectively, and the optically pure N-protected aminoalkylphosphinic acids (e.g., **43**) can be recovered by treatment with aqueous HCl followed by extraction. Alternatively, the parent aminoalkylphosphinic acids **33** can be produced by removal of the N-protecting group with 45% HBr in acetic acid followed by treatment with propylene oxide of the resulting hydrobromide salt **46**.

If desired, aminoalkylphosphinic acids may be readily oxidized to the phosphonic acid derivatives.[56,63] In fact, some care needs to be taken to exclude air when working with or storing phosphinic acids to prevent this oxidation.

**Scheme 15** Resolution of Z- and Boc-Protected Aminoalkylphosphinic Acids by Crystallization of Diastereomeric 1-Phenylethanamine Salts[56,63]

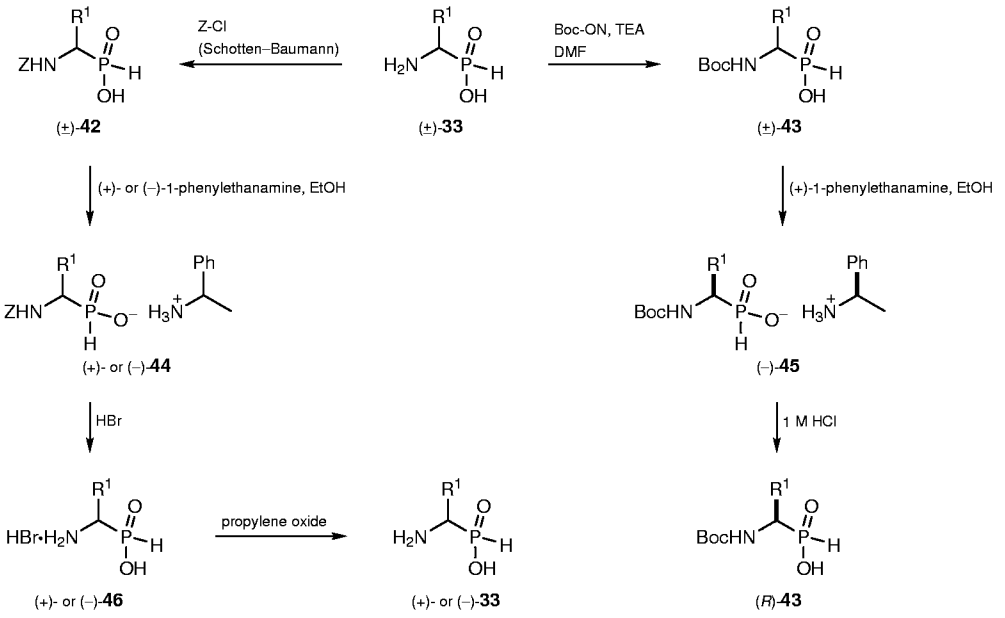


**Table 7** Resolution of Z- and Boc-α-Aminoalkylphosphinic Acids[56,63]

| $R^1$ | *N*-Acyl Group | α-MBA[d] Isomer | mp (°C) of **44** | $[\alpha]_D$ **44** or **45** | **33** | | Config | Ref |
|---|---|---|---|---|---|---|---|---|
| | | | | | mp (°C) | $[\alpha]_D$ | | |
| Me | Z | + | 145–146 | −12.8° (22 °C, *c* 5, EtOH) | 223–224[a] | −6.4°[a] (20 °C, *c* 2, $H_2O$) | R | [56] |
| Me | Z | − | 146–147 | +14.6° (22 °C, *c* 5, EtOH) | 224–225[a] | +7.0°[a] (20 °C, *c* 2, $H_2O$) | S | [56] |
| iPr | Boc | + | – | −28.1° | 125–125.5 | −39.8° | R | [63] |
| iPr | Z | + | 169 | −16.4° (25 °C, DMF/$H_2O$ 9:1) | 209 | −3.6° (25 °C, *c* 1.5, $H_2O$) | [*R*][a] | [56] |
| iPr | Z | − | 169 | +16.2° (25 °C, DMF/$H_2O$ 9:1) | 209 | +3.5° (25 °C, *c* 1.5, $H_2O$) | [*S*][a] | [56] |
| iBu | Z | + | 181–183 | −32° (22 °C, *c* 1, EtOH) | 233 | −23° (22 °C, *c* 1, $H_2O$) | *R*[a,b] | [56] |
| Bzl | Z | + | 190–193 | −47.6° (24 °C, *c* 1, EtOH) | 224 | −62.2° (25 °C, *c* 1, EtOH) | [*R*][a] | [56] |
| Bzl | Boc | + | – | | 92–93 | −26.9° | R | [63] |
| $CH_2CH_2SMe$ | Z | +[c] | 163–164 | −25° (22 °C, *c* 2, EtOH) | 232[c] | −30° (22 °C, *c* 1, $H_2O$) | [*R*][a] | [56] |
| $CH_2CH_2SMe$ | Z | −[c] | 163–164 | +25° (22 °C, *c* 2, EtOH) | 230[c] | +31° (22 °C, *c* 1, $H_2O$) | [*S*][a] | [56] |

[a] Absolute configuration not clarified in original article;[56] assignment shown is based on analogy to the other members of the series.
[b] Assignment from Dhawan and Redmore.[3]
[c] The original article does not correlate the optical isomer obtained with the 1-phenylethanamine enantiomer used; this assignment of the resolving agent is based on analogy to other members of the series.[56]
[d] α-MBA = α-methylbenzylamine (1-phenylethanamine).

**(±)-1-(Benzyloxycarbonylamino)-2-methylpropylphosphinic Acid (42, $R^1$ = iPr); Typical Procedure:**[56]
1-Amino-2-methylpropylphosphinic acid (**33**, $R^1$ = iPr; 27.5 g, 0.2 mol) was dissolved in $H_2O$ (100 mL), the soln was adjusted to pH 9.5 with 4 M NaOH and cooled to 0 °C. Z-Cl (34 g, 0.2 mol) was added over 1 h and the mixture was stirred for an additional 6 h while the pH was maintained at 9.0–9.5 by periodic addition of 4 M NaOH. The mixture was then allowed to warm to rt and washed with $Et_2O$. The aqueous soln was slowly added to a mixture of $H_2O$ (120 mL), concd HCl (80 mL), and ice (400 mL). The solid product was collected by filtration, dried, and recrystallized (EtOAc/light petroleum ether) to give the product, Z-Val$^P$ (**42**, $R^1$ = iPr); mp 110–111 °C; $^{31}P$ NMR ($D_2O$, δ): 18.6 ($J_{PH}$ = 540 Hz).

**1-(Benzyloxycarbonylamino)-2-methylpropylphosphinic Acid (+)-1-Phenylethanamine Salt (44, $R^1$ = iPr); Typical Procedure:**[56]
(+)-1-Phenylethanamine (15 g, 0.125 mol) in EtOH (75 mL) was added to a refluxing soln of (±)-Z-Val$^P$ (**42**, $R^1$ = iPr; 34 g, 0.125 mol) in EtOH (500 mL). The salt crystallized (22 g) and was recrystallized (abs EtOH 15 mL/g) to constant mp and optical rotation; mp 169 °C; $[\alpha]_D^{25}$ −16.4 (DMF/$H_2O$ 9:1).

**(−)-1-Amino-2-methylpropylphosphinic Acid [(−)-33, $R^1$ = iPr]; Typical Procedure:**[56]
The resolved salt [(−)-**44**, $R^1$ = iPr] was stirred with a slight excess of 45% HBr in AcOH at 0 °C for 1 h. Propylene oxide was added dropwise until precipitation started, then $Et_2O$ was added to complete the precipitation of (−)-**33**; mp 209 °C; $[\alpha]_D^{25}$ −3.6 (*c* 1, $H_2O$).

**(−)-(*R*)-1-(*tert*-Butoxycarbonylamino)-2-phenylethylphosphinic Acid (43, $R^1$ = Bzl); Typical Procedure:**[63]
A 100-mL round-bottomed flask was charged with 1-amino-2-phenylethylphosphinic acid (**33**, $R^1$ = Bzl; 12.89 g, 69.6 mmol), Boc-ON (18.86 g, 76.6 mmol), TEA (14.6 mL, 104 mmol), and DMF (21.0 mL), and the suspension was heated to 70 °C for 12 h. The excess TEA and DMF were removed under high vacuum. The residue was dissolved in $H_2O$ (300 mL), and the aqueous layer was washed with EtOAc (4 × 100 mL), acidified to pH 1 with 12 M HCl, and extracted with $CHCl_3$ (4 × 100 mL). The combined $CHCl_3$ extracts were dried ($Na_2SO_4$) and concentrated in vacuo to yield 9.23 g (46%) of the racemic carbamate **43** ($R^1$ = Bzl) as a yellow oil. This material was suspended in $Et_2O$ (50 mL) at reflux, and 50% (+)-1-phenylethanamine (3.92 g, 32.4 mmol) in $Et_2O$ was added. The white solid obtained was recrystallized ($Et_2O$, 4 ×) to give (−)-**45** ($R^1$ = Bzl); $[\alpha]_D^{20}$ −28.1 (EtOH). This salt was dissolved in 1 M HCl (100 mL), and the aqueous suspension was extracted with $CHCl_3$ (4 × 60 mL). The combined $CHCl_3$ layers were dried ($Na_2SO_4$) and concentrated in vacuo to yield the *R*-isomer of **43** ($R^1$ = Bzl) as a clear oil which could be crystallized from $Et_2O$; yield: 3.56 g (39% of theoretical yield); mp 125–125.5 °C; $[\alpha]_D^{22}$ −39.8 ($CHCl_3$); $^{31}P$ NMR (δ): 30.14 (major conformer), 29.52 (minor conformer).

*for references see p 525*

## 10.10.2 Compounds Containing Phospha-Peptide Linkages: P–N, P–O, and P–C Bonds

The three common tetrahedral phosphorus peptide bond replacements have both similarities and differences in the ways in which they are assembled and in their chemical and biological behavior. The phosphonamidates are the most analogous to peptides because they remain hydrogen-bond donors, but they are less stable toward hydrolysis than the other analogues. Phosphonates share some similarities with phosphonamidates in the way they are synthesized, the final products are more stable, and they may show similar or quite different biological activity in comparison to the phosphonamidates. The carbon-linked phosphinates are completely stable toward hydrolysis, generally highly active, yet assembled in a completely different fashion than the N- and O-linked analogues.

### 10.10.2.1 Synthesis of Phosphonamidate Peptide Analogues

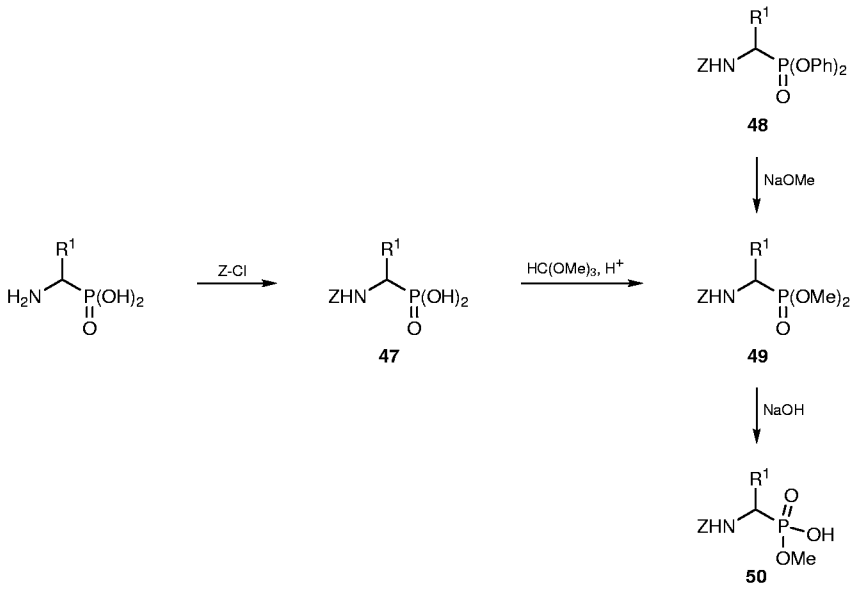
Phosphonamidates are usually prepared by coupling an activated phosphonate monoester with the amine of interest. Since most phosphonate amino acid analogues are prepared as their diesters, this procedure first requires selective cleavage to the monoester. Activation from the phosphonic acid oxidation level typically involves preparation of the phosphonochloridate, but it can also be accomplished by chlorination of an H-phosphinic acid analogue.

#### 10.10.2.1.1 Formation of Phosphonic Acid Monoesters

N-Protected α-aminoalkylphosphonic acid monoesters **50** are usually prepared by partial hydrolysis or cleavage of the diester **49** (Scheme 16);[64–67] alkaline hydrolysis does not normally result in cleavage of the second ester group, since the negatively charged phosphonate is resistant to further nucleophilic attack at phosphorus. The alkyl diester precursors **49**, if not synthesized directly in this form, are prepared by esterification of the diacids **47** with trimethyl orthoformate (or diazomethane)[65–68] or from the readily available diphenyl phosphonates **48**[26] by ester exchange in MeOH using methoxide[65] or KF/crown ether.[69,70]

Most of the $^{31}P$ NMR spectra of α-aminoalkylphosphonic acid derivatives having a carbamate group on the nitrogen (for example, **42** and **43**) are complicated by the presence of a minor peak (normally upfield) with an integrated intensity from 10–30% of the major peak. Extra peaks with similar relative intensities are also observed in many of the $^{1}H$ NMR spectra. The major and minor peaks coalesce when the sample is heated and are probably due to carbamate conformers.[71,72]

**Scheme 16** Preparation of N-Protected Aminoalkylphosphonic Acid Monomethyl Esters[64–67]




**(1*R*)-1-(Benzyloxycarbonylamino)-3-methylbutylphosphonic Acid (47, R¹ = iBu); Typical Procedure:**[66]
A slurry of (1*R*)-1-amino-3-methylbutylphosphonic acid[54] (4.84 g, 29.0 mmol), $NaHCO_3$ (4.89 g, 58.2 mmol), and $Na_2CO_3$ (6.14 g, 57.9 mmol) in 2 M NaOH (29.5 mL) was stirred mechanically and cooled in an ice water bath. To this mixture was added ZCl (95% pure, 4.45 mL, 29.6 mmol) by syringe. At 1-h intervals, two additional aliquots of ZCl (4.45 mL each, 29.6 mmol each) were added. After stirring at rt overnight, 2 M NaOH (10 mL) was added, and the mixture was partitioned between $H_2O$ and $Et_2O$. The aqueous portion was washed with $Et_2O$, acidified to litmus red with concd HCl, and extracted with EtOAc (× 3). The organic layer was washed with brine and dried (molecular sieves 4Å), and the solvent was evaporated to give **47** (R¹ = iBu) as a white foam; yield: 8.84 g (~100%); $[\alpha]_D^{23}$ −54 (*c* 1, 1 M NaOH); $^{31}P$ NMR (DMSO-$d_6$, δ): 21.78 (13%, minor conformer), 22.23 (87%, major conformer).

**Dimethyl (1*R*)-1-(Benzyloxycarbonylamino)-3-methylbutylphosphonate (49, R¹ = iBu); Typical Procedure:**[66]
Without further purification, the Z-protected diacid **47** was mixed with trimethyl orthoformate (50 mL, 457 mmol). The flask was equipped with a reflux condenser (without coolant) and placed in an oil bath at 110 °C. After 9 h, the volatile products and reactants were removed in vacuo, and the residue was purified by chromatography (0.5% EtOH/EtOAc) to yield **49** (R¹ = iBu) as a colorless oil; yield: 9.08 g (95%); $[\alpha]_D^{24}$ −36.4 (*c* 1, $CHCl_3$); $^{31}P$ NMR (120 MHz, δ): 27.99 (17%, minor conformer), 28.56 (83%, major conformer).

**Methyl (1*R*)-1-(Benzyloxycarbonylamino)-3-methylbutylphosphonate (50, R¹ = iBu); Typical Procedure:**[66]
A soln of **49** (R¹ = iBu; 9.08 g, 27.6 mmol) in MeOH (30 mL) and 2 M NaOH (40 mL) was stirred for 12 h at 21 °C. The mixture was then diluted with $H_2O$ (150 mL), washed with $CHCl_3$, acidified to litmus red with concd HCl, and extracted with $CHCl_3$. The organic layer was dried ($MgSO_4$) and evaporated to give **50** (R¹ = iBu); yield: 8.28 g (95%); mp 117–119 °C; $[\alpha]_D^{25}$ −43 (*c* 1, $CHCl_3$); $^{31}P$ NMR (81.7 MHz, δ): 28.6 (23%, minor conformer), 29.3 (77%, major conformer).

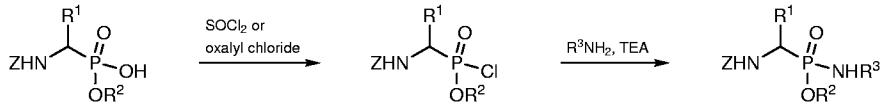
### 10.10.2.1.2 Synthesis of Phosphonamidates from Phosphonic Acid Monoesters

Protected-amino phosphonic acid monoesters **51** are usually activated by conversion to the phosphonochloridates **52**, then coupled with appropriate amine components such as C-protected amino acids or peptides to give phosphonoamidates **53** (Scheme 17). This procedure is in contrast to typical peptide coupling conditions, in which carboxylic acid chlorides are avoided because of their susceptibility to epimerization at the α-carbon. Since enolate-

*for references see p 525*

like conjugation is not possible for alkylphosphonate derivatives, epimerization is not an issue with phosphorus-containing peptide analogues. Although the conversion of the phosphonic acid into the phosphonochloridate is often performed using thionyl chloride,[64,65,67,73] a recent report suggested that better yields of phosphonamidate are obtained by using oxalyl chloride (2 equiv) in the presence of DMF (0.1 equiv).[69] The yield was further improved by adding AgCN (1.6 equiv) during the reaction of the phosphonochloridate with the amine.

**Scheme 17** Formation of Phosphonamidates from Phosphonic Acid Monoesters[64,65,67,72,73]




51 52 53

| $R^1$ | $R^2$ | Activating Agent | Amine Component | Yield (%) | $^{31}P$ NMR δ | Ref |
|---|---|---|---|---|---|---|
| H | Et | $SOCl_2$ | H-D-Trp-OEt | 36 | not reported | [72] |
| H | Me | $SOCl_2$ | H-Phe-OMe | 62 | 27.2, 27.9 | [64] |
| Me[a] | Me | $SOCl_2$ | H-Phe-OMe | 64 | 30.09, 30.90 | [67] |
| Bzl | Me | $(COCl)_2$, DMF | H-Ala-Pro-Phe-OMe | 75–90 | 33.7, 33.8, 33.9, 34.3 | [69] |
| Bzl | Ph | $SOCl_2$ | H-Ala-OMe | 77 | 26.76, 26.12, 25.36, 24.62 | [65] |
| Bzl | Me | $SOCl_2$ | H-Leu-Ala-OMe | 42 | 31.2, 30.8, 29.5, 29.0 | [73] |

[a] (*R*)-Isomer.

In a detailed study on phosphonate diester and phosphonamidate synthesis, Hirschmann, Smith, and co-workers reported that pyrophosphonate anhydrides may be produced as side products during conversion of phosphonate monoesters into phosphonochloridates.[72] They recommended adding the phosphonate monoester to a solution of the chlorinating agent (thionyl chloride or oxalyl chloride) to minimize formation of the less reactive anhydrides. They also found that addition of tertiary amines (e.g., TEA) to the phosphonochloridates, prior to addition of the alcohol or amine component, results in formation of a phosphonyltrialkylammonium salt that is more reactive than the corresponding phosphonochloridate and leads to better yields of the phosphonate esters and amides.

After conversion of the phosphonate monoester into a phosphonochloridate, or subsequently to a phosphonamidate, a new chiral center at phosphorus is created. Thus, after coupling to a chiral amine, even optically pure aminoalkylphosphonate monoesters produce a mixture of two diastereomers (racemic aminoalkylphosphonates result in four diastereomers). It is often possible to separate these diastereomers chromatographically. Of course, this is not necessary if the phosphorus ester will eventually be hydrolyzed.

The phosphonamide bond is fairly labile to acid-catalyzed hydrolysis, although this is not normally a serious problem unless the phosphorus ester is hydrolyzed to produce the phosphonamidic acid (see Section 10.10.3). Nevertheless, it is prudent to use a mild aqueous acid solution (e.g., sat. $KH_2PO_4$ or 2% $KHSO_4$) during reaction workups, rather than dilute mineral acids (e.g., 5% HCl or 5% $H_2SO_4$).

**Synthesis of Phosphonamidates 53; General Procedure:**

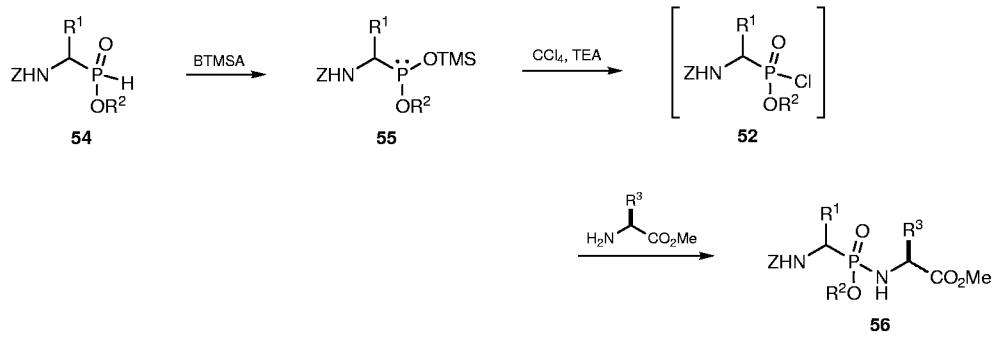
The following general procedure is derived from a number of the papers referenced above, as well as our own experience. The phosphonate monoester **51** (1 mmol) is dissolved in $CH_2Cl_2$ (5–10 mL). (Other inert solvents such as EtOH-free $CHCl_3$ may be used. Even if the phosphonate monoester is not completely soluble, the formation of the chloridate from the suspension is normally successful.) $SOCl_2$ (1.1–1.5 mmol) is added to the stirred soln at rt, and stirring is continued for 4–6 h. [For the oxalyl chloride procedure, the soln of the monoester is normally cooled to 0 °C before addition of DMF (0.1 mmol) and oxalyl chloride (2 mmol). The mixture is then stirred at rt for 2 h.] If desired, formation of the phosphonochloridates can be readily followed by $^{31}P$ NMR, since their chemical shifts are approximately 10 ppm higher than those of the corresponding phosphonate monoesters. By contrast, the chemical shift of the undesired pyrophosphonate anhydride byproducts are approximately 8 ppm lower

than the corresponding phosphonate monoesters. After formation of the phosphonochloridates, the solvent and volatile products are removed in vacuo. The resulting oily chloridates are redissolved in $CH_2Cl_2$ (2.5–5 mL), the soln is cooled to 0 °C, and the amine component (1–2 mmol, either as the free amine or the hydrochloride salt) and TEA (1.5–3 mmol) are added. (The amine and TEA are often dissolved together in a few mL of $CH_2Cl_2$ prior to addition, but a recent study[72] suggests that addition of the TEA before the amine is preferable.) The mixture is stirred at rt for 12–18 h, then worked up by washing successively with $H_2O$, sat. aq $KH_2PO_4$ (or 2% $KHSO_4$), aq $NaHCO_3$, and brine, drying ($Na_2SO_4$), and concentrating to dryness. The residue is then purified by chromatography (silica gel, 1–10% solns of MeOH, EtOH, or iPrOH in $CH_2Cl_2$ or $CHCl_3$ are commonly used.) Typical yields are 40–80%.

### 10.10.2.1.3 Synthesis of Phosphonamidates from H-Phosphinates

H-Phosphinates **54** may also serve as precursors for phosphonamidates **56** (Scheme 18).[63,74,75] In the simplest procedure, a N-protected H-phosphinate **54** (produced, for example, by the Baylis procedure,[56] see Section 10.10.1.2) is treated with a mixture of $CCl_4$, TEA, and the desired amino component. The H-phosphinates are converted in situ into phosphonochloridates **52**, which are then trapped by the amine component. Sampson and Bartlett[76] reported improved yields when the H-phosphinate was first treated with bis-(trimethylsilyl)acetamide (BTMSA) to produce the trivalent species **55**. They also noted that the reactions are generally cleaner when phosphinate ethyl esters are used instead of phosphinate methyl esters, presumably because the latter are more prone to nucleophilic demethylation.

**Scheme 18** Formation of Phosphonamidates from H-Phosphinates[63,74,75]




| $R^1$ | $R^2$ | $R^3$ | Yield (%) of **56** | Ref |
|---|---|---|---|---|
| iPr | Et | Me | 75 | [76] |
| iPr | Et | Bzl | 74 | [76] |
| Bzl | Et | iPr | 67 | [76] |
| Me | Bzl | H | 52[a] | [74] |
| Ph | Bzl | H | 65[a] | [74] |
| Ph | Bzl | Me | 48[a] | [74] |
| Ph | Bzl | iPr | 43[a] | [74] |

[a] BTMSA not used.

**Synthesis of Peptidyl Phosphonoamidates from H-Phosphinates; General Procedure:**[76]
In a dried, argon-filled, 5-mL round-bottomed flask fitted with a magnetic stir bar was dissolved the ethyl phosphinate (167 μmol) in dry MeCN (0.5 mL, deoxygenated by purging with argon for 5 min immediately prior to use). BTMSA (25 μL, 100 μmol) was added, followed 1 min later by the amino acid methyl ester hydrochloride (250 μmol), $CCl_4$ (0.5 mL, also deoxygenated), and TEA (58 mL, 418 μmol). The soln was stirred at rt for 30 min, cooled to 0 °C, and diluted with MeOH (1 mL). The solvent was removed, and analytically pure product was obtained by flash chromatography. For Z-Val$^P$(OEt)-Ala-OMe as product; yield: 75%; 4:4:1:1 mixture of diastereomers (Z-Val$^P$ starting material was racemic); $^{31}P$ NMR (δ): 28.69, 28.13, 27.99, 26.99.

*for references see p 525*

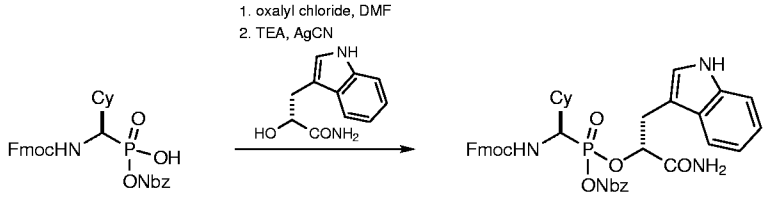
## 10.10.2.2 Synthesis of Phosphonate Peptide Analogues

Phosphonate peptides can be synthesized either by formation of the P–O bond or the O–C bond. The former coupling is typically carried out via reaction of an alcohol (e.g., an α-hydroxy acid) with an activated phosphonyl species in a manner similar to phosphonamidate synthesis (Section 10.10.2.1). The latter process employs the phosphonate anion as a nucleophile in a displacement reaction on an ester with a leaving group in the α-position.

### 10.10.2.2.1 Synthesis via Electrophilic, Activated Phosphonates

In addition to the conversion into phosphonochloridates as described in Section 10.10.2.1.2, phosphonate monoacids can also be activated for coupling via some of the popular reagents from standard peptide chemistry, such as BroP,[77] BOP,[78] and PyBOP.[79] One-pot activation and coupling of hydrogen phosphinates with α-hydroxy acids is also possible via the method outlined in Section 10.10.2.1.3.[76] The procedure outlined in Scheme 19 illustrates an optimized procedure for coupling a hindered phosphonate **57** and an α-hydroxy acid derivative via the phosphonochloridate.[72]

**Scheme 19** Formation of a Phosphonate Diester via the Phosphonochloridate[72]



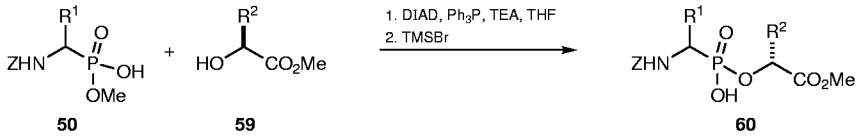

**D-2-{[L-Cyclohexyl(9-fluorenylmethoxycarbonylamino)methyl](4-nitrobenzyloxy)phosphoryloxy}-3-(1*H*-indol-3-yl)propanamide (58); Typical Procedure:**[72]

A soln of a (−)-4-nitrobenzyl [cyclohexyl(Fmoc-amino)methyl]phosphonic acid (**57**; 40.0 mg, 0.073 mmol) in $CH_2Cl_2$ (2 mL) was cooled to 0 °C, and dry DMF (6 μL, 0.01 mmol) and oxalyl chloride (13 μL, 0.147 mmol) were added. The mixture was stirred for 0.5 h at 0 °C, diluted with benzene (5 mL), and concentrated. The phosphonochloridate was redissolved in benzene (2 mL), and treated with TEA (20 μL, 0.147 mmol) followed by AgCN (29 mg, 0.217 mmol). After 5 min at rt, D-2-hydroxy-3-(1*H*-indol-3-yl)propanamide (22.4 mg, 0.011 mmol) was introduced, and the mixture was heated at reflux for 2 h, cooled to rt, filtered, and concentrated. The phosphonate diester was purified via flash chromatography (2% MeOH/$CH_2Cl_2$) to give a mixture of diastereomers (19.8 mg, 37%), which were separated by RP-HPLC. For the minor diastereomer (4.5 mg, amorphous solid): $[\alpha]_D$ +15 (*c* 0.47, EtOH).

### 10.10.2.2.2 Synthesis of Phosphonates via Mitsunobu Displacement

Phosphonate peptide analogues can be synthesized by formation of the O–C bond, either by alkylation with an α-halo ester[80] or by Mitsunobu coupling of the phosphonic acid **50** with an α-hydroxy ester **59** (Scheme 20).[81] Esterification by displacement of α-halo esters is restricted by substitution, and has so far only been applied to formation of the Gly analogues (glycolate esters).[80] The Mitsunobu process is more general, and is particularly convenient when the α-hydroxy ester **59** is the limiting component (note that the D-enantiomer is required to produce the L-product). Reaction times are increased with secondary alcohols and when β-branching is present, but the yields are generally high. The coupling reactions described by Campbell[81] were followed immediately by selective cleavage of the phosphonate methyl ester with bromotrimethylsilane to give the phosphonate monoester selectively (see Section 10.10.3.2.1).

**Scheme 20** Phosphonate Coupling via the Mitsunobu Protocol[81]




The Mitsunobu phosphonate coupling has been adapted to the solid-phase synthesis of peptidyl phosphonates as well.[82] In this environment, with excess phosphonate, even the more hindered, β-branched esters are formed efficiently. A series of phosphonate tripeptides have been synthesized and used to identify novel thermolysin inhibitors from a small library. (A number of other solid-phase syntheses of phosphonamidate-,[83] phosphonate-,[79,84] and phosphinate-peptides[85,86] have been described; however, in these instances the phosphorus bond forming steps were performed in solution, prior to solid-phase coupling.)

**Phosphonate Coupling To Give** 60 **Via the Mitsunobu Protocol; General Procedure:**[81]
To a soln of the methyl alkylphosphonate **50** (0.5 mmol), the alcohol **59** (0.75 mmol), and $Ph_3P$ (0.75 mmol) dissolved in anhyd THF (5 mL) were added DIAD (0.75 mmol) and TEA (5 mmol), followed by the α-hydroxy acid (1 mmol). Upon completion of the condensation reaction, TMSBr (1.5 mmol) was added and stirring continued for an additional 1 h. The mixture was diluted with $Et_2O$ (10 mL) and extracted with 5% $NaHCO_3$ (2 × 10 mL). The aqueous phase was washed with $Et_2O$ (3 × 10 mL), acidified to pH 2 with concd HCl, and extracted with EtOAc (3 × 10 mL). The EtOAc phase was dried ($MgSO_4$), filtered, and concentrated under reduced pressure. This material was suitable for additional synthetic steps or could be isolated in pure form as the 1-adamantanamine salt from $Et_2O$ or hexane. Characterization of (*R*,*S*)-Z-Val$^P$-(O)-D-Val-OMe (**60**; $R^1 = R^2 =$ iPr), isolated in 69% yield after 4 h reaction: $^{31}P$ NMR (δ): 24.23, 23.94.

## 10.10.2.3 Synthesis of Phosphinate Peptide Analogues

Peptides containing a $PO_2^-$—$CH_2$ linkage can be prepared using a variety of methods, including reactions with nucleophilic phosphorus, such as Arbuzov or conjugate addition reactions, and those in which the phosphorus serves as the electrophile.

### 10.10.2.3.1 Reactions Involving Nucleophilic Phosphorus

The most common method for the synthesis of phosphinopeptides is the addition of a nucleophilic, trivalent phosphorus species to a carbon electrophile, including conjugated double bonds, alkylating agents, imines, and carbonyl compounds. By analogy with the phosphite ester additions described above, the nucleophilic form of a phosphinic acid is the trivalent species, generated from the more stable, pentavalent PH derivative by deprotonation or by silylation (cf. Scheme 2).

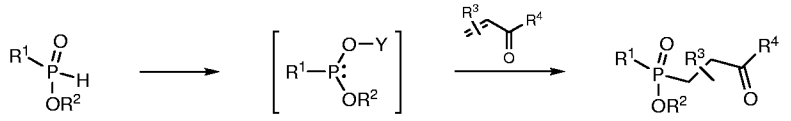
#### 10.10.2.3.1.1 Conjugate Addition of Alkyl Phosphonites to Acrylates

A convenient procedure for preparing dialkylphosphinic acids **62** involves addition of H-phosphinic acids and esters to conjugated double bonds via the silyl[87–89] or metal phosphonite **61**,[90–94] as illustrated in Scheme 21. The silyl phosphonite intermediates **61** (M = TMS) are typically formed either from phosphinic acids or esters using chlorotrimethylsilane or bis(trimethylsilyl)acetamide. The metal phosphonite intermediates **61** (Y = Li, Na, etc.) are prepared by deprotonation of the acids with a base such as sodium hydride, sodium methoxide, or lithium diisopropylamide. The conjugated double bonds are typically acrylic acids and esters substituted in the α-position with the appropriate amino acid side chain. After appropriate protecting group manipulations, additional amino acids

*for references see p 525*

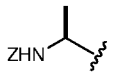
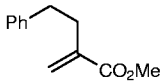
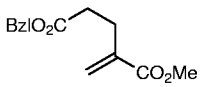
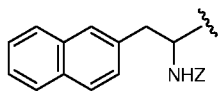
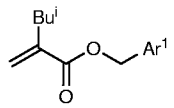
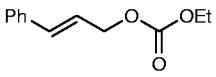
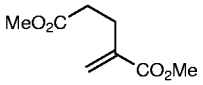
can be appended to the terminal carboxylate or amine under standard peptide coupling conditions. Yields for the conjugate addition reactions are generally good when run under oxygen-free conditions (Table 8).

**Scheme 21** Conjugate Addition of Alkyl Phosphonites[87–94]



**61** Y = Li, Na, TMS **62**

**Table 8** Conjugate Addition of Alkyl Phosphonites

| Entry | $R^1$ | $R^2$ | Silylating Agent or Base | Electrophile | Yield (%) | Ref |
|---|---|---|---|---|---|---|
| 1 | $(CH_2)_4Ph$ | Et | TMSCl | $CH_2$=C(Me)$CO_2Me$ | 93[a] | [87] |
| 2 | $(CH_2)_4Ph$ | Et | TMSCl | $CH_2$=$CHCO_2H$ | 61 | [87] |
| 3 | $(CH_2)_4Ph$ | H | TMSCl | cyclopent-2-enone | 77 | [87] |
| 4 | $(CH_2)_4Ph$ | H | TMSCl | (*E*)-PhCH=CHCHO | 88 | [87] |
| 5 | Me | Et | BTMSA | $CH_2$=C(Bzl)$CO_2Me$ | 81[a] | [87] |
| 6 | $CH_2NHZ$ | Me | BTMSA | $CH_2$=C(iBu)$CO_2Me$ | 87[a,b] | [78] |
| 7 |  | Me | TMSCl |  | 66[a,c] | [96] |
| 8 | $(CH_2)_2CO_2Bzl$ | H | TMSCl |  | 37[d] | [97] |
| 9 |  | Me | NaH |  | 88[a,e] | [98] |
| 10 |  | Me | NaOMe |  | 84[a,f] | [95] |

[a] Product is a mixture of diastereomers.
[b] $^{31}P$ NMR (δ): 50.4, 51.6.
[c] Yield given as a mixture of stereoisomers after phosphinic acid methylation with diazomethane.
[d] $^1H$ NMR available.
[e] $Ar^1$ = 3,4-dimethoxyphenyl.
[f] $^{31}P$ NMR (δ): 52.03, 52.84

**Methyl 2-[{[(Benzyloxycarbonylamino)methyl](methoxy)phosphoryl}methyl]-4-methylpentanoate (62, $R^1$ = $CH_2NHZ$; $R^2$ = Me; $R^3$ = iBu; $R^4$ = OMe) Via a Silyl Phosphonite Intermediate (Table 8, Entry 6); Typical Procedure:**[78]

To an oxygen-free soln of methyl (Z-aminomethyl)phosphinate (140 mg, 0.61 mmol) and methyl 2-isobutylacrylate (430 mg, 3.05 mmol, deoxygenated by three freeze–thaw cycles) in MeCN (0.4 mL) was added BTMSA (0.76 mL), and the mixture was stirred under argon at rt overnight. The mixture was diluted with MeOH (5 mL) and cooled to 0 °C, and ethereal diazomethane was added until a yellow color persisted. Glacial AcOH was added to consume excess diazomethane, and the soln was removed under reduced pressure. A standard extractive workup followed by chromatographic purification (EtOAc) produced the 1,4-addition product as a mixture of diastereomers; yield: 204 mg (87%).

**Dimethyl 2-[{[1-(Benzyloxycarbonylamino)ethyl](methoxy)phosphoryl}methyl]pentanedioate [62, $R^1$ = CH(Me)NHZ; $R^2$ = Me; $R^3$ = $CH_2CHCO_2Me$; $R^4$ = OMe] via a Sodium Phosphonite Intermediate (Table 8, Entry 10); Typical Procedure:**[95]

To a soln of methyl [1-(benzyloxycarbonylamino)ethyl]phosphinate (1.78 g, 6.9 mmol) in dry MeOH (15 mL) at 0 °C was added 2 M NaOMe in MeOH (3.8 mL). The mixture was stirred for 5 min and neat

dimethyl 2-methylenepentanedioate (1.1 mL, 6.9 mmol) was added dropwise. The soln was stirred at 0 °C for 15 min, then allowed to warm to rt and stirred an additional 4 h. The mixture was diluted with 1 M HCl (150 mL), extracted with $CH_2Cl_2$, dried ($MgSO_4$), and concentrated in vacuo. Purification of the residue by flash chromatography (MeCN/EtOAc 1:9) afforded the phosphinate dipeptide analogue as a colorless oil; yield: 2.5 g (84%); $^{31}P$ NMR (δ): 52.84, 52.03.

### 10.10.2.3.1.2 Addition of Alkyl Phosphonites to Carbonyl Compounds and Imines

In addition to the aforementioned electrophiles, trivalent phosphorus reagents can be added to imines,[29,33] aldehydes,[99–102] or isocyanates.[103] Chen and Coward[29] have reported a convenient method for the condensation of alkylphosphinic acids **64** with a *N*-acylimine **63** generated in situ, as depicted in Scheme 22. Analogous to the processes described above for preparation of α-aminoalkylphosphonic acids (see Section 10.10.1.1.1), this method employs a three component condensation reaction between an H-phosphinic acid, an aldehyde, and benzyl carbamate in acetyl chloride. The resulting products include orthogonally protected dipeptide mimetics, as shown by entries 7–9 in Table 9. In addition to serving as solvent, acetyl chloride serves two roles: as a dehydration agent in formation of the iminocarbamate **63** and to tautomerize the H-phosphinate to the trivalent, nucleophilic form.

**Scheme 22** Condensation of H-Phosphinic Acids with Imines and Isocyanates[29,33,103]

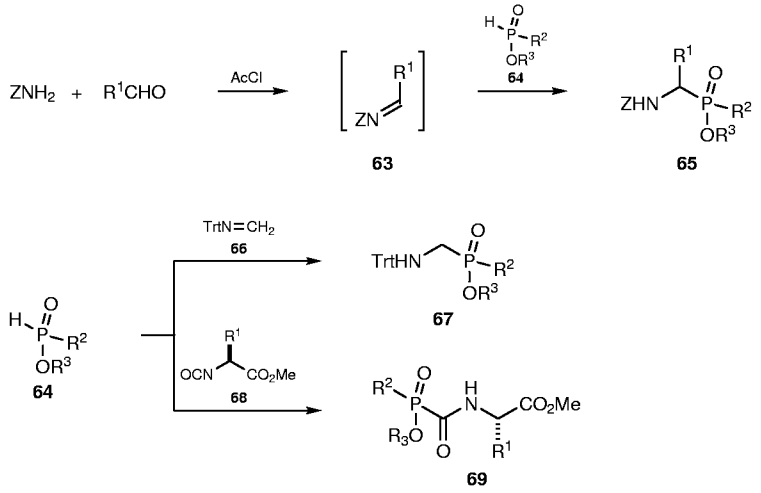


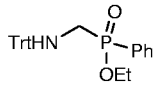
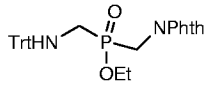
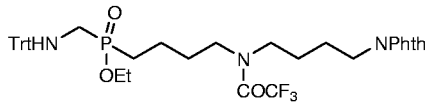
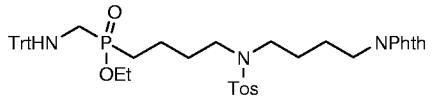

**Table 9** Condensation of H-Phosphinic Acids, Aldehydes, and Benzyl Carbamate in Acetyl Chloride[29]

| Entry | $R^2$ | $R^3$ | $R^1$ | Yield (%) | mp (°C) | $^{31}P$ NMR δ |
|---|---|---|---|---|---|---|
| 1 | Me | 1-Ada-$NH_3^+$ | H | 67 | 123–125 | 50.6 |
| 2 | Me | 1-Ada-$NH_3^+$ | iBu | 69 | 162–164 | 54.2 |
| 3 | Me | 1-Ada-$NH_3^+$ | Ph | 61 | 193–194 | 47.2 |
| 4 | Me | 1-Ada-$NH_3^+$ | 4-$MeOC_6H_4$ | 72 | 188–190 | 42.6 |
| 5 | Bu | H | H | 50 | 110–112 | 53.0 |
| 6 | $CH_2NPhth$ | H | Ph | 73 | 194–195 (dec.) | 36.6 |
| 7 | $CH_2CH_2CO_2Et$ | H | Me | 75 | 94–97 | 53.1 |
| 8 | $CH_2CH_2CO_2Et$ | H | Et | 48 | oil | 54.0 |
| 9 | $CH_2CH_2CO_2Et$ | H | 4-$MeOC_6H_4$ | 71 | 186–188 | 51.5 |

*for references see p 525*

By analogy to the Strecker-like synthesis of α-aminoalkylphosphonic acids (Section 10.10.1.1.1), alkyl phosphonites can be added to *N*-tritylimines **66** in refluxing toluene to form aminoalkyl(alkyl)phosphinates **67** (Scheme 22, Table 10).[33] The addition of a catalytic amount of Lewis acid, such as $BF_3 \cdot OEt_2$, dramatically improves the chemical yield for this process. Moreover, alkyl phosphonites can be added to acylating agents, such as alkyl isocyanates **68**, to form acyl(alkyl)phosphinates **69** (Scheme 22).[103]

**Table 10** Condensation of Alkylphosphinates with *N*-Methylenetritylamine[33]

| Entry | Product | Yield (%) | $^{31}$P NMR δ |
|---|---|---|---|
| 1 |  | 82 | 41.4 |
| 2 |  | 86 | 41.2 |
| 3 |  | 81 | 54.4, 54.0 |
| 4 |  | 70 | 54.5 |

**Ethyl 3-{[1-(Benzyloxycarbonylamino)ethyl](hydroxy)phosphoryl}propanoate (65, $R^1$ = Me; $R^2$ = $CH_2CH_2CO_2Et$; $R^3$ = H) (Table 9, Entry 7); Typical Procedure:**[29]

Acetaldehyde (15 mg, 0.34 mmol) was added to a stirring soln of ethyl 2-[hydro(hydroxy)phosphoryl]propanoate (56 mg, 0.33 mmol) and Z-$NH_2$ (51 mg, 0.34 mmol) in AcCl at 0 °C, and the mixture was stirred at 0 °C for 30 min, then at rt for 6 h. The soln was then removed in vacuo, the residue was partitioned between 5% $Na_2CO_3$ (20 mL) and $Et_2O$ (10 mL), and the aqueous soln was acidified to pH 2 with concd HCl and extracted with EtOAc (4 × 20 mL). The organic layers were combined, dried ($Na_2SO_4$), and concentrated to give 90 mg of syrup. Purification was carried out by ion-exchange chromatography on DEAE cellulose (MeOH/0.1 M $NH_4HCO_3$ buffer 1:1), followed by cation exchange on Dowex AG-8 resin ($H^+$ form); $^{31}$P NMR ($CD_3OD$, δ): 53.1.

**Ethyl Phenyl[(tritylamino)methyl]phosphinate (67, $R^2$ = Ph; $R^3$ = Et) (Table 10, Entry 1); Typical Procedure:**[33]

A soln of ethyl phenylphosphinate (18 mg, 0.11 mmol), *N*-methylenetritylamine (29 mg, 0.11 mmol), and $BF_3 \cdot OEt_2$ (1 drop) in dry toluene was stirred and heated to reflux for 8 h. Removal of the volatile components followed by chromatography (silica gel, EtOAc) afforded the product; yield: 39 mg (82%); mp 114–116 °C; $^{31}$P NMR ($CDCl_3$ δ): 41.4.

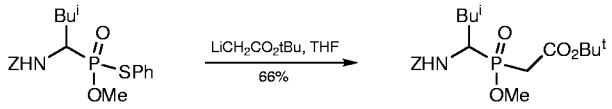
***N*-{[1-(Benzyloxycarbonylamino)-3-methylbutyl(methoxy)phosphoryl]carbonyl}-L-leucine Methyl Ester (69, $R^1$ = iBu; $R^2$ = CH(iBu)NHZ; $R^3$ = Me); Typical Procedure:**[103]

A soln of H-L-Leu-OMe (379 mg, 2.6 mmol) in dry THF (0.5 mL) was added over a 1 min period to a soln of trichloromethyl chloroformate (0.31 mL, 2.6 mmol) in dry THF (10 mL) at 22 °C. The mixture was stirred for 30 min and the solvent was removed in vacuo. To a stirring soln of methyl [1-(benzyloxycarbonylamino)-3-methylbutyl]phosphinate (195 mg, 0.65 mmol) in $CH_2Cl_2$ (2.8 mL) at 0 °C was added dry TEA (0.30 mL, 2.15 mmol) and freshly distilled TMSCl (0.25 mL, 1.96 mmol). After 30 min, dry TEA (0.161 mL, 1.16 mmol) and the crude Leu trichloromethyl carbamate/isocyanate (354 mg, 1.16 mol) in $CH_2Cl_2$ (0.5 mL) was added rapidly, and the mixture was stirred at 0 °C for 1.5 h. The mixture was partitioned between sat. $NH_4Cl$ and EtOAc, and the crude product isolated from the organic layer was purified by flash chromatography to afford the carbamoylphosphinate as a clear glass as an equimolar mixture of four diastereomers; yield: 273 mg (79%). An analytically pure sample was prepared by bulb-to-bulb distillation (>250 °C/0.05 Torr): $^{31}$P NMR (δ): 30.08, 29.96, 29.66, 29.63.

#### 10.10.2.3.2 Reactions Involving Electrophilic Phosphorus

A less common approach to the synthesis of phosphinates is the reaction of electrophilic phosphonates with carbon nucleophiles such as Grignard reagents or lithium enolates. For example, the phosphinic acid analogue **71** of the amino acid statine was synthesized by displacement of *tert*-butyl lithioacetate on a *S*-phenyl phosphonothioate **70** (Scheme 23).[104] The racemic diastereomers of the *S*-phenyl phosphonothioate were obtained in pure form, and the displacement of the phenylsulfanyl moiety was found to be stereospecific, although the stereocenter at phosphorus would later be lost on hydrolysis of the ester. A similar displacement reaction has been described using the phosphonochloridate.[71]

**Scheme 23** Addition of an Enolate to a *S*-Phenyl Phosphonothioate[104]




**70** **71**

***O*-Methyl *S*-Phenyl 1-(Benzyloxycarbonylamino)-3-methylbutylphosphonothioate (70):**[104]
Methyl 1-(benzyloxycarbonylamino)-3-methylbutylphosphinate (5.0 g, 15.9 mmol) was suspended in $CH_2Cl_2$ (10 mL) and cooled to 0 °C. $SOCl_2$ (1.27 mL, 17.5 mmol) was added and after 1 h the soln was warmed to rt. After 10 h at rt, the solvent was removed at reduced pressure and traces of HCl and $SO_2$ were removed by dilution and reevaporation from $CH_2Cl_2$. A soln of the residue in $CH_2Cl_2$ (100 mL) was cooled to 0 °C, TEA (6.65 mL, 47.7 mmol) and PhSH (4.9 mL, 47.7 mmol) were added, and the mixture was kept at rt for 12 h. The soln was washed three times with 2 M NaOH, $H_2O$, and brine, dried, filtered, and concentrated to leave a yellow semi-solid (5.45 g). Crystallization of this material ($CH_2Cl_2$/$iPr_2O$) afforded **70** as white crystals; yield: 3.62 g (56%); mp 142–143 °C. The diastereomers were separated by fractional crystallization ($CH_2Cl_2$/$iPr_2O$/$Et_2O$): diastereomer A: mp 142–144 °C; $^{31}P$ NMR (δ): 53.92; diastereomer B: mp 108–110 °C; $^{31}P$ NMR (δ): 54.43.

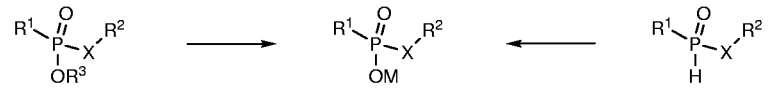
***tert*-Butyl 2-{[1-(Benzyloxycarbonylamino)-3-methylbutyl](methoxy)phosphoryl}acetate (71):**[104]
A soln of LDA was prepared in THF (40 mL) at −78 °C from $iPr_2NH$ (1.84 mL, 13.1 mmol) and 1.41 M BuLi in hexane (9.3 mL). tBuOAc (1.77 mL, 13.1 mmol) was added, followed, after 15 min, by a soln of *O*-methyl *S*-phenyl 1-(benzyloxycarbonylamino)-3-methylbutylphosphonothioate (**70**, diastereomer A; 1.61 g, 3.95 mmol) in THF (50 mL) precooled to −78 °C. The soln was stirred at −78 °C for 1 h and at 0 °C for 1 h, warmed to rt, and diluted with sat. $NH_4Cl$ (100 mL). The mixture was extracted with $CHCl_3$ (3 × 50 mL), the combined organic extracts were washed with twice with 2 M NaOH, $H_2O$, and brine, dried, filtered, and concentrated to give the phosphinate as an oily semi-solid (1.53 g). This material was crystallized ($CH_2Cl_2$/$iPr_2O$) to afford diastereomerically pure adduct as a white solid; yield: 1.08 g (66%); mp 113–114 °C; $^{31}P$ NMR (δ): 46.44. The other phosphinate diastereomer was prepared in the same fashion from diastereomer B of the phosphonothioate starting material; mp 89–90 °C; $^{31}P$ NMR (δ): 46.39.

## 10.10.3 Peptides Containing Deprotected Phosphorus

The target for most syntheses of phosphorus-containing peptide analogues has a P–OH group or its conjugate base (Scheme 24), although there are some exceptions. The acid is masked during the synthesis, either as an ester or as an H-phosphinate, and the final step of the synthesis is conversion into the acid or its salt by cleavage of the ester or oxidation of the H-phosphinate. Exceptions are electrophilic peptidyl phosphonates,[63,65] which are covalent inactivators of serine peptidases, some phosphinamides related to pepstatin, explored as potential inhibitors of pepsin,[66,105] and 4-nitrobenzyl esters developed as haptens for elucidation of catalytic antibodies for transacylation.[72] One report of dipeptidyl phosphonamidate methyl esters as HIV-1 protease inhibitors showed them to be very weakly bound (40–80 μM), albeit comparable to the corresponding phosphonamidate anions.[106]

*for references see p 525*

**Scheme 24** Final Step in the Synthesis of Phosphorus Peptide Analogues



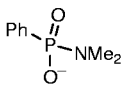
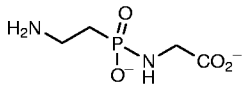
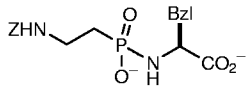
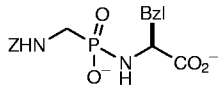
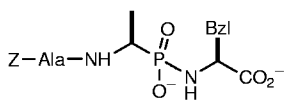
X = O, NH, $CH_2$; M = H, Li, Na

The phosphorus methyl ester is by far the most common protecting group used in this strategy, since it is easily introduced and a wide variety of methods exist for its removal. Cleavage of a benzyl ester by hydrogenolysis is also an attractive approach,[107] but has not been as widely used, both because of its lability toward nucleophiles as well as the common use of other protecting groups (particularly Z) that are removed by hydrogenolysis.

For the synthesis of simple peptides in which the phosphorus-containing amino acid analogue is at the "C-terminal" end of the peptide, acid-catalyzed cleavage of the diester has been used to produce the phosphonic diacid.[53,68] However, the strong acid needed to effect this transformation makes the strategy unattractive for more complex molecules and it has largely been superceded by the use of bromotrimethylsilane (see Section 10.10.3.2.1).

If there are no acid-sensitive groups in the molecule, the deprotected phosphonates (P–O) or phosphinates (P–C) may be isolated either as the acids or the corresponding salts. In contrast, since the P–N bond of phosphonamidic acids is readily hydrolyzed under acidic conditions, these analogues must be isolated as their phosphonamidate salts. Purification of the phosphonamidates is usually performed at pH 8.5–10 using ion-exchange chromatography or reverse-phase HPLC with the volatile triethylammonium hydrogencarbonate buffer. It is then necessary to add an equivalent of a nonvolatile base (e.g., LiOH) before lyophilization, otherwise removal of the volatile buffer leads to acidification and subsequent hydrolysis of the phosphonamidate. The stability of phosphonamidates varies widely, depending upon the exact structure, with half-lives from minutes to days at pH 7 (Table 11). Hydrolysis occurs via the zwitterionic species (e.g., $R^1PO_2^-–NH_2^+–R^2$);[108] hence, the phosphonamidate ester precursors are relatively stable.

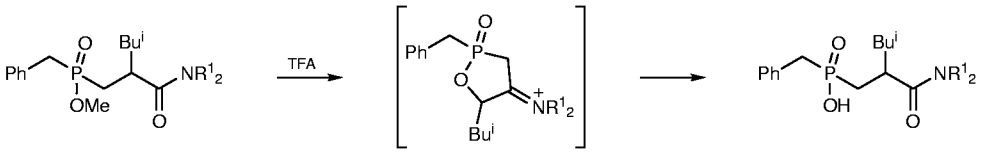
**Table 11** Stability of Phosphonamidates[64,71,107,109–111]

| Compound | pH | Temp (°C) | $t_{1/2}$ | Ref |
|---|---|---|---|---|
|  | 6.3<br>7 | 20.6<br>29.5 | 63 min<br>2 h | [109] |
|  | 7 | 25 | 3–6 h | [107] |
|  | 6.0<br>7.5 | 25<br>25 | 20% hydrolysis after 2.5 h<br>>2 days | [110] |
|  | 6.2<br>7.5<br>8.5 | 21<br>21<br>21 | 4 h<br>5–10 days<br>35 d | [64,111] |
|  | 7.0<br>7.5<br>8.5 | 21<br>21<br>21 | 5–10 min<br>10–15 min<br>160 min | [71] |

### 10.10.3.1 P–O Cleavage: Acid- and Base-Catalyzed Deprotection

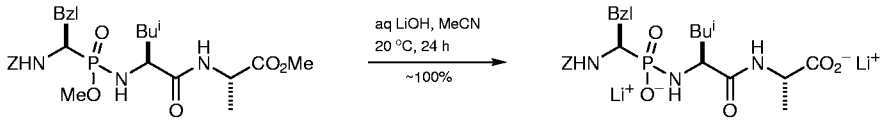
Acid-labile protecting groups are not used in the synthesis of phosphonamidate peptide analogues because of the sensitivity of the P–N linkage, and they are not commonly used with the phosphonates since orthogonal protection is often desired; moreover, anchimeric assistance from neighboring peptides makes even the P–O linkage susceptible to acid cleavage under some circumstances (Scheme 25).[112] However, acid-labile protecting groups, such as adamantyl[113] and benzhydryl[114] esters, have been employed for phosphinate derivatives, since they are the most robust. These esters are cleaved with trifluoroacetic acid, and the benzhydryl ester can also be cleaved by hydrogenolysis.

**Scheme 25** Anchimeric Assistance in Acid-Catalyzed Hydrolysis of Peptidyl Phosphinates[112]




For simple, symmetrical phosphonate diesters, treatment with aqueous hydroxide produces the monoester in good yield (see Section 10.10.2.1.1). However, for more complex, non-symmetrical phosphonate diesters, alkaline hydrolysis is not normally used, since it would give a mixture of the two possible monoesters. In contrast, the ester moiety of a phosphonamidate may be cleaved using aqueous hydroxide, since the amide nitrogen is a much poorer leaving group than the ester oxygen.[64,73,107,115] Thus, the thermolysin inhibitor **73** was produced by stirring the phosphonamidate **72** in a mixture of acetonitrile and aqueous LiOH for 24 hours at room temperature (Scheme 26).[73] Since carboxylate esters are more readily hydrolyzed than the corresponding phosphonate esters,[107] the dianion was formed under these conditions. Phosphinate esters may also be cleaved using aqueous hydroxide.[103]

**Scheme 26** Alkaline Hydrolysis of a Phosphonamidate Ester[73]




**72** **73**

**Z-Phe$^P$-Leu-Ala-OH (73); Typical Procedure:**[73]

A soln of the dimethyl ester **72** (30 mg, 0.054 mmol) in MeCN (0.11 mL) was treated with 2.75 M aq LiOH (0.59 mL, 0.162 mmol) and $H_2O$ (0.12 mL) at 20 °C for 24 h. After lyophilization, the crude salt was purified by chromatography (20-cm × 1-cm column of DEAE-Sephadex A-25 ($HCO_3^-$ form, using a linear gradient of 0–0.5 M triethylammonium hydrogencarbonate buffer pH 8.6, 300 mL). Fractions containing the desired compound (monitored by absorbance at 257 nm) were pooled, treated with 2.75 M LiOH (0.49 mL, 2.2 equiv based on $\varepsilon_{257}$ 340), lyophilized, and finally evaporated from MeOH (to remove traces of triethylammonium carbonate) to give a quantitative yield of the lithium salt of **73**; $^{31}$P NMR ($D_2O$, δ): 20.2 (major conformer), 19.5 (minor conformer).

### 10.10.3.2 O–C Cleavage by Nucleophilic Reagents

A number of methods for cleaving phosphorus esters proceed by nucleophilic attack on the alkyl group of the ester, rather than attack at the phosphorus. As expected for a bimolecular nucleophilic substitution reaction, these reactions are fastest for benzyl and methyl groups, and the rate drops rapidly as the steric hindrance around the alkyl group increases. Thus,

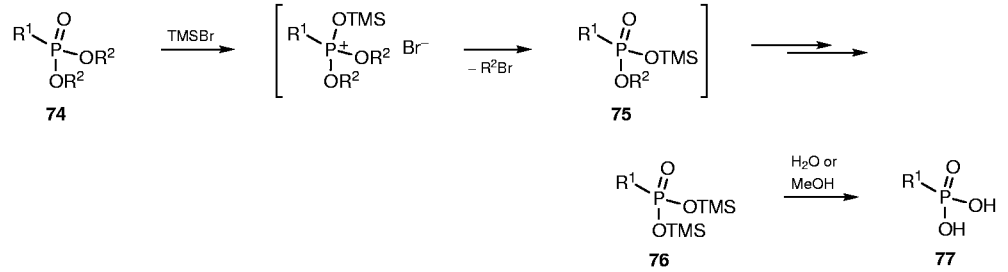
*for references see p 525*

these methods allow for selective deprotection of phosphorus methyl (or primary alkyl) esters in unsymmetrical phosphonate diesters. Two popular nucleophilic reagents, bromotrimethylsilane (Section 10.10.3.2.1) and lithium propanethiolate (Section 10.10.3.2.2), are discussed, but several others are also useful. Phosphorus methyl esters are rapidly cleaved when dissolved in *tert*-butylamine at room temperature.[116] The resulting *tert*-butylamine salt may be obtained by filtration or the volatile *tert*-butylamine may be removed. The phosphorus methyl ester can be cleaved selectively in the presence of a carboxylic acid methyl ester using this reagent.[116,117] Lithium or sodium azide and lithium or sodium iodide have also been used to cleave phosphorus alkyl esters.[72,118]

### 10.10.3.2.1 Bromotrimethylsilane

For conversion of simple phosphonic acid diesters **74** into the corresponding phosphonic diacids **77**, bromotrimethylsilane[119] is a mild and effective reagent (Scheme 27).[120,121] In analogy to the Arbuzov reaction, the mechanism involves attack by the phosphonyl oxygen on silicon, releasing bromide ion which then attacks the alkyl group on the phosphorus to give $R^2Br$ and a trimethylsilyl phosphonate **75**. Another round then produces the bis-(trimethylsilyl) phosphonate **76**. The silyl groups are readily removed by treatment with water or a simple alcohol. The yield for this process is normally quite high. Distilled TMSBr may be stored under an inert atmosphere at room temperature and used as needed. For acid-sensitive substrates, the reaction mixture may be saturated with 2-methylprop-1-ene or buffered with pyridine to react with any HBr that may be formed by hydrolysis.

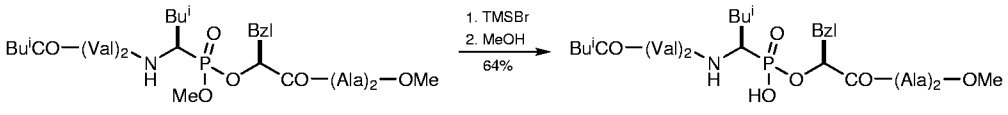
**Scheme 27** Cleavage of Phosphonate Diesters with Bromotrimethylsilane[119–121]




74 75 76 77

This method is compatible with many functional groups and shows considerable selectivity. Phosphonic acid esters may be cleaved in the presence of carboxylic acid esters, and phosphonate methyl esters are cleaved approximately 25 times faster than the isopropyl esters. For example, the phosphonate methyl ester of the hexapeptide analogue **78** was cleaved with TMSBr to give the aspartic peptidase inhibitor **79** in 64% yield (Scheme 28).[66]

**Scheme 28** Selective Cleavage of a Phosphonate Methyl Ester with Bromotrimethylsilane[66]




78 79

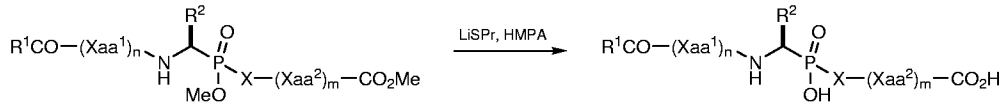
Phosphinate monoalkyl esters are also readily converted into the parent phosphinic acids using TMSBr.[66,104] However, attempted deprotection of a phosphonamidate monoester using this reagent was reported to result in cleavage of the P–N bond.[107]

**iBuCO-Val-Val-Leu$^P$-(O)Phe-Ala-Ala-OMe (79); Typical Procedure:**[66]
To a soln of 2-methylprop-1-ene in $CH_2Cl_2$ (5 mL, prepared by sparging with 2-methylprop-1-ene for 10 min) was added TMSBr (87 μL, 0.66 mmol). After 5 min, a soln of the phosphonate methyl ester **78** (100 mg, 0.13 mmol) in $CH_2Cl_2$ (1 mL) was added to the TMSBr soln and the reaction flask was sealed with a septum. After stirring at rt for 24 h, the flask was carefully vented with a small bore needle to release excess pressure. The solvent and volatile products were removed in vacuo and the residue was reevaporated twice from $CHCl_3$. MeOH (4 mL) was added, and the white solid that formed was collected and dried in vacuo to yield **79**; yield: 63 mg (64%); mp 225–227 °C; $^{31}$P NMR (DMSO-$d_6$, δ): 24.49.

### 10.10.3.2.2 Lithium Propanethiolate

In 1970, Bartlett and Johnson reported the use of lithium propanethiolate in HMPA as a reagent for dealkylation of hindered carboxylic methyl esters.[122] This reagent is also widely used for dealkylating phosphorus methyl esters in phosphonamidates (**80**, X = NH),[67] phosphonates (**80**, X = O),[67,71,73,103] and phosphinates[104] (Scheme 29). Because this reaction, too, involves nucleophilic attack on the alkyl group of the ester, unsymmetrical phosphonate diesters are cleaved preferentially at the less hindered methyl group. Thus, treatment of the phosphonate **81** with lithium propanethiolate for 1 hour at room temperature yielded the desired diacid **82** in 75% yield (Scheme 30).[73] This method is rapid and mild but the reactions must be done in a good fume hood due to the stench of the thiol. In addition, care must be taken in handling the HMPA, which is a cancer suspect agent.

**Scheme 29** Cleavage of Phosphonate, Phosphonamidate, and Carboxylate Methyl Esters Using Lithium Propanethiolate[67,71,73,103]

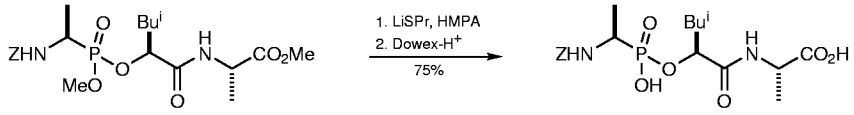



**80**

| Substrate | LiSPr (Equiv) | Isolated Form | Yield (%) | $^{31}$P NMR δ | Ref |
|---|---|---|---|---|---|
| Z-Ala$^P$-(O)Leu-Ala dimethyl ester | 2.5 | diacid | 75 | 25.9 | [73] |
| Z-Ala-Ala$^P$-(O)Phe dimethyl ester | 2.1 | dilithium salt | 37 | 20.5 | [67] |
| Z-Phe$^P$-(O)Leu dimethyl ester | 2.5 | diacid | 65 | 20.1, 19.8[a] | [103] |
| Z-Ala-Ala$^P$-(NH)Phe dimethyl ester | 3.1 | dilithium salt | 62[b] | 21.5 | [67] |
| Ac-Ala-Phe$^P$-(O)Leu-Ala dimethyl ester | 3.0 | dilithium salt | 83 | 19.1 | [71] |

[a] Mixture of epimers α to phosphorus.
[b] Crude yield, determined by $^{31}$P NMR.

**Scheme 30** Selective Cleavage of a Phosphonate Methyl Ester Using Lithium Propanethiolate[73]




**81** **82**

**Lithium Propanethiolate:**[122]

**CAUTION:** *Hexamethylphosphoric triamide (HMPA) is a substance suspected of carcinogenic potential in man. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

Under an inert atmosphere, freshly distilled PrSH (1.0 mL, 11 mmol) was added to a suspension of finely ground LiH (0.3 g, 38 mmol) in dry, oxygen-free HMPA (10 mL). After stirring for 1 h at rt, the soln was filtered under an inert atmosphere and stored under argon at 0 °C without removal of the excess thiol. The concentration of thiolate is typically 0.5–0.6 M. Contact with air should be avoided since the thiolate is readily oxidized.

*for references see p 525*

**Z-Ala$^P$-(O)Leu-Ala-OH (82); Typical Procedure:**[73]

**CAUTION:** *Hexamethylphosphoric triamide (HMPA) is a substance suspected of carcinogenic potential in man. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

To a soln of Z-Ala$^P$-(O)Leu-Ala dimethyl ester (**81**; 9.6 mg, 0.126 mmol) in dry, distilled HMPA (0.25 mL) was added 0.55 M LiSPr in HMPA (0.57 mL, 0.32 mmol) at 23 °C under $N_2$. After 1 h, $H_2O$ (2 mL) was added and the soln was extracted with $CHCl_3$ (4 × 1 mL) to remove the HMPA. The aqueous layer was lyophilized and purified on DEAE-Sephadex (A-25, $HCO_3^-$ form) (1-cm × 25-cm column, linear gradient of 0–0.3 M triethylammonium hydrogencarbonate buffer pH 8.6, 300 mL). Fractions (6 mL) were monitored by absorbance at 257 nm. Combination of fractions 26–39, lyophilization, ion exchange of the resulting white solid through Dowex 50W X8 resin ($H^+$ form) with $H_2O$/MeOH 1:1, and final lyophilization gave **82** as a white, hygroscopic solid; yield: 42 mg (75%).

## 10.10.3.3 Oxidation of H-Phosphinates

In contrast to phosphonic acids, H-phosphinic acids can be coupled with alcohols using carbodiimides to give H-phosphinates **83** (Scheme 31, Table 12).[63,123] These H-phosphinates **83** afford the corresponding phosphonic acids **84** directly by oxidation with $NaIO_4$[123] or by treatment with bis(trimethylsilyl)amide followed by carbon tetrachloride, triethylamine, and water.[63] Alternatively, as described above (Section 10.10.2.1.3), H-phosphinates may also be converted into the phosphonochloridates by oxidation with carbon tetrachloride and then coupled to nucleophiles other than water.[63,75,76]

**Scheme 31** Synthesis of Phosphonate Monoesters by Esterification and Oxidation of H-Phosphinic Acids[63,123]

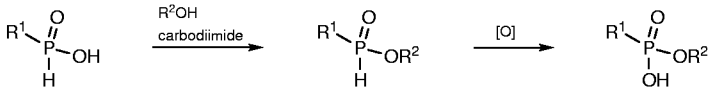
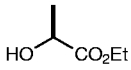
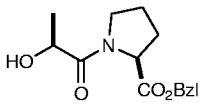
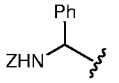
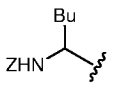



**Table 12** Synthesis of Phosphonate Monoesters by Esterification and Oxidation of H-Phosphinic Acids[123]

| Entry | $R^1$ | $R^2OH$ | Yield (%) | mp (°C) |
|---|---|---|---|---|
| 1 | $(CH_2)_4Ph$ | EtOH | 89 | 183–184 |
| 2 | $(CH_2)_4Ph$ | BzlOH | 76 | 181–184 |
| 3 | $(CH_2)_4Ph$ | iPrOH | 79 | 188–190 |
| 4 | $(CH_2)_4Ph$ | tBuOH | 88 | 188–191 |
| 5 | $(CH_2)_4Ph$ |  | 65 | 142–143 |
| 6 | $(CH_2)_4Ph$ |  | 84 | 172–174 |
| 7 | Ph | BzlOH | 64 | 257–259 |
| 8 | Cy | BzlOH | 76 | 215–217 |
| 9 |  | BzlOH | 82 | 163–164 |
| 10 |  | BzlOH | 82 | 154–156[a] |

[a] mp of free acid.

**N-{(2S)-2-[(Hydroxy)(4-phenylbutyl)phosphoryloxy]propanoyl}-L-proline Benzyl Ester, Adamantan-1-amine Salt (Table 12, Entry 6); Typical Procedure:**[123]

To a soln of 4-phenylbutylphosphinic acid (1.224 g, 6.18 mmol) and *N*-(L-2-hydroxypropanoyl)proline benzyl ester (1.80 g, 6.49 mmol) in dry THF (15 mL) at rt were added DCC (1.40 g, 6.80 mmol) and DMAP (0.076 g, 0.62 mmol). After stirring at rt for 2 h, the mixture was diluted with EtOAc (50 mL), filtered, and washed successively with 5% $KHSO_4$, sat. $NaHCO_3$, and brine, dried ($Na_2SO_4$), and concentrated to give the crude phosphinic monoester as a colorless oil. This oil was dissolved in dioxane (10 mL), treated with a soln of $NaIO_4$ (1.52 g, 7.11 mmol) in $H_2O$ (4 mL), and the resultant mixture was stirred at rt for 16 h. The mixture was filtered and partitioned between EtOAc and 2% $KHSO_4$ (75 mL each). The organic layer was washed with $H_2O$, dilute $NaHSO_3$, and brine, dried ($Na_2SO_4$), and concentrated to an oil. This oil was dissolved in $Et_2O$ (20 mL) and treated with a soln of adamantan-1-amine (1.00 g) in $Et_2O$ (5 mL). The resultant precipitate was collected and washed with $Et_2O$ to give the phosphonate monoester as the adamantan-1-amine salt; yield: 3.26 g (84%).


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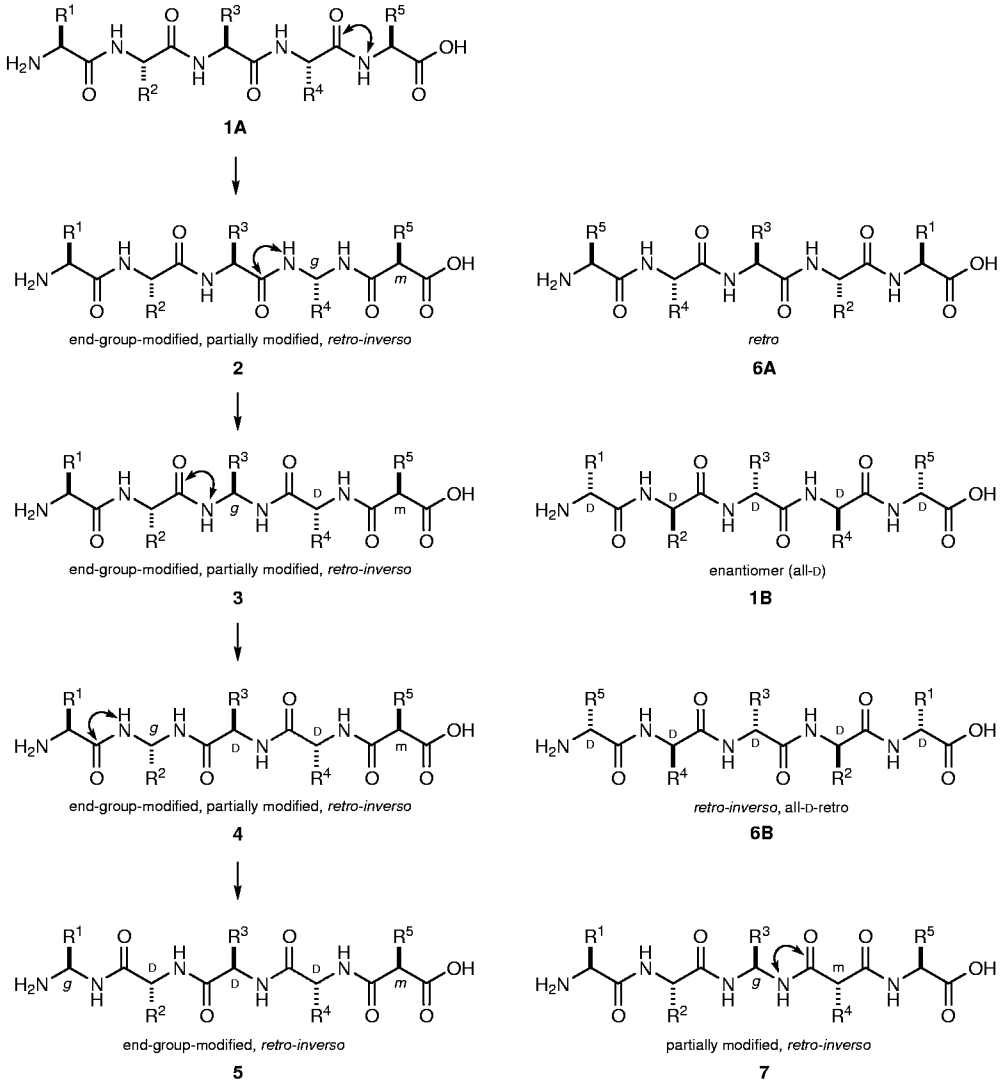
# 10.11 Synthesis of *retro-inverso*-Peptides

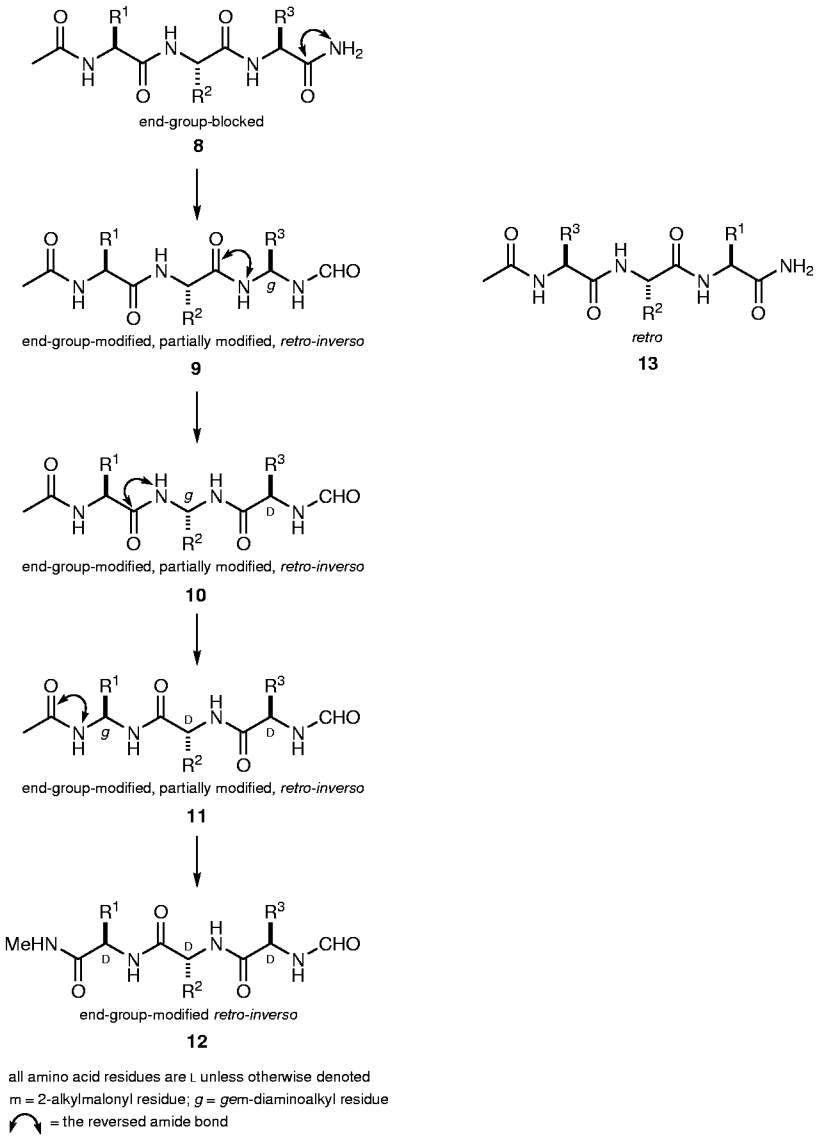
L. Scheibler and M. Chorev

The combination of the nonpalindromic nature of the peptide backbone and the chirality of the α-carbons in the amino acids comprising the oligomeric structure provides a unique opportunity for vast stereochemical and topochemical diversities. Earlier studies explored the interesting structural relationship between all-L-, *retro*-all-L-peptides, and their enantiomers (see reviews[1–3]). Schematically, a high degree of topochemical complementarity is obtained between parent cyclic peptides and their *retro-inverso* analogues.[1]

However, for the same structural transformations applied to linear peptides (e.g., **1A** and **8**; Scheme 1), the end groups cause inherent difficulties, which are absent in the end-to-end cyclic peptides. Since syntheses of linear enantiomers, their *retro*-isomers, and *retro-inverso*-isomers (e.g., **1A**, **6A**, **1B**, **6B**; Scheme 1) and their end-group-modified peptides (e.g., **8**, **12**, **13**; Scheme 1) follow the conventional protocols of peptide synthesis, they are not covered in this section.

**Scheme 1** Structural Isomers of a Linear Pentapeptide and a Linear Tripeptide






Structural isomers of pentapeptide **1A** and tripeptide **8** are represented in Scheme 1. For pentapeptide **1A**, its *retro*-peptide **6A**, enantiomer (all-D-peptide) **1B**, *retro-inverso*-peptide (all-D-*retro*-peptide) **6B**, partially modified, *retro-inverso*-peptide **7**, end-group-modified, partially modified, *retro-inverso*-peptides **2–4**, and end-group-modified, *retro-inverso*-peptide **5** are shown. For end-group-blocked tripeptide **8**, its *retro*-peptide **13**, end-group-modified, partially modified, *retro-inverso*-peptides **9–11**, and end-group modified *retro-inverso*-peptide **12** appear.

The *retro*-peptide bond is a true isosteric peptide bond surrogate and as such may offer an important tool to study topics such as the functional role of the peptide backbone in peptide hormone–receptor interactions, and modulation of metabolic stability and bioavailability. Partially modified *retro-inverso*-peptides (PMRI-peptides) (e.g., **2–4**, **7**; Scheme 1) result from a *retro-inverso* transformation of one or several peptide bonds in an amino- and carboxy-free peptide (e.g., **5**; Scheme 1). Evidently, partial or exhaustive *retro-inverso* transformations result in the introduction of two non-amino acid residues into the

*for references see p 549*

newly formed isomers, the *gem*-diaminoalkyl (*g*Xaa) and the 2-alkylmalonyl (*m*Xaa) residues at the amino and carboxy sides of the transformed sequence, respectively. Similarly, partial or exhaustive *retro-inverso* transformation of the amino- and carboxy-termini blocked peptides (e.g., **8**; Scheme 1), when initiated from the blocked C-terminus, lead to the formation of end-group-modified PMRI- (e.g., **9–11**; Scheme 1) and end-group-modified *retro-inverso*-peptides (e.g., **12**; Scheme 1). Again, PMRI modification of end-group-modified peptides will result in substituting the amino acid at the amino side of the transformed peptide bond with the corresponding *gem*-diaminoalkyl residue. Introduction of a PMRI transformation at the N-terminal peptide bond or at a peptide bond removed from the C-terminus will result in the incorporation of a *g*Xaa residue, with or without an *m*Xaa residue, respectively (not shown).

The introduction of *retro*-, *retro-inverso*-, and PMRI-peptides with free and blocked C- and N-termini has been successful in numerous biological systems such as neurotransmitters, inhibitors of proteases and protein kinases, sweeteners, antimicrobial peptides, hormones, adhesion molecules, antigenic epitopes, immuno-modulators, and immunological probes. Table 1 provides an exhaustive list of *retro*-, *retro-inverso*-, PMRI-, and end-group-modified *retro-inverso*-pseudopeptides derived from bioactive peptides.

**Table 1** Bioactive *retro*-, *retro-inverso*-, Partially Modified *retro-inverso*-, and End-Group-Modified *retro-inverso*-Peptides[4–71]

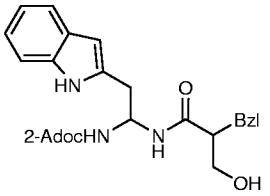
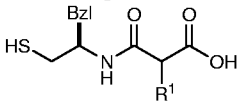
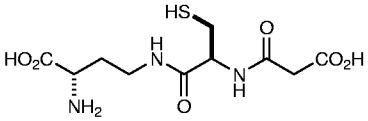
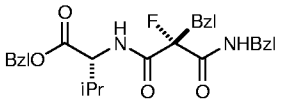
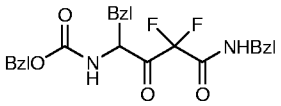
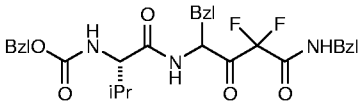
| Bioactive Peptide | Structure[a] | Ref |
|---|---|---|
| atrial natriuretic factor (ANF) | [Mpr$^7$,*g*Ser$^{25}$,(*RS*)-*m*Phe$^{26}$]*r*ANF(7–28)$NH_2$ | [4] |
| | [*g*Ser$^{25}$,(*RS*)-*m*Phe$^{26}$]*r*ANF(7–28)$NH_2$ | [4] |
| | [*g*Ser$^{25}$,(*RS*)-*m*Phe$^{26}$]*r*ANF(5–28)$NH_2$ | [4] |
| bombesin | H-*g*Asn-D-*r*Glu-D-*r*Trp-D-*r*Val-*r*Gly-D-*r*His-D-*r*Leu-(*RS*)-*m*Met-$NH_2$ | [5] |
| bradykinin potentiating peptide$_{9a}$ (BPP) | pGlu-Trp-Pro-Arg-Pro-*g*Lys-(*RS*)-*m*Phe-Ala-Pro-OH | [6] |
| | pGlu-Lys-*g*Phe-(*S*)-*m*Ala-Pro-OH | [7] |
| | HO-D-*r*Val-(*RS*)-*m*Trp-OH | [8] |
| | Bz-*g*Gly-D-*r*Val-(*RS*)-*m*Trp-OH | [8] |
| | PhNH-(*RS*)-*m*Phe-Gly-Trp-OH | [8] |
| | Bz-*g*Phe-*m*Gly-Trp-OH | [8] |
| | Bz-Phe-*g*Gly-(*RS*)-*m*Trp-OH | [8] |
| | Bz-*g*Gly-(L or D)-*r*Arg-(*RS*)-*m*Trp-OH | [8] |
| C-reactive protein (CRP): | | |
| CRP(27–30) | H-*g*Thr-(*RS*)-*m*Lys-Pro-Leu-OH | [9] |
| CRP(113–116) | H-*g*Gly-(*RS*)-*m*Lys-Pro-Arg-OH | [9] |
| CRP(200–203) | H-*g*Thr-(*RS*)-*m*Lys-Pro-Gln-OH | [9] |
| cecropin-melittin hybrid | $H_2N$-D-*r*Lys-D-*r*Trp-D-*r*Lys-D-*r*Leu-D-*r*Phe-D-*r*Lys-D-*r*Lys-D-*r*Ile-*r*Gly-D-*r*Ala-D-*r*Val-D-*r*Leu-D-*r*Lys-D-*r*Val-D-*r*Leu-H | [10] |
| | $H_2N$-D-*r*Lys-D-*r*Trp-D-*r*Lys-D-*r*Leu-D-*r*Phe-D-*r*Lys-D-*r*Lys-D-*r*Ile-D-*r*Glu-D-*r*Lys-D-*r*Val-*r*Gly-D-*r*Gln-*r*Gly-D-*r*Ile-*r*Gly-D-*r*Ala-D-*r*Val-D-*r*Leu-D-*r*Lys-D-*r*Val-D-*r*Leu-D-*r*Thr-D-*r*Thr-*r*Gly-D-*r*Leu-H | [10] |
| cholecystokinin (CCK) |  | [11] |
| | Boc-Tyr($SO_3Na$)-*g*Nle-*m*Gly-Trp-Nle-Asp-Phe-$NH_2$ | [12] |
| | Boc-Asp-Tyr($SO_3Na$)-*g*Nle-*m*Gly-Trp-Nle-Asp-Phe-$NH_2$ | [13] |
| | Boc-Tyr($SO_3Na$)-*g*Nle-*m*Gly-Trp-(Me)Nle-Asp-Phe-$NH_2$ | [12] |
| | Boc-Asp-Tyr($SO_3^-$)-Nle-Gly-Trp-Nle-*g*Asp-(*RS*)-*m*Phe-$NH_2$ | [14] |
| | Boc-Asp-Tyr($SO_3^-$)-Nle-Gly-Trp-Nle-*g*Asp-$COCH_2Bzl$ | [14] |
| | Boc-Asp-Tyr($SO_3^-$)-Nle-*g*Gly-D-*r*Trp-D-*r*Nle-D-*r*Asp-$COCH_2Bzl$ | [14] |
| | Boc-Asp-Tyr($SO_3^-$)-Nle-*g*Gly-D-*r*Trp-D-*r*Nle-D-*r*Asp-(*RS*)-*m*Phe-$NH_2$ | [14] |

**Table 1** (cont.)

| Bioactive Peptide | Structure[a] | Ref |
|---|---|---|
| | Boc-Tyr($SO_3^-$)-Nle-Gly-(L or D)-Trp-Nle-Asp-*g*Phe-CHO | [14] |
| dermorphin | H-Tyr-D-Ala-*g*Phe-*r*Gly-(*RS*)-*m*Tyr-Pro-Ser-$NH_2$ | [15] |
| | H-Tyr-D-Ala-Phe-*g*Gly-(*RS*)-*m*Tyr-Pro-Ser-$NH_2$ | [15] |
| | H-Tyr-D-Ala-*g*Phe-*m*Gly-Tyr-Pro-Ser-$NH_2$ | [15] |
| | H-Tyr-D-Ala-*g*Phe-*m*Gly-X [X = $NH_2$, NHBzl, D-NHCH(Me)Ph, NHAda] | [16] |
| | H-Tyr-D-Ala-Phe-*g*Gly-COX [X = H, Bzl, D-CH(Me)Ph, Ada] | [16] |
| | H-Tyr-D-Ala-*g*Phe-*r*Gly-COX [X = H, Bzl, D-CH(Me)Ph, Ada] | [16] |
| | H-Tyr-*c*[-D-$A_2$bu-Phe-*g*Phe-(*S* and *R*)-*m*Leu-] | [17] |
| | H-Tyr-*c*[-D-Glu-Phe-*g*Phe-(L and D)-*r*Leu-] | [17] |
| encephalitogenic epitope P87–99 | HO-D-*r*Val-D-*r*His-D-*r*Phe-D-*r*Phe-D-*r*Lys-D-*r*Asn-D-*r*Ile-D-*r*Val-D-*r*Thr-D-*r*Pro-D-*r*Arg-D-*r*Thr-D-*r*Pro-OH | [18] |
| enkephalin | H-Tyr-D-Ala-Gly-*g*Phe-D-*r*Xaa-CHO (Xaa = Leu, Met) | [19] |
| | H-Tyr-D-Ala-Gly-*g*Phe-(*RS*)-*m*Xaa-$NH_2$ (Xaa = Leu, Met) | [19] |
| | H-Tyr-Gly-*g*Gly-(*RS*)-*m*Phe-Leu-$NH_2$ | [20] |
| | H-Tyr-D-Ala-*g*Gly-(*RS*)-*m*Phe-Leu-$NH_2$ | [20,21] |
| | H-Tyr-*c*[-D-Glu-Gly-*g*Phe-D-*r*Leu-] | [22] |
| | H-Tyr-*c*[-D-$A_2$bu-Gly-*g*Phe-(*R* or *S*)-*m*Leu-] | [22] |
| | H-Tyr-*c*[-D-Glu-Gly-Phe-*g*Leu-] | [22] |
| | H-Tyr-*c*[-D-$A_2$bu-*g*Gly-(*R* or *S*)-*m*Phe-Leu-] | [21] |
| enkephalinase inhibitors | H-*g*Phe-(*RS*)-*m*Ala-OH | [23] |
| | H-("*S*" or "*R*")-*g*Phe-*m*Gly-OH | [23] |
| | *retro*-thiorphans:  $R^1$ = H, Me, iBu | [24] |
| | *retro*-kelatorphans: X-NOH-CO-CHBzl-NHCO-CHY-$CO_2$H (X = H; Y = H, Me, Bzl or X = Me; Y = H) | [25,26] |
| gastrin | Boc-Trp-Leu-*g*Asp-(*RS*)-*m*Phe-$NH_2$ | [27] |
| | Boc-Trp-*g*Leu-D-Asp-(*RS*)-*m*Phe-$NH_2$ | [27] |
| | Boc-Trp-*g*Nle-D-Asp-COX (X = $CH_2$Bzl, NHBzl) | [27] |
| | Boc-Trp-Leu-*g*Asp-COX (X = $CH_2$Bzl, NHBzl) | [27] |
| | XCO-D-*r*Asp-D-*r*Nle-D-*r*Trp-*r*βAla-OtBu (X = $CH_2$Bzl, NHBzl, OBzl) | [27] |
| glutathione |  | [28] |
| growth factor GHK | H-Gly-*g*His-(*RS*)-*m*Lys-OH | [29] |
| HIV protease inhibitor |  | [30] |
| |  | [31] |
| |  | [31] |
| Immunogenic probes: | | |
| CD4+ T-cell epitope: γ2a[b] 435–447 | $H_2$N-D-*r*Tyr-D-*r*Phe-D-*r*Met-D-*r*Tyr-D-*r*Ser-D-*r*Lys-D-*r*Leu-D-*r*Arg-D-*r*Val-D-*r*Gln-D-*r*Lys-D-*r*Ser-D-*r*Ala-Ac | [32] |

*for references see p 549*

**Table 1** (cont.)

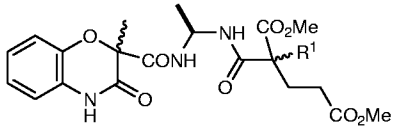
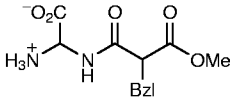
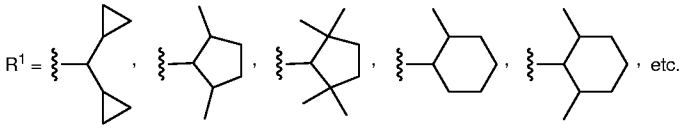
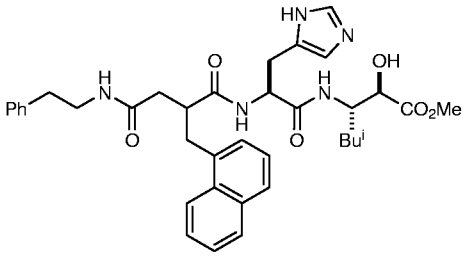
| Bioactive Peptide | Structure[a] | Ref |
|---|---|---|
| VP1(103–115) | $H_2N$-D-*r*Tyr-D-*r*Phe-D-*r*Ala-D-*r*Val-D-*r*Trp-D-*r*Lys-D-*allo*-*r*Ile-D-*r*Thr-D-*r*Tyr-D-*r*Lys-D-*r*Asp-D-*r*Thr-Ac | [33] |
| cytotoxic T-lymphocyte epitope | X-D-*r*Arg-*r*Gly-D-*r*Tyr-D-*r*Val-D-*r*Tyr-D-*r*Gln-*r*Gly-(*RS*)-*m*Leu-OH (X = OH, $NH_2$) | [34] |
| FcεRI α-chain-derived peptide | D-*r*C-D-*r*I-D-*r*Y-D-*r*K-D-*r*D-*r*G-D-*r*E-D-*r*A-D-*r*L-D-*r*K-D-*r*Y-D-*r*C (cyclic) | [35] |
| GP120 of HIV-1 antigen | HO-D-*r*Asp-D-*r*Asp-D-*r*Ala-D-*r*Leu-D-*r*Tyr-D-*r*Asp-D-*r*Asp-D-*r*Lys-D-*r*Asn-D-*r*Trp-D-*r*Asp-D-*r*Arg-D-*r*Ala-D-*r*Pro-D-*r*Gln-D-*r*Arg-D-*r*Cys-D-*r*Tyr-D-*r*Tyr-D-*r*Gln-H | [36] |
| histone H3(I[130]R-GERA[135]) analogues | $H_2N$-D-*r*Met-*r*Gly-*r*Gly-D-*r*Ile-D-*r*Arg-*r*Gly-D-*r*Glu-D-*r*Arg-(*RS*)-*m*Ala-OH | [37] |
| | $H_2N$-*r*Met-*r*Gly-*r*Gly-*r*Ile-*r*Arg-*r*Gly-*r*Glu-*r*Arg-(*RS*)-*m*Ala-OH | [37] |
| histone H3(28–45) | $H_2N$-D-*r*Ser-D-*r*Ala-D-*r*Pro-D-*r*Ala-D-*r*Thr-*r*Gly-*r*Gly-D-*r*Val-D-*r*Lys-D-*r*Lys-D-*r*Pro-D-*r*His-D-*r*Arg-D-*r*Tyr-D-*r*Arg-D-*r*Pro-*r*Gly-D-*r*Thr-Ac | [38] |
| IL-2 P15 | $H_2N$-D-*r*Arg-D-*r*Ala-D-*r*Asp-D-*r*Leu-D-*r*Asp-D-*r*Ala-D-*r*Pro-D-*r*Ala-D-*r*Leu-D-*r*Leu-D-*r*Trp-*r*Gly-D-*r*Leu-D-*r*Arg-D-*r*Val-OH | [39] |
| Ro52(277–291) | $H_2N$-Gly-D-*r*Leu-D-*r*Lys-D-*r*Lys-D-*r*Nle-D-*r*Leu-D-*r*Arg-D-*r*Thr-D-*r*Cys-D-*r*Ala-D-*r*Val-D-*r*His-D-*r*Ile-D-*r*Thr-(*RS*)-*m*Leu-OH | [38] |
| Ro60(304–324) | $H_2N$-D-*r*Val-D-*r*Cys-D-*r*Glu-D-*r*Lys-D-*r*Leu-D-*r*Cys-D-*r*Asn-D-*r*Glu-D-*r*Lys-D-*r*Leu-D-*r*Leu-D-*r*Lys-D-*r*Lys-D-*r*Ala-D-*r*Arg-D-*r*Ile-D-*r*His-D-*r*Pro-D-*r*Phe-D-*r*His-D-*r*Ile-Ac | [38] |
| VP1(135–158) of FMDV | X-*r*Gly-D-*r*Ser-*r*Gly-D-*r*Val-D-*r*Arg-*r*Gly-D-*r*Asp-D-*r*Phe-*r*Gly-D-*r*Ser-D-*r*Leu-D-*r*Ala-D-*r*Pro-D-*r*Arg-D-*r*Val-D-*r*Ala-D-*r*Arg-D-*r*Gln-Y [X = $NH_2$; Y = D-*r*Leu-D-*r*Cys-Ac, (*RS*)-*m*Leu-OH] | [40,41] |
| influenza matrix peptide (58–66) (HLA-A2 restricted epitope) | H-*g*Gly-(*RS*)-*m*Leu-Leu-Gly-Phe-Val-Phe-Thr-Leu-OH | [42,43] |
| | H-Gly-*g*Ile-(*RS*)-*m*Leu-Gly-Phe-Val-Phe-Thr-Leu-OH | [42,43] |
| | H-Gly-Ile-*g*Leu-*m*Gly-Phe-Val-Phe-Thr-Leu-OH | [42,43] |
| | H-Gly-Ile-Leu-*g*Gly-(*RS*)-*m*Phe-Val-Phe-Thr-Leu-OH | [42,43] |
| | H-Gly-Ile-Leu-Gly-*g*Phe-(*RS*)-*m*Val-Phe-Thr-Leu-OH | [42,43] |
| | H-Gly-Ile-Leu-Gly-Phe-*g*Val-(*RS*)-*m*Phe-Thr-Leu-OH | [42,43] |
| | H-Gly-Ile-Leu-Gly-Phe-Val-*g*Phe-(*RS*)-*m*Ala-Leu-OH | [42,43] |
| | H-Gly-Ile-Leu-Gly-Phe-Val-Phe-*g*Thr-(*RS*)-*m*Leu-OH | [42,43] |
| influenza nucleoprotein (CTL epitope) | H-Thr-Tyr-*g*Gln-(*RS*)-*m*Arg-Thr-Arg-Ala-Leu-Val-OH | [44] |
| inhibitors of integrin $\alpha_v\beta_3$ | HO-(L or D)-*r*Arg-*m*Gly-Asp-OH | [45,46] |
| | HO-(L or D)-*r*Arg-*m*Gly-Asp-X (X = Ser-OH, OBzl) | [46] |
| laminin-derived peptide | H-Tyr-Ile-Gly-*g*Ser-D-*r*Arg-CHO | [47] |
| | H-Tyr-*g*Ile-*r*Gly-D-*r*Ser-D-*r*Arg-CHO | [47] |
| luteinizing hormone-releasing hormone (LHRH) | pGlu-His-Trp-Ser-*g*Tyr-*m*Gly-Leu-Arg-Pro-Gly-$NH_2$ | [48] |
| | pGlu-His-Trp-Ser-*g*Tyr-*r*Gly-(*RS*)-*m*Leu-Arg-Pro-Gly-$NH_2$ | [48] |
| | pGlu-His-Trp-Ser-Tyr-Gly-Leu-Arg-*g*Pro-COEt | [48] |
| | *g*pGlu-D-*r*His-(*RS*)-*m*Trp-Ser-Tyr-Gly-Leu-Arg-Pro-Gly-$NH_2$ | [48] |
| | *g*pGlu-(*RS*)-*m*His-Trp-Ser-Tyr-Gly-Leu-Arg-Pro-Gly-$NH_2$ | [48] |
| | *g*pGlu-D-*r*His-D-*r*Trp-D-*r*Ser-D-*r*Tyr-*r*Gly-D-*r*Leu-D-*r*Arg-D-*r*Pro-COEt | [49] |
| | *g*pGlu-*r*Phe-D-*r*Trp-D-*r*Ser-D-*r*Tyr-*r*Gly-D-*r*Leu-D-*r*Arg-D-*r*Pro-*m*Gly-$NH_2$ | [49] |
| muramyl dipeptide |   $R^1$ = H, Me | [50] |
| neurotensin | H-Arg-Arg-Pro-Tyr-*g*Ile-(*RS*)-*m*Leu-OH | [51] |
| oxytocin | H-*g*Xaa-D-*r*Tyr-D-*r*Phe-Ac (Xaa = Met, Leu) | [52] |
| peptide sweetener |   | [53] |
| | HO-*r*Dap-(*RS*)-*m*Phe-OMe | [53] |

**Table 1** (cont.)

| Bioactive Peptide | Structure[a] | Ref |
|---|---|---|
| | H-Asp-"D"-*g*Ala-COR$^1$<br>$R^1$ =  , etc. | [54] |
| protein kinase C inhibitors | $H_2N$-D-*r*Arg-D-*r*Phe-D-*r*Ala-D-*r*Arg-D-*r*Lys-*r*Gly-D-*r*Xaa-D-*r*Leu-D-*r*Arg-D-*r*Gln-D-*r*Lys-D-*r*Asn-D-*r*Val-Ac (Xaa = Ala, Ser) | [55] |
| renin inhibitor |  | [56] |
| somatostatin | Ala—Gly—Cys—Lys—Asn—*g*Phe—(*RS*)-*m*Trp—Lys—Thr—Phe—Thr—Ser—Cys—OH | [57] |
| | Ala—Gly—Cys—Lys—Asn—Phe—*g*Trp—(*RS*)-*m*Lys—Thr—Phe—Thr—Ser—Cys—OH | [57] |
| | $H_2N$—*r*Phe—D-*r*Cys—D-*r*Phe—*r*Trp—D-*r*Lys—D-*r*Thr—D-*r*Cys—D-*r*Thr—H | [58] |
| | *c*[-Thz-Phe-D-Trp-Lys-Thr-*g*Phe-CO-] | [59] |
| | *c*[-*g*Sar-(*RS*)-*m*Phe-D-Trp-Lys-Thr-Phe-] | [59] |
| | *c*[-Pro-Phe-D-Trp-Lys-*g*Val-(*RS*)-*m*Phe-] | [59] |
| | *c*[-(*R* or *S*)-*m*Ala-Phe-D-Trp-Lys-Thr-*g*Phe-] | [59,60] |
| | *c*[-D-*r*(Me)Ala-D-*r*Phe-*r*Trp-D-*r*Lys-D-*r*Thr-D-*r*Phe-] | [61] |
| | *c*[-D-*r*Ala-D-*r*Phe-*r*Trp-D-*r*Lys-D-*r*Thr-D-*r*(Me)Phe-] | [61] |
| | *c*[-D-*r*Ala-D-*r*Phe-D-*r*Trp-*r*Lys-D-*r*Thr-D-*r*(Me)Phe-] | [61] |
| substance P | pGlu-Phe-*g*Phe-*m*Gly-Leu-Met-$NH_2$ | [62,63] |
| | *g*pGlu-(*RS*)-*m*Phe-Phe-Gly-Leu-Met-$NH_2$ | [63,64] |
| | pGlu-*g*Phe-(*RS*)-*m*Phe-Gly-Leu-Met-$NH_2$ | [65] |
| | $H_2N$-*r*Gln-*r*Gln-*r*Phe-*r*Phe-*r*Gly-*r*Leu-(*RS*)-*m*Met-$NH_2$ | [66] |
| | Boc-Ala-*g*Phe-D-*r*Phe-*m*Gly-Leu-Met-$NH_2$ | [67] |
| thrombospondin-1 | $H_2N$-D-*r*Lys-D-*r*Arg-D-*r*Phe-D-*r*Lys-D-*r*Gln-D-*r*Asp-*r*Gly-*r*Gly-D-*r*Trp-D-*r*Ser-D-*r*His-D-*r*Trp-D-*r*Ser-D-*r*Pro-D-*r*Trp-D-*r*Ser-D-*r*Ser-D-*r*Cys-Ac | [68] |
| thymopentin | H-*g*Arg-(*RS*)-*m*Lys-Asp-Val-Tyr-OH | [69] |
| | H-Arg-Lys-Asp-*g*Val-(*RS*)-*m*Tyr-OH | [69] |
| | H-Arg-Lys-*g*Asp-(*RS*)-*m*Val-Tyr-OH | [69] |
| | H-*g*Arg-(*RS*)-*m*Lys-*g*Asp-(*RS*)-*m*Val-Tyr-OH | [69] |
| | H-*g*Arg-D-*r*Lys-D-*r*Asp-D-*r*Val-(*RS*)-*m*Tyr-OH | [69] |
| | H-*g*Arg-(*RS*)-*m*Lys-Asp-OH | [69] |
| translocation peptide | HO-*r*Gly-D-*r*Arg-D-*r*Gln-D-*r*Val-D-*r*Lys-D-*r*Val-D-*r*Trp-D-*r*Phe-D-*r*Gln-D-*r*Asn-D-*r*Arg-D-*r*Arg-D-*r*Met-D-*r*Lys-D-*r*Trp-D-*r*Lys-D-*r*Lys-β-*r*Ala-β-*r*Ala-biotin | [70] |
| | cholesteryl-O-*r*Gly-$COCH_2S(CH_2)_2NH$-D-*r*Arg-D-*r*Gln-D-*r*Val-D-*r*Lys-D-*r*Val-D-*r*Trp-D-*r*Phe-D-*r*Gln-D-*r*Asn-D-*r*Arg-D-*r*Arg-D-*r*Met-D-*r*Lys-D-*r*Trp-D-*r*Lys-D-*r*Lys-β-*r*Ala-β-*r*Ala-biotin | [70] |
| tuftsin | H-*g*Thr-(*RS*)-*m*Lys-Pro-Arg-OH | [71] |

[a] The prefix *g*, *m*, or *r* preceding the standard three-letter notation for an amino acid residue denotes the following: *g*, *gem*-diaminoalkyl residue; *m*, 2-alkyl substituted malonyl residue; *r*, amino acid incorporated in the reversed orientation, namely NH ← CO.

*for references see p 549*

We have reviewed this class of *retro-* and *retro-inverso*-peptide analogues previously.[2,3,72] Since then, Fletcher and Campbell[73] have published a comprehensive review on PMRI-peptides. It also covers a wide range of synthetic methodologies related to the synthesis of substituted malonates and *gem*-diaminoalkyl residues.

This chapter focuses on the synthetic aspects unique to the preparation of PMRI-peptides, including those with non-amino acid residues (e.g., **2–5**, **7**, **9–11**; Scheme 1). These residues, *gem*-diaminoalkyl, and sometimes 2-alkylmalonic acid derivatives, are introduced as an integral part of the PMRI modification.

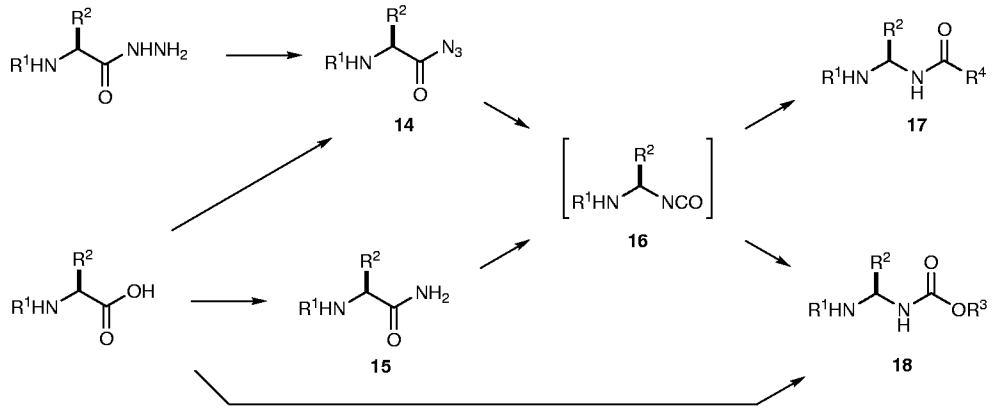
## 10.11.1 Synthesis in Solution

The early developed solution-based methodologies to prepare PMRI-peptides remain the most frequently used ones.[5,71,74–80] That is in spite of the successful adaptation of solid-phase peptide synthesis for the preparation of PMRI-peptides. Solution-based chemistry can generate suitably protected *gem*-diaminoalkyl and 2-alkylmalonyl moieties needed for the synthesis of PMRI-peptides by a variety of well-known reactions. The generated crude building blocks and the pseudopeptide units can be subjected to rigorous purification and characterization. Solid-phase synthesis of PMRI-peptides can be either by incorporation of precursors such as HO-Ala-(*RS*)-*m*Phe-(*R*)-Lys($N^{\varepsilon}$-Boc)-$NH_2$ or HO-*m*Gly-(*R*)-Phe-$NH_2$, to generate the PMRI unit on resin,[6,62] or by incorporation of the preformed PMRI unit PG-$Xaa^1\psi$[NHCO]$Xaa^2$-OH ($\psi$ = pseudopeptide bond) as a building block.[51,81] However, slow reaction rates, side reactions, and lack of compatibility between reaction conditions and the solid support complicate solid-phase synthesis of PMRI-peptides and prevent it from becoming the method of choice.

### 10.11.1.1 Preparation of *gem*-Diaminoalkyl Derivatives

Most of the pioneering syntheses of *gem*-diaminoalkyl derivatives are based on various rearrangement reactions leading to the formation of isocyanates. Organic chemistry offers a wide range of rearrangement reactions, but in practice only the Curtius and Hofmann rearrangements, of acyl azides **14** and acyl amides **15**, respectively, are utilized for the synthesis of PMRI-peptides (Scheme 2). The migrating group retains its configuration during the rearrangement, offering a means for the conversion of optically pure amino acids into the topographically complementary *gem*-diaminoalkyl derivatives. The isocyanate intermediates **16** can be trapped (Scheme 2) either with alcohols to produce *gem*-diaminoalkyl moieties **18** with urethane-type protecting groups, or, in a Goldschmidt–Wick-type reaction,[74] with an excess of a carboxylic acid to afford adducts **17** with the N,N′-diacylated *gem*-diaminoalkyl residue. Trapping the isocyanate with an N-protected amino acid afforded the *retro-inverso*-pseudopeptide unit.

**Scheme 2** Preparation of *gem*-Diaminoalkyl Derivatives by Curtius or Hofmann Rearrangement from Acyl Azides or Acyl Amides




$R^1$ = peptide chain or temporary protection group
$R^2$ = (protected) amino acid side chain

Goodman, Chorev, and co-workers[75,79,82] have carried out comparative studies on the different approaches to the Curtius rearrangement and optimized conditions to maximize the yield of the N,N′-diacylated *gem*-diaminoalkyl residue. A general procedure, which often produced the *gem*-diaminoalkyl derivative in good yields, is based on the Bergmann and Zervas stepwise degradation method (carbobenzoxy degradation) used for polypeptide sequencing. As depicted in Scheme 2, the method can be applied to both $N^{\alpha}$-protected amino acid residues leading to N,N′-diprotected *gem*-diaminoalkyl derivatives and to entire peptide chains resulting in the *N*-peptidyl,*N*′-alkyloxycarbonyl-*gem*-diaminoalkyl moiety.

This approach raises the following three questions: (a) How is the α-aminoacyl azide to be synthesized? (b) Which protecting groups are best? (c) Is it better to prepare free or peptide-bound *gem*-diaminoalkyl derivatives?

Goodman and Chorev[75] found that the required α-aminoacyl azides **14** are best prepared by reaction of the mixed anhydride of the amino acid with sodium azide. This method led to slightly better yields than the nitrosylation of *N*-formylaminoacyl hydrazide. Curtius rearrangement of the α-aminoacyl azide **14** yielded the isocyanate **16**, which was subsequently trapped as **17** or **18** as shown in Scheme 2. Comparable yields were obtained by nitrosylation with *tert*-butyl nitrite.[76] Other methods of acyl azide formation have rarely been employed for PMRI-peptide synthesis. Only Fincham et al.[11] reported the use of trimethylsilyl azide to synthesize an acyl azide en route to a PMRI-peptide.

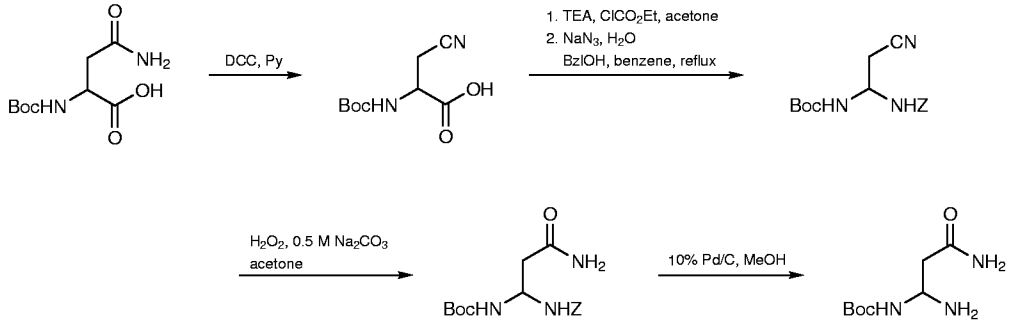
Comparison of different protecting groups on the yields of transforming a N-protected aminoacyl hydrazide to the corresponding N-protected *gem*-diaminoalkyl product led Goodman, Chorev, and co-workers[82] to the conclusion that acetyl (substituting for a peptidyl) is better than a urethane-type protection group (particularly Boc is disfavored over Z). They reasoned that urethane-type-protected isocyanate derivatives are more prone to form byproducts by heterolytic or displacement reactions. Therefore, the authors favor the conversion of the C-terminal amino acid residue of an entire peptide chain into the *gem*-diaminoalkyl derivative. However, other researchers have reported successful formation of N-protected *gem*-diaminoalkyl derivatives with Fmoc and Nps as reversible protecting groups.[77]

Another method, which generated in most cases good yields of N-protected *gem*-diaminoalkyl compounds, involves N-protected peptidyl or aminoacyl carboxamide derivatives. These are subjected to a Hofmann-type rearrangement with the mild oxidizing reagent iodobenzene bis(trifluoroacetate) (IBTFA).[83] {IBTFA is also known as phenyl iodosyl

*for references see p 549*

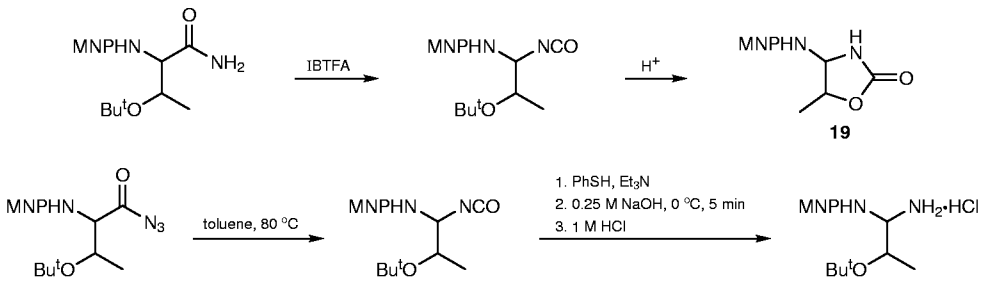
bis(trifluoroacetate) (PIFA or PIT), [bis(trifluoroacetoxy)iodo]benzene (TIB), phenyl-bis-(trifluoroacetato-*O*)iodine, and iodoso benzene 1,1-bis(trifluoroacetate).[83]} This route leads to the mono-N-acetylated, optically pure, *gem*-diaminoalkyl trifluoroacetates in good yields.[77] However, the use of IBTFA in the oxidative rearrangement of carboxamides containing Gln, Asn, and oxidation-labile amino acid residues such as Tyr, Trp, and Met is problematic. These residues have to be appropriately protected to avoid side reactions [e.g., Tyr(tBu), Trp($N^{in}$-For), or Met(O)].[3,77] Since the carboxamide function on the side chains of Asn and Gln are not compatible with the Curtius rearrangement[5] and obviously are reactive towards IBTFA, an alternative approach was developed by Cushman and co-workers (Scheme 3).[5] Dehydration of Boc-Asn-OH in the presence of DCC and pyridine yielded Boc-β-cyanoalanine which was transformed into the corresponding azide. Thermal rearrangement of the azide into the isocyanate in the presence of benzyl alcohol yielded Boc-NH-CH($CH_2CN$)-NHZ. Hydrolysis of the nitrile group in an aqueous solution of 0.5 M $Na_2CO_3$ and 30% $H_2O_2$ yielded the desired Boc-*g*Asn-Z.[5]

**Scheme 3** Synthesis of Boc-*g*Asn-H from Boc-Asn-OH[5]




Cleavage of the *tert*-butyl side-chain protecting group from MNP-Thr(tBu)-$NH_2$ during IBTFA treatment led to the formation of the corresponding 2-oxazolidone **19** as reported by Gazerro et al. (Scheme 4).[84] Therefore, a procedure avoiding the exposure of NMP-protected *gem*-diaminoalkyl derivatives containing acid-labile side chains to acidic conditions was developed. Higher yields of monoprotected *gem*-diaminoalkyl residue MNP-*g*Xaa-H were obtained from the basic hydrolysis of the corresponding *N*-MNP,*N'*-phenylthiocarbonyl *gem*-diaminoalkyl compounds. These phenylthiocarbonyl compounds were generated by trapping the isocyanates, formed in the thermal rearrangement of the *N*-MNP α-aminoacyl azide, with thiophenol (Scheme 4).[84]

**Scheme 4** Acid-Labile *N*-MNP-Protected Amino Acids Are Converted into the Corresponding *N*-MNP-Protected *gem*-Diaminoalkyl Derivatives To Avoid the Undesired Side Reaction Accompanying IBTFA-Mediated Conversion of Acyl Amide Terminated MNP-Thr(tBu)-$NH_2$ into MNP-*g*Thr(tBu)-H[84]

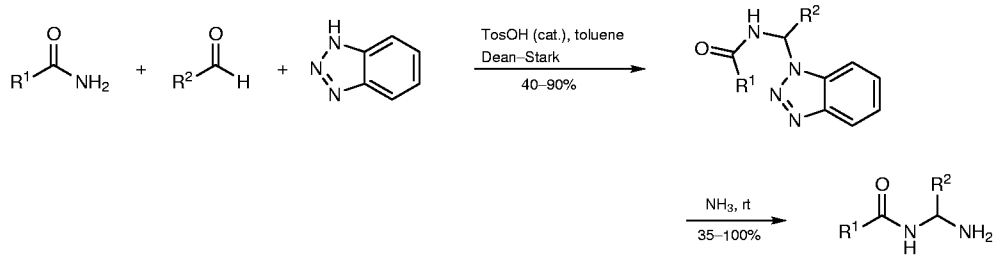


In general, monocarbamate-protected *gem*-diaminoalkyl derivatives, formed by the IBTFA-mediated rearrangement of *N*-carbamate-protected amino acid carboxamides, suffer from limited stability. Therefore, rearrangement of carboxamides to the corresponding *gem*-diaminoalkyl residues should be carried out preferentially on dipeptidyl carboxamides as precursors. The resultant *N*-aminoacyl *gem*-diaminoalkyl derivatives are more stable than the monocarbamate-protected *gem*-diaminoalkyls.[77] Importantly, the IBTFA-mediated rearrangement of optically active carboxamides occurs with complete retention of configuration at the migrating chiral center.[85,86]

In a one-pot procedure, treatment with diphenylphosphoryl azide (DPPA) transforms N-protected amino acid residues into the corresponding N-protected *gem*-diaminoalkyl derivatives.[87,88] Under the usual reaction conditions (elevated temperature), the acyl azide formed in situ undergoes a Curtius rearrangement to yield the corresponding isocyanate, which is trapped by the appropriate alcohol.[78] In addition, DeBons and Loudon[89] showed that, in the absence of a trapping reagent, the acyl azide can be isolated by careful control of the reaction conditions. Evidently, isocyanate trapping in a Goldschmidt–Wick-type reaction is only possible after complete consumption of the DPPA reagent or azide purification.

Other methods for the preparation of *gem*-diaminoalkyl residues have been reported. Some of them are directly applicable to synthesis of PMRI-peptides and some are only remotely related. Katritzky et al.[90] synthesized racemic, monoprotected diaminoalkyl derivatives by the Mannich reaction (Scheme 5). The diastereomeric mixtures of *N*-peptidyl *gem*-diaminoalkyl compounds were resolved to afford the diastereomerically pure *retro-inverso* surrogates.

**Scheme 5** Mannich Reaction Leading to (*R*,*S*)-*gem*-Diaminoalkyl Derivatives[90]




Interestingly, a host of methods for the preparation of orthogonally protected α-aminoglycine[91–93] and α-aminoalanine[94,95] have been reported. However, their applicability to the synthesis of PMRI-peptides is limited. Additional, less general methods have been extensively reviewed by Fletcher and Campbell.[73]

In conclusion, the IBTFA-mediated Hofmann rearrangement is the most convenient method for the synthesis of the N-acylated *gem*-diaminoalkyl derivatives and is certainly the most frequently used procedure for the generation of peptidyl *gem*-diaminoalkyl derivatives. However, as mentioned above, this method has limitations with regards to the choice of protecting groups and therefore requires careful consideration when applied to amino acids with reactive side chains.

***N*-(*tert*-Butoxycarbonyl)-*N'*-formyl-3-(methylsulfanyl)propane-1,1-diamine (For-"D"-*g*Met-Boc) [18, $R^1$ = For; $R^2$ = $(CH_2)_2SMe$; $R^3$ = tBu] by a Curtius Rearrangement with Isocyanate Trapping by an Alcohol; Typical Procedure:**[75]

*Method A; Mixed anhydride method:* To a stirred soln of For-D-Met-OH (1.77 g, 10 mmol) and NMM (1.11 mL, 10 mmol) in dry THF (20 mL) cooled to −15 °C, $ClCO_2Et$ (0.96 mL, 10 mmol) was added. After the mixture had stirred for 15 min, a soln of $NaN_3$ (1.30 g, 20 mmol) in $H_2O$ (10 mL) was added. The mixture was placed in an ice bath and allowed to stir for 30 min. The mixture was extracted with cold EtOAc (50 mL) and the organic layer was washed successively with cooled solns of brine, sat.

*for references see p 549*

$NaHCO_3$, and brine. The organic phase was dried ($MgSO_4$). Removal of the solvent under reduced pressure yielded the corresponding acyl azide **14** [$R^1$ = For; $R^2$ = $(CH_2)_2SMe$] as a clear oil (IR $\tilde{\nu}_{max}$: 2150 $cm^{-1}$).

The acyl azide was dissolved in dry toluene (10 mL) and heated to 80 °C under $N_2$. Effervescence ceased after 10 min; this indicated the full conversion of the azide into isocyanate **16** [$R^1$ = For; $R^2$ = $(CH_2)_2SMe$] (IR $\tilde{\nu}_{max}$: ca. 2260 $cm^{-1}$). Dry tBuOH (20 mL) was added. The mixture was kept for 6 h at 60 °C (complete disappearance of the IR band at $\tilde{\nu}$ 2260 $cm^{-1}$). The insoluble material was removed by filtration, and the filtrate was dried under reduced pressure. For-"D"-*g*Met-Boc [**18**, $R^1$ = For; $R^2$ = $(CH_2)_2SMe$; $R^3$ = tBu] was obtained as a solid residue, which was crystallized from $Et_2O$/hexane; yield: 1.64 g (66%); mp 97 °C; TLC (EtOAc) $R_f$ 0.43; $[\alpha]_D^{25}$ +2.5 (*c* 3, EtOH); IR (KBr) $\tilde{\nu}_{max}$: 1691, 1646, 1539 $cm^{-1}$.

*Method B; Nitrosyl chloride method:* To an ice-cold suspension of For-D-Met-$NHNH_2$ (3.84 g, 20 mmol) in dry THF (40 mL), a sat. soln of NOCl in dry THF was added until the red color persisted and the hydrazide had dissolved completely. The soln was diluted with ice-cold EtOAc and washed successively with ice-cold solns of sat. $NaHCO_3$ and brine. The organic phase was dried ($MgSO_4$). Removal of the solvent under reduced pressure yielded acyl azide **14** [$R^1$ = For; $R^2$ = $(CH_2)_2SMe$] as an oily residue (IR $\tilde{\nu}_{max}$: 2150 $cm^{-1}$). Acyl azide **14** [$R^1$ = For; $R^2$ = $(CH_2)_2SMe$] was treated as described above in method A. The physical and analytical data were the same as those described above.

**N-[N-(*tert*-Butoxycarbonyl)phenylalanyl]-N'-malonylethane-1,1-diamine (Boc-Phe-"L"-*g*Ala-*m*Gly-OH) [17, $R^1$ = COCH(Bzl)NHBoc; $R^2$ = Me; $R^4$ = $CH_2CO_2H$] by a Curtius Rearrangement with Isocyanate Trapping by a Goldschmidt–Wick-Type Reaction; Typical Procedure:**[75]

Boc-Phe-Ala-OH (3.36 g, 10 mmol) and NMM (1.11 mL, 10 mmol) were dissolved in THF (30 mL). The soln was cooled to −20 °C, and $ClCO_2Et$ (0.96 mL, 10 mmol) was added. After 15 min, a soln of $NaN_3$ (1.3 g, 20 mmol) in $H_2O$ (10 mL) was added, and the mixture was stirred for 30 min in an ice bath. The mixture was diluted with EtOAc, and washed with sat. solns of $NaHCO_3$ and brine. The organic phase was dried ($MgSO_4$) and the solvent was removed to yield the corresponding acyl azide **14** [$R^1$ = COCH(Bzl)NHBoc; $R^2$ = Me] (IR $\tilde{\nu}_{max}$: ca. 2132 $cm^{-1}$). The acyl azide was taken up in toluene (40 mL) and the mixture was heated under $N_2$ at 80 °C. A full rearrangement to isocyanate **16** [$R^1$ = COCH(Bzl)NHBoc; $R^2$ = Me] (IR $\tilde{\nu}_{max}$: ca. 2243 $cm^{-1}$), accompanied by strong effervescence, occurred within 10 min.

A soln of malonic acid (3.12 g, 30 mmol) in dioxane (12 mL) was added to the above mixture. Evolution of gas lasted for 5 min, after which no isocyanate could be detected by IR. After 30 min at 80 °C, the mixture was cooled to rt and the precipitate was collected and washed with toluene followed by hexane. Recrystallization of the precipitate from iPrOH gave the product as a white solid; yield: 1.85 g (47%); mp 199–201 °C; TLC ($CHCl_3$/MeOH/AcOH 3:1:1) $R_f$ 0.24; $[\alpha]_D^{22}$ +22.6 (*c* 2, DMF); IR (KBr) $\tilde{\nu}_{max}$: 1729, 1669 $cm^{-1}$.

**N-(Benzyloxycarbonyl)-N'-(*tert*-butoxycarbonyl)methanediamine [18, $R^1$ = Boc; $R^2$ = H; $R^3$ = Bzl] by a Curtius Rearrangement with Azide Formation by a DPPA Reagent; Typical Procedure:**[89]

In a dry 250-mL flask fitted with a stirrer and a drying tube were combined Boc-Gly-OH (1.75 g, 10 mmol), dry EtOAc (100 mL), and TEA (1.4 mL, 10 mmol), and the soln was cooled to 0 °C. Then, bis(4-nitrophenyl)phosphoryl azide (3.65 g, 10 mmol) was added and the mixture was stirred at 0 °C for 45 min. The precipitated triethylammonium bis(4-nitrophenyl) phosphate was removed by filtration and washed with $Et_2O$. The filtrate was immediately washed with ice water (2 × 50 mL) to remove the remaining salt, and dried ($MgSO_4$) for 1 h at 0 °C. The soln was filtered into a dry 250-mL flask and the solvent was removed in vacuo at <30 °C. Dry BzlOH was added (10.3 mL, 100 mmol) to the residual oil, and the flask was suspended in an oil bath at 75 °C. Vigorous gas evolution occurred within 2–3 min and was mostly over in 10 min. After 4.5 h, a short-path still was used to remove the solvent in vacuo. The semi-solid residue was freed of residual BzlOH by drying ($P_2O_5$) in vacuo at 56 °C to constant weight. The crude product was recrystallized; yield: 2.33 g (83%); mp 114–115 °C. (The aforementioned one-pot procedure is certainly an attractive alternative; however, the original reports[76,78] do not include a detailed experimental protocol.)

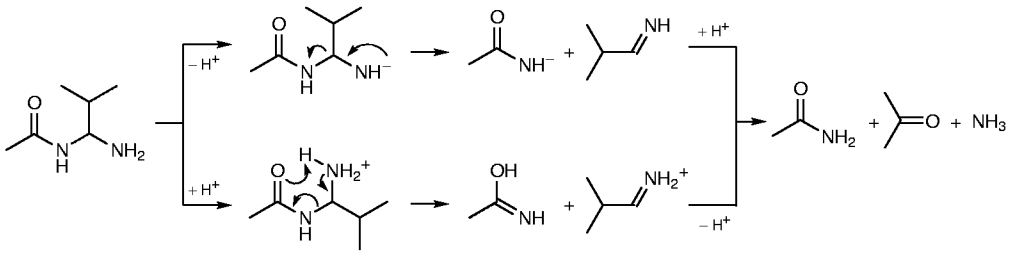
**N-[N-(*tert*-Butoxycarbonyl)phenylalanyl]ethane-1,1-diamine Trifluoroacetic Acid Salt (Boc-Phe-*g*Ala-H·TFA) by a Hofmann Rearrangement with IBTFA; Typical Procedure:**[77]

Boc-Phe-Ala-$NH_2$ (1.00 g, 3 mmol) was suspended in MeCN/$H_2O$ (6:4), and IBTFA (1.328 g, 3.09 mmol) was added. The resulting clear soln was stirred under a $N_2$ atmosphere for 3 h. Then MeCN was removed under reduced pressure, and the aqueous layer was lyophilized. The product was suspended and triturated in $iPr_2O$, and then filtered; yield: 1.18 g (93%); mp 142–143 °C; TLC ($CHCl_3$/acetone/AcOH 9:1:1) $R_f$ 0.43; $[\alpha]_D^{23}$ −16.8 (*c* 1, MeOH).

### 10.11.1.2 Coupling Procedures Leading to *gem*-Diaminoalkyl Derivatives

To our knowledge, there has so far been no comprehensive comparison of the wide array of available coupling procedures leading to *gem*-diaminoalkyl derivatives. However, due to the hydrolytic susceptibility of *gem*-diaminoalkyl derivatives (Scheme 6),[96] fast coupling methods seem to be advantageous.

**Scheme 6** Hydrolysis of Ac-*g*Val-H under Basic or Acidic Conditions[96]



The widely used activation of carboxylic acids with either HOBt or HOAt results in sufficiently short coupling times and can be easily adapted for the preparation of PMRI-peptides. However, according to Pallai et al.,[77] coupling efficiency is somewhat lower towards a *gem*-diaminoalkyl derivative than towards a normal amine and appears to be more sluggish per se. They further reported that the coupling reaction of mixed anhydride derived from carbonic and malonic acid residues does not yield the desired coupling product. Instead, an alkyl malonate is formed, in which the alkyl group is derived from the decomposed alkyl carbonate. In summary, slow fragment couplings of partial sequences to generate the pseudopeptide unit should be avoided. Instead, the PMRI-dipeptide unit should be assembled by the more efficient coupling of the appropriately protected building blocks, *gem*-diaminoalkyl and malonyl residues, followed by the standard elongation steps.

An obvious advantage of the Goldschmidt–Wick-type reaction in the synthesis of PMRI-peptides is the direct trapping of the in situ formed isocyanates by a carboxy-free component. This generates the bis-acylated *gem*-diaminoalkyl derivative in one step, and thus circumvents the coupling step between the newly formed monoacylated *gem*-diaminoalkyl moiety and the adjacent carboxy-free component. By this direct method Campbell et al.[97] were able to synthesize PMRI-dipeptides that are not readily available by acylation of the preformed less stable *gem*-diaminoalkyl derivative.

***N*-[*N*-(*tert*-Butoxycarbonyl)phenylalanyl]-*N'*-formyl-3-(methylsulfanyl)propane-1,1-diamine (Boc-Phe-"L"-*g*Met-For); Typical Procedure:**[75]

For-"D"-*g*Met-Boc (0.58 g, 2.33 mmol) was treated with 4 M HCl in dioxane (2 mL). After the mixture was stirred for 30 min at rt, the solvent was removed under reduced pressure. The residue was taken up in DMF (2 mL) and added to a stirred ice-cooled mixture of Boc-Phe-OH (0.66 g, 2.5 mmol) and NMM (0.26 mL, 2.33 mmol) in DMF (2 mL). To the resulting mixture, DCC (0.52 g, 2.5 mmol) was added, followed by HOBt (0.675 g, 5 mmol). The reaction was allowed to proceed for 1 h at ice-bath temperature, and an additional 2 h at rt. Then, the solvent was removed, and the residue was treated with EtOAc. The insoluble material was removed by filtration, and the filtrate was washed successively with brine, 2 M aq $NaHSO_4$, brine, sat. aq $NaHCO_3$, and brine. The organic phase was dried ($MgSO_4$) and the solvent was removed under reduced pressure. The solid crude product was chromatographed [silica gel (20 × 90 mm), EtOAc/hexane 1:1], and the compound thus obtained was crystallized (EtOAc/hexane) to yield a white solid; yield: 0.31 g (34%); mp 175 °C; TLC $R_f$ 0.45 (EtOAc); $[\alpha]_D^{25}$ +10.1 (*c* 3, TFE); IR (KBr) $\tilde{\nu}_{max}$: 1684, 1662, 1542 $cm^{-1}$.

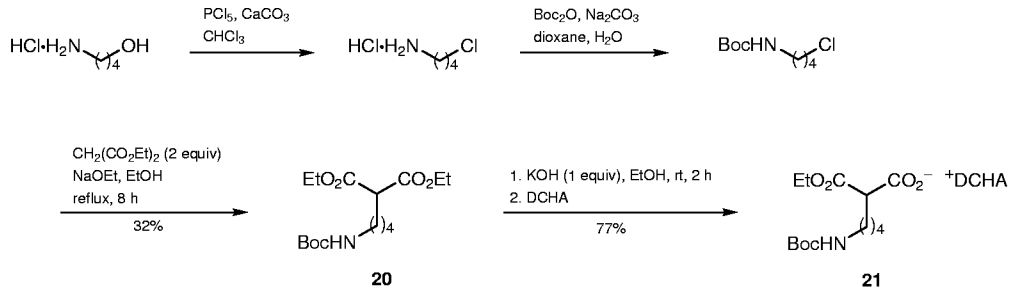
*for references see p 549*

### 10.11.1.3 Preparation of C2-Substituted Malonyl Derivatives

As mentioned above, monoprotected malonyl residues can be directly coupled to an amino or *gem*-diaminoalkyl derivative by conventional coupling reagents such as DCC/HOBt and BOP. The only standard coupling procedure, which is reported to generate mixed results, is the carbonate–mixed anhydride method. Pallai et al.,[77] Kametani et al.,[98] Gutman and Boltanski,[99] and Kim et al.[100] reported the formation of mainly side products, while others[101,102] described successful applications of this method. In conclusion, it seems to be judicious to avoid the mixed anhydride coupling of a malonyl residue to a *gem*-diaminoalkyl residue or an amino acid.

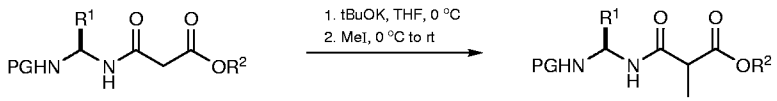
The subsequent paragraphs will focus on the preparation of C2-substituted malonyl derivatives for the incorporation into PMRI-peptides. In the classical method, C2 alkylation of a malonic acid diester is followed by partial hydrolysis to generate the C2-alkylated monoalkyl malonate (Scheme 7). The malonyl-derived topochemical replacements for Phe, Lys, Trp, Leu, Ala, His, and Met residues have been successfully prepared by this strategy.[48,57]

**Scheme 7** Synthesis of an Ammonium Salt of (*RS*)-HO-*m*Lys(Boc)-OEt[57]



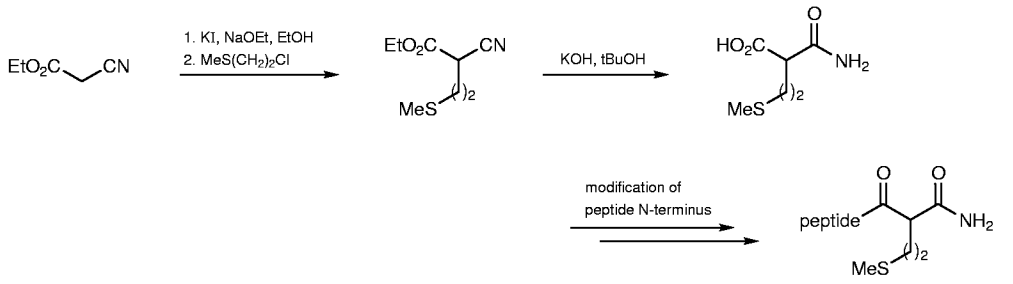

Campbell et al.[97] found that the yield of the Goldschmidt–Wick-type reaction was lower with C2-substituted monoalkyl malonates than with unsubstituted monoalkyl malonates. They therefore suggest selective C2 alkylation of a malonyl residue already incorporated into a PMRI-dipeptide (Scheme 8).[73] Unfortunately, no experimental details are available for what might potentially be an important synthetic manipulation.

**Scheme 8** Alkylation of Protected PMRI-Dipeptides[73,97]



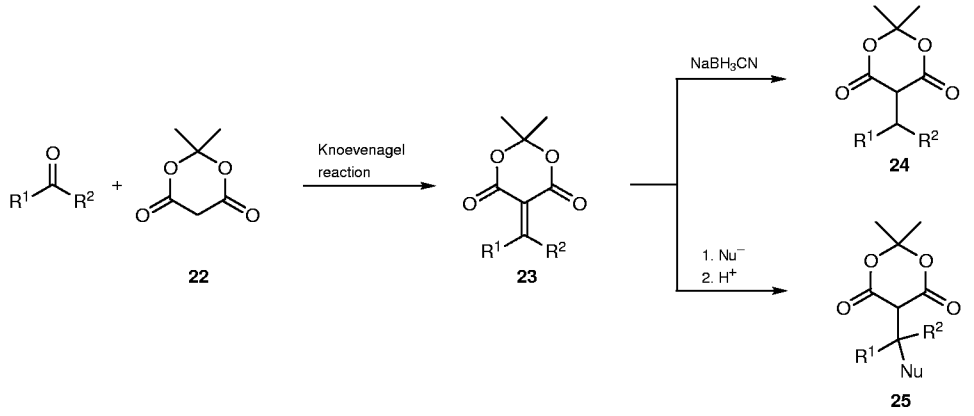

C2 Alkylation of cyanoacetates followed by hydrolysis results in C2-substituted malonamic acids suitable for incorporation at the C-terminus of PMRI-peptides. However, due to the higher acidity of the C2 protons, dialkylation at the C2 position may occur.[103] Cushman et al.[5] successfully applied this strategy for the synthesis of a terminal *m*Met-amide-modified bombesin derivative (Scheme 9).

**Scheme 9** C2 Alkylation of a Cyanoacetate and Its Incorporation into a PMRI-Peptide[5]




An attractive method to generate C5-monoalkyl-substituted Meldrum's acids consists of a two-step synthesis (Scheme 10). In the first step, a Knoevenagel reaction between Meldrum's acid (**22**) and an aldehyde (or sometimes a ketone) yields the C5-alkylidene derivative **23**. In the following step, the alkylidene derivative **23** is reduced, to generate the C5-monoalkyl-substituted Meldrum's acid **24**.[43,71,104,105] Alternatively, in the second step, Michael addition of a nucleophile to the conjugated C=C bond in the C5-alkylidene derivative generates a Meldrum's acid **25** monosubstituted on C5 by a more elaborated alkyl.[106]

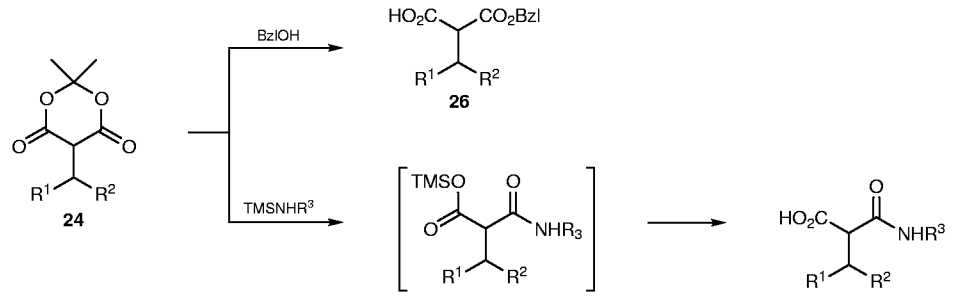
**Scheme 10** Two-Step Preparation of a C5-Substituted Meldrum's Acid. In the Second Step, the Alkylidene Derivative Can Either Be Reduced or Undergo Michael Addition with a Nucleophile[105,106]




Subsequently, alcoholysis of the Meldrum's acid derivative **24** with benzyl alcohol results in the corresponding monoester **26** of the malonic acid derivative (Scheme 11).[63] Ring opening with a silylated amine directly generates a PMRI surrogate.[80] Silylation of the amine prevents it from acting as a base, deprotonating and resulting in enolization of Meldrum's acid **24**, thereby deactivating it towards nucleophilic ring opening.

*for references see p 549*

**Scheme 11** Ring Opening of Derivatives of Meldrum's Acid[63,80]




Verdini and co-workers[71,81,107] were able to silylate the amine with *N,O*-bis(trimethylsilyl)-acetamide[108] (BTMSA) and to ring-open Meldrum's acid in a one-pot procedure without isolating the labile *N*-silyl derivative. Additional, less frequently employed methods for the synthesis of C2-substituted malonic acids are summarized in a review.[73]

None of the aforementioned procedures will generate optically pure monoprotected malonyl residues. This lack of chiral integrity is not a serious drawback, because the malonyl C2 position is inherently configurationally labile.[3] Malonamic derivatives are more optically stable than their corresponding malonic derivatives. Incorporation of malonyl residues into PMRI-peptides therefore results in an increase in the configurational stability of the peptide. However, other factors, such as the flanking amino acid sequences, may also affect the configurational stability.[5] RP-HPLC was shown to be useful for the separation of diastereomeric PMRI-peptides and the determination of epimerization half-lives.[3] Resolution and configurational assignment are important, since the two diastereomers often display significantly different biological activities, and only one diastereomer is formally topochemically similar to the parent peptide. For the NMR assignment of the absolute configuration of purified diastereomers, sequential NOEs between $NH_i$ and $C_\alpha H_{i-1}$ protons, and the NOE and coupling constants between vicinal protons H–N–$C_\alpha$–H can be used.[109] Only PMRI-peptides with a terminal malonic or malonamic acid moiety can be converted into the corresponding peptide by an IBTFA-mediated Hofmann rearrangement. Comparison of the chromatographic properties of the rearrangement-derived peptide with that of the same peptide, synthesized independently by conventional methods, allowed the unambiguous configurational assignment.[85,110]

Interesting solutions to the spontaneous epimerization of C2-substituted malonyl residues, to provide configurational stability, were explored independently by Dal Pozzo and Laurita[111] and Abouabdellah and Welch.[30] Replacement of the C2 hydrogen in C2-alkyl-substituted malonyl residues by either methyl[111] or fluorine[30] stabilized the configurational integrity of the C2 chiral center in the malonyl derivatives.

**DCHA·(*RS*)-HO-*m*Lys(Boc)-OEt (21) by C2 Alkylation of Malonic Acid; Typical Procedure:**[57]
$H_2N(CH_2)_4OH$ (1.9 mL, 20 mmol) was treated with $PCl_5$ (5.41 g, 26 mmol) in $CHCl_3$ in the presence of $CaCO_3$ (1.0 g, 10 mmol) to give $HCl \cdot H_2N(CH_2)_4Cl$, which was used without isolation. It was added to a soln of $Boc_2O$ (4.8 g, 22 mmol) in dioxane (20 mL) and $H_2O$ (20 mL) containing $Na_2CO_3$ (2.12 g). The mixture was intensively stirred overnight, and $Na_2CO_3$ was added as necessary to maintain a basic pH. On completion of the reaction, the soln was washed several times with $Et_2O$. The $Et_2O$ layer was washed with dil $NaHSO_4$ and brine, and dried ($MgSO_4$). $Et_2O$ was removed under reduced pressure to give $BocNH(CH_2)_4Cl$ as a clear oil; yield: 3.39 g (82%); TLC $R_f$ 0.58 ($CHCl_3$); TLC $R_f$ 0.84 ($CHCl_3$/MeOH/AcOH 85:10:5).

**CAUTION:** *Sodium reacts violently or explosively with a wide range of substances! Appropriate safety precautions and procedures should be adopted when handling and disposing of sodium metal.*

For the alkylation of diethyl malonate, Na metal (ca. 40 mmol) was dissolved in EtOH (15 mL) (freshly distilled from Mg), and diethyl malonate (6.07 mL, 40 mmol) and $BocNH(CH_2)_4Cl$, prepared as above, were added dropwise. The mixture was refluxed for 8 h, the solvents were concentrated under reduced pressure, and an aqueous workup was carried out. Removal of the organic solvent under reduced pressure gave an oil, which contained significant amounts of diethyl malonate and impurities. The product was purified by flash chromatography [silica gel (5 × 16 cm), $CHCl_3$]. The fractions were analyzed by TLC ($CHCl_3$). This procedure gave diester **20** as a pale clear oil; yield: 1.63 g (32%); TLC ($CHCl_3$) $R_f$ 0.90; TLC ($CHCl_3$/MeOH/AcOH 85:10:5) $R_f$ 0.79.

Diester **20** (2.81 g, 8.47 mmol) was dissolved in EtOH, and KOH (8.47 mmol) was added. The mixture was stirred for 2 h at rt. EtOH was then removed under reduced pressure, and the residue was distributed between 10% aq $NaHCO_3$ and $iPr_2O$. The aqueous layer was collected, acidified, and then extracted several times with $Et_2O$. The organic layers were pooled, washed with 2 M aq $NaHSO_4$ and brine, and then dried ($MgSO_4$). Removal of $Et_2O$ under reduced pressure gave an oil. DCHA (1.694 mL) was added to the oil in hexanes. Rapid cooling to −15 °C and trituration followed heating. This procedure was repeated several times to give **21** as a solid, which was isolated by filtration from hexanes; yield: 3.15 g (77%); mp 88–91 °C; TLC $R_f$ 0.60 ($CHCl_3$/MeOH/AcOH 85:10:5); TLC $R_f$ 0.52 ($CHCl_3$/MeOH 9:1).

**(*RS*)-2,2-Dimethyl-5-[4-(trifluoroacetamido)butyl]-1,3-dioxane-4,6-dione [24, $CHR^1R^2 = (CH_2)_4NHTfa$] by C5 Alkylation of Meldrum's Acid (22); Typical Procedure:**[71]

$(Tfa)_2O$ (29.4 mL, 210 mmol) was added dropwise over 2 h to a soln of $H_2N(CH_2)_3CH(OEt)_2$ (32.2 g, 200 mmol) and DMAP (25.6 g, 210 mmol) in $CH_2Cl_2$ (600 mL), stirred under $N_2$ at 0 °C. The mixture was allowed to reach rt and was stirred for 1 h. The solid was filtered off, the filtrate was washed with $H_2O$ and dried ($Na_2SO_4$), and the solvent was removed under reduced pressure. This gave $Tfa\text{-}NH(CH_2)_3\text{-}CH(OEt)_2$ as a colorless oil (47.8 g, 186 mmol), which was suspended in 1 M HCl (500 mL) and stirred at rt until a homogeneous soln was obtained (about 15 min). The soln was brought to pH 6 by addition of solid $NaHCO_3$, and then the mixture was heated on a steam bath for 15 min. After the mixture had cooled to rt, the solvent was partially removed under reduced pressure to give a volume of ca. 200 mL. The aqueous soln was extracted several times with $CH_2Cl_2$, the combined organic extracts were dried ($Na_2SO_4$), and the solvent was removed under reduced pressure; this yielded an oily mixture (27 g) containing about 60% of the aldehyde $Tfa\text{-}NH(CH_2)_3CHO$ (by HPLC). The oil was added to a soln of Meldrum's acid (**22**; 12.1 g, 84 mmol) and $NaBH_3CN$ (3.72 g, 59 mmol) in DMF (60 mL). After the mixture had stirred at rt for 1 h, $H_2O$ (200 mL) was added and the pH was adjusted to 4.5 with 1 M HCl. The resulting precipitate of **24** [$CHR^1R^2 = (CH_2)_4NH\text{-}Tfa$] was collected by filtration, washed with $Et_2O$, and dried in vacuo; yield: 17.5 g (28%); mp 131–132 °C; RP-HPLC [C18, 0% A (10% MeCN in 0.1% aq TFA) to 40% B (0.1% TFA in MeCN) in 20 min]: 18.74 min.

**Benzyl (*RS*)-2-Benzylmalonate (26, $CHR^1R^2$ = Bzl) by Formation and Alcoholysis of C5-Substituted Meldrum's Acid 24 ($CHR^1R^2$ = Bzl); Typical Procedure:**[63]

Concd $H_2SO_4$ (2 mL) was added dropwise to a stirred and cooled (0 °C) suspension of 2-benzylmalonic acid (20 g, 100 mmol) in $Ac_2O$ (50 mL). Dropwise addition of dry acetone (12.5 mL, 170 mmol) over 10 min followed. The mixture was allowed to react at rt for 1 h and was then cooled in an ice bath. Crushed ice (50 g) was added to the reaction with vigorous stirring. The precipitate was collected by filtration, washed thoroughly with cold $H_2O$, and dried ($P_2O_5$) in vacuo. The crude product (*RS*)-2,2-dimethyl-5-benzyl-1,3-dioxane-4,6-dione [(*RS*)-**24**, $CHR^1R^2$ = Bzl] was recrystallized from MeOH/heptane; yield 10.5 g (45%); mp 78–79 °C; TLC ($CHCl_3$/MeOH 9:1) $R_f$ 0.70; TLC (EtOAc/hexane 1:1) $R_f$ 0.38. To a stirred soln of this C5-substituted Meldrum's acid **24** ($CHR^1R^2$ = Bzl; 9 g, 38.5 mmol) in toluene (7 mL) was added BzlOH (8 mL, 77 mmol), and the mixture was kept under $N_2$ at 70 °C for 12 h. Removal of solvent under reduced pressure gave an oily residue, which was subjected to column chromatography [silica gel (2.0 × 50 cm), EtOAc/hexane 0% to 10% EtOAc]. Removal of the solvent under reduced pressure gave oily monobenzyl ester **26** ($CHR^1R^2$ = Bzl); yield: 9.5 g (90%); TLC $R_f$ 0.24 ($CHCl_3$/MeOH 9:1); TLC $R_f$ 0.60 ($CHCl_3$/MeOH/AcOH 17:2:1).

**(*RS*)-HO-*m*Lys(Tfa)-D-Thr(tBu)-$NH_2$ by Ring Opening of Meldrum's Acid with an Amine; Typical Procedure:**[71]

H-D-Thr(tBu)-$NH_2$ (3.5 g, 20 mmol) and BTMSA (8 mL, 40 mmol) were dissolved in $CH_2Cl_2$ (90 mL) and refluxed for 6 h. To the soln cooled to rt was added Meldrum's acid derivative (*RS*)-**24** [$CHR^1R^2 = (CH_2)_4NHTfa$; 5.6 g, 18 mmol)], prepared as described above. After stirring for 18 h, the mixture was extracted with 5% citric acid in $H_2O$, and dried ($Na_2SO_4$). The solvent was removed in vacuo; this gave an oil that was triturated in acetone/$H_2O$; yield: 5.2 g (66%); HPLC [C18, 0% A (10% MeCN in 0.1% TFA) to 40% B (0.1% TFA in MeCN) in 20 min]: 14.5 min, 16.2 min (corresponding to the two diastereomers); both peaks showed identical EI-MS spectra.

*for references see p 549*

## 10.11.2 Solid-Phase Synthesis of Partially Modified *retro-inverso* Peptides

The introduction of solid-phase synthesis to PMRI-peptides is a result of the systematic and continuous efforts of Verdini and co-workers.[6,51,62,81,84] Unfortunately, these studies are reported only in communications and symposium proceedings, and therefore detailed experimental protocols are not available. Nevertheless, the methodological significance of these studies and their potential applicability justifies a summary in this section.

### 10.11.2.1 On-Resin Generation of *gem*-Diaminoalkyl Moieties

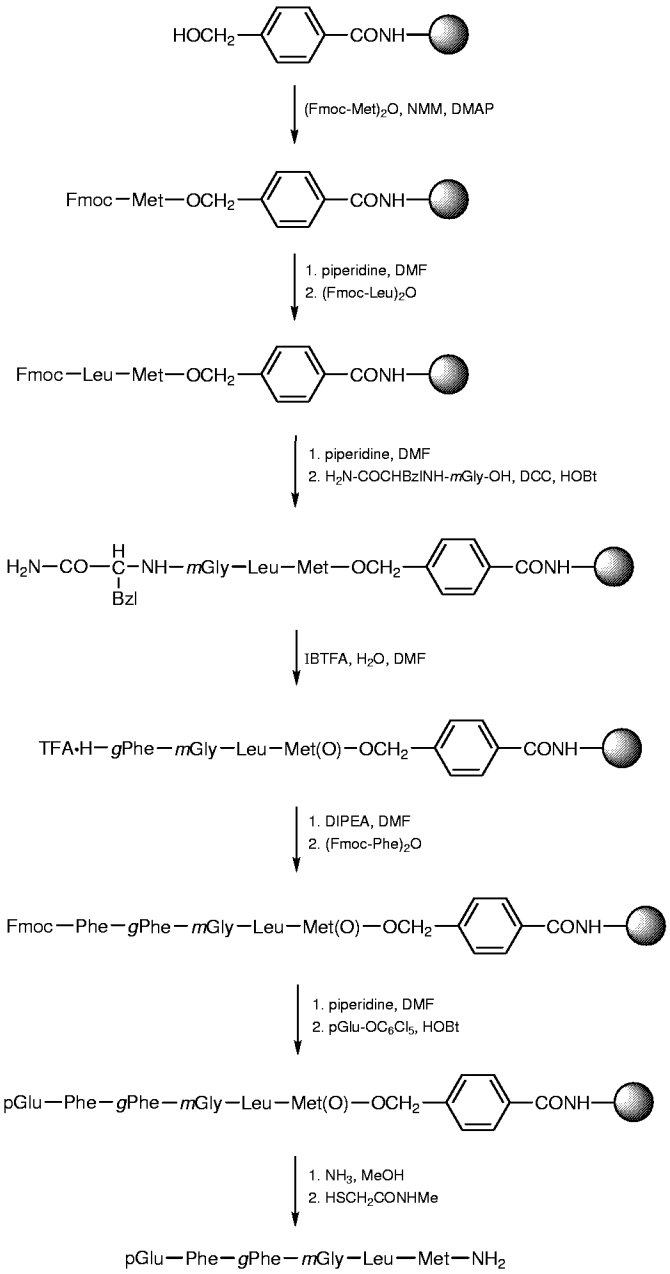
The first approach to the solid-phase synthesis of PMRI-peptides included the incorporation of pseudopeptides such as malonyl-D-phenylalanylamide or $H_2N$-D-*r*Lys(Boc)-(*RS*)-*m*Phe-Ala-OH, precursors of PMRI-units, into resin-bound dipeptides or amino acids.[6,62] These pseudopeptide precursors were prepared in solution by methods described above. Subsequent on-resin IBTFA-mediated conversion of the resin-bound carboxamide group into the corresponding ammonium trifluoroacetate compound completed the generation and incorporation of the PMRI modification into the resin-bound pseudopeptide. The subsequent elongation of the peptide at the free amino group of the N-terminal *gem*-diaminoalkyl residue was carried out in the standard manner. Since water needs to be used as a cosolvent in the IBTFA-mediated rearrangement, the choice of the solid support is crucial.[85] In their synthesis of a PMRI analogue of substance P, pGlu-Phe-*g*Phe-*m*Gly-Leu-Met-$NH_2$, Verdini and co-workers[62] employed Sheppard's polar polydimethylacrylamide-based 4-(hydroxymethyl)benzamide resin.[112] Treatment with methanolic ammonia cleaved the anticipated peptide from the resin (Scheme 12). However, for the synthesis of the PMRI-bradykinin potentiating peptide ($BPP_{9a}$), pGlu-Trp-Pro-Arg-Pro-*g*Lys-(*RS*)-*m*Phe-Ala-Pro-OH, these authors exploited a different poly(dimethylacrylamide)-based resin, functionalized by 2-[4-(hydroxymethyl)phenoxy]acetamide,[112] which allowed TFA-mediated acidolysis[6] (Scheme 13). These resins remain permeable and well swollen in the presence of water.

In the synthesis of PMRI substance P, the slow, DCC/HOBt-mediated incorporation of the pseudodipeptide suffered from extensive formation of the Leu-Met piperazine-2,5-dione; this led to about 20% loss of peptide from the resin. Two cycles of IBTFA-mediated (fourfold molar excess) rearrangement were carried out. The first one-hour cycle was followed by a second overnight cycle.[62]

In the synthesis of PMRI-$BPP_{9a}$, the incorporation of the pseudotripeptide amide was undertaken to overcome the enhanced generation of Ala-Pro piperazine-2,5-dione (ca. 50%) from the resin-bound Ala-Pro dipeptide. The overall yield of this synthesis was 25%.[6]

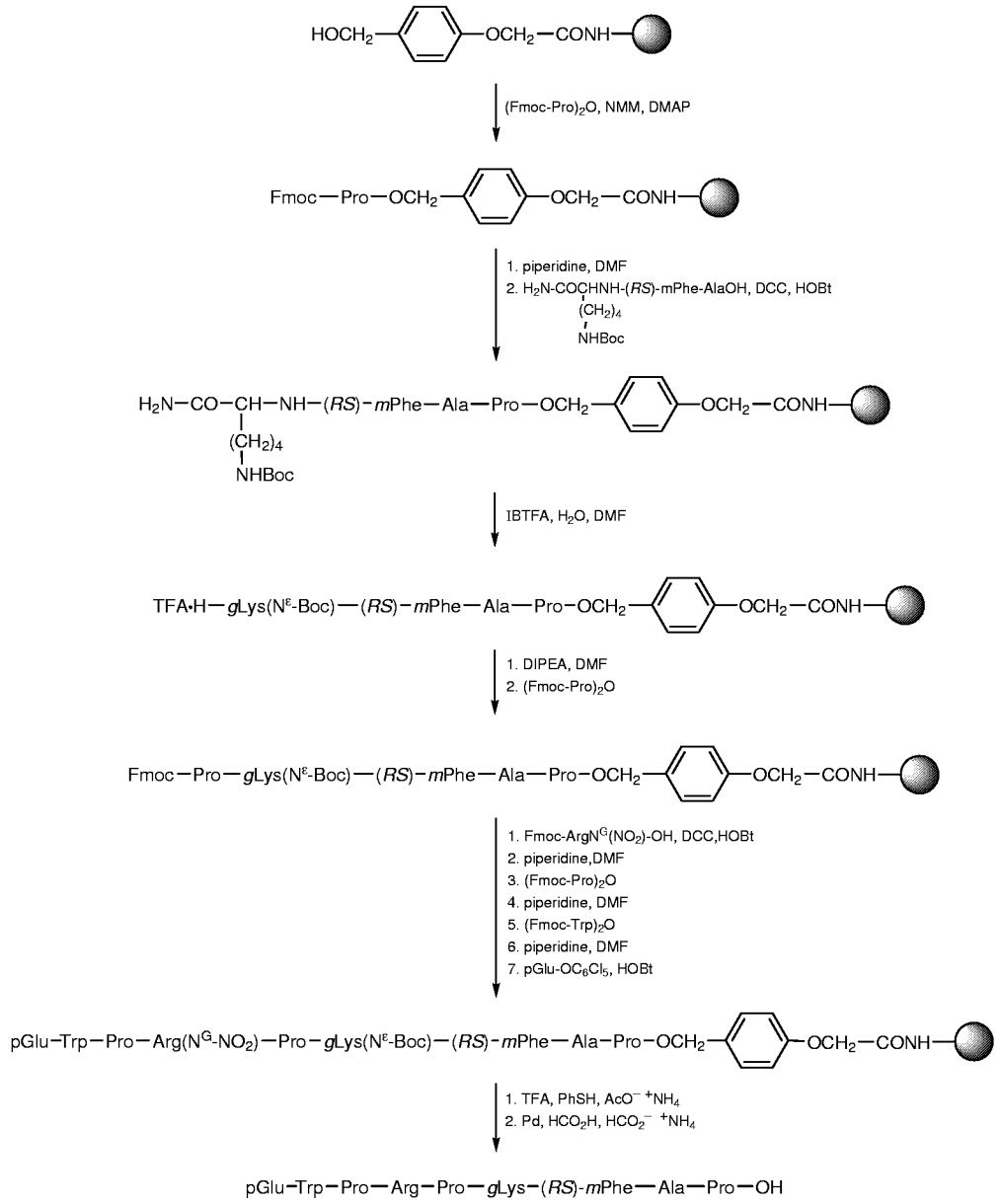
In both syntheses, conservation of the original side-chain topology, as presented by the extended conformation (all-L-peptide), was achieved by subjecting a D-aminoacylamide to the IBTFA-mediated rearrangement.[6,62] Combination of the D configuration and reversed incorporation of the D-amino acid result in the generation of a *gem*-diaminoalkyl with the required "L" configuration.

**Scheme 12** Solid-Phase Synthesis of [pGlu$^6$,*g*Phe$^8$,*m*Gly$^9$]-Substance P(6–11)[62]




*for references see p 549*

**Scheme 13** Solid-Phase Synthesis of [*g*Lys$^6$,(*RS*)-*m*Phe$^7$,Ala$^8$]-Bradykinin Potentiating Peptide$_{9a}$[6]

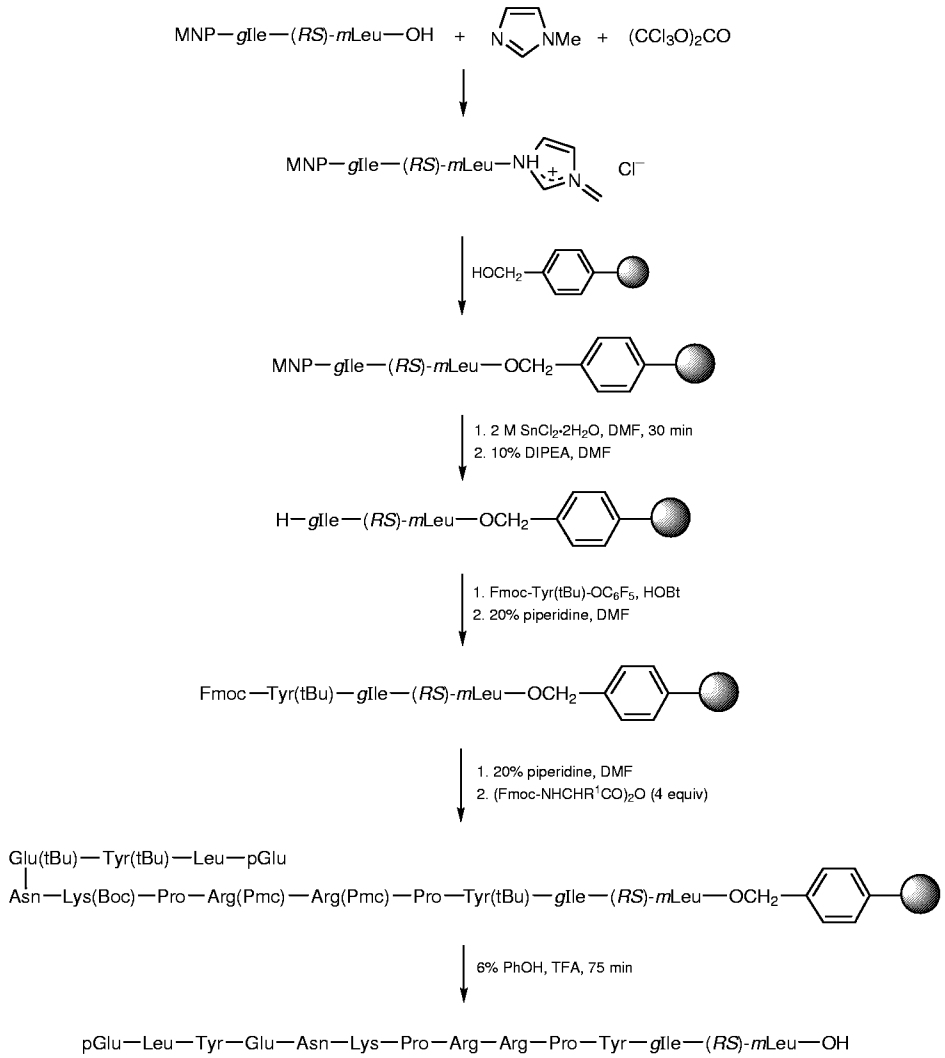



### 10.11.2.2 Synthesis by Incorporation of Partially Modified *retro-inverso* Dipeptide Synthons

The previously described solid-phase syntheses of PMRI-peptides suffer from the restrictions imposed by the oxidative nature of the on-resin IBTFA-mediated rearrangement.[6,62] The more recent solid-phase synthesis of PMRI-neurotensin peptides[51] incorporated a few important innovations: (a) The easily prepared *N*-MNP-monoprotected *gem*-diaminoalkyl hydrochlorides are stable upon storage. The MNP group is conveniently removed by mild reduction. (b) The facile preparation of 2-substituted malonamic acids from the corresponding Meldrum's acid derivative. (c) The use of kieselguhr-supported hydroxymethyl

poly(dimethylacrylamide) resin (Pepsyn KA). (d) The use of triphosgene and *N*-methylimidazole to generate in situ a highly active coupling agent, carbonyl bis-(*N*-methylimidazolium) dichloride, required for the efficient anchoring of the PMRI-dipeptide to the resin-bound hydroxymethyl group through the C-terminal malonyl moiety. The residual hydroxymethyl groups were blocked by acetylation (Scheme 14).

**Scheme 14** Synthesis of [*g*Ile$^{12}$,(*RS*)-*m*Leu$^{13}$]-Neurotensin[51]

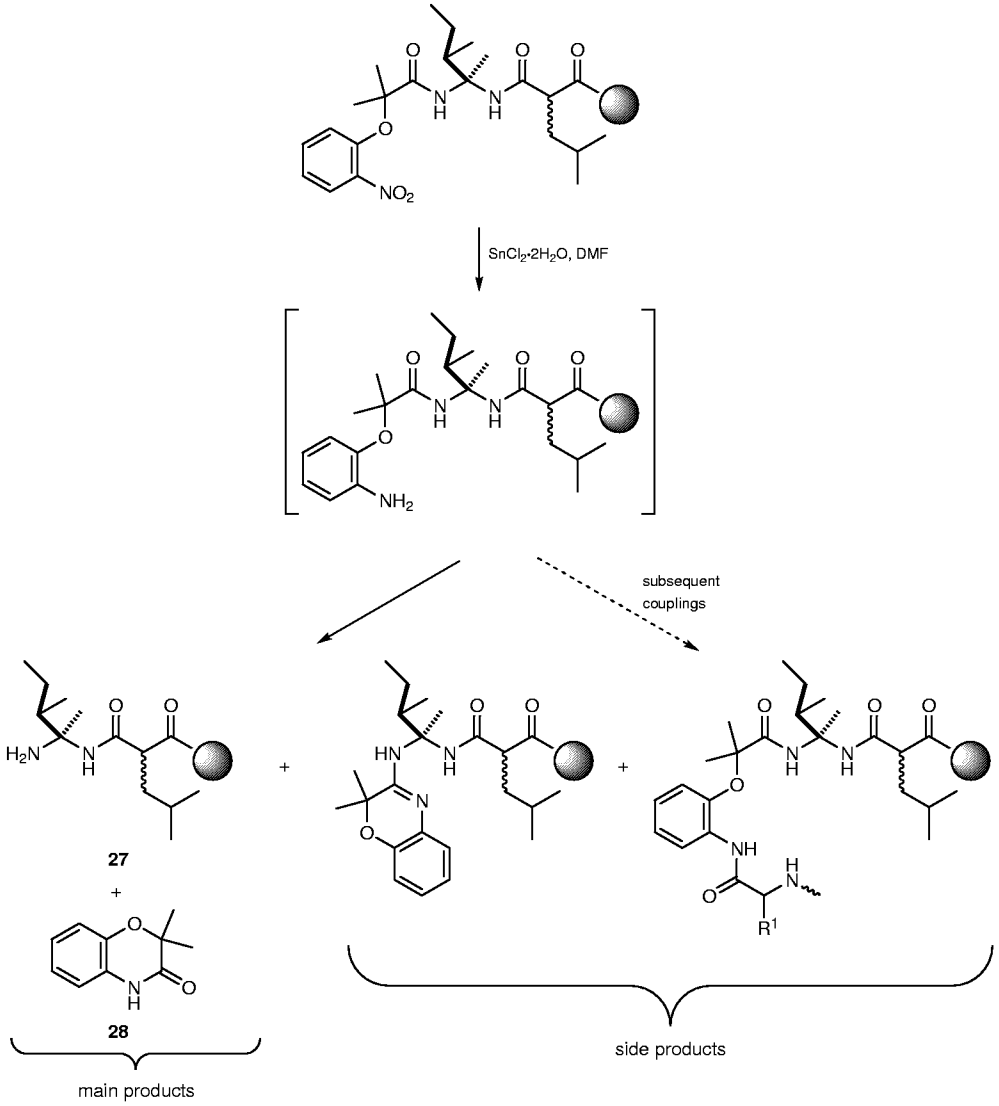



Anchoring the MNP-protected PMRI-dipeptide to the resin by employing short preactivation of fourfold excess of the PMRI-dipeptide with triphosgene/*N*-methylimidazole in dichloromethane led to 50–70% coupling yields. The susceptibility of the C-terminal malonyl residue led to Pmc being chosen for Arg side-chain protection. Deprotection of the N-terminal MNP protecting group, by reduction with $SnCl_2 \cdot 2H_2O$ in DMF, generated the free resin-bound PMRI-dipeptide **27** and 2,2-dimethyl-2*H*-1,4-benzoxazin-3(4*H*)-one (**28**) as the main products (Scheme 15).[81] However, incomplete release of the MNP protecting group resulted in the *N*-[2-(2-aminophenoxy)-2-methylpropionyl]-substituted PMRI resin-bound dipeptide, which became a source of minor side products generated upon subsequent chain

*for references see p 549*

elongation steps (Scheme 15).[51] Lower yields in this synthetic route could also be attributed to the formation of the six-membered amidine ring, which blocks the N-terminus of the resin-bound PMRI-dipeptide irreversibly (Scheme 15).[51]

**Scheme 15** Main Products and Side Products Generated upon Deprotection of MNP from the Resin-Bound MNP-Protected PMRI-Dipeptide MNP-*g*Ile-(*RS*)-*m*Leu-Resin[51,81]




In conclusion, progress has been made towards the development of a general and robust solid-phase synthesis of PMRI-peptides. Nevertheless, the published data suggest that additional investigations should be performed to further optimize this synthetic approach.

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# 10.12 Synthesis of Peptides Based on β-Amino Acids

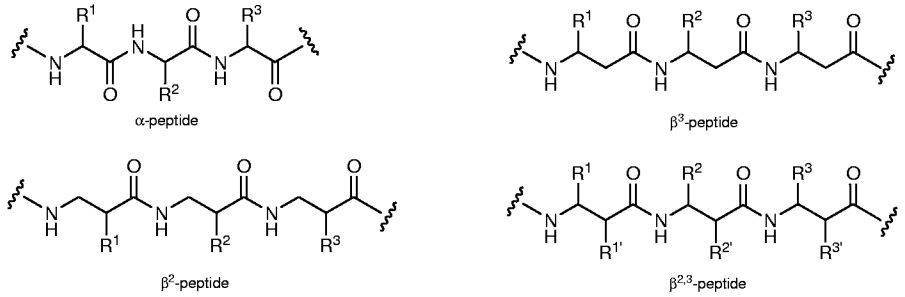
J. L. Matthews

The incorporation of β-amino acids into peptides has become an area of increasing interest. Approaches have varied from the incorporation of a sole β-amino acid in a standard peptide built from α-amino acids to the synthesis of peptides built entirely from β-amino acids. This latter category has become known as β-peptides and some spectacular differences between α-peptides and β-peptides have been uncovered.[1]

The surprising discovery that small oligo-β-peptides exhibit extraordinary tendencies to form stable secondary structures has led to rapid developments in the chemistry of these peptides. The earliest work in the field revolved around the synthesis of β-peptide polymers (the so-called nylon-3 derivatives).[2] Polymerization of β-amino acids led to polymers of undefined length. It was noted they could form stable structures but it proved impossible to gain any concrete information about the nature of those structures at that time. The developments in the synthesis of β-peptide oligomers of predefined length and advances in methods (i.e., NMR spectroscopy and X-ray crystallography) for the 3D characterization of such compounds has led to the discovery of new helical structures found to be adopted by a variety of β-peptides.[1,3–7]

The large amount of work devoted to the synthesis of oligo-β-peptides has created a need for some standardization in nomenclature. The proposals of Seebach[5] will be used throughout this section, i.e. the β-peptides will be named according to their substitution pattern with the superscript representing the position of the substituent on the peptide backbone (Scheme 1); the prefix H to the three-letter code for amino acids has been used throughout to denote a homoamino acid, e.g. HLeu is homoleucine, HVal is homovaline, etc.

**Scheme 1** Possible Peptide Substitution Patterns



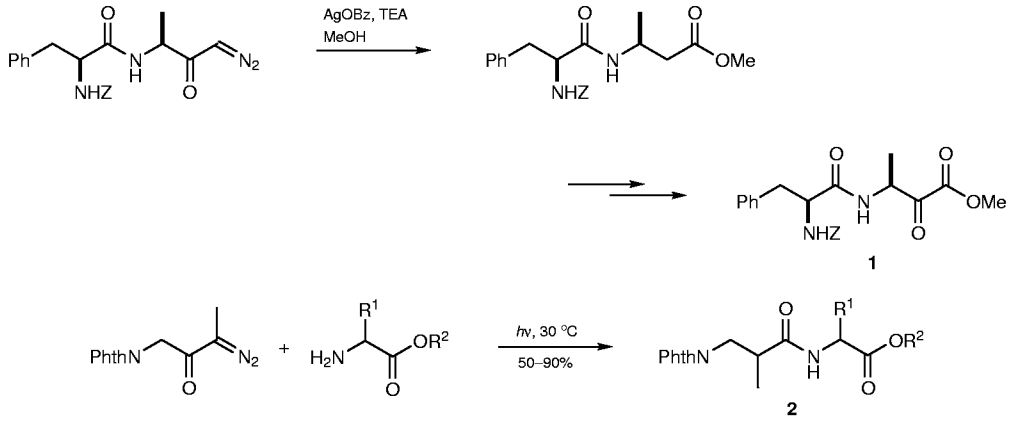

In comparison to α-peptides, the chemistry of β-peptides can be described as being at an early stage and hence there are relatively few methods described for the synthesis of these compounds. As interest in these compounds grows, there will inevitably be many more possibilities described.

## 10.12.1 Peptides Containing a Single β-Amino Acid

The synthesis of a peptide containing a single β-amino acid residue can be regarded as a simple task involving substituting an α-amino acid by a β-amino acid in the course of a standard peptide coupling sequence. Given the wealth of information available about peptide synthesis, it is inappropriate to consider this option in this section and hence, attention is confined to the incorporation of a β-amino acid into an α-peptide by the use of the Arndt–

Eistert reaction with concomitant peptide bond formation.[8] This strategy has been used in an approach to β-amino-α-oxo esters **1**[9] and the formation of dipeptides **2**[10] that possess an alkyl group at the α-position of the β-amino acid residue (Scheme 2).

**Scheme 2** Synthesis of β-Amino-α-oxo Esters and Dipeptides Using β-Amino Acids[9,10]



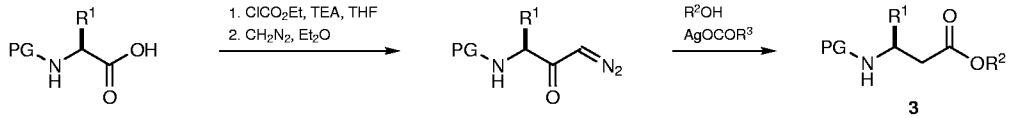

1

2

## 10.12.1.1 Synthesis by Silver Alkanoate Activation

The use of the Arndt–Eistert protocol for the homologation of phthaloyl-protected α-amino acids was first established in the early 1950s.[11] Since then the method has been extended to cover carbamate protected α-amino acids and it has become an exceptionally useful method for the synthesis of β-amino acid derivatives **3**[12] (Scheme 3).

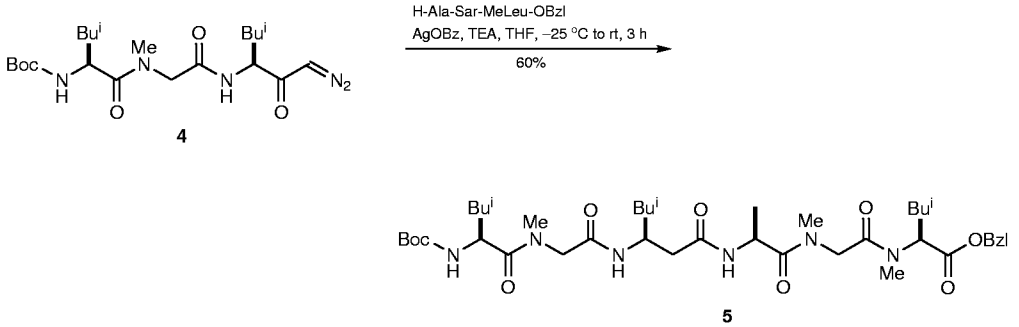
**Scheme 3** Synthesis of β³-Amino Acids by Arndt–Eistert Homologation[11,12]




3

Esters can be formed by the use of various alcohols as solvent with silver benzoate while silver trifluoroacetate in THF/$H_2O$ leads to the formation of free β-amino acids. This approach[8] has been used to form peptide **5** (Scheme 4). In this case, the alcohol solvent is replaced by an alternative nucleophile dissolved in THF. Hence, concomitant peptide coupling is achieved when the nucleophile is a peptide with a free N-terminus.

**Scheme 4** Synthesis of an α-Peptide Containing One β³-Amino Acid[8]




4

5

*for references see p 568*

**Boc-Leu-Sar-β³-HLeu-Ala-Sar-MeLeu-OBzl (5):**[8]

**CAUTION:** *Diazomethane is a detonator and highly toxic. Appropriate safety precautions should be taken when using this reagent.*

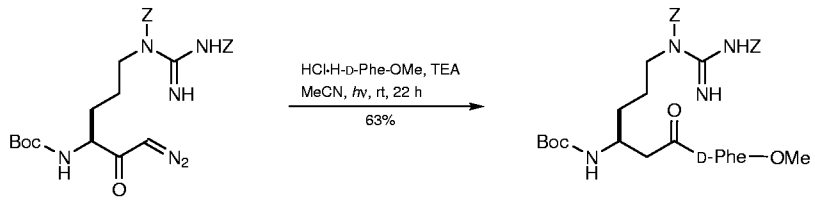
TEA (1.39 mL, 10 mmol) and $ClCO_2Et$ (0.953 mL, 10 mmol) were added to a soln of Boc-Leu-Sar-Leu-OH (4.16 g, 10 mmol) in THF (50 mL, distilled from Na/K) at −15 °C under argon. After 15 min, the resulting soln was allowed to warm to −5 °C and then treated with a soln of diazomethane in $Et_2O$ until the yellow color of $CH_2N_2$ persisted for a longer period of time. The mixture was allowed to warm to rt and stirred for an additional 3 h. Excess diazomethane was destroyed under vigorous stirring by the addition of a small amount of AcOH. After aqueous workup by extraction with sat. $NaHCO_3$, sat. $NH_4Cl$, and sat. NaCl, the organic phase was dried ($MgSO_4$) and concentrated to dryness. Chromatography (silica gel, EtOAc/hexane 2:1) gave diazo ketone **4**; yield: 3.76 g (86%).

A soln of silver benzoate (16 mg, 0.07 mmol) in TEA (0.215 mL, 1.54 mmol) was added to a soln of **4** (238 mg, 0.541 mmol) and H-Ala-Sar-MeLeu-OBzl (528 mg, 1.4 mmol) in THF (10 mL) at −25 °C under argon with exclusion of light. The mixture was warmed to rt over 3 h and some $Et_2O$ was added. After work up by extraction with 1 M HCl and sat. NaCl, sat. $NaHCO_3$ and sat. NaCl, the organic phase was dried ($MgSO_4$), and concentrated to dryness. The residue was purified by chromatography (silica gel, EtOAc) to give the protected peptide **5**; yield: 257 mg (60%).

## 10.12.1.2 Synthesis by Light Activation

Disadvantages of the silver-promoted method for Arndt–Eistert homologation are the large excess of nucleophile required and also the requirement for the use of the free amine during peptide formation procedures.[13] Activation of the diazo ketone by UV light has been shown to be suitable for this type of procedure and good yields are obtained using only 1.2 equivalents of a hydrochloride salt of an amino ester in the presence of triethylamine (Scheme 5).[14]

**Scheme 5** Synthesis of β³-Amino Acids Using UV Activation[14]




**6** **7**

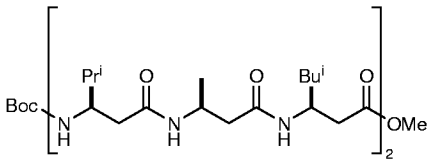
**Boc-β³-HArg(Z,Z)-D-Phe-OMe (7):**[14]

The diazo ketone **6** (~8.2 mmol, 80% purity) was dissolved in MeCN (150 mL) and a soln of HCl·H-D-Phe-OMe (2.1 g, 9.8 mmol) and TEA (1.4 mL, 9.8 mmol) in MeCN (200 mL) was added at rt. The mixture was stirred and irradiated (300 nm) under $N_2$ at 30 °C. After a few minutes, evolution of $N_2$ was observed. After a further 22 h, the resulting precipitate was filtered, washed with ice-cold MeCN and dried under vacuum to give Boc-β³-HArg(Z,Z)-D-Phe-OMe (**7**) (55%). Concentration of the filtrate under reduced pressure yielded a residue, which was dissolved in $CH_2Cl_2$. This soln was washed successively with $H_2O$, 1 M citric acid, $H_2O$, and brine. After drying ($Na_2SO_4$) and concentration further **7** (8%) was obtained: mp 166–167 °C.

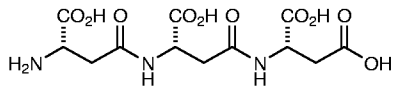
## 10.12.2 $\beta^3$-Peptides

Early work on β-peptides centered on the use of $\beta^3$-amino acids incorporated into polymers (see Section 10.12.5). Since then methods have been developed by Seebach and co-workers for the synthesis of $\beta^3$-peptides of a predefined length.[3,4] The synthesis of $\beta^3$-peptides can be considered as a relatively easy task since the problems of racemization often associated with α-peptide formation do not need to be considered. The rapidly developing interest in β-peptides has really evolved from the discovery that short β-peptide oligomers (~6 amino acids) such as **8** (Scheme 6) adopt an extremely stable helical conformation in solution. The stability of this helix is governed by the nature and position of the substituent. The synthesis of the $\beta^3$-amino acids has been achieved in many ways. Seebach's work has concentrated on the use of $\beta^3$-amino acids carrying proteinogenic side chains and hence the most logical synthetic approach[15] was to begin with α-amino acids and to convert them into the $\beta^3$-amino acids using the Arndt–Eistert protocol as described in Section 10.12.1.1. His group has also produced oligo-β-aspartates using commercially available α-protected Asp derivatives.[16] In earlier work, Kajtár et al.[17] demonstrated that oligo-β-aspartates **9** could be prepared using DCC as the coupling reagent.

**Scheme 6** Some $\beta^3$-Peptides




**8** Boc-[$\beta^3$-HVal-$\beta^3$-HAla-$\beta^3$-HLeu]$_2$-OMe



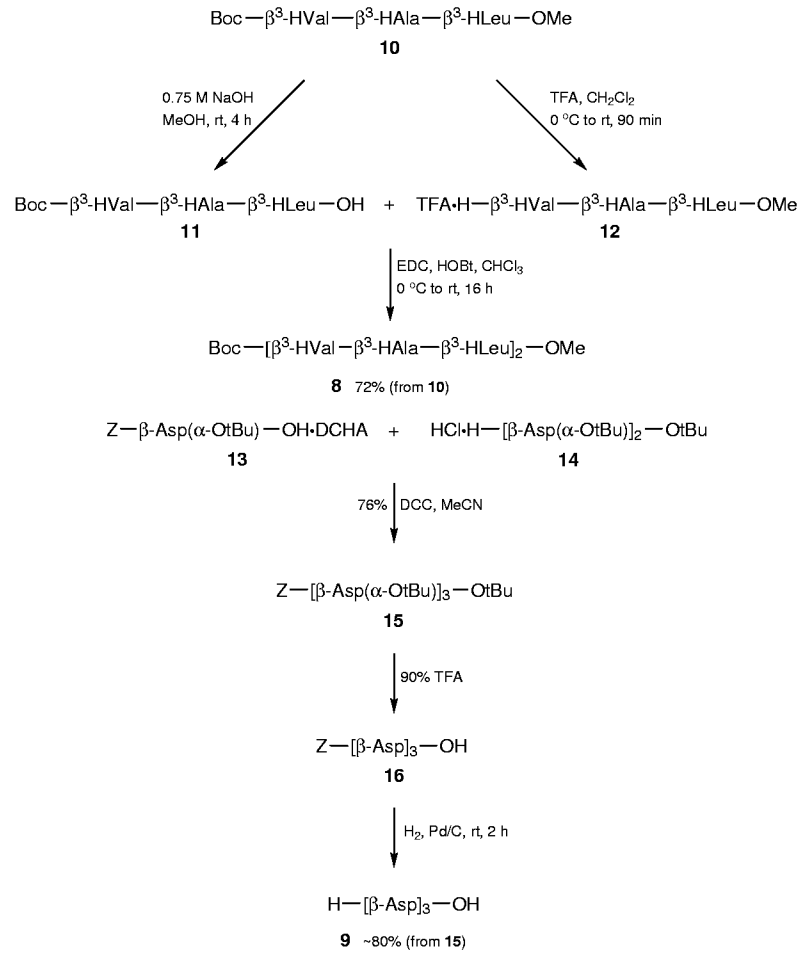

**9** H-[β-Asp]$_3$-OH

### 10.12.2.1 Peptide Coupling via an Activated Ester

Given the lack of a chiral α-carbon in $\beta^3$-amino acids, peptide coupling can be expected to proceed smoothly under most conditions. To date, one method has received almost all the attention and this is presumably because its use[3,4,16] is found to result in high yields of pure $\beta^3$-peptides (e.g., **8**, Scheme 6). The use of the water-soluble reagent EDC together with HOBt has been found to be the most effective carbodiimide-based coupling strategy (Scheme 7). The protecting groups most frequently used are Boc protection for the N-terminus and either methyl or benzyl esters for the C-terminus. The method has been found to be appropriate for use in a range of solvents according to the solubility of the peptide segments to be coupled.

*for references see p 568*

**Scheme 7** Synthesis of $\beta^3$-Peptides[3,17]




**Boc-($\beta^3$-HVal-$\beta^3$-HAla-$\beta^3$-HLeu)$_2$-OMe (8):**[3]

Boc-$\beta^3$-HVal-$\beta^3$-HAla-$\beta^3$-HLeu-OMe (**10**; 390 mg, 0.85 mmol) was dissolved in $CH_2Cl_2$ (0.5 M) and cooled to 0 °C. An equal volume of TFA was added and the mixture was allowed to warm to rt under $N_2$ for 90 min. The resulting soln was concentrated under reduced pressure and dried under high vacuum to yield TFA·H-$\beta^3$-HVal-$\beta^3$-HAla-$\beta^3$-HLeu-OMe (**12**).

Further Boc-$\beta^3$-HVal-$\beta^3$-HAla-$\beta^3$-HLeu-OMe (**10**; 390 mg, 0.85 mmol) was dissolved in MeOH (1.2 M) and treated with aq 0.75 M NaOH (1.2 equiv) at rt. After 4 h, the pH was adjusted to 2–3 using 1 M HCl and then extracted with EtOAc. Drying ($MgSO_4$) and concentration yielded Boc-$\beta^3$-HVal-$\beta^3$-HAla-$\beta^3$-HLeu-OH (**11**) as a white solid.

The crude TFA salt **12** was dissolved in $CHCl_3$ (1.0 M) and cooled to 0 °C. TEA (5 equiv) and a soln of the crude acid **11** in DMF (1.0 M) were added successively. Subsequently, HOBt (1.2 equiv) and EDC (1.2 equiv) were added portionwise. The mixture was allowed to warm to rt and stirring was continued for 16 h. The mixture was concentrated and the residue dried under high vacuum. Stirring the resulting residue in MeOH for 20 min resulted in a white precipitate that was collected by filtration and washed successively with MeOH and MeOH/$H_2O$ (1:1); yield: 480 mg (72%). The product **8** was dried under high vacuum over KOH; mp 240 °C (dec); $[\alpha]_D^{rt}$ −34.6 (*c* 1, TFE).

**H-(β-Asp)$_3$-OH (9):**[17]

Z-β-Asp(α-OtBu)-OH·DCHA (**13**; 19.97 g, 38.5 mmol) was dissolved in MeCN (115 mL) and a soln of HCl·H-[β-Asp(α-OtBu)]$_2$-OtBu (**14**; 18.52 g, 38.5 mmol) in MeCN (100 mL) was added. After stirring for 2 h at rt, the DCHA chloride was removed by filtration and DCC (7.94 g, 38.5 mmol) was added at 0 °C. After stirring for a further 2 h, the mixture was allowed to stand at rt overnight. The DCU was removed by filtration, and the solvent removed under reduced pressure. The remaining oil was dissolved

in $Et_2O$ (100–150 mL) and the soln was washed successively with 0.5 M $H_2SO_4$ (3 × 50 mL), 10% aq $NaHCO_3$ (3 × 50 mL), and $H_2O$ (50 mL), then dried ($Na_2SO_4$), and concentrated under reduced pressure. Crystallization ($Et_2O$/petroleum ether) yielded Z-[β-(Asp(α-OtBu)]$_3$-OtBu (**15**); yield: 21.0 g (76%): mp 93–95 °C, $[\alpha]_D^{rt}$ −22.1 (*c* 1, DMF). Compound **15** was then dissolved in 90% TFA (~20 mL per mmol) and stirred at 0 °C for 1 h and at rt for 2 h. The resulting soln was concentrated and the residue taken up in dry $Et_2O$. The solid **16** which precipitated was filtered, washed with $Et_2O$ and dried under vacuum over concd $H_2SO_4$ and then KOH. Carboxylic acid **16** was dissolved in MeOH (~40 mL per mmol), Pd/C was added and the mixture was placed under an atmosphere of $H_2$ at rt and stirred for ca. 2 h. After removal of the catalyst by filtration, the soln was concentrated and the resulting solid **9** was dried under vacuum over concd $H_2SO_4$ and then KOH. Approximate yield: 80% (from **15**); $[\alpha]_D^{rt}$ +16.1 (*c* 1, 0.5 M HCl).

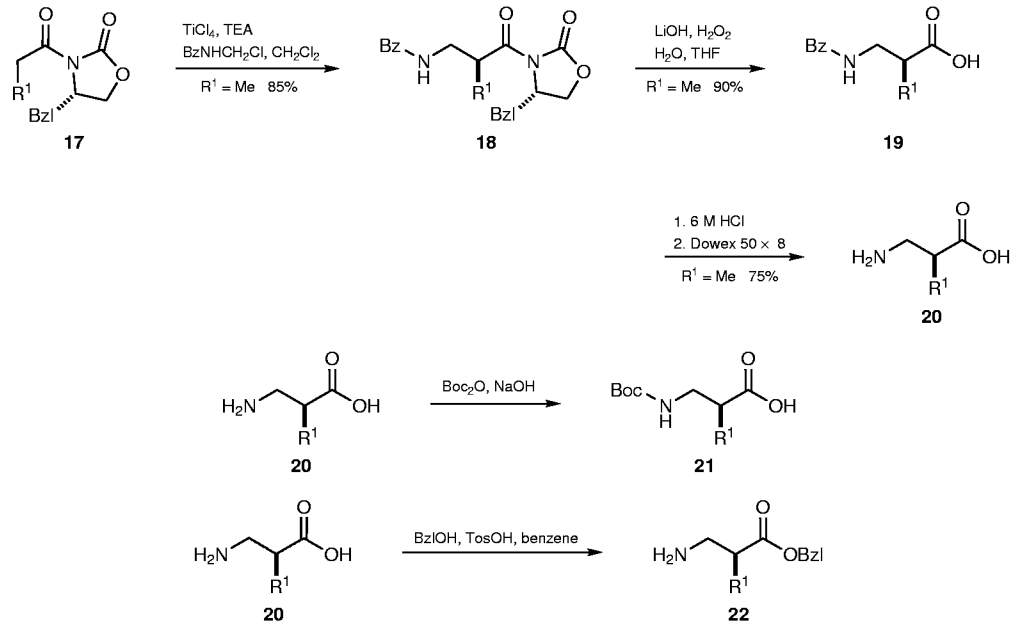
## 10.12.3 β²-Peptides

Similarly to their β³-counterparts, short β²-peptides have been shown to exhibit extraordinary tendencies to adopt stable secondary structures. Again, interest has, to date, focused on those β²-amino acids which bear protein side chains.[5,7] This interest has resulted in the development of methodology based on Evans' work[18] for the synthesis of these β-amino acids bearing a substituent on C2.

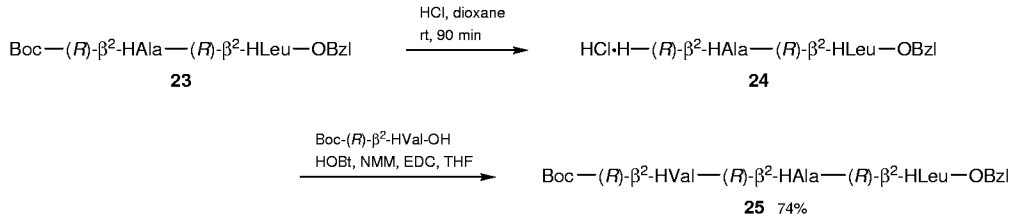
### 10.12.3.1 Synthesis via Evans Methodology

Although the racemization of the α-carbon can now be considered a potential problem, the synthesis of β²-peptides has been achieved in the same way as seen for β³-peptides. As the β²-amino acids cannot be prepared from the analogous α-amino acids, Seebach and co-workers[5,7] opted to use Evans oxazolidinone chemistry to produce enantiomerically pure β²-amino acids. Alkylation of 3-acyloxazolidin-2-ones **17** with *N*-(chloromethyl)benzamide yielded the products **18** with diastereomeric ratios between 93:7 and 99:1 (Scheme 8). Removal of the chiral auxiliary ($LiOH/H_2O_2$) and debenzoylation (refluxing acid) was followed by ion-exchange chromatography to yield the free β²-amino acids **20** which were converted by standard means into Boc **21** or benzyl ester **22** derivatives for peptide synthesis.

**Scheme 8** Synthesis of β²-Amino Acids and β²-Peptides[5,7]




*for references see p 568*




Peptide coupling was achieved using the standard combination of EDC and HOBt. Epimerization during the coupling step was found to be considerably reduced when TEA was replaced by the weaker base NMM.[7]

**H-(*R*)-β²-HAla-OH (20, R¹ = Me); Typical Procedure:**[7]

3-Propanoyl-4-benzyloxazolidin-2-one (**17**, R¹ = Me; 4.7 g, 20 mmol) was dissolved in $CH_2Cl_2$ (0.25 M) and cooled to 0 °C. $TiCl_4$ (2.20 mL, 20 mmol) was added slowly such that the internal temperature did not rise above 5 °C and the resulting yellow soln (or suspension) was stirred for 10 min. Again keeping the temperature below 5 °C, TEA (2.80 mL, 20 mmol) was added and the dark red soln was then stirred for 45 min at 0 °C. *N*-(Chloromethyl)benzamide (4.07 g, 24 mmol) was added and the mixture stirred at 0 °C for a further 60 min. Sat. $NH_4Cl$ was added and the organic layer was washed with 1 M HCl. The aqueous phase was then extracted with $CH_2Cl_2$. The combined organic layers were dried ($MgSO_4$) and concentrated. Recrystallization (EtOAc/hexane) gave **18** (R¹ = Me) as white needles; yield: 6.26 g (85%); mp 120–121 °C; $[\alpha]_D^{rt}$ +27.0 (*c* 0.89, $CHCl_3$).

The acyloxazolidinone **18** (R¹ = Me; 3.87 g, 10.6 mmol) was dissolved in THF/$H_2O$ 4:1 (0.2 M) and cooled to 0 °C. 30% aq $H_2O_2$ (4.30 mL, 42 mmol) and $LiOH \cdot H_2O$ (711 mg, 17 mmol) were added and the soln was stirred at 0 °C for 1 h. The mixture was treated with sat. $NaHSO_3$ and washed with $CH_2Cl_2$. The aqueous phase was acidified to pH 1–2 with 6 M HCl and extracted with EtOAc. The combined EtOAc phases were dried ($MgSO_4$) and concentrated under reduced pressure. Recrystallization (EtOAc/hexane) gave white crystals of **19** (R¹ = Me); yield: 1.96 g (90%); mp 111–112 °C; $[\alpha]_D^{rt}$ −25.2 (*c* 1.23, $CHCl_3$); $R_f$ 0.44 (EtOAc/hexane/AcOH 5:5:0.5). The chiral auxiliary was recovered from the $CH_2Cl_2$ phase.

The N-benzoyl-protected β-amino acid **19** (R¹ = Me; 1.04 g, 5 mmol) was dissolved in concd HCl/AcOH/$H_2O$ 2:1:1 (0.25 M) and refluxed at 110 °C for 2 d. The mixture was diluted with $H_2O$ and washed with $Et_2O$. The aqueous phase was concentrated and the crude HCl salt of the amino acid was purified by ion-exchange chromatography (Dowex 50 × 8, strongly acidic, elution with 1% aq $NH_3$ soln). Recrystallization from MeOH gave H-(*R*)-β²-HAla-OH (**20**, R¹ = Me); yield: 385 mg (75%): mp 185–186 °C; $[\alpha]_D^{29}$ −11.8 (*c* 1, 1 M HCl).

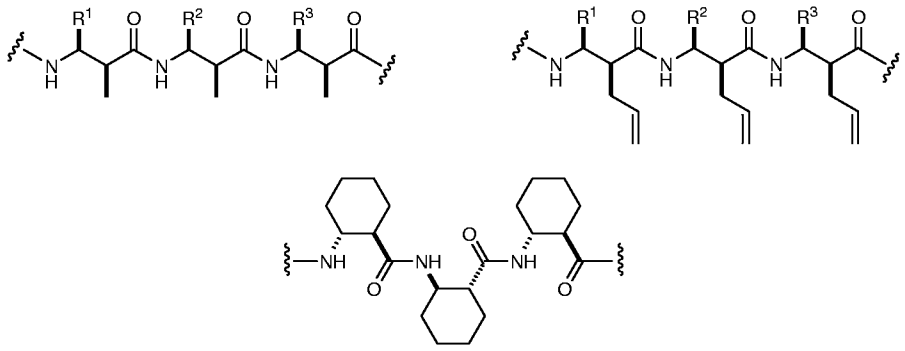
**Boc-(*R*)-β²-HVal-(*R*)-β²-HAla-(*R*)-β²-HLeu-OBzl (25):**[7]

Boc-(*R*)-β²-HAla-(*R*)-β²-HLeu-OBzl (**23**; 243 mg, 0.58 mmol) was dissolved in HCl saturated dioxane and stirred for 90 min at rt. Concentration under reduced pressure and drying under high vacuum yielded the crude HCl salt **24**. This salt was dissolved in THF (5 mL) and cooled to 0 °C. Boc-(*R*)-β²-HVal-OH (148 mg, 0.64 mmol), HOBt (95 mg, 0.70 mmol), NMM (0.20 mL, 1.8 mmol), and EDC (123 mg, 0.64 mmol) were added successively. After stirring at 0 °C for 1 h, the mixture was stirred at rt for 18 h. The mixture was diluted with EtOAc and then washed successively with 0.5 M HCl, sat. $K_2CO_3$, and sat. NaCl. The organic layer was dried ($Na_2SO_4$) and concentrated under reduced pressure. Flash column chromatography ($Et_2O$) yielded **25** as a white solid; yield: 231 mg (74%); mp 157–159 °C; $[\alpha]_D^{rt}$ −69.2 (*c* 0.95, $CHCl_3$).

## 10.12.4 β²,³-Peptides

The synthesis of β-peptides bearing substituents at both C2 and C3 atoms has seen the most widespread interest. Stable secondary structures have once again been observed in these classes of β-peptides. They can be divided into three classes: (1) those bearing protein side chains;[7] (2) those carrying a cycloalkyl ring;[19–21] and (3) those with allyl substituents[22] (Scheme 9).

**Scheme 9** Some $\beta^{2,3}$-Peptides

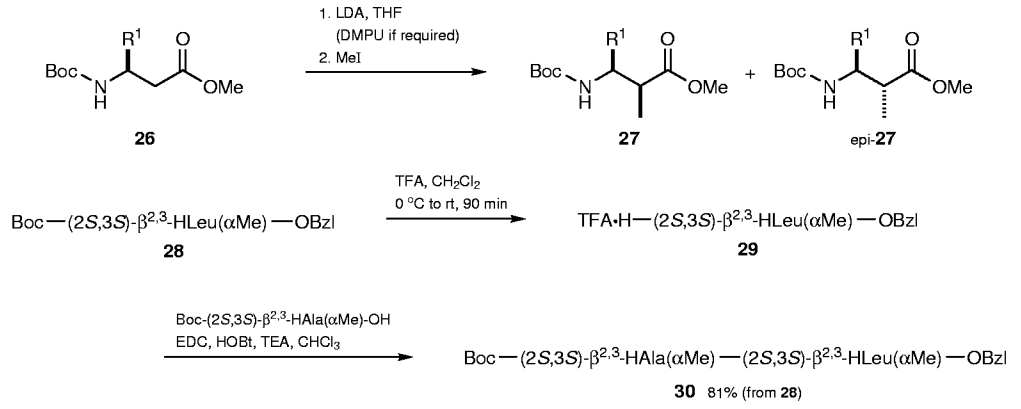


## 10.12.4.1 $\beta^{2,3}$-Peptides Bearing Protein Amino Acid Side Chains

### 10.12.4.1.1 Synthesis via Alkylation of $\beta^3$-Amino Acids

The synthesis of $\beta^{2,3}$-peptides bearing protein side chains has been achieved[7] starting from α-amino acids. Arndt–Eistert homologation led to the $\beta^3$-amino acids **26** and alkylation[23] at the α-carbon resulted in the formation of $\beta^{2,3}$-substituted amino acids **27** in high diastereomeric purity (Scheme 10). The ratio of diastereomers present has been controlled[7] by the use of additives: when a lithium salt (LiCl or LiBr) is added, the (2*S*,3*S*)-isomer is formed as the major diastereomer, but when DMPU is added, the (2*R*,3*S*)-isomer is the major isomer formed. Flash chromatography successfully isolates either isomer in a diastereomerically pure form. Care is needed to avoid epimerization during base-catalyzed hydrolysis of the methyl esters prior to subsequent peptide coupling and the direct conversion of the methyl esters into the benzyl esters via titanate-mediated transesterification is the preferred option.[24]

**Scheme 10** Synthesis of $\beta^{2,3}$-Peptides with Protein Side Chains[7]




**Boc-(2*S*,3*S*)-$\beta^{2,3}$-HLeu(αMe)-OMe (27, R¹ = iBu); Typical Procedure:**[7]

**CAUTION:** *Iodomethane is a suspected carcinogen. Inhalation, ingestion, or absorption through the skin may be fatal. Appropriate safety precautions and procedures should be taken during all stages of its handling and disposal.*

LiCl (0.29 g, 6.87 mmol) was suspended in THF (7 mL) and cooled to −78 °C. $iPr_2NH$ (0.72 mL, 5.04 mmol) and BuLi in hexane (3.36 mL, 5.04 mmol) were added and the mixture was stirred for 15 min.

*for references see p 568*

A soln of Boc-(3*S*)-β$^3$-HLeu-OMe (**26**, R$^1$ = iBu; 0.593 g, 2.29 mmol) in THF (5 mL) was added slowly over 20 min and the resulting mixture was stirred at −78 °C for 1 h. MeI (0.57 mL, 9.16 mmol) was added and the mixture was stirred for 18 h as the temperature rose slowly to −55 °C. It was then quenched with sat. $NH_4Cl$ and diluted with $Et_2O$. The organic layer was washed successively with sat. $NaHCO_3$, sat. $NH_4Cl$, and sat. NaCl, dried ($MgSO_4$), and concentrated under reduced pressure. Flash column chromatography (pentane/EtOAc 5:1) yielded Boc-(2*S*,3*S*)-β$^{2,3}$-HLeu(αMe)-OMe (**27**, R$^1$ = iBu) as an oil as the major diastereomer; yield: 0.348 g (56%); $R_f$ 0.49 (pentane/EtOAc 5:1); $[\alpha]_D^{rt}$ −43.0 (*c* 1, $CHCl_3$); and Boc-(2*R*,3*S*)-β$^{2,3}$-HLeu(αMe)-OMe (epi-**27**, R$^1$ = iBu) as a waxy solid as the minor diastereomer; yield: 0.117 g (19%).

**Boc-(2*R*,3*S*)-β$^{2,3}$-HLeu(αMe)-OMe (epi-27, R$^1$ = iBu); Typical Procedure:**[7]

**CAUTION:** *Iodomethane is a suspected carcinogen. Inhalation, ingestion, or absorption through the skin may be fatal. Appropriate safety precautions and procedures should be taken during all stages of its handling and disposal.*

BuLi in hexane (30.9 mL, 39.0 mmol) was added to a soln of $iPr_2NH$ (5.56 mL, 39.0 mmol) and DMPU (8.56 mL, 70.8 mmol) in THF (120 mL) at −78 °C. After 15 min, a soln of Boc-(3*S*)-β$^3$-HLeu-OMe (**26**, R$^1$ = iBu; 4.6 g, 17.7 mmol) in THF (18 mL) was added over 10 min and the mixture was stirred at −78 °C for a further 1 h. MeI (4.41 mL, 70.8 mmol) was then added slowly and the resulting mixture was stirred at −78 °C for a further 3 h. It was then quenched with sat. $NH_4Cl$ and diluted with $Et_2O$. The organic layer was washed successively with sat. $NaHCO_3$, sat. $NH_4Cl$, and sat. NaCl, dried ($MgSO_4$), and concentrated under reduced pressure. Flash column chromatography (pentane/$Et_2O$ 6:1) yielded the major diastereomer, Boc-(2*R*,3*S*)-β$^{2,3}$-HLeu(αMe)-OMe (epi-**27**, R$^1$ = iBu) as a waxy solid; yield: 2.72 g (57%); $R_f$ 0.40 (pentane/EtOAc 5:1); mp 41–43 °C, $[\alpha]_D^{rt}$ −43.0 (*c* 1, $CHCl_3$); and the minor diastereomer, Boc-(2*S*,3*S*)-β$^{2,3}$-HLeu(αMe)-OMe (**27**, R$^1$ = iBu) as an oil; yield: 1.29 g (27%).

**Boc-(2*S*,3*S*)-β$^{2,3}$-HAla(αMe)-(2*S*,3*S*)-β$^{2,3}$-HLeu(αMe)-OBzl (30):**[7]

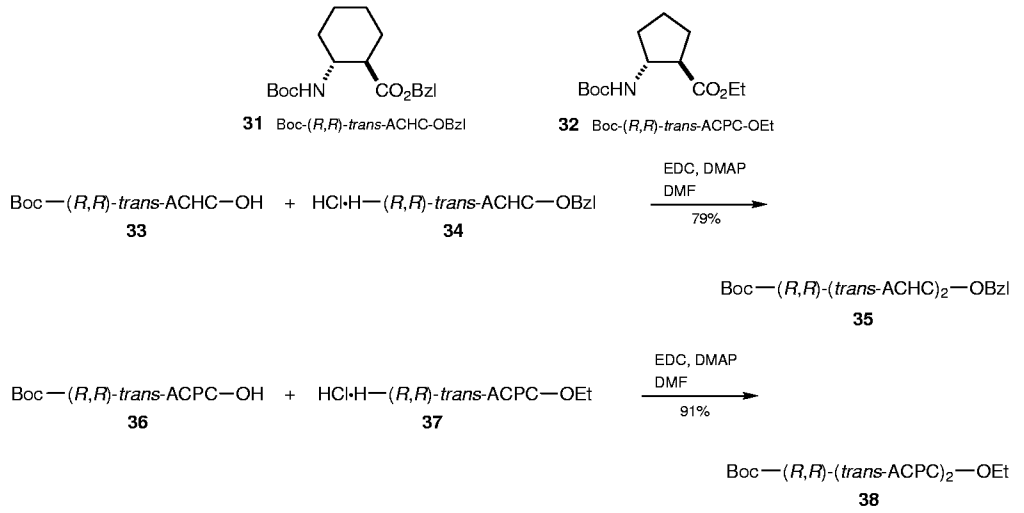
Boc-(2*S*,3*S*)-β$^{2,3}$-HLeu(αMe)-OBzl (**28**; 743 mg, 2.13 mmol) was dissolved in $CH_2Cl_2$ (0.5 M) and cooled to 0 °C. An equal volume of TFA was added and the mixture was stirred at rt for 90 min. Concentration under reduced pressure and drying under high vacuum yielded the crude TFA salt **29**. This salt **29** was dissolved in $CHCl_3$ and washed with 1 M NaOH (3 ×). The aqueous phase was extracted with $CHCl_3$ and the combined organic layers were dried ($MgSO_4$) and concentrated under reduced pressure. The resulting free amino ester was dissolved in $CHCl_3$ (0.4 M) and cooled to 0 °C. A soln of Boc-(2*S*,3*S*)-β$^{2,3}$-HAla(αMe)-OH (463 mg, 2.13 mmol) in $CHCl_3$ (0.4 M), TEA (5 equiv), HOBt (1.2 equiv), and EDC (1.2 equiv) were added successively. After stirring at 0 °C for 1 h, the mixture was stirred at rt for 18 h. The mixture was diluted with $CHCl_3$ and then washed successively with 1 M HCl, sat. $K_2CO_3$ and sat. NaCl. The organic layer was dried ($Na_2SO_4$) and concentrated under reduced pressure. Flash column chromatography (pentane/EtOAc 7:2) afforded **30**; yield: 758 mg (81%); mp 156–158 °C; $R_f$ 0.29 (pentane/EtOAc 7:2); $[\alpha]_D^{rt}$ −19.8 (*c* 1, $CHCl_3$).

## 10.12.4.2 β$^{2,3}$-Peptides from Cycloalkyl-Substituted Amino Acids

### 10.12.4.2.1 Coupling of Cycloalkyl-Substituted Amino Acids

The synthesis of β$^{2,3}$-peptides bearing cycloalkyl substituents has been developed by Gellman and co-workers.[19–21] These oligopeptides exhibit a strong tendency to form stable, helical secondary structures. The cyclohexyl-substituted peptide forms a helix identical to that seen with the β$^3$- and β$^{2,3}$-peptides bearing protein side chains.[1] However, the cyclopentyl-substituted peptide exhibits a novel helix which has a surprisingly small diameter.[21] These β$^{2,3}$-peptides were constructed by synthesis and resolution of the appropriate cycloalkyl-amino acids[25–29] (Scheme 11) and then coupled together using standard EDC-based peptide coupling procedures.

**Scheme 11** Synthesis of $\beta^{2,3}$-Peptides Bearing Cycloalkyl Substituents[29]




**(+)-(*S,S*)- and (−)-(*R,R*)-*trans*-2-(Benzoylamino)cyclohexanecarboxylic Acid by Preferential Crystallization:**[41]

(±)-*trans*-2-(Benzoylamino)cyclohexanecarboxylic acid[27] (2.0 g, 8.1 mmol) and (±)-1-aminopropan-2-ol (0.63 mL, 8.1 mmol) were dissolved in 40% MeOH/$H_2O$ (20 mL). Additional (±)-*trans*-2-(benzoylamino)cyclohexanecarboxylic acid (0.5 g, 2.0 mmol) was added and dissolved in the soln by heating. The soln was allowed to cool for a few min, then a few crystals of (+)-(*S,S*)-*trans*-2-(benzoylamino)cyclohexanecarboxylic acid were added. After 1 h, additional crystals were present in soln and were collected by suction filtration. Additional (±)-*trans*-2-(benzoylamino)cyclohexanecarboxylic acid (0.5 g, 2.0 mmol) was added to the filtrate and dissolved in the soln by heating and adding more MeOH as needed. The soln was allowed to cool for a few min, then a few crystals of (−)-(*R,R*)- *trans*-2-(benzoylamino)cyclohexanecarboxylic acid were added. After 1 h, additional crystals were present in soln and were collected by suction filtration. This procedure of adding racemate and seeding with crystals of one enantiomer or the other was repeated several times. The crystals with the same optical rotation were combined and recrystallized once from 95% EtOH. (±)-*trans*-2-(Benzoylamino)cyclohexanecarboxylic acid (4 g) gave the (−)-(*R,R*)-enantiomer; yield: 1.63 g (41%); $[\alpha]_D^{23}$ −45 (*c* 0.2, EtOH); and the (+)-(*S,S*)-enantiomer; yield: 0.395 g (10%); $[\alpha]_D^{23}$ +45 (*c* 0.2, EtOH).

**(−)-(*R,R*)-*trans*-2-(*tert*-Butoxycarbonylamino)cyclohexanecarboxylic Acid [Boc-(*R,R*)-*trans*-ACHC-OH, 33]:**[29]

Crystals of (−)-(*R,R*)-*trans*-2-(*tert*-butoxycarbonylamino)cyclohexanecarboxylic acid (0.5 g, 2.02 mmol) were suspended in 12 M HCl (55 mL). The soln was refluxed 48 h, and the crystals eventually dissolved. After the soln cooled to rt, the soln was cooled to 0 °C for 1 h in an ice bath. This resulted in the precipitation of benzoic acid, which was collected by suction filtration. The filtrate was concentrated on a rotary evaporator to a few mL, and dioxane/$H_2O$ (2:1, 30 mL), $K_2CO_3$ (2.8 g, 20 mmol), and $Boc_2O$ (0.441 g, 2.02 mmol) were added. The reaction was stirred 5 h. The pH of the reaction was adjusted to 2 with 12 M HCl, and the resulting soln was then extracted with EtOAc (3 ×, 150 mL total). The organic extracts were dried ($MgSO_4$), concentrated, and dried under vacuum; yield: 0.411 g (84%). Crystals were grown by slow evaporation of a soln of the product in 1,2-dichloroethane: mp 154–155 °C; $[\alpha]_D^{23}$ −22.8 (*c* 0.3, $CHCl_3$).

**(−)-Benzyl (*R,R*)-2-(*tert*-Butoxycarbonylamino)cyclohexanecarboxylate [Boc-(*R,R*)-*trans*-ACHC-OBzl, 31]:**[29]

Boc-(*R,R*)-*trans*-ACHC-OH (**33**; 0.2 g, 0.82 mmol) was dissolved in dry benzene (20 mL) (**CAUTION:** *carcinogen!*). BzlBr (0.1 mL, 0.82 mmol) and DBU (0.12 mL, 0.82 mmol) were added, and the soln was refluxed under $N_2$ for 6 h. White solid precipitated from the soln during this time. The mixture was cooled to rt. The soln was then filtered and the filtrate was concentrated to obtain a tan solid. The product was purified by column chromatography (silica gel, $CHCl_3$/EtOAc 70:1, $R_f$ 0.31); yield: 0.228 g (83%), which was recrystallized (heptane); mp 103 °C.

*for references see p 568*

**(−)-Benzyl (*R*,*R*)-2-[(*R*,*R*)-2-(*tert*-Butoxycarbonylamino)cyclohexylcarbonylamino]cyclohexanecarboxylate [Boc-(*R*,*R*)-(*trans*-ACHC)$_2$-OBzl, 35]:**[29]

Boc-(*R*,*R*)-*trans*-ACHC-OBzl (**31**; 0.047 g, 0.14 mmol) was dissolved in 4 M HCl/dioxane (0.5 mL) and stirred for 1 h. The solvent was then removed under a stream of $N_2$, and the residue **34** was dried under vacuum. Boc-(*R*,*R*)-*trans*-ACHC-OH (**33**; 0.034 g, 0.14 mmol) and DMAP (0.023 g, 0.19 mmol) were added followed by DMF (1 mL). EDC (0.059 g, 0.31 mmol) was added, and the reaction was stirred for 48 h under $N_2$. Solvent was removed under a stream of $N_2$, and the residue was further dried under vacuum. To this residue was added 1 M HCl (~3 mL) and the solid that did not dissolve was isolated by suction filtration and washed with additional 1 M HCl. The solid was dried under vacuum; yield: 0.051 g (79%). Fluffy crystals were grown by vapor diffusion of heptane into a soln of **35** in 1,2-dichloroethane; mp 195 °C.

**(−)-Ethyl (*R*,*R*)-2-[(*R*,*R*)-2-(*tert*-Butoxycarbonylamino)cyclopentylcarbonylamino]cyclopentanecarboxylate [Boc-(*R*,*R*)-(*trans*-ACPC)$_2$-OEt, 38]:**[29]

Boc-(*R*,*R*)-*trans*-ACPC-OEt[29] (**32**; 0.35 g, 1.4 mmol) was dissolved in MeOH (20 mL) and $H_2O$ (7 mL) and LiOH·$H_2O$ (0.57 g, 13.6 mmol) were added. The resulting soln was cooled to 0 °C, and 30.8% aq $H_2O_2$ (0.75 mL, 6.8 mmol) was added. The reaction was placed into a cold room at 5 °C, and stirred for 20 h. While the reaction was still cold, aq $Na_2SO_3$ (2.7 g in 12 mL of $H_2O$, 21.4 mmol) was added. The soln was moved out of the cold room, and the MeOH was removed on a rotary evaporator. The pH of the resulting soln was adjusted to 2 with 3 M HCl, and the aqueous soln was then extracted with $CH_2Cl_2$ (3 × 25 mL). The organic extracts were dried ($Na_2SO_4$), concentrated, and put under vacuum to yield **36** as a white solid; yield: 0.21 g (67%). Boc-(*R*,*R*)-*trans*-ACPC-OEt (**32**; 0.23 g, 0.9 mmol) was dissolved in 4 M HCl/dioxane (2 mL). The soln was stirred for 1 h, then the solvent was removed under a stream of $N_2$, and the residue further dried under vacuum to give HCl·H-(*R*,*R*)-*trans*-ACPC-OEt (**37**). Boc-(*R*,*R*)-*trans*-ACPC-OH (**36**) was dissolved in DMF (3 mL) and transferred via cannula to the flask containing HCl·H-(*R*,*R*)-*trans*-ACPC-OEt (**37**). DMAP (0.13 g, 1.1 mmol) was added, followed by EDC (0.38 g, 1.98 mmol). The reaction was stirred for 36 h under $N_2$. The solvent was removed under a stream of $N_2$ and the residue was further dried under vacuum. To this residue was added 1 M HCl (a few mL), and the resulting white solid collected by suction filtration, and dried under vacuum to afford **38** as a white solid; yield: 0.3 g (91%). Fluffy crystals were obtained by vapor diffusion of heptane into a soln of **38** in 1,2-dichloroethane; mp 180–181 °C (dec).

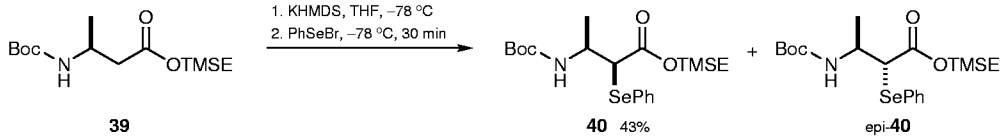
## 10.12.4.3 $\beta^{2,3}$-Peptides Bearing Allyl Substituents

The synthesis of $\beta^{2,3}$-peptides bearing allyl substituents has been achieved[22] by the direct modification of a $\beta^3$-peptide or the preparation of a $\beta^{2,3}$-amino acid and then peptide coupling. This novel approach makes use of radical-based chemistry to introduce the allyl substituent from a phenylselenyl-substituted peptide. The allyl substituent has been introduced both prior to the coupling of the amino acids and also after the coupling of several amino acids to form a selenylated oligopeptide.

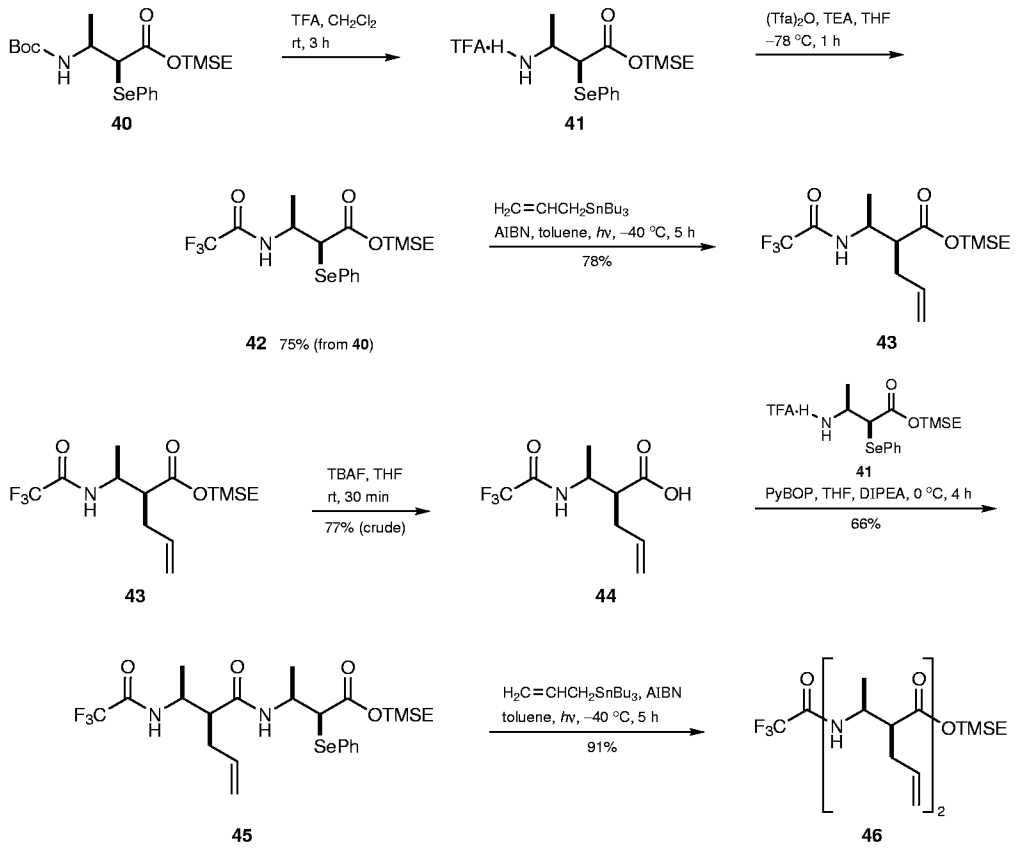
### 10.12.4.3.1 Iterative Free Radical C-Allylation Reactions

The modification of $\beta^3$-amino acids by α-selenylation and then free radical allylation leads to the formation of $\beta^{2,3}$-amino acids bearing an α-allyl group (Scheme 12). Hanessian and coworkers[22] initially used this approach to produce $\beta^{2,3}$-amino acids which were then subjected to suitable coupling procedures. Interestingly, the coupling method chosen made use of PyBOP and hence demonstrates that other peptide coupling procedures can be used in the synthesis of β-peptides.

**Scheme 12** Synthesis of $\beta^{2,3}$-Peptides Bearing α-Allyl Substituents[22]




**39** **40** 43% epi-**40**




**(2*S*,3*S*)-3-(*tert*-Butoxycarbonylamino)-2-(phenylselenyl)butanoic Acid 2-(Trimethylsilyl)ethyl Ester [40, Boc-(2*S*,3*S*)-β$^{2,3}$-HAla(αSePh)-OTMSE]:**[22]

A 0.5 M soln of KHMDS in toluene (6.2 mL, 3.12 mmol) was added dropwise to a soln of Boc-(*S*)-β$^3$-HAla-OTMSE (**39**; 315 mg, 1.04 mmol) in THF (10 mL) at −78 °C under argon. The resulting mixture was stirred at −78 °C for 30 min and then −20 °C for 30 min. A soln of PhSeBr (244 mg, 1.04 mmol) in THF (10 mL) at −78 °C was added to the dianion at −78 °C and this was stirred for 30 min. The resulting mixture was poured into a mixture of $Et_2O$ and 10% aq HCl. The $Et_2O$ layer was washed, dried, and concentrated and the product was purified by flash column chromatography (hexane/EtOAc 9:1) to give **40**; yield: 213 mg (43%); $[\alpha]_D^{rt}$ −69.7 (*c* 0.64, $CH_2Cl_2$), and a mixture of **40** and epi-**40** (92 mg, 19%).

**Tfa-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-OTMSE (43):**[22]

TFA (0.22 mL, 2.86 mmol) was added to a soln of Boc-(2*S*,3*S*)-β$^{2,3}$-HAla(αSePh)-OTMSE (**40**; 134 mg, 0.28 mmol) in $CH_2Cl_2$ (5 mL). The resulting mixture was stirred at rt for 3 h under argon. The solvent was removed and the crude product **41** was dissolved in THF (5 mL) and cooled to −78 °C. $(Tfa)_2O$ (0.08 mL, 0.57 mmol) and TEA (0.16 mL, 1.15 mmol) were added and the mixture was stirred at −78 °C for 1 h. The reaction was quenched with 10% aq HCl and the mixture extracted with $Et_2O$. Drying and concentration gave a crude product **42** that was purified by flash column chromatography (hexane/EtOAc 9:1). This gave **42** as a single isomer; yield: 96 mg (75%). Compound **42** was then treated with allyltributylstannane (0.13 mL, 0.42 mmol) and a catalytic amount of AIBN in toluene (2 mL) at −40 °C under argon. The mixture was irradiated with a 275-W sunlamp for 5 h. Removal of the solvent and flash column chromatography (hexane/EtOAc 100:0 to 9:1) gave **43**; yield: 55 mg (78%); $[\alpha]_D^{rt}$ −49.3 (*c* 0.27, $CH_2Cl_2$).

**Tfa-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-OTMSE (46):**[22]

A soln of Tfa-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-OTMSE (**43**; 54 mg, 0.16 mmol) in THF (5 mL) and 1.0 M TBAF in THF (0.48 mL, 0.48 mmol) was stirred at rt for 30 min. $Et_2O$ and 10% aq HCl were added to the mixture. The organic layer was separated, washed successively with $H_2O$ and brine and dried ($MgSO_4$). Concentration gave crude **44** (29 mg, 77%) which was used without further purification.

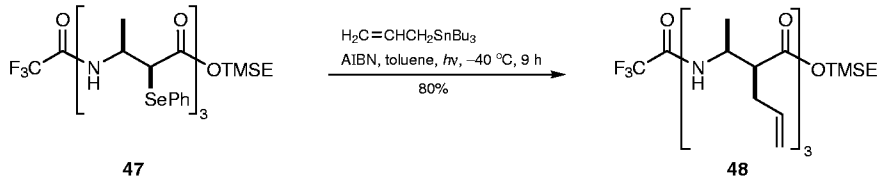
*for references see p 568*

TFA (0.085 mL, 1.10 mmol) was added to a soln of **40** (52.6 mg, 0.11 mmol) in $CH_2Cl_2$ (2 mL) and the mixture was stirred at rt for 3 h under argon. The solvent was removed and the crude product **41** was added to a soln of **44** (27 mg, 0.11 mmol) and PyBOP (95 mg, 0.18 mmol) in THF (3 mL) at 0 °C under argon. DIPEA (0.85 mL, 0.49 mmol) was added and the resulting mixture was stirred at 0 °C for 4 h. $Et_2O$ and 10% aq HCl were added to the mixture. The organic layer was separated, washed successively with $H_2O$ and brine and dried ($MgSO_4$). Concentration and flash column chromatography (hexane/EtOAc 8:2) gave compound **45** as a single isomer; yield: 41 mg (66%). Compound **45** was then treated with allyltributylstannane (0.042 mL, 0.14 mmol) and a catalytic amount of AIBN in toluene (2 mL) at −40 °C under argon. The mixture was irradiated with a 275 W sunlamp for 5 h. Removal of the solvent and flash column chromatography (hexane/EtOAc 100:0 to 9:1 to 8:2) gave **46**; yield: 28 mg (91%); $[\alpha]_D^{rt}$ −38.1 (*c* 0.16, $CH_2Cl_2$).

### 10.12.4.3.2 One-Pot Triallylation Reaction

In a fascinating development of the methodology described in Section 10.12.4.3.1, Hanessian and co-workers[22] have shown that it is also possible to take a tri-selenylated peptide and to substitute the three phenylselenyl groups with three allyl groups in a one-pot procedure (Scheme 13). In each case the selenyl groups are replaced with almost total retention of configuration.

**Scheme 13** Synthesis of β$^{2,3}$-Peptides Bearing α-Allyl Substituents Using a One-pot Triallylation Strategy[22]



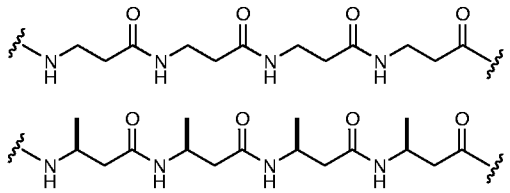

47 48

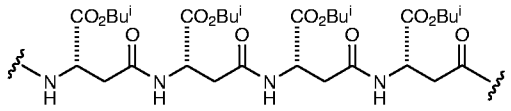
**Tfa-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-(2*S*,3*S*)-β$^{2,3}$-HAla(αallyl)-OTMSE (48):**[22]
Tfa-[(2*S*,3*S*)-β$^{2,3}$-HAla(αSePh)]$_3$-OTMSE (**47**; 33 mg, 0.034 mmol), allyltributylstannane (0.05 mL, 0.16 mmol) and a catalytic amount of AIBN were dissolved in $CH_2Cl_2$ (1 mL) and irradiated with a 275 W sunlamp at −40 °C under argon for 9 h. Concentration and flash column chromatography (hexane/EtOAc 100:0 to 9:1 to 7:3 to 6:4) gave **48**; yield: 16 mg (80%); $[\alpha]_D^{rt}$ −18.7 (*c* 1.15, $CHCl_3$).

## 10.12.5 β-Peptide Polymers

Polymeric β-peptides of undefined length (Scheme 14) were amongst the earliest examples of β-peptides that were studied. These so-called nylon-3 derivatives proved to be of considerable interest and there were many investigations into the secondary structures they adopted.[2,3] Opinions varied but it now seems clear that they too adopt helical secondary structures.[1,30]

**Scheme 14** Some β-Peptide Polymers



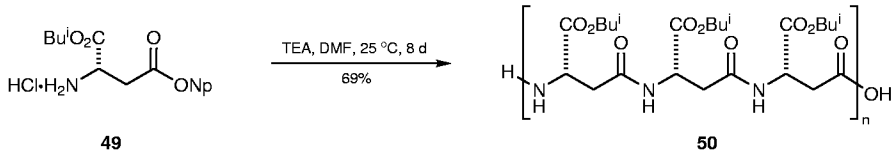


The homo-polymer that results from the coupling of many Asp residues has also been shown to be of interest as a biodegradable polymer.[31] Although most interest here involves the use of the mixed α-/β-peptide, the pure β-peptide also has potential uses in the field.

### 10.12.5.1 Polymerization of α-Protected Aspartic Acid

A poly[β-Asp(α-OiBu)]$_n$ (**50**) was prepared by Yuki et al.[32] by the polycondensation of the β-4-nitrophenyl ester of α-isobutyl-L-aspartate [HCl·H-Asp(β-ONp)-α-OiBu] (**49**) (Scheme 15). Some of the data recorded for the optical rotations of these polymers (and the monomer after hydrolysis of samples of the polymer) suggests that some racemization may have taken place during polymerization although the authors believe this not to be the case. For example, the addition of HOBt to the polymerization reaction appeared to result in an increased yield of 80% but the $[\alpha]_D$ value dropped from approximately −15 in most cases to −7 under these conditions. Co-polymerization of HBr·H-Asp(β-ONp)-α-OiBu and β-Ala (β-HGly) under the same conditions yielded a co-polymer with a composition of 53 mol% β-Ala and 47 mol% Asp(α-OiBu).

**Scheme 15** Synthesis of Poly[β-Asp(α-OiBu)]$_n$[32]



**49** **50**

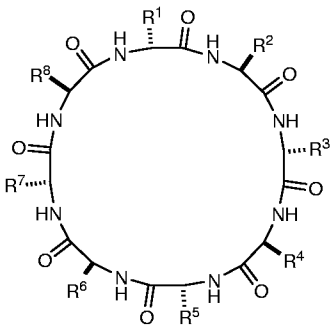
**Poly-[β-Asp(α-OiBu)]$_n$ (50):**[32]
HCl·H-Asp(β-ONp)-α-OiBu (**49**; 3.0 mmol) was placed in a dry glass ampoule which was repeatedly evacuated and flushed with $N_2$. DMF (2.4 mL) and TEA (3.0 mmol) were added and the ampoule was sealed and warmed to 25 °C for 8 d. The polymer **50** was precipitated by the addition of EtOH, filtered, and washed with EtOH and then dried at 50 °C under vacuum; yield: 69%; $[\alpha]_D^{30}$ −15 (*c* 0.5, $CHCl_3$).

## 10.12.6 Cyclo-β-peptides

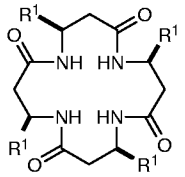
Cyclic α-peptides have seen applications in many areas of chemistry. It is therefore natural that interest in cyclo-β-peptides also exists. In addition to examples of cyclic α-peptides with interesting biological activity, there have been examples of cyclo-α-peptides **51** which can act as nanotubes and ion transporters (Scheme 16).[33,34] Although little evidence has yet been published for biological activity of β-peptides, Seebach and co-workers[35] have shown that the cyclic β-peptides **52** do exhibit some of the "tube-like" properties associated with the cyclic α-peptides built from alternating (*R*)- and (*S*)-amino acids.

*for references see p 568*

**Scheme 16** Comparison of Cyclopeptides[33–35]



**51** *c*[-(*R*)-Xaa-(*S*)-Xaa-]$_4$

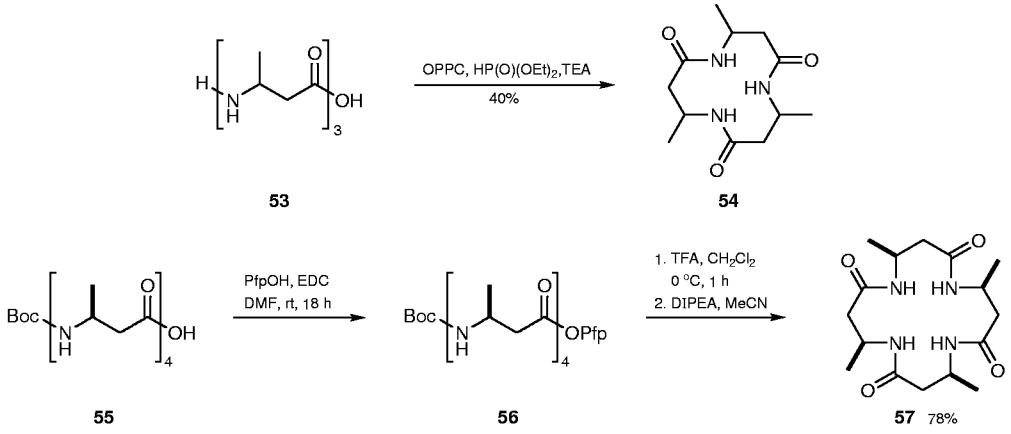


**52** *c*[-(*S*)-β$^3$-Xaa-]$_4$

## 10.12.6.1 Synthesis via a Pentafluorophenyl Ester

In 1986, Drey and co-workers[36] synthesized the first example of a cyclic oligo-β-peptide. They made use of an anhydride method based on *o*-phenylenephosphochloridite (OPPC) and found that cyclization of a sample of (±)-(β$^3$-HAla)$_3$-OH (**53**) led to the formation of the cyclic tripeptide **54** in 40% yield (Scheme 17). However, cyclization of the analogous enantiomerically pure peptide under identical conditions failed to produce any cyclic product. Their work showed that the conditions needed to be altered (by using an excess of diethyl phosphite) in order to ensure the starting material dissolved in the reaction mixture. More problems with solubility were encountered when dealing with the cyclic product. In this context, they noted that the enantiomerically pure *c*[-(*S*)-β$^3$-HAla-]$_3$ was insoluble in a variety of apolar and more polar solvents.

**Scheme 17** Synthesis of Cyclo-β-peptides[16,36]




**53** **54**

**55** **56** **57** 78%

Subsequently, Seebach and co-workers[16] put the insoluble nature of the products to good use when they exploited procedures similar to those developed by Schmidt[37] for the cyclization of oligopeptides. They made use of pentafluorophenyl esters as a method of activation with subsequent cleavage of the Boc-protecting group and then dropwise addition of the activated peptide into a hot solution of diisopropylamine in acetonitrile. This method resulted in precipitation of the cyclic β$^3$-peptides, derived from several linear oligopeptides, as white powders. Filtering the warm reaction mixture and drying the precipitate under high vacuum led to the pure products in high yields. In all cases, the cyclo-β-peptides exhibit a

high degree of insolubility in most solvents although the addition of a large excess of lithium chloride to a suspension of cyclo-β-peptide in THF does give a solution, albeit of low concentration, of these compounds.[16] Some structures of the cyclic β-peptides have recently been solved by powder diffraction methods. These studies have shown that the high degree of insolubility can be attributed to the formation of tube-like structures within the crystals.[35]

***c*[(*S*)-β$^3$-HAla]$_4$ (57):**[16]
Boc-[(*S*)-β$^3$-HAla]$_4$-OBzl (345 mg, 0.63 mmol) was dissolved in TFE (0.1 M) and a catalytic amount of 10% Pd/C was added. The apparatus was evacuated and flushed with $H_2$ (3 ×) and the mixture was then stirred under $H_2$ for 18 h. Subsequent filtration through Celite and concentration under reduced pressure yielded the crude carboxylic acid **55**, which was used without further purification.
The crude carboxylic acid **55** was dissolved in DMF and cooled to 0 °C. Pentafluorophenol (1.1 equiv) and EDC (1.5 equiv) were added and the resulting mixture was stirred at rt for 18 h. The resulting mixture was concentrated under reduced pressure. The residue was dissolved in $CHCl_3$ and washed successively with 1 M HCl and $H_2O$. The organic phase was dried ($MgSO_4$) and concentrated. The resulting pentafluorophenyl ester **56** was used without further purification.
Boc-[(*S*)-β$^3$-HAla]$_4$-OPfp (**56**) was dissolved in a mixture of $CH_2Cl_2$ and TFA and stirred at 0 °C for 1 h. The mixture was concentrated under reduced pressure to yield the crude TFA salt, which was dissolved in MeCN (0.01 M) for use in the cyclization procedure.
DIPEA (1.5 equiv) was dissolved in MeCN (0.003 M) and warmed to 70 °C. The soln of TFA·H-[(*S*)-β$^3$-HAla]$_4$-OPfp was added dropwise by syringe pump (10 mL·h$^{-1}$). The cyclopeptide precipitated as a white solid during the reaction. After cooling to rt, the cyclopeptide was collected by filtration, washed with MeCN and dried under high vacuum for 18 h. This resulted in the formation of *c*[(S)-β$^3$-HAla]$_4$ (**57**) as a white solid; yield: 168 mg (78%); mp 300–301 °C (dec).

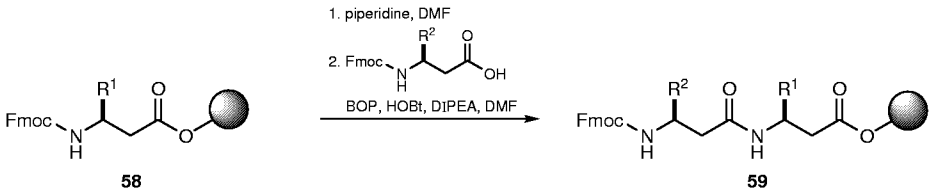
## 10.12.7 Applications of Solid-Phase Peptide Synthesis to β-Peptides

Solid-phase peptide synthesis has become widely used for the preparation of peptides built from α-amino acids of varying sizes and complexity, and also in the recent synthetic approaches to peptide libraries. It has been recognized that the use of solid-phase protocols for the synthesis of β-peptides is likely to make them more attractive lead compounds in drug discovery. Although still at an early stage, work has begun to develop suitable protocols for automated β-peptide synthesis.

### 10.12.7.1 Fmoc Strategy

Seebach and co-workers[38,39] have demonstrated that the coupling of β-amino acids can be achieved in a solid-phase protocol with the use of Fmoc-protected β-amino acids **58** linked to a chlorotrityl resin (Scheme 18). They demonstrated that while β$^3$-amino acids can be coupled in a straightforward manner, the analogous β$^2$-amino acids are a little more problematic, with HPLC analysis of the products showing some impurities.

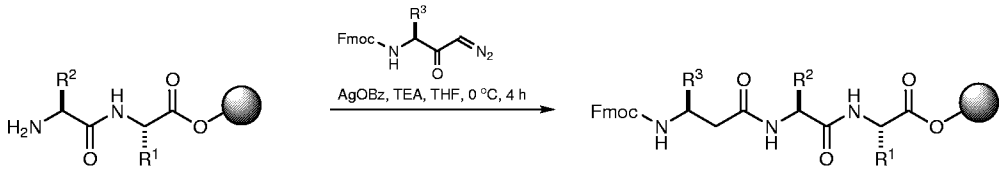
**Scheme 18** Coupling of β-Amino Acids on a Solid Phase[38,39]




*for references see p 568*

Marti et al.[40] have used a similar approach involving Fmoc-protection. However, they have demonstrated that it is possible to use Fmoc-protected diazo ketones derived from α-amino acids together with a peptide on a solid support with a free N-terminus in a silver-promoted Arndt–Eistert procedure. Hence, homologation and peptide coupling are achieved in one step. This approach has led to the synthesis of α-peptides containing one $\beta^3$-amino acid (Scheme 19) and also, if the homologation procedure is used repetitively, to $\beta^3$-peptides.

**Scheme 19** Solid-Phase β-Peptide Synthesis Using Arndt–Eistert Homologation[40]




60 61

**Solid-Phase Synthesis of $\beta^3$-Peptides: Coupling of Fmoc-$\beta^3$-Amino Acids Using BOP:**[38]
2-Chlorotrityl chloride resin (150 mg, 1.05 mmol $Cl^-$/g) was swelled in $CH_2Cl_2$ (2 mL) for 10 min. A soln of Fmoc-$\beta^3$-HLeu-OH (41 mg, 0.11 mmol) in $CH_2Cl_2$ (2 mL) and DIPEA (0.77 mL, 4.4 mmol) were then added successively and the suspension was mixed under argon for 4 h to give **58** ($R^1$ = iBu). Subsequently, the resin was filtered and washed successively with $CH_2Cl_2$/MeOH/DIPEA (17:2:1, 3 × 3 min), $CH_2Cl_2$ (3 × 3 min), DMF (2 × 3 min), $CH_2Cl_2$ (3 × 3 min) and MeOH (2 × 3 min). The substitution of the resin determined on a 3 mg sample by measuring the absorbance of the dibenzofulvene–piperidine adduct at 300 nm was 0.42 (74%). The Fmoc group was removed using 20% piperidine in DMF (4 mL, 2 × 20 min) under argon bubbling. The resin was then filtered and washed with DMF (6 × 3 min). For the coupling step a soln of the Fmoc-$\beta^3$-amino acid (3 equiv), BOP (3 equiv), and HOBt (3 equiv) in DMF (2 mL) and DIPEA (9 equiv) were added successively to the resin and the suspension was mixed for 1 h under argon. Monitoring of the coupling reaction was performed with TNBS instead of the classical ninhydrin test because Fmoc-deprotected $\beta^3$-peptide resin fails to give a blue color with ninhydrin. In the case of incomplete coupling (a positive TNBS test), the suspension was allowed to react for a further 1 h. After the removal of the Fmoc protecting group from **59**, the resin was washed successively with DMF (6 × 3 min), $CH_2Cl_2$ (3 × 3 min), $Et_2O$ (3 × 3 min), and dried under high vacuum for 3 h. The peptide was then cleaved from the resin using 2% TFA in $CH_2Cl_2$ (2 mL, 5 × 15 min) under argon. The solvent was removed and the oily residue was triturated in $Et_2O$ to give the crude peptide as a white solid; purification was achieved using HPLC.

**Solid-Phase Synthesis of $\beta^3$-Peptides: Reaction of Fmoc-Protected Diazo Ketones with Concomitant $\beta^3$-Peptide Formation on a Solid Support:**[40]
The $N^\alpha$-deprotected peptide-resin (Wang-O or Rink-NH) (0.25 mmol) **60** was suspended in DMF/THF (1:1, 20 mL/g resin) and Fmoc-Xaa-$CHN_2$ (1.0 mmol, 4 equiv) was added. The yellow suspension was cooled to 0 °C and a soln of silver benzoate (20 mg, 0.35 equiv) in TEA (0.32 mL) was added. The resulting mixture was stirred at 0 °C for 4 h under the exclusion of light. The peptide-resin **61** was washed with DMF/$CH_2Cl_2$, a soln of sodium diethyldithiocarbamate in DMF (~5%), DMF, and then $CH_2Cl_2$, and subsequently dried under high vacuum.

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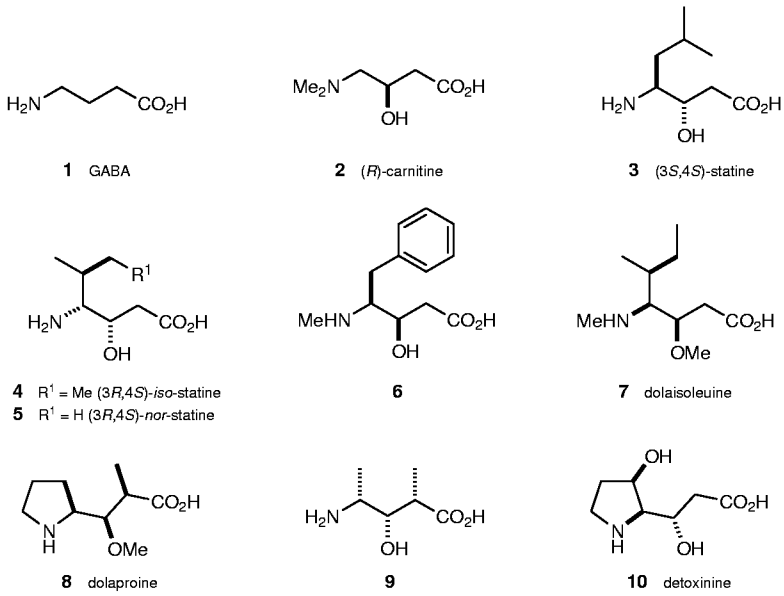
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# 10.13 Synthesis of Peptides Based on γ-Amino Acids

P. Jouin

In addition to the 20 common amino acids which serve as building blocks for proteins, living organisms produce various amino acids not found in proteins such as γ-amino acids, which have important biological properties (as regulatory molecules, intermediary metabolites, storage compounds, and defensive weapons). γ-Aminobutyric acid (GABA, **1**), derived from L-glutamate, is a major physiological inhibitor of neurotransmission in the brain. Unlike GABA and with the exception of its N,N-dimethylated β-hydroxy analogue carnitine (**2**), which is a cofactor of fatty acid metabolism in mammals, most of the γ-amino acids substituted at the β-position by a hydroxy moiety are part of pseudopeptides or appear as main-chain modifications in peptidomimetics of potential therapeutic value (Scheme 1).

**Scheme 1** Naturally Occurring γ-Amino Acids



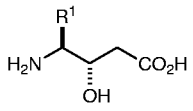

**1** GABA
**2** (*R*)-carnitine
**3** (3*S*,4*S*)-statine
**4** R¹ = Me (3*R*,4*S*)-*iso*-statine
**5** R¹ = H (3*R*,4*S*)-*nor*-statine
**6**
**7** dolaisoleuine
**8** dolaproine
**9**
**10** detoxinine

The discovery by Umezawa in 1970 of the new pepsin inhibitor pepstatin [Iva-L-Val-L-Val-(3*S*,4*S*)-statyl-L-Ala-(3*S*,4*S*)-statine] produced by *Streptomyces*[1] has opened the way to the development of numerous, efficient aspartyl protease inhibitors.[2] (Note: Iva = isovaleryl.) Ten years after the identification of statine (**3**), the γ-amino-β-hydroxy acid **4** derived from Ile was identified in the immunosupressive didemnin B of marine origin.[3,4] This residue presents an *anti* configuration of its γ-amino and β-hydroxy groups,[5] as opposed to the *syn* configuration in statine. Natural and synthetic *nor*-didemnin B, with the Val-derived γ-amino-β-hydroxy residue **5**, has also been reported.[6,7] The *N*-methyl phenylalanine derived γ-amino-β-hydroxy acid **6** of *anti* configuration was found in the recently isolated multidrug-resistant reversing agent hapalosin.[8] The methylated derivatives dolaisoleuine (**7**) and dolaproine (**8**) were also found in the antimitotic pentapeptide dolastatin 10.[9,10] The Ile-derived residue dolaisoleuine (**7**) is methylated at both the γ-amino and β-hydroxy positions and the proline derived dolaproine residue (**8**) at the β-hydroxy and α-methylene positions. The γ-amino-β-hydroxy acid motif is also encountered in a variety of other naturally occurring compounds such as 4-amino-3-hydroxy-2-methylpentanoic acid (**9**), in the complex structure of the antibiotic bleomycin B2,[11] and detoxinine (**10**) in detoxin D1.[12]

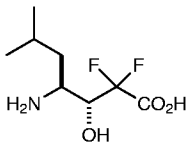
## 10.13.1 γ-Amino-β-hydroxy Acids

The intense interest in γ-amino-β-hydroxy acid related peptides has led to a great deal of effort devoted to their synthesis. Statine analogues Ahppa (**11**), Achpa (**12**), and Dahoa (**13**) (Scheme 2), which correspond to the replacement of the chain equivalent of leucine in statine by those of phenylalanine, cyclohexylalanine, and lysine, respectively, were used to synthesize inhibitors of pepsin, renin, and penicillinopepsin.[13,14] With the objective of developing highly specific inhibitors, C2-substituted analogues have also been introduced into short, substrate-derived inhibitors, e.g. the α,α-difluoro statine **14** in renin inhibitors,[15] and $N^{\alpha}$-substituted statine derivatives like **15** in HIV-1 protease inhibitors.[16]

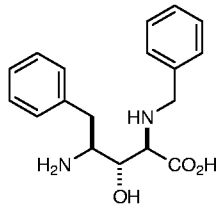
**Scheme 2** Statine Analogues Used as Inhibitors of Aspartyl Protease Synthesis




**11** $R^1$ = Bzl (3*S*,4*S*)-Ahppa
**12** $R^1$ = $CH_2Cy$ (3*S*,4*S*)-Achpa
**13** $R^1$ = $(CH_2)_4NH_2$ (3*S*,4*S*)-Dahoa




**14**



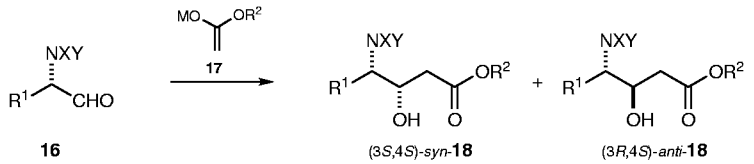

**15**

The biological activity of these peptides depends strongly on the relative and absolute stereochemistries of the incorporated γ-amino-β-hydroxy residues. Most synthetic pathways to γ-amino-β-hydroxy acids start from the corresponding α-amino acid which is homologated by two carbon atoms. A great number of diverse synthetic approaches have appeared in the literature during the past 20 years, which use chiral starting materials and/or stereoselective transformations to gain more flexibility with respect to the C4 chain variability. The C4 chain structure is either part of the starting material or introduced under diastereoselective control at a key step in the synthesis. Many published procedures, which suffer from containing a great number of chemical steps and a subsequently poor overall yield, are not described herein.

### 10.13.1.1 Aldol Condensation of Amino Aldehydes with Ester Enolates

The aldol reaction between a chiral α-amino aldehyde **16** and an acetate derived enolate **17** creates a new stereogenic center and two possible diastereomers. Several different methods for the synthesis of statine derivatives following an aldol reaction have been reported; most of them lead to a mixture of the (3*S*,4*S*)- and (3*R*,4*S*)-diastereomers **18** (Scheme 3), which have to be separated by laborious chromatographic methods.[17–21] Two distinct approaches for stereochemical control have been used: substrate control and reagent control.

**Scheme 3** Aldol Condensation of Amino Aldehydes with Ester Enolates




**16** (3*S*,4*S*)-*syn*-**18** (3*R*,4*S*)-*anti*-**18**

*for references see p 588*

### 10.13.1.1.1 Substrate Control in the Aldol Reaction

The addition of acetate-derived, achiral lithium enolates to monoprotected α-amino aldehydes is controlled by chelation, and leads to a modest stereochemical preference in favor of the 3,4-*syn* configuration (Table 1, entry a).[18] The formation of the 3,4-*syn*-product is enhanced by the use of acetate-derived silyl ketene acetals and the addition of titanium(IV) chloride or tin(IV) chloride to the reaction mixture (Table 1, entries b and c).[22,23] The same enolates add stereoselectively to *N,N*-dibenzyl α-amino aldehydes but with diastereomeric ratios in favor of the Felkin–Ahn 3,4-*anti*-product (Table 1, compare entries a and d, and b and f).[22,24] Reverse stereocontrol is observed in the presence of a Lewis acid such as titanium(IV) chloride, but the yield is low (Table 1, entry e).[24]

**Table 1** Substrate Control in the Aldol Reaction[18,22–24][a]

| Entry | Aldehyde **16** ($R^1$ = iBu) | | Reagent **17** | | Additive | dr (*syn*-**18**/*anti*-**18**) | Yield (%) of **18** | Ref |
|---|---|---|---|---|---|---|---|---|
| | X | Y | $R^2$ | M | | | | |
| a | Boc | H | Et | Li | none | 55:45 | 73 | [18] |
| b | | | Et | $SiMe_3$ | $SnCl_4$ | 91:9 | 60 | [22] |
| c | iPrOCO | H | Me | $SiMe_3$ | $TiCl_4$ | 94:6 | 89 | [23] |
| d | Bzl | Bzl | Et | Li | none | 7:93 | 74 | [24] |
| e | | | Ph | $SiMe_3$ | $TiCl_4$ | 95:5 | 25 | [24] |
| f | | | Bzl | $SiMe_3$ | $EtAlCl_2$ | 1:99 | 94 | [22] |

[a] See Scheme 3.

**Methyl (3*S*,4*S*)-3-Hydroxy-4-[(isobutyryloxy)amino]-6-methylheptanoate (*syn*-18, $R^1$ = iBu; $R^2$ = Me; X = iPrOCO; Y = H)**[23]

A 1 M soln of $TiCl_4$ in $CH_2Cl_2$ (0.065 mL) was added to a mixture of (*S*)-**16** ($R^1$ = iBu; X = iPrOCO; Y = H; 8.7 mg, 0.043 mmol), 4-Å molecular sieves (4 mg), and [(1-methoxyvinyl)oxy](trimethyl)silane (**17**, $R^2$ = Me, M = $SiMe_3$; 19 mg, 0.130 mmol) in $CH_2Cl_2$ (0.8 mL) at −78 °C. The mixture was stirred for 30 min at the same temperature, and then quenched with $NaHCO_3$ soln and filtered through a pad of Celite. The Celite layer was washed (EtOAc), and the combined filtrates were dried and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, hexane/EtOAc 4:1) to give a mixture of *syn*- and *anti*-**18** ($R^1$ = iBu; $R^2$ = Me; X = iPrOCO; Y = H) as a colorless oil; yield: 10.6 mg (89%); *syn*/*anti* 94:6. These diastereomers were separated by column chromatography (silica gel, hexane/EtOAc 10:1); *syn*-**18**: $[\alpha]_D^{20}$ −43.2 (*c* 1.07, $CHCl_3$); *anti*-**18**: mp 63–65 °C; $[\alpha]_D^{20}$ −29.8 (*c* 0.650, $CHCl_3$).

### 10.13.1.1.2 Reagent Control in the Aldol Reaction

In the reaction of chiral enolates (Scheme 4) with chiral aldehydes, the intrinsic diastereofacial selectivities of the two components either match or mismatch. Reagent control can be obtained only if the enolate selectivity overcomes the aldehyde selectivity.

Reaction of *N,N*-dibenzyl α-amino aldehydes with menthone derived thioester boron enolates **20** or *ent*-**20**, obtained from **19** or *ent*-**19**, respectively, gives either the 3,4-*syn*- or the 3,4-*anti*-adduct with very high diastereoselectivity (Table 2) by changing the chiral boron ligand configuration (Scheme 5).[25] Davies' chiral iron acyl enolate **21** undergoes diastereoselective aldol reactions with homochiral *N,N*-dibenzyl α-amino aldehydes.[26] Using the boron enolate chiral condensation methodology of Evans with carbamate protected α-amino aldehydes, the 3,4-*syn*-[27] or the inverse 3,4-*anti*-adduct[28] is obtained, starting from either (4*S*)-4-isopropyl-3-[(methylsulfanyl)acetyl]-1,3-oxazolidin-2-one (**22**) or (4*R*,5*S*)-4-methyl-3-[(methylsulfanyl)acetyl]-5-phenyl-1,3-oxazolidin-2-one (**23**), respectively, as the chiral reagent. The use of the lithium enolate of the readily available (1*S*)-2-hydroxy-1,2,2-triphenylethyl acetate (**24**) gives predominantly the *syn*-adduct (dr 93:7), isolated in an optically pure form in 50–60% overall yield after transesterification and flash chromatography.[29] The C3 statine epimer is obtained with lower diastereoselectivity (dr 3:1) due to the mismatched pair of aldehyde and oppositely configured (*R*) chiral enolate.[30]

**Scheme 4** Chiral Reagents for the Aldol Condensation

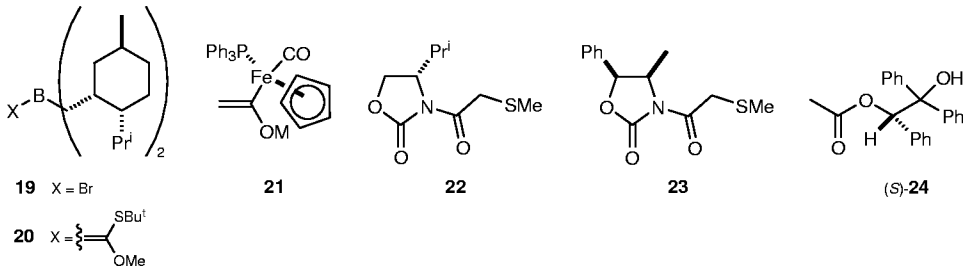
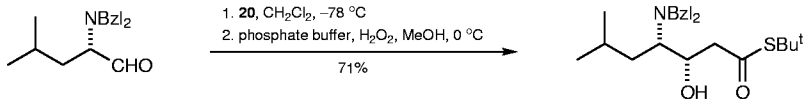



**Table 2** Reagent Control in the Aldol Condensation[25–28,30][a]

| Aldehyde **16** ($R^1$ = iBu) | | Reagent | Additive | dr (syn/anti) | Yield (%) of Adduct | Ref |
|---|---|---|---|---|---|---|
| X | Y | | | | | |
| Bzl | Bzl | **20** | TEA | 97.5:2.5 (**25**) | 71 | [25] |
| | | *ent*-**20** | TEA | 0:100 | 78 | [25] |
| | | **21** | BuLi/$Et_2AlCl$ | 0:100 | 71 | [26] |
| Boc | H | **22** | DIPEA/$Bu_2BOTf$ | 95:5 | 28 | [27] |
| | | **23** | DIPEA/$Bu_2BOTf$ | 0:100 | 14 | [28] |
| | | **24** | LDA | 93:7 | 48 | [30] |

[a] See Scheme 4.

**Scheme 5** Reagent Control in the Aldol Condensation[25]



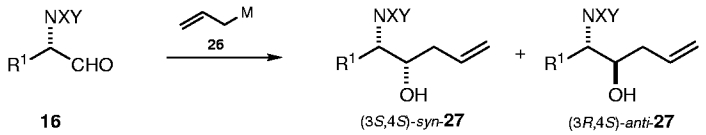

***S*-(*tert*-Butyl) (3*S*,4*S*)-4-(Dibenzylamino)-3-hydroxy-6-methylheptanethioate (*syn*-25):**[25]
To a cooled (0 °C) soln of *tert*-butyl thioacetate (132 mg, 1.0 mmol) in $Et_2O$ (4.16 mL) was added dropwise under an argon atmosphere a 0.4 M soln of **19** (X = Br; 3.7 mL, 1.5 mmol) in $CH_2Cl_2$, followed by TEA (0.22 mL, 1.6 mmol), the resulting compound **20** was used in situ. The reaction was stirred at 10 °C for 2.5 h before cooling to −78 °C. *N*,*N*-Dibenzylamino aldehyde **16** ($R^1$ = iBu; X = Y = Bzl; 383 mg, 1.3 mmol) in $CH_2Cl_2$ was then added dropwise. After being stirred for 18 h at −78 °C, the mixture was quenched with phosphate buffer (1 mL) and allowed to warm to rt. The solvent was removed under reduced pressure and the residue was dissolved in MeOH (10 mL) and phosphate buffer (3 mL). 30% $H_2O_2$ (3 mL) was then added at 0 °C, and the mixture was stirred at rt for 45 min. MeOH was removed under reduced pressure, and the crude mixture was extracted twice with $CH_2Cl_2$. The combined organic extracts were washed ($H_2O$ and brine), dried ($Na_2SO_4$), and the solvent was removed under reduced pressure. The crude products were analyzed by $^{13}C$ NMR spectroscopy in order to determine the dr (*syn*/*anti* 97.5:2.5). The crude products were chromatographed (silica gel, hexane/$Et_2O$ 6:1) to give the pure tBu thioester derivative *syn*-**25** as a colorless oil; yield: 408 mg (71%); $[\alpha]_D^{25} = -28.3$ (*c* 3.03, $CHCl_3$).

## 10.13.1.2 Allylation of Amino Aldehydes

Since the allyl function represents a latent carboxy unit, diastereomerically pure homoallylic alcohols *syn*-**27** or *anti*-**27**, obtained by condensation of allylic metals **26** with chiral α-amino aldehydes or acetals (Scheme 6), have been used to prepare the corresponding γ-amino-β-hydroxy acids (Scheme 7).

*for references see p 588*

**Scheme 6** Allylation of Amino Aldehydes or Acetals




Addition of *N*-carbamate α-amino aldehyde to the Grignard reagent **26** (M = MgBr) affords the corresponding 1,2-amino alcohols with good *syn* diastereoselectivity (Table 3, entry a).[31,32] The chelation-controlled reaction of allyl trimethylsilane with carbamate protected α-amino aldehydes is best achieved in the presence of a Lewis acid [e.g., tin(IV) chloride] to give preferentially the *syn*-isomer (Table 3, entry b).[33,34] Configurationally stable *N,N*-dibenzyl α-amino aldehydes react with allylmagnesium chloride with moderate control of diastereoselectivity (Table 3, entry c).[24] The degree of diastereoselectivity is increased by titanation (Table 3, entry d) and reversal of diastereoselectivity is achieved with Lewis acids (Table 3, entry e).[24]

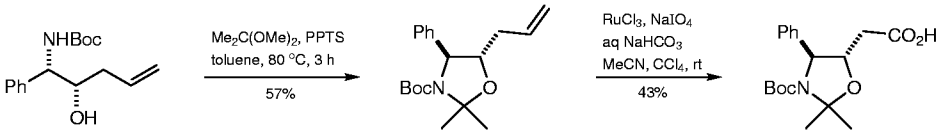
**Table 3** Addition of Allylmetal Reagents to α-Amino Aldehydes[24,32,33][a]

| Entry | Aldehyde **16** ($R^1$ = iBu) | | Reagent **26** | Additive | dr (*syn*-**27**/*anti*-**27**) | Yield (%) of **27** | Ref |
|---|---|---|---|---|---|---|---|
| | X | Y | M | | | | |
| a | Boc | H | MgBr | none | 87:13 | 53 | [32] |
| b | | | TMS | $SnCl_4$ | 95:5 | 68 | [33] |
| c | Bzl | Bzl | MgCl | none | 28:72 | 82 | [24] |
| d | | | $Ti(NEt_2)_3$ | none | 3:97 | 81 | [24] |
| e | | | TMS | $SnCl_4$ | 90:10 | 78 | [24] |

[a] See Scheme 6.

Pure homoallylic alcohols are converted into γ-amino-β-hydroxy acids by protection as the TBDMS ether or *N,O*-acetonide followed by degradative oxidation of the terminal alkene using a catalytic amount of ruthenium(III) chloride and sodium periodate (Scheme 7).[31–33,35]

**Scheme 7** Conversion of Homoallylic Alcohols into β-Hydroxy Acids[31]




***tert*-Butyl (4*S*,5*S*)-5-Allyl-2,2-dimethyl-4-phenyl-1,3-oxazolidine-3-carboxylate (28):**[31]
To a soln of *syn*-**27** ($R^1$ = Ph; X = Boc; Y = H; 2.8 g, 10 mmol) and 2,2-dimethoxypropane (12.5 g, 120 mmol) in anhyd toluene (50 mL) was added a cat. amount of PPTS, and the mixture was stirred at 80 °C for 3 h. The solvent was then removed under reduced pressure and the product purified by flash column chromatography (silica gel, hexane/EtOAc 19:1) to afford the pure product **28** as a light yellow oil; yield: 1.8 g (57%); $[\alpha]_D$ +23.0 (*c* 2.6, $CHCl_3$); FAB-MS *m/z*: 318 $[M+H]^+$.

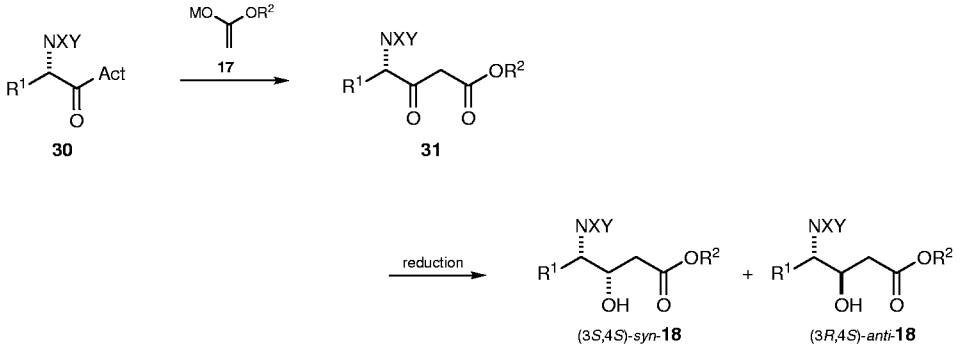
**[(4*S*,5*S*)-3-(*tert*-Butoxycarbonyl)-2,2-dimethyl-4-phenyl-1,3-oxazolidin-5-yl]acetic Acid (29):**[31]
To a stirred soln of oxazolidine **28** (1.4 g, 4.4 mmol) in MeCN (10 mL) was added a suspension of $NaHCO_3$ (2.4 g, 28.6 mmol) in $H_2O$ (20 mL) at rt. $NaIO_4$ (5.17 g, 24.2 mmol) was then added in small portions to the mixture which was stirred for 10 min. The mixture was treated with $RuCl_3 \cdot xH_2O$ (165 mg, 0.8 mmol), and stirring was continued at rt for 48 h. The heterogeneous mixture was diluted with $H_2O$ (100 mL) and filtered through a sintered funnel, and the residue was washed with $H_2O$ (2 × 25 mL). The combined filtrate was extracted with $CH_2Cl_2$ (50 mL), and the aqueous layer was cooled to 0 °C and acidified to pH 4 with aq 10% HCl soln. This acidic soln was then extracted with $CHCl_3$ (3 × 100 mL). The combined organic extracts were dried ($Na_2SO_4$) and concentrated to a volume of 50 mL. The

product was obtained by double base and acid extraction. The combined $CHCl_3$ extracts were washed (brine) and dried ($Na_2SO_4$). After the solvent was removed in vacuo, the crude product was dissolved in hot pentane and left at 0 °C overnight to yield **29** as a white solid; yield: 0.6 g (43%); mp 121–123 °C; $[\alpha]_D$ +26.2 (*c* 0.95, $CHCl_3$); FAB-MS *m/z*: 336 $[M+H]^+$.

## 10.13.1.3 Reduction of γ-Amino-β-oxo Esters

The use of the aldol condensation to synthesize γ-amino-β-hydroxy acids suffers from several drawbacks, such as possible racemization either of the initial carboxy group during its transformation to the α-amino aldehyde, or during any subsequent manipulations of the sensitive CHO moiety. An alternative route involves the direct activation of the carboxy group of the N-protected α-amino acids **30** followed by alkylation with enolates **17** to produce the corresponding β-oxo esters **31**.[36–52] The β-oxo esters can be selectively reduced under various conditions to produce diastereomeric mixtures of the target compounds **18** but with retention of the chirality of the initial α-amino acid (Scheme 8).

**Scheme 8** β-Oxo Ester Route to α-Amino-β-hydroxy Acids




Act = activating group

### 10.13.1.3.1 Diastereoselectivity during Borohydride Reduction

Common achiral reducing agents such as borohydrides produce the *anti*-isomer **18** as the major product of the reduction of the carbamate protected γ-amino-β-oxo esters **31** (Table 4, entries a–e).[36,38–41,50] The *syn*-isomer is accessible with high diastereoselectivity by reduction of the corresponding *N,N*-dibenzyl β-oxo esters using sodium borohydride stereoselective reduction under nonchelation control (Table 4, entry f).[45,52]

**Table 4** Reduction of β-Oxo Esters[36,38,41,44,45][a]

| Entry | β-Oxo Ester **31** ($R^1$ = iBu) | | | Reagent | dr (*syn*-**18**/*anti*-**18**) | Yield (%) of **18** | Ref |
|---|---|---|---|---|---|---|---|
| | $R^2$ | X | Y | | | | |
| a | tBu | Z | H | $NaBH_4$/MeOH | 20:80 | 73 | [38] |
| b | Et | Boc | H | $LiBH_4$/THF | 37:63 | 87 | [41] |
| c | Et | Boc | H | $Zn(BH_4)_2/Et_2O$ | 25:75 | 72 | [41] |
| d | Et | Boc | H | $KBH_4$/EtOH | 13:87 | 80 | [41] |
| e | tBu | Fmoc | H | $KBH_4$/EtOH | 25:75 | 50 | [36] |
| f | Et | Bzl | Bzl | $NaBH_4$/MeOH | 90:10 | 84 | [45] |
| g | Et | Boc | H | $H_2/RuBr_2$[(*R*)-BINAP] | 1:99 | 97 | [44] |

[a] See Scheme 8.

*for references see p 588*

***tert*-Butyl (3*S*,4*R*,5*S*)-4-{[(9H-Fluoren-9-ylmethoxy)carbonyl]amino}-3-hydroxy-5-methylheptanoate (*anti*-18, R¹ = sBu; R² = tBu; X = Fmoc; Y = H):**[36]

$SOCl_2$ (10 equiv, 3.65 mL) and DMF (3 drops) were added to a stirred suspension of Fmoc-Ile-OH (1.77 g, 5 mmol) in $CH_2Cl_2$ (10 mL). After stirring at 35–40 °C for 1.5 h, the excess $SOCl_2$ was removed under reduced pressure. The oily product was kept under high vacuum for 0.5 h where it crystallized. It was then dissolved in THF (8 mL), cooled to −85 °C, and added to a soln of lithium *tert*-butoxycarbonylmethanide, which was prepared at this same temperature by mixing a commercially available 1.5 M soln of LDA in cyclohexane (13.3 mL, 20 mmol) with a soln of tBuOAc (2.95 mL, 22 mmol) in THF (7 mL). After 25 min, the clear, yellow soln was quenched with 0.5 M HCl (50 mL) and extracted with $Et_2O$ (3 × 50 mL). The organic phase was dried ($Na_2SO_4$) and the solvent removed. To avoid crystallization, the intermediate was immediately redissolved in dry EtOH (20 mL), cooled to 0 °C, and reduced with $KBH_4$ (1.08 g, 20 mmol) for 3 h. Then, EtOH (20 mL) and glacial AcOH (20 mL) were added with stirring until all the remaining hydride was consumed. Insoluble material was filtered off, the solvent was removed, and the residue was dissolved in EtOAc/$H_2O$ (2:1, 100 mL). The organic phase was separated and the aqueous phase re-extracted with EtOAc (100 mL). The combined organic layers were dried ($Na_2SO_4$) and concentrated under reduced pressure. After flash chromatography [silica gel (280 g), EtOAc/$Et_2O$/2-methylpentane 3:1:7] the combined product fractions were concentrated to dryness to yield pure *anti*-**18** (R¹ = sBu; R² = tBu; X = Fmoc; Y = H) which crystallized overnight (EtOAc/2-methylpentane); yield: 1.72 g (78%); mp 135–136 °C; $[\alpha]^{20}_D$ −3.6 (*c* 0.25, EtOAc); FAB-MS *m/z*: 454.3 $[M+H]^+$.

### 10.13.1.3.2 Asymmetric Hydrogenation of β-Oxo Esters

Hydrogenation of β-oxo esters by BINAP–ruthenium(II) complexes produces the corresponding γ-amino-β-hydroxy acids with very high stereoselectivity.[44,46] For example, hydrogenation of the *N*-Boc derivative **31** (R¹ = iBu; R² = Et; X = Boc; Y = H) in the presence of $RuBr_2$[(*R*)-BINAP] leads to (3*S*,4*R*)-Boc-statine almost exclusively and in quantitative yield (Table 4, entry g).[44]

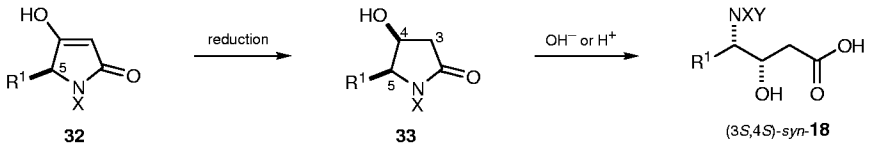
**Ethyl (3*S*,4*R*)-4-[(*tert*-Butoxycarbonyl)amino]-3-hydroxy-6-methylheptanoate (18, R¹ = iBu; R² = Et; X = Boc; Y = H):**[44]

A soln of **31** (R¹ = iBu; R² = Et; X = Boc; Y = H; 20.0 g, 59.7 mmol) in degassed, anhyd EtOH (40 mL) was placed in a 80-mL Schlenk tube and degassed by three freeze–thaw cycles. $RuBr_2$[(*R*)-BINAP] (100 mg, 0.113 mmol) was then added to this soln under argon, and the catalyst was dissolved with the aid of an ultrasonicator. The resulting light brown soln was degassed by two freeze–thaw cycles. This was then transferred by means of a cannula to a glass vessel placed in a 100-mL stainless steel autoclave. $H_2$ was pressurized to 100 atm, and the soln was stirred at 20 °C for 6 d. The solvent was removed under reduced pressure, and the remaining solid subjected to column chromatography [silica gel (Fuji Davison BW820 MH, 200 g), EtOAc/hexane 1:1] to give **18** (R¹ = iBu; R² = Et; X = Boc; Y = H); yield: 19.5 g (97%); HPLC analysis (Senshu Pak ODS −1251 −SH; $H_2O$/MeCN 55:45): $t_R$ (*anti*-**18**) 18.6 min; $t_R$ (*syn*-**18**) 15.8 min; dr (*anti*/*syn*) >99:1.

## 10.13.1.4 4-Hydroxy Lactam Intermediates

A major method for the preparation of γ-amino-β-hydroxy acids of *syn* configuration is the stereocontrolled reduction of enantiomerically pure tetramic acids **32**, followed by alkaline or acid hydrolysis of the resulting N-protected 4-hydroxy lactams **33** (Scheme 9).[53–57]

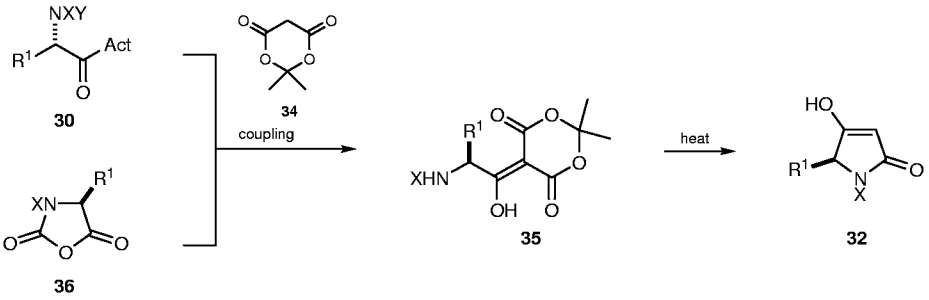
**Scheme 9** Tetramic Acid Route to γ-Amino-β-hydroxy Acids[53–57]



### 10.13.1.4.1 Chiral Tetramic Acids via Meldrum's Acid Condensation

The original method for the synthesis of tetramic acids (the Dieckmann condensation) leads to partial or complete racemization at C5.[58,59] Although racemization can be minimized under strictly controlled conditions,[57] the condensation of *N*-carbamate protected amino acids and Meldrum's acid (**34**) followed by thermal cyclization produces enantiomerically pure tetramic acids **32**. Since the appearance of this method in 1987, several different sets of activating conditions have been used for this condensation, with the original isopropenyl chlorocarbonate (IPCC)/DMAP cocktail[53] being replaced by DCC/DMAP[56] or the use of preactivated, urethane-protected α-amino acid *N*-carboxy anhydrides **36** (UNCAs).[60] The unstable intermediate **35** is directly converted into the corresponding tetramic acid **32** by heating in an organic solvent (Scheme 10).[53,56,60] The crude tetramic acid can generally be used directly for the subsequent reduction step. Alternatively, the tetramic acid is purified by basic extraction followed by its recovery under acidic conditions.

**Scheme 10** Meldrum's Acid Condensation to Chiral Tetramic Acids[53,56,60]




***tert*-Butyl (2*S*)-3-Hydroxy-2-isobutyl-5-oxo-2,5-dihydro-1*H*-pyrrole-1-carboxylate (32, R¹ = iBu; X = Boc):**[53]

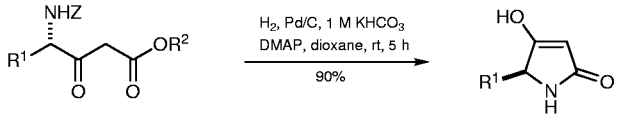
To a soln of dried Boc-Leu-OH (4.62 g, 20 mmol) in $CH_2Cl_2$ (100 mL), Meldrum's acid (**34**;3 g, 21 mmol), and DMAP (5.6 g , 46.1 mmol) were added. The mixture was cooled to −5 °C, and a soln of IPCC (2.6 mL, 21.5 mmol) in $CH_2Cl_2$ (10 mL) was added dropwise with stirring during 0.5 h. After 2 h, cold 5% aq $KHSO_4$ was added with vigorous stirring. The separated organic layer was washed (brine), dried ($Na_2SO_4$), and concentrated to give the crude product **35** [$R_f$ 0.60 (MeOH/EtOAc/AcOH 5:93:2)], which was redissolved in EtOAc (100 mL), brought to reflux, and maintained at reflux temperature for 0.5 h. After cooling to rt, the soln was extracted with 5% aq $NaHCO_3$ and the aqueous phase was acidified to pH 2 with powdered citric acid. The aqueous phase was then extracted with EtOAc. This extract was washed ($H_2O$), dried, and the solvent removed to dryness to provide tetramic acid **32** (R¹ = iBu; X = Boc) as a colorless oil which could be neither crystallized nor further purified by chromatography; yield: 4.5 g (88%, from Boc-Leu-OH); $R_f$ 0.38 (MeOH/EtOAc/AcOH 5:93:2); $[\alpha]^{D}_{20}$ +101 (*c* 1, MeOH).

### 10.13.1.4.2 Chiral Tetramic Acids via Cyclization of γ-Amino-β-oxo Esters

Enantiomerically pure, unprotected tetramic acids are accessible by hydrogenolytic deprotection of 4-(benzyloxycarbonyl)amino-3-oxocarboxylic esters **31** (X = Z; Y = H) followed by ring closure (Scheme 11).[55] In the absence of *N*-carbamate protection, the next reduction step is not totally stereoselective.[54]

*for references see p 588*

**Scheme 11** β-Oxo Ester Route to Chiral Tetramic Acids[55]




**31** X = Z; Y = H **32** X = H

**(5*S*)-Isopropylpyrrolidine-2,4-dione [(*S*)-32, R¹ = iPr; X = H]:**[55]
A soln of ethyl (4*S*)-5-methyl-3-oxo-4-[(phenylacetyl)amino]hexanoate [(*S*)-**31**, R¹ = iPr; R² = Et; X = Z; Y = H; 16.9 g, 52.5 mmol] and DMAP (1.6 g, 13.1 mmol) in dioxane/1 M aq $KHCO_3$ (105 mL:150 mL) was hydrogenated with 1.1 atm $H_2$ in the presence of 5% Pd/C (2.6 g) at rt for 5 h. After filtration and removal of the dioxane, the residue was acidified with 2.5 M $H_2SO_4$ (41 mL). The aqueous soln was extracted with $CHCl_3$, the combined organic extracts were dried ($MgSO_4$), and concentrated under reduced pressure. The residue was recrystallized from EtOAc/hexane affording (*S*)-**32** (R¹ = iPr; X = H); yield: 6.7 g (90%); mp 145 °C; $[\alpha]_D^{20}$ −80.7 (*c* 0.97, $CH_2Cl_2$).

### 10.13.1.4.3 Stereoselective Reduction of Tetramic Acids

The stereoselectivity of the reduction of tetramic acids is dependent on the substitution at the *N*-atom.[54] Sodium borohydride reduction or $PtO_2$ catalyzed hydrogenation of *N*-carbamate or *N*-acyl tetramic acids gives exclusively the corresponding *cis*-4-hydroxypyrrolidin-2-ones (Table 5, entries a, b).[54] In the case of N-unsubstituted tetramic acids, reduction by borohydrides gives only partial diastereofacial selectivity (Table 5, entry c); however, the stereoselectivity can be enhanced by employing catalytic hydrogenation with the Adams catalyst (Table 5, entry d).[54,55,57]

**Table 5** Reduction of Chiral Tetramic Acids[53,54][a]

| Entry | Tetramic Acid **32** (R¹ = iBu) | Reagent | dr (*syn*-**33**/*anti*-**33**) | Yield (%) of **33** | Ref |
|---|---|---|---|---|---|
| a | X = Boc | $NaBH_4$ | 100:0 | 88 | [53] |
| b | X = Ac | $NaBH_4$ | 100:0 | 81 | [54] |
| c | X = H | $NaBH_4$ | 68:32 | 79 | [54] |
| d | X = H | $H_2/PtO_2$ | 86:14 | 76 | [54] |

[a] See Scheme 11.

***tert*-Butyl (2*S*,3*S*)-3-Hydroxy-2-isobutyl-5-oxopyrrolidine-1-carboxylate (33, R¹ = iBu; X = Boc):**[53]
A soln of the crude 1,5-dihydro-2*H*-pyrrol-2-one **32** (R¹ = iBu; X = Boc) dissolved in $CH_2Cl_2$/AcOH (10:1, 110 mL) was cooled in an ice bath and stirred vigorously while being treated portionwise with $NaBH_4$ (1.4 g, 37.0 mmol) for 1 h. The mixture was maintained for a further 4 h at the same temperature. The mixture was then poured into cold $H_2O$ and the organic layer was washed ($H_2O$), dried ($Na_2SO_4$), and concentrated to dryness. Chromatography of the residue (silica gel, hexane/EtOAc 1:1) yielded the pyrrolidin-2-one *syn*-**33** (R¹ = iBu; X = Boc); yield: 3.6 g (70%, from Boc-Leu-OH); mp 90–91 °C (hexane); $[\alpha]_D^{20}$ +63 (*c* 1, MeOH).

## 10.13.2 Other γ-Amino Acids

Based on the structure of biologically active pseudopeptides and molecular modeling studies, γ-amino-β-hydroxy acids which are functionalized at their γ-amino, β-hydroxy and/or α-methylene positions have been incorporated into various molecular constructions. Original methods have been developed for their preparation.

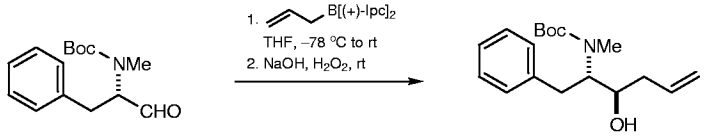
### 10.13.2.1 Synthesis of N/O-Methylated γ-Amino-β-hydroxy Acids

N-Methylated and N,O-dimethylated γ-amino-β-hydroxy acids are present in natural compounds of marine origin, e.g. hapalosin[8] and dolastatin 10.[9]

### 10.13.2.1.1 Synthesis from *N*-Methyl Amino Acids or Amino Aldehydes

N-Methylated γ-amino-β-hydroxy acids are accessible by the usual synthetic sequences, i.e. aldol condensation or γ-amino-β-oxo ester reduction, starting from the corresponding N-methylated α-amino acids, but are obtained with low diastereoselectivity.[61–63] Alternatively, Brown allylboration of the *N*-Boc-*N*-Me amino aldehyde **16** ($R^1$ = Bzl, X = Boc, Y = Me) gives the allylic N-methylated intermediate **27** in 64% yield and 90% de (Scheme 12).[64] Oxidative cleavage of the alkenol is performed using the two-step ozonolysis and sodium chlorite oxidation sequence.

**Scheme 12** Brown Allylboration of *N*-Boc-*N*-Me Amino Aldehyde[64]




**16** $R^1$ = Bzl; X = Boc; Y = Me *anti*-**27** $R^1$ = Bzl; X = Boc; Y = Me

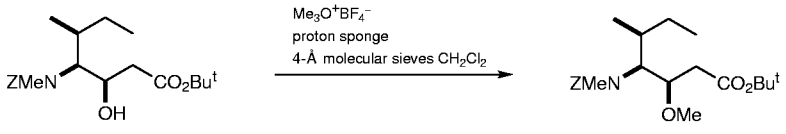
***tert*-Butyl (1*S*,2*R*)-1-Benzyl-2-hydroxypent-4-enyl(methyl)carbamate (27, $R^1$ = Bzl; X = Boc; Y = Me):**[64] To a soln of (+)-*B*-methoxydiisopinocamphenylborane (7.93 g, 24.1 mmol) in THF (65 mL) at −78 °C was slowly added allylmagnesium bromide (23.6 mL in $Et_2O$, 23.6 mmol), and the mixture was stirred at −78 °C for 15 min and then at rt for 1 h. The soln was cooled further to −78 °C, and a soln of amino aldehyde **16** ($R^1$ = Bzl; X = Boc; Y = Me; azeotroped with toluene under high vacuum at less than 25 °C; 5.50 g, 20.9 mmol) in THF (40 mL) was added slowly. The soln was stirred at −78 °C for 2 h and then for 1.5 h at rt. Oxidation of the boronate with aq 3 M NaOH (21 mL) and aq 30% $H_2O_2$ (9 mL) was allowed to proceed overnight at 25 °C. Most of the THF was removed under reduced pressure before extraction into EtOAc (3 × 200 mL) with brine (300 mL). Much isopinocamphenyl alcohol was removed under high vacuum at 90 °C. Flash chromatography (silica gel, EtOAc/hexane, graded elution with up to 50% EtOAc) afforded **27** ($R^1$ = Bzl; X = Boc; Y = Me) as a colorless oil; yield: 4.07 g (64%); $[\alpha]_D^{22}$ −26 (*c* 0.067, $CHCl_3$); FAB-HRMS *m/z*: 306.2071 $[M+H]^+$, calcd for $C_{18}H_{28}NO_3$, 306.2069.

### 10.13.2.1.2 Methylation of N-Protected γ-Amino-β-hydroxy Acids

Standard methylation conditions for γ-amino-β-hydroxy acids and esters give predominantly O-methylation or elimination products.[63,65] The N-methylated derivative is obtained by double protection of the amino and hydroxy groups with formaldehyde to give the oxazolidine, which in turn is reduced with $NaBH_3CN$/TFA to give the N-methylated derivative in high yield.[65] This two-step procedure is reminiscent of that developed by Freidinger for the preparation of *N*-Fmoc-*N*-Me amino acids.[66]

O-Methylation of N-methylated derivatives under basic conditions is disappointing,[50,63] but improved results are obtained using trimethyloxonium tetrafluoroborate (Scheme 13), or boron trifluoride-diethyl ether complex with diazomethane.[67] Conversely, methylation of the proline derived β-hydroxy acid using sodium hydride and iodomethane, as well as using trimethyloxonium tetrafluoroborate, gives the dolaproine unit (**8**) of dolastatin 10 in good yield.[68,69]

**Scheme 13** O-Methylation under Acidic Conditions[67]




**18** $R^1$ = sBu; X = Z; Y = Me **37**

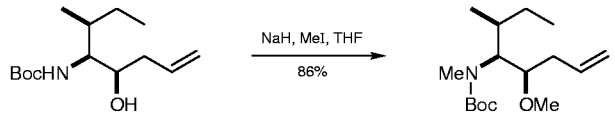
*for references see p 588*

***tert*-Butyl (3*R*,4*S*,5*S*)-4-{[(Benzyloxy)carbonyl](methyl)amino}-3-methoxy-5-methylheptanoate (37):**[67]
To a soln of ester **18** ($R^1$ = sBu; X = Z; Y = Me; 1.3 g, 3.42 mmol) in dichloroethane (50 mL) under $N_2$ at rt, were added 4-Å molecular sieves (1 g), followed by a proton sponge (1.85 g, 8.63 mmol) and $Me_3OBF_4$ (1.35 g, 9.13 mmol). The mixture was stirred at rt for 20 h. The soln was then filtered and the oily solid (2.45 g) obtained upon removal of solvent under reduced pressure was subjected to column chromatography (silica gel, hexane/EtOAc/acetone 8:1:1), giving **37** as an oil; yield: 1.17 g (87%); $[\alpha]_D^{25}$ −11 (*c* 0.23, $CHCl_3$); FAB-HRMS *m/z*: found, 394.2592 $[M+H]^+$; calcd for $C_{22}H_{36}NO_5$, 394.2594.

### 10.13.2.1.3 Methylation of Allylic Derivatives

Direct N,O-dimethylation of the allylic derivative **27** under basic conditions (Scheme 14) provides efficient access to the (3*R*,4*S*,5*S*) dolaisoleuine unit of dolastatin 10.[63,70]

**Scheme 14** Methylation under Basic Conditions[63]




**27** $R^1$ = sBu; X = Boc; Y = H **38**

***tert*-Butyl (1*S*,2*R*)-2-Methoxy-1-[(1*S*)-1-methylpropyl]pent-4-enyl(methyl)carbamate (38):**[63]
To a soln of **27** ($R^1$ = sBu; X = Boc; Y = H; 1.23 g, 5.05 mmol) in THF (20 mL) at −5 °C and under an inert atmosphere, were added successively MeI (5 mL, 80 mmol) and NaH (80% dispersion in mineral oil; 0.3 g, 10.1 mmol) over a 1-h period. The mixture was stirred for 1 h at −5 °C and then additionally for 1 h at 5 °C, prior to the portionwise addition of an excess of MeI (5 mL) and NaH (0.45 g, 15 mmol). After stirring overnight at rt, the mixture was diluted with $Et_2O$ and hydrolyzed with cold 5% $KHSO_4$. The organic layer was washed ($H_2O$, aq 5% sodium thiosulfate, and then sat. brine), dried ($MgSO_4$), filtered, and concentrated under reduced pressure. Purification by flash chromatography (EtOAc/hexane 5:95) furnished **38** as a colorless oil; yield: 1.18 g (86%); $[\alpha]_D^{20}$ −24 (*c* 1, MeOH); FAB-MS *m/z*: 272 $[M+H]^+$.

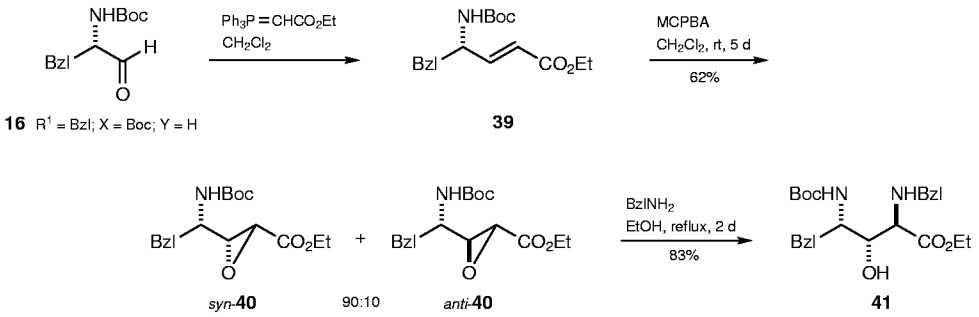
## 10.13.2.2 Synthesis of α-Substituted γ-Amino-β-hydroxy Acids

Potent inhibitors of aspartyl proteases have been prepared by incorporating hydroxymethylene transition-state analogues modified in the α-position. The α-methylene moiety is either functionalized by a heteroatom[16,71,72] or it is dihalogenated.[15,73,74] Alkyl substituents in that position are found in several residues present in compounds of natural origin, such as dolaproine (**8**) in dolastatin 10 and 4-amino-3-hydroxy-2-methylpentanoic acid (**9**) in bleomycin D (Scheme 1). This position is also substituted in synthetic, conformationally constrained six-membered-ring analogues.[75–77]

### 10.13.2.2.1 α-Heterosubstituted γ-Amino-β-hydroxy Acids

One strategy for the synthesis of α-heterosubstituted γ-amino-β-hydroxy acids is based on a two-step sequence, including the diastereoselective epoxidation of an α,β-unsaturated carboxylic ester (obtained by Wittig olefination of the corresponding protected α-amino aldehyde), followed by the regio- and stereospecific ring-opening of the epoxide with a heteronucleophile (Scheme 15).[16]

**Scheme 15** Synthesis of α-Heterosubstituted Statine Analogues[16]



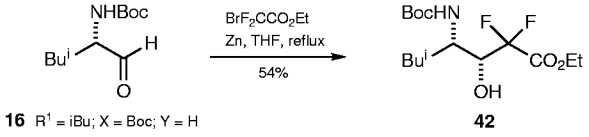

**Ethyl (2*S*/*R*,3*R*/*S*,4*S*)-4-[(*tert*-Butoxycarbonyl)amino]-2,3-epoxy-5-phenylpentanoate (40):**[16]
To a soln of **39** (3 g, 6.6 mmol) in $CH_2Cl_2$ (30 mL) was added MCPBA (3.42 mg, 10.9 mmol), and the mixture was stirred at rt for 5 d. After removal of the solvent, the residue was chromatographed (silica gel, toluene/EtOAc 4:1) to give **40** as a mixture of diastereomers which was used in the synthesis of **41** without further resolution; yield: 1.37 g (62%); dr (*syn*-**40**/*anti*-**40**) 9:1.

**Ethyl (2*R*,3*R*,4*S*)-2-(Benzylamino)-4-[(*tert*-butoxycarbonyl)amino]-3-hydroxy-5-phenylpentanoate (41):**[16]
$BzlNH_2$ (740 μL, 6.8 mmol) was added to a soln of **40** (1.60 g, 3.40 mmol) in EtOH (6 mL) and the soln was kept at 60 °C for 2 d. The solvent was removed and the residue was chromatographed (silica gel, toluene/EtOAc 2:1) to give **41** as a syrup; yield: 1.24 g (83%).

## 10.13.2.2.2 α,α-Dihalogenated γ-Amino-β-hydroxy Acids

Electron-withdrawing fluorine atoms are introduced on the methylenic α-carbon by the Reformatsky reaction of Boc-leucinal with ethyl bromodifluoroacetate in the presence of activated zinc dust with no diastereoselection. Under thermodynamic control, the *syn*-isomer is obtained almost exclusively (Scheme 16).[15] The use of additives such as diethylaluminium chloride–silver(II) acetate enhances the chemical yield of the reaction, but also presents the disadvantage of being nonstereoselective.[78]

**Scheme 16** Synthesis of an α,α-Dihalogenated Statine Analogue[15]



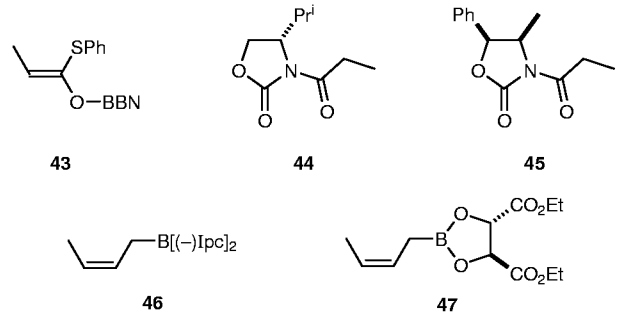

**Ethyl (3*R*,4*S*)-4-[(*tert*-Butoxycarbonyl)amino]-2,2-difluoro-3-hydroxy-6-methylheptanoate (42):**[15]
To a stirred suspension of activated Zn dust (2.21 g, 33.88 mmol) in THF (17 mL) under argon in an oil bath at 75 °C was added ethyl bromodifluoroacetate (203 mg, 1.0 mmol). After 1 min, a soln of Boc-L-leucinal (2.91 g, 13.55 mmol) and additional ethyl bromodifluoroacetate (6.40 g, 31.52 mmol) in THF (13 mL) was slowly added. After complete addition, the resulting mixture was allowed to reflux for an additional 30 min. The mixture was then cooled and partitioned between $CH_2Cl_2$ and aq 1 M $KHSO_4$. The organic phase was dried ($MgSO_4$) and then concentrated. The residue was chromatographed (silica gel, hexane/EtOAc 4:1) to give *syn*-**42**; yield: 1.05 g (54%); $[\alpha]_D$ −11 (*c* 0.615, $CHCl_3$).

*for references see p 588*

### 10.13.2.2.3 α-Alkylated γ-Amino-β-hydroxy Acids

The control of the three consecutive asymmetric centers in α-alkylated γ-amino-β-hydroxy acids is achieved by aldol condensation of a chiral aldehyde and a chiral reagent (Scheme 17 and Table 6), e.g. boron enolate **43**,[50,79] oxazolidinones of Evans' type **44**[80] or **45**,[68,69,80] or Brown's or Roush's crotylorganoboron reagents **46**[81] and **47**,[81,82] respectively.

**Scheme 17** Chiral Enolates for α-Alkylated γ-Amino-β-hydroxy Acids




**43** **44** **45**

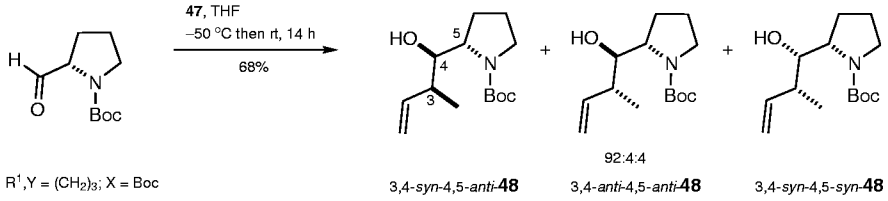
**46** **47**

**Table 6** Stereochemical Control in the Synthesis of α-Alkylated γ-Amino-β-hydroxy Acids[50,68,79–81]

| Aldehyde **16** | | | Reagent | Additive | de | Yield (%) | Ref |
|---|---|---|---|---|---|---|---|
| X | Y | R$^1$ | | | | | |
| Boc | H | Me | **43** | none | >90 | 62 | [79] |
| | | | **44** | $Bu_2BOTf$ | 100 | 100 | [80] |
| Z | $(CH_2)_3$ | | **44** | $Bu_2BOTf$ | 96 | 53 | [80] |
| Boc | $(CH_2)_3$ | | **43** | none | 85 | 64 | [50] |
| | | | **45** | $Bu_2BOTf$ | 90 | 60 | [68] |
| | | | **46** | none | 98 | 50 | [81] |
| | | | **47** | none | 92 | 68 | [81] |

Starting from Boc-prolinal, double stereodifferentiation, where both substrate and reagent match, allows diastereoselective access to the homoallylic alcohol 3,4-*syn*-4,5-*anti*-**48** in the synthesis of dolaproine (**8**) (Scheme 18).[81]

**Scheme 18** (*Z*)-Crotylboration of Boc-prolinal[81]




**16** R$^1$,Y = $(CH_2)_3$; X = Boc

3,4-*syn*-4,5-*anti*-**48** 3,4-*anti*-4,5-*anti*-**48** 3,4-*syn*-4,5-*syn*-**48**

**(*Z*)-Crotylboration of Boc-prolinal with Chiral (*Z*)-Crotylboronate 47; Synthesis of *tert*-Butyl 2-(1-Hydroxy-2-methylbut-3-enyl)pyrrolidine-1-carboxylate (48):**[81]

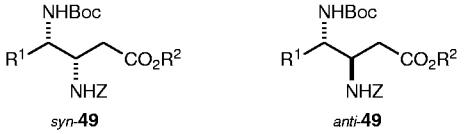
To a soln of Boc-Pro-H (398 mg, 2 mmol) in dry THF (5 mL) containing 4-Å molecular sieves (0.1 g) was added at −50 °C a 1 M soln of (*Z*)-crotylboronate **47** in THF (2.2 mL, 2.2 mmol). After being allowed to warm to rt, the mixture was stirred for 14 h, filtered through a short pad of silica gel, and concentrated. The residue was then purified by flash chromatography (silica gel, EtOAc/cyclohexane 1:4) to afford, by order of elution, compounds 3,4-*syn*-4,5-*syn*-**48**: {$R_f$ 0.52 (EtOAc/pentane 20:80); $[\alpha]_D$ −72 (*c* 1, MeOH)}; 3,4-*anti*-4,5-*anti*-**48**: {$R_f$ 0.30 (EtOAc/pentane 20:80); $[\alpha]_D$ −77 (*c* 1, MeOH)}; and 3,4-*syn*-4,5-*anti*-**48**: {$R_f$

0.24 (EtOAc/pentane 20:80); $[\alpha]_D$ −71 (*c* 1, MeOH)] in a 4:4:92 ratio; combined yield: 0.35 g (68%). The relative configurations were assigned after cyclization of each adduct into the corresponding oxazolidinone (see Section 10.13.4.2.1).

### 10.13.2.3 Synthesis of β,γ-Diamino Acids

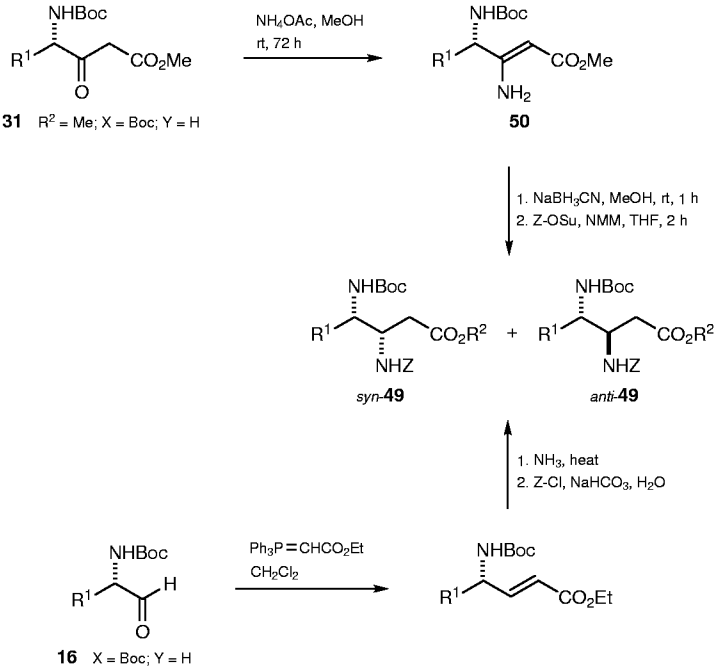
Structural studies of complexes between aspartyl proteases and statine-based inhibitors indicate that the hydroxy group of statine displaces a water molecule present at the active site, and forms hydrogen bonds with the two aspartic carbonyl groups which are essential to the mechanism of hydrolysis.[2] The replacement of this hydroxy moiety by an amine (Scheme 19) has been investigated in order to increase the binding strength by electrostatic interaction with the aspartic residues.[83]

**Scheme 19** α,β-Diamino Analogues of Statine




β,γ-Diamino analogues **49** of statine are prepared stereoselectively starting from the *O*-methyl hydroxamate derivative of N-protected statine. The reaction sequence involves the formation of a β-lactam intermediate obtained by internal cyclization under Mitsunobu conditions.[84] Alternatively, direct amination of either a β-oxo ester **31** followed by reduction of the resulting enamine **50**,[85] or by reduction of the corresponding α,β-unsaturated ester,[83] gives an enantiomeric mixture of the corresponding unprotected β-amine, which is protected by a carbamate prior to chromatographic separation (Scheme 20).

**Scheme 20** Amination of Unsaturated Intermediates[83,85]




*for references see p 588*

**Methyl (3*S*,4*S*)- and (3*R*,4*S*)-3-{[(Benzyloxy)carbonyl]amino}-4-[(*tert*-butoxycarbonyl)amino]-6-methylheptanoate (*syn*-/*anti*-49):**[85]

β-Oxoester **31** ($R^1$ = iBu; $R^2$ = Me; X = Boc; Y = H; 4.34 g, 15.1 mmol) was left in MeOH (35 mL) containing $NH_4OAc$ (23 g, 20 mol equiv) at rt for 72 h. The MeOH was then removed and $H_2O$ was added. Extraction with $Et_2O$ gave, upon drying (azeotroping with toluene), the enamino-ester **50** ($R^1$ = iBu) as an oil which solidified on standing and was used without further purification. Ester **50** was dissolved in MeOH (35 mL) containing AcOH (1.5 mL), and then treated with $NaBH_3CN$ (4.7 g, 5 mol equiv). After 1 h at rt the solvent was removed from the soln and the residue was taken up in THF (60 mL), and treated with NMM (5 mL) and Z-OSu (3.76 g, 15.1 mmol). After 2 h the solvent was removed from the soln, EtOAc was added and the soln was washed with 0.3 M $KHSO_4$ (2 ×), $H_2O$, and brine. Solvent removal (azeotroping with toluene) gave a colorless semisolid; yield: 6.4 g (ca. 100%). Dissolution in hot EtOAc (10 mL) gave, on cooling, needles of pure *anti*-**49**; yield: 1.5 g (23.5%); mp 142–143 °C; FAB-MS *m/z*: 423 $[M+H]^+$; FAB-HRMS *m/z*: found, 423.2495 $[M+H]^+$; calcd for $C_{22}H_{34}N_2O_6$, 422.2408. The mother liquor from crystallization was purified by flash chromatography (silica, EtOAc/petroleum ether 35:65) to give pure *syn*-**49** as an oil; yield: 2.5 g (39.2%); FAB-MS *m/z*: 423 $[M+H]^+$.

## 10.13.3 Peptides Containing γ-Amino-β-hydroxy Acids

The introduction of a γ-amino-β-hydroxy residue into a larger molecule with either peptide or ester linkages employs general methods used for peptides or depsipeptides composed of α-amino acids and α-hydroxy acids. Stepwise solution synthesis and solid-phase synthesis using either automated or manually operated instruments have been extensively described, starting from *N*-Boc or *N*-Fmoc protected γ-amino-β-hydroxy acids. In addition, a fragment condensation strategy is applicable to C-terminal γ-amino-β-hydroxy acid segments since these residues are not susceptible to racemization under activating conditions.[49] This property also permits macrocyclization at the C-terminus.[64]

## 10.13.4 Stereochemical Determination

γ-Amino-β-hydroxy acids possess two or more stereogenic centers. The diastereomeric purity is generally established using HPLC analysis but enantiomeric purity as well as relative and absolute configuration determination rely on several different chemical and analytical methodologies. In most cases, the γ-amino and β-hydroxy groups are well-suited for the formation of five-membered rings. These heterocycles are then analyzed by $^1H$ NMR spectroscopy to establish their relative configurations. Single-crystal X-ray diffraction analysis of several of these heterocycles has confirmed the stereochemical assignments. The published data for one compound has subsequently been used as a reference for identification of a stereoisomer isolated from a different synthetic pathway.

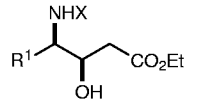
### 10.13.4.1 Determination of Enantiomeric Purity

Just like the common amino acids, the enantiomeric purity of γ-amino-β-hydroxy acids is established by formation of diastereomeric derivatives and their analysis by HPLC or NMR spectrometric techniques. In addition to derivatization of the γ-amino group, the β-hydroxy moiety of this class of compounds may also possibly be derivatized.

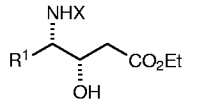
### 10.13.4.1.1 γ-Amino Derivatization

To evaluate the enantiomeric purity of (*S*,*S*)/(*R*,*R*)-statine, racemic mixtures and enantiomerically enriched samples are analyzed by HPLC after derivatization with 2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranosyl isothiocyanate (GITC, **51**; Scheme 21 and Table 7).[49] The GITC method also allows the determination of the enantiomeric composition of ethyl esters of statine analogues.[44,49] Another assay to determine the optical purity of statine has been developed, and uses L-Glu-NCA to produce dipeptides with different retention times in HPLC.[18]

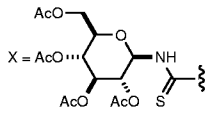
**Scheme 21** Statine Derivatives Formed with GITC[44,49]




(3*R*,4*R*)-*syn*-**18** $R^2$ = Et; Y = H




(3*S*,4*S*)-*syn*-**18** $R^2$ = Et; Y = H




**51**

**Table 7** Separation of Enantiomeric Statine Derivatives as their Diastereomeric Thioureine Derivatives Formed with GITC[49][a]

| GITC derived Statine **18** ($R^2$ = Et; X = **51**; Y = H) | $t_1$ (*R*,*R*) [min] | $t_2$ (*S*,*S*) [min] | α[b] | Ref |
|---|---|---|---|---|
| $R^1$ = iBu | 11.6 | 14.7 | 1.35 | [49] |
| $R^1$ = Bzl | 14.3 | 16.7 | 1.23 | [49] |

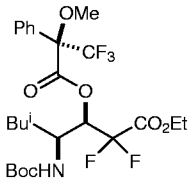
[a] See Scheme 21.

[b] Separation factor $\alpha = (t_2 - t_0)/(t_1 - t_0)$; t = elution time, $t_0$ = void time. Chromatographic conditions: Rsil silica gel (Altech, 5 μm); column: 4.6 × 250 mm; eluant: hexane/propan-2-ol (93:7); flow rate: 1.5 mL · $min^{-1}$; UV detection ($\lambda_{max}$): 254 nm.

### 10.13.4.1.2 β-Hydroxy Derivatization

The enantiomeric purity of α,α-difluoro statine analogues is determined via their diastereomeric Mosher's esters **52** and **53** by reaction with (*S*)-(+)-α-methoxy-α-(trifluoromethyl)phenylacetyl chloride. The $^{19}F$ NMR signals from the Tfm groups are well-resolved singlets and permit evaluation of their enantiomeric ratio (Scheme 22).[15]

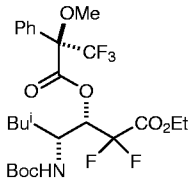
**Scheme 22** $^{19}F$ NMR Data of Mosher's Ester Derivatives of α,α-Difluoro Statine Analogues[15]




$\delta^{19}F$ 90.6

$\delta^{19}F$ 4.65, 5.43

(3*R*,4*S*)-*syn*-**52**




$\delta^{19}F$ 89.9

$\delta^{19}F$ 4.73, 5.28

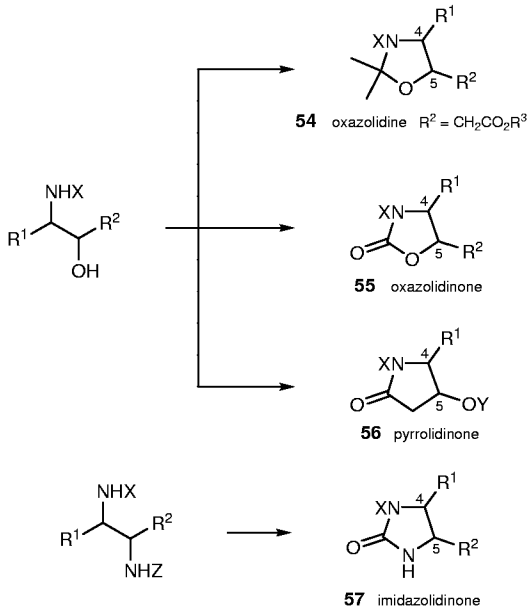
(3*S*,4*R*)-*syn*-**53**

*for references see p 588*

### 10.13.4.2 Absolute Configurations of γ-Amino-β-hydroxy Acids

The formation of oxazolidines **54** or oxazolidinones **55** is currently utilized to assign the absolute stereochemistry of diastereomers of 1,2-amino alcohols, based on $^1$H NMR analysis of the H4 and H5 protons of these heterocycles. In the case of γ-amino-β-hydroxy acids, the internally cyclized pyrrolidinone **56** is also suitable for determination of the relative configurations of the γ-amino and β-hydroxy groups (Scheme 23).

**Scheme 23** Five-Membered Heterocycles Derived from 1,2-Amino Alcohols and 1,2-Diamines




#### 10.13.4.2.1 Stereochemistry Determined from Oxazolidinones

The γ-amino-β-hydroxy acid derived oxazolidinones **55** are prepared from the corresponding N-unprotected γ-amino-β-hydroxy ester derivatives by reaction with phosgene,[19,39] carbonyl diimidazole,[41] or benzyl chloroformate.[86] Alternatively, cyclization is obtained from the *N*-carbamate protected derivatives, i.e. from the *N*-isopropenyloxycarbonyl derivatives under heating,[38] or from the *N*-Boc or *N*-Z derivatives under basic conditions.[68,81,87] By analogy, the β,γ-diamino acid analogue is converted into the imidazolidinone **57** by treatment of the unprotected compound with phosgene.[83,88]

It is possible to distinguish a 4,5-*trans*-disubstituted oxazolidinone or imidazolidinone from its 4,5-*cis*-analogue by using $^1$H NMR analysis. Indeed, both H4 and H5 protons resonate systematically at lower field for the *cis*-isomer than for the *trans*-isomer, and higher $J_{4,5}$ coupling constants are observed for the *cis*-oxazolidinones (Table 8). In addition, nuclear Overhauser effect (NOE) enhancement experiments confirm the *cis* or *trans* relationship of the H4 and H5 protons.

**Table 8** $^1$H NMR Determination of Oxazolidinone (**55**) and Imidazolidinone (**57**) Configurations[19,81,82,87,88][a]

| Oxazolidinone **55** | | | Config | $\delta_{H4}$ (ppm) | $\delta_{H5}$ (ppm) | $J_{4,5}$ (Hz) | Ref |
|---|---|---|---|---|---|---|---|
| X | $R^1$ | $R^2$ | | | | | |
| H | iBu | $CH_2CO_2H$ | *cis* | | 5.1 | 8.8 | [19] |
| | | | *trans* | | 4.5 | 5.0 | |
| H | iBu | allyl | *cis* | 3.9 | 4.6 | | [87] |
| | | | *trans* | 3.6 | 4.2 | | |
| $(CH_2)_3$ | | Me-3-allyl | *cis* | 3.4 | 3.7 | 6.9 | [82] |
| | | | *trans* | 3.3 | 3.6 | 4.5 | |
| H | iPr | Me-3-allyl | *cis* | | 4.3 | 7.0 | [81] |
| | | | *trans* | | 4.0 | 3.8 | |
| Imidazolidinone **57** | | | | | | | |
| X | $R^1$ | $R^2$ | | | | | |
| H | $CH_2Cy$ | $CH_2CO_2H$ | *cis* | 3.9 | 4.1 | 7.8 | [88] |
| | | | *trans* | 3.4 | 3.7 | 5.4 | |

[a] See Scheme 23.

### 10.13.4.2.2 Stereochemistry Determined from Pyrrolidinones

The pyrrolidinone analogues **56** of γ-amino β-hydroxy acids are obtained by reduction of the corresponding tetramic acid derivatives (see Section 10.13.1.4) or by cyclization of the corresponding γ-amino-β-hydroxy esters after deprotection of the γ-amino group.[16,20,21,50,78,84,89]

The relative stereochemistry at the C4 and C5 positions of pyrrolidinones is established by $^1$H NMR analysis (Table 9). The H4 proton of the *cis*-isomer appears downfield with respect to the H4 proton of the *trans*-isomer. The *cis* or *trans* relationship between H4 and H5 is confirmed by a lower $J_{4,5}$ coupling constant in the *trans* arrangement and a significant NOE effect in the *cis* arrangement.

**Table 9** $^1$H NMR Determination of Pyrrolidinone (**56**) Configurations[20,53,54,67][a]

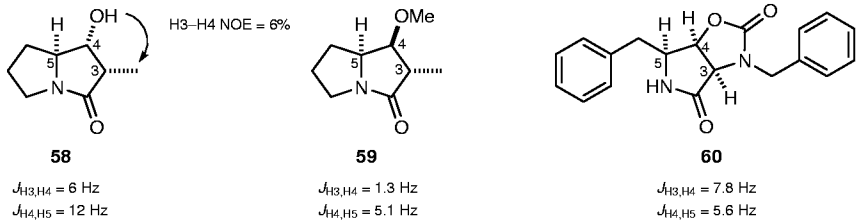
| Pyrrolidinone **56** | | | Config | $\delta_{H4}$ (ppm) | $J_{4,5}$ (Hz) | Ref |
|---|---|---|---|---|---|---|
| $R^1$ | X | Y | | | | |
| iBu | Boc | H | *cis* | 4.30 | 6.5 | [53] |
| $(CH_2)_3$ | | H | *cis* | 4.20 | | [54] |
| | | | *trans* | 4.03 | | |
| sBu | Me | Ac | *cis* | 3.57 | 5.1 | [20] |
| | | | *trans* | 3.51 | 1.8 | |
| sBu | Me | Me | *cis* | 4.04 | 7 | [67] |
| | | | *trans* | 3.68 | 1.7 | |

[a] See Scheme 23.

Formation of the rigid, bicyclic lactams **58–60** facilitates the determination of the relative configurations at the C3 positions in the case of α-monosubstituted γ-amino β-hydroxy acids by $^1$H NMR spectroscopy[16,62,82] (Scheme 24). The *cis*/*trans* stereochemistry of the lactam derived from the α,α-difluoride derivative cannot be assigned on the basis of the $J_{4,5}$ coupling constant; in this case, $^{13}$C NMR spectra[15] or NOE effects[78] are more informative. The signal from the C5 atom of the *cis*-isomer appears upfield relative to that of the *trans*-isomer.

*for references see p 588*

**Scheme 24** Bicyclic Lactams Derived from α-Substituted γ-Amino-β-hydroxy Acids





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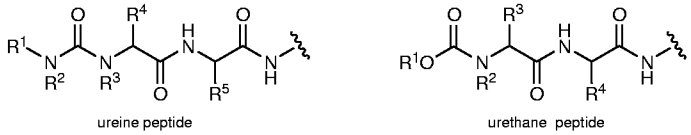
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# 10.14 Synthesis of Peptides Containing Ureine and Urethane Linkages

M. E. Wilson and J. S. Nowick

Ureine and urethane groups find many uses in peptide and medicinal chemistry. Ureine (*N*-alkyl urea) groups have been exploited as pharmacophores in peptide-based inhibitors[1] and as turn inducers in peptidomimetics.[2] Urethane (carbamate) groups have been used as structural elements in cyclic peptides[3] and as pharmacophores in bioactive peptides.[4] This chapter covers synthetic methods for introducing ureine and urethane linkages into peptides or peptidomimetics that contain a peptide bond. The resulting compound classes, ureine peptides and urethane peptides, are shown in Scheme 1. Urethane-based groups generally used for the protection of peptides are not discussed.

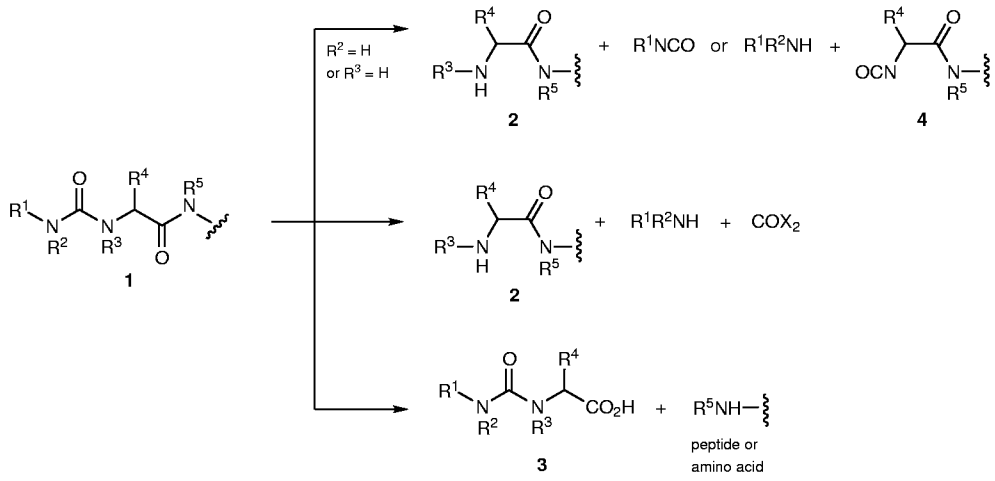
**Scheme 1** Ureine and Urethane Peptides




## 10.14.1 Ureine Peptides

Ureine peptides **1** have been synthesized by three general methods (Scheme 2). The first involves reacting either an $N^{\alpha}$-free peptide **2** with an isocyanate ($R^1NCO$) or a peptide isocyanate **4** with an amine ($R^1R^2NH$) to give a urea. The second involves condensing an active carbonic acid derivative ($COX_2$) with an $N^{\alpha}$-free peptide **2** and another amine ($R^1R^2NH$) to give a urea. The third involves attaching a urea-containing segment, (e.g., ureine amino acid **3**) to a peptide or amino acid using standard peptide coupling methods.

**Scheme 2** Methods for Constructing Ureine Peptides and Peptidomimetics



### 10.14.1.1 Ureine Linkages from Amines and Isocyanates

The reaction of amines with isocyanates provides a general route to ureine peptides with at least one hydrogen on the ureine group. The isocyanates may be available commercially or from in-house stock (Section 10.14.1.1.1), or may be prepared[5–7] from either the corresponding amines and phosgene or triphosgene (Section 10.14.1.1.2) or by the Curtius rearrangement (Section 10.14.1.1.3). The addition of primary amines to isocyanates proceeds smoothly at room temperature. Secondary amines and amines deactivated by steric constraints, conjugation, or electron-withdrawing groups may require mild heating or extended reaction times. When amine salts are used, tertiary amines, such as TEA, NMM, or DIPEA, are added. Common solvents used for this coupling include EtOAc, $CH_2Cl_2$, DMF, and THF. Peptide functional groups that have not been protected during coupling include carboxylic acids and the side chains of His, Arg, and Trp. Thiols, phenols, and alcohols are generally protected.

#### Safety and Ecological Aspects

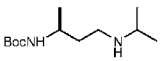
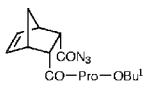
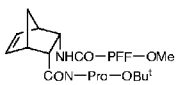
Isocyanates are hazardous. Prolonged exposure to their vapors has been associated with chronic airway disorders.[8] Phosgene is very hazardous and should be handled only in a well-ventilated fume hood.[9] Triphosgene and sodium azide are toxic.

#### 10.14.1.1.1 Available Isocyanates

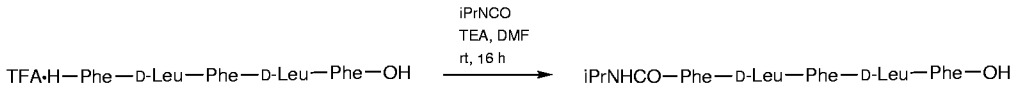
The reaction of $N^{\alpha}$-free peptides with available isocyanates is a particularly convenient route to ureine peptides. Although isocyanates may be obtained commercially or from in-house stock, this approach is limited by the paucity of available isocyanates; alkyl (Table 1, entry 1),[10] aryl,[11] and halogenated (Table 1, entry 2)[12] isocyanates are generally available, but peptide and amino acid isocyanates are not. Scheme 3 illustrates how Higgins et al.[10] prepared ureine peptide **6** by treating peptide TFA salt **5** with commercially available isopropyl isocyanate in the presence of TEA (Table 1, entry 1). Terminal primary ureine groups ($R^1NHCONH_2$) can be introduced by reacting an $N^{\alpha}$-free peptide with cyanic acid (formed in situ from KNCO; Table 1, entry 3).[13,14]

*for references see p 604*

**Table 1** Ureine Peptides Produced by Coupling Amines with Isocyanates[10,12 14 16,18,21 23 25,26]

| Entry | Section | R-NCO Component or Precursor | Amine Component | Coupling Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|---|---|---|
| 1 | 10.14.1.1.1 | iPrNCO | TFA·H-Phe-D-Leu-Phe-D-Leu-Phe-OH (**5**) | DMF. TEA, rt, 16 h | iPrNHCO-Phe-D-Leu-Phe-D-Leu-Phe-OH (**6**) | 81 | [10] |
| 2 | 10.14.1.1.1 | $ClCH_2CH_2NCO$ | H-Tyr-Gly-Gly-Phe-Leu-OH | DMF. TEA, rt, 1 h | $ClCH_2CH_2NHCO$-Tyr-Gly-Gly-Phe-Leu-OH | 81 | [12] |
| 3 | 10.14.1.1.1 | KNCO | HCl·H-Trp-Met-Asp-Phe-$NH_2$ | $H_2O$/DMF, AcOH, 48 h NMM, rt | $H_2NCO$-Trp-Met-Asp-Phe-$NH_2$ | 70 | [14] |
| 4 | 10.14.1.1.2 | 4-aminoheptane | TFA·H-Val-Asp(Pyr)-Asp-Leu-OH | $CH_2Cl_2$. NMM. rt, 16 h | (1-propylbutyl)-NHCO-Val-Asp(Pyr)-Asp-Leu-OH | ~ 70 | [16] |
| 5 | 10.14.1.1.2 | HCl·H-Phe-Leu-OMe (**7**) | *N*-ethylaniline | $CH_2Cl_2$, rt, 20 h | PhN(Et)CO-Phe-Leu-OMe (**9**) | 93 | [18] |
| 6 | 10.14.1.1.2 | HCl·H-Leu-OAl | *c*[-Trp(2-Br,5-OH)-(Me)-Leu-Leu-Ala-Lys-]-N$^{\alpha}H_2$ | THF, rt, 24 h | *c*[-Trp(2-Br,5-OH)-(Me)Leu-Leu-Ala-Lys-]-N$^{\alpha}$HCO-Leu-OAl | – | [21] |
| 7 | 10.14.1.1.2 | H-Tyr-Asn-Phe-Ala-Thr-Nle-resin |  | $CH_2Cl_2$, rt, 1 h | BocNHCH($CH_3$)$CH_2CH_2$. N(iPr)-CO-YNFAT-Nle-resin | – | [23] |
| 8 | 10.14.1.1.3 | Z-Lys(Boc)-Pro-Val-Gly-Lys(Boc)-$N_3$ | AcOH·H-Lys(Boc)-Arg-Arg-$NH_2$ | DMF. TEA, rt, 8 h | Z-K(Boc)-PVG-NH-CH-[($CH_2$)$_4$-NHBoc]-NHCO-K(Boc)-RR-$NH_2$ | 66 | [25] |
| 9 | 10.14.1.1.3 |  **10** | TFA·H-Pro-Phe-Phe-OMe | $CH_2Cl_2$, TEA, rt, 18 h |  **12** | 78 | [26] |

**Scheme 3** Addition of an Available Isocyanate to an $N^{\alpha}$-Free Peptide to Produce a Ureine Peptide[10]

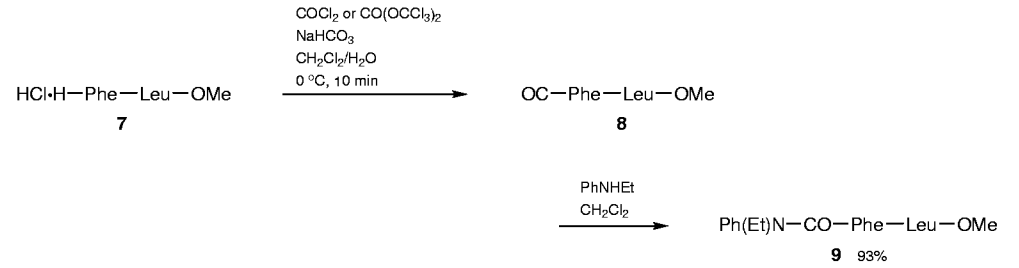



**iPrNHCO-Phe-D-Leu-Phe-D-Leu-Phe-OH (6):**[10]
To a rapidly stirring soln of **5** (100 mg, 0.12 mmol) and TEA (34 μL, 0.25 mmol) in dry DMF (5 mL) was added isopropyl isocyanate (18 μL, 0.19 mmol). The soln was stirred at rt for 16 h; then the solvent was removed. The residue was dissolved in MeOH (1 mL) and 1 M HCl (15 mL) was added with rapid stirring. The precipitate that formed was collected, washed with $H_2O$, and recrystallized ($EtOH/H_2O$) to give the ureine peptide **6**; yield: 75 mg (81%); FAB-MS *m/z* for $C_{43}H_{59}N_6O_7$ $[M+H]^+$: calcd. 771, found 771.

### 10.14.1.1.2 Isocyanates Prepared with Phosgene or Triphosgene

Isocyanates that are not commercially available can be prepared by treating amines with phosgene ($COCl_2$) or triphosgene [$CO(OCCl_3)_2$].[15] Non-peptidic isocyanates (Table 1, entry 4)[16] and some amino acid ester isocyanates[17] can be prepared by bubbling phosgene gas into a hot (~100 °C) suspension of the corresponding amine hydrochloride in toluene over several hours. These conditions are not compatible with acid-sensitive functional groups, and dipeptide esters are not converted into isocyanates, but instead yield hydantoins. Nowick and co-workers[18] developed an improved procedure for preparing amino acid ester isocyanates and peptide ester isocyanates, which involves adding a solution of phosgene or triphosgene to a cold (0 °C) suspension of amino acid ester or peptide ester hydrochloride in a biphasic mixture of dichloromethane and saturated aqueous sodium bicarbonate. Scheme 4 illustrates the application of this method to generate OC-Phe-Leu-OMe (**8**) and its subsequent reaction with *N*-ethylaniline to give ureine dipeptide **9** (Table 1, entry 5). Rapid stirring of the reaction mixture is essential to minimize symmetrical urea formation or, in the case of peptide substrates, epimerization at the N-terminal residue.[19] Alternative procedures involving the addition of pyridine (Table 1, entry 6)[20,21] or DIPEA[22] have also been reported. Limal et al.,[23] for example, generated peptide isocyanates on a polymer support using a solution of triphosgene and DIPEA in dichloromethane (Table 1, entry 7).

**Scheme 4** Formation of a Peptide Isocyanate Using Phosgene or Triphosgene and its Reaction with an Aniline Derivative to Produce a Ureine Peptide[18]




**OC-Phe-Leu-OMe (8):**[18]
To a 25-mL one-necked, round-bottomed flask, equipped with a mechanical stirrer, was added **7** (164 mg, 0.50 mmol), $CH_2Cl_2$ (10 mL), and sat. $NaHCO_3$ (10 mL). The mixture was stirred in an ice-bath (0 °C) for 10 min; then the stirring was stopped, and phosgene (520 μL of a 1.93 M soln in toluene, 1 mmol) or triphosgene (soln of 104 mg in 1 mL of $CH_2Cl_2$, 0.35 mmol) was added in a single portion. The mixture was stirred rapidly (ca. 600 rpm) for 10 min at 0 °C; then the layers were separated, and the aqueous phase was extracted with $CH_2Cl_2$ (3×10 mL). The combined organic layers were dried ($Na_2SO_4$), filtered, and concentrated to give the crude isocyanate **8** as a white solid; yield: 159 mg (quant); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3417, 2266 (–N=C=O), 1741, 1684 $cm^{-1}$.

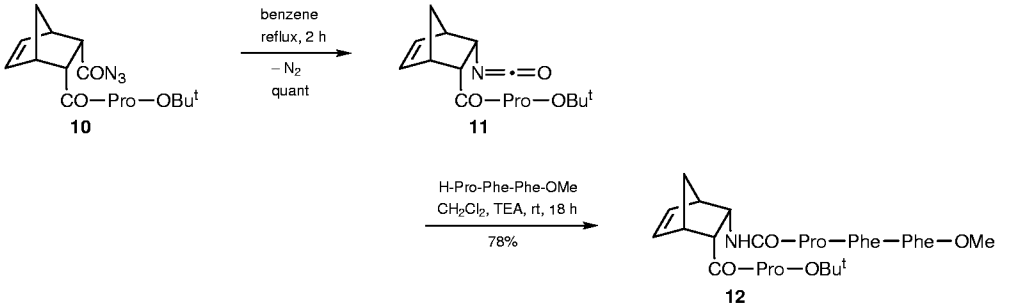
*for references see p 604*

**Ph(Et)NCO-Phe-Leu-OMe (9):**[18]
A soln of crude isocyanate **8** (159 mg, 0.50 mmol) and *N*-ethylaniline (125 μL, 0.99 mmol) in $CH_2Cl_2$ (5 mL) was stirred under $N_2$ for 20 h at rt. The solvent was removed and the residue was recrystallized (EtOAc/hexanes) to give the ureine dipeptide **9** as white crystals; yield: 204 mg (93 % from **7**) mp 98–100 °C; IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3427, 1741, 1674, 1643 $cm^{-1}$; HRMS *m/z* for $C_{25}H_{34}N_3O_4$ $[M+H]^+$: calcd. 440.2549, found 440.2547.

### 10.14.1.1.3 Isocyanates Prepared by the Curtius Rearrangement

Isocyanates used for the preparation of ureine peptides can also be obtained through the Curtius rearrangement, in which an acyl azide ($R^1CON_3$) rearranges to yield an isocyanate ($R^1NCO$) and $N_2$.[24] Acyl azides are readily prepared from carboxylic acids and undergo Curtius rearrangement at elevated temperatures (e.g., in refluxing benzene, toluene, or chloroform) to give isocyanates in high yield. The thermal instability of peptide acyl azides, which are intermediates in the azide peptide coupling method (Section 3.1), can give rise to ureine peptide side products, which form by the reaction of amines with peptide isocyanate Curtius rearrangement products. To study this side reaction, Inouye et al.[25] allowed peptide acyl azides to warm up to room temperature in the presence of $N^{\alpha}$-free peptides to afford ureine peptides (e.g., Table 1, entry 8). North and co-workers[26] installed a norbornene-based turn inducer within a short peptide using the Curtius rearrangement of acyl azide **10** to generate isocyanate **11**, which they then reacted with an $N^{\alpha}$-free tripeptide to obtain ureine peptide **12** (Scheme 5; Table 1, entry 9).

**Scheme 5** Preparation of an Isocyanate by Curtius Rearrangement and its Reaction with an $N^{\alpha}$-Free Peptide to Produce a Ureine Peptide[26]




**Isocyanate 11:**[26]
Isopropenyl chloroformate (0.87 mL, 8.0 mmol) was added to a mixture of 3-{[2-(*tert*-butoxycarbonyl)-1-pyrrolidinyl]carbonyl}bicyclo[2.2.1]hept-5-ene-2-carboxylic acid (2.5 g, 7.3 mmol) and TEA (2 mL) in dry THF (25 mL) at –20 °C. The soln was warmed to –10 °C and aq $NaN_3$ (1.19 g, 18.3 mmol) was added. The mixture was stirred for 1 h and allowed to reach rt. The mixture was diluted with $H_2O$ (30 mL) and extracted with EtOAc (3×30 mL). The organic phase was washed with aq $Na_2CO_3$ (30 mL), $H_2O$ (30 mL), and brine (30 mL), dried ($MgSO_4$), filtered, and concentrated to give 1.5 g (57 %) of acyl azide **10** as a white solid: mp 107–109 °C; $[\alpha]_D^{22}$ –19.5 (*c* 1, $CHCl_3$); IR $\tilde{\nu}_{max}$: 2138, 1729, 1647 $cm^{-1}$. Acyl azide **10** (1.5 g, 4.1 mmol) was dissolved in anhyd benzene (15 mL), and the soln was refluxed for 2 h. The solvent was removed to afford the isocyanate **11** as a white solid; yield : 1.4 g (quant); mp 61–62 °C; $[\alpha]_D^{22}$ –3.2 (*c* 1, $CHCl_3$); IR $\tilde{\nu}_{max}$: 2263, 1735, 1647 $cm^{-1}$; HRMS (EI) *m/z* for $C_{18}H_{24}N_2O_4$ $[M^+]$: calcd. 332.1736, found 332.1736.

**Ureine Peptide 12:**[26]
To a cold (0 °C) soln of isocyanate **11** (1.15 g, 3.5 mmol) and TFA · H-Pro-Phe-Phe-OMe (2.4 g, 4.5 mmol) in $CH_2Cl_2$ (10 mL) was added TEA (1 mL). The soln was stirred for 18 h at rt; then it was washed with 0.5 M aq HCl (5 mL), sat. $Na_2CO_3$ (5 mL), $H_2O$ (5 mL), and dried ($MgSO_4$). The soln was filtered, and the solvent was removed to give a yellow solid, which was triturated with EtOAc to give the ureine peptide **12** as a white solid; yield: 2.05 g (78 %); mp 60–61 °C; $[\alpha]_D^{23}$ –95.1 (*c* 1, $CHCl_3$); IR $\tilde{\nu}_{max}$: 3407, 3313, 1734, 1636 $cm^{-1}$; HRMS (CI, $NH_3$) *m/z* for $C_{42}H_{54}N_5O_8$ $[M+H]^+$: calcd. 756.397, found 756.397.

### 10.14.1.2 Ureine Linkages by Coupling Amines with Carbonic Acid Derivatives

A second method for obtaining ureine peptides involves linking an $N^{\alpha}$-free peptide with another amine (peptidic or otherwise) using carbonic acid derivatives $COX_2$. This process occurs through the sequential displacement of X by one amine ($R^1R^2NH$) to form an active carbamate ($R^1R^2NCOX$), which reacts with another amine ($R^3R^4NH$) to form a urea ($R^1R^2NCONR^3R^4$). Primary and secondary amines react with carbonic acid derivatives to afford active carbamates; while active carbamates of primary amines ($R^1NHCOX$) decompose in the presence of base or heat to yield isocyanates, active carbamates of secondary amines ($R^1R^2NCOX$) are generally stable. The carbonic acid derivatives discussed in this section are listed in Table 2, and include phosgene, CDI, and 4-nitrophenyl chloroformate and related aryl chloroformates.

The conditions for reacting amines with active carbamates are similar to those used for isocyanate coupling. Typical solvents for the coupling reaction include $CH_2Cl_2$, DMF, dioxane, THF, pyridine, and chloroform. Tertiary amine bases, e.g. TEA, DIPEA, or NMM, are added to the reaction to generate the free base of the amine or to neutralize acid byproducts. While most primary amines and secondary amines add to activated carbamates at room temperature, sterically congested secondary amines or amines rendered less nucleophilic through electrostatic effects may require heating, extended reaction times, or the addition of catalysts, e.g. HOBt.[27] Peptide functional groups not protected during the coupling reaction include carboxylic acids and the side chain of Trp. The amine/carbonic acid derivative method complements the amine/isocyanate method in two respects: (1) the reactive intermediates in the carbonic acid derivative method are easier to handle and store than the reactive intermediates in the isocyanate method, and (2) the carbonic acid derivative method offers a route to obtaining tetrasubstituted ureas, which are not possible by the isocyanate method.

#### Safety and Ecological Aspects

Carbamoyl chlorides, CDI, 4-nitrophenyl carbamates, and the compounds used to prepare them, e.g. phosgene and 4-nitrophenyl chloroformate, are toxic.

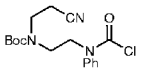
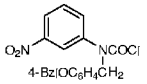
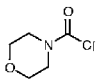
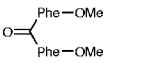
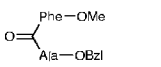
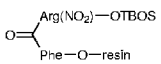
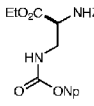
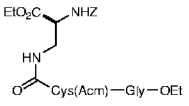
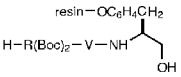
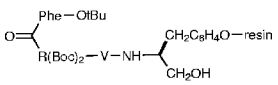
#### 10.14.1.2.1 Carbamoyl Chlorides

N,N-Disubstituted carbamoyl chlorides ($R^1R^2NCOCl$) are obtained by the condensation of secondary amines with phosgene or phosgene equivalents using the same procedures described for making isocyanates (Section 10.14.1.1.2). The carbamoyl chlorides shown in Table 2, entries 1 and 2, for example, were prepared using phosgene[28] and diphosgene ($CCl_3OCOCl$),[29] respectively. Some dialkylcarbamoyl chlorides can also be purchased.

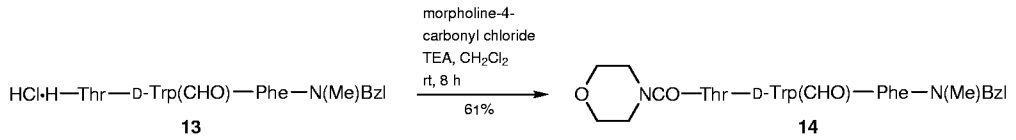
The N-terminus of peptides can be modified by reaction with dialkylcarbamoyl chlorides to afford N-terminal ureas: Nowick and co-workers[28] introduced a urea-linked molecular scaffold at the N-terminus of H-Phe-Ile-Leu-O-resin (Table 2, entry 1); Nugiel et al.[29] installed a novel Pro-Tyr replacement at the N-terminus of the dipeptide H-Ile-Leu-OBzl (Table 2, entry 2); and Albericio and Barany[30] treated H-Gly-Val-O-resin with dimethylcarbamoyl chloride to install an N-terminal dimethylaminocarbonyl group. Scheme 6 illustrates how Matsuo and co-workers[31] prepared ureine peptide **14** by reacting peptide hydrochloride **13** with morpholine-4-carbonyl chloride in the presence of NMM (Table 2, entry 3). Because the reaction of carbamoyl chlorides with amines generates hydrogen chloride, amine bases such as pyridine, DIPEA, NMM, or TEA are usually added. Alternatively, the HCl byproduct can be removed by refluxing the reaction mixture (Table 2, entry 2). Most primary amines react readily with *N,N*-dialkylcarbamoyl chlorides to give trisubstituted ureas in good yields. Secondary amines, however, are slow to react with *N,N*-dialkylcarbamoyl chlorides; these reactions often require high concentrations and heat to yield the tetrasubstituted ureas.

*for references see p 604*

**Table 2** Ureine Peptides Produced by Coupling of Amines using Carbonic Acid Derivatives[28,29,31,33,34,36,37,40]

| Entry | Section | Amine Component 1 or Active Carbamate | Amine Component 2 | Coupling Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|---|---|---|
| 1 | 10.14.1.2.1 |  | H-Phe-Ile-Leu-O-resin | DMF, TEA. rt, 24 h | $NCCH_2CH_2N(Boc)CH_2CH_2N(Ph)$-CO-Phe-Ile-Leu-O-resin | – | [28] |
| 2 | 10.14.1.2.1 |  | H-Ile-Leu-OBzl | THF. reflux |  | 85 | [29] |
| 3 | 10.14.1.2.1 |  | HCl·H-Thr-D-Trp(CHO)-Phe-N(Me)Bzl (**13**) | $CH_2Cl_2$, TEA. rt, 8 h | (morpholin-4-ylcarbonyl)-Thr-D-Trp(CHO)-Phe-N(Me)Bzl (**14**) | 61 | [31] |
| 4 | 10.14.1.2.2 | HCl·H-Phe-OMe | HCl·H-Phe-OMe | $CH_2Cl_2$, CDI. NMM. rt, 8 h |  | – | [33] |
| 5 | 10.14.1.2.2 | TosOH·H-Ala-OBzl (**15**) | HCl·H-Phe-OMe | $CH_2Cl_2$, CDI. NMM. rt, 8 h |  **16** | 74 | [33] |
| 6 | 10.14.1.2.2 | H-Phe-O-resin | H-Arg($NO_2$)-OTBOS | CDI. $CH_2Cl_2$ |  | – | [34] |
| 7 | 10.14.1.2.3 |  **18** | TFA·H-Cys(Acm)-Gly-OEt | dioxane/ $CHCl_3$, TEA. rt to 40 °C. 24 h |  **19** | 69 | [36] |
| 8 | 10.14.1.2.3 | NpOCO-Phe-OtBu |  | DMF, DIPEA, rt, 1 h |  | – | [37] |
| 9 | 10.14.1.2.3 | PhOCO-Leu-D-Trp-βAla-OEt | $iPrNH_2$ | $CHCl_3$, TEA. rt to 55 °C | iPrNH-CO-Leu-D-Trp-βAla-OEt | 92 | [40] |
| 10 | 10.14.1.2.3 | PhOCO-Leu-D-Trp-βAla-OEt | 3-$ClPhNH_2$ | $CHCl_3$, TEA. rt to 55 °C | 3-ClPhNH-CO-Leu-D-Trp-βAla-OEt | 8 | [40] |

**Scheme 6** Reaction of an $N^{\alpha}$-Free Peptide with a Dialkylcarbamoyl Chloride To Yield a Ureine Peptide[31]

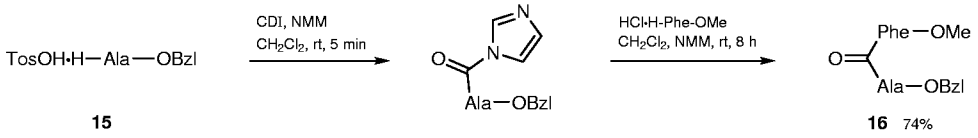



**(Morpholin-4-ylcarbonyl)-Thr-D-Trp(CHO)-Phe-N(Me)Bzl (14):**[31]
To a soln of **13** (1.0 g, 1.6 mmol) and TEA (0.40 g, 3.9 mmol) was added morpholine-4-carbonyl chloride (0.28 g, 1.9 mmol). The soln was stirred at rt for 8 h; then it was concentrated to give a residue, which was partitioned between EtOAc and $H_2O$. The layers were separated and the organic layer was washed with dil aq $NaHCO_3$, 0.5 M aq HCl, brine, and dried ($Na_2SO_4$). After filtration, the solvent was removed, and the residue was purified by column chromatography ($CHCl_3$/MeOH, 25:1) to afford the ureine tripeptide **14** as an amorphous solid; yield: 0.67 g (61 %); $[\alpha]_D^{25}$ +33.6 (*c* 1.1, DMF); IR (Nujol) $\tilde{\nu}_{max}$: 3400, 3280, 1710, 1660 (sh), 1640, 1630 $cm^{-1}$.

### 10.14.1.2.2 Carbonyl Diimidazole

CDI[32] is an easily handled carbonic acid derivative, which can be used to condense an $N^{\alpha}$-free peptide with another amine (peptidic or otherwise) to obtain a ureine peptide. Sequential displacement of imidazole in CDI by either the same or two different amines yields symmetrical and nonsymmetrical ureas, respectively. This transformation is generally accomplished by adding the amines sequentially to a solution of CDI in $CH_2Cl_2$ or DMF at room temperature. When amine salts are used, bases such as TEA, NMM, or DIPEA are added to the reaction mixture. Peña and co-workers,[33] for example, prepared symmetrical and nonsymmetrical ureas by sequentially adding amino acid salts to a solution of CDI in dichloromethane in the presence of NMM (Scheme 7; Table 2, entries 4 and 5). The ease of handling CDI makes it ideal for the solid-phase synthesis of ureine peptides (Table 2, entry 6).[34] While most amines add once to CDI to form the intermediate imidazole carbamate in good yield, addition of the second equivalent of amine may be difficult, particularly for secondary amines. A recent breakthrough involving activation of the imidazole group by N-alkylation provides an improved method for synthesizing tetrasubstituted ureas and may be useful in these cases.[35]

**Scheme 7** Ureine Peptide Obtained by the Sequential Displacement of Imidazole in CDI[33]




**MeO-Phe-CO-Ala-OBzl (16):**[33]
To a soln of CDI (1.25 g, 7.70 mmol) in $CH_2Cl_2$ was added **15** (2.45 g, 7.00 mmol) in $CH_2Cl_2$ and NMM (3.0 mL, 2.5 mmol). The mixture was stirred for 5 min under $N_2$; then HCl·H-Phe-OMe (1.50 g, 7.00 mmol in $CH_2Cl_2$) and NMM (3.0 mL, 2.5 mmol) were added. The mixture was stirred for 8 h at rt; then it was washed with dil aq $NaHCO_3$, $H_2O$, dried ($Na_2SO_4$), filtered, and concentrated to give a viscous oil, which was purified by column chromatography (EtOAc/hexane 3:1) to give the ureine **16** as a viscous oil; yield: 2.0 g (74 %).

### 10.14.1.2.3 Aryl Carbamates

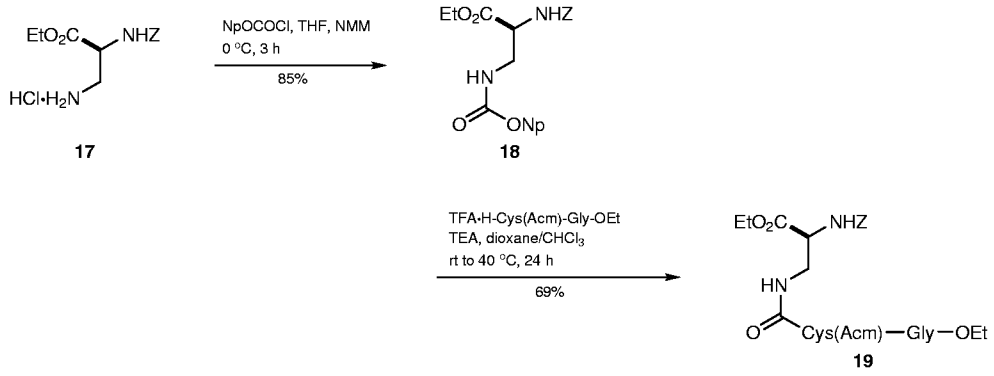
4-Nitrophenyl carbamates ($R^1R^2NCO_2Np$) and related aryl carbamates, e.g. phenyl-, dinitrophenyl-, trichlorophenyl-, and pentafluorophenyl carbamates, provide an additional route by which an $N^{\alpha}$-free peptide can be condensed with another amine (peptidic or

*for references see p 604*

otherwise) to obtain a ureine peptide. These carbamates are obtained by reacting amines with the parent chloroformate. 4-Nitrophenyl carbamates, for example, are prepared by reacting amines (or their acid salts) with 4-nitrophenyl chloroformate (NpOCOCl) at reduced temperature (−5 to 0 °C) in the presence of a mild base (e.g., NMM or 2,4,6-collidine) in dichloromethane, DMF, or THF. Functional groups that have active hydrogens, e.g. carboxylic acids, hydroxyls, amines, guanidines, indoles, and thiols, are generally protected during carbamate formation. 4-Nitrophenyl carbamates and related aryl carbamates of primary amines decompose in the presence of stronger bases (e.g., TEA and DIPEA) to yield isocyanates. Accordingly, these carbamates are used as isocyanate equivalents, which are much easier to handle and store than isocyanates. 4-Nitrophenyl carbamates and related aryl carbamates of secondary amines are much less reactive and are employed less frequently.

4-Nitrophenyl carbamates, which are the most commonly used type of aryl carbamates, react with primary and secondary amines in the presence of TEA, DIPEA, or NMM to give ureas. Common solvents used for this reaction include DMF, dichloromethane, chloroform, THF, and DMSO. These coupling reactions generally proceed well at room temperature or with mild heating. Thus, Calcagni et al.[36] reacted 4-nitrophenyl carbamate **18** with TFA · H-Cys(Acm)-Gly-OEt in the presence of TEA to obtain ureine peptide **19** (Scheme 8; Table 2, entry 7). Bradley and co-workers[37] coupled *N*-(4-nitrophenyloxycarbonyl)Phe-OtBu to a resin-bound dipeptide in their solid-phase synthesis of microbial alkaline protease inhibitor analogues (e.g., Table 2, entry 8). Similarly, they reacted $N^{\alpha}$-(4-nitrophenyloxycarbonyl)-$N^{\varepsilon}$-(9-fluorenylmethoxycarbonyl)-D-Lys allyl ester with polymer-bound Tyr(tBu) to make the naturally occurring ureine peptide oscillamide and its analogues.[38]

**Scheme 8** Preparation of a 4-Nitrophenyl Carbamate and its Reaction with an $N^{\alpha}$-Free Peptide to Produce a Ureine Peptide[36]




Phenyl carbamates ($R^1HNCO_2Ph$), obtained from the reaction of primary amines with phenyl chloroformate (PhOCOCl), also react with amines to give ureine peptides (Table 2, entries 9 and 10). These coupling reactions are promoted by using DMSO as the solvent.[39] While phenyl carbamates react well with good nucleophiles (e.g., isopropylamine, Table 2, entry 9), their reaction with poor nucleophiles generally gives lower yields (cf. 3-chloroaniline, Table 2, entry 10).[40] The major side product in entry 10 was a peptide hydantoin. Gray et al.[41] also observed hydantoin formation in peptides activated at their N-terminus as 2,4-dinitrophenyl carbamates. Thus, hydantoin formation severely limits N-terminal modification of peptides using N-terminal aryl carbamates or isocyanates.

**Z-NHCH($CO_2$Et)$CH_2NHCO_2$Np (18):**[36]
To a cold (0 °C) stirring soln of **17** (6.1 g, 20 mmol) in THF (80 mL) was added dropwise over 30 min a soln of 4-nitrophenyl chloroformate (4.1 g, 20 mmol) and NMM (4.1 g, 40 mmol) in THF. After 3 h at 0 °C

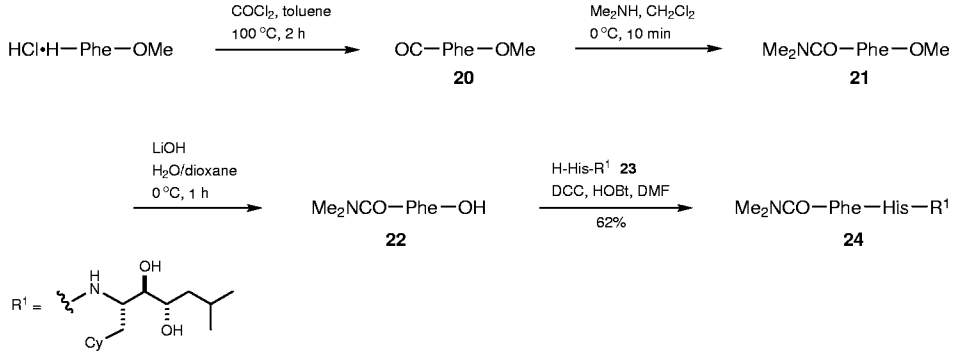
the soln was filtered to remove the precipitate, and the filtrate was concentrated. The residue was dissolved in $CHCl_3$, washed with 1 M $KHSO_4$, sat. $Na_2CO_3$, $H_2O$, and concentrated to give a residue, which was purified by chromatography ($CHCl_3$/MeOH, 97:3) to afford the 4-nitrophenyl carbamate **18** as a white solid; yield: 7.4 g (85 %); $[\alpha]_D^{25}$+35.0 (*c* 1, $CHCl_3$); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3435, 1750–1720 $cm^{-1}$.

**Z-NHCH($CO_2$Et)$CH_2$NHCO-Cys(Acm)-Gly-OEt (19):**[36]
To a cold (0 °C) stirring soln of TFA · H-Cys(Acm)-Gly-OEt (9.3 g, 24 mmol) in dioxane/$CHCl_3$ (140 mL, 2:1) was slowly added a soln of TEA (2.4 g, 24 mmol) in dioxane/$CHCl_3$ (20 mL, 2:1). After 5 min a soln of 4-nitrophenyl carbamate **18** (6.8 g, 16 mmol) in dioxane/$CHCl_3$ (40 mL, 2:1) was added in portions. The soln was heated to 40 °C for 24 h; then the solvent was removed and the residue was dissolved in EtOAc, washed with 0.5 M aq HCl, sat. aq $Na_2CO_3$, $H_2O$, and dried ($Na_2SO_4$). The soln was filtered and concentrated to give a residue, which was recrystallized (EtOAc/$Et_2O$) to give the ureine tripeptide **19**; yield: 6.2 g (69%); $[\alpha]_D^{25}$ −31.0 (*c* 1, MeOH); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3360 (br), 1740–1710, 1665, 1555 $cm^{-1}$.

### 10.14.1.3 Coupling Urea-Containing Segments to Amino Acids or Peptides

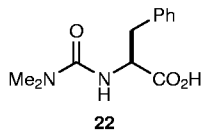
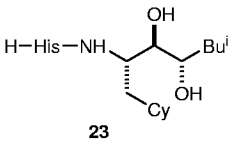
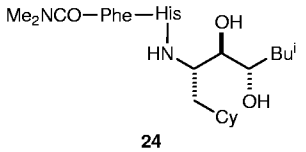
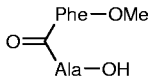
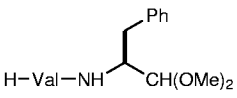
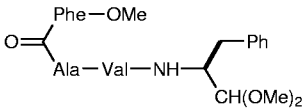
Ureine peptides can be prepared by attaching urea-containing segments to peptides or amino acids (Scheme 2). The urea-containing segments, which may be ureine amino acids, ureine peptides, or other urea-containing compounds, can be prepared using the previous two methods (Sections 10.14.1.1 or 10.14.1.2), and linked to a peptide or amino acid using standard peptide coupling methods (see Vol. E 22a, Section 3). The merits of this approach are twofold: (1) the parent peptide or amino acid component is not subjected to the reaction conditions required to generate the ureine group, and (2) the synthesis is convergent. The most common application of this method involves three steps: (1) installation of a ureine group at the N-terminus of a peptide ester or amino acid ester, (2) deprotection of the C-terminus of the urea-containing peptide or amino acid, followed by (3) coupling the urea-containing peptide or amino acid to the N-terminus of a peptide or amino acid. Scheme 9 illustrates how Plattner and co-workers[42] used this method to prepare renin inhibitor **24** (Table 3, entry 1). Treatment of HCl · H-Phe-OMe with excess phosgene in hot toluene, generated isocyanate **20** (Section 10.14.1.1.2), which reacted with dimethylamine to generate urea **21**. Saponification of the methyl ester in **21** afforded ureine amino acid **22**, which reacted with peptidomimetic **23** in the presence of DCC and HOBt to give ureine peptide **24** (Table 3, entry 1). A similar approach was used by Peña and co-workers[33] to synthesize a tetrapeptide protease inhibitor (Table 3, entry 2). A "one-pot" approach, which combines the first two steps, was recently reported by Weiberth.[43] Another recent development for generating ureine amino acid building blocks is the direct conversion of Fmoc-protected amino acids into ureine amino acids.[44] Both of these methods should provide a good source of ureine amino acid building blocks for the synthesis of ureine peptides.

**Scheme 9** Preparation of a Ureine Amino Acid Followed by Peptide Coupling To Produce a Ureine Peptide[42]




*for references see p 604*

**Table 3** Preparation of Ureine Peptides by Coupling Ureine-Containing Fragments to Amino Acids or Peptides

| Ureine Fragment | Parent Peptide or Amino Acid | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  **22** |  **23** |  **24** | 62 | [42] |
|  |  |  | 60 | [33] |

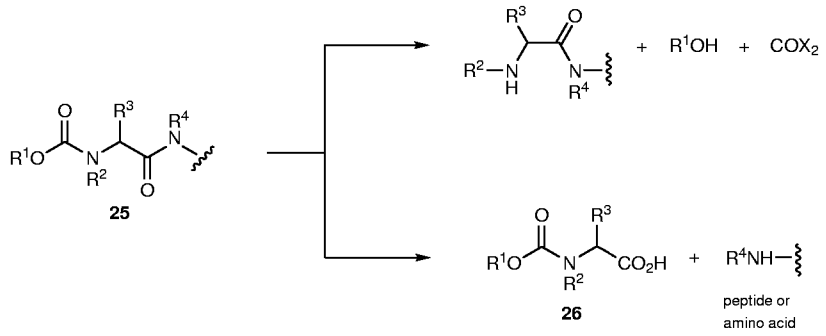
**(Dimethylaminocarbonyl)phenylalanine (22):**[42]
A suspension of HCl·H-Phe-OMe (7.0 g, 32 mmol) in toluene (125 mL) was heated (100 °C) while phosgene gas was bubbled into the mixture. Sparging with phosgene was continued until the mixture became homogeneous (ca. 2 h). After an additional 15 min, the solvent was removed, and benzene was added and removed several times to obtain crude isocyanate **20**. To a cold (0 °C) soln of isocyanate **20** (6.5 g, 31 mmol) was slowly added a soln of $Me_2NH$ (1.4 g, 31 mmol) in $CH_2Cl_2$ (5 mL). After 10 min at 0 °C, the mixture was partitioned between 0.5 M aq HCl and $CH_2Cl_2$. The organic layer was washed with aq $NaHCO_3$, dried ($MgSO_4$), and concentrated to afford a residue, which was triturated with hexanes to give the urea **21** as a solid; yield: 6.4 g (81 %); mp 79–80 °C. To a cold (0 °C) soln of urea **21** (0.91 g, 3.6 mmol) in dioxane (15 mL) was added a soln of LiOH (0.174 g, 4.15 mmol) in $H_2O$ (7.5 mL). After stirring for 1 h at 0–5 °C, the mixture was diluted with cold $H_2O$ and extracted twice with $Et_2O$. The aqueous portion was acidified with 6 M aq HCl and extracted with $Et_2O$. The organic fraction was washed with brine and concentrated to give the ureine amino acid **22** as a viscous oil; yield: 0.75 g (81%).

**(2*S*,3*R*,4*S*)-2-[(Dimethylaminocarbonyl-phenylalanyl-histidyl)amino]-1-cyclohexyl-3,4-dihydroxy-6-methylheptane (24):**[42]
To a stirred cold (0 °C) soln of ureine amino acid **22** (0.078 g, 0.33 mmol) in dry DMF (2 mL) was added a soln of peptidomimetic **23** (dihydrochloride, 0.14 g, 0.30 mmol) in DMF (5 mL). HOBt (0.054 g, 0.40 mmol) and DCC (0.068 g, 0.33 mmol) were added sequentially, and the soln was stirred for 2.5 h at 0 °C. The mixture was allowed to warm to rt. After 12 h the soln was filtered, and the filtrate was removed to give a residue, which was distributed between $CHCl_3$ and sat. $NaHCO_3$. The organic phase was washed sequentially with sat. $NaHCO_3$, brine, and dried ($Na_2SO_4$). The soln was filtered, and the solvent was removed to give a residue, which was purified by chromatography (5 % MeOH in $CHCl_3$) to afford the ureine peptide **24** as a solid; yield: 0.11 g (62 %); mp 108–110 °C.

## 10.14.2 Urethane Peptides

Urethane peptides **25** have been synthesized by two general methods (Scheme 10). The first involves condensing an active carbonic acid derivative ($COX_2$) with an $N^{\alpha}$-free peptide and an alcohol ($R^1OH$) to give a urethane peptide. The second involves the coupling of a urethane-containing segment (e.g., urethane amino acid **26**) to a peptide or amino acid using standard peptide coupling methods.

**Scheme 10** Methods for Constructing Urethane Peptides and Peptidomimetics



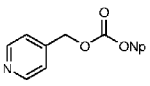
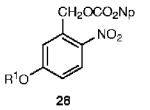
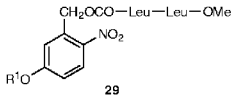

### 10.14.2.1 Urethane Linkages by Coupling Amines and Alcohols with Carbonic Acid Derivatives

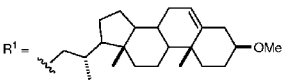
The reaction of $N^{\alpha}$-free peptides with active carbonic esters ($R^1OCOX$) generates urethane peptides. Common active carbonic esters include chloroformates (X = Cl) and 4-nitrophenyl carbonates (X = ONp). These compounds can either be prepared from alcohols and active carbonic acid derivatives ($COX_2$) or may be available commercially. Chloroformates are readily prepared by treating alcohols with phosgene using the same procedures described for preparing isocyanates (Section 10.14.1.1.2). 4-Nitrophenyl carbonates are prepared by treating alcohols with 4-nitrophenyl chloroformate using the same procedures described for preparing 4-nitrophenyl carbamates (Section 10.14.1.2.3). Functional groups containing active hydrogens, e.g. carboxylic acids, amines, thiols, and phenols are generally protected during formation of active carbonic esters.

$N^{\alpha}$-Free peptides react readily with chloroformates or 4-nitrophenyl carbonates at room temperature under the same conditions used for coupling amines with isocyanates (Section 10.14.1.1) or activated carbamates (Section 10.14.1.2). Elliott et al.,[45] for example, prepared a series of N-terminal urethane tetrapeptides by treating $N^{\alpha}$-free peptides with commercially obtained chloroformates in DMF in the presence of DIPEA (e.g., Table 4, entry 1). Similarly, Albericio and Barany[30] treated H-Gly-Val-O-resin with methyl chloroformate to obtain the corresponding N-terminal methoxycarbonyl derivative (Table 4, entry 2). Okada and coworkers[46] installed an N-terminal isonicotinyloxycarbonyl (iNoc) group onto a pentapeptide using a 4-nitrophenyl carbonate (Table 4, entry 3). Scheme 11 illustrates how Watanabe and Iwamura[47] linked H-Leu-Leu-OMe to a steroid skeleton using 4-nitrophenyl carbonate **28** (Table 4, entry 4).

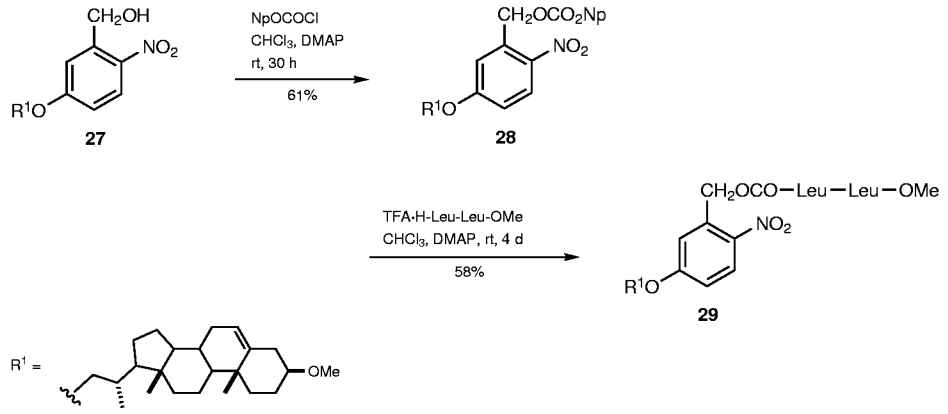
*for references see p 604*

**Table 4** Peptides Containing a Urethane Linkage Prepared by Coupling of Amines and Alcohols Using Carbonic Acid Derivatives

| Entry | Carbonic Acid Derivative | Parent Peptide or Amino Acid | Coupling Conditions | Product | Yield (%) | Ref |
|---|---|---|---|---|---|---|
| 1 | 2-adamantyl chloroformate | TFA·H-Trp-Lys-(Tac)-Asp(OBzl)-(Me)Phe-$NH_2$ | DMF, DIPEA, rt, 23 h | (2-Adoc)-Trp-Lys(Tac)-Asp(OBzl)-(Me)Phe-$NH_2$ | 85 | [45] |
| 2 | methyl chloroformate | H-Gly-Val-O-resin | $CH_2Cl_2$, TEA, rt, 2 d | MeOCO-Gly-Val-O-resin | – | [46] |
| 3 |  | TFA·H-Gln-D-Val-Val-Ala-Ala-pNA | DMF, NMM, rt, 8 h | iNoc-Gln-D-Val-Val-Ala-Ala-pNA | 52 | [46] |
| 4 |  **28** | TFA·H-Leu-Leu-OMe | $CHCl_3$, DMAP, rt, 4 d |  **29** | 58 | [47] |



**Scheme 11** Formation of a Urethane Linkage in a Peptide by Coupling an $N^{\alpha}$-Free Peptide with a 4-Nitrophenyl Carbonate[47]




**4-Nitrophenyl Carbonate** 28:[47]
A soln of alcohol **27** (98 mg, 0.20 mmol), 4-nitrophenyl chloroformate (46 mg, 0.23 mmol), and DMAP (40 mg, 0.33 mmol) in $CHCl_3$ (10 mL) was stirred for 30 h under argon. The mixture was washed with 1 M aq HCl, aq $NaHCO_3$, brine, and dried ($MgSO_4$). The soln was filtered and concentrated to give a residue, which was purified by gel-permeation chromatography to afford the 4-nitrophenyl carbamate **28** as white crystals after recrystallization (hexane/$CHCl_3$); yield: 80 mg (61%); mp 153–155 °C (dec); HRMS *m/z* for $C_{37}H_{47}N_2O_9$ $[M+H]^+$: calcd. 663.3282, found 663.3304.

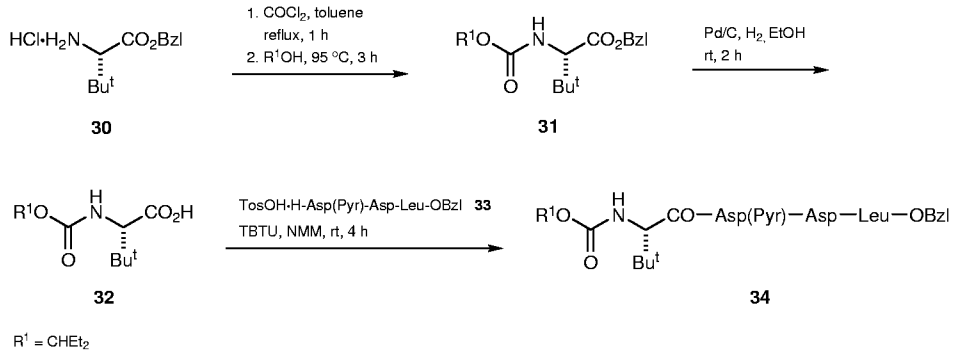
**Urethane Peptide** 29:[47]
A soln of 4-nitrophenyl carbamate **28** (45 mg, 0.068 mmol), TFA · H-Leu-Leu-OMe (31 mg, 0.083 mmol), and DMAP (13 mg, 0.11 mmol) in $CHCl_3$ (5 mL) was stirred for 4 d under argon. The mixture was washed with 1 M HCl, aq $NaHCO_3$, brine, and dried ($MgSO_4$). The soln was filtered and concentrated to give a residue, which was purified by gel-permeation chromatography to afford the urethane peptide **29** as white crystals; yield 31 mg (58%); mp 79–81 °C. UV $\lambda_{max}$ (MeOH): 309 nm (ε 12400).

## 10.14.2.2 Coupling Urethane-Containing Segments to Amino Acids or Peptides

The coupling of urethane-containing amino acids with $N^{\alpha}$-free peptides is the most common route to urethane peptides. This method is the basis of standard peptide coupling using Fmoc- or Boc-protected amino acids (see Vol. E 22a, Sections 3 and 4). Scheme 12 illustrates how Moss et al.[16] used this method to prepare urethane peptide **34**. Treatment of H-Val(3-Me)-OBzl (**30**) with phosgene generated the corresponding isocyanate (Section 10.14.1.1.2), which reacted with pentan-3-ol to give urethane **31**. Removal of the benzyl group in **31** and coupling of the resulting urethane amino acid **32** with peptide **33** using TBTU then gave urethane peptide **34**.

*for references see p 604*

**Scheme 12** Coupling a Urethane-Containing Amino Acid to a Peptide to Obtain a Urethane Peptide[16]




**Urethane Amino Acid** 32:[16]

Ester **30** (92 mg, 0.42 mmol) was treated with a soln of phosgene in toluene (1.93 M, 4 mL). The suspension was refluxed for 1 h; then the phosgene was removed by bubbling $N_2$ through the reaction mixture. Pentan-3-ol (180 μL, 1.65 mmol) was added, and the mixture was stirred at 95 °C for 3 h. The mixture was diluted with EtOAc (20 mL), washed with sat. $NaHCO_3$, brine, and dried ($MgSO_4$). The soln was filtered and concentrated to afford 90 mg of urethane **31** as a thick colorless liquid. A suspension of **31**, 10 % Pd/C (5 mg) in EtOH (6 mL) was stirred under an atmosphere of $H_2$ for 2 h. The mixture was filtered and concentrated to give the urethane amino acid **32** as a thick oil of sufficient purity for use in the subsequent coupling reaction; yield: 65 mg (65 % from **30**); FAB-MS *m/z* for $C_{12}H_{24}NO_4$ $[M+H]^+$: calcd. 246, found 246.

**Urethane Peptide** 34:[16]

To a soln of urethane amino acid **32** (245 mg, 1.00 mmol) in dry MeCN (2.5 mL) were added TBTU (321 mg, 1.00 mmol) and NMM (101 mg, 1.00 mmol). After ca. 5 min this soln was added to a soln of tripeptide **33** (677 mg, 1 mmol) in dry MeCN (2.5 mL) containing NMM (202 mg, 2 mmol). The mixture was stirred at rt for ca. 4 h; then it was poured into a mixture of EtOAc (50 mL) and sat. $NaHCO_3$ (50 mL). The organic phase was washed with sat. $NaHCO_3$, 1 M HCl (2×50 mL), and brine (50 mL), dried ($MgSO_4$), filtered, and concentrated to give urethane peptide **34**.


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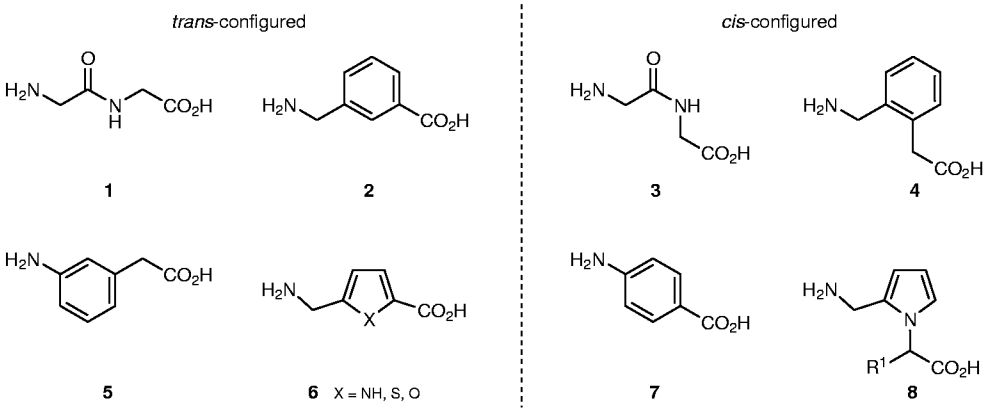
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# 10.15 Synthesis of Peptides Containing an Aromatic Spacer

D. Tourwé, J. Van Hemel, and J. Piron

The benzene ring has been proposed as an isosteric replacement in a dipeptide to enforce either the *trans* **1**[1] or the *cis* conformation **3**[2,3] (Scheme 1). Similarly, 2-(aminomethyl)pyrrole-1-acetic acid (**8**, $R^1 = H$) has been proposed as a *cis* peptide bond mimic,[4] having the same number of atoms between the amino and carboxylic acid functions as in a dipeptide. Several other amino- and carboxy-substituted aromatic structures have been used as spacers in peptides: 2-, 3-, and 4-aminobenzoic acids (Abz, e.g., **7**), 2-, 3-, and 4-(aminomethyl)benzoic acids (Amb, e.g., **2**), 2-, 3-, and 4-(aminophenyl)acetic acids (APha, e.g., **5**), 2- (**4**), 3-, and 4-(aminomethylphenyl)acetic acid (Ampa), (aminomethyl)pyrrole-, -thiophene-, and -furancarboxylic acids **6**, (aminomethyl)pyrrole- **8** and -thienylacetic acids, and aminobiphenylcarboxylic acids.

**Scheme 1** The Use of Aromatic Rings to Enforce the *trans* or *cis* Peptide Bond



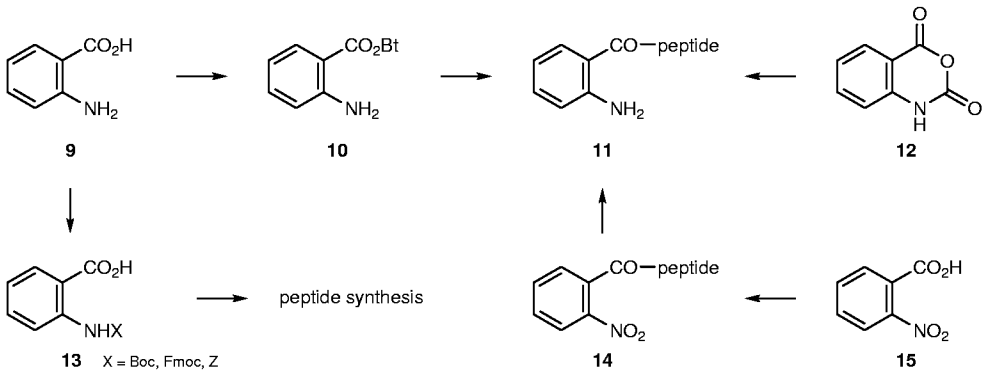

The aromatic spacers are either simply used to replace a part of the peptide stucture, or as templates to induce a specific conformation in the peptide.

## 10.15.1 Aminobenzoic Acid Peptides

2-Aminobenzoic acid (**9**, anthranilic acid, *o*-Abz) (Scheme 2), has been frequently attached to the N- or C-terminal position of peptides for use in internally quenched fluorimetric studies of enzyme substrates.[5–9] Compound **9** is also a constituent of naturally occurring cyclic peptides.[10–12] X-ray diffraction data of Boc-*o*-Abz-Gly-OMe show an intramolecular NH···O=C hydrogen bond in the *o*-Abz part that enforces a turn structure.[13] This intramolecular hydrogen-bonded turn conformation has also been deduced from the NMR spectra of the linear peptide H-D-Phe-Tyr-Tyr-*o*-Abz-Lys-Val-Phe-Trp-$NH_2$[14] and the cyclic peptide *c*[-*o*-Abz-Gly-]$_2$, for which a bowl-shaped conformation with fast ring inversion has been observed.[15] However, in *c*[-*o*-Abz-$Gly_4$-Phe-] the NMR evidence has been interpreted in terms of absence of the intramolecular hydrogen bond in the *o*-Abz part, but that both NH and CO groups are involved in the stabilization of a γ- and a β-turn. The corresponding cyclic peptide with *m*-Abz shows a considerable higher flexibility.[16] Cyclotripeptide analogues containing *o*-Abz and *o*-(Me)Abz have been prepared. The X-ray diffraction data of

the *N*-methyl analogue indicate the presence of three *cis* peptide bonds and an angle of 63.3° between the (Me)Abz aromatic ring and the best plane of the ring backbone.[17] In the case of the corresponding unmethylated Abz analogue, no cyclotripeptide form was isolated, but rather tautomeric cyclol and dehydrocyclol structures[18–20] were formed. These structures were not observed for the analogues containing the flexible βAla residue, thereby indicating that the rigid structure of *o*-Abz favors transannular interactions.

**Scheme 2** Couplings of Abz (Exemplified for *o*-Abz)




By theoretical conformational analysis the *meta*-isomer (*m*-Abz) has been shown to provide distances between the amino and the carboxy functions that are in the range of those accessible to a *trans* dipeptide.[21,22] The conformational flexibility is quite different, however, and turn structures are excluded.[23]

In the *para*-isomer (*p*-Abz) the distance between the amino and carboxy groups corresponds to a somewhat shortened version of a *trans*-dipeptide,[24] its geometry corresponding to a fixed *cis*-Gly-Gly dipeptide (Scheme 1).

Cyclic peptides containing multiple Abz residues adopt bowl-shape conformations; the rigid amino acid provides a cavity in the cyclic peptide that may be exploited for molecular recognition experiments. Thus, *c*[-Xaa-*m*-Abz-]$_3$ binds phosphomonoesters,[25] while serine protease activity is observed in *c*[-Ser-*m*-Abz-His-*m*-Abz-Asp-*m*-Abz-Ala-*m*-Abz-] following binding of the substrate in the cavity.[26]

All three isomers have been used in biologically active peptides, as replacements of part of the peptide sequence,[22,27] to induce a turn conformation,[14,28,29] and as spacers to construct double-stranded peptides,[30] whereby considerable differences in activity were observed among them. For instance, the replacement of the Met-Gly dipeptide fragment in Ac-CCK-7 by *m*-Abz resulted in an equipotent compound, whereas replacement with the *para*-isomer produced loss of potency but increased selectivity.[22] *m*-Abz was used to replace the Gly residue in RGDS-derived peptide mimetics and to provide the correct distance between the Arg side-chain guanidinium group and the Asp carboxylate.[31,32] Potent and selective antagonism of GPIIb/IIIa was observed, which could not be obtained with either *o*- or *p*-Abz. In contrast, in the inhibititor of p21 *ras* farnesyltransferase, H-Cys-Val-Ile-Met-OH, *p*-Abz was a better replacement for the central dipeptide than its *meta*-isomer.[23] An increased hydrophobicity was obtained in these inhibitors by introducing a 2-phenyl or a 2-naphthyl substituent in *p*-Abz.[33,34]

*for references see p 631*

### 10.15.1.1 Synthesis of Aminobenzoyl Peptides

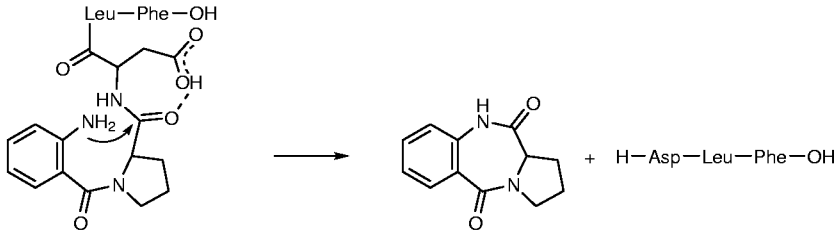
Typical properties of Abz are the weak nucleophilicity of the amino function and the reduced reactivity of the carboxy group, owing to the resonance over the aromatic ring. These characteristics can cause difficulties in peptide bond formation. Advantage can be taken, however, of these properties by the use of N-unprotected Abz in peptide synthesis. Thus, crystalline -OBt **10** (Scheme 2) and -OSu esters of *o*- and *p*-Abz were isolated and coupled to peptide fragments.[24,35] The corresponding -ONp or -OPhth esters were too unreactive. Coupling reactions of N-unprotected *o*-Abz to aliphatic amines using silicon tetrachloride as coupling agent have been reported, but were not applied to peptides.[36,37] In a combinatorial synthesis of 1,4-benzodiazepine-2,5-diones, a set of N-unprotected ring-substituted *o*-Abz analogues has been coupled to resin-bound secondary amino acids using EDC.[38,39] In this application, HATU gave poor results. *o*-Abz can conveniently be coupled as isatoic anhydride (**12**), the equivalent of an NCA amino acid, in the presence of DMAP,[18,19,40] thus generating a free N-terminal *o*-Abz amide or peptide **11**.

Alternatively, the nitro-substituted benzoic acid **15** can be incorporated into the peptide sequence as in **14**, followed by reduction of the nitro to the amino function to give **11**.

The Boc-, Z-, and Fmoc-protected derivatives **13** have been prepared. Using standard coupling protocols they can be applied to conventional solution[8,23,34] or solid-phase[8,32,41–43] peptide synthesis.

The synthesis of peptides containing *o*-Abz can be carried out without any particular problem. However, it was observed that peptides having an N-terminal *o*-Abz-Pro- sequence cyclize at ambient temperature in acidic medium, resulting in a fragmentation of the peptide chain with concomitant release of a [1,4]benzodiazepine **16** (Scheme 3).[5,43] This cyclization is intramolecularly catalyzed by a following Asp residue. In the case of other peptide sequences this reaction does not occur under normal synthetic conditions, since either acid or base catalysis and elevated temperatures are required.[39,41,43]

**Scheme 3** Acid-Catalyzed Cleavage in an *o*-Abz-Pro-Asp Sequence[43]




**16**

#### 10.15.1.1.1 Acylation with Unprotected Aminobenzoic Acid Active Esters

Active OBt esters of *o*-Abz and of *p*-Abz have been prepared by activation of H-Abz-OH with DCC.[24] These esters react with amino acid esters within 10 min to 1 h, whereas the corresponding -ONp esters require 1–10 days. The *ortho*-isomer was consistently more reactive than the *para*-isomer. The -OSu and -OOBt active esters of *o*-Abz are reported to be more reactive than the -OPhth esters.[35] Reaction of *o*-Abz-OSu with H-$Ac_3$c-OH resulted in 65% coupling after 24 h.[24]

**H-*o*-Abz-OBt; Typical Procedure:**[24]
The compound was obtained in 90% yield by coupling *o*-Abz with HOBt using DCC in dioxane as solvent (0.5 M) with addition of pyridine (2 equiv), and was recrystallized (iPrOH); mp 134.5–135.5 °C.

**H-*p*-Abz-OBt; Typical Procedure:**[24]
This compound was obtained in 96% yield using the procedure described above; mp 156–157 °C.

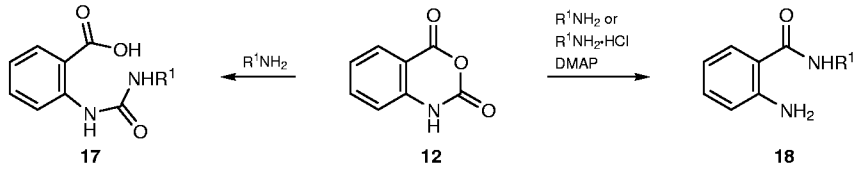
**H-*o*-Abz- and H-*p*-Abz-Xaa Esters; General Coupling Procedure:**[24]
The H-Abz-OBt active ester (1 mmol) was added to a mixture of the amino acid ester hydrochloride (1 mmol) and TEA (0.14 mL) in DMF (3 mL) at rt. After 10 min (H-*o*-Abz-OBt) to 1 h (H-*p*-Abz-OBt), aq $NaHCO_3$ was added. The precipitate was collected, washed ($H_2O$), and recrystallized.

H-*p*-Abz-Phe-OMe; yield: 99%; mp 148–150 °C; $[\alpha]_D^{21}$ –98.6 (*c* 0.5, DMF).

### 10.15.1.1.2 Acylation with Isatoic Anhydride

The use of DMAP in the reaction of isatoic anhydride (**12**) with nucleophiles completely eliminates the competing reaction pathway leading to ureidobenzoic acids **17**, and allows for mild reaction conditions to afford amides **18** (Scheme 4).[40] The yields obtained using isatoic anhydride or its *N*-methyl analogue are around 40–68%.[18,36] A solid-phase procedure has been described without the use of DMAP, but requiring overnight heating at 55 °C in DMF.[42] Substituted isatoic anhydrides can be prepared by treatment of the corresponding anthranilic acid with trichloromethyl chloroformate, or by treatment of the phthalic anhydride with azidotrimethylsilane.[36]

**Scheme 4** Reactions of Isatoic Anhydride[40]




**H-*o*-Abz- or H-*o*-(Me)Abz-Peptide Esters; General Coupling Procedure:**[18]
To a stirred soln of isatoic anydride (**12**) or *N*-methylisatoic anhydride (28.8 mmol) and DMAP (0.48 g, 3.93 mmol) in dry DMF (25 mL), the dipeptide methyl ester hydrochloride (32.0 mmol) and NMM (3.23 g, 32.0 mmol) in dry DMF (25 mL) were added at 0 °C under $N_2$. After 2 h at 0 °C and 15 h at 5 °C, the mixture was partitioned between EtOAc and $H_2O$ and the organic layer washed with brine. Drying and evaporation afforded a residue which was purified by column chromatography (silica gel, $CHCl_3$/$Et_2O$ 9:1).

H-*o*-Abz-Phe-Pro-OMe (**18**, $R^1$ = Phe-Pro-OMe); yield: 6.3 g (50%); oil; $[\alpha]_D^{20}$ –60.0 (*c* 1, $CHCl_3$).

H-*o*-(Me)Abz-Phe-Pro-OMe (**18**, $NH_2$ = HMe; $R^1$ = Phe-Pro-OMe); yield: 5.2 g (40%); mp 110–112 °C; $[\alpha]_D^{20}$ –66.0 (*c* 1, $CHCl_3$).

### 10.15.1.1.3 Coupling of Nitrobenzoic Acid

Nitrobenzoic acid chloride can be coupled to peptides in excellent yield, followed by reduction of the nitro function to the amine using either catalytic (Pd or Pt) hydrogenation[34,44–46] or tin(II) chloride.[33] This latter procedure has been applied to the solid phase.[41,42] A carbodiimide/HOBt coupling methodology has also been reported.[34]

**H-*o*-Abz(5-Me)-$(Gly)_4$-OEt:**[44]
A soln of 5-methyl-2-nitrobenzoic acid (3.54 g, 19.6 mmol) in $SOCl_2$ (50 mL) and a few drops of DMF was stirred at rt for 2 h and taken to dryness under reduced pressure. The resulting crude acid chloride was dissolved in DMF (100 mL) and the soln added over 1 h to a soln of H-$(Gly)_4$-OEt (5.55 g, 15.6 mmol) and TEA (5.4 mL, 39 mmol) in DMF (10 mL), maintained at 0 °C. After stirrring at 0 °C for 1 h and at rt overnight, DMF was removed. Then, $H_2O$ (400 mL) was added and the mixture was stirred and filtered. The solid was washed successively with 1 M HCl, $H_2O$, 5% $NaHCO_3$, $H_2O$, and dried. It was then triturated in MeOH (150 mL), filtered after addition of $Et_2O$ (100 mL), washed ($Et_2O$), and dried under vacuum; yield: 6.17 g (90%); mp 248–250 °C (dec.). A sample of 0.30 g was recrystallized (boiling MeOH); yield: 0.27 g; mp 250–253 °C.

*for references see p 631*

This 2-nitrobenzoyl-peptide **14** (2.49 g, 5.7 mmol) was dissolved in hot DMF (400 mL) and the clear soln cooled to rt. Then, 5% Pd/C (1.5 g) was added and the mixture was hydrogenated in a Parr apparatus overnight. Filtration followed by evaporation of DMF under reduced pressure gave a white solid, which was triturated in hot EtOAc (300 mL). The mixture was concentrated at 70 °C to ca. 30 mL and left at rt overnight. Then, $Et_2O$ (300 mL) was added, and the white solid was filtered on a Büchner, washed ($Et_2O$), and dried; yield 2.04 g (88%); mp 196–198 °C. A sample was crystallized (EtOAc); the white crystals had the same melting point.

### 10.15.1.1.4 Coupling with Urethane-Protected Aminobenzoic Acid

All Boc- or Fmoc-protected Abz isomers are currently commercially available. In solution phase synthesis, active ester coupling as well as coupling using a variety of agents have been reported. Coupling of Z-*p*-Abz-OTcp to a vasopressin fragment has been reported to yield 84% of the benzoylated compound after 2 days.[47] Acylation with Z-*m*- and Z-*p*-Abz-OSu esters were reported in minutes[48] to 18 hours.[5,49] As coupling agents, propanephosphoric acid anhydride (71%),[13] mixed anhydride,[36] 2-chloro-4,6-dimethoxy-1,3,5-triazine,[31] and DCC/HOBt[6,23,28,50] have been reported.

Solid-phase synthesis of Abz-containing peptides has been reported using standard Boc or Fmoc methodology, without any particular difficulty.[22,25,26,29,30,32,43]

**Z-*o*-Abz-OH (13, X = Z):**[51]
H-*o*-Abz-OH (1 g, 7.3 mmol) was dissolved in NaOH soln and treated with Z-Cl (6 g, 35.3 mmol), and the pH was kept basic by addition of 1 M NaOH. The oily layer was separated and treated with $Et_2O$. The precipitated oil solidified on standing in 2 M HCl. Crystallization (AcOH) yielded **13** (X = Z); yield: 1.2 g (61%); mp 140–141 °C.

**Z-*m*-Abz-OH:**[48]
H-*m*-Abz-OH (2.94 g, 21 mmol) was suspended in MeOH (50 mL) and treated with Z-Cl (3.0 mL, 21 mmol). The mixture was sonicated for 10 min. Then, $H_2O$ was added, resulting in the crystallization of the product; yield: 3.90 g (67%); mp 218–219 °C.

**Z-*p*-Abz-OH:**[49]
To a soln of NaOH (8 g, 0.2 mol) and H-*p*-Abz-OH (27.4 g, 0.2 mol) in $H_2O$ (300 mL) at 0–5 °C, Z-Cl (34 g, 0.2 mol) was added dropwise and with stirring (the pH of the mixture was kept at 8–9 by occasional addition of aq 2 M NaOH). After 2 h the mixture was acidified to pH 2–3 with concd HCl. The precipitate was isolated by filtration and recrystallized (BuOH); yield: 40 g (82%); mp 224–226 °C.

**Boc-*o*-Abz-OH (13, X = Boc):**[28]
The pH value of a soln of H-*o*-Abz-OH (4 g, 30 mmol) in DMF (20 mL) was adjusted to 8 by adding TEA. After adding $(Boc)_2O$ (9 g, 41 mmol), the mixture was stirred at rt overnight. The solvent was evaporated and the thick oily residue was dissolved in EtOAc (50 mL). The organic phase was washed with ice-cold 0.02 M $H_2SO_4$ (3×20 mL), then with $H_2O$ (3×20 mL). The EtOAc soln was dried ($Na_2SO_4$) and concentrated under reduced pressure. The thick oily residue became crystalline when kept in a refrigerator; yield: 5.7 g (80%); mp 91–93 °C. Compound **13** (X = Boc) was recrystallized (aq EtOH 1:1); mp 149–150 °C.[9]

Similar procedures in $H_2O$/tBuOH with NaOH as base (62%)[7] and in dioxane/$NaHCO_3$ (63%)[13] have been reported.

**Boc-*m*-Abz-OH:**[36]
This compound was prepared as described below for the *para*-isomer, but starting from H-*m*-Abz-OH (5.0 g, 36.5 mmol), 5% $Na_2CO_3$ soln (25 mL), and $(Boc)_2O$ (19.9 g, 91.5 mmol). After 65 h the mixture was worked up to yield a white solid; yield: 7.48 g (86%).

In a similar procedure (4.5 h reaction time) the compound was recrystallized (MeOH/$CH_2Cl_2$); yield: 1.02 g (60%); mp 196–197 °C. The Boc-protected 2- and 5-methyl-*m*-Abz analogues were prepared similarly.[22]

**Boc-*p*-Abz-OH:**[36]
H-*p*-Abz-OH (10.0 g, 72.9 mmol), 5% $Na_2CO_3$ soln (50 mL), and dioxane (40 mL) were added to a 500-mL round-bottomed flask equipped with a magnetic stir bar and an addition funnel. The soln was cooled in an ice-water bath and a soln of $(Boc)_2O$ (23.8 g, 109 mmol) in dioxane (40.0 mL) was added dropwise. The ice-water bath was removed, the mixture was warmed to rt, stirred for 24 h, and then cooled again

with an ice-water bath. An additional portion of $Boc_2O$ (1.0 equiv) in dioxane (20 mL) was added dropwise. The ice-water bath was removed, and the mixture was allowed to stir for 2 days. The solvent was removed in vacuo, and $H_2O$ (150 mL) was added to the resulting white solid. The pH was adjusted to ca. 2 with 1 M HCl, and the aqueous layer was extracted (EtOAc). The organic layer was dried ($MgSO_4$), filtered, and concentrated to give a white solid. The solid was triturated with hexanes and dried to give the product as a white solid; yield: 14.8 g (86%).

**Boc-*o*-Abz-OSu:**[7]
EtOAc (9.0 mL) was added to a soln of Boc-*o*-Abz-OH (6.31 g, 26.6 mmol) and HOSu (30.1 mmol) in dioxane (28 mL). DCC (30.2 mmol) was portionwise added to the soln, which was kept at 0 °C. The mixture was stirred at 4 °C overnight. Several drops of AcOH were added and after 30 min in the cold the solid was removed by filtration and washed (EtOAc). The solvent from the combined filtrates was removed. The residue was crystallized (iPrOH) to give the product; yield: 6.55 g (74%).

**Z-*m*-Abz-OSu:**[48]
Z-*m*-Abz-OH (0.30 g, 1.1 mmol), HOSu (0.30 g, 2.6 mmol), and EDC (0.36 g, 1.9 mmol) were combined in DMF (10 mL). After stirring for 12 h, the mixture was treated with $H_2O$ and EtOAc. The organic layer was separated, washed (0.1 M HCl, then 1 M $NaHCO_3$), and concentrated to afford the product; yield: 0.36 g (89%).

Coupling of the product to a monosubstituted ethylenediamine in DMF or in iPrOH/MeCN (1:1) was completed within several min.[48]

**Z-*p*-Abz-OSu:**[49]
Z-*p*-Abz-OH (40 g) was transformed into the active ester using DCC (30.2 g) and HOSu (17.0 g) in dioxane at 0 °C, and the product recrystallized (DME); yield: 35 g (65%); mp 235–236 °C.
Coupling of this ester to γ-fluoroglutamic for 18 h at rt in DME gave the product in a 70% yield.[49]

**Active Ester Coupling:**
**Boc-*o*-Abz-Pro-$NH_2$:**[5]
Z-Pro-$NH_2$ (500 mg, 2 mmol) was hydrogenated in MeOH using 5% Pd/C and the resulting H-Pro-$NH_2$ was allowed to react with Boc-*o*-Abz-OSu (0.69 g, 2 mmol) in DMF (10 mL) for 20 h at rt. The mixture was concentrated in vacuo and the residue was suspended in EtOAc, washed successively with 5% $NaHCO_3$, $H_2O$, and 0.5 M HCl, and dried ($Na_2SO_4$). The organic solvent was removed and the final product was triturated with $Et_2O$ to give the product; yield: 0.5 g (82%); mp 86–89 °C.

**Carbodiimide Coupling:**
**Boc-*o*-Abz-Leu-Gln-Pro-NHEt:**[28]
A soln containing H-Leu-Gln-Pro-NHEt·HBr (2.3 g, 5 mmol) in DMF (10 mL) was cooled to 0 °C. Then, the pH value of the soln was adjusted to 8 by adding TEA. After addition of HOBt (750 mg, 5.5 mmol) and DCC (1.1 g, 5.5 mmol), a soln containing Boc-*o*-Abz-OH (1.3 g , 5.5 mmol) in DMF (5 mL) was slowly added dropwise. The resulting mixture was stirred at rt for 48 h. The precipitated DCU was filtered off and the soln was concentrated under reduced pressure. The residue was dissolved in EtOAc (50 mL) and the soln washed with 10% citric acid (3×20 mL), $H_2O$ (3×20 mL), sat. $NaHCO_3$ (3×20 mL), and finally with sat. NaCl (20 mL). The EtOAc layer was dried ($Na_2SO_4$) and concentrated under reduced pressure. The oily residue was thoroughly triturated with $Et_2O$ and the pale yellow, powder-like precipitate was filtered and washed ($Et_2O$) to give the product; yield: 2.4 g (80%); mp 119–121 °C.

## 10.15.1.2 Synthesis of Peptidyl-Aminobenzoic Acid

The weak nucleophilicity of the Abz aromatic amine requires prolonged reaction times or a strong activation of the amino acids to be coupled to Abz. Direct couplings of amino acid mixed anhydrides to N-unprotected *m*-Abz[31] and *p*-Abz have been reported.[52–54] Similarly, amino acid NCAs were coupled to the hydrochlorides of *o*-Abz and *p*-Abz as well as to their ethyl esters in DMF at room temperature for 40 h (81–85%).[55] Originally, it was reported that the amide bond involving the arylamine was unstable under saponification conditions.[52,53] However, later experiments reported successful saponifications of Abz-containing peptide esters.[34,54,55] The instability during Na/liquid $NH_3$ cleavage was confirmed.[47,54]

Slow couplings (48 h) to the aromatic amino group of Abz-peptides are reported and the use of DMAP catalysis results in racemization.[11] Fmoc amino acid chlorides can be coupled in good yield without any racemization.[11] Similarly, Fmoc-Gly-Cl can be coupled to *o*-, *m*-,

*for references see p 631*

and *p*-Abz.[30] Also DCC,[25,26,45,46,55] symmetrical[24] and mixed anhydrides,[36] BOP, BrOP, and PyBrOP[56] coupling methods may be exploited in both solution and solid-phase syntheses. A segment coupling of pGlu-His-Trp-Ser-Tyr-$N_3$ to H-*o*-Abz-Leu-Gln-Pro-NHEt (yield 52%) was completed in 2 h at 0 °C.[28]

### 10.15.1.2.1 Acylation of Unprotected Aminobenzoic Acid

Coupling of Z-Arg($NO_2$)-OH to *m*-Abz using isobutyl chloroformate (DMF, 2.5 h, rt) gives a 31% yield.[31] Also, coupling of Z-Gly-OH or Boc-Gly-OH to *p*-Abz using this same mixed anhydride method results in only 49% yield of Z-Gly-*p*-Abz-OH[52] or 44% of Boc-Gly-*p*-Abz-OH, respectively.[54] These data indicate the need for a stronger -$CO_2H$ activation, such as in Ala NCA, which gave a much higher (85%) yield.[55]

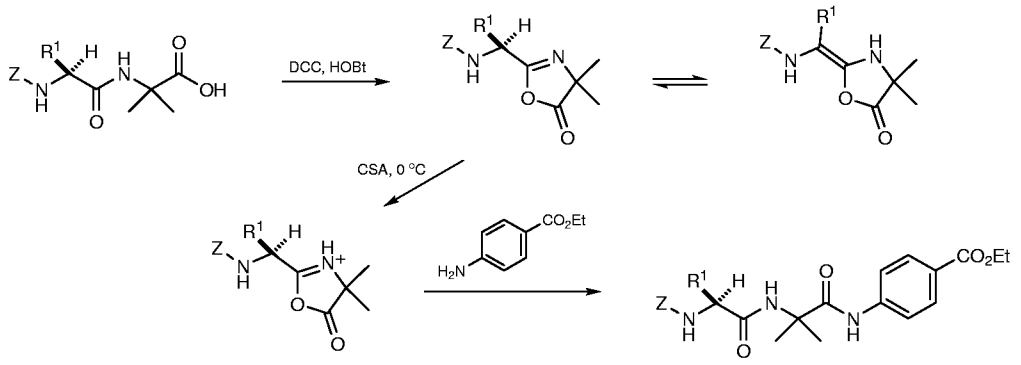
**H-Ala-*o*-Abz-OH · HCl:**[55]
L-Ala · NCA (1.15 g, 0.01 mmol) and H-*o*-Abz-OH (1.73 g, 0.01 mmol) were dissolved in DMF (5 mL) by shaking for some min. The soln, protected from moisture ($CaCl_2$), was left at rt for 40 h, during which time $CO_2$ evolved. After filtration and concentration (35 °C/2 Torr) the residue was dissolved in EtOH. The hydrochloride crystallized on addition of $Et_2O$ and petroleum ether, and was washed ($Et_2O$) and dried ($P_2O_5$); yield: 2.05 g (85%); mp 220–228 °C. Recrystallization from iPrOH/$Et_2O$ or EtOH/$Et_2O$ with addition of petroleum ether gave crystals (80% recovery); mp 228–230 °C, $[\alpha]_D^{23}$ −40.0 (*c* 1, 1 M HCl).

**H-Ala-*p*-Abz-OH · HCl:**[55]
This compound was obtained in 81% yield as described above for H-Ala-*o*-Abz-OH · HCl; recrystallization was from EtOH/$Et_2O$ with addition of petroleum ether; mp 262–265 °C; $[\alpha]_D^{23}$ +28.3 (*c* 1, 1 M HCl).

### 10.15.1.2.2 Coupling of Hindered Amino Acids to Aminobenzoic Acid Esters

Coupling of $C^{\alpha}$-tetrasubstituted amino acids, such as Aib, to Abz presents an extra challenge. Whereas Z-Gly-OH and Z-Ala-OH coupled to H-*p*-Abz-OEt using CDI activation in 95 and 73% yield, respectively, Z-Aib-OH failed to acylate under these conditions.[57] The isobutyl mixed anhydride gave the desired product in 53% yield, but with a urethane side product. Only the mixed anhydride method with methanesulfonyl chloride resulted in 52% yield without any side product after 12 h. However, this coupling was completed in 65–66% yield within 2 h using BroP or PyBroP/DMAP.[56] Coupling of dipeptides containing a C-terminal Aib residue to H-*p*-Abz-OEt using DCC/HOBt resulted in extensive racemization at the N-terminal residue (Scheme 5).[58] Addition of a catalytic amount of camphorsulfonic acid completely suppressed racemization by activation of the intermediate oxazol-5(4*H*)-one so that coupling could be performed at 0 °C.

**Scheme 5** Coupling of Z-Xaa-Aib-OH to H-*p*-Abz-OEt[58]




CSA = camphorsulfonic acid

**Z-Aib-*p*-Abz-OEt:**[56]
To a soln of Z-Aib-OH (3.92 g, 16.5 mmol), H-*p*-Abz-OEt (2.48 g, 15 mmol), and BroP (6.40 g, 16.5 mmol) in $CH_2Cl_2$ (15 mL), DIPEA (2.8 mL, 16.5 mmol), and DMAP (2.02 g, 16.5 mmol) were added. The mixture was stirred at rt for 2 h and concentrated; the residue was redissolved in EtOAc (20 mL), washed with 5% $KHSO_4$ (3×), brine, 5% $NaHCO_3$ (3×), brine, and dried ($Na_2SO_4$), filtered, and concentrated. Purification was performed by column chromatography (silica gel, hexane/EtOAc 6:4); yield: 4.19 g (66%); mp 126–127 °C.

**Suppression of Epimerization by Camphor-10-sulfonic Acid:**[58]
**Z-Ile-Aib-*p*-Abz-OEt:**
To a soln of Z-Ile-Aib-OH (70 mg, 0.200 mmol) in dry $CH_2Cl_2$ (1 mL) at 0 °C was added DCC (41 mg, 0.199 mmol) and, after 15 min, camphor-10-sulfonic acid (7 mg, mmol) and H-*p*-Abz-OEt (37 mg, 0.224 mmol). The reaction was stirred overnight at rt. The precipitated DCU was filtered and $CH_2Cl_2$ was added to the filtrate. The organic phase was extracted with 1 M NaOH (2×), 6 M HCl, and sat. NaCl, dried ($K_2CO_3$), and concentrated to dryness. The residue was dissolved in a small amount of EtOAc, filtered, and crystallized by addition of $Et_2O$ and petroleum ether; yield: 84 mg (85%); mp 79–80 °C; $[\alpha]_D$ +46.4 (*c* 1.1, EtOH).

## 10.15.2 (Aminophenyl)acetic Acid Peptides

The *meta*-isomer of (aminophenyl)acetic acid (*m*-APha, **5**) can be considered as a constrained analogue of a *trans* Gly-Gly dipeptide (Scheme 1). In comparison with 2-(aminomethyl)benzoic acid (*m*-Amb, **2**) the benzene ring is shifted in the chain, which can have important effects on the activity of the pseudopeptide.[59] This was illustrated for inhibitors of farnesyltransferase, where both *m*-APha and *m*-Amb have been used as spacers linking the biologically important Cys and Met residues, the *m*-Amb analogue being an order of magnitude more potent than the corresponding *m*-APha analogue.[59] In that application, α-benzyl-2-(aminophenyl)acetic acid was also used as a mimic for a Gly-Phe dipeptide. Similarly, an alkyl group was introduced at the benzylic position of Boc-*p*-APha-OH through an enolate alkylation reaction.[60] *o*-APha has been used to construct the cyclic peptide *c*[-*o*-APha-Pro-] with the aim of studying its conformation by CD.[50] The APha residue has also been incorporated into inhibitors of GPIIb/IIIa[61] and antihypertensive peptides.[62]

### 10.15.2.1 Synthesis of (Aminophenyl)acetyl Peptides

The *m*- and *p*-APha isomers are commercially available. The synthesis of Boc-*o*-APha-OH and its ethyl and *tert*-butyl esters by carboxylation of the dianion of Boc-*o*-methylaniline has been briefly mentioned.[63] Whereas the carboxy group in APha is expected to react normally, the aromatic amino function is expected to be less nucleophilic owing to the delocalization of the nitrogen lone pair. However, deactivation should be less significant than in *o*- and *p*-Abz. Only a few hints at this conclusion may be found in the literature. Indeed, coupling of Boc-*p*-APha-OSu to an aliphatic primary amine was conducted in DMF for 2 h, affording the amide in a 75% yield.[64]

**Z-*o*-APha-OH:**[50]
H-*o*-APha-OH (3.8 g, 25 mmol) was reacted with Z-Cl under Schotten–Baumann conditions. After acidification with concd HCl, the product was obtained as a crystalline solid; yield: 3.5g (49%); mp 111–112 °C after recrystallization (MeOH).

**Z-*p*-APha-OEt:**[48]
This oily compound was prepared similarly in quantitative yield.

*for references see p 631*

**Boc-*m*-APha-OH:**[61]
This compound was prepared similarly to Boc-*p*-APha-OH described below, from H-*m*-APha-OH, but in $CH_2Cl_2$ using DIPEA and overnight stirring; yield: 22%; mp 105 °C.

**Boc-*p*-APha-OH:**[60]
A soln of H-*p*-APha-OH (2 g, 13.24 mmol) in a mixture of dioxane (26 mL), $H_2O$ (26 mL), and $Na_2CO_3$ (1.4 g in 13 mL $H_2O$) was stirred and cooled in an ice bath. $(Boc)_2O$ (3.12 g, 14.57 mol) was added in one portion, and stirring was continued at rt for 4 h. Dioxane was removed in vacuo and the aqueons layer was chilled, EtOAc was added, and acidified to pH 4 with dil $KHSO_4$. The EtOAc layer was extracted and the product was purified by chromatography (EtOAc/hexane/AcOH 1:1:0.1) to afford Boc-*p*-APha-OH; yield: 2.5 g (76%); mp 153–154 °C.

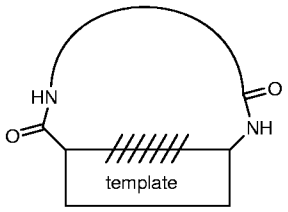
Boc protection of *p*-APha using Boc-ON in TEA/DMF has also been performed.[65]

**Boc-*p*-APha-OSu:**[64]
Boc-*p*-APha-OH (0.42 g, 1.67 mmol) was dissolved in DMF (20 mL) and treated with EDC (0.6 g, 2 mmol) and HOSu (0.35 g, 3 mmol). After 2 h, addition of iced $H_2O$ resulted in the precipitation of a solid. The product was dried in vacuo over KOH; yield: 0.40 g (69%); mp 169–170 °C.

## 10.15.3 (Aminomethyl)benzoic Acid Peptides

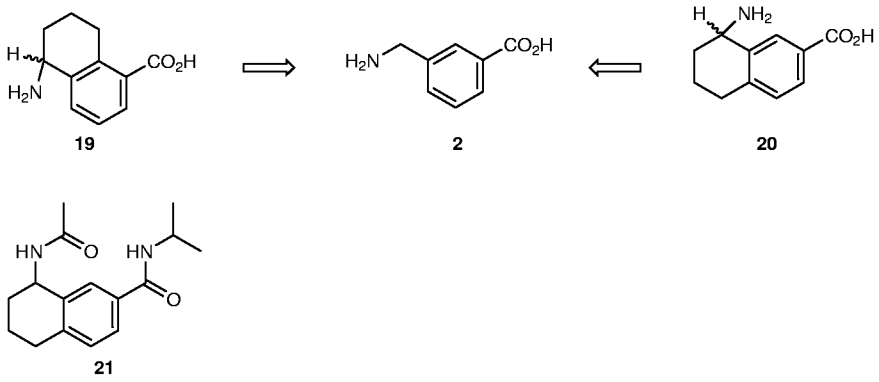
2-(Aminomethyl)benzoic acid (*m*-Amb, **2**) (Scheme 1) represents a constrained mimetic for a Gly-Gly dipeptide, with an alternative positioning of the aromatic ring compared to *m*-APha (**5**).[1] *m*-Amb has been used to replace the Gly-Gly segment in Leu-enkephalin. However, no biological data were reported for H-Tyr-*m*-Amb-Phe-Leu-OH.[1] *o*-Amb, as well as *m*-Amb, have been used as dipeptide mimics in GH secretagogues. Whereas the *m*-Amb analogue was very potent, the *o*-Amb analogue showed decreased activity.[66] In *ras* farnesyltransferase tetrapeptides, *m*-Amb was used as a hydrophobic replacement of the central dipeptide, resulting in potent inhibitors.[23,59,67,68] The *p*-Amb isomer showed a 17-fold reduced potency.[23] Molecular modeling was used to select *m*-Amb as a suitable template to force a tetrapeptide into a loop conformation (Scheme 6), which mimics well that of the binding region in an antibody.[69] *m*-Amb was also used to construct a synthetic antigen[70] and selected to induce the correct conformation in the cyclic ligands for GPIIb/IIIa *c*[-Xaa-Arg-Gly-Asp-*m*-Amb-].[71] Extensive conformational studies using NMR, X-ray diffraction analysis, and molecular modeling on these cyclic peptides indicate that the *m*-Amb residue indeed restricts the conformation to well-defined, fairly rigid macrocycles, which resemble the overall rectangular shape of cyclic hexapeptides.[72]

**Scheme 6** *m*-Amb as a Template to Induce a Loop Conformation in a Cyclic Peptide[69,72]



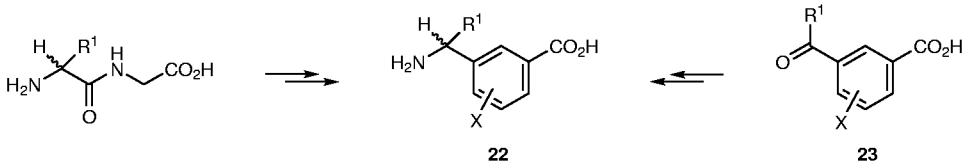

Further constraint of the aminomethyl chain has been performed by incorporating it into 5,6,7,8-tetrahydronaphthoic acids **19** and **20** (Scheme 7).[59,61,73] X-ray diffraction and NMR analysis of the *N*-acetyl-*N'*-isopropylamide **21** revealed that the compound induces a turn structure without the formation of an intramolecular hydrogen bond. However, molecular modeling studies indicate that this amino acid derivative prefers a partially extended conformation.[71,72]

**Scheme 7** Further Constraint of the *m*-Amb Residue by Incorporation into a Tetrahydronaphthoic Acid Ring[59,61,73]




Substituents have been introduced at the aminomethyl carbon (**22**, $R^1$ = Me, Ph) (Scheme 8)[71] in order to mimic the side chain of an amino acid, as well as on the aromatic ring (2-F,[74] 4- or 6-Cl, I, Me, or OMe,[61] 5-$NO_2$[70]). A full experimental description for the preparation of 5-(1-aminoethyl)-2-cyclohexylmethoxybenzamide (**22**, X = 2-$OCH_2Cy$; $R^1$ = Me) has been published.[75]

**Scheme 8** A Substituted *m*-Amb Residue

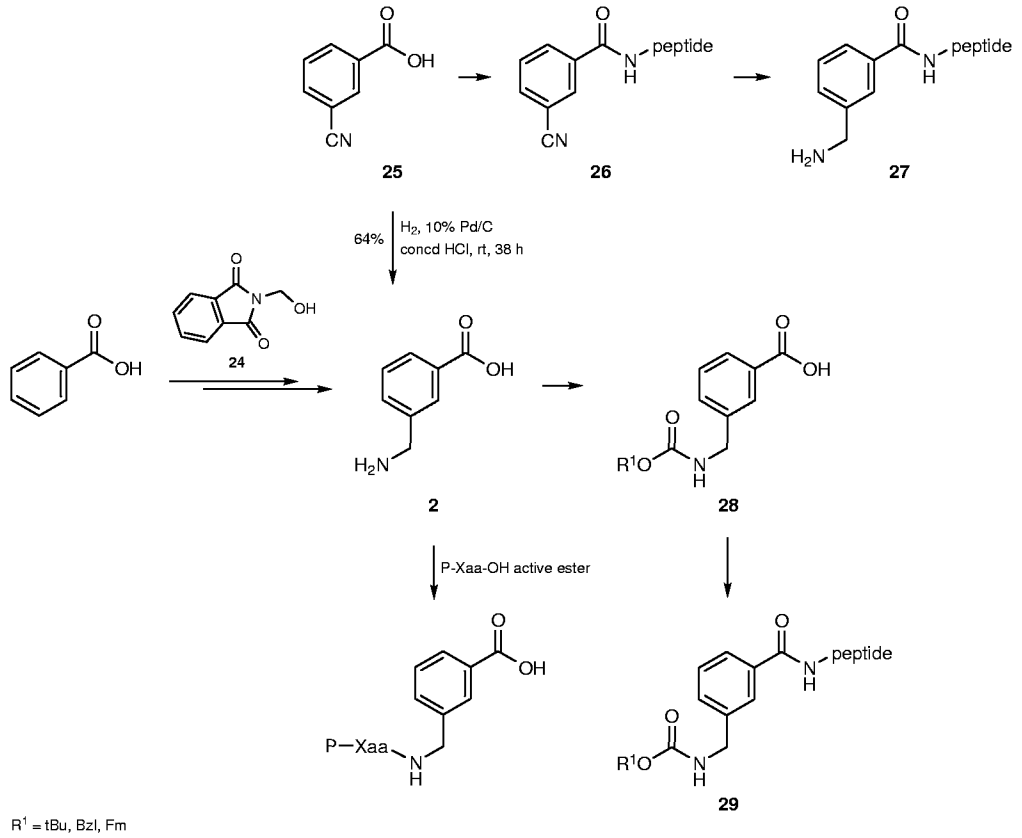



## 10.15.3.1 Synthesis of (Aminomethyl)benzoic Acid Peptides: Construction and Protection of (Aminomethyl)benzoic Acid

*p*-Amb and its Fmoc derivative are commercially available. *m*-Amb (**2**) was prepared by amidomethylation of benzoic acid using *N*-(hydroxymethyl)phthalimide (**24**, Scheme 9) followed by phthaloyl deprotection, with a yield of 36% over two steps.[69] Similarly, an amidoalkylation of various 2- or 4-substituted benzoic acids using *N*-(hydroxymethyl)dichloroacetamide provided the 4- or 6-substituted *m*-Amb analogues in comparable yields.[61] More conveniently, the *meta*- (**27**, **29**) and *para*-Amb peptides were obtained by catalytic hydrogenation of the corresponding cyanobenzoic acid isomers (e.g., **25**), either after coupling to an amino acid or peptide segment **26**,[31] or before coupling (via **2**).[71,76–78] In the latter case, Boc, Fmoc, or Z amino protection, as in **28**, was used before coupling to an amino acid or peptide. Acylation of the amino group of an unprotected Amb has been performed using OPfp or OSu active esters.[70,77,78]

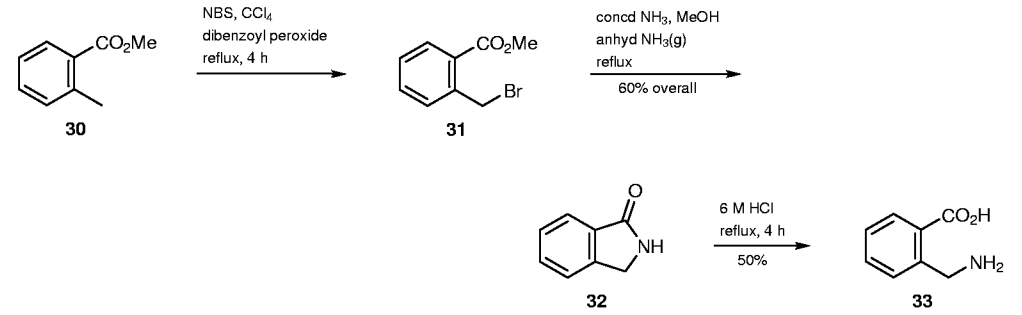
*for references see p 631*

**Scheme 9** Preparations of *m*-Amb



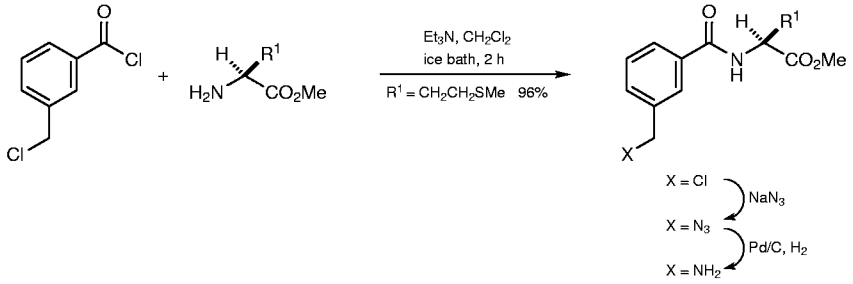

The *o*-Amb isomer **33** has been prepared from methyl 2-methylbenzoate (**30**) by benzylic bromination to give **31**, followed by ammonia substitution and hydrolysis of the resulting phthalimidine **32** (Scheme 10).[61,79]

**Scheme 10** Synthesis of *o*-Amb from Methyl 2-Methylbenzoate[67,79]




Alternatively, a 3- or 4-(chloromethyl)benzoic acid precursor is coupled to an amino acid or peptide, followed by azide substitution and catalytic hydrogenation (Scheme 11).[23,67]

**Scheme 11** Construction of *m*-Amb through a 3-(Chloromethyl)benzoic Acid Derivative[23,67]




### 10.15.3.1.1 3- and 4-(Aminomethyl)benzoic Acid From Cyanobenzoic Acid

**H-*m*-Amb-OH (2):**[71]

3-Cyanobenzoic acid (**25**; 10.0 g, 68 mmol) was dissolved in EtOH (200 mL) by heating to 35–50 °C in a water bath. Concd HCl (6.12 mL, 73 mmol) was added, and the soln was transferred to a 500-mL $N_2$-flushed round-bottom flask containing 10% Pd/C catalyst (1.05 g). The suspension was stirred under an atmosphere of $H_2$ for 38 h, filtered through a sintered glass funnel, and washed thoroughly with $H_2O$. Then, EtOH was removed under reduced pressure, and the remaining aqueous soln was diluted to 250 mL with additional $H_2O$. Then, $Et_2O$ (250 mL) was added, and the suspension was transferred to a separatory funnel. Upon vigorous shaking all solids dissolved, and the aqueous soln was extracted ($Et_2O$), concentrated under reduced pressure to a volume of 150 mL, and lyophilized to give **2**; yield: 8.10 g (64%).

A similar procedure, using $PtO_2$ as the catalyst in THF/iPrOH soln and HCl, has been used.[77]

**H-*p*-Amb-OH:**[76]

This compound was prepared from 4-cyanobenzoic acid (14 g, 0.09 mol) using a Raney Ni catalyzed hydrogenation; after crystallization ($H_2O$), a pink-colored product was obtained; yield: 11.5 g (80%); mp 347–350 °C.

### 10.15.3.1.2 2-(Aminomethyl)benzoic Acid From Phthalimidine

**2-Aminomethylbenzoic Acid γ-Lactam (Phthalimidine; Isoindolin-1-one) (32):**[79]

A mixture prepared by adding NBS (114 g, 0.64 mol) and dibenzoyl peroxide (1.29 g, 5 mmol) to a soln of methyl 2-methylbenzoate (**30**; 90 g, 0.66 mol) in $CCl_4$ (350 mL) was heated under reflux for 4 h. After cooling and filtration of the succinimide, the solvent was removed at the water pump. The residue (125 g), consisting of **31,** was dissolved in MeOH (500 mL). To this soln was added concd $NH_3$ (150 mL) and the system was brought to reflux. Anhyd $NH_3$ was bubbled through. After cooling, the volatiles were removed at the water pump. The solid residue was washed with $H_2O$ and then with $Et_2O$. The phthalimidine **32** was recrystallized ($H_2O$); yield: 48 g (60%); mp 155–156 °C.

**H-*o*-Amb-OH·HCl (33):**[61]

Lactam **32** (3.5 g, 26 mmol) was refluxed in 6 M HCl for 4 h. Then, the mixture was treated with activated charcoal filtered, and concentrated, to dryness. The residue was triturated with acetone and filtered to give **33**; yield: 2.4 g (50%); mp 233 °C.

### 10.15.3.1.3 N-Protection of (Aminomethyl)benzoic Acid

**Fmoc-*m*-Amb-OH (28, $R^1$ = Fm):**[69]

H-*m*-Amb-OH (**2**; 0.75 g, 5 mmol) was dissolved in $H_2O$ (5 mL) (the pH was adjusted to 8 by the addition of TEA). To this soln at 0 °C was added Fmoc-OSu (1.5 g, 4.5 mmol) in MeCN (5 mL). The pH of the soln was monitored and kept at ~8 by addition of TEA. With the pH constant, the mixture was allowed to stir at rt for 3 h. The mixture was then filtered, and the solvent was removed under reduced pressure. The isolated residue was added to vigorously stirred 1 M HCl. The resulting precipitate was collected by filtration, dissolved in EtOAc, and washed with several portions of 1 M HCl. After removal of solvent under reduced pressure, the product was purified by recrystallization (EtOAc/hexane) to give Fmoc-*m*-Amb-OH; yield: 1.27 g (68%); mp 200–201 °C.

*for references see p 631*

**Boc-*m*-Amb-OH (28, $R^1$= tBu):**[71]
H-*m*-Amb·HCl-OH (3.0 g, 16.0 mmol) was dissolved in $H_2O$ (60 mL). To this soln was added a soln of Boc-ON (4.33 g, 17.6 mmol) in acetone (60 mL), followed by TEA (5.56 mL, 39.9 mmol). The soln turned yellow, and the pH was adjusted to 9 by adding additional TEA (1.0 mL, 7.2 mmol). The soln was stirred overnight at rt. Then, acetone was removed under reduced pressure and the remaining aqueous layer was washed with $Et_2O$ (3×), acidified to pH 2 with 2 M HCl, and extracted with EtOAc (3×). The combined organic layers were washed with $H_2O$ (3×), dried ($MgSO_4$), and concentrated to dryness under reduced pressure. The solid material was recrystallized (EtOAc/hexane) to give Boc-*m*-Amb-OH; yield: 2.58 g (64%); mp 123–125 °C.

**Boc-*p*-Amb-OH:**[80]
$(Boc)_2O$ (19.1 g, 87.3 mmol) was added dropwise over a period of 1 h to H-*p*-Amb-OH (12.0 g, 79.4 mmol) in a soln of NaOH (3.18 g, 79.5 mmol), $H_2O$ (20 mL), and *t*-BuOH (30 mL). The mixture was stirred for 18 h. Then, $H_2O$ (200 mL) was added to the mixture which was extracted with hexanes (3×75 mL). The aqueous layer was cooled by adding ice (100 g) and acidified with $H_2SO_4$ to pH 2. The cold aq layer was quickly extracted with EtOAc (2×100 mL). The organic layers were combined and dried, and the solvent was removed in vacuo to leave a white powder; yield: 18.31 g (92%); mp 158.5 °C.

A similar procedure reported a yield of 63%; mp 166 °C.[81]

**Z-*m*-Amb-OH (28, $R^1$ = Bzl):**[1]
Z-OSu (249 mg, 1.1 mmol) in dioxane (2.5 mL) was added to a stirred soln of H-*m*-Amb-OH (151 mg, 1.0 mmol) and TEA (0.28 mL) in $H_2O$ (2.5 mL). After 4–5 h the mixture was filtered, acidified with 1 M HCl, and the precipitate collected and washed with $H_2O$ (284 mg, yield 99%). Derivative **28** ($R^1$ = Bzl) was recrystallized (EtOAc/petroleum ether); mp 138.5–140 °C.

**Z-*p*-Amb-OH:**[76]
H-*p*-Amb-OH (2 g, 13 mmol) was dissolved in 1.25 M NaOH at 4 °C. Z-Cl (3.5 mL, 20 mmol) was added with stirring, which was continued overnight at 4 °C. A white solid precipitate was collected by filtration, triturated with 10% HCl, and dried in vacuo; yield: 2 g (54%).

### 10.15.3.1.4 (Aminomethyl)benzoic Acid from (Chloromethyl)benzoic Acid

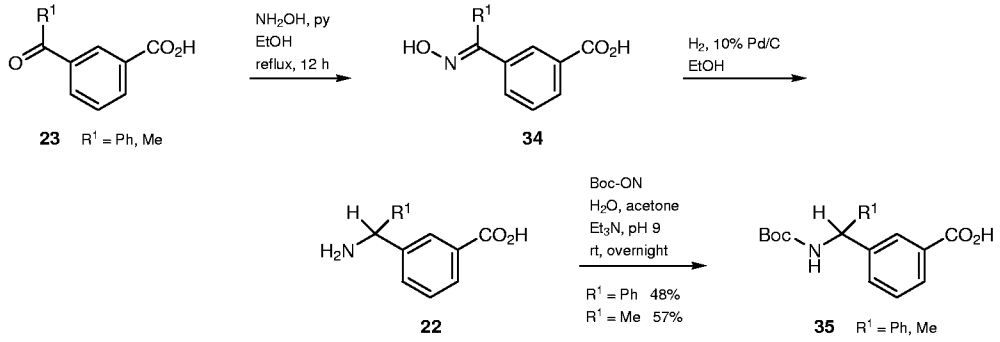
**[3-(Chloromethyl)benzoyl]-Met-OMe:**[82]
To a soln of H-Met-OMe·HCl (2.0 g, 10 mmol) and 3-(chloromethyl)benzoyl chloride (2.08 g, 11.0 mmol) in $CH_2Cl_2$ (50 mL) was slowly added TEA (3.07 mL, 22.0 mmol) at ice-bath temperature for 2 h. The mixture was washed with 0.5 M HCl (2×50 mL), brine (2×50 mL), and $H_2O$ (2×50 mL). The organic phase was dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by flash column chromatography (EtOAc/hexanes 3:7) to give the desired product; yield: 3.03 g (96%); mp 82–83 °C.

**[3-(Azidomethyl)benzoyl]-Met-OMe:**[82]
A suspension of [3-(chloromethyl)benzoyl]-Met-OMe (1.58 g, 5.0 mmol) and $NaN_3$ (1.3 g, 20.0 mmol) in DMSO (40 mL) was stirred at 80 °C for 7 h. The mixture was diluted with $CH_2Cl_2$ (100 mL), washed with brine (70 mL, 2×) and $H_2O$ (70 mL, 2×), and dried ($MgSO_4$). The solvent was removed under reduced pressure to give a slightly yellow residue. The residue was submitted to column chromatography (silica gel, EtOAc/hexanes 3:7) to provide the product azide; yield: 1.45 g (91%); mp 48–49 °C.

**H-*m*-Amb-Met-OMe:**[82]
A suspension of the above azide (1.29 g, 4.0 mmol) and 5% Pd/C (0.2 g) in MeOH (40 mL) was stirred under a $H_2$ atmosphere (1 atm) for 2 days at rt. The catalyst was removed by filtration through Celite (1.5 g), and the solvent was removed under reduced pressure. $H_2O$ (5 mL) was added to the residue and evaporated (2×) to afford the desired product as a foam; yield: 1.12 g (94%). This amine derivative was used immediately in the next coupling reaction.

### 10.15.3.1.5 α-Substituted 3-(Aminomethyl)benzoic Acid Derivatives

**Scheme 12** Synthesis of α-Substituted *m*-Amb Analogues




**(*R,S*)-Boc-*m*-Amb(α-Ph)-OH** (35, **R¹ = Ph**):[61,71]
A soln of 3-benzoylbenzoic acid (**23**, R¹ = Ph; 2.0 g, 9 mmol), hydroxylamine hydrochloride (2.00 g, 29 mmol), and pyridine (2.00 mL, 25 mmol) in EtOH was refluxed for 12 h (Scheme 12). After the usual extractive workup, a white solid was obtained (2.41 g). The product **34** (R¹ = Ph) still contained traces of pyridine, but it was used in the next step without further purification. The crude product (2.00 g, ~8 mmol) was dissolved in EtOH (200 mL), and 10% Pd/C (2.00 g) and concd HCl (1.3 mL, 16 mmol) were added. The mixture was hydrogenated (1550 Torr) for 1 h. The catalyst was filtered off, and the mixture was concentrated. Upon trituration of the residue with $Et_2O$ and drying under vacuum, **22** (R¹ = Ph) was obtained as a solid; yield: 2.12 g (97%).
The amino acid hydrochloride (1.00 g, 4 mmol) was converted into its Boc derivative by a procedure similar to those reported for **28**. Compound **35** (R¹ = Ph) was recrystallized (EtOH/hexane); yield: 0.6 g (48%); mp 190–192 °C.

**(*R,S*)-Boc-*m*-Amb(α-Me)-OH** (35, **R¹ = Me**):[61,71]
This compound was prepared from 3-acetylbenzoic acid (**23**, R = Me; 0.50 g, 3 mmol) using the procedure described above; yield: 0.3 g (57%). Compound **35** (R¹ = Me) was recrystallized (EtOAc/hexane); mp 116–118 °C.

## 10.15.3.2 Synthesis of (Aminomethyl)benzoyl Peptides

Originally, it was reported that coupling of Z-Tyr(OBzl)-*m*-Amb-OH to H-Phe-Leu-OBzl using DCC/HOBt produced an impure compound, leading to the conclusion that care may be necessary in the selection of coupling procedures involving the aromatic carboxy group.[1] However, this conclusion was not confirmed in more recent literature, where active esters (-ONp,[1] -OPfp,[77] -OSu[70]), DIC,[80] and HBTU[78,83] couplings were reported in solution synthesis, and standard conditions were described in solid-phase syntheses using BOP/HOBt[69] or HBTU[61,83] activation.

**Active Ester Coupling:**[77]

**Boc-Asp(OBzl)-*m*-Amb-D-Abu-MeArg($Z_2$)-Gly-OH:**
To a soln of Boc-Asp(OBzl)-*m*-Amb-OH (2.29 g, 5 mmol) and HOTfp (1.01 g, 5.5 mmol) in THF (15 mL) was added DCC (1.03 g, 5 mmol), and the mixture was stirred overnight. DCU was filtered off and rinsed with THF, and the solvent was removed under reduced pressure. To the residue was added a soln of H-D-Abu-MeArg($Z_2$)-Gly-OH (3.56 g, 5 mmol) in DMF (10 mL), followed by DIPEA (2.1 mL, 12 mmol). After stirring at rt for 6 h, 5% citric acid (50 mL) was added followed by EtOAc (80 mL). The organic phase was separated, washed with 1% citric acid and brine, dried ($MgSO_4$), and taken to dryness. The residue was triturated with $Et_2O$/petroleum ether to give the desired acid; yield: 4.8 g (92%); $[\alpha]_D^{25}$ −27.9 (*c* 0.76 , MeOH).

*for references see p 631*

**DIC Coupling:**[80]

**Boc-*p*-Amb-Phe-OMe:**
A soln of H-Phe-OMe·HCl (3.90 g, 18.0 mmol) in $H_2O$ (40 mL) was treated with a soln of $K_2CO_3$ (5.0 g, 36.2 mmol) in $H_2O$ (15 mL). The resulting aqueous soln was extracted with EtOAc (3×50 mL). The EtOAc layers were combined, dried ($MgSO_4$), and concentrated under reduced pressure. To the oily residue, $CH_2Cl_2$ (50 mL) and Boc-*p*-Amb-OH (4.69 g, 18.7 mmol) in DMF (5 mL) were added, followed by DIC (2.82 mL, 18.0 mmol). The soln was stirred for 18 h under argon, $CH_2Cl_2$ (50 mL) was added, and the mixture filtered through Celite. The filtrate was washed with sat. aq $NaHCO_3$ (2×50 mL) and sat. aq $NH_4Cl$ (2×50 mL). The $CH_2Cl_2$ phase was concentrated in vacuo to give the desired ester as a solid (5.7 g) which was purified by flash chromatography ($CH_2Cl_2$/EtOAc 3:1); yield: 2.72 g (52%); mp 129.5–131 °C.

### 10.15.3.3 (Aminomethyl)benzoic Acid: Coupling of Amino Acids or Peptides to (Aminomethyl)benzoic Acid

The amino group in Amb has a normal reactivity since it is not conjugated with the benzene ring. The OSu and OPfp active esters are coupled to C-unprotected *m*-Amb in excellent yield.[1,70,77,78] Also, DCC[67,68] and mixed anhydride[23] couplings to carboxy-protected Amb have been reported.

**Active Ester Coupling:**[77]

**Fmoc-Asp(OtBu)-*m*-Amb-OH:**
Fmoc-Asp(OtBu)OPfp (17.33 g, 30 mmol) was added to the crude *m*-Amb [obtained through catalytic reduction of 3-cyanobenzoic acid (5.88 g, 40 mmol)], followed by DIPEA (11.5 mL, 66 mmol). After stirring the soln at rt for 5 h, 5% citric acid (200 mL) was added, and the soln was extracted (EtOAc, 2×). The combined extracts were washed (brine), dried ($MgSO_4$), and taken to dryness to give a solid which was washed ($Et_2O$/petroleum ether) and dried; yield: 16.3 g (100%); mp 160–163 °C; $[\alpha]_D^{25}$ −14.3 (*c* 0.82, MeOH).

**DCC Coupling:**[68,82]

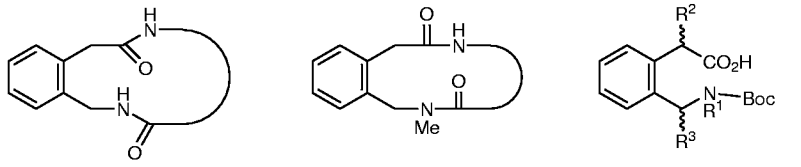
**Z-Cys(Z)-*m*-Amb-Met-OMe:**
H-*m*-Amb-Met-OMe (1.12 g, 3.8 mmol) and Z-Cys(Z)-OH (1.7 g, 4.4 mmol) were dissolved in $CH_2Cl_2$ (50 mL). In an ice bath, DCC (0.91 g, 4.4 mol) and HOBt (0.65 g, 4.8 mmol) were added to this soln, which was stirred overnight at rt. The precipitate was removed and the filtrate was washed with 0.5 M HCl, sat. $NaHCO_3$, and $H_2O$, and dried. After removal of the solvent, the residue was purified by flash chromatography (silica gel, EtOAc/hexane 2:3) to give a pale yellow solid; yield: 2.01 g (75%); mp 103–104 °C.

## 10.15.4 (Aminomethylphenyl)acetic Acid Peptides

2-(Aminomethylphenyl)acetic acid (**4**, *o*-Ampa) (Scheme 1) was originally used as an exocyclic residue in cephalosporin derivatives.[84] Its geometry corresponds to a *cis* Gly-Gly peptide bond mimic. It was therefore incorporated into a cyclic somatostatin analogue, *c*[-*o*-Ampa-Phe-D-Trp-Lys-Thr-].[2,3] NMR studies showed a strong intramolecular hydrogen bond in the *o*-Ampa part, which corresponds to a δ-turn-like conformation **36** (Scheme 13).[85] The aromatic moiety can rapidly flip from an orientation above the plane of the peptide ring to below.[3] N-Methylation changes the conformation to a βVI-turn[86] mimic **37**.[87] A benzyl substituent can be introduced at either benzylic position to provide *cis* Phe-Gly and *cis* Gly-Phe mimics **38**.[87] The (Me)Gly-Phe mimic, when incorporated into the somatostatin cyclic hexapeptide sequence, resulted in a potent analogue.[88] Energy calculations indicated only a limited conformational constraint for this peptidomimetic.[21] The corresponding lactam (tetrahydroisoquinolin-3-one) has been used as a γ-turn mimic, but with loss of receptor

affinity.[89] *m*-Ampa and *p*-Ampa containing somatostatin analogues also showed loss of activity.[90] *o*-Ampa has also been used in cyclic peptide inhibitors for GPIIb/IIIa, but with less success than *m*-Amb.[61]

**Scheme 13** Conformations of *o*-Ampa and Me-*o*-Ampa Peptides[85]



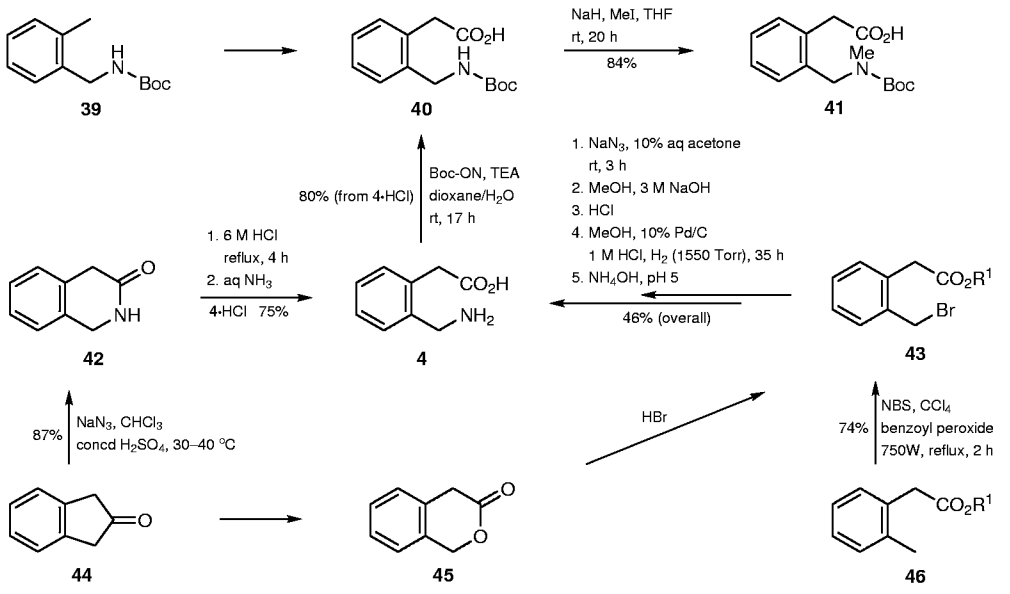
**36** δ-turn-type conformation **37** βVI-turn-type conformation **38** $R^1$ = H, Me; $R^2$ or $R^3$ = Bzl

## 10.15.4.1 Synthesis of (Aminomethylphenyl)acetic Acid Isomers

Boc-*o*-Ampa-OH (**40**) was used in solution or solid-phase peptide synthesis using the usual coupling conditions without any difficulty.[2,88,90]

A variety of methods leading to *o*-Ampa (**4**) have been described (Scheme 14). Indan-2-one (**44**) can be converted into the lactone **45** by a Baeyer–Villiger oxidation using MCPBA,[91,92] followed by treatment with hydrobromic acid to give the bromomethyl ester **43**.[93,94] The latter compound can be more conveniently obtained by radical bromination of (2-methylphenyl)acetic ester **46**.[95] Further azide displacement and reduction yields *o*-Amp, which is then Boc-protected to **40**. Substitution of the bromide by hexamethylenetetramine, followed by hydrolysis, also leads to *o*-Ampa.[96] An alternative pathway uses a Schmidt rearrangement on indan-2-one (**44**) to the lactam **42**, which is hydrolyzed to *o*-Ampa.[61,84,97] A rapid pathway to **40** is the carboxylation of the dianion derived from *N*-Boc-2-methylbenzylamine **39** (yield 90%).[98] Reaction of the same dianion with $(Boc)_2O$ resulted in the corresponding Boc-*o*-Ampa-OtBu ester.[63] N-Methylation of Boc-*o*-Ampa-OH (**40**) to give **41** has been performed.[87,88] Reduction of (2-cyanophenyl)acetic acid was reported, but in view of its lengthy synthesis this is not a preferred method.[84]

**Scheme 14** *o*-Ampa Syntheses



*for references see p 631*

### 10.15.4.1.1 2-(Aminomethylphenyl)acetic Acid From (2-Methylphenyl)acetic Acid

**Methyl [2-(Bromomethyl)phenyl]acetate (43):**[95]
A mixture of methyl (2-methylphenyl)acetate (**46**; 82.0 g, 0.5 mmol), NBS (89.0 g, 0.5 mmol), benzoyl peroxide (1.0 g), and $CCl_4$ (800 mL) was heated under reflux for 2 h while irradiating with a 750 W light source. The precipitated succinimide was then filtered off, the solvent removed, and the residue distilled under reduced pressure to give **43**; yield: 90.1 g (74%).

**H-*o*-Ampa-OH (4):**[95]
A mixture of **43** (90.1 g, 0.371 mol) and $NaN_3$ (26.0 g, 0.40 mol) in 10% aq acetone (750 mL) was stirred for 3 h at rt. The solvent was removed under reduced pressure and the residue was treated with $Et_2O$ (300 mL) and $H_2O$ (200 mL). The $Et_2O$ soln was dried and concentrated, yielding the crude azido ester (74.8 g). This compound was dissolved in MeOH (150 mL), the soln cooled in an ice bath, and treated with 3 M NaOH in MeOH (150 mL). The mixture was left for 1 h at rt, concentrated to dryness, and the residue was dissolved in $H_2O$. The aqueous soln was acidified and the precipitate was collected by filtration, dried, and crystallized (EtOAc/hexane) to give [2-(azidomethyl)phenyl]acetic acid; yield: 49.5 g (70%); mp 116–118 °C.
A mixture of the above acid (9.6 g, 0.050 mmol), 10% Pd/C (25 g) in MeOH (150 mL), and 1 M HCl (50 mL) was hydrogenated at 1550 Torr for 35 h. This mixture was filtered, concentrated under reduced pressure to an approximate volume of 30 mL, and extracted ($Et_2O$). Concentration of the $Et_2O$ soln gave 1–2 g of starting material. The aqueous soln, the pH of which was adjusted to 5 by addition of dil $NH_4OH$, was cooled in an ice bath. The precipitate was filtered, washed with iced $H_2O$, MeOH, and $Et_2O$, and dried ($P_2O_5$) to give **4**; yield: 5.4 g (65%); mp 179–181 °C (dec).

### 10.15.4.1.2 2-(Aminomethylphenyl)acetic Acid From δ-Lactams

**[2-(Aminomethyl)phenyl]acetic Acid δ-Lactam (42):**[84]
To a chilled suspension of freshly prepared indan-2-one (**44**; 46 g, 0.35 mol) and $NaN_3$ (27 g, 0.42 mol) in $CHCl_3$ (75 mL) was added dropwise concd $H_2SO_4$ (75 mL) at such a rate as to maintain the temperature at 30–40 °C under stirring. After the addition was completed and heat evolution ceased, the stirring was continued for an additional 3 h. The mixture was then poured onto crushed ice (1 kg). The $CHCl_3$ layer was separated and the aqueous layer was extracted with $CHCl_3$ (3×200 mL). The $CHCl_3$ layer was combined with the $CHCl_3$ extracts. The resulting soln was dried ($Na_2SO_4$) and concentrated to dryness to give pale yellow prisms of **42**; yield: 44.5 g (87%); mp 156–159 °C.
An alternative procedure using TMS-$N_3$ under identical conditions has also been used.[61] The lactam was purified by sublimation (145 °C/1 Torr), followed by crystallization (benzene); yield: 45%; mp 149–150 °C.

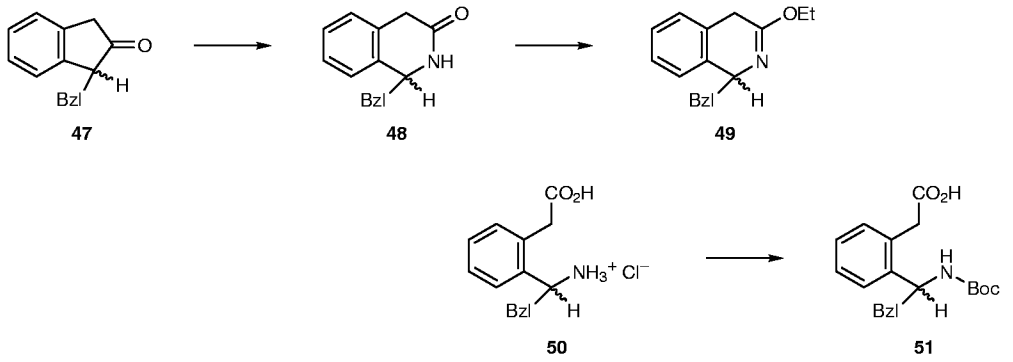
**H-*o*-Ampa-OH (4):**[84]
A mixture of of *o*-Ampa δ-lactam **42** (43.4 g, 0.30 mol) in 6 M HCl (140 mL) was refluxed for 4 h, treated with active carbon (1 g), and concentrated to dryness. The residual oil was triturated with acetone (500 mL) to afford H-*o*-Ampa-OH·HCl (**4**·HCl); yield: 45 g (75%); mp 163–164 °C. The hydrochloride was treated with aq $NH_3$ to give the free amino acid; mp 180–182 °C.

**Boc-*o*-Ampa-OH (40):**[2]
To a soln of H-*o*-Ampa·HCl (12 g, 0.059 mol) in dioxane/$H_2O$ (1:1, 200 mL), TEA (13 mL) and Boc-ON (17.5 g, 0.070 mol) were added. The mixture was stirred at rt for 17 h and concentrated. The residue was dissolved in 5% aq $NaHCO_3$ and the soln extracted ($Et_2O$). The aqueous phase was acidified to pH 3 with $KHSO_4$ and then extracted (EtOAc). The organic phase was dried ($MgSO_4$) and concentrated. The resulting solid was crystallized (cyclohexane/EtOAc); yield: 10.37–11.86 g (70–80%); mp 115.5–117 °C.

### 10.15.4.1.3 Preparation of α- and δ-Substituted 2-(Aminomethylphenyl)acetic Acid

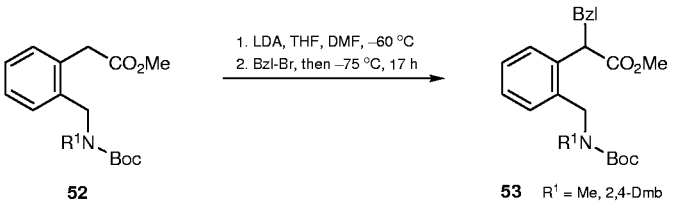
The introduction of a substituent at the aminomethyl carbon (δ-position) was performed starting from the corresponding 1-benzylindan-2-one (**47**), followed by a Schmidt rearrangement to the lactam **48** (yield 55%) (Scheme 15). Ring opening of this compound to **50** is more difficult than that of the unsubstituted compound **42** to **4**, and was achieved through the lactim ether **49** (yield 80%). However, during Boc protection of **50** (yield 20%) the compound mainly cyclized back to **48**.[87]

**Scheme 15** Preparation of δ-Substituted *o*-Ampa[87]




The introduction of a benzyl substituent at the phenylacetic acid methylene carbon (α-position) through the formation of an enolate required double nitrogen protection (see **52**) to avoid lactam formation during this step (Scheme 16). In addition to Boc incorporation, this was either achieved by the use of a 2,4-dimethoxybenzyl temporary protection[99] or by N-methylation.[88] In solid-phase peptide synthesis the 2,4-dimethoxybenzyl group is removed concomitantly with the Boc group during TFA treatment.[99]

**Scheme 16** Preparation of α-Substituted *o*-Ampa[88,89]




**Boc-*o*-(Me)Ampa-OH (41):**[100]
Boc-*o*-Ampa-OH (**40**; 13.5 g, 0.05 mol) was dissolved in dry THF (150 mL) and the soln cooled to 0 °C. MeI (25.5 mL, 0.39 mol) was added, followed by portionwise addition of NaH (6.62 g, 0.57 mol of a 60% dispersion). The mixture was stirred at rt for 20 h, and then poured onto ice. Most of THF was then removed by evaporation and the aqueous phase was first extracted with $Et_2O$ (100 mL), then brought to pH 3 with 3% HCl, and extracted again with EtOAc (3×200 mL). The combined EtOAc phases were washed ($H_2O$), dried ($MgSO_4$), and concentrated to give a yellow solid which was recrystallized (petroleum ether/EtOAc); yield: 11.7 g (84%); mp 97.5–98 °C, FAB-MS *m/z* 280 $[M+1]^+$.

**[(*R,S*)-Boc(Me)N-*o*-Ampa(α-Bzl)-OMe] (53, R¹ = Me):**[100]
Boc(Me)N-*o*-Ampa-OH (**41**; 6.5 g, 23.4 mmol) was dissolved in dry DMF (85 mL), and MeI (3.7 g, 26.4 mmol) and $CsHCO_3$ (4.5 g, 23.4 mmol) were added. After stirring for 17 h at rt, an extraction procedure as described above for **39** gave the methyl ester **52** ($R^1$ = Me) as an oil; yield: 6.3 g (93%). This ester (6 g, 20 mmol) was dissolved in dry DMF (150 mL) and added dropwise to a soln of LDA [prepared from LDA (2.28 g, 22.6 mmol) and 1.6 M BuLi in hexane (15 mL) in dry THF] at −60 °C. After 2 h at −60 °C, a soln of Bzl-Br (3.66 g, 21.5 mmol) in THF (50 mL) was added dropwise. The mixture was stirred at −75 °C for 17 h, then AcOH (1.3 mL) was added and the soln was concentrated in vacuo. The residue was taken up in $CHCl_3$, washed with $NaHCO_3$ (6%) and brine, dried ($MgSO_4$), and concentrated. The resulting oil was purified by chromatography (silica gel, hexane/$CHCl_3$); yield: 6.6 g (87%).

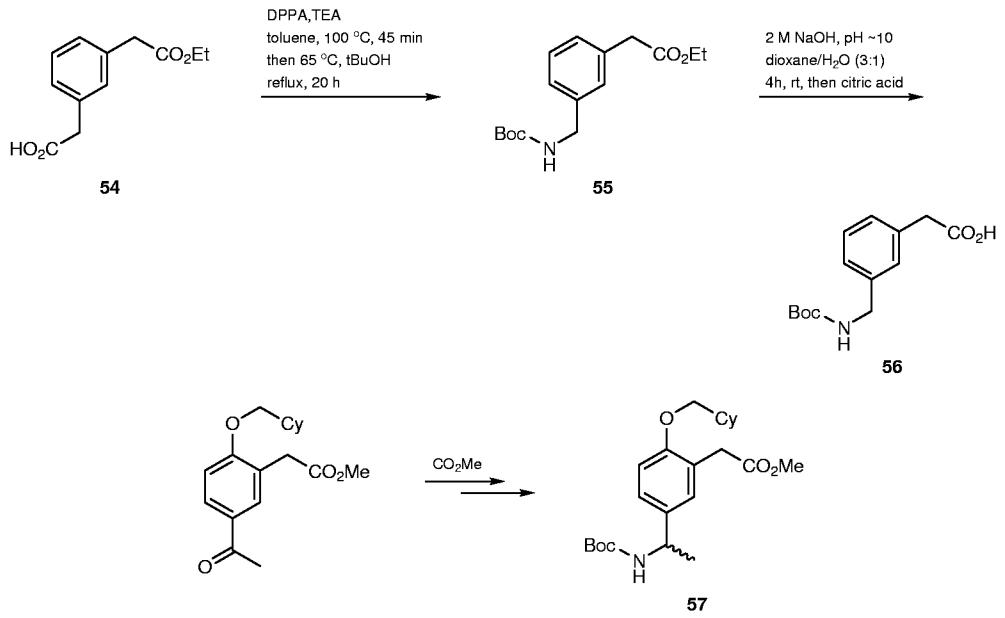
**[(*R,S*)-Boc(Me)N-*o*-Ampa(α-Bzl)-OH]:**[100]
Saponification of **53** ($R^1$ = Me; 1.66 g, 4.3 mmol) in dioxane/$H_2O$ with 20% NaOH for 17 h at rt gave the corresponding acid; yield: 1.33 g (84%).

*for references see p 631*

### 10.15.4.1.4 Synthesis of 3-(Aminomethylphenyl)acetic Acid

Boc-*m*-Ampa-OH (**56**) has been prepared by Curtius rearrangement of the monoester of *m*-phenylenediacetic acid **54** (Scheme 17).[90] It can probably also be obtained more rapidly through the *m*-bromomethyl precursor as described for the *o*- and *p*-isomers. A detailed description of the preparation of [5-(1-aminoethyl)-2-(cyclohexylmethoxy)phenyl]acetic acid methyl ester **57** has been given.[75]

**Scheme 17** *m*-Ampa Synthesis[90]




***m*-Phenylenediacetic Acid Monoethyl Ester (54):**[90]
To a mixture of *m*-phenylenediacetic acid (6.60 g, 34.0 mmol) and its diethyl ester (5.0 g, 20.0 mmol) were added EtOH (2.6 mL), concd HCl (1 mL) and $Bu_2O$ (9 mL). After refluxing for 22 h at 140 °C, the solvent was removed, benzene (10 mL) was added, and the diacid allowed to precipitate. The latter was removed by filtration (1.60 g, 8.3 mmol) and washed with benzene (2×10 mL). The combined filtrates were extracted with sat. $NaHCO_3$ and dried ($MgSO_4$), giving the diester (4.80 g, 13.2 mmol). The aqueous layer was then acidified with 10% HCl and extracted ($Et_2O$). The organic phase was dried ($MgSO_4$) and concentrated, giving the monoethyl ester as an oily product which was used without further purification; yield: 5.48 g (84% from the diacid).

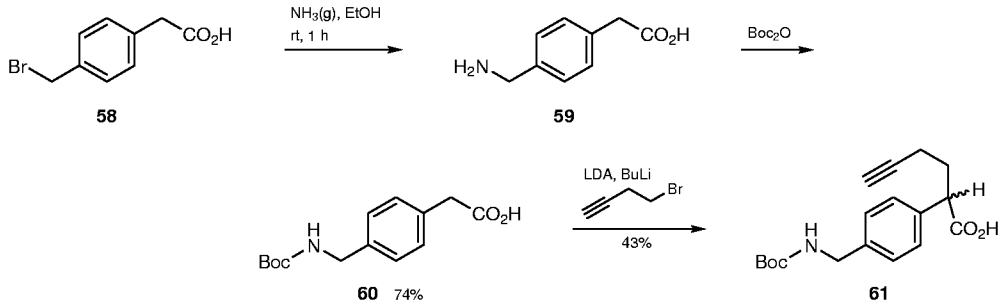
**Boc-*m*-Ampa-OEt (55):**[90]
To a soln of monoester **54** (4.70 g, 21.2 mmol) dissolved in dry toluene (20 mL), TEA (2.15 g, 21.2 mmol) and DPPA (5.83 g, 21.2 mmol) were added. After refluxing for 45 min at 100 °C, during which vigorous $N_2$ production occurred, the mixture was cooled to 60 °C and dry tBuOH (7.83 g, 105.9 mmol) dissolved in dry toluene (5 mL) was added. The soln was refluxed at 115 °C for 20 h. Then, toluene (50 mL) was added and the organic soln was washed with sat. $NaHCO_3$ and 10% citric acid. The organic layer was dried ($MgSO_4$) and concentrated. The crude mixture was purified by chromatography (silica gel, $CHCl_3$/MeOH 98:2) to give an oily product; yield: 3.29 g (53%).

**Boc-*m*-Ampa-OH (56):**[90]
Boc-*m*-Ampa-OEt (**55**; 3.60 g, 12.3 mmol) was dissolved in dioxane/$H_2O$ (3:1, 60 mL), which was brought to pH ~10 with 2 M NaOH. After 4 h at rt (the reaction was followed by TLC), dioxane was removed, the aqueous layer was extracted with $CHCl_3$ and then acidified with 10% citric acid, followed by extraction with $Et_2O$. After drying ($MgSO_4$) and concentration, the acid **56** was obtained as an amorphous solid which was used without further purification; yield: 3.12 g (96%).

### 10.15.4.1.5 Synthesis of 4-(Aminomethylphenyl)acetic Acid

A rapid synthesis of the *p*-isomer starts from commercially available [4-(bromomethyl)-phenyl]acetic acid (**58**, Scheme 18). A direct conversion into the aminomethyl compound **59** using ammonia has been described,[101] as well as a longer one through the azide intermediate.[102] A longer pathway using the Curtius rearrangement as described above for the *m*-isomer has also been reported.[90] The introduction of a substituent at the α-carbon was achieved by treating **60** with LDA, BuLi, and 4-bromobut-1-yne (as the electrophile) to give **61** in 43% yield.[101]

**Scheme 18** *p*-Ampa Synthesis



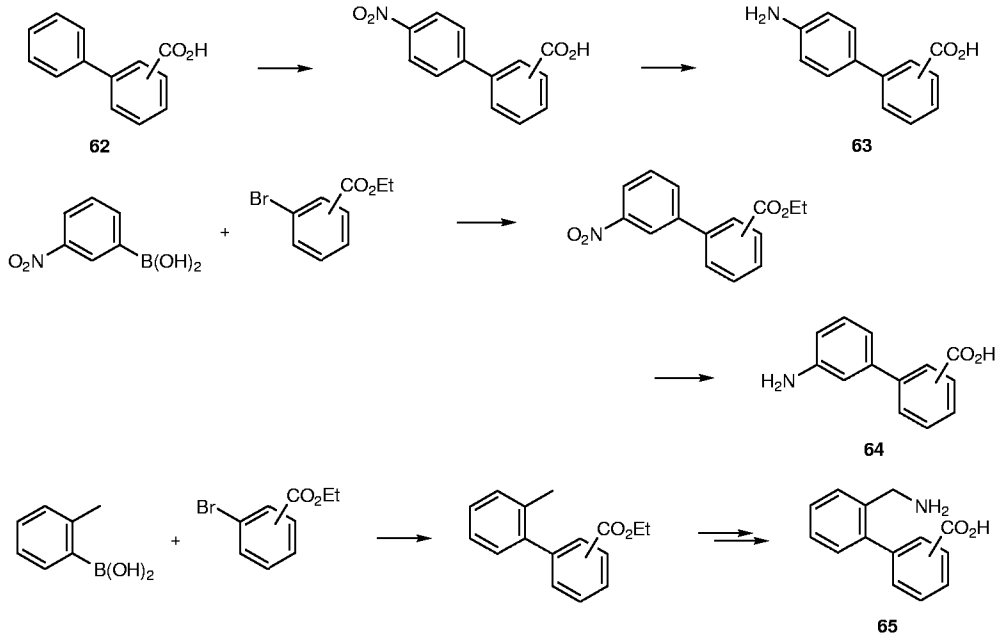

**Boc-*p*-Ampa** (60):[101]
The bromo acid **58** (1.61 g, 7.03 mmol) was added in small portions to EtOH (170 mL) previously saturated with anhyd $NH_3$. The mixture was stirred at rt for 1 h. The volatile components were removed under reduced pressure and the resulting solid was treated with aq 1 M NaOH (21 mL). $H_2O$ was removed and the resulting solid was dissolved in dioxane/$H_2O$ (1:1, 34.5 mL). Then, 1 M NaOH (6.97 mL) was added and the mixture was cooled to 0 °C. After $(Boc)_2O$ (1.67 g, 7.66 mmol) had been added in portions the mixture was allowed to warm to rt and stirred for 12 h. Dioxane was removed, and the aqueous layer was chilled and acidified to pH 3 with dil $KHSO_4$, and extracted (EtOAc). The crude product was purified by chromatography (EtOAc/hexane 1:2) to give **60**; yield: 1.37 g (74%); mp 110 °C.

## 10.15.5 (Aminomethyl)biphenylcarboxylic Acid Peptides

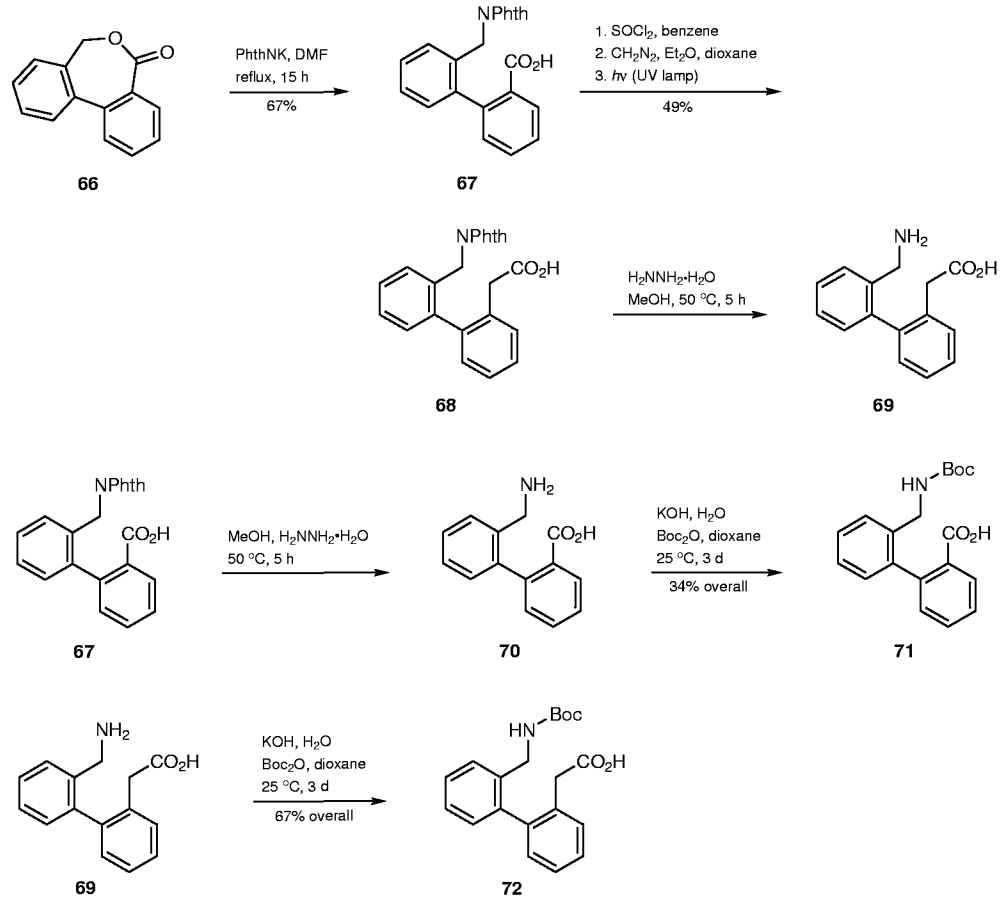
The 2′-(Aminomethyl)biphenyl-2-carboxylic acids **65** and **70** (Schemes 19 and 20) were designed as β-turn mimics.[103] When incorporated into a dimeric cyclic hexapeptide analogue, the slow atropoisomerism of the biphenyl unit allows the chromatographic separation of three isomers. X-ray diffraction analysis and NMR studies were performed, which indicated for one of the isomers a conformation not characterized by the double β-turn structure of cyclic hexapeptides.[103] Similarly, the homologous compound 2′-(aminomethyl)biphenyl-2-acetic acid (**69**), when incorporated into a dimeric hexapeptide analogue, showed three atropoisomers.[104] In larger rings these moieties still display slow atropoisomerism, but evidence for a β-turn/β-sheet conformation is present.[105] 2′-Aminobiphenyl-3-carboxylic acid and 3′-(aminoethyl)biphenyl-2-propanoic acid have been shown to nucleate β-sheet structures.[106,107] Various amino- and aminomethyl-biphenylcarboxylic acid isomers **63**, **64**, **65**, **69** have been used for the construction of libraries of partially peptidic vitronectin antagonists.[108] Cys-[4′-aminobiphenyl-3-carboxylic acid] is a potent and selective inhibitor of farnesyltransferase, in contrast to its 4-carboxylic acid analogue.[109] Since the biphenyl structure cannot take up a β-turn conformation, these data indicate that a turn conformation is not required for FTase inhibition. 5-Aminobiphenyl-2-carboxylic acid (or 4-amino-2-phenylbenzoic acid) has been used as farnesyl- or geranylgeranyltransferase inhibitors, but rather these compounds are considered as *p*-Abz analogues.[33,34]

*for references see p 631*

**Scheme 19** General Pathways towards Substituted Aminobiphenylcarboxylic Acids[108]




**Scheme 20** Synthesis of 2′-(Aminomethyl)biphenyl-2-carboxylic Acid and -acetic Acid



### 10.15.5.1 Synthesis of Derivatives of (Aminomethyl)biphenylcarboxylic Acids

4′-Aminobiphenyl-2- and -4-carboxylic acids **63** were prepared by nitration of biphenyl-2- and -4-carboxylic acids **62** (yields 14 and 36%, respectively, the 2′-$NO_2$ isomer being the major compound), followed by catalytic hydrogenation (Scheme 19).[108] The isomeric 3′-aminobiphenyl-2-, -3-, and -4-carboxylic acids **64** and the 2′-aminomethyl homologues **65** were prepared by Suzuki coupling.[108]

2′-(Aminomethyl)biphenyl-2-carboxylic acid (**70**) and -acetic acid (**69**) have been prepared from dibenzo-substituted oxepin-2-one **66** by potassium phthalimide ring opening. Further Arndt–Eistert homologation of **67** provided **68**.[104]

**Dibenzo[*c*,*e*]oxepin-5(7*H*)-one** (66):[104]
Sublimed diphenic anhydride (22.4 g, 0.1 mol) was suspended in dry DMF (100 mL) and cooled to 0 °C. $NaBH_4$ (4.0 g , 0.1 mol) was added over 10 min. After stirring for 1 h at rt the mixture was poured into 6 M HCl (100 mL) and diluted with $H_2O$ (300 mL). The precipitate was filtered off, washed ($H_2O$), dried, and purified by sublimation (140 °C/0.4 Torr); yield: 19.1 g (91%); mp 132–134 °C.

**2′-(Phthalimidomethyl)biphenyl-2-carboxylic Acid** (67):[104]
Dibenzo[*c*,*e*]oxepin-5(7*H*)-one (**66**; 10.5 g, 50 mmol) and potassium phthalimide (10.0 g, 50 mmol) were refluxed in dry DMF (100 mL) for 15 h. After cooling to rt the precipitate was filtered off, suspended in AcOH (50 mL), and stirred for 1 h. Then, $H_2O$ (200 mL) was added and the precipitate was filtered off. A crystalline compound was obtained by recrystallization (MeOH); yield: 12.0 g (67%); mp 197–199 °C.

**2′-(Phthalimidomethyl)biphenyl-2-acetic Acid** (68):[104]

**CAUTION:** *Diazomethane is highly toxic and irritating. It is also a detonator and appropriate precautions must be taken when using this reagent.*

2′-(Phthalimidomethyl)biphenyl-2-carboxylic acid (**67**; 10.0 g, 28 mmol), $SOCl_2$ (20 mL), and dry benzene (50 mL) were stirred under reflux for 1 h. Solvent and excess of $SOCl_2$ were removed in vacuo. The residue was taken up with dry benzene (20 mL) and concentrated again, leaving an oil which solidified on treatment with petroleum ether (10 mL); mp 123–125 °C. Then the petroleum ether was decanted and the acid chloride from **67** was dissolved in dry dioxane (100 mL). This soln was added with a dropping funnel to a soln of diazomethane in $Et_2O$ [prepared from *N*-nitrosomethylurea (10.0 g)] at 0 °C and stirred overnight at rt. After removing $Et_2O$ and excess diazomethane, the residue was diluted with dioxane (200 mL) and $H_2O$ (20 mL). This soln was irradiated with a mercury UV lamp until $N_2$ evolution stopped. The solvent was removed in vacuo, the remaining oil was dissolved in EtOAc (100 mL), and extracted with sat. $NaHCO_3$ (4×300 mL). The aqueous soln was acidified with HCl and extracted with $CH_2Cl_2$ (2×50 mL). The organic layer was dried ($MgSO_4$) and removal of the solvent gave **68**; yield: 5.1 g (49%); mp 105 °C (dec).

**2′-{[(*tert*-Butoxycarbonyl)amino]methyl}biphenyl-2-carboxylic Acid** (71):[105]
A soln of 2′-(phthalimidomethyl)biphenyl-2-carboxylic acid (**67**; 30 mmol) in MeOH/$H_2NNH_2 \cdot H_2O$ (10:1, 110 mL) was heated to 50 °C for 5 h to give **70**. A precipitate (phthaloyl hydrazide) was filtered off and the filtrate concentrated. Then, KOH (1.7 g) in $H_2O$ (50 mL) was added, followed by a soln of $(Boc)_2O$ (6.6 g, 30 mmol) in dioxane (50 mL). The mixture was stirred 3 days at 25 °C. The solvent was concentrated to 50 mL and cooled to 0 °C, $Et_2O$ (50 mL) was added, and the mixture acidified to rouge red with HCl. After phase separation, the aqueous layer was extracted with $Et_2O$ (50 mL). The combined organic layers were dried ($MgSO_4$) and concentrated to dryness. Recrystallization of the crude product ($CH_2Cl_2$/petroleum ether) afforded **71**; yield: 3.3 g (34%); mp 129–132 °C.

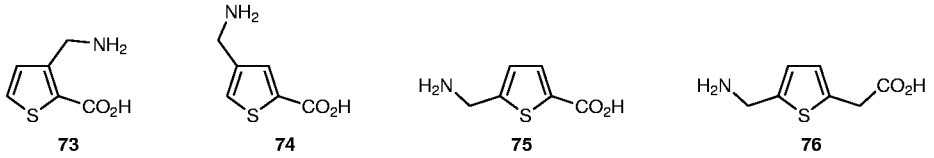
**2′-{[(*tert*-Butoxycarbonyl)amino]methyl}biphenyl-2-acetic Acid** (72):[105]
This compound was synthesized as described above for **71**, with 2′-(phthalimidomethyl)biphenyl-2-acetic acid (**68**; 11.04 g) as the starting material. Concentration of the $Et_2O$ layers yielded an oily material; yield: 6.8 g (67%). The DCHA salt was used for the elemental analysis.

*for references see p 631*

## 10.15.6 (Aminomethyl)thienylcarboxylic Acid Peptides

The various (aminomethyl)thiophene-2-carboxylic acid isomers **73–75** (Scheme 21) have been used in peptidomimetic GH secretagogues with slight loss of potency.[66] 5-(Aminomethyl)thiophene-2-carboxylic acid (**75**) was used as an alternative for *m*-Amb to provide potent cyclic peptide GPIIb/IIIa receptor antagonists, as were the corresponding furan, pyrrole, and pyridine analogues.[74] 5-(Aminomethyl)thiophene-2-acetic acid (**76**) was designed to investigate its β-turn-inducing properties. However, in a cyclic peptide, no evidence for this property was found. In contrast, a conformation was induced that is different from those observed in cyclopentapeptides of comparable ring size.[110]

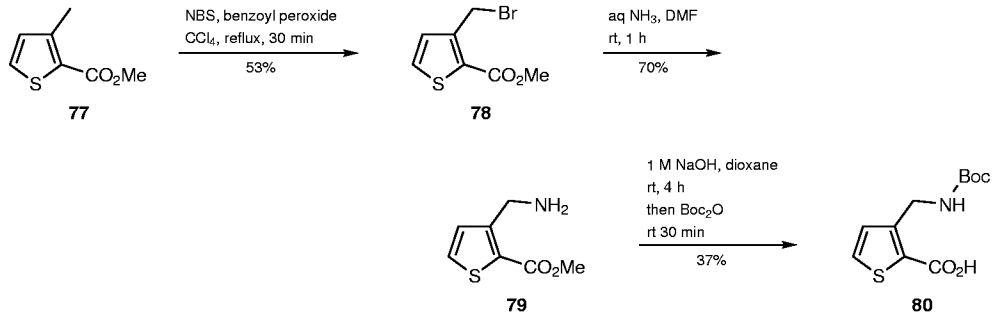
**Scheme 21** (Aminomethyl)thienylcarboxylic and -acetic Acids




### 10.15.6.1 Synthesis of Derivatives of (Aminomethyl)thienylcarboxylic Acids

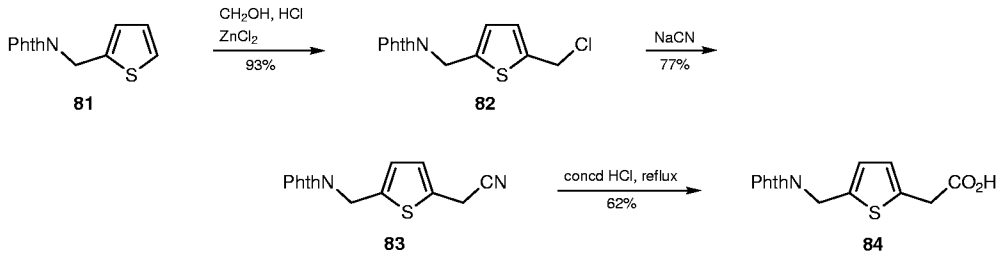
The (aminomethyl)thiophene-2-carboxylic acid isomers were prepared from the corresponding methylthiophene-2-carboxylic acid esters, as exemplified for the 3-(aminomethyl) isomer **79** (Scheme 22).[66] The 4-(aminomethyl) isomer was prepared in the same way, but with a low yield for the bromination step (7%).[66,111] For the 5-(aminomethyl) isomer the bromomethyl compound was reacted with sodium azide, followed by catalytic reduction.[74] The corresponding furan and pyridine analogues were prepared similarly.

**Scheme 22** Synthesis of 3-(Aminomethyl)thiophene-2-carboxylic Acid[66]




The 5-(aminomethyl)thiophene-2-acetic acid isomer **76** was prepared by chloromethylation in position 2 of thiophene followed by nucleophilic displacement with phthalimide to afford **81**, and a second chloromethylation in position 5 to give **82** (Scheme 23). Subsequent reaction with cyanide gave **83**, the hydolysis of which afforded the phthalide-protected amino acid **84** which was coupled to H-Ala-Ile-Gly-OMe using propanephosphoric anhydride, followed by hydrazine N-deprotection.[110]

**Scheme 23** Synthesis of 5-(Aminomethyl)thiophene-2-acetic Acid




**Methyl 3-Methylthiophene-2-carboxylate (77):**[111]
3-Methylthiophene-2-carboxylic acid (10.0 g, 70.3 mmol) was suspended in MeOH (14 mL). Concd $H_2SO_4$ (0.75 mL, 14.1 mmol) was added and the mixture was heated to reflux for 4 h, cooled to rt, poured into $H_2O$ (400 mL), and extracted with $CH_2Cl_2$ (3×70 mL). The combined organic layers were washed with sat. $NaHCO_3$ (150 mL) and dried ($MgSO_4$). The solvent was removed in vacuo. The crude product was purified by flash chromatography (silica gel, EtOAc/heptane 1:8); yield: 5.40 g (49%).

**Methyl 3-(Bromomethyl)thiophene-2-carboxylate (78):**[111]
A mixture of **77** (4.00 g, 26.1 mmol), NBS (4.64 g, 26.1 mmol), and benzoyl peroxide (0.32 g, 1.31 mmol) in $CCl_4$ (160 mL) was heated to reflux for 30 min. The solvent was removed in vacuo. The crude product was purified by flash chromatography (silica gel, EtOAc/heptane 1:8); yield: 3.16 g (53%).

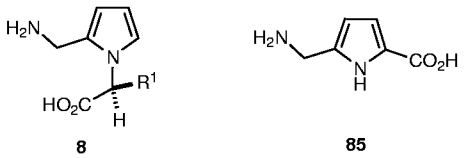
**Methyl 3-(Aminomethyl)thiophene-2-carboxylate (79):**[111]
A soln of **78** (1.20 g, 5.1 mmol) in DMF (10 mL) was added to a soln of 25% aq $NH_3$ (18 mL, 240 mmol) in DMF (50 mL). The mixture was stirred for 1 h and then the solvent was removed in vacuo. The crude product was purified by flash chromatography (silica gel, $CH_2Cl_2$/MeOH/25% aq $NH_3$ 100:10:1); yield: 610 mg (70%).

**3-{[(*tert*-Butoxycarbonyl)amino]methyl}thiophene-2-carboxylic Acid (80):**[111]
1 M NaOH (6.1 mL, 6.1 mmol) was added to a soln of **79** (500 mg, 2.92 mmol) in dioxane (6 mL) and $H_2O$ (3 mL). The mixture was stirred for 4 h at rt. A soln of $(Boc)_2O$ (700 mg, 3.2 mmol) in dioxane (5 mL) was added. The mixture was stirred for 30 min. 5% $KHSO_4$ (150 mL) was added and the mixture was extracted with EtOAc (5×40 mL). The combined organic layers were washed ($H_2O$, 150 mL) and dried ($MgSO_4$). The solvent was removed in vacuo. The product was crystallized (EtOAc/heptane) to give **80**; yield: 275 mg (37%); MS found 257.071864, calcd 257.07215.

## 10.15.7 (Aminomethyl)pyrrolecarboxylic Acid Peptides

5-(Aminomethyl)pyrrole-2-carboxylic acid (**85**, Scheme 24) was used with good results as a spacer in the synthesis of cyclic peptide GPIIb/IIIa receptor antagonists.[74] The 1,2-substituted pyrrole **8** represents a *cis* peptide bond surrogate in which the $R^1$ side chain can easily be varied. However, no application in biologically relevant peptides have been reported to date.

**Scheme 24** 2-(Aminomethyl)pyrrolecarboxylates



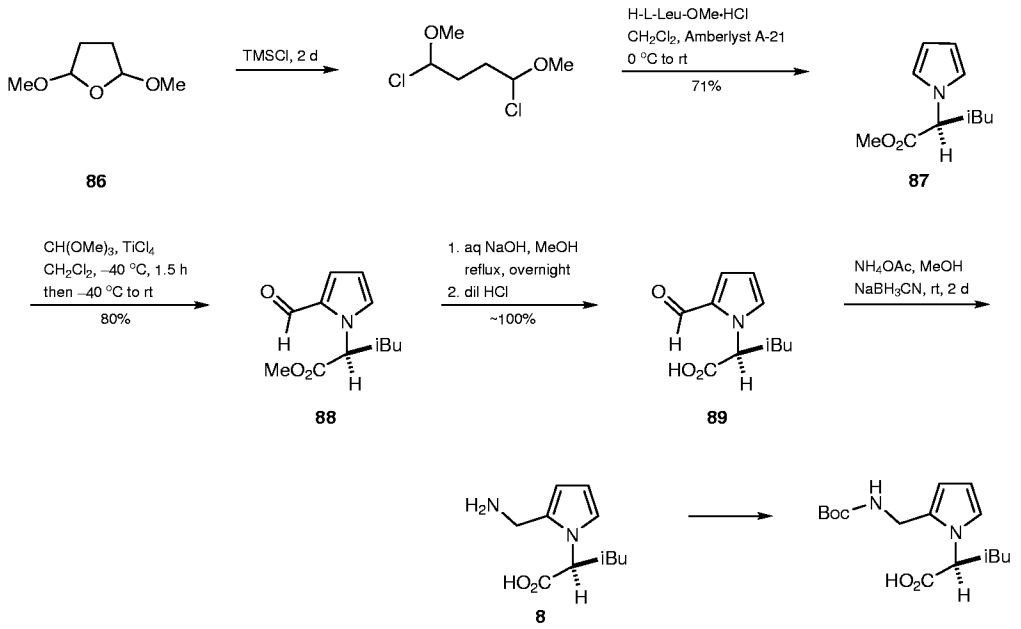

*for references see p 631*

## 10.15.7.1 Synthesis of Derivatives of (Aminomethyl)pyrrolecarboxylic Acids

5-(Aminomethyl)pyrrole-2-carboxylic acid (**85**) was prepared from methyl 5-formylpyrrole-2-carboxylate by conversion to the oxime and subsequent reduction, Boc protection, and saponification.[74]

The synthesis of the *cis* peptide bond surrogate **8** is shown in Scheme 25.[4]

**Scheme 25** Preparation of 2-(Aminomethyl)pyrrole-1-acetic Acid Derivatives[112–114]




**Methyl (2*S*)-4-Methyl-2-(1*H*-pyrrol-1-yl)pentanoate (87):**[112]

H-L-Leu-OMe·HCl (1.82 g, 10 mmol) in dry $CH_2Cl_2$ (50 mL) was stirred for 10 min. Amberlyst A-21 resin (2.5 g) was added. Then, 1,4-dichloro-1,4-dimethoxybutane (2.26 g, 12 mmol) [obtained from 2,5-dimethoxytetrahydrofuran (**86**) by treatment with TMSCl for 2 days] was added dropwise at 0 °C under $N_2$. After 30 min, 5 g of the resin were again added to the mixture. The mixture was allowed to warm to rt. After overnight stirring, the mixture was filtered on a fine-sintered glass filter and the filtrate taken to dryness. The crude product was purified by TLC-mesh column chromatography (hexane/EtOAc 7:3); yield: 1.39 g (71%); $R_f$ 0.48.

**Methyl (2*S*)-2-(2-Formyl-1*H*-pyrrol-1-yl)-4-methylpentanoate (88):**[112]

To trimethyl orthoformate (0.127 g, 1.2 mmol) in dry $CH_2Cl_2$ (5 mL) was added $TiCl_4$ (0.21 g, 1.2 mmol) dropwise at –40 °C under $N_2$. The pyrrole derivative **87** (0.195 g, 1 mmol) in dry $CH_2Cl_2$ (2 mL) was then added dropwise over 5 min. After the mixture was stirred at –40 °C for 1.5 h, it was allowed to warm to rt. The soln was quenched with $H_2O$ (1 mL) and extracted with $Et_2O$. The organic layer was dried ($MgSO_4$) and taken to dryness. The crude product was separated by TLC-mesh column chromatography (hexane/EtOAc 7:3) to give the 3-formyl isomer (10%) and the 2-formyl isomer **88**; yield: 0.179 g (80%).

**(2*S*)-2-(2-Formyl-1*H*-pyrrol-1-yl)-4-methylpentanoic Acid (89):**[112]

To compound **88** (0.112 g, 0.5 mmol) in MeOH (5 mL) was added aq NaOH soln [1.2 equiv, NaOH (24 mg) $H_2O$ (1 ml)]. After the mixture was refluxed overnight, it was cooled to rt and then poured into $H_2O$ (10 mL). This soln was washed with $Et_2O$ and the aqueous layer acidified with dil HCl. The organic layer was dried ($MgSO_4$), filtered, and concentrated to give a pale yellowish solid; yield: 0.105 g (ca. 100%); mp 74–77 °C.

**(2*S*)-2-[2-(Aminomethyl)-1*H*-pyrrol-1-yl]-4-methylpentanoic Acid (8, R$^1$ = iBu):**[113]

To a soln of **89** (R$^1$ = iBu; 767 mg, 3.5 mmol) and $NH_4OAc$ (2.120 g, 27.5 mmol) in MeOH (9.2 mL), $NaBH_3CN$ (113 mg, 1.8 mmol) was added under stirring. The resulting soln was stirred at rt for 2 days

and the solvent was evaporated in vacuo. Then, $H_2O$ (100 mL) was added and the mixture was washed with EtOAc (3×). The aqueous layer was freeze dried to remove the $NH_4OAc$ to give **8**; yield: 466 mg (64%); mp 195 °C (dec.).

The **8** ($R^1$ = Me) Ala analogue was prepared in a similar way and crystallized (MeOH/EtOAc); mp 170 °C.

For the preparation of the Gly analogue **8** ($R^1$ = H) pyrrole-2-carboxaldehyde was reacted with bromoacetic acid followed by reduction with $NaCNBH_3$ as described above; mp 170–172 °C.[114]


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# 11 Combined Side-Chain- and Main-Chain-Modified Peptides

C. Toniolo

Sections 9 and 10 deal with side-chain- or main-chain-modified peptides, respectively. In Section 11 combined side-chain- and main-chain-modifications of peptides are discussed. More specifically, two families of peptidomimetics are treated in detail.

The first family is represented by peptides based on α,β-dehydro α-amino acids in which a C=C bond confers rigidity and electronic Y-conjugation on the system (Section 11.1). Peptides rich in these amino acids are widespread in nature, particularly as antibiotics, and possess interesting conformational properties.[1–11] In addition to the *E* and *Z* configurational isomers, occurring when two different atoms (substituents) on $C^{\beta}$ are present, fully-extended, characteristically flat conformations or β-turns and $3_{10}$/α-helices may be induced by α,β-dehydro α-amino acids.

In Section 11.2 peptides based on a variety of aromatic heterocycles are discussed.[12–21] Some of them are natural products with bioactivities ranging from antibiotics to double stranded DNA intercalators. Heterocyclic systems with five or six atoms, characterized by a single or a combination of N, O, and S heteroatoms, are described. In selected cases reduced, nonaromatic rings are also covered.


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# 11.1 Synthesis of Peptides Based on α,β-Didehydro-α-amino Acids

R. M. Joshi and V. S. Chauhan

Dehydro-α-amino acids are those α-amino acids containing one or more double bonds not involved in the aromatic nucleus, such as the benzene, indole, or imidazole ring. The most common of these are the α,β-didehydro-α-amino acids, which are also constituents of several naturally occurring compounds. Extensive early chemical studies on α,β-didehydro-α-amino acids were carried out by Bergmann and co-workers[1] and continued later in the 1960s by Greenstein and associates.[2] This field of research received little attention for a long time except in studies dealing with chemical fragmentation of peptides and proteins at the level of Cys and Ser residues. However, due to improved methods of isolation and characterization, a large number of α,β-didehydropeptides have been identified. The last two decades have witnessed vigorous activity and interest in α,β-didehydro-α-amino acids and their incorporation in peptides, resulting in several new methods for their synthesis.[3–7]

A large number of α,β-didehydro-α-amino acids have been identified as constituents of relatively low molecular weight cyclic compounds from microbial sources. However, the presence of α,β-didehydroalanine in bacterial as well as in mammalian histidine ammonia lyase and in phenylalanine ammonia lyase shows that the occurrence of α,β-didehydro-α-amino acids is not limited to small molecules alone.[8] These residues are incorporated in natural sequences by posttranslation modification. α,β-Didehydro-α-amino acids have also been postulated to be precursors in the biosynthesis of several heterocyclic metabolites including penicillin and cephalosporin.[9] Other well-known compounds containing α,β-didehydro-α-amino acids are nisin[10,11] (a food preservative[12]), subtilin (a broad spectrum antibiotic),[13] and some of the metabolites isolated from *Streptomyces* strains such as griseoviridin.[14]

The conformational flexibility of the didehydropeptide backbone as well as the specific side chain of the didehydro residue is expected to be restricted by the $C^{\alpha}=C^{\beta}$ double bond. The presence of $sp^2$ hybridized carbon atoms at both α and β positions, the extended conjugation on account of the π system, the restriction of the side chain due to the double bond, and, moreover, the increased planarity of the residue as a whole, are some of the special features that have made dehydropeptides attractive targets for conformational and biological activity studies. The presence of dehydro amino acid residues in peptides confers increased resistance to enzymatic degradation. Dehydro amino acid residues have been introduced in several bioactive peptide sequences in order to obtain highly active agonist and antagonist analogues. Indeed, this modification has become one of the most promising tools to probe the structure–function relationship in biologically active peptides. This is reflected in dehydro analogues of some of the peptide hormones such as angiotensin,[15] bradykinin,[16] dermorphin,[17] enkephalin,[18] and gramicidin S.[19]

In nomenclature, the use of the capital Greek letter delta (Δ), prefixed to the three letter symbol for each amino acid residue is well accepted, e.g. ΔAla is used for α,β-didehydroalanine. The configuration of the α,β-double bond is designated by a superscript *E* or *Z*, e.g. $\Delta^{Z}$Phe represents (*Z*)-α,β-didehydrophenylalanine. *Z*-α,β-Didehydroamino acids have the larger β-substituents *cis* to the amino group. In contrast, the *E*-α,β-didehydroamino acids have the larger group *cis* to the carbonyl group. In this section the abbreviation DHA has been used for a generic α,β-didehydroamino acid, and unless stated otherwise Δ represents the *Z*-isomer.

The theoretical and crystallographic studies are mostly limited to ΔPhe and ΔAla peptides. The average $C^{\alpha}=C^{\beta}$ double bond length in a ΔPhe residue from crystal structure data is 1.33 Å. The $C^{\alpha}-C'$ bond length is 1.50 Å, slightly less than the corresponding distance of 1.53 Å in saturated amino acid residues. The value of 1.23 Å for the $C'=O$ bond in the ΔPhe residue, on the other hand, is slightly larger than the value found in Phe residues

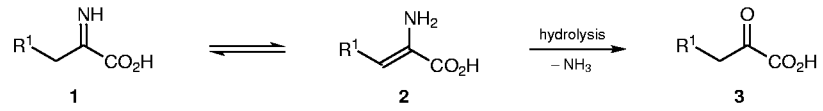
(1.203 Å).[20] The N–$C^{\alpha}$ bond distance (1.42 Å) is somewhat shorter than the saturated (1.45 Å) counterpart. The average $C^{\beta}$–$C^{\gamma}$ bond length is 1.45 Å. These values of bond lengths correspond well with the styrene system. The shortening of the N–$C^{\alpha}$, $C^{\alpha}$–C′, and $C^{\beta}$–$C^{\gamma}$ bond lengths and the somewhat larger value of the C=O double bond may be due to a partial conjugation involving the $C^{\alpha}$=$C^{\beta}$ double bond and the peptide bond. The bond angle N–$C^{\alpha}$–C′ is in general smaller than the expected value of 120° from an $sp^2$ hybridized carbon atom, whereas the corresponding values of N–$C^{\alpha}$=$C^{\beta}$ and $C^{\alpha}$=$C^{\beta}$–$C^{\gamma}$ are larger. By simple model building the following backbone torsion angles for the most favorable conformation for ΔPhe residues are observed: (ϕ,ψ) ~ (60°,30°), (−60°, −30°), (−60°, 150°), and (60°, −150°).

Studies in our laboratory and from several others have shown that in small model peptides containing a single ΔPhe, a type-II β-turn structure is generally observed.[21,22] In longer peptides containing one or more than one ΔPhe residues, helical structure are stabilized. In this regard the ΔPhe residue appears to induce conformational preferences similar to another well studied nonprotein amino acid residue, α-aminoisobutyric acid (Aib).[23] More recently, some synthetic model peptides containing ΔLeu,[24–29] ΔAla,[30–34] ΔAbu,[35,36] and ΔVal[37,38] have been reported in the literature.

## 11.1.1 α,β-Didehydro-α-amino Acids

As free DHAs are highly unstable compounds that readily undergo hydrolysis, they have generally been synthesized as their *N*-acyl derivatives or carboxylic esters and amides. For this reason DHAs occur in nature primarily as *N*-acyl derivatives or as part of a peptide sequence. Free DHAs **2** (Scheme 1) have not been characterized; most probably they exist as imines **1** and consequently they readily undergo hydrolysis to give the corresponding α-oxo acid **3** and ammonia.

**Scheme 1** Hydrolysis of α,β-Didehydro-α-amino Acids




1 2 3

The weak basicity and low nucleophilicity of the amino group in DHAs may be due to resonance stabilization of the system in which the nonbonded electrons of the nitrogen atom are delocalized into the carbon chain. This effect is further enhanced in the case of aromatic amino acids due to extended conjugation with the ring structure. Consequently, incorporation of DHAs in peptides through standard coupling reactions is more difficult and, therefore, in most cases they are introduced into peptide chains as part of a small dehydropeptide. The carboxy group of a DHA residue is sufficiently reactive to make peptide bond formation possible by the usual coupling procedures. One of the earliest syntheses of DHAs was the classical Erlenmeyer synthesis involving the condensation of *N*-acylglycine with aldehydes and ketones in the presence of a suitable base.[39] Several other general procedures for the preparation of DHAs have been described. This section will first describe the general methods of synthesis in detail, followed by some of the special methods applicable to specific DHAs.

*for references see p 659*

## 11.1.1.1 General Methods of Synthesis of α,β-Didehydro-α-amino Acids

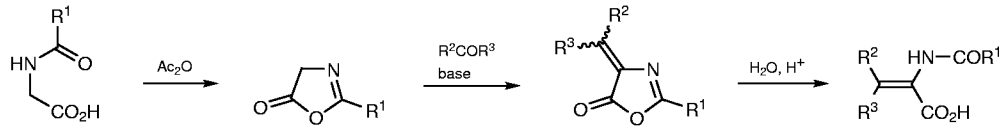
### 11.1.1.1.1 *N*-Acyl α,β-Didehydro-α-amino Acids Via Azlactones [Oxazol-5(4*H*)-ones]

One of the oldest methods for the preparation of DHAs is the ring opening of unsaturated azlactones by attack of a nucleophile on the carbonyl group. Several methods for the synthesis of unsaturated azlactones (Azl) have been developed over the years and the chemistry of these important intermediates has been reviewed.[1,2] Following are the methods of their synthesis that have remained in use during the last few decades.

#### 11.1.1.1.1.1 Erlenmeyer Synthesis

The preparation of the first unsaturated azlactone was reported in 1883 by Plochl,[40] who condensed benzaldehyde with hippuric acid in presence of acetic anhydride. This approach was later used by Erlenmeyer,[41] who extended the procedure to include other aldehydes and also established the usefulness of azlactones as intermediates in the synthesis of DHAs. The method involves the condensation of an *N*-acylglycine **4** with aldehydes and ketones in the presence of acetic anhydride and anhydrous sodium acetate (Scheme 2).[41–45] Other catalysts such as copper(II) acetate,[46] lead acetate,[47,48] potassium carbonate,[49] or potassium hydrogen carbonate[50] have also been used. The reaction proceeds via formation of an azlactone **5**, which then condenses with the appropriate aldehyde or ketone to give unsaturated azlactone **6**. Reaction of **6** with a nucleophile such as $^-$OH, $^-$OR, or $^-$NHR leads to the corresponding *N*-acyl-DHA derivatives **7**. Reaction with the sodium salt of an amino acid gives a DHA containing dipeptide acid.[51]

**Scheme 2** Synthesis of α,β-Didehydro-α-amino Acid by the Erlenmeyer Method[41,45]




4 5 6 7

The Erlenmeyer condensation is best carried out with aromatic aldehydes. Ketones and aliphatic aldehydes generally give poor yields. Doherty et al.[51] demonstrated that small dehydropeptides with a C-terminal Gly, when treated with an aromatic aldehyde at 40 °C in the presence of acetic anhydride and sodium acetate, gave the corresponding azlactones which on alkaline hydrolysis give peptides with two DHA residues. Thus, the reaction of Ac-ΔPhe-Gly-OH and benzaldehyde yielded an azlactone, which upon alkaline hydrolysis was converted into the doubly unsaturated peptide. Condensation with 4-hydroxybenzaldehyde yielded peptides incorporating a ΔTyr residue (Table 1). This method has been utilized by Pieroni et al.[52] and by Chauhan[53] to prepare a number of model peptides with two consecutive ΔPhe residues. The *N*-acyl-DHA can be coupled with other amino acids and peptides to form N-terminal dehydropeptides, but attempted hydrolytic deacylation yields the corresponding α-oxo acid and the amide. Disadvantages of this method are that in general drastic reaction conditions are required and that DHAs with carbamic acid protecting groups can not be prepared by this route.

**Table 1** Dehydropeptides Synthesized via Azlactone Intermediates[2,52,60,61]

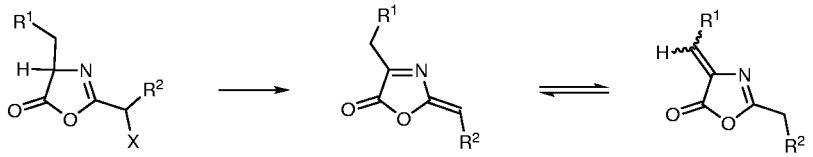
| Dehydropeptide | Yield (%) | mp (°C) | $[\alpha]_D$ | Ref |
|---|---|---|---|---|
| Ac-ΔPhe-ΔTyr-Glu-OH | 35 | | −100 | [52] |
| Z-Gly-ΔPhe-Glu-OH | | 177–179 | −29.6 | [2] |
| Ac-ΔPhe-Pro-ΔPhe-Ala-OMe | 63 | 185–187 | −74.5 | [60] |
| Ac-ΔPhe-Ala-ΔPhe-NHMe | 75 | 203–205 | −55 | [61] |
| Ac-ΔPhe-ΔPhe-Ala-OH | 60 | 207–208 | | [52] |

**Ac-ΔPhe-Azl (6, $R^1$ = Me; $R^2$ = Ph; $R^3$ = H):**[54]
A mixture of Ac-Gly-OH (15.43 g, 132 mmol), redistilled benzaldehyde (21.5 mL, 210 mmol), anhyd NaOAc (7.0 g, 85.4 mmol), and $Ac_2O$ (32 mL, 336 mmol) was refluxed with occasional stirring for 1 h. The mixture was kept in refrigerator overnight. The solid mass of yellow crystals was stirred with cold $H_2O$, filtered, and washed with $H_2O$. Recrystallization (hot $CCl_4$) gave Ac-ΔPhe-Azl; yield: 16.66 g (68%); mp 150–151 °C.

### 11.1.1.1.1.2 Bergmann–Stern Synthesis

Bergmann and Stern[55] found that dehydroazlactones can be obtained from treatment of α-haloacyl α-amino acids with acetic anhydride in presence of a base such as sodium acetate or pyridine. They showed that halo-azlactones of type **8** undergo spontaneous dehydrohalogenation giving a pseudoazlactone **9** that equilibrates with its tautomeric azlactone **10** (Scheme 3).

**Scheme 3** Synthesis of α,β-Didehydro-α-amino Acids by the Bergmann–Stern Method[55]




Thus, the *N*-chloroacetyl derivatives of Phe[55] and Leu,[51] for example, when warmed with acetic anhydride, readily form the requisite unsaturated azlactone by the elimination of one molecule each of water and hydrogen chloride. A series of DHA derivatives of Phe, Tyr, Leu, Nle, Val, and Ala were prepared by this method. However, this method has not been used routinely to prepare DHA derivatives.

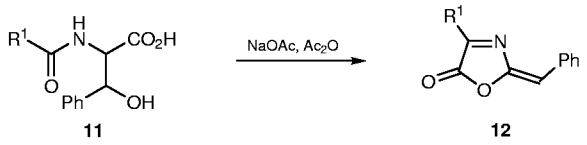
**Ac-ΔLeu-OH:**[51]
*N*-Chloroacetyl-Leu-OH (40 g, 0.222 mol) was stirred with $Ac_2O$ (60 mL) in a water bath at 60 °C for 1.5 h. On fractionation in vacuo, Ac-ΔLeu-Azl (**10**, $R^1$ = iPr; $R^2$ = H) was distilled at 68–69 °C/0.15 Torr, as a colorless liquid. The yield was only ~18 g. The azlactone was characterized by its transformation into the corresponding acid and its amide. When Ac-ΔLeu-Azl (5 g) was kept at 0 °C with $H_2O$ (12 mL), large crystals of Ac-ΔLeu-OH formed; yield: 2.5 g (50%); mp 155–157 °C.

### 11.1.1.1.1.3 Azlactone Formation from β-Hydroxy-α-amino Acids

Another method of obtaining unsaturated azlactones is through β-phenylserine [Ser(β-Ph)] derivatives. This method was initially introduced by Bergmann.[56] The method involves conversion of β-phenylserine derivatives **11** into the corresponding ΔPhe azlactones **12** by treatment with an acetic anhydride/sodium acetate solution (Scheme 4). This has remained as one of the most popular methods of synthesis.

*for references see p 659*

**Scheme 4** Formation of Azlactones from β-Hydroxy-α-amino Acids[56]




11 12

In this case, the amino acid is already oxidized at the β-position and elimination of the hydroxy group, as the acetate, generates the double bond. Reaction of the unsaturated azlactone and the free base of an amino acid ester produces the dipeptides containing ΔPhe. This approach has been extensively used in the preparation of a series of ΔPhe containing peptides[57–63]
(Table 2).

**Table 2** Some Examples of ΔPhe Azlactones[57–59,62,63,164]

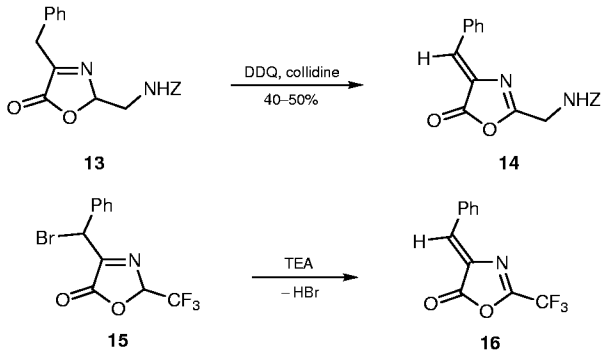
| Starting Material | Reaction Conditions | Product | Yield (%) | mp (°C) | Ref |
|---|---|---|---|---|---|
| Boc-Leu-DL-Ser(β-Ph)-OH | NaOAc, $Ac_2O$, rt | Boc-Leu-ΔPhe-Azl | 83 | 123–125 | [59] |
| Boc-Gly-DL-Ser(β-Ph)-OH | NaOAc, $Ac_2O$, rt | Boc-Gly-ΔPhe-Azl | 84 | 120–121 | [62] |
| Boc-Val-DL-Ser(β-Ph)-OH | NaOAc, $Ac_2O$, rt | Boc-Val-ΔPhe-Azl | 88 | 114–115 | [57] |
| Boc-Phe-DL-Ser(β-Ph)-OH | NaOAc, $Ac_2O$, rt | Boc-Phe-ΔPhe-Azl | 70 | 132–133 | [58] |
| Fmoc-Orn-DL-Ser(β-Ph)-OH | NaOAc, $Ac_2O$, rt | Fmoc-Orn-ΔPhe-Azl | 90 | 142–144 | [164] |
| Boc-Pro-DL-Ser(β-Ph)-OH | NaOAc, $Ac_2O$, rt | Boc-Pro-ΔPhe-Azl | 63 | syrup | [63] |

**Boc-Ala-ΔPhe-Azl:**[62]
Boc-Ala-Ser(β-Ph)-OH (7 g, 19.9 mmol) was reacted with anhyd NaOAc (2.12 g, 25.8 mmol) and distilled $Ac_2O$ (54.9 mL) for 24 h at rt. The mixture was poured over crushed ice. The resultant precipitate was washed with 10% $NaHCO_3$ and $H_2O$, and finally recrystallized (acetone/$H_2O$); yield: 74%; mp 103–105 °C.

### 11.1.1.1.1.4 Oxidation of Saturated Azlactones

Several attempts have been made, especially in recent years, to oxidize saturated to unsaturated azlactones.[64] The oxidation of saturated azlactone **13** with DDQ or *o*-chloranil in the presence of collidine as a weak base affords the unsaturated azlactone **14** in 40–50% yield (Scheme 5). Breitholle and Stammer[65] reported the preparation of unsaturated azlactone **16** by dehydrobromination of the bromopseudoazlactone **15** (Scheme 5). The pseudoazlactones derived from Ala and Abu tend to dibrominate, but careful reaction conditions allowed their isolation in acceptable yields.[65] This procedure is applicable to the preparation of derivatives of ΔAla, ΔAbu, ΔVal, and ΔLeu.[66]

**Scheme 5** Oxidation of Saturated Azlactones[64,65]




13 14

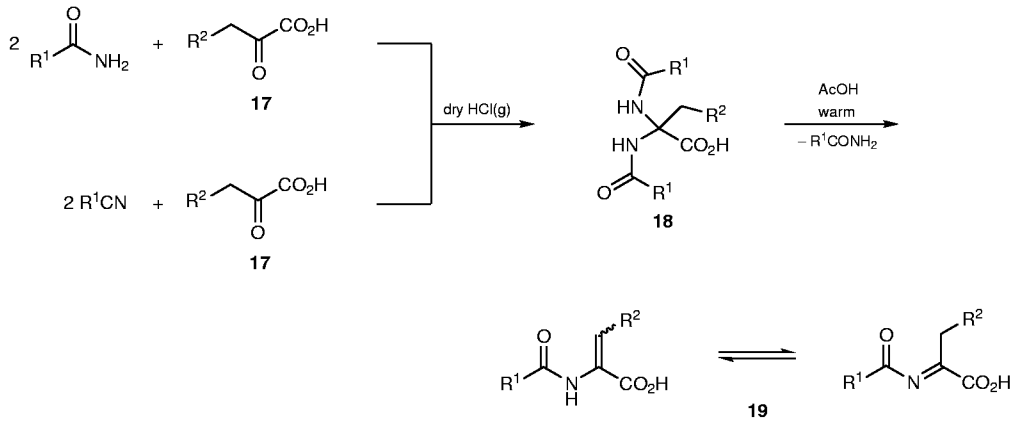
15 16

**Z-Pro-ΔPhe-Azl:**[64]
To a soln of Z-Pro-Phe-OH (2 g, 5 mmol) in dry THF (15 mL) at rt was added DCC (1.2 g, 6 mmol). The mixture was allowed to stand in a refrigerator overnight, DCU (2.1 g) was then filtered off and the solvent removed in vacuo. The residual oil was dissolved in $Et_2O$, a small amount of hexane was added and the soln was allowed to stand in refrigerator to give Z-Pro-Phe-Azl; yield: 1.7 g (89%); mp 92–94 °C. To a soln of Z-Pro-Phe-Azl (0.756 g, 2 mmol) in dry DME (20–30 mL) was added DDQ (0.454 g, 2 mmol) and collidine (0.244 g, 2 mmol). The mixture was stirred at rt for 6 days, the precipitate was filtered and the filtrate was concentrated in vacuo. The residual oil was dissolved in EtOAc and the soln was washed with 1 M HCl, sat. $NaHCO_3$, sat. brine, and dried ($Na_2SO_4$). The solvent was removed in vacuo and the residual oil was purified by chromatography (silica gel, 60–200 mesh, $Et_2O$/petroleum ether 1:1) giving a yellow oil which was crystallized ($Et_2O$/petroleum ether) to give Z-Pro-ΔPhe-Azl; yield: 0.364 g (48%); mp 93–94.5 °C; $[\alpha]_D^{29}$ −69 (*c* 1, THF); IR ($CDCl_3$) $\tilde{\nu}_{max}$: 1800, 1715, 1645 $cm^{-1}$.

### 11.1.1.1.2 Synthesis of α,β-Didehydro-α-amino Acid Derivatives by Condensation of Amides or Nitriles with α-Oxo Acids

An alternative to the azlactone procedure for the preparation of short-chain dehydropeptides **19** is offered by the direct condensation of an α-oxo acid **17** on heating with one equivalent of a carboxylic acid amide or by the treatment of an α-oxo acid **17** with a nitrile in the presence of dry HCl gas (Scheme 6). If the former reaction proceeds with the condensation of two molecules of amide per molecule of α-oxo acid, then the corresponding α,α-bis(acylamino) aliphatic acid **18** is formed, which on warming with acetic acid results in partial deamidation with formation of the corresponding dehydropeptide **19**.

**Scheme 6** Condensation of Amides or Nitriles with α-Oxo Acids

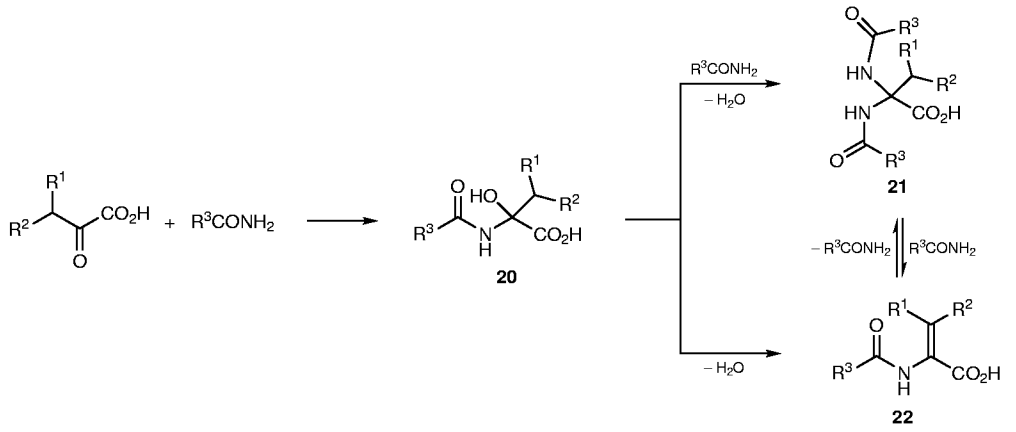



#### 11.1.1.1.2.1 Condensation of Amides

When a mixture of acetamide and pyruvic acid was heated under reduced pressure a mixture of α,α-bis(acetamido)propanoic acid **21** and α-acetamido acrylic acid **22** was formed (Scheme 7).[67]

*for references see p 659*

**Scheme 7** Condensation of Acetamide and Pyruvic Acid Derivatives[67]



Phenylpyruvic acid, benzoformic acid and α-oxoglutaric acid have also been employed as starting materials.[68] The reaction between an amide and an oxo acid proceeds through the primary addition of the amide to the carbonyl group giving **20**, followed either by direct replacement of the hydroxy group with another amide residue to form the bis(acylamido)-propanoic acid **21**, or by elimination of water with formation of an α,β-unsaturated intermediate **22** (N-acylated α-aminoacrylic acid) to which a second molecule of amide might add. Several modifications in reaction conditions have yielded improved experimental procedures,[69–76] some of which have been reviewed by Greenstein and Winitz.[2]

If benzyl carbamate is used as the amide component in the reaction with α-oxo acids **23** then the $N^{\alpha}$-benzyloxycarbonyl-DHA **24** is obtained directly (Scheme 8). Shin and et al.[77–84] widely exploited the above method in the synthesis of various α,β-didehydropeptides (Table 3). In presence of 3 M thionyl chloride and acetyl chloride, **24** gave the *N*-carboxy-DHA anhydride ΔNCA **25**, which could be conveniently converted into dehydropeptides.[77–84] Compared with the common saturated *N*-carboxy-α-amino acid anhydrides (NCAs), ΔNCAs were found to be stable at room temperature for several months.

**Scheme 8** Synthesis of an *N*-Carboxy α,β-Didehydro-α-amino Acid Anhydride[77–84]

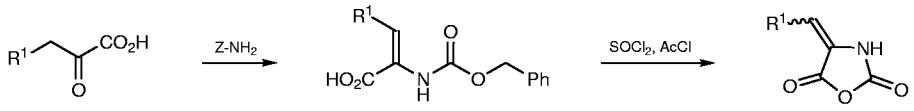


**Table 3** Dehydropeptides Prepared by Shin's Method[80,82,84]

| Compound | Yield (%) | mp (°C) | $[\alpha]_D$ | Ref |
|---|---|---|---|---|
| Ac-ΔAbu-Glu-OMe | 75 | 88–90 | −16.5 | [84] |
| TFA-ΔLeu-Phe-OMe | 89 | 103–104 | −15.6 | [84] |
| Boc-ΔPhe-Leu-OMe | 75 | 99–101 | −41.8 | [84] |
| Boc-ΔLeu-Ser-OMe | 60 | syrup | −11.9 | [84] |
| Z-Ala-ΔAbu-OMe | 52 | 137–138 | −10.5 | [84] |
| Boc-Leu-ΔPhe-OMe | 76 | 91–93 | 41.5 | [84] |
| Phth-Gly-ΔAbu-Phe-OMe | 58 | 175–176 | −24.9 | [84] |
| Phth-Leu-ΔLeu-Phe-OMe | 56 | 53–55 | −7.0 | [84] |
| Boc-Leu-ΔLeu-Ala-OMe | 77 | 126–127 | −60.8 | [84] |
| Boc-Leu-ΔPhe-Gly-Ala-OMe | 80 | 82–84 | −80.0 | [80] |
| Boc-Gly-ΔLeu-Gly-ΔLeu-OMe | 72 | syrup | | [82] |

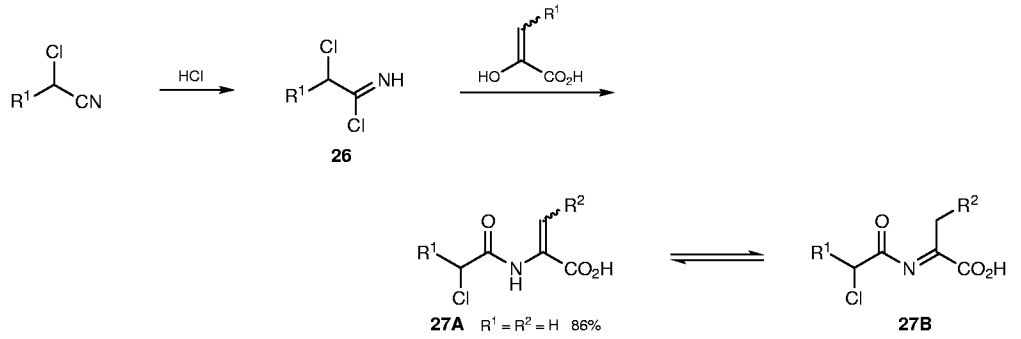
**Z-ΔIle-OH:**[84]
A soln of 3-methyl-2-oxopentanoic acid (65 g, 0.5 mol) and Z-$NH_2$ (83.1 g, 0.55 mol) in the presence of TosOH (0.172 g, 0.10 mmol) in dry benzene (800 mL) was refluxed for 6 h, removing $H_2O$ by means of a $H_2O$ separator. The soln was extracted with sat. aq $NaHCO_3$ (3×) and the combined aqueous extracts were acidified to pH 1–2 with concd HCl. The deposited crystals were collected and recrystallized (benzene) (**CAUTION:** *carcinogen!*) to give Z-ΔIle-OH as colorless needles; yield: 60%; (*E*/*Z*) mixture 3:2; mp 141–142 °C.

**ΔIle-NCA:**[84]
A soln of Z-ΔIle-OH (20 mmol) in AcCl (3 mL) and $SOCl_2$ (9 mL) was stirred at rt for 2 h. After removal of the solvent under reduced pressure, dry $CCl_4$ (30 mL) was added to the crystalline residue and the resulting soln was concentrated. This procedure was repeated three times. The residual crystals were recrystallized (cyclohexane) to give ΔIle-NCA as colorless needles; yield: 90%; (*E*/*Z*) mixture 3:2; mp 100–101 °C. The mixture was recrystallized repeatedly to give only the *E*-isomer; mp 122–123 °C.

### 11.1.1.1.2.2 Condensation of Nitriles

The reaction of α-oxo acids with nitriles in the presence of an acid catalyst gives acylamino acrylic acid derivatives, which are then used to prepare α,β-didehydropeptides.[85,86] The reaction proceeds with a high degree of purity and is less time consuming. The reaction takes place initially to give the imino chloride **26**, which subsequently reacts with the enolic form of the oxo acid. Several α-haloacyl-DHAs **27** have been prepared (Scheme 9) from which the corresponding α-aminoacyl-DHA can be prepared by ammonolysis in aqueous ammonia.

**Scheme 9** Synthesis of α,β-Didehydro-α-amino Acid Derivatives from Nitriles[85,86]




In the presence of water and excess nitrile α,α-bis(acetylamino) carboxylic acids are formed in very good yields, which can be converted into acylamino acrylic acids in a subsequent step.

***N*-(Chloroacetyl)-α,β-didehydroalanine (27A, $R^1 = R^2 = H$):**[86]
Redistilled $ClCH_2CN$ (0.1–0.15 g, 1.3–2 mmol ) was mixed with redistilled pyruvic acid (10.6 g, 0.12 mol), the mixture chilled and saturated with dry $H_2$ gas. $ClCH_2CO$-ΔAla-OH was obtained as a solid mass of crystals after keeping the mixture for 1 h at 5 °C. The mixture was allowed to stand in ice for another 12 h and then washed on a suction filter with dry $Et_2O$; yield: 14 g (86%); mp 156 °C.

This procedure may be employed for the synthesis of most of the α-haloacyl-DHAs. Ammonolysis with aq $NH_3$ yields the respective α-aminoacyl-DHA. More specifically, α-haloacyl-DHA (20 g) was dissolved in 28% aq $NH_3$ (200 mL) and the mixture kept at 37 °C for 36 h. The soln was then concentrated in vacuo at 35 °C, the residue dissolved in $H_2O$ (50 mL) and the resulting soln treated with abs EtOH (200 mL). The dehydropeptide was obtained as solid mass, which was filtered, washed with alcohol and $Et_2O$, and dried ($P_2O_5$).

*for references see p 659*

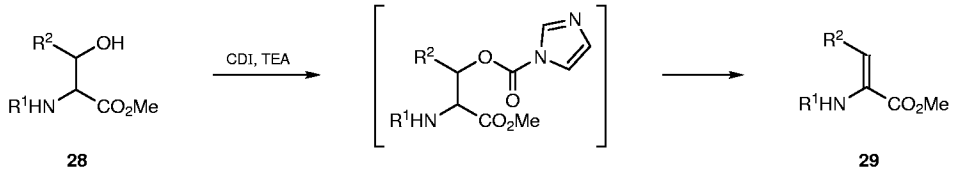
### 11.1.1.1.3 Synthesis of α,β-Didehydro-α-amino Acids by Elimination Reactions

Amino acids with appropriate β-substituents have frequently been employed as DHA precursors, particularly in the form of peptide chain subunits. Most commonly used derivatives for such elimination reactions are β-hydroxy- and β-sulfanyl-α-amino acids. Amino acids with β-halogen substituents[87–90] are also used, but these require more drastic elimination conditions and, therefore, are not preferred. Mannich bases of monoalkyl acylamino malonates have also found use as precursors of ΔAla derivatives.[91,92] The major advantage of this approach is that the elimination reaction can be carried out as the final step in the synthesis of unsaturated peptides.

#### 11.1.1.1.3.1 Elimination Reactions from β-Hydroxy-α-amino Acid Derivatives

The elimination of water from β-hydroxy-α-amino acid derivatives **28** is a particularly valuable method for the synthesis of DHAs **29**, when the corresponding hydroxy compounds are readily accessible, such as, those of Ser, Thr and Ser(β-Ph).[93] Some of the reagents recommended for the elimination of water are disuccinimidyl carbonate,[94,95] 1,1′-carbonyldiimidazole,[96] or a base and acetic anhydride.[97] Reaction of 1,1′-carbonyldiimidazole with arylidene Ser esters gives the corresponding DHA derivatives.[98] In the, presence of triethylamine with terminally protected Ser and Thr residues in an inert solvent this reagent affords the ΔAla and $\Delta^{Z}$Abu acid derivatives in good yields[96] (Scheme 10). This is perhaps one of the simplest and cleanest ways to prepare ΔAla and ΔAbu derivatives.

**Scheme 10** β-Elimination of β-Hydroxy-α-amino Acid Derivatives[96]




**28** **29**

**Didehydroamino Acids:**[96]
A soln of the substrate (a terminally protected Ser or Thr derivative) (5 mmol), CDI (0.81 g, 5 mmol) and TEA (0.7 mL, 5 mmol) in dry THF (20 mL) was stirred at rt for 4–6 h. The solvent was evaporated in vacuo and the residue chromatographed (silica gel 60, benzene/acetone 8:1) to give DHA derivatives which crystallized (EtOAc/petroleum ether or $Et_2O$/petroleum ether mixtures). Characterization of the DHA derivatives is given in Table 4.

**Table 4** Dehydropeptides Prepared by Dehydration of β-Hydroxy-α-amino Acids[96]

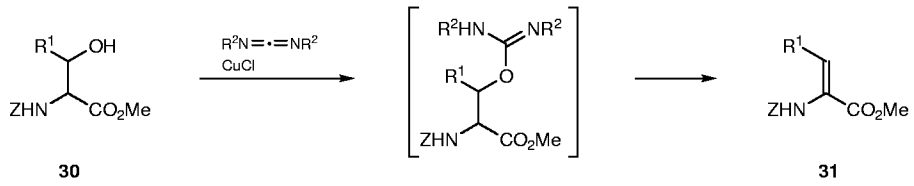
| Compound | Yield (%) | mp (°C) | Ref |
|---|---|---|---|
| Z-ΔAla-OMe | 85 | 36–38 | [96] |
| Boc-ΔAla-OMe | 83 | oil | [96] |
| $\Delta^{Z}$Abu-OMe | 76 | 66–68 | [96] |
| Boc-ΔAbu-OMe | 71 | 71–73 | [96] |
| Z-Gly-ΔAla-OMe | 68 | 73–74 | [96] |
| Z-Ser- ΔAla-OMe | 65 | oil | [96] |
| Z-Val- ΔAla-OMe | 71 | 116–118 | [96] |

### 11.1.1.1.3.2 β-Elimination Reactions from Serine Side-Chain Derivatives

In 1963 Photaki[99] introduced an efficient base-catalyzed elimination method using *O*-tosyl-Ser derivatives to generate ΔAla residues. Alternative β-leaving groups for Ser *O*-sulfonate have been reported, such as benzylsulfonyl ($SO_2Bzl$),[100,101] methylsulfonyl ($SO_2Me$),[102] diphenylphosphoryl [$(PhO)_2P{=}O$],[99] and *tert*-butoxycarbonyl.[103] However, in these β-elimination methods, due to the side reactions encountered, sometimes poor yields and inseparable mixtures are obtained. The possible side reactions, reviewed by Schmidt et al.[5] include oxazoline, aziridine, and hydantoin formation, and nucleophilic substitution.

These problems were solved by Miller,[104] who developed a direct isourea-mediated β-elimination process to prepare DHA derivatives. The reaction involves treatment of a Ser (or Thr) peptide **30** with a carbodiimide (1.1 equivalents) in the presence of copper(I) chloride as a catalyst to give a ΔAla (or ΔAbu) derivative **31**, respectively, in good yield (Scheme 11).

**Scheme 11** Isourea Mediated β-Elimination Reactions[104]




**30** **31**

However, aqueous, alcoholic, and carboxylic acid solvents and similarly reactive substituents must be avoided since they would competitively consume the carbodiimide. The advantage of this method is the ability to use water-soluble reagents to facilitate the reaction workup. α,β-Didehydropeptides containing ΔAla[29–34] and ΔAbu[35,36] residues have been synthesized in our laboratory by this method (Table 5). In another method reported by Ogura et al.[105] direct β-elimination of Ser derivatives was carried out by treatment with an equimolar amount of DSC and tertiary amine under mild conditions.

**Table 5** α,β-Didehydropeptides Prepared by Carbodiimide-Mediated β-Elimination Reactions[32,34,35]

| Compound | Yield (%) | mp (°C) | $[\alpha]_D^{25}$ | Ref |
|---|---|---|---|---|
| Boc-Ala-ΔAla-NHMe | 80 | 138–139 | −69.4 | [34] |
| Boc-Val-ΔAla-NHMe | 87 | syrup | −75.3 | [34] |
| Boc-Phe-ΔAla-NHMe | 77 | hygroscopic solid | −58.2 | [34] |
| Boc-Phe-ΔAla-OMe | 87 | 89–91 | | [32] |
| Boc-Ala-$\Delta^E$Abu-NHMe | 30 | 172–173 | −98.2 | [35] |
| Boc-Ala- $\Delta^Z$Abu-NHMe | 45 | 188–190 | −98.6 | [35] |
| Boc-Val- $\Delta^E$Abu-NHMe | 35 | 169–171 | −84.2 | [35] |
| Boc-Val- $\Delta^Z$Abu-NHMe | 55 | 179–180 | −83.8 | [35] |
| Boc-Phe $\Delta^E$Abu-NHMe | 40 | 188–190 | −89.2 | [35] |
| Boc-Phe $\Delta^Z$Abu-NHMe | 60 | 201–203 | −88.8 | [35] |
| Boc-$\Delta^E$Abu-Ala-NHMe | 40 | 172–173 | −58.5 | [35] |
| Boc-$\Delta^Z$Abu-Ala-NHMe | 60 | 170–172 | −58.4 | [35] |
| Boc-$\Delta^E$Abu-Val-NHMe | 45 | 173–174 | −40.5 | [35] |
| Boc-$\Delta^Z$Abu-Val-NHMe | 55 | 178–179 | −41.0 | [35] |
| Boc-$\Delta^E$Abu-Leu-NHMe | 40 | 202–203 | −88.2 | [35] |
| Boc-$\Delta^Z$Abu-Leu-NHMe | 60 | 208–210 | −85.3 | [35] |

**Boc-Ala-ΔAla-OMe:**[34]
To a soln of Boc-Ala-Ser-OMe (3 g, 10.34 mmol) in dry and peroxide free MeCN (15 mL) were added DCC (2.35 g, 11.37 mmol) and CuCl (0.3 g, 3.45 mmol) and the mixture was stirred at rt. The progress of the reaction was followed by TLC (the ΔAla compound moved higher than the Ser compound on TLC).

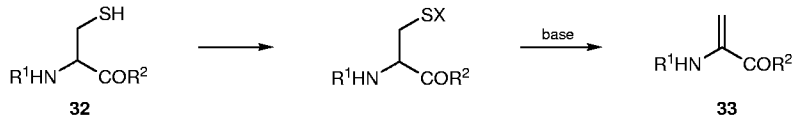
*for references see p 659*

After 6 h the solvent was removed to yield a green colored oily residue, which was dissolved in EtOAc and cooled. The precipitated DCU was filtered off and the organic layer was washed repeatedly with an 1% EDTA soln, until all green color was removed. The colorless organic layer was then washed with $H_2O$, dried (anhyd $Na_2SO_4$) and evaporated to give Boc-Ala-ΔAla-OMe in the form of a white amorphous solid; yield: 2.39 g (85%); $[\alpha]_D^{25}$ −55.4 (*c* 1.0, MeOH).

### 11.1.1.1.3.3 β-Elimination Reactions from Cysteine Side-Chain Derivatives

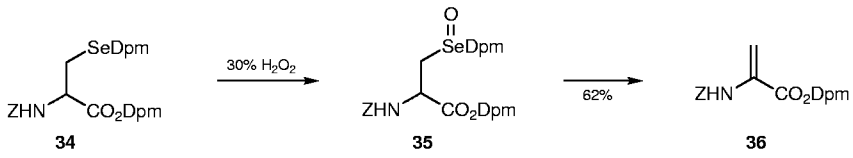
N-Acylated Cys derivatives **32** on treatment with silver(I) oxide or silver(I) carbonate,[106] mercury(II) oxide,[107] iron salts,[108] or DCC[109] undergo a β-elimination reaction to yield ΔAla derivatives **33**. The various β-leaving groups that have been reported include sulfoxide,[110] sulfone,[111] *S*,*S*-dimethylsulfonium [$Me_2^+S$],[112] 2,4-dinitrophenylsulfanyl[113] and *S*-dimethylphosphinothionyl[114] (Scheme 12).

**Scheme 12** α,β-Didehydro-α-amino Acid Derivatives from Cysteine Derivatives



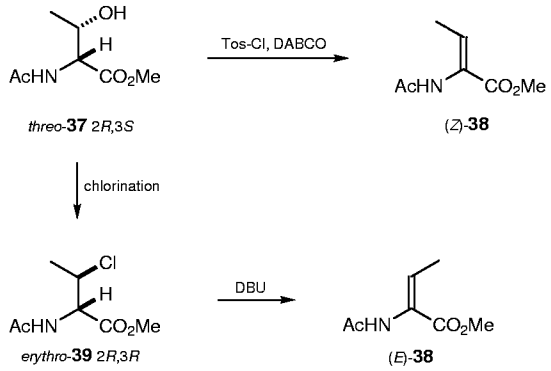

Reactions of *S*-alkyl or *S*-acyl Cys derivatives in basic media are often accompanied by β-elimination side reactions.[99,114,115] The simplest and most satisfactory method for the preparation of α,β-didehydropeptides is β-elimination from sulfonium salts of β-alkylsulfanyl-α-amino acid units.[112,116–118] Rich and Tam[118,119] alkylated the sulfide group with methyl fluorosulfonate and then performed the elimination reaction with triethylamine in a water-free solvent to prepare peptides containing ΔAla, ΔAbu, or ΔVal units. The procedure is not suitable for amino acids containing side-chain groups which can be alkylated such as Lys, Met, or His. In another type of elimination reaction, oxidation of selenocysteine (SeC) derivative **34** produces the ΔAla derivative **36** with greater ease through its sulfoxide-like selenoxide **35** (Scheme 13).[120] These intermediate oxides are so unstable that elimination reaction exclusively gives the DHA compounds.[121]

**Scheme 13** β-Elimination of a SeC Derivative[120]




Similar procedures can be used to prepare ΔAbu (both *E*- and *Z*-isomers) from Thr derivatives. Srinivasan et al.[89] found that β-elimination of the Thr derivative, Ac-(2*R*,3*S*)-Thr-OMe **37** (*threo* type), gave only the stable *Z*-isomer **38** upon O-tosylation and subsequent elimination by DABCO as a base (Scheme 14). The underlying mechanism for this reaction may be a *trans* E2-elimination. (2*S*,3*R*)-2-Acetamido-3-chlorobutanoic acid methyl ester (*erythro*) **39**, derived from the Thr *threo* form by chlorination with inversion of configuration at the β-carbon, yields predominantly the *E*-isomer **38** by brief treatment with DBU as a base. However, a prolonged reaction time and use of DABCO as a base causes a significant amount of isomerization to the *Z*-isomer.

**Scheme 14** β-Elimination Reaction of a Thr Derivative[89]

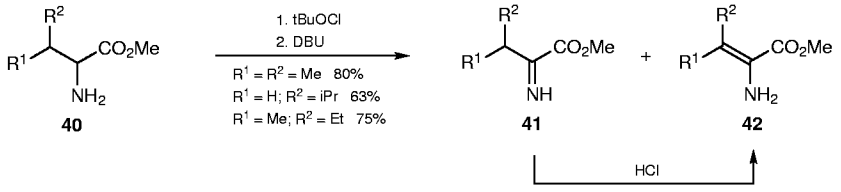



**Diphenylmethyl 2-[(Benzyloxycarbonyl)amino]-2-propenoate (36):**[121]
Z-SeC(Dpm)-ODpm (200 mg, 0.3 mmol) was dissolved in acetone (10 mL) and the soln cooled in an ice bath. To this soln was added 30% $H_2O_2$ (0.5 mL) and the mixture stirred for about 45 min at 4 °C. Then, the mixture was diluted with ice-cold $H_2O$ (50 mL) and extracted with EtOAc. The organic layer was separated, washed with $H_2O$ (3 × 25 mL), dried and concentrated in vacuo. The resultant oil, revealed two compounds by TLC (benzene/EtOAc 9:1) and was column chromatographed (silica gel). Z-ΔAla-ODpm was eluted first (hexane/benzene 1:3), then the starting material was eluted (benzene). Z-ΔAla-ODpm crystallized from EtOAc/hexane and characterized by IR and NMR; yield: 74 mg (62%); mp 78.5–79.5 °C. However, if the mixture was kept at rt for 15 min in the presence of $H_2O_2$ prior to workup, the starting material was completely converted into the ΔAla derivative.

### 11.1.1.1.4 α,β-Didehydro-α-amino Acid Derivatives by N-Chlorination/Dehydrochlorination

The method of N-chlorination of amino acid esters **40** and *N*-acylamino acid esters with *tert*-butyl hypochlorite and subsequent elimination of hydrogen chloride is of general use in the preparation of DHA derivatives **41/42**[122–124] (Scheme 15).

**Scheme 15** N-Chlorination/Dehydrochlorination to Yield α,β-Didehydro-α-amino Acid Derivatives[122–124]




The amino acid ester **40** initially gives a mixture of the imine **41** and enamine **42** which is treated with hydrogen chloride in diethyl ether to afford the enamine hydrochloride **42**·HCl. The *N*-chloroamide initially formed from the acylamino acid ester reacts, depending on the solvent, to give either the α-acylamino-α-alkoxy carboxylate or the acylimino compound. Treatment of the former with acid or of the latter with a strong base results in the formation of the *N*-acyl-DHA derivative.[125–129] This method is useful for all but sulfur containing amino acids and possibly those having aromatic rings such as Tyr and Trp that may be chlorinated by hypochlorite.

It was also reported that *N*-acyl-*N*-hydroxyamino acids are readily synthesized from α-bromo esters[130] and dehydrated to form the dehydro compounds. A number of N-terminal Gly dipeptides having ΔAla, ΔAbu, ΔNva, and ΔPhe at the C-terminus have been prepared in this manner.[131–133]

*for references see p 659*

**HCl·H-ΔVal-OMe**[38]

A soln of HCl·H-Val-OMe (2.0 g, 11.9 mmol) in $CHCl_3$ (30 mL) was treated with sat. $Na_2CO_3$ (30 mL). The organic layer was washed with $H_2O$, dried (anhyd $Na_2SO_4$), filtered, and concentrated in vacuo. The residue was taken up in dry $Et_2O$ (30 mL) and tBuOCl (1.4 mL, 11.9 mmol) was added dropwise over a period of 1 h in an ice bath in the dark. The mixture was concentrated in vacuo, the residue taken up in $CHCl_3$, washed successively with 0.5 M HCl and $H_2O$, dried (anhyd $Na_2SO_4$), and concentrated to yield *N*-chloro-Val-OMe which was dissolved in dry $Et_2O$ (30 mL). DBU (1.8 mL, 11.9 mmol) was added to the soln obtained, DBU·HCl was filtered from the mixture after 3 h and the filtrate concentrated in vacuo. HCl·H-ΔVal-OMe was obtained as an oily residue; yield: 1.2 g (85%); bp 76 °C.

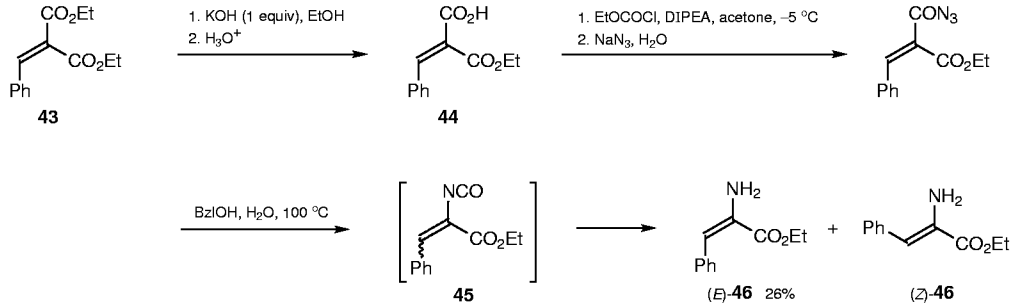
## 11.1.1.2 Synthesis of Aromatic α,β-Didehydro-α-amino Acids

Of the three aromatic DHAs, ΔPhe, ΔTyr, and ΔTrp, ΔPhe is the most frequently used and well studied. The main synthetic strategy used to prepare ΔPhe is through the azlactone method.[1,2] Most of the methods to prepare azlactones of ΔPhe have already been discussed (*vide supra*). The *Z*-isomer of ΔPhe is obtained by these methods, whereas the *E*-isomer is prepared by special methods;[134] namely by isomerization of the *Z*-isomer in saturated HBr,[135–137] by photochemical isomerization,[138] or from 2-phenyloxazolinonium perchlorate.[139]

### 11.1.1.2.1 Synthesis of α,β-Didehydrophenylalanine Derivatives

Stammer[140] and co-workers have introduced a method (Scheme 16) by which the ΔPhe isomer with *E* configuration can be prepared.

**Scheme 16** Synthesis of a $\Delta^E$Phe Derivative[140]




Compound **44** has the presumed *Z* configuration, since it is obtained by a sterically controlled hydrolysis of the diester **43**. The *E* configuration of **46** then follows from rearrangement of azide to isocyanate **45**, which occurs with retention of configuration.

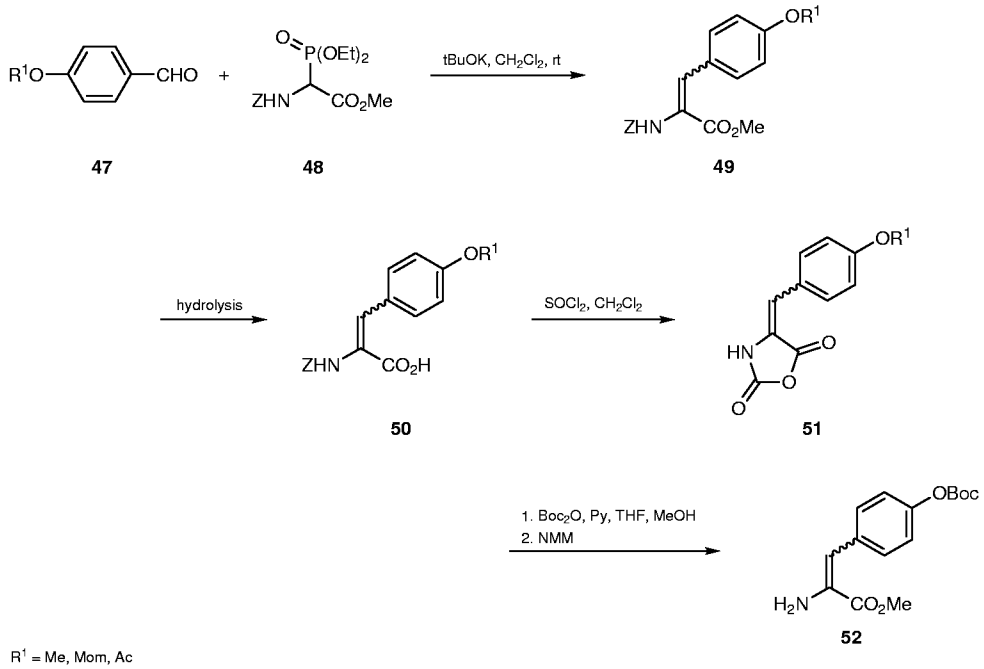
**H-$\Delta^E$Phe-OEt [(*E*)-46] and H-$\Delta^Z$Phe-OEt [(*Z*)-46]:**[140]

To ethyl hydrogen benzylidenemalonate (**44**; 5.00 g, 22.7 mmol) dissolved in dry acetone (30 mL) was added DIPEA (4.35 mL, 25.0 mmol). The soln was chilled to −5 °C (ice/salt/water bath) and ethyl chloroformate (2.4 mL, 25.0 mmol) in dry acetone (10 mL) was added dropwise, while maintaining the temperature at −5 °C. After 30 min, $NaN_3$ (3.25 g, 50.0 mmol) in $H_2O$ (15 mL) was added dropwise, while maintaining the temperature at or below −5 °C. The mixture was stirred for an additional 30 min, after which it was poured into an ice-cold mixture of toluene (75 mL) and $H_2O$ (25 mL). The aqueous layer was extracted with toluene (2 × 50 mL). The organic layers were combined, dried ($MgSO_4$), and concentrated in vacuo to give an oil, which was taken up in dry toluene (15 mL) and added quickly to BzlOH (30.0 g) at 100 °C. The soln was heated for an additional 5 min and then cooled immediately in an ice bath. The toluene was removed in vacuo and the resultant oil was dissolved in petroleum ether (40 mL) and $Et_2O$ (15 mL). Extraction with $H_2O$ (300 mL) caused separation into three layers. The lower aqueous layer was removed, and $Et_2O$ was added to induce mixing of the two remaining layers. $H_2O$ (300 mL) was added and the procedure was repeated four times till TLC showed the absence of BzlOH. The organic phase was dried ($MgSO_4$) and concentrated in vacuo to give 5.84 g of a yellow oil. Column chromatography (silica gel 60, 220 g, $Et_2O$/petroleum ether 1:10) afforded oily (*E*)-**46** (2.16 g) and oily (*Z*)-**46** (1.90 g), both of which solidify upon standing. Recrystallization of crude (*E*)-**46** [$Et_2O$ (20 mL)/petroleum ether (30 mL)] gave pure (*E*)-**46**; yield: 1.82 g (26%); mp 61–62 °C.

### 11.1.1.2.2 Synthesis of α,β-Didehydrotyrosine Derivatives

The azlactone of ΔTyr can also be prepared by condensation of a C-terminal Gly peptide with 4-hydroxybenzaldehyde.[2] Pieroni et al.[141] used this method to prepare peptides containing ΔTyr. Another method for preparing ΔTyr is from *N*-carboxy-dehydrotyrosine anhydride (ΔTyr-NCA) derivatives introduced by Shin et al.[142] In this method *N*-benzyloxycarbonyl-2-(diethoxyphosphinyl)glycine methyl ester (**48**) was reacted with 4-substituted benzaldehyde **47** in the presence of potassium *tert*-butoxide in $CH_2Cl_2$ at room temperature to give Z-ΔTyr($R^1$)-OMe **49** [$R^1$ = Me, Mom, Ac] (Scheme 17). In order to synthesize ΔTyr-NCA derivatives **51**, compound **50** was subjected to cyclization with $SOCl_2$ in $CH_2Cl_2$. Compound **51** was treated with $Boc_2O$ in the presence of a few drops of pyridine in THF; then methanol was added. The resulting solution was made basic (pH 10) with NMM in order to remove the protecting group from Boc-ΔTyr-OMe in 80% yield. On the other hand, in the case of using Z-Cl instead of $Boc_2O$, a similar reaction gives the Z-ΔTyr-OMe in 84% yield.

**Scheme 17** Synthesis of ΔTyr Derivatives[142]




**Z-ΔTyr(Mom)-OMe** (49, **$R^1$ = Mom**):[142]
Into a soln of tBuOK (3.1 g, 28 mmol) in dry $CH_2Cl_2$ (40 mL) was added a soln of *N*-Z-2-(diethoxyphosphophenyl)glycine methyl ester (10 g, 28 mmol), with stirring at −60 °C. After stirring for 15 min, the resulting soln was further stirred with 4-(methoxymethyl)benzaldehyde (4.65 g, 28 mmol) below −60 °C. After allowing this soln to stand at rt for 12 h, the soln was concentrated under reduced pressure to give a residual syrup, which was dissolved in EtOAc (150 mL). The resultant soln was washed successively with sat. aq $NH_4Cl$ and $H_2O$ and finally dried ($Na_2SO_4$). Removal of the solvent under reduced pressure gave a viscous syrup, which was then purified by column chromatography (silica gel, hexane/EtOAc 20:1) to give Z-Δ$^Z$Tyr(Mom)-OMe; yield: 63%.

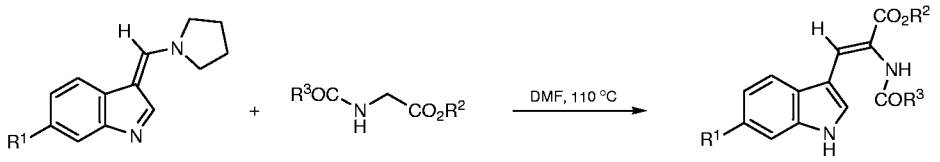
**H-ΔTyr(Mom)-NCA** (51, **$R^1$ = Mom**):[142]
A soln of Z-ΔTyr(Mom)-OH [**50**, $R^1$ = Mom; obtained by hydrolysis of (*Z*)-**49** with 1 M LiOH in dioxane] (0.2 g, 0.56 mmol) and $SOCl_2$ (2 mL, 27 mmol) in dry $Et_2O$ (2 mL) was stirred below 0 °C for 1.5 h. The soln was concentrated under reduced pressure and the residue was dissolved in $CCl_4$ (6 mL) and again concentrated. The procedure was repeated three times. Finally, the crystals collected were washed with warm $CHCl_3$ and then recrystallized (dioxane) to give H-ΔTyr(Mom)-NCA as colorless prisms; yield: 17%; mp 148–149 °C.

*for references see p 659*

### 11.1.1.2.3 Synthesis of α,β-Didehydrotryptophan Derivatives

Attempts to synthesize ΔTrp derivatives by the well-known Erlenmeyer–Plochl method using an azlactone result in very poor yields.[39] However, using an altogether different strategy Moriya et al.[143] synthesized ΔTrp derivatives with *Z* configuration in much better yields. In this method condensation of 3-(aminomethylene)-3*H*-indole **53** with *N*-acylglycine esters **54** was used to synthesize *N*-acyl-substituted ΔTrp derivatives **55** (Scheme 18).

**Scheme 18** Synthesis of ΔTrp Derivatives[143]




$R^1$ = H, Me; $R^2$ = Me, Et, tBu; $R^3$ = H, Me, Ph

The greatest yield of single isomer (ΔTrp) was obtained when *N*-formylglycine methyl ester (**54**, $R^2$ = Me; $R^3$ = H) was condensed with 3-[(1-pyrrolidinyl)methylene]-3*H*-indole (**53**, $R^1$ = H) in DMF at 110–120 °C for 4–6 hours. Treatment of HCO-ΔTrp-OMe (**55**, $R^1$ = H; $R^2$ = Me; $R^3$ = H) with excess 20% methanolic hydrogen chloride at room temperature afforded HCl·H-ΔTrp-OMe in an excellent yield which was used in the synthesis of peptides.

**HCO-ΔTrp-OMe (55, $R^1$ = $R^3$ = H; $R^2$ = Me):**[143]
A mixture of 3-[(1-pyrrolidinyl)methylene]-3*H*-indole (**53**, $R^1$ = H; 7.3 g, 0.037 mol) and HCO-Gly-OMe (**54**, $R^2$ = Me; $R^3$ = H; 5.0 g, 0.048 mol) in DMF (50 mL) and MeOH (6 mL) was stirred at 110 °C for 6 h. The mixture was diluted with EtOAc (300 mL) and poured into 1% HCl (250 mL). The organic layer was washed with $H_2O$, dried ($MgSO_4$), and then concentrated under reduced pressure. The residue was crystallized (MeOH) to give pale yellow prisms of HCO-ΔTrp-OMe; yield: 3.0 g (55% ); mp 196–198 °C.

**HCl·H-ΔTrp-OMe:**[143]
Dry methanolic HCl (20%, 50 mL) was added to HCO-ΔTrp-OMe (**55**, $R^1$ = $R^3$ = H; $R^2$ = Me; 2.44 g, 0.01 mol) at 0 °C and the mixture was stirred at rt for 2 h. $Et_2O$ (100 mL) was added to the mixture to give colorless fine crystals of HCl·H-ΔTrp-OMe; yield: 2.40 g (95%); mp 194–195 °C.

## 11.1.1.3 Synthesis of Aliphatic α,β-Didehydro-α-amino Acids

As discussed in Section 11.1.1.2.3 most of the aliphatic DHAs such as ΔAla, ΔVal, ΔLeu, ΔIle, ΔAsp, ΔGlu, and ΔOrn have been prepared either by β-elimination, N-chlorination/dehydrochlorination, or from *N*-carboxy α,β-didehydroamino acid anhydrides which, in turn, are prepared by condensation of an α-oxo acid with benzyl carbamate.

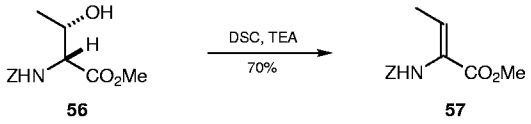
### 11.1.1.3.1 Synthesis of α,β-Didehydroalanine and α,β-Didehydroaminobutanoic Acid Peptides

Saturated amino acids such as Ser, Cys, Ala, and α,β-diaminopropanoic acid are usually utilized as precursors of ΔAla. Of the various methods discussed in the previous sections, that reported by Miller[104] (with a carbodiimide) was adopted as the method of choice in our laboratories, and the laboratories of others, to prepare peptides containing ΔAla.[29–34] A detailed experimental procedure has already been described in Section 11.1.1.1.3.2; this method was chosen for its low cost, simplicity, and availability of pure products in good yields.

Oxidation of SeC[144] derivatives also generates ΔAla derivatives. Hoffmann degradation of α,β-diaminopropanoic acid,[145] Bergmann's reaction[146] of condensation of amides with pyruvic acid, N-hydroxylation followed by dehydration,[147] and N-chlorination/dehydrochlorination[122–124] have also been used for the synthesis of ΔAla derivatives.

Analogously to ΔAla from Ser, ΔAbu can be derived from Thr by using essentially the same methods.[99–104] However, because of the possibility of *E*- and *Z*-isomers with ΔAbu, either stereoselective synthesis or separation of isomers must be taken into consideration. Srinivasan et al.[89] reported detailed stereochemical studies on the β-elimination of Thr derivatives. Lee et al.[122] confirmed the stereospecific formation of ΔAbu from *O*-tosyl Thr derivatives: the *Z*-isomer from the *threo* type and the *E*-isomer from the *erythro* type. If DSC is used for direct elimination, only the Z-isomer **57** is formed from Z-Thr-OMe **56** in 70% yield (Scheme 19).

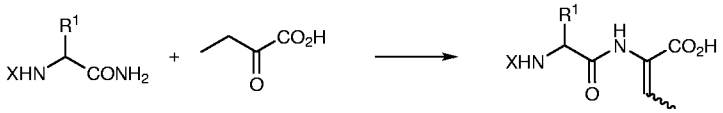
**Scheme 19** Synthesis of a ΔAbu Derivative[122]




**56** **57**

In contrast, the isourea-mediated β-elimination method from Thr peptides gave a mixture of *Z*- and *E*-isomers in a 3:2 ratio.[104] Similarly to ΔAla, peptides containing ΔAbu can also be synthesized by N-hydroxylation/dehydration[147] or by N-chlorination/dehydrochlorination.[122–124] Another method, reported by Smelka et al.[148] involves the synthesis of dipeptides **60** with a C-terminal ΔAbu by condensing the $N^{\alpha}$-protected α-amino acid amides **58** with the α-oxo acid **59** (Scheme 20). The ratio of *Z*/*E*-isomers is 10:1. We have used Miller's method to prepare peptides containing ΔAbu.[35]

**Scheme 20** Condensation of α-Amino Acid Amides with α-Oxo Acids to give ΔAbu Peptides[148]




**58** **59** **60**

X = Z, Tfa

**Boc-Ala-ΔAbu-OMe:**[35]
Boc-L-Ala-Thr-OMe (5 g, 16.4 mmol) was dissolved in dry MeCN (25 mL). DCC (3.39 g, 16.4 mmol) followed by CuCl (0.49 g, 4.92 mmol) were added to the soln and the mixture was stirred at rt for 6 h. When TLC showed absence of starting material, the solvent was removed in vacuo and EtOAc was added to the green oily residue so obtained. On cooling, the precipitated DCU was filtered off and the filtrate was washed repeatedly with 1% EDTA soln till the organic layer was colorless. It was finally washed with $H_2O$, dried ($Na_2SO_4$), and concentrated to yield Boc-Ala-ΔAbu-OMe as a white solid; yield: 4.32 g (92%); mp 168 °C.

### 11.1.1.3.2 Synthesis of α,β-Didehydrovaline, α,β-Didehydroleucine, and α,β-Didehydroisoleucine Peptides

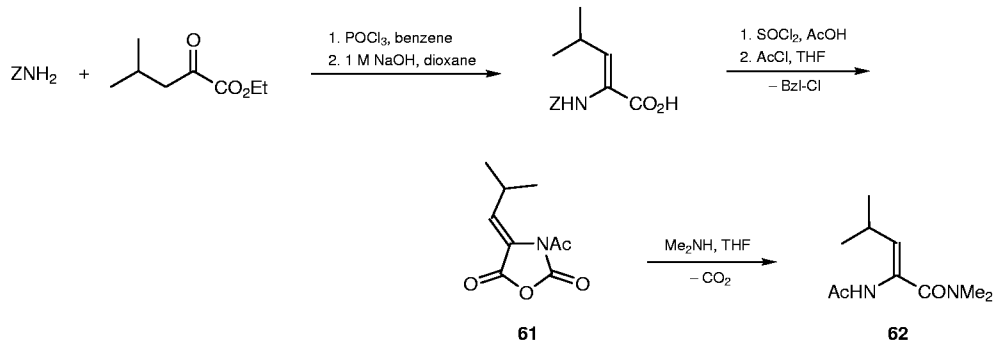
ΔVal residues can be synthesized by the condensation method or by the N-chlorination/dehydrochlorination method.[77,149] In the condensation method, either benzylcarbamate or a benzyloxycarbonylamino acid amide is condensed with 3-methyl-2-oxobutanoic acid in benzene using 4-toluenesulfonic acid as a catalyst to obtain Z-ΔVal or Z-Xaa-ΔVal dipeptides.[150] In this case it is not possible to elongate the chain from the N-terminus because the

*for references see p 659*

Z group can only be deprotected under strong acidic conditions or by catalytic hydrogenation. Indeed, neither of these methods is suitable for DHA-containing peptides. The other method for the synthesis of ΔVal peptides is the N-chlorination/dehydrochlorination, which is more useful and has been exploited in our laboratory to prepare model peptides.[38]

Similarly to ΔVal derivatives, ΔLeu methyl ester was prepared from Leu methyl ester in 90% yield using *tert*-butyl hypochlorite as the chlorinating reagent and DBU as the base for dehydrochlorination. A number of peptides have been prepared using this approach.[24–28] The other method reported by Shin et al.[84] involves a one-pot procedure using N-carboxy-*Z*-α,β-didehydroleucine anhydride **61** (ΔLeu-NCA) to give **62** (Scheme 21);[29,84] a ΔNCA can also be used to prepare ΔIle derivatives.[84]

**Scheme 21** Synthesis of a ΔLeu Derivative from *N*-Carboxy-α,β-didehydroleucine Anhydride[29,84]



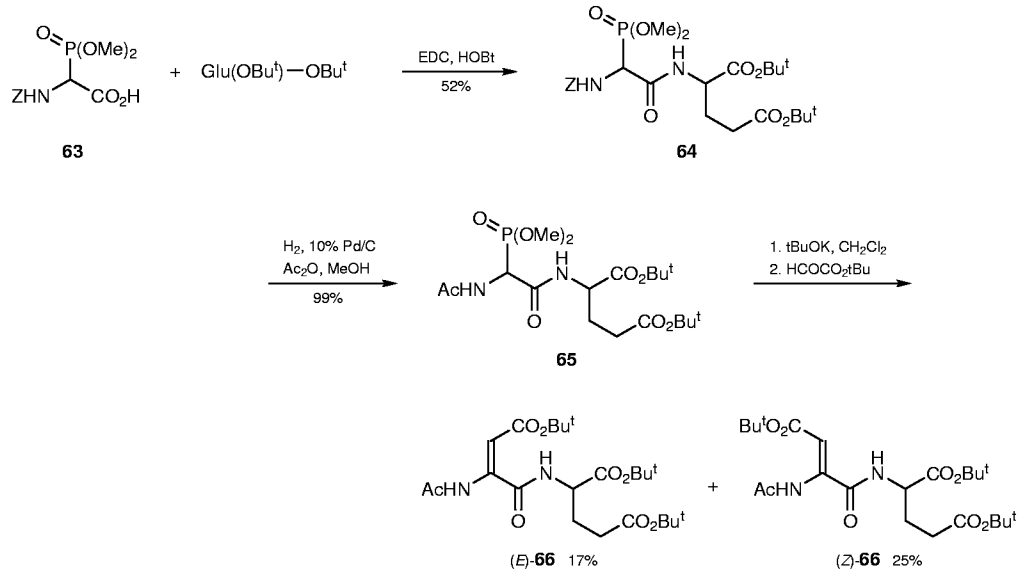

**61** **62**

**Boc-Met-ΔLeu-Phe-OMe:**[29]
To a soln of Boc-Met-OH (0.634 g, 2.54 mmol) in dry THF (4.2 mL) were added at −10 °C DCC (0.525 g, 2.54 mmol) and, after stirring for 20 min, ΔLeu-NCA[84] (0.329 g, 2.12 mmol) and dry pyridine (0.17 mL, 2.12 mmol). Stirring was continued at −10 °C for 2 h and at rt overnight. Then H-Phe-OMe·HCl (0.64 g, 2.97 mmol), dry THF (4.2 mL), TEA (0.42 mL, 2.97 mmol), and NMM (0.33 mL, 2.97 mmol) were added to the mixture, which was stirred for 2 h and the solvent removed under reduced pressure. The residue was dissolved in chilled EtOAc and DCU was filtered off. The filtrate was similarly worked up to give an oil (1.49 g), which was column chromatographed (silica gel, $CH_2Cl_2$ and $CH_2Cl_2/Et_2O$ 9:1). The nearly homogeneous product (0.614 g) was further chromatographed by preparative thin layer chromatography ($CH_2Cl_2$/EtOAc 7:3) to give pure title tripeptide; yield: 0.51 g (46%); mp 118 °C.

### 11.1.1.3.3 Synthesis of α,β-Didehydroglutamic Acid and α,β-Didehydroaspartic Acid Peptides

Analogous to ΔLeu and ΔIle, ΔGlu has been synthesized from its *N*-carboxy-α,β-didehydroglutamic acid anhydride (ΔGlu-NCA).[78] ΔAsp esters were synthesized via the general route of Schmidt et al.,[151] who have prepared a number of DHAs and dehydropeptides by the condensation of aldehydes with *N*-acyl-2-(dialkoxyphosphinyl)glycine esters in a Wittig–Horner type alkenation. According to this method, Ac-ΔAsp(OtBu)-Glu(OtBu)-OtBu **66** was synthesized by the pathway outlined in Scheme 22.[152] Racemic **63** was coupled to H-Glu(OtBu)-OtBu with water-soluble carbodiimide (EDC) and HOBt to give the phosphonodipeptide **64** as a mixture of diastereomers. Catalytic hydrogenolysis of this mixture in the presence of $Ac_2O$ gave the N-acetylated dipeptide **65**. The *E*- and *Z*-isomers of Ac-ΔAsp(OtBu)-Glu(OtBu)-OtBu **66** were formed by condensing the ylide of **65**, which was generated with tBuOK in dichloromethane, with *tert*-butyl glyoxylate. The isolated ratio of *E*/*Z* for **66** was 2:3. The two isomers could be readily distinguished by $^1H$ NMR, since it has been shown[132] that for Ac-ΔAsp derivatives the chemical shift of the *E*-isomer olefinic hydrogen appears downfield of that of the *Z*-isomer.

**Scheme 22** Synthesis of ΔAsp Peptides[152]




**Di-*tert*-butyl *N*-[2-(Acetamido)-2-(dimethoxyphosphinyl)acetyl]glutamate (65):**[152]
H-L-Glu(OtBu)OtBu·HCl (5.39 g, 18 mmol) and (±)-2-[(benzyloxycarbonyl)amino]-2-(dimethoxyphosphinyl)acetic acid **63** (5.39 g, 17 mmol) were dissolved in THF (25 mL) in a round-bottomed flask equipped with a stirrer bar. To this soln, NMM (3.64 g, 36 mmol) and HOBt (2.3 g, 17 mmol) were added. The mixture was cooled in an ice bath and after 30 min EDC (3.44 g, 18 mmol) was added. The mixture was stirred at 0 °C for 1 h and then overnight at rt. THF was removed under reduced pressure and the residue was taken up in EtOAc. The mixture was washed with 10% citric acid, 1 M $NaHCO_3$, and sat. brine and dried (anhyd $MgSO_4$). Removal of the EtOAc followed by drying under high vacuum gave an oil, which was purified by column chromatography (silica gel, EtOAc/hexane 2:1) to give di-*tert*-butyl *N*-{2-[(benzyloxycarbonyl)amino]-2-(dimethoxyphosphinyl)acetyl}glutamate **64** as a mixture of diastereomers; yield: 4.94 g (52%). The benzyloxycarbonyl mixture of diastereomers (4.7 g, 8.41 mmol) was dissolved in MeOH (7 mL). To this soln 10% Pd/C (0.7 g) and $Ac_2O$ (6.6 mL) were added. The mixture was hydrogenolyzed for 3 h under a $H_2$ pressure of 3 atm and then filtered. The MeOH was removed, and the residue obtained was dried under high vacuum to give **65** as an oil; yield: 3.9 g (99%).

**Ac-ΔAsp(OtBu)-Glu(OtBu)-OtBu (66):**[152]
tBuOK (0.06 g, 0.5 mmol) was placed in a round-bottomed flask and cooled in a dry ice/iPrOH bath. To this mixture was added $CH_2Cl_2$ (1 mL). While this soln was stirred well, Wittig reagent **65** (0.24 g, 0.5 mmol) in $CH_2Cl_2$ (4 mL) was added slowly. To this mixture *tert*-butyl glyoxylate (0.26 g, 2 mmol) in $CH_2Cl_2$ (1 mL) was added dropwise. The bath was allowed to warm up slowly to −10 °C(~2 h) and then stirring was continued for an additional 2 h. The $CH_2Cl_2$ was removed and the residue taken in EtOAc. This mixture was extracted with $H_2O$, washed with sat. $NH_4Cl$, and then dried ($MgSO_4$). Removal of EtOAc gave a yellow oil which was purified by medium-pressure C18 reverse-phase column chromatography(MeOH/$H_2O$ 2:1) to give **66** as a mixture of *E*- and *Z*-isomers; 0.13 g (55%). The isomers were separated by column chromatography (silica gel, EtOAc/hexane 1:2) to give 0.04 g (17%) of the *E*-isomer and 0.06 g (25%) of the *Z*-isomer as white solids. Ac-$\Delta^E$Asp(OtBu)-Glu(OtBu)-OtBu; (*E*)-**66**: mp 124–125 °C; $[\alpha]_D$ +11.7 (*c* 0.15, $CHCl_3$). Ac-$\Delta^Z$Asp(OtBu)-Glu(OtBu)-OtBu; (*Z*)-**66**: mp 116–117 °C; $[\alpha]_D$ +11.5 (*c* 0.15, $CHCl_3$).

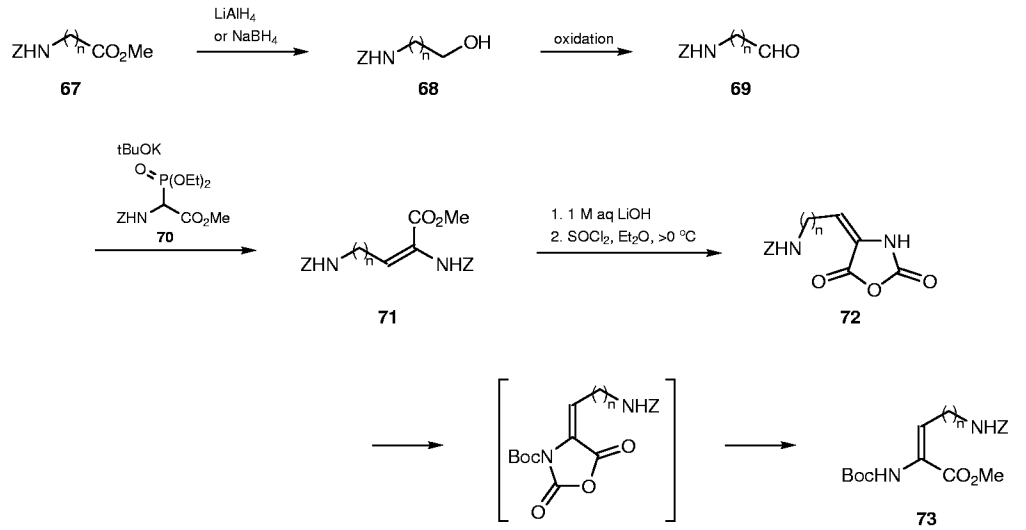
### 11.1.1.3.4 Synthesis of α,β-Didehydroornithine and α,β-Didehydrolysine Peptides

ΔOrn and ΔLys derivatives were synthesized by cyclization of $N^{\alpha}$,$N^{\omega}$-bisbenzyloxycarbonyl-α,β-didehydroornithine and -α,β-didehydrolysine **73**, derived from Z-protected amino aldehydes and *N*-benzyloxycarbonyl-2-(diethoxyphosphinyl)glycine esters (Scheme 23).[81] According to the method reported by Schmidt et al.,[151] advantage was take of the reaction of the Wittig–Horner reagent with an appropriate α-amino aldehyde. To prepare the desired

*for references see p 659*

α-amino aldehydes, the starting *N*-Z-protected amino alcohols **68**, obtained by acylation of 3-aminopropan-1-ol with Z-Cl or by reduction of methyl 4-(benzyloxycarbonyl)aminobutanoate **67** with $NaBH_4$ or $LiAlH_4$, respectively, were subjected to oxidation. Subsequently, condensation of an equimolar amount of the α-amino aldehyde **69** with Z-(diethoxyphosphinyl)Gly-OMe **70** was carried out in the presence of tBuOK for 3 hours to give the desired Z-ΔOrn(Z)-OMe and Z-ΔLys(Z)-OMe **71**. The corresponding free acids were obtained by hydrolysis of **71** with 1 M aq LiOH at room temperature for 3 hours. For the synthesis of ΔNCAs **72** of both ΔOrn and ΔLys, compounds **71** was subjected to an intramolecular cyclization with $SOCl_2$ in $Et_2O$ below 0 °C for 30 minutes and then at room temperature for 1 hour to give ΔOrn(Z)-NCA and ΔLys(Z)-NCA **72**, respectively (Scheme 23). However, in our laboratory attempts to prepare ΔLys-containing tuftsin analogues by the above method did not give good results. Therefore, for the synthesis of ΔLys we used the chlorination/dehydrohalogenation method.[153]

**Scheme 23** Synthesis of ΔOrn and ΔLys Derivatives[81,151]




**H-ΔLys(Z)OMe:**[153]

HCl·H-Lys(Z)OMe (3.00 g, 12 mmol) was taken in $Et_2O$ (50 mL) and treated with sat. $Na_2CO_3$ (50 mL). The $Et_2O$ layer was separated and washed once with distilled $H_2O$ and dried (anhyd $Na_2SO_4$). The H-Lys(Z)-OMe was transferred to a flask covered with dark paper and treated with tBuOCl (1.3 mL, 11 mmol) in cold (4 °C) and the reaction monitored by TLC. On completion of the reaction, the solvent was removed under reduced pressure, the oily residue was dissolved in $CHCl_3$(30 mL), washed with 0.5 M HCl (10 mL), $H_2O$ (10 mL), and then dried (anhyd $Na_2SO_4$). All the above operations were performed in the dark. The $CHCl_3$ was removed and the residue redissolved in dry $Et_2O$ (20 mL). This soln was treated dropwise with DBU (1.7 mL, 11.4 mmol) in dry $Et_2O$ (10 mL) at rt and stirred for 1.5 h; the reaction was followed by TLC. When the reaction was complete, the precipitated DBU·HCl was filtered off to give H-ΔLys(Z)-OMe; yield: 2.5 g (82%).

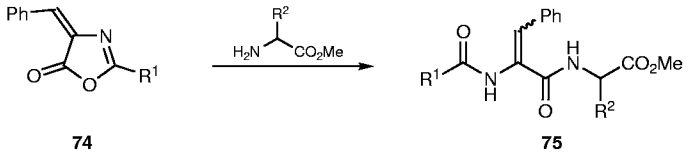
**Boc-Leu-ΔLys(Z)-OMe:**[153]

A soln of Boc-Leu-OH (1.0 g, 7 mmol) and NMM (0.76 mL, 7 mmol) in dried THF (15 mL) was cooled to −5 °C in an ice–salt bath and treated with isobutyl chloroformate (0.9 mL, 7 mmol). After stirring for 20 min, a cooled soln of H-ΔLys(Z)-OMe (2.0 g, 7 mmol) was added and the resulting mixture stirred for 15 min at low temperature, the mixture was then brought to rt and stirred for 4 h. On completion of the reaction, the solvent was removed and the residue taken up in EtOAc, washed with sat. $NaHCO_3$, $H_2O$, 5% citric acid, and brine, and dried (anhyd $Na_2SO_4$). The solvent was removed in vacuo and the oily product so obtained was purified by column chromatography (silica gel, petroleum ether/$CHCl_3$ 2:8) to give Boc-Leu-ΔLys (Z)-OMe as an oil; yield: 2.86 g (81%); $[\alpha]_{30}$ −7.0 (*c* 0.75, MeOH).

## 11.1.2 α,β-Didehydropeptides

As discussed above, DHAs have low basicity and nucleophilicity. Therefore when DHAs are used as the amino component, coupling is usually slow and inefficient. Aromatic DHAs are even more unreactive as the amino component in the synthesis of α,β-didehydropeptides. Various coupling methods, such as those involving an acid chloride,[83] mixed anhydride,[154,155] DCC, and water-soluble carbodiimide with HOBt,[156] have been used in the acylation of enamino acid esters, but with low yields. Azlactones have permitted the incorporation of DHAs into the peptide chain. Although unsaturated azlactones are relatively stable in cold water, their reaction with amines are quite facile. Treatment of an unsaturated azlactone **74** with the alkali salt of an α-amino acid or α-amino acid methyl ester proceeds with the opening of the azlactone ring leading to the formation of a peptide bond as in **75** (Scheme 24). This reaction scheme was subsequently employed for the synthesis of a notable series of α,β-didehydropeptides.[57–63]

**Scheme 24** Synthesis of α,β-Didehydropeptides from Azlactones




**74** **75**

When $N^{\alpha}$-blocked DHAs are used as acid components, the usual coupling procedures, such as the mixed anhydride,[83] *N*-carboxyanhydride,[155] and DCC or EDC/HOBt,[154] are applicable, the mixed anhydride being most efficient of all. Coupling of $N^{\alpha}$-blocked α,β-didehydroamino acid with α,β-didehydroamino acid esters, though possible (acid chloride method), is inefficient. The double bond in DHAs and their esters, is a good acceptor of nucleophiles under basic conditions, as in such residues it is in conjugation with the carbonyl group of the carboxylic moiety. However, dimerization of the imino tautomer under basic conditions has been reported. DHAs also undergo polar, photochemical,[156] and peroxide-catalyzed radical additions.[157] Thus, for a successful synthesis of α,β-didehydropeptides, it is important to decide when and where the dehydroamino acid residues are to be incorporated into the peptide sequence and to adopt a suitable procedure accordingly. While ΔLeu and ΔPhe peptides have been shown to be quite stable towards acids and bases, ΔAla-containing peptides tend to polymerize.

### 11.1.2.1 Synthesis of α,β-Didehydropeptides in Solution

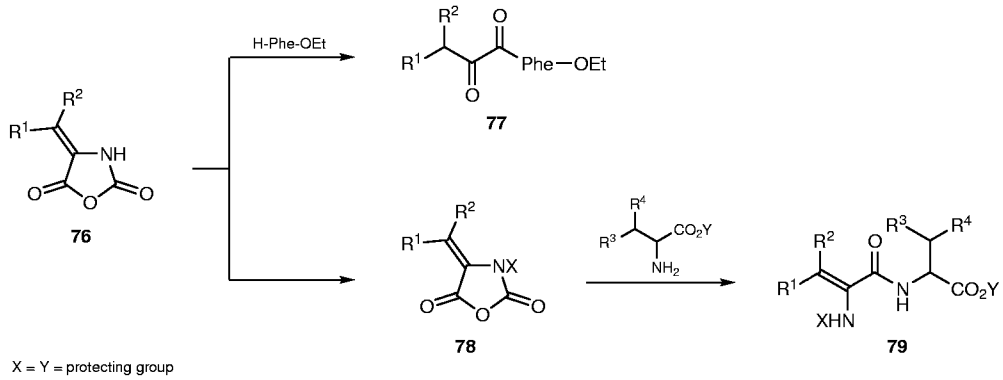
In general α,β-didehydropeptides are synthesized by activation of the carboxy group of *N*-acylated DHAs. The desired peptides are obtained via the mixed anhydride[83] in modest to good yields using DCC alone or in combination with DMAP or HOBt. Various activation methods have been investigated for the coupling of ΔPhe and ΔAla dipeptides. In addition to the azlactone method discussed earlier, ΔPhe tripeptides can also be prepared in very good yields using diphenylphosphoryl azide. For the synthesis of the difficult to access ΔAla peptides, activation with DCC/HOBt proves to be the best method. Generally, irrespective of the activating method, the *E* configuration of the starting DHA is not retained and the *Z*-derivative is obtained almost exclusively.

*N*-Carboxy-α,β-didehydroamino acid anhydrides **76** have been employed with success in the synthesis of peptides[77–83] (Scheme 25). The ΔNCA formed from the reaction of (*Z*)-DHA with thionyl chloride does not form the desired peptides on direct opening with amines

*for references see p 659*

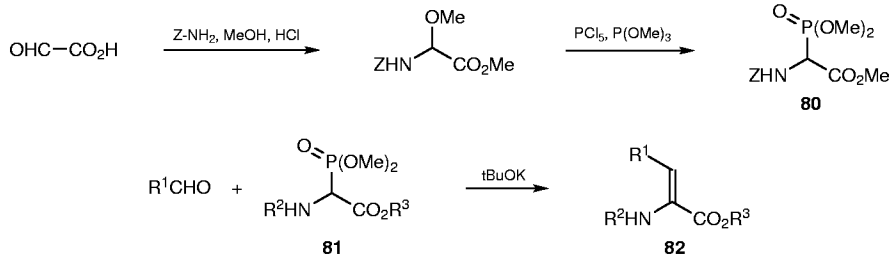
or amino acid esters; instead, the α-oxocarboxamides **77** are formed. However, when these ΔNCAs are protected at nitrogen or are acylated by amino acid chlorides or amino acid anhydrides to give compounds **78**, they can be employed with success in the construction of peptides. Ring opening with amino acid esters, dipeptides, or even tripeptides then furnishes the corresponding dehydropeptides **79** in high yields.

**Scheme 25** Synthesis of α,β-Didehydropeptides from ΔNCAs[77–83]




The frequently observed difficulties in the activation of α,β-didehydropeptides, especially in the case of ΔAla, can be avoided by peptide construction using the corresponding β-substituted amino acids and subsequent β-elimination in the peptide.[158–161] A further possibility is the phosphonate condensation, where the corresponding, modified phosphorylglycine esters can react with aldehydes to yield α,β-didehydropeptides[151,162,163] (Scheme 26).

**Scheme 26** Synthesis of α,β-Didehydropeptides by Condensation of Phosphorylglycine Esters and Aldehydes[151,162,163]




*N*-(Benzyloxycarbonyl)dimethoxyphosphorylglycine methyl ester **80** can be prepared in kilogram quantities in the laboratory from glyoxalic acid hydrate (Scheme 26). Phosphorylglycine esters with other N- and O-protecting groups can also be prepared easily. *N*-(Acyl)alkoxyphosphorylglycine esters **81** condense with aldehydes in the presence of base to give *N*-acyl-DHA esters **82** in excellent yields (Scheme 26). The most successful procedure is the condensation in dichloromethane at 70 °C with addition of potassium *tert*-butoxide. Under these conditions the *Z*-isomers are formed preferentially. Ketones do not react. Mesoxalic ester alone condenses with formation of ΔAsp derivatives. The reactions with aromatic, heterocyclic, or very sensitive aliphatic aldehydes gives a yield in the range of 80–95%.

### 11.1.2.2 Synthesis of α,β-Didehydropeptides on Solid Support

Very few reports exist in the literature on the synthesis of α,β-didehydropeptides on solid support. In most cases the DHA is first synthesized in solution as a dipeptide unit and then introduced into the growing peptide sequence. Dehydro analogues of bradykinin,[127] dermorphin,[164] and bombesin[164] are some of the examples where the synthesis was performed by the solid-phase method. Coupling of the amino acids was carried out by DCC/HOBt. Coupling times were variable, depending on the amino acids. α,β-Didehydrodipeptides usually take a slightly longer time than amino acids. Catalytic hydrogenation can not be used to deprotect side-chain functional groups or to cleave the peptide from the resin.

## 11.1.3 Bioactive Peptides Containing α,β-Didehydro-α-amino Acids

The incorporation of DHAs into natural bioactive peptides has become an interesting objective. As discussed above, the introduction of DHAs provides the peptides with a unique profile: rigidity, increased hydrophobicity, electrophilic reactivity, restricted orientation of β-substituents, and resistance to enzymatic degradation. By taking advantage of these properties, many DHA-containing analogues of bioactive peptides have been synthesized in recent years.[21,22] Replacement of an amino acid in a peptide hormone could lead to elucidation of the structural requirements of receptor interactions if the substitution was made at or close to the active or binding sites of hormones. Many peptide hormone analogues containing DHAs have been synthesized and their biological activity reported. In this section some of these examples are described.

### 11.1.3.1 Synthesis and Structure–Function Analysis of Bioactive Peptides Containing α,β-Didehydro-α-amino Acids

**Enkephalin**

Enkephalin [H-Tyr-Gly-Gly-Phe-Leu(or Met)-OH] is found in the brain as an opiate-like neuropeptide. English and Stammer[165,166,167] reported the synthesis of a highly potent [ΔAla$^2$, ΔPhe$^4$,Met$^5$]-enkephalin amide that was fivefold more potent than the saturated analogue. Shimohigashi and Stammer[19,128,168,169] prepared a series of unsaturated enkephalins: [ΔAla$^2$]-, [ΔAla$^2$-,ΔAla$^3$], [ΔAla$^2$,ΔPhe$^4$]-, and [ΔAla$^2$,ΔLeu$^5$]-enkephalins. All the peptides were synthesized by solution-phase methods. The ΔPhe moiety was produced either by DDQ oxidation of a saturated azlactone[56] or by spontaneous dehydration and azlactonization of the dipeptide Boc-Gly-Ser(β-Ph)-OH using NaOAc in $Ac_2O$.[63] The ΔAla moiety was prepared from Ser in the final protected pentapeptide by the standard tosylation and β-elimination (using DEA) reactions.[99] ΔLeu was introduced by the N-chlorination/dehydrochlorination method. These analogues exhibit almost the full receptor activity. It has been shown that [ΔPhe$^4$]- and [ΔLeu$^5$]-enkephalins have unchanged δ-enkephalin receptor selectivity. Furthermore, it has been suggested that the phenyl ring of ΔPhe$^4$, oriented in the *Z* configuration, is important for the interaction with the δ-receptor.

**Angiotensin**

[Sar$^1$,Ala$^8$]-Angiotensin II, H-Sar-Arg-Val-Tyr-Ile-His-Pro-Ala-OH; is an antagoinst of angiotensin II (AII).[170] The Ala residue in position 8 was replaced by ΔAla, ΔVal, ΔPhe, and ΔAun (α,β-didehydro-α-aminoundecanoic acid), and the agonist and antagonist behaviors of each dehydropeptide was evaluated for inhibition of aldosterone biosynthesis using isolated cells. [Sar$^1$,ΔAla$^8$]-AII was the most active AII inhibitor, being almost 35-fold more active than saralasine. [Sar$^1$,ΔVal$^8$]-AII was less active than [Sar$^1$,ΔAla$^8$]-AII, and [Sar$^1$,ΔAun$^8$]-AII was a mixed agonist–antagonist. However, [Sar$^1$,ΔPhe$^8$]-AII was an agonist.

*for references see p 659*

**Bradykinin**

Fisher et al.[127] synthesized the unsaturated analogues of the vasodilator peptide hormone bradykinin, H-Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg-OH, (BK). The [ΔPhe]-BK analogues were synthesized by the solid-phase technique from the Aoc-(Tos)Arg-resin with incorporation of ΔPhe into the growing peptide chain as the dipeptide Boc-Gly-ΔPhe-OH using DCC as the coupling reagent. Complete coupling required at least 20 hours.[127] [ΔPhe$^5$]-BK showed a high biological activity in its blood pressure lowering effects, being 23-fold more potent than BK. On the other hand, [ΔPhe$^8$]-BK was less potent than BK and [ΔPhe$^5$,ΔPhe$^8$]-BK had effect comparable to those of BK. It has been suggested that the high potency may be the result of increased resistance to enzymatic degradation and perhaps of conformational changes.

**Dermorphin**

The opioid heptapeptide amide dermorphin, H-D-Tyr-D-Ala-Phe-Gly-Tyr-Pro-Ser-$NH_2$, is the first example of a peptide from nonbacterial origin containing a D-amino acid. The Phe$^3$ and/or Tyr$^5$ residues in dermorphin and its hexapeptide and pentapeptide amides were replaced by ΔPhe in order to examine the effect on opioid activity.[18,171,172] The peptides were synthesized by the solution-phase method. The ΔPhe moiety was obtained through simultaneous dehydration and azlactonization of Z-D-Ala-(βϕ)Ser-OH. The hexa- and heptapeptides containing ΔPhe in position 3 and/or 5 displayed low potency. The unsaturation at positions 3 and 5 was detrimental to μ activity. The dramatic decrease of potency of both penta- and hexapeptides confirms a trend previously observed for the corresponding dermorphin tetrapeptides. The dramatic drop in activity shown by these peptides, where a single configuration was present at the unsaturated moiety strongly, suggests that an important interaction between the specific receptor site and the aromatic ring is no longer allowed.

**Substance P**

Seven analogues of substance P (H-Arg-Pro-Lys-Pro-Gln-Gln-Phe-Phe-Gly-Leu-Met-$NH_2$) were synthesized;[175] the Phe residue at position 5 was replaced by ΔPhe and the Gly residue in position 9 was replaced by Ala, Val, Sar, and Aib.[175] All the SP analogues were synthesized by the segment condensation approach in solution. The DHA residue was incorporated as a dipeptide sequence that was obtained as Boc-Phe-ΔPhe-Azl from Boc-Phe-Phe-Ser(β-Ph)-OH by dehydration and azlactonization. The hexapeptide series (fragment 6–11) showed significant activity.[19] Octapeptides [ΔPhe$^{5,8}$,Sar$^9$]-SP (4–11) were found to be the most potent analogues both *in vitro* and *in vivo* systems.[19] Even though turn structures have been described for substance P as the active conformations, a substantial retention of bioactivity in [ΔPhe$^{5,8}$,Sar$^9$]-SP (4–11) may suggest that a helical type of backbone could allow a meaningful ligand–receptor interaction.[173,174]

**LHRH[Phe$^5$,ΔAla$^6$]**

The des-Gly$^{10}$[Phe$^5$,ΔAla$^6$]-analogue of the luteinizing hormone releasing hormone (LHRH; Glp-His-Trp-Ser-Tyr-Gly-Leu-Arg-Pro-Gly-$NH_2$) was synthesized by the solution-phase method.[175] The LHRH sequence contains a number of amino acid side-chain functional groups that are usually protected in the course of the solution-phase synthesis. Due to the high reactivity of ΔAla, a synthetic strategy was adopted which avoided a drastic deprotection step. The analogues containing ΔAla were found to be inactive.[176] The solution structure of LHRH has been a subject of debate. Circular dichroism and other techniques have indicated ordered structures, at least in parts of the peptide sequence. However, a recent two-dimensional NMR study has shown that LHRH does not have any preferred conformation in solution. Although the exact nature of the conformational constraints brought in by DHA substitution is not yet clear, it may be that the introduction of ΔAla in position 6 would not be favorable for the LHRH molecule to adopt the conformation required for bioactivity.

**Melanostatin**

The C-terminal tripeptide tail of oxytocin H-L-Pro-Leu-Gly-$NH_2$ (melanocyte-stimulating hormone release inhibiting factor, MIF or melanostatin) is implicated as having a direct effect on the central nervous system. The ΔLeu analogue of melanostatin, [$\Delta Leu^2$-MIF] was prepared in view of the earlier work on the active conformation of melanostatin in which a β-turn was implicated.[177] ΔLeu was prepared by the N-chlorination/dehydrochlorination reaction and then coupled with Boc-Pro-OH.

**Chemotactic Peptides**

Chemotactic *N*-formylpeptides are involved in defense mechanisms against bacterial infections through binding with a specific receptor located on the neutrophil membranes. *N*-Formyl-Met-Leu-Phe methyl ester (fMLP-OMe), representing the reference model compound for formylpeptide receptor, is the most extensively studied member of this group of bioactive peptides. Several DHA analogues of chemotactic peptides with ΔPhe and ΔLeu have been prepared by different groups.[29,178–182] All the peptides were synthesized by the solution-phase method. ΔPhe was introduced by the usual azlactone method. Boc-Met-ΔLeu-Phe-OMe was obtained by following the one-pot method of Shin[142] starting from ΔLeu-NCA.[29,151] It has been observed that promotion of a β-turn is likely to be the reason for the highly selective, hypertensive potency of the chemotactic peptide analogue HCO-Met-ΔLeu-Phe-OMe in stimulating neutrophils to β-glucosaminidase release and superoxide production. The ability of the dehydropeptides to induce chemotaxis is, however, under discussion. Another chemotactic peptide analogue HCO-Met-Leu-ΔPhe-Phe-OMe that assumes an "open turn" conformation with its ΔPhe residue at the turn position $i+2$ is also active.

**Gramicidin S**

Gramicidin S (GS) is a cyclic decapeptide antibiotic with the primary structure *c*[-Val-Orn-Leu-D-Phe-Pro-]$_2$-GS has a β-sheet conformation with two type-II β-turns at the sequences -D-Phe-Pro-. This conformation is stabilized by four intramolecular hydrogen bonds between the Val and Leu residues. Gramicidin S analogues containing dehydroalanine were synthesized in solution and found to be only partially active.[183,184] The ΔPhe analogues, where the $Phe^{4,4'}$ residues were replaced by ΔPhe, were synthesized by the solution-phase method, by introducing the ΔPhe residue by spontaneous dehydration and azlactonization of Boc-Leu-Ser(β-Ph)-OH.[185] This resulted in the reinforcement of the β-sheet/β-turn structure, which is the most important structural element for GS bioactivity. [$\Delta Phe^{4,4}$]-Gramicidin S exhibited very strong antibacterial activities against Gram-positive bacteria.


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# 11.2 Synthesis of Peptides Based on Aromatic Heterocycles

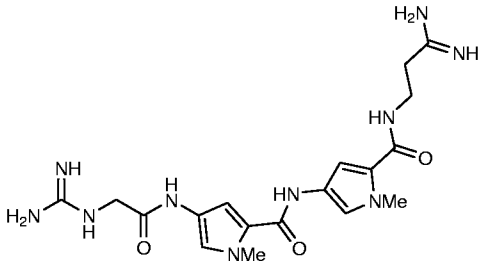
C. P. Miller and P. Wipf

Physicochemical and biological properties of peptide sequences are prominently altered by the introduction of aromatic heterocycles into the peptide backbone. The added rigidity and resistance toward metabolic changes clearly are major incentives for medicinal chemists to experiment with a broad range of heterocyclic building blocks in the quest for biologically active lead structures. Cyclohydration of Ser, Thr, and Cys residues to give oxazoles and thiazoles is a common biosynthetic modification in prokaryotic microorganisms. Many of the more subtle structural and biological effects of heterocycle–peptide hybrids are still unexplored, and in the future more systematic investigations will be necessary to determine the specific effects of the heterocyclic moiety of the peptide sequence. Therefore the ready synthetic availability of peptides with heterocyclic substructures is a crucial issue, and since much of the traditional heterocyclic chemistry uses conditions that are incompatible with the delicate functionality and stereochemistry of peptide strands, there is a continued demand for improved synthetic methodology in this area.

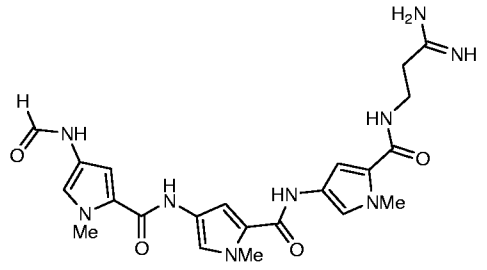
## 11.2.1 Pyrrole-Based Peptides

The pyrrole segment in peptide backbone sequences has been largely identified with the important natural products netropsin,[1] kikumycin B,[2] antehelvencin,[3] and distamycin[4] (Scheme 1). These natural products share the feature of one or more 2-acyl-4-amino-1-methylpyrrole subunits (along with one or two basic end pieces). Due to the important utility of these compounds as anticancer,[5] antibiotic,[6] and antiviral[7] agents, as well as their use as tools for base-pair specific recognition of double-stranded DNA,[8,9] pyrrole-based peptides have attracted considerable attention from the chemical and biological communities.

**Scheme 1** Netropsin and Distamycin: Important Natural Products Containing a Polypyrrole Segment[1,4]




netropsin

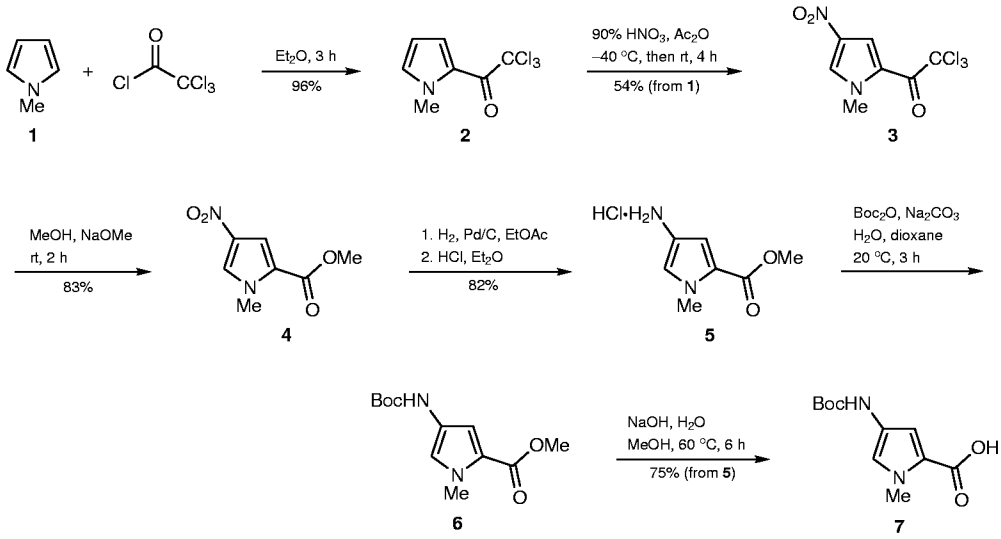



distamycin

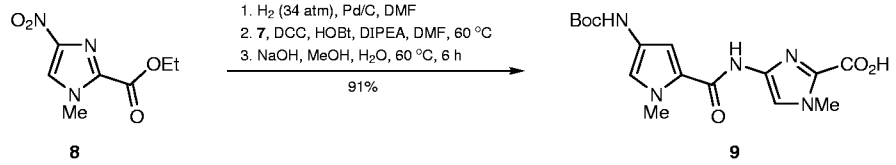
*for references see p 690*

The synthesis of a pyrrole segment common to netropsin and distamycin is shown in Scheme 2.[10] Friedel–Crafts acylation of 1-methylpyrrole (**1**) followed by nitration at C4 provides **3** in 54% yield. After a haloform reaction, hydrogenolysis, N-protection with $Boc_2O$, and saponification, the pyrrolecarboxylic derivative acid **7** was obtained in 30% overall yield from **3**. This monomer is readily chain-extended to the pyrrole–imidazole derivative **9** (Scheme 3).[10] Furthermore, solid-phase synthesis with this and related pyrrole-containing building blocks leads to polyamides that have recently been used in the recognition of a 16 base-pair sequence in the minor groove of DNA.[11]

**Scheme 2** Synthesis of 4-(*tert*-Butoxycarbonylamino)-1-methylpyrrole-2-carboxylic Acid: A Heterocyclic Subunit for Several Naturally Occurring Oligopeptides[10]




**Scheme 3** Synthesis of a Pyrrole–Imidazole Heterodimer Peptide[10]




**1-Methyl-2-(trichloroacetyl)pyrrole (2):**[10]

To a well-stirred soln of trichloroacetyl chloride (1 kg, 5.5 mol) in $Et_2O$ (1.5 L) in a 12-L flask was added dropwise over a period of 3 h a soln of 1-methylpyrrole (**1**; 0.45 kg, 5.5 mol) in anhyd $Et_2O$ (1.5 L). The soln was stirred for an additional 3 h, and the mixture was quenched by the dropwise addition of a soln of $K_2CO_3$ (400 g) in $H_2O$ (1.5 L). The layers were separated, and the ether layer was concentrated under reduced pressure to provide **2** as a yellow crystalline solid sufficiently pure to be used without further purification; yield: 1.2 kg (~96%).

**1-Methyl-4-nitro-2-(trichloroacetyl)pyrrole (3):**[10]

To a cooled (−40 °C) soln of 1-methyl-2-(trichloroacetyl)pyrrole (**2**; 1.2 kg, 5.1 mol) in $Ac_2O$ (6 L) in a 12-L flask equipped with a mechanical stirrer was added fuming $HNO_3$ (440 mL) over a period of 1 h while a temperature of −40 °C was maintained. The mixture was carefully allowed to warm to rt and stirred for an additional 4 h. The mixture was cooled to −30 °C and iPrOH (6 L) was added. The soln was stirred at −20 °C for 30 min, during which time a white precipitate formed. The soln was allowed to stand for 15 min and the resulting precipitate collected by filtration to provide **3**; yield: 0.8 kg (54% from **1**).

**Methyl 1-Methyl-4-nitropyrrole-2-carboxylate (4):**[10]
To a soln of **3** (800 g, 2.95 mol) in MeOH (2.5 L) in a 4-L Erlenmeyer flask equipped with a mechanical stirrer was added dropwise a soln of NaH (60% dispersion in mineral oil) (10 g, 0.25 mol) in MeOH (500 mL). The mixture was stirred at rt for 2 h, and the reaction was quenched by the addition of concd $H_2SO_4$ (25 mL). The mixture was then heated to reflux and allowed to slowly cool to rt as **4** crystallized as white needles, which were collected by vacuum filtration and dried; yield: 450 g (83%; 47% from **1**).

**Methyl 4-Amino-1-methylpyrrole-2-carboxylate Hydrochloride Salt (5):**[10]
Methyl 1-methyl-4-nitropyrrole-2-carboxylate (**4**; 450 g, 2.44 mol) was dissolved in EtOAc (8 L). A slurry of 10% Pd/C (40 g) in EtOAc (800 mL) was then added and the mixture was stirred under a slight positive pressure of $H_2$ (ca. 1.1 atm) for 48 h. Pd/C was removed by filtration through Celite and washed with EtOAc (1 × 50 mL) and the volume of the mixture was reduced to ca. 500 mL; cold $Et_2O$ (7 L) was added and HCl gas gently bubbled through the mixture. The precipitated amine hydrochloride was then collected by filtration to give a white powder; yield: 380 g (82%).

**Methyl 4-(*tert*-Butoxycarbonylamino)-1-methylpyrrole-2-carboxylate (6):**[10]
Methyl 4-amino-1-methylpyrrole-2-carboxylate hydrochloride (**5**; 340 g, 1.78 mol) was dissolved in 10% aq $Na_2CO_3$ (1 L) in a 3-L flask equipped with a mechanical stirrer; $Boc_2O$ (400 g, 1.83 mol) slurried in dioxane (500 mL) was added over a period of 30 min while a temperature of 20 °C was maintained. The reaction was allowed to proceed for 3 h and was determined complete by TLC; the mixture was cooled to 5 °C for 2 h and the resulting white precipitate collected by filtration. The product, which was contaminated with $Boc_2O$, was used without further purification for the next step.

**4-(*tert*-Butoxycarbonylamino)-1-methylpyrrole-2-carboxylic Acid (7):**[10]
Crude methyl 4-(*tert*-butoxycarbonylamino)-1-methylpyrrole-2-carboxylate (**6**) from the previous step was dissolved in MeOH (700 mL); 2 M NaOH (700 mL) was added and the soln was heated at 60 °C for 6 h. The mixture was cooled to rt and washed with $Et_2O$ (4 × 1 L), the pH of the aqueous layer lowered to ca. 3 with 10% $H_2SO_4$, and the mixture was extracted with EtOAc (4 × 2000 mL). The combined EtOAc extracts were dried ($Na_2SO_4$) and concentrated under reduced pressure to provide a tan foam. The foam was dissolved in $CH_2Cl_2$ (500 mL) and petroleum ether (2 L) added, and the resulting slurry was concentrated. The residue was redissolved and concentrated three additional times to provide **7** as a fine white powder; yield: 320 g (75% from **5**).

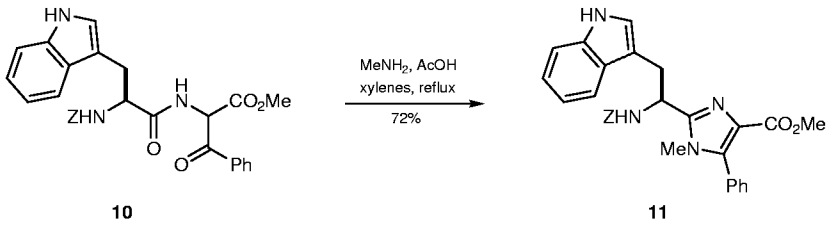
**4-({[4-(*tert*-Butoxycarbonylamino)-1-methyl-1*H*-pyrrol-2-yl]carbonyl}amino)-1-methyl-1*H*-imidazole-2-carboxylic Acid (9):**[10]
To a soln of Boc-pyrrole acid **7** (3.0 g, 12.5 mmol) in DMF (10 mL) was added HOBt (2.0 g, 14.8 mmol, 1.2 equiv) followed by DCC (3.0 g, 14.8 mmol, 1.2 equiv). The soln was stirred for 24 h and the DCU removed by filtration. Separately, to a soln of ethyl 1-methyl-4-nitroimidazole-2-carboxylate (2.9 g, 14.8 mmol) in DMF (5 mL) was added 10% Pd/C (1 g), and the mixture was hydrogenated in a Parr bomb apparatus (34 atm of $H_2$) for 2 h. The catalyst was removed by filtration through Celite and the filtrate immediately added to the OBt ester soln. An excess of DIPEA (4 mL) was then added and the mixture stirred at 60 °C for 8 h. The mixture was then added dropwise to a stirred soln of ice water and the resulting precipitate collected by filtration to provide crude ester. To this compound dissolved in MeOH (12 mL) was added 1 M NaOH (12 mL), and the resulting mixture was stirred at 60 °C for 6 h. Excess MeOH was removed in vacuo to give a brown powder; yield: 4.1 g (91%).

## 11.2.2 Imidazole-Based Peptides

The synthesis of peptidergic imidazoles can be readily accomplished through the same type of *N*-β-oxo amide intermediates used to prepare oxazoles described in Section 11.2.3. The formation of the peptide-based imidazole is accomplished by heating the *N*-β-oxo amide with ammonia or another primary amine.[12] An example of this reaction is shown in Scheme 4.[12] $C^{\alpha}$-Acylation of the Schiff base of Gly produced the α-amino-3-oxo ester, which was then acylated with Z-Trp-OH to generate **10**. The condensation between methylamine and **10** was best carried out in refluxing toluene and acetic acid while azeotropically removing water from the reaction mixture. The reaction reportedly does not work well in cases when the alkyl group on the amine is larger than a methyl.

*for references see p 690*

**Scheme 4** A Peptide-Based Imidazole from the Reaction of an *N*-β-Oxo Amide and Methylamine[12]




10 11

The preparation of an imidazole-pyrrole dipeptide segment for DNA binding studies is shown in Scheme 3[10] and described in Section 11.2.1.

**Methyl 2-[(*S*)-1-(Benzyloxycarbonylamino)-2-(1*H*-indol-3-yl)ethyl]-1-methyl-5-phenylimidazole-4-carboxylate (11):**[12]

A suspension of **10** (12.0 g, 23.4 mmol) in xylenes (200 mL) was treated with AcOH (10 mL) and 3.5 M $MeNH_2$ in THF (10 mL) and the mixture was refluxed for 1 h with azeotropic removal of $H_2O$. The soln was allowed to cool; additional 3.5 M $MeNH_2$ in THF (5 mL) and AcOH (10 mL) were added and the reflux continued for an additional 4 h. After cooling, the soln was washed once with brine, dried ($MgSO_4$), filtered, and concentrated in vacuo. Purification (silica gel, EtOAc/hexanes 1:1) followed by concentration and recrystallization gave imidazole **11**; yield: 8.6 g (72%); mp 142–144 °C; $[\alpha]_D^{25}$ +13.4 (*c* 1, DMF).

## 11.2.3 Oxazole-, Dihydrooxazole-, and Oxazolidine-Based Peptides

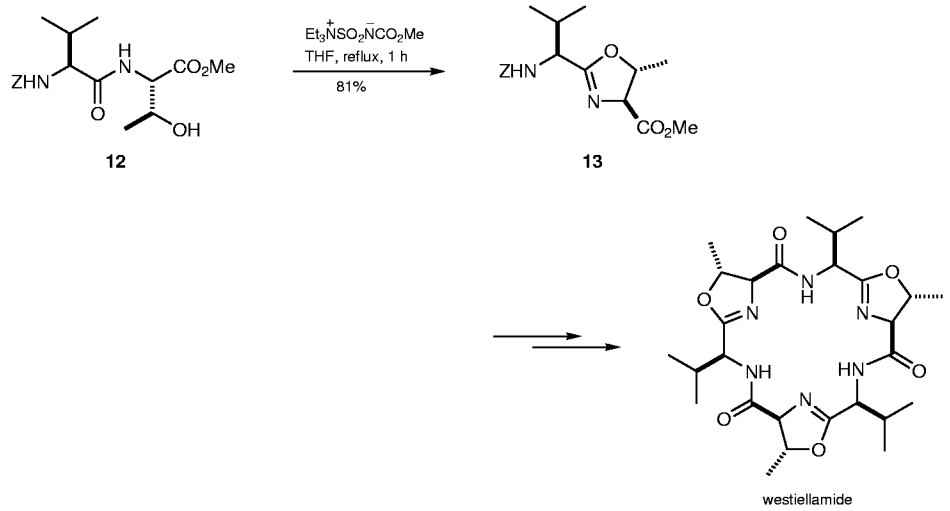
### 11.2.3.1 Peptide-Based Dihydrooxazoles as Precursors to Peptide-Based Oxazoles

The synthesis of oxazoles from the oxidation of dihydrooxazoles (oxazolines) has seen wide utility as a method for incorporation of heterocycles into the backbone of peptides (see also Vol. E 22 b, Section 6.8.5.2.3). The preparation of many peptidergic oxazoles is consequently limited to the methods available for the synthesis of the required dihydrooxazole precursors. The cyclodehydration reactions of peptides containing Thr or Ser residues result in a 5-methyl- or 5-unsubstituted 4,5-dihydrooxazole, respectively. Several methods have been described wherein the Ser or Thr hydroxy group is initially activated as a mesylate or tosylate and subsequently heated in the presence of a base.[13,14] Alternatively, a Ser hydroxy group may be converted into a chloride which can then be cyclized by heating the β-halo peptide with a base.[15,16] Of greater general utility has been the use of dehydrating reagents such as thionyl chloride,[17,18] DEAD/$Ph_3P$ (Mitsunobu conditions),[19,20–23] the Burgess reagent [methyl *N*-(triethylammoniosulfonyl)carbamate],[24–28] and DAST as well as Deoxo-Fluor.[29] These reagents have the advantage of allowing the cyclodehydration to occur in a single step. In the case of Thr residues, the displacement results in a chiral inversion at the β-carbon and the resulting dihydrooxazole has *cis* stereochemistry at the C4 and C5 positions.[30] For the synthesis of oxazoles from dihydrooxazoles, the stereochemistry is lost upon oxidation. Whatever method is employed for dihydrooxazole formation should be general and result in minimal formation of other side products such as β-elimination of the alcohol,[31] competing aziridine[25,32] or β-lactam formation,[33] or peptide racemization,[34] and the reagent and conditions should be compatible with various peptide protecting groups. The Burgess reagent method appears to be the most generally suitable for effecting these transformations because: (1) it leads to minimal racemization at sensitive residues adjacent to the dihydrooxazole;[34] (2) the reaction does not result in the formation of β-lactam or aziridines;[22,25] (3) little or no β-elimination side products are obtained;[25] and (4) the method appears to be compatible with all common peptide protecting groups.

A PEG-supported Burgess reagent has been reported which provides increased yields of dihydrooxazoles as well as increased reagent stability.[35,36]

In Scheme 5, the synthesis of a dihydrooxazole derived from the dipeptide segment of Z-Val-aThr-OMe is demonstrated.[37] Simple heating of the protected dipeptide **12** in THF in the presence of 1 equivalent of the Burgess reagent results in a good yield of the desired dihydrooxazole **13**. In this particular case, the unnatural amino acid allothreonine (aThr) was used so that the cyclodehydration would yield the desired *trans*-dihydrooxazole. The resulting dipeptide dihydrooxazole was used in the first total synthesis of the silver-sequestering marine natural product westiellamide.[37–39] Westiellamide is a representative member of a class of marine cyclopeptide alkaloids containing multiple dihydrooxazole, oxazole, dihydrothiazole, and thiazole rings.[40–42] The conformational properties of marine cyclopeptide alkaloids, as well as the use of five-membered heterocycles as peptide mimetics and scaffolds favoring β-turn conformations, are of considerable interest.[42–44]

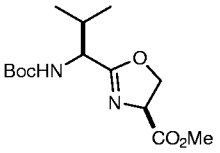
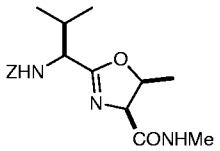
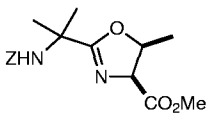
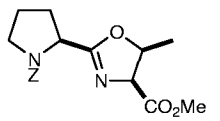
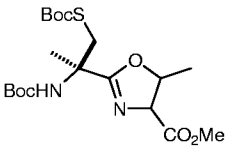
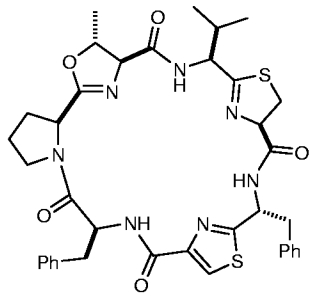
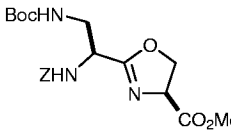
**Scheme 5** Peptide-Based Dihydrooxazoles via Cyclodehydration Mediated by the Burgess Reagent[37]




Further examples for the cyclodehydration of β-hydroxy-α-amino acid containing peptides to the corresponding dihydrooxazoles with Burgess reagent are given in Table 1.

*for references see p 690*

**Table 1** Cyclodehydration of Peptide Segments Containing Ser or Thr Residues to Dihydrooxazoles Using the Standard Burgess or the Polymer-Linked Burgess Reagent[25,35,45–47]

| Substrate[a] | Conditions[b] | Product | Yield (%) | Ref |
|---|---|---|---|---|
| Boc-Val-Ser-OMe | Burgess reagent, THF, 60 °C |  | 74 | [25] |
| Z-Val-Thr-NHMe | Burgess reagent, THF, 60 °C |  | 64 | [25] |
| Z-Aib-Thr-OMe | PEG-supported Burgess reagent, dioxane/THF, 85 °C |  | 88 | [35] |
| Z-Pro-Thr-OMe | PEG-supported Burgess reagent, dioxane/THF, 85 °C |  | 85 | [35] |
| Boc-(*R*)-(αMe)Cys-(Boc)-D,L-Thr-OMe | Burgess reagent, THF, 65 °C |  | 70 | [45] |
| *c*[-Pro-aThr-Val-ψ[C(=S)-NH]Ser-Phe-Ser(thz)-Phe-] | Burgess reagent, THF, 65 °C |  | 61[c] | [46] |
| Z-Dap(Boc)-Ser-OMe | Burgess reagent, THF, 70 °C |  | 63 | [47] |

[a] thz = thiazolyl.
[b] Burgess reagent = $Et_3N^+SO_2N^-CO_2Me$.
[c] Lissoclinamide 4.

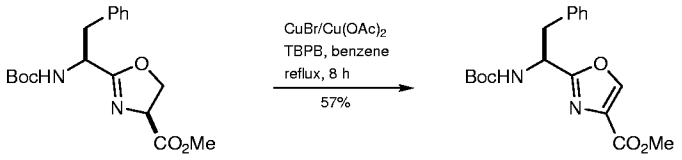
**Methyl (4*S*,5*R*)-2-[(*S*)-1-(Benzyloxycarbonylamino)-2-methylpropyl]-5-methyl-4,5-dihydrooxazole-4-carboxylate (13):**[37]

A 0.2 M soln of the dipeptide **12** in THF was treated with the Burgess reagent (1.1 equiv) and heated at reflux for 1 h. The mixture was allowed to cool, concentrated onto silica gel and chromatographed (EtOAc/hexanes, 1:1) to give the *trans*-dihydrooxazole **13** as a foam; yield: 81%; $[\alpha]_D^{23}$ +47.6 (*c* 0.2, $CH_2Cl_2$).

### 11.2.3.2 Oxidation of Peptide-Based Dihydrooxazoles to Peptide-Based Oxazoles

A broad range of conditions for oxidizing peptide-based dihydrooxazoles to oxazoles have been developed. The most useful methods have applied $BrCCl_3$,[48] PhSeCl,[49] $NiO_2$,[50] Cu(II) salts,[51] or mixtures of Cu(I) and Cu(II) salts.[52] While often $NiO_2$ has been used for small-scale transformations, its general utility has been limited because of reagent expense, toxicity, and modest and/or variable yields of desired products. The application of Cu(II)-mediated protocols has enjoyed modest success and the recent description of mixed Cu(I)/Cu(II) systems has shown special promise. The dehydrogenation of dihydrooxazoles to oxazoles proceeds in fair to good yields and is relatively simple to carry out. In Scheme 6 this method is demonstrated for the oxidation of dihydrooxazole **14** to oxazole **15**.[52]

**Scheme 6** Peptide-Based Oxazoles from Peptide-Based Dihydrooxazoles via Oxidation with Cu(I)/Cu(II)/Peroxide[52]




**14** **15**

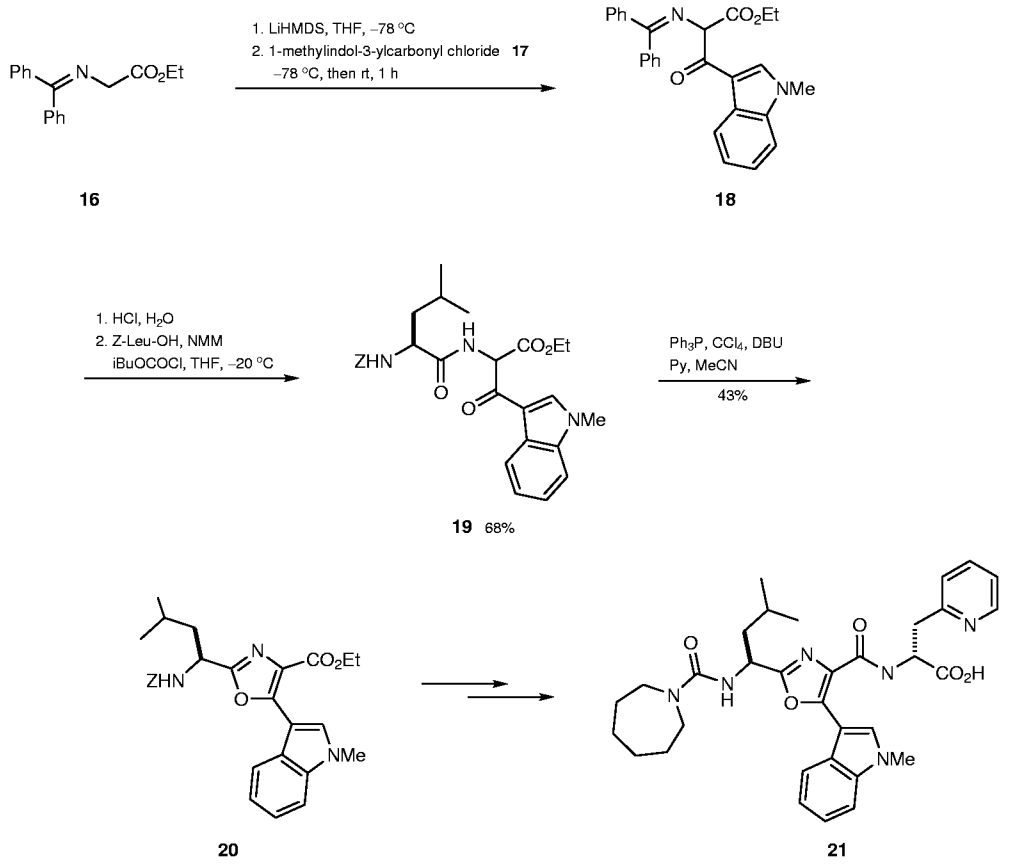
**Methyl 2-[(*S*)-1-(*tert*-Butoxycarbonylamino)-2-phenylethyl]oxazole-4-carboxylate (15):**[52]
To neat dihydrooxazole **14** (0.400 g, 1.15 mmol) in a 50-mL round-bottomed flask were added CuBr (0.182 g, 1.27 mmol) and $Cu(OAc)_2$ (0.230 g, 1.27 mmol). The flask was evacuated and then filled with argon. The evacuation/argon addition was repeated twice. Benzene (**CAUTION:** *carcinogen!*) was syringed into the flask and stirring initiated (dihydrooxazole concentration at 0.16 M). The mixture was warmed to 60 °C, and TBPB (0.335 g, 1.73 mmol) was added gradually over 15 min. The mixture was then heated to reflux for 8 h. After cooling to rt, EtOAc (10 mL) was added, and the mixture was washed with buffered 10% $NH_4OH$ soln to remove all the copper salts. The aqueous layer was extracted with EtOAc (3 × 10 mL), the combined organic extracts were dried ($Na_2SO_4$), and the crude product, obtained after concentration under reduced pressure, was subjected to flash chromatography (silica gel, hexane/EtOAc 80:20) which gave **15** as a colorless oil; yield: 0.227 g (57%); $[\alpha]_D^{23}$ −16.0 (*c* 2.8, $CHCl_3$).

### 11.2.3.3 Peptide-Based Oxazoles from Glycine Acylation and Cyclodehydration

An approach which has particularly general appeal for the synthesis of substituted oxazoles involves the direct $C^{\alpha}$-acylation of a Gly ester.[12] The method has the advantage of allowing for a large number of substituents at the C5 position of a C-terminal peptide-based oxazole, because any acid chloride can presumably be used for the acylation of the Gly ester. Additionally, the same intermediate is useful for the preparation of thiazoles and imidazoles, *vide supra*. In fact, the versatility of the method has allowed its application in industrial processes.[53,54] Recently, for example, the glycine acylation methodology was used in the synthesis of endothelin antagonists incorporating a substituted oxazole as a heterocycle surrogate for an amide bond (Scheme 7).[55] The Gly Schiff base **16**[56] was treated with lithium hexamethyldisilazanide in THF at −78 °C. The resulting ester anion was then reacted with 1-methylindol-3-ylcarbonyl chloride (**17**, prepared from the carboxylic acid) and the product **18** of this reaction was used without purification for the subsequent steps. The amine terminus of **18** was deprotected by hydrolysis of the imine with $HCl/H_2O$ and coupled with Z-Leu-OH using the mixed anhydride method[57] to give the oxo amide **19** which was subsequently cyclodehydrated under Appel conditions[58] to form the desired oxazole **20**. This dipeptide surrogate was then converted through numerous steps into the desired product **21**, which was evaluated for endothelin activity.

*for references see p 690*

**Scheme 7** A Peptide-Based Oxazole via $C^{\alpha}$-Acylation of a Glycine Anion[55]




***N*-(Benzyloxycarbonyl)leucyl-2-(1-methylindol-3-ylcarbonyl)glycine Ethyl Ester (19):**[55]

1-Methylindole-3-carboxylic acid (1 g, 5.7 mmol) was suspended in $CH_2Cl_2$ (10 mL); oxalyl chloride (1.05 mL, 12 mmol) was added (gas evolves), and the resultant soln was stirred at rt for 90 min. The solvents were removed under reduced pressure, and the residue was dried under vacuum to remove traces of reagent from the crude acid choride **17**, which was used without further purification. Simultaneously, *N*-(diphenylmethylene)glycine ethyl ester (**16**; 2.6 g, 9.7 mmol) was dissolved in THF (15 mL), and the soln was cooled to −78 °C. 1 M LiHMDS in THF (10 mL) was added dropwise, and the resultant yellow slurry was stirred at −78 °C for 30 min. The slurry was transferred via cannula to a soln of the acid chloride **17** in THF (10 mL) at −78 °C. After the addition was complete, the mixture was allowed to warm to rt and stirring was continued for 2 h. The mixture was then quenched with 2 M HCl (120 mL) and allowed to stir at rt for 1 h. The solvent was removed and the resultant aqueous soln was extracted with EtOAc (2 × 10 mL). The organic phases were re-extracted with 2 M HCl and then discarded, and the combined aqueous phases were concentrated. The resultant slurry was treated with EtOH (20 mL) and filtered. The filtrate was concentrated in vacuo to give the product as a pinkish solid which was used without further purification. (This material may be stored at −20 °C as the salt, but is unstable as the free base.) Z-Leu-OH (2.65 g, 10 mmol) was dissolved in THF (15 mL), and the soln was cooled to −20 °C. NMM (1.11 mL, 10 mmol, 1.0 equiv) was added, followed by the dropwise addition of isobutyl chloroformate (1.30 mL, 10 mmol). After the addition was complete, the mixture was stirred at −20 °C for 30 min. The cooling bath was removed, and the above salt was dissolved in DMF (20 mL) and added to the mixed anhydride. NMM (1.2 mL, 11 mmol) was added dropwise via syringe pump over a 1 h period. After the addition was complete, the mixture was allowed to stir at rt for 15 h. The solvents were removed in vacuo, and the residue was taken up in EtOAc. The organic layer was washed sequentially with sat. $NaHCO_3$, 0.33 M $H_3PO_4$, and brine, dried ($MgSO_4$), and concentrated under reduced pressure to give an orange oil which was purified by flash chromatography (silica gel, EtOAc/hexane 1:2 to 1:1) to give **19** as a yellowish oil; yield: 3.45 g (68% overall).

**Ethyl 2-[(*S*)-1-(Benzyloxycarbonylamino)-3-methylbutyl]-5-(1-methylindol-3-yl)oxazole-4-carboxylate (20):**[55]

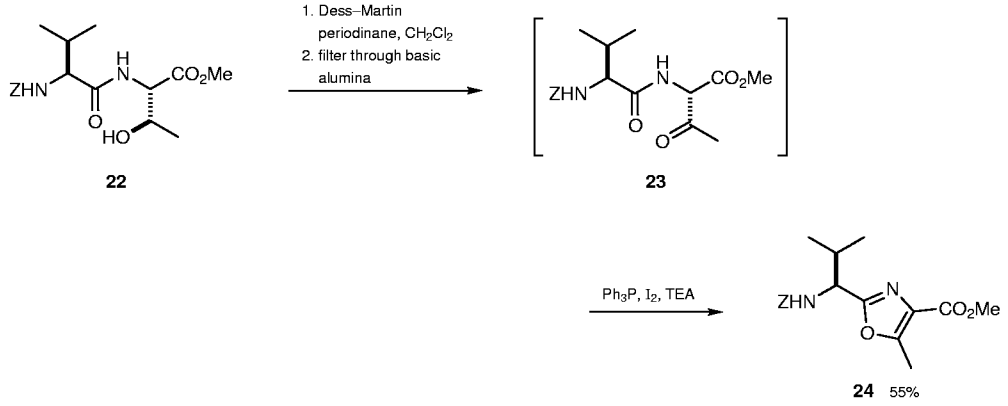
Oxo ester **19** (500 mg, 1.0 mmol) was dissolved in MeCN/pyridine (1:1, 3 mL); $CCl_4$ (0.2 mL, 2.0 mmol), DBU (0.5 mL, 3.3 mmol), and $Ph_3P$ (285 mg, 1.08 mmol) were added sequentially, and the resultant soln was allowed to stir at rt for 18 h. The solvents were removed under reduced pressure, and the residue was dissolved in EtOAc. The soln was washed with sat. $NaHCO_3$, 0.33 M $H_3PO_4$, and brine, dried ($MgSO_4$), and concentrated to give a yellowish oil which was purified by flash chromatography (silica gel, 25% EtOAc/hexane) to give **20**; yield: 208 mg (43%).

## 11.2.3.4 Peptide-Based Oxazoles by Oxidation and Cyclodehydration of Threonine and β-Phenylserine Residues

A complementary pathway to the preparation of oxazoles by dehydrogenation of dihydrooxazoles derived from β-hydroxy α-amino acids involves the cyclodehydration of an *N*-β-oxo amide to form the oxazole directly. Conceptually, this method is related most closely to the Robinson–Gabriel oxazole synthesis.[59,60] The primary challenge of this method is found in the synthesis of the ketone or aldehyde within the peptide fragment. Any oxidation of alcohol precursors requires that the method be mild and compatible with various peptide functionalities, and fragmentation of the backbone or retro-Claisen reactions are very facile in peptide derivatives. This is particularly an issue in the side-chain oxidation of serine; the resulting aldehyde is quite unstable and difficult to isolate. Another problem is the use of active, yet mild and selective cyclodehydrating conditions because overly harsh reagents attack backbone amide groups and must therefore be avoided.

The synthesis of peptides containing 5-methyloxazoles or 5-phenyloxazoles can be accomplished efficiently by the oxidation of peptide fragments containing threonine or β-phenylserine residues using the Dess–Martin periodinane reagent [1,1,1-triacetoxy-1,1-dihydro-1,2-benziodoxol-3(1*H*)-one 1-oxide].[61,62] In Scheme 8, this method is demonstrated for the synthesis of the oxazole **24** derived from the dipeptide Z-Val-Thr-OMe (**22**).[63] After oxidation with Dess–Martin reagent, the β-oxo ester **23** thus formed can then be cyclodehydrated very efficiently through the use of $Ph_3P/I_2$/TEA to provide the oxazole **24**. Often it is not necessary to isolate the ketone; the entire oxidation reaction mixture can be filtered through a plug of basic alumina (under a nitrogen atmosphere) into a solution of the $Ph_3P/I_2$/TEA reagent combination.

**Scheme 8** Synthesis of a Peptide-Based 5-Methyloxazole from a Dipeptide Containing a Threonine Residue via an Oxidation–Cyclodehydration Sequence[63]

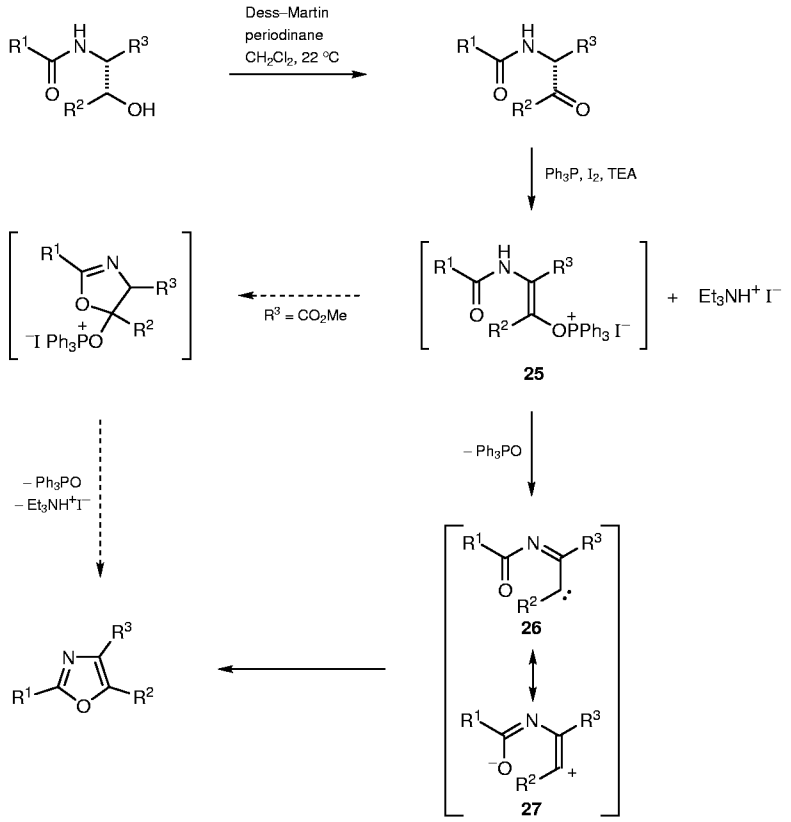



The proposed mechanism for this reaction sequence includes a carbene intermediate and is shown in Scheme 9.[63] Although the addition–elimination sequence of **25** (dashed arrows) cannot be completely ruled out, it would require a disfavored 5-*endo-trig* dig cyclization.[63,64]

*for references see p 690*

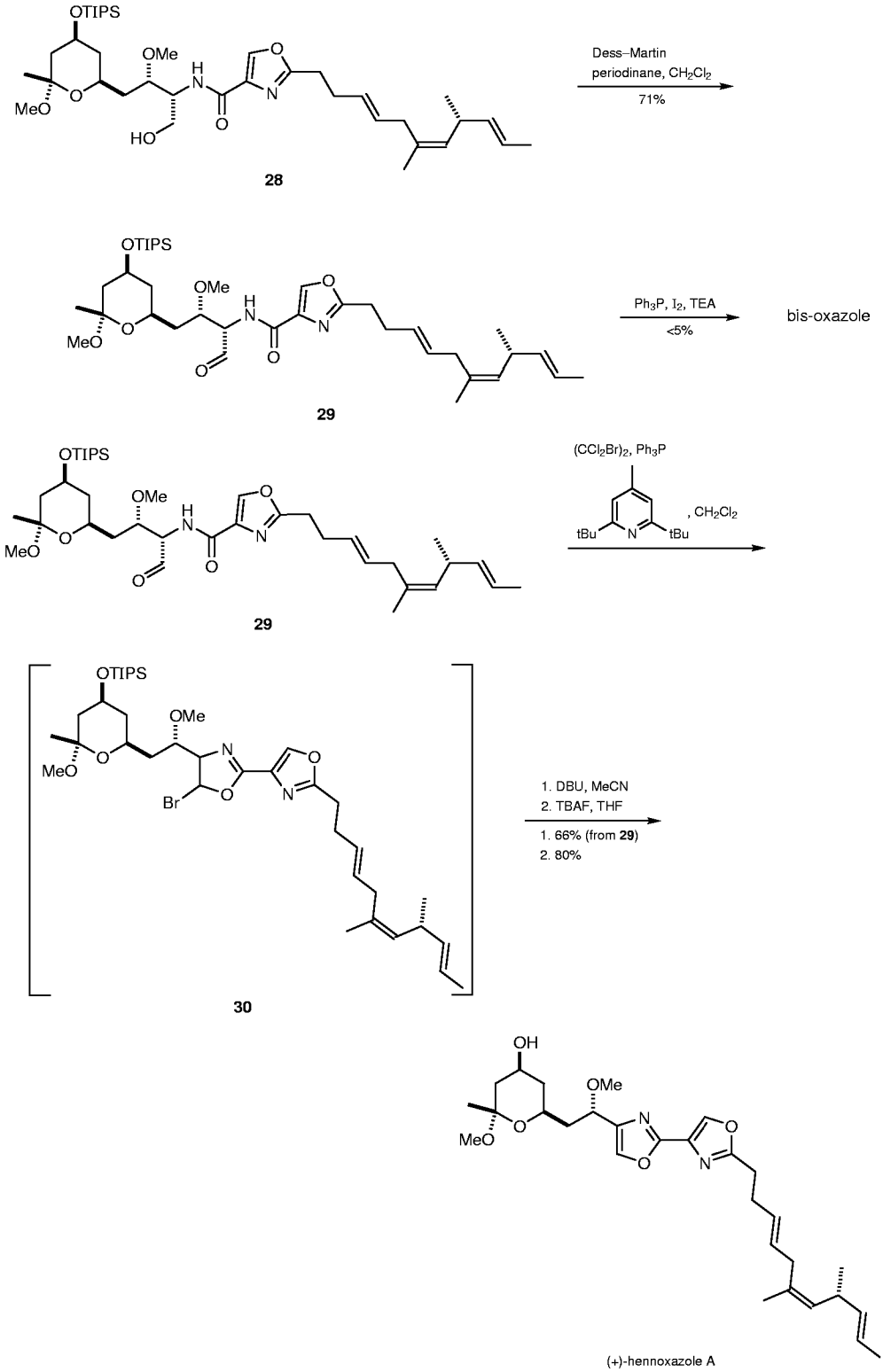
Additionally, this oxazole synthesis can also be used when the group at C4 is not likely to stabilize a negative charge, which lends further support to the elimination–addition sequence via intermediate **26** and its resonance structure **27**.

**Scheme 9** Proposed Mechanism for Peptide-Based Oxazole Formation via Dehydration of an *N*-β-Oxo Amide[63]




Attempts to synthesize 5-unsubstituted oxazoles by oxidation and cyclodehydration of serine residues are generally low yielding. However, a modification of this protocol that provides C5-unsubstituted oxazoles from hydroxymethylene amides via bromodihydrooxazoles was recently developed by Wipf and Lim and successfully applied in total syntheses of hennoxazole A and phorboxazole A (Scheme 10).[65–67] Oxidation of primary alcohol **28** with Dess–Martin periodinane provided the labile aldehyde **29** in 71% yield. Under standard conditions for oxazole formation,[63] decomposition of the acid-labile ketal function in **29** was a major concern. In contrast, the milder cyclodehydrating agent derived from treatment of triphenylphosphine with 1,2-dibromo-1,1,2,2-tetrachloroethane in the presence of the hindered base 2,6-di-*tert*-butyl-4-methylpyridine provided the bromodihydrooxazole **30** in satisfactory yield. After dehydrohalogenation with DBU in acetonitrile and removal of the triisopropylsilyl ether protecting group with TBAF, the marine natural product hennoxazole A was obtained in 37% overall yield from hydroxy amide **28**. This reaction sequence demonstrates that peptide-based oxazole formation is possible even in highly functionalized, acid-, base-, and oxidation-sensitive substrates.

**Scheme 10** Synthesis of a 5-Unsubstituted Oxazole-Based Peptide from Hydroxymethylene Amides via Bromodihydrooxazoles[65–67]




*for references see p 690*

Examples of the cyclodehydration of peptide segments containing serine or threonine residues or the corresponding oxo amides to oxazoles are given in Table 2, Section 11.2.3.5.

**Methyl 2-[(*S*)-1-(Benzyloxycarbonylamino)-2-methylpropyl]-5-methyloxazole-4-carboxylate (24):**[63]
A mixture of Z-Val-Thr-OMe (**22**; 0.20 g, 0.55 mmol) and Dess–Martin periodinane (0.28 g, 0.66 mmol) in $CH_2Cl_2$ (5 mL) was stirred for 1 h, and filtered under argon through a bed of basic alumina (1.5 cm, activity 1) into a flask containing a freshly prepared soln of $Ph_3P$ (0.30 g, 1.10 mmol), $I_2$ (0.28 g, 1.10 mmol), and TEA (0.31 mL, 2.20 mmol) in $CH_2Cl_2$ (5 mL). The alumina filter cake was washed with $CH_2Cl_2$ (2 × 5 mL). After 15 min, the black mixture was transferred to a separatory funnel and diluted with $Et_2O$ (100 mL) and washed with concd $Na_2S_2O_3$ (30 mL). After vigorous extraction of the aqueous layer for a few minutes (in order to insure complete reduction of any remaining $I_2$), the $Et_2O$ layer was washed with sat. $NaHCO_3$ (20 mL) and dried ($Na_2SO_4$). Purification (silica gel, EtOAc/hexanes 1:3) afforded **24** as a crystalline white solid; yield: 0.103 g (55%); mp 110–110.5 °C; $[\alpha]_D^{22}$ −35.9 (*c* 0.4, $CH_2Cl_2$).

**(2*S*,4*S*,6*S*)-6-[(2*S*)-2-{2′-[(3*E*,6*Z*,8*R*,9*E*)-6,8-Dimethylundeca-3,6,9-trienyl]-2,4′-bioxazolyl-4-yl}-2-methoxyethyl]-2-methoxy-2-methyl-tetrahydropyran-4-ol [(+)-Hennoxazole A]:**[65,66]
To a soln of alcohol **28** (11 mg, 16 μmol) in $CH_2Cl_2$ (2 mL) was added Dess–Martin periodinane (21 mg, 49 μmol). After 30 min at 21 °C, the mixture was directly subjected to column chromatography (silica gel, hexanes/EtOAc 5:1 to 3:1) to give the aldehyde **29**; yield: 7.8 mg (71%).

To a soln of aldehyde **29** (7.8 mg, 11 μmol), $Ph_3P$ (92 mg, 0.35 mmol), and 2,6-di-*tert*-butyl-4-methylpyridine (0.22 g, 1.07 mmol) in $CH_2Cl_2$ (4 mL) was added $(CCl_2Br)_2$ (114 mg, 0.35 mmol) at 0 °C. After 30 min at 0 °C, DBU (0.16 g, 1.05 mmol) was added followed by MeCN (5 mL). After 2 h at 21 °C, the solvent was removed and the residue was chromatographed (silica gel, hexanes/EtOAc 10:1 to 5:1) to afford O-TIPS-protected hennoxazole A as an oil; yield: 5.0 mg (66%).

A soln of this TIPS ether (5 mg, 7.4 μmol) in THF (1 mL) was treated with 1.0 M TBAF in THF (5 drops) at 22 °C. The mixture was stirred for 1 h, concentrated and chromatographed on (silica gel, hexanes/EtOAc 3:1 to 1:2) to give (+)-hennoxazole A as an oil; yield: 3.2 mg (80%); $[\alpha]_D^{21}$ +50 (*c* 0.16, $CHCl_3$).

## 11.2.3.5 Peptide-Based Oxazoles by Rhodium-Mediated Carbenoid Insertion of Diazocarbonyl Compounds into an Amide N–H Bond Followed by Cyclodehydration

This method uses the same type of β-oxo intermediate as described in Section 11.2.3.4 and converges with it in the cyclodehydration step with $Ph_3P/I_2$/TEA. The method of forming the β-oxo intermediate, involving the rhodium-mediated carbenoid insertion of an α-diazo-β-oxo ester, however, is novel and allows for the introduction of a variety of substituents at the 5-position of the oxazole. This methodology has recently been used for the synthesis of the natural product (+)-nostocyclamide.[68,69] The synthesis of the oxazole dipeptide fragment of the marine cyclopeptide (+)-nostocyclamide is shown in Scheme 11.[70] The rhodium-catalyzed N–H insertion reaction of Z-alaninamide **31** with methyl diazoacetoacetate resulted in the formation of the *N*-β-oxo amide **32** which was subsequently dehydrated to **33** by the $Ph_3P/I_2$/TEA protocol discussed previously for Section 11.2.3.4. The dipeptide oxazole fragment was then coupled to thiazole-containing fragments and cyclized to the final product (+)-nostocyclamide.

**Scheme 11** Synthesis of a Peptide-Based Oxazole via Rhodium-Catalyzed N–H Insertion of an α-Diazo-β-oxo Ester into $N^{\alpha}$-(Benzyloxycarbonyl)alaninamide Followed by Cyclodehydration[70]

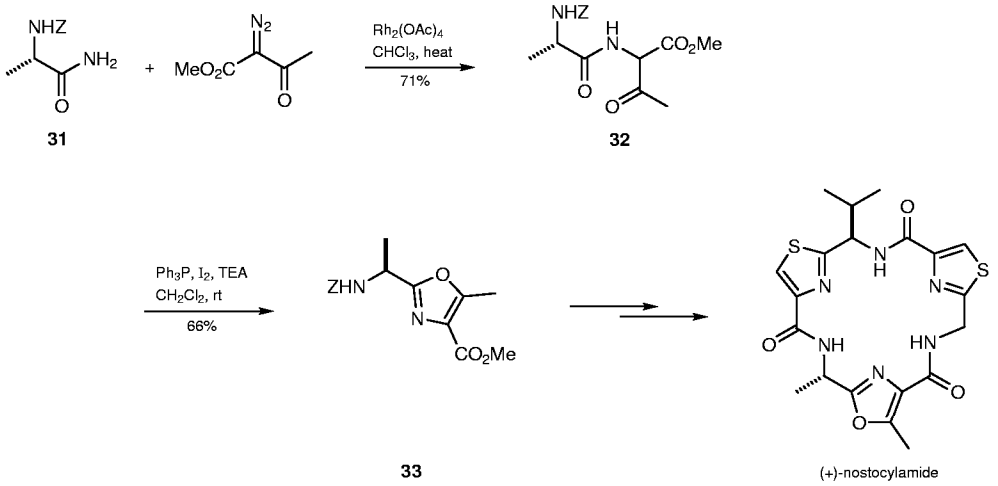
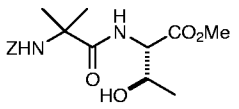
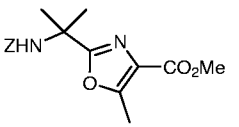
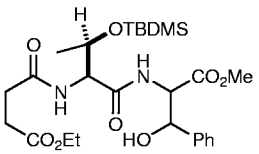
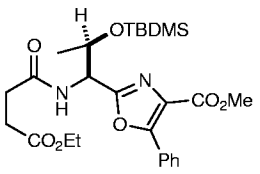
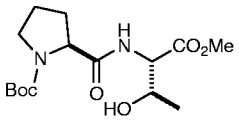
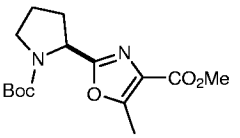
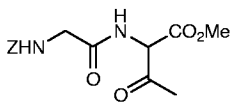
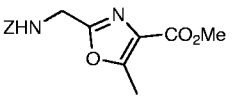
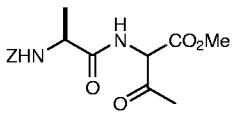
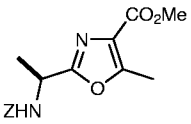
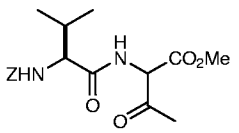
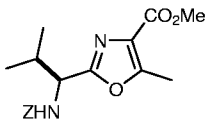


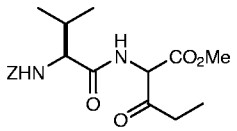
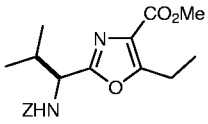
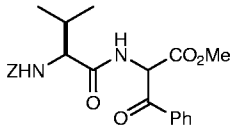
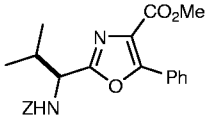

Table 2 provides an overview of peptide-based oxazole syntheses according to Sections 11.2.3.4 and 11.2.3.5.

**Table 2** Cyclodehydration of Peptide Segments Containing Ser or Thr Residues or the Corresponding *N*-β-Oxo Amides to Oxazoles[63,70,72]

| Substrate | Conditions[a] | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | A |  | 58 | [63] |
|  | A |  | 71 | [63] |
|  | A |  | 68 | [72] |
|  | B |  | 56 | [70] |
|  | B |  | 66 | [70] |
|  | B |  | 65 | [70] |

*for references see p 690*

**Table 2** (cont.)

| Substrate | Conditions[a] | Product | Yield (%) | Ref |
|---|---|---|---|---|
|  | B |  | 88 | [70] |
|  | B |  | 31 | [70] |

[a]Conditions: A: 1. Dess–Martin periodinane; 2. $Ph_3P$, $I_2$, TEA; B: $Ph_3P$, $I_2$, TEA.

**Methyl 2-Diazo-3-oxobutanoate:**[70]
A soln of methyl 3-oxobutanoate (4.38 g, 37.8 mmol) and 4-acetamidobenzenesulfonyl azide[71] (10.0 g, 41.6 mmol) in MeCN was cooled to 0 °C and treated with a dropwise addition of TEA (15.8 mL, 113.4 mmol). After stirring at rt for 16 h the mixture was concentrated and the resultant solid was triturated ($Et_2O$/petroleum ether). The filtrate was concentrated in vacuo and purified by flash chromatography (silica gel, $Et_2O$/petroleum ether 1:4) to yield the desired product as a light oil; yield: 91%.

**Methyl (*S*)-2-[[2-(Benzyloxycarbonylamino)propanoyl]amino]-3-oxobutanoate (32):**[70]
A soln of methyl 2-diazo-3-oxobutanoate (1.0 g, 7.0 mmol) in dry $CHCl_3$ was added dropwise over 6 h to a boiling soln of $N^{\alpha}$-(benzyloxycarbonyl)-L-alaninamide (**31**; 1.11 g, 5 mmol) and $Rh(OAc)_2$ (2 mol%) in EtOH-free $CHCl_3$. The mixture was refluxed for a further 30 min, allowed to cool, concentrated and purified by column chromatography (silica gel, EtOAc/petroleum ether 1:1) to afford **32** as an inseparable mixture of diastereomers; yield: 1.19 g (71%); mp 122–123 °C (colorless needles, EtOAc/petroleum ether).

**Methyl 2-[(*S*)-1-(Benzyloxycarbonylamino)ethyl]-5-methyloxazole-4-carboxylate (33):**[70]
TEA (4.1 equiv) and a soln of **32** (~3 mmol, 1 equiv) in dry $CH_2Cl_2$ (10 mL) were added sequentially to a stirred soln of $Ph_3P$ (2.0 equiv) and $I_2$ (2 equiv) in dry $CH_2Cl_2$ (40 mL) at rt under $N_2$. The mixture was stirred until TLC analysis indicated that the reaction was complete, concentrated, purified by flash chromatography (silica gel, EtOAc/petroleum ether) to yield the desired product **33**; yield: 66%; mp 125.5–126 °C (colorless needles, EtOAc/petroleum ether); $[\alpha]_D^{16.5}$ −34.4 (*c* 1.0, $CHCl_3$).

## 11.2.4 Thiazole-, Dihydrothiazole-, and Thiazolidine-Based Peptides

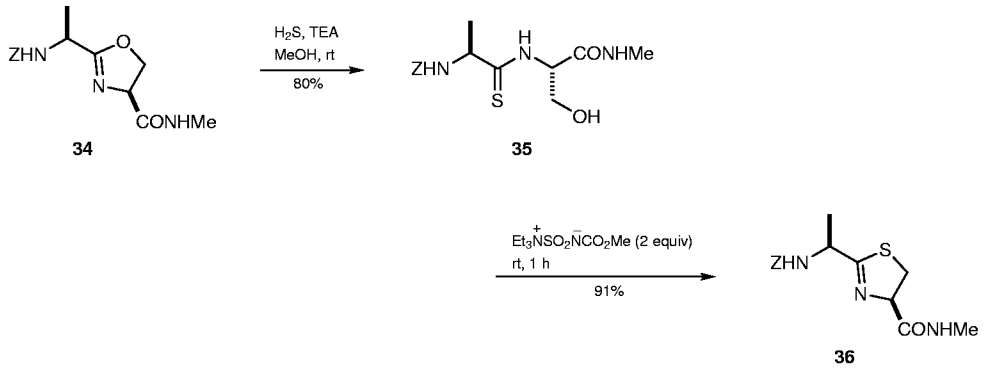
To a large extent, similar methods are used for the preparation of thiazole-, dihydrothiazole-, and thiazolidine-based peptides as presented in Section 11.2.3 for the preparation of the corresponding oxazole- and dihydrooxazole-based peptides. However, the oxidation of dihydrothiazole and thiazolidines to the aromatic heterocycles is considerably more facile than for the analogous oxygen-containing substrates.

### 11.2.4.1 Peptide-Based Dihydrothiazoles as Precursors to Peptide-Based Thiazoles: The Interconversion of Peptide-Based Dihydrooxazoles to Peptide-Based Dihydrothiazoles

The synthesis of thiazoles from dihydrothiazoles (thiazolines) has experienced broad application as a method for incorporating thiazoles into the backbone of peptides (see also Vol. E 22b, Section 6.8.5.2.2). The ability to incorporate thiazoles into the backbone of peptides by oxidation of dihydrothiazole precursors is consequently limited to the methods available for the synthesis of the required dihydrothiazole precursors. The most straightforward approach would allow for a peptide containing natural amino acids to serve as a precursor to

the dihydrothiazole. Indeed, the biosynthesis of dihydrothiazoles is believed to occur as a result of the attack of a Cys sulfur onto the amide carbonyl of the immediately preceding amino acid.[73,74] In fact, this strategy has been also been used in the laboratory to synthesize dihydrothiazoles but the conditions which must be employed (strong acid or $TiCl_4$, neat) may not be conducive to the typical peptide environment with racemizable α-centers.[75,76] In contrast, the nonbiosynthetic approach wherein a thioamide cyclizes to displace the β-alcohol of a Ser or Thr residue can be carried out under mild dehydrative conditions such as the Mitsunobu reaction[21,22] or through use of the Burgess reagent.[22] The Burgess reagent has proven to be particularly well suited to this reaction since the intramolecular $S_N2$-substitution gives consistently good yields with a minimum of racemization at the α-carbon immediately adjacent to the thioamide.[77,78] This approach has been used for the synthesis of dihydrothiazoles with both hydrogen-substitution as well as methyl-substitution at the 5-position. The difficulties inherent in this protocol are primarily limited to the ability to form the desired thioamides at the amino acid carbonyl adjacent to the Ser or Thr residue. These peptide-based thioamides can be prepared by treating the amide containing segment with Lawesson's reagent,[79,80] by thioacylation,[81,82] or in a sequence-specific manner by the interconversion of peptide-based dihydrooxazoles to peptide-based dihydrothiazoles through the opening of dihydrooxazole rings with $H_2S$ to form the desired ψ[C(=S)-NH]Ser (or ψ[C(=S)-NH]Thr) residues.[77] With the latter method, whenever the dihydrooxazole can be formed stereo- and chemoselectively (see Section 11.2.3.1), the dihydrothiazole can also be formed. The conditions of the overall conversion are mild enough to avoid significant racemization even in peptides bearing relatively sensitive amino acids.[78,83] In Scheme 12, the conversion of the dihydrooxazole **34** derived from the dipeptide Z-Ala-Ser-NHMe into the dihydrothiazole **35** is demonstrated. The heterocycle interconversion process takes place in good overall yield and no racemization at either chiral center could be detected.[84]

**Scheme 12** Regioselective, Epimerization-Free Interconversion of Peptide-Based Dihydrooxazoles to Peptide-Based Dihydrothiazoles[84]




**Z-Ala-ψ[C(=S)-NH]Ser-NHMe (35):**[84]

**CAUTION:** *Hydrogen sulfide is a poison by inhalation, is extremely flammable and can ignite on contact with a range of metal oxides. Handle in a fume hood.*

A soln of the dihydrooxazole **34** (0.10 g, 0.33 mmol) in MeOH (5 mL) and TEA (1 mL) was saturated with $H_2S$ gas and stirred at rt for 3 h, concentrated, and chromatographed (silica gel, EtOAc) to yield the thioamide **35** as a foam; yield: 0.089 g (80%); $[\alpha]_D^{23}$ −74 (*c* 1.35, $CH_2Cl_2$).

**(4*R*)-1-[(*S*)-2-(Benzyloxycarbonylamino)ethyl]-*N*-methyl-4,5-dihydrothiazole-4-carboxamide (36):**[84]
A soln of thioamide **35** (0.029 g, 0.085 mmol) in THF (2 mL) was flushed with $N_2$ and treated with the Burgess reagent (0.040 g, 0.17 mmol) and the mixture stirred at rt for 1 h, concentrated, and chromatographed (silica gel, EtOAc) to afford the dihydrothiazole **36** as a foam; yield: 0.025 g (91%); $[\alpha]_D^{23}$ −8.02 (*c* 0.86, $CH_2Cl_2$).

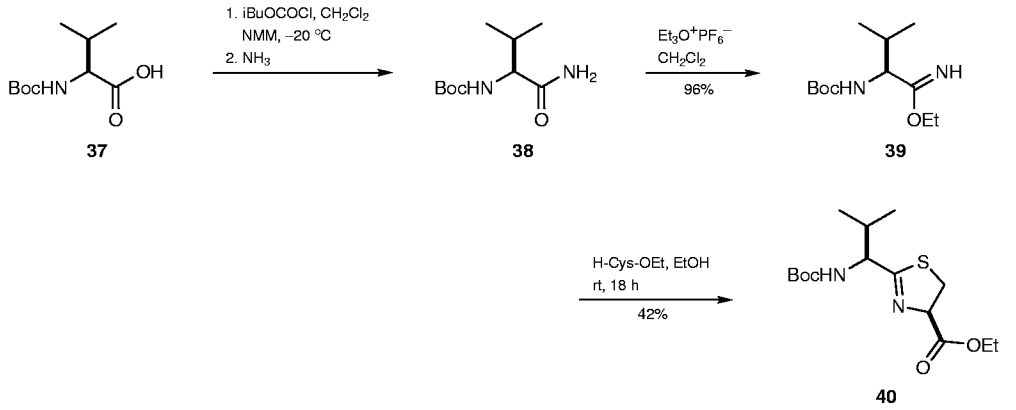
*for references see p 690*

## 11.2.4.2 Peptide-Based Dihydrothiazoles from Condensation of an Imidate and a β-Amino Thiol

A common method of dihydrothiazole synthesis involves the condensation of a Cys residue with an imidate.[85] This method, however, suffers from some drawbacks: (1) It does not readily lend itself to the synthesis of dihydrothiazoles substituted with groups other then a hydrogen at the C5 position because Cys residues substituted at the β-position are not commercially available. (2) The method requires the preparation of an imidate from a peptide C-terminus, which may require several steps and sometimes leads to racemization at the carbon next to the imidate carbon. (3) The dihydrothiazole cannot be introduced into a complete peptide fragment as in Section 11.2.4.1.

The synthesis of a dipeptide dihydrothiazole by way of an imidate is shown in Scheme 13.[52] The Boc-protected amino acid **37** (in this case Val) was converted into the primary amide **38** by the mixed anhydride coupling with ammonia. The amide was then directly converted into the imidate **39** using triethyloxonium hexafluorophosphate which was subsequently condensed with H-Cys-OEt to provide the dihydrothiazole **40**.

**Scheme 13** Synthesis of a Dipeptide Dihydrothiazole by Condensation of an Imidate with Ethyl Cysteinate[52]




**Boc-Val-$NH_2$ (38):**[52]

To a stirred soln of Boc-L-Val-OH (**37**; 0.727 g, 3.35 mmol) in $CH_2Cl_2$ (15 mL) at −30 °C was added NMM (0.405 mL, 3.69 mmol) followed by isobutyl chloroformate (0.482 mL, 3.69 mmol). The mixture was cooled to −20 °C over 30 min, and gaseous $NH_3$ was bubbled through for 15 min. After warming to rt over 1 h, the mixture was diluted with $H_2O$ and the layers were separated. The aqueous layer was extracted with $CH_2Cl_2$, and the combined organic extracts were dried ($Na_2SO_4$) and concentrated under vacuum to give the desired product **38** (yield not given): $[\alpha]_D^{23}$ +0.30 (*c* 2.67, EtOH).

***N*$^{\alpha}$-(*tert*-Butoxycarbonyl)valinamide Ethyl Imidate (39):**[52]

A suspension of Boc-Val-$NH_2$ (**38**; 2.0 g, 9.26 mmol) in dry $CH_2Cl_2$ (23 mL) was cooled to 0 °C, treated with triethyloxonium hexafluorophosphate (2.55 g, 10.19 mmol), and the resulting soln was stirred at rt for 18 h. The mixture was then washed with sat. $NaHCO_3$, dried ($Na_2SO_4$), and concentrated in vacuo to give **39** as a colorless oil which was used without further purification for the next step; yield: 2.17 g (96%).

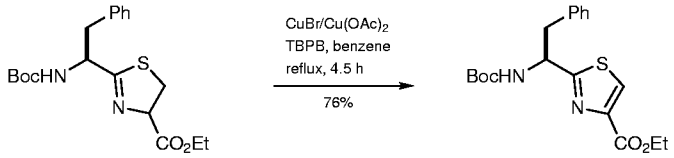
**Ethyl (4*R*)-2-[(*S*)-1-(*tert*-Butoxycarbonylamino)-2-methylpropyl]-4,5-dihydrothiazole-4-carboxylate (40):**[52]

To a soln of the crude imidate **39** (2.17 g, 8.89 mmol) in anhyd EtOH (28 mL) was added H-Cys-OEt (1.32 g, 8.89 mmol) and the resulting soln stirred at rt for 18 h. The reaction solvent was removed and the residue chromatographed (silica gel, EtOAc/hexanes 15:85) which gave **40** as a colorless oil; yield: 0.75 g (42%); $[\alpha]_D^{23}$ −64.3 (*c* 1.1, $CHCl_3$).

### 11.2.4.3 Oxidation of Peptide-Based Dihydrothiazoles to Peptide-Based Thiazoles

The oxidation of dihydrothiazoles to thiazoles can be carried out in a similar fashion as described for the oxidation of dihydrooxazoles to oxazoles (see Section 11.2.3.2). The primary difference between the oxidation of the two heterocycles is that the oxidation of dihydrothiazoles to thiazoles is typically a higher yielding transformation and several general reaction conditions are capable of delivering satisfactory results. The Cu(I)/Cu(II) system described in Section 11.2.3.2 is reported to give good yields and an example is shown in Scheme 14.[52]

**Scheme 14** Peptide-Based Thiazoles from Peptide-Based Dihydrothiazoles by Oxidation with a Cu(I)/Cu(II)/Peroxide System[52]




41 42

**Ethyl 2-[(*S*)-1-(*tert*-Butoxycarbonylamino)-2-phenylethyl]thiazole-4-carboxylate (42):**[52]
To the dihydrothiazole **41** (0.300 g, 0.794 mmol) in a 50-mL round-bottomed flask were added CuBr (0.126 g, 0.873 mmol) and $Cu(OAc)_2$ (0.158 g, 0.873 mmol). The flask was evacuated and then filled with argon. The evacuation/argon addition was repeated twice. Benzene (**CAUTION:** *carcinogen!*) was syringed into the flask and stirring initiated (dihydrothiazole concentration 0.16 M). The mixture was warmed to 60 °C, and TBPB (0.200 g, 1.03 mmol) was added gradually over 15 min. The mixture was then heated to reflux for 8.0 h. After cooling to rt, EtOAc (10 mL) was added, and the mixture was washed with buffered 10% $NH_4OH$ to remove all the copper salts. The aqueous layer was extracted with EtOAc (3 × 10 mL), the combined organic extracts were dried ($Na_2SO_4$), and the crude product, obtained after concentration, was subjected to flash chromatography (silica gel, hexane/EtOAc 85:15) which gave **42** as a colorless oil; yield: 0.227 g (76%); $[\alpha]_D^{23}$ + 5.1 (*c* 1.8, $CHCl_3$).

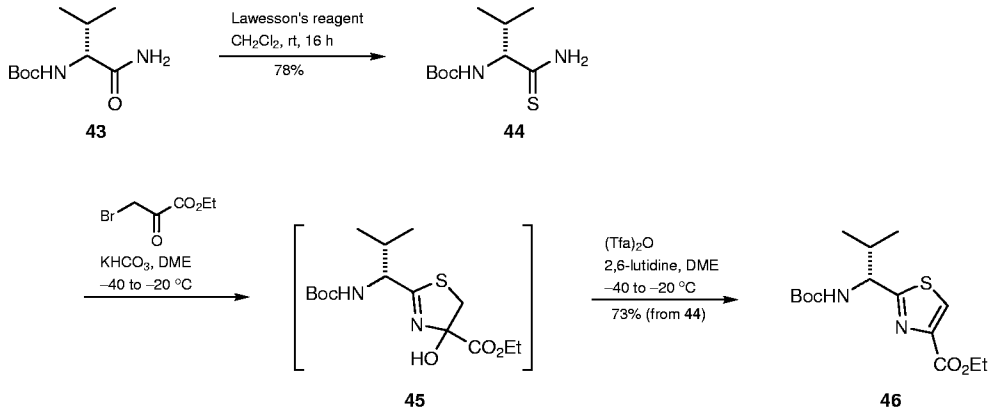
### 11.2.4.4 Peptide-Based Thiazoles via the Condensation of β-Halo-α-oxo Esters with Thioamides: A Modified Hantzsch Synthesis

The Hantzsch synthesis of thiazoles has been applied extensively to the synthesis of peptidergic thiazoles. The original Hantzsch synthesis typically employs simple heating a bromopyruvate with the thioamide in a protic solvent such as ethanol.[86] A potential problem with this approach occurs when the carbon α to the thioamide is chiral. At elevated temperatures, racemization at the chiral center can become extensive. This has led to the investigation of low temperature conditions which can obviate this problem.[87–91] The modified Hantzsch synthesis conditions employ a two-step condensation where the thioamide is allowed to react with a bromopyruvate in the presence of base followed by low-temperature activation of the aminal hydroxy group with a suitable electrophile and base-induced elimination. The conditions must be suitably mild due to the ease of epimerization at the acidic center adjacent to the dihydrothiazole. The most recent and reportedly mild method for the synthesis of peptide-based thiazoles using the Hantzsch synthesis is shown in Scheme 15.[69]

The thioamide **44** was prepared in 78% yield from the treatment of Boc-D-Val-$NH_2$ (**43**) with Lawesson's reagent in $CH_2Cl_2$. Thioamide **44** was then treated with $KHCO_3$ and ethyl bromopyruvate from −40 °C to −20 °C in order to form the intermediate hydroxy-dihydrothiazole **45** (not isolated) which was subsequently dehydrated with $(Tfa)_2O$ and 2,6-lutidine at −20 °C to form the dipeptide thiazole **46** in 73% yield based on thioamide. The product was ascertained to be essentially one enantiomer (>99% ee by chiral HPLC).

*for references see p 690*

**Scheme 15** A Peptide-Based Thiazole via a Modified Hantzsch Synthesis[69]




**Boc-D-Val-ψ[C(=S)-NH]-$NH_2$ (44):**[69]
A soln of Boc-(*R*)-Val-$NH_2$ (**43**; 0.63 g, 2.90 mmol) and Lawesson's reagent (0.61 g, 1.50 mmol) in dry $CH_2Cl_2$ (50 mL) was stirred for 16 h at rt. The mixture was concentrated and purified by flash chromatography (silica gel, $Et_2O$/petroleum ether 1:2) to afford the desired thioamide **44** as a pale yellow solid; yield: 0.53 g (78%). A small portion was recrystallized (EtOAc/petroleum ether) to afford the product as colorless prisms; mp 101–103 °C; $[\alpha]_D^{22.5}$ +44 (*c* 1.02, $CHCl_3$).

**Ethyl 2-[(*R*)-1-(*tert*-Butoxycarbonylamino)-2-methylpropyl]thiazole-4-carboxylate (46):**[69]
$KHCO_3$ (0.75 g, 7.45 mmol) and ethyl bromopyruvate (1.0 mL, 7.97 mmol) were added sequentially to a stirred soln of the thioamide **44** (0.43 g, 1.86 mmol, 1.0 equiv) in DME (4 mL) at –40 °C. The mixture was warmed to –20 °C, stirred for 16 h and filtered through a pad of Celite. The pad was washed with $Et_2O$ (4 mL) and the combined filtrates were concentrated. The residue containing **45** was dissolved in DME (4 mL), cooled to –40 °C and a soln of $(Tfa)_2O$ (0.86 mL, 6.09 mmol, 3.3 equiv) and 2,6-lutidine (1.50 mL, 12.9 mmol, 6.9 equiv) in DME (2 mL) at –20 °C was added. The mixture was warmed to –20 °C over 10 min, stirred at –20 °C for 30 min, concentrated, and partitioned between $CHCl_3$ (40 mL) and $H_2O$ (40 mL). The aqueous layer was further extracted with $CHCl_3$ (20 mL) and the organic extracts were combined, washed with $H_2O$ (40 mL), dried ($Na_2SO_4$), concentrated, and purified by flash chromatography (silica gel, EtOAc/petroleum ether 1:6) to afford **46** as colorless needles; yield: 0.49 g (73%); mp 118.5–119 °C (EtOAc/petroleum ether); $[\alpha]_D^{19}$ +41.6 (*c* 1.06, $CHCl_3$).

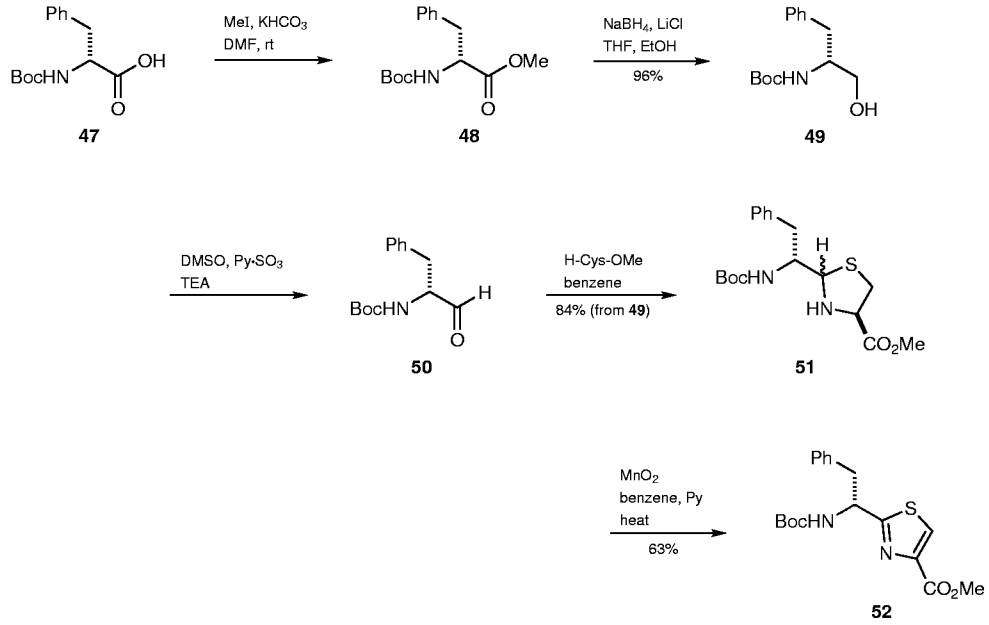
## 11.2.4.5 Peptide-Based Thiazoles from the Synthesis and Oxidation of Peptide-Based Thiazolidines

A useful method for the synthesis of peptide-based thiazoles consists of the chain extension of a peptide segment with a thiazolidine-4-carboxylic acid unit. The N-protected amino acid or peptide is converted into a C-terminal aldehyde and then condensed with a Cys ester in order to form the peptide-based thiazolidine. Subsequent oxidation with $MnO_2$ provides the thiazole.[92] The C-terminal ester can then be hydrolyzed, if desired, and the peptide sequence extended by segment condensations. The difficulties in this method are primarily related to the number of steps required to form the desired heterocycle, as well as the difficulty of obtaining thiazoles substituted with functionalities other than hydrogen at the C5 position. This protocol is reported to proceed without significant racemization and has been used in a number of successful syntheses of thiazole-containing marine cyclopeptides.[93–96]

The experimental conditions for this method are demonstrated in Scheme 16 for the synthesis of the Boc-D-Phe-Cys-OMe related thiazole derivative **52**.[92] Starting with the N-protected amino acid Boc-D-Phe-OH (**47**), methyl ester **48** was obtained by alkylation with MeI and $KHCO_3$ in DMF. The methyl ester **48** was then reduced to the amino alcohol **49** with $NaBH_4$/LiCl, followed by oxidization to the amino aldehyde **50** (without appreciable

racemization) under Parikh–Doering conditions.[97] The amino aldehyde was condensed with H-Cys-OMe to form the thiazolidine **51** as a mixture of diastereomers. The final product **52** was obtained from the thiazolidine by the use of chemically activated $MnO_2$ in 63% yield and with >98% ee.

**Scheme 16** Synthesis of a Peptide-Based Thiazole via the Formation and Oxidation of a Peptide-Based Thiazolidine[92]




**Boc-D-Phe-OMe** (48):[92]
A soln of Boc-D-Phe-OH (**47**; 66.33 g, 0.25 mol) in DMF (400 mL) was treated with pulverized $KHCO_3$ (50 g, 0.5 mol), followed by MeI (25 mL, 0.4 mol). The mixture was stirred at rt for 4–5 h. $H_2O$ (1 L) was added, and the mixture was extracted with EtOAc/benzene (1:1, 3 × 200 mL) (**CAUTION:** *carcinogen!*). The organic extracts were successively washed with $H_2O$ (2 × 200 mL), 5% aq $Na_2SO_3$ (200 mL), and sat. aq NaCl (200 mL), and dried ($Na_2SO_4$). Removal of the solvent gave **48** which was used for the next step without further purification.

***N*-(*tert*-Butoxycarbonyl)-D-phenylalaninol** (49):[92]
Methyl ester **48** (ca. 0.25 mol) was dissolved in THF (350 mL) under argon, and anhyd LiCl (21.2 g, 0.5 mol) and then $NaBH_4$ (18.9 g, 0.5 mol) were added. After addition of EtOH (700 mL), the mixture was stirred at rt overnight. The mixture was cooled with ice water, adjusted to pH 4 by the gradual addition of 10% aq citric acid (ca. 250 mL), and concentrated. $H_2O$ (700 mL) was added to the residue, which was extracted with $CH_2Cl_2$ (3 × 600 mL) and dried ($Na_2SO_4$). Removal of the solvent gave the amino alcohol **49**; yield: 60.32 g (96%); mp 96–97 °C (EtOAc/hexane); $[\alpha]_D^{23}$ +28.3 (*c* 1, MeOH).

***N*-(*tert*-Butoxycarbonyl)-D-phenylalaninal** (50):[92]
To a soln of **49** (25.13 g, 0.1 mol) and TEA (30.4 g, 0.3 mol) in $CH_2Cl_2$ (300 mL) cooled to −10 °C was added in one portion sulfur trioxide–pyridine complex (47.7 g, 0.3 mol) in DMSO (300 mL). The mixture was stirred vigorously for 10 min at 10–20 °C, poured into ice/sat. NaCl (900 mL), and extracted with cold $Et_2O$ (400 mL, then 2 × 200 mL). The organic extracts were washed with cold 10% aq citric acid (200 mL) and cold sat. NaCl (2 × 200 mL), dried ($Na_2SO_4$), and taken to dryness to give **50**, which was used for the next step without further purification.

**Methyl (4*R*)-2-[(*R*)-1-(*tert*-Butoxycarbonylamino)-2-phenylethyl]thiazolidine-4-carboxylate** (51):[92]
To a stirred soln of **50** (0.2 mol) in benzene (240 mL) (**CAUTION:** *carcinogen!*) was added a 1 M H-Cys-OMe soln (240 mL, 0.24 mol). The mixture was stirred overnight at rt. The concentrated residue was purified by column chromatography (EtOAc/hexane 1:4 to 1:2) to give the thiazolidine **51**, which was used without further purification for the next step; yield: 62.3 g (84% from **49**); $^1H$ NMR and TLC revealed that **51** was a mixture of epimers.

*for references see p 690*

**Methyl 2-[(*S*)-1-(*tert*-Butoxycarbonylamino)-2-phenylethyl]thiazole-4-carboxylate** (52):[92]
To chemically activated $MnO_2$ (391 g, 4.5 mol) suspended in benzene (1.2 L) (**CAUTION:** *carcinogen!*) containing pyridine (18 mL) was added at 55 °C a soln of the thiazolidine **51** (0.18 mol) from the previous step in benzene (600 mL). The mixture was stirred at 55 °C for 2–4 h (not specified for this specific example). After filtration, the insoluble material was washed with benzene. The filtrate was concentrated in vacuo, and the residue was purified by recrystallization ($Et_2O$/hexane) or by column chromatography (silica gel, either benzene/$Et_2O$ 9:1 or EtOAc/hexane 5:2 to 4:1) to give the thiazole **52**; yield: 39.86 g (63%); mp 103–105 °C; $[\alpha]_D^{22}$ −8.8 (*c* 1, MeOH).

## 11.2.5 Oxadiazole-Based Peptides

The 1,2,4-oxadiazole moiety has been used extensively as an ester or amide bioisostere.[98–101] However, relatively few actual peptide derivatives have been reported. The first example of 1,2,4-oxadiazoles and 1,3,4-oxadiazoles for amide replacement in X-Gly dipeptidomimetics has been reported.[102]

### 11.2.5.1 Peptide-Based 1,2,4-Oxadiazoles by the Reaction of Amino Acid Anhydrides with Amidoximes

The synthesis of the peptide surrogate Boc-Phe(1,2,4-oxadiazole)-OEt **55** is shown in Scheme 17.[102] Boc-Phe-OH (**53**) was converted into its symmetric anhydride **54** by stirring with DCC at 0 °C. The anhydride (without isolation) was then reacted with ethyl amino(hydroxyimino)acetate[103] in refluxing pyridine. The product, 1,2,4-oxadiazole **55**, was thus obtained in 72% yield and 93% ee. In general, this method gives good to excellent enantiomeric excesses (Table 3).[102] The use of a mild base such as pyridine is critical for the suppression of unwanted epimerization at the α-carbon attached to the heterocycle.

**Scheme 17** Synthesis of a Dipeptide 1,2,4-Oxadiazole[102]

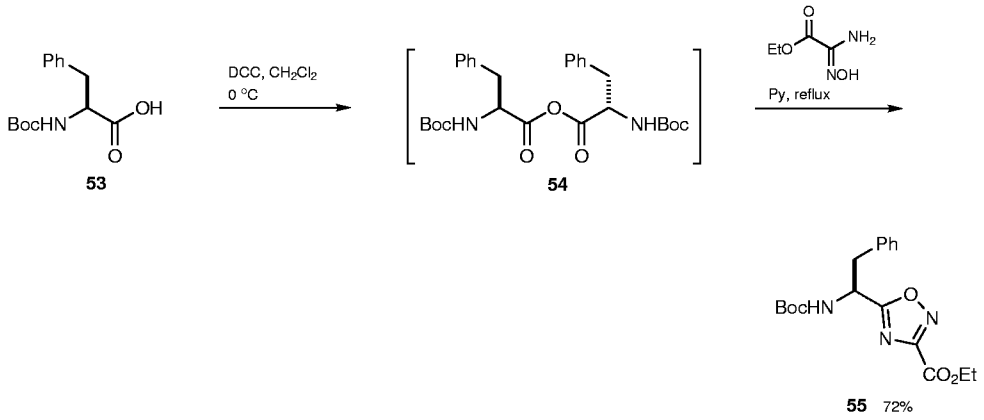
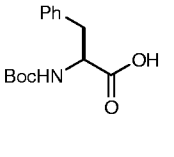
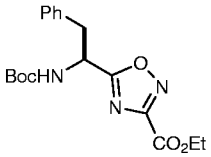
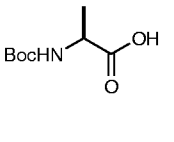
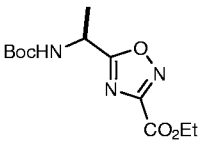
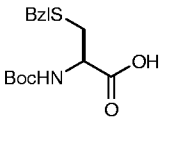
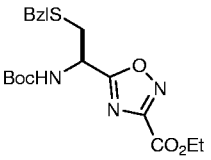
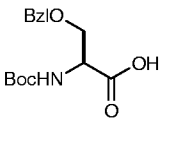
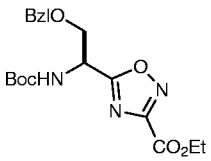
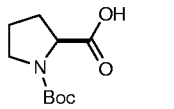
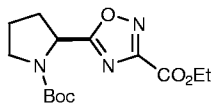
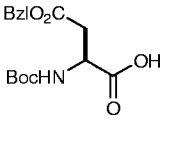
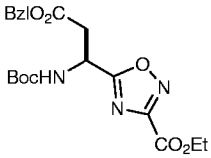
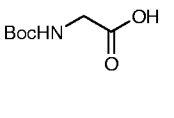
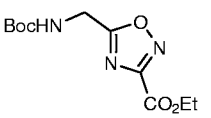


**Table 3** Peptide-Based 1,2,4-Oxadiazoles by the Reaction of Amino Acid Anhydrides with Amidoximes[102]

| Substrate[a] | Product | ee (%) | Yield (%) | Ref |
|---|---|---|---|---|
|  |  | 93 | 72 | [102] |
|  |  | >95 | 76 | [102] |
|  |  | –[b] | 81 | [102] |
|  |  | 72 | 54 | [102] |
|  |  | >95 | 38 | [102] |
|  |  | >95 | 66 | [102] |
|  |  | | 63 | [102] |

[a] Conditions: 1. DCC, $CH_2Cl_2$; 2. $EtO_2CC(NOH)NH_2$, pyridine, reflux.
[b] Not determined.

**Ethyl 5-[(*S*)-1-(*tert*-Butoxycarbonylamino)-2-phenylethyl]-1,2,4-oxadiazole-3-carboxylate (55):**[102]
A soln of Boc-Phe-OH (**53**; 0.614 g, 2.32 mmol) in $CH_2Cl_2$ was treated with DCC (0.239 g, 1.16 mmol) and stirred at 0 °C for 1 h. The resulting DCU was filtered off and the filtrate was concentrated. The residue was redissolved in pyridine, and a soln of ethyl amino(hydroxyimino)acetate[103] (102 mg, 0.77 mmol) in pyridine was added dropwise. After the addition the mixture was heated to reflux for 2 h. $H_2O$ was added and then the solvent removed. The residue was partitioned between $CH_2Cl_2$ and $H_2O$. The organic layer was washed with sat. aq $NaHCO_3$, aq HCl (pH 2), $H_2O$, brine, dried ($MgSO_4$), filtered, and concentrated. The residue was purified by column chromatography (silica gel, pentane/EtOAc 8.5:1.5) to afford **55**; yield: 0.20 g (72%): mp 75–76 °C ($Et_2O$/hexane); $[\alpha]_D$ –29.4 (*c* 1.0, MeOH); 93% ee (ee was determined by running a sample on a Chiracel OD-H column; mobile phase: hexane/iPrOH/DEA 90:10:0.1, 0.5 mL·min$^{-1}$).

*for references see p 690*

### 11.2.5.2 Peptide-Based 1,3,4-Oxadiazoles by the Preparation and Reaction of Diacylhydrazines

The 1,3,4-oxadiazole moiety, in analogy to the 1,2,4-oxadiazole discussed in Section 11.2.5.1, has been used extensively as an ester or amide bioisostere, but also has only recently been applied as an amide replacement in actual peptide segments.[104–107] The synthesis of the peptide surrogate 1,3,4-oxadiazole derivative **60** is shown in Scheme 18.[102] The N-protected amino acid Boc-Ala-OH (**56**) was coupled with ethanol to form the ester **57** which was subsequently reacted with hydrazine to form the amino acid hydrazide **58**.[108] The hydrazide **58** was reacted with ethyl oxalyl chloride at −30 °C to room temperature to provide the diacylhydrazide **59**. This intermediate was subsequently dehydrated with thionyl chloride in refluxing toluene to form the desired 1,3,4-oxadiazole **60** in >95% ee. Although the overall yields are only moderate, the reported enantioselectivities of the final compounds are very good (Table 4).[102]

**Scheme 18** Synthesis of a Heterocyclic Dipeptidomimetic Incorporating an 1,3,4-Oxadiazole[102]

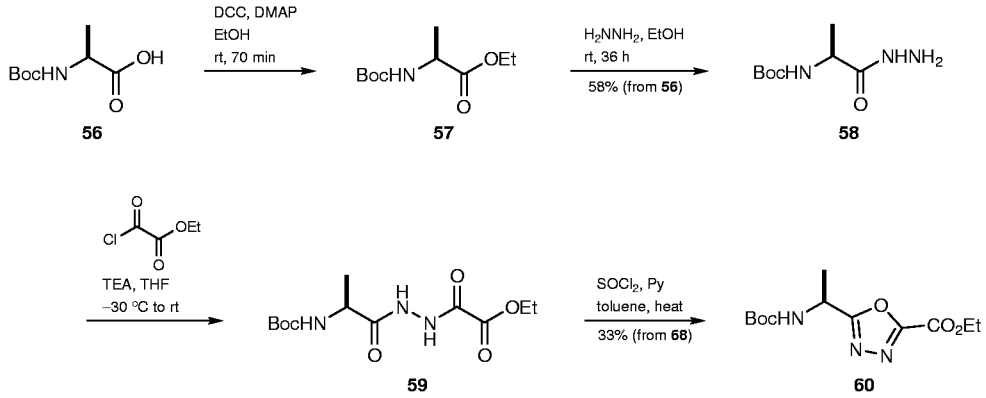



**Table 4** Peptide-Based 1,3,4-Oxadiazoles by the Preparation and Reaction of Diacylhydrazines[102]

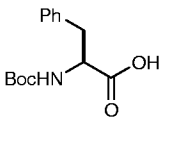
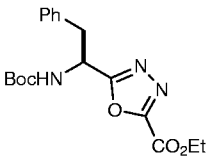
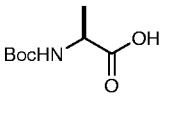
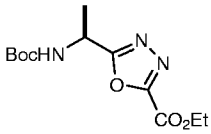
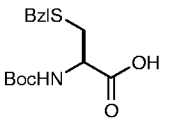
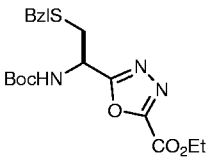
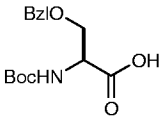
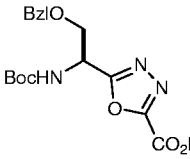
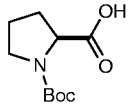
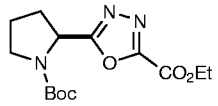
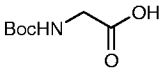
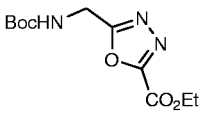
| Substrate[a] | Product | ee (%) | Yield (%) | Ref |
|---|---|---|---|---|
|  |  | >95 | 30 | [102] |
|  |  | >95 | 17 | [102] |
|  |  | –[b] | 6 | [102] |

**Table 4** (cont.)

| Substrate[a] | Product | ee (%) | Yield (%) | Ref |
|---|---|---|---|---|
|  |  | –[b] | 20 | [102] |
|  |  | >95 | 6 | [102] |
|  |  | | 17 | [102] |

[a] Conditions: 1. (a) DCC, DMAP, EtOH, $Et_2O$, rt; (b) $H_2NNH_2 \cdot H_2O$, EtOH, rt; 2. $ClCOCO_2Et$, TEA, THF, –30 °C to rt; 3. (a) $SOCl_2$, pyridine, $Et_2O$, 0 °C; (b) toluene, reflux.
[b] Not determined.

**Boc-Ala-NHNH$_2$ (58):**[102]
A soln of Boc-Ala-OH (**56**; 0.500 g, 2.64 mmol), DCC (0.600 g, 2.91 mmol), DMAP (0.0349 g, 0.28 mmol), and EtOH (0.366 mL, 6.28 mmol) in $Et_2O$ were stirred together at rt for 70 min. The DCU was filtered off and the filtrate extracted with 1 M $NaHSO_4$, sat. aq $NaHCO_3$, and $H_2O$, dried ($MgSO_4$), filtered, and concentrated to yield **57**. The residue was redissolved in EtOH and treated with hydrazine hydrate (0.416 mL, 8.56 mmol) at rt for 36 h. The solvent was removed. Recrystallization (EtOAc/pentane) of the residue provided pure **58**; yield: 0.312 g (58%); mp 95.5–96.5 °C; $[\alpha]_D$ –25.4 (*c* 1.01, MeOH).

**Boc-Ala-NHNHCO$_2$Et (59):**[102]
Ethyl oxalyl chloride (0.24 mL, 2.1 mmol) was added to a chilled (–30 °C) mixture of the hydrazide **58** (0.407 g, 2.0 mmol) and TEA (0.29 mL, 2.1 mmol) in THF. The soln was gradually allowed to warm to rt and stirred for 3 h. The triethylammonium hydrochloride was filtered off, and the filtrate was concentrated. The residue was partitioned between $Et_2O$ and $H_2O$. The organic layer was extracted with sat. aq $NaHCO_3$ and 1 M $NaHSO_3$, dried ($MgSO_4$), filtered, and concentrated. The crude product **59** was used in the next step without further purification.

**Ethyl 5-[(*S*)-1-(*tert*-Butoxycarbonylamino)ethyl]-1,3,4-oxadiazole-2-carboxylate (60):**[102]
To a soln of **59** (all of the material from the previous step) and pyridine (0.39 mL, 4.79 mmol) in dry $Et_2O$ was added $SOCl_2$ (0.15 mL, 2.40 mmol) under $N_2$ and the mixture was allowed to stir at 0 °C for 2 h. The resulting salt was filtered off and the filtrate concentrated without heating. The residue was redissolved in toluene and the soln heated to reflux for 12 h. Purification by column chromatography (silica gel, pentane/EtOAc 7:3) yielded **60**; yield: 0.188 g (33% overall yield for the two steps); mp 66–67 °C ($CH_2Cl_2$/pentane); $[\alpha]_D$ –43.5 (*c* 1.0, $CHCl_3$); >95% ee (ee was determined by running a sample on a Chiracel OD-H column using a mobile phase consisting of hexane/iPrOH/DEA 90:10:0.1, 0.5 mL·min$^{-1}$).

## 11.2.6 Triazole-Based Peptides

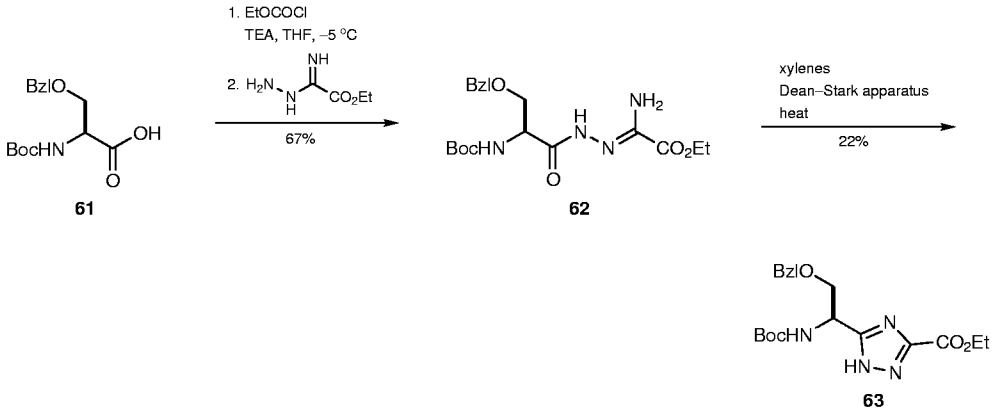
The 1,2,4-triazole moiety, like the 1,2,4-oxadiazole and 1,3,4-oxadiazole discussed in Section 11.2.5, has been used extensively as an ester or amide bioisostere but has rarely been inserted into peptide segments.[109–112] 1,2,4-Triazoles are complementary to the 1,2,4- and 1,3,4-oxadiazoles due to their ability to donate a hydrogen bond as well as accept a hydrogen bond.

*for references see p 690*

### 11.2.6.1 Peptide-Based 1,2,4-Triazoles by the Preparation and Reaction of Acylamidrazones

The synthesis of the peptide-based triazole **63** is shown in Scheme 19.[102] Coupling of Boc-Ser(Bzl)-OH (**61**) with ethyl oxamidrazonate[113,114] proceeded under mixed anhydride conditions[57] using ethyl chloroformate as the activating reagent. The product acylamidrazone **62** thus formed was simply heated in xylenes under Dean–Stark conditions to form the desired triazole **63**.

**Scheme 19** Synthesis of a Dipeptide 1,2,4-Triazole[102]




Unlike 1,2,4- and 1,3,4-oxadiazoles, the 1,2,4-triazole heterocycle has the possibility of existing in different tautomeric forms (Scheme 20). X-ray crystal analysis of the peptide-based triazole **63** showed only one tautomer.[102] Additionally, the $^{13}C$ NMR spectrum for this compound indicated only one tautomeric form since the signals due to C3 and C5 appeared as sharp peaks. This is not the case for some other peptidergic triazoles (Table 5) where peaks corresponding to C3 and C5 were broad and of low intensity or not visible. Differences in tautomeric preferences have been explained by the ability of some triazole-containing peptides to form intermolecular hydrogen bonds and thus stabilize one tautomer over the other.

**Scheme 20** Tautomeric Forms of Ethyl 5-[(*R*)-2-(Benzyloxy)-1-(*tert*-butoxycarbonylamino)ethyl]-1*H*-1,2,4-triazole-3-carboxylate[102]

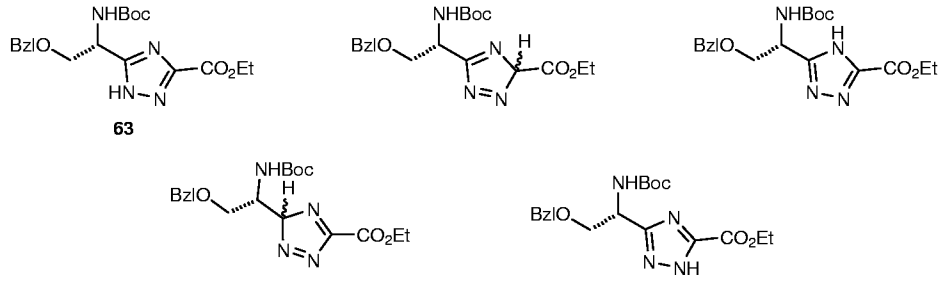
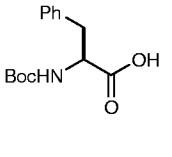
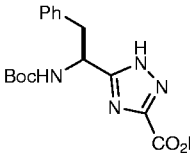
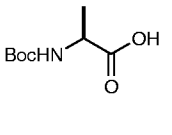
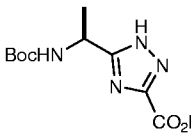
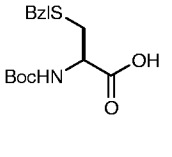
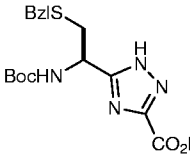
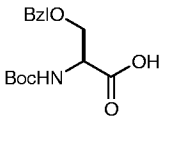
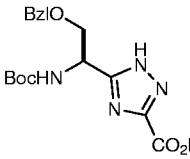
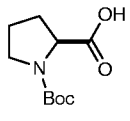
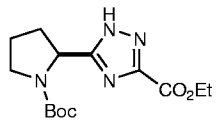
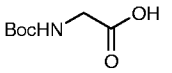
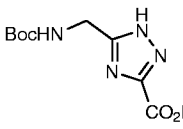


**Table 5** Synthesis of Peptide-Based 1,2,4-Triazoles by the Preparation and Reaction of Acylamidrazones[102]

| Substrate[a] | Product[b] | ee (%) | Yield (%) | Ref |
|---|---|---|---|---|
|  |  | >95 | 52 | [102] |
|  |  | >95 | 39 | [102] |
|  |  | | 13 | [102] |
|  |  | >95 | 15 | [102] |
|  |  | >95 | 26 | [102] |
|  |  | | 28 | [102] |

[a] Conditions: 1. $ClCO_2Et$, TEA, THF, −5 °C; 2. $EtO_2CC(NH)NHNH_2$, THF, rt; 3. xylenes, heat.

[b] All triazoles other than those containing Pro and Ser residues were mixtures of triazole tautomers as determined by $^{13}C$ line-broadening at C3 and C5 of the triazoles.

**Boc-Ser(Bzl)-NHNH=($NH_2$)$CO_2$Et (62):**[102]

A soln of Boc-Ser(Bzl)-OH (**61**; 0.174 g, 0.59 mmol) and TEA (0.098 mL, 0.70 mmol) in dry THF was cooled to −5 °C and treated with ethyl chloroformate (0.068 mL, 0.70 mmol). The soln was stirred at −5 °C for 30 min and the resulting triethylamine hydrochloride was filtered off. The filtrate was treated with a soln of ethyl oxamidrazonate (0.059 g, 0.45 mmol) in THF. Filtration of the mixture afforded pure **62**; yield: 0.124 g (67%); mp 179–180 °C; $[\alpha]_D$ +26.2 (*c* 0.97, DMF).

**Ethyl 5-[(*R*)-2-(Benzyloxy)-1-(*tert*-butoxycarbonylamino)ethyl]-1*H*-1,2,4-triazole-3-carboxylate (63):**[102]

A soln of acylamidrazone **62** (0.113 g, 0.28 mmol) in xylenes (mixture of isomers) was heated in an oil bath at 190 °C for 5 h using a Dean–Stark apparatus. The soln was concentrated and the residue chromatographed (silica gel, EtOAc/pentane 3:2) to afford **63**; yield: 24 mg (22%).

*for references see p 690*

## 11.2.7 Tetrazole-Based Peptides

The use of 1,5-tetrazoles in peptide sequences has been investigated particularly for these heterocycles perceived ability to mimic a *cis*-amide bond in the peptide backbone (Scheme 21).[115–118] The incorporation of the tetrazole moiety is typically performed on an N-protected dipeptide fragment. If a larger peptide containing the fragment is desired, then the dipeptide fragment can be subsequently deprotected and the peptide extended by normal coupling procedures. Protecting groups must be chosen, however, which do not require the use of strong base for removal, or exposure to any base for extended periods of time, due to the ease of racemization at the carbon α to the tetrazole. In addition, the conditions for the formation of the tetrazole have to be suitably mild in order to prevent racemization at the α-carbon during the actual tetrazole synthesis.

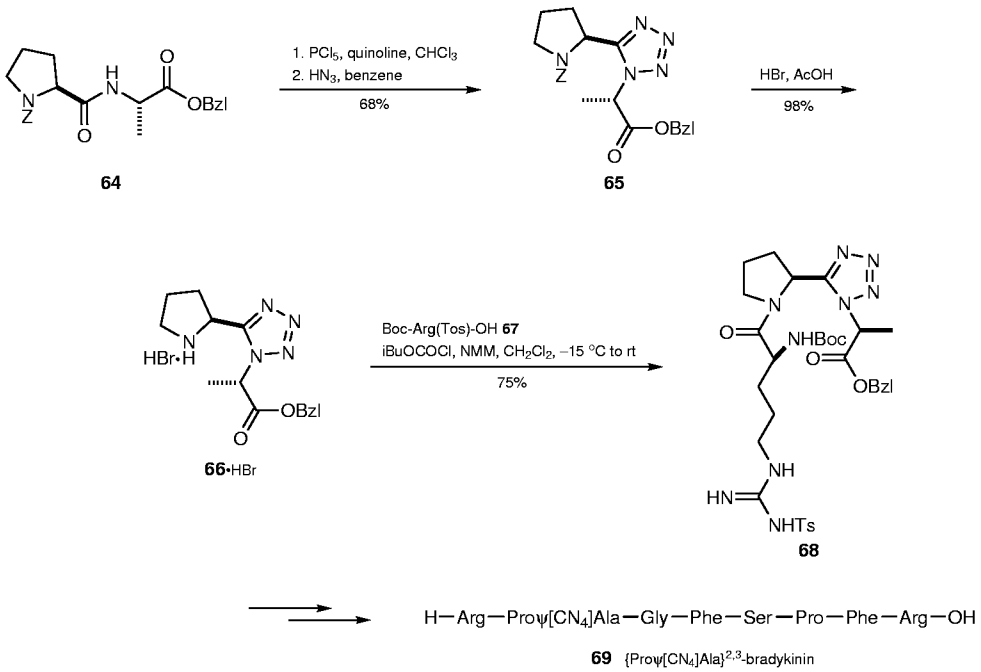
**Scheme 21** Isosteric Replacement of a *cis*-Amide Bond with a 1,5-Substituted Tetrazole[115–118]




### 11.2.7.1 Peptide-Based Tetrazoles from the Reaction of Imidoyl Chlorides and Hydrazoic Acid

Mild conditions that minimize racemization of the tetrazole containing subunit were developed during the synthesis of bradykinin analogues; this strategy also allowed for the incorporation of the heterocyclic segment into bradykinin.[119] The synthesis of the tetrazole-containing dipeptide **65** is shown in Scheme 22.[119] A solution of the dipeptide Z-Pro-Ala-OBzl (**64**) and $PCl_5$ and quinoline in $CHCl_3$ (the selection of the base is important: bases with higher $pK_a$ values can result in increased racemization; however, using no base resulted in complete racemization) was then treated with hydrazoic acid in benzene and allowed to stir at room temperature. The product **65** was isolated in 68% yield with >95% ee. The tetrazole-containing dipeptide was then deprotected at the N-terminus using HBr/AcOH to yield the dipeptide HBr salt of **66**. Immediate acylation of the dipeptide **66** was required in order to prevent the formation of the piperazinedione, which is favored due to the *cis*-conformation of the tetrazole amide bond surrogate. Boc-Arg(Tos)-OH (**67**) was activated via the mixed anhydride method (iBuOCOCl, NMM)[57] and reacted with **66** to yield the tripeptide **68** in 75% yield. This tripeptide was then converted through a number of coupling steps into the bradykinin analogue **69**.

**Scheme 22** Synthesis of a 1,5-Tetrazole Amide Surrogate and Incorporation into the Bradykinin Sequence[119]




**Benzyl (*S*)-2-[(*S*)-5-[1-(Benzyloxycarbonyl)pyrrolidin-2-yl]-1*H*-tetrazol-1-yl]propanoate (65):**[119]
To a stirred soln of $PCl_5$ (0.21 g, 1 mmol) in $CHCl_3$ (5 mL) was added quinoline (0.26 g, 2 mmol) at rt (a white precipitate formed). The mixture was stirred for 20 min and then the dipeptide **64** (0.41 g, 1 mmol) was added in portions at such a rate that the temperature stayed below 20 °C. After 30 min at 20 °C (a clear soln had formed), a benzene soln (**CAUTION:** *carcinogen!*) of hydrazoic acid (3 mL) was added. The mixture was stirred at rt for 1 h. The solvent was removed and the residue was partitioned between EtOAc and $H_2O$ (30 mL of each). The organic layer was washed with 1 M HCl (2 × 15 mL), 1 M $NaHCO_3$ (2 × 15 mL), $H_2O$ (2 × 15 mL), and sat. NaCl (30 mL). The soln was dried ($Na_2SO_4$), concentrated, and the residue chromatographed (silica gel, $CH_2Cl_2$/acetone 30:1) to afford the product **65** as a white crystalline solid; yield: 68%; mp 97.5–98 °C; $[\alpha]_D^{25}$ −15.9 (*c* 0.5, MeOH).

**Benzyl (*S*)-2-[(*S*)-5-Pyrrolidin-2-yl-1*H*-tetrazol-1-yl]propanoate Hydrobromide Salt (66·HBr):**[119]
A soln of **65** (0.97 g, 2 mmol) in AcOH (2 mL) was treated (stirring) with 30% HBr in AcOH (5 mL). After 20 min at rt, the soln was poured into $Et_2O$ (50 mL, precooled to −10 °C) with vigorous stirring. The resulting precipitate was treated with petroleum ether (bp 30–60 °C) (20 mL), and the mixture was kept at 0 °C for 15 min and then filtered. The solid was washed with $Et_2O$/petroleum ether (1:1, 2 ×) and dried in vacuo to give **66**·HBr as a nonhygroscopic salt; yield: 0.75 g (98%); mp 152–153 °C; $[\alpha]_D^{25}$ −44.8 (*c* 1, MeOH).

**Benzyl (*S*)-2-(5-{(*S*)-1-[$N^{\alpha}$-(*tert*-Butoxycarbonyl)-$N^{\omega}$-tosylarginyl]pyrrolidin-2-yl}-1*H*-tetrazol-1-yl)propanoate (68):**[119]
A soln of **67** (0.813 g, 1.9 mmol) in $CH_2Cl_2$/DMF (1:1, 5 mL) was cooled to −15 °C and treated with NMM (0.21 mL, 1.9 mmol), followed by isobutyl chloroformate (0.26 mL, 1.9 mmol). The mixture was stirred for 10 min; then solid **66**·HBr (0.732 g, 1.9 mmol) was introduced, followed by addition of NMM (0.21 mL, 1.9 mmol) at such a rate that the temperature stayed below −10 °C. After being stirred at −10 °C for 1 h, the mixture was warmed up slowly to rt and stirred overnight. The solvents were removed and the residue taken up in EtOAc (50 mL) and washed with 1 M $NaHSO_4$ (3 × 20 mL), 1 M $NaHCO_3$ (3 × 20 mL), $H_2O$ (2 × 20 mL), and sat. NaCl (20 mL). The soln was dried ($Na_2SO_4$), concentrated, and the residue was purified by chromatography (silica gel, $CH_2Cl_2$/acetone 3:1) to yield **68** isolated as an amorphous solid; yield: 1.01 g (75%).

*for references see p 690*

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# 12 Peptides Incorporating Secondary Structure Inducers and Mimetics

C. Toniolo

The most stable elements of secondary structure of peptides and proteins are turns, helices, and extended conformations. Within each of these 3D-structures the most commonly found representatives are β-turns,[1–4] α-helices, and antiparallel β-sheet conformations, respectively. γ-Turns,[5] $3_{10}$-helices, poly(Pro)$_n$ helices, and β-sheet conformations with a parallel strand arrangement have also been observed, although less frequently. Among the many types of β-turns classified, type-I, type-II, and type-VI are the most usual, all being stabilized by an intramolecular $i \leftarrow i+3$ (backbone)C=O···H–N(backbone) H-bond and characterized by either a *trans* (type-I and type-II) or a *cis* (type-VI) conformation about the internal peptide bond. In the type-I β-turn a helical $i+1$ residue and a quasi-helical $i+2$ residue are found, whereas in the type-II β-turn the $i+1$ residue is semi-extended and the $i+2$ residue is also quasi-helical but left-handed. This latter corner position may be easily occupied by the achiral Gly or a D-amino acid residue.

There has been an increasing interest in the development of rigidified or constrained scaffolds to mimic or stabilize the secondary structural features of peptides and proteins that are thought to play an important role in molecular and chiral recognitions and in biological activity, but which, in the parent molecules, are often highly flexible. These templates are designed to have a higher avidity for the partner molecule by overcoming the inherent entropic cost paid for regular structure formation upon binding. These peptidomimetics are potential conformational probes, thus enabling bioactivity to be more effectively correlated with 3D-structure, and prospective therapeutics overcoming drawbacks such as proteolytic susceptibility and poor bioavailability, typical of peptides.

Sections 12.1 and 12.2 discuss the synthetic aspects of peptidomimetics of turn elements (β- and γ-turns, respectively).[6–15] Peptides incorporating helix and β-sheet inducers and mimetics[10,15–21] are covered in Sections 12.3 and 12.4, respectively.

Sugar amino acids represent an outstanding class of conformationally constrained templates that have been extensively exploited in building up a number of peptidomimetic compounds.[22] They have emerged as an important family of synthetic monomers leading to a variety of de novo oligomeric libraries with architecturally appealing 3D-structures and useful properties. Based on these hybrid starting materials, mimetics of peptides containing β- and γ-amino acids and their higher homologues have been reported (Section 12.5).

Peptide nucleic acids represent analogues of nucleic acids that, in contrast to the latter, contain neither carbohydrate nor phosphate residues, but have uncharged peptidomimetic backbones.[23–26] The heterocyclic base (purine or pyrimidine) is covalently bound to the backbone amino group through an acyl linkage. The geometry of the backbone and its relative flexibility confer on peptide nucleic acids an ability to mimic, with striking exactness, the spatial structure of carbohydrate–phosphate backbones of nucleic acids. Peptide nucleic acids, which hybridize strongly, with sequence specificity, to complementary RNA/DNA, are described in Section 12.6.


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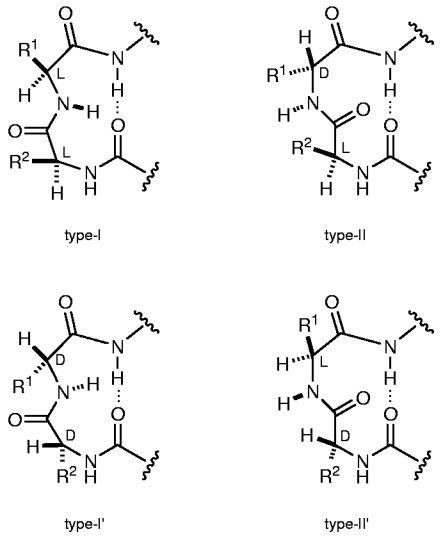
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# 12.1 Synthesis of Peptides Incorporating β-Turn Inducers and Mimetics

M. KAHN and M. EGUCHI

The β-turn is a common structural element in peptides and proteins. The idealized β-turn, first recognized by Venkatachalam,[1] is typified by the definition: four consecutive residues with a strictly defined stereochemistry, which are joined by a hydrogen bond between the C=O of the first turn residue ($i$) and the NH of the last residue ($i+3$), as illustrated in Scheme 1 for the four most classical types.

**Scheme 1** Classical Types of β-Turns[1]




| Bend Types | Dihedral Angles of Two Central Residues (°) | | | |
|---|---|---|---|---|
| | $\phi_2$ | $\psi_2$ | $\phi_3$ | $\psi_3$ |
| I | −60 | −30 | −90 | 0 |
| I′ | +60 | +30 | +90 | 0 |
| II | −60 | +120 | +80 | 0 |
| II′ | +60 | −120 | −80 | 0 |

In practice, turns seen in X-ray diffraction elucidated protein structures often fail to satisfy such strict criteria. Both structural variation and measurement error can lead to nonideal geometries, and this complexity has given rise to a variety of working definitions. The most common strategy defines a chain site as a turn when the $C^{\alpha}(i)\cdots C^{\alpha}(i+3)$ distance is less than 7 Å and the residues involved are not in a helix.

Distance-dependent strategies have the advantage of being easy to apply, and they serve to identify both β-turns as well as more gradual turns. However, the chosen cutoff distance is always a compromise between overinclusion and overexclusion. Ideally, a definition less sensitive to minor variations in the choice of parameters would be preferable.

β-Turns are generally located at the surface of proteins.[2–6] This topographical tendency is a consequence of the hydrogen-bonding requirements of backbone polar groups. Residues found in turns are predominantly polar.[3] Thus, turns are highly polar secondary structures owing to the combined contributions of both backbone and side-chain groups. The surface localization of turns in proteins, and the predominance of potentially reactive functional groups in their side chains, has led to the suggestion that turns function as recognition sites

*for references see p 736*

which trigger complex immunologic, metabolic, genomic, and endocrinologic regulatory mechanisms. Additionally, turns have been implicated as nucleation sites of protein folding[7–9] and as markers for the posttranslational events of phosphorylation,[10] glycosylation,[11] enzymatic processing,[12] and degradation.[13]

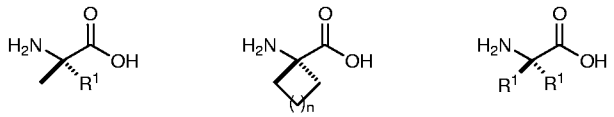
## 12.1.1 β-Turn Inducers

Modification of the peptide backbone, including $C^{\alpha}$-alkyl α-amino acids, α,β-didehydro α-amino acids, and $N^{\alpha}\leftrightarrow C^{\alpha}$ and $C^{\alpha}\leftrightarrow C^{\alpha}$ cyclized α-amino acids and dipeptides, may influence the conformational properties of the resulting pseudopeptide molecules while preserving the mandatory side-chain substituents.

### 12.1.1.1 $C^{\alpha}$-Alkyl α-Amino Acids

$C^{\alpha}$-Alkyl α-amino acids have a structure in which the $C^{\alpha}$-hydrogen atom is replaced by an alkyl group. The effect of this modification is that rotation around the $N-C^{\alpha}$ (ϕ) and $C^{\alpha}-C(O)$ (ψ) bonds is severely restricted. Approximately 70% of the conformational space available to Gly is precluded by the addition of one methyl group (Ala). The addition of a second methyl group (Aib, α-aminoisobutyric acid; **1**, $R^1$ = Me) (Scheme 2) eliminates an additional 20% of the conformational space available to Gly.[14–16] Aib, a naturally occurring amino acid observed in channel-forming peptides, is the most extensively investigated of the $C^{\alpha}$-alkyl α-amino acids.[17] The conformational space favored by Aib residues includes regions of both the left- and right-handed α- and $3_{10}$-helices.[18,19] This residue has been incorporated into numerous bioactive peptides including enkephalin,[20,21] angiotensin,[22] and bradykinin[23] in an attempt to obtain potent analogues and to define their bioactive conformations.

**Scheme 2** Chemical Structures of Various $C^{\alpha}$-Alkyl α-Amino Acids




**1** $R^1$ = Me, Et, iPr, iBu, Bzl **2** n = 0–4 **3** $R^1$ = Et, Pr

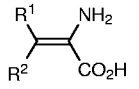
Toniolo et al.[24] have reviewed the conformational preferences for several additional $C^{\alpha}$-methyl amino acids including isovaline ($C^{\alpha}$-methyl-α-aminobutyric acid; **1**, $R^1$ = Et), $C^{\alpha}$-methylvaline (**1**, $R^1$ = iPr), $C^{\alpha}$-methylleucine (**1**, $R^1$ = iBu), and $C^{\alpha}$-methylphenylalanine (**1**, $R^1$ = Bzl). These preferences have been determined experimentally (either by X-ray diffraction crystallography, or by IR absorption, CD, and $^1$H NMR spectroscopic techniques) and via computational energy calculations. The consensus is that peptides (tripeptides and longer) are folded in β-turn conformations, and rarely adopt fully extended ($\phi = 180°$, $\psi = 180°$) structures. Similarly, α-aminocycloalkanecarboxylic acids **2** generally prefer to adopt a β-turn structure.[25] Interestingly, the higher homologues of Aib, $C^{\alpha,\alpha}$-diethylglycine (**3**, $R^1$ = Et) and $C^{\alpha,\alpha}$-dipropylglycine (**3**, $R^1$ = Pr; Scheme 2) typically exhibit fully extended structures.

Hinds et al.[26] used a helicogenic $C^{\alpha}$-alkylproline derivative[27] in the peptide Ac-Tyr-Pro-Tyr-Asp-Val-Pro-Asp-Tyr-Ala-OH, which is derived from an immunogenic sequence in influenza hemagglutinin, specifically replacing the first Pro residue in the sequence. They found increased binding to a pair of monoclonal antibodies which recognize this site.[26] The insertion of $C^{\alpha}$-alkyl α-amino acids into peptide systems is discussed in Section 10.3.

### 12.1.1.2 α,β-Didehydro-α-amino Acids

α,β-Didehydro α-amino acid residues **4** (Scheme 3) are often found in naturally occurring peptides of microbial and fungal origin.[28–32] In small model peptides containing a single dehydrophenylalanine (ΔPhe), a type-II β-turn is generally observed,[33–41] whereas incorporation of more than one ΔPhe residue stabilizes helical structures.[42–48] Although less information is available, it is clear that the conformational behavior of the ΔLeu residue is quite similar to ΔPhe. In ΔLeu peptides where the peptide backbone was long enough to form an $1 \leftarrow 4$ intramolecular hydrogen bond, the ΔLeu residue nucleated and occupied the $i+2$ position of the turn.[49–51] Interestingly, ΔAla peptides containing multiple consecutive ΔAla residues adopt a fully extended conformation ($2.0_5$-helix) in solution and in the crystal state.[52] The incorporation of α,β-didehydro α-amino acids into peptides is discussed in Section 11.1.

**Scheme 3** Chemical Structures of Some α,β-Didehydro α-Amino Acids

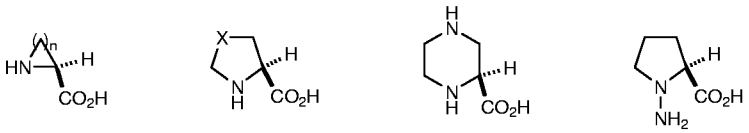


**4** ΔAla $R^1 = R^2 = H$
$\Delta^Z$Leu $R^1 = iPr$; $R^2 = H$
$\Delta^Z$Phe $R^1 = Ph$; $R^2 = H$

### 12.1.1.3 $N^{\alpha} \leftrightarrow C^{\alpha}$ and $C^{\alpha} \leftrightarrow C^{\alpha}$ Cyclized α-Amino Acids and Dipeptide Units

Proline (**5**, n = 3) (Scheme 4) is the only naturally occurring mammalian $N^{\alpha} \leftrightarrow C^{\alpha}$ cyclized α-amino acid, which has several important consequences. The backbone conformation of Pro is severely restricted. The backbone dihedral φ-angle is limited to a small range around $\phi = -65°$;[53–56] however, the ψ-angle is able to populate either the α-helical region ($\psi$ = ca. −40°) or the β-sheet region ($\psi$ = ca. +150°). Additionally, the bulky ring constrains the conformation of adjacent residues. Another unique property of Pro is its ability to form *cis*-peptide bonds, which occurs approximately 5% of the time in globular proteins. This is because the tertiary amide bond lends to facile (energy of activation ca. 25 kcal·mol$^{-1}$) *cis-trans* amide-bond rotation, as the difference in energy between these conformers is significantly less (ca. 2 kcal·mol$^{-1}$) than a normal peptide bond (ca. 12 kcal·mol$^{-1}$).[55] A large proportion of *cis*-Xaa-Pro dipeptides in globular proteins are found as type-VI turns (idealized φ, ψ angles are VIa1: −60, 120°; −80, 0°; VIa2: −120, 120°; −60, 0°; VIb: −135, 135°; −75, 160°). Additionally, a number of Pro analogues have been described including aziridine-2-carboxylic acid (**5**, n = 1), azetidine-2-carboxylic acid (**5**, n = 2), pipecolic acid (**5**, n = 4), oxaproline (oxazolidine-4-carboxylic acid; **6**, X = O), thioproline (thiazolidine-4-carboxylic acid; **6**, X = S), piperazine-2-carboxylic acid (**7**), and 1-aminoproline (Scheme 4). The synthesis of peptides incorporating these derivatives is discussed in Section 9.2. "Azaproline" (**8**, Scheme 4) analogues have also been incorporated into peptides and can induce β-turn conformations. The synthesis of azaproline analogues is discussed in Section 10.4.

**Scheme 4** Chemical Structures of Various Cyclized α-Amino Acids

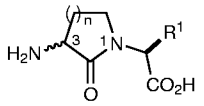


**5** n = 1–4 **6** X = O, S **7** **8**

*for references see p 736*

In 1980, Freidinger et al.[57] reported the use of a conformationally constrained $C^{\alpha}(i) \leftrightarrow N^{\alpha}(i+1)$ cyclized dipeptide unit to induce a β-turn. Lactam **9A** (Scheme 5) was incorporated into the peptide LHRH, replacing the $Gly^6$-$Leu^7$ residues in the sequence, providing a superpotent agonist with 8.9 times the in vitro biological activity of LHRH itself. Based upon computational analysis, it was assumed that the induced conformation closely resembled an idealized type-II′ β-turn from $Tyr^5$-$Arg^8$. Subsequent conformational analysis using $^1H$ NMR spectroscopy[58] was in general agreement with the proposed structure. The same group subsequently reported[59] a hexapeptide analogue of substance P, which contained the enantiomer **9B** of opposite chirality. This analogue did not bind significantly to the substance P receptor; however, it was found to be a potent agonist of eledoisin, which, like substance P, is a member of the tachykinin family. It was suggested that the dipeptide unit might be stabilizing a β-turn around the $Ile^8$-$Met^{11}$ sequence important for the biological activity of eledoisin. Interestingly, replacement of **9B** with **9A** in this peptide provided a compound which was inactive with respect to both substance P and eledoisin. The synthesis of the γ-lactams can be readily extended to provide compounds **10A** and **10B** (Scheme 5) which were used to replace the $Gly^2$-$Gly^3$ sequence in Met-enkephalinamide. Conformational analyses of these molecules, coupled with their low biological activity, were used to argue against a $Tyr^1$-$Phe^4$ type-II or type-II′ β-turn being involved in the biologically active conformation of Met-enkephalin. One caveat with this interpretation is that this inducer occupies space beyond the normal peptide sequence, which could cause steric interference.

**Scheme 5** Cyclized Dipeptide Units




| Compound | n | $R^1$ | Configuration |
|---|---|---|---|
| **9A** | 1 | iBu | 3*S* |
| **9B** | 1 | iBu | 3*R* |
| **10A** | 1 | H | 3*S* |
| **10B** | 1 | H | 3*R* |
| **11** | 2 | iBu | 3*S* |

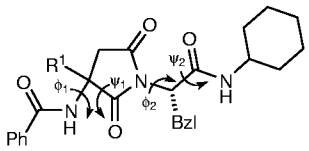
In a separate study, compounds **10A** and **10B** were incorporated into the gastrin peptide sequence, replacing the $Tyr^{12}$-$Gly^{13}$, to provide a potent analogue.[60] These data were used to imply an $Ala^{11}$-$Trp^{14}$ β-turn in the bioactive conformation of gastrin.

An interesting caveat relating to conclusions drawn concerning the bioactive conformation of peptides containing β-turn inducers is represented by the work of Valle et al.[61] They replaced the $Leu^2$-$Gly^3$ sequence in the neurotransmitter tripeptide H-Pro-Leu-Gly-$NH_2$ with **10B**, leading to an analogue with a $10^4$ increase in biological potency. Although the crystal structure of the prototypical peptide contains a type-II β-turn in the X-ray diffraction structure,[62] the analogue incorporating **10B** did not.

The six-membered-ring δ-lactam analogue **11** (Scheme 5) incorporated into the $Leu^3$-$Sar^4$ sequence in the cyclic undecapeptide sequence of the immunosuppressive agent cyclosporin[63] gave a compound with poor activity. Extensive conformational analysis[64] suggested a very similar type-II′ β-turn, and the poor biological activity was attributed to unfavorable steric interaction between the δ-lactam ring and the receptor.

Obrecht et al.[65] generated both enantiomers of $C^{\alpha}$-alkyl Glu and $C^{\alpha}$-alkyl Asp imides [$C^{\alpha}(i) \leftrightarrow N^{\alpha}(i+1)$ cyclization] as turn inducers (see Scheme 6 for $C^{\alpha}$-alkyl Asp succinimide peptides **12**).

**Scheme 6** $C^{\alpha}$-Alkyl Asp Succinimide Peptides[65]

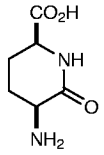


12

| $R^1$ | Dihedral Angles (°) | | | | | |
|---|---|---|---|---|---|---|
| | Diastereomer | $\phi_1$ | $\psi_1$ | $\phi_2$ | $\psi_2$ | Turn Type |
| Me | (*R*,*S*) | −58 | +120 | +56 | +29 | type II |
| iBu | (*R*,*S*) | −57 | +125 | +63 | +19 | type II |

Kemp and McNamara[66] designed and synthesized lactam **13** (Scheme 7) to stabilize the backbone conformation of a rare β-turn involving a *cis*-peptide bond found between the $(i+1)$- and $(i+2)$-residues in a few cyclic peptide turns via $C^{\alpha}(i) \leftrightarrow C^{\alpha}(i+1)$ cyclization.[67] However, its incorporation into a bioactive peptide has never been reported.

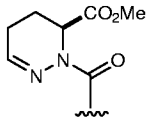
**Scheme 7** The *cis*-Peptide Bond β-Turn Inducer[66]



13

More recently, Xi et al.[68] described the synthesis of conformationally rigidified dipeptides incorporating 2-acyl-2,3,4,5-tetrahydropyridazine-3-carboxylic acids **14** and hypothesized, although did not experimentally test, that these dipeptides could be utilized to enforce β-turns (Scheme 8).

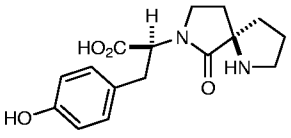
**Scheme 8** 2-Acyl Derivatives of 2,3,4,5-Tetrahydropyridazine-3-carboxylic Acids[68]



14

Spiro lactam **15** (Scheme 9), a dipeptide mimic with $C^{\alpha}(i) \leftrightarrow N^{\alpha}(i+1)$ cyclization, was used to replace the Pro-Tyr segment of an immunogenic peptide.[69] Comprehensive conformational analysis led to the conclusion that a type-II β-turn conformation was adopted in solution.

**Scheme 9** A Spiro Lactam β-Turn Inducer[69]

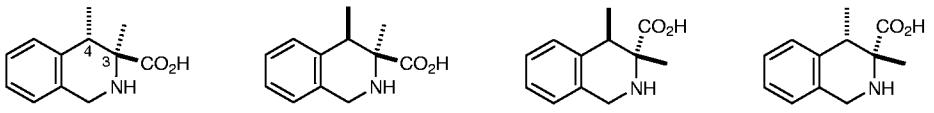


15

A series of 1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid derivatives **16** (Scheme 10) as inducers of β-turns has also been described.[70]

*for references see p 736*

**Scheme 10** 1,2,3,4-Tetrahydroisoquinoline-3-carboxylic Acid Derivatives[70]




(3*S*,4*S*)-**16** (3*S*,4*R*)-**16** (3*R*,4*R*)-**16** (3*R*,4*S*)-**16**

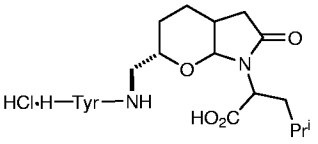
Using X-ray diffraction crystallographic analysis, it has been concluded that incorporation of an azaproline residue (**8**, see Scheme 4) into a peptide, in contrast to Pro itself, should destabilize the formation of any type of β-turn with the residue following it, but could accommodate a folded structure with a pseudo type-VI β-turn with the preceding residue.[71]

## 12.1.2 β-Turn Mimetics and Their Incorporation into Peptide Sequences

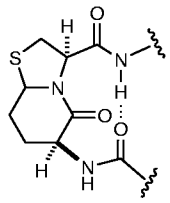
In the design of reverse-turn mimetic systems there are a number of concerns and criteria which need to be addressed. β-Turns comprise a rather diverse group of structures. β-Turns are classified according to the ϕ- and ψ-angles of the $(i+1)$- and $(i+2)$-residues. In addition to a number of experimentally detected turn types (I, I′, II, II′, III, III′, VIa, VIb), the $C^{\alpha}(i)\cdots C^{\alpha}(i+3)$ distance varies from 4 to 7 Å. From this cursory discussion it should be readily apparent that no one structure can accurately mimic this diversity of turns. The interaction of the amino acid side chains with their complementary receptor groups is the critical determinant of biological specificity. A successful peptidomimetic must appositely position the appropriate functional groups on a relatively rigid framework. Therefore, an idealized mimetic design should incorporate the ability to accurately display critical pharmacophoric information in the same manner in which it is presented in native reverse turns. This is a far from trivial synthetic problem, in that it requires the stereo- and enantiocontrolled introduction of a minimum of four noncontiguous asymmetric centers. Furthermore, the nonpeptidic character of these molecules promises to best overcome the inherent problems of peptides. Synthetic expediency is a major concern that should not be lightly regarded, particularly at an early stage when the delineation of structure–activity relationship is critical and requires the preparation and evaluation of a series of related structures.[72]

One of the earliest detailed proposals for turn mimetics was presented by Farmer,[73] who suggested that a disubstituted cyclohexane framework could mimic these structures. Currie et al.[74] in 1982 described a bicyclic rigid β-bend analogue **17** (Scheme 11) of enkephalin that exhibited weak opiate activity. No further work on this system has been reported. Similarly, Nagai and Sato[75,76] have developed the bicyclic dipeptide skeleton **18** as a β-turn mimic, and have incorporated it into analogues of the bioactive peptide gramicidin S (**19**) at the β-turns in place of the two D-Phe-Pro sequences to generate gramicidin S analogue **20** (Scheme 11). Preparation of the protected bicyclic β-turn peptidomimetic **31** is outlined in Scheme 12 (experimental details given below).[77] Feigel[78] synthesized the aromatic sulfone derivative **21** to enforce hydrogen bonding between antiparallel peptide strands **22** in a manner similar to a β-turn **23** (Scheme 11). He proposed that compound **21** may be used as an organic substitute simulating a pair of amino acids preferring the $i+1$ and $i+2$ positions of reverse-turn peptides. These units all suffer a common limitation, in that they are basically only rigid spacers for nucleating secondary structures from turns, and cannot readily be suitably functionalized in order to investigate the critical roles that turns are hypothesized to play in a wide array of molecular recognition events.

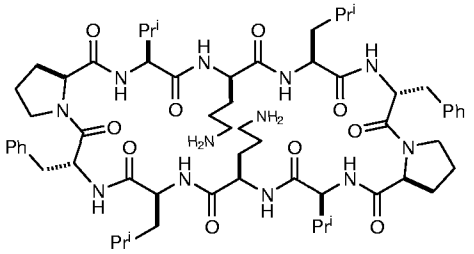
**Scheme 11** β-Turn Peptidomimetics and Their Incorporation into Peptide Sequences[74–76,78]



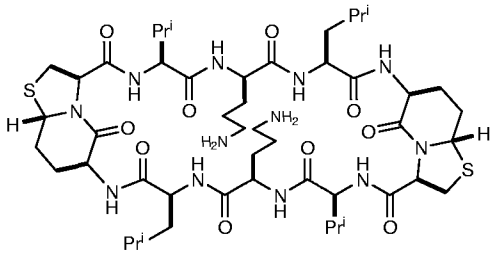
17



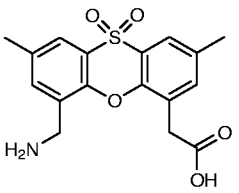
18



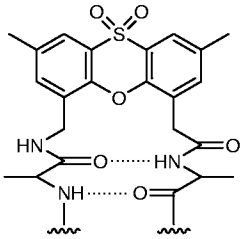
19



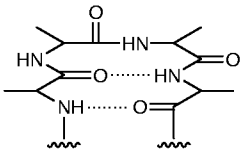
20



21



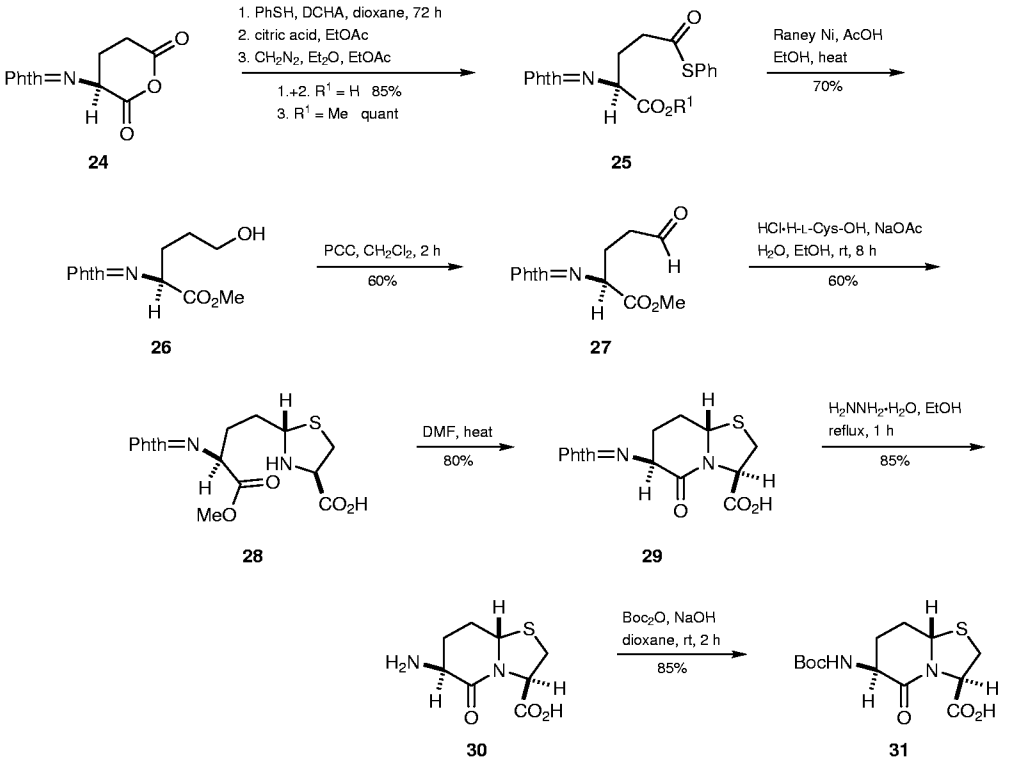
22



23

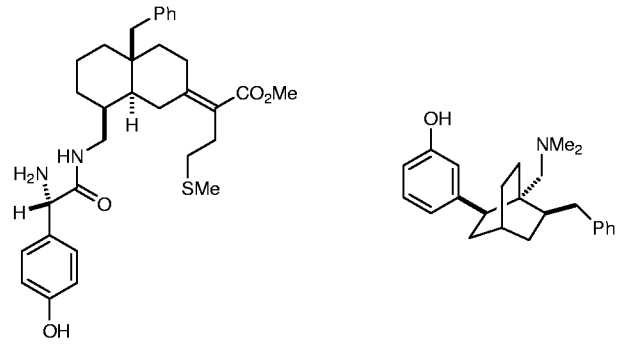
*for references see p 736*

**Scheme 12** Preparation of a Bicyclic β-Turn Peptidomimetic[77]



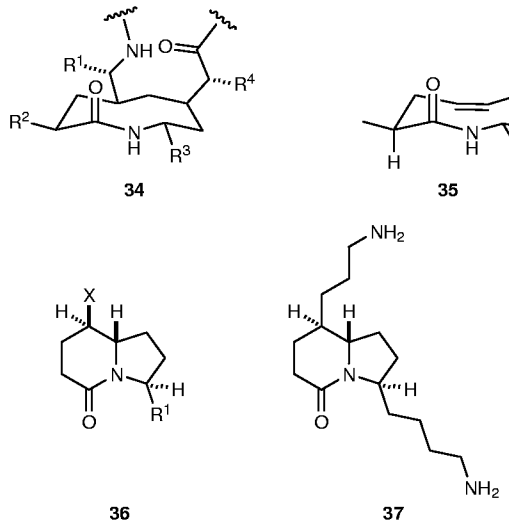

Compounds **32** and **33** (Scheme 13), reported in the 1980s,[79,80] were designed as mimetics of the pentapeptide Met-enkephalin. Both compounds had very poor biological activity, which could be attributable to a number of features.

**Scheme 13** Enkephalin Mimetics[79,80]



Our initial design led us to the nine-membered-ring lactam **34** (Scheme 14).[81] Preliminary molecular modeling suggested that this template could accurately mimic an idealized type-I or type-II β-turn. Although an idealized β-turn is in essence a pseudocyclic ten-membered ring, initially it was felt that the nine-membered-ring skeleton would be a better fit, in that several amide bonds had been replaced with slightly longer carbon–carbon linkages. In our initial efforts, we had not completely come to grips with the complexity of dealing with a fairly diverse family of structures, which could not be adequately mimicked by one particular template. The synthesis of the model system proceeded smoothly utilizing the procedure of Wilson and Sawicki;[82] however, it became painfully apparent that the stereocontrolled introduction of the (*i*)- and (*i*+3)-substituents constituted a significant problem. All attempts to functionalize the alkene in **35** were either unsuccessful or resulted in transannular participation of the amide nitrogen leading to indolizidinones of type **36** (Scheme 14). We were able to capitalize on this mode of reactivity to synthesize indolizidinone **37**, a mimic of the immunosuppressant tripeptide (Lys-Pro-Arg), which is a competitive antagonist of its endogenous progenitor tuftsin (Thr-Lys-Pro-Arg).[83]

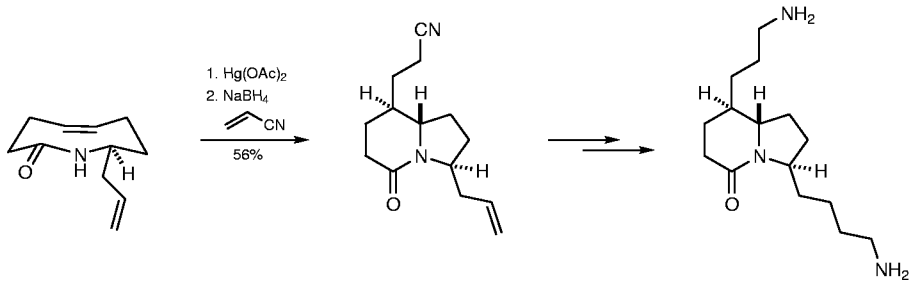
**Scheme 14** Nine-Membered-Ring Lactam and Indolizidinone-type β-Turn Peptidomimetics[81]




The synthesis of indolizidinone **37**[84] (Scheme 15) involved a transannular amidomercuration of lactam **38**. Subsequent stereospecific reductive trapping with acrylonitrile afforded the key indolizidinone **39** in 56% yield.[85] Further manipulation provided **37** which antagonized, in a dose-dependent fashion, the stimulatory effect of tuftsin in an opsonized sheep red blood cell assay.

*for references see p 736*

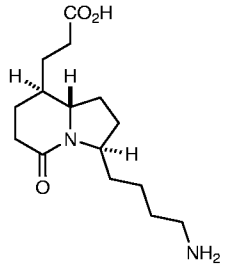
**Scheme 15** Preparation of an Indolizidinone-Based Tuftsin Antagonist[84]




38 39 37

A similar strategy was utilized to construct indolizidinone **40** (Scheme 16), a designed mimic of the neurotoxic loop of erabutoxin b, a nicotinic acetylcholine receptor antagonist.[86]

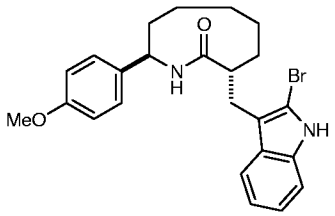
**Scheme 16** An Indolizidinone-Based Nicotinic Acetylcholine Receptor Antagonist[86]




40

Preparation of a mimetic of the cyclic peptide jaspamide was reported in 1988.[87] Compound **41** (Scheme 17) was designed to mimic the type-II β-turn of the peptide, which contained the proposed pharmacophoric phenol and 2-bromoindole functional groups. This analogue displayed interesting insecticidal activity against the tobacco budworm, *Heliothis vireseins.*

**Scheme 17** A Jaspamide-Based β-Turn Mimetic[87]

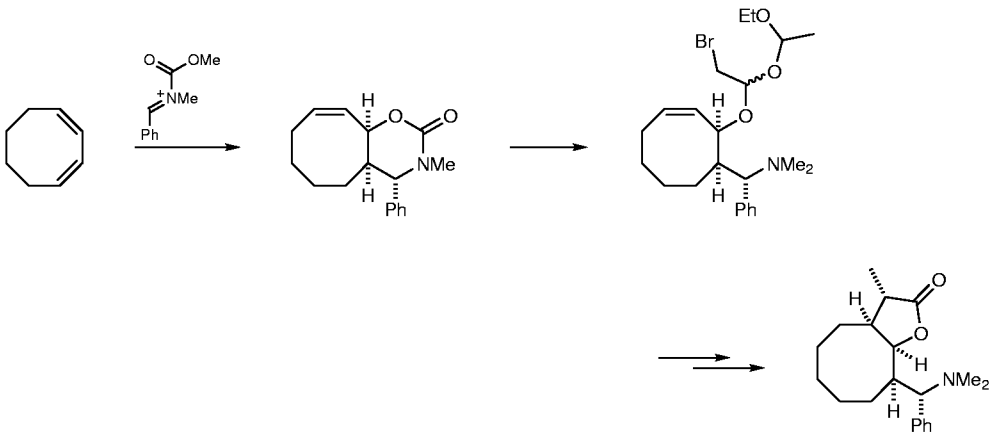



41

The idealized mimetic **34** (Scheme 14) was designed to incorporate the ability to introduce functionality at each of the positions $R^1$, $R^2$, $R^3$, and $R^4$, and both an amino and carboxy terminus for coupling to peptide chains. The stereospecific and regiospecific synthesis of a molecule which contains six nonadjacent asymmetric centers proved to be a difficult task. Model studies for the synthesis were performed.[88] The salient feature of this synthesis was an acylimminium cyclization[89] coupled with a tin hydride induced radical cyclization (Scheme 18).[90] Unfortunately, we were not able to readily relate the introduction of the

(*i*+1)- and (*i*+2)-residues (the front side) of the turn with this strategy. An alternative approach to effect the introduction of these groups suffered from poor stereocontrol.[81]

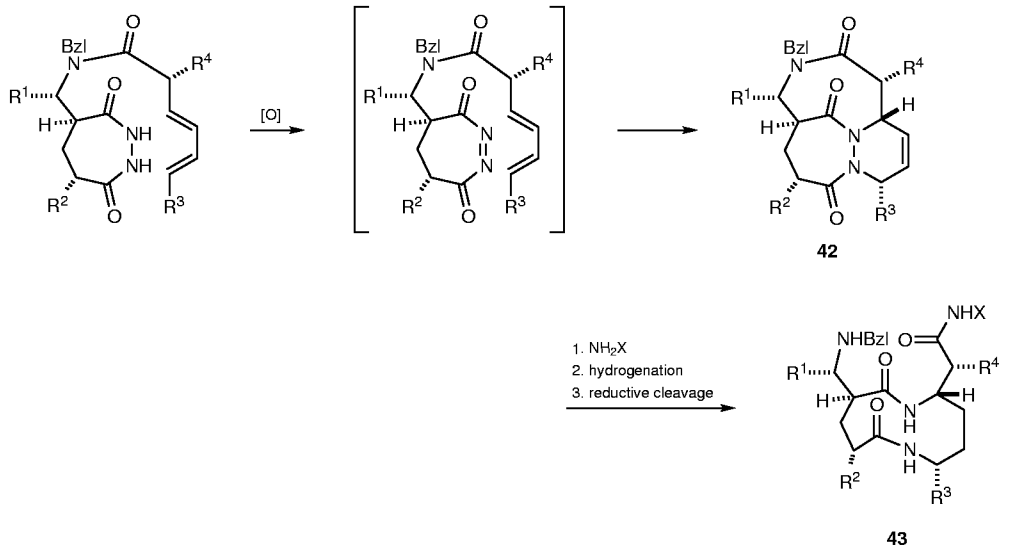
**Scheme 18** Model Studies for the Synthesis of the Idealized Mimetic[88]




Recognizing the limitations of our first generation turn mimetic, the second generation system **43** was designed based upon a synthetic strategy which provided for a more controlled elaboration of the requisite functionality.[91] As initially envisioned, the key synthetic step was an oxidative intramolecular cycloaddition of an azodicarbonyl system (Scheme 19).[92,93] Subsequent nucleophilic rupture of the tricyclic lactam **42**, followed by reductive cleavage of the diacylhydrazide, would provide the desired eleven-membered ring. The choice of an eleven-membered ring at this stage was principally decided on the basis of the desire to maintain the proper distances and spatial relationships among the $R^1$ to $R^4$ substituents. Although a slightly larger ring system than our first generation system, the additional amide linkage imparted similar flexibility/rigidity characteristics on this template. Additionally, we envisioned, and later showed, that the bicyclic precursor could also serve as a useful turn prosthetic unit.

*for references see p 736*

**Scheme 19** Synthetic Scheme for the Preparation of β-Turn Peptidomimetics[92,93]

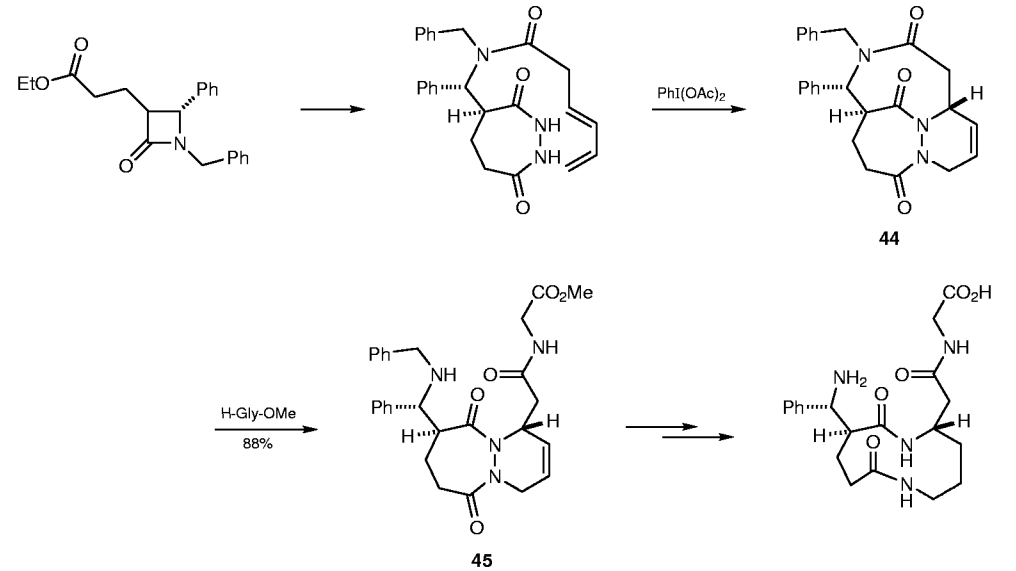



42

43

Computational analysis indicated that the basic ring structure **43** provides an excellent foundation for the development of a variety of β-turn mimetics.[94] The nucleophilic opening of the tricyclic system **42** was initially a point of consternation. However, we were convinced that the inherent strain introduced by the unusual bridging topology of the eight-membered ring would allow for selective cleavage of the desired amide linkage. The model study for this sequence is shown in Scheme 20.[91]

**Scheme 20** Synthetic Scheme for the Model β-Turn Peptidomimetic[91]




44

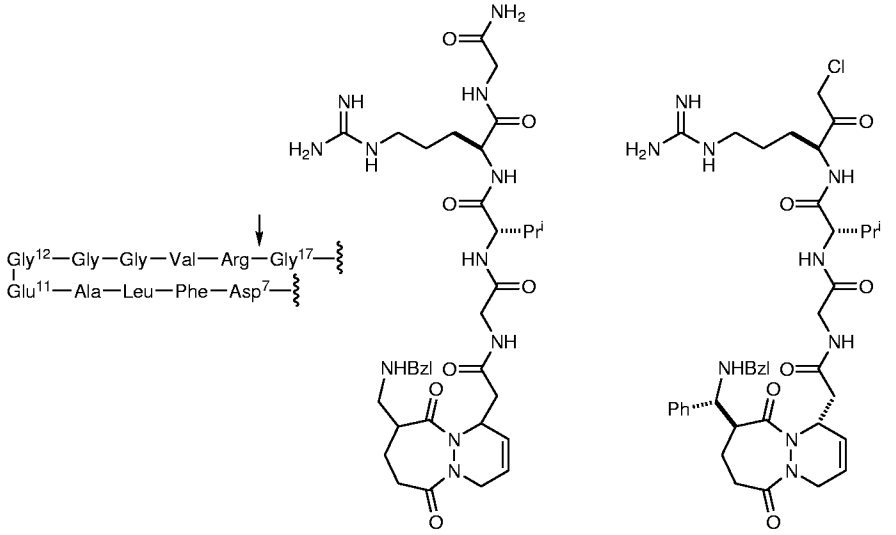
45

Gratifyingly, the intramolecular oxidative cycloaddition proceeded smoothly (10 min in refluxing $CHCl_3$) and in essentially quantitative yield to provide tricyclic lactam **44**.[91] The IR absorption stretching frequency observed at 1785 $cm^{-1}$ attested to the strain and lack of amide resonance in this novel ring system. Nucleophilic ring opening proceeded readily with

methyl glycinate to afford bicyclic lactam **45** in 88% yield after chromatography. Based on the facility of the reaction, we anticipated that the tricyclic lactam could be utilized as the activated ester component in solid-phase peptide synthesis. In the event, we demonstrated that the synthesis of peptide chimeras (i.e., peptide turn mimetics incorporated into oligopeptides) was readily accomplished using solid-phase protocols.[95] SPPS could be performed on a variety of resins (PAM, MBHA, etc.) using either Boc or Fmoc strategies. Cleavage from the polystyrene support could be effected by several procedures (aminolysis, reduction, HF treatment) depending on the resin type and the desired outcome of the C-terminus. Isolated and purified yields ranged from 33 to 69%.

Peptide chimeras of this type have proven extremely valuable for the examination of an array of recognition events. In conjunction with molecular modeling, peptidomimetic substrates and inhibitors of human thrombin were designed and synthesized to evaluate our proposed structure for the thrombin-bound conformation of fibrinopeptide A (Scheme 21).[96]

**Scheme 21** Amino Acid Sequence of a Segment of Fibrinopeptide A and Its β-Turn Peptidomimetics[96]




A similar strategy was used to examine the potential role of a reverse turn as a recognition element adjacent to the cleavage site of substrates of HIV protease.[97] A series of inhibitors was prepared, the synthesis of which involved the solution coupling of the statine-like transition state mimic to the β-turn mimetic to provide **46** (Scheme 22). Subsequent sodium in ammonia reduction provided analogue **47**. One of the compounds was a reasonably potent inhibitor of protease activity ($IC_{50} = 2.6 \times 10^{-8}$ M) (Table 1).

*for references see p 736*

**Scheme 22** Preparation of HIV Protease Inhibitors with β-Turn Peptidomimetics[97]

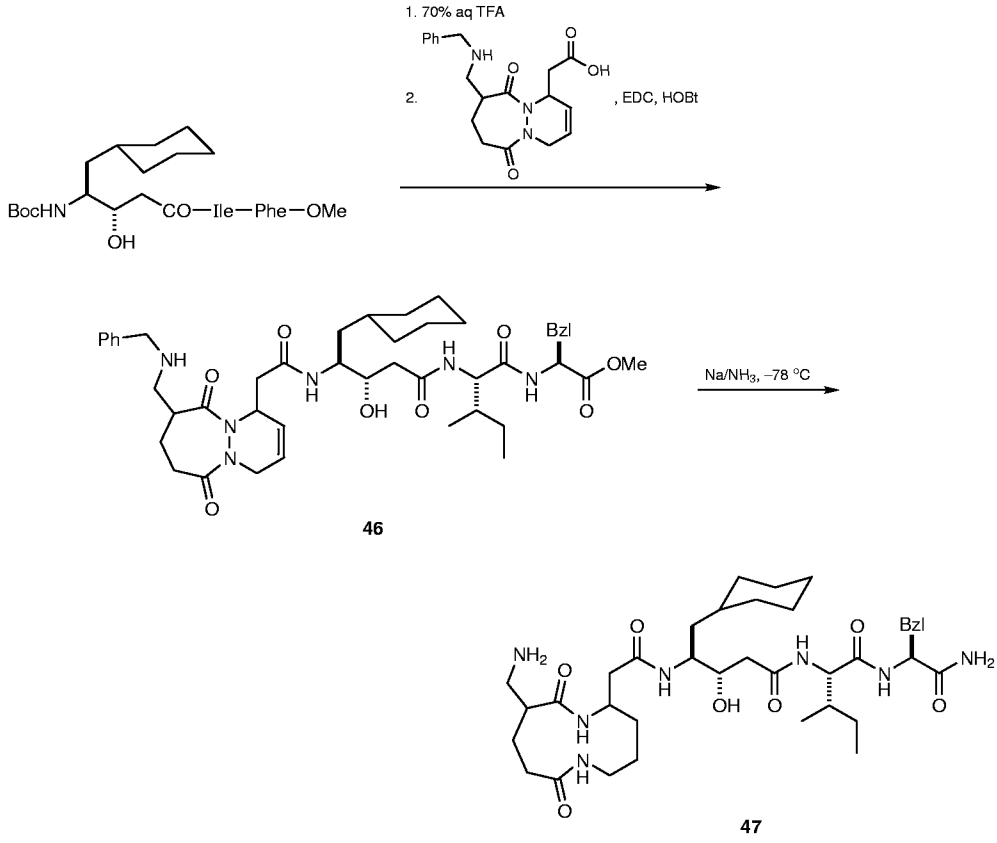
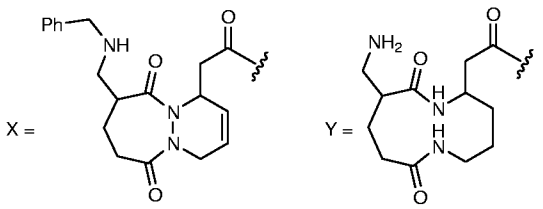



**Table 1** Inhibition of HIV Protease Activity[97]

| Inhibitor[a] | $IC_{50}$[b] (M) | Ref |
|---|---|---|
| X-Achpa-Ile-Phe-OMe (**46**) | $3.6 \times 10^{-4}$ | [97] |
| X-Sta-Ile-Phe-OMe | $1.4 \times 10^{-4}$ | [97] |
| X-Sta-Ile-Phe-$NH_2$ | $2.1 \times 10^{-5}$ | [97] |
| X-Ahppa-Ile-Phe-OMe[c] | $3.5 \times 10^{-4}$ | [97] |
| Y-Achpa-Ile-Phe-$NH_2$ (**47**) | $2.6 \times 10^{-8}$ | [97] |

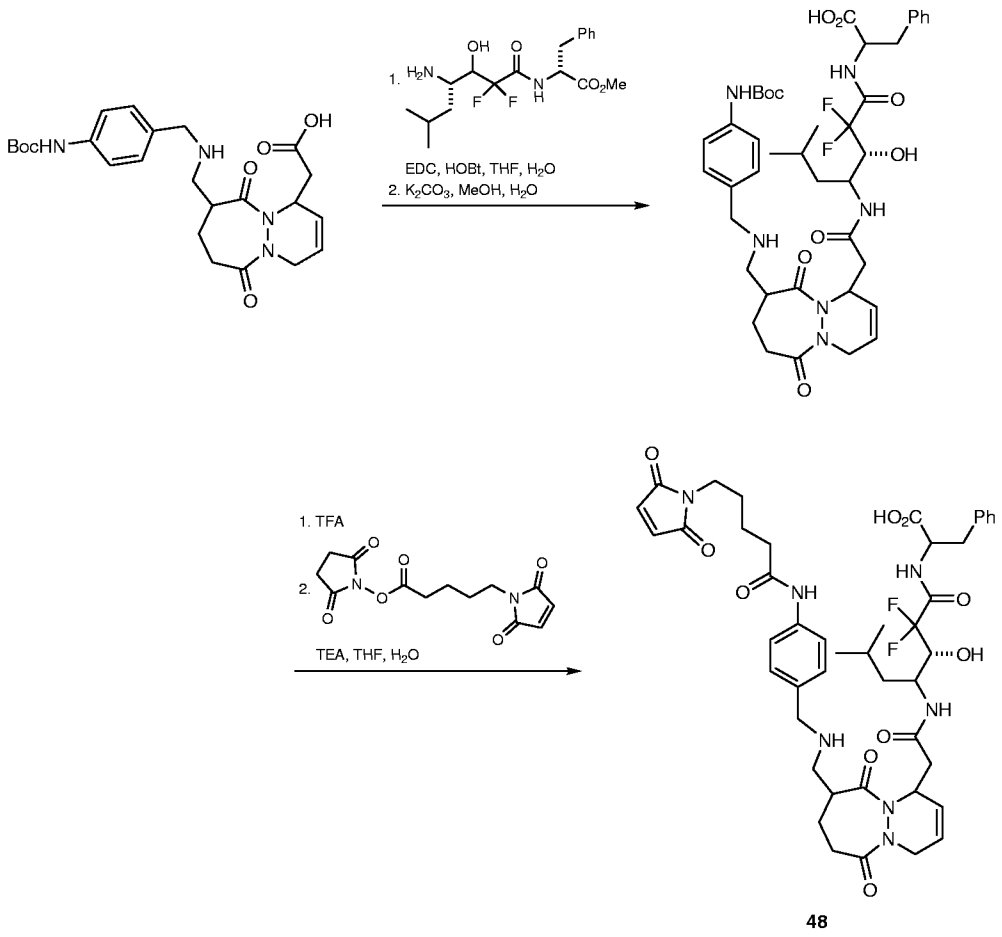
[a] 


[b] Estimated from plots of percent inhibition vs inhibitor concentration.

[c] Ahppa = (3*S*,4*S*)-4-amino-3-hydroxy-5-phenylpentanoic acid.

An approach for the generation of secondary structure specific abzymes, which utilized a chimeric construction as the hapten, has been described.[98] The hapten was designed to elicit a catalytic antibody which could potentially site-selectively cleave a peptide bond. The synthesis of the hapten **48** is outlined in Scheme 23.

**Scheme 23** Preparation of a Hapten with a β-Turn Peptidomimetic[98]

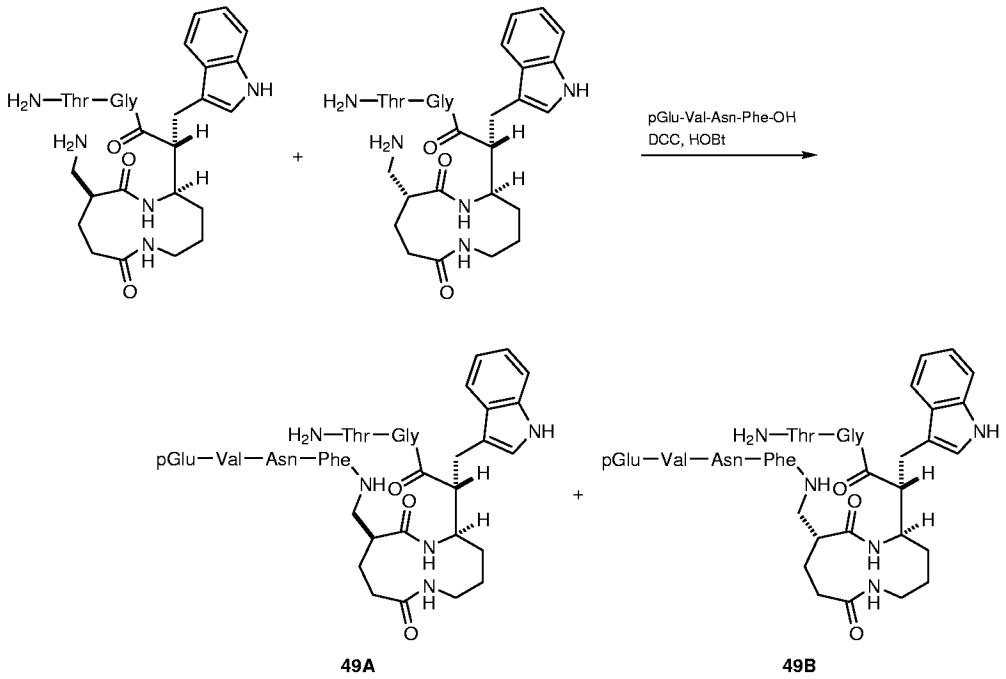



48

Using a similar synthetic strategy, mimetics of the hypertrehalosemic hormone, a decapeptide (pGlu-Val-Asn-Phe-Ser-Pro-Gly-Trp-Gly-Thr-$NH_2$) which controls energy metabolism in insects, were constructed, replacing the proposed bioactive turn region -Ser-Pro-Gly-Trp- with a turn mimetic. Two diastereomeric mimetics **49A** and **49B** (Scheme 24) were synthesized and evaluated for biological activity. Only mimetic **49B** displayed significant hyperglycemic activity in the bioassay system.[99] Stereoselective introduction of the hydroxymethyl group to the (*i*)-position of this β-turn template precursor has also been successfully carried out as shown in Schemes 25 and 26 (experimental details given below).[100]

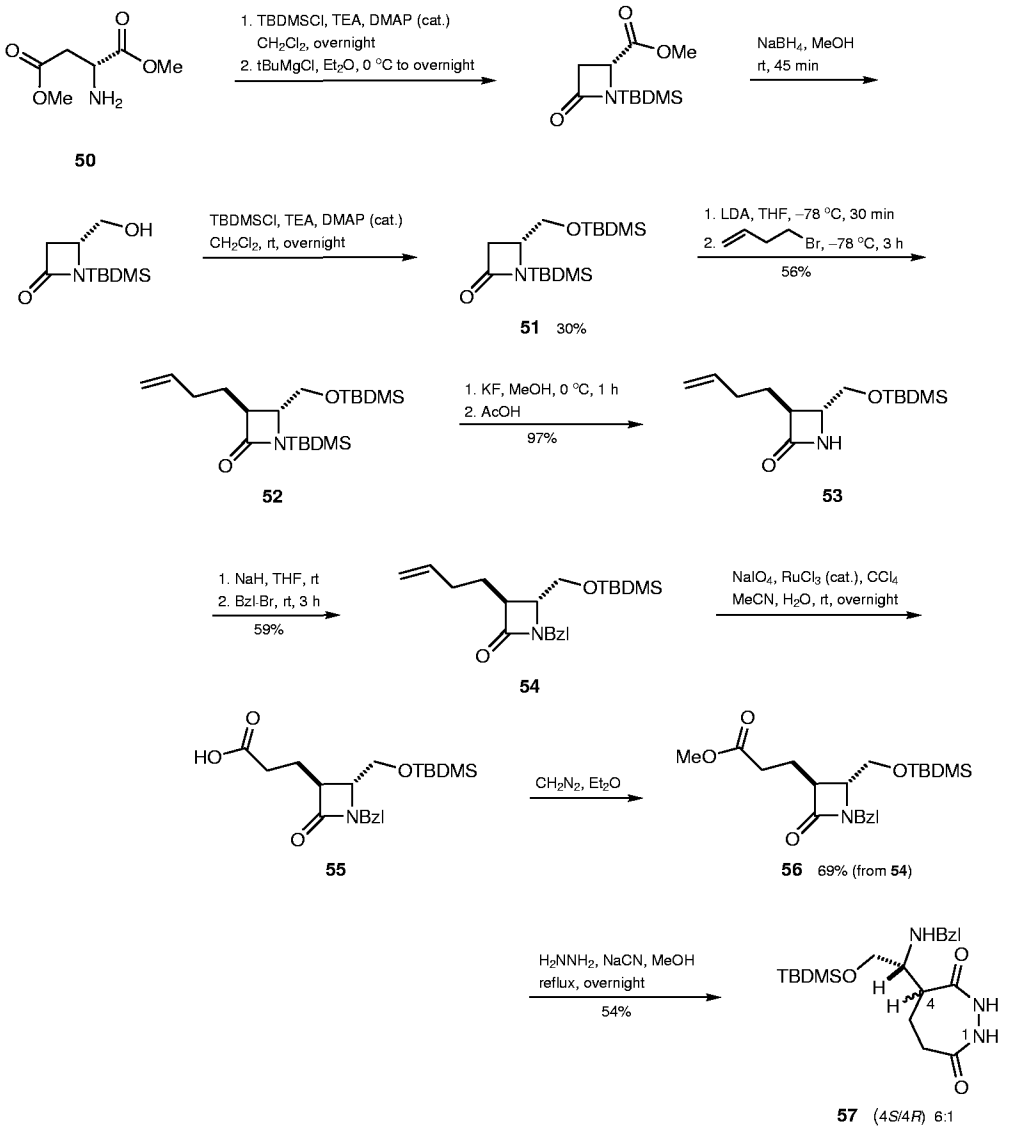
*for references see p 736*

**Scheme 24** Preparation of Hypertrehalosemic Hormone β-Turn Peptidomimetics[99]



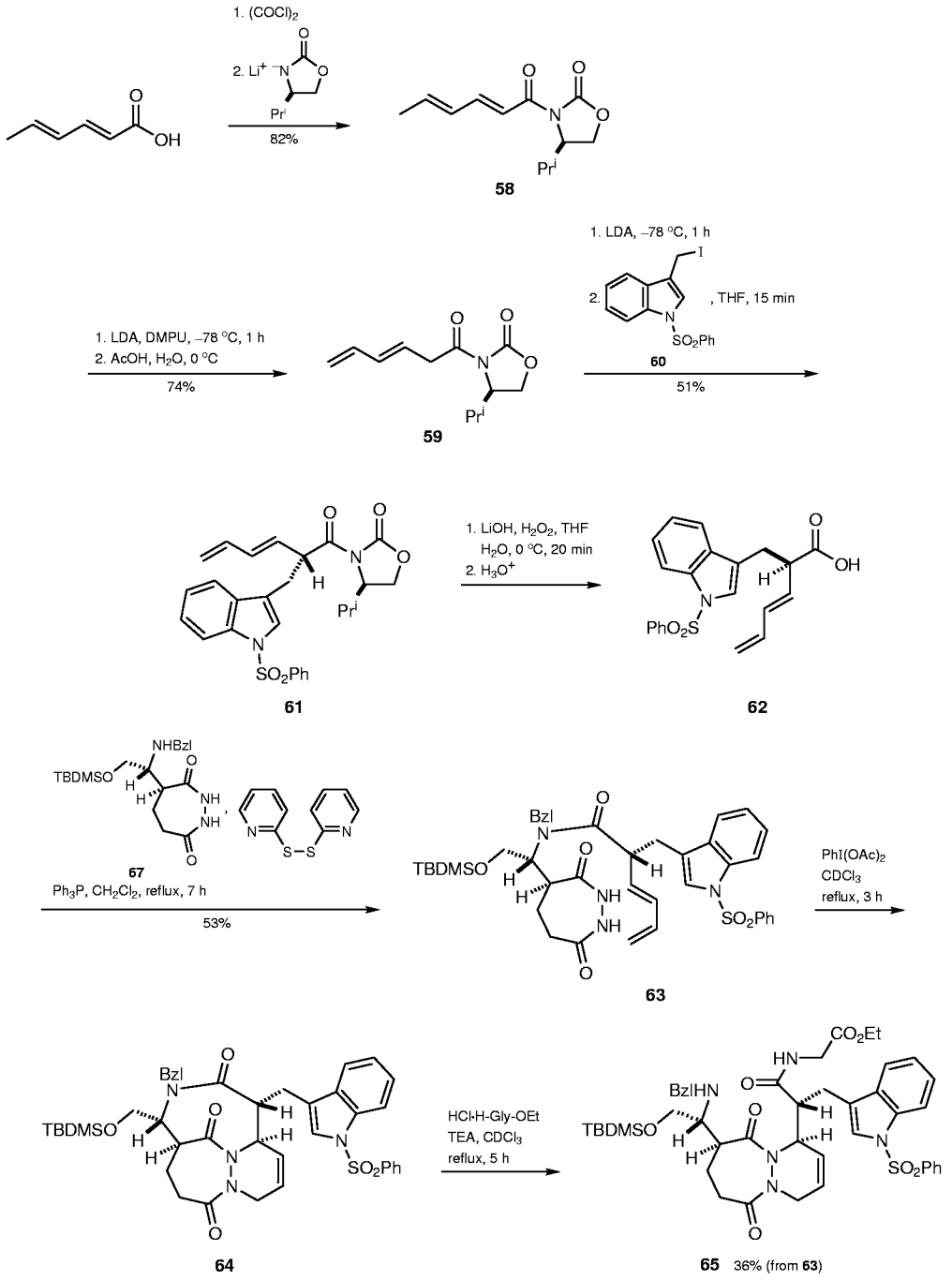

**49A** **49B**

**Scheme 25** Preparation of the Diacylhydrazide Intermediate[100]



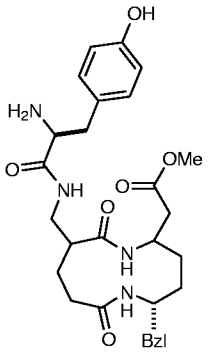

*for references see p 736*

**Scheme 26** Preparation of a β-Turn Peptidomimetic via Intramolecular Cycloaddition[100]



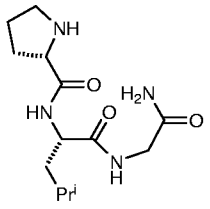

Similarly, a turn mimetic analogue of Leu-enkephalin (Scheme 27) was prepared.[101] The compound exhibited significant in vivo analgesia despite relatively low affinity in a binding assay.

**Scheme 27** A Leu-enkephalin β-Turn Peptidomimetic[101]

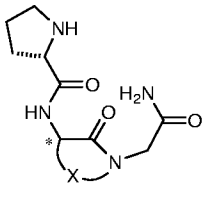


The endogenous tripeptide H-Pro-Leu-Gly-$NH_2$ (**66**, Scheme 28) has a modulatory effect on the CNS dopamine receptor and is believed to adopt a type-II β-turn conformation based upon NMR spectroscopic[102] and X-ray diffraction[62] investigations. The biologically active conformation was probed by introducing various constraints **67–72**.[103–106] The initial analogue **67**,[107] which incorporated a γ-lactam at the Leu-Gly position, showed a promising 10,000-fold enhancement of the binding of the dopamine agonist 2-amino-6,7-dihydroxy-1,2,3,4-tetrahydronaphthalene compared with the linear tripeptide. Surprisingly, variation in the size and chirality of the lactam ring **67–72** did not affect potency. The 5,5- and 6,5-bicyclic thiazolidine lactam systems **73** and **74**, respectively, originally designed to mimic a type-II′ β-turn,[76] showed only a modest (2- to 5-fold) increase in activity compared with the linear tripeptide. Compound **73B** did not show any activity. Conformational analysis of **73A** and **74A** using a random conformational search[108] with the MM2 force field[109] indicated that both **73A** and **74A** possess nearly ideal $\psi_2$ and $\phi_3$ torsional angles (ideal angles are −120° and −80°, respectively). The synthesis of **74A** starting from oxazolidin-5-one **75** is outlined in Scheme 29 (experimental details given below).[106]

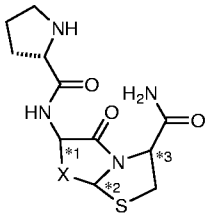
**Scheme 28** H-Pro-Leu-Gly-$NH_2$ and Some of Its β-Turn Peptidomimetics[103–107]



**66**



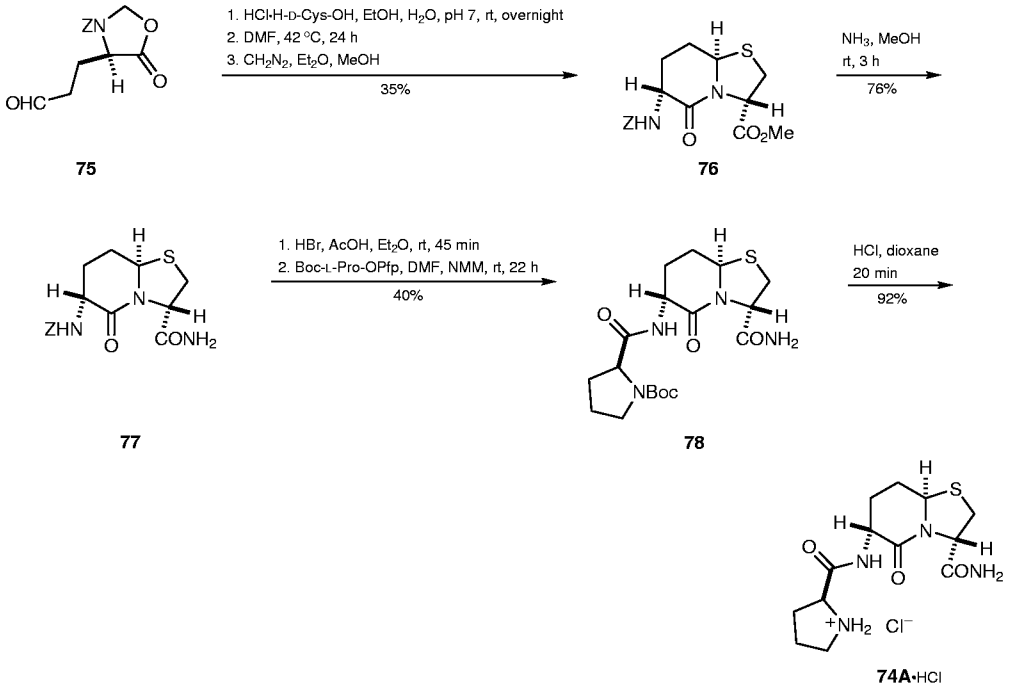
**67** X = $(CH_2)_2$; * = *R*
**68** X = $CH_2$; * = *R*
**69** X = $CH_2CO$; * = *R*
**70** X = $CH_2O$; * = *R*
**71** X = $(CH_2)_3$; * = *R*
**72A** X = $(CH_2)_4$; * = *R*
**72B** X = $(CH_2)_4$; * = *S*



**73A** X = $CH_2$; * = *R,R,S*
**73B** X = $CH_2$; * = *S,S,R*
**74A** X = $(CH_2)_2$; * = *R,R,S*
**74B** X = $(CH_2)_2$; * = *S,S,R*

*for references see p 736*

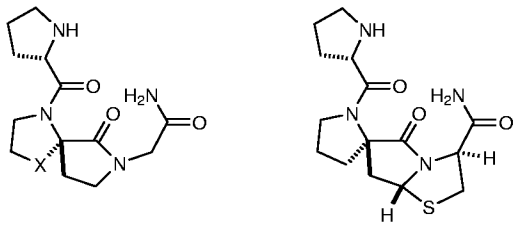
**Scheme 29** Preparation of a H-Pro-Leu-Gly-$NH_2$ β-Turn Peptidomimetic[106]




The (*R*)-4,4- and the 5,4-spirolactam systems were also used to induce a β-turn in the H-Pro-Leu-Gly-$NH_2$ sequence (i.e., **79** and **80**, Scheme 30).[103] Analysis by X-ray crystallography indicated that the structures form a type-II β-turn.[104] Previous studies[110] with a series of neurokinin antagonists had shown that the extended conformation is favored for the (*R*)-4,4-spirolactam system; however, the (*S*)-4,4-spirolactam demonstrated a type-II′ β-turn conformation.

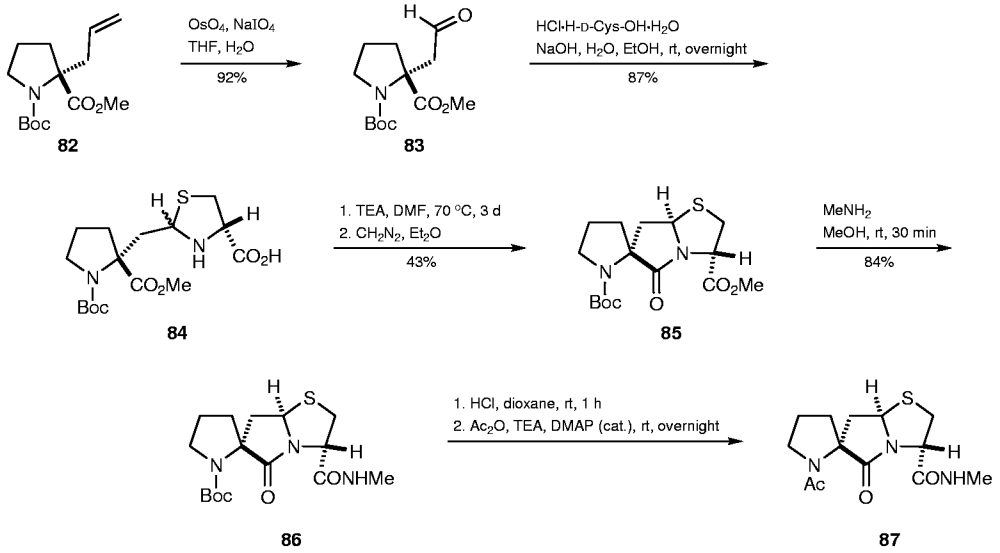
Further constrained analogues in this series were prepared by combining the bicyclothiazolidine lactam and spirolactam to form a spirobicyclic system **81** (Scheme 30).[111] The preparation of the spirobicyclic template **87** has been accomplished as shown in Scheme 31 (experimental details given below). Molecular modeling and NMR spectroscopic studies in $CDCl_3$ suggested the formation of a type-II β-turn, through restriction to almost ideal values of three out of the four torsion angles ($\phi_2$, $\psi_2$, and $\phi_3$). However, in DMSO, the mimetic exhibited a high temperature coefficient for the amide hydrogen, indicating that the β-turn is not stable, presumably due to flexibility at $\psi_3$. The modulatory activity of this analogue was tested and found to enhance the binding of 2-amino-6,7-dihydroxy-1,2,3,4-tetrahydronaphthalene to the dopamine receptor by only 40% at 1 μM concentration compared to 26% at 1 μM concentration for the linear H-Pro-Leu-Gly-$NH_2$ (**66**). The authors claim that these results clearly demonstrate that the bioactive conformation of **66** is a type-II β-turn; however, the failure to improve on the initially less constrained compound **67** (Scheme 28) by introducing additional constraints leads one to believe that, although the β-turn conformation may be important, the addition of buttressing groups beyond the peptide backbone may be interfering with binding.

**Scheme 30** H-Pro-Leu-Gly-$NH_2$ β-Turn Peptidomimetic Spirolactams[103,111]



**79** X = $CH_2$
**80** X = $(CH_2)_2$
**81**

**Scheme 31** Preparation of a H-Pro-Leu-Gly-$NH_2$ β-Turn Peptidomimetic Spirolactam[111]



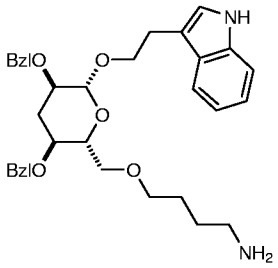

**82** **83** **84** **85** **86** **87**

Baca et al.[112] made use of Nagai and Sato's[76] type-II′ thiazolidine lactam β-turn mimetic **18** (Scheme 11) to replace the type-I′ β-turn between $Gly^{16}$ and $Gly^{17}$ in HIV-1 protease. The β-turn-mimetic-containing enzyme dimerized similarly to the native enzyme. It was fully active, possessed the same substrate specificity as the native enzyme, and showed enhanced thermal stability.

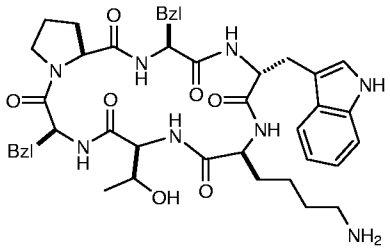
Nicolaou and co-workers[113] utilized the 3-deoxy-β-D-glucose **88** (Scheme 32) as a scaffold to synthesize the first nonpeptide analogue of the receptor-recognizing β-turn ($Phe^7$, $Trp^8$, $Lys^9$, $Thr^{10}$) of somatostatin analogue **89** (Scheme 32). This mimetic binds to the pituitary gland somatostatin receptor with an $IC_{50}$ value of 1.3 μM. Interestingly, in a functional assay this mimetic displayed agonist activity at 3 μM, a feature that is not frequently observed with peptidomimetics.

*for references see p 736*

**Scheme 32** Glucose β-Turn Peptidomimetic and Somatostatin Analogue[113]



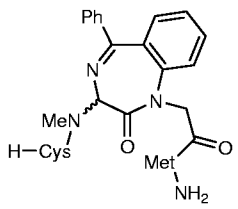

88



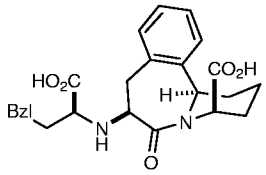

89

An interesting application involved the design of a benzodiazepine peptidomimetic **90** (Scheme 33) to inhibit the enzyme *Ras* farnesyl transferase. The benzodiazepine system was intended to mimic the proposed β-turn in the -Cys-Ala-Ala-Xaa- sequence of the enzyme substrate.[114] It had an $IC_{50}$ <1 nM, although subsequent X-ray diffraction crystallography has shown that the natural sequence is bound in an extended strand conformation. Interestingly, compound **91** (Scheme 33), a structurally related lipophilic, conformationally restricted mimic of the tripeptide -Phe-His-Leu-, is a potent inhibitor of angiotensin-converting enzyme.[115] In this case, by analogy to other metalloprotease–substrate interactions,[116] the enzyme-bound substrate presumably adopts an extended structure.

**Scheme 33** Benzodiazepine and Benzazepine β-Turn Peptidomimetics[114,115]



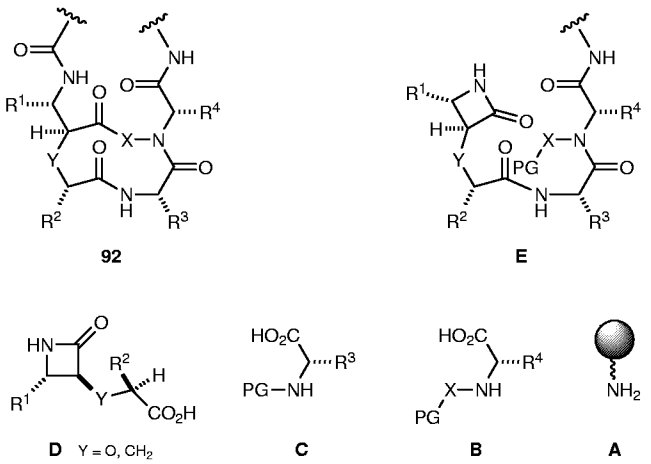

90




91

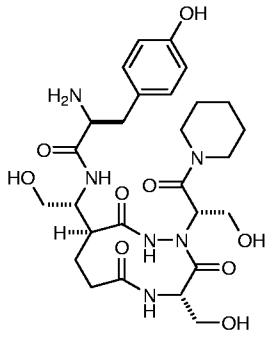
In order to capture the modular component nature and automated facility of solid-phase peptide synthesis, the system depicted in Scheme 34 was developed.[72,117] The synthesis of the reverse-turn mimetic **92** can be performed in solution; however, it is designed to be, and is fully compatible with, solid-phase synthesis protocols. In essence, it involves the coupling of the first modular component **B** to the amino terminus of a growing peptide chain **A**. Coupling of the second component **C**, removal of the protecting group, and subsequent coupling of the third modular component **D** provides the nascent β-turn **E** (Scheme 34). The critical step in this sequence involves the use of an azetidinone as an activated amide to effect the macrocyclization reaction.[118] Upon nucleophilic opening of the azetidinone by nucleophile X, a new amino terminus is generated for continuation of the synthesis. An important feature of this scheme is the ability to alter the X group linker in **92**, both with regard to length and degree of rigidity or flexibility. The requisite stereogenic centers are readily derived, principally from the "chiral pool". The synthesis allows for the introduction of natural or unnatural amino acid side-chain functionality in either L or D configuration.

**Scheme 34** Retrosynthetic Strategy for the Facile Preparation of β-Turn Peptidomimetics[72,117]




We have used our third generation mimetic system to explore structure–function relationships among molecules of the immunoglobulin gene superfamily. Immunoglobulins are constructed from a series of antiparallel β-pleated sheets connected by loops.[119,120] The specificity of these molecules is determined by the sequence and size of the canonical hypervariable complementarity determining regions (CDRs).[121,122] The variable light chain CDR-2 sequence canonically exists in a reverse-turn conformation.[123,124] Therefore, we designed and synthesized a reverse-turn mimetic **93** (Scheme 35), which incorporates the sequence -Tyr-Ser-Gly-Ser-Ser-.[125] Importantly, it displayed similar binding properties to the cellular reovirus receptor and to MAb 9BG5, and had the same inhibitory effect on cell proliferation as did the native antibody.

**Scheme 35** Third Generation β-Turn Peptidomimetic[125]

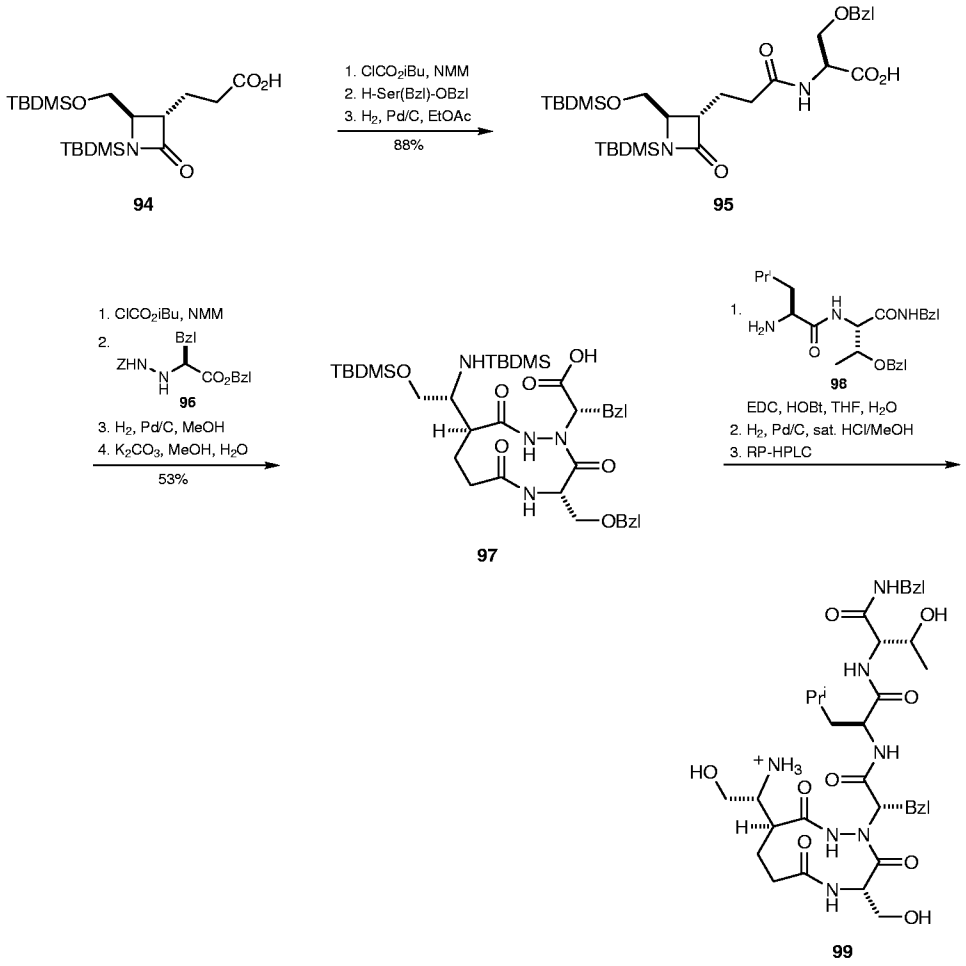



We have also designed a mimetic of the CDR-2 like region of human CD4, a 55 kDa glycoprotein primarily found on the cell surface of the helper class of T cells. CD4 binds the human immunodeficiency virus glycoprotein (HIV gp120) with high affinity ($K_d$ ca. 1–4 nM), and is an important route of cellular entry for the virus. Extensive mutagenesis[126,127] and peptide mapping experiments[128] have shown that the region of amino acids 40–55 within the CDR-2 like domain of CD4 is critical for gp120 binding. X-ray diffraction crystallographic analysis showed that residues $Gln^{40}$ through $Phe^{43}$ reside on a highly surface-exposed β-turn connecting the C′ and C″ β-strands. A third generation mimetic of this region **99** was designed and synthesized according to Scheme 36.[129] Azetidinone **94** was prepared in six steps

*for references see p 736*

(22% yield) from H-D-Asp(OMe)-OMe in a fashion analogous to procedures described by Salzmann et al.[130] and Williams et al.[131] Mixed anhydride coupling of **94** to H-Ser(Bzl)-OBzl and subsequent hydrogenolytic cleavage of the benzyl ester provided acid **95** in 88% yield. Coupling of the benzyloxycarbonyl-protected hydrazino derivative of phenylalanine **96**[132] to **95** using the procedure of Tung et al.[133] proceeded smoothly. Reductive closure[118] and saponification afforded the ten-membered-ring β-turn mimetic **97** in 53% yield. Dipeptide **98** was coupled to **97** using an aqueous carbodiimide protocol. Hydrogenolytic deprotection of the side-chain protecting groups and purification by C18 reversed-phase HPLC provided homogeneous **99**. NMR spectroscopic and molecular modeling analyses confirmed that the ten-membered-ring system closely mimics the conformation of the β-turn loop in question (*vide supra*). Importantly, this small molecule mimic abrogates the binding of HIV-1 (IIIB) gp120 to $CD4^+$ cells at low micromolar levels and reduces syncytium formation 50% at 250 $\mu g \cdot mL^{-1}$.[129]

**Scheme 36** Preparation of a β-Turn Peptidomimetic of the CDR-2 Like Region of Human CD4[129]



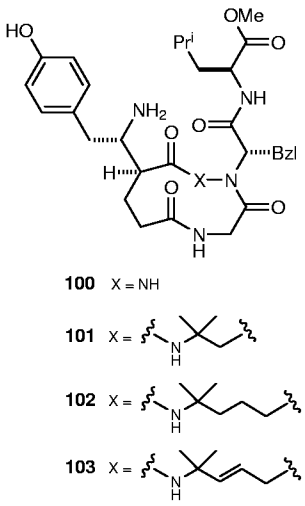

More recently, we have been addressing a problem described previously concerning the relationship between morphine and enkephalin. The inherent mobility of the enkephalin framework, its rapid degradation in vivo,[134] and the existence of multiple receptor subtypes[135,136] have hampered the assessment of its bioactive conformations. Conformationally

constrained peptides or peptidomimetics[80,101,137,138] should facilitate this task. Several turn conformations have been proposed based upon computational models,[139–142] X-ray diffraction crystallography,[143,144] and spectroscopic studies.[145–147]

In 1976 Bradbury et al.[148] proposed a β-bend model stabilized by an intramolecular hydrogen bond between the NH of $Phe^4$ and the C=O of $Tyr^1$, which produces a spatial disposition between the $Phe^4$ aromatic ring and the tyramine segment of $Tyr^1$ analogous to that existing between the corresponding moieties in the potent morphine analogue oripavine. Further support for the relevance of this conformation was provided by energy calculations on the potent [D-$Ala^2$, $Met^5$]-enkephalin analogue, the lowest energy conformer[149–151] of which contained a folded structure with a β-turn centered on residues 2 and 3. However, evidence contrary to the biological significance of a β-turn has been presented by Freidinger,[152] and additionally by Schiller and DiMaio[153] in their analyses of the conformations of 13- and 14-membered rigid cyclic analogues.

To further examine this hypothetical bioactive conformation, a family of β-turn mimetics **100–103** has been synthesized (Schemes 37 and 38, experimental details for **100** given below).[154] The lowest energy conformer of the ten-membered-ring system **100** is an excellent mimic of an idealized type I′ β-turn (6 atom rms deviation 0.22 Å) and displays good overlap with the critical $Phe^4$ aromatic ring and tyramine moieties of the morphine analogue *endo*-6,14-etheno-7-(1-hydroxy-1-methyl-3-phenylpropyl)-6,7,8,14-tetrahydrothebaine, yet it is essentially devoid of biological activity. Only the 14-membered-ring analogue **103**, which has a rather expanded loop structure, demonstrates any binding activity at the μ-receptor, although even this was minimal. The results of this investigation can be interpreted as casting significant doubt on the biological relevance of a β-turn conformation for enkephalin. It is hoped that this type of systematic approach to the synthesis of constrained reverse-turn analogues, in conjunction with multiple peptide synthetic strategies, will clarify the situation as to the biological significance of these and other proposed receptor-bound reverse-turn conformations.

**Scheme 37** Enkephalin β-Turn Peptidomimetics[154]




**100** X = NH

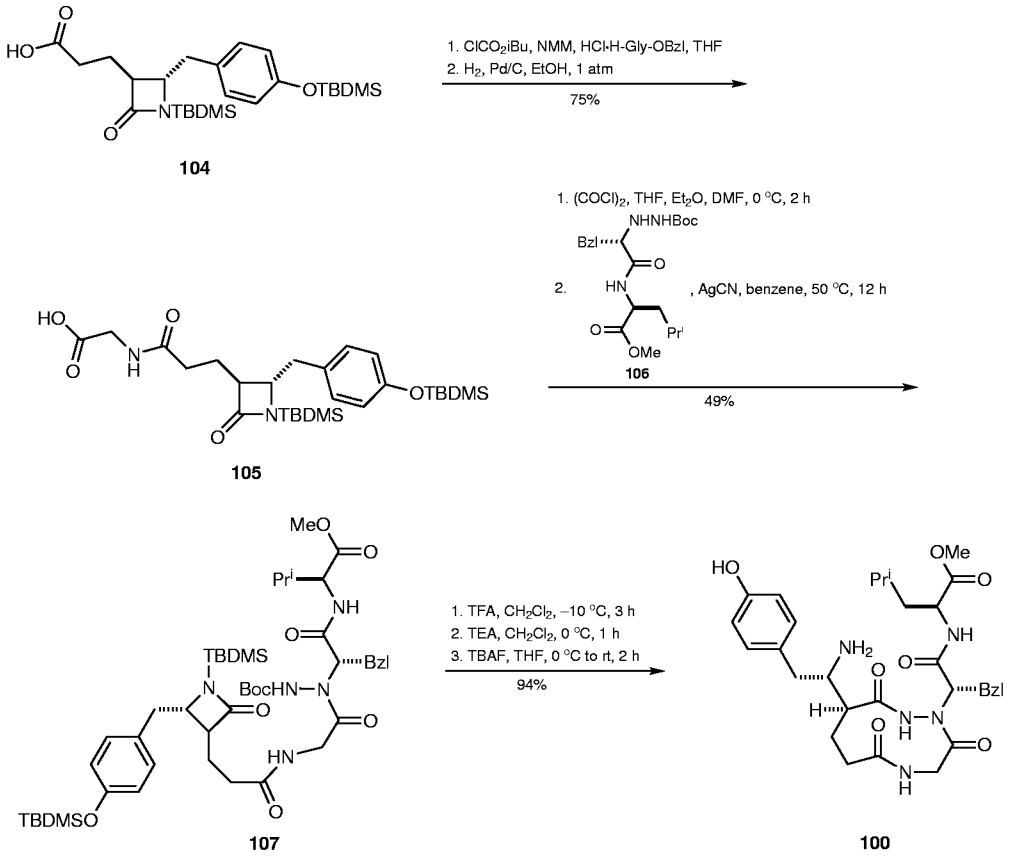
**101** X = 

**102** X = 

**103** X = 

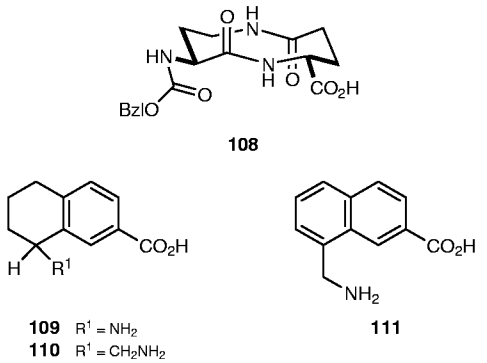
*for references see p 736*

**Scheme 38** Preparation of an Enkephalin-Based with the β-Turn Peptidomimetic[154]



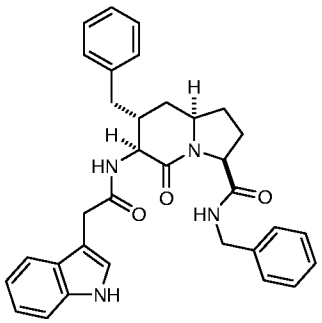

Kemp and Stites[155] prepared the diazacyclodecanedione **108** (Scheme 39) as a template to promote β-turn formation. Similarly, Mutter and co-workers[156] described a series of constrained naphthoic acid derivatives **109–111** (Scheme 39) and their subsequent incorporation into peptide chains.

**Scheme 39** Diazacyclodecanedione β-Turn Peptidomimetic and Naphthoic Acid Derivatives[155,156]




Hanessian et al.[157] again utilized the indolizidinone skeleton to mimic the proposed β-turn in the -Trp-Phe-Gly-Leu- sequence of the undecapeptide substance P. The compound **112** (Scheme 40) had no affinity for the NK1 receptor and low affinity for the recombinant human NK2 receptor.

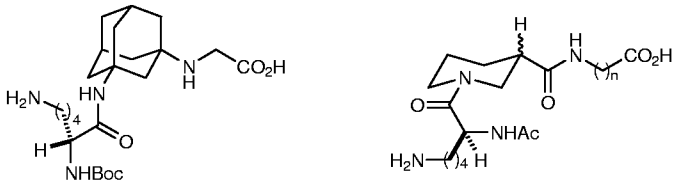
**Scheme 40** The Substance P-Based β-Turn Peptidomimetic[157]



**112**

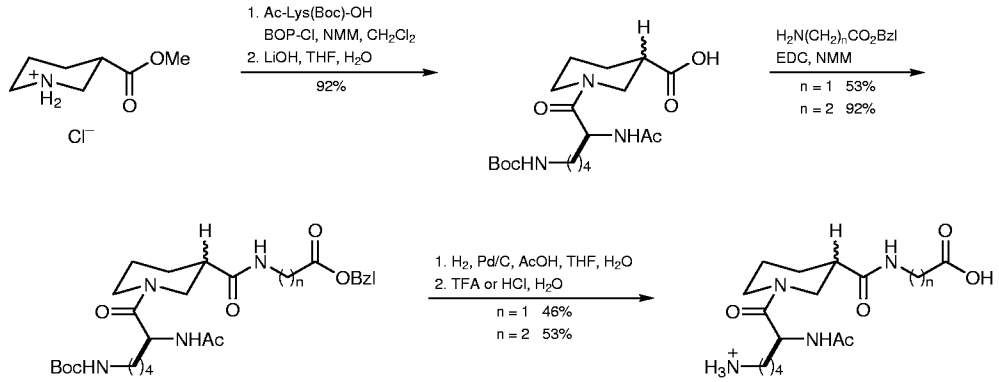
A limited series of antagonists to the fibrinogen binding site on the platelet glycoprotein gpIIb/IIIa has been designed[158] which incorporate either adamantane or nipecotic acid derivatives as β-turn mimetics. Compounds **113** and **114** (Scheme 41) had modest affinity; however, compound **115** was significantly more potent ($IC_{50} = 0.074\,\mu M$) and displayed 86% inhibition of platelet aggregation at 50 μM. The synthesis of the nipecotic acid-based peptides is depicted in Scheme 42.[158]

**Scheme 41** β-Turn Peptidomimetics as Platelet Glycoprotein gpIIb/IIIa Antagonists[158]



**113** **114** n = 1
**115** n = 2

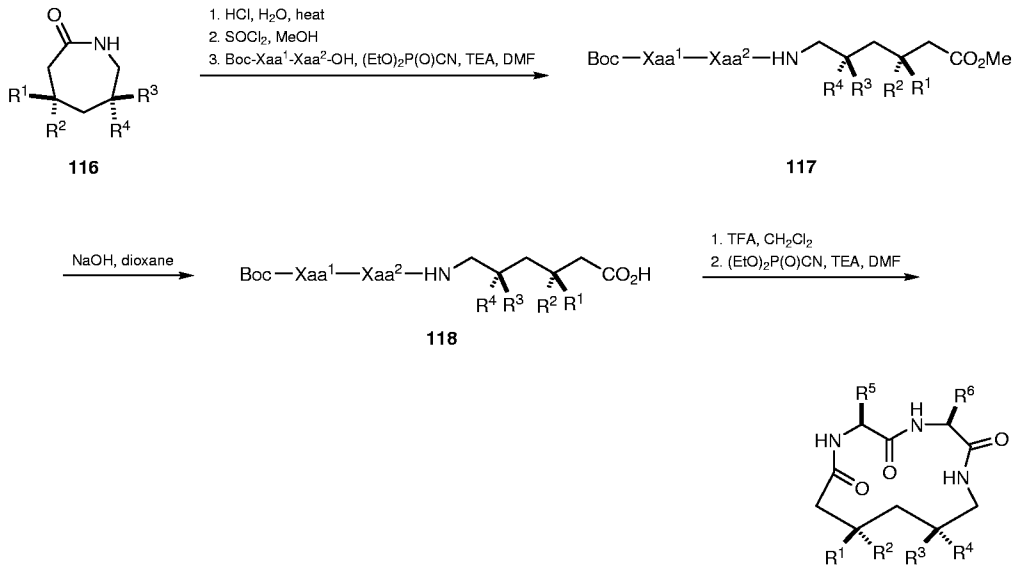
**Scheme 42** Preparation of β-Turn Peptidomimetics as gpIIb/IIIa Antagonists Based on the Nipecotic Acid Template[158]




Substituted 6-aminohexanoic acid linkers were utilized by Aubé et al.[159] to enforce either a type-I or type-II β-turn, depending on the substitutents at the $R^1$ to $R^4$ positions, in structure **119** (Scheme 43).

*for references see p 736*

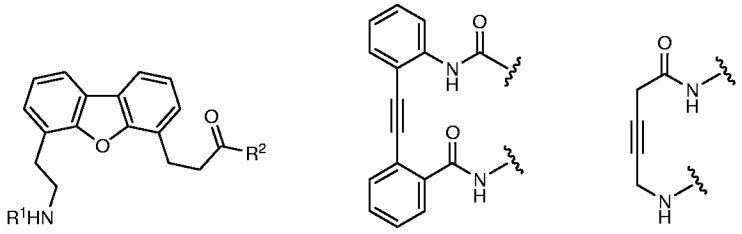
**Scheme 43** Preparation of Cyclic Peptides as β-Turn Peptidomimetics[159]



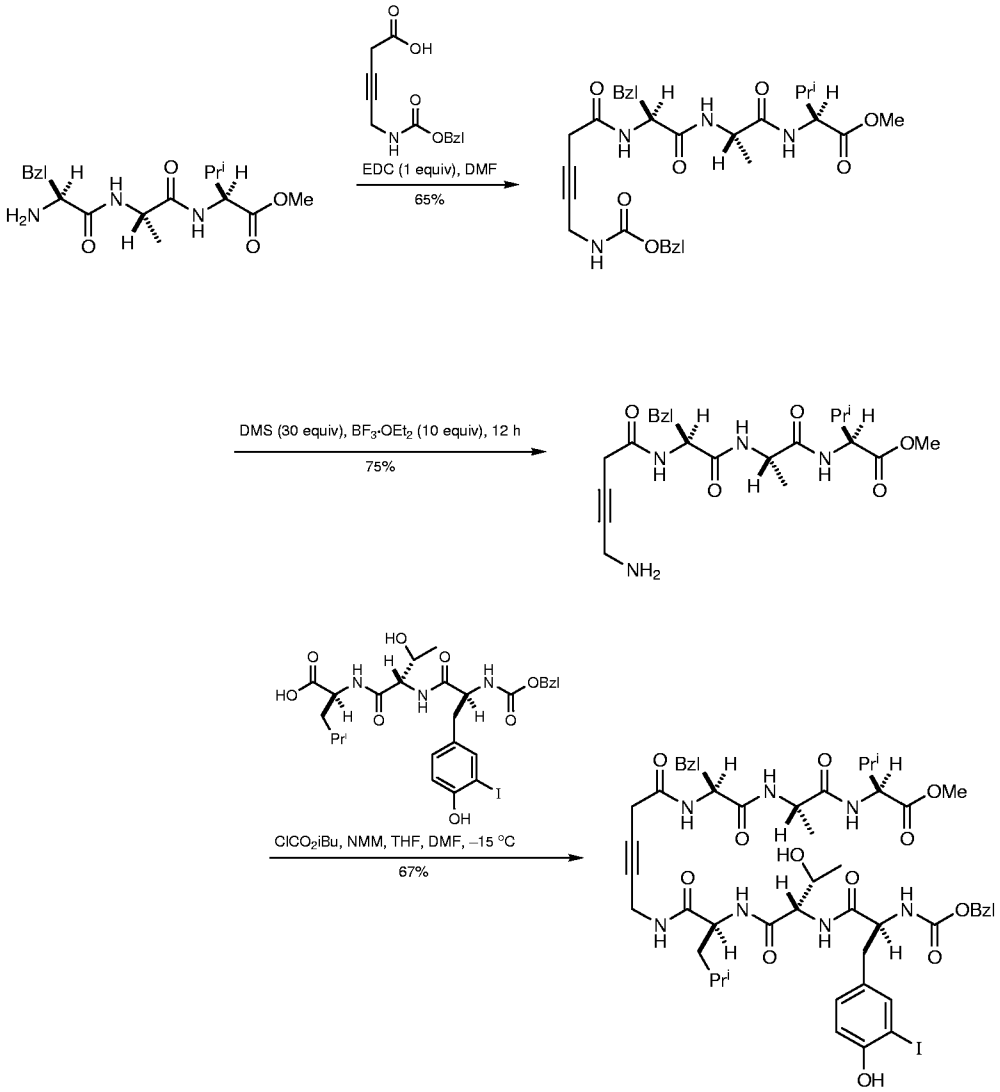

| $R^1$ | $R^2$ | $R^3$ | $R^4$ | $R^5$ | $R^6$ | $Xaa^1$ | $Xaa^2$ | Yield (%) of **117** | Yield (%) of **118** | Yield (%) of **119** | Ref |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Me | H | H | H | H | H | Gly | Gly | 65 | 80 | 35 | [159] |
| H | Me | H | Me | Me | H | Ala | Gly | 96 | 67 | 51 | [159] |
| H | Me | Me | H | Me | H | Ala | Gly | 81 | 43 | 71 | [159] |
| H | Me | H | H | Me | H | Ala | Gly | 74 | 83 | 42 | [159] |
| Me | H | H | H | Me | H | Ala | Gly | 85 | 77 | 51 | [159] |
| Me | H | Me | H | Me | H | Ala | Gly | 75 | 60 | 63 | [159] |
| Me | H | H | Me | Me | H | Ala | Gly | 80 | 61 | 71 | [159] |
| Me | H | H | Me | Me | Me | Ala | Ala | 71 | 89 | 59 | [159] |

The use of rigid-turn templates to initiate antiparallel β-sheet structures has been investigated by Kelly et al.[160] and Kemp and Li[161] utilizing a dibenzofuran **120** and diphenylacetylene **121** template, respectively (Scheme 44). Gani et al.,[162] utilizing an analogous but less constrained analogue **122**, were able to prepare model oligopeptides (Scheme 45).

**Scheme 44** β-Turn Mimetics to Initiate Antiparallel β-Sheet Structures[160–162]



**Scheme 45** Preparation of an Antiparallel β-Sheet Peptide Incorporating a β-Turn Mimetic[162]

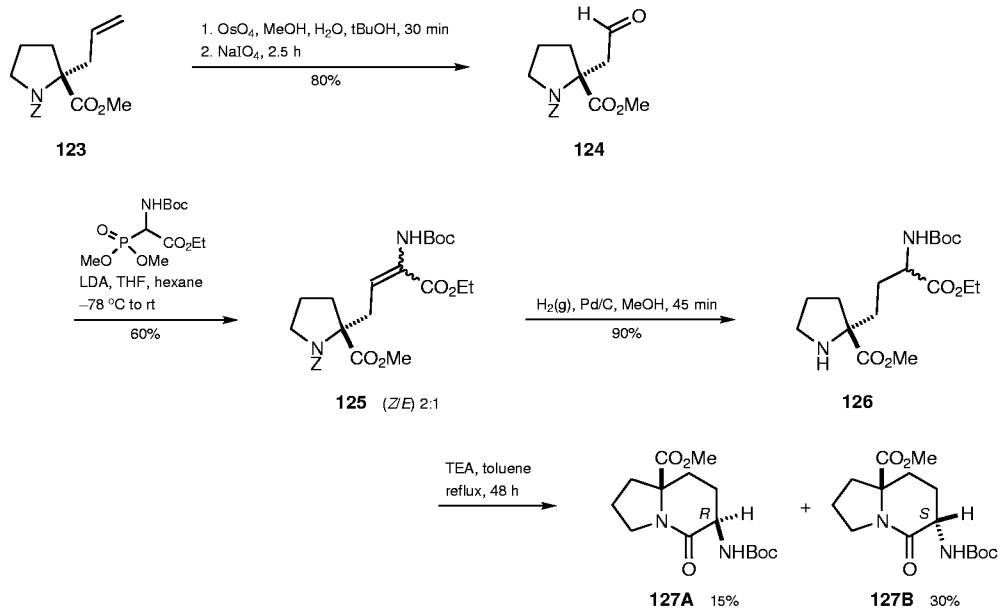



The preparation of the diastereomeric bicyclic lactams **127A** and **127B** (Scheme 46, experimental details given below) has been described and their use as potential mimics of the unusual type-VI (-Xaa-Pro-) β-turn was implied.[163]

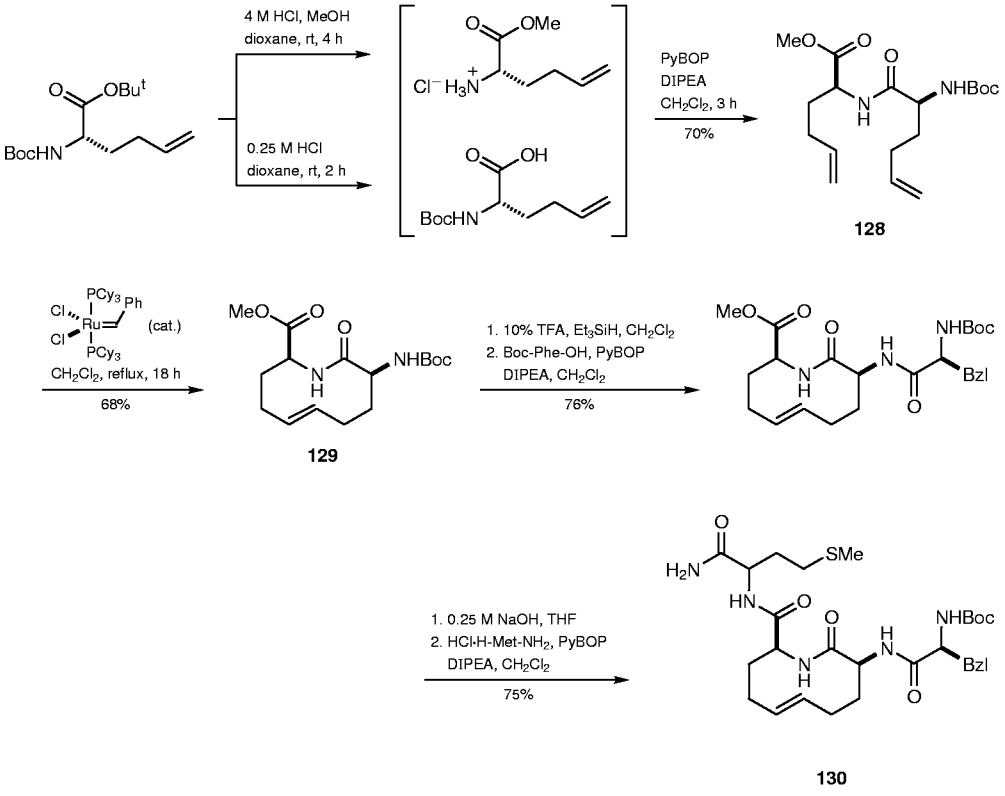
Utilizing the Grubbs' alkene metathesis reaction, Katzenellenbogen et al.[164] described the synthesis of the ten-membered-ring lactam β-turn mimetic **129** (Scheme 47, some experimental details given below) and prepared a mimetic **130** of the neuropeptide substance P, and in particular of the four C-terminal amino acid sequence -Phe[8]-Gly-Leu-Met[11]-. Compound **130** was unable to inhibit the binding of radiolabeled substance P.

*for references see p 736*

**Scheme 46** Preparation of Bicyclic Lactams as Type-VI β-Turn Peptidomimetics[163]



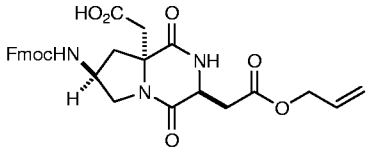

**Scheme 47** Formation of a Lactam β-Turn Peptidomimetic and Its Use in Preparation of the Substance P Mimetic[164]



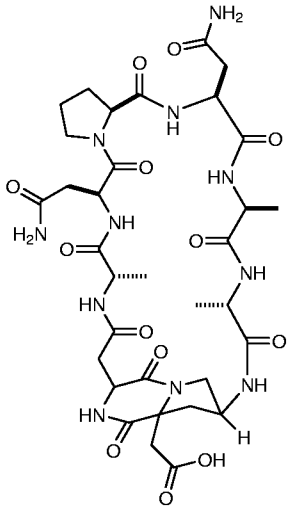

The immunodominant portion of the malaria circumsporozoite surface protein contains a tetrapeptide motif (-Asn-Pro-Asn-Ala-), tandemly repeated almost 40 times. Bisang et al.[165] incorporated the β-turn mimetic **131** into the cyclic template **132** (Scheme 48). BALB/c mice

immunized with a multiple-antigen peptide containing four copies of the template-bound loop conjugated to a single universal T-cell epitope produced antibodies that bound *Plasmodium falciparum* sporozoites in immunofluorescence assays.

**Scheme 48** Bicyclic β-Turn Peptidomimetic and Its Incorporation into a Cyclic Peptide[165]



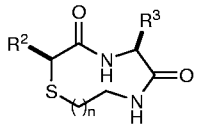
**131**



**132**

A simple and efficient synthesis of a large library of chiral substituted heterocycles of the type **133** (Scheme 49) as β-turn mimetics containing the side chains for $R^2$ and $R^3$ of the parent peptide has been reported.[166]

**Scheme 49** Macrocyclic β-Turn Peptidomimetics[166]



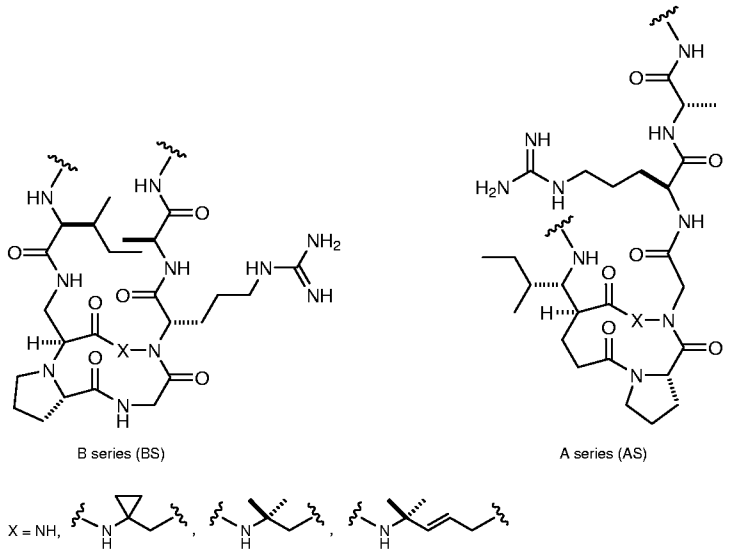
**133**

n = 1, 2

Antibodies to the principal neutralizing determinant in the HIV gp120 V3 loop prevent infection. The -Gly-Pro-Gly-Arg- sequence from residues 312 to 315 is found in 85–90% of HIV isolates and is believed to exist in reverse-turn conformation.[117] To determine the optimum construction of our B-cell epitope, a small library of restricted-turn constructs was screened against the V3 directed MAb 50.1. Based upon our previous analysis, both the B (*before* binding) and A (*after* binding) type constructions (Scheme 50) were incorporated into the constrained B-cell epitope library.

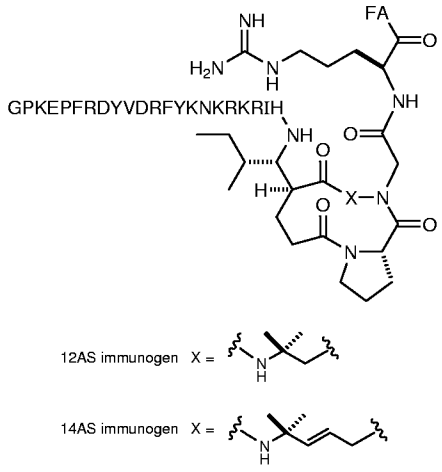
*for references see p 736*

**Scheme 50** HIV gp120 V3 Loop β-Turn Peptidomimetics[117]



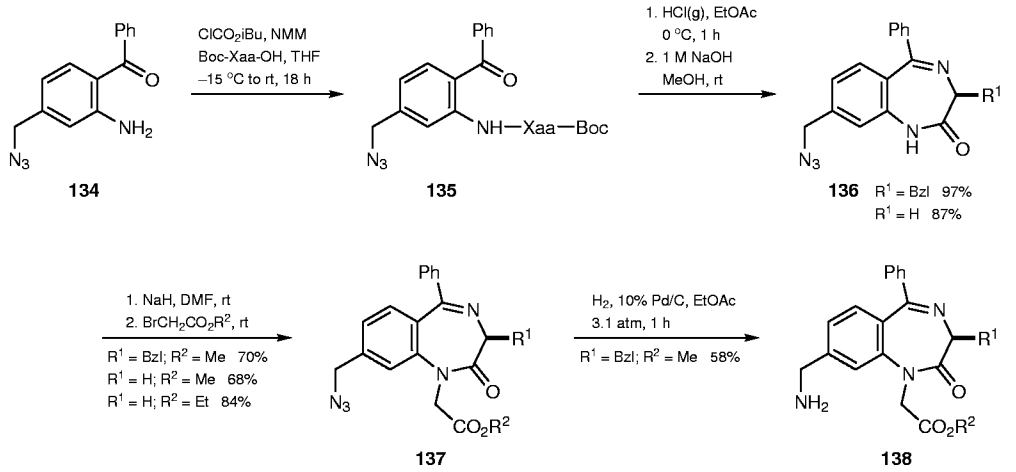

The library was screened using ELISA for the ability to inhibit the binding of gp120 MAb 50.1. From this assay, two constructions of 12AS and 14AS were selected as candidates for incorporation into full-length tandem T–B immunogens. In particular, we synthesized analogues of the linear peptide immunogen CLTB 108 (Connaught Laboratories T cell B cell epitope 108, GPKEPFRDYVDRFYKNKRKRIHIGPGRAF) which incorporated either the 12-membered-ring (12AS) or 14-membered-ring (14AS) reverse-turn templates at the crown of the V3 loop (Scheme 51) and elicited antisera in guinea pigs against these constructions. Gratifyingly, the response was very specific for the native protein epitope in gp120. The 12AS tandem construction elicited exceptionally high antititer reactivity against gp160 in both incomplete Freunds adjuvant and alum. In alum, the response was fourfold higher than the response of the linear CLTB-36.[117]

**Scheme 51** Immunogen CLTB 108 β-Turn Peptidomimetics[117]

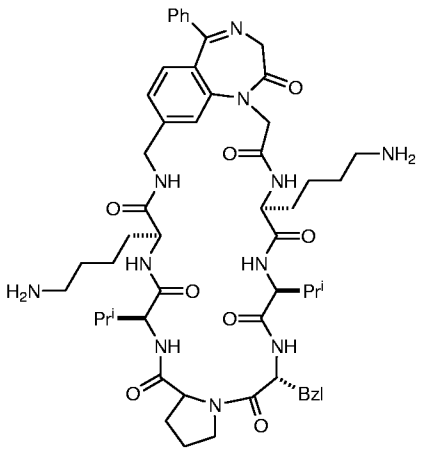



Ripka et al.[167,168] suggested that benzodiazepines of the type **138** (Scheme 52, experimental details given below) could serve as a mimetic of a β-turn. This mimetic was incorporated into a cyclic "octapeptide"-containing analogue of the antibiotic peptide gramicidin S to provide **139** (Scheme 53).

**Scheme 52** Preparation of Benzodiazepines as β-Turn Peptidomimetics[167,168]

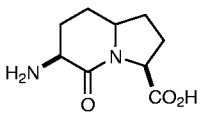



**Scheme 53** Incorporation of a Benzodiazepine into Gramicidin S[167]

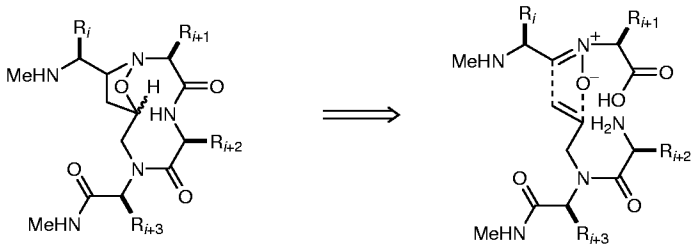



Several groups have described congeners of the thiazabicyclo[4.3.0]nonanone system **18** (Scheme 11) originally synthesized by Nagai and Sato.[76] Lubell and Lombart,[169] as well as the groups of Hanessian,[157] González-Muñiz,[170] Revesz,[171] and Robl,[172] described approaches to the azabicyclo[4.3.0]nonanone system **140** (Scheme 54). Robl[172] also described the synthesis of the [5.4.0] system. Hermkens et al.[173] used an intramolecular [3+2] cycloaddition to construct systems of the type **141** (Scheme 54).

**Scheme 54** Azabicyclic and Oxazabicyclic β-Turn Peptidomimetics[157,169–173]



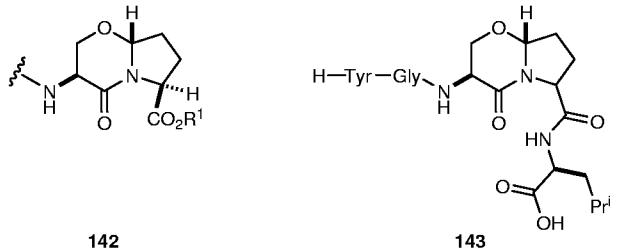

*for references see p 736*



141

Schofield et al.[174] described their approach to the oxazabicyclo[4.3.0]nonanone **142** (Scheme 55), as part of an approach to mimic Leu-enkephalin. Template **142** was incorporated into the pentapeptide sequence, replacing the -Gly-Phe- sequence, to provide **143**. No biological data were reported.

**Scheme 55** Oxazabicyclic β-Turn Peptidomimetic and the Enkephalin Analogue Containing the Peptidomimetic[174]



142 143

**(2*S*)-2-(1,3-Dioxo-1,3-dihydro-2*H*-isoindol-2-yl)-5-oxo-5-(phenylsulfanyl)pentanoic Acid (25, R¹ = H):**[77]
To Phth-L-Glu-OH anhydride (**24**; 5.2 g, 20 mmol) in dioxane (20 mL), PhSH (1.32 g, 12 mmol) and DCHA (1.90 g, 10 mmol) were added under stirring. Stirring was continued for 72 h. Exhaustive removal of the solvent under reduced pressure at 50 °C afforded a viscous, slightly yellow oily residue, which was triturated with $Et_2O$ (50 mL) and kept overnight in a refrigerator. The solid formed was collected by filtration. Drying gave the DCHA salt of the product; yield: 4.7 g (85%); mp 147–156 °C.
The DCHA salt (4.5 g, 8.2 mmol) was treated with 10% citric acid (40 mL) and EtOAc (20 mL), and the soln was stirred vigorously. After separation of the organic layer, the aqueous layer was again extracted with EtOAc (30 mL). The combined organic layers were washed with $H_2O$ (20 mL) and dried ($Na_2SO_4$). Removal of the solvent after removal of the dessicant afforded a viscous oily product, which crystallized slowly on standing; yield: 3.0 g (quantitative); mp 182–191 °C; $[\alpha]_D^{23}$ −36.0 (*c* 1.0, DMF).

**Methyl (2*S*)-2-(1,3-Dioxo-1,3-dihydro-2*H*-isoindol-2-yl)-5-oxo-5-(phenylsulfanyl)pentanoate (25, R¹ = Me):**[77]

**CAUTION:** *Diazomethane is highly toxic and irritating. All operations with diazomethane should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

Compound **25** (R¹ = H; 2.7 g, 7.3 mmol) was suspended in EtOAc (20 mL). Ethereal $CH_2N_2$ soln was added dropwise to the suspension until complete dissolution of solid material occurred and the solution turned slightly yellow. Concentration of the mixture afforded a viscous oil; yield: 2.9 g (quantitative).

**Methyl (2*S*)-2-(1,3-Dioxo-1,3-dihydro-2*H*-isoindol-2-yl)-5-hydroxypentanoate (26):**[77]
Raney Ni (90 g) suspended in EtOH (300 mL) with a small amount of AcOH (0.6 mL) was refluxed on a $H_2O$ bath. Compound **25** (R¹ = Me; 6 g, 16 mmol) dissolved into EtOH (50 mL) was dropped into the refluxing suspension in the course of 20 min. The mixture was monitored by TLC (silica gel, $CHCl_3$/MeOH 99:1), and refluxing was stopped when the starting material had disappeared. After cooling, the Raney Ni was filtered off and washed with EtOH (30 mL), and the combined filtrate and washings were concentrated under reduced pressure. The oily residue was dissolved in $CHCl_3$, passed through a column of Florisil to remove the insoluble material, and then subjected to chromatography (silica gel, 200 g; $CHCl_3$/MeOH). The fractions showing the TLC spot of the desired product ($R_f$ 0.25, silica gel; $CHCl_3$/MeOH 99:1) were collected, and concentrated to give oily product; yield: 3.1 g (70%).

**Methyl (2*S*)-2-(1,3-Dioxo-1,3-dihydro-2*H*-isoindol-2-yl)-5-oxopentanoate (27):**[77]
To a stirred soln of PCC (1.2 g, 5.57 mmol) in $CH_2Cl_2$ (7.5 mL) was added compound **26** (1.0 g, 3.6 mmol) dissolved in $CH_2Cl_2$ (6 mL) under a $N_2$ atmosphere. Stirring was continued for 2 h, while the disappearance of the starting material was monitored by TLC (silica gel, $CHCl_3$/MeOH 98:2). Addition of anhyd $Et_2O$ (30 mL) resulted in the formation of a dark brown precipitate. The supernatant was decanted off, and the sticky precipitate was triturated twice with $Et_2O$. The combined $Et_2O$ layers were passed through a column of Florisil and the column was washed with $Et_2O$. UV absorbing fractions were collected and concentrated to give an oily residue; yield: 0.6 g (60%).

**(2*S*,4*R*)-2-[(3*S*)-3-(1,3-Dioxo-1,3-dihydro-2*H*-isoindol-2-yl)-4-methoxy-4-oxobutyl]thiazolidine-4-carboxylic Acid (28):**[77]
Anhyd NaOAc (500 mg) was added to a soln of HCl·H-L-Cys-OH (425 mg, 2.7 mmol) in degassed $H_2O$ (6 mL) under a $N_2$ atmosphere. Compound **27** (600 mg, 2.18 mmol) dissolved in degassed EtOH (10 mL) was added under stirring. Stirring at rt was continued for 8 h under a $N_2$ atmosphere. About 10 min after starting the reaction, the mixture became turbid. If not turbid, slight warming (ca. 40 °C) is advisable. The mixture was diluted with $H_2O$ (30 mL) and left several h at rt. The precipitate was then filtered, washed with a small amount of $H_2O$, and dried. The dried crude product was powdered with a mortar. A small volume of $CHCl_3$ was added to dissolve the unreacted starting material, and the soln was filtered. The dried product [yield: 500 mg (60%)] was used for the next step without further purification.

**(3*R*,6*S*,8a*S*)-6-(1,3-Dioxo-1,3-dihydro-2*H*-isoindol-2-yl)-5-oxohexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxylic Acid (29):**[77]
The thiazolidine derivative **28** (1.0 g, 2.6 mmol) was dissolved in DMF (20 mL) and heated under a $N_2$ atmosphere in an oil bath at 90 °C. The progress of the reaction was monitored by TLC (silica gel, BuOH/AcOH/$H_2O$ 4:1:2). When the starting material had disappeared, heating was stopped. Evaporation of DMF from the mixture afforded an oily residue, all of which crystallized on seeding of a few crystals. A small volume of EtOAc (5 mL) was added, triturated, and kept in a refrigerator for more than 1 h to allow for crystallization. Crystals were filtered and washed with cold EtOAc and dried to give product **29**. Combined filtrate and washings were extracted with sat. $NaHCO_3$, acidified with 10% citric acid, and extracted with EtOAc. The organic layer was washed with sat. NaCl, and dried ($Na_2SO_4$). Removal of the solvent and recrystallization of the residue afforded further crystals of product **29**; combined yield: 730 mg (80%); mp 266–268 °C (dec); $[\alpha]_D^{24}$ −249 (*c* 1.0, DMF); IR (KBr) $\tilde{\nu}_{max}$: 3280, 1780, 1760, 1720, 1650 $cm^{-1}$.

**(3*R*,6*S*,8a*S*)-6-Amino-5-oxohexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxylic Acid (30):**[77]

**CAUTION:** *Hydrazine hydrate is a poison upon ingestion and by intravenous routes. It is an experimental carcinogen and is a corrosive irritant to the eyes, skin, and mucous membranes. Appropriate safety precautions and procedures should be taken during all stages of its handling and disposal.*

Compound **29** (300 mg, 0.87 mmol) was dissolved in EtOH (4 mL) with slight heating. A soln of $H_2NNH_2 \cdot H_2O$ (210 µL, 4.3 mmol) in EtOH (8 mL) was then added. The mixture was refluxed for 1 h. It was concentrated under reduced pressure, $H_2O$ (5 mL) was added to the residue, the pH was adjusted to 4.0, and the whole was kept in a refrigerator for 30 min. After removal of the precipitate, the filtrate was passed through a column of Dowex-I ($OH^-$ form) and the column was washed with $H_2O$ until the pH was neutral. The column was eluted with 1 M AcOH and ninhydrin positive fractions were collected and concentrated; yield: 85%; mp >280 °C; $[\alpha]_D^{25}$ −273.8 (*c* 1.0, $H_2O$).

**(3*R*,6*S*,8a*S*)-6-[(*tert*-Butoxycarbonyl)amino]-5-oxohexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxylic Acid (31):**[77]
Amine **30** (108 mg, 0.50 mmol) was dissolved in 0.2 M NaOH (2.5 mL). $Boc_2O$ (120 mg, 0.55 mmol), dissolved in dioxane (2.5 mL), was added to the soln in an ice bath. The mixture was stirred at rt for 2 h. $H_2O$ (15 mL) was added to the mixture, which was then acidified with 10% citric acid and extracted with EtOAc (50 mL). The organic layer was washed with 10% citric acid and sat. NaCl, and dried ($Na_2SO_4$). The soln was filtered and concentrated, affording an oily residue, which was then triturated with an EtOAc/pentane mixture in a strongly cooled bath of dry ice/EtOH. By pipetting off the supernatant, the product **31** was obtained; yield: 134 mg (85%); mp 151–157 °C; $[\alpha]_D^{24}$ −214 (*c* 1.0, MeOH).

**(4*R*)-4-(*tert*-Butyldimethylsiloxymethyl)-1-(*tert*-butyldimethylsilyl)azetidin-2-one (51):**[100]
To a mixture of HCl·H-D-Asp(OMe)-OMe (**50**; 2.00 g, 10.1 mmol), TBDMSCl (1.68 g, 11.1 mmol), and DMAP (62 mg, 0.51 mmol) in $CH_2Cl_2$ (50 mL) TEA (3.24 mL, 23.3 mmol) was added slowly at rt and the mixture was allowed to stir overnight. The mixture was washed with aq $NH_4Cl$, sat. $Na_2CO_3$, and brine, then dried ($Na_2SO_4$) and concentrated under reduced pressure.

*for references see p 736*

The residue (2.75 g) was dissolved in $Et_2O$ (50 mL). This soln was cooled to 0 °C and 2.0 M tBuMgCl in $Et_2O$ (5.24 mL, 10.5 mmol) was added dropwise. The mixture was allowed to warm to rt overnight with stirring. The reaction was cooled to 0 °C and sat. $NH_4Cl$ was added. The mixture was stirred for 30 min. The mixture was added to $H_2O$ and the organic layer was separated. The aqueous layer was extracted with $Et_2O$ (2 × 30 mL). The combined organic layers were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure.
The residue (1.47 g) was dissolved in MeOH (60 mL). This soln was added with stirring at rt to $NaBH_4$ (1.14 g, 30.1 mmol) in a flask equipped with a reflux condenser. The mixture started to reflux due to the exothermic reaction during the addition and ceased in ca. 20 min. After 45 min in total, the mixture was cooled to 0 °C and aq $NH_4Cl$ was added. The mixture was extracted with $CH_2Cl_2$ (3 × 50 mL). The combined organic extracts were dried ($Na_2SO_4$) and the volatiles were removed under reduced pressure. The residue (1.30 g) was dissolved in $CH_2Cl_2$ (30 mL). To this soln TBDMSCl (1.00 g, 6.63 mmol) and DMAP (37 mg, 0.30 mmol) were added. TEA (1.10 mL, 7.87 mmol) was added slowly and the mixture was allowed to stir overnight at rt. The mixture was washed with aq $NH_4Cl$ and brine, dried ($Na_2SO_4$), and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 9:1) provided **51** as a colorless liquid; yield: 1.01 g (30%, overall from **50**); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1732 $cm^{-1}$.

**(3*S*,4*R*)-3-(But-3-enyl)-4-(*tert*-butyldimethylsiloxymethyl)-1-(*tert*-butyldimethylsilyl)azetidin-2-one (52):**[100]
To a soln of $iPr_2NH$ (0.62 mL, 4.41 mmol) in THF (20 mL) was added 2.5 M BuLi in hexane (1.77 mL, 4.42 mmol) at 0 °C dropwise. The mixture was cooled to −78 °C and azetidinone **51** (970 mg, 2.94 mmol) in THF (10 mL) was added slowly. After 30 min of stirring, 4-bromobut-1-ene (0.90 mL, 8.83 mmol) was added and the mixture was allowed to stir for 3 h at −78 °C. Sat. $NH_4Cl$ was added and the mixture was warmed to rt. $H_2O$ was added and the mixture was extracted with $Et_2O$ (2 × 40 mL). The combined organic extracts were washed with brine, dried ($Na_2SO_4$), and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 19:1) provided **52** as a colorless liquid; yield: 632 mg (56%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1730 $cm^{-1}$.

**(3*S*,4*R*)-3-(But-3-enyl)-4-(*tert*-butyldimethylsiloxymethyl)azetidin-2-one (53):**[100]
To a soln of **52** (600 mg, 1.56 mmol) in MeOH (16 mL) cooled in an ice bath, $KF \cdot 2H_2O$ (162 mg, 1.72 mmol) was added. The mixture was allowed to stir at 0 °C for 1 h, then AcOH (91 μL, 1.59 mmol) was added. The volatiles were removed under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 4:1) provided azetidinone **53** as a colorless liquid; yield: 409 mg (97%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3460, 1756 $cm^{-1}$.

**(3*S*,4*R*)-1-Benzyl-3-(but-3-enyl)-4-(*tert*-butyldimethylsiloxymethyl)azetidin-2-one (54):**[100]
In a round-bottomed flask was placed an 80% dispersion of NaH in mineral oil (60 mg, 2.00 mmol). Then, the NaH was washed successively with hexane and THF, and subsequently dispersed in THF (4 mL). To this mixture was added azetidinone **53** (360 mg, 1.34 mmol) in THF (10 mL) slowly at rt. After 10 min, BzlBr (0.17 mL, 1.47 mmol) was added dropwise and the mixture was allowed to stir for 3 h at rt. The mixture was cooled to 0 °C and aq $NH_4Cl$ was added. The organic layer was separated and the aqueous layer was extracted with $Et_2O$ (2 × 20 mL). The combined organic layers were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 9:1) provided **54** as a colorless liquid; yield: 283 mg (59%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1740 $cm^{-1}$.

**Methyl (2*R*,3*S*)-1-Benzyl-2-(*tert*-butyldimethylsiloxymethyl)-4-oxoazetidine-3-propanoate (56):**[100]

**CAUTION:** *Diazomethane is highly toxic and irritating. All operations with diazomethane should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

To a soln of **54** (180 mg, 0.50 mmol) in a mixture of $CCl_4$ (1 mL) and MeCN (1 mL) were added $H_2O$ (1.5 mL) and $NaIO_4$ (439 mg, 2.05 mmol). To this biphasic soln was added $RuCl_3$ (21 mg, 0.10 mmol). The mixture was allowed to stir overnight at rt and then taken up in EtOAc (10 mL) and $H_2O$ (5 mL). The organic layer was separated. The aqueous layer was saturated with NaCl and extracted with EtOAc (2 × 10 mL). The combined organic layers were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure. The residue **55** was dissolved in $Et_2O$ (4 mL). To this soln was added an ethereal soln of $CH_2N_2$ until the mixture remained a pale yellow. The volatiles were removed under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 9:1) provided **56** as a colorless liquid; yield: 135 mg (69%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1740 $cm^{-1}$.

**(4*RS*)-4-[(1*R*)-1-(Benzylamino)-2-(*tert*-butyldimethylsiloxy)ethyl]-1,2-diazepane-3,7-dione (57):**[100]

**CAUTION:** *Cyanide salts can be absorbed through the skin and are extremely toxic. Appropriate safety precautions and first aid procedures should be adopted when handling cyanide salts.*

**CAUTION:** *Hydrazine is a highly toxic and potentially explosive compound. All operations with hydrazine should be conducted in a well-ventilated hood and behind a safety shield.*

To a soln of methyl ester **56** (130 mg, 0.33 mmol) in MeOH (5 mL) were added NaCN (16 mg, 0.33 mmol) and anhyd $H_2NNH_2$ (1.05 mL, 32.8 mmol). The mixture was allowed to reflux overnight with stirring. The volatiles were removed under reduced pressure and the residue was partitioned between $CH_2Cl_2$ and brine. The organic layer was dried ($Na_2SO_4$) and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, $CHCl_3$/MeOH 39:1) provided **57** as a ca. 6:1 mixture of the 4*S*/4*R*-diastereomers (inseparable by TLC) as a viscous yellow oil; yield: 70 mg (54%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3380, 1786, 1726, 1626 $cm^{-1}$; MS (*m/z*): calcd, 391.6; found, 391.4.

**(4*R*)-3-[(2*E*,4*E*)-Hexa-2,4-dienoyl]-4-isopropyloxazolidin-2-one (58):**[100]

To a soln of (2*E*,4*E*)-hexa-2,4-dienoic acid (2.0 g, 17.8 mmol) in $CH_2Cl_2$ (40 ml) was added $(COCl)_2$ (3.11 ml, 35.7 mmol) dropwise at rt, and the mixture allowed to reflux with stirring for 1.5 h. The majority of the volatiles were then removed at atmospheric pressure and the remainder under reduced pressure, and the residue dissolved in THF (25 mL). In a separate flask, (4*R*)-4-isopropyloxazolid-2-one (17.8 mmol) was dissolved THF (60 mL), cooled to 0 °C, and 2.5 M BuLi in hexane (7.11 mL, 17.8 mmol) was added slowly. The mixture was stirred at 0 °C for 30 min and was then further cooled to −78 °C. The soln of acid chloride in THF was transferred to this flask slowly and the mixture was allowed to warm to rt overnight with stirring. The mixture was added to aq. $NH_4Cl$ and the organic layer separated. The aqueous layer was extracted with $Et_2O$ (2×50 mL) and the combined organic layers were washed with 5% $NaHCO_3$ and brine, dried ($MgSO_4$), and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 17:3) provided *N*-acyloxazolidin-2-one **58** as a clear oil; yield: 82%; IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1770, 1678, 1632 $cm^{-1}$; MS (*m/z*): calcd, 223.3; found, 223.

**(4*R*)-3-[(3*E*)-Hexa-3,5-dienoyl]-4-isopropyloxazolidin-2-one (59):**[100]

To a soln of $iPr_2NH$ (0.54 mL, 3.86 mmol) in THF (8 mL) was added 2.5 M BuLi in hexane (1.54 mL, 3.85 mmol) dropwise at 0 °C. The mixture was cooled to −78 °C. DMPU (1.4 mL, 11.58 mmol) was added and the mixture was allowed to stir for 30 min at −78 °C. *N*-Acyloxazolidinone **58** (287 mg, 1.29 mmol) in THF (5 mL) was added slowly and the mixture was allowed to stir at −78 °C. After 1 h, the mixture was poured into vigorously agitated aq AcOH at 0 °C. The organic layer was separated and the aqueous layer was extracted with $Et_2O$ (2 × 20 mL). The combined organic layers were washed with 5% $NaHCO_3$ and brine, dried ($MgSO_4$), and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 17:3) provided the deconjugated *N*-acyloxazolidinone **59** as a clear oil; yield: 212 mg (74%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1780, 1698 $cm^{-1}$; MS (*m/z*): calcd, 223.3; found, 224.

**(4*R*)-4-Isopropyl-3-[(2*R*,3*E*)-2-{[1-(phenylsulfonyl)-1*H*-indol-3-yl]methyl}hexa-3,5-dienoyl]oxazolidin-2-one (61):**[100]

To a soln of $iPr_2NH$ (0.14 mL, 1.02 mmol) in THF (4 mL) was added 2.5 M BuLi in hexane (0.41 mL, 1.02 mmol) dropwise at 0 °C. The mixture was cooled to −78 °C and *N*-acyloxazolidinone **59** (208 mg, 0.93 mmol) in THF (4 mL) was added. The mixture was allowed to stir at −78 °C for 1 h, at which time iodide **60** (378 mg, 0.95 mmol) in THF (2.5 mL) was added. The dry ice/acetone bath was replaced with an ice bath and, after 15 min, sat. $NH_4Cl$ was added. The mixture was warmed to rt and added to $H_2O$. The organic layer was separated and the aqueous layer was extracted with $Et_2O$ (2 × 15 mL). The combined organic layers were washed with brine, dried ($MgSO_4$), and concentrated under reduced pressure. Flash chromatography of the residue (silica gel, hexane/EtOAc 17:3) provided compound **61** as a viscous clear oil [yield: 236 mg (51%); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1780, 1698, 1650 $cm^{-1}$; MS (*m/z*): calcd, 492.6; found, 492], along with the (2′*S*)-diastereomer [yield: 14 mg (3%)].

**(2*R*,3*E*)-2-{[1-(Phenylsulfonyl)-1*H*-indol-3-yl]methyl}hexa-3,5-dienoic Acid (62):**[100]

Imide **61** (800 mg, 1.62 mmol) was dissolved in THF (20 mL) and $H_2O$ (0.6 mL) was added. The mixture was cooled to 0 °C and 1 M LiOH (3.6 mL, 3.6 mmol) and 30% $H_2O_2$ (0.36 mL, 3.7 mmol) were added. After the mixure was stirred for 20 min at 0 °C, 1.5 M $Na_2SO_3$ (0.3 mL) was added. The diastereomeric mixture of **62** was added to $H_2O$ and washed with $CH_2Cl_2$. The separated layer was acidified with 10% HCl and extracted with $CH_2Cl_2$. The organic extracts were washed with brine, dried ($Na_2SO_4$), and concentrated under reduced pressure to provide a mixture of diene acids **62** and (4*R*)-4-isopropyloxazolidin-2-one in a ca. 6:1 ratio; yield: not reported.

*for references see p 736*

**(2*R*,3*E*)-*N*-Benzyl-*N*-{(1*R*)-2-(*tert*-butyldimethylsiloxy)-1-[(4*S*)-3,7-dioxo-1,2-diazepan-4-yl]ethyl}-2-{[1-(phenylsulfonyl)-1*H*-indol-3-yl]methyl}hexa-3,5-dienamide (63):**[100]

Coupling of diene acid **62** with diacylhydrazide **57** (500 mg, 1.28 mmol) was carried out in $CH_2Cl_2$ (10 mL) to which were also added at rt 2,2′-dipyridyl disulfide (313 mg, 1.42 mmol) and $Ph_3P$ (373 mg, 1.42 mmol). The mixture was allowed to reflux for 7 h with stirring. The volatiles were then removed under reduced pressure, and the remainder was subjected to flash chromatography (hexane/EtOAc 3:2, then hexane/EtOAc 1:1); yield of major diastereomer (lower $R_f$): 53%; yellowish white solid; mp 105–110 °C; yield of minor diastereomer (higher $R_f$): 9%.

**Ethyl *N*-{(2*R*)-2-{(1*R*,9*S*)-9-[(1*R*)-1-(Benzylamino)-2-(*tert*-butyldimethylsiloxy)ethyl]-6,10-dioxo-1,4,7,8,9,10-hexahydro-6*H*-pyridazino[1,2-*a*][1,2]diazepin-1-yl}-3-[1-(phenylsulfonyl)-1*H*-indol-3-yl]-propionyl}glycinate (65):**[100]

To a soln of dienamide **63** (490 mg, 0.65 mmol) in $CDCl_3$ (16 mL) was added $PhI(OAc)_2$ (274 mg, 0.85 mmol) at rt. The mixture was allowed to reflux for 3 h with stirring to provide **64**. After cooling, HCl·H-Gly-OEt (457 mg, 3.27 mmol) and TEA (0.91 mL, 6.55 mmol) were added. The mixture was allowed to reflux for 5 h with stirring. The volatiles were removed under reduced pressure to provide crude **65**, which was subjected to flash chromatography (hexane/EtOAc 1:1) to give a yellow solid; yield: 36%; mp 100–105 °C; IR ($CHCl_3$) $\tilde{\nu}_{max}$: 1780, 1740, 1674, 1630 $cm^{-1}$.

**Methyl (3*S*,6*R*,8a*R*)-6-[(Benzyloxycarbonyl)amino]-5-oxohexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxylate (76):**[106]

**CAUTION:** *Diazomethane is highly toxic and irritating. All operations with diazomethane should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

Oxazolidin-5-one derivative **75** (0.41 g, 1.5 mmol) was dissolved in EtOH (8 mL) and treated with a soln of HCl·H-D-Cys-OH (0.39 g, 2.2 mmol) in $H_2O$ (5 mL). This mixture was neutralized ($NaHCO_3$) and then stirred overnight at rt under argon. EtOH and $H_2O$ were removed under reduced pressure and the resulting solid was dispersed in dry DMF. This mixture was heated to 42 °C under an argon atmosphere. The mixture was kept at this temperature for 24 h after which time the DMF soln was filtered and the solvent was removed from the filtrate under high vacuum. The residue was dissolved in MeOH and treated with an $Et_2O$ soln of $CH_2N_2$. The obtained product was chromatographed (silica gel column, hexane/EtOAc 3:2) to give **76** as an oil; yield: 0.19 g (35%); $[\alpha]_D$ +140.6 (*c* 0.67, MeOH); FABMS *m/z*: $[M+H]^+$ 365.

**(3*S*,6*R*,8a*R*)-6-[(Benzyloxycarbonyl)amino]-5-oxohexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxamide (77):**[106]

Protected methyl ester **76** (0.14 g, 0.38 mmol) was dissolved in a soln of MeOH saturated with ammonia. This mixture was stirred for 3 h at rt, at which time the MeOH was removed under reduced pressure. The resulting residue was chromatographed (silica gel column, EtOAc/hexane 5:2) to give **77** as a white solid; yield: 0.1 g (76%); mp 162–163 °C; $[\alpha]_D$ +143.4 (*c* 1.0, MeOH); FABMS *m/z*: $[M+H]^+$ 350.

**(3*S*,6*R*,8a*R*)-6-[(2*S*)-1-(*tert*-Butoxycarbonyl)prolylamino]-5-oxohexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxamide (78):**[106]

Protected amide **77** (90 mg, 0.26 mmol) was suspended in $Et_2O$ (10 mL). This suspension was treated with HBr in AcOH (4 mL). The reaction was stirred for 45 min at rt under argon. Excess HBr and AcOH were removed under reduced pressure, with the last traces of AcOH being removed from the residue by azeotropic distillation with toluene (3 ×). The resulting ammonium bromide salt was obtained as a colored solid after being dried under high vacuum overnight. This material was dissolved in dry DMF (10 mL). NMM (33 μL, 0.3 mmol) was added to the soln, and the resulting mixture was stirred for 10 min under argon. Boc-L-Pro-OPfp (100 mg, 0.26 mmol) was added to the mixture, which then was stirred at rt for 22 h. The DMF was removed under reduced pressure, and the resulting residue was dissolved in EtOAc. The EtOAc soln was washed with 10% citric acid and sat. NaCl, and dried ($MgSO_4$). The EtOAc was removed under reduced pressure, and the residue was chromatographed (silica gel column, MeOH/EtOAc 2:98) to give **78** an oil; yield: 43 mg (40%); $[\alpha]_D$ +35.5 (*c* 0.76, MeOH); FAB-MS *m/z*: $[M+H]^+$ 413.

**(3*S*,6*R*,8a*R*)-5-Oxo-6-[(2*S*)-prolylamino]hexahydro-5*H*-thiazolo[3,2-*a*]pyridine-3-carboxamide Hydrochloride (74A·HCl):**[106]

Amide **78** (20 mg, 0.05 mmol) was treated with 4 M HCl in dioxane under argon for 20 min, at which time the dioxane soln was frozen and lyophilized to give the HCl salt of **74A** as a highly hygroscopic, white solid; yield: 16 mg (92%); $[\alpha]_D$ +35.5 (*c* 0.7, MeOH); HRMS–FAB (*m/z*): calcd for $C_{13}H_{21}N_4O_3S+H^+$, 313.1334; found, 313.1337. The product was homogeneous by HPLC analysis (C18, 1% aq HCl, 3 mL·$min^{-1}$, $t_R$ = 42.4 min).

**Methyl (2*R*)-1-(*tert*-Butoxycarbonyl)-2-(2-oxoethyl)pyrrolidine-2-carboxylate (83):**[111]

**CAUTION:** *Osmium tetroxide is an inhalation hazard and can cause temporary blindness. Handle with extreme caution!*

The fully protected derivative **82** (390 mg, 1.45 mmol) was dissolved in THF/$H_2O$ (4:1, 25 mL). This soln was stirred under $N_2$, and $OsO_4$ (20 mg, 0.079 mmol) was added. The reaction turned dark brown. After 5 min, $NaIO_4$ (780 mg, 3.65 mmol) was added in three batches over a 10-min period. The reaction turned light yellow, and stirring of the mixture was continued for 5 h. The reaction was diluted with $Et_2O$ (20 mL) and $H_2O$ (10 mL). The aqueous layer was extracted with $Et_2O$ (3 × 25 mL). The $Et_2O$ layers were combined, washed with $H_2O$, and dried ($MgSO_4$). Removal of solvent under reduced pressure gave a tan oil which turned black on standing. This material was chromatographed (1 × 45 cm silica gel flash column, EtOAc/hexane 1:3); aldehyde **83** was isolated as a colorless oil; yield: 360 mg (92%); $[\alpha]_D$ +35.4 (*c* 1.6, $CHCl_3$); FAB-MS *m/z*: $[M+H]^+$ 272.

**(2*RS*,4*S*)-2-[(2*R*)-1-(*tert*-Butoxycarbonyl)-2-(methoxycarbonyl)pyrrolidin-2-ylmethyl]thiazolidine-4-carboxylic Acid (84):**[111]

HCl·H-D-Cys-OH·$H_2O$ (123 mg, 0.70 mmol) was dissolved in $H_2O$ (2 mL). NaOH (1 equiv) in $H_2O$ (1 mL) was added followed by a soln of **83** (200 mg, 0.7 mmol) in EtOH (3 mL). The reaction was stirred overnight at rt, then diluted with $H_2O$ (5 mL) and extracted with EtOAc (3 × 25 mL). The organic layers were combined and dried ($MgSO_4$). The solvent was removed under reduced pressure to give pure diastereomeric mixture **84**; yield: 87%; mp 88–90 °C; FAB-MS *m/z*: $[M+H]^+$ 375.

**Methyl (2*R*,3′*S*,7a′*R*)-1-(*tert*-Butoxycarbonyl)-5′-oxotetrahydrospiro{pyrrolidine-2,6′-pyrrolo-[2,1-*b*]thiazole}-3′-carboxylate (85):**[111]

**CAUTION:** *Diazomethane is highly toxic and irritating. All operations with diazomethane should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

The thiazolidine mixture **84** (0.94 g, 2.51 mmol) was dissolved in dry DMF (100 mL) under $N_2$. TEA (0.35 mL, 2.51 mmol) was added, and the soln was heated to 70 °C with stirring under $N_2$ for 3 d. The soln turned a golden yellow color. The solvent was removed under reduced pressure to give a dark yellow oil which was dissolved in $Et_2O$ (25 mL). An excess of $CH_2N_2/Et_2O$ soln was added until the evolution of gas ceased. Removal of solvent under reduced pressure gave a yellow oil which was dissolved in EtOAc (100 mL), washed with 10% citric acid and sat. NaCl soln, and dried ($MgSO_4$). The solvent was removed under reduced pressure to give a yellow oil which was chromatographed (2.5 × 45 cm silica gel flash column, $Et_2O$/hexane 1:1). The product **85** was isolated as a colorless oil; yield: 380 mg (43%); $[\alpha]_D$ +115.2 (*c* 1.8, $CHCl_3$); FAB-MS *m/z*: $[M+H]^+$ 357.

**(2*R*,3′*S*,7a′*R*)-1-(*tert*-Butoxycarbonyl)-*N*-methyl-5′-oxotetrahydrospiro[pyrrolidine-2,6′-pyrrolo-[2,1-*b*]thiazole]-3′-carboxamide (86):**[111]

Spirobicyclic ester **85** (90 mg, 0.25 mmol) was dissolved in a sat. soln of $MeNH_2$ in MeOH (10 mL). The soln was stirred for 30 min at rt, then the solvent and excess $MeNH_2$ were removed under reduced pressure to give a light yellow residue. This material was chromatographed (1.5 × 45 cm silica gel flash column, EtOAc/hexane 3:1). The *N*-methylamide **86** was isolated as an oil; yield: 75 mg (84%); $[\alpha]_D$ +158.1 (*c* 1.0, MeOH); FAB-MS *m/z*: $[M+H]^+$ 356.

**(2*R*,3′*S*,7a′*R*)-1-Acetyl-*N*-methyl-5′-oxotetrahydrospiro[pyrrolidine-2,6′-pyrrolo[2,1-*b*]thiazole]-3′-carboxamide (87):**[111]

*N*-Methylamide **86** (40 mg, 0.11 mmol) was $N^{\alpha}$-deprotected in 4 M HCl in dioxane (5 mL) at rt for 1 h. The solvent was removed under reduced pressure. The residue was taken up in $CH_2Cl_2$, whereupon the solvent was again removed under reduced pressure. The residue was dried under reduced pressure overnight. It was then dissolved in $Ac_2O$ (10 mL) and cooled in an ice bath, whereupon TEA (1 equiv) and DMAP (cat.) were added. The soln was stirred at rt overnight. The solvent was removed under reduced pressure, and the residue was chromatographed (1.5 × 45 cm silica gel flash column, $CH_2Cl_2$/MeOH). The product **87** was isolated as an oil; yield: not reported; $[\alpha]_D$ +139.2 (MeOH); FAB-MS *m/z*: $[M+H]^+$ 298.

***N*-(3-{(2*S*)-2-[4-(*tert*-Butyldimethylsiloxy)benzyl]-1-(*tert*-butyldimethylsilyl)-4-oxoazetidin-3-yl}-propionyl)glycine (105):**[154]

Azetidin-2-one **104** (202 mg, 0.423 mmol) was dissolved in freshly distilled THF (3 mL) under argon and cooled to 0 °C. NMM (50 μL, 0.444 mmol) and isobutyl chloroformate (58 μL, 0.444 mmol) were added sequentially. The reaction was stirred for 15 min at 0 °C. To the resulting soln were added HCl·H-Gly-OBzl (90 mg, 0.444 mmol) and additional NMM (50 μL, 0.444 mmol). The reaction was warmed to rt over the course of 3 h. The mixture was diluted to 100 mL with EtOAc, washed with sat. aq $NH_4Cl$

*for references see p 736*

(20 mL), sat. aq $NaHCO_3$ (20 mL), and brine (20 mL), and dried ($Na_2SO_4$). The volatiles were removed under reduced pressure, and the residue was chromatographed (EtOAc/hexanes 40:60) to provide a clear oil (250 mg). The residue was dissolved in EtOH (3 mL) and stirred with a catalytic amount of 10% Pd/C for 12 h under 1 atm of $H_2$(g). The solids were removed by vacuum filtration and the volatiles were removed under reduced pressure to give **105**; yield: 170 mg (75% from **104**).

**Methyl *N*-[(2*S*)-2-{2-(*tert*-Butoxycarbonyl)-1-[(3-{(2*S*)-2-[4-(*tert*-butyldimethylsiloxy)benzyl]-1-(*tert*-butyldimethylsilyl)-4-oxoazetidin-3-yl}propionyl)aminoacetyl]hydrazino}-3-phenylpropionyl]-L-leucinate (107):**[154]

**CAUTION:** *Benzene is carcinogenic!*

Azetidinone adduct **105** (62 mg, 0.116 mmol) was azeotroped with benzene (3 × 4 mL), dissolved in THF/$Et_2O$ (2:8, 1 mL), and cooled to 0 °C under argon. Oxalyl chloride (11 μL, 0.127 mmol) was added, followed by DMF (3 μL, 0.039 mmol). The reaction was stirred for 2 h at 0 °C and 45 min at rt. Volatiles were removed under reduced pressure and the residue was dissolved in freshly distilled benzene (1 mL). Hydrazide **106** (61 mg, 0.151 mmol) was added to the soln, followed by AgCN (31 mg, 0.232 mmol). The reaction was stirred at 50 °C for 12 h, diluted to 10 mL with EtOAc, and filtered through a pad of Celite. The volatiles were removed under reduced pressure and the residue was chromatographed (EtOAc/hexane 1:1) to give the mimetic precursor **107**; yield: 52 mg (49%); $R_f$ 0.4 (EtOAc/hexane 6:4).

**Methyl *N*-[(2*S*)-2-{(9*S*)-9-[(1*S*)-1-Amino-2-(4-hydroxyphenyl)ethyl]-3,6,10-trioxo-1,2,5-triazadecan-2-yl}-3-phenylpropionyl]-L-leucinate (100):**[154]

**CAUTION:** *Trifluoroacetic acid is corrosive and severely irritating to skin, eyes, and mucous membranes. Inhalation can destroy tissue and be fatal. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

Azetidinone mimetic precursor **107** (13 mg, 0.014 mmol) was dissolved in freshly distilled $CH_2Cl_2$ (2.5 mL) and cooled to −10 °C. Anhyd TFA (500 μL) was added and the reaction was stirred at −10 °C for 3 h. The volatiles were removed under reduced pressure, and the residue was dried under high vacuum for 30 min, and subsequently dissolved in $CH_2Cl_2$ (2.5 mL). The reaction was cooled to 0 °C and TEA (200 μL) was added. The soln was stirred at 0 °C for 1 h. Volatiles were removed under reduced pressure and the residue was dissolved in freshly distilled THF (2 mL). The soln was cooled to 0 °C and TBAF (50 mg) was added. The reaction was stirred to rt over the course of 2 h. The volatiles were removed under reduced pressure and the residue was chromatographed (MeOH/$CH_2Cl_2$ 6:94) to give the 10-membered-ring 4→1 β-turn mimetic **100**; yield: 8 mg (94%); FAB-MS *m/z*: 596.

**Methyl (3*R*,5*S*)-6-{[*N*-(*tert*-Butoxycarbonyl)-L-alanylglycyl]amino}-3,5-dimethylhexanoate (117, $R^1 = R^3 = Me$; $R^2 = R^4 = H$; $Xaa^1$ = Ala; $Xaa^2$ = Gly); Typical Procedure:**[159]

A soln of lactam **116**[175] ($R^1 = R^3 = Me$; $R^2 = R^4 = H$; 200 mg, 1.42 mmol) in concd HCl (2 mL) and $H_2O$ (6 mL) was maintained at gentle reflux for 5 h. The solvent was removed under reduced pressure to afford crude 3,5-dimethylaminohexanoic acid (not characterized). $SOCl_2$ (1 mL) was added to MeOH (4 mL) at 0 °C and stirred for 20 min. To this soln at 0 °C was added the crude aminohexanoic acid in MeOH (4 mL), and the mixture was stirred overnight at rt. The solvent was then removed under reduced pressure to afford methyl (3*R*,5*S*)-6-amino-3,5-dimethylhexanoate hydrochloride.

The crude methyl ester, Boc-L-Ala-Gly-OH (420 mg, 1.7 mmol), and diethyl cyanophosphonate (0.23 mL, 1.56 mmol) were dissolved in DMF (5 mL). To this soln was added TEA (0.45 mL, 3.12 mmol), and then the mixture was stirred for 5 h. EtOAc (200 mL) was added and the soln was washed with sat. $NaHCO_3$ (25 mL), 10% aq citric acid (25 mL), and $H_2O$ (25 mL). The organic layer was dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by flash column chromatography (EtOAc) to give **117** ($R^1 = R^3 = Me$; $R^2 = R^4 = H$; $Xaa^1$ = Ala; $Xaa^2$ = Gly) as white crystals; yield: 444 mg (75%); mp 104 °C; $[\alpha]_D$ +12.8 (*c* 0.125, $CH_2Cl_2$); IR (KBr) $\tilde{\nu}_{max}$: 3325, 3270, 1725, 1680, 1620 $cm^{-1}$.

**(3*R*,5*S*)-6-{[*N*-(*tert*-Butoxycarbonyl)-L-alanylglycyl]amino}-3,5-dimethylhexanoic Acid (118, $R^1 = R^3 = Me$; $R^2 = R^4 = H$; $Xaa^1$ = Ala; $Xaa^2$ = Gly); Typical Procedure:**[159]

0.5 M NaOH (3 mL) was added to a soln of ester **117** ($R^1 = R^3 = Me$; $R^2 = R^4 = H$; $Xaa^1$ = Ala; $Xaa^2$ = Gly; 429 mg, 1.07 mmol) in dioxane (3 mL) at rt. After stirring for 3 h at rt, the mixture was carefully acidified with 2 M HCl to pH 2 at 0 °C. The soln was extracted with EtOAc and dried ($MgSO_4$). After filtration, the solvent was removed under reduced pressure. The residue was purified by flash column chromatography ($CH_2Cl_2$/MeOH 20:1) to give **118** ($R^1 = R^3 = Me$; $R^2 = R^4 = H$; $Xaa^1$ = Ala; $Xaa^2$ = Gly) as white crystals; yield: 251 mg (60%).

**(6*S*,10*R*,12*S*)-6,10,12-Trimethyl-1,4,7-triazacyclotridecane-2,5,8-trione** (119, **$R^1 = R^3 = R^5 = Me$; $R^2 = R^4 = R^6 = H$); Typical Procedure:**[159]

**CAUTION:** *Trifluoroacetic acid is corrosive and severely irritating to skin, eyes, and mucous membranes. Inhalation can destroy tissue and be fatal. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

Acid **118** (240 mg, 0.62 mmol) was dissolved in $CH_2Cl_2$ (3 mL) and TFA (3 mL), and the soln was stirred for 2 h at rt. The soln was then concentrated under reduced pressure. The crude product was dissolved in DMF (140 mL), and $NaHCO_3$ (254 mg, 3.1 mmol) was added. To the mixture was added diethyl cyanophosphonate (0.48 mL, 3.1 mmol). After stirring overnight, DMF was evaporated carefully under reduced pressure. The residue was purified by column chromatography ($CH_2Cl_2$/MeOH 25:1) to give white crystals; yield: 105 mg (63%); mp 303 °C (dec); $[\alpha]_D$ +29.6 (*c* 0.23, MeOH); IR (KBr) $\tilde{\nu}_{max}$: 3280, 1655–1630 $cm^{-1}$; CIMS *m/z*: $[M+1]^+$ 270, $[M]^+$ 269; HRMS (*m/z*): calcd for $C_{13}H_{24}N_3O_3$, 270.1818; found, 270.1810.

**Methyl (2*R*)-1-(Benzyloxycarbonyl)-2-(2-oxoethyl)prolinate** (124):[163]

**CAUTION:** *Osmium tetroxide is an inhalation hazard and can cause temporary blindness. Handle with extreme caution!*

To a soln of alkene **123** (1.0 g, 3.3 mmol) in MeOH/$H_2O$ (2:1, 60 mL) was added 2.5% $OsO_4$ in tBuOH (0.25 mL). The mixture was stirred for 30 min until a dark brown color developed, and then was treated with $NaIO_4$ (2.06 g, 9.9 mmol) in small portions over a 30-min period under vigorous stirring, which resulted in a white precipitate. After an additional 2-h stirring period, the soln was poured onto $H_2O$ (150 mL) and extracted with EtOAc (5 ×). The organic extracts were dried ($MgSO_4$) and concentrated under reduced pressure to give aldehyde **124** as a brown oil; yield: 0.8 g (80%); $[\alpha]_D^{25}$ −20.6 (*c* 1.0, $CHCl_3$); IR (KBr) $\tilde{\nu}_{max}$: 1739, 1705 $cm^{-1}$.

**Methyl (2*R*)-1-(Benzyloxycarbonyl)-2-{(*E*)- and (*Z*)-3-[(*tert*-butoxycarbonyl)amino]-4-ethoxy-4-oxobut-2-enyl}prolinate [(*E*)- and (*Z*)-**125**]:**[163]
To a soln of LDA (1.51 mmol) in THF/hexane (1:3, 4.0 mL) at −78 °C was added dropwise a soln of ethyl [(*tert*-butoxycarbonyl)amino](dimethoxyphosphoryl)acetate (0.537 g, 1.73 mmol) in THF (3.0 mL). The temperature was allowed to warm to −10 °C to dissolve the precipitate. The resulting orange soln was again cooled to −78 °C and then treated with a soln of aldehyde **124** (0.440 g, 1.44 mmol) in THF (3.0 mL). The mixture was warmed to rt and stirred for an additional 2 h. The solvent was removed under reduced pressure and the residue was dissolved in EtOAc. The orange layer was then washed with $H_2O$, dried ($Na_2SO_4$), filtered, and concentrated under reduced pressure to give a crude brown oil. Chromatography (silica gel column, hexane/EtOAc 3:1) afforded the alkene as a 2:1 mixture of *Z*/*E*-isomers; combined yield: 60%; *Z*-isomer: $[\alpha]_D^{25}$ −18.8 (*c* 0.7, $CHCl_3$); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3423, 1726, 1709, 1691 $cm^{-1}$; *E*-isomer: $[\alpha]_D^{25}$ −18.2 (*c* 0.8, $CHCl_3$); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3423, 1726, 1707, 1693 $cm^{-1}$.

**Methyl (2*R*)-2-{(3*RS*)-3-[(*tert*-Butoxycarbonyl)amino]-4-ethoxy-4-oxobutyl}prolinate** (126):[163]
The *E*/*Z* mixture of alkenes (*E*)-**125** and (*Z*)-**125** (52.5 mg, 0.11 mmol) was dissolved in MeOH (2.0 mL) and $N_2$ was bubbled through. To this soln was added 10% Pd/C (8 mg), and the mixture was flushed again with $N_2$. The soln was stirred for 45 min under a $H_2$ atmosphere before being filtered through Celite to remove the catalyst. The solvent was then removed under reduced pressure to give **126** as a clear oil; yield: 35.5 mg (90%); $[\alpha]_D^{25}$ −17.2 (*c* 0.6, $CHCl_3$).

**Methyl (6*R*,8a*R*)- and (6*S*,8a*R*)-6-[(*tert*-Butoxycarbonyl)amino]-5-oxohexahydroindolizine-8a(1*H*)-carboxylate** (127A **and** 127B):[163]
A soln of **126** (0.550 g, 1.54 mmol) in toluene (20 mL) and TEA (0.8 mL) was refluxed for 48 h under a $N_2$ atmosphere. The solvent was removed under reduced pressure to give a yellow oil (0.5 g). Chromatography (silica gel, hexane/EtOAc 1:2) provided the bicyclic lactams **127A** and **127B** as crystalline solids; yield of **127A**: 0.07 g (15%); mp 89–92 °C; $[\alpha]_D^{25}$ −20.0 (*c* 0.6, $CHCl_3$); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3428, 1735, 1698, 1651 $cm^{-1}$. Yield of **127B**: 0.140 g (30%); $[\alpha]_D^{25}$ −18.2 (*c* 1.0, $CHCl_3$); IR ($CHCl_3$) $\tilde{\nu}_{max}$: 3433, 1736, 1709, 1649 $cm^{-1}$.

**Methyl (2*S*,5*E*,9*S*)-9-[(*tert*-Butoxycarbonyl)amino]-10-oxo-1,2,3,4,7,8,9,10-octahydroazecine-2-carboxylate** (129):[164]
To degassed $CH_2Cl_2$ (500 mL) at reflux was added benzylidenedichlorobis(tricyclohexylphosphine)-ruthenium (35 mg, 0.042 mmol). To this light purple soln at reflux was added, via cannula, a soln of **128** in

*for references see p 736*

degassed $CH_2Cl_2$ (10 mL). The purple color immediately faded to orange, and the reaction was allowed to reflux for 18 h. The mixture was then exposed to air until black, and the solvent was removed under reduced pressure to afford a black solid. Flash chromatography (acetone/hexanes 2:8) afforded **129**; yield: 62.2 mg (68%); mp 132–133 °C (EtOAc/hexanes); FAB-MS *m/z*: $[M+H]^+$ 327.2.

**(3*S*)-8-(Azidomethyl)-3-benzyl-5-phenyl-1,3-dihydro-2*H*-1,4-benzodiazepin-2-one (136, R¹ = Bzl); Typical Procedure:**[168]

A soln of Boc-Phe-OH (19.1 g, 72 mmol) in THF was cooled in a salt/ice bath to −15 °C and treated with NMM (7.3 g, 73 mmol). Isobutyl chloroformate (9.7 g, 72 mmol) was then slowly added to the mixture. The soln was stirred for 5 min and then the amino(azidomethyl)benzophenone **134** (15.13 g, 60 mmol) was added. The mixture was allowed to warm to rt and stirred overnight. The soln was diluted with $H_2O$ and extracted into $CH_2Cl_2$. The organic extract was washed with $H_2O$, 0.1 M NaOH, and brine, then dried ($MgSO_4$). The solvent was removed to give **135** as a thick yellow oil (36 g).

The oil was dissolved in EtOAc (500 mL) and cooled to 0 °C. Then, HCl (g) was bubbled into the soln for a period of 1 h. The solvent was removed under vacuum at rt to give a solid. The solid was dissolved in MeOH (100 mL), and the soln was made slightly basic (pH 8) by the addition of 1 M NaOH and stirred overnight at rt. Most of the solvent was removed and the residue was extracted into $CH_2Cl_2$. The extract was washed with brine and dried. The solvent was removed and the residue was chromatographed (EtOAc/hexane 3:7) to give the benzodiazepinone **136** (R¹ = Bzl) as a solid; yield: 22.0 g (97%); mp 84–86 °C; $[\alpha]_D^{25}$ +82.2 (*c* 0.6, MeOH); DCI-MS ($CH_4$): $[M+H]^+$ 382.2; HRMS (*m/z*): calcd for $C_{23}H_{19}N_5O$, 381.1589; found, 381.1589.

**Methyl [(3*S*)-8-(Azidomethyl)-3-benzyl-2-oxo-5-phenyl-2,3-dihydro-1*H*-1,4-benzodiazepin-1-yl]acetate (137, R¹ = Bzl; R² = Me); Typical Procedure:**[168]

Under $N_2$, NaH (60% dispersion in oil; 0.42 g, 0.01 mol) was washed several times with hexane and decanted. To a slurry of the washed NaH in dry DMF (25 mL) was added, in small portions, benzodiazepinone **136** (R¹ = Bzl; 3.8 g, 0.01 mol) and the soln was stirred at rt for 30 min. Then, $BrCH_2CO_2Me$ (1.6 g, 0.011 mol) was added dropwise and the soln was allowed to stir at rt for 1 h. The soln was diluted with $H_2O$ (200 mL) and extracted into $CH_2Cl_2$. The organic extract was washed with $H_2O$ and brine, and then dried. The solvent was removed at rt and the residue was chromatographed (EtOAc/hexane 1:1) to give **137** (R¹ = Bzl; R² = Me) as a foam; yield: 3.0 g (70%); mp 62–65 °C; $[\alpha]_D^{25}$ +44.9 (*c* 0.6, MeOH); DCIMS ($NH_3$): $[M+H]^+$ 454.5; HR-MS (*m/z*): calcd for $C_{26}H_{23}N_5O_3$, 453.1801; found, 453.1811.

**Methyl [(3*S*)-8-(Aminomethyl)-3-benzyl-2-oxo-5-phenyl-2,3-dihydro-1*H*-1,4-benzodiazepin-1-yl]acetate (138, R¹ = Bzl; R² = Me); Typical Procedure:**[168]

A soln of (azidomethyl)benzodiazepinone **137** (R¹ = Bzl, R² = Me; 2.6 g, 5.7 mmol) in EtOAc (150 mL) was hydrogenated at 3.1 atm in a Parr hydrogenator with 10% Pd/C (1 g) for 1 h. The soln was filtered through Celite and the filtrate was extracted with 1 M aq HCl (2 × 200 mL). The acid extract was cooled in an ice bath and made basic with the dropwise addition of 50% aq NaOH. The resulting solid was extracted into EtOAc (2 × 200 mL) and washed with $H_2O$ and brine. After drying, the solvent was removed to give the amino ester **138** (R¹ = Bzl; R² = Me) as a foam; yield: 1.4 g (58%); mp 76–80 °C; $[\alpha]_D^{25}$ +43.0 (*c* 0.6, MeOH); DCI-MS ($NH_3$): $[M+H]^+$ 428.5; HRMS (*m/z*): calcd for $C_{26}H_{25}N_3O_3$, 427.1896; found, 427.1895.


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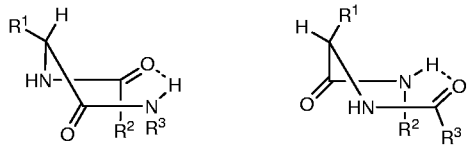
# 12.2 Synthesis of Peptides Incorporating γ-Turn Inducers and Mimetics

M. KAHN and M. EGUCHI

γ-Turns are defined by the existence of a hydrogen bond between the carbonyl oxygen of the first residue (*i*) and the NH of the (*i*+2)th residue, thereby generating a pseudocyclic seven-membered ring.[1,2] The hydrogen bonds in γ-turns include a large percentage of relatively weak ones;[3] however, if these are included, γ-turns may be said to be relatively abundant in peptides and proteins.[4,5]

For L-amino acids γ-turns can be divided into two subgroups, the so-called inverse (or equatorial) (ϕ,ψ values generally between −95 to −70°, 45 to 75°) and classic (or axial) (ϕ,ψ values generally between 70 to 95°, −75 to −45°) γ-turns (Scheme 1).[6,7] They occur as one of two possible diastereomers (in the same way that type I and I′ or type II and II′ β-turns are related).

**Scheme 1** The Inverse and Classic γ-Turn for L-Amino Acids




$C_7$ equatorial inverse γ-turn

$C_7$ axial classic γ-turn

Of the two, classic ones are far less common; however, those that do exist are frequently found at the loop end of β-hairpins.[8] Inverse γ-turns generally do not lead to peptide chain reversal and are frequently situated at either the end of α-helices or within strands of β-sheets or adjacent to certain loop motifs.[9] They are generally well conserved during evolution and some are found at key positions within proteins.

It has been suggested that γ-turns are present in the solution structures of several peptides, and furthermore implicated in their bioactive conformations[10] including bradykinin,[11] substance P,[12] cyclic somatostatin analogues,[13] cyclolinopeptide,[14] and the δ-opioid receptor bound conformation of enkephalin.[15] Yet, despite the fact that γ-turns are frequently hypothesized to represent important features of secondary structure[16] based upon computational,[17] IR absorption,[18] NMR spectroscopic,[19,20] and X-ray diffraction crystallographic determinations,[21] verification of the role of this predicted secondary structural element remains a difficult, but nonetheless critical step.

Conformationally restricted mimetics of reverse turns limit the potential array of conformations adopted by the peptide backbone and thus allow one to potentially validate the presence of reverse-turn pharmacophores. Therefore, there is significant interest in the design and synthesis of γ-turn inducers and mimetics.

## 12.2.1 Peptides Incorporating γ-Turn Inducers and Mimetics

In the design of any type of turn mimetic systems, there are a number of concerns and criteria that need to be addressed. The interaction of the amino acid side chains with their complementary receptor groups is the critical determinant of biological specificity. A successful inducer or mimic must limit the possible conformational arrays and thereby correctly

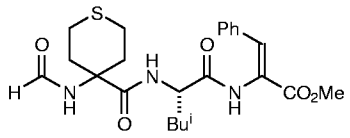
*for references see p 757*

position the appropriate functional groups on a relatively rigid framework. In addition, synthetic expediency, nonpeptidic character, and the facility of incorporation into peptide sequences are criteria that are worth considering.

The definition that we will utilize for a turn inducer is: an inducer is a chemical moiety that when incorporated into a peptide sequence has a rigidifying influence such that a stronger preference for a turn conformation is exhibited by the modified peptide compared to its unmodified counterpart.

One motif which has been utilized to induce β- and γ-turn conformations is an α,β-unsaturated didehydro-α-amino acid. α,β-Unsaturated amino acids have been found to occur naturally in small peptides[22] of microbial origin and in some proteins.[23] Several active analogues of naturally occurring peptides have been synthesized with a didehydroamino acid incorporated. Didehydrophenylalanine (ΔPhe) has been the most frequently chosen residue for introduction. ΔPhe was introduced into position 4 of enkephalin (Tyr-D-Ala-Gly-$\Delta^Z$Phe-Met-$NH_2$) providing a fivefold increase in potency compared to the saturated analogue and increased proteolytic resistance.[24,25] Bradykinin (Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg) analogues with $\Delta^Z$Phe replacements at the 5, 8, and both the 5 and 8 positions all possessed activity greater than or equal to the parent peptide and were less prone to enzymatic degradation.[26] Although didehydroamino acids can adopt ϕ,ψ angles which are compatible with the $i+1$ and $i+2$ positions of β-turns, they are also conformationally compatible with the $i+1$ position of a γ-turn as well.[27] The secondary structure element induced by the didehydroamino acid unit appears to be quite sequence and context dependent. For example, the peptide HCO-Thp-Leu-$\Delta^Z$Phe-OMe (**1**) (Scheme 2; Thp = 4-aminotetrahydrothiopyran-4-carboxylic acid), an analogue of the chemotactic tripeptide HCO-Met-Leu-Phe-OMe, adopts a type-I β-turn conformation stabilized by a weak H-bond between the carbonyl oxygen of the formyl group and the $\Delta^Z$Phe NH in the crystal state.[28] However, $^1$H NMR analysis in $CDCl_3$ suggested that a γ-turn with the Thp residue in the $i+1$ position of a γ-turn was preferred.

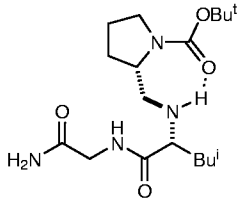
**Scheme 2** The Chemotactic Tripeptide Analogue HCO-Thp-Leu-$\Delta^Z$Phe-OMe[28]




**1**

Other modified amino acids have been used as γ-turn inducers. For example, Toniolo et al.[21] utilized a reduced peptide isostere (-$CH_2NH$-) moiety and incorporated it into the neurotransmitter tripeptide -Pro-Leu-Gly-. The X-ray diffraction structure of the N-terminal Boc- or Z-protected tripeptide Boc(or Z)-Proψ[$CH_2NH$]Leu-Gly-$NH_2$ **2** provides unequivocal proof of the existence of a γ-turn (Scheme 3).

**Scheme 3** The γ-Turn in Boc-Proψ[$CH_2NH$]Leu-Gly-$NH_2$[21]

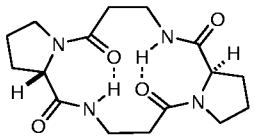



**2**

Kishore and Balaram[29] described the spectroscopic characterization of the protected tripeptide Boc-Cys-Ala-Cys-NHMe that was stabilized by an intramolecular disulfide bridge. NMR data confirmed the presence of an 11-membered disulfide loop, which enforced a γ-turn conformation. A major limitation of this type of approach is the requirement that two of the three residues in the γ-turn must be cysteine and therefore only one functional group can be presented in the turn.

Pavone et al.[30] synthesized the cyclic tetrapeptide *c*[-Pro-βAla-Pro-βAla-] (**3**) using classical solution methods and cyclization of the free tetrapeptide in dilute dichloromethane solution using DCC. In the crystal structure H-bonds between the CO and NH groups of the two β-alanine residues are part of an inverse γ-turn in which the Pro residue occupies the $i+1$ position (Scheme 4). Solution studies in $CD_3CN$ show that the molecule adopts a conformation consistent with that observed in the X-ray diffraction structure.[30] *Per force*, the act of peptide cyclization enforces a turn conformation upon the peptide backbone. In the case of cyclo-tetrapeptides incorporating exclusively α-amino acids, generally two γ-turns have been found. For cyclic pentapeptides, they generally contain one γ- and one β-turn.

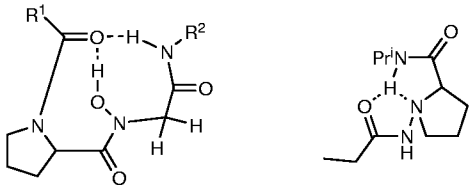
**Scheme 4** The Two γ-Turns in *c*[-Pro-βAla-Pro-βAla-][30]



3

Marraud et al.[31] demonstrated that incorporation of an *N*-hydroxy amide into model dipeptides induced a γ-turn-like structure **4**, as do hydrazino peptides **5** (Scheme 5).

**Scheme 5** Incorporation of an *N*-Hydroxy Amide and a Hydrazine into Model Dipeptides to Induced γ-Turn-Like Structures[31]



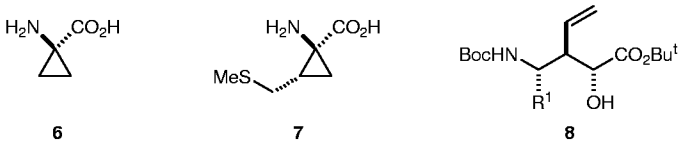
4 5

2,3-Methano amino acids **6** and **7** (Scheme 6), which have rigidly defined $\chi^1$ orientations, have been utilized by several groups and shown to induce γ-turn structures. Incorporation of (2*S*,3*S*)-*c*-Met **7** into the anti-opiate peptide H-Phe-Met-Arg-Phe-$NH_2$ (with replacement of the Met residue) using solid-phase synthesis provided an analogue whose lowest energy conformation involved a γ-turn centered around the *c*-Met residue (Scheme 6).[32]

Reetz et al.[33] described the stereoselective synthesis of partially protected α-hydroxy-γ-amino acids **8** (Scheme 6) as potential pseudo-γ-turn mimetics. X-ray diffraction structure analysis as well as NMR studies in $CDCl_3$ showed that these compounds display a pseudocyclic eight-membered ring arising from an intramolecular H-bond between the amide carbonyl ($i-1$) and the alcohol hydroxy group ($i$). However, the ability of such a H-bond to induce or stabilize a pseudo-γ-turn structure in an aqueous environment is doubtful.

*for references see p 757*

**Scheme 6** 2,3-Methano Amino Acids and Partially Protected α-Hydroxy-γ-amino Acids[32,33]



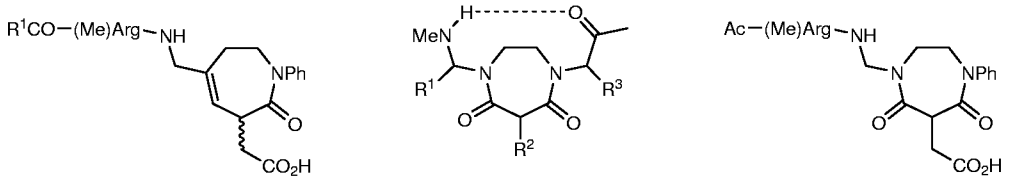

6 7 8

Further discussion and additional references to the synthesis of peptides containing dide-hydro-α-amino acids, reduced peptide isosteres, *N*-hydroxy amides and hydrazides, and $C^{\alpha}$-tetrasubstituted amino acids including 2,3-methano amino acids can be found in Sections 11.1, 10.7, 10.8, and 10.3, respectively.

### 12.2.1.1 Synthesis of γ-Turn Inducers and Mimetics

A *trans*-alkene γ-turn mimetic was used to restrain this conformation at the Asp residue in the RGD sequence,[34,35] which is known to inhibit the binding of fibrinogen to its receptor (gp-IIb/IIIa), thus resulting in inhibition of platelet aggregation and thrombi formation. NMR spectroscopic, X-ray diffraction, and molecular modeling studies[36] of the cyclic peptide Ac-*c*[-Cys-(Me)Arg-Gly-Asp-Pen-]-$NH_2$ had shown in the RGD region an extended Gly and a γ-turn at the Asp residue. Two γ-turn mimetic containing RGD analogues **9** (Scheme 7) were synthesized and found to be potent platelet aggregation inhibitors in vitro and retained nanomolar affinity for the isolated human gp-IIb/IIIa receptor.[34] The activity of these mimetics is comparable to that of the cyclic peptide mentioned above and more than two orders of magnitude higher than that of the corresponding linear peptide. A retro-amide γ-turn mimetic **10** (Scheme 7) was also incorporated into the RGD peptide sequence to enforce the γ-turn conformation at the Asp residue. However, the retro-amide mimetic **11** (Scheme 7) exhibited a significant reduction in binding to the human gp-IIb/IIIa receptor and in vitro platelet antiaggregatory activity.[35,37]

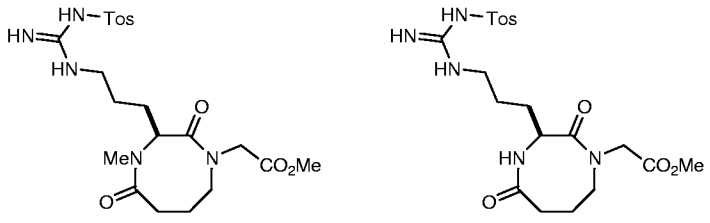
**Scheme 7** γ-Turn and Retro-amide γ-Turn Mimetic Containing RGD Analogues[34,35,37]




9 $R^1$ = Me, Ph 10 11

Bean et al.[38] have also designed, synthesized, and investigated the conformations of two diazocine γ-turn mimetics **12** and **13** (Scheme 8).

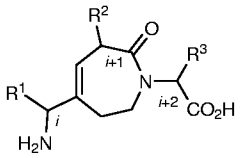
**Scheme 8** Diazocine γ-Turn Mimetics[38]




12 13

Huffman et al.[16] have also incorporated a *trans*-alkene γ-turn mimetic into a series of enkephalin analogues **14** (Scheme 9) with complete loss of biological activity and into an HIV-1 protease inhibitor replacing the -Asn-Tyr-Pro- sequence in **15** corresponding to the $P_2$-$P_1$-$P_1'$ positions of the peptide substrate giving **16**[39,40] (Scheme 10).

**Scheme 9** Incorporation of a *trans*-Alkene γ-Turn Mimetics in a Series of Enkephalin Analogues[16]

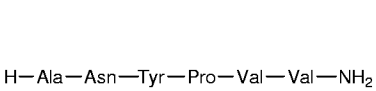



**14**

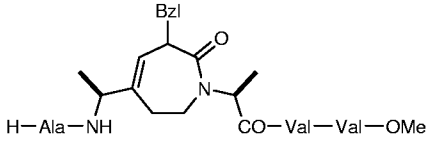
| Name[a] | $R^1$ | $R^2$ | $R^3$ |
|---|---|---|---|
| γ(Gly-Gly-Phe) | H | H | Bzl |
| γ(Gly-D/L-Phe-Leu)[b] | H | Bzl | iBu |
| γ(DL-Tyr-Gly-Gly)[c] | 4-$HOC_6H_4CH_2$ | H | H |

[a] γ Refers to the replacement of the three amino acid residues in parentheses by the seven-membered ring lactam.
[b] D/L Indicated a single stereoisomer (*R* or *S*) of undetermined stereochemistry.
[c] DL Indicated a mixture of stereoisomers (*R* and *S*).

**Scheme 10** Incorporation of a *trans*-Alkene γ-Turn Mimetics in an HIV-1 Protease Inhibitor[39,40]



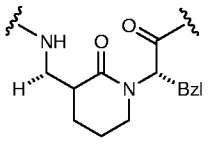

**15** $K_m$ = 17700 μM




**16** $K_i$ = 147 ± 4 μM and 372 ± 17 μM (two diastereomers)

A six-membered lactam ring **17** (Scheme 11) was incorporated into bradykinin in which a γ-turn formed between residues 6–8 had been hypothesized to be one potential bioactive conformation.[10] One of the diastereomers was found to show moderate (micromolar) affinity for the bradykinin receptor despite lacking the proper side chains of $Ser^6$ and $Pro^7$. The result tends to support the presence of a reverse turn (either a γ-turn or, more probably, a β-turn) at this site.

**Scheme 11** Six-Membered Lactam Ring Incorporated into Bradykinin[10]

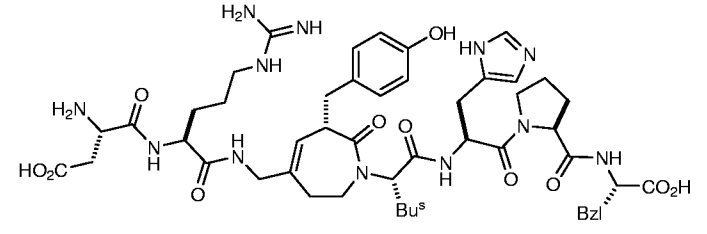



**17**

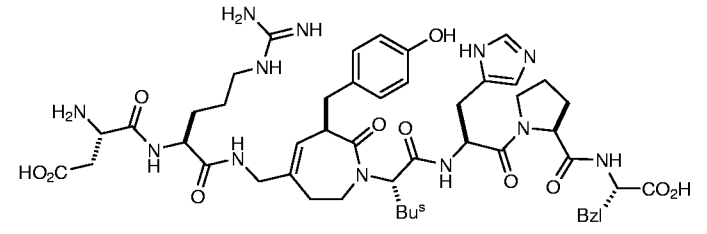
*for references see p 757*

A series of angiotensin II (AII) (Asp-Arg-Val-Tyr-Ile-His-Pro-Phe) peptide analogues **18–21** incorporating γ-turn mimetics to replace amino acid residues 3–5 were synthesized by Schmidt et al.[41] (Scheme 12).

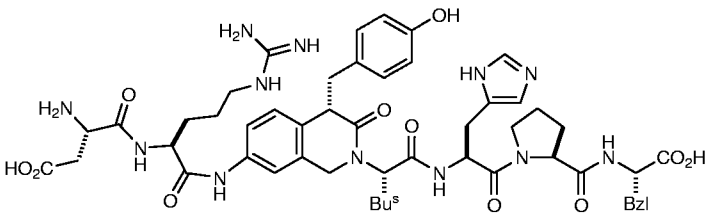
**Scheme 12** Angiotensin II (AII) (Asp-Arg-Val-Tyr-Ile-His-Pro-Phe) Peptide Analogues[41]



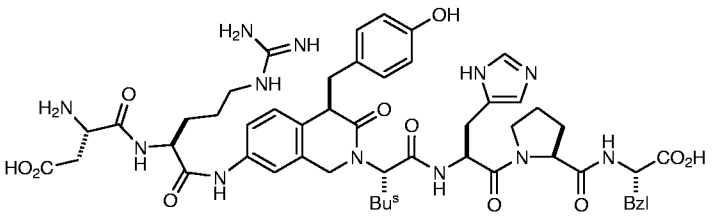
**18**



**19**



**20**

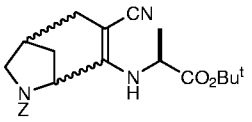


**21**

Interestingly, the two diastereomers **18** and **19** are equipotent to AII in binding to the $AT_1$ receptor. In a functional assay, using contraction of rabbit aorta rings, one of these diastereomers is a full agonist, but has approximately 300-fold less potency that AII. Both diastereomers **20** and **21** were devoid of activity.

Kemp and Carter[42] described the synthesis of the conformationally restricted γ-turn mimetic *tert*-butyl *N*-{6-(benzyloxycarbonyl)-3-cyano-6-azabicyclo[3.2.1]oct-3-en-4-yl}alaninate (**22**) (Scheme 13). No conformational analysis or biological evaluation of this system has been reported.

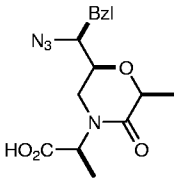
**Scheme 13** *tert*-Butyl *N*-{6-(Benzyloxycarbonyl)-3-cyano-6-azabicyclo[3.2.1]oct-3-en-4-yl}alaninate[42]




22

An inverse γ-turn mimetic, containing a 3-morpholinone ring **23** (Scheme 14) as the key structural feature was prepared enantiospecifically.[43] Molecular mechanics calculations revealed that the torsional angles of the second residue of the mimetic were effectively constrained in an inverse γ-turn-like conformation.

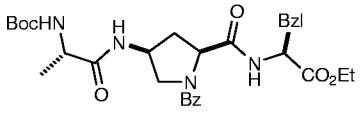
**Scheme 14** An Inverse γ-Turn Mimetic Containing a 3-Morpholinone Ring[43]




23

Curran et al.[44] synthesized analogues of $N^{\alpha}$-benzoyl-*cis*-4-aminoproline as a γ-turn mimetic. This unit was incorporated to provide a tripeptide analogue **24** (Scheme 15). FTIR absorption and NMR analyses in chloroform were consistent with the presence of a γ-turn conformation.

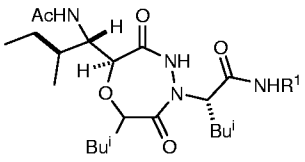
**Scheme 15** A Tripeptide Containing a $N^{\alpha}$-Benzoyl-*cis*-4-aminoproline Analogue[44]




24

Quite recently, we have described the seven-membered ring γ-turn mimetic **25**[45] (Scheme 16) designed to mimic a proposed bioactive conformation in the cyclic pentapeptide malformin family. The synthesis is depicted in Scheme 17. The compounds were tested in a relevant bioassay, but were devoid of activity.

**Scheme 16** Seven-Membered Ring γ-Turn Mimetic of Malformin[45]




25

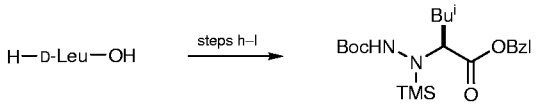
*for references see p 757*

**Scheme 17** Synthesis of a γ-Turn Mimetic of Malformin[45]



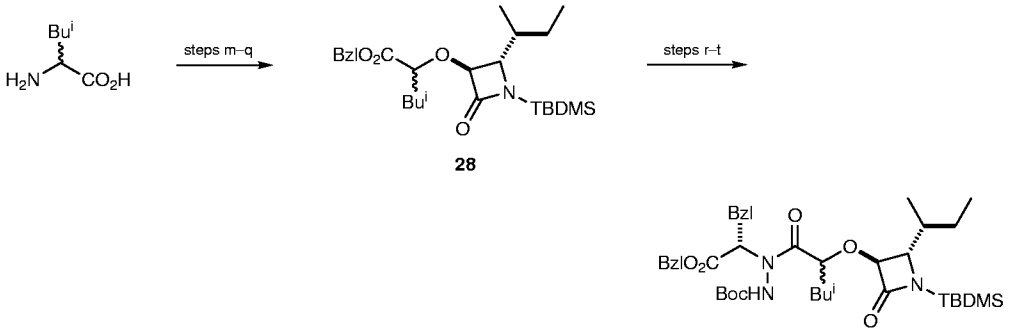

**26**

| Step | Reaction Conditions | Yield (%) | Ref |
|---|---|---|---|
| a | iBuOCOCl, NMM, $Et_2O$ | – | [45] |
| b | $CH_2N_2$ | 58 | [45] |
| c | BzOAg, TEA | 95 | [45] |
| d | HCl | – | [45] |
| e | TBDMSCl, DMAP, TEA | – | [45] |
| f | tBuMgCl | 42 | [45] |
| g | LDA, oxaziridine | 65 | [45] |




**27**

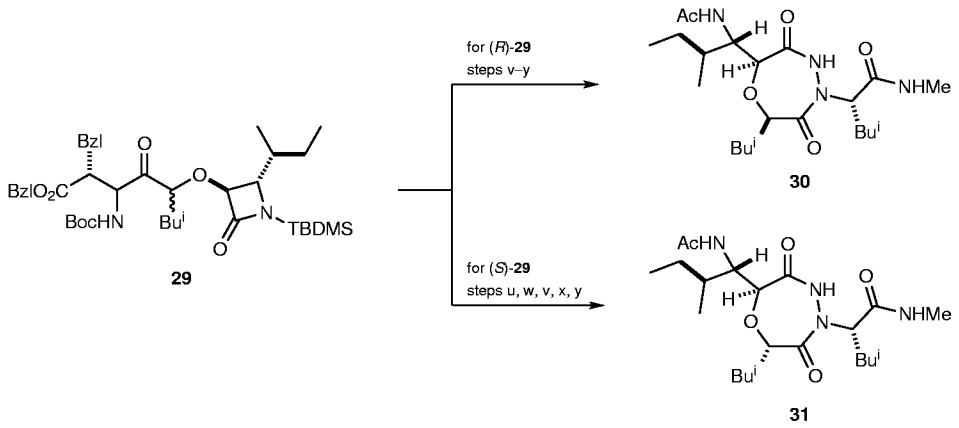
| Step | Reaction Conditions | Yield (%) | Ref |
|---|---|---|---|
| h | $NaNO_2$, $H_2SO_4$, $H_2O$ | 76 | [45] |
| i | $Cs_2CO_3$, MeOH, $H_2O$ | – | [45] |
| j | BzlBr, DMF | 86 | [45] |
| k | $Tf_2O$, 2,6-lutidine, *tert*-butyl carbazate, $CH_2Cl_2$ | 25 | [45] |
| l | BTMSA | – | [45] |




**28**

**29**

| Step | Reaction Conditions | Yield (%) | Ref |
|---|---|---|---|
| m | $NaNO_2$, $H_2SO_4$, $H_2O$ | 79–93 | [45] |
| n | $Cs_2CO_3$, MeOH, $H_2O$ | – | [45] |
| o | BzlBr, DMF | 77–86 | [45] |
| p | $Tf_2O$, 2,6-lutidine | – | [45] |
| q | **26**, NaHMDS, THF, 18-crown-6 | 35–44 | [45] |
| r | $H_2$, $Pd(OH)_2/C$ | 100 | [45] |
| s | cyanuric fluoride, pyridine | – | [45] |
| t | **27**, benzene | 46 | [45] |



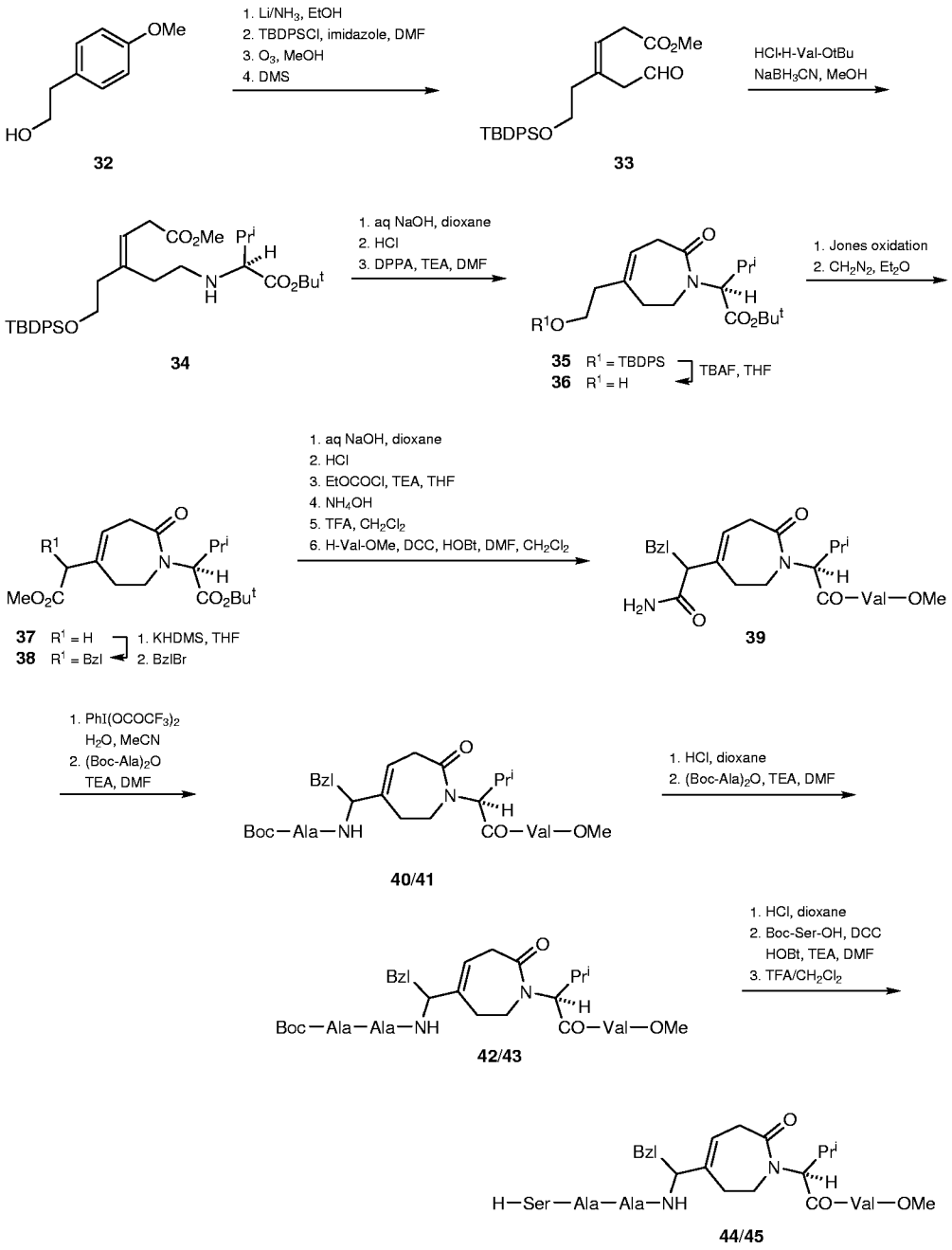

| Step | Reaction Conditions | Yield (%) | Ref |
|---|---|---|---|
| u | TBAF, THF | – | [45] |
| v | TFA/anisole (9:1) | – | [45] |
| w | $Ac_2O$, Py | 51 | [45] |
| x | $H_2$, $Pd(OH)_2/C$ | – | [45] |
| y | EDC, HOBt, $HCl \cdot H_2NMe$ | 70 | [45] |

A γ-turn mimetic HIV-1 protease inhibitor **44/45** was synthesized as shown in Scheme 18.[39,40]

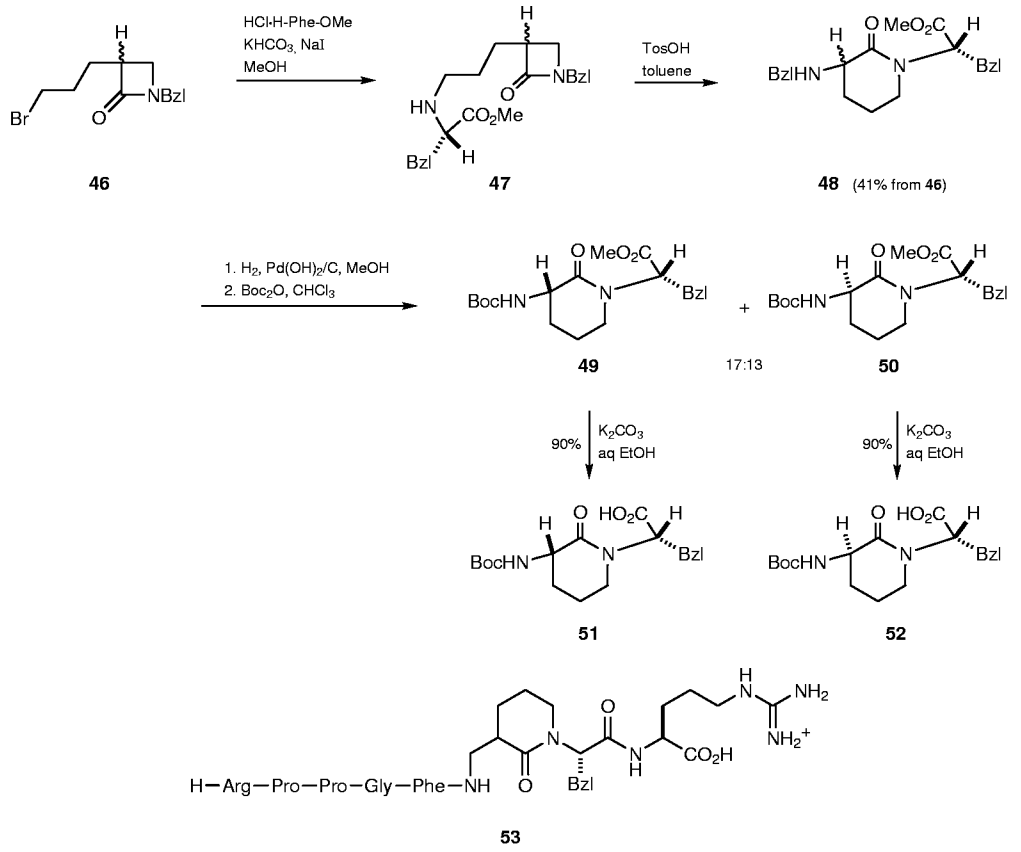
Scheme 19 describes the synthesis of a bradykinin analogue that incorporates a γ-turn mimetic.[10,46]

*for references see p 757*

**Scheme 18** Synthesis of a γ-Turn Mimetic of an HIV-1 Protease Inhibitor[39,40]



**Scheme 19** Synthesis of a Bradykinin Analogue Incorporating a γ-Turn Mimetic[10,46]




**Synthesis of Compounds 44 and 45:**[39,40]

**Methyl 4-{2-[(*tert*-Butyldiphenylsilyl)oxy]ethyl}-6-oxohex-3-enoate (33):**

Liquid $NH_3$ (520 mL) was condensed in a three-necked flask, which was fitted with a cold finger and overhead stirrer and kept at −78 °C with a dry ice/iPrOH bath. A soln of **32** (50 g, 329 mmol) in THF (140 mL) was added to the reaction flask followed by the addition of small pieces of Li wire (10.3 g, 1.48 mol) over a period of about 15 min. The mixture was stirred for an additional 30 min at −78 °C and then quenched by the slow addition of abs EtOH (400 mL). After the reaction was completely quenched (white color), it was allowed to warm to rt overnight to allow most of the $NH_3$ to evaporate. The residue was partitioned between $H_2O$ and $Et_2O$, and the organic layer was collected, dried ($MgSO_4$), filtered, and concentrated at reduced pressure. The residue, which still contained some $H_2O$, was dissolved in $CH_2Cl_2$, dried ($MgSO_4$), filtered, and concentrated under reduced pressure to give crude diene as an oil (40.3 g) contaminated with about 20–25% of the over-reduced cyclohexene product.

The crude diene was dissolved in DMF (150 mL) and treated at rt with imidazole (44.8 g, 658 mmol) and TBDPSCl (85.6 mL, 329 mmol). The resulting soln was stirred at rt for 2 d (though the reaction was essentially complete after a few hours), after which time it was diluted with 10% EtOAc/hexane, washed (3 ×) with $H_2O$, dried ($MgSO_4$), filtered, and concentrated to give crude protected diene still contaminated with ene side product.

The above crude protected diene (~164 mmol) was dissolved in MeOH/$CH_2Cl_2$ (4:1; 500 mL), the resulting soln was cooled to −78 °C and treated with $O_3$ until the starting material disappeared by TLC. The mixture was then reduced with DMS (25 mL) and slowly brought to rt, where it was stirred for 18 h. The mixture was concentrated under reduced pressure and the residue purified by flash chromatography (silica gel, 10 × 20 cm column, 15% EtOAc/hexane) to give **33** contaminated with its corresponding saturated aldehyde; yield: 26.77 g (38%).

*for references see p 757*

**N-[(*E*)-3-[2-[(*tert*-Butyldiphenylsilyl)oxy]ethyl]-5-(methoxycarbonyl)-3-pentenyl]valine *tert*-Butyl Ester (34):**
The aldehyde **33** (36.81 g, 86.7 mmol), contaminated with the saturated aldehyde, was dissolved in MeOH (300 mL) and treated at −20 °C with a mixture of H-Val-OtBu (9.69 g, 55.9 mmol) and HCl·H-Val-OtBu (11.7 g, 55.9 mmol) followed by $NaBH_3CN$ (6.0 g, 95.4 mmol). The resulting mixture was stirred at rt for 24 h. The mixture was concentrated under reduced pressure, the residue dissolved in EtOAc, washed with 5% $Na_2CO_3$, dried ($MgSO_4$), filtered, and evaporated at reduced pressure. The residue was purified by flash chromatography (silica gel, 10 × 22 cm column, 10% EtOAc/hexane) to give **34**, contaminated with the saturated amino diester; yield: 30.53 g (61%).

**1-[(*S*)-1-(*tert*-Butoxycarbonyl)-2-methylpropyl]-5-[2-[(*tert*-butyldiphenylsilyl)oxy]ethyl]-1,3,6,7-tetrahydro-2*H*-azepin-2-one (35):**
The amino diester **34** (30.5 g, 52.5 mmol) was dissolved in dioxane (130 mL) and the resulting soln was treated with 1 M NaOH (63 mL) at rt for 4 h. The mixture was then treated with 3 M HCl (63 mL) and concentrated under reduced pressure. The residue was taken up in toluene and re-concentrated (3 ×) and the residue dried in vacuo. The resulting α-amino ester monoacid was dissolved in dry DMF (1800 mL), cooled to 0 °C, and treated sequentially with TEA (22 mL, 157.5 mmol) and DPPA (22.6 mL, 105 mmol). The mixture was brought to rt and stirred for 48 h. The mixture was concentrated under reduced pressure and the residue purified by flash chromatography (silica gel, 10 × 24 cm column, 15–20% EtOAc/hexane) to give **35**, contaminated with the saturated α-amino ester; yield: 19.83 g (69%).

**1-[(*S*)-1-(*tert*-Butoxycarbonyl)-2-methylpropyl]-5-(2-hydroxyethyl)-1,3,6,7-tetrahydro-2*H*-azepin-2-one (36):**
The protected alcohol **35** (19.83 g, 36.1 mmol) was dissolved in THF (75 mL) and treated with 1 M TBAF in THF (72 mL) at rt for 5 h. The mixture was concentrated and the residue was purified using flash chromatography (silica gel, 10 × 22 cm, 60–100% EtOAc/hexane) to give **36** still partially contaminated with the saturated alcohol. Further purification by column chromatography (silica gel, 4 × 50 cm, 75% EtOAc/hexane) gave pure **36**; yield: 7.63 g (68%).

**1-[(*S*)-1-(*tert*-Butoxycarbonyl)-2-methylpropyl]-5-[(methoxycarbonyl)methyl]-1,3,6,7-tetrahydro-2*H*-azepin-2-one (37):**

**CAUTION:** *Diazomethane is highly toxic and irritating. All operations with diazomethane should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

The alcohol **36** (4.91 g, 15.8 mmol) in acetone (60 mL) was treated with Jones reagent (11 mL) at 0 °C for 1 h. The mixture was quenched with iPrOH, diluted with sat. brine, and extracted with EtOAc. The combined organic extracts were dried ($MgSO_4$) and concentrated under reduced pressure. The residue was then dissolved in $CH_2Cl_2$ and treated with excess ethereal $CH_2N_2$ at 0 °C. After the excess $CH_2N_2$ was decomposed, the mixture was concentrated. Purification of the crude residue by flash chromatography (silica gel, 5 × 20 cm, 25–35% EtOAc/hexane) gave **37**; yield: 3.50 g (65%).

**1-[(*S*)-1-(*tert*-Butoxycarbonyl)-2-methylpropyl]-5-[(*R*/*S*)-1-(methoxycarbonyl)-2-phenylethyl]-1,3,6,7-tetrahydro-2*H*-azepin-2-one (38):**
A soln of **37** (1.15 g, 3.39 mmol) in dry THF (10 mL) was cooled to −78 °C and treated with 1 M KHMDS in THF (3.5 mL) and stirred for 20 min. BzlBr (806 μL, 6.78 mmol) was added and the mixture was stirred at −78 °C for 10 min and then slowly warmed to rt. The reaction was quenched with 10% $NH_4Cl$, diluted with $H_2O$, and extracted with EtOAc. The combined organic extracts were dried ($MgSO_4$) and concentrated under reduced pressure. The residue was purified by flash chromatography (silica gel, 3 × 18 cm, 25% EtOAc/hexane) to give **34** as an inseparable set of diastereomers; yield: 528 mg (36%).

**N-[2-[5-[(*R*/*S*)-1-Carbamoyl-2-phenylethyl]-2,3,6,7-tetrahydro-2-oxo-1*H*-azepin-1-yl]-3-methylbutanoyl]valine Methyl Ester (39):**
A soln of **38** (528 mg, 1.23 mmol) in dioxane (4 mL) was treated with 1 M NaOH (1.4 mL) at rt for 24 h. The mixture was acidified with 3 M HCl and concentrated under reduced pressure. The residue was dissolved in EtOAc, washed with $H_2O$, dried ($MgSO_4$), and concentrated to give the crude acid product (544 mg), which was used without purification.

A soln of the crude acid in THF (14 mL) was cooled to −20 °C and treated with TEA (686 μL, 4.92 mmol) and ethyl chloroformate (470 μL, 4.92 mmol) and stirred for 30 min. A mixture of concd $NH_4OH$ (2 mL) and THF (10 mL) was added and the mixture was stirred at −20 °C for 30 min and left standing at 5 °C for 18 h. The mixture was then poured into ice-cold 3 M HCl and extracted with EtOAc. The combined organic extracts were dried ($MgSO_4$) and concentrated under reduced pressure. Partial purification of the residue by flash chromatography (silica gel, 3 × 20 cm, $CHCl_3$/MeOH/AcOH 96:4:0.1) gave carboxamide (553 mg) as a mixture of diastereomers contaminated with ethyl carbamate.

The carboxamide was dissolved in a mixture of 80% TFA/$CH_2Cl_2$ (12 mL) and stirred at rt for 1 h. The reaction was then concentrated under reduced pressure, re-concentrated from toluene, and then dried in vacuo. This acid was dissolved in DMF (5 mL) and the pH of the resulting soln adjusted to ~7 with TEA. This soln was then treated with H-Val-OMe (520 mg, 3.96 mmol), HOBt (250 mg, 1.85 mmol), and DCC (279 mg, 1.35 mmol). After stirring for 44 h, the mixture was filtered and concentrated under reduced pressure. The residue was purified by flash chromatography (silica gel, 3 × 20 cm, $CHCl_3$/MeOH 98:2) to give **39** as an inseparable mixture of diastereomers; yield: 486 mg (84%).

***N*-[2-{5-[(*R/S*)-1-{[(*tert*-Butoxycarbonyl)alaninyl]amino}-2-phenylethyl]-2,3,6,7-tetrahydro-2-oxo-1*H*-azepin-1-yl}-3-methylbutanoyl]valine Methyl Ester (40/41):**

The soln of **39** (486 mg, 1.03 mmol) in MeCN/$H_2O$ (4:1, 10 mL) was treated at rt with [bis(trifluoroacetoxy)iodo]benzene (500 mg, 1.16 mmol) for 4 h. The mixture was concentrated under reduced pressure and then evaporated from $CHCl_3$/toluene. A soln of Boc-Ala-OH (1.17 g, 6.18 mmol) in $CH_2Cl_2$ (50 mL) was treated with DCC (607 mg, 2.94 mmol) at 0 °C for 30 min and then filtered and concentrated to give $(Boc\text{-}Ala)_2O$. The crude amine salt from above was dissolved in DMF (10 mL), brought to pH 8 with TEA, and treated with a soln of $(Boc\text{-}Ala)_2O$ in DMF at rt for 18 h. After concentration under reduced pressure, the residue was first purified by flash chromatography (silica gel, 3 × 20 cm, 45–60% EtOAc/hexane) and then by gravity chromatography (silica gel, 2.5 × 50 cm, 50–60% EtOAc/hexane) to give the two diastereomers **36** and **37**. Compound **36** yield: 201 mg (32%); FAB-MS *m/z*: 615 $[M+H]^+$. Compound **37** yield: 142 mg (22%); FAB-MS *m/z*: 615 $[M+H]^+$.

***N*-[2-{5-[(*R/S*)-1-{[(*tert*-Butoxycarbonyl)alaninylalaninyl]amino}-2-phenylethyl]-2,3,6,7-tetrahydro-2-oxo-1*H*-azepin-1-yl}-3-methylbutanoyl]valine Methyl Ester (42):**

The protected pseudopeptide **40** (117 mg, 0.190 mmol) was treated with 4 M HCl in dioxane (5 mL) at rt for 1.5 h. The mixture was concentrated under reduced pressure evaporated from toluene, and dried in vacuo. The residue was dissolved in DMF (4 mL), brought to pH 8 with TEA, and treated with a soln of $(Boc\text{-}Ala)_2O$ (2 equiv) in DMF at rt for 24 h. After concentration under reduced pressure, the residue was purified by flash chromatography (silica gel, 3 × 20 cm, 50–100% EtOAc/hexane) to give **42**; yield: 61 mg (47%).

***N*-[2-{5-[(*R/S*)-1-{[(*tert*-Butoxycarbonyl)alaninylalaninyl]amino}-2-phenylethyl]-2,3,6,7-tetrahydro-2-oxo-1*H*-azepin-1-yl}-3-methylbutanoyl]valine Methyl Ester (43):**

The protected pseudopeptide **41** (66 mg, 0.107 mmol) was treated with 4 M HCl in dioxane (5 mL) at rt for 1.5 h. The mixture was concentrated at reduced pressure, evaporated from toluene, and dried in vacuo. The residue was dissolved in DMF (4 mL), brought to pH 8 with TEA, and treated with a soln of $(Boc\text{-}Ala)_2O$ (2 equiv) in DMF at rt for 96 h. After concentration under reduced pressure, the residue was purified by flash chromatography (silica gel, 2 × 20 cm, 75–90% EtOAc/hexane) to give **43**; yield: 50 mg (68%).

***N*-{2-[2,3,6,7-Tetrahydro-2-oxo-5-{(*R/S*)-2-phenyl-1-[(serinylalaninylalaninyl)amino]ethyl}-1*H*-azepin-1-yl]-3-methylbutanoyl}valine Methyl Ester (44):**

The protected peptide **42** (61 mg, 0.089 mmol) was treated with 4 M HCl in dioxane (3 mL) at rt for 2 h. The mixture was concentrated at reduced pressure, evaporated (2 ×) from DMF/toluene, and dried in vacuo. The residue was dissolved in DMF (4 mL), neutralized (pH ~7.5) with TEA, treated with Boc-Ser-OH (55 mg, 0.267 mmol), HOBt (40 mg, 0.294 mmol), and DCC (55 mg, 0.267 mmol), and stirred at rt for 20 h. After concentration under reduced pressure, the residue (protected peptide) was treated with TFA/$CH_2Cl_2$ (2:1, 6 mL) at rt for 4 h. After concentration the residue was purified first by partition chromatography (G-25, 25 × 1000 mm, eluted with BuOH/AcOH/$H_2O$ 4:1:5, upper phase) and then twice by preparative HPLC (5 μm Apex-ODS, Jones chromatography; $H_2O$/MeCN 55:45, 0.1% TFA; $H_2O$/MeCN 65:35, 0.1% TFA; UV detection at 220 nm) to give **44** (5.2 mg); HRMS-FAB *m/z*: calcd for $C_{34}H_{52}N_6O_8$, 672.3846; found, 672.3858; HPLC $t_R$ 2.31 min (5 μm Apex-ODS, Jones Chromatography; $H_2O$/MeCN 65:35, 0.1% TFA; UV detection at 200 nm); HPLC $t_R$ 6.73 min (5 μm Apex-ODS; gradient: A, $H_2O$/0.1% TFA; B, MeCN/0.1% TFA; 90–50% A during 20 min, hold 10 min; UV detection at 220 nm); TLC $R_f$ 0.45 (silica gel, BuOH/AcOH/$H_2O$ 4:1:1); $R_f$ 0.61 (silica gel, BuOH/AcOH/$H_2O$/EtOAc 1:1:1:1).

***N*-{2-[2,3,6,7-Tetrahydro-2-oxo-5-{(*R/S*)-2-phenyl-1-[(serinylalaninylalaninyl)amino]ethyl}-1*H*-azepin-1-yl]-3-methylbutanoyl}valine Methyl Ester (45):**

The protected peptide **43** (50 mg, 0.073 mmol) was treated with 4 M HCl in dioxane (3 mL) at rt for 2 h. The mixture was concentrated under reduced pressure, evaporated (2 ×) from DMF/toluene and dried in vacuo. The residue was dissolved in DMF (5 mL), neutralized (pH ~7.5) with TEA, treated with Boc-Ser-OH (45 mg, 0.219 mmol), HOBt (33 mg, 0.241 mmol), and DCC (45 mg, 0.219 mmol), and stirred at rt for 72 h. After concentration under reduced pressure the residue was purified by flash chromatography

*for references see p 757*

(silica gel, 2 × 17 cm, 5% MeOH/$CHCl_3$) to give protected peptide (28 mg, 68%). The protected peptide was treated with TFA/$CH_2Cl_2$ (1:1, 4 mL) at rt for 4 h and then concentrated under reduced pressure. The residue was purified by preparative HPLC (5 μm Apex-ODS, Jones Chromatography; $H_2O$/MeCN 65:35, 0.1% TFA; UV detection at 220 nm) to give **45** (4.5 mg). HRMS-FAB *m/z*: calcd for $C_{34}H_{52}N_6O_8$, 672.3846; found, 672.3831; HPLC $t_R$ 1.85 min (5 μm Apex-ODS, Jones Chromatography; $H_2O$/MeCN 65:35, 0.1% TFA; UV detection at 220 nm); HPLC $t_R$ 6.42 min (5 μm Apex-ODS, gradient: A, $H_2O$/0.1% TFA; B, MeCN/0.1% TFA; 90–50% A during 20 min, hold for 10 min; UV detection at 220 nm); TLC $R_f$ 0.43 (silica gel, BuOH/AcOH/$H_2O$ 4:1:1); $R_f$ 0.60 (silica gel, BuOH/AcOH/$H_2O$/EtOAc 1:1:1:1).

**Synthesis of** 53:[10,46]

**1-Benzyl-3-(3-bromopropyl)-2-azetidinone** (46)**:**

To a soln of 5-bromovaleric acid (3.62 g, 20.0 mol) in $CH_2Cl_2$ (40 mL) was added dropwise oxalyl chloride (3.49 mL, 40.0 mmol). The mixture was allowed to reflux with stirring. After 1.5 h, the majority of the volatiles was evaporated at atmospheric pressure and the remainder was removed in vacuo. The residue was dissolved in $CH_2Cl_2$ (40 mL) and cooled to −78 °C. 1,3,5-Tribenzyl-hexahydro-1,3,5-triazine (2.86 g, 8.0 mmol) was dissolved in $CH_2Cl_2$ (20 mL) in a separate flask, to which $BF_3 \cdot OEt_2$ (2.95 mL, 24.0 mmol) was added dropwise at rt. After stirring for 10 min, this mixture was transferred to the flask containing the soln of the acid chloride. TEA (8.36 mL, 60.0 mmol) in $CH_2Cl_2$ (10 mL) was added slowly to this mixture. The reaction was allowed to warm from −78 °C to rt overnight with stirring. The volatiles were removed in vacuo and the residue was dissolved in EtOAc (60 mL). The organic layer was washed with 1 M HCl and brine, dried ($Na_2SO_4$), and concentrated in vacuo to provide **46** as a yellow oil; yield: 4.97 g (88%).

**δ-Lactam-Phe-OMe** (48)**:**

To a mixture of MeCN (72 mL) and $H_2O$ (8 mL) were added azetidinone **46** (0.80 g, 2.83 mmol), HCl · H-Phe-OMe (1.83 g, 8.48 mmol), $KHCO_3$ (0.85 g, 8.49 mmol), and NaI (2.55 g, 17.0 mmol). The mixture was allowed to reflux for 40 h with stirring. The MeCN was removed in vacuo and $CH_2Cl_2$ (120 mL) was added. The mixture was washed successively with 1 M HCl, $H_2O$, 5% $NaHCO_3$, and brine, dried ($Na_2SO_4$), and concentrated in vacuo to yield 0.86 g of a mixture of the reaction product of azetidinone **47** with H-Phe-OMe, a small amount of δ-lactam-Phe-OMe and the starting azetidinone **46**. This mixture was dissolved in toluene (50 mL), to which TosOH · $H_2O$ (160 mg, 0.84 mmol) was added. The mixture was allowed to reflux for 20 h with stirring. The reaction was washed with 5% $Na_2CO_3$ and brine, dried ($Na_2SO_4$), and concentrated in vacuo. Flash chromatography of the residue (silica gel, EtOAc) provided δ-lactam-Phe-OMe **48** as a mixture of two diastereomers (yellow oil); yield: 445 mg (41%).

**Boc-δ-lactam-Phe-OMe** (49,50)**:**

To a soln of δ-lactam-Phe-OMe **48** (400 mg, 1.05 mmol) in $CHCl_3$ (12 mL) was added $Boc_2O$ (345 mg, 1.58 mmol). This mixture was allowed to reflux for 2 h with stirring. The volatiles were removed in vacuo. The residue was dissolved in MeOH (30 mL) in a flask, to which Pearlman's catalyst $Pd(OH)_2/C$ (1.00 g), was added. A $H_2$-filled balloon was connected to the vessel and the reaction was allowed to stir at rt overnight. The reaction was filtered through Celite and the volatiles were removed in vacuo. Flash chromatography of the residue (silica gel, hexane/EtOAc 1:1) provided pure Boc-δ-lactam-Phe-OMe diastereomer **49** (140 mg, 34%), the pure diastereomer **50** (108 mg, 26%), and a mixture of the two (82 mg, 20%). Each of them was a yellow oil. The following reactions were carried out separately on the individual diastereomers. For **49**: IR ($CHCl_3$) $\tilde{\nu}_{max}$ ($cm^{-1}$): 3500, 1744, 1708, 1635; MS *m/z*: calcd, 390.5; found, 390. For **50**: IR ($CHCl_3$) $\tilde{\nu}_{max}$ ($cm^{-1}$): 3500, 1742, 1706, 1634; MS *m/z*: calcd, 390.5; found, 390.

**Boc-δ-lactam-Phe-OH** (51,52)**:**

To a mixture of EtOH (5 mL) and $H_2O$ (5 mL) was added Boc-δ-lactam-Phe-OMe **49** (120 mg, 0.31 mmol) and $K_2CO_3$ (64 mg, 0.46 mmol). The mixture was allowed to reflux for 2 h with stirring, then acidified with 10% HCl and extracted with $CH_2Cl_2$ (2 × 20 mL). The combined organic extracts were washed with brine, dried ($Na_2SO_4$) and concentrated in vacuo to provide Boc-δ-lactam-Phe-OH **51** as a white solid; yield: 104 mg (90%); mp 70–72 °C; MS *m/z*: calcd, 376.5; found, 376.3.

Hydrolysis of the other diastereomer **50** was performed in the same manner to provide **52**; mp 67–69 °C; MS *m/z*: calcd, 376.5; found, 376.2.

**Synthesis of** 53**:**

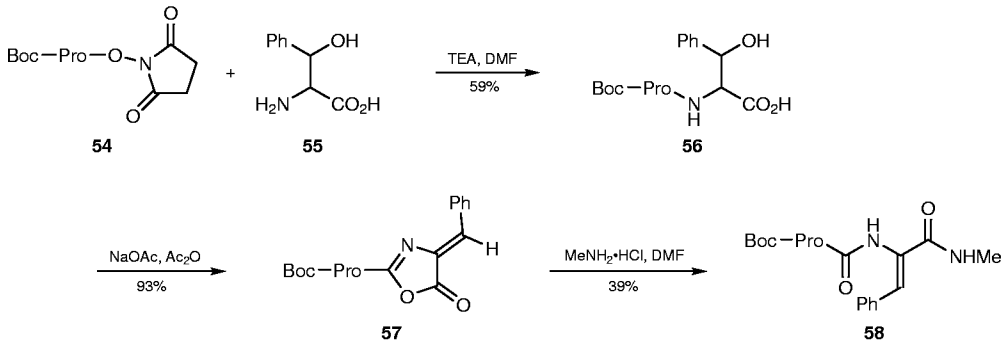
Mimetics **51** and **52** were coupled to Boc-Arg(Tos)-Pam-resin by preactivation (5 min, 0–5 °C, DMF) with BOP reagent. Appropriate protected amino acids were incorporated by double couplings of their respective symmetrical anhydrides or, for Boc-Arg(Tos), as its HOBt ester. After couplings judged incomplete by qualitative ninhydrin assay, the growing peptide chain was capped with $Ac_2O$. The peptide was removed from the resin using anhyd HF/anisole (9:1) at 0–5 °C, and purified by reverse-phase flash chromatography [bonded phase ethyl ($C_2$)] followed by HPLC (C18 column, $H_2O$/MeCN gradient in 0.1% TFA) to separate the two diastereomers of **53**; first diastereomer $t_R$ 16 min (corre-

sponding to elution at 42% MeCN); second diastereomer $t_R$ 19.9 min (corresponding to elution at 48% MeCN). The $^1H$ and PDMS data for the two diastereomers were consistent with their structures.

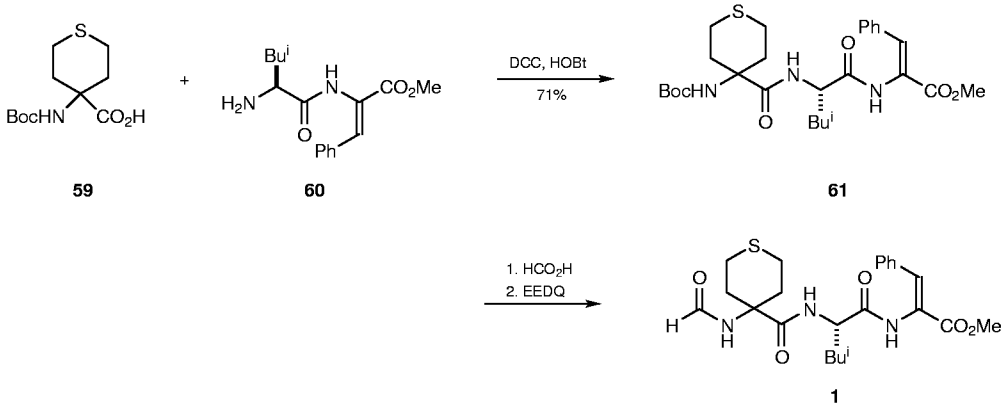
### 12.2.1.2 Synthesis of Peptides Containing γ-Turn Inducers and Mimetics

In the synthesis of peptides containing γ-turn inducers and mimetics a key factor to consider is the ease of preparing and incorporating the inducing or mimetic moiety. In the examples described in Section 12.2.1.2, incorporation involves minor modification of standard peptide synthetic protocols. In the case of ΔPhe, this component is relatively easy to prepare in a protected form that is fully compatible with standard Boc chemistry (Schemes 20 and 21, and Section 11.1).

**Scheme 20** Synthesis of a ΔPhe Model Dipeptide[47]




**Scheme 21** Synthesis of a ΔPhe-Containing Chemotactic Tripeptide Analogue[28]




**Synthesis of a ΔPhe Model Dipeptide:**[47]

**Boc-Pro-DL-Ser(βPh)-OH (56):**

Boc-Pro-OSu (**54**) (1.70 g, 5.70 mmol) and H-DL-Ser(βPh)-OH (**55**) (1.29 g, 5.70 mmol) were suspended in DMF (16 mL). TEA (1.60 mL, 2 equiv) was added and the reaction stirred for 48 h. The soln was filtered to remove a fine particulate and the filtrate concentrated in vacuo, leaving a black oil, which was dissolved in sat. $NaHCO_3$ (60 mL) and washed with EtOAc (2 × 40 mL). The aqueous soln was cooled to 4 °C and acidified with solid citric acid monohydrate to pH 3. The acidic soln was washed with EtOAc (2 × 50 mL). The organic soln then was washed with sat. brine (2 × 50 mL) and dried (anhyd $Na_2SO_4$). Removal of the solvent left a yellow foam, one spot by TLC with appropriate NMR parameters; yield: 1.28 g (59%).

**Boc-Pro-$\Delta^Z$Phe Azlactone (57):**

Boc-Pro-DL-Ser(βPh)-OH (**56**; 0.6 g, 1.70 mmol) and NaOAc (0.23 g, 0.5 equiv) were dissolved in $Ac_2O$ (10 mL), stirred for 24 h and the solvent removed. The residue was dissolved in ice-cold $H_2O$ (60 mL) and chilled for 30 min. The aqueous soln was treated with sat. $NaHCO_3$ (30 mL) for 30 min at rt and

*for references see p 757*

extracted with EtOAc (2 × 75 mL). The EtOAc soln was then washed with distilled $H_2O$ and sat. brine (2 × 50 mL each). The organic phase was dried (anhyd $Na_2SO_4$), and the solvent removed, leaving a yellow oil, pure by TLC, and showing an appropriate NMR spectrum; yield: 0.53 g (93%).

**Boc-Pro-$\Delta^Z$Phe-NHMe (58):**
To Boc-Pro-$\Delta^Z$Phe azlactone **57** (0.3 g) were added HCl·$H_2$NMe and DMF (10 mL). The pH was adjusted to 8.5 with TEA (checked by moistened narrow-range pH paper), and the reaction was allowed to proceed for 48 h. Removal of the solvent yielded an oil that was dissolved in $CH_2Cl_2$, and extracted with 0.2 M HCl, 5% $NaHCO_3$, and distilled $H_2O$ (2 × 40 mL each), and dried ($Na_2SO_4$). Removal of the solvent and crystallization (EtOAc/$Et_2O$) gave the desired dipeptide, based on spectral characterization; yield: 0.13 g (39%); mp 220–224 °C.

**Synthesis of HCO-Thp-Leu-$\Delta^Z$Phe-OMe:**[28]
**Boc-Thp-Leu-$\Delta^Z$Phe-OMe (61):**
To chilled a soln of Boc-Thp-OH (**59**) (0.323 g, 1.24 mmol) and HOBt (0.334 g, 2.48 mmol) in dry DMF (2.5 mL) was added DCC (0.255 g, 1.24 mmol). The mixture was kept at −4 °C for 15 min, and at rt for 20 min, and then it was mixed with a soln of TFA·H-Leu-$\Delta^Z$Phe-OMe (**60**·TFA) (0.5 g, 1.24 mmol) in dry DMF (2.5 mL) neutralized with NMM and cooled to −5 °C. The mixture was stirred at rt for 24 h, filtered from the precipitated DCU, and the solvent removed in vacuo. The remaining residue was taken up with EtOAc, washed with 2% citric acid, $H_2O$, 5% aq $NaHCO_3$, $H_2O$, and dried. The solvent was removed to give a residue that was chromatographed (silica gel, hexane/EtOAc 7: 3) to afford the title compound; yield: 0.47 g (71%); mp 153–154 °C (EtOAc/hexane); $[\alpha]_D$ −49 (*c* 1.0, $CHCl_3$); $R_f$ 0.62 (hexane/EtOAc 1:1); IR (KBr) $\tilde{\nu}_{max}$ ($cm^{-1}$): 3330, 3289, 1729, 1691, 1676; Anal. Calcd for $C_{27}H_{39}N_3O_6S$: C, 60.76; H, 7.37; N, 7.87. Found: C, 60.90; H, 7.47; N, 7.64.

**HCO-Thp-Leu-$\Delta^Z$Phe-OMe (1):**
Boc-Thp-Leu-$\Delta^Z$Phe-OMe (**61**; 0.273 g, 0.51 mmol) was dissolved in 99% formic acid (2.1 mL) and stirred at rt for 24 h. After removal of the excess of formic acid in vacuo, the residue was dissolved in dry $CHCl_3$ (2.1 mL) and EEDQ (0.151 g, 0.61 mmol) was added. The soln was stirred at rt for 5 h and the product was precipitated by addition of hexane. The solid obtained was washed several times with $Et_2O$ and dried to give pure title tripeptide; yield: 0.200 g (85%); mp 112–118 °C (dec) (EtOH); $[\alpha]_D$ −21 (*c* 1, $CHCl_3$); $R_f$ 0.63 (EtOAc); IR (KBr) $\tilde{\nu}_{max}$ ($cm^{-1}$): 3508, 3289, 3213, 1723, 1667, 1660, 1628; Anal. Calcd for $C_{23}H_{31}N_3O_5S \cdot C_2H_5OH$: C, 59.14; H, 7.34; N, 8.28. Found: C, 59.08; H, 7.28; N, 8.19.

**Synthesis of *c*[-βAla-Pro-βAla-Val-]:**[30]
**Z-Pro-βAla-OtBu:**
To a soln of Z-Pro-OH (10.97 g, 44 mmol) and NMM (5.30 mL, 48 mmol) in dry THF (150 mL) was added dropwise, at −10 °C under stirring, iBuOCOCl (6.34 mL, 48 mmol) diluted in $CHCl_3$ (10 mL). After 20 min a soln of H-βAla-OtBu (7.98 g, 44 mmol) and NMM (5.30 mL, 48 mmol) in $CHCl_3$ (150 mL) was added dropwise to the mixture over 20 min. After stirring for 2 h at −10 °C and overnight at rt, the salt precipitated during the reaction was filtered off and the soln was concentrated to dryness. The crude material was dissolved in $CHCl_3$ and the soln was washed with 10% citric acid, sat. $NaHCO_3$, and $H_2O$. After dehydration ($Na_2SO_4$) and removal of the solvent, the desired product was obtained as an oil, pure by TLC; yield: 13.75 g (83%); $R_f$ 0.57 ($CHCl_3$/MeOH 98:2) and 0.81 (BuOH/AcOH/$H_2O$ 4:2:1).

**H-Pro-βAla-OtBu:**
Z-Pro-βAla-OtBu (13.70 g, 36 mmol) was hydrogenated in MeOH (320 mL) over 10% Pd/C (3.87 g) at rt under stirring for 7 h. After removal of the catalyst by filtration and concentration of the filtrate, H-Pro-βAla-OtBu was obtained as a white powder pure, by TLC; yield: 8.70 g (99%); $R_f$ 0.29 (BuOH/AcOH/$H_2O$ 4:2:1).

**Z-βAla-Pro-βAla-OtBu:**
A well-stirred soln of Z-βAla-OH (8.01 g, 35.9 mmol) in dry THF (120 mL), containing NMM (4.30 mL, 39.5 mmol), was chilled at −10 °C under stirring. iBuOCOCl (5.20 mL, 39.5 mmol), diluted in $CHCl_3$ (10 mL), was added dropwise in ~10 min under stirring. After 30 min a soln of H-Pro-βAla-OtBu (8.70 g, 35.9 mmol) and NMM (4.30 mL, 39.5 mmol) in $CHCl_3$ (120 mL) was added dropwise over 30 min. The mixture was stirred for 2 h at −10 °C and overnight at rt. The solvent was removed in vacuo and the crude material was purified by flash chromatography (silica gel, acetone/hexane) to provide Z-βAla-Pro-βAla-OtBu as an oil; yield: 13.65 g (85%); $R_f$ 0.54 (acetone/hexane 8:2) and 0.81 (BuOH/AcOH/$H_2O$ 4:2:1).

**H-βAla-Pro-βAla-OtBu:**
Z-βAla-Pro-βAla-OtBu (13.60 g, 30.4 mmol) was hydrogenated in MeOH (300 mL) over 10% Pd/C (3.16 g) at rt under stirring for 7 h. After removal of the catalyst by filtration and concentration of the

filtrate, H-βAla-Pro-βAla-OtBu was obtained as an oil, pure by TLC; yield: 9.05 g (95%); $R_f$ 0.31 (BuOH/AcOH/$H_2O$ 4:2:1).

**Z-Val-βAla-Pro-βAla-OtBu:**
To a soln of Z-Val-OH (4.0 g, 16.0 mmol) and NMM (2.0 mL, 17.6 mmol) in dry THF (300 mL) was added dropwise at −10 °C under stirring iBuOCOCl (2.3 mL, 17.6 mmol) diluted in $CHCl_3$ (5 mL). After 30 min a soln of H-βAla-Pro-βAla-OtBu (5.0 g, 16 mmol) and NMM (2.0 mL, 17.6 mmol) in $CHCl_3$ (200 mL) was added dropwise to the mixture over 20 min. After stirring for 2 h at −10 °C and overnight at rt, the salt precipitated during the reaction was filtered off and the soln was concentrated to dryness. The crude material was purified by flash chromatography (silica gel, acetone/hexane) to give Z-Val-βAla-Pro-βAla-OtBu as an oil; yield: 7.52 g (86%); $R_f$ 0.50 (acetone/hexane 8:2) and 0.66 (acetone).

**Z-Val-βAla-Pro-βAla-OH:**
The OtBu ester group was removed by treatment of Z-Val-βAla-Pro-βAla-OtBu (7.5 g, 13.7 mmol) with 1 M HCl in AcOH (6 mL) for 1 h at rt under stirring. The soln was then concentrated to dryness and the residue washed with benzene and dried in vacuo over KOH pellets, giving the pure product as an oil; yield: 6.0 g (85%); $R_f$ 0.71 (BuOH/AcOH/$H_2O$ 4:2:1).

**H-Val-βAla-Pro-βAla-OH:**
Z-Val-βAla-Pro-βAla-OH (2.5 g, 5.09 mmol) was hydrogenated in MeOH (200 mL) over 10% Pd/C (542 mg) for 2 h at rt. After removal of the catalyst by filtration, concentration of the filtrate gave the desired product as a white powder, pure by TLC; yield: 1.65 g (91%); $R_f$ 0.36 (BuOH/AcOH/$H_2O$ 4:2:1).

***c*[-L-Val-βAla-L-Pro-βAla-]:**
H-Val-βAla-Pro-βAla-OH (0.280 g, 0.78 mmol) was dissolved in DMF (20 mL). $CH_2Cl_2$ (180 mL) was added and the mixture was chilled at −10 °C. DCC (0.352 g, 1.72 mmol) in $CH_2Cl_2$ (10 mL) was added dropwise over 10 min at −10 °C under stirring. After 20 min, NMM in $CH_2Cl_2$ was added until the pH was ~7.5–8. The mixture was stirred at −10 °C for 2 h and for 96 h at rt. The solvent was removed in vacuo. The residue was dissolved in $CH_2Cl_2$ (40 mL), $H_2O$ (40 mL) was added, and the mixture was shaken. The $CH_2Cl_2$ layer was separated and concentrated and the $H_2O$ soln was decanted. These operations were repeated several times. The $H_2O$ solns were combined and taken to dryness. The crude material was purified on a Sephadex G-10 column (900 × 20 mm, $H_2O$, flow rate of 3 mL·$h^{-1}$). The pooled fractions containing the desired product afforded the cyclic peptide; yield: 92 mg (35%); $R_f$ 0.54 (BuOH/AcOH/$H_2O$ 4:2:1). Analytical HPLC (reverse phase, C18 column, MeCN/$H_2O$, 0.1% TFA, flow rate 1 mL·$min^{-1}$) confirmed the purity of the product. Recrystallization (EtOH) gave crystals; FAB-MS *m/z*: 339 $[M+H]^+$.


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# 12.3 Synthesis of Peptides Incorporating Helix Inducers and Mimetics

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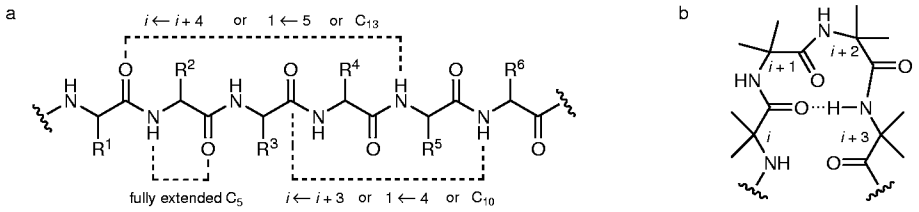
Helices constitute the most abundant secondary structural element in proteins. About a third of the residues in globular proteins are α-helices.[1] In addition to α-helices, $3_{10}$-helices represent another principal structure.[2] Much has been learnt about helices since the discovery of the α-helix by Pauling in 1951.[3] In this section, an attempt is made to present some of the strategies that have been used to induce and stabilize peptide helices. In making helices, it is important to consider all factors, including the helix propensities of individual amino acids and sequence-dependent phenomena as well as synthetic strategies for inducing and stabilizing helices.

The helical conformation is defined by the dihedral angles ϕ and ψ, corresponding to rotations about the backbone N–$C^{\alpha}$ and $C^{\alpha}$–C′ bonds, respectively. The ϕ and ψ angles for an L-residue in a right-handed α-helix are typically −57 and −47°, respectively.[4] The less-stable left-handed α-helix is rarely encountered[5] and will not be included in the methods described here.

A helix is also characterized by a repeating pattern of hydrogen bonds that stabilizes it. The α-helix is defined by hydrogen bonds from a peptide N–H group to a peptide C=O group that is four residues along the peptide chain ($i \leftarrow i + 4$) or ($1 \leftarrow 5$) (Scheme 1a). The loop that is consequently formed consists of 13 atoms, or 3.6 residues. The mean length of α-helices in proteins is about 10 residues, or slightly less than three turns.[1]

A helix is referred to as an $n_r$ helix, where $n$ is the number of residues per turn and $r$ is the number of atoms in the ring formed by the hydrogen bond. Thus, an α-helix is also designated as a $3.6_{13}$-helix. The α-helix turn is also denoted $C_{13}$, according to the number of atoms comprising the hydrogen-bonded loop.

**Scheme 1** Conformations Defined by Hydrogen-Bonding Patterns and a β-Turn Conformation




The most common variation to the α-helix is the $3_{10}$-helix, which, as indicated by its name, is composed of 3 (in reality 2.6) residues per turn, with 10 atoms forming the hydrogen-bonded ($C_{10}$) loop.[2,6] The $3_{10}$-turn has also been called a type III β-turn (Scheme 1b).[7] The $3_{10}$-turn is characterized by an ($i \leftarrow i + 3$) or ($1 \leftarrow 4$) hydrogen-bonding pattern, with ϕ and ψ angles of −57 and −30°, respectively.[2] As there are fewer atoms making up each turn, $3_{10}$-helices are more tightly wound than an α-helix. The $3_{10}$-turn occurs in single turns, in longer helical segments, and at the ends of α-helices in proteins.[1,2] The $3_{10}$-helix is also induced by some nonproteinogenic amino acids such as Aib (α-aminoisobutyric acid).[2]

Protein helices generally adopt variations of the usual helix geometries depending on the environment.[1] The methods described in Sections 12.3.1 and 12.3.2 will refer only to the synthesis of peptides incorporating inducers and mimetics for the α- and $3_{10}$-helices.

*for references see p 789*

## 12.3.1 Helix Inducers

It is now well accepted that the residues making up the peptide sequence, their position at the beginning, middle, and ends of helices, the context in which they occur relative to other residues, as well as peptide length and environment, all influence helix induction and stability. In forming peptide helices, careful planning of the sequence is essential and all these factors have to be taken into account. Section 12.3.1.1 presents many of these factors to consider for inducing helix formation without the use of synthetic appendages.

### 12.3.1.1 Helix Induction Resulting from the Peptide Sequence

#### 12.3.1.1.1 Proteinogenic Amino Acids

Helix propensities have been measured by different groups for all 20 proteinogenic amino acids using statistical surveys of protein helices,[8,9] host–guest analysis,[10] small model peptides and protein fragments,[11–14] site-directed mutagenesis,[15] and calculation studies.[16,17] The results from each method are different, although some correlations can be made.

Of all the amino acids, Ala has the highest tendency to adopt an α-helical conformation based on the Chou and Fasman rules for secondary structure prediction[8] and host–guest determination of helical propensities.[18] This has also been demonstrated in studies by Marqusee et al. on short alanine-based peptides which were found to form stable monomeric peptide helices in water.[19]

After Ala, residues with long side chains exhibit higher tendencies to form helices due to possible position-dependent, side-chain interactions that are helix stabilizing, such as salt bridges and hydrogen bonding. However, aromatic or β-branched side chains act as helix breakers in monomeric peptides due to constraints on side-chain rotational entropy when in an α-helical conformation.[11,14,20]

Of all the residues, Pro appears to have the least preference for occurring within a helix. It is generally known as a helix breaker.[8,21,22] Not only does it lack the amide proton necessary for hydrogen bonding in the helix, but its bulky pyrrolidine ring[23] sterically restricts the conformation of the residue preceding it in the sequence, thereby making it difficult for that residue to adopt an α-helical conformation. However, the pyrrolidine ring does not interfere with the formation of hydrogen bonds involving the Pro carbonyl oxygen and the residues of the turn of the α-helix that follow Pro. Therefore, Pro can frequently be found at the N-terminus of an α-helix. With the ϕ angle locked in place, it also favors turns, making it more predominant at the $i+1$ position.

Gly is another helix breaker and is often found at the C-terminus of a helix. The lack of a side chain in the α-carbon of Gly not only deprives it of helix-promoting side-chain interactions, but also increases its backbone conformational flexibility, favoring a random coil conformation.

Measured s values (helix propagation factor values) obtained by various groups[10,12,24–31] for all the amino acids are listed in Table 1. Chakrabartty and Baldwin[28] have also compared free-energy changes and determined rank orders of helix propensities of all the amino acids as measured in various experimental peptide systems.

**Table 1** s Values for the Protein Amino Acids[10,12,24–31]

| Amino Acid | s Values | | | |
|---|---|---|---|---|
| | Scheraga[a] | Baldwin[b,e,g] | Stellwagen[c,e,f] | Kallenbach[d,e,f] |
| Ala | 1.07 | 1.54 | 1.81 | 2.19 |
| Arg | 1.03 | 1.04 | 1.94 | n.r. |
| Asn | 0.78 | 0.29 | 0.23 | 0.73 |
| Asp ($Asp^-$) | 0.68 | 0.29 (0.29) | 0.23 | n.r. |
| Cys | 0.99 | 0.33 | 0.43 | n.r. |
| Glu ($Glu^-$) | 0.97 | 0.63 (0.43) | 0.73 | n.r. |
| Gln | 0.98 | 0.53 | 0.58 | 1.20 |
| Gly | 0.59 | 0.05 | 0.05 | 0.57 |
| His ($His^+$) | 0.69 | 0.36 (0.06) | 0.37 | n.r. |
| Ile | 1.14 | 0.42 | 0.43 | 1.02 |
| Leu | 1.14 | 0.92 | 1.03 | 1.55 |
| Lys ($Lys^+$) | 0.94 | (0.78) | 1.29 | n.r. |
| Met | 1.20 | 0.60 | 0.79 | 1.41 |
| Phe | 1.09 | 0.28 | 0.79 | n.r. |
| Pro | 0.19 | ~0.001 | n.r. | n.r. |
| Ser | 0.76 | 0.36 | 0.29 | 0.86 |
| Thr | 0.82 | 0.13 | 0.18 | 0.79 |
| Trp | 1.11 | 0.29–0.36 | 0.58 | n.r. |
| Tyr | 1.02 | 0.37–0.50 | 0.43 | n.r. |
| Val | 0.95 | 0.22 | 0.18 | 0.93 |

[a] Obtained from peptides of undefined sequence at 0 °C by host–guest studies using the Zimm–Bragg algorithm and allowing the nucleation parameter to vary.[10]

[b] Values obtained using Lifson–Roig theory modified to include either N-capping or charged group helix macrodipole interactions; measured at pH 7.0 in 1.0 M NaCl, 0 °C for uncharged residues and $Lys^+$, and 10 mM NaCl for Asp, Arg, Glu, and His.[24–28]

[c] Obtained in KCl solutions at pH 7.0 and 0 °C using the Lifson–Roig algorithm and assuming $\sigma = 0.003$.[29,30]

[d] Obtained at pH 7.0 in 0.01 M KF at 3 °C, analyzed by modification of Zimm–Bragg theory to account for $i \leftarrow i + 4$ side-chain interactions.[12,31]

[e] From data compiled by Chakrabartty and Baldwin.[28]

[f] n.r. = not reported.

[g] Amino acids and values in brackets indicate s values for the ionized amino acid, e.g. ($Asp^-$) (0.29) for when the side chain of Asp is in the negatively charged carboxylate form.

#### 12.3.1.1.2 Nonproteinogenic α-Amino Acids

Early studies of peptides isolated from microbial sources and possessing antibiotic activity led to the discovery of some α-amino acids not normally found in proteins. The peptides containing these nonproteinogenic, but natural α-amino acids exhibit helical structures which act as channels for transmembrane ion transport.[32] Formation of α- or $3_{10}$-helices in these peptides was ascribed to the unique geometry of these residues and their present use in the induction of these conformations is now widely established.

##### 12.3.1.1.2.1 $C^{\alpha,\alpha}$-Disubstituted Glycines

For further information on $C^{\alpha,\alpha}$-disubstituted glycines, see also Section 10.3. α-Aminoisobutyric acid (Aib, **1**) (Scheme 2), also known as $C^{\alpha}$-methylalanine, is the prototype of the class of residues termed $C^{\alpha,\alpha}$-disubstituted glycines. It is also the nonproteinogenic α-amino acid with the highest helix propensity. Aib has two methyl groups on the α-carbon. The extra methyl group severely limits the possible $\phi$ and $\psi$ values, such that allowed regions of

*for references see p 789*

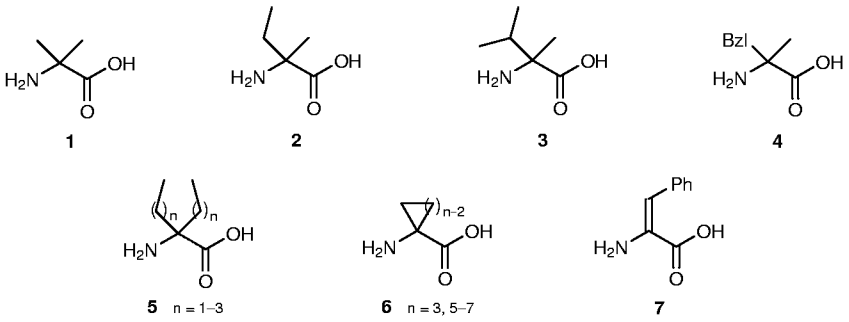
conformational space for Aib are restricted to $\phi$ ~60 ± 20° and $\psi$ ~30 ± 20°.[33] Both α- and $3_{10}$-helices are accessible with these angles and, due to the inherent symmetry in Aib, both left and right-handed helices can be obtained. The resulting handedness then depends on the chirality of protein residues in the sequence: left-handed for D-residues and right-handed for L-residues.

Factors which determine the formation of either α- or $3_{10}$-helices include peptide length, percentage of Aib residues present, amino acid sequence, and experimental conditions. In solution, it is believed that a longer peptide chain (eight residues or more) promotes helicity.[34,35] Short Aib-rich peptides (50% or more) favor a $3_{10}$-helix, while those with 30–35% Aib are predominantly α-helical.[35–37] While an α-helix is generally more stable, a $3_{10}$-helix has an extra hydrogen bond which can be satisfied in solvents of low polarity.[33,34] As the peptide length increases, the inherent stability of the α-helix compensates for the lost hydrogen bond and predominates in solution. There is also a strong α-helix stabilization in aqueous solution, arising mainly from solute–solvent electrostatic interactions.[38]

In the solid state, long Aib-rich peptides (more than eight residues) form a mixed $3_{10}$/α-helix, depending on the solvent of crystallization.[35,37] The enhanced electrostatic interaction and improved packing in a $3_{10}$-helix makes it very easy to crystallize. These are but empirical generalizations and it is the balance of these factors, along with the steric requirements of the amino acid side chains, which ultimately determines the type of helix that will be formed.

The helix-inducing effect of Aib has been compared to that of a number of other $C^{\alpha,\alpha}$-disubstituted glycines (Scheme 2) when incorporated in peptides. The helix-inducing ability of isovaline (Iva, **2**) compares well to that of Aib and despite its chirality, it similarly exhibits a screw-sense indifference.[39,40] $C^{\alpha}$-Methylvaline [(αMe)Val, **3**],[40,41] $C^{\alpha}$-methylleucine [(αMe)Leu], and $C^{\alpha}$-methylphenylalanine [(αMe)Phe, **4**][40,42] also have helix-inducing abilities similar to that of Aib. Their helical propensities are higher than their unmethylated counterparts with L-(αMe)Val showing a clear-cut preference for the right-handed screw sense.

**Scheme 2** Nonproteinogenic Amino Acids




Conformational preferences of homopeptides of Aib, residues with symmetrical longer chain α-carbon substituents **5** and those with cyclized side-chain residues **6**, have also been studied by Toniolo and Benedetti.[43] Since unfavorable steric interactions are more significant in the oligo(Aib)$_n$ α-helix than in the $3_{10}$-helix, poly(Aib) peptides were found to prefer a $3_{10}$-helical conformation in both the crystal and solution ($CHCl_3$) states. As side-chain length is increased helix stability decreases, due to unfavorable main-chain–side-chain and side-chain–side-chain interactions. X-ray diffraction, as well as IR absorption and NMR investigations, reveal that for oligo(Deg)$_n$ and oligo(Dpg)$_n$ the fully extended $C_5$ conformation (Scheme 1) is preferred. This is confirmed by the substitution of Deg (**5**, n = 1), Dpg (**5**, n = 2), and Dbug (**5**, n = 3) into Boc-Ala-Xaa-Ala-OMe peptides, which shows that $C^{\alpha,\alpha}$-dialkylglycine-containing peptides rule out the helical conformation and instead show a pref-

erence for the extended $C_5$ form.[44] Cyclic $Ac_nc$ residues **6** (n = 3, 5–7) have long side chains which are "tied up" in a ring. Residue $Ac_3c$ (**6**, n = 3) exhibits a marked preference for the conformational space where $\phi,\psi \approx \pm 90°,0°$, that is, for position $i+2$ of type I(I′) and type II(II′) β-bends. It can also be accommodated in distorted type III(III′) β-bends and $3_{10}$-helices ($\phi,\psi \approx \pm 60°,30°$). Similarly Aib (**1**), $Ac_5c$, $Ac_6c$, and $Ac_7c$ (**6**, n = 5–7, respectively), exhibit preference for regular type II β-bends and $3_{10}$-helices, as seen in crystal structures and in solution ($CHCl_3$). Examination of long $Ac_5c$ and $Ac_6c$ (**6**, n = 5 and 6, respectively) homopeptides reveals that the cyclopentyl ring adopts either an envelope or half-chair conformation, while the cyclohexyl ring has almost a perfect chair conformation with the amino substituent positioned axially.

#### 12.3.1.1.2.2 α,β-Didehydrophenylalanine

For further information on α,β-didehydrophenylalanine, see also Section 11.1. Like Aib, α,β-didehydrophenylalanine (ΔPhe, **7**; Scheme 2) is a modified amino acid found in microbial sources. The presence of an $sp^2$ α-carbon atom and a C=C bond gives this residue an unusual geometry which has been used to promote helical conformations in numerous peptides.

Short peptides three to five residues in length that contain one or two ΔPhe residues adopt a β-bend[45–48] and $3_{10}$-helical folding. For example, crystal structure studies of a three-residue peptide containing two ΔPhe residues, Ac-ΔPhe-Ala-ΔPhe-NHMe,[49] and a five-residue peptide with one ΔPhe residue, Boc-Leu-Phe-Ala-ΔPhe-Leu-OMe,[50] show the presence of two consecutive β-bends, corresponding to an incipient $3_{10}$-helix in both peptides. Such didehydropeptides exist in both screw senses and the spacing between ΔPhe residues does not seem to affect the type of helix formed.[51,52]

As the length of the peptide increases, the α-helical conformation becomes favored, as can be shown for the heptapeptide Boc-Gly-ΔPhe-Ala-Phe-Leu-ΔPhe-Ala-NHMe[53] and octapeptide Boc-Val-ΔPhe-Phe-Ala-Leu-Ala-ΔPhe-Leu-OMe,[54] although some heterogeneity in conformation is observed in highly polar solvents such as DMSO.

As is the case with Aib, peptides rich in ΔPhe, as in tetra- and heptapeptides with alternating ΔPhe[55] or in a nonapeptide with three ΔPhe residues,[56] adopt $3_{10}$-helical conformations despite the peptide main-chain length (for two review articles, see refs[57,58]).

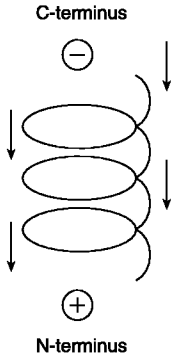
#### 12.3.1.1.3 Helix End Capping

Early studies on helix propagation had proposed a "central residue initiation" mechanism, in which helices were believed to form outwards in two directions starting from central residues with high helix-forming propensities. The Chou and Fasman predictions[8,59,60] worked well for predicting centrally located residues with high tendencies to form helices. They, however, did not explain the early observation that certain proteins contain centrally located residues deemed to be "helix indifferent" or that amino acids with charged side chains seem to occur very frequently at the ends of helices.

It is now known that amino acids with high propensities for the middle of helices are not necessarily favored at the ends. Polar and charged amino acids are more prevalent at the helix termini, whereas the middle of an α-helix contains predominantly hydrophobic residues.[61] This is rationalized by what is now known as the helix dipole model,[62–65] first proposed by Blagdon and Goodman (Figure 1).[62]

*for references see p 789*

**Figure 1** Helix Dipole Model[62]




An α-helix is characterized by main-chain hydrogen bonds between successive amide hydrogen donors and carbonyl oxygen acceptors located four residues previously in the sequence.[3] These hydrogen bonds are almost parallel to the helix axis and all point in the same direction, so the individual dipoles are aligned head-to-tail. In the α-helix, the first four main-chain NH groups at the N-terminus are not hydrogen bonded and carry a partial positive charge. Thus, the N-terminus is the positive end of the helix macrodipole. In the same way, the last four carbonyl groups at the C-terminus are bereft of hydrogen-bonding partners and carry a partial negative charge, making the C-terminus the negative pole of the helix.

In accordance with the helix dipole model, the charges imparted to the ends of the helix by the dipole necessitate that the positively charged side chains of amino acids are preferred at the C-terminus and negative charged side chains at the N-terminus. Thus, Asp and Glu residues prefer the N-terminus while Lys, Arg, and His are frequently found at the C-terminus of peptides. Conversely, the presence of a free, protonated amino group ($NH_3^+$) at the N-terminus or deprotonated carboxy group ($COO^-$) at the C-terminus is helix destabilizing. The unfavorable interaction with the dipole can be eliminated by "capping" the charges with an acetyl (0 charge) or succinyl (−1 charge) group on the N-terminus[66,67] and amidating the carboxy group at the C-terminus.

While the helix is held in place by backbone hydrogen bonds, the first four NH hydrogens and last four CO oxygens have no hydrogen-bonding partners for lack of another turn at the ends. Presta and Rose postulated in 1988[68] that a necessary condition for helix formation would be the presence of residues flanking the helix termini whose side chains can hydrogen bond with the four termin NH and CO groups that are lacking intrahelical partners. Thus, residues whose side chains can hydrogen bond to NH groups (Asn, Asp, Gln, Glu, Ser, Thr, and His) are desired at the N-terminus and residues that can hydrogen bond to CO groups (Asn, Arg, Gln, Lys, Ser, Thr, His) are preferred at the C-terminus. There is also a notable preference for negatively charged side chains in the first turn and positively charged ones in the last turn due to stabilization arising from interaction with the helix dipole.

A number of studies have concentrated on determining the propensity of amino acids to be found at helix termini by the examination of protein sequences[69–73] and mutation of short peptides.[74–79] The following labels have been used for helices and their flanking residues: N″-N′-(N-cap)-N1-N2-N3-...-C3-C2-C1-(C-cap)-C′-C″ where N1, N2, and N3, and C1, C2, and C3 are the residues on the N- and C-termini, respectively. The residue immediately preceding the N-terminal side of the helix is known as the N-cap, while the residue immediately following the C-terminal side of the helix is the C-cap.[61] Capping residues have nonhelical $\phi$ and $\psi$ angles.

A number of groups have summarized the positional preferences for all 20 proteinogenic amino acids in α-helices.[61,71,78] The positional preferences determined by Richardson and Richardson are in Table 2.[61]

**Table 2** Amino Acid Preferences for Specific Positions in Helices[61][a,b]

| Average Percentage Composition (%) | Amino Acid | N″ | N′ | N-cap | N1 | N2 | N3 | N4 | N5 | Mid | C5 | C4 | C3 | C2 | C1 | C-cap | C′ | C″ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9.0 | Ala | 13 | 16 | 10 | 24 | 31 | 19 | 20 | 20 | **106** | **23** | 21 | 13 | 27 | 21 | 15 | 18 | 12 |
| | | 0.7 | 0.8 | 0.5 | 1.2 | 1.6 | 1.0 | 1.1 | 1.4 | **1.8** | **1.8** | 1.3 | 0.7 | 1.4 | 1.1 | 0.8 | 1.0 | 0.7 |
| 3.8 | Arg | 3 | 4 | 3 | 6 | 7 | 3 | 11 | 7 | 33 | 5 | 5 | 6 | **17** | 8 | 7 | 11 | 8 |
| | | 0.4 | 0.5 | 0.4 | 0.7 | 0.9 | 0.4 | 1.5 | 1.2 | 1.3 | 1.0 | 0.8 | 0.8 | **2.1** | 1.0 | 0.9 | 1.4 | 1.1 |
| 4.5 | Asn | 11 | 9 | **34** | 7 | 7 | 7 | 0 | 9 | 28 | 4 | 5 | 8 | 9 | 11 | 15 | 8 | 13 |
| | | 1.2 | 1.0 | **3.5** | 0.7 | 0.7 | 0.7 | 0 | 1.2 | 0.9 | 0.6 | 0.6 | 0.8 | 0.9 | 1.2 | 1.6 | 0.9 | 1.5 |
| 6.0 | Asp | 17 | 10 | **27** | 10 | **33** | **28** | 4 | 6 | 38 | 6 | 5 | 7 | 9 | 5 | 9 | 17 | 17 |
| | | 1.4 | 0.8 | **2.1** | 0.8 | **2.6** | **2.2** | 0.3 | 0.6 | 1.0 | 0.7 | 0.5 | 0.6 | 0.7 | 0.4 | 0.7 | 1.4 | 1.4 |
| 2.4 | Cys | 3 | 4 | 3 | 4 | 6 | 3 | 5 | 6 | 11 | 0 | 3 | 1 | 6 | 8 | 2 | 4 | 2 |
| | | 0.6 | 0.8 | 0.6 | 0.8 | 1.2 | 0.6 | 1.1 | 1.6 | 0.7 | 0 | 0.7 | 0.2 | 1.2 | 1.6 | 0.4 | 0.8 | 0.4 |
| 3.6 | Gln | 7 | 7 | 3 | 5 | 6 | 12 | 9 | 8 | 30 | 5 | 1 | 10 | 12 | **16** | 7 | 10 | 8 |
| | | 1.0 | 0.9 | 0.4 | 0.7 | 0.8 | 1.5 | 1.3 | 1.4 | 1.3 | 1.0 | 0.2 | 1.3 | 1.6 | **2.1** | 0.9 | 1.4 | 1.1 |
| 5.6 | Glu | 11 | 8 | 5 | **27** | **24** | **40** | 6 | 8 | 30 | 9 | 7 | 19 | 20 | 10 | 4 | 9 | 8 |
| | | 1.0 | 0.7 | 0.4 | **2.2** | **2.0** | **3.3** | 0.5 | 0.9 | 0.8 | 1.1 | 0.7 | 1.6 | 1.7 | 0.8 | 0.3 | 0.8 | 0.7 |
| 8.7 | Gly | 29 | 25 | **33** | 6 | 16 | 12 | 6 | 8 | 29 | 6 | 8 | 2 | 3 | 4 | **72** | 21 | 10 |
| | | 1.6 | 1.4 | **1.8** | 0.3 | 0.9 | 0.6 | 0.4 | 0.6 | 0.5 | 0.5 | 0.5 | 0.1 | 0.2 | 0.2 | **3.9** | 1.2 | 0.6 |
| 2.1 | His | 5 | 4 | 5 | 3 | 3 | 3 | 6 | 3 | 14 | 7 | 7 | 5 | 8 | **15** | 6 | 5 | 4 |
| | | 1.2 | 0.9 | 1.1 | 0.7 | 0.7 | 0.7 | 1.5 | 0.9 | 1.0 | 2.4 | 1.9 | 1.1 | 1.8 | **3.4** | 1.3 | 1.2 | 1.0 |
| 5.1 | Ile | 9 | 14 | 2 | 10 | 8 | 4 | 11 | 7 | 41 | 9 | 14 | 15 | 4 | 8 | 8 | 11 | 7 |
| | | 0.9 | 1.3 | 0.2 | 0.9 | 0.7 | 0.4 | 1.1 | 0.9 | 1.2 | 1.3 | 1.6 | 1.4 | 0.4 | 0.7 | 0.7 | 1.1 | 0.7 |
| 7.7 | Leu | 15 | 20 | 3 | 15 | 5 | 10 | **39** | 13 | 63 | 13 | 19 | **31** | 13 | 11 | 11 | 14 | 8 |
| | | 0.9 | 1.2 | 0.2 | 0.9 | 0.3 | 0.6 | **2.6** | 1.1 | 1.2 | 1.2 | 1.4 | **1.9** | 0.8 | 0.7 | 0.7 | 0.9 | 0.5 |
| 6.7 | Lys | 14 | 16 | 10 | 9 | 14 | 12 | 11 | **21** | 49 | 13 | 12 | **31** | **27** | **28** | 18 | 16 | 17 |
| | | 1.0 | 1.1 | 0.7 | 0.6 | 1.0 | 0.8 | 0.8 | **1.9** | 1.1 | 1.4 | 1.0 | **2.2** | **1.9** | **2.0** | 1.3 | 1.2 | 1.3 |
| 1.8 | Met | 1 | 7 | 3 | 1 | 4 | 4 | 6 | 5 | 18 | 7 | **9** | 4 | 5 | 4 | 3 | 3 | 0 |
| | | 0.3 | 1.8 | 0.8 | 0.3 | 1.0 | 1.0 | 1.7 | 1.7 | 1.5 | 2.7 | **2.8** | 1.0 | 1.3 | 1.0 | 0.8 | 0.8 | 0 |
| 3.8 | Phe | 9 | 9 | 2 | 4 | 7 | 5 | 14 | 6 | 34 | 10 | **19** | 14 | 2 | 6 | 4 | 1 | 9 |
| | | 1.2 | 1.1 | 0.2 | 0.5 | 0.9 | 0.6 | 1.9 | 1.0 | 1.3 | 1.9 | **2.9** | 1.8 | 0.3 | 0.7 | 0.5 | 0.1 | 1.2 |
| 4.5 | Pro | 6 | 9 | 8 | **25** | 5 | 4 | 1 | 2 | 10 | 2 | 0 | 0 | 2 | 0 | 7 | 17 | 13 |
| | | 0.7 | 1.0 | 0.8 | **2.6** | 0.5 | 0.4 | 0.1 | 0.3 | 0.3 | 0.3 | 0 | 0 | 0.2 | 0 | 0.7 | 1.9 | 1.5 |
| 6.9 | Ser | 22 | 17 | **34** | 11 | 12 | 6 | 6 | 12 | 27 | 5 | 6 | 8 | 23 | 25 | 11 | 9 | 12 |
| | | 1.6 | 1.2 | **2.3** | 0.7 | 0.8 | 0.4 | 0.4 | 1.1 | 0.6 | 0.5 | 0.5 | 0.6 | 1.6 | 1.7 | 0.8 | 0.7 | 0.9 |

*for references see p 789*

**Table 2** (cont.)

| Average Percentage Composition (%) | Amino Acid | N″ | N′ | N-cap | N1 | N2 | N3 | N4 | N5 | Mid | C5 | C4 | C3 | C2 | C1 | C-cap | C′ | C″ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6.1 | Thr | 4 | 10 | 21 | 11 | 9 | 13 | 6 | 6 | 43 | 4 | 6 | 9 | 12 | 13 | 4 | 10 | **25** |
| | | 0.3 | 0.8 | 1.6 | 0.8 | 0.7 | 1.0 | 0.5 | 0.6 | 1.0 | 0.5 | 0.6 | 0.7 | 0.9 | 1.0 | 0.3 | 0.8 | **2.1** |
| 1.3 | Trp | 3 | 2 | 1 | 6 | 5 | 4 | **8** | 3 | 14 | 2 | 5 | 1 | 1 | 0 | 0 | 1 | 7 |
| | | 1.1 | 0.7 | 0.3 | 2.1 | 1.7 | 1.4 | **3.1** | 1.4 | 1.5 | 1.1 | 2.1 | 0.4 | 0.4 | 0 | 0 | 0.4 | 2.7 |
| 3.4 | Tyr | 13 | 3 | 6 | 13 | 3 | 9 | 4 | 1 | 19 | 6 | 5 | 8 | 2 | 9 | 6 | 6 | 3 |
| | | 1.9 | 0.4 | 0.8 | 1.8 | 0.4 | 1.2 | 0.6 | 0.2 | 0.8 | 1.3 | 0.8 | 1.1 | 0.3 | 1.2 | 0.8 | 0.9 | 0.5 |
| 7.1 | Val | 10 | 15 | <u>1</u> | 17 | 9 | 16 | 21 | 9 | 56 | 4 | 17 | 19 | 10 | 10 | <u>3</u> | 9 | 14 |
| | | 0.7 | 1.0 | <u>0.1</u> | 1.1 | 0.6 | 1.1 | 1.5 | 0.8 | 1.2 | 0.4 | 1.4 | 1.3 | 0.7 | 0.7 | <u>0.2</u> | 0.6 | 1.0 |
| total | | 205 | 209 | 214 | 214 | 214 | 214 | 194 | 160 | 650 | 140 | 174 | 211 | 212 | 212 | 212 | 200 | 197 |
| chain termini | | 9 | 5 | | | | | | | | | | | | | | 12 | 15 |

[a] Reprinted from (Richardson; Richardson. Science), Copyright (1988), p 1648, with permission from The American Association for the Advancement of Science.
[b] Values represent observed occurrence of residues in the specific position out of a number listed at the bottom. The lower entry is the relative preference value: the ratio of observed occurrences to the expected number based on average percentage composition. Boldface values are higher and underscored values lower than expectation by more than three standard deviations.

The positional preferences of amino acids, for either the N- or C-terminus of peptides, have varied when reported on by different groups. The differences in rankings may be due to differences in methods used to determine the ranking. Statistical methods apply different definitions and assumptions, while mutation studies suffer from inherent differences in environments, as well as short- and long-range interactions. Despite the differences, however, certain general observations have been made.

Gly and Pro, known as "helix breakers", are frequently found at the ends of helices. Pro can be better described as a "helix initiator", rather than a "helix breaker", since it is very commonly found at the N1 position of protein helices. Pro may stabilize the first turn of a helix because, with only one rotatable angle, it loses less entropy than other amino acids in forming a regular structure. Presta and Rose[68] noted that in native protein helices, side chains are frequently seen hydrogen bonded to the backbone NH of N2 and especially N3, but never to N1. Since Pro does not have an NH group, it would be ideal in N1.[61] In contrast, Gly has a very strong preference for the C-cap position. Gly at C-cap of a helix can satisfy two successive backbone CO groups while turning the chain in a direction that prevents continuation of the helix.

In general, the stability of a helix is closely correlated to the ability of groups at helix ends to form hydrogen bonds. This is exemplified by the helix "capping box" motif, where two reciprocal main-chain–side-chain hydrogen bonds are formed between the N-cap residue and an amino acid $i + 3$ residues away.[61,68] More recently, Aurora and Rose[80] noted that a hydrophobic interaction is always associated with hydrogen-bonded caps and identified three N-terminal and four C-terminal capping motifs.

#### 12.3.1.1.4 Helix Stabilization by Side-Chain Interactions

Side-chain interactions involving salt bridges, hydrogen bonding, and aromatic interactions have been found by a number of workers to be important for helix stabilization. Bierzynski et al.[81] studied the 13-residue C-peptide [Lys-Glu-Thr-Ala-Ala-Ala-Lys-Phe-Glu-Arg-Gln-His-Hse(lactone)] and found that the helix it forms is at least 1000 times more stable than would be predicted using the then known calculated host–guest s values[17,82] and the Zimm–Bragg equation[83] (which assumes that the s value of an amino acid is independent of its position in the helix). Later studies revealed several side-chain interactions responsible for this stability: the interaction of $(Glu^2)^-$ with the helix dipole,[84] a $(Glu^2)^-$ salt bridge with $(Arg^{10})^+$,[85] and an aromatic interaction between the imidazolium ring of $His^{12}$ and the ring of $Phe^8$.[86] This latter interaction was found to be based on the $i$, $i + 4$ spacing between these residues, and, while it is not specific for the protonated form of His, the aromatic interaction is two times stronger with $His^+$ than with $His^0$.[87] This example underscores the importance of diverse mechanisms of helix stabilization that are concurrently present in protein sequences. It is possible that there are other side-chain interactions still to be identified.

The existence of electrostatic interactions between oppositely charged residues and hydrogen bonding between side chains agrees with the observations in protein helices that (1) helix probability correlates with the frequency of occurrence of oppositely charged residues spaced $i$, $i + 4$ apart in proteins[88] and (2) there is a strong tendency for nearby, oppositely charged, side chains to point toward each other.[89] In the case of C-peptide, the side-chain interactions were also evident in the crystal structure of RNase A.

Studies of de novo designed peptides have explored the effect of different orientations and spacings between oppositely charged residues such as Glu···Lys,[90,91] Asp···Arg and Glu···Arg,[92] His···Asp,[93] and Gln···Asp.[94] Although different host peptides and residue pairings were used, some generalizations could be made: (1) hydrogen bonds between side chains and interactions of charged side chains with the helix dipole both contribute importantly to helix stabilization, (2) preference for $i \leftarrow i + 4$ interactions is greater or equal

*for references see p 789*

to $i \leftarrow i + 3$ interactions, (3) because of helix dipole interactions, A-B orientations are often favored over B-A orientations, where A is an acidic residue and B is a basic residue, and (4) no helix-stabilizing salt bridges are found between $(i, i + 1)$ or $(i, i + 2)$ spaced residues.

### 12.3.1.2 Other Factors Affecting Helix Stability

Helix stability is affected not only by the peptide composition, but by the environment as well. Factors to consider for improving helicity in a given peptide are hydrophobic interactions, solvent, temperature, pH, and ionic strength.

Early studies of peptides from helical regions of proteins failed to detect any helicity in experiments conducted at 25 °C.[95,96] Despite the ensuing expectation that short peptides cannot form stable helices in water, Brown and Klee[97] in 1971 first noted that the C-peptide and S-peptide of RNase A form a partial helix in water at 1 °C. Bierzynski et al.[81] also studied the S-peptide and found α-helix formation at 10 °C. It is now known that helices are more stable at lower temperatures, undergoing thermal unfolding to the unordered coil form as the temperature is raised.

Further studies of the C-peptide showed that a helix is very strongly affected by pH, due to the presence of ionizable side chains. Helical stabilization occurs when favorable interactions can occur between ionized side chains and the helix dipole, and also when ionization can promote hydrogen bonding between side chains. In the case of C-peptide, two ionized groups (with p$K_a$s near 3.5 and 6.5) promote helix formation and helix content follows a bell-shaped curve, with a maximum near pH 5.[81] Increasing concentrations of salt (NaCl) also increases conformational stability.[98]

Following the discovery by Goodman and Listowsky[99] of its helix-enhancing effect, TFE has become the solvent of choice for promoting helix formation in peptides. Addition of up to 30% TFE to a peptide with marginal helix propensity can lead to complete helix formation.[100] The exact mechanism of helix induction by TFE is still unclear, although it has been studied by a number of groups.[101–104] Other alcohols also show helix-stabilizing abilities.[103,105]

The presence of bulky, β-branched side chains can be helix promoting or destabilizing depending on the environment.[100] The role of hydrophobic residues in helix stabilization has been studied in Ala-based peptides[106] as well as through Monte Carlo calculations.[107] The positioning of hydrophobic residues is also important. In amphiphilic helices, hydrophobic residues repeat approximately every three to four residues, such that one side of the helix is hydrophilic and one side hydrophobic. The amphiphilicity makes the peptide susceptible to helix formation in the presence of lipid–water interfaces.[108,109]

Increasing the length of a peptide chain increases helicity. Short peptides exhibit helical character only if they contain more than seven amino acid residues.[110–113] The 13-residue C-peptide of RNase A forms 50–60% helix in 0.1 M NaCl, pH 5.2, at 3 °C[114] and the Baldwin 16-residue Ala-based peptide, in which no side-chain stabilizing interactions are present, forms as much as 80% helix in 1.0 M NaCl, pH 7, at 1.0 °C.[19] Merutka and Stellwagen[115] investigated peptides stabilized by $i \leftarrow i + 4$ Glu···Arg interactions and reported a complete α-helix under optimal helix-forming conditions by increasing peptide length to 27 residues.

The above results have been confirmed experimentally[29,30,116–118] and through calculation studies[108] on model peptides subjected to changes in temperature, pH, salt, and solvent concentrations.

It is now believed that consideration of all these factors in concert can lead to a shift in the helix–coil equilibrium toward the desired helical conformation.

## 12.3.2 Helix Mimetics

Helix induction can be augmented by mimicking natural mechanisms using a variety of synthetic expedients. In general, these approaches seek to either stabilize or mimic helix nucleation sites or to naturally or artificially secure side-chain linkages. In most cases, these transformations make use of additional synthetic procedures beyond standard peptide synthesis methodologies to juxtapose amino acids or essential elements thereof in a manner that mimics a protein helix.

### 12.3.2.1 Synthetic Templates

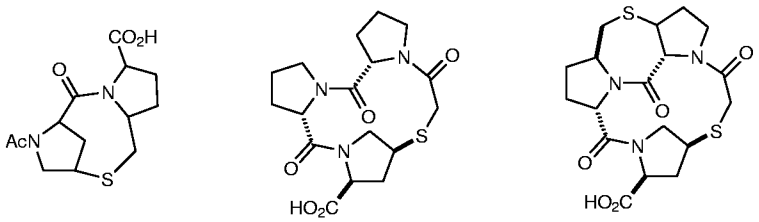
Helix initiation (represented by the helix nucleation parameter σ) in a random coil conformation is the slowest and energetically least favored step, whereas subsequent growth of the helix nucleus (represented by the helix propagation factor, s) is rapid and relatively favored.[83]

The formation of an α-helix can be favored by nucleation with appropriate templates. Starting with a rigid template, three carbonyl groups are positioned in the proper 3D-orientation as would be found in the first turn of an α-helix. Once the orientation of the hydrogen bond acceptors is set, the attached residues are easily induced to fold over the template through the hydrogen bonding between amide groups in the peptide backbone. In such a case, the helix initiation factor takes on a value close to one, instead of the typical $10^{-3}$ to $10^{-4}$ range. The amount of helix induced in the appended peptide then depends only on the helix propagation tendency of the residues in the sequence.[83] It has been demonstrated that attachment of a template to short peptides can lead to significant helix stabilization in TFE (4–6 residues)[119] and water (6–7 residues).[120,121]

#### 12.3.2.1.1 Kemp's Templates: Restricted Analogues of Oligoprolines

Noting that Pro often plays the role of "helix initiator" and is often found at the beginning of helices, Kemp and co-workers have designed a series of helix-nucleating templates based on constrained di-[122–124] or triprolines[125–128] **8–10** (Scheme 3). These templates can be easily incorporated at the N-terminus of any sequence.

**Scheme 3** Kemp's Di- and Triproline Templates[122–128]



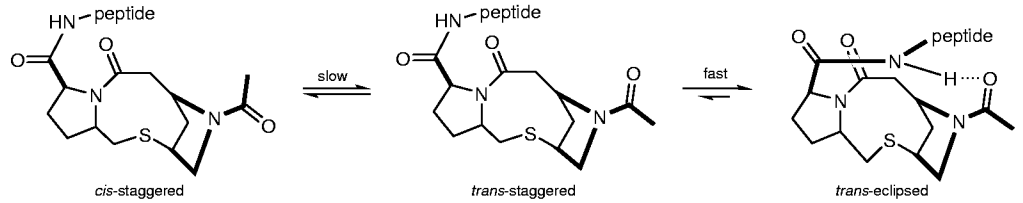

**8** Ac-Hel[1]-OH **9** **10**

The best characterized of these templates Ac-Hel[1]-OH (**8**)[122,123] consists of an Ac-$(Pro)_2$ moiety constrained by a $S{-}CH_2$ bridge between the $Pro^1$ γ-carbon and $Pro^2$ δ-carbon. The template adopts three major conformations in solution corresponding to relative orientations of the acetyl group and the carbonyl bearing the conjugated peptide. Only the *trans*-eclipsed form is helix nucleating. In the crystal state and in a chloroform solution, the template as the free acid exists only as the *cis*-staggered conformation. In other solvents there is an equilibrium between the *cis*-staggered and *trans*-staggered states, and the *trans*-

*for references see p 789*

staggered state is believed to be in rapid equilibrium with the *trans*-eclipsed state (Scheme 4). Changes in the population of the different conformations can be detected by the ratio of integrated intensities in the $^1$H NMR spectrum. As attachment of a peptide to the template induces formation of the *trans* form, the amount of helix in the conjugated peptide is correlated with the ratio of *trans/cis* forms or the ratio of the helix nucleating to the non-nucleating form.

**Scheme 4** Equilibria between Three Conformations of the Diproline Template[123]



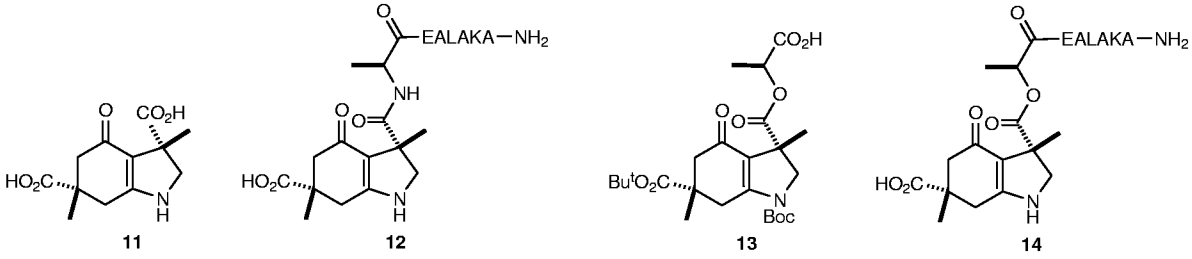

Helicity has been induced into a hexameric Ala peptide by attaching it to **8** [to form Ac-Hel$^1$-(Ala)$_6$-OH].[129,130] The effects of temperature, salts, sodium chloride, urea, and TFE concentration on this resulting helix have been studied in water by CD and NMR.[129] There is a negligible temperature dependence of the *trans/cis* conformational ratio in water over a temperature range of 10–60 °C. Addition of 3 M sodium chloride to water doubles the *trans/cis* conformational ratio, consistent with salt stabilization of the helix dipole. In pure water only 36% of Ac-Hel$^1$-(Ala)$_6$-OH is present in the *trans*-eclipsed form. Increase in TFE concentration from 0 to 10 mol% at 25 °C increases the population of the *trans*-eclipsed state. Further increments of TFE had no significant effect. The CD spectrum of Ac-Hel$^1$-(Ala)$_6$-OH in water was uncharacteristic of a helix, although addition of 20 mol% TFE induced the expected ellipticity at 222 nm. Analysis of coupling constants and nuclear Overhauser effects derived from NMR spectra in water/deuterium oxide (9:1) with and without TFE indicated the presence of an α-helix in the appended peptide, although a rapidly equilibrating $3_{10}$-helix could not be ruled out and fraying at the C-terminus was observed. For an α-helix to form, the amide NH of Ala$_1$ must hydrogen bond to solvent, making way for Ala$_{2-4}$ to hydrogen bond to the template.

**Ac-Hel$^1$-Ala-OtBu; General Procedure for Ac-Hel$^1$-(Ala)$_n$-OtBu (n = 1–6):**[130]
To a soln of Ac-Hel$^1$-OH (**8**;[122] 12.7 mg, 40.7 μmol, 1.0 equiv) in DMF (2 mL) at 0 °C under $N_2$ was added with stirring H-Ala-OtBu (46 μmol, 1.1 equiv) and HOBt (6.5 mg, 1.0 equiv), followed by DIPEA (8.0 μL, 1.1 equiv) and EDC (10.2 mg, 53.1 μmol, 1.3 equiv). After 1 h at 0 °C and 3 h at 23 °C the DMF was removed in vacuo, and the resulting residue was dissolved in $CH_2Cl_2$ (10 mL). The soln was extracted with 0.1 M HCl (3 × 3 mL), sat. $NaHCO_3$ (3 × 3 mL), and brine (1 × 4 mL), and then dried ($MgSO_4$), filtered, and concentrated to give a crude solid [yield: 14.8 mg (85%)] which was purified by flash chromatography ($CHCl_3$/MeOH 19:1) to give Ac-Hel$^1$-L-Ala-OtBu as an amorphous solid; yield: 13.8 mg (80%); MS *m/z*: 425 [M$^+$]; HRMS: calcd for $C_{20}H_{31}N_3O_5S$, 425.1984; found, 425.19842.

### 12.3.2.1.2 Bartlett's Templates: Derivatives of 3,6-Dimethyl-4-oxo-2,3,4,5,6,7-hexahydro-1*H*-indole-3,6-dicarboxylic Acid

Nonpeptidic scaffolds may also be used to position carbonyl groups for the nucleation of helices. In the present case, three successive carbonyl groups are positioned in 3,6-dimethyl-4-oxo-2,3,4,5,6,7-hexahydro-1*H*-indole-3,6-dicarboxylic acid (**11**) (Scheme 5) to mimic those in the first turn of an α-helix.[120] Peptide H-Glu-Ala-Leu-Ala-Lys-Ala-$NH_2$ was attached to **11** through a linking residue and helix induction was examined by CD and NMR.

**Scheme 5** Bartlett's Templates: Derivatives of 3,6-Dimethyl-4-oxo-2,3,4,5,6,7-hexahydro-1*H*-indole-3,6-dicarboxylic acid[120]




N-Terminal peptide attachment to give **12** showed little helix formation due to the possible onset of a nonproductive hydrogen bond from the linking amide group to the template. On the other hand, when the peptide was attached to the ester-linked *S,S*-derivative **13** and deprotected to give **14**, strong helix induction was observed. In this case, the ester group cannot hydrogen bond to the template, making way for the following residues of the appended peptide to hydrogen bond to the template. CD spectroscopy showed **14** to be 32–50% helical at 23 °C in pH 5.2 aqueous buffer and 49–77% at 0 °C, compared to only 8% at 0 °C for the nontemplated control peptide. Addition of 50% TFE gave maximum helical character, which was only slightly better than that in aqueous buffer.

The observed ellipticities below 250 nm are characteristic of an α-helix and different from a $3_{10}$-helix, according to the distinction described by Toniolo et al.[131] The CD spectrum indicates that two turns of a helix can be readily detected by this measure, in contrast to previous expectations.[132,133] The molar ellipticity shows no temperature dependence. It is, however, affected by pH as protonation of the template carboxylate and the Glu side chain leads to loss of the favorable dipole interactions and salt bridges, respectively.

NMR characterization of **14** in water/deuterium oxide (9:1), at 4 °C, pH 4.2 confirmed a helical conformation by the observation of one of four possible $d_{\alpha N}(i, i + 3)$ NOEs. Unfortunately, the difficulty posed by overlapping of β-carbon protons prevented further characterization of other medium-range NOE connectivities. Low temperature coefficients over a temperature range of 4–62 °C and coupling constants close to 4 Hz (4 °C) further established substantial helical character in the appended peptide. At 25 °C, fraying at the C-terminus is noted, consistent with thermal unfolding with increased temperatures.

**[(3*S*,6*S*)-3,6-Dimethyl-4-oxo-2,3,4,5,6,7-hexahydro-1*H*-indole-6-carboxylic acid]-L-Lactyl-Glu-Ala-Leu-Ala-Lys-Ala-NH$_2$; General Procedure for the Attachment of a Peptide to 2-({[6-(*tert*-Butoxycarbonyl)-3,6-dimethyl-4-oxo-2,3,4,5,6,7-hexahydro-1*H*-indol-3-yl]carbonyl}oxy)propanoic Acid:**[120]

To a soln of the methyl ester of **13**[120] (0.032–0.114 mmol) in 25% aq MeOH (1–2 mL) was added 1 M LiOH (1.2 equiv). The resulting mixture was stirred at rt for 1–2 h and then concentrated under reduced pressure. The residue was dissolved in $H_2O$ (15 mL) and carefully acidified to pH 2 with 1 M HCl. The resulting soln was extracted with $CH_2Cl_2$ (3 × 15 mL), and the combined organic fractions were dried ($MgSO_4$) and concentrated in vacuo to give the monocarboxylic acid; yield 61–88%.

Fmoc-EALAKA-Rink-resin (1.25 equiv) was allowed to swell in DMF (1.5 mL) for 1 h. The DMF was removed by filtration and the resin washed with additional DMF (1.5 mL). The resin was then treated with 20% piperidine/DMF (2 × 1.5 mL) for 10 min to remove the Fmoc protecting group, filtered, and washed with additional DMF (10 × 1.5 mL). A soln of the carboxylic acid prepared above, EDC (1 equiv), and HOBt (1 equiv) in DMF (1.5 mL) was added to the resin and the slurry was stirred for 1 h. The resin was filtered, washed with DMF (2 × 1.5 mL) followed by $CH_2Cl_2$ (2 × 1.5 mL), and stored in vacuo overnight.

*for references see p 789*

The resin was treated with cleavage reagent R [TFA/PhSMe/EDT/PhOMe (9:5:3:2, 5 mL)] under $N_2$ for 2.5 h. The mixture was filtered through glass wool into cold $Et_2O$, and the soln was stored at 0 °C overnight. The fluffy precipitate that formed was collected on a fine frit, washed with cold $Et_2O$, and stored in vacuo overnight. The peptide conjugate was purified by RP-HPLC using a two-solvent system [solvent A: $H_2O$/MeCN/TFA (90:10:0.1), solvent B: $H_2O$/MeCN/TFA (10:90:0.1)] with programmed elution (0 min, 100% A; 30 min, 50% A; 40 min, 100% B; 50 min, 50% B; 60 min, 100% A) on a Vydac C18 preparative column, monitored at 304 nm. FAB-HRMS: calcd for $C_{41}H_{66}N_9O_{14}$, 907.4729; found, 907.4725.

### 12.3.2.1.3 Tetrahydronaphthalene-Based Templates

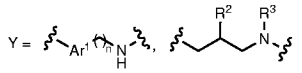
Tetrahydronaphthalene-based compounds (Scheme 6) have been reported as N- and C-cap templates that induce a 30–40% increase in helix content in attached nine-residue peptides.[134,135] CD studies show that whereas Boc-$(Ala)_2$-Aib-LKAEIAQKL-$NH_2$ exhibits 50% helix in water, **15**-LKAEIAQKL-$NH_2$ exhibits 87% helix.

There are few examples of C-terminal helix mimetics and, while their utility has been considered uncertain,[136] successful helix induction has been described using tetrahydronaphthalene compounds represented by **16**.[135] Templated peptide **17**, for example, demonstrates 81% helix by CD while control peptide **18** exhibits only 50% helix under the same conditions (Scheme 7). For the preparation of tetrahydronaphthalene templates and their attachment to peptides, see refs.[134,135]

**Scheme 6** N- and C-cap Templates: Tetrahydronaphthalene Compounds[134,135]

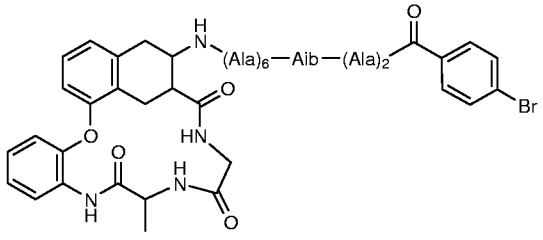


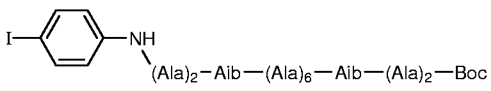

$R^1$ = polypeptide; X = O, S; $Ar^1$ = aromatic ring




n = 0, 1; $R^2$ = $R^3$ = H, alkyl

**Scheme 7** Templated and Control Peptides[135]

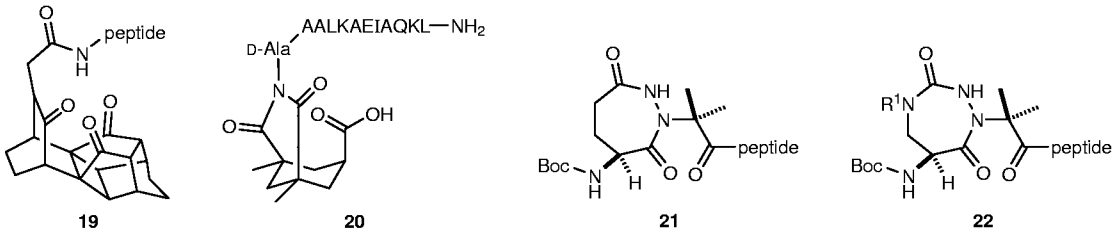


#### 12.3.2.1.4 Other Templates

Templates other than Kemp's, Barlett's, and tetrahydronaphthalene-based templates (Sections 12.3.2.1.1–12.3.2.1.3) have been reported for helix induction. Müller et al.[137] have described cage compound **19** for use as a N-terminal helix initiator (Scheme 8). Template **19** is a rigid molecule that locks three carbonyl groups in the required orientation for hydrogen bonding to an appended peptide.[137] CD spectroscopy shows an increase in helicity from 40% in an *N*-Boc-protected peptide to 70% in the corresponding templated nine-residue peptide H-Ala-Aib-$(Ala)_4$-Aib-$(Ala)_2$-NH-$C_6H_4I$. X-ray diffraction crystallography confirmed an α-helix. Similarly, attachment of Kemp's triacid derivative **20** to the N-terminus of H-D-Ala-AALKAEIAQKL-$NH_2$ is reported to give an α-helix-like CD spectrum, indicating nearly 60% helicity in water.[137] Compounds **21** and **22** have been reported to be both N- and C-terminal templates.[138] Synthetically very simple N-terminal templates based on $C^{\alpha}$-methylated α-amino acids enhance the helicity of peptides to a small, but significant degree in water.[139]

**Scheme 8** Other Templates[137,138]




**19** **20** **21** **22**

### 12.3.2.2 Covalent Main-Chain Links

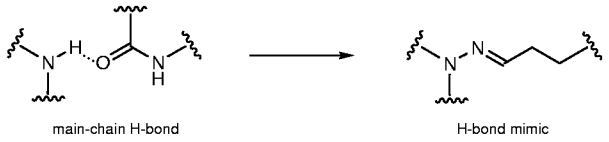
Most protein conformations are formed as a result of hydrogen bonding between main-chain peptide bonds.[140] In helices these bonds are both stabilizing and structure defining. Intramolecular hydrogen bonds, however, compete with water which disrupts the structure. Short peptides that do not have the benefit of long-range stabilizing interactions, such as those present in proteins, normally undergo rapid equilibrium between folded and unfolded states. These arise from the almost simultaneous formation and unraveling of hydrogen bonds. For this reason, covalent hydrogen bond mimics have been conceived and developed.[141,142] Using these mimics, peptides have been constrained to various stable conformations such as helices[121,142] and loops,[142] simply by variation of the position of the mimic in the amino acid sequence.

#### 12.3.2.2.1 Hydrogen Bond Mimic: α-Helix Nucleation from the N-Terminus

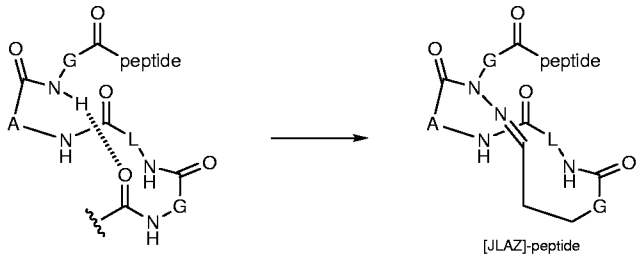
The best developed hydrogen bond mimic is the hydrazone link (Scheme 9), which can be inserted into peptides during solid-phase synthesis.[121] In α-helices, substitution of the N-terminal $i \leftarrow i + 4$ hydrogen bond (N–H···O=C–NH) with the hydrazone link (N–N=CH–$CH_2$–$CH_2$) leads to a 13-membered ring, corresponding to the basic unit of an α-helix (Scheme 10).[5] The result is an α-helix nucleation site (NucSite) which, analogous to templates, increases the value of the helix initiation parameter σ, thereby allowing the peptide to overcome the highly unfavorable entropic barrier to helix formation. Nucleation sites conformationally restrict an extra main-chain carbonyl oxygen by replacing it with the link, thereby assuring that the only stable conformation that can be propagated is an α-helix.

*for references see p 789*

**Scheme 9** Substitution of a Main-Chain Hydrogen Bond with a Hydrazone Covalent Hydrogen Bond Mimic[121]

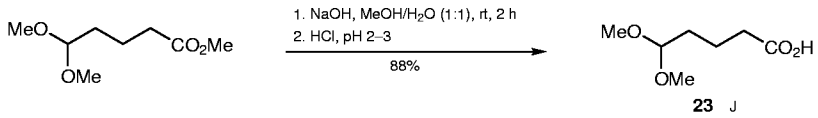



**Scheme 10** An $i \leftarrow i + 4$ Hydrogen Bond is Replaced with the Hydrazone Link To Form an α-Helix Nucleation Site[5]

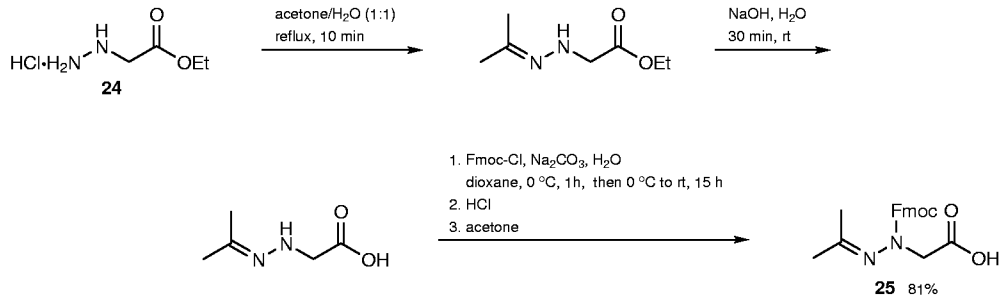



The hydrazone link is formed by the reaction of two linkers, J (5,5-dimethoxypentanoic acid, **23**) and [1-(9-fluorenylmethoxycarbonyl)-2-(isopropylidene)hydrazino]acetic acid (**25**). J (**23**) is obtained by hydrolyzing the corresponding methyl ester, which is commercially available or can be prepared by following the method of Stevens and Lee[143] (Scheme 11). Compound **25** is prepared by protecting commercially available ethyl hydrazinoacetate hydrochloride (**24**) (Scheme 12).[121]

**Scheme 11** Synthesis of Linker J[121]



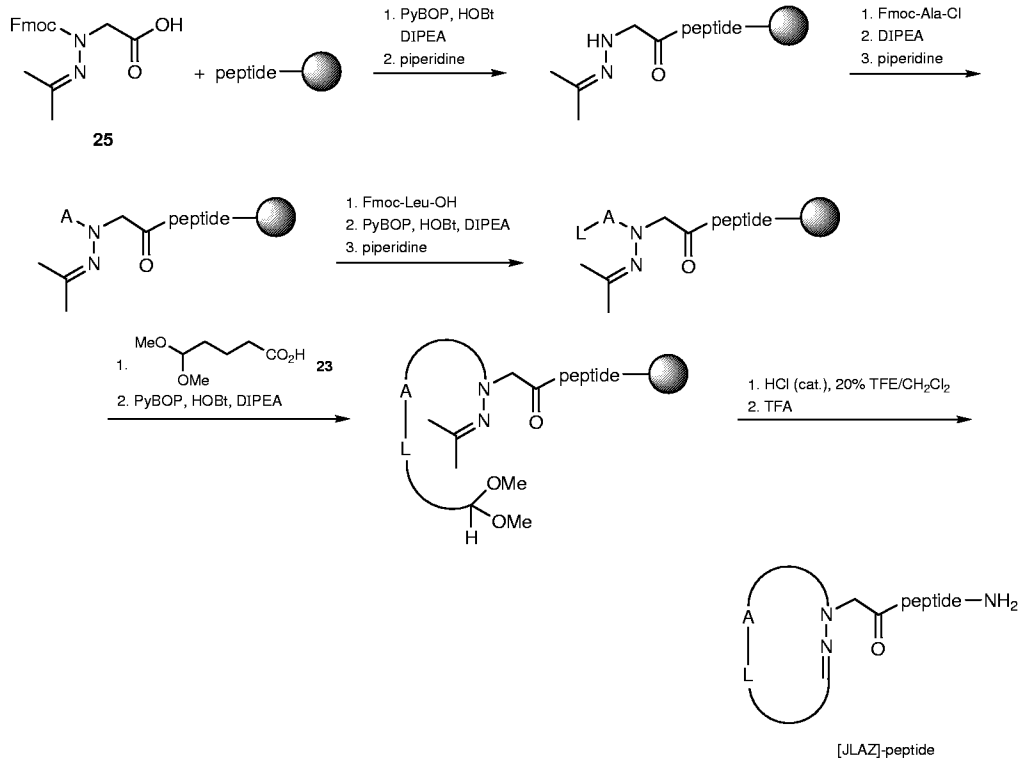

**Scheme 12** Synthesis of Linker [1-(9-Fluorenylmethoxycarbonyl)-2-(isopropylidene)hydrazino]acetic Acid[121]




A solid-phase method has been devised for incorporating J (**23**) and [1-(9-fluorenylmethoxycarbonyl)-2-(isopropylidene)hydrazino]acetic acid (**25**) into a peptide using standard Fmoc chemistry (Scheme 13).[121] For α-helix nucleation, J (**23**) and **25** are introduced at the N-terminus $i, i + 4$ apart. An Fmoc amino acid chloride is used for coupling to **25**. Nucleation sites are formed on the solid phase in a rapid cyclization reaction with a catalytic amount of hydrogen chloride in 20% TFE/$CH_2Cl_2$ as solvent. After cyclization, the nucleated peptide is cleaved from the resin with TFA. The resulting nucleated peptide is stable for long periods in water at physiological pH and temperature.[121]

When the hydrazone link is substituted for a putative $i \leftarrow i + 4$ hydrogen bond in essentially random coiled Ac-GLAGAEAAKA-$NH_2$ to form [JLAZ]-AEAAKA-$NH_2$, depicted in Scheme 10, it induces a substantial α-helix in water at 22 °C.[121] The α-helix extends through the full length of the constrained peptide as demonstrated with 2D NMR experiments using deuterated Ala analogues.[121] Nucleation sites can encompass different amino acids;[121,144] [JLPZ] induces about 50% more α-helix than [JLAZ].[121]

**Scheme 13** Solid-Phase Synthesis of a Nucleation Site Peptide[121]




**5,5-Dimethoxypentanoic Acid (J, 23):**[121]

J was prepared by adding methyl 5,5-dimethoxypentanoate[144] (10 g, 56.8 mmol) to a soln of NaOH (3.028 g, 75.7 mmol) in MeOH/$H_2O$ (1:1, 100 mL) and stirring at rt for 2 h. The mixture was concentrated, the residue dissolved in a mixture of $H_2O$/EtOAc (1:2, 150 mL) and cooled to 0 °C. The mixture was vigorously stirred while 1 M HCl was added until the mixture was at pH 2–3. The organic and aqueous layers were immediately separated and the aqueous layer extracted with ice-cold EtOAc (100 mL). The combined organic layers were washed with ice-cold brine (100 mL), dried ($MgSO_4$), filtered, and concentrated to give a colorless oil; yield 8.2 g (88%); Anal. Calcd for $C_7H_{14}O_4$: C, 51.8; H, 8.7. Found: C, 50.6; H 8.9; FAB-MS *m/z*: $[M + Na]^+$ calcd, 185.0790; found, 185.0793. J is stable for > 1 year at −20 °C.

**[1-(9-Fluorenylmethoxycarbonyl)-2-(isopropylidene)hydrazino]acetic Acid [Fmoc-Z(Isopropylidene)-OH, 25]:**[121]

Ethyl hydrazinoacetate hydrochloride (**24**, 15.4 g, 100 mmol) was dissolved in acetone/$H_2O$ (1:1, 300 mL). The mixture was refluxed for 10 min and then cooled to rt. NaOH (8.4 g, 210 mmol) was added and the mixture stirred at rt for 30 min to hydrolyze the ester. $Na_2CO_3$ (10.6 g, 100 mmol) was dissolved in $H_2O$ (50 mL) and added to the mixture which was then cooled to 0 °C. A soln of Fmoc-Cl (25.9 g, 100 mmol) in dioxane (50 mL) was added dropwise with stirring to the mixture over a period of 1 h. Stirring was continued for 15 h with warming to rt. The mixture was concentrated and the residue suspended in ice-cold EtOAc (200 mL). This mixture was then vigorously stirred and acidified to pH 3 or lower by adding 1 M HCl. Proceeding rapidly to minimize acid-catalyzed hydrolysis of the propylidene linkage, the two phases were separated and the aqueous phase extracted with EtOAc (200 mL). The EtOAc extracts were combined, extracted with brine, and dried ($MgSO_4$). After filtration and con-

*for references see p 789*

centration to dryness, the product was dissolved in acetone (100 mL), refluxed for 10 min, and again concentrated to dryness. Crystallization (EtOAc/hexane) gave **25**; yield: 31.4 g (81%); mp 146–148 °C; Anal. Calcd for $C_{20}H_{20}N_2O_4$: C, 68.17; H, 5.72; N, 7.95. Found: C, 67.97; H, 5.71; N, 7.85; FAB-MS: $[M+H]^+$ calcd, 353.1422; found, 353.1511. Compound **25** is stable when stored dry at 4 °C.

**JLAZ(Isopropylidene)-Ala-Glu(tBu)-Ala-Ala-Lys(Boc)-Ala-Rink-Resin:**[121]
This compound was assembled on Rink resin (100 mg, 0.046 mmol) sealed in a 2 × 2 cm polypropylene mesh packet.[121,145,146] The packet was placed in a 30-mL Nalgene wide mouth polypropylene bottle for washings and all reaction steps except for the coupling steps, which were carried out in a capped 2-mL microtube. Wash solns were added to the packet in the bottle, shaken for 1 min on a platform shaker, and poured off. Reaction solns were shaken with the resin packet for the prescribed time and poured off. A standard coupling cycle involved the following steps: (1) The resin was swollen in its packet by shaking with $CH_2Cl_2$ (5 mL) and then was washed with DMF (5 mL). (2) The Fmoc protecting group was cleaved from the protected peptide-resin with 20% piperidine/DMF (5 mL) for 10 min. The resin was washed with DMF (4 × 5 mL), iPrOH (5 mL), $CH_2Cl_2$ (5 mL), and NMP (5 mL). (3) All coupling reactions with one exception were carried out by adding Fmoc amino acid (0.092 mmol), Fmoc-Z(isopropylidene)-OH (**25**, 32.5 mg, 0.092 mmol), or J (17 mg, 0.092 mmol), and HOBt (12.4 mg, 0.092 mmol), PyBOP (47.9 mg, 0.092 mmol), and DIPEA (0.032 mL, 0.184 mmol) in NMP (1 mL) to the peptide-resin and shaking for 30 min. Fmoc-Z(isopropylidene)-OH (**25**) was coupled to the peptide-resin using the protocol just described. After coupling, the resin was washed with $CH_2Cl_2$ (2 ×) and air-dried. A small amount of resin was removed from the resin packet and tested for primary amine by the Kaiser ninhydrin test. The coupling step was repeated if necessary. The exception was for the coupling (step 3) of Fmoc-Ala-Cl[147] to Z(isopropylidene)-peptide. Fmoc-Ala-Cl was dried under vacuum before use to remove traces of HCl which would hydrolyze the Z(isopropylidene) moiety. The order of addition is critical for this reaction. Fmoc-Ala-Cl (30 mg, 0.092 mmol) was dissolved in NMP (1 mL), added to the resin packet, and briefly shaken before adding DIPEA (0.016 mL, 30.4 mg, 0.092 mmol). Shaking was continued for 15 min. The Kaiser ninhydrin test is not applicable for this reaction. Following the addition of J, the resin packet was washed with $CH_2Cl_2$ (4 × 5 mL) and air-dried.

**[JLAZ]-Ala-Glu-Ala-Ala-Lys-Ala-$NH_2$:**[121]
For cyclization, the dry peptide-resin packet from the previous step was shaken for 15 min in a soln of 20% TFE/$CH_2Cl_2$ (15 mL) that had been premixed with a small quantity of 4 M HCl in dioxane (0.02 mL, 5.3 mM HCl final). The resin packet was washed with $CH_2Cl_2$ (4 × 5 mL). The slight pink coloring of the resin was used as an indication that sufficient acid had been added for cyclization. To cleave the peptide from the resin, the cyclic peptide-resin was washed with $CH_2Cl_2$ (3 ×), dried, then added to 10% TFA/$CH_2Cl_2$ (15 mL) in a capped polypropylene bottle and shaken for 15 min. The soln containing the cleaved protected peptide was removed and the packet treated again in the same manner. The acidic solns were combined and concentrated to leave a residue that was treated with TFA (2 mL) for 15 min. The resin packet was also treated with TFA (2 mL). (For peptides with protecting groups, the cleavage can generally be carried out with 95% TFA/$H_2O$: the addition of free thiol scavengers and silanes should be avoided since they destroy the hydrazone link.) The peptide was precipitated from the combined acid solns by adding cold $Et_2O$ (30 mL). The precipitate was centrifuged, washed with cold $Et_2O$, dried, dissolved in 10% MeCN/$H_2O$ (1 mL), and lyophilized.

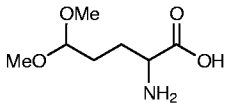
The crude product was analyzed by HPLC on a Merck Lichrosorb analytical C18 column (10 μm, 4.6 × 250 mm). Products were eluted with a 100% $H_2O$ to 100% MeCN gradient in 0.1% TFA over 15 min at 2 mL·min$^{-1}$. The major product was purified by HPLC on a Vydac preparative C18 column (201TP1022, 2.2 × 25 cm, 10 μm) by eluting with a 100% $H_2O$ to 60% MeCN gradient in 0.1% TFA over 40 min at 5 mL·min$^{-1}$. Fractions were lyophilized to give the product; yield: 19.2 mg (47%, based on initial resin loading). The product was identified by 2D NMR spectroscopy[121] and FAB-MS: $[M+H]^+$ calcd, 896; found, 896; FAB-HRMS *m/z*: $[M+Cs]^+$ calcd, 1027.3977; found, 1027.4019. An NMR peak diagnostic of nucleation site formation is the signal for the hydrazone proton $^1H$ NMR (10% $D_2O/H_2O$, pH 2.9, 22 °C, δ): 7.18 (dd, 1H).

### 12.3.2.2.2 N-Terminal Extension of a Nucleated Helix

To expand the utility of the hydrogen bond mimic, J (**23**) has been modified to allow for extension from the N-terminus of a hydrazone-linked cyclic peptide.[148] This allows not only the use of the hydrogen bond mimic in preparing internal loops, but also the positioning of a helix nucleation site between two peptides to form supersecondary structures. To accomplish this, an α-amino group can be added to J to give J′ (**26**) (Scheme 14). Incorporation of linker J′ into a nucleation site using the procedure applied to **23** allows it to mimic an N-cap formed

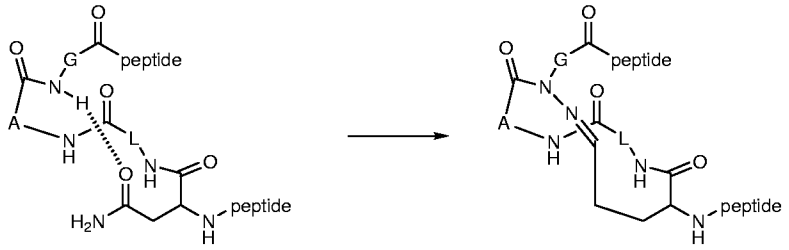
between the side-chain carbonyl oxygen of Asn and the main-chain amide proton three residues along the chain, $i \leftarrow i + 3$, which frees the N-terminal amino group for extension (Scheme 15).[148] The preponderance of Asn at the N-cap position engaged in hydrogen bonding with main-chain amide protons has been reported by several groups.[61,69] Forming Ac-[J′LAZ]AEAAKA-$NH_2$ results in helix induction in the appended peptide showing that an extension from the N-terminus of a nucleation site is feasible.[148] The procedure for J′ synthesis remains to be published and more detailed NMR studies remain to be done.

**Scheme 14** Structure of J′




**26** J′

**Scheme 15** α-Helix Nucleation between Two Peptides[148]




## 12.3.2.3 Covalent Side-Chain Links

In contrast to nucleation at the helix N-terminus via templates and nucleation sites, helix stabilization has been achieved by various researchers through macrocyclization. In this method, covalent cross-links between two side chains or a side chain and the N- or C-terminus are introduced to stabilize the desired helical conformation. Such links include amide bonds and disulfides formed between residues positioned $i$, $i + 4$ or $i$, $i + 7$ residues apart. Although side-chain linkage results in large rings, N-terminal as well as C-terminal extensions can contribute to the stabilization of a helix and propagate the helix in both directions. In addition, several groups have utilized multiple side-chain cross-links, arranged sequentially or in overlapping fashion, to improve stabilization.[149–153] Side-chain links block one face of the helix which should be taken into account when preparing helical peptides for receptor binding.

### 12.3.2.3.1 Lactams

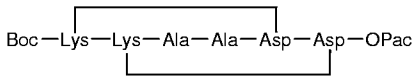
Felix et al.[154] were the first to establish side-chain-to-side-chain cyclization of a peptide on solid phase by lactamization. Lactam bridging of side chains of Lys and Glu or Asp have since been shown to stabilize an α-helical conformation in analogues of human growth hormone releasing factor[155] and parathyroid hormone,[156] amongst others. Osapay and Taylor[149,150] expanded the utility of the method by inserting three lactam bridges in a 21-residue peptide to achieve very high helix induction. Three heptads of sequence Lys-Leu-Lys-Glu-Leu-Lys-$Xaa^1$ ($Xaa^1$ = Glu, Asp) were prepared and cyclized on Kaiser oxime resin, then subjected to solution-phase segment condensations to obtain the H-(Lys-Leu-Lys-Glu-Leu-Lys-$Xaa^1$)$_3$-OH peptide cyclized between the 3–7, 10–14, and 17–21 residues. CD measurements in aqueous phosphate buffer gave calculated helix values of 69% when $Xaa^1$ = Asp, 21% when $Xaa^1$ = Glu, and 13% for the corresponding uncyclized peptide.

*for references see p 789*

Measurements in 50% TFE showed almost identical spectra for the cyclic peptides (88% helix), while the linear peptide remained significantly less helical. It has been concluded that linking the side chains of $Lys^{i}$ and $Asp^{i+4}$ is strongly helix stabilizing, while linking $Lys^{i}$ and $Glu^{i+4}$ is weakly helix stabilizing. In another application, lactam bridges had been placed in the 14-residue peptide Ac-EXEALKKEXEALKK-$NH_2$ between residues 3–7 and 10–14, where X was substituted for Ile, Val, and Ala.[151] X substitutions had been made to determine the effect of decreasing the hydrophobicity of the nonpolar face on helical content and stability. The lactam bridges positioned between residues could engage in salt bridging in the linear analogue. It has been found that in all cases peptides with covalent side-chain links are substantially more helical. Increase in the hydrophobicity of the nonpolar face of a helix leads to greater dimerization of the peptide. NMR analysis conducted in $H_2O/D_2O$ (9:1) showed a predominantly α-helical conformation in the lactam-bridged peptides, although weak $d_{\alpha N}(i, i+2)$ NOEs have been noted, indicative of a possible shift to a $3_{10}$-helix at the C-terminus.[151]

Insertion of two overlapping $i$, $i + 4$ lactam bridges in the terminally protected hexapeptide Boc-Lys-Lys-Ala-Ala-Asp-Asp-OPac results in a highly conformationally constrained structure **27** (Scheme 16).[152] CD spectra showed a rigid structure that exhibited only partial melting at high temperature, while NMR studies in TFE-$d_3$/$H_2O$ revealed a predominantly α-helical structure, as judged by observed NOEs.

**Scheme 16** A Terminally Protected Hexapeptide with Two Lactam Bridges[152]




**27**

The effect of ring size, spacing, and positioning of lactam bridges has been studied by Hodges and co-workers[157] using a series of 14-residue peptides. Glu and Lys spaced $i$, $i + 4$ apart gives better helix induction than a spacing of $i$, $i + 3$. A lactam bridge with a Glu···Lys orientation leads to greater helix induction than Lys···Glu lactams in aqueous phosphate solutions, although both could be induced to form 90% helices in 50% TFE.

**$c^{3,12}$[$Ala^{15}$]-GRF(1–29)-$NH_2$; General Procedure for Lactam Bridge Formation between Side Chains:[154] Coupling Protocol for the Solid-Phase Peptide Synthesis:[154]**

Peptides were prepared by SPPS carried out on a Vega Model 1000S semi-automated synthesizer. One cycle of the synthesis consisted of the following operations (15–20 mL solvent per gram of resin): (1) 1% DMS/$CH_2Cl_2$ (1 × 1 min); (2) 50% TFA/$CH_2Cl_2$ with 1% DMS (1 × 1 min); (3) 1% DMS/$CH_2Cl_2$ (1 × 1 min); (4) 50% TFA/$CH_2Cl_2$ with 1% DMS (1 × 20 min); (5) $CH_2Cl_2$ (2 × 1 min); (6) MeOH (1 × 1 min); (7) $CH_2Cl_2$ (1 × 1 min); (8) 10% DIPEA/$CH_2Cl_2$ (1 × 5 min); (9) $CH_2Cl_2$ (2 × 1 min); (10) MeOH (2 × 1 min); (11) $CH_2Cl_2$ (2 × 1 min); (12) repeat steps 8–10; (13a) addition of Boc-Xaa-OH/$CH_2Cl_2$ (2.5 equiv, 5 min); (13b) addition of DCC/$CH_2Cl_2$ (2.5 equiv, 60 min); (13c) addition of 1.5% DIPEA/$CH_2Cl_2$ (15 min); (14) $CH_2Cl_2$ (1 × 1 min); (15) repeat steps 13a–c and 14; (16) MeOH (2 × 1 min); (17) $CH_2Cl_2$ (2 × 1 min). One exception to this protocol was the coupling with Boc-Arg-(Tos)-OH, which was carried out in DMF/$CH_2Cl_2$.

**Preparation of Boc-Arg(Tos)-BHA-Resin:[154]**

BHA resin (60 g, 0.7 mequiv·$g^{-1}$, 42 mequiv) was washed with $CH_2Cl_2$ (900 mL), MeOH (900 mL), $CH_2Cl_2$ (900 mL), and 25% TEA/$CH_2Cl_2$ (900 mL) three times. Boc-Arg(Tos)-OH (35.95 g, 84 mmol, 2 equiv) was added and reacted for 24 h. The resultant Boc-Arg(Tos)-BHA-resin was washed with DMF, $CH_2Cl_2$, MeOH, and $CH_2Cl_2$. Amino acid analysis of an aliquot revealed a substitution of 0.342 mmol·$g^{-1}$. The resin was acetylated for 2 h with 50% $Ac_2O$/pyridine (300 mL) at 25 °C, washed with $CH_2Cl_2$, MeOH, and then again with $CH_2Cl_2$, and dried in vacuo to give Boc-Arg(Tos)-BHA-resin; yield: 70 g (49%).

**Preparation of [Asp$^3$(OFm),Lys$^{12}$(Fmoc),Ala$^{15}$]-GRF(1–29)-BHA-Resin:**[154]
After deprotection and neutralization of the Boc-Arg(Tos)-BHA-resin (30 g, 0.342 mmol·g$^{-1}$, 0.26 mmol), the next residue, Boc-Ser(Bzl)-OH, was coupled according to the protocol described above. The resultant peptide-resin (32.43 g) was divided and the synthesis was continued with a 20.62-g portion (0.332 mmol·g$^{-1}$, 6.84 mmol). The other trifunctional amino acids were protected as follows: Boc-Arg(Tos), Boc-Asp(OCy), Boc-Glu(OBzl), Boc-Lys[Z(2-Cl)], Boc-Ser(Bzl), Boc-Thr(Bzl), Boc-Tyr[Bzl(2,6-Cl$_2$)], Boc-Asp$^3$(OFm), and Boc-Lys$^{12}$(Fmoc). Exceptions to the protocol were for the coupling of Boc-Gln-OH (positions 24 and 16) and for the coupling immediately following Gln (positions 23 and 15) in which the symmetrical anhydride (3 equiv) was used in DMF (double coupled). In special cases, in which a third coupling was required, the HOBt ester was prepared using Boc-Xaa-OH (4 equiv), HOBt (4.36 equiv), and DCC (4 equiv) preactivated in DMF (50 mL) at 0 °C for 30 min, filtered (removal of DCU), and diluted with toluene to 500 mL. Coupling proceeded for 2 h and then 1.5% DIPEA was added and the mixture reacted for 15 min longer.

The intermediate, Boc-[Ala]$^{15}$-GRF(13–29)-BHA-resin was dried in vacuo (38.25 g, 6.84 mmol) and a portion (9.63 g, 1.72 mmol) was removed and stepwise solid-phase synthesis continued as described in the protocol above. After the incorporation of Lys$^{12}$(Fmoc), step 8 was replaced with 0.6% DIPEA/$CH_2Cl_2$. The coupling with Boc-Asn-OH (2.63 g, 6 equiv) was carried out by preactivation with HOBt (1.53 g, 6.6 equiv) and DCC (2.34 g, 6 equiv). The intermediate Boc-[Asp$^3$(OFm),Lys$^{12}$(Fmoc),Ala$^{15}$]-GRF(3–29)-BHA-resin (12.0 g, 1.72 mmol) was dried in vacuo and a portion (7.75g, 1.11 mmol) was removed for the last two cycles of the synthesis.

**Preparation of $c^{3,12}$[Ala$^{15}$]-GRF(1–29)-NH$_2$:**[154]
Following the assemblage of Boc-[Asp$^3$(OFm),Lys$^{12}$(Fmoc),Ala$^{15}$]-GRF(1–29)-BHA-resin, a portion of the resin (1.70 g, 0.243 mmol) was washed with $CH_2Cl_2$ (1 × 1 min), and DMF (1 × 1 min), deprotected with 20% piperidine/DMF (1 × 1 min and 1 × 20 min), and washed with DMF (1 × 1 min), MeOH (1 × 1 min), $CH_2Cl_2$ (1 × 1 min), and MeOH (1 × 1 min) to give Boc-[Asp$^3$,Lys$^{12}$,Ala$^{15}$]-GRF(1–29)-BHA-resin. A portion (250 mg, 35.7 μmol) was cyclized by reaction with BOP reagent (47.5 mg, 107 μmol, 3.0 equiv) in DMF (20 mL) containing 1.5% DIPEA (6 equiv) for 2 h. A negative Kaiser ninhydrin test was observed and the peptide-resin was washed and dried. A portion (200 mg, 28.6 μmol) was cleaved with HF (~5 mL) containing EDT (200 μL) at 0 °C for 2 h. Removal of HF was followed by washing with EtOAc, extraction with TFA (3 × 5 mL), concentration, and trituration with $Et_2O$ to give the crude product; yield: 112 mg. It was determined that the crude product contained 19 mg of $c^{3,12}$[Ala$^{15}$]-GRF(1–29)-NH$_2$; yield: 14% (based on analytical HPLC).

The crude product (75 mg) was suspended in 0.5% TFA/$H_2O$ (5 mL), stirred, centrifuged, filtered, and applied on two ODS-3 columns (1 × 50 cm) connected in series; eluants: (A) $H_2O$ (0.5% TFA) and (B) MeCN (0.25% TFA); linear gradient 15–45% B in 120 min, flow rate 4 mL·min$^{-1}$. Fractions were collected at 1-min intervals. Fractions 120–122 were pooled and lyophilized to give pure product (5.6 mg); yield: 7.4 mg (8.2%). The product was shown to be homogeneous by analytical HPLC using a Lichrosorb RP-8 column; eluant 0.1 M $NaClO_4$ (pH 2.5)/MeCN. Amino acid analysis (6 M HCl, 110 °C, 72 h): Asx, 3.00; Thr, 1.04; Ser, 2.96; Glx, 2.06; Ala 4.13; Val, 1.02; Met, 0.96; Ile, 1.92; Leu, 4.07; Tyr, 2.04; Phe, 0.97; Lys, 1.95; Arg, 2.88. FAB-MS *m/z*: calcd for $C_{150}H_{246}N_{44}O_{41}S$, 3354.9; found, 3355.3.

### 12.3.2.3.2 Disulfides

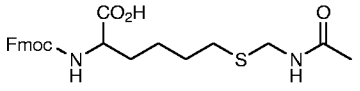
It is generally recognized that disulfides are an important structure-defining feature of proteins. The utility of disulfides in stabilizing the helical conformation has been demonstrated by Pease et al.[158] by making hybrids of the bee venom peptide apamin and S-peptide from RNase A. Apamin contains two disulfide bridges which stabilize an α-helix. The structure of apamin has been used as a scaffold to induce helicity in S-peptide by replacing the non-Cys residues of apamin with the sequence of S-peptide. Examination of the activity of the hybrid peptide showed that disulfides stabilize the helical segments of S-peptide completely, with 2D NMR showing that the hybrid adopts a stable helical conformation identical to that of apamin.

Cys residues spaced *i*, *i* + 3 apart have been covalently linked to obtain cyclic peptides which fold into a $3_{10}$-helix. Balaram et al.[159] synthesized Boc-*c*[-Cys-Pro-Aib-Cys-]-NHMe, wherein a disulfide bond links the two Cys residues. Pro and Aib residues were used to restrict conformational flexibility. NMR and X-ray diffraction studies revealed a highly folded, compact formation of an incipient $3_{10}$-helix, with the S–S bridge lying approximately parallel to the helix axis.

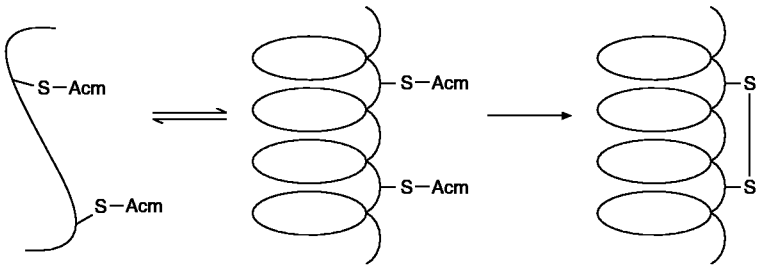
*for references see p 789*

Jackson et al.[160] used a disulfide link bridging *i* and *i* + 7 residues to stabilize two turns of an α-helix. Instead of Cys residues, D- and L-forms of *S*-6-[(acetamidomethyl)sulfanyl]-*N*-(9-fluorenylmethoxycarbonyl)norleucine (**28**) (Figure 2) can be used. The D- and L-enantiomers of **28** are synthesized via conversion of the ε-amino group of D- and L-Boc-Lys-OH into a pyridinium salt according to the method of Katritzky,[161] followed by displacement with (4-methoxyphenyl)methanethiol, then deprotection with TFA, and reprotection with Acm and Fmoc groups. Peptides are synthesized on an automated synthesizer on Fmoc-aminomethyl-Rink-resin using HBTU/Fmoc-Xaa-OH activation in the presence of HOBt and DIPEA. Cleavage and deprotection are accomplished by reagent K.[160] Jackson et al.[160] reported that the cyclized peptides containing the disulfide bridge exhibited high helicity in water at 0 and 60 °C, as measured by CD. The synthesis of **28** has also been reported by Phelan et al.[162] In this study, the cyclized peptides exhibited a smaller effect and the NMR data indicated distortion of the helix where the D-enantiomer of **28** occurs in the sequence.[162]

**Figure 2** Stablization of Two α-Helical Turns by an *i*, *i* + 7 Disulfide Bridge Formed by Oxidation of *S*-6-[(Acetamidomethyl)sulfanyl]-*N*-(9-fluorenylmethoxycarbonyl)norleucine[160]




**28**



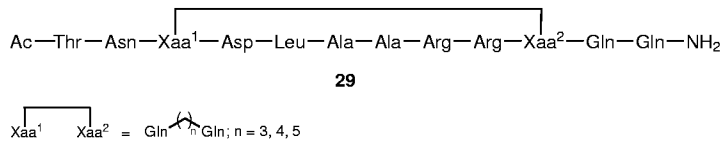

**General Procedure for Disulfide Linking of Side Chains:**[162]
Linear peptides containing **28** were synthesized by standard Merrifield and Fmoc chemistry methods. Cyclization was achieved by simultaneous removal of the Acm protecting group and oxidation. The Acm-protected peptide (~16 mg) was dissolved in $H_2O$ (1.5 mL) containing 10% AcOH and then diluted to a total volume of 50 mL with TFE to give a final concentration of ca. 200 μM. A 6 mM soln of $I_2$ (12 mL) [80 mg dissolved in AcOH (3 mL) and diluted to 50 mL with TFE] was added in 1-mL portions over the course of 10 h while progress of the reaction was monitored by HPLC. When the starting material had been consumed, the reaction was diluted with $H_2O$ and lyophilized, followed by purification of the crude oxidized material. Complete oxidation was verified by the Ellman test.[163]

### 12.3.2.3.3 Alkanediyl Links

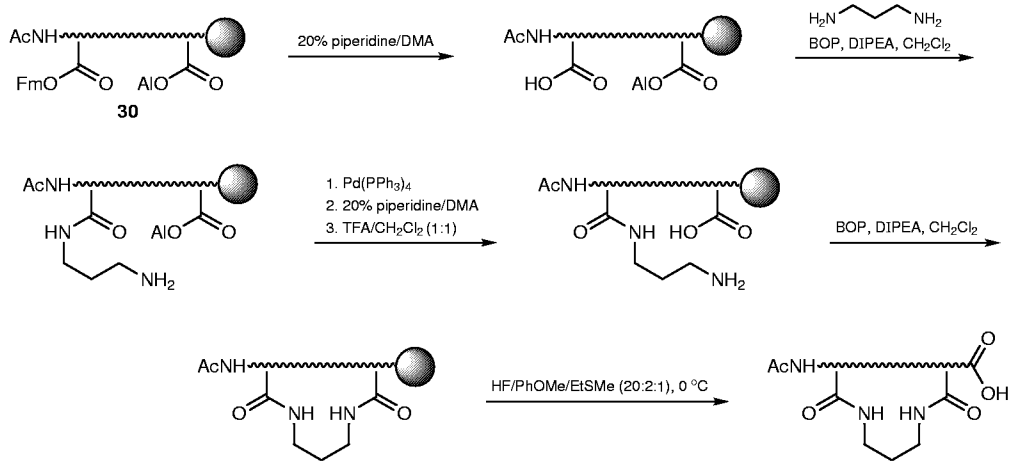
Links between side-chain nitrogen atoms have been prepared.[162] Gln residues spaced *i*, *i* + 7 apart are linked by an alkanediyl chain between the two side-chain nitrogens. Peptide **29** (Scheme 17) has been shown to be helical between residues $Asn^2$ and $Gln^{10}$. CD spectra suggested formation of 84% helix when n = 3 and up to 100% helix when n = 5, and the linear control showed 20% helicity.[162] Peptide **29** is prepared from the corresponding protected linear peptide **30**, where $Xaa^1$ = Glu(OFm) and $Xaa^2$ = Glu(OAl), by removal of the OFm group from $Glu^3$, coupling with the appropriate alkanediamine, removal of the allyl ester from $Glu^{10}$, and cyclization (Scheme 18).[162] Formation of an α-helix rather than the $3_{10}$-helix in the cyclized peptide has been established by 2D NMR.[162] Comparison of **29** with a

corresponding disulfide-linked peptide has shown much higher helix induction with alkanediyl links compared to disulfide links.[162]

**Scheme 17** Sequence of a Peptide That is Helical between Asparagine and Glutamine Residues[162]




**Scheme 18** Formation of an Alkanediyl Tether[162]




Tandem pentanediyl tethers have been used to generate helices in an HIV gp41 peptide.[153] Introduction of one link markedly increases helicity over a linear control. Two links arranged sequentially increase the helical content more dramatically, with the CD spectrum showing characteristics of an α-helix.

**Synthesis of Peptides with Alkanediyl-Linked Side Chains; General Procedure:**[162]
The peptide was synthesized on *p*-MBHA resin (4.25 g, 0.57 mequiv·g$^{-1}$, 2.42 mmol) using standard coupling protocol with Boc-Xaa-OH (3 equiv), BOP (3.3 equiv), and NMM (3.3 equiv) in $CH_2Cl_2$ (and DMA, if needed for solubility) at rt for 1 h. Acetyl N-capping was carried out by treatment with $Ac_2O$ (5 mL) in 3% TEA/$CH_2Cl_2$ at rt for 20 min. The resin was dried and weighed (4.41 g, 0.22 mequiv·g$^{-1}$). The OFm side-chain C-protecting group was removed from the peptide-resin (0.51 g, 0.11 mmol) with 20% piperidine/DMA for 20 min. The resulting piperidinium salt was neutralized by washing with 1% TFA/$CH_2Cl_2$ (2 ×). The free side-chain carboxylic acid group was coupled to propane-1,3-diamine (0.12 mL, 1.44 mmol) with BOP (0.40 g, 0.90 mmol) and DIPEA (0.17 mL, 0.98 mmol) in $CH_2Cl_2$ for 1 h, followed by addition of DMA and continued coupling for an additional 45 min. The Glu(OAl) ester was deprotected with $Pd(PPh_3)_4$ (0.21 g, 0.18 mmol) in 20% piperidine/DMA for 1.5 h. Piperidine was then removed by washing with 1% TFA/$CH_2Cl_2$ (2 ×). The resulting amino- and carboxylic acid containing peptide-resin was cyclized with BOP (0.32 g, 0.72 mmol) and DIPEA (0.13 mL, 0.75 mmol) in $CH_2Cl_2$ for 3.5 h. When, a Kaiser ninhydrin test gave a purple color, the coupling was repeated with BOP (0.44 g, 0.99 mmol) and DIPEA (0.19 mL, 1.09 mmol) in $CH_2Cl_2$ for 3 d.

The peptide-resin was washed and dried in preparation for cleavage. The peptide was cleaved from the resin with anhyd HF (10 mL) at 0 °C for 1 h with PhOMe (1 mL) and EtSMe (0.5 mL) as scavengers. The resin was washed with $Et_2O$ (2 ×), EtOAc, and again with $Et_2O$. The free peptide was then extracted from the resin with sequential washes of 10% AcOH, glacial AcOH, MeCN, 10% AcOH, and $H_2O$. The combined solns were lyophilized and the residue purified by RP-HPLC. Yields were improved by using mono-*N*-Boc alkanediamines and 2% DVB cross-linker.[162]

*for references see p 789*

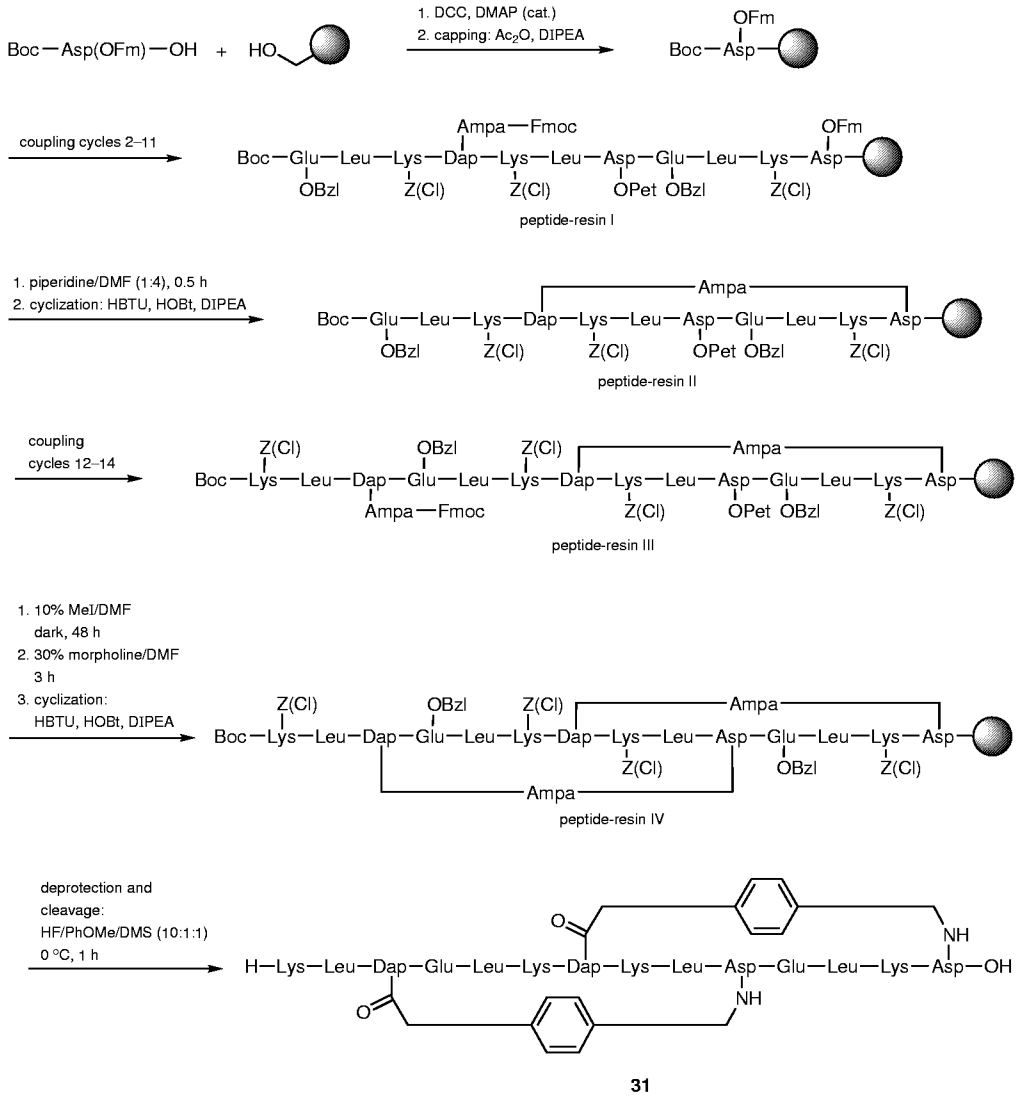
#### 12.3.2.3.4 [(4-Aminomethyl)phenyl]acetic Acid Bridged Peptides: Single or Overlapping *i*, *i* + 7 Side-Chain Bridges

While exploring alternative *i*, *i* + 7 side-chain bridges, Yu and Taylor[164] linked an Ampa residue by amide bond formation to the side chain of a Dap residue at the N-terminal, *i* position, and an Asp residue at the C-terminal, *i* + 7 position. CD studies indicated that peptide helicity can be improved from an estimated 12% for an acyclic control peptide to 47% in aqueous phosphate buffer at 25 °C. This is similar to that of a comparable peptide with two *i*, *i* + 4 Lys···Asp bridges (41% helix).[164]

The utility of the Ampa side-chain link has been extended by placing two such linkers into a 14-residue sequence, in a cross-linked fashion, resulting in a bicyclic peptide.[165] These overlapping side-chain bridges impose an even more rigid constraint for helix induction. The latter approach uses an orthogonal protection strategy required to prepare such bicyclic peptides, e.g. **31** (Scheme 19). The standard Boc/Bzl peptide chain extension strategy[166] has been used in combination with the base-labile Fmoc and OFm protecting groups for the amine and carboxylic acid functional groups, respectively, which must be selectively cleaved to form the first side-chain bridge. This follows the strategy reported by Felix et al.[154] The last level of selective protection that is necessary to make a bicyclic peptide is provided by the masked base-labile 2-(2-pyridyl)ethyl ester protecting group (Pet).[167,168] This group protects the Asp side chain in position 10 during the first cyclization step, and allows subsequent activation by methylation for base cleavage so that the second cyclization can be performed. To adapt the method on solid support, the reported conditions of the methylation reaction used to activate this group to base cleavage had been improved.[165]

The shape of the CD spectrum, at pH 7.0 and 25 °C, of the bicyclic peptide indicates a high degree of helical structure although the intensities of the negative maxima are less than expected for an α-helical peptide.[165] NMR studies remain to be done to characterize the helix that is stabilized by the bis(Ampa)-bridged peptide.

**Scheme 19** Synthesis of a Bis(Ampa)-Bridged Peptide[165]




**31**

Pet = 2-(2-pyridyl)ethyl ester

**Bis(Ampa)-Bridged Peptide 31:**[165,169]

**Fmoc-Ampa-OH:**[165]

A soln of 4-(bromomethyl)phenylacetic acid (6.41 g, 28.0 mmol) in acetone (200 mL) was added dropwise to a stirred soln of 2 M $NH_3$/EtOH (800 mL) at rt. This mixture was stirred for a further 1 h and the solvents were then removed. To the residual solid 1 M NaOH (28.0 mmol) was added followed by the removal of the $H_2O$. The residue was dissolved in further $H_2O$ (200 mL) and the concentration was repeated to remove any remaining ammonia. The solid was dried in vacuo and dissolved in $H_2O$ (150 mL). A soln of DIPEA (5.0 mL, 28.6 mmol) and Fmoc-OSu (9.45 g, 28.03 mmol) in MeCN (150 mL) was added and the mixture was stirred at rt for 2 h. Most of the MeCN was removed, and the residue was then poured into 1.5 M HCl (200 mL). The solid precipitate was collected by filtration, washed with $H_2O$, and dried in vacuo. The crude product was purified by flash chromatography (silica gel, hexanes/$CHCl_3$/MeOH/AcOH), and then recrystallized (EtOH/$Et_2O$); yield: 5.3 g (49%); mp 191–193 °C; Anal. Calcd for $C_{24}H_{21}NO_4$: C, 74.4; H, 5.5; N, 3.6. Found: C, 73.9; H, 5.2; N, 3.5.

*for references see p 789*

**Boc-Dap[(Fmoc)-Ampa]-OH:**[165]
Fmoc-Ampa-OH (2.32 g, 6.0 mmol) was dissolved in anhyd dioxane/EtOAc (1:1, 100 mL). To the mixture HOSu (0.81 g, 7.0 mmol) and DCC (1.34 g, 6.5 mmol) were added. The mixture was stirred at rt for 20 h, and DCU was then removed by filtration. The solvents were removed and the crude active ester product (Fmoc-Ampa-OSu) was used without further purification. Boc-Dap-OH[170,171] (1.43 g, 7 mmol) was dissolved in 2.5% $NaHCO_3$ (60 mL) soln. A soln of the Fmoc-Ampa-OSu in dioxane (60 mL) was added, and this mixture was stirred at rt for 10 h. The dioxane was then removed, and the residue was acidified with 5% citric acid and extracted with $CH_2Cl_2$. The organic layer was washed with citric acid and $H_2O$, dried ($Na_2SO_4$), and concentrated. The crude product was then purified by flash chromatography (silica gel, $CHCl_3$/MeOH/AcOH). The eluant was removed and the residue was dissolved in $CH_2Cl_2$ (500 mL). This soln was washed with $H_2O$, dried ($Na_2SO_4$), and concentrated; yield: 2.41 g (70%); mp 110 °C (sublimes); Anal. Calcd for $C_{32}H_{35}N_3O_7$: C, 67.0; H, 6.2; N, 7.3. Found, C, 66.5; H, 6.2; N, 7.1.

**Boc-Asp(OPet)-OBzl:**[165]
Boc-Asp-OBzl (16.2 g, 50 mmol) and 2-(2-pyridyl)ethanol (6.2 g, 50 mmol) were dissolved in $CH_2Cl_2$ (150 mL). The soln was cooled to 0 °C, DCC (11.33 g, 55 mmol) and DMAP (0.122 g, 1.0 mmol) were added, and the mixture was stirred overnight at rt. The DCU was removed by filtration, and the $CH_2Cl_2$ soln was washed with 5% citric acid, $H_2O$, 5% $NaHCO_3$, and then again with $H_2O$. The $CH_2Cl_2$ soln was then dried ($Na_2SO_4$) and concentrated. Purification by precipitation from EtOAc and hexanes gave the desired compound; yield: 16.7 g (78%); $^1$H NMR ($CDCl_3$, δ): 8.56 (d, 1H), 7.60 (t, 1H), 7.32 (s, 5H), 7.2–7.1 (m, 2H), 5.80 (d, 1H), 5.14 (q, 2H), 4.65–4.55 (m, 1H), 4.50–4.30 (m, 2H), 3.10–2.95 (m, 3H), 2.85–2.70 (m, 1H), 1.44 (s, 9H).

**Boc-Asp(OPet)-OH:**[165]
Boc-Asp(OPet)-OBzl (12.86 g, 30 mmol) was dissolved in MeOH (250 mL) in a round-bottomed flask with a magnetic stir bar at rt. The air was completely displaced with $N_2$ and 10% Pd/C (1.5 g) was added. The soln was flushed with $N_2$ for another 5 min, and then a slow stream of $H_2$ was passed through the stirred flask for 2 h, after which time TLC indicated that the reaction was complete. The catalyst was removed by filtration and the solvent was removed by concentration. Dissolution followed by precipitation from an EtOAc/hexanes system allowed recovery of the desired product; yield: 8.9 g (88%); $^1$H NMR (DMSO-$d_6$, δ): 8.48 (d, 1H), 7.75–7.63 (m, 1H), 7.22–7.18 (m, 2H), 7.10 (d, 1H), 4.37 (t, 2H), 4.30–4.20 (m, 1H), 3.02 (t, 2H), 2.8–2.5 (m, 2H), 1.36 (s, 9H).

**Solid-Phase Peptide Synthesis:**[165]
The sequence of solid-phase assembly and cyclization reactions used to prepare the bis(Ampa)-bridged peptide **31** is summarized in Scheme 19. All solid-phase procedures were performed in a standard, custom-built, 200-mL capacity, manual peptide synthesis reaction vessel.[165]
*(a) Loading of the C-Terminal Residue: Preparation of Boc-Asp(OFm)-Resin:* Boc-Asp(OFm)-OH was attached to hydroxymethylated, 1% cross-linked, polystyrene–divinylbenzene copolymer beads (100–200 mesh, 1.05 mequiv·g$^{-1}$) as follows: The hydroxymethyl resin (3.0 g, 3.15 mequiv) was swollen in $CH_2Cl_2$ (100 mL) for 30 min. Boc-Asp(OFm)-OH (2.46 g, 10 mmol) was added to the swollen resin, and the mixture was cooled to 0–5 °C. DCC (2.2 g, 10.5 mmol) and DMAP (20 mg, 0.16 mmol) were then added, and the mixture was shaken at 0 °C for 1 h and then overnight at rt. The substituted resin was then washed with $CH_2Cl_2$ (3 × 100 mL), DMF (3 × 100 mL), iPrOH (2 × 100 mL), and $CH_2Cl_2$ (2 × 100 mL). The resin was then thoroughly dried in vacuo, and the Boc-Asp(OFm)-resin substitution level was estimated by picric acid titrations performed in triplicate on 20–30-mg samples;[149] yield: 0.23–0.28 mequiv·g$^{-1}$ [80–90%, based on Boc-Asp(OFm)-OH]. Unreacted hydroxy groups on the solid support were then blocked by acetylation: the Boc-Asp(OFm)-resin (3.5 g) was swollen in $CH_2Cl_2$ (60 mL), and $Ac_2O$ (4.0 mL, 42 mmol) and DIPEA (2.0 mL, 14.3 mmol) were added. This mixture was shaken at rt for 3 h, drained, and washed thoroughly with $CH_2Cl_2$ prior to peptide-chain elongation.
*(b) The Peptide-Chain Elongation Cycle:* The sequence of washings, deprotection, and coupling reaction conditions used for peptide-chain elongation were as follows: The $N^{\alpha}$-Boc-protected peptide-resin was washed with $CH_2Cl_2$ (3 × 100 mL × 1 min) and freshly prepared 30% TFA/$CH_2Cl_2$ (1 × 80 mL × 1 min), then shaken with the 30% TFA soln at rt for 30 min. The resin was then drained and washed with $CH_2Cl_2$ (3 × 100 mL × 1 min), 5% DIPEA/$CH_2Cl_2$ (2 × 100 mL × 2 min), and DMF (3 × 100 mL × 1 min), and then drained. For the coupling reaction, Boc-Xaa$^1$-OH (3 equiv), HBTU (3 equiv), HOBt (3 equiv), and DIPEA (9 equiv) were added to DMF/$CH_2Cl_2$ (2:1) to give a total volume of 80 mL. The mixture was adjusted to pH 9 by adding additional DIPEA if necessary. The mixture was shaken for 1–2 h until coupling was judged complete by the Kaiser ninhydrin test performed on small quantities of the peptide-resin material periodically removed from the reaction vessel. After completion of the coupling reaction, the peptide-resin was washed with DMF (3 × 100 mL × 1 min), iPrOH (2 × 100 mL × 1 min), $CH_2Cl_2$ (3 × 100 mL × 1 min), and then again with DMF (2 × 100 mL × 1 min).

*(c) Peptide-Resin I:* Fully protected peptide-resin I (Scheme 19), incorporating residues 4–14 of peptide **31**, was prepared from the Boc-Asp(OFm)-resin (3.5 g, ~0.88 mequiv) by peptide-chain extension in the N-terminal direction, using the standard elongation cycle described above. The following L-amino acid derivatives were used in the coupling cycles: Boc-Lys[Z(Cl)]-OH, Boc-Leu-OH, Boc-Glu(OBzl)-OH, Boc-Dap[(Fmoc)Ampa]-OH, and Boc-Asp(OPet)-OH.
*(d) First Cyclization: Peptide-Resin II:* The first cyclization reaction on the solid support was performed after selective cleavage of the side-chain Fmoc and OFm protecting groups on residues $Dap^7$ and $Asp^{14}$, respectively, of peptide **31**. Peptide-resin I was washed with 20% piperidine/DMF (1 × 100 mL × 1 min), and then shaken in 20% piperidine/DMF (100 mL) for 30 min. The selectively deprotected peptide-resin was then washed with DMF (3 × 100 mL × 1 min), iPrOH (2 × 100 mL × 1 min), $CH_2Cl_2$ (3 × 100 mL × 1 min), and DMF (2 × 100 mL × 1 min). The peptide-resin was drained, and HBTU (3 equiv), HOBt (3 equiv), and DIPEA (6 equiv) were added to DMF/$CH_2Cl_2$ (2:1) to give a total volume of 80 mL. This mixture was adjusted to pH 9 by adding additional DIPEA if necessary. The cyclization reaction was continued for about 12 h, at which time a negative Kaiser ninhydrin test indicated that it was complete. The product, peptide-resin II (Scheme 19), was then washed with DMF (3 × 100 mL × 1 min), iPrOH (2 × 100 mL × 1 min), $CH_2Cl_2$ (3 × 100 mL × 1 min), and again with DMF (2 × 100 mL × 1 min).
*(e) Second Cyclization: Peptide-Resins III and IV:* The monocyclic, protected peptide-resin II was extended in the N-terminal direction to complete the 14-residue peptide sequence by coupling Boc-Dap[(Fmoc)Ampa]-OH, Boc-Leu-OH, and then Boc-Lys[Z(Cl)]-OH, using the standard chain elongation cycle described above. The product, peptide-resin III (Scheme 19), was then methylated on the pyridine ring nitrogen of the OPet protecting group on $Asp^{10}$ (activating the OPet ester to base cleavage), as follows: Peptide-resin III was shaken in the dark in 10% MeI/DMF (100 mL) at rt for 48 h. The methylated peptide-resin was then washed with DMF (3 × 100 mL × 1 min), followed by 30% morpholine/DMF (1 × 100 mL × 1 min), and shaken in 30% morpholine/DMF (100 mL) at rt for 3 h. The selectively deprotected peptide-resin was then washed with DMF (3 × 100 mL × 1 min), iPrOH (2 × 100 mL × 1 min), $CH_2Cl_2$ (3 × 100 mL × 1 min), and again with DMF (2 × 100 mL × 1 min). The second cyclization, giving peptide-resin IV (Scheme 19), was then performed using the same procedures as those described above for the first cyclization.
*(f) Deprotection and Cleavage:* Protected bicyclic peptide-resin IV was subjected to Boc deprotection procedures in the standard chain elongation cycle, and then thoroughly dried in vacuo over $P_2O_5$. In a typical procedure, an aliquot of this resin (500 mg) was subjected to HF deprotection and cleavage using a standard HF line. The peptide-resin was stirred under $N_2$ in HF/PhOMe/DMS (10:1:1, ~12 mL) and maintained at 0 °C for 1 h. The reaction was then terminated by purging the HF and DMS into a KOH trap with a steady flow of $N_2$ over approx. 30 min at 0 °C. The solid residue was thoroughly washed with ice-cold $Et_2O$ (10 × 20 mL) and collected in a 10–20 mm porosity Buchner funnel. The solid material was then extracted into 10% aq AcOH and lyophilized giving the crude peptide; yield 120 mg. The bis-(Ampa)-bridged peptide **31** was readily identified as the major peak on analytical C18 RP-HPLC of the crude, lyophilized powder.[164] This peptide was then purified to apparent homogeneity by a single passage through a semi-preparative C18 RP-HPLC column, using a 30-min gradient of 30 to 50% B in A (A: 0.1% TFA in 10% MeCN/$H_2O$; B: 0.1% TFA in 90% MeCN/$H_2O$) at 3 mL·min$^{-1}$. Overall yield of the purified peptide from deprotection and cleavage of the peptide-resin IV (500 mg; Scheme 19): 20 mg; ES-MS *m/z*: $[M+H]^+$ calcd, 1903.3; found, 1903.0 ± 0.6; Amino acid analysis: Asx 1.96 (2), Glx 1.92 (2), Leu 4.00 (4), Lys 3.82 (4), Dap 1.92 (2), Ampa 1.85 (2).

### 12.3.2.4 Metal-Chelated Side-Chain Cross-links

Metal chelation of amino acid side chains in peptides is analogous to the covalent cross-linking of side chains. Researchers have drawn on the examples of metal ion stabilization of folded states of natural proteins[172,173] to develop methods for stabilizing the helical conformation not only in synthetic proteins, but in short peptides as well. Side-chain cross-linking through ligation with metal ions[174–176] and metal complexes[177,178] has been demonstrated to promote helicity by restricting the available conformational states of the random coil form. Entropic stabilization of the unfolded form thus shifts the equilibrium from a random, unfolded macromolecule to a folded one. Factors promoting helix formation through metal chelates include the presence of amino acids with high affinity for metal binding (i.e., proteinogenic amino acids such as His and Cys), the right spacing of the pair of ligands in the peptide chain (one that would support one turn of a helix), and an amino acid sequence which is amenable to adopting the helical conformation. This last factor is dem-

*for references see p 789*

onstrated in proteins wherein bidentate metal-chelation sites involved in α-helical conformations have been found to commonly occur in motifs[172] such as His-Xaa$^1$-Xaa$^2$-Xaa$^3$-His,[179–185] Asp-Xaa$^1$-Xaa$^2$-Xaa$^3$-His,[172] His-Xaa$^1$-Xaa$^2$-Xaa$^3$-Asp,[172] and Cys-Xaa$^1$-Xaa$^2$-Xaa$^3$-His.[186] Effective metal-chelation requires a three amino acid spacer (two or four amino acid spacers are not observed). His is the most important metal-chelating residue,[187,188] this property arises from the presence of the imidazole ring. Replacement of His by Asp reportedly results in weaker chelating sites.[172] In using Cys/His combinations, $Zn^{2+}$ is the preferred metal ion since $Cu^{2+}$ oxidizes free thiols.[189] The extent of helical induction also depends on the metal ion affinity for the ligand, and the compatibility of metal ion geometry and coordination sphere with the α-helical conformation.[190,191] A major limitation of using metal ion chelation is the instability of the metal complexes in biological systems.[138]

#### 12.3.2.4.1 Chelation with Metal Ions

Ghadiri and Choi studied helix nucleation of peptides by cross-linking His and Cys side chains with various metal ions.[176] Peptides Ac-Ala-Glu-Ala-Ala-Ala-Lys-Glu-Ala-Ala-Ala-Lys-Xaa$^1$-Ala-Ala-Ala-Xaa$^2$-Ala-$NH_2$, where Xaa$^1$ = Cys and Xaa$^2$ = His, or Xaa$^1$ = Xaa$^2$ = His, have been designed to ensure high helix propensity. These peptides have been synthesized by solid-phase synthesis and CD analysis of buffered aqueous solutions has been performed with increasing amounts of copper sulfate, cadmium(II) chloride, zinc(II) chloride, nickel(II) chloride, potassium tetrachloropalladate(II), and potassium tetrachloroplatinate(II). Greater helix induction is seen with the His/His combination than with the Cys/His combination. Helicity in the peptide where Xaa$^1$ = Cys and Xaa$^2$ = His increased from 54%, without a metal ion, to 90% on addition of $Cd^{2+}$, but $Zn^{2+}$ had no effect on the helical content. The peptide where Xaa$^1$ = Xaa$^2$ = His demonstrated up to 90% helix with $Cu^{2+}$. Helix induction with increasing concentration of the metal ion is $Ni^{2+} > Cd^{2+} > Cu^{2+} = Zn^{2+}$, even though these metals have the same binding affinity for the peptide (Xaa$^1$ = Xaa$^2$ = His). The square-planar geometry of the palladium and platinum complexes, on the other hand, are helix destabilizing.

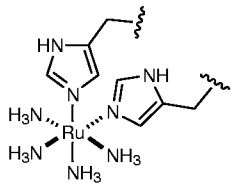
**Helix Induction in Peptides by Addition of Metal Ions:**[176]
Peptides Ac-Ala-Glu-Ala-Ala-Ala-Lys-Glu-Ala-Ala-Ala-Lys-Xaa$^1$-Ala-Ala-Ala-Xaa$^2$-Ala-$NH_2$ (Xaa$^1$ = Cys; Xaa$^2$ = His, or Xaa$^1$ = Xaa$^2$ = His) were synthesized by the standard solid-phase protocol using Boc chemistry. When Xaa$^1$ = Cys and Xaa$^2$ = His 90% helix induction was obtained with $2.2 \times 10^{-6}$ M peptide and $7.3 \times 10^{-4}$ M $CdCl_2$ in $H_2O$ and adjusting the soln to pH 6.7 at 4 °C. When Xaa$^1$ = Xaa$^2$ = His the same amount of helix was induced with $1.8 \times 10^{-6}$ M soln of peptide and $7.4 \times 10^{-5}$ M $CuSO_4$ in 5 mM $Na_2B_4O_7$, pH 6.4, at 0 °C.

#### 12.3.2.4.2 Chelation with Metal Complexes

For the stabilization of α-helices, Ghadiri and Fernholz[177] explored the use of a ruthenium metal complex, which is less susceptible to ligand exchange. Formation of metal complex **32**, *cis*-$[Ru(NH_3)_4L_2]^{3+}$ where $L_2$ are the side chains of two His residues spaced *i*, *i* + 4 apart, arises from the high affinity of the precursor, *cis*-$[Ru(NH_3)_4(OH_2)_2]^{2+}$, for these amino acid residues. *cis*-$[Ru(NH_3)_4(H_2O)_2]^{2+}$ is obtained by reduction of *cis*-$[Ru(NH_3)_4Cl_2]Cl$ in the presence of zinc amalgam in a degassed Tris buffer, pH 7. The latter is prepared following the method of Pell et al.[192]

Chelation of **32** with the 17-residue peptides **33** and **34** (Scheme 20) results in up to 80 and 50% helix, respectively, at 21 °C, as measured by CD spectroscopy. Without the metal complex, **33** is 45% helical, while uncomplexed **34** exhibits the CD spectrum of a random coil structure.[177] Helix induction results from the loss of conformational flexibility of the peptides upon complexation.

**Scheme 20** A Ruthenium Metal Complex and the 17-Residue Peptides Which it Stabilizes[177]



**32**

Ac—AEAAAKEAAAKHAAAHA—$NH_2$

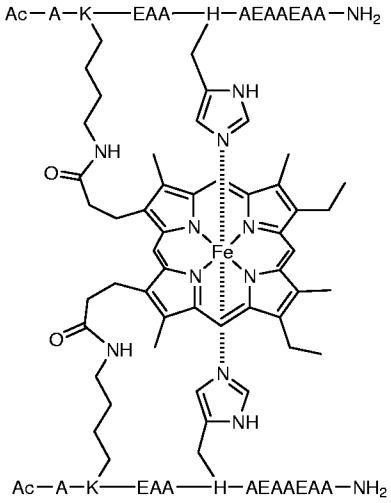
**33**

Ac—AEAAAKHAAAHEAAAKA—$NH_2$

**34**

Another type of coordination with a metal complex for helix induction has been demonstrated by Benson et al.[105] using iron(III) mesoporphyrin IX. Two identical 13-residue long peptides Ac-AKEAAHAEAAEAA-$NH_2$ are linked to iron(III) mesoporphyrin IX via amide formation between two heme propionyl groups and the ε-amino groups of two Lys side chains giving **35** (Scheme 21). Starting from completely random coil conformations, at least 52% helix induction in 100% $H_2O$ is achieved by coordination of the His residue in each peptide with iron, resulting in a peptide-sandwiched mesoheme **35**. It has been noted that positive ellipticity contributions from heme in the far-UV CD spectra of iron(III) mesoporphyrin IX result in underestimation of helical content. Maximal helicity (97%) is attained with 20–25% TFE. Other groups have also studied model hemoproteins containing α-helical peptides.[178,193–198]

**Scheme 21** Peptide-Sandwiched Mesoheme[105]




**35**

**Metal Complex** 35:[105]

Peptide Ac-AKEAAHAEAAEAA-$NH_2$ was synthesized using Boc chemistry on MBHA resin (100–200 mesh, 1% DVB, substitution level 0.53 mmol·$g^{-1}$). Purification was carried out by HPLC on a 2.2-cm Vydac C18 column using a gradient of MeCN/$H_2O$ in 0.1% TFA (5 to 40% over 40 min) at a flow rate of 8 mL·$min^{-1}$. Mesoporphyrin IX was converted into the iron(III) complex by refluxing with excess $Fe(OAc)_2$ in AcOH.[199] The bis(4-nitrophenyl) ester of iron(III) mesoporphyrin IX was subsequently prepared using the method of Collman et al.[200]

An aliquot of Ac-AKEAAHAEAAEAA-$NH_2$ in $H_2O$ was placed in a preweighed 1.5-mL microcentrifuge tube and lyophilized to dryness. The dried powder (15 mg, 11.7 μmol) was dissolved in dry DMSO (100 μL) and 29 μM bis(*p*-nitrophenyl) ester of iron(III) mesoporphyrin IX in DMSO (100 μL) was added, followed by DIPEA (45 μL, 22 equiv). The soln was warmed to 40 °C until HPLC indicated no further reaction (ca. 3 h). It was then diluted to 1.0 mL with 50 mM NaOAc, pH 6, and purified by sequential ion-exchange chromatography (Sephadex CM50; 50 mM NaOAc, pH 6), RP-HPLC (Vydac

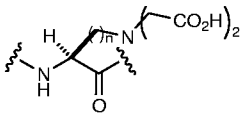
*for references see p 789*

1.0-cm C4 column using a gradient of MeCN in 10 mM $NH_4OAc$, pH 6.0, 10 to 40% over 40 min at 2.0 mL·min$^{-1}$), and size-exclusion chromatography (Sephadex G25, fine); FAB-MS *m/z*: $[M+H]^+$ 3145.

### 12.3.2.4.3 Incorporation of Metal-Binding Ligands in a Peptide Chain

In the absence of naturally occurring amino acids that can bind metals, unnatural amino acids whose side chains could act as metal-binding ligands have been used. Hopkins and coworkers[174,175] have incorporated homologous amino acids containing amino diacetic acid side chains (Adaa) in peptides to study the structural requirements for metal ion helix stabilization. Adaa amino acids **36** can be prepared with different numbers of side-chain carbon atoms to vary the tether length between the peptide backbone and the actual metal binding site (Scheme 22). Procedures for the syntheses of these unnatural amino acids have been described.[175]

**Scheme 22** Unnatural Amino Acids with Different Tether Lengths[174,175]



**36**

n = 1–4

Helix induction is obtained when Adaa$^3$ amino acids **36** (n = 3) are arranged with *i*, *i* + 4 spacing as in the peptide Ac-Adaa$^3$-$(Ala)_3$-Adaa$^3$-$[(Ala)_4$-Glu-Lys$]_3$-$NH_2$, which, on complexation with $Cd^{2+}$, improves helical content by roughly 43% at 25 °C.[174] Long Adaa side chains (e.g., such as in **36**, n = 2 or 3) are able to form chelates more effectively with a number of metal ions ($Co^{2+}$, $Cu^{2+}$, $Ni^{2+}$, $Zn^{2+}$, $Cd^{2+}$) and to promote a helical conformation, while a short side chain (e.g., such as in **36**, n = 1) can induce helicity only when complexed with the largest metal ion, $Cd^{2+}$. The largest helix induction has been obtained with an 11-residue peptide, Ac-Adaa$^3$-$(Ala)_3$-Adaa$^3$-$(Ala)_4$-Glu-Lys-$NH_2$, which was converted from < 5% helix into about 80% helix in the presence of $Cd^{2+}$ at 4 °C.[174] The type of helix induced (α, $3_{10}$, etc.) remains to be determined.

**Ac-Adaa$^3$-$(Ala)_3$-Adaa$^3$-$[(Ala)_4$-Glu-Lys$]_3$-$NH_2$:**[175]
Standard peptide synthesis protocol using Boc chemistry was followed. Synthesis commenced by automated methods on MBHA resin until one of the Adaa residues was incorporated. At this point the resin was removed from the automated synthesizer and couplings were continued manually on approx. 10-μmol scale. Side-chain protecting groups used were Lys[Z(2-Cl)] and Glu(OBzl). General protocol: MBHA resin (~0.32 mmol·g$^{-1}$ substitution) was allowed to swell in $CH_2Cl_2$ (1 mL × 10 min × 3). Attachment of each amino acid was accomplished as follows, using 1 mL of solvent for all steps: (1) prewash: 50% TFA/$CH_2Cl_2$ (1 × 1.5 min); (2) deprotection: 50% TFA/$CH_2Cl_2$ (1 × 30 min); (3) wash: $CH_2Cl_2$ (6 × 1.5 min); (4) neutralization: 10% TEA/$CH_2Cl_2$ (2 × 2 min); (5) wash: $CH_2Cl_2$ (6 × 1.5 min); (6) coupling: Boc amino acid (3–6 equiv) or Boc-Adaa$^3$-OH (3 equiv) was added to the resin. After 1.5 min, a soln of DCC in $CH_2Cl_2$ (equimolar to the Boc amino acid) was added at 25 °C and the mixture was shaken for 2 h (completion of deprotection and coupling steps was monitored by the analytical Kaiser ninhydrin test); (7) wash: $CH_2Cl_2$ (6 × 1.5 min); (8) repeat steps 2–6 for all amino acid residues; (9) acetylation of N-terminus: $Ac_2O$/DMF/TEA (0.4:1.5:0.7) for 30 min at 25 °C; (10) cleavage: the resin-bound peptide was dried at 25 °C (0.1 Torr) in a Teflon vessel of an HF cleavage apparatus. PhOMe (~6 drops) was added and anhyd HF (~5 mL, predried with $CoF_3$) was distilled into the cold (−78 °C) reaction vessel. After 1 h at 0 °C with stirring, the volatiles were removed in vacuo. The residue was washed with EtOAc (3 × 1 mL). The remaining residue was extracted with 20% aq AcOH (3 × 1 mL) and filtered from the resin. The combined aq AcOH extracts were concentrated to dryness to afford the crude peptide. The crude cleaved peptide was dissolved in $H_2O$ and centrifuged to remove insoluble residues, then purified by HPLC on an analytical (0.46 × 25 cm) macrosphere 5-μm C18 column, using a linear gradient of 18 to 35% MeCN/$H_2O$ (constant 0.1% TFA) over 50 min with UV detection at 214 nm. The purified peptide was at least 95% pure as analyzed by HPLC; FAB-MS (thioglycerol/DMSO/HCl as a matrix) *m/z*: $[M+H]^+$ calcd, 2358, found, 2357. Helix induction was observed by CD spectroscopy

upon titration of the peptide solution, 6–14 μM, pH 7.9 (200 mM aq sodium borate), 25 °C, with 200 μM solns of metal ions.


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# 12.4 Synthesis of β-Sheet Peptides and Proteins Incorporating Templates

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The incorporation of templates into peptides or proteins is becoming increasingly common as a means to control or influence the conformation of the attached α-amino acid residues.[1–12] Small peptides are typically unfolded unless a preorganizing template is integrated into the sequence to yield α-helical or β-sheet structures, although there are exceptions to this general rule.[5] Templates are used to propagate a certain conformational bias onto the peptide substructure.[3] It is now reasonably straightforward to incorporate templates into peptides or proteins using either solution-phase or solid-phase strategies, and in some cases combinations of these approaches.[1–4,6–12] This chapter serves to overview the methodology that has been developed in our laboratory to incorporate templates into β-sheet polypeptide sequences. Numerous other laboratories, only a few of which are cited below, have also developed elegant chemistry for the incorporation of templates into peptide sequences.

These synthetic methods build from and take advantage of the pioneering work of Merrifield[13,14] for the development of solid-phase peptide synthesis, the seminal work of Hirschmann,[15] Kaiser,[16,17] and De Grado[17] who developed segment couplings of protected peptides as a means of synthesizing proteins, and the assembly of proteins from the ligation of largely unprotected peptide fragments originally envisioned by Kemp,[18] and elegantly developed by Kent,[19] and Tam,[20] as well as by Muir[19,21] and Dawson.[19,22] The majority of the methods outlined in this chapter have already been incorporated into published manuscripts, where the focus is not synthetic methodology.[1–12] Herein, we present three classes of synthetic methodologies that have been utilized to incorporate a wide variety of templates into β-sheet peptides and proteins.

## 12.4.1 β-Sheet Peptides Incorporating Templates via Solution-Phase Synthesis

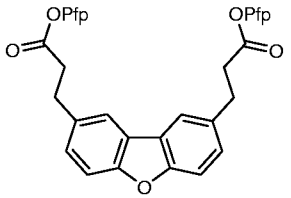
### 12.4.1.1 Incorporation of the Dibenzofuran-2,8-dipropanoic Acid Template

The dibenzofuran-2,8-dipropanoic acid template was designed to covalently position two similarly oriented peptides (N-to-C) at an average chain separation of approximately 10 Å so that they could recognize a peptide guest between the strands via the formation of a three-stranded antiparallel β-sheet structure.[1,12] Guest binding is highly selective. In addition, it has also been discovered that the receptors dimerize without the guest peptide under certain conditions and self-assemble into complex quaternary structures such as protofilaments, filaments, fibrils, and ribbons. The morphology of the assemblies can be controlled by pH, presumably by taking advantage of quaternary structure dependent p$K_a$ perturbations.[1]

The synthesis of the template, dibenzofuran-2,8-dipropanoic acid is described in detail elsewhere.[1] Briefly, iodination of dibenzofuran gave 2,8-diiododibenzofuran which was subjected to a tandem palladium-catalyzed Heck cross-coupling reaction with ethyl acrylate yielding the bis-α,β-unsaturated ethyl ester. Saponification and hydrogenation of the diester afforded the symmetric dibenzofuran-2,8-dipropanoic acid. The diacid was activated for segment condensation by converting it into 2,8-bis[3-oxo-3-(pentafluorophenoxy)propyl]dibenzofuran (**1**) (Scheme 1) utilizing a DCC-mediated esterification with pentafluorophenol. The overall yield of **1** from dibenzofuran was 48%.

*for references see p 805*

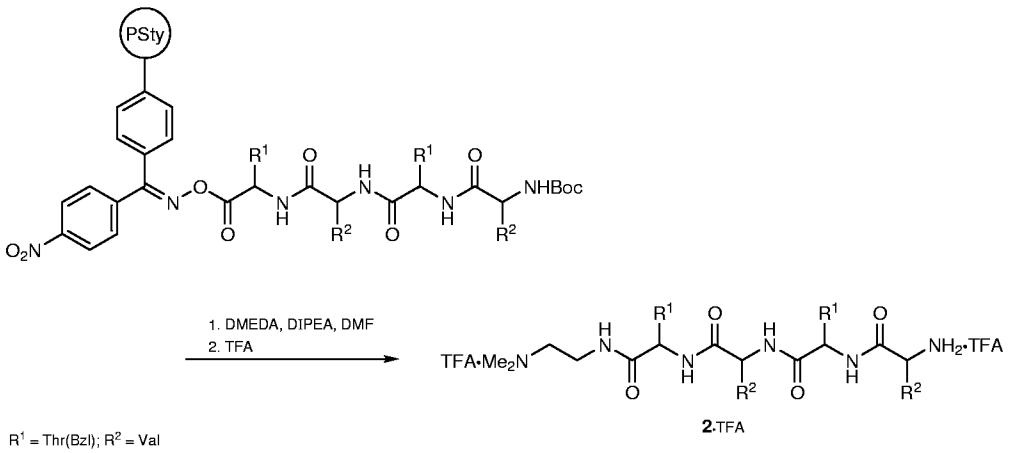
**Scheme 1** Chemical Structure of 2,8-Bis[3-oxo-3-(pentafluorophenoxy)propyl]dibenzofuran[1]




1

The side-chain-protected peptide strands used in the synthesis of the peptidomimetic incorporating the 2,8-dibenzofuran-based template were assembled on oxime solid-phase resin using the Boc strategy. The Thr-Val-Thr-Val peptide was synthesized by attaching the carboxy group of Boc-Thr(Bzl)-OH to the oxime resin (Scheme 2). The peptide was then elongated in a stepwise manner to afford Boc-Val-Thr(Bzl)-Val-Thr(Bzl)-oxime-resin using the methodology[2,23] outlined in detail below. Cleavage of the side-chain-protected peptide from the oxime resin was achieved with the nucleophile *N,N*-dimethylethylenediamine (DMEDA).[24] The Boc protecting group was removed using a 1:3 TFA/$CH_2Cl_2$ solution and the peptide was purified by RP-HPLC (with 0.1% TFA in the aqueous and organic phases) on a C18 column affording the trifluoroacetate salt **2**.

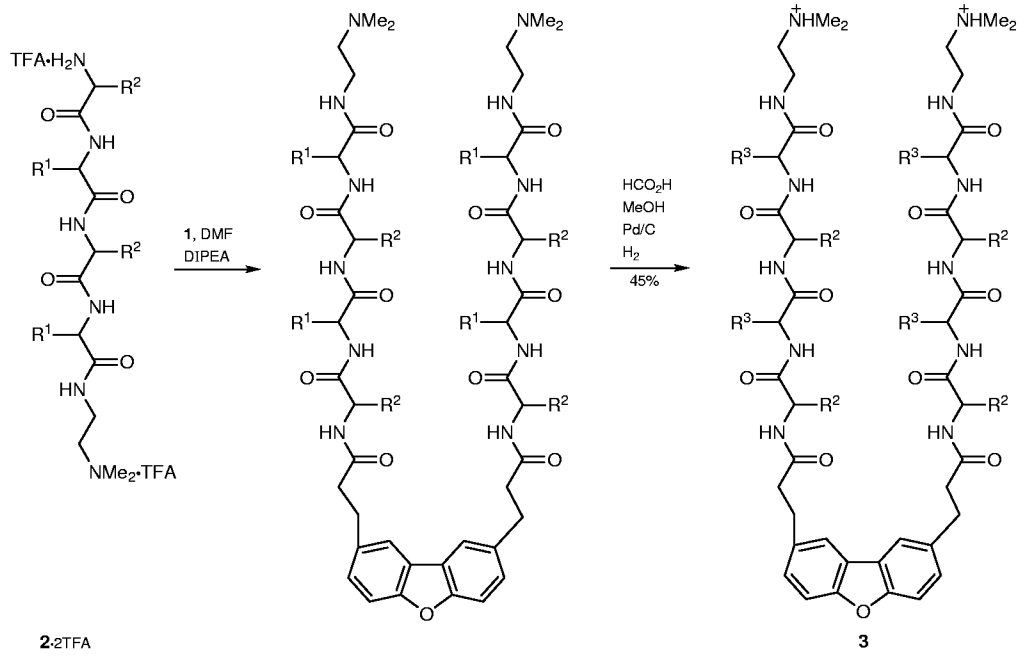
**Scheme 2** Synthesis of Side-Chain-Protected Peptides Employing the Oxime Resin[2]




2·TFA

Peptidomimetic **3** was synthesized by reacting 2,8-bis[3-oxo-3-(pentafluorophenoxy)-propyl]dibenzofuran (**1**) with the side-chain-protected peptide segment 2TFA·H-Val-Thr(Bzl)-Val-Thr(Bzl)-DMEDA (**2**·TFA) in DMF (Scheme 3).[1] The reaction was initiated by the addition of sufficient DIPEA to make the coupling solution slightly basic (pH 7–9) when applied to hydrated pH paper. The time course of the coupling reaction was monitored by RP-HPLC. Side-chain deprotection of the Bzl protecting groups was accomplished by catalytic hydrogenation in the presence of 10% Pd/C in a mixture of formic acid and methanol (approx. 2:1).[25] Chromatography (RP-HPLC) of the crude product gave peptidomimetic **3** in >98% purity.

**Scheme 3** Synthesis of a Peptidomimetic Incorporating the 2,8-Dibenzofuran-Based Template[1]




$R^1$ = Thr(Bzl); $R^2$ = Val; $R^3$ = Thr

**Synthesis of Side-Chain-Protected Peptides (e.g., 2) Employing the Oxime Resin:**[2]
Oxime resin having a loading of 0.5 mmol·g$^{-1}$ was placed in a fritted reaction vessel, solvated, and washed with $CH_2Cl_2$ (15 mL·g$^{-1}$ resin). The C-terminal amino acid was attached to the resin using Boc-protected amino acid (1.6 equiv) and DIC in $CH_2Cl_2$ containing 10% DMF while shaking the resin for 12–24 h. Pivalic anhydride (10 equiv) in $CH_2Cl_2$ was added and agitated for 6–12 h to cap the unreactive oxime groups ($Ac_2O$ should not be used owing to the transacetylation that can occur during peptide synthesis which frees the oxime functionality and results in deletion sequences). The remaining amino acids were attached to the resin-bound peptide using the following procedure: (1) wash resin with DMF (2 × 1 min), $CH_2Cl_2$/iPrOH (3:1; 2 × 1 min), and $CH_2Cl_2$ (3 × 1 min); (2) prewash resin with 35% TFA and 1% PhSMe in $CH_2Cl_2$ (1 × 1 min); (3) remove Boc group with 35% TFA and 1% PhSMe in $CH_2Cl_2$ (1 × 30 min) with vigorous shaking; (4) wash with $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (2 × 1 min); (5) couple the preactivated Boc-protected amino acid (3 equiv) [15 min preactivation with BOP (3 equiv)] with DIPEA (6 equiv) in $CH_2Cl_2$ containing 10% DMF. [NOTE: The quantities of amino acid, BOP, and DIPEA were doubled for the coupling of the third amino acid to increase the coupling rate relative to the competing intramolecular reaction, i.e. piperazine-2,5-dione (DKP) formation.] Preneutralization of the resin before coupling was also avoided to prevent intramolecular cyclization of the peptide which cleaves it from the resin and liberates the oxime functionality which can react during subsequent amino acid coupling cycles affording deletion peptides. The pH of the coupling mixtures was always checked to ensure that the soln was basic when applied to prehydrated pH paper. If not, more DIPEA was added. After 3–6 h of shaking, a Kaiser ninhydrin test was performed to evaluate the extent of coupling. If positive, step 5 was repeated or, if the N-terminus appeared unreactive after a few cycles, the free amino groups were acetylated with $Ac_2O$ (10 equiv) in $CH_2Cl_2$ for 1–2 h. The peptide was removed from the resin by using approx. 3–20 equiv of a primary or secondary amine or $NH_4OAc$ in the presence of DIPEA in DMF. The peptide was precipitated from DMF by injecting the DMF soln into a 25-fold excess of $H_2O$. After filtration, the fragment was dried under reduced pressure and Boc deprotected using 35% TFA in $CH_2Cl_2$ for 30 min. After removing the TFA and $CH_2Cl_2$ under reduced pressure, the peptide was precipitated in $H_2O$, dried, and characterized by MALDI-MS. The side-chain protected peptide may or may not need purification based on analytical HPLC analysis.

**Synthesis of Peptidomimetic 3:**[1]
To an oven dried, 50-mL round-bottom flask was added **2**·2TFA (196 mg, 0.22 mmol) and 2,8-bis-[3-oxo-3-(pentafluorophenoxy)propyl]dibenzofuran (**1**; 50 mg, 0.08 mmol). Anhyd DMF (20 mL) was

*for references see p 805*

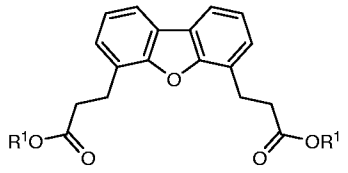
added with stirring and a 10-μL aliquot was analyzed by C18 RP-HPLC. The coupling reaction was initiated by the addition of DIPEA (81 μL, 0.45 mmol) to neutralize the TFA salt of the peptide (pH monitored using hydrated pH paper). Additional DIPEA was added in μL quantities, if necessary, to yield a soln pH of 7–9. The reaction was monitored by HPLC to determine the endpoint (ca. 4 h). Once the reaction was complete, the peptide was precipitated by diluting the reaction into $H_2O$ (200 mL). The peptide was isolated by filtration (0.22-μm nylon membrane) and dried under reduced pressure to give crude **1**-[Val-Thr(Bzl)-Val-Thr(Bzl)-DMEDA]$_2$; yield: 95 mg (67%).

A 25-mL round-bottom flask was charged with **1**-[Val-Thr(Bzl)-Val-Thr(Bzl)-DMEDA]$_2$ (180 mg, 102 μmol) dissolved in MeOH (2 mL) and formic acid (approx. 3 mL). The flask was flushed with $N_2$ and 10% Pd/C (102 mg) was added. The $N_2$ atmosphere was then replaced with an atmosphere of $H_2$. The reaction was monitored by C18 RP-HPLC. After 4–5 h, the MeOH was removed under reduced pressure and the thick, heterogeneous soln was diluted with $H_2O$ (10 mL). The pH of the soln was adjusted to approx. 7 and the catalyst filtered using a 0.22-μm acetate membrane. The soln was analyzed by analytical C18 RP-HPLC and purified using semi-preparative C18 RP-HPLC. The MeCN in the column fractions was removed under reduced pressure and the sample lyophilized to afford the side-chain deprotected peptidomimetic **3**; yield: 58 mg (45%); ESI-MS *m/z*: $[M+H]^+$ calcd. 1253.5; found 1254.0.

### 12.4.1.2 Incorporation of the Dibenzofuran-4,6-dipropanoic Acid Template

The aromatic diacid template dibenzofuran-4,6-dipropanoic acid (**4**, $R^1$ = H) (Scheme 4), when flanked by hydrophobic α-amino acid residues, was envisioned to form a 14-membered ring, hydrogen-bonded, hydrophobic cluster structure, nucleating parallel β-sheet folding.[2] The hydrophobic cluster is composed of the dibenzofuran ring system of the template **4** ($R^1$ = H), oriented perpendicular to the plane of the β-sheet, interacting with the side chains of the flanking hydrophobic α-amino acid residues. The synthesis of **4** ($R^1$ = H) has been described previously.[7,26] Briefly, dibenzofuran was lithiated at the 4- and 6-positions with *sec*-butyllithium, then reacted with iodine to afford 4,6-diiododibenzofuran. The latter intermediate was subjected to a tandem Heck reaction with ethyl acrylate to afford the 4,6-bis-α,β-unsaturated ethyl ester which was saponified and hydrogenated to give **4** ($R^1$ = H). Diacid **4** ($R^1$ = H) was converted into the bis(pentafluorophenyl) active ester (**4**, $R^1$ = Pfp) (Scheme 4) via a DCC-mediated esterification.

**Scheme 4** Chemical Structure of the Dibenzofuran-4,6-dipropanoic Acid Template[2]



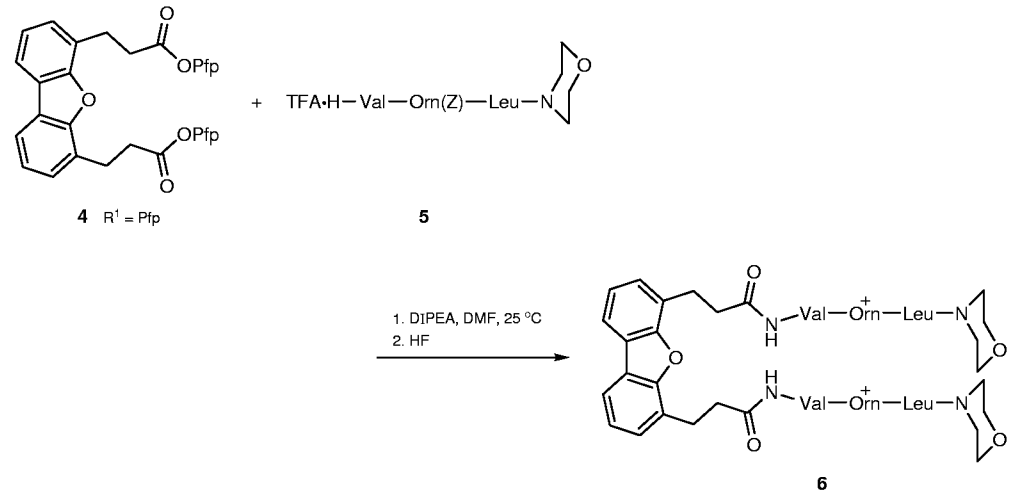

**4**

$R^1$ = H, Pfp

The peptide segments to be coupled to the template were produced by solid-phase peptide synthesis using the 4-nitrobenzophenone oxime resin.[16] The protected peptides were cleaved from the resin by nucleophilic displacement with ammonium acetate to yield the C-terminal primary amide or with a primary or secondary amine in the presence of the non-nucleophilic base (DIPEA) to yield the secondary or tertiary amide, respectively, in analogy to the reaction shown in Scheme 2.[1,16,24,27] The $N^{\alpha}$-Boc- and side-chain protected peptide was precipitated in water and characterized by MALDI-MS. The Boc group was removed by TFA and the resulting peptide analyzed by HPLC to determine whether further purification of the segments was necessary. The side-chain protected peptides (for an example see tripeptide **5** in Scheme 5) were then reacted with the bis(pentafluorophenyl) ester **4** ($R^1$ = Pfp), ensuring that enough DIPEA was added to afford a basic solution (pH 7–9) on hydrated pH paper. The extent of the coupling reaction was monitored by HPLC, noting the dis-

appearence of the limiting reagent **4** ($R^1$ = Pfp). The side-chain-protected peptidomimetic was deprotected by catalytic transfer hydrogenation in formic acid[25] or by the high HF method,[28] to give crude peptidomimetic **6** which was purified by HPLC and characterized by MALDI-MS.

**Scheme 5** Synthesis of a Peptidomimetic Incorporating the 4,6-Dibenzofuran-Based Template[2]




**Synthesis of Peptidomimetic 6:**[2,7]

**CAUTION:** *Hydrofluoric acid is highly toxic and corrosive, and severely irritating to the respiratory system and eyes. All operations should be performed in a well-ventilated fume hood using appropriate safety precautions and procedures.*

A 25-mL round-bottomed flask was charged with the side-chain-protected peptide [H-Val-Orn(Z)-Leu-morpholide] trifluoroacetate salt (**5**; 0.2 mmol), dibenzofuran-4,6-propanoyl pentafluorophenyl ester (**4**, $R^1$ = Pfp; 0.07 mmol), and DIPEA (0.15 mmol) in DMF (10–20 mL; use minimal volume for solubility). Additional DIPEA was added in μL quantities, as needed, to raise the pH of the mixture to 8–9, as monitored by moistened pH paper. The trifluoroacetate salt segment condensation reaction was carried out at 25 °C for the minimal time required based on the HPLC monitored loss of the limiting reagent **4** ($R^1$ = Pfp), the ultimate disappearance of the mono-coupled product, and the appearance of a major peak representing the desired bis-coupled peptide. The side-chain protected peptidomimetic was precipitated in a large volume of $H_2O$ (500–700 mL), filtered, and dried under reduced pressure. The resulting crude product was side-chain deprotected using catalytic transfer hydrogenation with formic acid or high HF.[28] Neither method appears to be decidedly superior. Peptidomimetic **6** was then purified by HPLC and its authenticity confirmed by MALDI-MS *m/z*: $[M+H]^+$calcd 1103.4; found 1103.7.

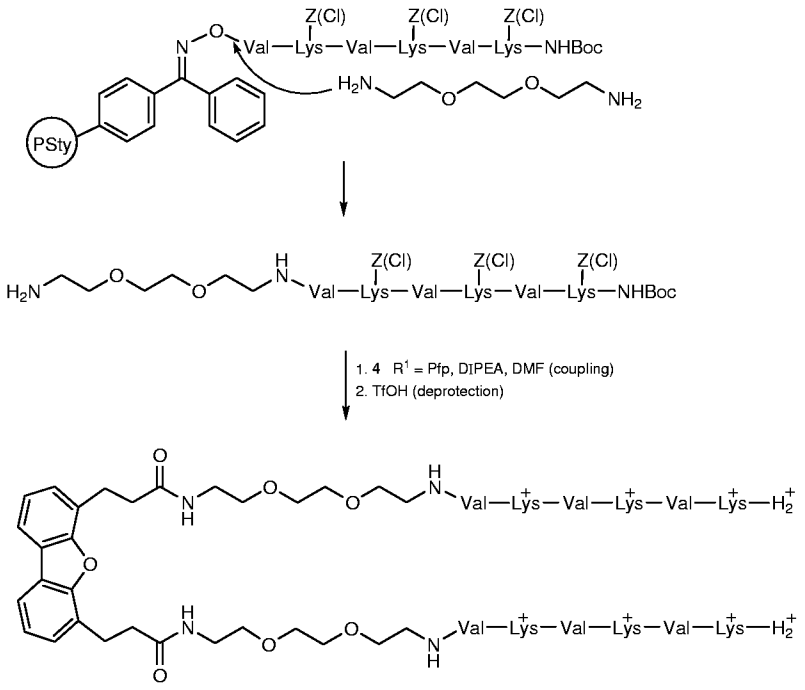
## 12.4.1.3 Incorporation of the Dibenzofuran-4,6-dipropanoic Acid Template with Intervening Ethylene Glycol Units Between the Template and Peptide Strands

The nucleation of parallel β-sheet structure via metal ion chelation in aqueous solution was investigated by incorporating an ethylene glycol spacer between a dibenzofuran-4,6-dipropanoic acid template and the two peptide strands (Scheme 6).[7] The flexibility of the ethylene glycol spacer allows the peptide to sample a variety of largely unordered conformations in the absence of metal ions. Upon binding of palladium(II), nickel(II), or copper(II) ions, the peptidomimetic adopts a well-defined β-sheet conformation as discerned from biophysical and spectroscopic studies (i.e., CD, IR absorption, and $^1$H NMR spectrometry).[7] The synthesis of the dibenzofuran-4,6-dipropanoic acid template was discussed

*for references see p 805*

above along with the bis(pentafluorophenyl) ester analogue **4** ($R^1$ = Pfp) (Scheme 4), which is one of the reactive components used in the segment condensation reaction. The other component comes from solid-phase peptide synthesis, which was carried out employing the 4-nitrobenzophenone oxime resin (see Section 12.4.1.2). The resin-bound protected peptides were cleaved from the oxime resin by nucleophilic displacement with excess ethylene glycol terminal bis-amine. One amino group of the ethylene glycol bis-amine attacks the oxime ester linkage liberating the peptide from the resin and affording a peptide having the ethylene glycol substructure linked at the C-terminus via an amide bond (Scheme 6). The other end of the ethylene glycol substructure bears an amino group (at what is formally the C-terminus of the peptide). The primary amine at the N-terminus of the peptide remains protected by a Boc group. The side-chain protected peptide was precipitated from DMF by injection into water. After filtration, the peptide was dried in vacuo and characterized by MALDI-MS. When analytical HPLC revealed less than 95% purity, the side-chain-protected peptide was purified by preparative C18 RP-HPLC. The side-chain-protected peptide, bearing a free amino group at the C-terminus and a Boc group at the N-terminus, was then reacted with the bis(pentafluorophenyl) ester **4** ($R^1$ = Pfp). The completeness of the coupling reaction was monitored by HPLC noting the disappearance of bis(pentafluorophenyl) ester **4** ($R^1$ = Pfp), and the mono-coupled intermediate. The *N*-Boc terminal protecting group was generally removed with TFA, and the peptidomimetic was side-chain deprotected by the low acidity trifluoromethanesulfonic acid method[28,29] to give peptidomimetic **7** (Scheme 6).

**Scheme 6** Incorporation of an Ethylene Glycol Spacer Between the Template and the Peptide Strands[7]




**7**

**Synthesis of Peptidomimetic 7:**[7]

The side-chain protected peptide was first synthesized on 4-nitrobenzophenone oxime resin utilizing Boc chemistry (see Section 12.4.1.2). Briefly, the first amino acid residue was attached to the resin using DIC activation. Subsequent couplings of residues to the resin-bound peptide were accomplished employing the BOP reagent for carboxy activation of the α-amino acids avoiding preneutralization of the resin-bound peptides for reasons discussed above. A six-fold excess of amino acid is utilized for the third

coupling to prevent competitive intramolecular piperazine-2,5-dione (DKP) formation which is a problem with the oxime resin owing to the activated nature of the resin linkage. The protected peptide having an N-terminal Boc group was cleaved from the oxime resin using an ethylene glycol bis-amine (3 equiv) as the nucleophile in DMF. The peptide fragment was precipitated into a large amount of $H_2O$ to remove the free bis-amine. After filtration, the peptide was dried under reduced pressure and the crude peptide characterized by MALDI-MS. The peptide was purified by RP-HPLC when this proved necessary as evaluated by analytical HPLC.

A 25-mL round-bottomed flask was charged with Boc-Lys[Z(Cl)]-Val-Lys[Z(Cl)]-Val-Lys[Z(Cl)]-Val-$NH(CH_2)_2O(CH_2)_2O(CH_2)_2NH_2$ (0.2 mmol), pentafluorophenyl ester **4** ($R^1$ = Pfp; 0.07 mmol), and DIPEA (0.15 mmol) in DMF (10–20 mL). Additional DIPEA was added in μL quantities, if necessary, to raise the pH of the reaction to pH 8–9 as monitored by wet pH paper. The basic mixture was stirred at 25 °C for 6–12 h. The reaction was monitored by HPLC to follow the loss of the reagent **4** ($R^1$ = Pfp) and the appearance of a single peptide coupled intermediate. After the reaction was complete, DMF was removed under reduced pressure. The protected crude peptide was subjected to 25–30% TFA treatment in $CH_2Cl_2$ for 30 min to remove the N-terminal Boc group. TFA and $CH_2Cl_2$ were then removed under reduced pressure. Excess TFA was eliminated azeotropically with hexane (removed under reduced pressure). The protected crude peptide was then side-chain deprotected by using the low-acidity TfOH method employing a soln of a *m*-cresol/dimethylsulfide/TFA/TfOH (1:3:5:1) mixture at 0 °C for 3–4 h. The side-chain deprotected crude peptide was precipitated with cold $Et_2O$ (150 mL). The crude peptide was washed with cold $Et_2O$ (2 × 50 mL), redissolved in acetate buffer at pH 4, and lyophilized. The final peptide was then purified by RP-HPLC and its authenticity confirmed by MALDI-TOF *m/z*: $[M+H]^+$calcd 1936.3; found 1936.8.

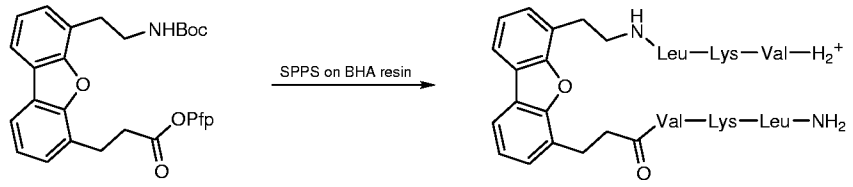
## 12.4.2 β-Sheet Peptides and Proteins Incorporating Templates via Solid-Phase Peptide Synthesis

### 12.4.2.1 Incorporation of the 4-(2-Aminoethyl)dibenzofuran-6-propanoic Acid Template

Previous studies by our laboratory reveal that the template 4-(2-aminoethyl)dibenzofuran-6-propanoic acid is capable of adopting a 15-membered ring intramolecularly hydrogen bonded conformation consistent with that required for antiparallel β-sheet structure.[3] In addition, a variety of studies suggests that the dibenzofuran substructure of this residue adopts a perpendicular orientation relative to the β-sheet plane, allowing the aromatic ring to interact with the hydrophobic side chains of the flanking α-amino acid residues. Structural variations on the template demonstrate that both hydrogen bonding and hydrophobic clustering are important for the nucleation of antiparallel β-sheet structure in aqueous solution. Herein, we present the synthetic methodology used to incorporate this template into a peptide chain by solid-phase peptide synthesis. We will briefly outline the synthesis of the template and then discuss in detail how the peptidomimetic is prepared. The synthesis of the template has been published previously.[30] Briefly, metallation and silylation of the 4-position of dibenzofuran, followed by metallation and iodination of the 6-position leads to an unsymmetrical molecule which is subjected to a Heck reaction with ethyl acrylate. The trimethylsilyl group is replaced with iodine employing iodine monochloride, and the resulting aryl halide subjected to a second Heck reaction with benzyl acrylate. After hydrogenation, dibenzofuran bears a propanoic acid residue at the 4-position and a propanoic acid ethyl ester at the 6-position. A Curtius rearrangement in *tert*-butyl alcohol with diphenylphosphoryl azide yields the Boc derivative at the 4 position, which is utilized in the synthesis of peptidomimetic **9** by way of its pentafluorophenyl ester derivative **8** (Scheme 7).[3,30]

*for references see p 805*

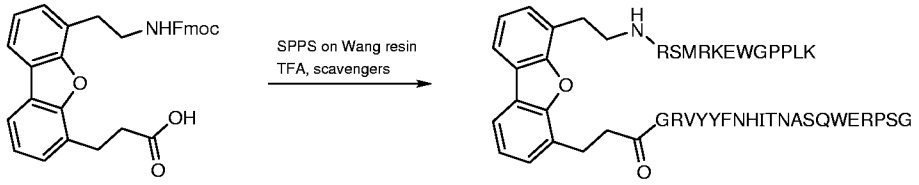
**Scheme 7** Synthesis of Peptidomimetics Incorporating the 4-(2-Aminoethyl)dibenzofuran-6-propanoic Acid Template[3,30]




**8** **9**

In addition to incorporating the 4-(2-aminoethyl)dibenzofuran-6-propanoic acid template into small peptides where a reverse turn is desired, we have also recently incorporated this template into a mini-protein called the PIN WW domain. WW domains have a three-stranded antiparallel β-sheet structure that mediates intracellular protein–protein interactions.[31] Substitution of this β-turn mimetic into loop 1 of the PIN WW domain leads to a folded, three-stranded, antiparallel β-sheet structure with a stability indistinguishable from that of the all α-amino acid sequence. The template-incorporated PIN WW domain (**11**) was synthesized by an Fmoc-based solid-phase peptide synthesis strategy (Scheme 8), utilizing *N*-Fmoc-protected 4-(2-aminoethyl)dibenzofuran-6-propanoic acid **10**.[11] The synthesis of **10**, similar to that of **8**, has been published.[30]

**Scheme 8** Synthesis of the Template-Incorporated WW Domain[11]




**10** **11** PIN WW

**Synthesis of Peptidomimetic 9:**[3]

Boc-Leu-OH was loaded onto BHA resin (0.66 mmol·$g^{-1}$) by shaking DIC-activated amino acid (1.6 equiv) with BHA resin for 24 h in $CH_2Cl_2$. The resin was washed with DMF (2 × 1 min), $CH_2Cl_2$ (1 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), and $CH_2Cl_2$ (4 × 1 min). The completion of the coupling was monitored by the Kaiser ninhydrin test. If the result was slightly positive (blue in color), the remaining unreacted amino groups on the resin were capped by acetylation with $Ac_2O$. The following cycle was used for each coupling of a new residue: TFA prewash (50% TFA, 1% PhSMe × 1 min); TFA deprotection (50% TFA, 1% PhSMe × 50 min); wash steps $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (4 × 1 min); preneutralization, 12% DIPEA (1 × 1 min); neutralization, 12% DIPEA (1 × 9 min); coupling [amino acid (3 equiv), BOP (3 equiv), DIPEA (4 equiv) in $CH_2Cl_2$ containing 10–15% DMF for 2–8 h], DMF (2 × 1 min); wash steps $CH_2Cl_2$ (1 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), and $CH_2Cl_2$ (4 × 1 min). The couplings of Boc-Lys[Z(Cl)]-OH and Boc-Val-OH preceding template incorporation were monitored by the Kaiser ninhydrin test. A solution of pentafluorophenyl active ester **8** (3 equiv) and DIPEA (3 equiv) in $CH_2Cl_2$ was employed with overnight shaking to incorporate the template, 4-(2-aminoethyl)dibenzofuran-6-propanoic acid, into the peptide. The Boc group was removed using standard procedures and Boc-Leu-OH, Boc-Lys[Z(Cl)]-OH, and Boc-Val-OH were coupled in that order as prescribed above. After the completion of the α-amino acid sequence incorporating the template, the resin-bound peptides were treated with TFA to remove the Boc group on Val, dried under reduced pressure, and finally cleaved from the resin and the side-chain deprotected by HF.[28] The crude peptidomimetic **9** was purified by preparative HPLC as described earlier. The structure of peptidomimetic **9** was confirmed by MALDI-TOF-MS *m/z*: $[M+H]^+$ calcd 964.3; found 964.1.

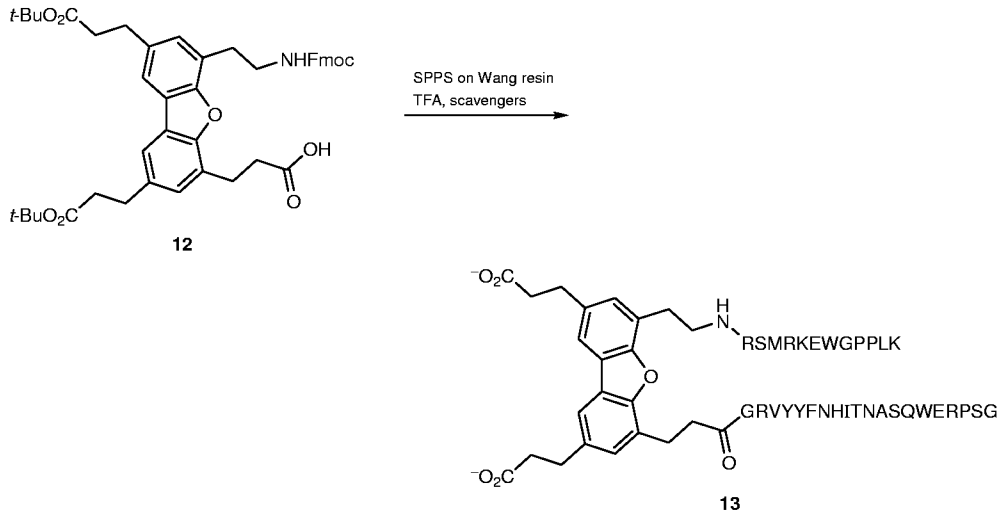
**Synthesis of Peptidomimetic 11:**[11]
The synthesis was performed on preloaded Wang resin (0.1 mmol) using standard Fmoc-based SPPS chemistry. The coupling reactions of the amino acids (10 equiv) were carried out in NMP (10 mL), employing the activating agents HBTU (10 equiv) and HOBt (10 equiv) in the presence of DIPEA (10 equiv) for 30 min. After the coupling was completed, the resin was washed with DMF (3 × 1 min). Removal of the N-terminal Fmoc protecting group was achieved with 20% piperidine in DMF (5 mL) for 20 min followed by washes with NMP (3 × 1 min). The template **10** (1.2 equiv) was incorporated using the activating agents HBTU/HOBt (1.2 equiv) in NMP (2 mL) in the presence of DIPEA (2.2 equiv, 2 M in NMP), over a 24-h period. Peptide cleavage and side-chain deprotection were carried out by gently agitating the crude peptide-resin in Reagent K (Reagent K: 82.5% TFA, 5% m-cresol, 5% PhSMe, 5% $H_2O$, and 2.5% EDT; 1 mL per 0.025 mmol of crude peptide on the resin) for 2 h at rt. The crude peptide **11** was precipitated in tBuOMe, centrifuged, redissolved in DMF, and purified by RP-HPLC.

### 12.4.2.2 Incorporation of the 4-(2-Aminoethyl)-2,8-bis(2-carboxyethyl)-dibenzofuran-6-propanoic Acid Template

Recently, peptidomimetics have been incorporated into small proteins to help understand the basis for their folding and stability.[11,12] Despite successful syntheses of several template-based proteins, there are problems associated with their aqueous solubility. Hydrophilic templates that have the potential to alleviate this problem are clearly desirable. The *N*-Fmoc-protected 4-(2-aminoethyl)-2,8-bis-(2-carboxy-*tert*-butyl)dibenzofuran-6-propanoic acid template **12** is a more soluble analogue of β-turn mimetic **10**, created by adding 2-carboxyethyl substituents at positions 2 and 8 on the dibenzofuran ring. These two polar substituents enhance the aqueous solubility of the template without interfering with hydrogen-bonded hydrophobic cluster formations thought to be critical for nucleating β-hairpin formation. Comparisons of WW domains incorporating residues $N^{\alpha}$-deprotected **10** and $N^{\alpha}$- and $C^{\gamma}$-deprotected **12** reveal that the latter template improves solubility without a decrease in the thermodynamic stability of the resulting β-sheet structure.[12] The synthesis of template **12** has been published previously.[12] Briefly, the synthesis commences with electrophilic aromatic iodination of dibenzofuran at positions 2 and 8 and conversion of the iodine atoms into the much less reactive trimethylsilyl groups by lithiation followed by treatment with trimethylsilyl chloride. Oxygen-directed metalation at the 4- and 6-positions followed by halogenation with iodine affords 4,6-diiododibenzofuran with trimethylsilyl groups at positions 2 and 8. Positions 4 and 6 were differentially functionalized, in analogy to the approach used for **10**. The trimethylsilyl groups were transformed to reactive iodide functionalities with iodochloride. The aryl iodides at the 2- and 8-positions were transformed into *tert*-butyl acrylates utilizing a tandem Heck reaction. Saponification and hydrogenation, which also enable the conversion of the Z-protecting group into an Fmoc group, afford **12**. This template was incorporated into the PIN WW domain via a Fmoc-based solid-phase peptide synthesis strategy to afford **13** (Scheme 9).

*for references see p 805*

**Scheme 9** Synthesis of the Template-Incorporated WW **13**[12]

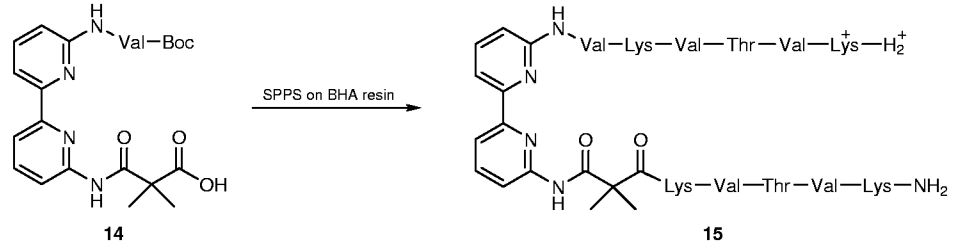



**Synthesis of Peptidomimetic 13:**[12]
The synthesis was performed on preloaded Wang resin (0.1 mmol) using standard Fmoc-based SPPS chemistry. The coupling reactions of the amino acids (10 equiv) were carried out in NMP (10 mL), employing the activating agents HBTU (10 equiv) and HOBt (10 equiv) in the presence of DIPEA (10 equiv) for 30 min. After the coupling was completed, the resin was washed with DMF (3 × 1 min). Removal of the N-terminal Fmoc protecting group was achieved with 20% piperidine in DMF (5 mL) for 20 min followed by washes with NMP (3 × 1 min). The template **12** (1.2 equiv) was incorporated using the activating agents HBTU/HOBt (1.2 equiv) in NMP (2 mL) in the presence of DIPEA (2.2 equiv, 2 M in NMP) employing a reaction time of 24 h. Peptide cleavage, side-chain deprotection, and template deprotection were accomplished by gently agitating the crude peptide-resin in Reagent K (Reagent K: 82.5% TFA, 5% m-cresol, 5% PhSMe, 5% $H_2O$, and 2.5% EDT; 1 mL per 0.025 mmol of crude resin-bound peptide) for 2 h at rt. Crude peptide **13** was precipitated in tBuOMe, centrifuged, redissolved in DMF, and purified by RP-HPLC.

### 12.4.2.3 Incorporation of the 6,6′-Bis(acylamino)-2,2′-bipyridine-Based Template

The concept of using conformationally restricted molecules to nucleate β-sheet formation where metal binding initiates folding by further restricting the conformation has been investigated by our laboratory[4] and others. Metal-binding sites are common in protein structures and in some cases the metals are required for folding. We have synthesized a metal-binding template to nucleate antiparallel β-sheet folding.[4] This substructure (e.g., **14** in Scheme 10), when coordinated to copper(II), is designed to replace the $i+1$ and $i+2$ residues of a β-turn. Crystal structures of the template bound to copper(II) reveal that the bipyridine ring system adopts a cisoid conformation with a nearly perfect square-planar geometry about the copper ion wherein the amide or imidate ligands (the amide is converted into an imidate at higher pH) and the pyridine rings are roughly coplanar. The synthesis of this metal-binding nucleator **14** begins with 6,6′-diamino-2,2′-bipyridine.[32] Attachment of the 2,2′-dimethylmalonate by an amide bond to the bipyridine utilizes 2-chloro-1-methylpyridinium iodide activation to afford the malonate ethyl ester.[4] Boc-Val-OH was coupled to the remaining anilino nitrogen via the BOP activation method. This residue was chosen owing to its high β-sheet propensity. Saponification of the malonate ester protecting group afforded template **14** with an *N*-Boc group at one end and a carboxylic acid at the other, ready to be utilized in solid-phase synthesis of antiparallel β-sheet forming peptides, e.g. **15**.[3]

**Scheme 10** Synthesis of Peptidomimetics Incorporating the 6,6′-Bis(acylamino)-2,2′-bipyridine-Based Template[3,4]




**Synthesis of Peptidomimetic 15:**[3]
Boc-Lys[Z(Cl)]-OH was loaded onto BHA resin (0.7 mmol·g$^{-1}$) by shaking the BOP-activated Lys derivative (3.0 equiv) with BHA resin for 24 h in $CH_2Cl_2$ containing 10% DMF (5 equiv DIPEA). The peptide was elongated by sequentially coupling the following protected residues Boc-Val-OH, Boc-Thr(OBzl)-OH, Boc-Val-OH, and Boc-Lys[Z(Cl)]-OH to resin-bound Lys using standard solid-phase synthesis procedures (see Section 12.4.2.1). The bipyridine-based template **14** was incorporated into the polypeptide sequence by preactivating **14** (2 equiv) with DIC (2 equiv) and HOBt (2 equiv) in $CH_2Cl_2$/DMF (9:1) at 0 °C for 20 min. The HOBt ester of **14** was then added to the resin along with DIPEA (1.1 equiv) and a minimal amount of additional $CH_2Cl_2$ (just enough to form a resin slurry). The following residues were coupled sequentially to the resin-bound peptidomimetic, again using the procedures outlined in Section 12.4.2.1; Boc-Lys[Z(Cl)]-OH, Boc-Val-OH, Boc-Thr(OBzl)-OH, Boc-Val-OH, and Boc-Lys[Z(Cl)]-OH. Peptidomimetic **15** was deprotected and cleaved from the resin employing the high HF procedure,[28] and purified by RP-HPLC. The integrity of the synthesis was confirmed by MALDI-TOF-MS *m/z*: $[M+H]^+$ calcd 1510.9; found 1511.1.

## 12.4.3 β-Sheet Peptides Incorporating Templates via a Combination of Solution- and Solid-Phase Peptide Synthesis

### 12.4.3.1 Incorporation of the Dibenzofuran-4,6-dipropanoic Acid Template

The aromatic diacid residue dibenzofuran-4,6-dipropanoic acid **4** ($R^1$ = H) is known to nucleate parallel β-sheet structures in aqueous solution.[2] This template propagates its conformational bias onto the attached peptide chains through a 14-membered ring hydrogen bond and a hydrophobic cluster wherein the aromatic ring interacts with the side chains of the flanking hydrophobic α-amino acid residues.[2,7] Together these conformational constraints are enough to induce folding in peptides that are prepared via solution-phase segment condensation reactions (see above).[2] This synthetic approach affords parallel β-sheet forming peptidomimetics substituted with identical peptide sequences on each propanoic acid arm. Dibenzofuran-4,6-dipropanoic acid-based parallel β-sheet forming peptidomimetics having different peptides attached to each propanoic acid substructure of the template have also been prepared.[2]

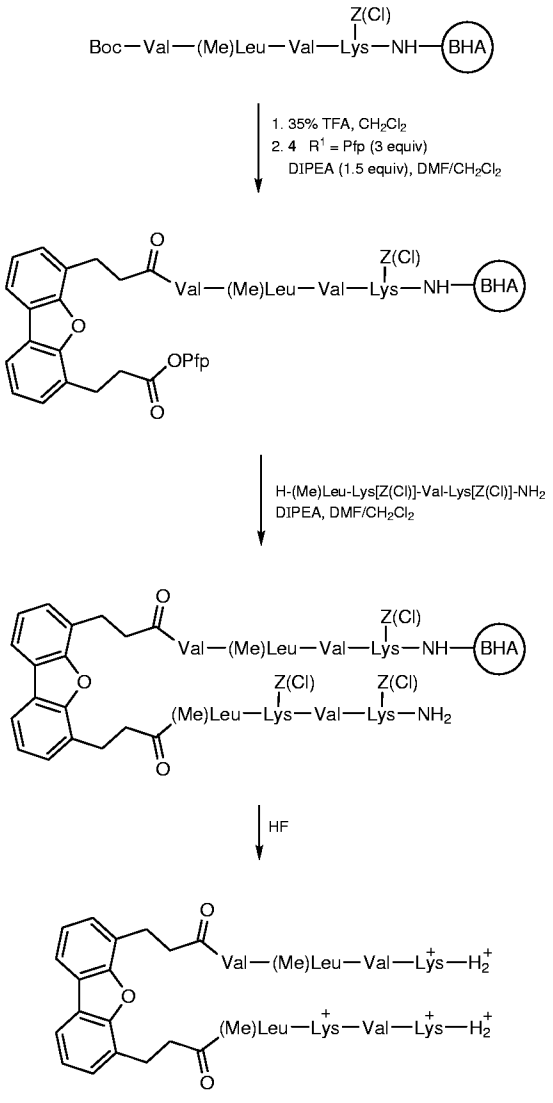
Using a solution segment condensation reaction analogous to that described above it is possible to obtain a maximal 50% yield of a monosubstituted template using one equivalent of peptide nucleophile. The monosubstituted product can be purified away from the unsubstituted and disubstituted templates, and subjected to a second coupling reaction with a different peptide nucleophile affording an unsymmetrical peptidomimetic, albeit in modest yield.

A better, higher yielding approach for the preparation of templates attached to different peptides employs a combination of solid-phase and solution-phase couplings (Scheme 11).[2] A peptide chain is synthesized on a low-loading benzhydrylamine resin and reacted with a bifunctional electrophile such as the bis(pentafluorophenyl) ester **4** ($R^1$ = Pfp) (low resin

*for references see p 805*

loading is necessary to avoid two resin-bound peptide chains from reacting with the active ester to afford a symmetrically substituted peptidomimetic). The remaining active ester group on the template-terminated resin-bound peptide is attacked by a second side-chain protected peptide bearing a C-terminal amide group and a nucleophilic free α-amino group. This second peptide was prepared separately on oxime resin as outlined above. The approach described within provides a means to make medium-sized peptide chains (e.g., **16**) that have different peptides oriented in a parallel fashion relative to the template (in this case N-to-C).[2]

**Scheme 11** Synthesis of Peptidomimetics Incorporating the 4,6-Dibenzofuran-Based Template To Give Asymmetrical Parallel β-Sheet Sequences[2]




**16**

**Synthesis of Peptidomimetic 16:**[2]
Asymmetric parallel β-sheet sequences incorporating template **4** ($R^1$ = H) were synthesized on BHA resin and reacted with the bifunctional electrophile **4** ($R^1$ = Pfp). The remaining active ester on the template-terminated resin-bound peptide was reacted with the N-terminus of a different side-chain protected peptide prepared on the oxime resin, as shown in Scheme 11.

*Synthesis of Strand 1:* The peptide H-Val-(Me)Leu-Val-Lys[Z(Cl)]-resin corresponding to strand 1 was synthesized on low-loading BHA resin (0.2 meq·$g^{-1}$) using the traditional SPPS strategy outlined above.[14] Briefly, the BHA resin was pre-swollen in $CH_2Cl_2$ for 10 min and washed with 12% DIPEA in $CH_2Cl_2$ (2 × 5 min) and $CH_2Cl_2$ (4 × 1 min) prior to use. Boc-Lys[Z(Cl)]-OH was loaded onto the resin by shaking the amino acid (1.2 equiv; based on 0.2 mmol·$g^{-1}$ of the resin) and DIC in $CH_2Cl_2$ for 24 h. The resin was washed with $CH_2Cl_2$ (2 × 1 min), DMF (2 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), and $CH_2Cl_2$ (4 × 1 min). The remaining unreacted amino groups of the peptidyl-resin were acetylated by reaction with $Ac_2O$ (30 equiv) in $CH_2Cl_2$. The following cycle was employed for the sequential coupling of Boc-Val-OH, Boc-(Me)Leu-OH, and Boc-Val-OH to the growing strand: TFA prewash (1 × 1 min); TFA deprotection (1 × 50 min), $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), $CH_2Cl$ (2 × 1 min), iPrOH (1 × 1 min), and $CH_2Cl_2$ (4 × 1 min); preneutralization, 12% DIPEA (1 × 1 min); neutralization, 12% DIPEA (1 × 9 min); coupling [Boc-amino acid (3 equiv), BOP (3 equiv), DIPEA (4 equiv) in $CH_2Cl_2$ containing 10–15% DMF for 2–5 h], DMF (2 × 1 min), iPrOH (1 × 1 min), $CH_2Cl_2$ (2 × 1 min), iPrOH (1 × 1 min), and $CH_2Cl_2$ (4 × 1 min). The completion of each coupling step was monitored by the Kaiser ninhydrin test. If the result was positive, the coupling reaction was repeated until the Kaiser test became negative. The bis(pentafluorophenyl) ester **4** ($R^1$ = Pfp) was employed to incorporate template **4** ($R^1$ = H) into the peptide. The coupling was performed by shaking the preneutralized resin with bis-active ester **4** ($R^1$ = Pfp; 3 equiv) utilizing DIPEA (2 equiv) in $CH_2Cl_2$ containing 10% DMF. Completion of the coupling step was again monitored by the Kaiser ninhydrin test.

*Synthesis of Strand 2:* The peptide, H-(Me)Leu-Lys[Z(Cl)]-Val-Lys[Z(Cl)]-$NH_2$, was synthesized on oxime resin (~0.5 mmol·$g^{-1}$ loading) using an experimental approach identical to that described above. $NH_4OAc$ (10 equiv based on the resin loading) in the presence of DIPEA (2 equiv) was used to cleave strand 2 from the oxime resin yielding the peptide amide. The Boc group at the N-terminus of strand 2 was then removed with a 35% TFA soln in $CH_2Cl_2$ (30 min) to yield the side-chain-protected segment containing a free terminal amino group. The oxime-derived peptide having a free amino group (TFA salt) was purified by preparative C18 HPLC using a linear 0–50% gradient of MeCN/TFA/$H_2O$ (94.5:0.1:5) in MeCN/TFA/$H_2O$ (5:0.1:94.5) over 20 min, employing UV detection at 230 nm. The chemical structure was confirmed by MALDI-TOF-MS.

*Attaching Different Parallel Strands to the Template:* Peptide strand 2 (1.2 equiv) was coupled to the remaining propanoic acid arm of the template (via a pentafluorophenyl ester) utilizing DIPEA (3 equiv) in DMF for 24 h with shaking at 25 °C. This is a segment condensation reaction between strand 1-residue 4 (0.2 mmol·$g^{-1}$) that is covalently linked to BHA resin and strand 2 which is free in soln as the nucleophilic component. The resin-bound peptide was then washed with warm DMF (4 × 1 min) to remove excess nucleophile, $CH_2Cl_2$ (2 × 1 min), dried under reduced pressure, and finally cleaved from the resin and side-chain deprotected employing the high HF method.[28] The crude peptide was purified by preparative HPLC as described above resulting in a 10% overall yield. (Yields approaching 20% were observed for two other nonapeptides prepared in this fashion and a yield of 30% was achieved for a tridecapeptide analogously prepared.) The primary structure was confirmed by MALDI-TOF-MS *m/z*: $[M+H]^+$ calcd 1246.8; found 1247.2.


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[30] Bekele, H.; Nesloney, C. L.; McWilliams, K. W.; Zacharias, N. M.; Chitnumsub, P.; Kelly, J. W., *J. Org. Chem.*, (1997) **62**, 2259.
[31] Sudol, M., *Prog. Biophys. Mol. Biol.*, (1996) **65**, 113.
[32] Schneider, J. P.; Topgi, R. S.; Kelly, J. W., *Synth. Commun.*, (1992) **22**, 1033.

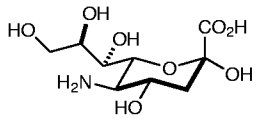
# 12.5 Synthesis of Peptides with Sugar Amino Acids

E. Gräfin von Roedern, M. A. Born, M. Stöckele, and H. Kessler

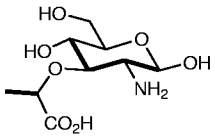
## 12.5.1 Sugar Amino Acids as Peptidomimetics

Sugar amino acids (Saa)[1] are carbohydrate scaffolds that carry an amino and a carboxy function.[2] Due to their structural and functional diversity, no detailed nomenclature for this class of amino acids has yet been established. Saa are found in nature as subunits of oligosaccharides (neuraminic acid **1**) in cell walls of bacteria (muraminic acid **2**) and in some antibiotics **3**–**6** (Scheme 1).[2–8]

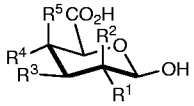
**Scheme 1** Naturally Occurring Sugar Amino Acids[3–8]



**1** neuraminic acid



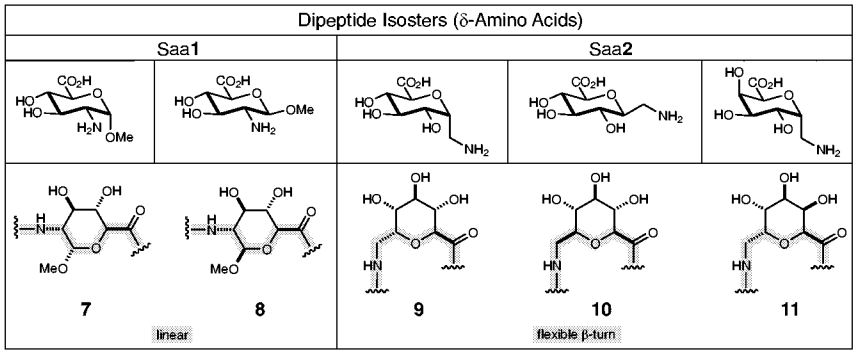
**2** muraminic acid



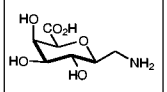
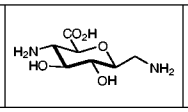
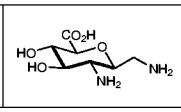
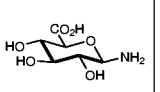
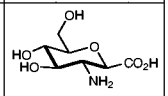
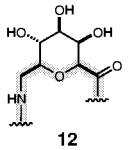
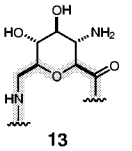
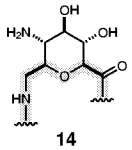
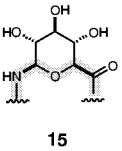
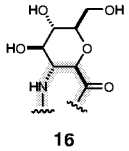
| | | $R^1$ | $R^2$ | $R^3$ | $R^4$ | $R^5$ | Ref |
|---|---|---|---|---|---|---|---|
| glucosaminuronic acid | **3** | $NH_2$ | H | OH | OH | H | [3] |
| galactosaminuronic acid | **4** | $NH_2$ | H | OH | H | OH | [4] |
| mannosaminuronic acid | **5** | H | $NH_2$ | OH | OH | H | [7] |
| 4-amino-4-deoxyglucurononic acid | **6** | OH | H | OH | $NH_2$ | H | [5,6] |

The study of Saa has so far concentrated on their use as biopolymer building blocks to mimic oligo- and polysaccharide structures via amide bond linkages.[2,9–43] In 1994 the first example of a Saa as a new type of peptidomimetic [replacement of amino acid(s) by Saa] was reported,[44] which was later extended to the Saa construction kit **7**–**16**, and others, a versatile tool for the manipulation of peptide properties[45–51] (Scheme 2).

**Scheme 2** Sugar Amino Acid Construction Kit[44–48]

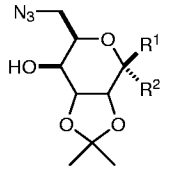



*for references see p 819*

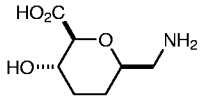
| Dipeptide Isosters (δ-Amino Acids) | | | γ-Amino Acid | β-Amino Acid |
|---|---|---|---|---|
| Saa2 | | | Saa3 | Saa4 |
|  |  |  |  |  |
|  **12** |  **13** flexible β-turn |  **14** |  **15** β-turn |  **16** γ-turn |

Since 1994 the syntheses of many other Saa **17–26** have been published, thereby expanding this diverse class of peptidomimetics for further incorporation into peptides (Scheme 3).[52–70]

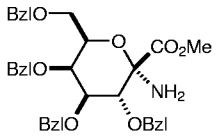
**Scheme 3** Further Sugar Amino Acids Building Blocks[52,53,61–70]



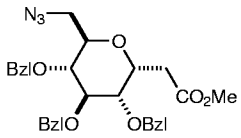
**17** $R^1 = H$; $R^2 = CH_2CO_2Me$
$R^1 = CH_2CO_2Me$; $R^2 = H$



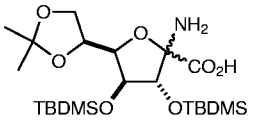
**18**



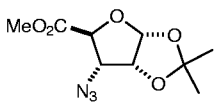
**19**



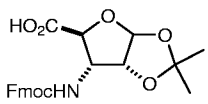
**20**



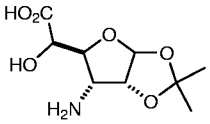
**21**



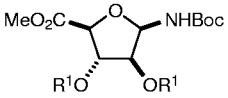
**22**



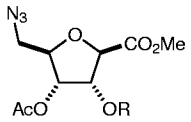
**23**



**24**



**25** $R^1 = H$, Bzl



**26**

Carbohydrates represent an attractive option for scaffolding as they contain well-defined and readily convertible substituents with a rigid pyran ring[71–77] or more flexible furan ring.[78] Carbohydrates are among the major classes of biomolecules and have the potential for almost unlimited structural variation.[79] The molecular diversity of carbohydrates offers a valuable tool for drug discovery in the areas of biologically significant oligosaccharides, glycoconjugates, and molecular scaffolding by investigating their structural and functional impact. Glycosylation of peptides and proteins is a key biological process, for example, to increase hydrophilicity. The modification of the naturally occurring O/N-glycosidic bond remains of interest as a method to increase the enzymatic stability.[80,81] The synthesis of Saa

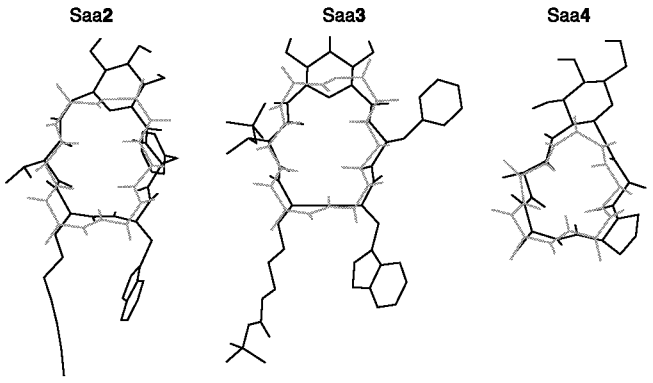
makes it possible to exploit these features and make them accessible for peptide chemistry.[44]

Novel properties of peptides can be obtained by incorporation of Saa. Peptides are characteristically highly flexible molecules whose 3D structure is strongly influenced by their environment. Their often unordered conformation in solution complicates their use in determining the receptor bound bioactive structure.[82,83] Conformational constraints,[84–88] cyclization,[89] and/or replacement of the peptide backbone or parts of it[62,90–92] can provide information on the required conformation for biological activity.

Scheme 2 shows a construction kit for predetermined constrained local conformations in synthetic peptides containing a series of Saa [we use this three-letter code analogous to the amino acid three-letter code, e.g. Fmoc-Saa**2**$(Bzl)_3$-OH indicates the N-terminally protected Saa No. 2 with a free carboxy end and benzylated hydroxy groups as side-chain modifications]. These units have possibilities as mimetic structures for both amino acids and dipeptide isosteres. They contain a six-membered ring with all substituents in equatorial positions, except for the methoxy group in α-Saa**1** and the galactose Saa**2** building block. The chair conformation renders them very stable and rigid, consequently allowing the conformational restriction introduced into peptides to be predicted. α-Saa**1** and β-Saa**1** constrain a linear peptide conformation, whereas the others act as turn mimetics. The turn diameter decreases from Saa**2** to Saa**4**. Saa**2** and Saa**3** serve as β-turn mimetics and Saa**4** as a γ-turn mimetic.[45] Incorporation of these turn mimetics into cyclic peptides has an effect on the turn pattern in these compounds.

The conformational influence of Saa**2**, Saa**3**, and Saa**4** on the peptide backbone of three cyclic peptides **27**–**29** (Figure 1) has been investigated in more detail by NMR spectroscopy, distance geometry (DG), and subsequent molecular dynamic (MD) calculations.[45]

**Figure 1** Overlay of Averaged and Minimized Structures of Cyclic Peptides on Their Idealized βII′/βII′- or βII′/γ-Turn Arrangement[45]a,b




[a] Reprinted from (Graf von Roedern; Lohof; Hessler; Hoffmann; Kessler, *Journal of the American Chemical Society*, Vol. 118), Copyright (1996), p 10156, with permission from The American Chemical Society.

[b] Black = averaged and minimized structures; gray = idealized βII′/βII′-turn arrangement (**27** and **28**) or βII′/γ-turn arrangement (**29**).

The pharmacophoric and the hydrophilic and lipophilic profiles of the Saa scaffolds can be manipulated by the substitution pattern of the carbohydrate functionalities, e.g. Saa with nonfunctionalized hydroxy groups are more hydrophilic compared to benzylated Saa which can act as mimetic for the Phe residue.[93]

Recently, several cyclic Saa-containing peptides and cyclic Saa structures were described.[65,94–96]

*for references see p 819*

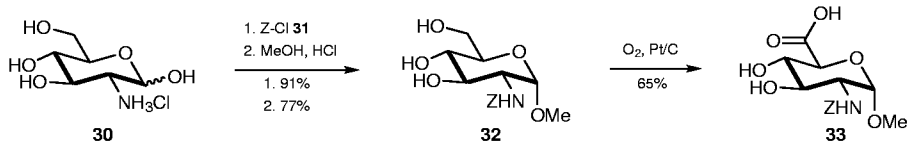
## 12.5.1.1 Sugar Amino Acid Building Blocks

The synthesis of Saa can be accomplished in a few steps from readily available carbohydrates, i.e. glucose, glucosamine, galactose, etc. The amino function can be introduced as an azide, a cyanide, or a nitromethane equivalent followed by reduction. These functional groups may even serve as protecting groups during the synthesis. The carboxy group is introduced directly as $CO_2$, a hydrolyzable cyanide, or a selectively oxidizable alcohol.

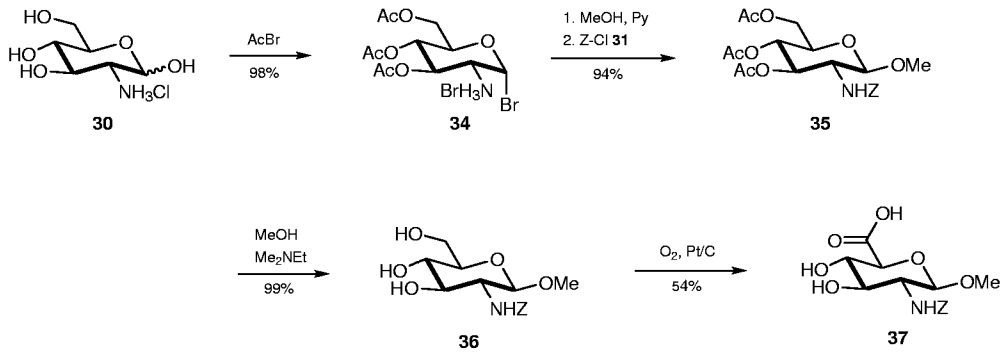
### 12.5.1.1.1 Synthesis of 2-Amino-2-deoxy-1-*O*-methyl-α/β-D-glucopyranuronic Acid

Z-α-Saa**1**-OH (**33**) is synthesized as described by Heyns and Paulsen[2] starting from glucosamine **30** in an overall yield of 45% (Scheme 4). The amino group is protected using Z-Cl (**31**) in 91% yield. The α-methyl glucoside **32** is obtained via the classical acid-catalyzed Fischer glycosylation using MeOH.[97] The primary hydroxy group of **32** is subsequently oxidized using 10% Pt/C and oxygen. The β-anomer Z-β-Saa**1**-OH (**37**) is prepared from glucosamine **30** in 49% yield[98,99] (Scheme 5). The β-methyl glucoside is obtained via treatment of bromide **34** with MeOH in pyridine and further Z-protection. Deacetylation of **35** by methanolysis and subsequent selective oxidation of the primary alcohol of **36** using Pt/C and oxygen affords Z-β-Saa**1**-OH (**37**), which can be employed in standard solution-phase peptide synthesis. Since both derivatives are *O*-glycosides prolonged exposure to acidic conditions, like TFA, leads to degradation.

**Scheme 4** Synthesis of Z-α-Saa**1**-OH[2,45]




**Scheme 5** Synthesis of Z-β-Saa**1**-OH[45,98,99]




**Z-α-Saa1-OH (33):**[2,45,100]

Dry glucosamine hydrochloride (**30**; 64.5 g, 0.3 mol) and $NaHCO_3$ (94.5 g, 1.125 mol) were dissolved in $H_2O$ (0.9 L) in a 4-L round-bottomed flask. Under vigorous stirring 96% Z-Cl (**31**; 75 mL, 500 mmol) was added and the soln stirred for 6 h. The foamy soln was filtered through a fritted glass funnel and washed with cold $H_2O$ (~300 mL). The dry filter cake was washed with $CHCl_3$ (400 mL) and $Et_2O$ (400 mL). After drying under reduced pressure the Z-derivative of **30** (colorless powder) was obtained; yield: 85 g (91%); mp 206 °C (dec).

This latter compound (16 g, 51 mmol) was dissolved in 1% HCl and dry MeOH (1 L) which was heated to 45–50 °C in a gas-tight container for 11 d. The mixture was neutralized with $PbCO_3$ until no more $CO_2$ evolution was observed. The inorganic salts were filtered off and the solvent was removed to

dryness. The residue was recrystallized with $H_2O$ (70 mL) to give **32** as colorless needles; yield: 12.8 g (77%); mp 161 °C (dec).

Compound **32** (6 g, 18.3 mmol) was dissolved in $H_2O$ (100 mL) and stirred with 10% Pt/C (1.0 g, ca. 50% $H_2O$) at 90 °C under a stream of $O_2$. The gas was pumped through the soln within a closed apparatus using an air balloon as a refillable gas reservoir and purified by passing through 4 M NaOH. After 5 h another portion of 10% Pt/C (1.0 g, ca. 50% $H_2O$) was added. After completion of the oxidation (oxidation was monitored by TLC) the catalyst was filtered off and the filtrate was washed with $CH_2Cl_2$ (2 × 30 mL), reduced to 50 mL, and heated to 80 °C. The pH was adjusted to 2 by addition of concd HCl. The product **33** crystallized upon cooling as colorless needles; yield: 4.04 g (65%); mp 184 °C; $R_f$ 0.53 (MeCN/$H_2O$ 4:1).

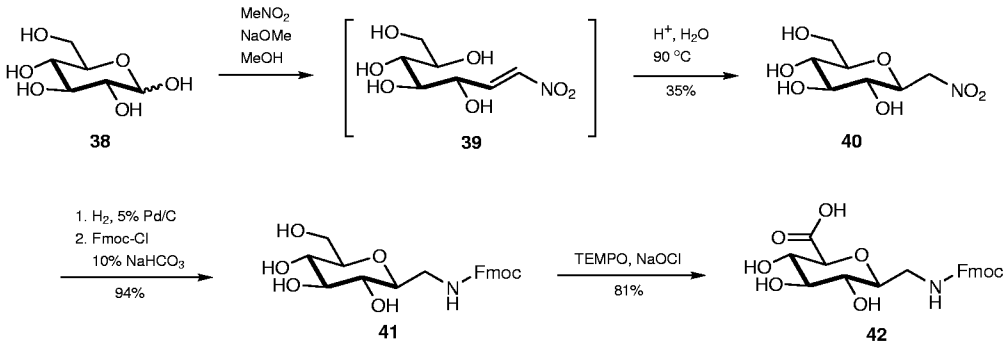
**Z-β-Saa1-OH (37):**[45,98,99]

Dry glucosamine hydrochloride (**30**; 60 g, 0.28 mol) was added to AcBr (100 mL, 1.35 mol) and stirred under $N_2$ in the dark for 3 d. The resulting pink lumps were filtered off, minced with a spatula, and washed with small amounts of dry $Et_2O$. The solid was dried to give pink crystals of **34** (123 g, ~98%), which were dissolved in dry MeOH (1 L). Dry pyridine (22 mL) was added, and the soln was stirred for 6 h. The solvent was removed and the residue dried in vacuo. The residue was redissolved in $H_2O$ (200 mL) and added dropwise to a soln, which was vigorously stirred, made as follows: to EtOAc (1.4 L) a soln of $NaHCO_3$ (94 g, 1.12 mol) in $H_2O$ (1 L) was added, followed by 50% Z-Cl (**31**; 100 mL, 293 mmol) in toluene. After the evolution of $CO_2$ stopped, the organic phase was separated and washed with 0.5 M HCl (3 × 100 mL) and $H_2O$ (3 × 100 mL), then dried ($MgSO_4$). The solvent was removed to give **35** as a colorless solid; yield: 115 g (94%); mp 230 °C (dec). To a soln of **35** (62 g, 0.14 mol) in abs MeOH (200 mL) was added $Me_2NEt$ (10 mL) and the soln was stirred for 24 h. The solvents were removed in vacuo and the residue recrystallized ($H_2O$) to give **36** as colorless needles; yield: 44 g (99%). The solid **36** (10.0 g, 31 mmol) was dissolved in $H_2O$ (100 mL) and stirred with 10% Pt/C (5.0 g, ca. 50% $H_2O$) at 90 °C under a stream of $O_2$. The gas was pumped through the soln within a closed apparatus using an air balloon as a refillable gas reservoir and purified by passing through 4 M NaOH. The pH of the mixture was maintained between 7 and 8 by addition of 10% aq $NaHCO_3$. Additional catalyst (2 g) was added after 15 h and 30 h and after 50 h the catalyst was removed by filtration. The filtrate was neutralized by addition of acidic ion-exchange resin (Amberlyst 15, strongly acidic, $H^+$ form, 20 g) and stirred for 10 min, filtered, and concentrated in vacuo to 30 mL. The product **37** crystallized at 4 °C as colorless needles; yield: 5.2 g (54%); mp 142 °C (dec); $[\alpha]_D^{20}$ −35.7 (*c* 1.0, MeOH).

### 12.5.1.1.2 Synthesis of 7-Amino-2,6-anhydro-7-deoxy-L-*glycero*-L-*gulo*-heptonic Acid

Saa**2** can be prepared as the Fmoc derivative, Fmoc-Saa**2**-OH (**42**), for SPPS or as Z-Saa**2**-OMe for solution-phase peptide synthesis. The -$CH_2NH_2$ equivalent is introduced as $MeNO_2$ via nucleophilic aldol reaction. The linear condensation product **39** is cyclized by acid-catalyzed intramolecular ring closure to yield β-D-glucopyranosylnitromethane (**40**) which, because of its polarity, is isolated using basic ion-exchange resin. The nitro compound is reduced with hydrogen using 10% Pd/C under pressure (reduction under atmospheric pressure takes about three times as long) and subsequently protected with Fmoc-Cl to give **41** (Scheme 6). The primary hydroxy group can be selectively oxidized to **42** using sodium hypochlorite under TEMPO catalysis, although the pH has to be maintained below 10.0 (otherwise the Fmoc group will be cleaved).

**Scheme 6** Synthesis of Fmoc-Saa**2**-OH[45,93,101]




*for references see p 819*

**β-D-Glucopyranosylnitromethane (40):**[101]
Anhyd glucose (**38**; 250 g, 1.39 mol) was dissolved in anhyd DMSO (1 L) in a 10-L Erlenmeyer flask and anhyd MeOH (500 mL) was added. Subsequently anhyd $MeNO_2$ (500 mL) and a soln of Na (62.5 g) in abs MeOH (1750 mL) was added and the mixture was stirred for at least 24 h at rt (the mixture warms up in the beginning and solid material starts to precipitate). Then, BuOH (1250 mL) was added and the mixture was stirred for 1 h at 0 °C. The solid material was filtered off with a sintered glass filter (G3, diameter 20–25 cm) under slight suction and washed with iPrOH (500 mL) and $Et_2O$ (500 mL). The residue was dried by suction to give large quantities of a slightly yellow powder, which contains the intermediate **39** and unreacted glucose (all TLC were performed on silica gel 60, mobile phase MeCN/$H_2O$ 4:1). The powder was dissolved in $H_2O$ (1 L) and applied to a column of cation-exchange resin (Amberlite IR-120, $H^+$ form, thoroughly washed with MeOH and $H_2O$ before use, 10 × 30 cm, 1 kg, $H_2O$, 2.5 L). The red aqueous soln was concentrated under reduced pressure to 1 L and heated for 30 h at 90 °C. The cooled soln was passed over a column of anion-exchange resin [Amberlite IRA-120, $OH^-$ form, conditioned before use with 5 M NaOH (2 L) and washed thoroughly with $H_2O$, 10 × 50 cm, 2 kg]. The column was eluted with $H_2O$ (~25 L) until no glucose was detectable by TLC. Compound **40** was then eluted with $H_2O$ saturated with $CO_2$ (dry ice was constantly added onto the column) and the fractions with the product were collected (~15 L, TLC control). $H_2O$ was removed under reduced pressure and the residue was redissolved and concentrated five times with toluene. The residue was recrystallized (dissolved in minimal boiling MeOH and crystallized at 0 °C overnight). The crude product, 72 g (23.2% yield, based on glucose), contained 25% of α-product (α- and β-products are not separable by preparative TLC, but are determined by $^1H$ NMR in $D_2O$). By two further recrystallization steps from MeOH three fractions of product were obtained: 31 g (10% yield, ~2% of α-side-product), 11 g (3.4% yield, ~25% α-side-product), 28 g (9% yield, ~50% α-side-product). In every crystallization step the amount of the α-side-product is lowered by ~70% while about 25–40% of the material remains in the mother liquor. The main fraction has the following analytical data: mp 175–176 °C; $^1H$ NMR (500.13 MHz, $D_2O$, δ): 5.02 (dd, 1H, $H^l$-1), 4.77 (dd, 1H, $H^h$-1), 4.16 (ddd, 1H, H-2), 3.92 (dd, 1H, $H^l$-7), 3.76 (dd, 1H, $H^h$-7), 3.61 (dd, 1H, H-4), 3.52 (ddd, 1H, H-6), 3.48 (dd, 1H, H-5), 3.44 (dd, 1H, H-3).

***N*-(9-Fluorenylmethoxycarbonyl)-β-D-glucopyranosylmethylamine (41):**[101]
β-D-Glucopyranosylnitromethane (**40**; 11.16 g, 50 mmol, 2% α) was dissolved in a mixture of $H_2O$ (50 mL) and MeOH (100 mL). Pd/C catalyst (2.5 g, 5% Pd, 50% $H_2O$) was added and hydrogenated with $H_2$ (45 bar pressure) in a 0.5-L autoclave under stirring for 20 h at rt (hydrogenation at atmospheric pressure is also feasible, but takes approximately 3 d). After the reaction was over (monitoring by TLC) the catalyst was filtered off. $H_2O$ (150 mL) was added to the filtrate and MeOH was removed under reduced pressure. In the aqueous soln of β-D-glucopyranosylmethylamine, $NaHCO_3$ (5.04 g, 60 mmol) was dissolved and THF (200 mL) was added. To this soln, cooled to 0 °C, a soln of Fmoc-Cl (14.3 g, 55 mmol) in THF (100 mL) was added. The resulting soln was stirred for 30 min at 0 °C. After completion of the reaction (monitoring by TLC), cation-exchange resin (Amberlyst 15, $H^+$ form, 24 g) was added and the soln stirred for 10 min. The cation-exchange resin was filtered off and washed with EtOH. $H_2O$ (200 mL) was added to the filtrate and the organic solvents were removed under reduced pressure. During precipitation of the product the mixture was cooled to 4 °C. The product was filtered off, washed with a small amount of $H_2O$ and washed slowly with hexane (200 mL), and dried in vacuo to give a colorless, TLC-pure powder; yield: 19.60 g (94%); mp 115–118 °C (dec); $R_f$ 0.36 ($CHCl_3$/MeOH 4:1).

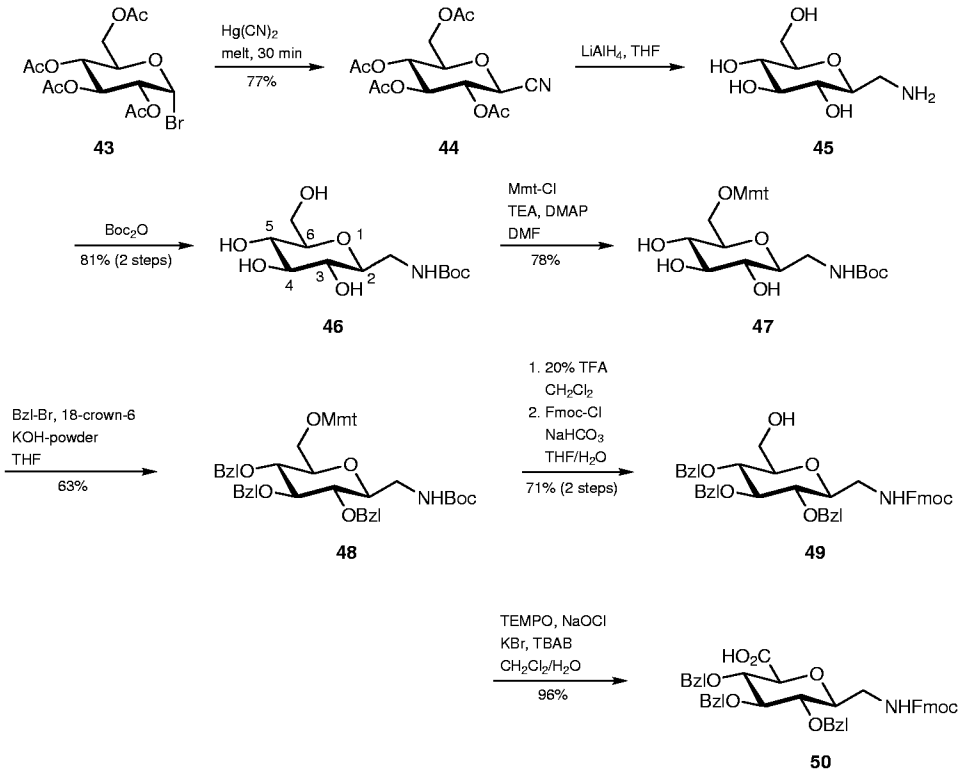
**2,6-Anhydro-7-deoxy-7-[(9-fluorenylmethoxycarbonyl)amino]-L-*glycero*-L-*gulo*-heptonic Acid (42):**[101]
*N*-(9-Fluorenylmethoxycarbonyl)-β-D-glucopyranosylmethylamine (**41**; 10.4 g, 25 mmol), NaBr (1 g), and TEMPO (50 mg)[102–104] were suspended in $H_2O$ (500 mL). A small amount of THF was added so that a clear soln was obtained. The soln was cooled to 0 °C and the pH was monitored during the reaction. To this was added dropwise with stirring 12% aq NaOCl (50 mL) such that the pH never exceeded 10. When all the NaOCl soln was added the pH was adjusted to 10 by dropwise addition of 1 M NaOH. The reaction was completed when pH 10 was stable without any further addition of NaOH (TLC control). Then a cation-exchange resin (Amberlyst 15, $H^+$ form) was added until the pH was 1.5. The resin was filtered off and washed with $H_2O$. The product was extracted from the filtrate with EtOAc (7 × 50 mL). The combined organic phases were washed with sat. brine (50 mL) and $H_2O$ (2 × 50 mL). The organic soln was not dried with inorganic salts. EtOAc was removed under reduced pressure and the product was redissolved and concentrated with toluene. After treating the resulting syrup with $Et_2O$ the crystalline product was collected and washed with $Et_2O$ and pentane, and dried in vacuo to give a colorless, TLC-pure, powder; yield: 8.71 g (81%); mp 169–170 °C (dec); $R_f$ 0.50 (MeCN/$H_2O$ 4:1).

### 12.5.1.1.3 Synthesis of 7-Amino-2,6-anhydro-3,4,5-tri-*O*-benzyl-7-deoxy-L-*glycero*-L-*gulo*-heptonic Acid

2,3,4,6-Tetra-*O*-acetyl-α-D-glucopyranosyl bromide (**43**) reacts with $Hg(CN)_2$ in a melt[10] in only 30 minutes and in good yield to give 2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranosyl cyanide (**44**), followed by reduction to β-D-glucopyranosylmethylamine (**45**) with $LiAlH_4$. This pathway is much more efficient than alternative routes such as the synthesis of **44** by reaction of pentaacetylglucose with trimethylsilyl cyanide[105] or the synthesis of compound **45** by reduction of β-D-glucopyranosylnitromethane.[45]

Boc protection of **45** was realized in situ under standard conditions[106] with $Boc_2O$. Reaction of Mmt-Cl (1.1 equiv) with *N*-(*tert*-butoxycarbonyl)-β-D-glucopyranosylmethylamine (**46**) in DMF solution overnight at room temperature in the presence of catalytic amounts of DMAP and TEA (1.5 equiv) produced *N*-(*tert*-butoxycarbonyl)-6-*O*-(4-methoxytrityl)-β-D-glucopyranosylmethylamine (**47**) in good yield. Compound **48** was obtained in 63% yield by treating **47** in THF solution overnight at room temperature with Bzl-Br (3.3 equiv), KOH powder (5.4 equiv), and catalytic amounts of 18-crown-6 under argon. After simultaneous cleavage of the Mmt ether and Boc deprotection, the intermediate was Fmoc protected with Fmoc-Cl to yield 2,3,4-tri-*O*-benzyl-*N*-(9-fluorenylmethoxycarbonyl)-β-D-glucopyranosylmethylamine (**49**) in 71% over two steps. Finally, TEMPO oxidation in a two-phase system ($H_2O$/$CH_2Cl_2$) with TBAB as phase-transfer catalyst led to the final product **50** in excellent yields.

**Scheme 7** Synthesis of Fmoc-Saa**2**$(Bzl)_3$-OH[95]




**2,3,4,6-Tetra-*O*-acetyl-β-D-glucopyranosyl Cyanide (44):**[10,95]
2,3,4,6-Tetra-*O*-acetyl-α-D-glucopyranosyl bromide (**43**; 81.59 g, 198.4 mmol) and $Hg(CN)_2$ (100 g, 395.8 mmol) were combined and pulverized. The mixture was melted at 85 °C under argon as described by Fuchs et al.[10] The reaction time was 30 min. The crude product **44** was purified by flash chromatography (Kieselgel 60, hexane/EtOAc 9:1 → 4:1 → 7:3 → 3:2 stepwise); yield: 54.3 g (77%); $R_f$ 0.44 (hexane/EtOAc 1:1).

*for references see p 819*

***N*-(*tert*-Butoxycarbonyl)-β-D-glucopyranosylmethylamine (46):**[95]

2,3,4,6-Tetra-*O*-acetyl-β-D-glucopyranosyl cyanide (**44**; 94 g, 0.263 mol) was dissolved in abs THF (600 mL). A suspension of $LiAlH_4$ (37 g, 0.973 mol) in abs THF (400 mL) was added at 0 °C to the clear soln. During the reaction, the soln was diluted with abs THF (1.3 L) to keep the mixture fluid. After 4 h the reaction was complete. Excess $LiAlH_4$ was hydrolyzed by addition of MeOH and finally of $H_2O$. The solvent was concentrated to about 400 mL. 25% aq $NH_4OH$ (350 mL) was added to the suspension until the gray precipitate became white and flaky. The precipitate was centrifuged and washed several times with $H_2O$ (total 3.5 L), and subsequently concentrated to 300 mL. The product **45** was Boc protected in situ without further purification or isolation in the following way:

To the alkaline, aqueous soln was added $Boc_2O$ (67 g, 0.307 mol) in THF (250 mL). The soln was stirred overnight at rt. The white precipitate was centrifuged and washed with MeOH. The solvents were removed and the product **46** was purified by flash chromatography (Kieselgel 60, $CHCl_3$/MeOH 9:1 → 4:1 → 7:3). The pure product was lyophilized in dioxane; yield: 62.4 g (81% over two steps); $R_f$ 0.69 (MeCN/$H_2O$ 4:1).

***N*-(*tert*-Butoxycarbonyl)-6-*O*-(4-methoxytrityl)-β-D-glucopyranosylmethylamine (47):**[95]

*N*-(*tert*-Butoxycarbonyl)-β-D-glucopyranosylmethylamine (**46**; 8.65 g, 29.5 mmol) was dissolved in abs DMF (100 mL). To the clear soln, Mmt-Cl (10.05 g, 32.5 mmol), TEA (7.4 mL, 53.4 mmol), and DMAP (0.28 g, 2.3 mmol) were added and the mixture was stirred overnight at rt. The yellow soln was poured into ice water. The aqueous soln was extracted several times with $CH_2Cl_2$. The organic layer was dried ($MgSO_4$) and the solvent removed. Purification by flash chromatography (Kieselgel 60, $CHCl_3$/MeOH 19:1) gave **47** as a white solid; yield: 12.9 g (78%); $R_f$ 0.41 ($CHCl_3$/MeOH 9:1).

**2,3,4-Tri-*O*-benzyl-*N*-(*tert*-butoxycarbonyl)-6-*O*-(4-methoxytrityl)-β-D-glucopyranosylmethylamine (48):**[95]

*N*-(*tert*-Butoxycarbonyl)-6-*O*-(4-methoxytrityl)-β-D-glucopyranosylmethylamine (**47**; 7.54 g, 13.3 mmol) was dissolved in abs THF (50 mL) under an argon atmosphere. 18-Crown-6 (0.44 g, 1.7 mmol), KOH powder (4.0 g, 71.3 mmol), and, after 15 min, Bzl-Br (5.3 mL, 44.6 mmol) were added and the white suspension was stirred overnight under argon at rt. THF was removed, the residue was dissolved in EtOAc, extracted with $H_2O$, dried ($Na_2SO_4$), and concentrated. Flash chromatography gave **48**; yield: 7.0 g (63%); $R_f$ 0.33 (hexane/EtOAc 4:1).

**2,3,4-Tri-*O*-benzyl-*N*-(9-fluorenylmethoxycarbonyl)-β-D-glucopyranosylmethylamine (49):**[95]

2,3,4-Tri-*O*-benzyl-*N*-(*tert*-butoxycarbonyl)-6-*O*-(4-methoxytrityl)-β-D-glucopyranosylmethylamine (**48**; 1.54 g, 1.84 mmol) was dissolved in 20% TFA (100 mL) with a splash of $H_2O$ and stirred for 1 h. Then, the solvent and TFA were removed by coevaporating with toluene (3 ×). The residue was dissolved in THF (30 mL) and a soln of Fmoc-Cl (0.75 g, 2.90 mmol) in THF (5 mL) was added. $NaHCO_3$ (0.2 g, 2.38 mmol) in $H_2O$ (5 mL) was added at 0 °C and the soln slowly brought to rt. After 1 h, sat. $NaHCO_3$ (10 mL) was added followed by the addition of Fmoc-Cl (0.3 g, 1.16 mmol) in a small amount of THF (5 mL). After another hour, the reaction was complete (TLC control). The soln was acidified with Amberlite IR 120. The $H^+$ ion-exchange resin was filtered off and washed with EtOAc. The aqueous layer was extracted with EtOAc (4 ×). The organic layers were combined and the solvent was removed. The residue was purified by flash chromatography (Kieselgel 60, hexane/EtOAc 4:1 → 2:1 → 1:1 stepwise) to give **49**; yield: 0.89 g [71% (over 2 steps)]; $R_f$ 0.37 (hexane/EtOAc 1:1).

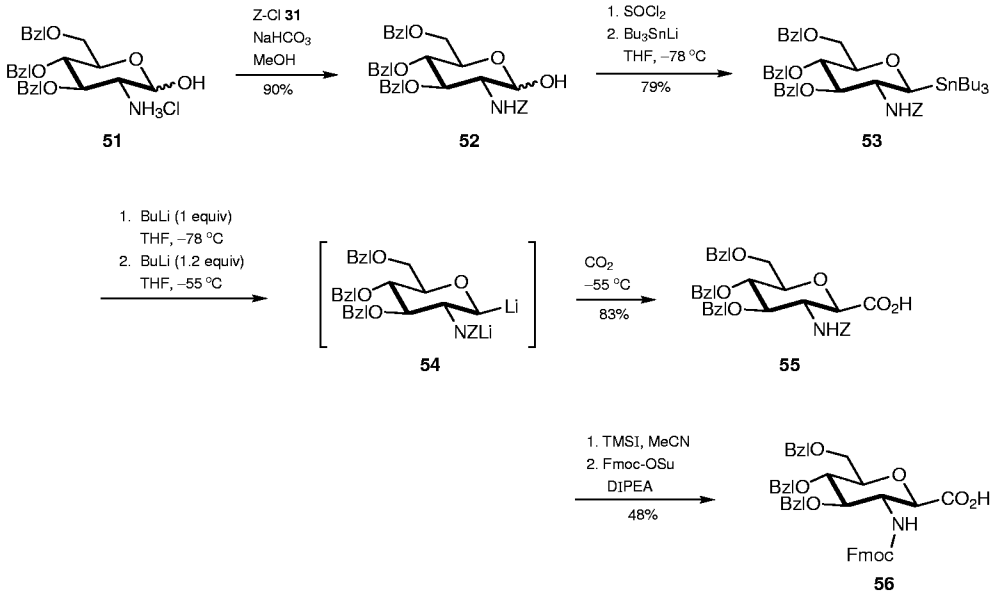
**2,6-Anhydro-3,4,5-tri-*O*-benzyl-7-deoxy-7-[(9-fluorenylmethoxycarbonyl)amino]-L-*glycero*-L-*gulo*-heptonic Acid (50):**[95]

2,3,4-Tri-*O*-benzyl-*N*-(9-fluorenylmethoxycarbonyl)-β-D-glucopyranosylmethylamine (**49**; 0.42 g, 0.61 mmol) was dissolved in $CH_2Cl_2$ (2 mL) and cooled in an ice bath. TEMPO (16 mg, 0.1 mmol) and then a mixed soln of sat. aq $NaHCO_3$, KBr (20 mg, 0.17 mmol), and TBAB (25 mg, 78 μmol) were added. Under cooling, a soln of NaOCl (13%, 0.92 mL), sat. $NaHCO_3$ (0.67 mL), and brine (1.35 mL) was added so that the pH value never exceeded pH 10. After 75 min, the reaction was complete (TLC control). The soln was diluted with $CH_2Cl_2$ and $H_2O$ for better phase separation. The aqueous layer was acidified with Amberlite IR 120 and extracted several times with EtOAc. The EtOAc and the $CH_2Cl_2$ solns were combined, concentrated, and the residue purified by flash chromatography (Kieselgel 60, hexane/EtOAc 2:1 then $CHCl_3$/MeOH 39:1). The product **50** was lyophilized in dioxane to obtain a white powder; yield: 0.41 g (96%); $R_f$ 0.60 ($CHCl_3$/MeOH 4:1).

### 12.5.1.1.4 Synthesis of 3-Amino-2,6-anhydro-4,5,7-tri-*O*-benzyl-3-deoxy-D-*glycero*-D-*gulo*-heptonic Acid

The synthesis of Fmoc-Saa**4**$(Bzl)_3$-OH (**56**) (Scheme 8) followed a procedure published for the stereoselective *C*-glycosidation of 2-acetamido-2-deoxy-D-glucose.[107–110] Starting from D-glucosamine, the partially benzylated sugar **51** was obtained in two steps according to the procedure reported by Inch and Fletcher.[111] The amino group was subsequently protected with Z-Cl to give **52** in 90% yield. Chlorination of the anomeric hydroxy group provided the α-chloro compound, which was treated with tributyltin lithium to afford **53** in 79% yield.[107–110] The generation of the glycosyl dianion **54** was accomplished in two separate temperature steps: first, deprotonation of the urethane nitrogen at −78 °C using 1 equivalent of BuLi; secondly, transmetalation at −55 °C using 1.2 equivalents of BuLi. The dianion **54** was visualized by a deep red color of the solution and was subsequently trapped by carbon dioxide to afford the carboxylic acid **55** in 83% yield. For the application of Saa**4** to solid-phase peptide synthesis, **55** was transformed into the Fmoc derivative **56**. TFA/thioanisole[112] or catalytic hydrogenation on Pd/C[113] was not selective for the removal of the Z group of **55**. The best result for cleaving the Z group was obtained by using iodotrimethylsilane in MeCN.[114] However, the $C^7$-*O*-benzyl ether of **55** was cleaved to some extent. While the amount of side product was temperature independent, the yield was optimized by varying the reaction time. The crude reaction mixture was treated with Fmoc-OSu[115] to afford **56** in 48% yield.

**Scheme 8** Synthesis of Fmoc-Saa**4**$(Bzl)_3$-OH[45,107,108,110]




**3,4,6-Tri-*O*-benzyl-2-[(benzyloxycarbonyl)amino]-2-deoxy-D-glucosamine** (52)**:**[45]
50% Z-Cl (**31**; 4.9 mL, 14.4 mmol) in toluene diluted in $CHCl_3$ (100 mL) was slowly added to a soln of the partially benzylated D-glucosamine **51** (7.0 g, 14 mmol) and $NaHCO_3$ (6.0 g, 71 mmol) in MeOH (200 mL) at 0 °C. After 3 h, additional $NaHCO_3$ (6.0 g, 71 mmol) and Z-Cl (2.5 mL, 7.2 mmol) were added and the suspension was stirred overnight at rt. The mixture was concentrated, suspended in EtOAc (1 L), filtrated and concentrated again. The crude product was crystallized (EtOAc/hexanes) to give **52** as colorless crystals; yield: 8.03 g (90%); mp 192–193 °C (hexanes/EtOAc); $R_f$ 0.39 (hexanes/acetone 1:2); $[\alpha]_D^{20}$ +57.7 (*c* 1.06, $CHCl_3$); FAB-MS *m/z*: 566 $[M - H_2O + H]^+$.

*for references see p 819*

**Tributyl{3,4,6-tri-*O*-benzyl-2-[(benzyloxycarbonyl)amino]-2-deoxy-β-D-glucopyranosyl}stannane (53):**[45]
Compound **52** (7.0 g, 12 mmol) was treated with $SOCl_2$ (100 mL) at rt for 30 min, concentrated to dryness, and the product redissolved and concentrated with dry $CHCl_3$ (20 mL). The yellow solid was dissolved in dry THF (100 mL) and added within 20 min to a soln of $Bu_3SnLi$[116] in THF (~3 equiv) at −78 °C. The mixture was stirred for 1 h at −78 °C, quenched with sat. $NH_4Cl$ (10 mL), warmed to rt, diluted with $H_2O$, and extracted with EtOAc (2 × 700 mL). The combined organic layers were dried ($MgSO_4$) and concentrated to afford a yellow oil. Purification was achieved by flash chromatography (hexanes/EtOAc, gradient 1:0 to 4:1) to give **53** as a colorless oil; yield: 8.10 g (79%); $R_f$ 0.91 (hexane/acetone 3:1); $[\alpha]_D^{20}$ −2.7 (*c* 1.0, $CHCl_3$); FAB-MS *m/z*: $[M - Bu + H]^+$ for $C_{47}H_{63}NO_6{}^{120}Sn$; calcd, 801.31; found, 801.

**2,6-Anhydro-4,5,7-tri-*O*-benzyl-3-[(benzyloxycarbonyl)amino]-3-deoxy-β-D-*glycero*-D-*gulo*-heptonic Acid (55):**[45]
Stannane **53** (8.1 g, 9.5 mmol) was dissolved in dry THF (60 mL). At −78 °C, 1.6 M BuLi in hexane (5.9 mL, 9.5 mmol) was added within 10 min. The mixture was then warmed to −55 °C, BuLi (7.1 mL, 11 mmol) was added within 5 min while the color of the soln changed to deep red. $CO_2$ was pumped for 15 min through the mixture, which was quenched after 1 h with 10% $KHSO_4$ (50 mL) at −55 °C, warmed to rt and extracted with EtOAc (2 × 500 mL). Purification was achieved by flash chromatography (hexanes/acetone 1:0 and 1:1, and $CHCl_3$/MeOH/AcOH 50:7:3) to afford **55** as a colorless solid; yield: 4.77 g (83%); mp 155 °C; $R_f$ 0.59 ($CHCl_3$/MeOH 1:3); $[\alpha]_D^{20}$ +15.1 (*c* 1.0, $CHCl_3$); FAB-MS *m/z*: 634 $[M + Na]^+$.

**2,6-Anhydro-4,5,7-tri-*O*-benzyl-3-deoxy-3-[(9-fluorenylmethoxycarbonyl)amino]-D-*glycero*-D-*gulo*-heptonic Acid (56):**[45]
Carboxylic acid **55** (1.0 g, 1.6 mmol) was dissolved in dry MeCN (10 mL) at rt in a Falcon tube. TMSI[115] (0.49 mL, 3.60 mmol) was added and the reaction stirred for 12 min, treated with MeOH (100 μL), diluted with THF (10 mL), and treated with Fmoc-OSu[115] (720 mg, 2.0 mmol). The pH of the soln was maintained at 7 to 8 by addition of DIPEA. The completion of the reaction was followed by HPLC [30 → 90% MeCN in $H_2O$, 0.1% TFA, 30 min: $t_R$ (amine) 20.4, $t_R$ (urethane) 26.4 min]. After 14 h and 24 h additional Fmoc-OSu was added. Purification was achieved by flash chromatography [acetone/hexanes 1:3, and acetone/hexanes (+0.1% TFA) gradient 1:2 to 2:1] to give **56**; yield: 546 mg (48%); $R_f$ 0.79 ($CHCl_3$/MeOH 1:3); $t_R$ 20.9 min (58 → 72% MeCN/$H_2O$, 0.1% TFA, 30 min); $[\alpha]_D^{20}$ +13.8 (*c* 0.92, THF).

### 12.5.1.2 Peptides with Sugar Amino Acid Building Blocks

The synthesis of the Saa-peptide conjugates in solution follows standard solution-phase synthesis using, for example, IIDQ[117] or EDC as the coupling agent. The reaction is checked by TLC (ninhydrin and/or 10% $H_2SO_4$ in MeOH dip). The product is usually purified by flash chromatography.

Due to the functionality of Saa, these building blocks can be synthesized as Fmoc derivatives and can then be employed under standard SPPS conditions using the Fmoc/tBu strategy and TBTU/HOBt[118] as the coupling reagent. For best results the hydroxy groups are protected as acetates, silyl ethers, or benzyl ethers. Protections can be removed if desired. The Saa building block is generally placed near or in the middle of the peptide sequence, thereby utilizing its structure-inducing effect. Cyclization is performed in solution either by TBTU/HOBt or DPPA/$NaHCO_3$[119] activation.

More recently, we turned to HATU/HOAt/2,4,6-collidine in DMF as coupling conditions to avoid racemization on the sugar ring. It turned out that this coupling proceeds with high yields (up to 90%) with stereochemical purity. Protected Saa can also be cyclized with DIC/HOAt/NMM; however, for unprotected Saa very low yields were achieved with this method.

If a Saa with free hydroxy groups is used the cyclization is performed with TBTU/HOBt by adding 1 equivalent of $H_2O$ per hydroxy group to the linear precursor to prevent intermolecular acylation. The peptide is then added dropwise to a coupling solution of TBTU/

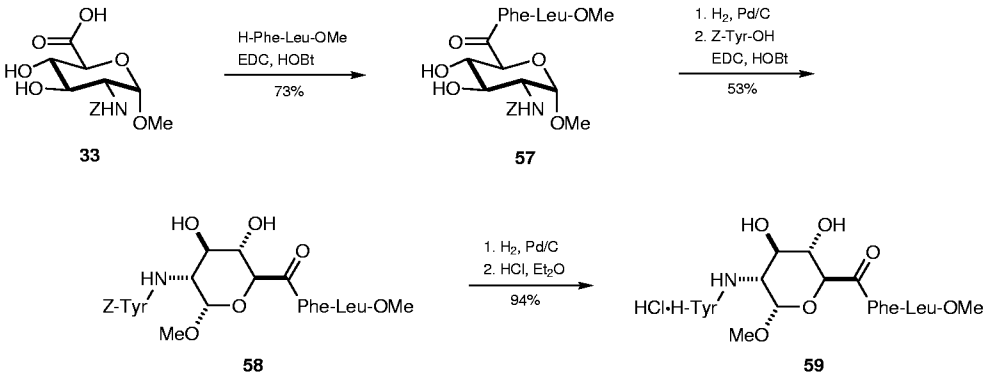
HOBt/DIPEA in NMP or DMF over a period of 1 hour. Due to the susceptibility of the α-hydrogen next to the carboxy function of uronic acids to elimination, the pH should not exceed 10.0.

The azide group can be used as a temporary protecting group of the N-terminus, and can be reduced either in solution or in SPPS with DTT or Pd/C and hydrogen, when required.

### 12.5.1.2.1 Solution-Phase Synthesis of Peptides Based on 2-Amino-2-deoxy-1-*O*-methyl-α-D-glucopyranuronic Acid

Incorporation of α-Saa**1** by solution-phase synthesis, used to demonstrate its influence on the conformation of a linear peptide, has been performed for the Leu-enkephalin analogue H-Tyr-α-Saa**1**-Phe-Leu-OMe (**59**) in which the Saa replaces the Gly-Gly dipeptide segment of the natural sequence H-Tyr-Gly-Gly-Phe-Leu-OH (Scheme 9).[45]

**Scheme 9** Solution-Phase Peptide Synthesis of HCl·H-Tyr-α-Saa**1**-Phe-Leu-OMe[45]




In enkephalin the dipeptide Gly-Gly serves as a spacer between the messenger amino acid Tyr, which is essential for activity, and the address sequence Phe-Leu responsible for selectivity.[120,121]

**Z-α-Saa1-Phe-Leu-OMe (57):**[45]
H-Phe-Leu-OMe (2 mmol) was dissolved in THF (5 mL) and cooled to 0 °C. Z-α-Saa**1**-OH (**33**; 0.68 g, 2.0 mmol), HOBt (0.3 g, 2.0 mmol), and EDC·HCl (0.40 g, 2.1 mmol) were added. The pH was adjusted to 7 by dropwise addition of NMM. After 10 h the solvent was removed and the residue dissolved in EtOAc (100 mL). The soln was washed with 0.5 M HCl (3 × 20 mL), 5% aq $NaHCO_3$ (3 × 20 mL), and $H_2O$ (3 × 20 mL). The organic layer was dried ($MgSO_4$) and concentrated to give **57** as a colorless solid; yield: 0.90 g (73%); $R_f$ 0.29 (toluene/EtOAc 1:5).

**Z-Tyr-α-Saa1-Phe-Leu-OMe (58):**[45]
Z-α-Saa**1**-Phe-Leu-OMe (**57**; 0.62 g, 1.0 mmol) was dissolved in MeOH (5 mL) and 10% Pd/C catalyst (50 mg) was added. The mixture was stirred for 1 h under $H_2$. The catalyst was filtered off and the solvent was removed in vacuo. The residue was dissolved in a mixture of DMF (5 mL) and THF (5 mL), then cooled to 0 °C. Z-Tyr-OH (0.32 g, 1.0 mmol), HOBt (150 mg, 1.0 mmol), and EDC·HCl (0.20 g, 1.1 mmol) were added. By dropwise addition of NMM the pH was adjusted to 7. After 10 h the solvent was removed and the residue was chromatographed (silica gel, $CHCl_3$/MeOH 12:1) to afford **58** as a colorless solid; yield: 0.41 g (53%); $R_f$ 0.20 ($CHCl_3$/MeOH 12:1).

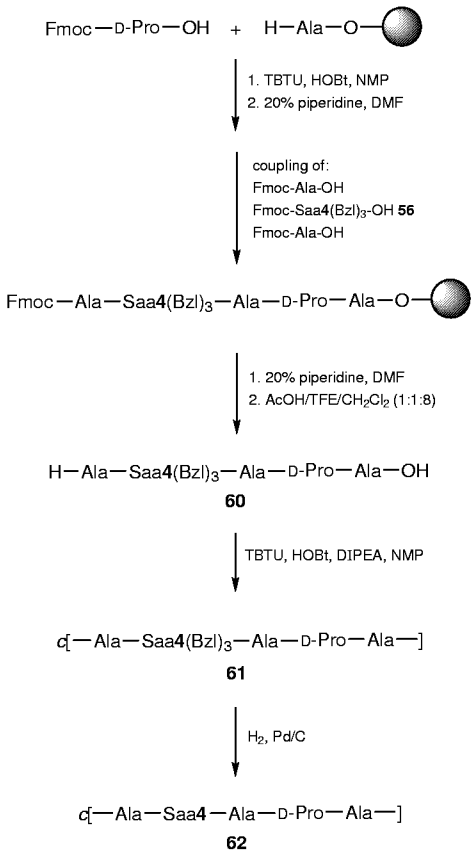
**HCl·H-Tyr-α-Saa1-Phe-Leu-OMe (59):**[45]
Z-Tyr-α-Saa**1**-Phe-Leu-OMe (**58**; 155 mg, 0.2 mmol) was dissolved in MeOH (5 mL) and 10% Pd/C (50 mg) was added. The mixture was stirred for 1 h under $H_2$. The catalyst was filtered off, sat. HCl in $Et_2O$ (0.5 mL) was added, and the solvent was removed in vacuo to give **59** as a colorless solid; yield: 112 mg (94%); $R_f$ 0.08 ($CHCl_3$/MeOH 8:1).

*for references see p 819*

### 12.5.1.2.2 Solid-Phase Synthesis of Peptides Based on 3-Amino-2,6-anhydro-3-deoxy-D-*glycero*-D-*gulo*-heptonic Acid

The synthesis of peptides containing Saa can also be carried out by standard SPPS using Fmoc/tBu protection and TBTU/HOBt activation (Scheme 10).[45]

**Scheme 10** Solid-Phase Peptide Synthesis of *c*[-Ala-Saa**4**-Ala-D-Pro-Ala-][45]




***c*[-Saa4-Ala-D-Pro-Ala-Ala-] (62):**[45]

Fmoc-Ala-OH (0.2 g, 0.7 mmol) was coupled to the 2-chlorotritylchloride resin (0.4 g, 1.25 mmol Cl/g resin) to initiate the synthesis. Then, the resin was Fmoc deprotected and Fmoc-D-Pro-OH (0.24 g, 0.7 mmol), Fmoc-Ala-OH (0.22 g, 0.7 mmol), and Fmoc-Saa**4**-$(Bzl)_3$-OH (**56**; 0.32 g, 0.46 mmol) were used for chain elongation. The TBTU/HOBt method was used for activation. The peptide was cleaved from the resin with an AcOH/TFE/$CH_2Cl_2$ (1:1:8) solvent mixture to give crude H-Ala-Saa**4**$(Bzl)_3$-Ala-D-Pro-Ala-OH (**60**); yield: 265 mg (82%), which was purified by preparative HPLC (40 → 70% MeCN/$H_2O$, 0.1% TFA, 30 min, $t_R$ 13.6 min); FAB-MS *m/z*: 810 $[M+Na]^+$, 788 $[M+H]^+$.

H-Ala-Saa**4**$(Bzl)_3$-Ala-D-Pro-Ala-OH (**60**; 52 mg) was dissolved in NMP (100 mL) and added slowly to a soln of TBTU (70 mg), HOBt (58 mg), and DIPEA (pH ca. 7.5) in NMP (300 mL). After 3 h at rt $H_2O$ (30 mL) was added, the soln was concentrated and the product *c*[-Ala-Saa**4**$(Bzl)_3$-Ala-D-Pro-Ala-] (**61**) purified by preparative HPLC (40 → 70% MeCN/$H_2O$, 0.1% TFA, 30 min, $t_R$ 20.7 min); FAB-MS *m/z*: 792 $[M+Na]^+$.

The cyclic peptide **61** was dissolved in $H_2O$/AcOH (1:1, 30 mL) and hydrogenated for 24 h in the presence of 5% Pd/C (30 mg). The product **62** was purified by preparative HPLC (5 → 40% MeCN/$H_2O$, 0.1% TFA, 30 min, $t_R$ 12.5 min); yield: 23 mg (68%).

[1] We use sugar amino acid as a functional, succinct classification term, although a plethora of terms have been proposed in the literature for compounds derived from Saa. These include saccharide-peptide hybrids, glycosamino acids, peptidosaccharides, saccharopeptides, amide-linked carbohydrates, and carbopeptoids, although the latter compounds most often do not have a peptoid functionality. In some cases Saa are linked to each other, in other cases to amino acids.
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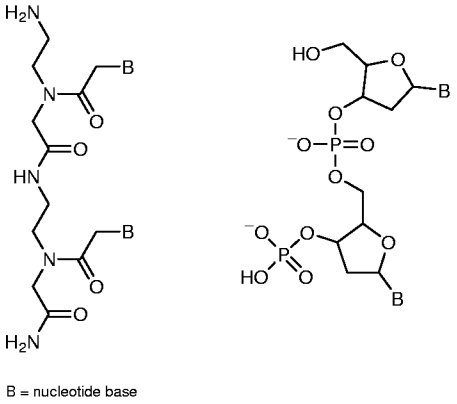
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# 12.6 Peptide Nucleic Acids

A. Püschl and P. E. Nielsen

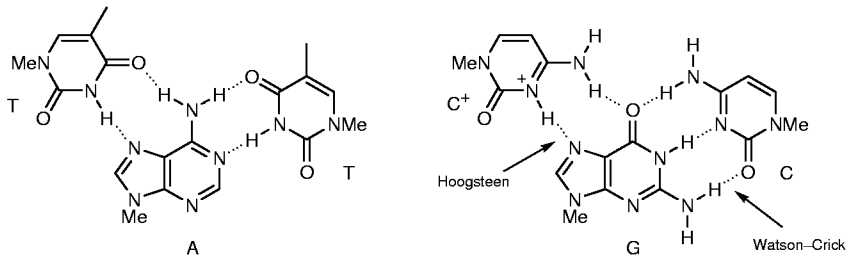
Peptide nucleic acid (PNA) is a DNA mimic built on a pseudo peptide (polyamide backbone) (Scheme 1).[1,2] Thus, PNA may be described as a peptide that behaves like a DNA molecule.[3] This dual identity makes PNA of interest in several areas of chemistry, biology, and medicine, including basic molecular recognition, origin of life, genetic diagnostics, and gene therapeutic drug development.[4–8]

**Scheme 1** Structures of Peptide Nucleic Acid (Left) and DNA (Right)[1,2]




PNA oligomers hybridize to complementary DNA, RNA, or PNA oligomers with high sequence specificity and the resulting duplexes are in general thermally more stable than the corresponding oligonucleotide duplexes. Furthermore, homopyrimidine PNAs bind to sequence complementary homopurine DNA or RNA oligomers by forming $(PNA)_2$–DNA (or RNA) Watson–Crick- and Hoogsteen-type triplexes (Scheme 2) of very high thermal stability (a decamer triplex typically exhibits a melting temperature of 70 °C).

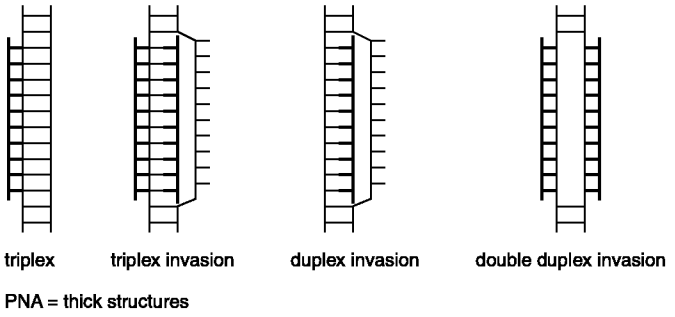
**Scheme 2** Triplex Recognition




The high stability of $(PNA)_2$–DNA triplexes also explains why homopyrimidine PNAs, when targeting double-stranded DNA, usually do not bind in the conventional triplex mode. Rather, PNA invades the double-stranded DNA target to form an internal $(PNA)_2$–DNA triplex within the complementary homopurine DNA strand leaving the homopyrimidine DNA strand as a single-stranded P-loop (Scheme 3).[9] Certain homopurine PNAs that form especially stable duplexes are also able to invade double-stranded targets, albeit with significantly lower efficiency than triplex-forming PNAs.[10] In some cases conventional triplex complexes have also been detected with cytosine-rich homopyrimidine PNAs, but these complexes appear much less stable than the triplex invasion complexes.[11,12] Finally, using

modified adenine (diaminopurine) and thymine (thiouracil) nucleobases, it is possible to construct pseudo-complementary PNAs that, despite sequence complementarity are not able to bind to each other, but have retained their ability to bind to complementary DNA targets.[13] Such PNAs can bind to double-stranded DNA targets by double duplex invasion, which greatly reduces the restrictions on sequence choice.

**Scheme 3** Peptide Nucleic Acid Binding Modes[9]




PNAs bound to double-stranded DNA targets by triplex invasion inhibit the binding of proteins to these targets. This property has been exploited for development of genome analysis methods, and is also the basis for the potential development of PNAs as antigene therapeutic drugs.

The efficient and specific binding of PNA to RNA targets has been exploited in efforts to develop PNA antisense drugs. Initial experiments showed that, despite PNAs inability to initiate RNase H cleavage, it can efficiently inhibit translation in vitro, especially when targeted to the translation initiation sequence (AUG) initiation region.[14–16] However, only recently have these results been followed up by experiments with cells in culture and a few animal studies. There is good reason to expect a renewed interest in PNA antisense applications after the recent reports of greatly improved methods of cellular uptake of PNA. These methods include the use of PNA-peptide conjugates as well as PNA–DNA chimera and complexes delivered via cationic liposomes.[17,18] Furthermore, the recent reports that even "naked" PNAs exhibit antisense effects in bacteria,[19] nerve cells,[20] and even upon injection into the brain of rats[20,21] should refocus the interest in PNA.

Finally, several PNA based methods for use in genetic diagnostics have been developed. Of particular interest are methods for fluorescence in situ hybridization (FISH), modulation of polymerase chain reaction (PCR) analyses, and detection of genetic material by sensors and mass spectrometry.[8]

Apart from the above applications, PNA is also of interest in its own right. This quite unique DNA mimic has inspired many chemists to search for similar molecules both by making derivatives and analogues of the original PNA structure, but also by searching for novel structural motifs.

Finally, the fact that a peptide-like molecule is able to carry genetic information has triggered speculations that perhaps the prebiotic genetic material before RNA or DNA could indeed have been such a molecule. This idea has so far been supported by a number of experiments relating to chemical information transfer between PNA and the biological genetic materials RNA and DNA.[22,23]

*for references see p 834*

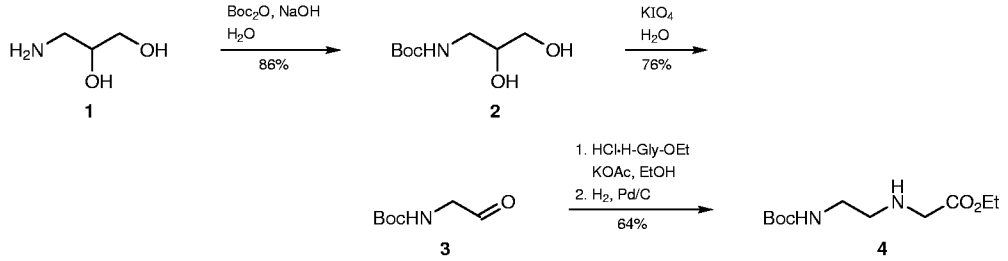
## 12.6.1 Peptide Nucleic Acid Monomers

The four standard PNA monomers [adenine (A), cytosine (C), guanine (G), and thymine (T)] used in solid-phase synthesis are commercially available. However, three additional monomers used routinely are not. The syntheses of these monomers are given in Sections 12.6.1.1–12.6.1.4 together with the synthesis of the *N*-[2-(*tert*-butoxycarbonylamino)-ethyl]glycine backbone.

### 12.6.1.1 Synthesis of the *N*-[2-(*tert*-Butoxycarbonylamino)ethyl]glycine Backbone

Ethyl *N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (**4**) can be purchased or easily prepared from 3-aminopropane-1,2-diol (**1**) as shown in Scheme 4.[24] Any nucleobase can then be coupled to this backbone. Alkaline hydrolysis then gives the PNA Boc monomer. The temperature should be kept below 120 °C during distillation of **3** to avoid decomposition. If **3** is used in excess compared to HCl·H-Gly-OEt, dialkylation will occur.

**Scheme 4** Synthesis of Ethyl *N*-[2-(*tert*-Butoxycarbonylamino)ethyl]glycinate[24]




**1** **2** **3** **4**

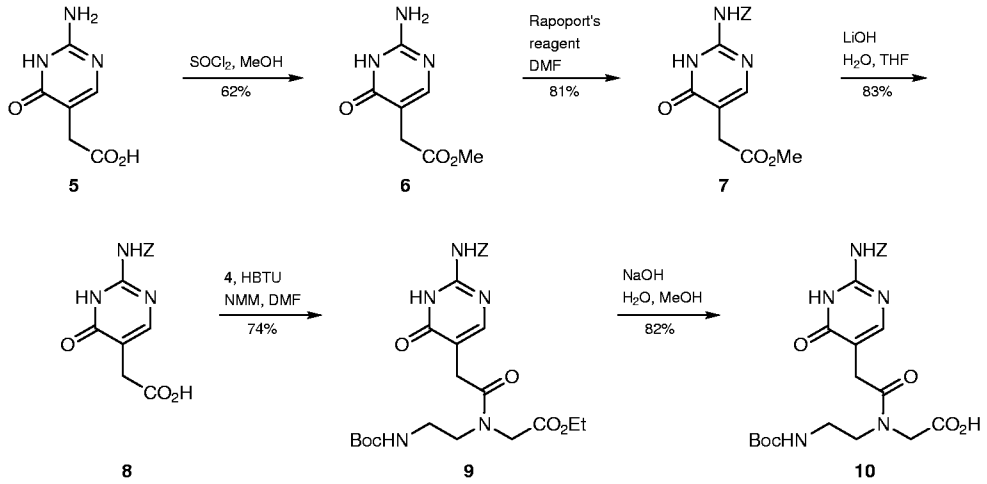
**(*tert*-Butoxycarbonylamino)acetaldehyde (3):**[24]
To 3-aminopropane-1,2-diol (**1**; 80.0 g, 0.88 mol), dissolved in $H_2O$ (1.5 L) and cooled to 0 °C, was added $Boc_2O$ (230 g, 1.05 mol) in one portion. The mixture was heated to rt with stirring on a water bath. The pH was maintained at 10.5 by continuous addition of NaOH (70.2 g, 1.76 mol) in $H_2O$ (480 mL). After the addition of NaOH was completed, the mixture was stirred overnight at rt. Subsequently, EtOAc (1 L) was added to the mixture followed by cooling to 0 °C. The pH was adjusted to 2.5 with 4 M HCl under vigorous stirring and the two phases were separated. The aqueous phase was washed thoroughly with EtOAc (8 × 0.5 L). The combined organic phases were reduced to 1.5 L by concentration under reduced pressure and washed successively with half-sat. $KHSO_4$ (1.5 L) and brine (1 L). The organic phase was dried ($MgSO_4$) and the solvent removed under reduced pressure to give **2**; yield: 145.3 g (86%). The product, which could be made to solidify by evaporation from $CH_2Cl_2$ and subsequent cooling, was not purified further prior to use in the synthesis of **3**. 3-(*tert*-Butoxycarbonylamino)-propane-1,2-diol (**2**; 20.76 g, 0.109 mol) was suspended in $H_2O$ (150 mL). $KIO_4$ (24.97 g, 0.109 mol) was added and the mixture was stirred at rt for 2 h under $N_2$ followed by filtration. Then, the $H_2O$ phase was extracted with $CHCl_3$ (6 × 250 mL). The organic phase was dried ($MgSO_4$) and the solvent removed to afford crude **3** as an oil which was Kugelrohr distilled (80 °C/0.2 mbar) to give **3** as a semi-crystalline solid; yield: 13.19 g (76%). (*tert*-Butoxycarbonylamino)acetaldehyde prepared this way is stable for at least 3 months when stored at −20 °C.

**Ethyl *N*-[2-(*tert*-Butoxycarbonylamino)ethyl]glycinate (4):**[24]
A soln of (*tert*-butoxycarbonylamino)acetaldehyde (**3**; 10.0 g, 62.8 mmol), HCl·H-Gly-OEt (8.77 g, 62.8 mmol), and KOAc (12.33 g, 126 mmol) was hydrogenated (10% Pd/C) in EtOH (625 mL) until the $H_2$ uptake had ceased (2 h). A total of 1.38 L (62.8 mmol) had been consumed. The mixture was filtered and the solvent removed under reduced pressure. The residue was suspended in $H_2O$ and the pH was adjusted to 8 by addition of 1 M NaOH under vigorous stirring. The aqueous phase was extracted with $CH_2Cl_2$ (5 ×). The combined organic phases were dried ($Na_2SO_4$), filtered, and concentrated under reduced pressure to give crude **4** as an oil; yield: 17.03 g. The crude product was Kugelrohr distilled (110 °C/0.2 mbar) to afford **4** as a colorless liquid; yield: 9.88 g (64%).

### 12.6.1.2 Synthesis of the Pseudoisocytosine Peptide Nucleic Acid Monomer

Triplex formation with cytosine containing PNAs (or oligonucleotides) requires N3 protonation of cytosine (Scheme 2) and is therefore only efficient at acidic pH (pH <6). However, substituting pseudoisocytosine for cytosine in the Hoogsteen strand of bis-PNAs in triplex recognition gives pH independent binding, because pseudoisocytosine mimics the protonated cytosine. The synthesis of pseudoisocytosine is shown in Scheme 5.[25]

**Scheme 5** Synthesis of the Pseudoisocytosine Peptide Nucleic Acid Monomer[25]




**Isocytosin-5-ylacetic Acid (5):**[25]
NaOMe (41.9 g, 0.78 mol) was dissolved in dry MeOH (200 mL) and Gn-Cl (49.4 g, 0.52 mol) was added. The mixture was stirred for 10 min under $N_2$ at rt. A soln of methyl α-formylsuccinate (30.0 g, 0.17 mol) in dry MeOH (100 mL) was added to the mixture, which was subsequently refluxed under $N_2$ for 3 h and then stirred at rt overnight. The mixture was filtered, and the filter cake was washed once with MeOH. The collected filtrate and washings were concentrated under reduced pressure. The resulting residue was dissolved in $H_2O$ (80 mL) and the soln was acidified with concd HCl to pH 4.2. After being stirred at 0 °C the mixture was filtered and the precipitate washed once with $H_2O$. The soln was then freeze-dried; yield: 28.29 g (97%).

**Methyl Isocytosin-5-ylacetate (6):**[25]
$SOCl_2$ (3.6 mL, 46 mmol) was mixed with stirred MeOH (210 mL) at −40 °C under $N_2$. Isocytosin-5-ylacetic acid (**5**; 7.0 g, 41 mmol) was added and the mixture was stirred at rt for 1 h, at 60 °C for 3 h, and finally at rt overnight. The mixture was evaporated to dryness and the residue was dissolved in sat. aq $NaHCO_3$ (80 mL) giving a foamy precipitate. The pH was adjusted to 6.5 by addition of 4 M HCl and the suspension was stirred for 1 h. The precipitate **6** was collected by filtration, washed with $H_2O$, recrystallized ($H_2O$), and then freeze-dried; yield: 4.66 g (62%); FAB-MS *m/z*: 184.

**Methyl [$N^2$-(Benzyloxycarbonyl)isocytosin-5-yl]acetate (7):**[25]
Methyl isocytosin-5-ylacetate (**6**; 9.5 g, 52 mmol) was dissolved in dry DMF (95 mL) and the soln was stirred at 0 °C under $N_2$. *N*-Benzyloxycarbonyl-*N'*-methylimidazolium triflate (Rapoport's reagent; 37.99 g, 104 mmol) was added slowly and the mixture was stirred at 0 °C for 30 min, and then at rt overnight. $CH_2Cl_2$ (800 mL) was added and the resultant mixture was washed with half-sat. $NaHCO_3$ (2 × 400 mL), half-sat. $KHSO_4$ (2 × 400 mL), and brine (400 mL). The organic phase was dried ($MgSO_4$), filtered, and concentrated under reduced pressure. The residue was recrystallized (MeOH); yield: 13.32 g (81%); FAB-MS *m/z*: 318.

**[$N^2$-(Benzyloxycarbonyl)isocytosin-5-yl]acetic Acid (8):**[25]
Methyl [$N^2$-(benzyloxycarbonyl)isocytosin-5-yl]acetate (**7**; 5.2 g, 16 mmol) was suspended in THF (52 mL) and cooled to 0 °C. 1 M LiOH (49 mL, 49 mmol) was added and the mixture was stirred at 0 °C for 25 min. Additional 1 M LiOH (20 mL, 20 mmol) was added and the mixture was stirred at 0 °C for 90 min. The product **8** was precipitated by acidifying to pH 2 with 1 M HCl, collected by filtration, washed with $H_2O$ (1×), and dried; yield: 4.12 g (83%); FAB-MS *m/z*: 304.

*for references see p 834*

**Ethyl *N*-{[*N*$^2$-(Benzyloxycarbonyl)isocytosin-5-yl]acetyl}-*N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (9):**[25]

[*N*$^2$-(Benzyloxycarbonyl)isocytosin-5-yl]acetic acid (**8**; 2.0 g, 6.6 mmol) was transferred to a flask equipped with a stirring bar and a septum through which a flow of $N_2$ was applied. Dry DMF (20 mL) and NMM (2.2 mL, 19.8 mmol) were added. The mixture was cooled to 0 °C, and ethyl *N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (**4**; 1.8 g, 7.3 mmol) and HBTU (3.0 g, 7.9 mmol) were added. The mixture was stirred under $N_2$ for 4 h followed by addition of $CH_2Cl_2$ (100 mL). The organic phase was washed with half-sat. $NaHCO_3$ (2 × 75 mL), half-sat. $KHSO_4$ (2 × 75 mL), and brine (75 mL), then dried ($MgSO_4$), filtered, and concentrated under reduced pressure. The residue was dissolved in EtOAc (50 mL), the soln was stirred at 0 °C for 10 min and filtered through Celite which was washed with EtOAc. The collected filtrate and washings were concentrated to a volume of 10 mL. $Et_2O$ (100 mL) was added and the resultant soln was stirred at rt overnight. The precipitated product **9** was collected by filtration, washed once with $Et_2O$, and dried; yield: 2.6 g (74%); FAB-MS *m/z*: 532.

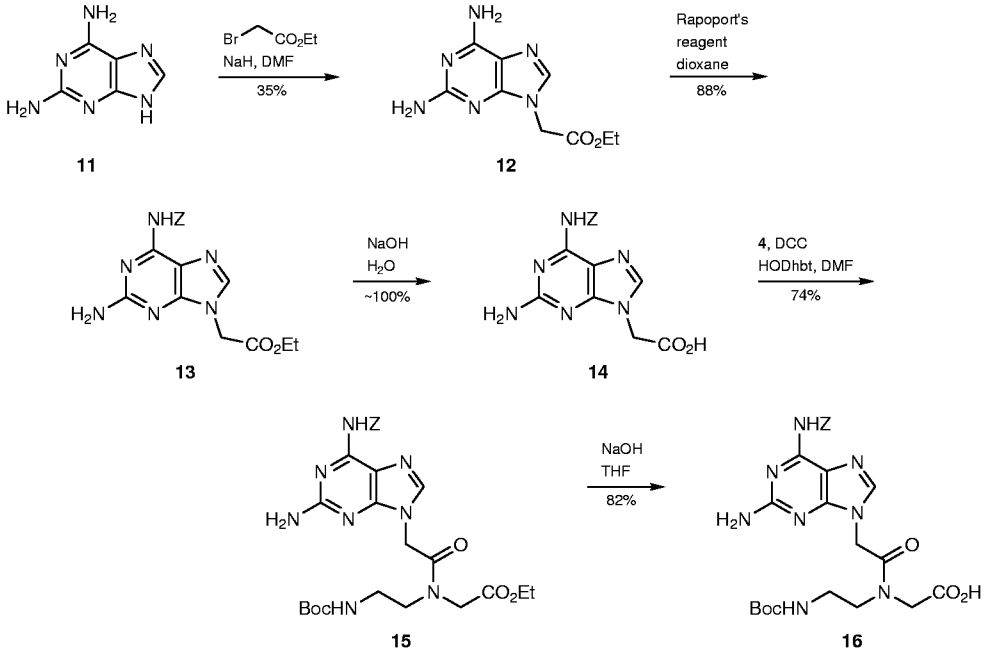
***N*-{[*N*$^2$-(Benzyloxycarbonyl)isocytosin-5-yl]acetyl}-*N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycine (10):**[25]

Ethyl *N*-{[*N*$^2$-(benzyloxycarbonyl)isocytosin-5-yl]acetyl}-*N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (**9**; 1.6 g, 3.0 mmol) was dissolved in MeOH (16 mL) by gentle heating. The soln was cooled to 0 °C and 2 M NaOH (23 mL) was added. The mixture was stirred at rt for 75 min and then cooled to 0 °C again. The pH was adjusted to 1.7 and the product was collected by filtration, washed once with $H_2O$, and dried; yield: 1.24 g (82%); FAB-MS *m/z*: 504.

## 12.6.1.3 Synthesis of the 2,6-Diamino-9*H*-purine Peptide Nucleic Acid Monomer

2,6-Diamino-9*H*-purine can form three hydrogen bonds with thymine as compared to only two for adenine. Therefore, PNA oligomers in which adenines have been replaced with 2,6-diamino-9*H*-purine bind with increased affinity and specificity to DNA and RNA.[26] The 2,6-diamino-9*H*-purine PNA monomer is therefore very useful for a variety of applications requiring strong hybridization. Once this monomer has been incorporated into a PNA oligomer, capping is omitted since the 2-amino group is unprotected. Synthesis of the 2,6-diamino-9*H*-purine monomer is described in Scheme 6.

**Scheme 6** Synthesis of the 2,6-Diamino-9*H*-purine Peptide Nucleic Acid Monomer[26]



**Ethyl (2,6-Diaminopurin-9(9*H*)-yl)acetate (12):**[26]
To a suspension of finely ground 2,6-diamino-9*H*-purine (**11**; 10.0 g, 66.6 mmol) in dry, degassed DMF (400 mL) 60% NaH in oil (2.93 g, 73 mmol) was added in portions. After 2 h, $BrCH_2CO_2Et$ (14.46 g, 86.6 mmol) was added and the reaction was stirred for a further 90 min. The mixture was then filtered through Celite and the clear orange soln was reduced in volume to 100 mL, then, abs EtOH (100 mL) was added. The soln, maintained at −5 °C overnight, produced light yellow crystals (10.25 g). Recrystallization (EtOH) gave **12**; yield: 5.50 g (35%); mp 168–169 °C; FAB-MS *m/z*: 237.1.

**Ethyl [2-Amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetate (13):**[26]
To a soln of ethyl (2,6-diaminopurin-9(9*H*)-yl)acetate (**12**; 7.00 g, 29.6 mmol) in dry, degassed dioxane (250 mL) was added freshly prepared *N*-benzyloxycarbonyl-*N'*-methylimidazolium triflate (Rapoport's reagent; 16.30 g, 44.5 mmol) and the reaction was stirred for 20 h. The dioxane was removed under reduced pressure and the yellow viscous oil was triturated with EtOAc/$Et_2O$ (1:4) until a solid product resulted. Crystallization (5% MeOH/EtOAc) afforded **13**; yield: 9.60 g (88%); mp 185–187 °C; high-resolution FAB-MS *m/z*: calcd. 371.1468; found 371.1468.

**[2-Amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetic Acid (14):**[26]
Ethyl [2-amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetate (**13**; 3.00 g, 8.1 mmol) was added to 2 M NaOH (30 mL). Dissolution occurred after 1 h. The reaction was then cooled to 0 °C and the product precipitated by acidifying to pH 2.5 with 2 M HCl. The solid product was filtered, washed with $H_2O$, and dried affording **14**; yield: 2.82 g (~100%); mp >260 °C; IR (KBr) $\tilde{\nu}$: 3330, 3095, 1750, 1630, 1590, 1410 $cm^{-1}$; high-resolution FAB-MS *m/z*: calcd. 343.1155; found 343.1158.

**Ethyl *N*-{[2-Amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetyl}-*N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (15):**[26]
A soln of [2-amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetic acid (**14**; 2.00 g, 5.84 mmol), HODhbt (1.05 g, 6.44 mmol), and DCC (1.33 g, 6.45 mmol) in dry DMF (25 mL) was stirred at rt for 1 h after which ethyl *N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (**4**; 1.50 g, 6.45 mmol) was added. The reaction was stirred for 16 h. DCU was removed by filtration and DMF removed under reduced pressure. $CH_2Cl_2$ (50 mL) was added and the organic layer was washed with sat. $NaHCO_3$ (3 ×), sat. $KHSO_4$ (2 ×), and finally brine. After drying ($MgSO_4$) and removal of the solvent, the resulting white foam was crystallized (EtOH/$H_2O$) to afford **15**; yield: 2.40 g (74%).

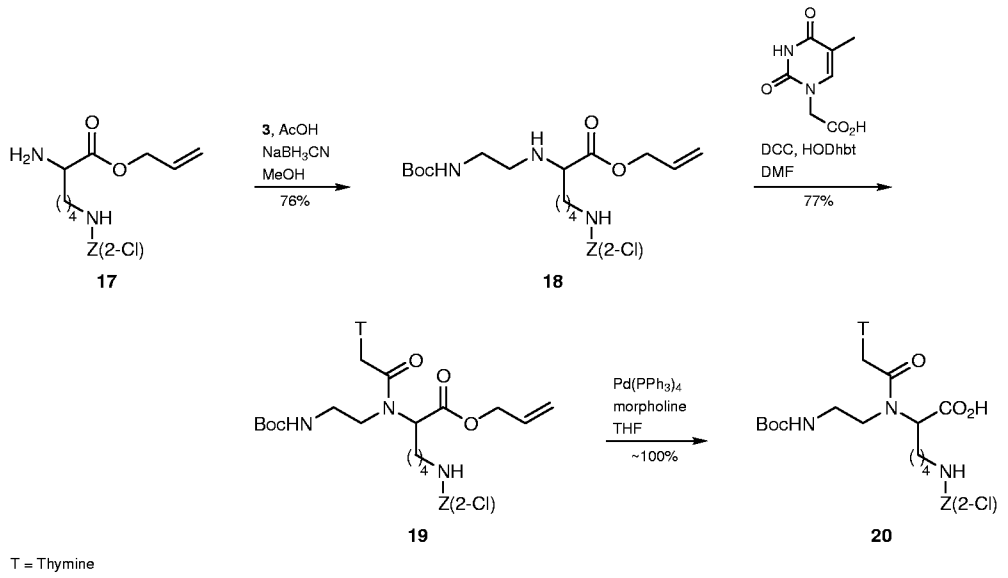
***N*-{[2-Amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetyl}-*N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycine (16):**[26]
A soln of ethyl *N*-{[2-amino-6-(benzyloxycarbonylamino)purin-9(9*H*)-yl]acetyl}-*N*-[2-(*tert*-butoxycarbonylamino)ethyl]glycinate (**15**; 2.0 g, 3.59 mmol) in 2 M NaOH (20 mL) and THF (15 mL) was stirred for 1 h, after which most of the THF was removed under reduced pressure and the aqueous layer cooled to 0 °C. The product was precipitated by adjusting the pH to 4 with 4 M HCl, yielding a solid, which was washed extensively with $H_2O$. Drying under high vacuum gave the title compound **16**; yield: 1.60 g (82%); high-resolution FAB-MS *m/z*: calcd. 543.2315; found 543.2313.

## 12.6.1.4 Synthesis of the D-Lysine Peptide Nucleic Acid Monomer

The Gly residue of the PNA backbone can be replaced with other α-amino acids without dramatically affecting the hybridization properties of the PNA oligomer.[27–29] This replacement allows the introduction of chemical functionalities in the PNA and also chirality into the backbone. One of the most useful of these α-amino acid derived analogues is the PNA monomer containing the D-Lys side chain, which confers increased solubility and binding affinity (Scheme 7).[28,29] The carboxylic acid group is protected as an allyl ester during synthesis to avoid any possible base-catalyzed racemization in the final step.[27] Preactivation of these monomers with HATU or HBTU in the presence of DIPEA leads to extensive racemization.[30] These monomers can be coupled with less than 1% racemization by DIC/HOBt after neutralization of the resin (5% DIPEA/$CH_2Cl_2$).

*for references see p 834*

**Scheme 7** Synthesis of the D-Lysine Peptide Nucleic Acid Monomer[28]




**Allyl 2-Amino-6-[(2-chlorobenzyloxycarbonyl)amino]hexanoate (17):**[28]
$Cs_2CO_3$ (3.30 g, 10.12 mmol) was added to a stirred soln of Boc-D-Lys[Z(2-Cl)]-OH (4.00 g, 9.64 mmol) in DMF (35 mL). After 15 min, allyl bromide (1.53 g, 12.65 mmol) was added and the mixture was stirred overnight and then filtered through Celite. The filtrate was concentrated, and the residue was dried under reduced pressure. Sat. $NaHCO_3$ (100 mL) was added, and the mixture was extracted with EtOAc (200 mL). The organic phase was washed with brine (2 × 100 mL), dried ($Na_2SO_4$), and concentrated under reduced pressure to give Boc-D-Lys[Z(2-Cl)]-OAl; yield: 4.40 g (100%); $R_f$ 0.50 (EtOAc/hexane 1:1). Then, TFA (7.4 mL) was added at 0 °C to a soln of Boc-D-Lys[Z(2-Cl)]-OAl (4.39 g, 9.65 mmol) in $CH_2Cl_2$ (7.4 mL). The soln was stirred at rt for 60 min and then neutralized with sat. $NaHCO_3$ (150 mL). The mixture was extracted with $CH_2Cl_2$ (200 mL) and EtOAc (200 mL). The combined extracts were dried ($Na_2SO_4$), and the solvent removed to produce a clear oil; yield: 3.27 g (96%); $R_f$ 0.44 ($CH_2Cl_2$/MeOH 9:1).

**Allyl 2-{[2-(*tert*-Butoxycarbonylamino)ethyl]amino}-6-[(2-chlorobenzyloxycarbonyl)amino]hexanoate (18):**[28]
A soln of **17** (3.20 g, 9.01 mmol) and (*tert*-butyloxycarbonylamino)acetaldehyde (**3**; 1.44 g, 9.01 mmol) in MeOH (60 mL) was stirred at 0 °C for 5 min. $NaBH_3CN$ (340 mg, 5.41 mmol) followed by AcOH (0.62 mL, 10.81 mmol) were added and the soln was stirred at rt for 1 h. The soln was concentrated under reduced pressure, sat. $NaHCO_3$ (50 mL) was added, and the mixture extracted with EtOAc (200 mL). The organic phase was washed with brine (50 mL), dried ($Na_2SO_4$), and the solvent was removed under reduced pressure. The product was further purified by flash chromatography (EtOAc/hexane 3:1) to give an oil; yield: 3.39 g (76%); $R_f$ 0.26 (EtOAc/hexane 3:1).

**Allyl 2-{[2-(*tert*-Butoxycarbonylamino)ethyl][(thymin-1-yl)acetyl]amino}-6-[(2-chlorobenzyloxycarbonyl)amino]hexanoate (19):**[28]
DCC (2.22 g, 10.80 mmol) was added to a stirred soln of thymin-1-ylacetic acid (1.99 g, 10.80 mmol) and HODhbt (1.76 g, 10.80 mmol) in dry DMF (23 mL) at rt. Stirring was continued at rt for 45 min. Compound **18** (3.36 g, 6.75 mmol), dissolved in dry DMF (23 mL), was added and the soln was stirred overnight. DCU was removed by filtration, and the DMF was removed in vacuo. The remaining oil was dissolved in EtOAc (200 mL), and the organic phase was washed with half-sat. $NaHCO_3$ (50 mL) and brine (2 × 50 mL). The organic phase was dried ($Na_2SO_4$) and concentrated under reduced pressure to give an oil (5.44 g), which was purified twice by flash chromatography (EtOAc/hexane 5:1 and 10% MeOH/$CH_2Cl_2$); yield: 3.64 g (77%); $R_f$ 0.29 (EtOAc).

**2-{[2-(*tert*-Butoxycarbonylamino)ethyl][(thymin-1-yl)acetyl]amino}-6-[(2-chlorobenzyloxycarbonyl)-amino]hexanoic Acid (20):**[28]
$Pd(PPh_3)_4$ (59 mg, 0.051 mmol) was added to a soln of **19** (3.41 g, 5.13 mmol) and morpholine (4.44 mL, 50.9 mmol) in THF (51 mL). The soln was stirred at rt for 45 min and then concentrated under reduced

pressure. The resulting white foam (4.27 g) was dissolved in EtOAc (200 mL) and extracted with 10% citric acid (2 × 100 mL). The combined $H_2O$ phases were extracted with EtOAc (200 mL). The combined organic phases (400 mL) were washed with brine (50 mL) and dried ($Na_2SO_4$). The solvent was removed under reduced pressure to give a foam which was dissolved in EtOAc (30 mL) and then slowly added to hexane (300 mL) under stirring at 0 °C; yield: 3.06 g (~100%); high-resolution FAB-MS *m/z*: calcd. 624.2436; found 624.2422; $[\alpha]_D^{24}$ 19.1 (*c* 3, MeOH).

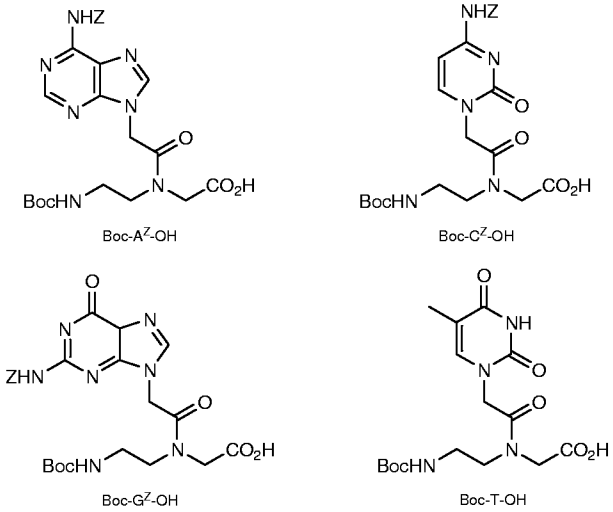
## 12.6.2 Peptide Nucleic Acid Oligomers

PNA oligomers have been synthesized by either Boc[31–33] or Fmoc[34] methods on polymer beads. The Fmoc method has also been used in array synthesis of PNAs.[8,35] The synthesis of PNA/DNA chimera requires base-labile protection groups on the PNA nucleobases and will not be discussed herein.[8,36–38]

### 12.6.2.1 *tert*-Butoxycarbonyl/Benzyloxycarbonyl Method

The four standard protected PNA monomers (A = adenine; C = cytosine; G = guanine; and T = thymine) are shown in Scheme 8.

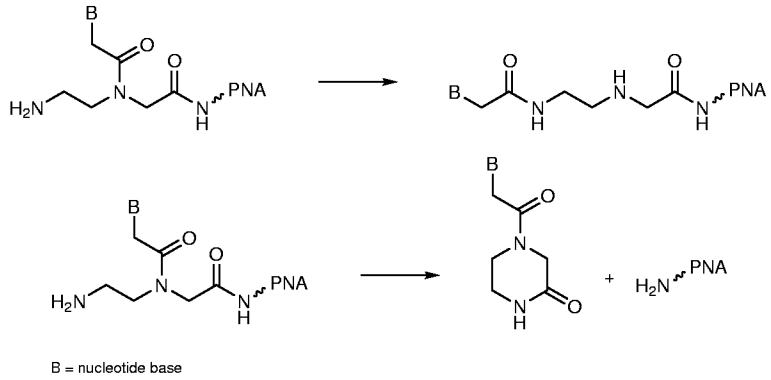
**Scheme 8** The Four *tert*-Butoxycarbonyl/Benzyloxycarbonyl-Protected Peptide Nucleic Acid Monomers




The synthesis of PNA oligomers is essentially similar to standard Boc peptide chemistry with a few exceptions. The protocol has been optimized using an automated synthesizer.[32,33] A polystyrene based MBHA resin is used to produce the PNA as a C-terminal amide (a Lys can be coupled as the first residue to increase the water solubility of the PNA). In the first step the Boc group is cleaved with TFA (the deprotection is complete within 3 min). A scavenger (*m*-cresol) is added during cleavage to protect the nucleobases from alkylation. Originally, the resin was neutralized with 5% DIPEA/$CH_2Cl_2$ prior to the addition of the monomers as Pfp esters or using HOBt/DIC. However, two side reactions can take place in PNA synthesis (Scheme 9) if the amino group is left unprotected.

*for references see p 834*

**Scheme 9** Possible Side Reactions in Peptide Nucleic Acid Synthesis

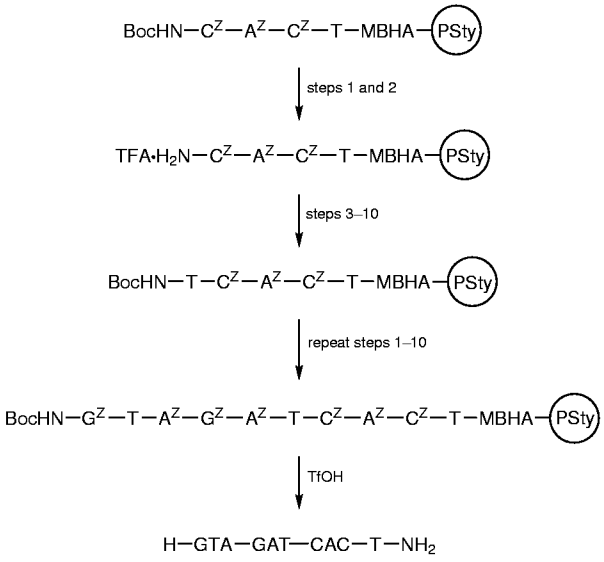



B = nucleotide base

It was found that neutralization of the resin in situ (using 2 equiv of DIPEA) in combination with HBTU or HATU gives the highest coupling yields (the concentration of the activated monomers was 0.08 M). In addition, exploitation of solvent mixtures (DMF/$CH_2Cl_2$ or DMF/pyridine) gives higher coupling yields. It is important to use a small excess of PNA monomer compared to HATU to avoid guanidinium formation. A capping step is included to ease purification. In addition, it is very important to use only a small amount of acetic anhydride and to wash the resin carefully after the capping step. Since acetic anhydride can acylate guanine, a DMF/piperidine treatment is included to reverse this base modification.

A procedure for Boc-PNA manual synthesis is given below. If the coupling is incomplete (Kaiser test) double coupling is performed. The synthesis cycle is shown in Scheme 10 and the procedure is given in Table 1.

**Scheme 10** Peptide Nucleic Acid Synthesis by the *tert*-Butoxycarbonyl/Benzyloxycarbonyl Method[32,33]




G = guanine; A = adenine; C = cytosine; T = thymine

**Table 1** Peptide Nucleic Acid Synthesis Protocol[32,33]

| Step | Time (min) |
|---|---|
| TFA/*m*-cresol (95:5; 2 ×) | 3 |
| $CH_2Cl_2$ (5 ×), DMF (5 ×) | 1 |
| Preactivate | 1 |
| Coupling | 16 |
| DMF (5 ×) | 1 |
| Kaiser test | 1 |
| Capping | 1 |
| DMF (5 ×), $CH_2Cl_2$ (5 ×) | 1 |
| DMF/piperidine (9:1) | 1 |
| DMF (5 ×), $CH_2Cl_2$ (5 ×) | 1 |

Reagents: Stock solns of 0.26 M in NMP of each monomer are prepared and kept at 4 °C. Gentle heating (55 °C) or shaking overnight is necessary to dissolve Boc-$G^Z$-OH. DMF and NMP (HPLC grade) are stored in the dark over 4-Å sieves to ensure that they are free of amines.

**Boc-T-MBHA-PSty:**[32,33]
MBHA-PSty (300 mg) was swelled in $CH_2Cl_2$ overnight. The resin was neutralized with 5% DIPEA/$CH_2Cl_2$ (3 min) and then washed with $CH_2Cl_2$ (10 ×). Boc-T-OH (23 mg, 60 μmol) was dissolved in NMP. DIPEA (19.5 μL, 112 μmol) was added. Then, this soln was added to HATU (19 mg, 50 μmol) suspended in pyridine (100 μL) in an Eppendorf tube. The tube was vortexed (1 min) and the clear yellow soln was added to the resin. The glass reactor was stirred on an orbital shaker for 2 h and the coupling mixture was then drained out. The resin was washed with DMF (5 ×) and $CH_2Cl_2$ (5 ×). Unreacted amino groups were capped ($Ac_2O$/pyridine/NMP 1:2:2, 1.5 mL; 10 min). The resin was washed (5 × DMF, 5 × $CH_2Cl_2$) and then dried in a desiccator. By coupling Fmoc-Gly-OH on a small portion (30 mg) of this resin, the loading was determined to be 0.14 mmol · $g^{-1}$.

**H-GTA-GAT-CAC-T-$NH_2$:**[32,33]

**CAUTION:** *Both TFA and* m-*cresol are very poisonous. Initial experiments suggest that* m-*cresol can be substituted by anisole during the repetitive Boc cleavage procedure.*

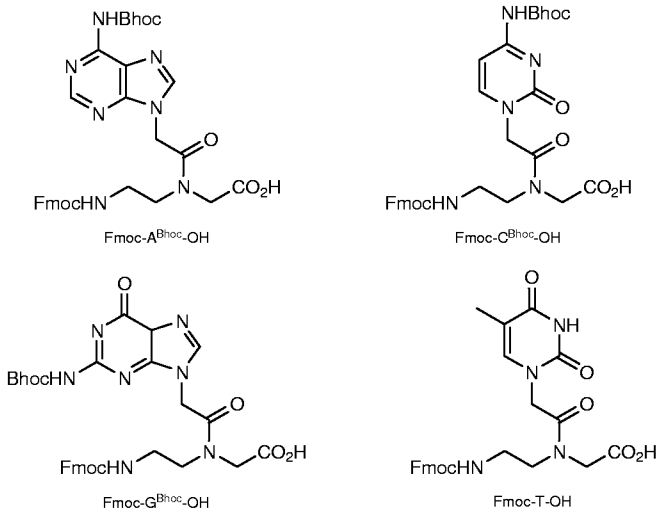
Boc-T-MBHA-PSty (50 mg, loading approx. 0.14 mmol · $g^{-1}$, 7 μmol) was swelled in $CH_2Cl_2$ in a small glass reactor overnight. TFA/*m*-cresol (95:5) was added (3 × 3 min). The resin was washed with $CH_2Cl_2$ (5 ×) and DMF (5 ×). Meanwhile, an activated soln of Boc-$C^Z$-OH was prepared: HATU (19 mg, 50 μmol) was suspended in pyridine (100 μL) in an Eppendorf tube. A Boc-$C^Z$-OH soln (200 μL, 52 μmol) was added and then DIPEA (19.5 μL, 112 μmol). The tube was vortexed (1 min) and the clear yellow soln was added to the resin. The glass reactor was shaken on an orbital shaker for 16 min and the coupling mixture was then drained out. The resin was washed with DMF (5 ×) and a small amount of beads were removed for Kaiser test. This test produced a yellow color (i.e., complete coupling). Any unreacted amino groups were capped with capping soln (1 mL × 1 min). The resin was washed (5 × $CH_2Cl_2$, 5 × DMF), treated with DMF/piperidine (9:1; 1 min), and then washed again (5 × $CH_2Cl_2$, 5 × DMF). The cycle (using the appropriate nucleobase monomers) was carried out 9 times until the synthesis was complete. The last Boc group was left on. The resin was washed with DMF and $CH_2Cl_2$, and dried 30 min in a desiccator. The resin was treated with TFA/*m*-cresol (3 min) and the PNA was deprotected and cleaved from the resin with TFA/TfOH/*m*-cresol/PhSMe (300 μL:100 μL:50 μL:50 μL; 90 min). The resin was washed with TFA (2 × 300 μL). The cleavage mixture and TFA were combined in a centrifuge tube and the PNA was then precipitated by addition of $Et_2O$ (10 mL). After cooling at −78 °C (15 min) the tube was centrifuged and the $Et_2O$ phase was decanted off. This procedure was repeated three times to produce a white pellet that was dried under reduced pressure. Purity by analytical HPLC (260 nm): 84%. This value corresponds to an average coupling yield of 98%. MALDI-MS *m/z*: calcd. 2725; found 2723.

### 12.6.2.2 9-Fluorenylmethoxycarbonyl/Benzhydryloxycarbonyl Method

The Fmoc synthesis of PNAs was introduced by Thompson et al.[34] and has further largely been developed by Nielsen and Egholm.[8] The synthesis was performed on a DNA synthesizer. The four monomers are shown in Scheme 11. The nucleobases are protected with the benzhydryloxycarbonyl (Bhoc) group. This protecting group is cleaved with TFA in less than a minute.

*for references see p 834*

**Scheme 11** The Four 9-Fluorenylmethoxycarbonyl/Benzhydryloxycarbonyl Protected Peptide Nucleic Acid Monomers




The synthesis cycle is in principle the same as for Boc chemistry, except that the temporary protecting group is cleaved with 20% piperidine (2 min) instead of TFA, and that the DMF/piperidine (9:1) treatment is unnecessary. The synthesis is carried out on a PEG-PS resin (purchased as Fmoc-PAL-PEG-PS or Fmoc-XAL-PEG-PS).

There has only been one direct comparison of the two [Boc/Z (Section 12.6.2.1) and Fmoc/Bhoc] methods:[39] The same resin (PEG-PS) and the same synthesizer have been used (the handle was BHA for the Boc method and PAL for the Fmoc method). The sequence was H-$T_{12}$-Lys-$NH_2$, and the Boc/Z method out-performed the Fmoc/Bhoc method. The Boc/Z method was then further optimized by capping with Rapoport's reagent (10% in NMP, 1 min) instead of acetic anhydride. Using this method a model PNA 17-mer, H-CGG-ACT-AAG-TCC-ATT-GC-Gly-$NH_2$ with a crude purity of >90% (i.e., average coupling yield >99.5%), was synthesized. The oligomer was cleaved from the resin with TfOH/TFA/*m*-cresol (1:8:1). It is interesting to note that the PEG-PS resin, as opposed to Tentagel, is not degraded by 10% TfOH.

## 12.6.3 Peptide Nucleic Acid–Peptide Conjugates

PNA peptide conjugates have been prepared for radioactive labeling of the peptide segment (e.g., the "Kemptide").[40] Peptides have also been attached to PNAs in order to transport the PNAs into cells.[20,41–43] A special issue is the attachment of Cys to PNAs when using Fmoc oligomerization. In the Fmoc strategy, use of Cys(Trt) should be avoided since the thiol group will scavenge the benzhydryloxycarbonyl protecting group (Bhoc) during TFA cleavage.[44] Instead Boc-Cys(StBu) proves to be useful.

**H-CysGGGTT-$NH_2$:**[44]
H-GGGTT-PAL-PEG-PS was prepared as described in Section 12.6.2.2 using the Fmoc/Bhoc method. Boc-Cys(StBu) (4 equiv), HATU (4 equiv), DIPEA (4 equiv), and 2,4,6-collidine (4 equiv) were pre-activated in NMP (0.1 M) and then added to the resin. After 20 min the resin was drained and washed with NMP/$CH_2Cl_2$ (1:1; 3 ×). The resin was treated with TFA/*m*-cresol (4:1) for 2 h and the crude oligomer was precipitated by addition of $Et_2O$. Reduction of the disulfide bond was carried out by dissolving the PNA (0.5 mg) in Tris·HCl buffer (200 µL, 5 mM, pH 7.5) containing DTT (20 mM) and the soln was left overnight at rt. MALDI-MS *m/z*: 1527.5.

## 12.6.4 Labeled Peptide Nucleic Acid Oligomers

For many applications of PNAs (e.g., in hybridization assays or to study cellular uptake) it is necessary to attach a label to the PNA.[8] This can be done with a radioactive or a non-radioactive label.

### 12.6.4.1 Radioactive Labeling

PNAs have been radioactive labeled by $^{32}P$ and $^{14}C$.

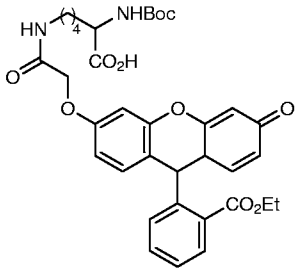
Koch et al.[40] developed a method in which a peptide sequence (H-Leu-Arg-Arg-Ala-Ser-Leu-Gly-OH, the "Kemptide") was synthesized either at the N-terminal or the C-terminal residue of the PNA. This peptide is a substrate for protein kinase A, which phosphorylates the critical Ser residue. This phosphorylation was carried out using [$\gamma$-$^{32}P$]ATP. Kozlov et al.[45] made 2′-deoxycytidine-3′-phosphoroimidazolide to react with the N-terminus of a PNA in solution. T4 polynucleotide kinase then phosphorylated the 5′-hydroxy group of the nucleotide by [$\gamma$-$^{32}P$]ATP.

Gray et al.[46] radiolabeled a PNA using [$^{14}C$]formaldehyde in a reductive formylation reaction to specifically methylate the exocyclic amino groups of the nucleotide bases.

### 12.6.4.2 Non-radioactive Labeling

The N-terminal amino group or a Lys residue can easily be used to label the PNA. This can be carried out in solution after purification, or more conveniently, while the PNA is still attached to the resin. Fluorescein and rhodamine are the most common fluorophore labels, while biotin is the most common affinity tag.[8] Fluorescein or rhodamine are usually coupled to the amino group as -OSu esters or isothiocyanates. 4,4′-Dimethoxytrityl-protected biotin and Piv-protected fluorescein have also been coupled to the N-terminus of a PNA as their 1-phenylpyrazolin-5-one carboxylate esters.[47] In 1997 a new fluorescein-conjugated Lys monomer (**21**, Scheme 12) was described.[48]

**Scheme 12** Fluorescein Monomer[48]




**21**

It was possible to incorporate this monomer into PNA oligomers to produce fluorescent products. In contrast to fluorescein, the photophysical properties of this monomer were independent from pH 3 to 8. Incorporating the normally double negatively charged fluorescein with only one O-linker moiety between the chromophore and the PNA resulted in lowering the melting temperature by 12 °C.[49] It is therefore recommended to use at least two O-linkers (in this case the O-linker is 8-amino-3,6-dioxaoctanoic acid) between the PNA and the label.[8] Rhodamine labeling to a PNA attached to a PEG-PS support was recently shown to cause problems.[50] Such a labeling required a large excess of carboxytetramethylrhodamine OSu ester probably because the resin absorbed the fluorophore. However, addition of sulfo-rhodamine to the resin prior to the labeling gave good yields of labeled PNA.

*for references see p 834*

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